Thermal dimensions in the long-range Ising model

Analytic coefficients through fifth order and
numerical sixth- to eighth-order coefficients

Research calculation record

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Abstract

We determine the thermal scaling dimension of the long-range Ising model in the expansion parameter $\epsilon = 2s - d$. The analytic result through $C_5(d)$ is expressed in an explicitly defined master basis, with its graph weights, subtraction prescription, regulator expansions and finite special-function representations derived below. At $d = 1$, the result uses only fixed Gamma/polygamma values and ordinary generalized hypergeometric values; the proof of the $\mathcal{K}$ reduction, the logarithmic-wheel reduction and the five-value evaluation of $P_5(1)$ are included in full. At $d = 2$ the primitive periods factor in the same scalar values, and the nonprimitive masters $\mathcal{K}_2$, $\mathcal{L}_2$ and Z_2 are evaluated in closed form, so that $C_5(2)$ is an explicit expression in those values, $\Gamma(\frac{1}{4})$, π, $\log 2$, Catalan's constant and $\zeta(3)$. At $d = 3$ the remaining rank-two structure is identified without claiming an unavailable scalar summation. For generic $0 < d < 4$ the masters $\mathcal{L}_d$ and Z_d are reduced to at most three samples of finitely many hypergeometric functions with at most two summation indices at unit argument and their first and second derivatives in the rim exponent, using a proof that the top Fourier modes of products of balanced kernels vanish; $\mathcal{K}_d$ is reduced, under stated analytic inputs, to one Laurent coefficient of such a sample, and, by a second route, to a polygamma part plus one two-fold Mellin–Barnes integral. A finite representation of a reduced analytic expression in nine special-function values gives $C_6(1) = 14331.68703275805$; exact sums over all six-loop graphs at increasing quadrature degree, which do not use that expression, give $14331.6870328(5)$, in agreement to 1.0×10^{-8}. Numerical finite-forest calculations give $C_6(2) = 362.366 \pm 0.004$, $C_6(3) = 33.5754 \pm 0.0005$, $C_7(1) = -322550 \pm 41$, $C_7(2) = -3876.1 \pm 2.1$, $C_7(3) = -220.35 \pm 0.13$, $C_8(1) = (8.060 \pm 0.017) \times 10^6$, $C_8(2) = 45990 \pm 33$ and $C_8(3) = 1602.6 \pm 3.9$. The integration kernels, regulator-dependent fixed-point extraction, local subtractions, sampling designs, covariance propagation and convergence checks supporting these estimates are retained. Numerical allowances are empirical, not rigorous intervals or bounds on truncating the epsilon expansion. Part V records a second observable built from the same graphs and master functions, the bulk one-point coefficient a_{ϕ^2} of the auxiliary bulk field through ϵ^6 at general d, from a calculation supplied by the author whose checks are internal to this framework. This note records one common graph/renormalization calculation; it is not an independent field-theoretic certification of its higher-loop results.

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Part I

Conclusions, conventions and renormalization

1 Scope and current coefficients

The model is the long-range Ising model: spins $\sigma_i = \pm 1$ on the sites of $\mathbb{Z}^d$ with ferromagnetic energy

$$\mathcal{H}_{\rm lat} = -\sum_{i<j} J(|i-j|)\,\sigma_i\sigma_j, \qquad J(r) \sim r^{-(d+s)},$$

in the critical long-range regime above $s = d/2$. Its continuum description is a Gaussian field with covariance proportional to $|p|^{-s}$ in momentum space and to $|x|^{-(d-s)}$ in position space, normalized to coefficient one, with a quartic interaction of engineering dimension $2s - d = \epsilon$. The observable is the scaling dimension of the thermal operator $\mathcal{E} = [\phi^2]$:

$$\epsilon = 2s - d, \qquad \Delta_\phi = \frac{d-\epsilon}{4}, \qquad \Delta_{\mathcal{E}}(d,\epsilon) = \sum_{n\geq 0} C_n(d)\epsilon^n. \tag{1}$$

The general-dimensional long-range regime is fixed $0 < d < 4$ and sufficiently small positive ϵ; noninteger dimensions and intermediate meromorphic integrals have the continuation prescriptions stated in the relevant sections. This is a local expansion near $s = d/2$, not an unresummed approximation over the entire interacting regime.

Order	$d = 1$	$d = 2$	$d = 3$
C_0	$1/2$	1	$3/2$
C_1	$-1/6$	$-1/6$	$-1/6$
C_2	0.657131263981	0.308065413582	0.181221785406
C_3	-3.719856322902	-0.874850012653	-0.353581520433
C_4	45.42488010356	5.048868431	1.228063545
C_5	-731.13947662454	-38.833645896	-5.850351421
C_6	14331.68703275805	362.366 ± 0.004	33.5754 ± 0.0005
C_7	-322550 ± 41	-3876.1 ± 2.1	-220.35 ± 0.13
C_8	$(8.060 \pm 0.017)10^6$	45990 ± 33	1602.6 ± 3.9

The first six rows use the special-function evaluations, not the noisier lower-order checks produced during high-loop integration; their closed forms are collected in Section 1.1. The entry for $C_6(1)$ is the finite special-function representation of Section 1.2. It evaluates a reduced analytic expression that is taken as an input and is not derived in this note (Section 1.3). The independent deterministic sum over all graphs at quadrature degree 22 (Section 22) gives 14331.6870328(5). It has no sampling error; its uncertainty is the empirical convergence-and-rounding allowance of Section 22.7, and the representation lies 1.0×10^{-8} below its unrounded value 14331.68703276838. For the remaining entries of the last three rows, the allowances above and the sampling standard errors below are *different* quantities:

Coefficient	Underlying estimator	Sampling standard error
$C_6(2)$	362.36647176375726	0.0007997307
$C_6(3)$	33.57542664131543	0.0001185495235
$C_7(1)$	−322549.8888129889	10.08791295
$C_7(2)$	−3876.1260927949925	0.5244496691
$C_7(3)$	−220.35448047392794	0.0318188006
$C_8(1)$	8059766.723129955	4209.576151
$C_8(2)$	45990.00263614446	8.216820958
$C_8(3)$	1602.6052229663408	0.9516489064

Each allowance is the output of a stated rule applied to the sampling standard error and, where they exist, to measured convergence steps. The table gives the rule for each entry and when it was fixed: before the production run, during it but before its result was known, or after the result. "Four SE" means four sampling standard errors; a half-split estimate, called a half-stream estimate in Sections 28 and 31, uses only the first or only the second half of the replicates of every random stream.

Entry	Allowance	Rule	Fixed
$C_6(1)$, graph sum	$\pm 4.9 \cdot 10^{-7}$	the last change with quadrature degree, plus the difference of the two cutoffs, a floating-point allowance and the lower-loop convergence (Section 22.7)	before production
$C_6(2)$	±0.004	four SE, rounded up to a multiple of 0.001 (Section 26.7)	before the result was known; rounding step refined from 0.01 afterwards
$C_6(3)$	±0.0005	four SE, rounded up to a multiple of 0.0001 (Section 26.8)	before production; rounding step refined from 0.001 afterwards
$C_7(1)$	±41	the largest of four SE, twice the last refinement step plus four of its SE, twice the degree six to seven check step plus four of its SE, and the half-split displacement, rounded up to two significant figures (Section 27.8)	before production
$C_7(2)$	±2.1	the larger of four SE and the largest half-split deviation, rounded up to a multiple of 0.1 (Section 28)	after the result (the rule of an earlier, lower-precision $C_7(2)$ calculation)
$C_7(3)$	±0.13	four SE, rounded up to a multiple of 0.01 (Section 29)	before the result was known
$C_8(1)$	±17000	as for $C_7(1)$, with the degree eight to nine check step, and with twice the last degree step of the exact loop-five input as a fifth candidate (Section 30.1)	before production; batch count raised afterwards
$C_8(2)$	±33	as for $C_7(2)$, rounded up to a multiple of 1 (Section 31)	after the result
$C_8(3)$	±3.9	four SE, rounded up to a multiple of 0.1 (Section 32)	after the result

For $C_7(1)$ the four candidates are 40.35, 39.50, 22.49 and 12.58, so the allowance is four sampling standard errors. The radial quadrature of $C_7(1)$ stops at degree six, and the rule adds no model of the truncation beyond it. Adding the measured degree six to seven step and a geometric tail at the last step ratio gives the post hoc estimate $C_7(1) \approx -322524 \pm 38$ (753); its central value lies inside the quoted interval. That tail model agrees with the lower-order checks at fifth and sixth

order but is not validated at seventh order, where the step ratio is larger. The result is therefore $C_7(1) = -322550 \pm 41$, and the truncation-aware value is a check (Section 27.9). For $C_8(1)$ the rule and the design were fixed before production, but the number of production batches was not. Under the stopping rule set first, nine batches give 8061480 ± 5300 with allowance ± 22000; after interim results of five to eight batches had been seen, all fourteen planned batches were completed (Section 30.3). At degree eight the radial quadrature is converged well inside the allowance. For $C_8(2)$ and $C_8(3)$ four sampling standard errors exceed the distances of both half-split estimates from the full estimate. The coverage of the allowance on $C_8(3)$ is conditional on a finite variance, for which Section 32.5 gives empirical evidence, not a bound.

Two eighth-order results are checked by a separately developed code. At $d = 2$ it shares the method and the per-graph input files of loops three to seven and gives $C_8(2) = 46006.1 \pm 13.4$ (sampling standard error), $+1.03$ combined standard errors from the estimator (Section 31.7). At $d = 3$ it shares the method and the seventh-order integrator and gives $C_8(3) = 1600.98 \pm 3.97$, -0.40 combined standard errors from the estimator (Section 32.7). The additional estimator digits are not certified physical digits. Numerical errors do not cover shared mistakes in the graph/renormalization framework, and are not bounds on the omitted orders in (1).

Part V (Section 34) adds a second observable built from the same graphs and master functions: the bulk one-point coefficient a_{ϕ^2} of the composite $:\Phi^2:$ of the auxiliary bulk field, through ϵ^6 at fixed $0 < d < 4$, which is the horizontal coordinate of the bootstrap bounds of [8]. It comes from a calculation supplied by the author with its own code. Its checks are internal to the master framework of this note, together with comparisons against this note's own inputs (Section 34.12); Section 34.1 states what it establishes.

1.1 Closed forms through fifth order

The entries C_0 to C_5 of the table are values of exact expressions, not statistical estimates. This subsection states the expressions together with every definition they need; each formula below repeats one derived later in the note, and symbols with more than one meaning in the note are listed in Section 1.4. At $d = 1$ every step is proved in this note (Part II). At $d = 2$ the closed form of $C_5(2)$ is proved in this note. At $d = 3$ the inputs of $C_5(3)$, namely Q_3, $P_4(3)$, $P_5(3)$, $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$, are evaluated numerically. The values of Q_3, $P_4(3)$ and $P_5(3)$ from their fixed formulas agree to 10^{-74} or better between evaluations at 90 and 100 working digits. Independent evaluations confirm Q_3 and $P_4(3)$ to about 10^{-79} and $2 \cdot 10^{-86}$ (a 100-digit spectral evaluation), $P_5(3)$ to about $2 \cdot 10^{-53}$ (a 60-digit integration of the Gamma kernels), and $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ to $1.2 \cdot 10^{-93}$ or better. Propagated into $C_5(3)$ these differences are below 10^{-54}, well below one unit in the last of the 50 digits printed for it. These agreements are not interval bounds. The spectral formula for $\mathcal{K}_d$ below is proved at $d = 2$ in Section 18.9; at other d its derivation uses input (M) of Section 21.1 and the extraction of $[\tau^2]$ under the trace, which is proved only at $d = 2$ (Section 21.12). The value of $\mathcal{K}_3$ does not depend on (M): the two-fold Barnes representation (588), derived without it, reproduces it to $4 \cdot 10^{-100}$. Section 1.3 lists what is proved, what is conditional and what is numerical.

Any dimension. With $q = d/4$, the polygamma functions $\psi_m(z) = d^{m+1} \log \Gamma(z)/dz^{m+1}$ and the combinations

$$A = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), \quad R = \psi_1(2q) - \psi_1(1),$$
$$U = \psi_2(q), \qquad\qquad V = \psi_2(1) + \psi_2(2q), \quad X = \psi_3(2q) - \psi_3(1).$$

the coefficients are

$$\boxed{\begin{aligned} &C_0(d)=\frac{d}{2}, && C_1(d)=-\frac{1}{6},\quad C_2(d)=\frac{A}{9},\\ &C_3(d)=\frac{A^2+R-8Q}{108},\\ &648C_4(d)=16P_4-A^3-8AQ-3AR+\frac{53}{2}U-3V. \end{aligned}}$$

$$\boxed{\begin{aligned} 46656C_5(d)=\;&-1536(P_5+\mathcal{K}_d)+384AP_4-576\mathcal{L}_d-1728Q^2\\ &+720A^2Q+1008QR+7152Z_d\\ &-15A^4-90A^2R+414AU-108AV-117R^2-75X. \end{aligned}}$$

Here Q_d, $P_4(d)$ and $P_5(d)$, written Q, P_4 and P_5 in the boxed formulas, are the primitive (zig-zag) periods with three, four and five loops, normalized by h_d^{L-1} with $h_d=\Gamma(d/2)/\pi^{d/2}$, and $\mathcal{K}_d$, $\mathcal{L}_d$ and Z_d are nonprimitive master data. All of them are spectral integrals. With

$$\begin{aligned} &d_0(d)=1, && d_\ell(d)=\frac{(2\ell+d-2)\Gamma(\ell+d-2)}{\Gamma(d-1)\Gamma(\ell+1)}\quad(\ell\geq 1),\\ &a_\ell=\frac{\ell}{2}+\frac{q}{2}, && b_\ell=a_\ell+q,\\ &t_\ell(\nu)=\Gamma(2q)\frac{\Gamma(a_\ell+\mathrm{i}\nu)\Gamma(a_\ell-\mathrm{i}\nu)}{\Gamma(b_\ell+\mathrm{i}\nu)\Gamma(b_\ell-\mathrm{i}\nu)}. \end{aligned}$$

and the trace

$$\langle f\rangle_d=\frac{1}{\pi}\sum_{\ell=0}^{\infty}d_\ell(d)\int_0^\infty f(\ell,\nu)\,d\nu.$$

the four-loop and three-loop periods are

$$3Q_d=\langle t_\ell^3\rangle_d,\qquad P_4(d)=\langle t_\ell^4\rangle_d.$$

and, with

$$\begin{aligned} w_1&=-\psi_0(q)+\frac{1}{2}\Re\{\psi_0(a_\ell+\mathrm{i}\nu)+\psi_0(b_\ell+\mathrm{i}\nu)\},\\ w_2&=\frac{1}{8}\Re\{\psi_1(a_\ell+\mathrm{i}\nu)-\psi_1(b_\ell+\mathrm{i}\nu)\}. \end{aligned}$$

the two wheel master data are

$$\boxed{\mathcal{L}_d=\langle 4t_\ell^4 w_1\rangle_d,\qquad Z_d=\left\langle t_\ell^3\left(w_1^2+\frac{w_2}{3}\right)\right\rangle_d.}$$

For $\mathcal{K}_d$ put $c_*=\psi_0(1)+\psi_0(2q)$ and, at $a=a_\ell$ and $b=b_\ell$,

$$\begin{aligned} p&=\Re\{\psi_0(a+\mathrm{i}\nu)+\psi_0(b+\mathrm{i}\nu)\},\quad h=\Im\{\psi_0(a+\mathrm{i}\nu)-\psi_0(b+\mathrm{i}\nu)\},\\ j&=\Re\{\psi_1(a+\mathrm{i}\nu)-\psi_1(b+\mathrm{i}\nu)\},\quad k=\Im\{\psi_1(a+\mathrm{i}\nu)+\psi_1(b+\mathrm{i}\nu)\},\\ r&=\Re\{\psi_2(a+\mathrm{i}\nu)+\psi_2(b+\mathrm{i}\nu)\},\quad E_2=-\frac{h^2}{2}-\frac{5j}{8}. \end{aligned}$$

Then

$$\boxed{\mathcal{K}_d=-\frac{AU}{4}+\frac{1}{4}\left\langle t_\ell^2\left[(c_*-p)E_2-\frac{hk}{2}-\frac{5r}{32}\right]\right\rangle_d.}$$

The master $\mathcal{K}_d$ is defined as the finite second-order coefficient of a subtracted two-loop master graph, and the spectral formula above is its evaluation. The two-fold Mellin–Barnes integral (588), derived in position space without that formula (Section 21.10), agrees to about 10^{-100} with the

proved values at $d = 1$ (76) and $d = 2$ (271) and with the residue series at $d = 3$ (Section 21.11). The trace forms converge directly for $\frac{2}{3} < d < 4$ (Q_d, Z_d), for $d > \frac{1}{2}$ (P_4, $\mathcal{L}_d$) and for $0 < d < 4$ ($\mathcal{K}_d$); elsewhere they are continued analytically.

The five-loop primitive needs the two-vector zig-zag spectrum. With $p = d/2 - 1$ (not the p of the formula for $\mathcal{K}_d$),

$$\tau_n^{(d)}(\nu) = (-1)^n \pi^{d/2} \frac{\Gamma(n/2 + d/8 + \mathrm{i}\nu)\Gamma(n/2 + d/8 - \mathrm{i}\nu)}{\Gamma(n/2 + 3d/8 + \mathrm{i}\nu)\Gamma(n/2 + 3d/8 - \mathrm{i}\nu)},$$
$$\rho_n^{(d)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3d/2+1}\Gamma(2p)n!} \left[(n+p)^2 + 4\nu^2\right] \frac{\Gamma(p + 2\mathrm{i}\nu)\Gamma(p - 2\mathrm{i}\nu)}{\Gamma(2\mathrm{i}\nu)\Gamma(-2\mathrm{i}\nu)}.$$

the periods with $L = 3, 4, 5$ loops are, for $2 < d < 4$,

$$\boxed{P_L(d) = h_d^{L-1} \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu)\, [\tau_n^{(d)}(\nu)]^{L+2} d\nu, \qquad L = 3, 4, 5.}$$

and, for $0 < d < 2$, where the first pair of Gamma poles crosses the real contour,

$$P_L(d) = h_d^{L-1} \Bigg\{ \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu) [\tau_n^{(d)}(\nu)]^{L+2} d\nu + \sum_{n=1}^{\infty} \frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3d/2} n! \Gamma(-p)} [\tau_n^{(d)}(-\mathrm{i}p/2)]^{L+2} \Bigg\}.$$

At removable singular values such as $d = 1, 2$ the Gamma ratios are taken in their joint analytic limit. The masters can be reduced further, to a few two-variable hypergeometric samples and their low-order parameter jets. For generic d this is a finite representation, not a reduction to scalar values.

One dimension. At $d = 1$ every coefficient is a fixed value. With $q = \frac{1}{4}$, $\bar{q} = \frac{3}{4}$ and

$$\boxed{\begin{array}{ll} \mathsf{A} = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), & \mathsf{D} = \psi_0(\bar{q}) - \psi_0(q), \\ \mathsf{T} = \psi_1(q), & \overline{\mathsf{T}} = \psi_1(\bar{q}), \\ \mathsf{U} = \psi_2(q), & \overline{\mathsf{U}} = \psi_2(\bar{q}). \end{array}}$$

whose values are, with Catalan's constant $\mathsf{G} = \sum_{n \geq 0} (-1)^n/(2n+1)^2$,

$$\begin{array}{cc} \mathsf{A} = A, & \mathsf{D} = \pi, \\ \mathsf{T} + \overline{\mathsf{T}} = 2\pi^2, & \mathsf{T} - \overline{\mathsf{T}} = 16\mathsf{G}, \\ \mathsf{U} + \overline{\mathsf{U}} = -112\zeta(3), & \overline{\mathsf{U}} - \mathsf{U} = 4\pi^3, \\ H = \dfrac{5\mathsf{T} - \overline{\mathsf{T}}}{6}, & \log 2 = \dfrac{\mathsf{A} - \mathsf{D}}{4}. \end{array}$$

and with $Q = \Gamma(\frac{1}{4})^4/\pi$, the coefficients are

$$\boxed{\begin{array}{ll} C_0 = \dfrac{1}{2}, & C_1 = -\dfrac{1}{6}, \quad C_2 = \dfrac{\mathsf{A}}{9}, \\ C_3 = \dfrac{\mathsf{A}^2 + (\mathsf{T} + \overline{\mathsf{T}})/6 - 8Q}{108}. & \end{array}}$$

$$\boxed{648 C_4 = 16 P_4 - \mathsf{A}^3 - \frac{\mathsf{A}}{2}(\mathsf{T} + \overline{\mathsf{T}}) - 8\mathsf{A}Q + \frac{365\mathsf{U} - 6\overline{\mathsf{U}}}{14}.}$$

$$\boxed{\begin{aligned}46656C_5 = &-1536P_5 + 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_- - 1728Q^2\\ &+16Q(45\mathsf{A}^2+383\mathsf{T}-64\overline{\mathsf{T}}) - 768\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})\\ &-15\mathsf{A}^4 - 15\mathsf{A}^2(\mathsf{T}+\overline{\mathsf{T}})\\ &-\frac{6\mathsf{A}}{7}(431\mathsf{U}+18\overline{\mathsf{U}}) - \frac{83}{4}(\mathsf{T}+\overline{\mathsf{T}})^2.\end{aligned}}$$

The periods are ordinary hypergeometric values:

$$P_4 = 13B_- - 3B_+, \qquad P_5 = 16\pi^{3/2}\big[-35X_0 + (22\sqrt{2}-13)X_+ + (22\sqrt{2}+13)X_-\big],$$

with

$$\begin{aligned}B_- &= \Gamma(\tfrac14)^4 F_-, \quad B_+ = \frac{64\pi^4}{\Gamma(\frac14)^4}F_+,\\ X_0 &= \mathcal{Y}_2, \qquad X_+ = c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5, \quad X_- = c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7.\\ F_- &= {}_4F_3\left(\begin{matrix}\frac14,\frac14,\frac12,\frac12\\ \frac34,\frac34,1\end{matrix};1\right), \quad F_+ = {}_4F_3\left(\begin{matrix}\frac34,\frac34,\frac12,\frac12\\ \frac54,\frac54,1\end{matrix};1\right),\\ c_8 &= \cos(\pi/8), \qquad s_8 = \sin(\pi/8),\\ \mathcal{Y}_r &= \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 {}_9F_8\left(\begin{matrix}1,1+\frac{r}{8},((r+2)/8)^{\times 7}\\ \frac{r}{8},((r+6)/8)^{\times 7}\end{matrix};-1\right),\\ &\quad r\in\{1,2,3,5,7\}.\end{aligned}$$

Here $(a)^{\times n}$ stands for n copies of the parameter a, and ${}_pF_q$ is the series of Section 1.4. At $r=2$ one upper and one lower parameter 1 cancel, so $\mathcal{Y}_2$ is an ${}_8F_7$ value. No elementary evaluation or minimality of these values is asserted.

Two dimensions. With $a = \log 2$,

$$\boxed{Q_2 = \frac16\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4 = \frac{\Gamma(1/4)^8}{24\pi^4}.}$$

$$\boxed{\begin{aligned}C_0(2) &= 1, \qquad C_1(2) = -\frac16, \quad C_2(2) = \frac{4a}{9},\\ C_3(2) &= \frac{4a^2}{27} - \frac{\Gamma(1/4)^8}{324\pi^4},\\ 648C_4(2) &= 16P_4(2) - 64a^3 - 32aQ_2 - 359\zeta(3),\\ 46656C_5(2) &= -1536\{P_5(2)+\mathcal{K}_2\} + 1536aP_4(2) - 576\mathcal{L}_2 - 1728Q_2^2\\ &\quad + (11520a^2 + 28608\mathsf{G})Q_2 - 3840a^4 - 21456a\zeta(3).\end{aligned}}$$

The two nonprimitive masters are $\mathcal{L}_2 = -2\log 2\, P_4(2)$, proved in Section 18.7 for the series definition of $\mathcal{L}_2$, and $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$, proved in Section 18.8 for its master-path definition. The formula for $C_5(2)$ below also uses $Z_2 = 4\mathsf{G}Q_2$ (265). The primitive periods reuse the one-dimensional values above (proved in Section 18.5):

$$\boxed{P_4(2) = \frac{B_-^2 - B_+^2}{4\pi^3}.}$$

$$\boxed{\begin{aligned}P_5(2) = \frac{8}{\pi}\big[&25X_0^2 - 28\sqrt{2}\,X_0(X_+ + X_-)\\ &+ 18(X_+^2 + X_-^2) + 28X_+X_-\big].\end{aligned}}$$

Hence

$$\boxed{\begin{aligned}972\,C_5(2) = &-32P_5(2)+56\log 2\,P_4(2)-36Q_2^2+\big(240\log^2 2+596\mathsf{G}\big)Q_2\\ &-1024\mathsf{G}^2+449\,\zeta(3)\log 2-80\log^4 2.\end{aligned}}$$

so $C_5(2)$ needs only $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the five values $\mathcal{Y}_r$.

Three dimensions. At $q=\frac{3}{4}$ the polygamma combinations give $A=2+4\log 2-\pi$ and $R=\frac{\pi^2}{3}-4$, so $C_2(3)=(2+4\log 2-\pi)/9$. The higher coefficients follow from the formulas for any dimension with the $d=3$ values of the masters, which reduce to fixed two-variable hypergeometric functions. No reduction of them to scalar ordinary ${}_pF_q$ values is known or asserted. Evaluated this way, $C_5(3)=-5.85035142075132722921794423049774430011099734637 83\ldots$ to the 50 digits shown.

1.2 A finite special-function representation of $C_6(1)$

The value of $C_6(1)$ in the table is a finite formula in nine special-function values. It comes from an analytic calculation in the package `C6_research_note/`. That calculation takes a reduced analytic expression for the six-loop coefficient as its input and converts each of its sectors into one value at unit argument; Appendix B gives the conversion. With $A=\pi+4\log 2$, $Q=\Gamma(\frac{1}{4})^4/\pi$ and Catalan's constant G,

$$\begin{aligned}2187\,C_6(1) = E &-\frac{476}{3}\pi^3(32\mathsf{G}+9\pi^2)-2176\big[(32\mathsf{G}+9\pi^2)L_3+128\,C_{212}\big]\\ &-4\pi^2\rho_4\,\mathcal{G}_4+192\pi\sqrt{290}\,\mathcal{G}_6+16\pi^3\rho_8\,\mathcal{G}_8-393\pi\sqrt{89}\,\mathcal{H}_W\\ &+24\pi^{3/2}\big(\rho_7\,\mathcal{G}_7+196\pi\sqrt{185}\,\mathcal{H}_7\big)+7680\pi^3\,\mathcal{H}_*,\end{aligned}\tag{2}$$

where E is the polynomial

$$\begin{aligned}E=&\frac{Q}{8}\Big[63A^3+120A\mathsf{G}+79A\pi^2+2013\pi^3+54204\zeta(3)\Big]+Q^2(29\pi-249A)+\frac{21}{640}A^5+\frac{7}{64}A^3\pi^2\\ &+\frac{1149}{64}A^2\pi^3+\frac{7863}{16}A^2\zeta(3)+192A\mathsf{G}(16\mathsf{G}+3\pi^2)+\frac{14853}{640}A\pi^4-\frac{544}{3}\mathsf{G}\pi^3\\ &+\frac{37631}{128}\pi^5+\frac{722279}{48}\pi^2\zeta(3)-\frac{805959}{8}\zeta(5).\end{aligned}$$

The nine values are the following.

- $L_3=\mathrm{Ls}_3(\frac{\pi}{2})$ and $C_{212}=\mathrm{Cl}_{2,1,2}(\frac{\pi}{2})$, a log-sine integral and a multiple Clausen value:

$$\mathrm{Ls}_3(\sigma)=-\int_0^\sigma \log^2\!\big|2\sin\tfrac{\theta}{2}\big|\,d\theta,\qquad \mathrm{Cl}_{2,1,2}(\sigma)=\sum_{n>m>k\geq 1}\frac{\cos(n\sigma)}{n^2mk^2}.$$

- Four Meijer G values at unit argument, in the convention of [22]:

$$\mathcal{G}_4=G^{2,3}_{5,5}\left(1\,\middle|\,\begin{matrix}(\frac{1}{2})^{\times 4},\alpha_4\\ 0^{\times 4},\alpha_4\end{matrix}\right),\qquad \mathcal{G}_6=G^{4,5}_{7,7}\left(1\,\middle|\,\begin{matrix}(\frac{1}{2})^{\times 6},\alpha_6\\ 0^{\times 6},\alpha_6\end{matrix}\right),$$

$$\mathcal{G}_8=G^{3,8}_{11,11}\left(1\,\middle|\,\begin{matrix}-\frac{1}{4},(\frac{1}{2})^{\times 8},\alpha_{8,+},\alpha_{8,-}\\ 0^{\times 8},\frac{3}{4},\alpha_{8,+},\alpha_{8,-}\end{matrix}\right),$$

$$\mathcal{G}_7=G^{3,8}_{10,10}\left(1\,\middle|\,\begin{matrix}-\frac{1}{4},(\frac{1}{2})^{\times 7},\eta_{7,+},\eta_{7,-}\\ 0^{\times 7},\frac{3}{4},\eta_{7,+},\eta_{7,-}\end{matrix}\right),$$

with the parameters

$$C_d = 96Q - 42A^2 - 18A\pi - 1227\pi^2, \qquad C_s = 60Q - \tfrac{105}{4}A^2 - 9A\pi - \tfrac{2469}{4}\pi^2,$$
$$\rho_4 = \sqrt{C_d^2 + C_s^2}, \qquad \alpha_4 = \frac{1}{\pi}\arctan\frac{C_s}{C_d}, \qquad \alpha_6 = \frac{1}{\pi}\arctan\frac{11}{13},$$
$$\rho_8 = \sqrt{2177203 - 1028322\sqrt{2}},$$
$$\alpha_{8,\pm} = \frac{1}{2\pi}\Big[\arctan\frac{556}{1233 - 417\sqrt{2}} \pm \mathrm{i}\,\mathrm{arcosh}\,\frac{1233 - 139\sqrt{2}}{\rho_8}\Big],$$
$$q_{7,2} = 259A - 49\pi, \qquad q_{7,1} = 481A - 539\pi, \qquad q_{7,0} = 407A - 1127\pi,$$
$$d_7 = \sqrt{q_{7,1}^2 - 4q_{7,2}q_{7,0}}, \qquad \rho_7 = \sqrt{(q_{7,2} - q_{7,0})^2 + q_{7,1}^2}, \qquad \eta_{7,\pm} = \frac{1}{\pi}\arctan\frac{q_{7,1} \pm d_7}{2q_{7,2}},$$

with principal values throughout.

- Three bivariate H values at unit arguments, $\mathcal{H}_W$, $\mathcal{H}_7$ and $\mathcal{H}_*$, that is, two-fold Mellin–Barnes integrals. The first is

$$\mathcal{H}_W = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac18-\mathrm{i}\infty}^{\frac18+\mathrm{i}\infty}\frac{N(s)\,\Gamma(t)\Gamma(t+\frac12)\Gamma(\frac12-t)\Gamma(1-t)}{(\frac12-s-t)^2}\,dt\,ds,$$
$$N(s) = \frac{\Gamma(s)^2\Gamma(\frac12-s)^3}{\Gamma(1-s)^2\Gamma(\frac12+s)\Gamma(s+\alpha_W)\Gamma(1-s-\alpha_W)},$$

with $\alpha_W = \pi^{-1}\arctan\frac58$. The other two are

$$\mathcal{H}_7 = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac{1}{16}-\mathrm{i}\infty}^{\frac{1}{16}+\mathrm{i}\infty}\frac{\mathcal{A}_7(s)\cot(2\pi t)}{(\frac12-t)(\frac12-s-t)}\,dt\,ds,$$
$$\mathcal{A}_7(s) = \frac{\Gamma(s)^3\Gamma(\frac12-s)^7\Gamma(\frac54-s)}{\Gamma(1-s)^3\Gamma(-s)\Gamma(\frac14-s)\prod_{b\in\{\beta_7,\bar\beta_7\}}\Gamma(s+b)\Gamma(1-s-b)},$$
$$\mathcal{H}_* = \frac{1}{(2\pi\mathrm{i})^2}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\int_{\frac14-\mathrm{i}\infty}^{\frac14+\mathrm{i}\infty}\frac{\Psi_+(s)\Psi_-(t)}{\Gamma(1-s-t)}\,dt\,ds,$$
$$\Psi_\sigma(s) = \frac{\Gamma(s)^3\Gamma(\frac12-s)^3}{\Gamma(\frac12+s)\prod_{b\in\{\beta_\sigma,\bar\beta_\sigma\}}\Gamma(s+b)\Gamma(1-s-b)},$$

where $\beta_7 = \pi^{-1}\arctan\big((13+\mathrm{i}\sqrt{139})/14\big)$ and, for $\sigma = \pm1$, $\beta_\sigma = \pi^{-1}\arctan\big(B_\sigma/2 + \mathrm{i}\sqrt{C_\sigma - B_\sigma^2/4}\big)$ with $B_\sigma = \frac{21}{20} + \sigma e_*$, $C_\sigma = \frac{\lambda_*}{10} + \sigma e_*\chi_*$, $e_* = \sqrt{40\lambda_* - 279}/20$, $\chi_* = 2(21\lambda_* - 150)/(40\lambda_* - 279)$, and λ_* is the root in $(15,16)$ of $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250 = 0$. The bar denotes the complex conjugate; all inverse tangents are principal (complex for β_7 and β_σ).

The formula evaluates to

$$C_6(1) = 14331.687032758054320445463255246217118317 32\ldots$$

The three bivariate values carry the three sectors of the reduced expression for which no evaluation in standard constants is known: the wheel sum W_0, the seventh-sector sum M_7 (608) and the pairing P_a of primitive hypergeometric functions (609) (Section 22.9). The count of nine describes how the terms are packaged, not the number of independent constants.

The reduced expression is an input: it is derived neither in this note nor in the package. Two of its Laurent coefficients were missing from its first version and were added by a correction; they contribute +659.279 to $C_6(1)$. The deterministic sum over all six-loop graphs at quadrature degree

22 does not use it and gives 14331.6870328(5). The representation lies 1.03×10^{-8} below its unrounded value, 14331.68703276838, which is 0.021 of its allowance; the comparison tests the reduced expression at the level of 3×10^{-11} relative. This confirms the reduced expression and its correction numerically; it does not prove them. The package encloses an earlier form of the reduced expression, (958), in an interval of width below 1.2×10^{-145} (Section B.8.5). That form reaches (604) only through exact identities of the package that this note does not reproduce, among them two inherited normalizations of the wheel periods; the later forms, reassembled from the same intervals with M_7 enclosed through the tail bound (956), have the same width. The enclosure therefore holds for the reduced expression conditionally on those identities, and it encloses the reduced expression, not $C_6(1)$. An independent review (`LRI_C6_closed_form_review/`) recomputed the input and the nine values with separate codes and found no error.

1.3 Status of the analytic results

The table separates what this note proves from what it takes as an input or confirms only numerically. "Proved here" means that the note contains the complete argument; "conditional" names the unproved input; "numerical" means agreement between independent evaluations, without an interval bound unless one is stated. The numerical coefficients $C_6(d)$ for $d = 2, 3$, $C_7(d)$ and $C_8(d)$ are sampling estimates with the allowances of Section 1 and are not listed.

Statement	Where	Status
One dimension		
Graph weights, subtraction and RG assembly of C_0–C_5	Secs. 2, 3, 13; App. A	proved here; every four- and five-loop residue is listed exactly
$\mathcal{K}$ (76)	Sec. 8.3	proved here
Wheel reduction of $\mathcal{L}$	Sec. 9.2	proved here
$P_5(1)$ in five fixed values (29)	Sec. 10.7	proved here
Graph integration of $C_5(1)$	Sec. 14	numerical: agrees with the closed form to $6 \cdot 10^{-13}$ with the span cutoff and $3 \cdot 10^{-12}$ with the maxgap cutoff, at degree 24; not a bound
$C_6(1) = 14331.68703275805\ldots$ (2)	Secs. 1.2, 22.9; App. B	conditional: the reduced expression (604) is taken as an input and derived neither here nor in `C6_research_note/`; its conversion to the nine values is derived in App. B, from three inherited elliptic-moment evaluations (911)–(913) and published identities; the interval of Section B.8.5 encloses an earlier form of the reduced expression, which reaches (604) only through package identities not reproduced here, among them two inherited wheel normalizations; confirmed numerically by the graph sum to $3 \cdot 10^{-11}$ relative
$C_6(1) = 14331.6870328(5)$, graph sum	Sec. 22	numerical: exact sum over all graphs at quadrature degree 22, with an empirical convergence allowance
Two dimensions		
$Q_2 = \Gamma(\frac{1}{4})^8/(24\pi^4)$	(192), (267)	proved here (complex Selberg integral, gamma reflection)
$Z_2 = 4\mathsf{G}Q_2$ (265)	Sec. 16.7	proved here: a complex Selberg integral for the two-screening integral (264), with the wheel jets (244)

Statement	Where	Status
$P_4(2) = (B_-^2 - B_+^2)/(4\pi^3)$ (333)	Sec. 18.5	proved here: finite-Fourier two-mode theorem and the elliptic moments of Section 9; the first derivation of b cites the mixed-moment integral and the Stieltjes identity of [25]; the second (Section 18.5.2) uses neither
$P_5(2)$, quadratic in X_0, $X_\pm$ (334)	Sec. 18.5	proved here: square-lattice factorization (342), the one-dimensional three-mode function (144) and exact residue algebra
$\mathcal{L}_2 = -2\log 2\, P_4(2)$ (270), for the series (256)	Sec. 18.7	proved here: elliptic lemmas of Section 9 and one contour integral, with the Stieltjes identity of [25] cited. Three further proofs avoid that identity: rim–spoke reflection (Section 18.7.6), Parseval pairings on the unit circle (Section 18.7.8) and the Fourier representation of the two modes (Section 18.7.9); the last two use the half-strip evaluation of the elliptic moments (Section 18.7.7)
$\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$ (271), for the master-path definition (238)	Secs. 18.8, 18.9.8	proved here: position-space proof (rotation lemma, sum of two squares, Laurent extraction) with the one-dimensional lemmas of Section 8.3; a second position-space proof of (377), by Riemann's bilinear relation (Section 18.9.8)
$\mathcal{K}_2$ also equals the series (260) and the spectral form (248)	Secs. 18.9.6, 18.9.7	proved here: regularization in the rim exponent, residues on the square lattice and periodization, with the one-dimensional lemmas of Section 8; for the series also the per-spin residue identity, proved at $d = 2$ with explicit arc bounds
Mixed-trace step (249), for $0 < \tau < \frac{1}{2}$	Sec. 18.9.5	proved here at $d = 2$, in the full-Mellin-line reading: the closed-form trace (390) with (377); at other d it is input (M) of Section 21.1
Closed form of $C_5(2)$ (272)	Sec. 17	proved here: exact algebra from the rows above and (223)
General dimension		
C_0–$C_5(d)$ (222), (223)	Sec. 16.3	proved by exact computation: exact symbolic residues of the 26 four-loop and 129 five-loop classes (certificate and exact audit in `lri_general_special_functions_update/`)
Spectral and series forms of Q_d, P_4, $\mathcal{L}_d$, Z_d (243), (246), (255)–(257)	Secs. 16.4.1, 16.6	proved here
Spectral and series forms of $\mathcal{K}_d$ (248), (260)	Secs. 16.4.2, 21.12	at $d = 2$ proved here (Section 18.9); at other d conditional on input (M) and on extracting $[\tau^2]$ under the trace
Finite-Fourier theorem of the primitive sector and finite evaluation rules for Q_d, $P_4(d)$, $P_5(d)$	Sec. 18	proved here for $\frac{1}{2} < q < 1$, then by continuation

Statement	Where	Status
Double residue series of $P_5(d)$ (281)	Sec. 17.4; checks in 17.7	proved here for $\frac{1}{2} < q < 1$; the extension by uniform convergence to $\frac{2}{5} < d < 4$ uses convergence orders whose elementary estimates are stated but not written out; checked in double precision at $d = 2, \frac{13}{5}, 3$
Finite Horn representation of $\mathcal{L}_d$, Z_d, $\mathcal{K}_d$ (298)	Sec. 17	proved here for $\frac{2}{3} < d < 2$, then by continuation of the complete combination
Finite Fourier theorem at a general rim exponent, top-mode theorem and its product form, residue functional	Secs. 20.1.1, 20.1.2, 20.1.3; status in 20.10	proved here, with complete proofs in Sec. 20.1. Reviewed independently in the form before the merge (`LRI_C5_general_d/review/VERIFICATION.md`): the finite Fourier theorem, the top-mode theorem in its wheel-only form, the residue functional and the block decomposition; the product form separately (`review/lemmaG-proof`). Not reviewed: the explicit constants of (465), (466), (467) (Sec. 21.6) and the contour of the residue-functional proof, changed at the merge
Sample formulas for $\mathcal{L}_d$, Z_d and the reduced $C_5(d)$	Secs. 20.2–20.7; checks in 20.9	sample formulas proved (Sections 20.3 and 20.4); reduced $C_5(d)$ derived (exact algebra, rederived symbolically in a separate review), conditional on (M), and on (H) on the jet circle, unless $\mathcal{K}_d$ comes from (588); confirmed numerically to about 10^{-48}
Constant (560) of the two-wheel sample and bilateral identity (561)	Sec. 21.2; status in 21.12	proved here from the case $r = 2$ of the top-mode theorem, the residue functional and the two-wheel convolution; not independently reviewed; constancy and value confirmed numerically
Growth theorem for the R family of $\mathcal{K}_d$	Secs. 21.4, 21.6; status in 21.12	proved here; parts (a) (sufficiency), (b) and (c) are the case $r = 3$ of the proof of the top-mode theorem, which was reviewed independently (Section 20.10); the sharpness of $\mathbb{H}_q$ and the explicit constants not independently reviewed
Sample formulas for $\mathcal{K}_d$	Secs. 21.7, 21.8; status in 21.12	given (M) and (J), proved on $\mathbb{H}_q$ under the conditions of Section 21.12; the residue functional and the three-node rule not independently reviewed; at $d = 2$, (M) is proved for real $0 < \tau < \frac{1}{2}$ (Section 18.9.5; complex τ is not addressed there) and (J) in Section 18.8.4; on the jet circle also conditional on (H), at every d
Barnes representation of $\mathcal{K}_d$ (588)	Sec. 21.10	derived here without (M), (J), (H); the statement re-derived by separate AI sessions; the printed derivation not reviewed
Three dimensions		
Determinant and rank-two identification of Q_3, $P_4(3)$, $P_5(3)$, and the complex-node variant (438)	Secs. 19.4, 19.5, 19.6	proved here

Statement	Where	Status
Specialization of $\mathcal{L}_d$ and Z_d to $d = 3$	Sec. 20.6; conditions in 20.10	derived here: exact algebra on two identities proved in Section 20.6 (the half-shift proof at a general rim exponent, with Dixon's and Dougall's sums); the identities re-derived by separate AI sessions, the printed derivation not reviewed; confirmed numerically to 10^{-51}
Values of Q_3, $P_4(3)$, $P_5(3)$ and $C_5(3)$	Sec. 19.10	numerical: independent evaluations agree to about 10^{-79} (Q_3), $2 \cdot 10^{-86}$ ($P_4(3)$) and $2 \cdot 10^{-53}$ ($P_5(3)$), which changes $C_5(3)$ by less than 10^{-54}; not interval bounds. $C_5(3)$ does not depend on (M), through (588)
No ${}_3F_2(1)$ or ${}_4F_3(1)$ relation for $\mathfrak{H}_3(0)$ of Q_3 (405) in a bounded search; no relation for $\mathsf{r}_3(3)$, $\mathcal{K}_3$	Secs. 19.10, 20.8, 21.9	numerical only: bounded catalogue and PSLQ searches in stated bases, not independence proofs; no search is stated for $P_4(3)$, $P_5(3)$
Bulk one-point coefficient (Part V)		
$a^{(2)}_{\phi^2}$, $a^{(3)}_{\phi^2}$	Sec. 34.6.1	derived there; agree with [8]
$a^{(4)}_{\phi^2}$–$a^{(6)}_{\phi^2}$ and the Gamma-factor form (820)	Secs. 34.3, 34.6	calculation supplied by the author (16 October 2026); checks internal to the master framework: the five- and six-vertex graphs are reconstructed from master jets and checked with other regulator directions, not integrated one by one; the crossed kite rests on the package's Mellin–Barnes representation
b_5 and c_5 separately (834), (835)	Secs. 34.5, 34.12	$3c_5 - b_5$ equals (229) exactly; the split agrees numerically with graph sums at $d = 1$ and with the seventh- and eighth-order productions at $d = 2, 3$
Map to $C_j(d)$ (829)	Sec. 34.4	exact algebra between two calculations with shared inputs; validates neither
Exact $a^{(n)}_{\phi^2}(1)$, $a^{(n)}_{\phi^2}(2)$ and the closed $d = 2$ map (862)	Secs. 34.9, 34.8	exact algebra on the proved $d = 1$, $d = 2$ masters, conditional on H_4 of the supplied calculation
Comparison with the numerical bootstrap	Sec. 34.11	exploratory: readings of published figures, no bootstrap problem solved; the spread of prescriptions is not an error bar

1.4 Notation and the logical route

In the one-dimensional coordinate prescription the covariance is $\langle\phi(x)\phi(0)\rangle_0 = |x|^{-(1-\epsilon)/2}$ and the interaction is $g\phi^4/4!$. The graph weights and MS recursion are fixed first. The one-dimensional master reductions then give the full analytic coefficient; the residue certificate of Appendix A lists the residue of every four- and five-loop class. The general-dimensional sections replace the one-dimensional integrals by normalized Schwinger and spectral integrals, derive the master jets and provide their finite-function representations; Sections 20 and 21 shorten those of the three nonprimitive masters to a few two-variable hypergeometric samples and their low-order parameter jets. The numerical sections retain a separate, regulator-dependent finite-scheme formulation and its complete error accounting. Part V uses the same graphs, subtractions and masters for a second observable, the bulk one-point coefficient. Appendix A lists the residue certificate, and Appendix B derives the special-function representation of $C_6(1)$ quoted in Section 1.2.

Symbol	Meaning
$C_n(d)$	Physical coefficient of ϵ^n in $\Delta_{\mathcal{E}}$.
b_L, c_L	L-loop beta and mass-anomalous-dimension coefficients; lowercase c_L is not C_L.
z	Analytic regulator, set equal to physical ϵ only after the required Laurent or Taylor data have been retained.
d, q	Spatial dimension and $q = d/4$ in the general-dimensional sections. In the five-index one-dimensional master $\mathcal{J}(a, b, c, d, e)$, d is only an edge exponent.
$\widehat{b}_L(z), \widehat{c}_L(z)$	Finite-scheme RG functions, whose z dependence cannot be dropped before fixed-point extraction.
$P_3(d), Q_d$	Primitive normalization $P_3(d) = 3Q_d$. At $d = 1$, $Q = \Gamma(1/4)^4/\pi$.
$\mathcal{K}_d, \mathcal{L}_d, Z_d$	Explicit nonprimitive master data; $\mathcal{K}_d$ is not an elliptic integral. Reduced at generic d in Sections 20 and 21.
$K(m), \mathbf{K}(k)$	Complete elliptic integrals in parameter and modulus conventions: $\mathbf{K}(k) = K(k^2)$. A complementary argument is not a derivative.
$\psi_m(z)$	$d^{m+1} \log \Gamma(z)/dz^{m+1}$; ψ_0 is digamma.
G	Catalan's constant; distinct from graph symbols and matrix-argument hypergeometric functions.
a_{ϕ^2}, $a^{(n)}_{\phi^2}(d)$, $a^{(n)}$	The bulk one-point coefficient of $:\Phi^2:$ and its coefficient of ϵ^n (Part V); not the squared defect OPE coefficient $\lambda^2_{\phi\phi\mathcal{E}}$.
(M)	Input of Section 21.1: the mixed trace $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$ as an identity of meromorphic functions of τ, with the trace (554) read on the full Mellin line and continued from its strip of convergence; proved at $d = 2$ for real $0 < \tau < \frac{1}{2}$ (Section 18.9.5).
(J)	Input of Section 21.1: the jet $J_t - J_t^\Gamma = \mathcal{K}_d t^2 + O(t^3)$ on the master path (238), whose orders t^0, t^1 are used explicitly; at $d = 2$ its low orders are rederived in Section 18.8.4.
(H)	Continuation hypothesis of Section 21.8: the continued three-node samples (581), (585) are jointly holomorphic in (t, τ) on a domain that contains the disk bounded by the jet circle and meets $\mathbb{H}_q$, off a closed discrete set.
$\mathbb{H}_q$	The half-plane $\operatorname{Re}\tau > \tau_*(q) = \max(0, 2q - 1, \frac{1}{2} - 2q)$ (571), on which the lattice sum of the R family converges absolutely (Section 21.4).
jet circle	The circle $\lvert\tau\rvert = r_\tau$ around the pole $\tau = 0$ on which the τ-jet of (586) is taken as a Cauchy integral; it does not lie in $\mathbb{H}_q$ (Section 21.8).
$[\tau^2]$ under the trace	The interchange of the Taylor coefficient $[\tau^2]$ at $\tau = 0$ with the trace, $[\tau^2]\langle t_\ell^2(R_\tau - S_\tau)\rangle_d = \langle t_\ell^2 E_2\rangle_d$, used to derive (248) from (238) (Sections 16.4.2, 21.3); justified only at $d = 2$ (Section 18.9.2).
Part III families	
b, $b_{\max}$, b_{spec}	The rim exponent of a balanced wheel with spoke exponent $d - b$ (242), complex in general; the critical wheel has $b = d/2$. Sections 20 and 21 write $\sigma = b/2$ and $g = (d-b)/2$ (g_b in Section 21). The spectral integral (460) represents W_r for $0 < \operatorname{Re} b < b_{\text{spec}} = d - \max(d, 1)/r$, and the finite Fourier theorem of Section 20.1.1 holds for $0 < \operatorname{Re} b < b_{\max}(r, d)$ (463), with $b_{\max} \le b_{\text{spec}}$. Not b_L, b_ℓ or the mode b of (340).

Symbol	Meaning
$W_r(d;b)$, $t_{\ell,b}$, N_b, Ξ_r	The r-wheel $W_r(d;b) = \langle t_{\ell,b}^r \rangle_d$ (242), with $t_{\ell,b}(\nu) = N_b\, g_\ell(\mathrm{i}\nu;b)$ for real b and the normalization N_b of (460); at $b = d/2$, $t_{\ell,b} = t_\ell$ and $N_{d/2} = \Gamma(2q)$. $\Xi_r(t)$ (304) and $\Xi_r(t;b)$ (462) are the periodized samples, and Ξ_2 (Sections 20.9, 21.2) is the case $r = 2$; Ξ_{R} (570) belongs to the R family, and Ξ_Γ (Section 24.1) is unrelated. The rim sample $\mathfrak{t}_3(b)$, first used in Section 19, is defined in (495).
$T_{\ell n}$, $T_{mn}(x)$	$T_{\ell n}$ (251), with $S_{\ell n}$, $a_{\ell n}$, $b_{\ell n}$ there (not S_τ, a_ℓ, b_ℓ), and $T_{\ell n}(b)$ at a rim exponent b (490). $T_{mn}(x) = c_m(x)c_n(x)$ (345) has the first index $m = \ell + n$, so that $T_{mn}(\frac{1}{2}) = T_{\ell n}$ at $d = 2$; $T_\ell(z)$, with one index, is a product kernel of Section 20.1.2.
J_t, $J_{2\tau}$, $C(\tau)$, R_τ, S_τ	The master path $J_t = J_d(\frac{d}{2}, \frac{d}{2}, \frac{d}{2}, d-t, \frac{d}{2})$ of the jet (237), its Gamma part J_t^Γ and $\mathcal{K}_d = [t^2](J_t - J_t^\Gamma)$ (238). With $t = 2\tau$, $C(\tau) = \Gamma(2q)\Gamma(\tau)/\Gamma(2q-\tau)$ and the kernels $R_\tau(s)$, $S_\tau(s)$ (249) form the mixed trace of input (M); the R and S families of $\mathcal{K}_d$ are those of R_τ and S_τ. Used in Sections 16.4.2, 18.8, 18.9 and 21. $C(\tau)$ is not a coefficient C_j, R_τ is not R, and S_τ is not $S_{\ell n}$.
μ_1, μ_2, η_m, θ_m, λ_m, γ_C	Polygamma combinations at the arguments $a_{\ell n}$ of (251): λ_m (252) and η_m, θ_m, μ_1, μ_2 (253), used in the series of Section 16.5. The λ_j of (274) and the η_r, θ_r, μ_1, μ_2 of (275) are the same combinations, with $m = \ell + n$; Sections 18.7.1, 20.5 and 20.4.4 continue them in the rim exponent. $\gamma_C(a) = \Gamma(a)/\Gamma(1-a)$ (Sections 15, 16.7, 18.7.6, 20.6).
$\mathcal{N}_d$, $\mathcal{N}_x = N_x$	$\mathcal{N}_d = \mathcal{K}_d + \frac{3}{8}\mathcal{L}_d - \frac{149}{32}Z_d$ (403), the only combination in which the nonprimitive masters enter $C_5(d)$. $N_x = \Gamma(1-x)/\Gamma(x)$ (345) is written $\mathcal{N}_x$ in Sections 20.1.3 and 20.6: one object with two names, unrelated to $\mathcal{N}_d$; it equals N_b at $d = 2$, $x = \frac{b}{2}$.
Symbols with more than one meaning; each is defined where it is used	
h	$h_d = \Gamma(d/2)/\pi^{d/2}$, and an $\Im\psi_0$ combination in the formula for $\mathcal{K}_d$ (both Section 1.1); the propagator exponent $(1-\epsilon)/2$ or $(1-z)/2$ (Sections 2, 3); the Gamma ratios $h_b(z)$ (Section 20.6) and $h_x(\sigma)$ (Section 18.7.9); further local meanings in Sections 17.4, 18.9.7, 19, 20.1, 22.9, 23, 25, 28 and 32.7.
p	an $\Re\psi_0$ combination in the formula for $\mathcal{K}_d$, and $p = d/2 - 1$ in the five-loop spectral formula (both Section 1.1); $p = D/2 - 1$ in Section 10; local regulators, momenta, edge powers, exponent sums, pole orders and auxiliary polynomials elsewhere, for example in Sections 18.7.6, 18.9.4, 19.5, 20.1.4 and 28.
B, B, $\mathcal{B}$, $\mathfrak{B}$	Euler's $B(u,v)$; $\mathsf{B}(u,v) = \int_{\mathbb{R}} \lvert x\rvert^{-u}\lvert 1-x\rvert^{-v}\,dx$ (Section 7); $\mathcal{B}(g)$ and $B(g)$, the same reduced beta function (Sections 2, 22, 23.1); $\mathcal{B}_L(z)$, the mass-insertion amplitude (Section 3); $B_\pm$, hypergeometric values (Section 1.1), not B_σ, $\sigma = \pm 1$, of Section 1.2; B_d (230); the residue weights $B_{r,k}(\sigma)$ (Section 20.1.3); the Bernoulli polynomials $B_n(x)$ (Sections 19.10, 21.8); $\mathcal{B}(x)$, a mode amplitude (349), and $\mathcal{B}_\tau$ (Section 18.9.5); local B_k, $B_i(n)$ (Sections 18.9.2, 18.9.7); $\mathfrak{B}_{\mathrm{kn}}$ (Section 23.1), unrelated to $\mathfrak{B}_c$ and the transforms $\mathfrak{B}_{1/4}$, $\mathfrak{B}_{1/8}$ of Appendix B.

Symbol	Meaning
H, H_7, $\mathcal{H}_7$, $\mathfrak{H}_7$	$H = 2\pi^2/3 + 8\mathsf{G}$ (Sections 1.1, 4); H_7, a ${}_7F_6$ value (Section 9); $\mathcal{H}_7$, a bivariate H function, with $\mathcal{H}_W$, $\mathcal{H}_*$ (Section 1.2); $\mathfrak{H}_7$, a two-variable Horn function (Section 19), with the Horn series $\mathfrak{H}[\cdots]$ (290) and the values $\mathfrak{H}_+$, $\mathfrak{H}_c$ (Section 20.4.3); (H), the continuation hypothesis (first block), and the half-plane $\mathbb{H}_q$; $H(a;v)$ (274); $\mathcal{H}_x$ (348); H_6^χ and $H_6(d)$, bare channels, not ${}_6F_5$ values (Sections 22, 23.1); the harmonic numbers H_n (Section 22.9, Appendix B); the Hardy space H^2 (Section 18.7.8); $H(\tau)$ (Section 18.9.7); H_{R}, H_{M}, H_{L} (Section 21.7); further local meanings in Sections 11, 14, 23 and 27.
G_4, $\mathcal{G}_4$, $\mathcal{G}_L$, G_n, G	$G_4 = \Gamma(\frac{1}{4})^4$ (Section 9); $\mathcal{G}_4$, $\mathcal{G}_6$, $\mathcal{G}_7$, $\mathcal{G}_8$, Meijer G values (Section 1.2); $\mathcal{G}_L(z)$, a Gamma prefactor (Section 25); $\mathcal{G}_x(s)$, a Gamma product (Sections 18.7.6, 18.7.9), and $\mathcal{G}_\alpha(n)$ (584); $G_n(z)$, coupling counterterms (Section 3); G_0, G_1, G_2 (236); $\mathsf{G} = \mathsf{G}$, Catalan's constant; $\mathfrak{G}(\tau)$, with $\mathfrak{F}(\tau)$ (75), not the graph set $\mathfrak{G}_L$ (Section 3); local $G_{\ell k}$, $G(\beta)$, $G_\ell(s)$ (Sections 17.4, 18.7.7, 18.9.3).
Q, Q_d, $\mathcal{Q}_3$	$Q = \Gamma(\frac{1}{4})^4/\pi = Q_1$; Q_d, the three-loop period, written Q in the general-dimensional formulas of Section 1.1; $Q = Q_3$ in Section 24; $\mathcal{Q}_3 = [\tau^3]\mathcal{Q}(\tau)^2$ (Section 18.8).
R, U, T, $\mathcal{T}$	polygamma combinations (R, U) and values (T, U) in Section 1.1; integrals or functions in Sections 5, 7 and 9, those of Section 9 reused in Section 18.7; U is also the allowance of Section 22.7; $\mathcal{R}$ has different meanings in Sections 3 and 24, and $\mathcal{R}_{\ell n}$, $\mathcal{R}_+$, $\mathcal{R}_{\mathrm{m}}$ are lattice sums (Sections 16.5, 20.3); the R family of $\mathcal{K}_d$ is that of R_τ (Part III families); R, the number of replicates (Sections 26.11, 31.7); U_G, U_0, U_F, Symanzik polynomials, and R, a parameter sum (Sections 23, 25, 28); $T_\tau(s)$, $T_{\mathsf{h}}(\eta)$ (Sections 18.9.3, 18.9.4) and T_{R}, T_{L} (Sections 21.4, 21.6); $\mathcal{T}_d$, the corrected tetrahedron (Section 15), against $\mathcal{T} = \sum_n a_n^4 h_n(8h_n^2 + \delta_n)$ in the wheel identities of Section B.4.
E	$\mathcal{E} = [\phi^2]$; E_2 in the formula for $\mathcal{K}_d$ (Section 1.1), unrelated to the polynomials E_m of Section 18; E, a polynomial in constants (Section 1.2); $E(a,b)$, the Euler integral of K (Sections 8, 18.9.4); $E_{\mathfrak{k}}$, $E^\sharp$ (Section 18.7.6); E_0, E_0^*, E_*^* (Sections 19, 19.5); E, the number of edges of a graph (Section 26); the line sets E_B, $E(W)$ (Sections 28, 32.7).
Z	$\mathcal{Z}_g$, Z_r and the counterterm coefficients $Z_n(z)$ (Section 3); Z_d, the wheel master, with $Z_2 = 4\mathsf{G}Q_2$ (265).
D, D	$\mathsf{D} = \pi$ (Section 1.1); D, the dimension in Section 10; $D = \pi\cot\pi q$ (459), with S, C there; D_4, D_5 and in general D_L, the combinations $3c_L - b_L$ (Sections 2, 15), with D_5^{prim} (Section 16.3). Not $3c_L - b_L$: D_3, D_5, polylogarithm values (Sections 22.9, B.7), and D_4, D_6, wheel sums (Section B.4); D_2, the two-loop kernel of Section 23.8; $\widetilde{D}_j$ (274); D_s, D_r (Sections 16.3, 16.4.1); the sine products $D_q(t)$, $D_b(t)$, D_T, D_{R} (Sections 18, 20.1, 20.1.2, 21.4); $\mathfrak{D}_L$ (Section 20.2); $D(\ell)$ (Section 18.9.7); local $\mathcal{D}(\tau)$, $\mathcal{D}_2$, $\mathcal{D}_3$ (Section 18.9.2), D_{F} (Section 18.7.9) and $\mathcal{D}_4(y)$ (Section B.8.2).
$\mathcal{K}$, K	$\mathcal{K}$ and $\mathcal{K}_d$, the nonprimitive master; PP and, in Sections 23–24, $\mathcal{K}$, the pole-part projector; K_G, the forest kernels of Sections 25–28; $K(m)$, $\mathbf{K}(k)$, elliptic integrals; $\boldsymbol{K}_{\leq 4}$, the loop one-to-four vector of Section 27; $K_\ell(s)$, the five-loop primitive kernel (303); K_5, the complete graph (Sections 16.3, 28.6); K_n, a Pochhammer ratio (Section 18.7.6) and the sums K_n (Section B.1); $\mathfrak{K}$, a compact set (Section 21.6).

Symbol	Meaning
$G_{L,d}$	the Gamma prefactor of Sections 25 and 26; its two normalizations differ by a constant, so the log-jets $a_{L,j}$ and ℓ_j agree for $j \geq 1$.
r_j, r_k	r_j, series coefficients (Section 16.5); r_k, s_k, Laurent coefficients (555), with $\mathsf{r}_{-1}, \dots, \mathsf{r}_2$ in (564) and $\mathsf{r}_3 = [\tau^3]\langle t_\ell^2 R_\tau \rangle_d$ (Section 21); $r_3 = 2Q_2$, the primitive three-loop residue (Section 23); r_1, r_2 (841), the coefficients of the ratio $b_0^{\Phi,-}/b_0^{\Phi,+}$ of bulk-to-defect coefficients (Section 34.6.1), not the r_1, r_2 of (564).
w, $\mathfrak{w}$	w_1, w_2, spectral weights (Sections 1.1, 20.4.4); the graph weights $w_4 = w_A$ and $w_2 = w_B$ (Sections 3, 22); $w_H = 3w_B - w_A$; $w(k) = \log \frac{k}{1-k^2}$ (Sections 9, 18.7.7); $w_{\mathcal{K}}$, the weight of (248) at $d = 2$ (Section 18.9.1); $w_j(\tau)$, the node weights (580); $\mathfrak{w}_{r,h}$, spin-tail weights (Section 19.10), and $\mathfrak{w}_{r,j}(b)$, node weights (484).
z, z_1, z_2	z, the analytic regulator; z_1, z_2, deformation coordinates (Sections 7, 16.3) or Mellin–Barnes variables (Section 21.10); z is also a local complex variable in proofs where the regulator does not enter or is called τ (Sections 18.7.6, 18.7.8, 18.9.2, 20.1), and $z_\ell = \frac{\ell}{2} + \mathrm{i}\nu$ (Section 20.1.4).
D_G, ϖ	local meanings in Sections 23 and 25 (D_G); Mellin–Barnes weights (Section 21.10), hypergeometric values (Section 22.9) and Binet's function $\varpi(t)$ (Section 18.7.6) (ϖ).
MOM	the finite zero-momentum scheme of Section 25.
a, $\mathfrak{a}$	$a = \log 2$ and $a_\ell = \frac{\ell}{2} + \frac{q}{2}$ (Section 1.1), with $a_\ell = \frac{\ell+\sigma}{2}$ at a rim exponent (460); the finite Laurent coefficients $a_{L,k}$ (Sections 3, 15); the series terms $a_n(t)$ (Section 8); the log-jets $a_{L,j}$ (Sections 25, 26); a_n in Section 22.9; $a^{(n)}$ and the amplitudes $\mathsf{a}_n(\epsilon)$ (Part V); $a = 8N_{\mathrm{e}}/\pi$, a mode at $d = 2$ (340); $a = a_\ell$ at fixed spin (Section 18.9.1); $a = \frac{1}{2} + \frac{z}{4}$ (Sections 23, 28, 31.7); the prefactors a_r^0, a_r^+, a_r^c (Sections 18.4, 20.4.3); a_G (Sections 23, 28.6). $\mathfrak{a}_3$, $\mathfrak{b}_3$, $\mathfrak{a}_4$, $\mathfrak{b}_4$, mode amplitudes (Section 19), continued in b by (543); $\mathfrak{a}(x)$, $\mathfrak{b}(x)$ and $\mathfrak{s}(x)$ (Section 18.7.6) are not these or the Horn values $\mathfrak{s}_0$, $\mathfrak{s}_{1/2}$ (Sections 20.2, 20.3).
A, $\mathcal{A}$	A and A_d (176), so $A_2 = 4\log 2$ and $A_3 = 2 + 4\log 2 - \pi$; $A = \mathsf{A} = A_1$ in Sections 1.1 and 1.2; the Horn blocks A_r^0, A_r^+ (321) (Sections 20.1.3, 20.4.3); local A_k, B_k and $A_i(m)$ (Sections 18.9.2, 18.9.7), not A_d. $\mathcal{A}_L(z)$, $\mathcal{A}_L(d, z)$, the bare amplitudes (Sections 3, 15), and $\mathcal{A}_L^\chi$ (Section 22); $\mathcal{A}_{\mathrm{s}}(t)$ (Section 10); $\mathcal{A}_{\ell k}$ (Sections 18, 17.4); $\mathcal{A}(x)$, a mode amplitude (349); $\mathcal{A}_\tau$ (Section 18.9.5); $\mathcal{A}(h, z)$ (691) and $\mathcal{A}(x, z)$ (Section 27); $\mathcal{A}_7(s)$ (Section 1.2).
b, c	b_L, c_L (first block), and the rim exponent b (Part III families); $b = 8M_{\mathrm{e}}/\pi$, a mode at $d = 2$ (340); $b = b_\ell$ at fixed spin (Section 18.9.1); $b = \frac{1}{2} + \frac{3z}{4}$ (Section 31.7); $b_0^{\Phi,\pm}$, bulk-to-defect coefficients (Section 34.6.1); c_j (274), so the c_5, c_6 of (280) are not loop coefficients; $c_n = (\frac{1}{2})_n/n!$ (Sections 18.5.2, 18.7, 18.9.7); $c_k(x)$ (345); c_* (247); $c_\tau = \cos \pi\tau$ (Sections 18.8, 18.9); the Fourier coefficients $c_{r,k}$ (Sections 18, 20.1.1) and $c_k(\tau)$ (Section 21.4); generic constants c, c_0 (Sections 18.9.2, 21.6).

Symbol	Meaning
C, S	$C_n(d)$ (first block); $C(\tau)$ (Part III families); $S = \sin \pi q$, $C = \cos \pi q$ (459) (Sections 20.2–20.7); the Horn blocks $C_r(t)$ (321), for example $C_3(t)$, not $C_3(d)$ (Sections 20.1.3, 20.4.3); (C), the condition (469); generic constants C, C_0, C_r, C_q, C_K, $C_{\mathfrak{K}}$ in bounds (Sections 18.7.6, 18.9.2, 20.1, 20.4.1, 21.6); C_d, C_s, C_σ, C_{212} of Section 1.2; the zonal polynomials $C_\kappa^{(2)}$ (Sections 19.6, 19.10); C (Part V); $S_{\ell n}$ (251), the periodized sums $S_r(t)$, S_{R} (Sections 18, 20.1, 21.4) and the sectors S_{pol}, S_6, S_8 (Section 22.9).
F	$F_4(v)$, $F_2(v)$, the bare amplitudes (Section 3); $F_1(d,z)$, $F_2(d,z)$ of (199) (Sections 15, 28, 31.7); the polygamma polynomials F_3, F_4, $F_{3,1}$, $F_{4,1}$ (254) (Sections 16.5, 20.3, 20.4.4); $F(\tau)$ and its coefficients F_k (555), so that $F_3 = \mathsf{r}_3 - \mathsf{s}_3$ in Section 21.3 is not the polynomial F_3; $F(s)$ (263); $F_{r,j}$ (Section 17); $F_\pm$, ${}_4F_3$ values (Section 1.1); local F_{L}, F_{R} (Section 18.7.6) and $F(\beta)$, $G(\beta)$, $L(\beta)$ (Section 18.7.7).
g	the coupling (Section 2); the Barnes kernels $g_\ell(s)$, $g_\ell(s;b)$ (Sections 16–21) and $\mathsf{g}_\ell(s)$ (Section 18.7.6); $g = (d-b)/2$, the spoke half-exponent (Section 20), written g_b in Section 21; the free exponents g_i of f_{g_i} (Section 20.1.2); $g(k) = \mathbf{K}(k')$ (Sections 9, 18.7.7); $g(m,n)$ (Section 18.9.7); g_{alt} (Section 34.5).
I	$I_m(z)$ (Section 3); the Euler moments $I_1, \dots, I_6$ (Sections 8, 18.8, 18.9.4, 18.9.8); I_1, I_3, I_5, I_7 (Section 10); the pairings I_{jk} (Section 22.9, Appendix B); identity matrices I_2, I_L (Sections 19.6, 19.9, 23); I, the number of model terms (Section 21.8); local I_ℓ, I_{L}, I_{R} and intervals I, I_{F} (Sections 20.1.2, 18.7.6, 18.7, 18.7.9).
J, J	$\mathcal{J}_m(z)$ (Section 3) and the master $\mathcal{J}(a,b,c,d,e)$ (first block); the jet J_d (237) and the master path J_t, $J_{2\tau}$ (Part III families); $J_0(q)$ (Section 15); J_1, J_3, J_5 (Section 10); $J_r(a;v)$ (274); the bare channels J_5^χ, $J_5(d)$ (Sections 22, 23.1); the Jacobians J, J_L, $J_{f,L}$ (Sections 26.13, 28.7); $J(\lvert i-j\rvert)$, the lattice coupling (Section 1); $\mathsf{J}_0(x)$, $\mathsf{J}_1(x)$, Barnes integrals (Section 18.7.6), not J_d of Section 34.6.1 at $d = 0, 1$.
P	the periods $P_3(d) = 3Q_d$, $P_4(d)$, $P_5(d)$ (first block); P_W, a wheel period (Sections 12, 16.4.1); the lattice weights $P_\pm$, P_{m} (Section 20.3) and P_{R}, P_{M}, P_{L} (Sections 21.6, 21.7); P_{10}, the retained ten primitive six-loop kernels (Sections 26.7, 26.12–26.14); P_a, a pairing (609); local P_f, P_{h} (Section 18.9.4).
q	$q = d/4$ (first block); the nome $q = e^{-\pi \mathbf{K}(k')/\mathbf{K}(k)}$ (Section 18.7.9); the coefficients q_n of $\mathsf{q}(z)$ (Section 18.7.8) and the sequence q_n of Section B.4; the parameters $q_{7,j}$ of Section 1.2.
s, ς	the long-range exponent s (Section 1); the Mellin variable s (Sections 16.4.2, 21.1); $s_b = \sin \frac{\pi b}{2}$, $\varsigma_b = \cos \frac{\pi b}{2}$ (Section 20); $s_\tau = \sin \pi\tau$ (Sections 18.8, 18.9); ς_f, sign vectors (Section 18.8); s_{K}, a parametric excess (Section 19.9).
s_G	$s_G(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S(x)$, the regulator logarithm (Section 26); $s_G = \sum_i e_i(e_i - 1)$, the leg count of a graph (Section 28).
t	the path parameter of J_t (238), not $t = d - e$ of (233); the periodization variable of $S_r(t)$, $\Xi_r(t)$ (Sections 18, 20, 21.4); the wheel eigenvalues t_ℓ, $t_{\ell,b}$ (Section 1.1, Part III families); a direction parameter in Section 34.6.1.

Symbol	Meaning
u, v, σ	u_j, the coefficients of g_* (Section 2); u, v, σ with τ (Section 8) or ϱ (Sections 18.8.4, 18.9.1), Taylor data of $\mathfrak{F}$, $\mathfrak{G}$; $u(b)$, $v(b)$, the weights of (496), with u_0, v_0 (Section 20.4.5); local u, v (Section 20.6); u_h (314); $v_m(x)$ (Section 18.7.9). $\sigma_i = \pm 1$, the spins (Section 1), and $\sigma = \pm 1$ (Section 1.2); $\sigma = b/2$ (Sections 20, 21), a free offset in Section 20.1.2; $\sigma = \operatorname{Im} t$ (Section 19.5); $\widehat{\sigma}$, standard errors (Part IV); further local meanings in Sections 7, 10, 15, 16.3, 16.7 and 23.
V, X, $\mathcal{X}$, X	V, X, polygamma constants (221); X_0, $X_\pm$, the one-dimensional values of Section 1.1, against X_0, X_h at $d = 3$ (421), with $X_0 = \Xi_4(0; \frac{3}{2})$ (Section 20.6); $\mathsf{X}_r(t)$ (Section 19); $\mathcal{X}_d(\tau)$ (555), with $\mathcal{X}_2(\tau)$ (Section 18.9.1); the samples $\mathcal{X}_0$, $\mathcal{X}_{1/2}$, $\mathcal{X}_T$ (Sections 20.2, 20.3), where the subscript labels a node, not a dimension; the model factors $X(m)$ (Section 21.8); V, V_0, elliptic moments (Sections 9, 18.7.7); $V_r(t)$, $V_r^{(0)}(t)$ (Sections 19, 19.5); the variances V_C, V_{pf}, V_Δ, $V_{f\Delta}$ (Section 28) and $\widehat{V}_G$, $\widehat{V}_{\mathrm{bench}}$ (Section 26.11); further local meanings in Sections 11, 20.1, 23 and 34.2.
x_0, m_0	x_0, a vertex fixed at the origin (Sections 3, 9, 10, 34.6.1); x_0, a Gamma ratio (583); m_0, an integer cutoff (Section 21.6), and m_0 of Section B.2.2; $\mathfrak{m}_0$, a moment (Section 18.7.9).
C_d, C_s, C_σ, d_7, $q_{7,j}$, ρ_4, ρ_7, ρ_8; W_0, M_7, P_a	parameters of the nine values of Section 1.2, used again in Section 22.9 and Appendix B: C_d, C_s, C_σ, B_σ, d_7, $q_{7,j}$, ρ_4, ρ_7, ρ_8, $\eta_{7,\pm}$ and λ_* are not $C_n(d)$, $B_\pm$, d_ℓ, q, $\rho_n^{(d)}$, η_m or λ_m. Section 1.2 uses the wheel sum W_0, the seventh-sector sum M_7 and the pairing P_a before their definitions (608), (609); W_0 is not a wheel W_r.
β	$\beta_{\mathrm{RG}}(g)$ (Section 2); α, β, γ, constants of the five-loop primitive (Sections 10.7, 18.5); $\beta_\ell = a_\ell + g$ (460); the free shifts β_i of (469) (Sections 20.1.2, 21.5); β_k, β_n, digamma sequences (Sections 18.7, 18.7.8); β_f (Section 18.8); β_7, β_σ (Section 1.2); the bubbles β_{ij} (Section 28).
γ	$\gamma_r(g)$ (Section 2); α, β, γ, constants of the five-loop primitive (Section 10.7); Euler's constant γ_E (Section 8), written γ in Section 18.7.9; γ_C (Part III families); the Fourier coefficients γ_m (Sections 20.1.1, 20.1.2, 21.6); γ, a divergent subgraph (Sections 28, 32.7).
η	η_m, polygamma combinations (Part III families); $\eta_r = 1/(8\pi r)$, the imaginary offset of the nodes (312) (Sections 20.1.3, 20.1.4); η and η_{R} (571), decay exponents of the growth bounds (Sections 18, 20.1, 20.1.1, 21.4); $\eta_{7,\pm}$, parameters of $\mathcal{G}_7$ (Section 1.2); η, the remainder of (728); a local real variable in Sections 18.7.7, 18.7.9, 18.8, 18.9.4, 18.9.8, 21.2 and 24.1.
θ	θ_m and the θ_r of (275), polygamma combinations (Part III families); $\theta = \theta_0 = 2q$ or $\theta_1 = 2q - 1$, the index of the measure block M_θ (Section 17.4); $\theta = -\frac{1}{4}$, a parameter of a complex Selberg integral (Section 15); $\theta_i(z)$, Stirling remainders (Section 21.6); θ_{cv}, a control coefficient (Section 26.14); elsewhere an angle.
ϑ	the auxiliary coordinate of the edge exponents α_e (Section 34.6.5); an angle, with $f_\vartheta = \mathbf{K}(\sin\vartheta)$ and $g_\vartheta = \mathbf{K}(\cos\vartheta)$ (Section 18.7.8), and a local angle in Section 18.8; $\vartheta_\tau(x)$, a Gamma ratio (Section 18.9.2); $\vartheta = 2\psi_1(q) - \psi_1(2q)$ (Section 21.10).
κ	$\kappa(d, z)$ (174); $\kappa_{\theta,j}$, κ_j^0 (280); $\kappa(S)$, a nominal endpoint power (Section 26.11); local exponents, partitions and constants in Sections 18.8, 18.7.6, 18.7.7, 19.6, 20.1.2, 20.2, 20.5 and 22.9. The exponent $\varkappa$ (Sections 20.1.2, 21.6) is a different letter.

Symbol	Meaning
λ	λ_m and λ_j, the same polygamma combinations, continued as $\lambda_j(x)$ (Part III families); λ_j, local to the wheel-tail paragraph of Section 19.10; λ_k, the Stirling coefficients (583); λ_e, the completion exponents (Sections 7, 16.3); the bare couplings λ_0 (Section 15), λ_B and λ (Sections 34.2, 34.5); λ_*, a cubic root (Sections 1.2, B.2.5); $\lambda^2_{\phi\phi\mathcal{E}}$, a defect OPE coefficient (Part V); scale factors in Part IV.
τ	the regulator of the master path, $t = 2\tau$, and of R_τ, S_τ, $C(\tau)$ (Part III families); $\tau_n(\nu)$, $\tau_n^{(d)}(\nu)$, the zig-zag spectra (Sections 1.1, 10, 15); τ, a Taylor coefficient of $\mathfrak{G}$ (Section 8), written ϱ in Section 18.9.1; $\tau_*(q)$ (571); local parameters in Sections 15, 16.7 and 18.7.6, and the sums τ_n of Section B.4.
ϕ, φ, Φ	the field ϕ (Sections 1.4, 34.2) and the bulk field Φ (Part V); $\phi(x) = \Gamma(\frac{1}{4}+x)/\Gamma(\frac{3}{4}+x)$ (Section 18.9.2); the angle ϕ (Section 18.7.9); $\phi(R, R')$ (Section 18.8); $\varphi_x(s)$ (345), $\varphi_\tau(y)$ (Section 18.9.8), $\varphi(z)$, $\bar\varphi(z)$ (501) and φ_m (Section 20.6); Φ_2 (Section 17.2), $\Phi_t(m)$, $\widehat\Phi$ (Sections 21.4–21.8), Φ_-, Φ_0, Φ_+ (Section 22.9) and $\Phi_{\le 4}$ (Section 32.7); local meanings in Sections 5, 18.7 and 20.1.4.
ψ, Ψ	ψ_m, the polygamma functions (first block), written $\psi = \psi_0$ in Sections 28, 34.6.1 and B.5; in the proof of the twofold-convolution lemma of Section 18.7.9, $\psi = \theta - \frac{\phi}{2}$ and $\psi_0 = \pi - \frac{\phi}{2}$ are angles; $\Psi_\sigma(s)$ (Section 1.2), $\Psi(\sigma)$ (Section 18.7.9), $\Psi_{r,m}(v; b)$ (Section 20.6) and $\Psi_t(n)$, $\widehat\Psi$ (Sections 21.4–21.8).
ω, Ω	$\omega(\rho)$, the constant of the Gamma-ratio bound (465) (Sections 20.1, 21.6), against ω_j, the sine shifts of the top-mode proof (Section 20.1.2); $\omega = w(\kappa)+\log 2$ (Section 18.7.7); ω_τ (Section 18.8); $\omega = \pm 1$ (Section 19); ω_b, ω_c (Appendix A); $\Omega_r(t)$, $\Omega(t; x)$, $\Omega_r(t; x)$, Ω_{R}, periodized wheels (Sections 18.5, 18.7, 20.6, 21.9); the simplex measure $d\Omega$ (Part IV).
sans-serif letters	local to the section that defines them, except A, D, T, U (Section 1.1), B and $\mathsf{G} = \mathsf{G}$. Part V uses H_j, e_j, f_j, I_j, S_n, w_n, L_n, J_d, Y, C, $\mathsf{K}_\times$, $\mathsf{K}_\parallel$, $\mathsf{c}_\times$, a_n, r_1, r_2, $\mathsf{F}_{\mathrm{sch}}$, $\mathsf{f}_k^{\mathrm{sch}}$, to keep them apart from h_d, $B_\pm$, ℓ_j, W_r, $\mathcal{L}_d$, $\mathcal{J}$, H and C_n. Part III uses its own in Sections 17–21, for example c_j (345), X_r, r_k, s_k, P_τ, F_i, E, N_q and the sequences x_m, y_j, z_n (486); Section 28 uses P_2, P_3, p_2, p_3, and Appendix B the matrices M, N. Clashes: the strip T (Section 18.7.7) and the transpose T have the glyph of $\mathsf{T} = \psi_1(q)$; the arrays $\mathsf{A}_{M,\alpha}$, $\mathsf{B}_{M,\alpha}$ (298) and $\mathsf{U}_\pm$, $\mathsf{V}_\pm$ (Section 18.4) are not A, B, U; J_0, J_1 (Section 18.7.6) are not J_d; a_j (Section 18.7.6) is not a_n; r_1, r_2 are not r_k; and P_x, Q_x (Section 18.7.9) are functions, not the pole sets P_T, P_τ (Sections 20.1.2, 21.4).

For reference,

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}, \quad B(u,v) = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}, \quad (a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}.$$

Our ordinary hypergeometric convention is

$${}_pF_q\left(\begin{matrix}a_1,\dots,a_p\\ b_1,\dots,b_q\end{matrix}; x\right) = \sum_{n=0}^{\infty} \frac{\prod_i (a_i)_n}{\prod_j (b_j)_n} \frac{x^n}{n!}.$$

For $p = q + 1$, positive real parametric excess ensures absolute convergence at $x = 1$ in the nonsingular cases used here [22]. Repeated parameters may not be cancelled along a deformation

unless they agree as functions of that deformation, not merely at its evaluation point. Contours, endpoint subtractions and continuation rules are part of the master definitions, not optional numerical conventions. In particular, $1/4 \mapsto d/4$ alone does not generalize a one-dimensional formula: $3d/4$ and $1 - d/4$ coincide only at $d = 1$.

Evidence and reproducibility. The derivations and definitions below specify the retained calculation. Published lower-order results and general mathematical identities are cited separately from this project's higher-loop claims. Algebraic or master-level agreement does not independently validate every graph. The public repository of this work contains the code, graph arrays, numerical inputs and records behind every quoted number; Section 33 gives the package map, in which package paths are relative to the folder `GFF/code/` of that repository, and the provenance of every merged block. These files supply reproducibility data, not unstated definitions in the formulas.

2 Renormalization-group derivation

Normalize the free covariance and perturbations by

$$\langle \phi(x)\phi(0)\rangle_0 = |x|^{-h}, \qquad h = \frac{1-\epsilon}{2}, \qquad S_{\rm int} = \int dx \left(\frac{g_0}{4!}\phi^4 + \frac{r_0}{2}\phi^2\right). \tag{3}$$

After subtraction of the critical local mass power divergence, the long-range kinetic term has no wave-function counterterm. This is the field-theory framework of Ref. [7]. In the mass-anomalous-dimension convention used here,

$$\beta_{\rm RG}(g) = -\epsilon g + \sum_{L\geq 1} b_L g^{L+1}, \qquad \gamma_r(g) = \sum_{L\geq 1} c_L g^L, \tag{4}$$

$$\beta_{\rm RG}(g_*) = 0, \qquad \Delta_{\mathcal{E}} = 1 - \nu^{-1} = \frac{1-\epsilon}{2} + \gamma_r(g_*). \tag{5}$$

Here b_4, c_4 are *loop coefficients*, whereas C_4, C_5 are coefficients in the physical ϵ expansion. None is Euler's beta function $B(u,v) = \Gamma(u)\Gamma(v)/\Gamma(u+v)$.

2.1 Lagrange inversion

Put

$$\mathcal{B}(g) = b_1 + b_2 g + b_3 g^2 + b_4 g^3 + b_5 g^4 + \cdots, \qquad \epsilon = g_* \mathcal{B}(g_*). \tag{6}$$

Then, for $n \geq 1$,

$$\boxed{C_n = -\tfrac{1}{2}\delta_{n1} + \frac{1}{n}[g^{n-1}]\frac{\gamma_r'(g)}{\mathcal{B}(g)^n}.} \tag{7}$$

This follows directly by applying Lagrange–Bürmann inversion to $\gamma_r(g_*(\epsilon))$. Equivalently, for $g_* = \sum_j u_j \epsilon^j$,

$$\begin{aligned} &u_1 = b_1^{-1}, && u_2 = -b_2 b_1^{-3}, \\ &u_3 = 2b_2^2 b_1^{-5} - b_3 b_1^{-4}, \\ &u_4 = -5b_2^3 b_1^{-7} + 5b_2 b_3 b_1^{-6} - b_4 b_1^{-5}, \\ &u_5 = 14 b_2^4 b_1^{-9} - 21 b_2^2 b_3 b_1^{-8} + (6b_2 b_4 + 3b_3^2) b_1^{-7} - b_5 b_1^{-6}. \end{aligned} \tag{8}$$

The coefficients through three loops in this normalization are

$$\begin{aligned} &b_1 = 3, && c_1 = 1, && b_2 = -6A, && c_2 = -A, \\ &b_3 = 6Q + \frac{87}{4}A^2 - \frac{3}{4}\pi^2 - 4\pi, && c_3 = \frac{7}{2}A^2 - \frac{1}{6}\pi^2 - \frac{4}{3}\pi. \end{aligned} \tag{9}$$

Consequently,

$$\boxed{C_4 = \frac{60A^3 - 8Ab_3 + 18Ac_3 + 3c_4 - b_4}{243},} \tag{10}$$

and

$$\boxed{\begin{aligned} C_5 = \frac{1}{729}\big[&336A^4 - 60A^2b_3 + 108A^2c_3 + b_3^2 - 3b_3c_3 \\ &- 10Ab_4 + 24Ac_4 + 3c_5 - b_5\big]. \end{aligned}} \tag{11}$$

At fifth order the new combination is $D_5 = 3c_5 - b_5$. The four-loop combination $D_4 = 3c_4 - b_4$, sufficient for C_4, is not sufficient by itself for C_5. Extending a four-loop fixed-point series without five-loop input would miss $D_5/729$.

3 The complete graph and subtraction prescription

This section fixes the full four- and five-loop calculation, including nonprimitive graphs. Write z for an auxiliary regulator, equal to ϵ on the physical regulator path, and set $h = (1 - z)/2$.

3.1 Ordered graphs and exact weights

For loop order L, let $\mathfrak{G}_L$ be the set of symmetric edge-multiplicity matrices on $0, \dots, L$ satisfying

$$\begin{aligned} &m_{ii} = 0, \quad m_{ij} \in \{0, 1, 2, 3\}, \quad \sum_{i<j} m_{ij} = 2L, \quad d_i := \sum_j m_{ij} \in \{2, 3, 4\}, \\ &\sum_{\substack{i \in S \\ j \notin S}} m_{ij} \geq 2 \quad \text{for every nonempty proper vertex subset } S. \end{aligned} \tag{12}$$

The cut condition expresses connectedness and one-particle irreducibility. Put $e_i = 4 - d_i$, so $\sum_i e_i = 4$, and use the rational weights

$$\boxed{w_4(m) = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_2(m) = \frac{w_4(m)}{12} \sum_i e_i(e_i - 1).} \tag{13}$$

The mass insertion can be represented by choosing two external half-edges at the same vertex of a quartic graph, which yields the second weight. All ordered vertex labels are retained. The perturbative vertex factorial cancels the sum over orderings; there is no additional factor $(L + 1)!$ in the following ordered integrals. The enumerated counts for $L = 1, \dots, 5$ are

$$|\mathfrak{G}_L| = 1,\ 6,\ 85,\ 1890,\ 59145. \tag{14}$$

These are not counts of isomorphism classes.

A zero-dimensional benchmark of the weights. The weights (13) can be checked without enumerating graphs. In zero dimensions put

$$M_j(g) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^j \exp\Big[-\frac{x^2}{2} - \frac{g}{24}(x^4 - 6x^2 + 3)\Big]\, dx;$$

the normal-ordered interaction removes tadpoles. With r coupled to $x^2/2$, the Legendre transformation gives the amputated one-particle-irreducible four-point function and the mass insertion,

$$\Gamma^{(4)} = 3\frac{M_0^2}{M_2^2} - \frac{M_4 M_0^3}{M_2^4}, \qquad \partial_r \Gamma^{(2)}\big|_{r=0} = \frac{M_0 M_4 - M_2^2}{2M_2^2}, \tag{15}$$

whose Gaussian moments give the exact series

$$\Gamma^{(4)} = g - \tfrac{3}{2}g^2 + \tfrac{15}{4}g^3 - \tfrac{105}{8}g^4 + \tfrac{893}{16}g^5 - \tfrac{8795}{32}g^6 + \tfrac{97987}{64}g^7 - \tfrac{10951025}{1152}g^8 + \tfrac{49985915}{768}g^9 - \cdots,$$
$$\partial_r\Gamma^{(2)} = 1 - \tfrac{1}{2}g + \tfrac{3}{4}g^2 - \tfrac{49}{24}g^3 + \tfrac{343}{48}g^4 - \tfrac{2875}{96}g^5 + \tfrac{83711}{576}g^6 - \tfrac{2763929}{3456}g^7 + \tfrac{33994099}{6912}g^8 - \cdots.$$

At every loop order the weights obey

$$\sum_{m\in\mathfrak{G}_L} w_4(m) = (-1)^L(L+1)!\,[g^{L+1}]\Gamma^{(4)}, \qquad \sum_{m\in\mathfrak{G}_L} w_2(m) = (-1)^L(L+1)!\,[g^L]\,\partial_r\Gamma^{(2)}. \tag{16}$$

A class G with automorphism group $\mathrm{Aut}(G)$ has $(L+1)!/|\,\mathrm{Aut}(G)|$ vertex labelings, so the class weights (754) sum to $(-1)^L[g^{L+1}]\Gamma^{(4)}$ and $(-1)^L[g^L]\partial_r\Gamma^{(2)}$. For $L = 1,\dots,5$ the ordered sums are 3, $\frac{45}{2}$, 315, $\frac{13395}{2}$, $\frac{395775}{2}$ and 1, $\frac{9}{2}$, 49, $\frac{1715}{2}$, $\frac{43125}{2}$; the five- and six-loop sums of the enumeration in the table after (597) are those of (16). A second enumerator, which builds the adjacency matrices row by row and tests connectedness and the cut condition of (12) by a bridge search, rather than building the graphs edge by edge and testing every cut, reproduced every ordered-label graph and every rational weight (13); its record is not part of the repository. The class inventories of the eighth-order calculation for $L = 1,\dots,8$ (Section 30.2), whose loops one to seven reproduce those of the seventh-order calculation, satisfy the class form exactly (`LRI_C8_HPC/gen/zero_dim_check.py`, with its output in `LRI_C8_HPC/inputs8/verification/logs/c8gen_fin_54591916.out`). The zero-dimensional integral is a combinatorial test of the weights, not an approximation to a critical exponent.

With $x_0 = 0, x_L = 1$, define

$$\begin{aligned} \mathcal{J}_m(z) &= \int_{0<x_1<\cdots<x_{L-1}<1} \prod_{i<j}(x_j - x_i)^{-hm_{ij}}\, d^{L-1}x, \\ I_m(z) &= \frac{\mathcal{J}_m(z)}{Lz}, \qquad \mathcal{A}_L(z) = \sum_m w_4(m) I_m(z), \qquad \mathcal{B}_L(z) = \sum_m w_2(m) I_m(z). \end{aligned} \tag{17}$$

The factor $1/(Lz)$ is the overall scale integral with the total vertex span cut off at one. Translation has been removed. The mass power divergence is treated by analytic continuation and critical mass subtraction, rather than by discarding the associated nontrivial graphs.

3.2 Minimal subtraction

The bare amplitudes are

$$F_4(v) = v + \sum_{L\ge1}(-1)^L\mathcal{A}_L(z)v^{L+1}, \qquad F_2(v) = 1 + \sum_{L\ge1}(-1)^L\mathcal{B}_L(z)v^L, \tag{18}$$

with $\mathcal{A}_1 = 3/z$, $\mathcal{B}_1 = 1/z$. Write

$$g_0 = \mu^z\mathcal{Z}_g(g,z), \quad r_0 = \mu^s r Z_r(g,z), \quad \mathcal{Z}_g = g + \sum_{n\ge2} G_n(z)g^n, \quad Z_r = 1 + \sum_{n\ge1} Z_n(z)g^n. \tag{19}$$

The coefficients G_n, Z_n contain only pole terms. If PP selects negative Laurent powers of z, finiteness of $F_4(\mathcal{Z}_g)$ and $Z_rF_2(\mathcal{Z}_g)$ gives

$$G_n = -\,\mathrm{PP}[g^n]F_4(\mathcal{Z}_{g,<n}), \qquad Z_n = -\,\mathrm{PP}[g^n]\,[Z_{r,<n}F_2(\mathcal{Z}_g)]. \tag{20}$$

The RG coefficients follow from the simple poles:

$$\boxed{b_L = L[z^{-1}]G_{L+1}, \qquad c_L = L[z^{-1}]Z_L.} \tag{21}$$

Equivalently, the pole equations are

$$z\mathcal{Z}_g + \beta_{\mathrm{RG}}\partial_g\mathcal{Z}_g = 0, \qquad \beta_{\mathrm{RG}}\partial_g Z_r + \gamma_r Z_r = 0. \tag{22}$$

They imply the useful recursions

$$G_n = \frac{\sum_{\ell=1}^{n-1} b_\ell (n-\ell) G_{n-\ell}}{(n-1)z}, \qquad Z_n = \frac{\sum_{\ell=1}^{n} [(n-\ell)b_\ell + c_\ell] Z_{n-\ell}}{nz}, \tag{23}$$

where $G_1 = Z_0 = 1$.

3.3 Primitive versus nonprimitive: the precise distinction

A primitive diagram has an overall ultraviolet divergence and no divergent proper subdiagram. A nonprimitive diagram contains subdivergences and must be evaluated together with its counterterms. For example, a two-edge bubble has the local integral

$$\int_0^\ell r^{-1+z}\, dr = \frac{1}{z} + \log \ell + O(z).$$

If it occurs inside a larger graph, its pole subtraction multiplies the quotient graph. In particular,

$$\left(-\frac{p}{z}\right)\left(\frac{q_{-1}}{z} + q_0 + O(z)\right) = -\frac{pq_{-1}}{z^2} - \frac{pq_0}{z} + O(1),$$

so the larger graph's new simple pole depends on the lower-loop *finite part* q_0. Primitive periods alone do not determine it. The forest formula implements these products systematically [21].

The phrase "remaining nonprimitive" used during the calculation also included the lower-loop fixed-point terms. Precisely,

$$C_5 = C_5^{\text{primitive}} + \frac{\mathcal{R} - 10Ab_4 + 24Ac_4 + 3c_5^{\text{nonprim}} - b_5^{\text{nonprim}}}{729},$$

where $\mathcal{R} = 336A^4 - 60A^2 b_3 + 108A^2 c_3 + b_3^2 - 3b_3 c_3$. The later separation into primitive, wheel-plus-assembly, and rest sectors is stated in Section 13. Those pieces are bookkeeping within the chosen normalization; the universal object is their sum.

3.4 Laurent orders needed for the fifth coefficient

Let $a_{L,k} = [z^k]\mathcal{A}_L$ and $d_{L,k} = [z^k]\mathcal{B}_L$. Expanding (20) gives

$$\begin{aligned}
b_4 &= 36Aa_{2,0} - 216a_{2,1} + 48a_{3,0} - 4a_{4,-1}, \\
c_4 &= 26Ad_{2,0} - 140d_{2,1} + 40d_{3,0} - 4d_{4,-1}, \qquad (24) \\
b_5 &= 5a_{5,-1} - 75a_{4,0} + 450a_{3,1} - 60Aa_{3,0} - 1350a_{2,2} + 585Aa_{2,1} - 5b_3 a_{2,0}, \\
c_5 &= 5d_{5,-1} - 65d_{4,0} + 325d_{3,1} - \frac{95}{2}Ad_{3,0} - \frac{2275}{3}d_{2,2} \\
&\quad + \frac{725}{2}Ad_{2,1} - \frac{10b_3 + 5c_3}{3}d_{2,0}. \qquad (25)
\end{aligned}$$

The lower-loop data must therefore be retained through

Loop order	Required Laurent powers in $\mathcal{A}_L, \mathcal{B}_L$
2	$z^{-2}, z^{-1}, 1, z, z^2$
3	$z^{-3}, z^{-2}, z^{-1}, 1, z$
4	$z^{-4}, z^{-3}, z^{-2}, z^{-1}, 1$
5	$z^{-5}, z^{-4}, z^{-3}, z^{-2}, z^{-1}$

The finite four-loop terms $a_{4,0}, d_{4,0}$ are new inputs relative to the computation of C_4. Simplifying C_4 alone cannot determine them.

What the higher-pole identities cannot detect. The pole equations (22) fix the higher poles of G_n and Z_n from the lower-order simple poles. They do not fix a new simple pole. For example, the shift

$$b_5 \longmapsto b_5 + \delta b_5, \qquad \mathcal{Z}_g \longmapsto \mathcal{Z}_g + \frac{\delta b_5}{5z}\, g^6$$

preserves $z\mathcal{Z}_g + \beta_{\mathrm{RG}}\partial_g \mathcal{Z}_g = 0$ through g^6: the three new terms are $\frac{1}{5}\delta b_5 g^6$, $-\frac{6}{5}\delta b_5 g^6$ and $\delta b_5 g^6$. It changes the physical coefficient by $\delta C_5 = -\delta b_5/729$. More generally $\delta C_5 = (3\delta c_5 - \delta b_5)/729$ when the lower loops are held fixed. Agreement of the higher poles with their renormalization-group values, used as a check in Sections 14 and 22.8 and at seventh and eighth order, therefore tests the lower-order inputs and the pipeline but cannot detect missing new simple-pole information. This caveat was written by OpenAI GPT-6 astra in the research archive behind the sixth-order packages (package `LRI_C6_D2_HPC/reference/`).

Part II

One-dimensional analytic coefficients through fifth order

4 Final one-dimensional formulas

We use

$$\epsilon = 2s - 1, \qquad \Delta_\phi = \frac{1-\epsilon}{4}, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + C_2\epsilon^2 + C_3\epsilon^3 + C_4\epsilon^4 + C_5\epsilon^5 + O(\epsilon^6).$$

Here $\mathcal{E} = [\phi^2]$ is the thermal operator. The covariance is $\langle\phi(x)\phi(0)\rangle_0 = |x|^{-(1-\epsilon)/2}$, and the interaction is $g\phi^4/4!$. Set

$$A = \pi + 4\log 2, \qquad Q = \frac{\Gamma(1/4)^4}{\pi}, \qquad H = \frac{2\pi^2}{3} + 8\mathsf{G}, \qquad \mathsf{G} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}. \tag{26}$$

Thus G is Catalan's constant. It is not a graph amplitude.

The complete reduction obtained from the graph calculation is

$$\boxed{\begin{aligned} 46656C_5 = {} & -1536(P_5 + \mathcal{K}) + 384AP_4 - 576\mathcal{L} - 1728Q^2 \\ & + 48Q(15A^2 + 149H + 7\pi^2) - 15A^4 - 30A^2\pi^2 \\ & - 828A\pi^3 - 21456A\zeta(3) - 83\pi^4. \end{aligned}} \tag{27}$$

The four constants $P_4, P_5, \mathcal{L}, \mathcal{K}$ are evaluated in Sections 8–10; their graph-normalized definitions and complete reductions are included. The preferred fixed-parameter presentation of P_5 is given next.

4.1 The five fixed hypergeometric values for P_5

For $r \in \{1, 2, 3, 5, 7\}$ define

$$\boxed{\begin{aligned} \mathcal{Y}_r = {} & \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 \\ & \times {}_9F_8\left(\begin{matrix} 1,\ 1+r/8,\ ((r+2)/8)^{\times 7} \\ r/8,\ ((r+6)/8)^{\times 7} \end{matrix}; -1\right). \end{aligned}} \tag{28}$$

Put $c_8 = \cos(\pi/8)$ and $s_8 = \sin(\pi/8)$. Then

$$\boxed{\begin{aligned} P_5 = 16\pi^{3/2}\Big\{ &-35\mathcal{Y}_2 \\ &+ (22\sqrt{2}-13)(c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5) \\ &+ (22\sqrt{2}+13)(c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7)\Big\}. \end{aligned}} \tag{29}$$

Every parameter is fixed and positive; all arguments are the ordinary -1. Each series has excess $3/2$ and is absolutely convergent. For $r = 2$ one matching numerator/denominator parameter 1 cancels, so

$$\mathcal{Y}_2 = \frac{\pi^{7/2}}{4}\,{}_8F_7\left(\begin{matrix}\frac{5}{4}, (\frac{1}{2})^{\times 7} \\ \frac{1}{4}, 1^{\times 6}\end{matrix}; -1\right). \tag{30}$$

Thus (29) uses four ${}_9F_8$ values and one ${}_8F_7$ value. It has no parameter derivative, Taylor-coefficient extraction, coalescence limit, extra summation, multivariate function, or unstated analytic-continuation sheet. Section 10.7 proves the identity using a regularized bilateral sum with exactly three Fourier harmonics. Inserting it in the polygamma form of C_5 makes the complete result an ordinary fixed-parameter hypergeometric expression. No elementary evaluation or minimality of this five-value basis is asserted.

4.2 The reflected-quarter polygamma basis

The following is the derivative-free form of every coefficient $C_0, \ldots, C_5$. The evaluation of $\mathcal{K}$ used here is proved in Section 8.3, and the wheel reduction in Section 9.2.

4.2.1 A basis adapted to $1/4$ and $3/4$

Put $q = 1/4$ and $\bar{q} = 1 - q = 3/4$. With the convention $\psi_m(z) = d^{m+1}\log\Gamma(z)/dz^{m+1}$, define

$$\boxed{\begin{aligned} &\mathsf{A} = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), && \mathsf{D} = \psi_0(\bar{q}) - \psi_0(q), \\ &\mathsf{T} = \psi_1(q), && \overline{\mathsf{T}} = \psi_1(\bar{q}), \\ &\mathsf{U} = \psi_2(q), && \overline{\mathsf{U}} = \psi_2(\bar{q}). \end{aligned}} \tag{31}$$

An overline here means evaluation at the reflected argument, not complex conjugation. There is no differentiation with respect to q in the coefficient formulas. At $q = 1/4$, the exact dictionary is

$$\begin{aligned} \mathsf{A} &= A, & \mathsf{D} &= \pi, \\ \mathsf{T} + \overline{\mathsf{T}} &= 2\pi^2, & \mathsf{T} - \overline{\mathsf{T}} &= 16\mathsf{G}, \\ \mathsf{U} + \overline{\mathsf{U}} &= -112\zeta(3), & \overline{\mathsf{U}} - \mathsf{U} &= 4\pi^3, \\ H &= \frac{5\mathsf{T} - \overline{\mathsf{T}}}{6}, & \log 2 &= \frac{\mathsf{A} - \mathsf{D}}{4}. \end{aligned} \tag{32}$$

These follow from the polygamma reflection and multiplication identities [32, 33]. An entirely quarter-argument version of A, apart from the reference value at 1, is

$$\mathsf{A} = \frac{4\psi_0(1) - 5\psi_0(q) + \psi_0(\bar{q})}{3} \qquad (q = 1/4).$$

We retain the gamma normalization Q and the ordinary periods P_4, P_5. For clarity, the wheel normalizations are

$$B_- = \Gamma(q)^4 F_-, \qquad B_+ = \frac{64\pi^4}{\Gamma(q)^4}F_+, \qquad P_4 = 13B_- - 3B_+.$$

If desired as a uniform notation, $Q = \Gamma(q)^4/\mathsf{D}$ and $B_+ = 64\mathsf{D}^4F_+/\Gamma(q)^4$ at $q = 1/4$. This is not an evaluation of Q or the periods in polygamma values alone.

4.2.2 The two master reductions

The proved polygamma evaluation and the digamma-prefactor wheel identity are

$$\boxed{\begin{aligned}\mathcal{K} &= \frac{1}{2}\left[\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})+\mathsf{AU}\right],\\ \mathcal{L} &= -\frac{3\mathsf{A}-\mathsf{D}}{6}P_4-\frac{2\mathsf{D}}{3}B_-.\end{aligned}} \tag{33}$$

Equivalently,

$$\mathcal{L} = -\frac{13\mathsf{A}-3\mathsf{D}}{2}B_- + \frac{3\mathsf{A}-\mathsf{D}}{2}B_+.$$

The first identity eliminates the entire six-${}_3F_2$ parameter derivative in the definition of $\mathcal{K}$. The second removes the parameter derivative in the definition of the logarithmic wheel but does not eliminate the two wheel periods $B_\pm$. Both are equalities to the original master definitions, not merely equalities among proposed right-hand sides.

4.2.3 C_0 through C_5

Writing $\Delta_{\mathcal{E}} = \sum_{n=0}^{5} C_n\epsilon^n + O(\epsilon^6)$, the additional coefficient formulas are

$$\boxed{\begin{aligned}&C_0 = \frac{1}{2}, && C_1 = -\frac{1}{6}, \quad C_2 = \frac{\mathsf{A}}{9},\\ &C_3 = \frac{\mathsf{A}^2+(\mathsf{T}+\overline{\mathsf{T}})/6-8Q}{108}.\end{aligned}} \tag{34}$$

$$\boxed{\begin{aligned}648C_4 = 16P_4 - \mathsf{A}^3 - \frac{\mathsf{A}}{2}(\mathsf{T}+\overline{\mathsf{T}}) - 8\mathsf{A}Q\\ + \frac{365\mathsf{U}-6\overline{\mathsf{U}}}{14}.\end{aligned}} \tag{35}$$

$$\boxed{\begin{aligned}46656C_5 = &-1536P_5 + 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_- - 1728Q^2\\ &+16Q(45\mathsf{A}^2+383\mathsf{T}-64\overline{\mathsf{T}}) - 768\mathsf{T}(3\mathsf{T}-\overline{\mathsf{T}})\\ &-15\mathsf{A}^4 - 15\mathsf{A}^2(\mathsf{T}+\overline{\mathsf{T}})\\ &-\frac{6\mathsf{A}}{7}(431\mathsf{U}+18\overline{\mathsf{U}}) - \frac{83}{4}(\mathsf{T}+\overline{\mathsf{T}})^2.\end{aligned}} \tag{36}$$

Section 8.3 proves the master identity used in this substitution. All explicit logarithmic, zeta, Catalan, and power-of-π coefficients in these expressions are written using fixed-argument ψ_0, ψ_1, ψ_2. The non-polygamma input consists of the gamma normalization Q, the two wheel periods, and P_5; their original definitions and normalizations have not changed.

For checking (36), the wheel term is exactly

$$384AP_4 - 576\mathcal{L} = 96(7\mathsf{A}-\mathsf{D})P_4 + 384\mathsf{D}B_-.$$

An exact symbolic check (`lri_psi_complete_update/verification/symbolic_checks.json`) compares each C_n with the formulas of Section 4, in particular (27), after substituting (32) and (33). In particular, no arithmetic inference is based on fitting decimal values of the periods.

5 The full-line Selberg formula used in the expansions

For

$$M(u,v,w)=\iint_{\mathbb{R}^2}|xy|^{-u}|(1-x)(1-y)|^{-v}|x-y|^{-w}\,dx\,dy,$$

let

$$U=S_2(1-u,1-v,-w/2),\qquad c=\cos(\pi w/2), \tag{37}$$

$$\theta_u=\pi(u+w/2),\quad \theta_v=\pi(v+w/2), \tag{38}$$

$$T=\pi(u+v+w),\qquad \Phi=\pi(u+v+w/2), \tag{39}$$

where

$$S_2(a,b,c)=\frac{\Gamma(a)\Gamma(b)\Gamma(a+c)\Gamma(b+c)\Gamma(1+2c)}{\Gamma(a+b+c)\Gamma(a+b+2c)\Gamma(1+c)}.$$

The real-line chamber sum is

$$\begin{aligned}M=U\Bigg[1-\frac{2c(\sin\theta_u+\sin\theta_v)}{\sin T}\\ +\frac{\sin\theta_v\sin(\pi v)+\sin\theta_u\sin(\pi u)+2c\sin\theta_u\sin\theta_v}{\sin T\sin\Phi}\Bigg].\end{aligned} \tag{40}$$

Apparent removable singularities are treated in the whole expression. The elementary Selberg evaluation is standard [13]; the real-line chamber weighting follows by splitting each integration variable into $(-\infty,0),(0,1),(1,\infty)$ and mapping the resulting chambers to the Selberg square; the trigonometric factors are fixed by analytic continuation of those chamber integrals. The formula is stated in full so that no separate chamber table is an input. This formula supplies every symmetric two-variable expansion used here.

6 Root and subtraction conventions

For an L-loop graph the rooted bare integral has the overall factor $2/(Lz)$, with the two root positions fixed at zero and one and their separation integrated with a unit infrared cutoff. Roots must not both lie in a proper superficially divergent collision set. In the code, the root-safety test includes power-divergent and non-one-particle-irreducible collision sets as well as logarithmic ones. This avoids spurious infrared poles in a rooted integral. The counterterm forest itself still contains the proper logarithmic one-particle-irreducible subgraphs; induced contractions that create tadpoles vanish in the stated prescription.

In a quotient graph, the roots are inherited from the parent graph. Choosing arbitrary new quotient roots before subtracting can change finite terms and is not a legitimate shortcut. Counterterms of proper subgraphs can be computed with their own safe roots, since their local pole coefficients are independent of that infrared choice. These distinctions were enforced when producing the residue certificate and the numerical comparisons.

7 Why the remaining diagrams need only three second-order expansions

The nonprimitive graph calculation requires only the following three second-order master expansions.

A vertex connected to two other vertices can be integrated using

$$\mathsf{B}(u,v)=\int_{\mathbb{R}}|x|^{-u}|1-x|^{-v}\,dx=\sqrt{\pi}\,\frac{\Gamma((1-u)/2)\Gamma((1-v)/2)\Gamma((u+v-1)/2)}{\Gamma(u/2)\Gamma(v/2)\Gamma((2-u-v)/2)}. \tag{41}$$

It replaces its two edges by one edge of power $u+v-1$. A block with two internal vertices and two boundary vertices reduces to

$$\mathcal{J}(a,b,c,d,e)=\iint_{\mathbb{R}^2}\frac{dx\,dy}{|x|^a|1-x|^b|y|^c|1-y|^d|x-y|^e}. \tag{42}$$

A generic chamber decomposition of (42) is a finite sum of ordinary ${}_3F_2(1)$ functions. The crucial further simplification is that the full five-loop coefficient does *not* require its arbitrary-parameter form: only its second-order regulator expansions at three critical index patterns occur.

Complete the five active indices to the ten edges on vertices $(0,1,x,y,\infty)$, ordered lexicographically:

$$\begin{aligned}\boldsymbol{\lambda}=(&3-a-b-c-d-e,\ a,\ c,\ b+d+e-1,\ b,\ d,\\ &a+c+e-1,\ e,\ 2-a-b-e,\ 2-c-d-e).\end{aligned} \tag{43}$$

Every completed vertex has incident power sum two. Real-line conformal changes of variables permit permutations of the five completed vertices, followed by fixing three positions to $0,1,\infty$. This gives an exact S_5 action on the five active indices. Zero-edge representatives factor into (41).

After this action and the zero-edge factorizations, the critical representatives needed are

$$T=(1/2,1/2,1/2,1/2,1/2),\quad A_0=(1/2,1/2,1/2,1/2,1),\quad B_0=(1/2,1/2,1/2,1,1/2). \tag{44}$$

The exceptional higher-connectivity graphs are reduced to these same patterns by the regulator-corrected moves in Section 11.

At five loops a nontrivial two-internal-vertex block is multiplied by at most two further simple-pole factors and the overall scale pole. An analytic remainder of total regulator degree at least three cannot then contribute to the simple pole. Consequently, the three second-order expansions below suffice. This order counting also covers the finite lower-loop terms required by the five-loop forests.

7.1 Regular tetrahedron pattern: a gamma-valued Hessian

At T, $\mathcal{J}=3Q$ and all first index derivatives vanish. On the balanced completed variations, the unique permutation-invariant quadratic form is proportional to $\sum_e(\delta\lambda_e)^2$. Evaluation along a symmetric Selberg path fixes it:

$$\boxed{\mathcal{J}=3Q+\frac{QH}{4}\sum_{e=1}^{10}(\delta\lambda_e)^2+O(3).} \tag{45}$$

Here $O(3)$ denotes analytic terms of total regulator degree at least three.

For an explicit check, put

$$M(u,v,w)=\mathcal{J}(u,v,u,v,w).$$

The full-line Selberg evaluation gives

$$M(1/2,1/2,1/2)=3Q,\qquad \nabla M=0,\qquad \frac{\nabla^2 M}{3Q}=H\begin{pmatrix}2&1&1\\1&2&1\\1&1&1\end{pmatrix}. \tag{46}$$

The appearance of Catalan's constant comes from $\psi_1(1/4)=\pi^2+8\mathsf{G}$ and $\psi_1(3/4)=\pi^2-8\mathsf{G}$. The Selberg formula used to establish these identities is recalled in Section 5 [13].

7.2 Pattern A_0: no new regular quadratic constant

Write

$$a = u + p, \quad c = u - p, \quad b = v + q, \quad d = v - q, \quad e = w, \qquad \sigma_0 = 2 - 2u - w, \quad \sigma_1 = 2 - 2v - w.$$

Subtracting the endpoint pole differences from the symmetric integral gives

$$\boxed{\mathcal{J} = M(u, v, w) + R(w)\left(\frac{p^2}{\sigma_0} + \frac{q^2}{\sigma_1}\right) + O(3),} \tag{47}$$

where, through the needed order,

$$R(w) = \pi^3 + 28\zeta(3) - \frac{A}{2}[\pi^3 + 28\zeta(3)](w - 1) + O((w - 1)^2). \tag{48}$$

In (47), the remainder has total scaling degree at least three when all local regulators are scaled together; higher terms from the endpoint pole residues are included in this notation. The diagonal collision residue is independent of p, q and cancels in the difference. The two endpoint residues yield $R(w) = \partial_x^2 \mathsf{B}(x, w)|_{x=(2-w)/2}$.

After the full endpoint pole parts have been removed, symmetry permits only two additional second-order constants, multiplying $p^2 + q^2$ and pq. Both vanish. To determine them, set $u = v = 1/2$, $w = 1 - t$, and use the two unique-star paths

$$p = t,\ q = 0: \quad \mathcal{J} = \mathsf{B}(1/2 + t, 1/2)\mathsf{B}(1 - t, 1 - t), \tag{49}$$

$$p = q = t/2: \quad \mathcal{J} = \mathsf{B}((1 + t)/2, (1 + t)/2)\mathsf{B}(1 - t, 1 - t). \tag{50}$$

Expansion of these exact gamma products through t^2 fixes both regular quadratic constants to zero. The symbolic verification code is not part of the public repository.

7.3 Pattern B_0: exactly one new mixed derivative

Use local coordinates

$$p_1 = 1 - d, \qquad p_2 = 2 - b - d - e, \tag{51}$$

$$z_1 = 3 - a - b - c - d - e, \qquad z_2 = 2 - c - d - e. \tag{52}$$

Equivalently,

$$d = 1 - p_1, \quad b = 1 + p_1 - e - p_2, \quad c = 1 + p_1 - e - z_2, \quad a = e - p_1 + p_2 + z_2 - z_1.$$

The base point has $p_1 = p_2 = z_1 = z_2 = 0$, $e = 1/2$. The only local pole hyperplanes are $p_1 = 0$ and $p_2 = 0$. A convenient complete pole part is

$$\mathcal{P} = \frac{2}{p_1}\mathsf{B}(e + p_2 + z_2 - z_1, 1 - p_2) + \frac{2}{p_2}\mathsf{B}(e, 1 + p_1 - e) - \frac{4}{p_1 p_2}. \tag{53}$$

After subtracting it, the function is jointly analytic in all five local coordinates.

On $z_2 = 0$, the y integration is a unique three-point star. On $z_1 = 0$, the completed graph has a zero edge. These exact restrictions are

$$U_2(z_1) = \mathsf{B}(1 + p_1 - e, 1 - p_1)\mathsf{B}(e + p_2 - z_1, 1 - p_2), \tag{54}$$

$$U_1(z_2) = \mathsf{B}(e - p_1 + p_2 + z_2, 1 + p_1 - e - z_2)\mathsf{B}(1 + p_1 - e - p_2, 1 - p_1), \tag{55}$$

$$U_0 = U_2(0) = U_1(0). \tag{56}$$

Analytic interpolation between the two restrictions leaves only one second-order mixed term. Explicitly,

$$\boxed{\begin{aligned}\mathcal{J} &= U_2(z_1) + U_1(z_2) - U_0 + z_1 z_2 \mathcal{K} \\ &\quad + \frac{2}{p_1}\big[\mathsf{B}(e+p_2+z_2-z_1, 1-p_2) - \mathsf{B}(e+p_2-z_1, 1-p_2) \\ &\qquad - \mathsf{B}(e+p_2+z_2, 1-p_2) + \mathsf{B}(e+p_2, 1-p_2)\big] + O(3).\end{aligned}} \tag{57}$$

The subtracted analytic difference vanishes on both $z_1 = 0$ and $z_2 = 0$, hence is divisible by $z_1 z_2$. Its coefficient at the base point is precisely the one constant $\mathcal{K}$. Any dependence of that coefficient on another local index raises the total degree to at least three and is not needed.

This explains why the many apparent ordinary-hypergeometric parameter derivatives in the remaining graph classes collapse to a single new constant rather than a large independent basis.

8 Evaluation of $\mathcal{K}$

8.1 From the original master to six Euler integrals

In (57), choose

$$p_1 = p_2 = z_1 = z_2 = t, \qquad e = 1/2.$$

Then

$$J_t = \mathcal{J}(1/2, 1/2, 1/2, 1-t, 1/2)$$

and the gamma interpolation becomes

$$J_t^{\Gamma} = \mathsf{B}(1/2, 1-t)^2 + \frac{2}{t}\left[\mathsf{B}(1/2, 1-t) - \mathsf{B}(1/2+2t, 1-t)\right]. \tag{58}$$

Thus

$$J_t - J_t^{\Gamma} = t^2 \mathcal{K} + O(t^3). \tag{59}$$

Integrating x in (42) gives the full-line elliptic star $F(y)$. Fold its three real-line intervals onto $(0, 1)$, obtaining

$$\begin{aligned}J_t = 4\int_0^1 [K(x) + K(1-x)]\big[&x^{-1/2}(1-x)^{t-1} \\ &+ x^{-t}(1-x)^{t-1} + x^{-1/2}(1-x)^{-t}\big]\, dx.\end{aligned} \tag{60}$$

This integral is initially convergent for small positive t; its regulated poles are handled explicitly by multiplication by t^2.

For the beta-weighted elliptic moment,

$$E(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} K(x)\, dx, \tag{61}$$

Euler integration gives

$$E(a, b) = \frac{\pi}{2} B(a, b)\, {}_3F_2\left(\begin{matrix}1/2, 1/2, a \\ 1, a+b\end{matrix}; 1\right). \tag{62}$$

A Thomae transformation gives the alternative form

$$E(a, b) = \frac{\pi}{2}\frac{\Gamma(b)^2}{\Gamma(b+1/2)^2}\, {}_3F_2\left(\begin{matrix}b, b, 1-a \\ b+1/2, b+1/2\end{matrix}; 1\right). \tag{63}$$

These follow from the standard beta transform and unity-argument hypergeometric identities [22]. Apply the appropriate form to each of the six terms in (60). Multiplication by t^2 gives exactly (68). Since

$$V_r(t) = t\mathsf{B}(1/2 + rt, 1 - t),$$

(59) proves the regular fourth-derivative formula (70).

For an additional check, the gamma interpolation alone has

$$[t^2]J_t^\Gamma = \frac{A^4}{6} - \frac{A^2\pi^2}{6} - \frac{A\pi^3}{3} - 12A\zeta(3) - \frac{\pi^4}{180}. \tag{64}$$

The endpoint-subtracted logarithmic coefficient of (60), minus this expression, is a second numerical route to $\mathcal{K}$.

8.2 An analytic representation of the regulated master

All derivatives in this section act on the displayed parameters; the argument of every hypergeometric function remains one. Define

$$f_1(t) = {}_3F_2\left(\begin{matrix}t, t, 1/2\\ 1/2+t, 1/2+t\end{matrix}; 1\right), \qquad f_2(t) = {}_3F_2\left(\begin{matrix}t, t, t\\ 1/2+t, 1/2+t\end{matrix}; 1\right), \tag{65}$$

$$f_3(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, t\\ 1, 1/2+t\end{matrix}; 1\right), \qquad f_4(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, t\\ 1, 1\end{matrix}; 1\right), \tag{66}$$

$$f_5(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, 1/2\\ 1, 3/2-t\end{matrix}; 1\right), \qquad f_6(t) = {}_3F_2\left(\begin{matrix}1/2, 1/2, 1-t\\ 1, 3/2-t\end{matrix}; 1\right). \tag{67}$$

Put

$$\begin{aligned}\widehat{J}(t) = 2\pi\Big\{&\frac{\Gamma(1+t)^2}{\Gamma(1/2+t)^2}[f_1(t) + f_2(t)]\\ &+ \frac{t\sqrt{\pi}\,\Gamma(1+t)}{\Gamma(1/2+t)} f_3(t) + t\Gamma(1+t)\Gamma(1-t) f_4(t)\\ &+ \frac{t^2\sqrt{\pi}\,\Gamma(1-t)}{\Gamma(3/2-t)}[f_5(t) + f_6(t)]\Big\},\end{aligned} \tag{68}$$

and, for $r = 0, 2$,

$$V_r(t) = 2\sqrt{\pi}\,\frac{\Gamma(1/4 - rt/2)\Gamma(1 + t/2)\Gamma(1/4 + (r-1)t/2)}{\Gamma(1/4 + rt/2)\Gamma(1/2 - t/2)\Gamma(1/4 + (1-r)t/2)}. \tag{69}$$

The new constant has the regular parameter-derivative evaluation

$$\boxed{\mathcal{K} = \frac{1}{24}\,\frac{d^4}{dt^4}\Big[\widehat{J}(t) - V_0(t)^2 - 2\big(V_0(t) - V_2(t)\big)\Big]\Big|_{t=0}.} \tag{70}$$

The bracket starts at order t^4. Every gamma factor is regular near zero, and the six hypergeometric series have parametric excess either $1/2$ or $1 - t$. Their defining series and the required derivatives therefore converge near the evaluation point [22]. No singular defining series has to be assigned a value at $t = 0$ in (70).

Two numerical evaluations that do not use the result of the proof below give

$$\mathcal{K} = 39.331569128259039057503187151597861523 04\ldots: \tag{71}$$

a 32-node Cauchy evaluation of the fourth derivative in (70) at 45 working digits, and the endpoint-subtracted logarithmic integral obtained from (60), which agree to about 4.1×10^{-33} (a 16-node

evaluation at 40 digits differs by 3.7×10^{-17}). The hypergeometric evaluation uses gamma-ratio asymptotics and Hurwitz-zeta tail summation, the integral explicitly subtracted endpoint expansions. At 110 working digits the endpoint integral agrees with the proved value (76) to 6×10^{-109} (`lri_psi_complete_update/verification/numerical_checks.json`). Equation (70) is an intermediate analytic representation; the following proof reduces this exact master to fixed polygamma values.

Conventions and special values. We use the derivative-order convention

$$\psi_m(z) = \frac{d^{m+1}}{dz^{m+1}} \log \Gamma(z), \qquad m = 0, 1, 2, \dots. \tag{72}$$

Thus $\psi_0 = \psi = \Gamma'/\Gamma$ is the digamma function, $\psi_1 = \psi'$ is trigamma, and $\psi_2 = \psi''$ is tetragamma. These are ordinary one-variable polygamma functions, not the multivariate functions sometimes denoted by uppercase Ψ_1, Ψ_2. In particular, a square such as $\psi_1(1/4)^2$ means an ordinary product, not $\psi_2(1/4)$. The standard series and rational-argument identities [32, 33] give

$$\begin{aligned}
\psi_0(1) &= -\gamma_E, & \psi_0(1/2) &= -\gamma_E - 2\log 2, \\
\psi_0(1/4) &= -\gamma_E - \frac{\pi}{2} - 3\log 2, & \psi_0(3/4) &= -\gamma_E + \frac{\pi}{2} - 3\log 2, \\
\psi_1(1/4) &= \pi^2 + 8\mathsf{G}, & \psi_1(3/4) &= \pi^2 - 8\mathsf{G}, \\
\psi_2(1/4) &= -2\pi^3 - 56\zeta(3), & \psi_2(3/4) &= 2\pi^3 - 56\zeta(3).
\end{aligned} \tag{73}$$

For example, for $m \geq 1$,

$$\psi_m(z) = (-1)^{m+1} m! \sum_{n=0}^{\infty} (n+z)^{-m-1}.$$

Splitting the positive odd integers into the residue classes 1 and 3 modulo four yields the displayed quarter-argument values. At order two, $\sum(4n+1)^{-3} + \sum(4n+3)^{-3} = 7\zeta(3)/8$ and their difference is $\pi^3/32$; the latter also follows from the reflection formula in [32]. The project constants therefore obey the exact dictionary

$$\boxed{\begin{aligned} A &= \psi_0(1/2) + \psi_0(1) - 2\psi_0(1/4), \\ H &= \psi_1(1/4) - \frac{\pi^2}{3}. \end{aligned}} \tag{74}$$

8.3 The compact evaluation and its proof

The proof starts from the original master path (59) and does not assume a numerical integer relation, the wheel reduction, or an unproved hypergeometric summation. The main step is a bilinear identity for a pair of ordinary ${}_3F_2(1)$ functions. The periodicity argument below includes the growth estimate needed to rule out a nonconstant periodic factor.

8.4 Statement and notation

Write $\ell = \log 2$, and define the parameter functions

$$\begin{aligned}
\mathfrak{F}(t) &= {}_3F_2\left(\begin{matrix} 1/2, 1/2, t \\ 1, 1 \end{matrix}; 1\right), \\
\mathfrak{G}(t) &= {}_3F_2\left(\begin{matrix} t, t, t \\ 1/2+t, 1/2+t \end{matrix}; 1\right).
\end{aligned} \tag{75}$$

These are f_4 and f_2, respectively, in the six-function formula. They are unrelated to the fixed wheel values $F_\pm$ or Catalan's constant G. We will prove

$$\boxed{\begin{aligned}\mathcal{K} &= 128\mathsf{G}^2 + 24\pi^2\mathsf{G} - 4\pi^3\ell - 28(\pi+4\ell)\zeta(3)\\ &= 2\psi_1(1/4)^2 - \pi^2\psi_1(1/4) + \frac{A}{2}\psi_2(1/4).\end{aligned}} \tag{76}$$

The two equalities have different roles: the first evaluates the original master, and the second is the rational-argument polygamma dictionary.

8.5 Reducing the six elliptic moments to two

Keep the *parameter* convention $K(x)$, and put

$$E(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}K(x)\,dx.$$

For real $0 < t < 1$, define

$$\begin{aligned} I_1 &= E(1/2,t), & I_2 &= E(t,1/2), & I_3 &= E(1-t,t),\\ I_4 &= E(t,1-t), & I_5 &= E(1/2,1-t), & I_6 &= E(1-t,1/2).\end{aligned}$$

Then (60) says $J_t = 4\sum_{j=1}^6 I_j$. Set $s_t = \sin(\pi t)$ and $c_t = \cos(\pi t)$. The four exact moment relations are

$$\boxed{\begin{aligned} I_1 &= c_tI_3 + s_tI_4, & I_5 &= s_tI_3 - c_tI_4,\\ I_2 &= s_tI_3, & I_6 &= s_tI_4.\end{aligned}} \tag{77}$$

Here is a contour proof, including the branch signs. Integrate $z^{-1/2}(1-z)^{t-1}K(z)$ on an upper-half-plane contour, using principal powers continued from $0 < z < 1$. The boundary values for $x > 1$ and $x < 0$ are [34, 23]

$$K(x+\mathrm{i}0) = x^{-1/2}\{K(1/x) + \mathrm{i}K(1-1/x)\}, \qquad K(x) = (1-x)^{-1/2}K\!\left(\frac{x}{x-1}\right).$$

The three real intervals contribute

$$-\mathrm{i}I_5 + I_1 - e^{-\mathrm{i}\pi t}(I_3 + \mathrm{i}I_4) = 0.$$

Real and imaginary parts give the first two relations. The analogous contour for $z^{-1/2}(1-z)^{t-1}K(1-z)$ gives

$$-I_5 - \mathrm{i}I_6 + I_2 - e^{-\mathrm{i}\pi t}I_4 = 0,$$

and hence the last two. At $z = 0$ the worst singularity is $z^{-1/2}\log z$; at $z = 1$ it is $(1-z)^{t-1}\log(1-z)$. The small arcs therefore vanish. The large arc is $O(R^{t-1}\log R)$ and vanishes for $t < 1$. These estimates justify the contours throughout the indicated interval; the resulting identities continue meromorphically in t.

Euler integration and the Thomae transformation already used in the main text give [22]

$$I_3 = \frac{\pi}{2}\frac{\Gamma(t)^2}{\Gamma(1/2+t)^2}\mathfrak{G}(t), \qquad I_4 = \frac{\pi^2}{2s_t}\mathfrak{F}(t).$$

Consequently,

$$\boxed{\begin{aligned} J_t = 2\pi(1 + c_t + 2s_t)&\frac{\Gamma(t)^2}{\Gamma(1/2+t)^2}\mathfrak{G}(t)\\ &+ 2\pi^2\left(2 + \tan\frac{\pi t}{2}\right)\mathfrak{F}(t).\end{aligned}} \tag{78}$$

This is an exact meromorphic reduction of the six-function master, not a truncation in t.

8.6 A contiguous relation with an explicit telescoping certificate

Write

$$a_n(t) = \frac{(1/2)_n^2 (t)_n}{(n!)^3}, \qquad \mathfrak{F}(t) = \sum_{n=0}^{\infty} a_n(t).$$

Direct rational algebra proves the termwise identity

$$\begin{aligned} t^2 a_n(t+1) - (2t^2 - 2t + 3/4) a_n(t) + (t-1)^2 a_n(t-1) \\ = \frac{(n+1)^3 a_{n+1}(t)}{n+t} - \frac{n^3 a_n(t)}{n+t-1}. \end{aligned} \tag{79}$$

For generic $\Re t < 0$ the right-hand side telescopes to zero: $a_n(t) = O(n^{\Re t - 2})$, so the upper boundary is $O(n^{\Re t})$, and the lower boundary vanishes. Apparent exceptional parameter values are handled by analytic continuation. Thus

$$\boxed{t^2 \mathfrak{F}(t+1) - (2t^2 - 2t + 3/4)\mathfrak{F}(t) + (t-1)^2 \mathfrak{F}(t-1) = 0.} \tag{80}$$

The defining series is holomorphic for $\Re t < 1$. This recurrence extends $\mathfrak{F}$ meromorphically to the plane, with possible poles only at positive integers.

8.7 The discrete Wronskian is constant, not merely periodic

The function $\mathfrak{F}(1-t)$ solves the same recurrence. Its discrete Wronskian with $\mathfrak{F}(t)$ is

$$W(t) = t^2 \{\mathfrak{F}(t+1)\mathfrak{F}(1-t) - \mathfrak{F}(t)\mathfrak{F}(-t)\}. \tag{81}$$

Equation (80) gives $W(t) = W(t-1)$. To determine this periodic function, first use the Thomae identity

$$\mathfrak{F}(1-t) = \frac{\Gamma(t)}{\Gamma(1-t)\Gamma(1/2+t)^2} \mathfrak{G}(t). \tag{82}$$

Since $\mathfrak{G}$ is analytic near zero and $\mathfrak{G}(0) = 1$, we have $\mathfrak{F}(1-t) = 1/(\pi t) + O(1)$. The apparent singularity of W at zero is removable and

$$W(0) = -\frac{1}{\pi^2}.$$

All possible poles of W are integers. Periodicity therefore makes each one removable, and W is entire.

A growth bound is essential here. The double Euler representation is

$$\mathfrak{F}(t) = \frac{1}{\pi^2} \int_0^1 \int_0^1 \frac{(1-uv)^{-t}}{\sqrt{uv(1-u)(1-v)}}\, du\, dv, \qquad \Re t < 1. \tag{83}$$

It follows by expanding $(1-uv)^{-t}$ and applying two beta integrals; absolute convergence at $u = v = 1$ holds precisely for $\Re t < 1$. In particular, $|\mathfrak{F}(x + \mathrm{i}y)| \le \mathfrak{F}(x)$. For $-1/2 \le \Re t \le 1/2$, the four functions $\mathfrak{F}(t), \mathfrak{F}(-t), \mathfrak{F}(t-1), \mathfrak{F}(-t-1)$ are uniformly bounded by $\mathfrak{F}(1/2)$, independently of $\Im t$. Solving (80) at t and $-t$ expresses $\mathfrak{F}(1+t)$ and $\mathfrak{F}(1-t)$ in terms of those bounded functions. Their rational coefficients are bounded for $|\Im t| \ge 1$. Hence

$$W(t) = O(1 + |\Im t|^2) \quad \text{uniformly in a full vertical strip of width one.}$$

An entire one-periodic function with this polynomial vertical growth is constant. Explicitly, its nth Fourier coefficient can be integrated on any horizontal translate of a period; its modulus is

bounded by a polynomial in $|y|$ times $e^{2\pi n y}$. Sending $y \to -\infty$ for $n > 0$ and $y \to +\infty$ for $n < 0$ kills every nonzero mode. We conclude that $W(t) \equiv -1/\pi^2$, or

$$\boxed{\mathfrak{F}(t)\mathfrak{F}(-t) - \mathfrak{F}(1+t)\mathfrak{F}(1-t) = \frac{1}{\pi^2 t^2}.} \tag{84}$$

This argument supplies both the normalization and the reason that a nonconstant periodic multiplier is impossible.

Apply (82) at t and $-t$, and use gamma reflection. The equivalent identity, analytic near zero, is

$$\boxed{\cos^2(\pi t)\,\mathfrak{G}(t)\mathfrak{G}(-t) + \pi^2 t^2\,\mathfrak{F}(t)\mathfrak{F}(-t) = 1.} \tag{85}$$

No numerical input has been used in this proof.

8.8 Only one combination of higher Taylor coefficients is needed

Write ordinary Taylor coefficients, without hidden factorials, as

$$\mathfrak{F}(t) = 1 + ut + vt^2 + O(t^3), \qquad \mathfrak{G}(t) = 1 + \sigma t^3 + \tau t^4 + O(t^5).$$

The absence of the first two terms in $\mathfrak{G} - 1$ follows immediately from its three upper parameters t, t, t. Extracting $[t^4]$ in (85) gives

$$2\tau + \frac{\pi^4}{3} + \pi^2(2v - u^2) = 0,$$

hence the exact elimination rule

$$\boxed{\tau + \pi^2 v = \frac{\pi^2 u^2}{2} - \frac{\pi^4}{6}.} \tag{86}$$

Neither v nor τ has to be evaluated individually. The two lower coefficients are elementary inverse-binomial sums:

$$\begin{aligned} u &= \sum_{n\ge1} \frac{\binom{2n}{n}^2}{16^n n} = 4\ell - \frac{8\mathsf{G}}{\pi}, \\ \sigma &= \sum_{n\ge1} \frac{16^n}{n^3\binom{2n}{n}^2} = 8\pi\mathsf{G} - 14\zeta(3). \end{aligned} \tag{87}$$

For completeness, both evaluations follow without an external summation assumption. Differentiate (83) at zero, set $u = \sin^2\theta$, $v = \sin^2\phi$ in that integral, and use

$$\int_0^{\pi/2} \log(1 - a\sin^2\phi)\,d\phi = \pi \log\frac{1+\sqrt{1-a}}{2}.$$

This elementary identity follows by differentiating in a and matching at $a = 0$. It gives

$$u = -\frac{16}{\pi}\int_0^{\pi/4} \log\cos x\,dx = 4\ell - \frac{8\mathsf{G}}{\pi}.$$

For σ, expand

$$(\arcsin x)^2 = \frac{1}{2}\sum_{n\ge1} \frac{4^n x^{2n}}{n^2\binom{2n}{n}}$$

and integrate with $x = \sin\theta$, using $\int_0^{\pi/2} \sin^{2n-1}\theta\,d\theta = 4^n/(2n\binom{2n}{n})$. The result is

$$\begin{aligned} \sigma &= 4\int_0^{\pi/2} \frac{\theta^2}{\sin\theta}\,d\theta = -8\int_0^{\pi/2} \theta\log\tan(\theta/2)\,d\theta \\ &= 8\pi\sum_{k\ge0}\frac{(-1)^k}{(2k+1)^2} - 16\sum_{k\ge0}\frac{1}{(2k+1)^3} = 8\pi\mathsf{G} - 14\zeta(3). \end{aligned}$$

In the last step one may use an Abel factor in the cosine series $\log\tan(\theta/2) = -2\sum_{k\ge0}\cos((2k+1)\theta)/(2k+1)$ and then take its limit. The endpoint logarithms are integrable, and integration yields absolutely convergent sums.

8.9 Extraction of the original master coefficient

The exact gamma interpolation in the main text gives

$$\mathcal{K} = [t^2]J_t - g_2, \qquad g_2 = \frac{A^4}{6} - \frac{A^2\pi^2}{6} - \frac{A\pi^3}{3} - 12A\zeta(3) - \frac{\pi^4}{180}.$$

Here $[t^2]J_t$ is a Laurent coefficient. No ordinary derivative at a pole is being taken. To expand (78), use

$$\frac{\pi\Gamma(1+t)^2}{\Gamma(1/2+t)^2} = \exp\left(4\ell t - \frac{\pi^2}{3}t^2 + 4\zeta(3)t^3 - \frac{7\pi^4}{90}t^4 + O(t^5)\right).$$

Direct coefficient extraction now yields

$$\begin{aligned}\mathcal{K} = {} & 4A\sigma + 4\tau + \pi^3 u + 4\pi^2 v - 32\ell^2\pi^2 - 8\ell\pi^3 \\ & + 112\ell\zeta(3) + \frac{2\pi^4}{3} + 28\pi\zeta(3).\end{aligned} \tag{88}$$

Substituting (86) first reduces this to

$$\mathcal{K} = 4A\sigma + 2\pi^2u^2 + \pi^3u - 32\ell^2\pi^2 - 8\ell\pi^3 + 28A\zeta(3).$$

The elementary values (87) then give exactly

$$\mathcal{K} = 128\mathsf{G}^2 + 24\pi^2\mathsf{G} - 4\pi^3\ell - 28A\zeta(3).$$

Finally (73) proves (76) and (33). Thus the compact evaluation is an analytic consequence of the original master definition. This completes the proof.

Scope of the proof. The proof evaluates the one-dimensional master constant defined by (70). It proves a local coefficient identity by means of a global parameter identity; it does not claim that the whole deformed master J_t is a gamma quotient, and it does not establish a corresponding identity in other dimensions.

Numerical checks of the identities. The proof uses no numerical input. Its identities were also tested numerically, separately from the proof. These tests are consistency checks, not premises of the proof, and none of them is an interval bound. Besides the 110-digit endpoint comparison quoted after (70), the record `lri_psi_complete_update/verification/numerical_checks.json` contains the following tests. A direct evaluation of $\mathfrak{F}$ and $\mathfrak{G}$ at $t = 0.2$ and 50 working digits satisfies the bilinear identity (85) to 2.6728×10^{-51}. At 75 working digits, with two settings of the asymptotic tail summation, the six-function master $\widehat{J}(t)/t^2$ of (68) and the two-function reduction (78) agree to better than 2×10^{-68} at $t = 0.2$, 0.37 and $0.12 + 0.07\mathrm{i}$, and (85) holds to better than 4×10^{-72} at the two real points. At the complex point the 75-digit bilinear residual is only 3×10^{-59}, and 6×10^{-48} with the longer tail, because the gamma-ratio tail cancels there. Rerun at 130 working digits, the residual has modulus 1.39×10^{-102} (`lri_psi_complete_update/verification/complex_bilinear_130.json`); the less accurate 75-digit values are kept in the record. At 110 digits the integral forms of u and σ given after (87) reproduce their closed forms to 8×10^{-112} and 7×10^{-111}. Finally, the coefficients $C_0, \ldots, C_5$ of Section 4 and their reflected-quarter forms agree in exact symbolic algebra (Section 4.2) and numerically at 70 working digits, with largest difference 9.3×10^{-69}, in C_5 (`lri_psi_complete_update/verification/coefficient_values.json`). The numerical comparison reuses the 240-digit values of $F_\pm$ (Section 9) from an earlier record, stored beside it as `wheel_inputs.json`, and recomputes P_5 from its gamma moment. None of these checks bears on the offset of the earlier full-graph control, which Section 14 explains.

9 The wheel periods P_4 and $\mathcal{L}$

9.1 The four-loop wheel: two ${}_4F_3(1)$ values

Let $G_4 = \Gamma(1/4)^4$, and define

$$F_- = {}_4F_3\left(\begin{matrix}1/4,1/4,1/2,1/2\\3/4,3/4,1\end{matrix};1\right), \qquad F_+ = {}_4F_3\left(\begin{matrix}3/4,3/4,1/2,1/2\\5/4,5/4,1\end{matrix};1\right).$$

Then

$$\boxed{P_4 = 13G_4F_- - \frac{192\pi^4}{G_4}F_+.} \tag{89}$$

It evaluates to 2229.52206107094196429452519974 85 .... It follows that the two-${}_4F_3$ expression of the fourth coefficient is

$$\boxed{648C_4 = 208G_4F_- - \frac{3072\pi^4}{G_4}F_+ - A^3 - A\pi^2 - 53\pi^3 - 1436\zeta(3) - 8AQ.} \tag{90}$$

For comparison with the earlier elliptic formulas, define

$$H_4 = {}_4F_3\left(\begin{matrix}1/2,1/2,1/2,1/2\\1,1,1\end{matrix};1\right), \quad H_7 = {}_7F_6\left(\begin{matrix}5/4,(1/2)^{\times 6}\\1/4,1,1,1,1,1\end{matrix};1\right).$$

The two elimination identities are

$$\begin{aligned} H_4 &= \frac{G_4}{4\pi^3}F_- - \frac{16\pi}{G_4}F_+, \\ H_7 &= \frac{G_4}{2\pi^4}F_- + \frac{32}{G_4}F_+. \end{aligned} \tag{91}$$

Indeed, write $\mathbf{K}(k) = K(k^2)$, $k' = \sqrt{1-k^2}$ and $N_\nu = \int_0^1 k^\nu \mathbf{K}(k)\mathbf{K}(k')\,dk$. Euler integration gives

$$N_\nu = \frac{\pi^2}{8}\frac{\Gamma(a)^2}{\Gamma(a+1/2)^2}{}_4F_3\left(\begin{matrix}1/2,1/2,a,a\\1,a+1/2,a+1/2\end{matrix};1\right), \qquad a = \frac{\nu+1}{2}. \tag{92}$$

The Landen substitution $k = (1-t)/(1+t)$, together with the mixed-to-square moment identity, gives $N_{-1/2} - N_{1/2} = 2N_0$ and $N_{-1/2} + N_{1/2} = 4\int_0^1 \mathbf{K}(k)^2\,dk$. Using $N_0 = \pi^3 H_4/8$ and $\int_0^1 \mathbf{K}(k)^2\,dk = \pi^4 H_7/32$ proves the stated identities. These elliptic-moment evaluations are discussed in [25].

9.1.1 From the wheel graph to the elliptic period

For the wheel edges $02, 03, 04, 12, 13, 14, 24, 34$, fix $x_0 = 0, x_1 = 1$. The two remaining outer-vertex integrations factor, giving

$$P_4 = \int_{\mathbb{R}} \frac{F(x)^2}{\sqrt{|x(1-x)|}}\,dx, \qquad F(x) = \int_{\mathbb{R}} \frac{du}{\sqrt{|u(1-u)(u-x)|}}.$$

Splitting at $u = 0, x, 1$ gives $F(x) = 4[K(x) + K(1-x)]$ for $0 < x < 1$. The transformations

$$F(x) = x^{-1/2}F(1/x) \quad (x > 1), \qquad F(x) = (1-x)^{-1/2}F\left(\frac{x}{x-1}\right) \quad (x < 0)$$

fold the tails. Hence

$$P_4 = 16\int_0^1 \frac{1+\sqrt{x}+\sqrt{1-x}}{\sqrt{x(1-x)}}[K(x) + K(1-x)]^2\,dx.$$

The ordered tetrahedron satisfies

$$\int_{0<x<y<1} \frac{dx\,dy}{\sqrt{xy(1-x)(1-y)(y-x)}} = \frac{1}{2}S_2\left(\frac{1}{2},\frac{1}{2},-\frac{1}{4}\right) = \frac{\Gamma(1/4)^4}{4\pi};$$

its full-line period is $3Q$. The three-loop calculation of [7], with the tetrahedron value corrected in [12], uses this normalization; the integral above is a Selberg integral [13].

9.1.2 The moment identities and their contour proof

Let $M = \int_0^1 \mathbf{K}(k)^2\,dk$, $N = N_0$, and $J = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')/k'\,dk$ with $k' = \sqrt{1-k^2}$. Euler integration of the complementary moment gives

$$\int_0^1 k^r \mathbf{K}(k')\,dk = \frac{\pi}{4}\frac{\Gamma((r+1)/2)^2}{\Gamma((r+2)/2)^2}, \quad r > -1.$$

Expanding the other $\mathbf{K}$ proves the formula for N_ν above. The square-moment evaluation is $M = \pi^4 H_7/32$ [25]. To see the mixed-to-square identity $J = 2M$, integrate $\mathbf{K}(z)^2$ around the upper semicircle from 1 to -1. Cauchy's theorem gives $-2M$ on the arc. The boundary-value Landen relation

$$e^{i\theta}\mathbf{K}(e^{2i\theta}) = \tfrac{1}{2}[\mathbf{K}(\cos\theta) + i\mathbf{K}(\sin\theta)], \quad 0 < \theta < \pi/2,$$

with the limit taken from inside the disk, makes the real part of that arc integral equal to $-J$; the imaginary part is odd under $\theta \mapsto \pi/2 - \theta$. Integrable logarithmic endpoint singularities are handled by an interior limit. Thus $J = 2M$.

The real Landen substitution $k = (1-t)/(1+t)$ preserves the mixed measure:

$$\mathbf{K}(k)\mathbf{K}(k')|dk| = \mathbf{K}(t)\mathbf{K}(t')dt.$$

The sum and difference of the half-integer weights transform to $2/t'$ and $2t/t'$, respectively. Therefore

$$N_{-1/2} + N_{1/2} = 2J = 4M, \qquad N_{-1/2} - N_{1/2} = 2N_0.$$

Since

$$N_{-1/2} = \frac{G_4}{16}F_-, \quad N_{1/2} = \frac{4\pi^4}{G_4}F_+, \quad N_0 = \frac{\pi^3}{8}H_4,$$

these prove both elimination identities. Folding the wheel integral gives $P_4 = 16(20M + 16N)$, and consequently the stated two-${}_4F_3$ result.

9.1.3 Four-loop counterterm combination

The residue certificate, including the wheel $b_4^{\text{wheel}} = -6P_4$, gives

$$3c_4 - b_4 = 6P_4 + 45AQ + \frac{405A^3 - 27A\pi^2 - 159\pi^3}{8} - 8\pi A - \frac{1077}{2}\zeta(3).$$

Inserting this in the fourth-order RG formula eliminates the intermediate πA terms and yields the compact C_4. This is an analytic reduction within the graph calculation, not an additional external benchmark.

$$B_- = G_4 F_-, \qquad B_+ = \frac{64\pi^4}{G_4}F_+, \qquad P_4 = 13B_- - 3B_+. \tag{93}$$

The derivative-free formula is

$$\boxed{\begin{aligned}\mathcal{L} &= -(5\pi + 26\log 2)B_- + (\pi + 6\log 2)B_+ \\ &= -\frac{13A - 3\pi}{2}B_- + \frac{3A - \pi}{2}B_+.\end{aligned}} \tag{94}$$

$$\boxed{\begin{aligned}\mathcal{L} = -\frac{2}{3}\Big\{&[\psi_0(1) - \psi_0(1/4)]P_4 \\ &+ [\psi_0(3/4) - \psi_0(1/4)]B_-\Big\}.\end{aligned}} \tag{95}$$

The two digamma differences are $\pi/2 + 3\log 2$ and π, respectively. Numerically, $\mathcal{L} = -5805.301629548802380141179877001 38\ldots$.

9.2 The logarithmic wheel: full contour proof

We give the contour proof of the logarithmic-wheel reduction. It proves (94) and (95), independently of the evaluation of $\mathcal{K}$. The symbols M, N, U, V, and R below denote local elliptic moments. They are distinct from the $V_r(t)$ of Section 8.

9.3 Conventions and the folded integral

In this section put $\ell = \log 2$ and use $B_\pm$ from (93). The elliptic integral uses the *modulus* convention:

$$\mathbf{K}(k) = \frac{\pi}{2}{}_2F_1(1/2, 1/2; 1; k^2), \qquad k' = \sqrt{1-k^2}, \qquad f(k) = \mathbf{K}(k), \quad g(k) = \mathbf{K}(k').$$

The prime in k' denotes the complementary modulus, not differentiation. In the parameter convention, $K(x) = \mathbf{K}(\sqrt{x})$; the full-line definition below uses it. Define

$$\begin{aligned} M &= \int_0^1 f(k)^2\,dk, & N &= \int_0^1 f(k)g(k)\,dk, \\ U_0 &= \int_0^1 \frac{f(k)^2}{k'}\,dk, & V_0 &= \int_0^1 \frac{f(k)g(k)}{k'}\,dk, \\ U &= \int_0^1 \frac{f(k)^2\log(kk')}{k'}\,dk, & V &= \int_0^1 \frac{f(k)g(k)\log(kk')}{k'}\,dk, \\ w(k) &= \log\frac{k}{1-k^2}, & R &= \int_0^1 f(k)^2 w(k)\,dk. \end{aligned}$$

All these integrals converge.

The wheel master is defined on the full line by

$$\mathcal{L} = \int_{\mathbb{R}} \frac{F(x)^2 \log|x(1-x)|}{\sqrt{|x(1-x)|}}\,dx, \qquad F(x) = \int_{\mathbb{R}} \frac{du}{\sqrt{|u(1-u)(u-x)|}}.$$

For $0 < x < 1$, $F(x) = 4[K(x) + K(1-x)]$. Folding the two tails gives

$$\mathcal{L} = 16\int_0^1 [K(x)+K(1-x)]^2 \Big[\frac{\log x + \log(1-x)}{\sqrt{x(1-x)}} + \frac{\log(1-x) - 2\log x}{\sqrt{1-x}} + \frac{\log x - 2\log(1-x)}{\sqrt{x}}\Big]\,dx.$$

Substitution $x = k^2$ and reflection of the second tail term give the useful identity

$$\frac{\mathcal{L}}{128} = U + V + \int_0^1 (f+g)^2 w\,dk. \tag{96}$$

9.4 Real Landen identities

Let $T(k) = (1-k)/(1+k)$. Standard Landen transformations [31, 34] give

$$f(T(k)) = \frac{1+k}{2}g(k), \qquad g(T(k)) = (1+k)f(k).$$

Consequently,

$$f(T)g(T)|T'| = fg, \qquad g(T)^2|T'| = 2f^2, \qquad f(T)^2|T'| = \tfrac{1}{2}g^2. \tag{97}$$

In particular, $\int_0^1 g^2\,dk = 2M$. Since

$$w(T(k)) = -w(k) - 2\ell,$$

we have

$$\int_0^1 fgw\,dk = -\ell N, \qquad \int_0^1 g^2 w\,dk = -2R - 4\ell M. \tag{98}$$

Combining these with (96) yields

$$\frac{\mathcal{L}}{128} = U + V - R - 4\ell M - 2\ell N. \tag{99}$$

It remains to evaluate just three logarithmic combinations U, V, R.

9.5 Evaluation of U by the Stieltjes transform

Use the standard identity (Wan, Theorem 1, equation (15)) [25]:

$$f(t)^2 = \frac{4}{\pi}\int_0^1 \frac{kf(k)g(k)}{1-t^2k^2}\,dk, \qquad 0 \le t < 1. \tag{100}$$

Integration with respect to dt gives

$$M = \frac{4}{\pi}\int_0^1 fg\ \operatorname{artanh} k\,dk.$$

Applying T and $\operatorname{artanh} T(k) = -\frac{1}{2}\log k$ therefore gives

$$\int_0^1 fg \log k\,dk = -\frac{\pi}{2}M. \tag{101}$$

Also $\log(1+T(k)) = \ell - \log(1+k)$, so

$$\int_0^1 fg \log(1+k)\,dk = \frac{\ell}{2}N. \tag{102}$$

Integrating (100) against $dt/\sqrt{1-t^2}$ proves $U_0 = 2N$, using the elementary integral $\int_0^{\pi/2}(1-k^2\sin^2\theta)^{-1}\,d\theta = \pi/(2k')$ and then complementing k.

For the logarithmic version, the tangent substitution gives

$$\int_0^{\pi/2} \frac{\log(\sin\theta\cos\theta)}{1-k^2\sin^2\theta}\,d\theta = -\frac{\pi}{2k'}\log\frac{(1+k')^2}{k'}.$$

Insert this into (100) and complement the outer variable:

$$\begin{aligned} U &= -2\int_0^1 \frac{kfg}{k'}\log\frac{(1+k')^2}{k'}\,dk \\ &= -2\int_0^1 fg\,[2\log(1+k) - \log k]\,dk. \end{aligned}$$

Equations (101)–(102) now prove

$$\boxed{U = -\pi M - 2\ell N.} \tag{103}$$

The integral interchanges are justified by positivity for the nonlogarithmic identities and by Tonelli applied to the negative logarithm for the last one; the resulting moments are finite.

9.6 Evaluation of R by a first-quadrant contour

Continue $\mathbf{K}(z)$ from its series at zero and put

$$W(z) = \operatorname{Log} z - \operatorname{Log}(1 - z^2),$$

with the logarithms analytic in the first quadrant and their indicated upper boundary values on the real axis. Integrate $\mathbf{K}(z)^2 W(z)$ around the first quadrant. The large arc vanishes because the integrand is $O(\log^3 |z|/|z|^2)$. The small arcs at 0 and 1 also vanish; their singularities are only integrable powers of logarithms. Thus

$$\int_0^\infty \mathbf{K}(x + i0)^2 W(x + i0)\, dx = i \int_0^\infty \mathbf{K}(iy)^2 W(iy)\, dy. \tag{104}$$

The reciprocal and imaginary-modulus connection formulas [31, 34] are

$$\mathbf{K}(x + i0) = \frac{1}{x}[f(1/x) + ig(1/x)] \quad (x > 1), \qquad \mathbf{K}(iy) = \frac{1}{\sqrt{1 + y^2}} f\left(\frac{y}{\sqrt{1 + y^2}}\right).$$

Furthermore $W(x+i0) = w(1/x)+i\pi$ for $x > 1$ and $\operatorname{Im} W(iy) = \pi/2$. The real part of the left-hand side of (104) is therefore

$$2R - \int_0^1 g^2 w\, dk - 2\pi N = 4R + 4\ell M - 2\pi N.$$

On the right-hand side, set $y = k/k'$; the real part is

$$-\frac{\pi}{2} \int_0^\infty \mathbf{K}(iy)^2\, dy = -\frac{\pi}{2} U_0 = -\pi N.$$

Equating these gives

$$\boxed{R = \frac{\pi}{4} N - \ell M.} \tag{105}$$

In particular, the extra πN term is a branch contribution; discarding the imaginary part of the reciprocal-modulus continuation would lose it.

9.7 Evaluation of V by an upper-semicircle contour

Integrate the same $\mathbf{K}(z)^2 W(z)$ over the upper half of the unit disk. The integral over the diameter from -1 to 1 is

$$2R + i\pi M.$$

On the semicircle, put $z = e^{2i\theta}$, $0 < \theta < \pi/2$. The boundary-value Landen identity, with continuation from the interior of the disk, reads

$$e^{i\theta} \mathbf{K}(e^{2i\theta}) = \frac{1}{2}[\mathbf{K}(\cos\theta) + i\mathbf{K}(\sin\theta)],$$

This follows by applying $\mathbf{K}(z) = (1+z)^{-1}\mathbf{K}(2\sqrt{z}/(1+z))$: as z approaches $e^{2i\theta}$ from inside the disk, its transformed modulus approaches $\sec\theta$ from the upper side. The reciprocal-modulus formula then fixes the displayed plus sign. Also,

$$W(e^{2i\theta}) = -\log(4 \sin\theta \cos\theta) + \frac{i\pi}{2}.$$

Write $f_\theta = \mathbf{K}(\sin\theta)$, $g_\theta = \mathbf{K}(\cos\theta)$. Then

$$\mathbf{K}(z)^2\, dz = \left[-f_\theta g_\theta + \frac{i}{2}(g_\theta^2 - f_\theta^2)\right] d\theta.$$

Terms containing $g_\theta^2 - f_\theta^2$ times a symmetric factor integrate to zero. The complete arc integral is therefore

$$V + 2\ell V_0 - \frac{i\pi}{2} V_0.$$

Cauchy's theorem implies, from its imaginary and real parts respectively,

$$V_0 = 2M, \qquad V = -2R - 4\ell M.$$

Together with (105), this proves

$$\boxed{V = -\frac{\pi}{2} N - 2\ell M.} \tag{106}$$

The endpoint limits are valid because the logarithmic singularities are integrable and the small indentations contribute zero.

9.8 Cancellation and reduction to the two ${}_4F_3$ values

Substituting (103), (105), and (106) into (99) gives

$$\boxed{\mathcal{L} = -(128\pi + 640\ell)M - (96\pi + 512\ell)N.} \tag{107}$$

To eliminate M, N, introduce

$$N_\nu = \int_0^1 k^\nu f(k) g(k)\, dk, \qquad \nu > -1.$$

Expanding f and using the beta moment of g yields

$$N_\nu = \frac{\pi^2}{8} \frac{\Gamma((\nu+1)/2)^2}{\Gamma((\nu+2)/2)^2} {}_4F_3\left(\begin{matrix} 1/2, 1/2, (\nu+1)/2, (\nu+1)/2 \\ 1, (\nu+2)/2, (\nu+2)/2 \end{matrix} ; 1 \right).$$

Hence $N_{-1/2} = B_-/16$ and $N_{1/2} = B_+/16$. Applying T to their sum and difference gives

$$N_{-1/2} + N_{1/2} = 2V_0 = 4M, \qquad N_{-1/2} - N_{1/2} = 2N.$$

Thus

$$M = \frac{B_- + B_+}{64}, \qquad N = \frac{B_- - B_+}{32}.$$

Substitution into (107) proves (94). The nonlogarithmic folded integral gives $P_4 = 320M + 256N = 13B_- - 3B_+$, which proves (95). No parameter-differentiation identity is required.

10 The five-loop primitive period

10.1 Primitive graph and normalization

A zig-zag decompletion has edges

$$02, 03, 04, 05, 13, 14, 15, 24, 25, 35.$$

Fixing $x_0 = 0, x_5 = 1$ and performing two independent star integrations gives

$$P_5 = \int_{\mathbb{R}^2} \frac{F(u)F(v)}{\sqrt{|v(1-u)(u-v)|}}\, du\, dv, \quad F(x) = \int_{\mathbb{R}} \frac{dt}{\sqrt{|t(1-t)(t-x)|}}. \tag{108}$$

This integral is convergent and has no subdivergence subtractions. The zig-zag class has automorphism order 2 and contributes $24P_5$ to b_5. The other two primitive classes share the product period $(3Q)^2 = 9Q^2$ and have automorphism orders 16 and 12. Their combined contribution is

$$\left(\frac{48}{16} + \frac{48}{12}\right) 9Q^2 = 63Q^2.$$

All primitive mass weights vanish because the four external legs are on distinct vertices. This yields $C_5^{\text{primitive}} = -(63Q^2 + 24P_5)/729$ in the present normalization.

10.2 Spectral representation and continuation to one dimension

The starting point is the conformal-triangle spectral representation of zig-zag graphs in [24], in particular its eigenvalues, spectral measure and two-point reduction, equations (5.23), (5.27) and (6.2). In the present loop convention its eigenvalue power is $k = L+2$, where L is the contribution's loop order in the quartic beta function.

Continuing the spectral formula to $D = 1$ requires more than substituting $D = 1$ into the density on the real spectral axis. A pole crossing contributes a discrete sum. Define

$$\tau_n(\nu) = (-1)^n\sqrt{\pi}\left|\frac{\Gamma(n/2+1/8+i\nu)}{\Gamma(n/2+3/8+i\nu)}\right|^2, \qquad n = 0, 1, \tag{109}$$

$$\sigma_j = (-1)^j\sqrt{\pi}\,\frac{\Gamma(j/2-1/8)\Gamma(j/2+3/8)}{\Gamma(j/2+1/8)\Gamma(j/2+5/8)}, \qquad j = 1, 2, \ldots. \tag{110}$$

For the zig-zag family, the one-dimensional period is

$$\boxed{P_L = \frac{1}{\pi^2}\int_0^\infty \nu\tanh(2\pi\nu)\left[\tau_0(\nu)^k + \tau_1(\nu)^k\right]\,d\nu + \frac{1}{2\pi^2}\sum_{j=1}^{\infty}(j-\tfrac{1}{2})\sigma_j^k, \qquad k = L+2.} \tag{111}$$

The first term alone is not the period.

10.2.1 Origin of the discrete sum

Here is the contour-continuation detail needed to fix its sign and normalization. Set $p = D/2-1$ and specialize the propagator parameter in [24] to $\beta = D/4$. The combined density is

$$\rho_n^{(D)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3D/2+1}\Gamma(2p)n!}\left[(n+p)^2+4\nu^2\right] \times \frac{\Gamma(p+2i\nu)\Gamma(p-2i\nu)}{\Gamma(2i\nu)\Gamma(-2i\nu)}. \tag{112}$$

The corresponding eigenvalue is

$$\tau_n^{(D)}(\nu) = (-1)^n\pi^{D/2}\frac{\Gamma(n/2+D/8+i\nu)\Gamma(n/2+D/8-i\nu)}{\Gamma(n/2+3D/8+i\nu)\Gamma(n/2+3D/8-i\nu)}.$$

For $D > 2$ the spectral formula uses the real ν contour. As D passes below 2, the poles $\nu = \pm ip/2$ exchange sides. Retaining the analytically continued contour gives a correction

$$-2\pi i\,\mathrm{Res}_{\nu=-ip/2}\,\rho_n^{(D)}(\nu)\tau_n^{(D)}(\nu)^k.$$

Before taking $D \to 1$, its coefficient is

$$\frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3D/2}n!\Gamma(-p)}\,\tau_n^{(D)}(-ip/2)^k. \tag{113}$$

The $n = 0$ crossing cancels identically. For $n \geq 1$, the limit of the prefactor in (113) is $(n-1/2)/(2\pi^2)$, while the eigenvalue becomes (110). This gives the second term of (111).

On the real axis, only $n = 0, 1$ remain in the pointwise $D \to 1$ limit, each with density $\nu\tanh(2\pi\nu)/\pi^2$. The fact that higher-spin densities vanish on this axis does not remove their crossed-pole residues. Losing these residues would already spoil the known three-loop tetrahedron normalization.

The continuation and residue reduction here specialize the spectral construction of [24]; they are not an independently published evaluation of this Ising coefficient.

10.3 Closing the contour and summing the gamma poles

Define

$$R_\alpha(x) = \frac{\Gamma(\alpha+x)\Gamma(\alpha-x)}{\Gamma(\alpha+1/4+x)\Gamma(\alpha+1/4-x)}. \tag{114}$$

With $x = i\nu$, closing the continuous spectral integral to the right expresses it as

$$P_L^{\text{cont}} = \pi^{k/2-1} \sum_{\Re x>0} \operatorname{Res}_x \left\{ x \tan(2\pi x) [R_{1/8}(x)^k + (-1)^k R_{5/8}(x)^k] \right\}. \tag{115}$$

One can use expanding rectangular contours avoiding the real poles. Gamma reflection and Stirling estimates give vanishing outer-contour contributions for $k > 4$.

There are two pole families. The tangent poles are $x_j = (2j-1)/4$, $j \geq 1$, where

$$R_{5/8}(x_j) = -R_{1/8}(x_j), \qquad \sqrt{\pi} R_{1/8}(x_j) = \sigma_j.$$

Their residues in (115) give exactly

$$-\frac{1}{2\pi^2} \sum_{j=1}^{\infty} (j - \tfrac{1}{2}) \sigma_j^k.$$

They cancel the discrete term of (111). The complete period is therefore given just by the gamma poles at $x = \alpha + n$, $n \geq 0$, for $\alpha = 1/8, 5/8$.

10.4 A Barnes integral with no unwanted tangent poles

Set

$$s = 2x - \frac{1}{4}, \qquad g(s) = \frac{\Gamma(s+\frac{1}{2})}{\Gamma(s+1)}, \qquad C(s) = \csc(\pi s), \qquad D(s) = \cot(\pi s).$$

Reflection and duplication imply

$$R_{1/8}(x) = g(s)\big[1 - \cot(\pi s/2)\big], \tag{116}$$

$$R_{5/8}(x) = g(s)\big[1 + \tan(\pi s/2)\big]. \tag{117}$$

The two gamma-pole families have become $s = 0, 1, 2, \ldots$. Define

$$S(s) = \tan(\pi s + \pi/4)\big\{[1 - \cot(\pi s/2)]^7 - [1 + \tan(\pi s/2)]^7\big\}. \tag{118}$$

Its unwanted tangent poles are at $s = 1/4 + j$. Their residues equal those of $16\sqrt{2}\,\csc[\pi(s-1/4)]$. Consequently the kernel

$$\mathcal{R}(s) = S(s) - 16\sqrt{2}\,\csc[\pi(s-1/4)]$$

has only integer poles on the right of $\Re s = -1/4$ and retains the desired residues. A rational trigonometric simplification gives

$$\boxed{\begin{aligned}\mathcal{R}(s) = {} & -128C + 160C^3 + 192C^5 - 128C^7 \\ & - 16DC - 400DC^3 + 192DC^5.\end{aligned}} \tag{119}$$

The SymPy check `lri_fixed_hypergeometric_update/public/kernel_identity.py` verifies this exactly after replacing $\tan(\pi s/2)$ by an indeterminate.

Closing a vertical contour to the right is clockwise. Also, $x\,dx = (s+1/4)\,ds/4$. Therefore

$$P_5 = -\frac{\pi^{5/2}}{4} \frac{1}{2\pi \mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 \mathcal{R}(s)\, ds. \tag{120}$$

No poles lie on this line. On rectangles avoiding the real poles, gamma-ratio asymptotics and the trigonometric factors justify the vanishing of the closing contributions. The remaining integer-pole sum converges absolutely.

10.5 The reflection that removes four derivative orders

For odd j, write

$$I_j = \frac{1}{2\pi \mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 C(s)^j \, ds, \tag{121}$$

$$J_j = \frac{1}{2\pi \mathrm{i}} \int_{-1/4-\mathrm{i}\infty}^{-1/4+\mathrm{i}\infty} (s+\tfrac{1}{4}) g(s)^7 D(s) C(s)^j \, ds. \tag{122}$$

The substitution $s \mapsto -1/2 - s$ preserves the vertical line. Gamma reflection and elementary trigonometry give

$$\boxed{g(-\tfrac{1}{2}-s) = -D(s)g(s), \qquad C(-\tfrac{1}{2}-s) = -\frac{C(s)}{D(s)}, \qquad D(-\tfrac{1}{2}-s) = \frac{1}{D(s)}.} \tag{123}$$

Together with $D^2 = C^2 - 1$, these imply

$$\boxed{I_7 = 0, \qquad I_5 = I_3, \qquad J_5 = 0, \qquad J_3 = J_1.} \tag{124}$$

For example, $I_7 = -I_7$. For I_3 the reflected integrand gives $I_3 = -I_7 + 2I_5 - I_3$, whence $I_3 = I_5$. Similarly, $J_5 = -J_5$ and $J_1 = -J_5 + 2J_3 - J_1$.

Substitution into (119)–(120) yields

$$\boxed{P_5 = \frac{\pi^{5/2}}{4}(128 I_1 - 352 I_3 + 416 J_1).} \tag{125}$$

The pole order is now at most three rather than seven. This is the precise mechanism behind the reduction from sixth to second parameter derivatives.

10.5.1 Summing the three remaining pole families

Define the analytic function

$$\mathcal{A}_{\mathrm{s}}(t) = \sum_{n=0}^{\infty} (-1)^n (n + \tfrac{1}{4} + t) \left[\frac{\Gamma(n+\frac{1}{2}+t)}{\Gamma(n+1+t)}\right]^7. \tag{126}$$

Termwise parameter derivatives through the needed order are absolutely convergent near zero. Using $\csc[\pi(n+t)] = (-1)^n \csc(\pi t)$ and $\cot[\pi(n+t)] = \cot(\pi t)$, closing to the right gives

$$\boxed{\begin{aligned} I_1 &= -\frac{\mathcal{A}_{\mathrm{s}}(0)}{\pi}, \\ I_3 &= -\frac{\mathcal{A}_{\mathrm{s}}''(0) + \pi^2 \mathcal{A}_{\mathrm{s}}(0)}{2\pi^3}, \\ J_1 &= -\frac{\mathcal{A}_{\mathrm{s}}'(0)}{\pi^2}. \end{aligned}} \tag{127}$$

Here the minus signs are the clockwise-contour signs. For the third formula, $\csc(\pi t)\cot(\pi t) = (\pi t)^{-2} - 1/6 + O(t^2)$ has no simple-pole term. Substitution into (125) gives

$$P_5 = \frac{4}{\sqrt{\pi}} \left[11 \mathcal{A}_{\mathrm{s}}''(0) - 26\pi \mathcal{A}_{\mathrm{s}}'(0) + 3\pi^2 \mathcal{A}_{\mathrm{s}}(0)\right]. \tag{128}$$

This intermediate identity follows by substituting the three residues into (125). The finite-periodization proof below eliminates all three Taylor operations from the final evaluation.

10.6 Derivation of the positive gamma integral

On the same vertical line, put $s = -1/4 + \mathrm{i}y$ and $G(y) = |\Gamma(1/4 + \mathrm{i}y)|^2$. Gamma reflection gives

$$g(s) = \frac{G(y)}{\sqrt{2}\,\pi}\,[\,\cosh(\pi y) - \mathrm{i}\sinh(\pi y)\,]. \tag{129}$$

Writing $z = \cosh(\pi y) - \mathrm{i}\sinh(\pi y)$, one has $C(s) = -\sqrt{2}/z$ and $D(s) = -\overline{z}/z$. Taking conjugate halves of the contour integral yields

$$I_1 = \frac{1}{8\pi^8}\int_0^\infty y G(y)^7\,[\,\sinh^3(2\pi y) - 3\sinh(2\pi y)\,]\,dy, \tag{130}$$

$$I_3 = -\frac{1}{2\pi^8}\int_0^\infty y G(y)^7\,\sinh(2\pi y)\,dy, \tag{131}$$

$$J_1 = \frac{1}{4\pi^8}\int_0^\infty y G(y)^7\,\sinh(2\pi y)\cosh(2\pi y)\,dy. \tag{132}$$

Inserting these into (125) and using $4\sinh^3 v = \sinh(3v) - 3\sinh v$ and $2\sinh v\cosh v = \sinh(2v)$ proves (133). This derivation fixes the coefficients $1, 13, 29$ and the factor $\pi^{-11/2}$ analytically; they were not obtained by fitting a numerical period.

$$\boxed{P_5 = \frac{1}{\pi^{11/2}}\int_0^\infty y|\Gamma(\tfrac{1}{4} + \mathrm{i}y)|^{14}[\sinh(6\pi y) + 13\sinh(4\pi y) + 29\sinh(2\pi y)]\,dy.} \tag{133}$$

The integrand is $O(y^2)$ at zero and $O(y^{-5/2}e^{-\pi y})$ at infinity by the Gamma-ratio asymptotics [27]; it is positive on the integration interval. This provides an independent convergent numerical representation of the same period.

10.7 Finite periodization and the five fixed values

10.7.1 The unilateral and bilateral sums

Use the meromorphic function

$$f(s) = \left(s + \frac{1}{4}\right)\left[\frac{\Gamma(s + \frac{1}{2})}{\Gamma(s+1)}\right]^7, \qquad \mathcal{A}_{\mathrm{s}}(t) = \sum_{n=0}^{\infty}(-1)^n f(n+t). \tag{134}$$

This is the same unilateral sum as (126). By taking out the $n = 0$ factors and using $(t + \frac{1}{4} + n)/(t + \frac{1}{4}) = (t + \frac{5}{4})_n/(t + \frac{1}{4})_n$,

$$\begin{aligned}\mathcal{A}_{\mathrm{s}}(t) = {} & \left(t + \frac{1}{4}\right)\left[\frac{\Gamma(t + \frac{1}{2})}{\Gamma(t+1)}\right]^7 \\ & \times {}_9F_8\left(\begin{matrix}1, t + \frac{5}{4}, (t + \frac{1}{2})^{\times 7} \\ t + \frac{1}{4}, (t+1)^{\times 7}\end{matrix}; -1\right).\end{aligned} \tag{135}$$

Consequently

$$\mathcal{Y}_r = \mathcal{A}_{\mathrm{s}}\left(\frac{r-2}{8}\right), \qquad r = 1, 2, 3, 5, 7. \tag{136}$$

The parameter sum difference in (135) is $3/2$; gamma-ratio asymptotics give absolute and locally uniform convergence away from poles. In particular all five evaluation points in (136) are nonsingular and have positive hypergeometric parameters [22, 27].

Now extend the index in (134) to all integers:

$$\Sigma(t) = \sum_{n\in\mathbb{Z}}(-1)^n f(n+t), \qquad \Xi(t) = \cos^7(\pi t)\,\Sigma(t). \tag{137}$$

These expressions are initially defined away from the half integers. Gamma reflection gives the two useful identities

$$\frac{\Gamma(-s)}{\Gamma(\frac{1}{2}-s)} = -\cot(\pi s)\frac{\Gamma(s+\frac{1}{2})}{\Gamma(s+1)}, \qquad f(-\tfrac{1}{2}-s) = \cot^7(\pi s) f(s). \tag{138}$$

The minus sign from the linear factor in f cancels the seventh power of the minus sign in the first identity. These are direct consequences of the standard gamma reflection formula [26]. Writing a negative summation index as $n = -m-1$ and using (138) shows, including the alternating sign, that

$$\boxed{\begin{aligned}\Sigma(t) &= \mathcal{A}_{\mathrm{s}}(t) - \tan^7(\pi t)\mathcal{A}_{\mathrm{s}}(\tfrac{1}{2}-t),\\ \Xi(t) &= \cos^7(\pi t)\mathcal{A}_{\mathrm{s}}(t) - \sin^7(\pi t)\mathcal{A}_{\mathrm{s}}(\tfrac{1}{2}-t).\end{aligned}} \tag{139}$$

Thus evaluating Ξ requires only two ordinary unilateral hypergeometric values, with no new special-function convention.

10.7.2 Why only three harmonics survive

We give the analytic growth argument explicitly, because periodicity alone would not imply a finite Fourier polynomial. For a compact set of t avoiding half integers, Stirling's ratio estimate and (138) imply

$$f(n+t) = O(|n|^{-5/2}) \qquad (n \to \pm\infty),$$

locally uniformly. Indeed the positive-index estimate follows from $\Gamma(s+1/2)/\Gamma(s+1) \sim s^{-1/2}$; reflection reduces the negative-index estimate to the same right-half-plane bound. Hence Σ is meromorphic. Its only possible poles are at half integers and have order at most seven. Multiplication by $\cos^7(\pi t)$ removes them. The same estimates after multiplication are locally uniform at a half integer, so the removal applies to the entire series, not just to individual summands. Therefore Ξ is entire.

Reindexing the absolutely convergent bilateral sum gives

$$\Sigma(t+1) = -\Sigma(t), \qquad \Xi(t+1) = \Xi(t).$$

For $0 \le x \le 1$ and $|y| \ge 1$, the reflection factors are uniformly bounded. The right-half-plane gamma estimate therefore also gives

$$|\Sigma(x+\mathrm{i}y)| \le C\sum_{n\in\mathbb{Z}}(1+|n+\mathrm{i}y|)^{-5/2} = O(|y|^{-3/2}). \tag{140}$$

It follows that

$$|\Xi(x+\mathrm{i}y)| = O\Big(e^{7\pi|y|}|y|^{-3/2}\Big). \tag{141}$$

Let $c_k = \int_0^1 \Xi(x)e^{-2\pi\mathrm{i}kx}\,dx$. Entirety and periodicity allow the contour to be shifted vertically. For $k \ge 4$, shift down to $\operatorname{Im} t = -Y$ and use (141) to obtain

$$|c_k| \le CY^{-3/2}e^{(7-2k)\pi Y} \longrightarrow 0.$$

Shifting upward gives $c_k = 0$ for $k \le -4$. Thus only the seven Fourier modes $-3,\dots,3$ are possible.

There is one more symmetry. Reflection and reindexing imply

$$\Sigma(-\tfrac{1}{2}-t) = \cot^7(\pi t)\Sigma(t), \qquad \Xi(\tfrac{1}{2}-t) = -\Xi(t).$$

In the second equation periodicity has been used after multiplying by $\cos^7(\pi t)$. If $\Xi(t) = \sum_{k=-3}^{3} c_k e^{2\pi\mathrm{i}kt}$, this gives $c_0 = 0$ and $c_{-k} = -(-1)^k c_k$. Consequently

$$\boxed{\Xi(t) = \alpha\cos(2\pi t) + \beta\sin(4\pi t) + \gamma\cos(6\pi t).} \tag{142}$$

The coefficients are real because (139) is real at nonsingular real t. This establishes the three-dimensional space analytically; the coefficients have not been fitted by assuming an unproved trigonometric ansatz.

10.7.3 Exact finite sampling and evaluation of the period

Set

$$\begin{aligned} X_0 &= \Xi(0) = \mathcal{Y}_2, \\ X_+ &= \Xi(\tfrac{1}{8}) = c_8^7\mathcal{Y}_3 - s_8^7\mathcal{Y}_5, \\ X_- &= \Xi(-\tfrac{1}{8}) = c_8^7\mathcal{Y}_1 + s_8^7\mathcal{Y}_7. \end{aligned} \tag{143}$$

Equation (142) gives exactly

$$\begin{aligned} \alpha &= \frac{X_0}{2} + \frac{X_+ + X_-}{2\sqrt{2}}, \\ \beta &= \frac{X_+ - X_-}{2}, \\ \gamma &= \frac{X_0}{2} - \frac{X_+ + X_-}{2\sqrt{2}}. \end{aligned} \tag{144}$$

These are fixed, finite evaluation points; no spacing tends to zero and there is no interpolation remainder.

For completeness, the link to the previously derived Barnes integrals can be made without a numerical differentiation step. The first equation in (139) gives $\Sigma(t) = \mathcal{A}_{\rm s}(t) + O(t^7)$ at zero, since $\mathcal{A}_{\rm s}(1/2 - t)$ is analytic there. Hence

$$\begin{aligned} \mathcal{A}_{\rm s}(0) &= \alpha + \gamma, \\ \mathcal{A}_{\rm s}'(0) &= 4\pi\beta, \\ \mathcal{A}_{\rm s}''(0) &= \pi^2(3\alpha - 29\gamma). \end{aligned} \tag{145}$$

Substitution of (145) into (128) gives

$$P_5 = 16\pi^{3/2}(9\alpha - 26\beta - 79\gamma).$$

Using (144) therefore yields

$$P_5 = 16\pi^{3/2}[-35X_0 + (22\sqrt{2} - 13)X_+ + (22\sqrt{2} + 13)X_-], \tag{146}$$

which proves (29). Differentiation is used only in the derivation of (128), not in the final fixed-value formula.

10.7.4 All five parameter arrays

For clarity, the repeating parameters and the shifted pair in the five functions are all displayed below. Each row also has the numerator parameter 1, as in (28).

r	shifted upper	seven equal uppers	shifted lower	seven equal lowers
1	9/8	3/8	1/8	7/8
2	5/4	1/2	1/4	1
3	11/8	5/8	3/8	9/8
5	13/8	7/8	5/8	11/8
7	15/8	9/8	7/8	13/8

The positive denominator parameters rule out singular defining series. The only cancellation used is the valid, fixed-parameter one at $r = 2$. The eighths arise from the choice of the exact sampling points $t = \pm 1/8$, not from a change of physical dimension or from a new regulator. There is no claim that the five individual values are independent or that their appearance is minimal.

The eighths in the parameters arise from the fixed sample nodes $t = 0, \pm 1/8$. They are not new regulators or a change in physical dimension. At 125 working digits the independently evaluated Gamma integral and the five-value expression differed by 3.63×10^{-121}; this is a numerical consistency check rather than an interval enclosure. The period begins

$$P_5 = 28599.293176562103536646477021422057490321246031264$$
$$16921356927935177776442703824905147018458419833882\ldots.$$

11 Restoring finite regulator corrections

The preceding expansions are inserted in the forest calculation, not merely in bare uniqueness integrals. The general forest framework is standard [21]; the concrete deformations and order-by-order corrections below are part of this calculation.

Let the original edge powers be $m_e(1-z)/2$ and introduce

$$a_e(z,d) = m_e(1-z)/2 + d\, v_e.$$

The chosen v_e have zero sum over the entire graph and over every proper logarithmically divergent subgraph. Therefore d changes shapes of the integrals but not their logarithmic degrees. At fixed d, write

$$C_G(z,d) = \sum_{k\geq 1} \frac{c_{G,k}(d)}{z^k}, \qquad r_G(d) = c_{G,1}(d). \tag{147}$$

The higher-pole identity is

$$L_G c_{G,k}(d) = \sum_{\gamma \subsetneq G} L_\gamma r_\gamma(d) c_{G/\gamma,k-1}(d), \qquad k \geq 2, \tag{148}$$

where the sum is over connected logarithmic one-particle-irreducible proper subgraphs. In particular, the simple residue belongs to the subgraph, not to the quotient.

Define the higher-pole part

$$H_G(z,d) = \frac{1}{L_G} \sum_{\gamma \subsetneq G} \frac{L_\gamma r_\gamma(d)}{z}\, C_{G/\gamma}(z,d). \tag{149}$$

Let $\mathfrak{F}'$ denote nonempty sets of disjoint proper logarithmic subgraphs; nested subgraphs are already included inside each C_γ. The physical residue is obtained from an integrable diagonal $d = z$ by

$$\boxed{\begin{aligned} r_G(0) = {} & -\operatorname{Res}_{z=0}\left[I_G(z,z) + \sum_{F\in\mathfrak{F}'} \Big(\prod_{\gamma\in F} C_\gamma(z,z) \Big) I_{G/F}(z,z) \right] \\ & -\operatorname{Res}_{z=0} H_G(z,z). \end{aligned}} \tag{150}$$

Crucially, $C_\gamma(z,z)$ means first form the fixed-d counterterm and then set $d = z$. It is not the counterterm obtained by setting $d = z$ before subtraction.

To prove (150), expand the fixed-d forest-subtracted amplitude. Its poles are $-C_G(z,d)$. On the diagonal, the coefficient of z^{-1} receives $-r_G(0)$ and also Taylor coefficients of the higher poles. Equation (148) identifies these additional terms as $-\operatorname{Res} H_G(z,z)$, giving the stated formula. This retains precisely the finite regulator corrections that were missing from bare uniqueness reductions.

11.1 Seven regulator-corrected graph classes

For six vertices use edge-word order

$$(01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45).$$

Class numbers are zero-based positions of lexicographically minimal edge words in the sorted 129-class list. The shifts used are:

Class	Nonzero v_e	Fixed roots
52	$v_{13} = -2,\ v_{23} = 2$	$0, 4$
56	$v_{14} = -2,\ v_{24} = 2$	$0, 4$
60	$v_{14} = -2,\ v_{45} = 2$	$0, 2$
79	$v_{23} = 2,\ v_{15} = -2$	$0, 4$
80	$v_{12} = -2,\ v_{13} = 2$	$0, 4$
91	$v_{14} = 2,\ v_{15} = -2,\ v_{25} = 2,\ v_{34} = -2$	$0, 4$
127	$v_{05} = -4,\ v_{14} = 2,\ v_{23} = 2$	$1, 2$

The exceptional four-loop class with word `0011101211` uses $v_{24} = 2, v_{34} = -2$, with roots $0, 1$. Its shift is antisymmetric under a graph automorphism, so the first derivative of its simple residue vanishes. Its higher-pole derivatives are still retained through (148).

Ordinary degree-three unique stars reduce six of the five-loop classes. Class 79 uses a two-internal-vertex conformal three-point block. On its diagonal both internal vertices have incident power sum two, and conformal covariance fixes the external dependence. With $h = (1 - z)/2$, its bare integral becomes

$$I^U_{79}(z) = \frac{2}{5z} M(h, h, h + 2z)\mathsf{B}(h, 1 - 3z)\mathsf{B}(h, 1 - 4z). \tag{151}$$

It is gamma-valued. The regulator-restoration terms in (150) use the tetrahedron Hessian (46). They give

$$r_{79} = 3Q\left[-\frac{13A^2}{20} + \frac{\pi^2}{12} + \frac{37H}{60}\right] = Q\left[-\frac{39A^2}{20} + \frac{89\pi^2}{60} + \frac{74\mathsf{G}}{5}\right]. \tag{152}$$

Dropping those restoration terms would not give this residue.

The complete resulting seven residues are recorded compactly below. Define

$$X = -\frac{25A^4}{12} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5} - \frac{11\pi^4}{900}, \tag{153}$$

$$Y = Q\left[-\frac{39A^2}{20} + \frac{89\pi^2}{60} + \frac{74\mathsf{G}}{5}\right]. \tag{154}$$

Then

$$r_{52} = Q\left[-\frac{3A^2}{2} + \frac{41\pi^2}{30} + \frac{76\mathsf{G}}{5}\right], \tag{155}$$

$$r_{56} = X - \frac{16}{5}\mathcal{K}, \tag{156}$$

$$r_{60} = -\frac{47A^4}{240} - \frac{11A^2\pi^2}{120} + \frac{19A\pi^3}{60} + \frac{43A\zeta(3)}{5} + \frac{3\pi^4}{400}, \tag{157}$$

$$r_{79} = Y, \tag{158}$$

$$r_{80} = X + \frac{4}{5}\mathcal{K}, \tag{159}$$

$$r_{91} = -\frac{79A^4}{80} - \frac{43A^2\pi^2}{360} + \frac{17A\pi^3}{12} + \frac{117A\zeta(3)}{5} - \frac{623\pi^4}{3600}, \tag{160}$$

$$r_{127} = Y. \tag{161}$$

These are unweighted counterterm residues. Their graph weights, words, and the residue expressions for all other required classes are in the residue certificate of Appendix A, and in machine-readable form in `lri_numerical_resolution/verification/inputs/residue_certificate.json`.

12 The six nonprimitive wheel-insertion topologies

A primitive graph has no ultraviolet-divergent proper subgraph. A nonprimitive graph requires subtraction of such subgraphs in addition to its overall pole. The six classes in this section contain a one-loop bubble or a four-loop wheel subgraph. They were evaluated separately from the remaining master-integral classes and must be included exactly once.

Using the edge-word order of Section 11, define $d_G = -5(3w_2 - w_4)/|\operatorname{Aut} G|$. The unweighted counterterm residues are

Class	Edge word	$\lvert\operatorname{Aut} G\rvert$	d_G	r_G
51	000110111111110	4	15/2	$-2AP_4/5 - 3\mathcal{L}/10$
78	000111101110111	2	15	$-2AP_4/5 - \mathcal{L}/10$
99	000120111111100	2	30	$-2AP_4/5 - \mathcal{L}/10$
113	000121101110110	1	60	$-2AP_4/5 + \mathcal{L}/10$
124	001110111112000	8	15/2	$-2AP_4/5 - \mathcal{L}/2$
128	001111011201100	4	15	$-2AP_4/5 - 3\mathcal{L}/10$

The table includes the subdivergence subtractions; these are not bare integrals. Weighted summation gives

$$(3c_5 - b_5)_{\text{wheel insertions}} = -54AP_4 - 9\mathcal{L}. \tag{162}$$

Since $b_4^{\text{wheel}} = -6P_4$ and $c_4^{\text{wheel}} = 0$, the lower-loop assembly supplies $60AP_4$. Thus

$$C_5^{\text{wheel+assembly}} = \frac{(-54AP_4 - 9\mathcal{L}) + 60AP_4}{729} = \frac{2AP_4 - 3\mathcal{L}}{243}. \tag{163}$$

The finite star expansion and the regulated wheel derivatives reduce the six residues to P_4 and $\mathcal{L}$. The test programs are not part of the public repository; the detailed derivation is given below. This sector is not the entire nonprimitive contribution.

12.1 Derivation of the residue table

12.1.1 First regulator derivative of the full-line star

Introduce

$$F_h(x) = \int_{\mathbb{R}} |u(1-u)(u-x)|^{-h}\, du.$$

For $0 < x < 1$, split the real line at $0, x, 1$ to obtain

$$\begin{aligned} F_h(x) = B(1-h,1-h)\Big[&x^{1-2h}{}_2F_1(h,1-h;2-2h;x) \\ &+ (1-x)^{1-2h}{}_2F_1(h,1-h;2-2h;1-x)\Big] \\ + B(1-h,3h-1)\Big[&{}_2F_1(h,3h-1;2h;x) + {}_2F_1(h,3h-1;2h;1-x)\Big]. \end{aligned} \tag{164}$$

At $h = 1/2$, the lower-parameter derivative terms cancel in the sum. Euler's transformation [23] gives

$$\partial_a {}_2F_1(a,b;1;x)\big|_{a=b=1/2} = -\frac{1}{2}\log(1-x)\,{}_2F_1(1/2,1/2;1;x).$$

Using this in (164), the beta-function derivative terms cancel as well, leaving

$$\boxed{\partial_h F_h(x)|_{h=1/2} = -\log|x(1-x)|F(x).} \tag{165}$$

The tails follow from $F_h(x) = x^{1-3h}F_h(1/x)$ for $x > 1$ and the corresponding negative-tail transformation.

In the opposite-outer-root representation of the wheel, put

$$P_W(z) = \int_{\mathbb{R}} |x(1-x)|^{-h} F_h(x)^2\, dx, \qquad h = (1-z)/2.$$

Equation (165) implies

$$\boxed{P'_W(0) = \frac{3}{2}\mathcal{L}.} \tag{166}$$

12.1.2 Edge derivatives and root changes

In this rooted wheel, the fixed roots are opposite outer vertices. The remaining variables are the center x and the two other outer vertices. Let D_eP denote the derivative with respect to an independent edge power at the point where all eight powers are $1/2$.

The completed wheel is the octahedral graph. For conformally balanced index variations, averaging over its edge-transitive automorphism group makes the first variation vanish. In the opposite-root representation, the allowed deformations obey zero total edge shift and zero sum of spoke shifts. It follows that the index gradient is constant separately on the four rim edges and the four spokes. The two root-to-center spokes differentiate the explicit factor $|x(1-x)|^{-1/2}$, so each contributes $-\mathcal{L}/2$. Equation (165) for one outer integrated vertex then fixes the rim derivative. Thus

$$\boxed{D_{\text{spoke}}P = -\frac{1}{2}\mathcal{L}, \qquad D_{\text{rim}}P = -\frac{1}{4}\mathcal{L}.} \tag{167}$$

Changing the fixed outer pair from opposite to adjacent gives, by the scale Jacobian,

$$P_W^{\text{adj}\,\prime}(0) = P_W^{\text{opp}\,\prime}(0) + 4D_{\text{rim}}P = \frac{1}{2}\mathcal{L}.$$

Only these first-derivative statements are needed; they are not identities for arbitrary higher index variations.

12.1.3 Bubble insertions on a rim or spoke

The full-line convolution coefficient is

$$\mathsf{B}(u,v) = \sqrt{\pi}\,\frac{\Gamma((1-u)/2)\Gamma((1-v)/2)\Gamma((u+v-1)/2)}{\Gamma(u/2)\Gamma(v/2)\Gamma((2-u-v)/2)}.$$

In particular, $\mathsf{B}(h,2h) = 2/z + A + O(z)$. For classes 99 and 113 a degree-three bubble endpoint can be integrated exactly. The rooted bare integral is

$$I_G(z) = \frac{2}{5z}\mathsf{B}(h,2h)P_e(z), \qquad P'_e(0) = P'_W(0) - D_eP.$$

The bubble counterterm contributes $-P_W(z)/z^2$ to the subtraction bracket. Its simple-pole extraction yields

$$r_G = -\frac{2}{5}AP_4 + \frac{1}{5}P'_W(0) + \frac{4}{5}D_eP.$$

The modified quotient edge is a spoke for class 99 and a rim for class 113. Equations (166)–(167) give their table entries.

12.1.4 Splitting the degree-four center

For classes 124 and 128 one needs the finite part of a star with one doubled edge. The local distributional expansion is

$$\begin{aligned}&\int_{\mathbb{R}} |u-y|^{-h}|u-z_1|^{-h}|u-x|^{-1+z}\,du\\&\quad= |x-y|^{-h}|x-z_1|^{-h}\left[\frac{2}{z}+A+2\log\frac{|x-y|\,|x-z_1|}{|y-z_1|}\right]+O(z).\end{aligned}\tag{168}$$

One derivation shifts the two single-edge powers from h to $h+z$, applies the unique star–triangle integral, and restores the finite logarithmic terms from the local bubble residue. Those logarithms cannot be dropped.

After subtraction,

$$r_G=-\frac{2}{5}AP_4+\frac{1}{5}P_W'(0)-\frac{4}{5}\mathcal{L}_{\triangle},$$

where $\mathcal{L}_{\triangle}$ inserts the logarithm in (168) into the wheel. For opposite neighbors, $\mathcal{L}_{\triangle}=\mathcal{L}$. For adjacent neighbors, (167) gives $\mathcal{L}_{\triangle}=3\mathcal{L}/4$. This proves classes 124 and 128.

12.1.5 A wheel inserted into the outer one-loop graph

Classes 51 and 78 have only the wheel as a proper ultraviolet-divergent subgraph. Invert about the degree-two root and fix the other root at one. The additional powers produced by inversion differ from their zero-regulator values by $O(z)$; their logarithms vanish at the only proper divergent collision, so they do not affect the constant term of the angular integral. One obtains

$$J_G(z)=\mathsf{B}(h+2z,1-4z)P_W^{\text{root pair}}(z)+O(z),\quad \mathsf{B}(h+2z,1-4z)=\frac{1}{2z}+A+O(z).$$

The wheel counterterm times the bubble quotient contributes only a double pole. Hence

$$r_G=-\frac{2}{5}AP_4-\frac{1}{5}P_W^{\text{root pair}\,\prime}(0).$$

The pair is opposite for class 51 and adjacent for class 78, proving the remaining entries.

13 Assembly of the full coefficient

Write

$$\beta(g)=-\epsilon g+\sum_{L\ge1}b_Lg^{L+1},\qquad \gamma_r(g)=\sum_{L\ge1}c_Lg^L.$$

The long-range kinetic term has no wave-function counterterm in this setup [7]. At the nonzero fixed point,

$$\Delta_{\mathcal{E}}=(1-\epsilon)/2+\gamma_r(g_*),\qquad \beta(g_*)=0.$$

With $b_1=3, c_1=1, b_2=-6A, c_2=-A$ and

$$b_3=6Q+\frac{87}{4}A^2-\frac{3}{4}\pi^2-4\pi,\qquad c_3=\frac{7}{2}A^2-\frac{1}{6}\pi^2-\frac{4}{3}\pi,$$

Lagrange inversion gives

$$\begin{aligned}729C_5=&\,336A^4-60A^2b_3+108A^2c_3+b_3^2-3b_3c_3\\&-10Ab_4+24Ac_4+3c_5-b_5.\end{aligned}\tag{169}$$

The primitive and wheel-insertion sectors are

$$C_5^{\text{primitive}} = -\frac{63Q^2 + 24P_5}{729}, \qquad C_5^{\text{wheel+assembly}} = \frac{2AP_4 - 3\mathcal{L}}{243}. \tag{170}$$

The result for all other subtracted terms, including their lower-loop assembly terms, is

$$\boxed{\begin{aligned} C_5^{\text{rest}} = &-\frac{5A^4}{15552} + \frac{5A^2Q}{324} - \frac{5A^2\pi^2}{7776} - \frac{23A\pi^3}{1296} - \frac{149A\zeta(3)}{324} \\ &+ \frac{298\mathsf{G}Q}{243} - \frac{8\mathcal{K}}{243} + \frac{4Q^2}{81} + \frac{319Q\pi^2}{2916} - \frac{83\pi^4}{46656}. \end{aligned}} \tag{171}$$

The coefficient of the only new master constant is $-8/243$. Adding the three sectors and using $H = 2\pi^2/3 + 8\mathsf{G}$ proves (27) by exact algebra.

The enumeration and weighting are fixed by the following graph construction. With $e_i = 4 - \sum_j m_{ij}$,

$$w_4 = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_2 = \frac{w_4}{12} \sum_i e_i(e_i - 1),$$

and an unlabeled L-loop class contributes

$$b_L = L(-1)^L \frac{w_4}{|\operatorname{Aut} G|} r_G, \qquad c_L = L(-1)^L \frac{w_2}{|\operatorname{Aut} G|} r_G.$$

The 129 five-loop classes are exhausted by 92 explicitly evaluated remaining classes, six wheel-insertion classes, three primitive classes, and 28 classes whose $3w_2 - w_4$ vanishes. The 26 non-wheel four-loop classes are also included. These disjoint sets cover the full enumeration; no unknown residue is hidden in (171).

14 Independent full-graph numerical control

The following control checks the assembled coefficient independently of special-function evaluation. Only the precision-preserving formulation and its converged comparison are needed here. The code and records are in the package `lri_numerical_resolution/` (Section 33).

The graph control uses the two coordinate cutoffs and meromorphic Chebyshev quadrature defined in Section 22, specialized to five loops: the cutoff "maxgap" requires the largest gap between consecutive ordered vertices to be below one, and the cutoff "span" requires the total vertex span to be below one. Below, N_{lower} is the quadrature degree of the lower-loop inputs. In particular its integrand is (600) with $L = 5$, and its radial moments are (601). The bare amplitudes are renormalized by (20) before applying (11). Thus the special-function period values are not inputs to this independent integration.

The combined bare simple pole is

$$H_{5,-1} = [z^{-1}](3\mathcal{B}_5 - \mathcal{A}_5).$$

Its exact sensitivity, with all lower-loop inputs held fixed, is

$$\delta C_5 = \frac{5}{729}\, \delta H_{5,-1}.$$

Indeed the new fifth-order contribution is $(3c_5 - b_5)/729$, while (25) gives $\delta(3c_5 - b_5) = 5\delta H_{5,-1}$. The program therefore forms the exact integer channel weights before integration; this removes 9,540 zero-weight ordered graphs from the 59,145-graph inventory, rather than subtracting two noisy channel sums only at the end. Quadrature weights, smooth-kernel evaluation, within-group

accumulation and final Laurent assembly are all precision-controlled. A same-degree comparison with the full two-channel calculation changes C_5 by less than 1.7×10^{-12}. The combined channel has 335 rather than 342 sector exponent groups, and exhaustive exact enumeration shows that the groups with exponent $a = -2$ disappear from it: its most negative exponent is $-\frac{3}{2}$. At degree 18, replacing only the binary64 five-loop kernel by the extended one, with the same lower-loop inputs, changes C_5 by -7.957×10^{-9} (span) and -7.965×10^{-9} (maxgap), most of the offset of the earlier version described after the table below. For an independent pole check, the renormalization-group equations predict the higher poles $H_{5,-p}$, $p \geq 2$, from the lower loops; the largest residuals are 1.96×10^{-11} (span) and 1.74×10^{-11} (maxgap). They are diagnostics only and do not certify the simple pole, which carries the new information.

With lower-loop degree $N_{\text{lower}} = 32$, the complete results are

Degree	Cutoff	Graph value of C_5	Graph value minus closed form
22	span	-731.1394766245375925	$+2.2334 \times 10^{-12}$
22	maxgap	-731.1394766245106192	$+2.9207 \times 10^{-11}$
24	span	-731.1394766245391892	$+6.3667 \times 10^{-13}$
24	maxgap	-731.1394766245368057	$+3.0201 \times 10^{-12}$

The comparison target is the special-function evaluation $-731.139476624539825853869126582527\ldots$. These are observed residuals, not rigorous bounds on either representation. The cutoff difference alone is not an accuracy certificate: a shared regular contribution can have a common error that cancels in that difference. The complete-channel and precision/degree comparisons are therefore separate checks. An earlier version of this control, whose quadrature kernel used binary64 arithmetic, gave -731.1394766152787 with the span cutoff, 9.26×10^{-9} above the closed form. The precision-preserving formulation removes that offset; the analytic coefficient was not changed. The signed error ledger of the earlier value is in `lri_numerical_resolution/verification/records/resolution_audit.json` (field `error_budgets`).

14.1 Further checks of the analytic coefficient

The following checks of the one-dimensional result were run with the analytic calculation and do not use the full graph integration above.

- *Exact algebra.* The two regular quadratic coefficients at pattern A_0 vanish; the fourth-order weighted sum reproduces the analytic $3c_4 - b_4$, including all gamma and zeta terms; every five-loop class is accounted for exactly once in the nonzero, primitive, wheel or zero-weight sets; and the rationally weighted residue table sums to (171), whose addition to the primitive and wheel sectors gives (27). No coefficient is inferred from a numerical integer-relation fit (`lri_numerical_resolution/verification/records/symbolic_audit.json`).

- *Seven subtracted graphs.* With extended precision in the quadrature weights, integrands and accumulation, seven classes were integrated anew, with the analytic residues used only as comparison targets. At interpolation degree 18 the numerical minus analytic residues are -1.02×10^{-9} (class 52), -4.95×10^{-9} (56), $+3.31 \times 10^{-10}$ (60), -2.25×10^{-9} (79), -5.19×10^{-10} (80), -5.82×10^{-11} (91) and $+2.27 \times 10^{-10}$ (127); at degree 22 those of classes 56 and 79 fall to -1.02×10^{-11} and -1.92×10^{-11}. The lower-loop inputs of these comparisons were evaluated as bare span-cutoff integrals. The checks cover gamma-valued and $\mathcal{K}$-dependent residues; their record is not part of the repository.

- *Fresh evaluation of the masters.* Evaluating all four masters without stored period values, at 65 working digits with 64 Cauchy nodes for the parameter derivatives, reproduces $C_5 = -731.13947662453982585386912658252723066 25\ldots$; a separate 28-digit run agrees to its

requested precision (record not part of the repository; the replay of Section 33 recomputes C_5 from the closed forms).

- *Regenerated certificate.* In the code delivery that preceded this note, the symbolic task regenerated all 26 non-wheel four-loop and 92 retained five-loop residues of Appendix A from the graph-class data, the master expansions and the forest rules, without reading a saved residue; all 118 comparisons were exact. That record is not part of the repository. Regenerating the table is stronger than adding the saved rows, but it uses the same master identities and reduction code.

These tests, like the graph integration, do not establish the physical five-loop coefficient independently of the shared graph and renormalization framework.

14.2 External lattice comparison: informative but inconclusive

The Monte Carlo data of Uzelac, Glumac and Aničić concern the same classical long-range chain, with $J(r) \propto r^{-(1+s)}$ [11]. Their Table II gives $y_t = \nu^{-1}$, which here is $1 - \Delta_{\mathcal{E}}$, at three couplings. An earlier comparison resummed the series of y_t through ϵ^5 in six Borel–Padé–Leroy forms: the approximants [3/2] and [2/3] of the Borel–Leroy transform $\sum_k c_k t^k/\Gamma(k+b+1)$, with $b = 0, 1, 2$, followed by the Laplace integral $\int_0^\infty e^{-t} t^b [L/M](\epsilon t)\, dt$. None of the six approximants has a positive real pole at these couplings. No Monte Carlo value was fitted.

s	ϵ	y_t of [11]	Resummation range
0.6	0.2	0.50 ± 0.02	0.51738–0.51774
0.7	0.4	0.51 ± 0.02	0.51518–0.51886
0.8	0.6	0.47 ± 0.02	0.50021–0.51231

The third row shows a tension, and it must not be omitted from the interpretation: at $s = 0.8$ all six resummed values lie above the Monte Carlo interval, the lowest by 0.010. The resummation spread is not a confidence interval. Changing C_5 by 20% in either direction still gives pole-free resummed values inside the Monte Carlo interval nearest to the expansion point, at $s = 0.6$, so these data do not isolate the fifth coefficient. With C_5 lowered by 20%, however, the three [3/2] approximants acquire positive real Borel poles at all three couplings, and only the three [2/3] values remain; with C_5 raised by 20%, all six stay pole-free. The closed-form $C_0, \ldots, C_5$ of Section 4 reproduce the three ranges to the digits shown: `lri_numerical_resolution/public/lattice_comparison.py` recomputes the ranges, the absence of positive Borel poles and the 20% statement. No simulation was run for this note.

Part III

Arbitrary-dimensional analytic coefficients and specializations

15 General-dimensional normalization and lower-order derivation

15.1 Dimension, regulator and normalization

In this section, d always denotes the spatial dimension, not the dummy exponent called d in some earlier one-dimensional subintegrals. Take

$$0 < d < 4, \qquad \epsilon = 2s - d, \qquad \zeta = \frac{s}{2} = \frac{d+\epsilon}{4}, \qquad \Delta_\phi = \frac{d-\epsilon}{4}. \tag{172}$$

The physical long-range kinetic term has $0 < \zeta < 1$; at fixed $d < 4$ this holds for sufficiently small positive ϵ. Noninteger d is defined by analytic continuation of the dimensionally parameterized integrals. Analytically continuing individual periods to $d = 4$ is useful as a check, but it is not the same as replacing this expansion by the short-range Wilson–Fisher expansion: the absence of a wave-function counterterm, used below, is a long-range statement [7, 12].

After tuning the relevant local mass term to the critical surface,

$$\Delta_{\mathcal{E}}(d,\epsilon) = d - \nu^{-1} = \frac{d-\epsilon}{2} + \gamma_r(g_*), \qquad \beta(g) = -\epsilon g + \sum_{\ell\geq 1} b_\ell(d) g^{\ell+1}, \qquad \gamma_r(g) = \sum_{\ell\geq 1} c_\ell(d) g^\ell. \tag{173}$$

We use a minimal-subtraction convention with $b_1 = 3$, $c_1 = 1$ and an exactly pole-only one-loop bubble. Write z for the analytic regulator and introduce the infrared covariance $(p^2+\mu^2)^{-(d+z)/4}$. The bare coupling normalization is

$$\lambda_0 = \mu^z \kappa(d,z) \mathcal{Z}_g(g,z), \qquad \kappa(d,z) = (4\pi)^{d/2} \frac{\Gamma((d+z)/2)}{\Gamma(1+z/2)}. \tag{174}$$

The parameter μ is an infrared regulator, not the coefficient of the physical ϕ^2 perturbation. A direct momentum integration gives

$$\kappa(d,z) \int \frac{d^d p}{(2\pi)^d} (p^2+1)^{-(d+z)/2} = \frac{2}{z}. \tag{175}$$

Hence the one-loop four-point and mass amplitudes are $3/z$ and $1/z$. The extra z-dependent normalization in (174) must be kept before extracting Laurent coefficients. Finite parts computed in this massive-Schwinger prescription must not be mixed with finite parts from the one-dimensional ordered-span prescription of Section 3.

15.2 The evaluated coefficients through the third order

Let

$$\begin{aligned} q &= d/4, & A_d &= \psi_0(2q) + \psi_0(1) - 2\psi_0(q), \\ R_d &= \psi_1(2q) - \psi_1(1), & M_d &= \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)} = -\frac{\Gamma(1-q)\Gamma(2q)^2}{q\Gamma(3q)}, \end{aligned} \tag{176}$$

and let $\mathcal{T}_d$ be the corrected tetrahedron beta-function coefficient α_{I_4} of [12]. Its independent integral definition is given in (188); put

$$Q_d = \frac{\mathcal{T}_d}{6}. \tag{177}$$

In particular $Q_1 = Q$, but Q_d is not defined by replacing the argument of the Gamma functions in the one-dimensional value Q. The coefficient formulas are

$$\boxed{\begin{aligned} &C_0(d) = \frac{d}{2}, && C_1(d) = -\frac{1}{6}, \quad C_2(d) = \frac{A_d}{9}, \\ &C_3(d) = \frac{A_d^2 + R_d - 8Q_d}{108} = \frac{A_d^2 + R_d}{108} - \frac{\mathcal{T}_d}{81}. \end{aligned}} \tag{178}$$

Here and below the polygamma convention is $\psi_m(z) = d^{m+1}\log\Gamma(z)/dz^{m+1}$. The corresponding three-loop MS coefficients in (174) are

$$\boxed{\begin{aligned} &b_1 = 3, && c_1 = 1, && b_2 = -6A_d, \quad c_2 = -A_d, \\ &b_3 = 6Q_d + \frac{87}{4}A_d^2 - \frac{9}{4}R_d + M_d, && c_3 = \frac{7}{2}A_d^2 - \frac{1}{2}R_d + \frac{1}{3}M_d. \end{aligned}} \tag{179}$$

The M_d term cancels from the physical $C_3(d)$, but it cannot be dropped from the RG functions used at fourth and fifth order.

At $d = 1$,

$$A_1 = A, \qquad R_1 = \frac{\pi^2}{3}, \qquad M_1 = -4\pi, \qquad Q_1 = \frac{\Gamma(1/4)^4}{\pi}. \tag{180}$$

This recovers (9) and the one-dimensional C_3 of Section 4. Equation (176) also displays a concrete obstruction to blind argument substitution: $3q$ and $1-q$ cancel in M_d only at $d = 1$. In addition, the reflected identity for R_d is not obtained by keeping $\psi_1(q) + \psi_1(1-q) = 2\pi^2$ away from $q = 1/4$.

15.2.1 Derivation from the published three-loop calculation

The three-loop beta functions of [7], Eqs. (2.13) and (2.14) there, are sums over graphs with coefficients α_X: α_D for the one-loop graph, α_S for the two-loop graph, and α_U, α_T and $\alpha_{I_1}, \ldots, \alpha_{I_4}$ for the seven three-loop graphs, each the normalized graph amplitude of Eq. (2.12) there. The tetrahedron coefficient α_{I_4} given in [7] was corrected in [12]; the corrected value is required here. For clarity, let $t_1 = \psi_1(1)$, $t_q = \psi_1(q)$, $t_2 = \psi_1(2q)$ and $J = J_0(q)$. The coefficients quoted in [12], specialized to one scalar, are

$$\begin{aligned} \alpha_U = \alpha_{I_2} &= -t_1 - t_q + 2t_2 + J, \\ \alpha_T &= \tfrac{1}{2}A_d^2 + \tfrac{1}{2}t_1 + t_q - \tfrac{3}{2}t_2 - J, \\ \alpha_{I_1} &= \tfrac{3}{2}A_d^2 + \tfrac{1}{2}(t_1 - t_2), \qquad \alpha_{I_3} = M_d/3, \qquad \alpha_{I_4} = 6Q_d. \end{aligned} \tag{181}$$

An explicit definition, useful for reproducing the intermediate cancellation, is

$$\begin{aligned} J_0(q) = \frac{1}{\Gamma(q)^2}\sum_{n=1}^{\infty} &\frac{\Gamma(n+2q)\Gamma(n+q)^2}{n\,n!\,\Gamma(2q+2n)} \\ &\times \left[2\psi_0(n+1) - \psi_0(n) - 2\psi_0(n+q) - \psi_0(n+2q) + 2\psi_0(2q+2n)\right]. \end{aligned} \tag{182}$$

After dividing the zero-momentum one-loop normalization out of the coupling, write $\alpha_S/\alpha_D^2 = -A_d + s_1 z + O(z^2)$. The expansion in [12] gives

$$s_1 = \frac{1}{4}A_d^2 + \frac{3}{4}t_1 + t_q - \frac{7}{4}t_2 - J. \tag{183}$$

The two-loop bare amplitudes are then

$$\mathcal{A}_2 = \frac{9}{z^2} + \frac{3A_d}{z} - 3s_1 + O(z), \qquad \mathcal{B}_2 = \frac{2}{z^2} + \frac{A_d}{2z} - \frac{s_1}{2} + O(z). \tag{184}$$

Consequently, at $z = 0$, the zero-momentum and MS variables obey

$$g_{\mathrm{ZM}} = g_{\mathrm{MS}} - 3s_1 g_{\mathrm{MS}}^3 + O(g^4), \qquad r_{\mathrm{ZM}} = r_{\mathrm{MS}}(1 - \tfrac{1}{2}s_1 g_{\mathrm{MS}}^2 + O(g^3)). \tag{185}$$

These finite transformations imply

$$\begin{aligned} b_3^{\mathrm{MS}} &= 3\alpha_T + 12\alpha_U + 12\alpha_{I_1} + 3\alpha_{I_3} + \mathcal{T}_d + 9s_1, \\ c_3^{\mathrm{MS}} &= \alpha_T + 2\alpha_{I_1} + \alpha_U + \alpha_{I_3}. \end{aligned} \tag{186}$$

Substitution yields (179): all J and t_q terms cancel. Alternatively, the physical coefficient can be checked directly before choosing the MS scheme:

$$\begin{aligned} C_3 &= -\frac{1}{81}\left[-12A_d^2 + 9s_1 + 9\alpha_U + 6\alpha_{I_1} + \mathcal{T}_d\right] \\ &= \frac{A_d^2 + R_d}{108} - \frac{\mathcal{T}_d}{81}. \end{aligned} \tag{187}$$

The cancellation is symbolic; it is not an integer-relation fit.

15.3 A specified tetrahedron at every dimension

The equal-index specialization of the corrected kite representation in [12], eqs. (3.8)–(3.11), is

$$\boxed{\begin{aligned} \mathcal{T}_d = \frac{2\Gamma(2q)^2}{\Gamma(q)^4} \int_{-q/3-\mathrm{i}\infty}^{-q/3+\mathrm{i}\infty} \frac{du}{2\pi\mathrm{i}} \int_{-q/3-\mathrm{i}\infty}^{-q/3+\mathrm{i}\infty} \frac{dv}{2\pi\mathrm{i}} \\ \times \frac{\Gamma(-u)^2\Gamma(-v)^2\Gamma(q+u)\Gamma(q+v)\Gamma(q+u+v)^2\Gamma(-u-v)}{\Gamma(q-u)\Gamma(q-v)\Gamma(2q+u+v)}. \end{aligned}} \tag{188}$$

Both contours run upwards. Every numerator Gamma argument has positive real part on these contours for $q > 0$, so no pole separation is left implicit. If $u = -q/3 + \mathrm{i}x$, $v = -q/3 + \mathrm{i}y$, the exponential part of Stirling's bound is $\exp[-\pi(|x| + |y| + |x+y|)]$; the double integral is absolutely convergent. There are no hidden regulator derivatives in this representation. It is a twofold Mellin–Barnes integral, not a claim of an ordinary single-variable hypergeometric evaluation for generic d.

For an independent geometric normalization, let $|e| = 1$ and

$$\begin{aligned} P_3^{\mathrm{raw}}(d) &= \int_{\mathbb{R}^d} d^d x \int_{\mathbb{R}^d} d^d y \, \left[x^2(x-e)^2y^2(y-e)^2(x-y)^2\right]^{-q}, \\ h_d &= \frac{\Gamma(d/2)}{\pi^{d/2}}, \\ \mathcal{T}_d &= 2h_d^2 P_3^{\mathrm{raw}}(d), \qquad Q_d = \frac{h_d^2}{3} P_3^{\mathrm{raw}}(d). \end{aligned} \tag{189}$$

The spatial integral defines the same analytic family; the MB formula provides a direct meaning when d is not an integer.

15.3.1 An exact two-dimensional Gamma evaluation

For $d = 2$ identify $\mathbb{R}^2$ with $\mathbb{C}$. Then

$$P_3^{\mathrm{raw}}(2) = \int_{\mathbb{C}^2} \frac{d^2z\, d^2w}{|z|\,|1-z|\,|w|\,|1-w|\,|z-w|}. \tag{190}$$

Write $\gamma_C(a) = \Gamma(a)/\Gamma(1-a)$ for real equal holomorphic and antiholomorphic parameters. The complex Selberg identity [6], with $n = 2$, $\sigma = \tau = 1/2$, $\theta = -1/4$, gives

$$\begin{aligned} P_3^{\mathrm{raw}}(2) &= 2\pi^2 \prod_{j=1}^{2} \frac{\gamma_C(1/2 + (j-1)\theta)^2 \gamma_C(j\theta)}{\gamma_C(1+j\theta)\gamma_C(\theta)} \\ &= \frac{\pi^2}{2}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4. \end{aligned} \tag{191}$$

The convergence inequalities hold strictly: $\theta > -1/2$, $\sigma+\theta = \tau+\theta = 1/4 > 0$, $\sigma+\tau+\theta = 3/4 < 1$, and $\sigma+\tau+2\theta = 1/2 < 1$. Thus this is a convergent integral evaluation, not a formal continuation of a divergent Selberg integral. Using $h_2 = 1/\pi$ yields

$$\boxed{\begin{aligned} \mathcal{T}_2 &= \left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4, \qquad Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4, \\ C_2(2) &= \frac{4\log 2}{9}, \qquad C_3(2) = \frac{4\log^2 2}{27} - \frac{1}{81}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4. \end{aligned}} \tag{192}$$

The fact that quarter-arguments reappear at $d = 2$ here is a consequence of the complex Selberg reduction, not a justification for replacing arguments in the $d = 1$ formulas.

15.4 Dimension-independent graph data and the d-dimensional kernel

Use precisely the finite matrix set in (12): $L+1$ vertices, $2L$ internal edges, $m_{ij} \in \{0,1,2,3\}$, internal degrees between two and four, and every nontrivial cut containing at least two edges. The weights w_4, w_2 are those in (13). In a full d-dimensional integral there is *no ordering of vertices on a line*. Therefore the labeled sum now includes the vertex factorial:

$$\boxed{\begin{aligned} \mathcal{A}_L(d,z) &= \frac{1}{(L+1)!}\sum_{m\in\mathfrak{G}_L} w_4(m)\mathcal{I}_m(d,z), \\ \mathcal{B}_L(d,z) &= \frac{1}{(L+1)!}\sum_{m\in\mathfrak{G}_L} w_2(m)\mathcal{I}_m(d,z). \end{aligned}} \tag{193}$$

Equivalently, sum one representative of each vertex-isomorphism class with weights $w_4/|\operatorname{Aut}_v(m)|$, $w_2/|\operatorname{Aut}_v(m)|$. Parallel-edge factorials are already present in w_4, w_2; Aut_v counts vertex permutations, not a second set of edge permutations.

Let the $2L$ individual internal lines, including parallel lines, be labeled $e = 1, \dots, 2L$. The first Symanzik polynomial and simplex measure are

$$U_m(x) = \sum_{\text{spanning trees } T}\prod_{e\notin T} x_e, \qquad d\Omega = \delta\!\left(1 - \sum_{e=1}^{2L} x_e\right)\prod_{e=1}^{2L} dx_e. \tag{194}$$

Put $a = (d+z)/4$ and $r(d,z) = \Gamma(2a)/\Gamma(1+z/2)$. The complete normalized graph kernel is

$$\boxed{\mathcal{I}_m(d,z) = \frac{r(d,z)^L\Gamma(Lz/2)}{\Gamma(a)^{2L}}\int_{x_e\ge 0} d\Omega\,\frac{\prod_{e=1}^{2L} x_e^{a-1}}{U_m(x)^{d/2}}.} \tag{195}$$

For a primitive graph it suffices to set $z = 0$ in the simplex factor when computing the residue. For nonprimitive graphs one must first meromorphically continue the integral and apply the subtraction recursion; the simplex integral at $z = 0$ need not converge.

To derive (195), apply a Schwinger parameter to every $(p^2+1)^{-a}$, perform the L Gaussian momentum integrals, and set the Schwinger parameters equal to λx_e. The powers of the common scale are

$$\lambda^{2La-1-dL/2} = \lambda^{Lz/2-1}, \qquad \int_0^\infty e^{-\lambda}\lambda^{Lz/2-1}d\lambda = \Gamma(Lz/2).$$

The factors $(4\pi)^{-Ld/2}$ cancel against (174). For the one-loop bubble $U = x_1 + x_2 = 1$ on the simplex, (195) gives exactly $2/z$, confirming both the factor of two and the vertex-factorial convention.

Power subgraphs and continuation. For fixed $0 < d < 4$ a proper unmarked two-point subgraph has a local power divergence, not an independent wave-function counterterm. We use the analytic critical-mass prescription of [7], as in the one-dimensional calculation: its local mass power divergence is removed by critical tuning and its analytic continuation is retained. In particular, graphs with triple lines or vacuum-attached pieces are not silently dropped. An ordinary domain from which to continue the raw Schwinger integrals is real $z > d/3$: for an induced subgraph with v vertices and e edges, the ultraviolet condition is $ez + d(2v - 2 - e) > 0$, and the worst tadpole-free two-vertex case is $e = 3$. Continuing to $z = 0$ implements the specified analytic regularization; subsequent logarithmic subdivergences are removed by MS. No cutoff on individual divergent simplex faces is to be substituted for this prescription.

L	Labeled matrices	Vertex classes	Primitive classes	Nonzero labeled $3w_2 - w_4$
1	1	1	1	0
2	6	2	0	3
3	85	8	1	55
4	1890	27	1	1470
5	59145	129	3	49605

These counts are for the displayed matrix set, not for a minimal nonprimitive master basis. Some classes can disappear or combine only after analytic continuation and subtraction. The inventory file `lri_general_special_functions_update/verification/graph_inventory.json`, written by `code/graph_inventory.py` of that package, prints all 167 classes through five loops, their exact rational weights, expanded edge lists and every monomial of U_m. The polynomials were checked independently against the matrix-tree determinant at exact integer edge parameters (record `verification/inventory_checks.json`). The identity

$$\frac{1}{(L+1)!}\sum_m w_i(m)\mathcal{I}_m = \sum_{[m]} \frac{w_i(m)}{|\operatorname{Aut}_v(m)|}\mathcal{I}_m \tag{196}$$

follows from orbit–stabilizer and was also checked in the enumeration.

15.5 An all-regulator two-loop master in ordinary hypergeometric functions

There are only two two-loop vertex classes. The one-vertex-reducible class has amplitude $(2/z)^2$; denote the other by $S_d(z)$. Their exact weights give

$$\mathcal{A}_2 = \frac{3}{z^2} + 3S_d(z), \qquad \mathcal{B}_2 = \frac{1}{z^2} + \frac{1}{2}S_d(z). \tag{197}$$

For the same $a = (d+z)/4$, put $b = (d+3z)/4$ and define

$$\begin{aligned} F_1(d,z) &= {}_3F_2\left(\begin{matrix} d/2, z/2, a \\ 1-z/2, a+1/2 \end{matrix}; \frac{1}{4}\right), \\ F_2(d,z) &= {}_3F_2\left(\begin{matrix} z, 2a, b \\ 1+z/2, b+1/2 \end{matrix}; \frac{1}{4}\right), \\ W(d,z) &= \frac{r(d,z)^2\Gamma(1+z)\Gamma(1-z/2)\Gamma(b)^2}{\Gamma(a)^2\Gamma(d/2)\Gamma(2b)}. \end{aligned} \tag{198}$$

Then the full meromorphic two-loop amplitude is

$$\boxed{S_d(z) = \frac{4F_1(d,z) - 2W(d,z)F_2(d,z)}{z^2}.} \tag{199}$$

The two defining ${}_3F_2$ series have argument $1/4$, strictly inside the convergence disk. Near $z=0$ their parameters and the Gamma factors in W are regular. Thus (199) determines all of the two-loop Laurent data needed at fourth and fifth order, not just the residue. The pole is explicit and the numerator is analytic.

15.5.1 Derivation and a residue check

Number the edges so that the Symanzik polynomial is $U=(x_1+x_2)(x_3+x_4)+x_3x_4$. Set $x_1+x_2=u$, $x_3=(1-u)v$, $x_4=(1-u)(1-v)$. The beta integral over the split of x_1,x_2 and then over u reduces the simplex to

$$\begin{aligned} S_d(z) =& \frac{r(d,z)^2\Gamma(z)}{\Gamma(a)^4}B(a,a)B(2a,z/2) \\ &\times \int_0^1 [v(1-v)]^{a-1}\,{}_2F_1\!\left(\frac{d}{2},\frac{z}{2};\frac{d}{2}+z;1-v+v^2\right)dv. \end{aligned} \tag{200}$$

Apply the Gauss connection formula at $1-v(1-v)$ [23]. Its two pieces have powers $[v(1-v)]^0$ and $[v(1-v)]^{z/2}$, respectively. For either piece, termwise beta integration uses

$$\frac{B(c+n,c+n)}{B(c,c)}=\frac{(c)_n^2}{(2c)_{2n}}=\frac{(c)_n}{4^n(c+1/2)_n}.$$

This gives exactly F_1,F_2. The first prefactor simplifies to $4/z^2$; the second to $-2W/z^2$. The derivation starts where the beta integrals converge and then continues meromorphically.

At $z=0$, $F_1=F_2=W=1$. The $n\ge 1$ terms of the defining series show $\partial_z F_2(d,0)=2\partial_z F_1(d,0)$, while $\partial_z \log W(d,0)=-A_d/2$. It follows directly that

$$4F_1-2WF_2=2+A_dz-s_1z^2+O(z^3),\qquad S_d(z)=\frac{2}{z^2}+\frac{A_d}{z}-s_1+O(z). \tag{201}$$

Thus the known two-loop pole is recovered without evaluating the intermediate hypergeometric derivatives separately. Derivatives appear here only in the proof of a Taylor identity; the all-regulator formula (199) is an ordinary fixed-argument evaluation.

15.6 Complete higher-order MS assembly

For all the following quantities use the same prescription (174)–(195). Define

$$a_{L,k}(d)=[z^k]\mathcal{A}_L(d,z),\qquad e_{L,k}(d)=[z^k]\mathcal{B}_L(d,z). \tag{202}$$

The letter e here is a mass-amplitude Laurent coefficient, not an external half-edge count e_i. The following finite recipe makes the general-dimensional C_4,C_5 representation unambiguous.

Set

$$\begin{aligned} F_4(v)&=v+\sum_{L=1}^{5}(-1)^L\mathcal{A}_Lv^{L+1}, & F_2(v)&=1+\sum_{L=1}^{5}(-1)^L\mathcal{B}_Lv^L,\\ \mathcal{Z}_g&=g+\sum_{n=2}^{6}G_n(z)g^n, & Z_r&=1+\sum_{n=1}^{5}Z_n(z)g^n. \end{aligned} \tag{203}$$

All G_n,Z_n are pure pole parts. Recursively impose

$$\begin{aligned} G_n&=-\operatorname{PP}[g^n]F_4(\mathcal{Z}_{g,<n}), & Z_n&=-\operatorname{PP}[g^n]\{Z_{r,<n}F_2(\mathcal{Z}_g)\},\\ b_L&=L[z^{-1}]G_{L+1}, & c_L&=L[z^{-1}]Z_L. \end{aligned} \tag{204}$$

Here PP retains negative powers of z. This is also an explicit forest subtraction: multiplication by lower counterterms is carried out before selecting a new simple pole. Keeping only primitive graphs would not implement (204).

Write $A = A_d$, $R = R_d$, $M = M_d$, $Q = Q_d$ temporarily in the formulas below. Expanding the recursion gives

$$\begin{aligned}
b_4 &= 36Aa_{2,0} - 216a_{2,1} + 48a_{3,0} - 4a_{4,-1},\\
c_4 &= 26Ae_{2,0} - 140e_{2,1} + 40e_{3,0} - 4e_{4,-1},
\end{aligned} \tag{205}$$

$$\begin{aligned}
b_5 &= 5a_{5,-1} - 75a_{4,0} + 450a_{3,1} - 60Aa_{3,0} - 1350a_{2,2} + 585Aa_{2,1} - 5b_3a_{2,0},\\
c_5 &= 5e_{5,-1} - 65e_{4,0} + 325e_{3,1} - \frac{95}{2}Ae_{3,0} - \frac{2275}{3}e_{2,2}\\
&\quad + \frac{725}{2}Ae_{2,1} - \frac{10b_3 + 5c_3}{3}e_{2,0}.
\end{aligned} \tag{206}$$

Every $a_{L,k}, e_{L,k}$ in these formulas is the coefficient of the explicit finite integral sum (193), with U defined in (194); there are no unspecified diagram labels. The code inventory is a convenience for enumerating this finite definition, not a replacement for it.

For completeness, the required orders are

Loop order	Highest Laurent power	Required analytic input
2	z^2	Full formula (199)
3	z^1	Master expansions through the stated order
4	z^0	Master finite parts and the primitive period
5	z^{-1}	Subtracted simple poles and primitive periods

The combinations of these finite data required by the physical coefficients are reduced in Section 16. They cannot be discarded merely because the primitive residues are known.

15.6.1 The coefficient formulas through fifth order

Lagrange inversion applied to $\epsilon = g_*(3 + b_2g_* + b_3g_*^2 + b_4g_*^3 + b_5g_*^4)$ gives

$$C_n(d) = -\frac{1}{2}\delta_{n1} + \frac{1}{n}[g^{n-1}]\frac{\gamma_r'(g)}{(3 + b_2g + b_3g^2 + b_4g^3 + b_5g^4)^n}. \tag{207}$$

Define $D_4(d) = 3c_4(d) - b_4(d)$ and $D_5(d) = 3c_5(d) - b_5(d)$. Substituting the evaluated (179) gives the collected general-d formulas

$$\boxed{243C_4(d) = D_4 - 51A^3 + 9AR - 48AQ - 2AM,} \tag{208}$$

$$\boxed{\begin{aligned}
729C_5(d) = D_5 + 10AD_4 - 6Ac_4 - \frac{5541}{16}A^4 - \frac{51}{4}A^2M - 162A^2Q + \frac{315}{8}A^2R\\
+ 6MQ - \frac{3}{4}MR + 36Q^2 - 18QR + \frac{27}{16}R^2.
\end{aligned}} \tag{209}$$

Equations (193), (195), and (205)–(206) are part of these formulas: D_4, c_4, D_5 are completely specified Laurent functionals, not asserted new elementary constants. Nevertheless, leaving them as integrals is a weaker endpoint than the finite hypergeometric reduction obtained at $d = 1$. In particular, a known value of D_4 would not by itself determine c_4 or the finite four-loop amplitudes needed for C_5.

An equivalent form, identical to (10) and (11) and convenient for checking signs, is

$$243C_4 = 60A^3 - 8Ab_3 + 18Ac_3 + 3c_4 - b_4,$$
$$729C_5 = 336A^4 - 60A^2b_3 + 108A^2c_3 + b_3^2 - 3b_3c_3 - 10Ab_4 + 24Ac_4 + 3c_5 - b_5. \tag{210}$$

All terms in (208)–(209) have been checked by exact symbolic algebra, including independent extraction of the simple-pole coefficients in (205)–(206) from the RG pole equations. Substituting the established one-dimensional RG data into the universal assembly reproduces the retained $C_4(1), C_5(1)$ formulas. That last check is an algebraic specialization, not a fresh numerical evaluation of the new Schwinger integrals at $d = 1$.

15.6.2 The combined five-loop channel survives in arbitrary d

The exact integrand weight used in the numerical resolution of the one-dimensional problem was combinatorial:

$$w_H = 3w_2 - w_4 = \frac{w_4}{4}\left[\sum_i e_i(e_i - 1) - 4\right]. \tag{211}$$

It therefore cancels the same 9540 of the 59145 labeled five-loop matrices in every dimension. With lower-loop data held fixed,

$$\boxed{\delta C_5(d) = \frac{5}{729}\,\delta[z^{-1}]\{3\mathcal{B}_5(d,z) - \mathcal{A}_5(d,z)\}.} \tag{212}$$

This does not remove the $a_{4,0}, e_{4,0}$ terms. It identifies the five-loop channel that should be integrated directly, before a large numerical subtraction, when performing a general-d evaluation.

15.7 General-dimensional primitive periods

The spectral construction extends the primitive periods, but not by keeping the one-dimensional five hypergeometric parameter lists. Let $P_L^{\mathrm{raw}}(d)$ be the conformal zig-zag decompletion with edge exponents $q = d/4$, two fixed points at distance one and the remaining $L - 1$ vertices integrated over $\mathbb{R}^d$. Normalize

$$P_L(d) = h_d^{L-1} P_L^{\mathrm{raw}}(d), \qquad h_d = \Gamma(d/2)/\pi^{d/2}. \tag{213}$$

Thus $P_3(d) = 3Q_d$, $P_4(1) = P_4$, and $P_5(1) = P_5$. The raw coordinate integrals at integer dimension and their continued spectral or parametric representations fix the graph definition.

Set $p = d/2 - 1$ and $k = L + 2$. The eigenvalue and measure obtained from [24], eqs. (5.23), (5.27), and (6.2), are

$$\tau_n^{(d)}(\nu) = (-1)^n \pi^{d/2} \frac{\Gamma(n/2 + d/8 + \mathrm{i}\nu)\Gamma(n/2 + d/8 - \mathrm{i}\nu)}{\Gamma(n/2 + 3d/8 + \mathrm{i}\nu)\Gamma(n/2 + 3d/8 - \mathrm{i}\nu)}, \tag{214}$$

$$\rho_n^{(d)}(\nu) = \frac{\Gamma(p)\Gamma(n+2p)(n+p)}{2\pi^{3d/2+1}\Gamma(2p)n!}\left[(n+p)^2 + 4\nu^2\right]\frac{\Gamma(p + 2\mathrm{i}\nu)\Gamma(p - 2\mathrm{i}\nu)}{\Gamma(2\mathrm{i}\nu)\Gamma(-2\mathrm{i}\nu)}. \tag{215}$$

For $2 < d < 4$ the ordinary real-axis representation is

$$\boxed{P_L(d) = h_d^{L-1} \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu)\,[\tau_n^{(d)}(\nu)]^{L+2} d\nu, \qquad L = 3, 4, 5.} \tag{216}$$

The angular quantum number n runs over an infinite tower. Its presence is one reason the finite $d = 1$ hypergeometric formula cannot be lifted by changing quarter-arguments alone.

For $0 < d < 2$, $-1 < p < 0$ and the first pair of Gamma poles crosses the real spectral contour. Continuing the contour as in the earlier one-dimensional derivation adds

$$P_L(d) = h_d^{L-1}\Bigg\{ \sum_{n=0}^{\infty} \int_0^{\infty} \rho_n^{(d)}(\nu)[\tau_n^{(d)}(\nu)]^{L+2} d\nu + \sum_{n=1}^{\infty} \frac{\Gamma(n+2p)(n+p)n(n+2p)}{2\pi^{3d/2} n!\Gamma(-p)} [\tau_n^{(d)}(-\mathrm{i}p/2)]^{L+2} \Bigg\}. \tag{217}$$

The $n = 0$ crossing vanishes identically. At removable singular parameter values such as $d = 1, 2$, take the joint analytic limit of the Gamma ratios, not the separate factors. There are no additional measure-pole crossings inside $0 < d < 2$; the eigenvalue Gamma arguments for the $n \geq 1$ residues stay away from their pole sets. The residue coefficient is $-2\pi\mathrm{i}\,\mathrm{Res}_{\nu=-\mathrm{i}p/2}(\rho\tau^{L+2})$, fixing its sign. At $d = 1$, formula (217) reduces to the continuous-plus-discrete formula (111) of Section 10. The numerical code `lri_general_special_functions_update/code/primitive_spectral.py` (record `verification/primitive_spectral_checks.json`) implements only (216) with $d > 2$; it does not silently substitute $d < 2$ into that real-axis code.

As a normalization check, at $d = 4$ one has $\tau_n = (-1)^n 4\pi^2/[(n+1)^2 + 4\nu^2]$. An elementary beta integration then gives the graph-period limits

$$P_3(4) = 6\zeta(3), \qquad P_4(4) = 20\zeta(5), \qquad P_5(4) = \frac{441}{8}\zeta(7). \tag{218}$$

Equivalently, $\mathcal{T}_4 = 12\zeta(3)$. These are checks of the period normalization, not assertions that the complete long-range RG functions have a regular short-range limit; M_d, for example, has a pole at $d = 4$.

15.7.1 Primitive weights and the product periods

The primitive classes at $L = 3, 4, 5$ have vertex automorphism orders 24, 8, and $(12, 16, 2)$ respectively. The four external legs are on distinct vertices, so their mass weights vanish. At five loops, the first two completed graphs have a three-vertex cut with two components of two vertices each. Fix the cut vertices at $0, e, \infty$; then the two remaining integrations decouple into two tetrahedron integrals. Since every completed vertex has four propagators of exponent $d/4$, the conformal normalization gives $P_3(d)^2 = 9Q_d^2$, independently of d. The third class is the zig-zag $P_5(d)$. Consequently the primitive pieces are

$$\boxed{\begin{aligned} &b_3^{\mathrm{prim}}(d) = 6Q_d, && b_4^{\mathrm{prim}}(d) = -6P_4(d), \\ &b_5^{\mathrm{prim}}(d) = 63Q_d^2 + 24P_5(d), && c_L^{\mathrm{prim}}(d) = 0 \quad (L = 3, 4, 5). \end{aligned}} \tag{219}$$

Their direct new-loop contributions to C_4, C_5 are therefore $2P_4(d)/81$ and $-[63Q_d^2 + 24P_5(d)]/729$. The lower-loop RG assembly and nonprimitive graphs remain essential. The inventory file `graph_inventory.json` supplies explicit completed edge lists and the cut vertices for the product factorizations, rather than identifying them only by topology names.

15.8 Numerical checks of the lower-order and primitive data

These checks were run by OpenAI GPT-6 astra in the general-dimensional calculation (package `lri_general_special_functions_update/`, scripts `tetrahedron.py`, `two_loop_master.py`, `primitive_spectral.py`, `symbolic_checks.py` and `inventory_checks.py` in `code/`; records of the same names in `verification/`). They are numerical consistency residuals, not interval enclosures.

Tetrahedron. A double-precision tensor trapezoidal quadrature of (188) used the contours $\Re u = \Re v = -q/3$, spacings 0.015 and 0.01, and imaginary cutoffs 8 and 10. The finer run differs from the exact values at $d = 1, 2, 4$ ((180), (192) and $\mathcal{T}_4 = 12\zeta(3)$) by about 3.65×10^{-13}, 6.43×10^{-14} and 5.63×10^{-15}. The resulting values, with $C_2(d)$ and $C_3(d)$ from (178), are (record `C0_to_C3_values.csv`)

d	$\mathcal{T}_d$	$C_2(d)$	$C_3(d)$
1	330.008919265	0.657131264	−3.719856323
1.5	142.350694846	0.427213506	−1.612227399
2	76.628287192	0.308065414	−0.874850013
2.5	46.354193318	0.233416481	−0.535556041
3	30.102615387	0.181221785	−0.353581520
3.5	20.493170714	0.142032326	−0.246028075

The rows at $d = 1, 2, 3$ agree with the table of Section 1. At $d = 1.5$ and 2.5 the record values of $\mathcal{T}_d$ (the full double-precision values in `C0_to_C3_values.csv`) agree with $6Q_d$ from the independent spectral code of `LRI_C5_general_d/references/reference_values.json` to 1.0×10^{-13} and 2.2×10^{-13} absolute (comparison made for this note). The rounded numerical values published with the corrected tetrahedron [12] are useful benchmarks, but their last printed digits are not used as inputs.

Two-loop master. At 75 working digits the two-${}_3F_2$ formula (199) was compared with the Euler integral (200) at $(d, z) = (1, 0.4)$, $(2, 0.3)$ and $(3, 0.35)$. The differences were 1.38×10^{-44}, below 2.2×10^{-58} and below 4.5×10^{-75}. Cauchy extraction of the analytic numerator of (199), with radii 0.025 and 0.02 and 72 and 80 nodes, gives the Laurent data through z^2 at $d = 1, 2, 3$; changing the settings changes the extracted coefficients by less than 2.0×10^{-69}. At $d = 1$ the finite parts in this massive-Schwinger prescription are $[z^0]\mathcal{A}_2 \simeq -55.190665$ and $[z^0]\mathcal{B}_2 \simeq -9.198444$. They differ from the ordered-span finite parts of Section 3, as they must; the two sets are not interchangeable (Section 15).

Primitive spectral formula. Gauss–Legendre quadrature of (216) after a map of the spectral half-line, with 192 and 320 nodes and Euler acceleration of the alternating angular sums, gives at $d = 3$ $P_3(3) \simeq 15.05130769366$, equal to $\mathcal{T}_3/2 = 3Q_3$ of the separate Mellin–Barnes calculation, and $P_5(3) \simeq 240.34031776$. The $P_4(3)$ run is an unextrapolated positive partial sum: 62.0583263456, 62.0583277437, 62.0583279184 and 62.0583279403 at 128, 256, 512 and 1024 angular modes, with no bound claimed for the omitted tail. All three agree with the high-precision values of Section 19.10 to the accuracy of the quadrature. The same code at $d = 4$ reproduces the four-dimensional periods $P_3 = 6\zeta(3)$ and $P_5 = \frac{441}{8}\zeta(7)$ within 4.3×10^{-13} and 3.4×10^{-12}. None of these is a determination of the complete $C_4(3)$ or $C_5(3)$.

Exact algebra and graph data. The symbolic check reconstructs the MS three-loop coefficients (179) from the zero-momentum ones, shows the cancellation of J_0, expands the Lagrange inversion through fifth order, derives (205)–(206) independently from the pole equations, and recovers the one-dimensional coefficient formulas by substitution. The inventory check verifies all 167 Kirchhoff polynomials by exact matrix-tree arithmetic, the orbit factors, the primitive mass weights, the two product factorizations of (219) and the 9540 zero combined weights of (211). No general-dimensional nonprimitive four- or five-loop integral is evaluated numerically in these checks.

16 General-dimensional master reduction and collected coefficients

16.1 Conventions and the collected answer

Here d is the physical spatial dimension, not the exponent-shift parameter d of Section 11. Set

$$q = \frac{d}{4}, \qquad \epsilon = 2s - d, \qquad \Delta_\phi = \frac{d-\epsilon}{4}, \qquad \Delta_{\mathcal{E}} = \sum_{j=0}^{5} C_j(d)\epsilon^j + O(\epsilon^6). \tag{220}$$

The physical perturbative range is fixed $0 < d < 4$ and sufficiently small positive ϵ. The endpoint $d = 4$ is not substituted into unrenormalized power-subgraph formulas. Define

$$\begin{aligned} &A = \psi_0(2q) + \psi_0(1) - 2\psi_0(q), \quad R = \psi_1(2q) - \psi_1(1), \\ &U = \psi_2(q), \qquad\qquad\qquad\quad V = \psi_2(1) + \psi_2(2q), \quad X = \psi_3(2q) - \psi_3(1). \end{aligned} \tag{221}$$

Here and below $Q = Q_d$, $P_4 = P_4(d)$ and $P_5 = P_5(d)$ have the normalization of (213): $P_3 = 3Q$. The three additional dimension-dependent functions are denoted by $\mathcal{K}_d, \mathcal{L}_d, Z_d$. Explicit series definitions are (256), (257) and (260). They are not free parameters. Sections 20 and 21 give shorter finite-sample evaluations of all three, and (553) is (223) written in them.

The complete collected coefficients are

$$\boxed{\begin{aligned} C_0(d) &= \frac{d}{2}, & C_1(d) &= -\frac{1}{6}, \quad C_2(d) = \frac{A}{9}, \\ C_3(d) &= \frac{A^2 + R - 8Q}{108}, \\ 648C_4(d) &= 16P_4 - A^3 - 8AQ - 3AR + \frac{53}{2}U - 3V. \end{aligned}} \tag{222}$$

$$\boxed{\begin{aligned} 46656C_5(d) = &-1536(P_5 + \mathcal{K}_d) + 384AP_4 - 576\mathcal{L}_d - 1728Q^2 \\ &+ 720A^2Q + 1008QR + 7152Z_d \\ &- 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X. \end{aligned}} \tag{223}$$

In particular, the extra Gamma factor associated with power-divergent mass subgraphs cancels from the explicit coefficients of both physical answers. It must nevertheless be retained until the graph and fixed-point assembly is complete.

Exact one-dimensional specialization. At $d = 1$, the dictionary is

$$\begin{aligned} R &= \frac{\pi^2}{3}, & U &= -2\pi^3 - 56\zeta(3), & V &= -16\zeta(3), \\ X &= \frac{14\pi^4}{15}, & Z_1 &= QH, & H &= \frac{2\pi^2}{3} + 8\mathsf{G}, \\ \mathcal{K}_1 &= \mathcal{K}, & \mathcal{L}_1 &= \mathcal{L}. \end{aligned} \tag{224}$$

Substitution into (222)–(223) reproduces the one-dimensional coefficients of Section 4 (for C_5, (27)) exactly. In particular, the $7152Z_d$ term becomes $48 \cdot 149QH$. Neither $Z_d = Q_dH_d$ with a guessed H_d, nor the one-dimensional polygamma formula for $\mathcal{K}$, is assumed away from $d = 1$.

16.2 The evaluated RG combinations

For comparison with the assembly formulas (208)–(210), define

$$\begin{aligned} M &= \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)} = -\frac{\Gamma(1-q)\Gamma(2q)^2}{q\Gamma(3q)}, \\ N &= \psi_0(-q) + \psi_0(3q) - \psi_0(1) - \psi_0(2q), \qquad D_j = 3c_j - b_j. \end{aligned} \tag{225}$$

For $0 < q < 1$, the last Gamma representation has only positive arguments. The recurrence $\psi_0(-q) = \psi_0(1-q) + 1/q$ likewise avoids a negative argument. The exact fourth-order results are

$$b_4 = -\frac{393}{4}A^3 - \frac{11}{2}AM - 54AQ + \frac{117}{4}AR - \frac{3}{2}MN - 6P_4 - \frac{165}{8}U + \frac{9}{4}V, \tag{226}$$

$$c_4 = -\frac{127}{8}A^3 - \frac{7}{6}AM - 3AQ + \frac{51}{8}AR - \frac{1}{2}MN - \frac{57}{16}U + \frac{3}{8}V, \tag{227}$$

$$D_4 = \frac{405}{8}A^3 + 2AM + 45AQ - \frac{81}{8}AR + 6P_4 + \frac{159}{16}U - \frac{9}{8}V. \tag{228}$$

Separate the fifth-order nonprimitive nonwheel part from its wheel and primitive contributions:

$$\begin{aligned} D_5^{\mathrm{rem}} = &-\frac{16347}{64}A^4 - \frac{57}{4}A^2M - \frac{1179}{4}A^2Q + \frac{3159}{32}A^2R - 3AMN \\ &- \frac{3657}{32}AU + \frac{189}{16}AV - 24\mathcal{K}_d - 6MQ + \frac{3}{4}MR + \frac{135}{4}QR \\ &- \frac{225}{64}R^2 - \frac{75}{64}X + \frac{447}{4}Z_d, \\ D_5 = D_5^{\mathrm{rem}} &- 54AP_4 - 9\mathcal{L}_d - 63Q^2 - 24P_5. \end{aligned} \tag{229}$$

These expressions determine the three combinations D_4, c_4 and D_5 left open in (208)–(209). Inserting them into (210), together with (179), gives (222)–(223); the M, MN and MR terms cancel. The script `lri_general_special_functions_update/code/general_special/exact_audit.py` checks this substitution in exact arithmetic. It computes b_3 and c_3 from the three-loop graph rows instead of taking (179) as input.

16.3 Why the graph reduction extends to general dimension

Normalize every coordinate integration with $h_d = \Gamma(d/2)/\pi^{d/2}$. The two-point convolution coefficient is

$$B_d(a,b) = \Gamma(d/2)\frac{\Gamma((d-a)/2)\Gamma((d-b)/2)\Gamma((a+b-d)/2)}{\Gamma(a/2)\Gamma(b/2)\Gamma((2d-a-b)/2)}. \tag{230}$$

It multiplies $|x-y|^{-(a+b-d)}$. It is symmetric under permutations of $a, b, 2d-a-b$. The star–triangle relation applies when its three absolute-distance exponents sum to $2d$. These are dimension-dependent Gamma identities, not replacements of special one-dimensional constants [24].

A reduced two-vertex master is

$$J_d(a,b,c,f,e) = h_d^2 \int_{\mathbb{R}^d} d^dx \int_{\mathbb{R}^d} d^dy \, \frac{1}{|x|^a|x-\hat{e}|^b|y|^c|y-\hat{e}|^f|x-y|^e}, \qquad |\hat{e}| = 1. \tag{231}$$

Integrals outside their ordinary convergence domain are defined by analytic continuation and the critical-mass prescription of [7] used in Section 15. Its conformal completion is a balanced K_5 with ordered edges $(01, 02, 03, 04, 12, 13, 14, 23, 24, 34)$ and exponents

$$\begin{aligned} \boldsymbol{\lambda} = (&3d-a-b-c-f-e,\ a,\ c,\ b+f+e-d,\ b,\ f, \\ &a+c+e-d,\ e,\ 2d-a-b-e,\ 2d-c-f-e). \end{aligned} \tag{232}$$

Each completed vertex has incident exponent sum $2d$; permutation of its five vertices gives the exact completion covariance. The needed Taylor data reduce to three critical patterns.

16.3.1 Pattern A: the Gamma-determined jet

For the critical point $(d/2, d/2, d/2, d/2, d)$ set

$$\begin{aligned} &t = d-e, \qquad p_0 = 2d-a-c-e, \quad p_1 = 2d-b-f-e, \\ &u = (a-c)/2, \quad v = (b-f)/2, \qquad \sigma = p_0 + p_1. \end{aligned} \tag{233}$$

An explicit pole subtraction is

$$\begin{aligned}\mathcal{P}_A = \frac{2}{p_0}B_d\left(\frac{d+t}{2}-u,d-t\right)+\frac{2}{p_1}B_d\left(\frac{d+t}{2}-v,d-t\right)\\ +\frac{2}{t}B_d(d-p_0,d-p_1)-\frac{4}{t}\left(\frac{1}{p_0}+\frac{1}{p_1}\right).\end{aligned} \tag{234}$$

The analytic remainder through total degree two is

$$\boxed{J_d-\mathcal{P}_A=\sigma\left(\frac{AR}{2}+\frac{U}{8}\right)+t\sigma\left(\frac{A^2R}{8}+\frac{AU}{16}-\frac{R^2}{8}\right)-\frac{AV}{8}\sigma^2+O(3).} \tag{235}$$

This is fixed without evaluating a new master. On each of $a+b+e=2d$, $c+f+e=2d$, and $a+b+c+f+e=3d$, a completion edge vanishes and star–triangle reduction gives a product of two copies of (230). After (234), the remainder is analytic. An analytic difference vanishing on these three distinct hyperplanes is divisible by their product. It consequently has no term of total degree at most two. Expanding the Gamma products therefore determines (235) uniquely. The exact interpolation in `lri_general_special_functions_update/code/general_special/derive_A.py` uses rational directions only; it is finite polynomial linear algebra, not numerical fitting.

16.3.2 Pattern B: one genuinely independent quadratic coefficient

At $(d/2,d/2,d/2,d,d/2)$ put the following local coordinates; p_1 here is not the p_1 of (233), which equals p_2 here: $p_1=d-f$, $p_2=2d-b-f-e$, $z_1=3d-a-b-c-f-e$, $z_2=2d-c-f-e$. Define

$$\begin{aligned}G_2&=B_d(d+p_1-e,d-p_1)B_d(e+p_2-z_1,d-p_2),\\ G_1&=B_d(e-p_1+p_2+z_2,d+p_1-e-z_2)B_d(d+p_1-e-p_2,d-p_1),\\ G_0&=B_d(d+p_1-e,d-p_1)B_d(e+p_2,d-p_2),\\ \Delta_B&=B_d(e+p_2+z_2-z_1,d-p_2)-B_d(e+p_2-z_1,d-p_2)\\ &\quad-B_d(e+p_2+z_2,d-p_2)+B_d(e+p_2,d-p_2).\end{aligned} \tag{236}$$

The needed jet is

$$J_d=G_2+G_1-G_0+\frac{2}{p_1}\Delta_B+\mathcal{K}_d z_1z_2+O(3). \tag{237}$$

The Gamma comparison agrees on $z_1=0$ and $z_2=0$. Completion symmetry and analyticity make the degree-two difference proportional to z_1z_2. Its normalization is fixed by the one-parameter path below. Its parameter t is not the $t=d-e$ of (233):

$$\begin{aligned}J_t&=J_d(d/2,d/2,d/2,d-t,d/2),\\ J_t^{\Gamma}&=B_d(d/2,d-t)^2+\frac{2}{t}\left[B_d(d/2,d-t)-B_d(d/2+2t,d-t)\right],\\ \mathcal{K}_d&=[t^2](J_t-J_t^{\Gamma}).\end{aligned} \tag{238}$$

This coefficient extraction is a *proof and normalization definition*; the final evaluation (260) has neither t nor a derivative. At $d=1$ it is exactly the one-dimensional master $\mathcal{K}$ of (57), not an unrelated constant with a matching numerical value.

16.3.3 Pattern T: tetrahedron symmetry

At $a=b=c=f=e=d/2$, all ten completion edges equal $d/2$. Balanced variations satisfy $\sum_{e\ni v}\delta\lambda_e=0$ at every vertex. Here e labels the ten completion edges; it is not the exponent e

of (231). There is no invariant linear form, and every invariant quadratic form on this balanced subspace is proportional to $\sum_e(\delta\lambda_e)^2$. Indeed the coefficients of a general permutation-invariant quadratic form depend only on whether two edges coincide, meet, or are disjoint; the balance equations reduce the latter two sums to the first. Thus

$$\boxed{J_d = 3Q_d + \frac{Z_d}{4}\sum_{e=1}^{10}(\delta\lambda_e)^2 + O(3).} \tag{239}$$

This argument applies in every dimension; the coefficient Z_d itself is not fixed by symmetry. A central-edge-only change by t has squared completion variation $6t^2$. A balanced three-wheel path has $18t^2$, which fixes the factors of three and nine used below.

16.3.4 Forest sum and six wheel rows

The graph set, exact Wick weights and subtraction prescription are those already specified in (193)–(195). Equivalently, a rooted logarithmic L-loop coordinate graph has common radial factor $2/(Lz)$ and edge exponents $m_{ij}(d-z)/2$. Its proper disjoint forests are subtracted recursively; nested forests occur inside the subgraph counterterms. The local counterterm is

$$Z_G = -\,\mathrm{PP}\left[I_G + \sum_{\varnothing\neq\mathcal{F}}\left(\prod_{\gamma\in\mathcal{F}} Z_\gamma\right) I_{G/\mathcal{F}}\right].$$

The shift method of Section 11, used on exceptional graphs, preserves the sum of edge deformations in every divergent subgraph. Only convolution, star–triangle, completion covariance and the jets above enter. Consequently its combinatorics is dimension independent. Critical base exponents scale with d; regulator slopes do *not* scale with d.

The residue certificate for these classes consists of the files `rows_4_0_27.json` and `rows_5_0_129.json` in `lri_general_special_functions_update/code/general_special/`. They list all 26 nonprimitive four-loop classes and the 92 nonzero five-loop classes that are neither wheels nor primitive. At five loops six further classes are wheels, three are primitive, and 28 have zero combined weight. The sum is 129 classes. For each class the files give the canonical word, the rational weights and the exact symbolic residue. They give (226)–(229); no period value is used to obtain these polynomial coefficients.

For completeness the six wheel residues are $-2AP_4/5 + (\alpha D_s + \beta D_r)/5$, where $D_s = \frac{1}{4}\partial_a P_W$ and $D_r = \frac{1}{4}\partial_b P_W$ are the spoke and rim derivatives at $a = b = d/2$ of the rooted four-wheel P_W of Section 16.4.1. The class index is the zero-based position in the five-loop list of `lri_general_special_functions_update/code/general_special/C5_wheel_nonprimitive/reduced_classes.json`. The data are:

Class index	combined weight	α	β
51	15/2	2	2
78	15	2	−2
99	30	2	−2
113	60	−2	2
124	15/2	6	−2
128	15	6	−6

Their sum is $-54AP_4 + 24D_s - 12D_r = -54AP_4 - 9\mathcal{L}_d$. The completion/product argument behind (219) gives $D_5^{\mathrm{prim}} = -63Q^2 - 24P_5$.

16.4 A common spectral kernel for all nonprimitive master data

Let

$$\begin{aligned} d_0(d) &= 1, & d_\ell(d) &= \frac{(2\ell+d-2)\Gamma(\ell+d-2)}{\Gamma(d-1)\Gamma(\ell+1)} \quad (\ell \ge 1), \\ a_\ell &= \frac{\ell}{2}+\frac{q}{2}, & b_\ell &= a_\ell + q, \\ t_\ell(\nu) &= \Gamma(2q)\frac{\Gamma(a_\ell+\mathrm{i}\nu)\Gamma(a_\ell-\mathrm{i}\nu)}{\Gamma(b_\ell+\mathrm{i}\nu)\Gamma(b_\ell-\mathrm{i}\nu)}. \end{aligned} \tag{240}$$

Here a_ℓ, b_ℓ are spin-dependent Gamma arguments, not the RG coefficients b_j of (173). At $d=1$, $(d_0, d_1, d_2, \ldots) = (1, 1, 0, \ldots)$; at $d=2$ the multiplicities are $(1, 2, 2, \ldots)$. At noninteger $d<1$ they are continued Gamma ratios, not positive integer multiplicities. For notational economy write

$$\langle f \rangle_d = \frac{1}{\pi}\sum_{\ell=0}^{\infty} d_\ell(d)\int_0^\infty f(\ell,\nu)\,d\nu. \tag{241}$$

This is the *one-vector wheel kernel*, not the two-vector zig-zag measure in (215). The primitive $P_5(d)$ keeps the latter representation and its stated continuation prescription.

For a balanced wheel with rim exponent b and spoke exponent $d-b$, the commuting H_α construction of [24] gives eigenvalue

$$t_{\ell,b}(\nu) = \Gamma(d/2)\frac{\Gamma((d-b)/2)}{\Gamma(b/2)}\left|\frac{\Gamma(\ell/2+b/4+\mathrm{i}\nu)}{\Gamma(\ell/2+d/2-b/4+\mathrm{i}\nu)}\right|^2, \qquad W_r(d;b) = \langle t_{\ell,b}^r\rangle_d. \tag{242}$$

One can also obtain it directly by applying the normalized convolution operator $h_d|x-y|^{-b}|y|^{b-d}$ to homogeneous harmonic functions. The spherical-harmonic addition theorem gives the factor $d_\ell/(2\pi)$ on the full Mellin line, hence d_ℓ/π on the positive half-line. In particular

$$3Q_d = \langle t_\ell^3\rangle_d, \qquad P_4(d) = \langle t_\ell^4\rangle_d. \tag{243}$$

At the graph-period check point $d=4$, $t_\ell = (\nu^2+(\ell+1)^2/4)^{-1}$ and $d_\ell = (\ell+1)^2$, giving $W_3 = 6\zeta(3)$ and $W_4 = 20\zeta(5)$.

16.4.1 The logarithmic wheel and the tetrahedron Hessian

Let a rooted four-wheel have four spokes of exponent a and four rims of exponent b, and let $P_W(a,b)$ be its period, normalized as in (242), so that $P_W(d-b,b) = W_4(d;b)$. At $a=b=d/2$ its completion is the octahedron. The completed exponents are four a's, four b's, two $-2d+3a+2b$'s, two $2d-a-2b$'s, and the three absent-edge exponents $2d-4a$, $4d-4a-4b$, 0. Their vertex sums equal $2d$. The octahedral symmetry has two edge orbits: 12 present and three absent. The total variation is zero, while the sum over the present edges is $8\,\delta a + 4\,\delta b$. Its invariant first derivative therefore obeys $\partial_a P_W = 2\partial_b P_W$, or $D_s = 2D_r$ when each derivative is divided by four. It follows that

$$\mathcal{L}_d = \frac{4}{3}(D_r - 2D_s) = -4D_r = \partial_b W_4(d;b)|_{b=d/2}, \qquad Z_d = \frac{1}{9}\partial_b^2 W_3(d;b)\Big|_{b=d/2}. \tag{244}$$

These derivatives are used only in the proof. For direct evaluation put

$$\begin{aligned} w_1 &= -\psi_0(q) + \frac{1}{2}\Re\{\psi_0(a_\ell+\mathrm{i}\nu)+\psi_0(b_\ell+\mathrm{i}\nu)\}, \\ w_2 &= \frac{1}{8}\Re\{\psi_1(a_\ell+\mathrm{i}\nu)-\psi_1(b_\ell+\mathrm{i}\nu)\}. \end{aligned} \tag{245}$$

Then

$$\boxed{\mathcal{L}_d = \langle 4t_\ell^4 w_1\rangle_d, \qquad Z_d = \left\langle t_\ell^3\left(w_1^2+\frac{w_2}{3}\right)\right\rangle_d.} \tag{246}$$

The first derivative $\langle 3t_\ell^3 w_1\rangle_d$ vanishes identically; it is a useful normalization and tail-control check.

16.4.2 A convergent single-kernel formula for the subtracted master

Set $c_* = \psi_0(1) + \psi_0(2q)$ and, at $a = a_\ell, b = b_\ell$, define the local abbreviations below. Here h is not h_d, j and k are not indices, and r is not the wheel order:

$$\begin{aligned} p &= \Re\{\psi_0(a+\mathrm{i}\nu) + \psi_0(b+\mathrm{i}\nu)\}, \quad & h &= \Im\{\psi_0(a+\mathrm{i}\nu) - \psi_0(b+\mathrm{i}\nu)\}, \\ j &= \Re\{\psi_1(a+\mathrm{i}\nu) - \psi_1(b+\mathrm{i}\nu)\}, \quad & k &= \Im\{\psi_1(a+\mathrm{i}\nu) + \psi_1(b+\mathrm{i}\nu)\}, \\ r &= \Re\{\psi_2(a+\mathrm{i}\nu) + \psi_2(b+\mathrm{i}\nu)\}, \quad & E_2 &= -\frac{h^2}{2} - \frac{5j}{8}. \end{aligned} \tag{247}$$

The complete subtraction collapses to

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}\left\langle t_\ell^2 \left[(c_* - p)E_2 - \frac{hk}{2} - \frac{5r}{32}\right]\right\rangle_d .} \tag{248}$$

Unlike the unsubtracted J_t of (238), this integral has no ultraviolet pole. Its large-spin summand is $O(\ell^{-3}\log\ell)$.

Here is a derivation that fixes its finite part. Write $t = 2\tau$ in (238) and $s = \mathrm{i}\nu$. Here and below s is the Mellin variable, not the long-range exponent s of (220). The mixed trace of two critical wheel operators and one operator of rim exponent $d - 2\tau$ is $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$, where (the letters are local: $C(\tau)$ is not a coefficient C_j, R_τ is not R of (221), and S_τ is not $S_{\ell n}$ of (251))

$$\begin{aligned} C(\tau) &= \frac{\Gamma(2q)\Gamma(\tau)}{\Gamma(2q-\tau)}, \\ R_\tau(s) &= \frac{\Gamma(a+s)\Gamma(b-\tau-s)}{\Gamma(a+\tau+s)\Gamma(b-s)}, \\ S_\tau(s) &= \left[\frac{\Gamma(a-\tau/4+s)\Gamma(a-\tau/4-s)\Gamma(b+s)\Gamma(b-s)}{\Gamma(b+\tau/4+s)\Gamma(b+\tau/4-s)\Gamma(a+s)\Gamma(a-s)}\right]^2 . \end{aligned} \tag{249}$$

At $d = 2$ this mixed-trace step is proved, for $0 < \tau < \frac{1}{2}$ and in its full-Mellin-line reading, in Section 18.9.5. Its reading for complex τ and its status at other d are stated as input (M) in Section 21.1. The two-wheel convolution is elementary: $W_2(d; b') = B_d(d - b', 2b')$. Taking $b' = d/2 - \tau$ gives the exact identity

$$J_{2\tau} = C(\tau)^2 \left[\frac{\Gamma(q-\tau/2)}{\Gamma(q+\tau/2)}\right]^4 + C(\tau)\langle t_\ell^2 (R_\tau - S_\tau)\rangle_d . \tag{250}$$

On the Mellin line, the real difference starts at order τ^2:

$$\Re(R_\tau - S_\tau) = E_2\tau^2 + \left[-pE_2 - \frac{hk}{2} - \frac{5r}{32}\right]\tau^3 + O(\tau^4).$$

Using $C(\tau) = \tau^{-1} + c_* + O(\tau)$ and $t = 2\tau$ yields the integral in (248). The coefficient of t^2 in the explicit Gamma term of (250) is

$$\frac{A^4}{6} - \frac{A^2 R}{2} + \frac{R^2}{8} + \frac{A(2V-U)}{12} - \frac{X}{48},$$

whereas that of J_t^Γ is

$$\frac{A^4}{6} - \frac{A^2 R}{2} + \frac{R^2}{8} + \frac{A(V+U)}{6} - \frac{X}{48}.$$

Their difference is exactly $-AU/4$. This derives (248) from the master definition (238), rather than from a numerical relation search, given the mixed-trace step and the extraction of $[\tau^2]$ under the trace. Both are proved at $d = 2$ (Sections 18.9.2 and 18.9.5). At other d the first is input (M) of Section 21.1, and the second is an unproved interchange of $[\tau^2]$ with the trace (Section 21.12); both are confirmed numerically in Section 21.11.

16.5 Explicit polygamma–hypergeometric series: no integrals or parameter derivatives

All quantities needed for the nonprimitive terms can now be written using positive-argument Gamma and polygamma functions and integer sums. For $\ell, n \geq 0$ define

$$\begin{aligned} T_{\ell n} &= \frac{(q)_{\ell+n}(1-q)_n}{(2q)_{\ell+n}\, n!}, & S_{\ell n} &= \frac{(2q)_{\ell+n}\, n!}{(3q)_{\ell+n}(1+q)_n}, \\ a_{\ell n} &= \ell + n + q, & b_{\ell n} &= \ell + n + 2q, \\ \chi_m(n;q) &= (-1)^m \psi_m(n+1-q) + \psi_m(q) - (-1)^m \psi_m(1-q). \end{aligned} \tag{251}$$

In terms of (240), $a_{\ell n} = 2a_\ell + n$ and $b_{\ell n} = a_\ell + b_\ell + n$; they are not a_ℓ, b_ℓ. Reflection gives $\chi_m(n;q) = \psi_m(q-n)$, but the positive-argument expression is preferable for evaluating it. Thus the natural pair $q = d/4$ and $1-q$ remains explicit; no trigonometric replacement is required.

For $m = 1, 2, 3, 4$ set (these λ_m are not the completion exponents λ_e of (232))

$$\begin{aligned} \lambda_m = \psi_{m-1}(a_{\ell n}) - \psi_{m-1}(b_{\ell n}) - (-1)^m \chi_{m-1}(n;q) \\ + (1 + (-1)^m)\psi_{m-1}(1) - \psi_{m-1}(n+1). \end{aligned} \tag{252}$$

The following are explicit polygamma combinations, not differentiation operators:

$$\begin{aligned} \eta_m &= \frac{1}{2}\{\psi_m(a_{\ell n}) + (-1)^m \chi_m(n;q)\}, & m &= 1, 2, 3, \\ \theta_m &= \frac{1}{4}\{\psi_{m+1}(a_{\ell n}) - (-1)^m \chi_{m+1}(n;q)\}, & m &= 1, 2, \\ \mu_1 &= \frac{1}{2}\{\psi_0(a_{\ell n}) - \psi_0(q) + \psi_0(n+1-q) - \psi_0(1-q)\}, \\ \mu_2 &= \frac{1}{4}\{\psi_1(a_{\ell n}) - \psi_1(q) + \psi_1(n+1-q) - \psi_1(1-q)\}. \end{aligned} \tag{253}$$

All subscripts ℓ, n on the right are suppressed in the following polynomials:

$$\begin{aligned} F_3 &= \frac{9}{2}\lambda_1^2 + \frac{3}{2}\lambda_2, \\ F_{3,1} &= 9\lambda_1\eta_1 + \frac{3}{2}\eta_2, \\ F_{3,2} &= 9\eta_1^2 + 9\lambda_1\theta_1 + \frac{3}{2}\theta_2, \\ F_4 &= \frac{32}{3}\lambda_1^3 + 8\lambda_1\lambda_2 + \frac{2}{3}\lambda_3, \\ F_{4,1} &= 32\lambda_1^2\eta_1 + 8\eta_1\lambda_2 + 8\lambda_1\eta_2 + \frac{2}{3}\eta_3. \end{aligned} \tag{254}$$

The subscripts $1, 2$ in these labels name the displayed polynomials; they are not a request to differentiate a special function.

The primitive wheel periods and the two corresponding nonprimitive master data are

$$\boxed{\begin{aligned} Q_d &= \frac{1}{3}\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^3 F_3, \\ P_4(d) &= -\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^4 F_4. \end{aligned}} \tag{255}$$

$$\boxed{\mathcal{L}_d = -\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^4 (4\mu_1 F_4 + F_{4,1}).} \tag{256}$$

$$\boxed{Z_d = \frac{1}{9}\sum_{\ell=0}^{\infty} d_\ell(d) \sum_{n=0}^{\infty} T_{\ell n}^3 \left[(9\mu_1^2 + 3\mu_2)F_3 + 6\mu_1 F_{3,1} + F_{3,2}\right].} \tag{257}$$

These are ordinary nonnegative-integer series at fixed dimension. They contain only Pochhammer symbols and $\psi_0, \ldots, \psi_3$ at the explicit arguments above.

For $\mathcal{K}_d$ define, for $j = 0, 1, 2, 3$,

$$r_j = \frac{1}{2}\lambda_{j+1} + \psi_j(b_{\ell n}) + \psi_j(n+1) - (1 - (-1)^j)\psi_j(1), \qquad c_* = \psi_0(1) + \psi_0(2q), \tag{258}$$

and the polynomial

$$\begin{aligned}\mathcal{R}_{\ell n} = {} & \frac{r_0 - c_*}{2}(\lambda_1^3 + 3\lambda_1\lambda_2 + \lambda_3) + \frac{3r_1}{4}(\lambda_1^2 + \lambda_2) - \frac{\lambda_1 r_2 + r_3}{4} \\ & - \frac{\lambda_1^4}{12} + \frac{\lambda_1^2\lambda_2}{8} + \frac{7\lambda_1\lambda_3}{24} + \frac{\lambda_2^2}{8} + \frac{\lambda_4}{12}.\end{aligned} \tag{259}$$

Then the explicit result is

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \sum_{\ell=0}^{\infty} d_\ell(d) \left\{\sum_{n=0}^{\infty}\left[\frac{M^2 S_{\ell n}^2}{8} - \frac{T_{\ell n}^2 \mathcal{R}_{\ell n}}{4}\right]\right\}.} \tag{260}$$

The square brackets must be formed before summing: they are the two pole families of the same convergent integral. The inner n sum is performed before the outer spin sum. Separating the two families or rearranging this as an unrestricted double sum is not justified; for example, absolute double summability fails in part of the physical range even though the stated iterated sum converges.

Domains and continuation. For $2/3 < d < 4$, the Q_d, Z_d series and their Barnes representations converge directly; the corresponding lower threshold for P_4 and $\mathcal{L}_d$ is $d > 1/2$. The formulas cover $d = 1, 2, 3$ without any continuation. The subtracted $\mathcal{K}_d$ spin integral and its ordered residue sum converge for $0 < d < 4$. Numerically, at $d = 0.8$ the literal bracket of (260) loses its digits to cancellation far out in the radial tail, so Section 21.11 evaluates the canceled form (289) there instead. For smaller positive dimensions in the other cases, use analytic continuation of the displayed special functions from their common convergence strip, equivalently the convergent normalized coordinate/Schwinger periods. Do not evaluate individual divergent spin integrals and assign them finite numbers. The analytically continued multiplicities at $d < 1$ provide cancellations that need not survive taking absolute values term by term.

Why these are hypergeometric-series data, rather than new graph names. The coefficient $d_\ell T_{\ell n}^r$ is a two-variable Horn-type hypergeometric term: its ratios under $\ell \mapsto \ell + 1$ and $n \mapsto n + 1$ are rational. Away from removable exceptional dimensions,

$$d_\ell(d) = \frac{(4q-2)_\ell (2q)_\ell}{(2q-1)_\ell\, \ell!}.$$

Together with $(q)_{\ell+n}^r/(2q)_{\ell+n}^r$ and $(1-q)_n^r/(n!)^r$, this prints the full parameter array of the underlying Kampé de Fériet-type series. The extra polygamma polynomials are explicit resonant weights; the weighted series is not falsely labeled as an unweighted ordinary ${}_pF_q$. Most importantly, (251)–(260) are executable definitions in standard Gamma and polygamma functions, with no unsupplied master value.

16.6 Residue proof of the series

Consider the even Barnes moment for one fixed spin,

$$I_r(a,g)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\left[\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(a+g+s)\Gamma(a+g-s)}\right]^r. \tag{261}$$

For the wheel at $b=d/2$, $a=a_\ell$ and $g=q$. Closing an upward contour to the right is clockwise, so the integral is the negative sum of residues at $s=a+n$. Put $s=a+n+u$. Then

$$\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(a+q+s)\Gamma(a+q-s)}=-\frac{T_{\ell n}}{\Gamma(2q)\,u}\exp\left(\sum_{m\ge1}\frac{\lambda_m}{m!}u^m\right). \tag{262}$$

This follows from Gamma reflection and

$$\Gamma(-n-u)=\frac{(-1)^{n+1}\Gamma(1-u)}{n!\,u\prod_{j=1}^{n}(1+u/j)}.$$

The regular logarithmic derivative of this last factor at order m is $(1+(-1)^m)\psi_{m-1}(1)-\psi_{m-1}(n+1)$, which is precisely the last line of (252). The second and third complete exponential Bell polynomials give F_3 and F_4. Their signs are opposite because the pole orders are three and four. This proves (255).

For a general balanced rim exponent put $x=b/2$, $g=d/2-x$. The normalized residue prefactor is

$$-\frac{(x)_{\ell+n}(1-g)_n}{(d/2)_{\ell+n}\,n!}.$$

Its logarithmic first and second b derivatives at $b=d/2$ are μ_1,μ_2. The corresponding first and second derivatives of λ_m are the already displayed η_m,θ_m. Differentiating the convergent residue identities in the proof and then substituting these *explicit* formulas yields (256)–(257). No differentiation remains in those evaluation formulas.

For the subtracted master there are two right-hand pole families. Write

$$g(s)=\frac{\Gamma(a+s)\Gamma(a-s)}{\Gamma(b+s)\Gamma(b-s)},\quad \ell(s)=\frac{g'(s)}{g(s)},\quad p(s)=\frac12\{\psi_0(a+s)+\psi_0(a-s)+\psi_0(b+s)+\psi_0(b-s)\}.$$

Here $\ell(s)$ is the logarithmic derivative of $g(s)$, not the spin ℓ. The holomorphic continuation of the weight in (248) is

$$F(s)=(c_*-p)\left(\frac{\ell(s)^2}{8}-\frac{5\ell'(s)}{16}\right)+\frac{\ell(s)p'(s)}{4}-\frac{5p''(s)}{32}. \tag{263}$$

Primes in this proof act on the elementary Gamma/polygamma kernel, not on an unevaluated master. At $s=a+n+u$,

$$\ell(s)=-u^{-1}+\sum_{m\ge1}\lambda_m\frac{u^{m-1}}{(m-1)!},\qquad p(s)=\frac{1}{2u}+\sum_{j\ge0}r_j\frac{u^j}{j!}.$$

Multiplying these finite jets through the required order gives the negative residue $-T_{\ell n}^2\mathcal{R}_{\ell n}/4$ of $\Gamma(2q)^2g(s)^2F(s)/4$. At the other family, $s=b+n+u$, $g(s)^2$ has a double zero but $F(s)=-1/(2u^3)+O(u^{-2})$. The resulting negative residue is $M^2S_{\ell n}^2/8$. This proves both terms and the sign in (260). The local residue polynomial was also checked independently by Cauchy circles around both families at integer and noninteger dimensions.

16.7 An additional exact two-dimensional master evaluation

The dimension-two balanced three-wheel is a complex Selberg integral. For b close to one take, in the complex Selberg identity [6], the parameters $n = 2$, $\sigma = b/2$, $\tau = 1 - b/2$ and $\theta = -b/4$. These names are local to this paragraph; they are not the σ of (233), the τ of (249) or the θ_m of (253). With $\gamma_C(x) = \Gamma(x)/\Gamma(1-x)$, its normalized value simplifies to

$$\boxed{W_3(2;b) = \frac{1}{2}\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}.} \tag{264}$$

All convergence inequalities hold in a neighborhood of $b = 1$: $0 < b < 4/3$ suffices for the displayed two-screening integral. At $b = 1$ its logarithmic derivative vanishes, and its logarithmic second derivative is

$$\frac{3}{4}\{\psi_1(1/4) - \psi_1(3/4)\} = 12\mathsf{G}.$$

Combining this with (244) gives the exact check

$$\boxed{Z_2 = 4\mathsf{G}\,Q_2, \qquad Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4.} \tag{265}$$

This is a derivation from a convergent integral identity, not an integer-relation fit to the numerical value of Z_2.

A second derivation of (264) and of (265), from Dixon's theorem and the two-node formula of Section 20.2 without the complex Selberg integral, is given in Section 20.6, (548)–(550); it holds for $0 < \mathrm{Re}\, b < \frac{4}{3}$.

16.8 Special-function coefficient checks

The one-dimensional specialization recovers the analytic formulas of Part II. The general master calculations give, at the precision relevant here,

d	$C_4(d)$	$C_5(d)$
2	5.048868431	−38.833645896
3	1.228063545	−5.850351421

These are master/assembly evaluations, not a second full-graph calculation in arbitrary dimension. The two-dimensional exact result $Z_2 = 4\mathsf{G}Q_2$ supplies an independent master check. The closed forms (270), (271) make the $d = 2$ entry of C_5 exact through (272).

Master values at $d = 1, 2, 3$. The six masters of (223) take the following values, correctly rounded to 30 significant digits:

Master	d	Value	Source
Q_d	1	55.001486544162982619920766 2385	(180)
	2	12.771381198646957657749763 6845	(267)
	3	5.017102564553855365 9085484569	(405)
$P_4(d)$	1	2229.52206107094196429452519975	(93)
	2	250.48731558186728553185175 6888	(333)
	3	62.058327943406672466463388 5841	(422)
$P_5(d)$	1	28599.29317656210353664647 0214	(29)
	2	1548.08264023312061513243907395	(334)
	3	240.34031775908354981636654 6388	(425)
$\mathcal{K}_d$	1	39.331569128259039057503187 1516	(76)
	2	3.518109138590252241649090 17031	(271)
	3	1.148594649527179336556646 31081	(260), record below
$\mathcal{L}_d$	1	−5805.30162954880238014117987700	(94)
	2	−347.24915312320113105847678 3075	(270)
	3	−62.710584968387669347296 1734335	records below
Z_d	1	764.93103019958857704016177 8769	(224)
	2	46.79258307232969665673882 58780	(265)
	3	8.82092489306009248553686732817	records below

At $d = 1$ and $d = 2$ every entry is a closed form of this note, evaluated at 60 digits; the entries agree with all values of these masters printed elsewhere in the note. At $d = 3$, Q_3, $P_4(3)$ and $P_5(3)$ are the fixed rational formulas of Section 19, evaluated at 90 digits with 160 summed terms and a Hurwitz-zeta tail expansion of order 64 (`LRI_C5_d3/inputs/general_d3/period_values.json`); Section 19.10 compares them with a less converged 55-digit run and with independent integrations of the Gamma kernels. $\mathcal{K}_3$ is the 430-digit ball-arithmetic evaluation of (260) (`LRI_C5_d3/wf_K3/K3_value.json`); at $d = 3$ that series rests on input (M) of Section 21.1 and on the extraction of $[\tau^2]$ under the trace (Section 21.12). The two-fold Barnes integral (588), which uses neither, reproduces it to $4 \cdot 10^{-100}$ (Section 21.11). $\mathcal{L}_3$ and Z_3 come from the $d = 3$ formulas of Section 20.6, which rest on the half-shift identities of Section 20.6 (records `LRI_C5_d3/wf_L3/L3_final_values.json`, about 158 digits, and `LRI_C5_d3/wf_Z3/Z3_values.json`, about 250 digits). An independent 100-digit evaluation of the spectral traces of Section 16.4, with the derivatives in b of (244) and the τ-expansion of (250) taken on Cauchy circles (`LRI_C5_d3/run_C_combined.json`), agrees with $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ to 1.2×10^{-93}, 2.9×10^{-98} and 1.5×10^{-96}, and with $3Q_3$ and $P_4(3)$ to 3.5×10^{-79} and 2.1×10^{-86}. These agreements are numerical, not interval bounds. Values at other dimensions (and Z_d to 36 digits) are in the validation tables of Sections 20.3.1 and 20.9; the first, lower-precision evaluations are described in the paragraph on controls below.

An independent route to $P_5(2)$. The spectral formula (216) has a regular limit $d \to 2$. Rename its angular index n (of (214) and (215)) to ℓ. At $d = 2$ one has $p = 0$ and $h_2 = 1/\pi$, the joint limit of the Gamma ratios in (215) gives the multiplicities $d_\ell(2) = 1, 2, 2, \ldots$, and $\tau_\ell^{(2)}(\nu) = (-1)^\ell \pi\, t_\ell(\nu)$, with $t_\ell(\nu)$ of (240) at $d = 2$: there $\Gamma(2q) = 1$ and $t_\ell(\nu) = |\Gamma(\ell/2 + \frac{1}{4} + \mathrm{i}\nu)/\Gamma(\ell/2 + \frac{3}{4} + \mathrm{i}\nu)|^2$. For $L = 5$,

$$P_5(2) = \frac{1}{2\pi} \sum_{\ell=0}^{\infty} (-1)^\ell d_\ell(2) \int_0^\infty (\ell^2 + 4\nu^2)\, t_\ell(\nu)^7 \, d\nu. \tag{266}$$

OpenAI GPT-6 astra evaluated (266) in double precision, in the package `lri_general_special_functions_update/` (script `code/general_special/spectral_evaluate.py`, record `verification/general_special/spectral_master_values.json`, entry `zigzag_d2`). With 192, 384 and 768 quadrature nodes it agrees with the table to 6×10^{-11}, and the same limit for $L = 3$ reproduces $P_3(2) = 3Q_2$. This route uses neither the square factorization of Section 18.5 nor the finite sampling of Section 18.

Controls of the master evaluations. These controls belong to the same general-dimensional calculation (folder `code/general_special/` of `lri_general_special_functions_update/`, records in `verification/general_special/`). The first values of Q_d, $P_4(d)$, $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$ came from the common spectral kernel of Section 16.4 (`spectral_evaluate.py`, record `spectral_master_values.json`). That evaluator is independent of the two-contour kite representation, which was used separately to check $\mathcal{K}_d$ after extracting its pinch residue (`kappa_mb.py`, record `kappa_mb_checks.json`). The common-kernel runs vary the spin cutoff (128, 256, 512), the number of quadrature nodes (384, 512, 768) and the number of terms of an angular-tail expansion. That expansion follows from centered Gamma ratios and even inverse powers of $\ell + (d-2)/2$; its fitted coefficients accelerate the integration and are not an ansatz for a master identity. At $d = 2$ and $d = 3$ the two-contour and common-kernel values of $\mathcal{K}_d$ agree to about 10^{-11} or better. The slower two-contour run at $d = 1$ is not assigned that accuracy; there the one-dimensional closed form is the comparison target. Local Cauchy residues test both families of (260) (`local_residue_checks.json`), and raw finite sums of the series converge towards the spectral values (`ordered_series_checks.json`); raw rectangular truncations are not assigned an error bound, and the iterated convention of (260) is kept. Exact SymPy calculations (`exact_audit.py`) check the three-loop benchmark, both fixed-point assemblies, the cancellation of M and N, both one-dimensional specializations, the tetrahedron quadratic normalization and the local $\mathcal{K}$ residue polynomial. The digits of these runs are superseded by the high-precision values of Section 16.8 (see also Sections 17.2, 19.10, 20.9 and 21.11); they remain an independent lower-precision route. They are master-level checks, not a second general-dimensional full-graph calculation, and not interval enclosures.

The one-dimensional polygamma form of $\mathcal{K}$ does not lift to general d. The combination $2\psi_1(q)^2 - 3R\psi_1(q) + A\psi_2(q)/2$, with A, R of (221), equals $\mathcal{K}$ at $d = 1$ but gives $25.3748796\ldots$ at $d = 2$, whereas $\mathcal{K}_2 = 3.5181091\ldots$ (record `kappa_mb_checks.json`).

17 Finite multivariable representations for the masters

This section writes the masters of (223) in standard function classes. For $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$ it reaches finite combinations of unweighted multivariable Horn series at unit arguments (Section 17.6 and the finite arrays after it). This is a function-class statement, not a reduction to a few ordinary ${}_pF_q$ values: the combinations are long, outside a common convergence strip only the complete combination may be continued in d, and the weighted double series of Section 16.5 remain preferable for numerical work. For the primitive $P_5(d)$ the section gives a derivative-free double residue series (Section 17.4); the shorter finite sampling of the primitives is Section 18, and at $d = 2$ all masters have closed forms (Section 17.2). Apart from those closed forms, the constructions of this section were written by OpenAI GPT-6 astra (package `lri_general_closed_update/`).

17.1 The two-dimensional normalization and its immediate consequences

At $d = 2$ the tetrahedron normalization (177), with $\mathcal{T}_2$ from (192), gives

$$\boxed{Q_2 = \frac{1}{6}\left[\frac{\Gamma(1/4)}{\Gamma(3/4)}\right]^4 = \frac{\Gamma(1/4)^8}{24\pi^4}.} \tag{267}$$

Indeed, reflection gives $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$, so the fourth power of the ratio is $\Gamma(1/4)^8/(4\pi^4)$ [26]. The factor $1/6$ is that of $Q_d = \mathcal{T}_d/6$ in (177); it must not be dropped. Numerically,

$$Q_2 = 12.771381198646957657749763684524008422037284985934 7\ldots, \qquad Z_2 = \frac{\mathsf{G}\,\Gamma(1/4)^8}{6\pi^4}. \tag{268}$$

The second identity is $Z_2 = 4\mathsf{G}Q_2$ of (265), written in the same Gamma normalization.

For $a = \log 2$, the dimension-two specialization of all coefficients is

$$\begin{aligned}
C_0(2) &= 1, & C_1(2) &= -\frac{1}{6}, \quad C_2(2) = \frac{4a}{9},\\
C_3(2) &= \frac{4a^2}{27} - \frac{\Gamma(1/4)^8}{324\pi^4},\\
648C_4(2) &= 16P_4(2) - 64a^3 - 32aQ_2 - 359\zeta(3),\\
46656C_5(2) &= -1536\{P_5(2) + \mathcal{K}_2\} + 1536aP_4(2) - 576\mathcal{L}_2 - 1728Q_2^2\\
&\quad + (11520a^2 + 28608\mathsf{G})Q_2 - 3840a^4 - 21456a\zeta(3).
\end{aligned} \tag{269}$$

The remaining dimension-two masters in this display are the fixed hypergeometric functions specified below, not new undetermined constants; the two nonprimitive ones, $\mathcal{K}_2$ and $\mathcal{L}_2$, are evaluated in closed form in Section 17.2. None of these identities is inferred from numerical values.

17.2 Closed forms of the two-dimensional nonprimitive masters

The two masters of (269) that are neither primitive periods nor Gamma values have closed forms in constants that are already present:

$$\boxed{\mathcal{L}_2 = -2\log 2\, P_4(2) = -\frac{A}{2}\, P_4(2) = -\frac{\log 2}{2\pi^3}(B_-^2 - B_+^2),} \tag{270}$$

$$\boxed{\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2 = \frac{1}{2}\Big[\frac{\{\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4})\}^2}{4} + AU\Big].} \tag{271}$$

Here $A = 4\log 2$ and $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$ are the values of (221) at $d = 2$, $B_\pm$ are the retained one-dimensional values of (330), and $\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4}) = 16\mathsf{G}$ by (73). The second form of (271) reads $\mathcal{K}_2 = \frac{1}{2}[\Phi_2 + AU]$ with $\Phi_2 = 64\mathsf{G}^2$. The one-dimensional result (76) has the same shape, $\mathcal{K} = \frac{1}{2}[\Phi_1 + AU]$ with $\Phi_1 = \psi_1(\frac{1}{4})\{3\psi_1(\frac{1}{4}) - \psi_1(\frac{3}{4})\}$ and the $d = 1$ values of A and U.

Status. Both identities are proved; no numerical identification enters either proof.

- (270) is proved in Section 18.7 for $\mathcal{L}_2$ defined by the series (256) at $d = 2$. The same proof establishes the two single-sum relations (356), which were first found numerically. The wheel-derivative reading $\partial_b W_4(2; b)|_{b=1} = -2\log 2\, W_4(2; 1)$ of (244), where $W_4(2; 1) = P_4(2)$ by (243), is the form in which $\mathcal{L}_d$ enters (229). It follows because the function $\mathcal{W}(x)$ of Section 18.7 equals $W_4(2; 2x)$ for $\frac{2}{5} \le x \le \frac{3}{5}$. This is the per-spin residue evaluation of (261) at rim exponent $2x$, written out at $d = 2$ as (360) in Section 18.7.6; it is also the case $d = 2$ of (479) with (539). Since the two functions agree on an interval, so do their derivatives. The wheel-derivative reading has also been proved directly, by a rim–spoke reflection identity (Section 18.7.6).
- (271) is proved in Section 18.8 for $\mathcal{K}_2$ defined by the master path (238). The proof starts from the position-space integral and uses neither (248) nor (260). The series (260) and the spectral form (248) are also proved to equal the right side of (271), in Section 18.9, which also proves the mixed-trace step at $d = 2$ and gives a second position-space proof of (377).

Three further proofs were found independently. Two of them prove (270) through the rim derivative (347) and the residue functional (350) of Section 18.7, but obtain the derivatives of the two modes differently: one evaluates the single sums of (352) by Hardy-space Parseval pairings on the unit circle (Section 18.7.8), the other differentiates a Fourier (Cauchy beta integral) representation of the two-mode coefficients (Section 18.7.9). The logarithmic elliptic identities they need are proved

in Section 18.7.7 by half-strip contours and the real Landen transformation, without Wan's Stieltjes identity. The third proves that the right side of (248) at $d = 2$ equals (271). It regularizes (250) in the rim exponent, evaluates the regulated trace by residues on the square lattice and periodizes the resulting single sums; it is written out in Sections 18.9.2–18.9.6. Its statements for (260) and (238) rest on the per-spin residue identity and on the mixed-trace step, which Sections 18.9.7 and 18.9.5 prove at $d = 2$.

Numerical confirmation. These checks are not proof steps. A 70-digit Euler–Maclaurin evaluation of the square-lattice factorization of (256), and of (260) through the limit $\theta \to 0$ of its factorized single sums at $z = e^{\mathrm{i}\theta}$, gives the residuals 4.5×10^{-69} for (270) and 2.0×10^{-70} for (271). The factorized single sums of (260) diverge individually, so the unrestricted double sum is not used; the limit $\theta \to 0$ of the lattice sum with weight $e^{\mathrm{i}(m-n)\theta}$ preserves the spin order and equals the ordered spin sum (part (c) of the proposition on the series and the strip limit, Section 18.9.7). An independent evaluation from the spectral kernel, with $\mathcal{L}_2 = \partial_b W_4(2;b)|_{b=1}$ as in (244) and $\mathcal{K}_2$ from the spectral form (248), gives 1.7×10^{-79} and 3.1×10^{-62}. To fifty digits,

$$\begin{aligned} P_4(2) &= 250.48731558186728553185175688791140716835571092358\ldots, \\ \mathcal{L}_2 &= -347.24915312320113105847678307522316920816760199487\ldots, \\ \mathcal{K}_2 &= 3.51810913859025224164909017030773952262267280947 10\ldots, \\ P_5(2) &= 1548.08264023312061513243907394578934038811023232 29\ldots. \end{aligned}$$

The five-loop coefficient. Inserting (270) and (271) into (269) and dividing by 48 gives (exact algebra)

$$\boxed{\begin{aligned} 972\,C_5(2) = {} & -32P_5(2) + 56\log 2\,P_4(2) - 36Q_2^2 + (240\log^2 2 + 596\mathsf{G})Q_2 \\ & - 1024\mathsf{G}^2 + 449\,\zeta(3)\log 2 - 80\log^4 2. \end{aligned}} \tag{272}$$

In the general-dimensional form (223) the two substitutions are $384AP_4 - 576\mathcal{L}_2 = 672AP_4$ and $-1536\mathcal{K}_2 = -768(\Phi_2 + AU)$ at $d = 2$. Thus $C_5(2)$ requires only $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, the two ${}_4F_3(1)$ values $F_\pm$ of (331) and the five fixed values (332) that enter (334). All of them already occur in the one-dimensional coefficients, and no multivariable value of Section 17.6 is needed at $d = 2$. Numerically (272) gives $C_5(2) = -38.83364589622536055293698\ldots$, in agreement with the table of Section 1.

Remaining gaps. There is no gap in the proofs of (270) and (271) as stated. The following related points are not settled in this note.

- Wan's Stieltjes identity (100), which enters the proof of Section 18.7 through $U_0 = 2N_{\mathrm{e}}$ and (101), is cited rather than reproved there, exactly as in Section 9.2. The proofs of Sections 18.7.6, 18.7.8 and 18.7.9 do not use it. The half-strip contours of Section 18.7.7 prove $U_0 = 2N_{\mathrm{e}}$ and reprove (103), (106) and (105) without it, and Section 18.5.2 reproves (340), hence (333), without it; among the proofs of (270) only the first still uses it, through (101).
- The derivation of (248) in Section 16.4 uses the mixed-trace representation $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$ stated before (250). At $d = 2$ it is proved for $0 < \tau < \frac{1}{2}$, in its full-Mellin-line (real-part) reading only, in Section 18.9.5; its reading for complex τ and its status at other d are input (M) of Section 21.1. The closed forms here do not use it.
- The four-exponent generalization of the rim–spoke reflection ratio of Section 18.7.6 has been checked numerically but not proved; it is not used.

17.3 A common residue dictionary

For nonnegative integers ℓ, n put $m = \ell + n$, $N = \ell + 2n$, and

$$\begin{aligned} T_{\ell n} &= \frac{(q)_{\ell+n}(1-q)_n}{(2q)_{\ell+n} n!}, \\ d_0(4q) &= 1, \qquad d_{j+1}(4q) = 4q\frac{(2q+1)_j(4q-1)_j}{(2q)_j(2)_j}, \quad j \geq 0. \end{aligned} \tag{273}$$

The second line is a nonsingular way to implement angular multiplicities. At $d = 1$ the upper parameter $4q - 1 = 0$ terminates the positive-spin series at $j = 0$, leaving just $\ell = 0, 1$. At $d = 2$ every positive-spin multiplicity is two. At noninteger $d < 1$ this is an analytic multiplicity, not a positive integer count of modes.

Define four useful classes of explicit scalar sequences:

$$\begin{aligned} \Delta_r(a, b; v) &= \psi_r(v + a) - \psi_r(v + b), \qquad r \geq 0, \\ J_r(a; v) &= \psi_r(v + a), \qquad r \geq 1, \\ H(a; v) &= \psi_0(v + a) - \psi_0(a), \\ c_j &= (-1)^{j+1}\psi_{j-1}(q) - \psi_{j-1}(1-q) + (1 + (-1)^j)\psi_{j-1}(1), \\ \widetilde{D}_j &= \Delta_{j-1}(q, 2q; m) + \Delta_{j-1}(1-q, 1; n), \qquad \lambda_j = c_j + \widetilde{D}_j. \end{aligned} \tag{274}$$

Here $j \leq 6$ suffices. All appearances of ψ_r with a summation index will be eliminated by finite Horn insertions below. Only c_j and the other fixed-argument prefactors retain polygamma notation.

For the lower wheel functions, define, with the indicated finite ranges,

$$\begin{aligned} \eta_r &= \frac{(-1)^r\psi_r(q) - \psi_r(1-q)}{2} + \frac{J_r(q; m) + J_r(1-q; n)}{2}, && 1 \leq r \leq 3, \\ \theta_r &= -\frac{(-1)^r\psi_{r+1}(q) + \psi_{r+1}(1-q)}{4} + \frac{J_{r+1}(q; m) + J_{r+1}(1-q; n)}{4}, && 1 \leq r \leq 2, \\ \mu_1 &= \frac{H(q; m) + H(1-q; n)}{2}, \\ \mu_2 &= \frac{J_1(q; m) + J_1(1-q; n) - \psi_1(q) - \psi_1(1-q)}{4}. \end{aligned} \tag{275}$$

The ordinary finite polynomials needed are

$$\begin{aligned} F_3 &= \frac{9}{2}\lambda_1^2 + \frac{3}{2}\lambda_2, & F_{3,1} &= 9\lambda_1\eta_1 + \frac{3}{2}\eta_2, \\ F_{3,2} &= 9\eta_1^2 + 9\lambda_1\theta_1 + \frac{3}{2}\theta_2, \\ F_4 &= \frac{32}{3}\lambda_1^3 + 8\lambda_1\lambda_2 + \frac{2}{3}\lambda_3, \\ F_{4,1} &= 32\lambda_1^2\eta_1 + 8\eta_1\lambda_2 + 8\lambda_1\eta_2 + \frac{2}{3}\eta_3. \end{aligned} \tag{276}$$

Thus the already proved wheel definitions can be written as

$$\boxed{\begin{aligned} Q_d &= \frac{1}{3}\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^3 F_3, \\ P_4(d) &= -\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^4 F_4, \\ \mathcal{L}_d &= -\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^4 (4\mu_1 F_4 + F_{4,1}), \\ Z_d &= \frac{1}{9}\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^3 \{(9\mu_1^2 + 3\mu_2)F_3 + 6\mu_1 F_{3,1} + F_{3,2}\}. \end{aligned}} \tag{277}$$

The subscripts on $F_{r,j}$ are labels of printed polynomials, not unperformed derivatives. Equations (273)–(277) are also the exact finite inputs for the unweighted Horn construction.

17.4 A double residue series for $P_5(d)$

This subsection completes the residue dictionary above for the remaining primitive (implementation `code/general_closed/primitive_series.py` of `lri_general_closed_update/`). Like (255) for Q_d and $P_4(d)$, the result is an explicit double series, with no Barnes integral and no parameter derivative. At $d = 1, 2, 3$ the coefficients use the fixed-value formulas (29), (334) and (425); for other d the stated evaluator is (319). The series is an independent representation and a check of them. The proof below uses objects defined in Section 18, which follows: the Barnes form (303) with the kernel (300), the measure block (324), the array $\mathcal{A}_{\ell k}$ and the values $G_{\ell k}$ of (309), the two-channel identity (325) and the polynomials E_m of (316).

In this subsection θ is $\theta_0 = 2q$ or $\theta_1 = 2q - 1$, and M_θ is the measure block of (324); this θ is not a wheel sequence θ_r of (275). At the shifted arguments $q + N$, $N = 0, 1, 2, \ldots$, the relation $\Gamma(x - N) = (-1)^N \Gamma(x)/(1 - x)_N$ gives

$$M_\theta(q + N) = \frac{\Gamma(\theta + q + N)\Gamma(\theta - q - N)}{\Gamma(q + N)\Gamma(-q - N)} = M_\theta(q)\, \frac{(\theta + q)_N (1 + q)_N}{(q)_N (1 + q - \theta)_N}. \tag{278}$$

In the second channel below only $\ell \geq 1$ contributes, so $N \geq 1$ there, and one uses

$$M_\theta(q + N) = M_\theta(q + 1)\, \frac{(\theta + q + 1)_{N-1} (2 + q)_{N-1}}{(q + 1)_{N-1} (2 + q - \theta)_{N-1}}, \qquad N \geq 1, \tag{279}$$

rather than the prefactor $M_\theta(q)$ of (278). For $\theta = 2q - 1$ that prefactor contains $\Gamma(3q - 1)$, which is singular at $q = \frac{1}{3}$; there, for $N \geq 1$, it multiplies the vanishing $(3q - 1)_N$. The value $M_{2q-1}(q)$ is itself singular at $q = \frac{1}{3}$, so the $\ell = 0$ terms of the second channel are omitted rather than weighted by zero. In the block notation of (296), $\ell = j + 1$ and $N - 1 = j + 2n$. The first channel uses both $\ell = 0$ and $\ell \geq 1$.

With c_j, $\widetilde{D}_j$ and Δ_r of (274) and $N = \ell + 2n$ of (273), put, for $1 \leq j \leq 6$,

$$\begin{aligned} \kappa_{\theta,j} &= \kappa_j^0 + 7\widetilde{D}_j + 2^j \{\Delta_{j-1}(\theta + q, q; N) + \Delta_{j-1}(1 + q, 1 + q - \theta; N)\}, \\ \kappa_j^0 &= \begin{cases} (7 - 2^{j+1})\, c_j, & j \text{ odd}, \\ 7c_j, & j \text{ even}. \end{cases} \end{aligned} \tag{280}$$

Thus $(\kappa_1^0, \ldots, \kappa_6^0) = (3c_1, 7c_2, -9c_3, 7c_4, -57c_5, 7c_6)$, the same in both channels. This follows from reflection and recurrence in the proof below, not from a fit. The $\kappa_{\theta,j}$ are unrelated to the normalization $\kappa(d, z)$ of (174). Let E_6 be the polynomial of (316), displayed in (317); $E_6(x_1, \ldots, x_6)$ is the coefficient of u^6 in $\exp(\sum_{j=1}^6 x_j u^j/j!)$. Then

$$\boxed{\begin{aligned} P_5(d) = \frac{1}{2\Gamma(2q)^2} \sum_{\ell,n \geq 0} (-1)^\ell d_\ell(d)\, T_{\ell n}^7 \Big[& M_{2q}(q + N)\, E_6(\kappa_{2q,1}, \ldots, \kappa_{2q,6}) \\ & + \ell(\ell + 4q - 2)\, M_{2q-1}(q + N)\, E_6(\kappa_{2q-1,1}, \ldots, \kappa_{2q-1,6}) \Big]. \end{aligned}} \tag{281}$$

Every ingredient is explicit and free of derivatives. The double series converges absolutely for $\frac{2}{5} < d < 4$; smaller positive dimensions are reached by analytic continuation of the same master.

Proof. Take $\frac{1}{2} < q < 1$ and start from the Barnes form (303), with $p = 2q - 1$ and a_ℓ, b_ℓ, $g_\ell(s)$ of (300); in this strip the contour separates the pole families. The line points upwards and is closed to the right, so each spin integral is minus the sum of its right residues. There are two families of

right poles: $s = a_\ell + n$, $n \geq 0$, of order seven, from $g_\ell(s)^7$, and the measure poles $s = (p+k)/2$, $k \geq 0$, from $\Gamma(p-2s)$. The second family must not simply be discarded term by term.

Measure poles. Since $\mathrm{Res}_{s=(p+k)/2}\,\Gamma(p-2s) = (-1)^{k+1}/(2\,k!)$ and $\Gamma(p+k)\Gamma(-p-k) = (-1)^{k+1}\pi/[(p+k)\sin\pi p]$, minus the residue at $s = (p+k)/2$, times the spin weight $(-1)^\ell d_\ell$, is $-\Gamma(2p+1)\sin(\pi p)\,\mathcal{A}_{\ell k}/(4\pi)$ before the overall factor $\Gamma(2q)^5/2$, with $\mathcal{A}_{\ell k}$ and $G_{\ell k} = g_\ell((p+k)/2)$ of (309). The multiplicity d_k in $\mathcal{A}_{\ell k}$ arises as $\Gamma(2p+k)(p+k)/k! = \Gamma(2p+1)d_k/2$, by (240) with $d-2 = 2p$. The factor is finite in the initial strip and independent of ℓ, k. Gamma reflection gives $G_{k\ell} = (-1)^{\ell-k}G_{\ell k}$ for integer ℓ, k (Section 18), and the power seven is odd, so

$$\mathcal{A}_{k\ell} = -\mathcal{A}_{\ell k}, \qquad \mathcal{A}_{\ell\ell} = 0, \qquad \sum_{\ell,k\geq 0} \mathcal{A}_{\ell k} = 0. \tag{282}$$

The last statement uses absolute convergence. For $\frac{1}{2} < q < 1$ and all $\ell, k \geq 0$, with $h = |\ell - k|$,

$$|\mathcal{A}_{\ell k}| \leq \mathrm{const}\cdot(1+\ell+k)^{q-3}\,h\,(1+h)^{-7q}. \tag{283}$$

Indeed, $d_\ell(d) = O((1+\ell)^{4q-2})$ by (240) and $4q-2 > 0$, so $d_\ell d_k \leq \mathrm{const}\cdot(1+\ell+k)^{8q-4}$; next, $|(\ell+p)^2 - (k+p)^2| = h(\ell+k+2p) \leq 2h(1+\ell+k)$; finally, Gamma-ratio estimates [27] bound the first ratio in $G_{\ell k}$ by a constant times $(1+\ell+k)^{-q}$ and, for $\ell \geq k$, the second by a constant times $(1+h)^{-q}$, and $|G_{\ell k}| = |G_{k\ell}|$ covers $\ell < k$. The right side of (283) is summable over $\ell, k \geq 0$: at fixed $\ell + k$ each value of h occurs at most twice, so the factor $h(1+h)^{-7q}$ summed over these pairs is at most $2\sum_{h\geq 0} h(1+h)^{-7q}$, finite since $7q > 2$, and $(1+\ell+k)^{q-3}$ is summable over $\ell + k$ since $q < 2$. These estimates justify the exchange $\ell \leftrightarrow k$, so the measure poles cancel by an exact involution; no divergent sum is discarded. (The periodized kernel (304) needs the finer statement that canceling pairs lie in the same congruence class; it is shown after (309).)

The two channels. Since $\Gamma(p+1\pm 2s) = (p\pm 2s)\Gamma(p\pm 2s)$,

$$[(\ell+p)^2 - 4s^2]\Gamma(p+2s)\Gamma(p-2s) = \Gamma(p+1+2s)\Gamma(p+1-2s) + \ell(\ell+2p)\Gamma(p+2s)\Gamma(p-2s).$$

After division by $\Gamma(2s)\Gamma(-2s)$ this is (325) at $V = 2s$, because $p+1 = 2q$ and $2p = 4q-2$. At $s = a_\ell + n + u$ one has $V = q + N + 2u$, so the two channels are $M_{2q}(q+N+2u)$ and $\ell(\ell+4q-2)M_{2q-1}(q+N+2u)$.

Local expansion. At $s = a_\ell + n + u$, (262) gives $g_\ell(s) = -T_{\ell n}\exp(\sum_{j\geq 1}\lambda_j u^j/j!)/(\Gamma(2q)u)$, with $\lambda_j = c_j + \widetilde{D}_j$ of (274); these extend the λ_m of (252) to $j \leq 6$. For either channel,

$$\log\frac{M_\theta(q+N+2u)}{M_\theta(q+N)} = \sum_{j\geq 1}\frac{u^j}{j!}\Big\{2^j\left[\psi_{j-1}(\theta+q+N) - \psi_{j-1}(q+N)\right] + (-2)^j\left[\psi_{j-1}(\theta-q-N) - \psi_{j-1}(-q-N)\right]\Big\}.$$

Write $\mathrm{ct}_m(x) = \frac{d^m}{dx^m}\pi\cot(\pi x)$, so that reflection reads $\psi_m(x) = (-1)^m\psi_m(1-x) - \mathrm{ct}_m(x)$ [26]. Applied to the two negative arguments, the polygamma part of the second bracket becomes $2^j\Delta_{j-1}(1+q, 1+q-\theta; N)$, and, since ct_m has period one, its cotangent part is $(-2)^j[\mathrm{ct}_{j-1}(-q) - \mathrm{ct}_{j-1}(\theta-q)]$. In both channels $\theta - q \equiv q \pmod 1$, and $\mathrm{ct}_m(-x) = (-1)^{m+1}\mathrm{ct}_m(x)$, so this equals $(-2)^j[(-1)^j - 1]\,\mathrm{ct}_{j-1}(q)$. It vanishes for even j. For odd j it is $2^{j+1}\mathrm{ct}_{j-1}(q) = -2^{j+1}c_j$, because then $c_j = \psi_{j-1}(q) - \psi_{j-1}(1-q) = -\mathrm{ct}_{j-1}(q)$. Adding $7\lambda_j$ from $g_\ell(s)^7$ gives exactly $\kappa_{\theta,j}$ of (280). Hence $g_\ell(s)^7 M_\theta(q+N+2u)$ equals $-(T_{\ell n}/\Gamma(2q))^7\,M_\theta(q+N)\,u^{-7}\exp(\sum_{j\geq 1}\kappa_{\theta,j}u^j/j!)$, and minus its residue at $u = 0$ is $(T_{\ell n}/\Gamma(2q))^7 M_\theta(q+N)E_6(\kappa_{\theta,1},\ldots,\kappa_{\theta,6})$, the sign coming from the seventh power. Summing over n and ℓ with the weights $(-1)^\ell d_\ell$ and multiplying by $\Gamma(2q)^5/2$ gives (281) for $\frac{1}{2} < q < 1$. The local expansion is a device of the proof; the final polynomial needs no coefficient extraction.

Convergence and continuation. For fixed ℓ the terms of (281) are $O(n^{-10q})$; for fixed n each channel is $O(\ell^{q-2})$; in the joint large-index region they are $O((\ell+n)^{-6q-2})$. These orders follow from

$d_\ell(d) = O(\ell^{4q-2})$, $T_{\ell n} = O((1+\ell+n)^{-q}(1+n)^{-q})$, $M_\theta(q+N) = O(N^{2\theta})$ and the boundedness of $\kappa_{\theta,j}$; the elementary estimates are not written out. Hence the series converges absolutely for $\frac{1}{10} < q < 1$, locally uniformly away from the removable parameter singularities treated by (279). Uniform convergence extends the identity from $\frac{1}{2} < q < 1$ to this range, and smaller positive dimensions follow by analytic continuation of the same master; no finite subtraction is added. □

The numerical checks of (281) are collected with the other checks of this section in Section 17.7.

17.5 Making the $\mathcal{K}_d$ cancellation explicit

The paired convention of (260) is essential. We expose its common non-summable part before passing to separate Horn terms. Put

$$\begin{aligned} x &= \pi\cot(\pi q), \qquad P = \pi^2, \\ R_\infty &= \frac{(P+x^2)^2}{2} = \frac{\pi^4}{2\sin^4(\pi q)}, \\ J_j &= J_j(2q;m) + J_j(1;n), \quad 1 \le j \le 3, \\ H &= H(2q;m) + H(1;n). \end{aligned} \tag{284}$$

Here P is just π^2 and is not a graph period. The first four fixed constants in (274) simplify to

$$(c_1,c_2,c_3,c_4) = \left(-x, -x^2 - \frac{2}{3}P, -2x^3 - 2Px, -6x^4 - 8Px^2 - \frac{28}{15}P^2\right). \tag{285}$$

In particular $c_1^3 + 3c_1c_2 + c_3 = 0$. This identity removes an otherwise standalone logarithmic large-index term.

The full remaining polynomial is

$$\begin{aligned} \mathcal{P} = {} & \frac{\widetilde{D}_1^4}{6} - \frac{2x\widetilde{D}_1^3}{3} + \frac{5}{4}\widetilde{D}_1^2\widetilde{D}_2 + \frac{3}{4}\widetilde{D}_1^2 J_1 - \left(\frac{13P}{12} + \frac{x^2}{4}\right)\widetilde{D}_1^2 \\ & - \frac{5x}{2}\widetilde{D}_1\widetilde{D}_2 + \frac{5}{12}\widetilde{D}_1\widetilde{D}_3 - \frac{3x}{2}\widetilde{D}_1 J_1 - \frac{1}{4}\widetilde{D}_1 J_2 + \left(\frac{4}{3}Px + x^3\right)\widetilde{D}_1 \\ & + \frac{1}{2}\widetilde{D}_2^2 + \frac{3}{4}\widetilde{D}_2 J_1 + \left(-\frac{11P}{12} + \frac{x^2}{4}\right)\widetilde{D}_2 - \frac{5x}{12}\widetilde{D}_3 - \frac{1}{24}\widetilde{D}_4 \\ & + H\left(\frac{1}{2}\widetilde{D}_1^3 - \frac{3}{2}x\widetilde{D}_1^2 + \frac{3}{2}\widetilde{D}_1\widetilde{D}_2 - P\widetilde{D}_1 - \frac{3}{2}x\widetilde{D}_2 + \frac{1}{2}\widetilde{D}_3\right) \\ & - \frac{P}{2}J_1 + \frac{x}{4}J_2 - \frac{1}{4}J_3. \end{aligned} \tag{286}$$

Every monomial contains at least one $\widetilde{D}_j$ or J_j; the possible H factor does not occur alone. Direct exact algebra in the residue polynomial gives $\mathcal{R}_{\ell n} = R_\infty + \mathcal{P}$.

Gauss summation [4] gives a second useful identity:

$$\begin{aligned} r(c) &= \frac{\Gamma(c)^2}{\Gamma(c-q)\Gamma(c+q)} = {}_2F_1\left(\begin{matrix} q, -q \\ c \end{matrix}; 1\right), \\ \delta(c) &= r(c) - 1 = -\frac{q^2}{c}\,{}_3F_2\left(\begin{matrix} 1, 1-q, 1+q \\ 2, c+1 \end{matrix}; 1\right). \end{aligned} \tag{287}$$

The second line follows by isolating the zeroth term and shifting the remaining index. Both series converge for the positive c used here. Using $S_{\ell n}$ and M of (260), gamma reflection shows

$$\frac{M^2 S_{\ell n}^2}{T_{\ell n}^2} = 2R_\infty\, r(m+2q)^2 r(n+1)^2. \tag{288}$$

Therefore the exact canceled form is

$$\boxed{\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}\sum_{\ell,n\geq 0} d_\ell(d) T_{\ell n}^2 \Big\{ R_\infty\big[(1+\delta(m+2q))^2(1+\delta(n+1))^2 - 1\big] - \mathcal{P}\Big\}.} \tag{289}$$

The bracket involving δ has exactly eight nonconstant monomials, with coefficients $\binom{2}{i}\binom{2}{j}$, $0 \leq i,j \leq 2$, $(i,j) \neq (0,0)$. Together with (286), this is a finite polynomial input to Horn functions; the two original divergent family sums have not been assigned independent values.

For $0 < q < 1/2$, the monomial sums obtained this way are absolutely convergent. In the joint large-index region the uncancelled base scales as $O((\ell+n)^{-2})$, and every printed monomial supplies at least one additional inverse power, up to harmless logarithms. At fixed n, the remaining angular bound is $O(\ell^{2q-2}\log\ell)$, which explains the upper endpoint $q = 1/2$. At fixed ℓ, the radial bound is summable for positive q. The endpoint and larger dimensions must be treated by continuation of the *complete finite combination*, not by separately summing divergent monomial series at argument one. The paired iterated series remains a direct alternative there.

17.6 The standard hypergeometric function and the finite parameter rules

For finite lists of parameter–vector pairs define the ordinary multivariable Horn-type generalized hypergeometric series

$$\mathfrak{H}\left[\begin{matrix}(a_\alpha, \boldsymbol{u}_\alpha)_\alpha \\ (b_\beta, \boldsymbol{v}_\beta)_\beta\end{matrix}; \boldsymbol{z}\right] = \sum_{\boldsymbol{n}\in\mathbb{N}_0^r} \frac{\prod_\alpha (a_\alpha)_{\boldsymbol{u}_\alpha\cdot\boldsymbol{n}}}{\prod_\beta (b_\beta)_{\boldsymbol{v}_\beta\cdot\boldsymbol{n}}} \prod_{i=1}^r \frac{z_i^{n_i}}{n_i!}. \tag{290}$$

All vectors used below have nonnegative integer entries. Repeated pairs are allowed. Thus each coefficient ratio on increasing a single index is an explicit rational function of the indices. This is the standard Horn/Gamma-ratio class, not a new LRI-specific function whose evaluation has been left unspecified. It includes generalized Lauricella-type arrays; the $\ell + 2n$ vectors in the primitive make the more general Horn notation convenient [3, 5].

A finite set of identities is sufficient to print all its parameters. For $a, b > 0$, nonnegative integer v and the indicated orders,

$$\begin{aligned}
J_r(a;v) &= (-1)^{r+1} r! \sum_{k\geq 0} \frac{1}{(v+a+k)^{r+1}}, && r \geq 1,\\
\Delta_r(a,b;v) &= (-1)^{r+1} r!(b-a) \sum_{j=0}^{r} \sum_{k\geq 0} \frac{1}{(v+a+k)^{r+1-j}(v+b+k)^{j+1}}, && r \geq 0,\\
H(a;v) &= v \sum_{k\geq 0} \frac{1}{(a+k)(a+v+k)},\\
\delta(a+v) &= -q^2 \sum_{k\geq 0} \frac{(1-q)_k(1+q)_k}{(2)_k(a+v)_{k+1}}.
\end{aligned} \tag{291}$$

The first three follow from polygamma series and a finite partial-fraction identity [32]; the last is (287). For a product, assign a *different* index k to every factor. The choices $0 \leq j \leq r$ are finite, so expanding any of the printed master polynomials produces finitely many unweighted series. There is no harmonic-number or polygamma factor left in their coefficients.

Every reciprocal affine power is encoded by one pair of parameter rows:

$$(a + \boldsymbol{v}\cdot\boldsymbol{n})^{-s} = a^{-s} \frac{(a)^s_{\boldsymbol{v}\cdot\boldsymbol{n}}}{(a+1)^s_{\boldsymbol{v}\cdot\boldsymbol{n}}}. \tag{292}$$

For the last identity in (291), use

$$\frac{1}{(a+v)_{k+1}} = \frac{1}{a}\frac{(a)_v}{(a+1)_{v+k}}. \tag{293}$$

An otherwise absent $k!$ is supplied by an upper pair $(1, \boldsymbol{e}_k)$ in (290), which cancels its standard denominator. This fixes all factorial conventions explicitly.

The polynomial factor v in a harmonic insertion, and the primitive's $\ell(\ell+4q-2)$, also require no Euler derivative. Expand any resulting power of an index into falling factorials:

$$n^s = \sum_{j=0}^{s} \left\{ {s \atop j} \right\} n^{\underline{j}}, \qquad (m+j)^{\underline{j}} = j!\frac{(j+1)_m}{(1)_m}. \tag{294}$$

Shift $n = m + j$ in each term. Each old parameter pair $(a, \boldsymbol{v})$ is shifted to $(a + v_n j, \boldsymbol{v})$, with a prefactor $(a)_{v_n j}$ in the numerator or denominator as appropriate. Multiply by $j! z_n^j$ and add the upper/lower pair $(j+1, \boldsymbol{e}_n)/(1, \boldsymbol{e}_n)$. The braces in (294) are Stirling numbers of the second kind, so this operation is a finite algebraic rule.

17.6.1 Explicit starting arrays

For any of the powers $r = 2, 3, 4, 7$, split the original double sum into $\ell = 0$ and $\ell = j + 1$. Before inserting weights, the zero-spin block has scalar prefactor one and rows

$$\begin{matrix} (1;1),\ (q;1)^{\times r},\ (1-q;1)^{\times r} \\ (2q;1)^{\times r},\ (1;1)^{\times r} \end{matrix}, \qquad z_n = 1. \tag{295}$$

The positive-spin block has scalar prefactor $4q/2^r$ and rows

$$\begin{matrix} (1;1,0), (1;0,1), (2q+1;1,0), (4q-1;1,0), (q+1;1,1)^{\times r}, (1-q;0,1)^{\times r} \\ (2q;1,0), (2;1,0), (2q+1;1,1)^{\times r}, (1;0,1)^{\times r} \end{matrix}, \qquad (z_j, z_n) = (1,1). \tag{296}$$

Here semicolons separate a scalar parameter from its index vector; these are precisely the pairs in (290).

17.6.2 An explicit parameter-array example

For illustration, it is enough to print one unambiguous insertion instead of thousands of nearly identical arrays. The zero-spin contribution of $T_{0n}^3 \Delta_0(q, 2q; n)$ alone is

$$-\frac{1}{2q}\mathfrak{H}\left[\begin{matrix} (1;1,0), (1;0,1), (q;1,0)^{\times 3}, (1-q;1,0)^{\times 3}, (q;1,1), (2q;1,1) \\ (2q;1,0)^{\times 3}, (1;1,0)^{\times 3}, (q+1;1,1), (2q+1;1,1) \end{matrix}; (1,1)\right]. \tag{297}$$

Its coefficient is obtained from $\Delta_0(q, 2q; n) = -q\sum_k [(n+q+k)(n+2q+k)]^{-1}$. There is no remaining derivative or finite-index harmonic sum in (297). All other entries follow the same printed rules. The two-variable example is deliberately elementary; products insert additional variables, up to the bounds in the following table.

17.7 Finite arrays, scope of continuation, and what is now closed

The scalar polynomial inputs for the array construction are precisely (276), (277), (286)–(289). They have finite degree. Applying (291)–(294) therefore gives a finite list of conventional parameter arrays for every master:

$$M(q) = M_{\text{scalar}}(q) + \sum_{\alpha=1}^{N_M} p_{M,\alpha}(q)\, \mathfrak{H}[\mathsf{A}_{M,\alpha}(q); \mathsf{B}_{M,\alpha}(q); \boldsymbol{z}_{M,\alpha}], \tag{298}$$

where M is one of $\mathcal{L}, Z, \mathcal{K}$. The primitive masters use the smaller finite-sample representation of Section 18; their residue series (the Q_d and $P_4(d)$ lines of (277), and (281)) are not converted into arrays here, and Section 18.6 compares the counts such a conversion gives. Only $M = \mathcal{K}$ has a separate scalar term, $-AU/4$. The coefficients $p_{M,\alpha}$ use rational functions of q, Gamma values and *fixed-argument* polygamma values through ψ_3. All parameter rows are explicit affine functions of q with rational coefficients, and the nonprimitive argument vectors contain only 1. No unspecified generic-index kite function or Feynman integral is included in (298).

The supplied direct compiler cancels identical parameter pairs and merges literally identical arrays but does not attempt contiguous relations or permutations of all auxiliary axes. Its nonminimal counts are

Master	Number of fixed Horn values	Maximum variables per value
$\mathcal{L}_d$	244	6
Z_d	318	6
$\mathcal{K}_d$	386	6

These counts quantify an important tradeoff: a finite unweighted function-class expression has been reached, but the mechanical expansion is *not shorter* than the weighted double series. It gives finite function-class closure, not a short scalar-hypergeometric evaluation. The counts assert neither minimality nor independence. A much shorter representation is obtained by a different route in Sections 20 and 21: periodization in the rim exponent and the top-mode theorem (473) reduce $\mathcal{L}_d$ and Z_d to the values and first and second rim-exponent derivatives of eight functions with at most two summation indices (the inventory after (553)), and, under the inputs stated in Section 21.1, $\mathcal{K}_d$ to a third-order τ-jet of three-node samples of three two-variable Horn functions (586). These forms keep parameter derivatives, so they lie outside the unweighted class of (298).

The array files, their scalar-parameter dictionary, and the compiler are included in `lri_general_closed_update/` under `code/general_closed/` and `verification/general_closed/horn_arrays/`. Each JSON row records the scalar prefactor, the full upper and lower parameter–vector lists with multiplicities, and the argument vector. The equations above specify a finite way of generating those same lists without needing the program; the files are a compact machine-readable certificate, not a replacement for a missing mathematical definition.

Continuation is part of the formula. There is a nonempty common absolute-convergence strip

$$\frac{1}{6} < q < \frac{1}{2}, \qquad \text{equivalently } \frac{2}{3} < d < 2, \tag{299}$$

for this finite construction. This follows from the wheel bounds and the canceled $\mathcal{K}$ bounds. The tail insertions have fixed sign or are finite signed sums of positive denominators there, and their absolute sums have the same bounds as the weights they replace. This justifies each finite expansion and the added summation indices in that strip.

For other $0 < d < 4$, use the unique dimensional analytic continuation of *the complete finite combination* from (299). At a removable exceptional value, evaluate that continued combination, not a separate pole part of each summand. In particular some individual $\mathcal{K}$ Horn series diverge at unity for $d \geq 2$, even though their continued combination is the finite physical master. This is not an extra free regulator, nor does it introduce an adjustable finite part; it is the continuation already fixed by the original subtracted master. It does mean that the result must not be advertised as separately convergent unity-argument series for every physical dimension. Direct paired summation or the original independent master integral is often numerically preferable at these points.

Thus (222)–(223), together with the finite parameter rules, give all coefficients through $C_5(d)$ in a specified standard multivariable hypergeometric class. Sections 20 and 21 shorten this considerably, at the price of low-order parameter derivatives; (553) is $C_5(d)$ in that form. A finite reduction to

ordinary ${}_pF_q$ values for generic d remains a different, stronger question. It is not asserted by these results, and the searches of Section 20.8, which found no such reduction of the remaining constants, are numerical and rule nothing out. At $d = 2$ such a reduction holds for the whole coefficient: Section 17.2 expresses $C_5(2)$ through ${}_4F_3(1)$, ${}_9F_8(-1)$ and ${}_8F_7(-1)$ values and fixed constants.

Checks of the finite multivariable construction. These checks belong to the package `lri_general_closed_update/` (scripts `check_new_identities.py`, `check_primitive_sums.py`, `check_horn_compiler.py` and `check_insertion_prefactors.py` in `code/general_closed/`; records `identity_checks.json`, `primitive_sum_checks.json`, `horn_compiler_checks.json` and `insertion_prefactor_checks.json` in `verification/general_closed/`). They are consistency checks, not interval bounds. The Gamma-ratio and eighth-power forms of Q_2 in (267), evaluated separately at 60 working digits, differ by about -4.98×10^{-60}. The same run checks the canceled polynomial (286), the normalization (288) and the shifted Gauss identity (287) at $q = 1/4, 3/10, 1/2, 3/4$ and several nonzero indices, with residuals below 2.0×10^{-58}. The Horn compiler is checked by 1038 exact rational comparisons of normalized coefficients at $q = 3/10$, over 519 sampled rows and including inserted tail indices, and by 12 falling-factorial shift identities. A further 336 exact rational checks cover the scalar prefactor of every type of tail insertion, and exact algebra verifies the polynomial E_6 and both dimension-two substitutions in (269). These checks verify the parameter encoding; they do not sum every continued multivariable endpoint function numerically.

For the double residue series (281) of Section 17.4: at 60 digits, Cauchy circles around poles of both channels reproduce the local residues, with largest normalized residual below 8.8×10^{-55}; the antisymmetry (282) was checked for unequal spins at $q = \frac{3}{5}$ and $\frac{3}{4}$; and the constants κ_j^0 agree with the unreduced reflection expressions. Double-precision sums of (281), with 112 spins, an Euler transform of the alternating spin sum and radial cutoffs 128 to 1024, were compared with spectral quadrature of (216) at 224 and 352 nodes:

d	residue-sum value of $P_5(d)$	residue sum minus 352-node control
2	1548.0826402331213	-4.43×10^{-11}
13/5	473.7956970211414	-6.37×10^{-12}
3	240.3403177590616	-2.77×10^{-11}

The displayed digits are outputs, not certified digits. Against the high-precision values of Section 16.8 the residue sums differ by $+6.8 \times 10^{-13}$ at $d = 2$ and -2.2×10^{-11} at $d = 3$. At $d = 1$ the raw radial series converges much more slowly: against a 60-digit evaluation of the Gamma integral its residual falls from 1.05×10^{-5} at cutoff 128 to 5.86×10^{-8} at cutoff 1024. No high-precision agreement is claimed there; the five-value formula (29) remains the evaluator at $d = 1$.

The rigorous parts of the construction are the algebraic normalization, the local residue identities, the involutive cancellation of the measure poles of $P_5(d)$ (Section 17.4) and the finite Horn conversion in its stated convergence domain, followed by the prescribed continuation; the numerical residuals corroborate these steps but are not interval bounds. The physical identification of the coefficients rests on the graph reduction and forest assembly of Sections 15 and 16; no independent general-dimensional five-loop graph calculation is claimed.

18 Finite periodization of the primitive sector

18.1 The three Barnes kernels

Put $q = d/4$, $p = 2q - 1$ and

$$\begin{aligned} a_\ell &= (\ell + q)/2, & b_\ell &= a_\ell + q, \\ g_\ell(s) &= \frac{\Gamma(a_\ell + s)\Gamma(a_\ell - s)}{\Gamma(b_\ell + s)\Gamma(b_\ell - s)}, & D_q(t) &= \sin(\pi t)\sin\big(\pi(t+q)\big). \end{aligned} \tag{300}$$

Use $d_\ell = d_\ell(4q)$ from (240). A version with no spurious $0/0$ at $d = 1, 2$ is

$$d_0 = 1, \qquad d_{k+1} = 4q\,\frac{(2q+1)_k(4q-1)_k}{(2q)_k(2)_k}, \qquad k \geq 0. \tag{301}$$

The normalized wheel periods are

$$W_r = \Gamma(2q)^r \sum_{\ell \geq 0} d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, g_\ell(s)^r, \qquad W_3 = 3Q_d, \quad W_4 = P_4(d). \tag{302}$$

For the five-loop primitive define

$$\begin{aligned} K_\ell(s) &= \big[(\ell+p)^2 - 4s^2\big]\frac{\Gamma(p+2s)\Gamma(p-2s)}{\Gamma(2s)\Gamma(-2s)} g_\ell(s)^7, \\ P_5(d) &= \frac{\Gamma(2q)^5}{2} \sum_{\ell \geq 0} (-1)^\ell d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, K_\ell(s). \end{aligned} \tag{303}$$

Initially take $1/2 < q < 1$ so that the contours separate the pole families without crossed-pole terms. The result below is then continued with the same prescription as the original primitive master. The kernels and normalization are the wheel and conformal-triangle kernels already specified in this note, based on the operator framework of [24]. No new renormalization convention is introduced.

For the proof only, periodize the kernels:

$$\begin{aligned} S_r(t) &= \sum_{\ell \geq 0} d_\ell \sum_{n \in \mathbb{Z}} g_\ell(a_\ell + n + t)^r, & r &= 3, 4, \\ S_7(t) &= \sum_{\ell \geq 0} (-1)^\ell d_\ell \sum_{n \in \mathbb{Z}} K_\ell(a_\ell + n + t), & & \\ \Xi_r(t) &= D_q(t)^r S_r(t), & r &= 3, 4, 7. \end{aligned} \tag{304}$$

The infinite sums in (304) are proof devices, not additional operations in the finite evaluation rule below.

18.2 Why finitely many Fourier modes suffice

18.2.1 Wheel poles, reflection and growth

For $r = 3, 4$ the only pole classes of S_r modulo one are $t = 0$ and $t = -q$, each of order at most r. Gamma reflection gives, for $n \geq 0$,

$$g_\ell(a_\ell + n + t) = -\frac{\sin\pi(q-t)}{\sin\pi t}\,\frac{\Gamma(\ell+n+q+t)\Gamma(n+1-q+t)}{\Gamma(\ell+n+2q+t)\Gamma(n+1+t)}. \tag{305}$$

Reflection, recurrence and the Gamma-ratio asymptotics used here are standard identities [26, 27]; the periodization argument is given explicitly rather than inferred from those identities alone.

On compact sets avoiding the pole classes, split the lattice into $n \geq 0$, $n \leq -\ell - 1$, and $-\ell \leq n \leq -1$. In the middle part put $n = -j - 1$, $\ell = j + k + 1$; then

$$g_\ell(a_\ell + n + t) = \frac{\Gamma(k+q+t)\Gamma(j+1-t)}{\Gamma(k+2q+t)\Gamma(j+q+1-t)}. \tag{306}$$

These bounds give locally uniform convergence, including after the possible pole factors are cleared. Consequently Ξ_r is entire. Evenness of $g_\ell(s)$ and the integer reindexing $n \mapsto -\ell - n$ give

$$\Xi_r(t+1) = \Xi_r(t), \qquad \Xi_r(-q-t) = \Xi_r(t). \tag{307}$$

For $t = x + \mathrm{i}y$, $0 \le x \le 1$, $|y| \ge 1$, the right-hand sum is bounded by a constant times

$$\sum_{\ell,n\ge 0} (1+\ell)^{4q-2}(1+|\ell+n+\mathrm{i}y|)^{-rq}(1+|n+\mathrm{i}y|)^{-rq}.$$

The middle expression (306) has the analogous bound with weight $(1+j+k)^{4q-2}$. Integral comparison, initially for $1/2 < q < 1$, gives

$$|S_r(x+\mathrm{i}y)| = O(|y|^{-2q(r-2)}), \qquad |\Xi_r(x+\mathrm{i}y)| = O(e^{2\pi r|y|}|y|^{-2q(r-2)}). \tag{308}$$

For example the endpoint $\ell \gg n, |y|$ has power $\ell^{q(4-r)-2}$, which is summable in this strip. This also controls the endpoints of the scaled integral, not just simultaneous large indices.

18.2.2 Five-loop measure poles cancel class by class

The extra measure poles of K_ℓ occur at $2s = \pm(p+k)$, $k \ge 0$. Their *weighted* right residues have the form

$$\begin{aligned} \mathcal{A}_{\ell k} &= (-1)^\ell d_\ell d_k[(\ell+p)^2 - (k+p)^2]G^7_{\ell k}, \\ G_{\ell k} &= \frac{\Gamma((\ell+k+3q-1)/2)}{\Gamma((\ell+k+5q-1)/2)}\frac{\Gamma((\ell-k+1-q)/2)}{\Gamma((\ell-k+1+q)/2)}. \end{aligned} \tag{309}$$

Gamma reflection [26], applied to the second ratio, gives $G_{k\ell} = (-1)^{\ell-k}G_{\ell k}$ and hence $\mathcal{A}_{k\ell} = -\mathcal{A}_{\ell k}$; the diagonal residue is zero. For the present purpose cancellation of the *total* measure-residue sum is not by itself sufficient. One must check the congruence classes. A right measure pole is at

$$t = \frac{q-1+k-\ell}{2} \pmod 1.$$

Exchanging k, ℓ changes this by an integer, so the canceling residues are in the same class. A left measure pole is at $t = (1-3q-k-\ell)/2 \pmod 1$, also unchanged by the exchange. Thus the extra poles cancel locally in the spin-summed periodization, not only after the final contour integration. Absolute convergence in the initial strip, which the bound (283) provides, permits the pairing. Coincident pole parameters are handled by continuation of the complete expression, not by keeping a separate singular residue.

After this cancellation only the two order-seven Gamma-pole classes remain; Ξ_7 is entire and satisfies (307). The measure contributes at most a factor of order $(1+\ell+n+|y|)^{4q}$ to the bound behind (308), so

$$|S_7(x+\mathrm{i}y)| = O(|y|^{-6q}), \qquad |\Xi_7(x+\mathrm{i}y)| = O(e^{14\pi|y|}|y|^{-6q}). \tag{310}$$

The endpoint $\ell \gg n, |y|$ now has power ℓ^{q-2}, again summable for $q < 1$.

18.2.3 Finite-dimensionality theorem

For an entire one-periodic function the Fourier coefficient is unchanged when its integration contour is translated vertically. Apply this to (308) or (310). For mode $k \ge r$, translation down by $\mathrm{i}Y$ bounds its absolute value by $Ce^{2\pi(r-k)Y}Y^{-\eta}$ with $\eta > 0$; it therefore vanishes, including the

boundary mode $k = r$. Upward translation deals with $k \le -r$. Reflection (307) pairs the remaining modes. Thus, exactly,

$$\boxed{\Xi_r(t) = \sum_{k=0}^{r-1} c_{r,k}(q)\cos\bigl(2\pi k(t+q/2)\bigr), \qquad r = 3, 4, 7.} \tag{311}$$

For real q the coefficients are real, since the meromorphic kernel is real on nonsingular real arguments. This is a proved finite Fourier space, not a fitted trigonometric ansatz. For the wheels the top coefficient also vanishes: $c_{3,2}(q) = c_{4,3}(q) = 0$ for $\frac{2}{r} < d < 4$, by (473), proved in Section 20.1.2 for a general rim exponent. The sums in (318) may therefore stop at $k = r - 2$ for $r = 3, 4$; the rules below remain exact as written. No such statement is made for $r = 7$, whose kernel is not a product of balanced factors.

18.3 A finite evaluation rule, with no derivatives

A stable fixed sampling grid is

$$\begin{aligned} &N_r = 2r - 1, \qquad \eta_r = \frac{1}{8\pi r}, \quad t_j = -\frac{q}{2} + \frac{j}{N_r} + \mathrm{i}\eta_r, \quad j = 0, \dots, r-1, \\ &Y_j = \Re\Xi_r(t_j). \end{aligned} \tag{312}$$

The nonzero imaginary part avoids every real pole class. It is a fixed number, not a regulator to be sent to zero. Cosine orthogonality gives

$$\boxed{\begin{aligned} c_{r,0} &= \frac{1}{N_r}\left(Y_0 + 2\sum_{j=1}^{r-1} Y_j\right), \\ c_{r,k} &= \frac{2}{N_r\cosh(k/(4r))}\left[Y_0 + 2\sum_{j=1}^{r-1} Y_j\cos\frac{2\pi kj}{N_r}\right], \quad 1 \le k < r. \end{aligned}} \tag{313}$$

Only finite sums appear here. For complex q, use all $2r - 1$ complex samples and the ordinary discrete Fourier transform, rather than using complex conjugation to reconstruct half of them.

The residue functional can also be printed without differentiating anything. For $1 \le h < r$ set

$$\begin{aligned} L_h(q) &= (-1)^{h+1}\psi_{h-1}(1-q) - \psi_{h-1}(q), \\ u_h(q) &= r\left[(1 + (-1)^h)\psi_{h-1}(1) - L_h(q)\right]. \end{aligned} \tag{314}$$

These particular polygamma combinations are elementary reflection values. If $c = \cot(\pi q)$, an implementation requiring no higher polygamma calls may use

$$\begin{aligned} &\frac{u_1}{r\pi} = -c, && \frac{u_2}{r\pi^2} = c^2 + \frac{4}{3}, && \frac{u_3}{r\pi^3} = -2c(1+c^2), \\ &\frac{u_4}{r\pi^4} = \frac{32}{15} + 8c^2 + 6c^4, && \frac{u_5}{r\pi^5} = -8c(1+c^2)(2+3c^2), \\ &\frac{u_6}{r\pi^6} = \frac{1024}{63} + 136c^2 + 240c^4 + 120c^6. \end{aligned} \tag{315}$$

Only $u_1, \dots, u_{r-1}$ are used. These follow from reflection and the integer polygamma values, and introduce no additional period constants.

Define finite polynomials E_m, only for $0 \le m \le 6$, by

$$E_0 = 1, \qquad E_m(x_1, \dots, x_m) = \frac{1}{m}\sum_{h=1}^{m} \frac{x_h}{(h-1)!} E_{m-h}(x_1, \dots, x_{m-h}). \tag{316}$$

This is a finite polynomial definition, not a parameter derivative. For orientation,

$$\begin{aligned} E_2 &= (x_1^2+x_2)/2, & E_3 &= (x_1^3+3x_1x_2+x_3)/6,\\ 720E_6 &= x_1^6+15x_1^4x_2+20x_1^3x_3+45x_1^2x_2^2+15x_1^2x_4 \\ &\quad +60x_1x_2x_3+6x_1x_5+15x_2^3+15x_2x_4+10x_3^2+x_6. \end{aligned} \tag{317}$$

Write

$$\begin{aligned} B_{r,k}(q) &= \frac{1}{2}\big[e^{\mathrm{i}\pi kq}E_{r-1}(u_1+2\pi\mathrm{i}k,u_2,\dots,u_{r-1}) \\ &\qquad + e^{-\mathrm{i}\pi kq}E_{r-1}(u_1-2\pi\mathrm{i}k,u_2,\dots,u_{r-1})\big], \\ \mathcal{E}_r(q;\boldsymbol{Y}) &= -\frac{\sum_{k=0}^{r-1}B_{r,k}(q)c_{r,k}}{(\pi\sin\pi q)^r}. \end{aligned} \tag{318}$$

Then the finite primitive formulas are

$$\boxed{Q_d=\frac{\Gamma(2q)^3}{3}\mathcal{E}_3,\qquad P_4(d)=\Gamma(2q)^4\mathcal{E}_4,\qquad P_5(d)=\frac{\Gamma(2q)^5}{2}\mathcal{E}_7.} \tag{319}$$

To check the sign and factors, close each Barnes contour to the right. Its orientation is clockwise, so its value is minus the sum of right residues. That sum is $\mathrm{Res}_{t=0}\,S_r(t)$. In $D_q(t)^{-r}$ the leading coefficient is $(\pi\sin\pi q)^{-r}t^{-r}$; the logarithmic coefficients of the analytic factor are exactly (314). Expanding a cosine then gives (318). Residues are used in this proof only; the final formula (319) contains finite arithmetic and fixed special-function values.

18.4 Explicit one- and two-variable functions for every sample

For finite upper/lower lists of parameter–integer-vector pairs use

$$\mathfrak{H}[\mathsf{U};\mathsf{V};\boldsymbol{z}]=\sum_{\boldsymbol{n}\ge0}\frac{\prod_{(a,v)\in\mathsf{U}}(a)_{v\cdot\boldsymbol{n}}}{\prod_{(b,w)\in\mathsf{V}}(b)_{w\cdot\boldsymbol{n}}}\prod_i\frac{z_i^{n_i}}{n_i!}. \tag{320}$$

This is the unweighted Horn convention already used in the note. Signed integer-vector entries are permitted; $(a)_{-m}$ means $\Gamma(a-m)/\Gamma(a)$, not a truncated product. Consecutive coefficient ratios are rational in the indices. There are no index-dependent polygammas in the following arrays. The function definition contains its usual series, just as the definition of ${}_pF_q$ does; the master formula has no additional outer infinite sum. This distinction is not a claim that a two-variable function has become a one-variable function.

For two variables put $\boldsymbol{e}=(1,0)$, $\boldsymbol{f}=(0,1)$, $\boldsymbol{c}=\boldsymbol{e}+\boldsymbol{f}$. Repeated rows are denoted by a superscript $\times r$. Define

$$\begin{aligned} A_r^0(t) &= a_r^0(t)\,{}_{2r+1}F_{2r}\left(\begin{matrix}1,(q+t)^{\times r},(1-q+t)^{\times r}\\ (2q+t)^{\times r},(1+t)^{\times r}\end{matrix};1\right),\\ a_r^0(t) &= \left[\frac{\Gamma(q+t)\Gamma(-t)}{\Gamma(2q+t)\Gamma(q-t)}\right]^r,\\ A_r^+(t) &= a_r^+(t)\mathfrak{H}[\mathsf{U}_+;\mathsf{V}_+;(1,1)],\\ a_r^+(t) &= 4q\left[\frac{\Gamma(q+1+t)\Gamma(-t)}{\Gamma(2q+1+t)\Gamma(q-t)}\right]^r,\\ C_r(t) &= a_r^c(t)\mathfrak{H}[\mathsf{U}_c;\mathsf{V}_c;(1,1)],\\ a_r^c(t) &= 4q\left[\frac{\Gamma(q+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+q-t)}\right]^r. \end{aligned} \tag{321}$$

The complete two-variable arrays are

$$\begin{aligned}
\mathsf{U}_+ &= \{(q+1+t,\boldsymbol{c})^{\times r},(1,\boldsymbol{e}),(2q+1,\boldsymbol{e}),(4q-1,\boldsymbol{e}),(1,\boldsymbol{f}),(1-q+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{V}_+ &= \{(2q+1+t,\boldsymbol{c})^{\times r},(2q,\boldsymbol{e}),(2,\boldsymbol{e}),(1+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{U}_c &= \{(2q+1,\boldsymbol{c}),(4q-1,\boldsymbol{c}),(1,\boldsymbol{e}),(1-t,\boldsymbol{e})^{\times r},(1,\boldsymbol{f}),(q+t,\boldsymbol{f})^{\times r}\},\\
\mathsf{V}_c &= \{(2q,\boldsymbol{c}),(2,\boldsymbol{c}),(1+q-t,\boldsymbol{e})^{\times r},(2q+t,\boldsymbol{f})^{\times r}\}.
\end{aligned} \tag{322}$$

For $r=3,4$ every sample is therefore

$$\boxed{\Xi_r(t)=D_q(t)^r\big[A_r^0(t)+A_r^+(t)+A_r^0(1-q-t)+A_r^+(1-q-t)+C_r(t)\big].} \tag{323}$$

The right tail gives the first two functions, evenness maps the left tail to $1-q-t$, and (306) gives the last function. Equation (301) then proves every Pochhammer row in (322). This is also a direct coefficient-by-coefficient proof of the sample formula.

18.4.1 The eight primitive sample blocks

The $r=7$ arrays are obtained from the same printed blocks by finite row insertions, as follows. Put

$$v=q+2t,\quad \theta_0=2q,\quad \theta_1=2q-1,\qquad M_\theta(v)=\frac{\Gamma(\theta+v)\Gamma(\theta-v)}{\Gamma(v)\Gamma(-v)}. \tag{324}$$

The identity underlying the two channels is

$$\big[(\ell+p)^2-V^2\big]\frac{\Gamma(p+V)\Gamma(p-V)}{\Gamma(V)\Gamma(-V)}=M_{2q}(V)+\ell(\ell+4q-2)M_{2q-1}(V). \tag{325}$$

For clarity, a "block" below includes its scalar prefactor, both lists and its argument vector.

Define a right insertion $\mathcal{R}_{\theta,w,h}$ on any block by multiplying its prefactor by $M_\theta(w)$ and adjoining the rows

$$\text{upper: } (\theta+w,h),(1+w,h),\qquad \text{lower: } (w,h),(1+w-\theta,h). \tag{326}$$

It multiplies its coefficient by $M_\theta(w+h\cdot\boldsymbol{n})$. This follows by reflecting the negative Gamma argument and using that $h\cdot\boldsymbol{n}$ is an integer. A middle insertion $\mathcal{C}_{\theta,w,h}$ multiplies the prefactor by $M_\theta(w)$ and adjoins instead

$$\text{upper: } (\theta+w,h),(\theta-w,-h),\qquad \text{lower: } (w,h),(-w,-h). \tag{327}$$

This is the same identity before reflecting the index difference.

Start with A_7^0,A_7^+,C_7. Let A^- be A_7^+ with a minus sign in its prefactor and arguments $(-1,1)$; let C^- be C_7 with a minus sign and arguments $(-1,-1)$. Let $\widehat{A}^-$ be obtained from A^- by multiplying the prefactor by $4q-1$, replacing its upper row $(4q-1,\boldsymbol{e})$ by $(4q,\boldsymbol{e})$ and its lower row $(2,\boldsymbol{e})$ by $(1,\boldsymbol{e})$. Define $\widehat{C}^-$ with the same replacements along $\boldsymbol{c}$. These replacements supply the factor $\ell(\ell+4q-2)$ in the second channel of (325).

There are three right blocks and two middle blocks:

$$\begin{aligned}
Z(t)&=\mathcal{R}_{\theta_0,v,2}A_7^0(t),\\
P_0(t)&=\mathcal{R}_{\theta_0,v+1,\boldsymbol{e}+2\boldsymbol{f}}A^-(t),\quad P_1(t)=\mathcal{R}_{\theta_1,v+1,\boldsymbol{e}+2\boldsymbol{f}}\widehat{A}^-(t),\\
C_0(t)&=\mathcal{C}_{\theta_0,v-1,\boldsymbol{f}-\boldsymbol{e}}C^-(t),\quad C_1(t)=\mathcal{C}_{\theta_1,v-1,\boldsymbol{f}-\boldsymbol{e}}\widehat{C}^-(t).
\end{aligned} \tag{328}$$

The single-variable vector in the first line is the integer 2. Duplication of those step-two Pochhammers turns $Z(t)$ into an ordinary ${}_{19}F_{18}(1)$. All other blocks have only two indices. The complete fixed sample is

$$\boxed{\begin{aligned}\Xi_7(t)=D_q(t)^7\big[&Z(t)+P_0(t)+P_1(t)\\&+Z(1-q-t)+P_0(1-q-t)+P_1(1-q-t)+C_0(t)+C_1(t)\big].\end{aligned}} \tag{329}$$

Equations (312)–(329) specify every parameter, argument and prefactor using finite lists; the JSON templates in `lri_general_periodization_update/verification/general_periodization/arrays/` are machine-readable copies, not missing definitions.

Convergence and continuation. The proofs above start at $1/2 < q < 1$. The right and middle Gamma-ratio estimates give direct absolute convergence of the displayed blocks for $q > 1/(2r)$ in a wheel and $q > 1/10$ in the primitive, always with $q < 1$. The endpoint in a positive-spin right block imposes the additional condition $q(4-r) < 1$ for a wheel and $q < 1$ for the primitive; these are automatic in the stated ranges. Thus the new Q, P_4, P_5 formulas are direct endpoint-series formulas for $d > 2/3, 1/2, 2/5$, respectively. For smaller positive dimensions use their continued complete combinations. Exceptional real parameter coincidences are dealt with by the same continuation; the off-real sampling points themselves are nonsingular. In particular $d = 1, 2, 3$ are valid specializations, though the retained five ordinary values remain a better presentation for $P_5(1)$.

18.5 A stronger result at two dimensions: reuse the one-dimensional values

In this subsection B_-, B_+ and X_0, X_+, X_- refer to the *retained one-dimensional* values, not to new functions of d:

$$\begin{aligned} &B_- = \Gamma(\tfrac14)^4 F_-, \quad B_+ = \frac{64\pi^4}{\Gamma(\frac14)^4} F_+, \\ &X_0 = \mathcal{Y}_2, \qquad\qquad X_+ = c_8^7 \mathcal{Y}_3 - s_8^7 \mathcal{Y}_5, \quad X_- = c_8^7 \mathcal{Y}_1 + s_8^7 \mathcal{Y}_7. \end{aligned} \tag{330}$$

For a self-contained dimension-two parameter dictionary, these are

$$\begin{aligned} &F_- = {}_4F_3\left(\begin{matrix}\frac14,\frac14,\frac12,\frac12\\ \frac34,\frac34,1\end{matrix};1\right), \quad F_+ = {}_4F_3\left(\begin{matrix}\frac34,\frac34,\frac12,\frac12\\ \frac54,\frac54,1\end{matrix};1\right), \\ &c_8 = \cos(\pi/8), \qquad\qquad s_8 = \sin(\pi/8), \end{aligned} \tag{331}$$

$$\begin{aligned} &\mathcal{Y}_r = \frac{r}{8}\left[\frac{\Gamma((r+2)/8)}{\Gamma((r+6)/8)}\right]^7 {}_9F_8\left(\begin{matrix}1, 1+\frac{r}{8}, ((r+2)/8)^{\times 7}\\ \frac{r}{8}, ((r+6)/8)^{\times 7}\end{matrix};-1\right), \\ &r \in \{1,2,3,5,7\}. \end{aligned} \tag{332}$$

At $r = 2$ one upper and lower parameter 1 cancel, giving an ordinary ${}_8F_7$; all parameter lists are fixed and positive rational. The new identities are

$$\boxed{P_4(2) = \frac{B_-^2 - B_+^2}{4\pi^3}.} \tag{333}$$

$$\boxed{\begin{aligned} P_5(2) = \frac{8}{\pi}\big[&25X_0^2 - 28\sqrt{2}\,X_0(X_+ + X_-) \\ &+ 18(X_+^2 + X_-^2) + 28X_+X_-\big]. \end{aligned}} \tag{334}$$

Thus no new hypergeometric value is needed for either two-dimensional primitive. The four-loop result uses the same two ${}_4F_3(1)$ values; the five-loop result uses exactly the preferred four ${}_9F_8(-1)$ and one ${}_8F_7(-1)$ with positive rational parameters. The formulas are quadratic rather than linear. By themselves they do not evaluate the two-dimensional nonprimitive sector; that evaluation is completed in Section 17.2, with the proofs in Sections 18.7 and 18.8.

18.5.1 Factorization of the wheel lattice

At $q = 1/2$ the multiplicities are $d_0 = 1$, $d_{\ell>0} = 2$. In a right half-lattice put $k = \ell + n$, $j = n$. Gamma reflection gives

$$g_\ell(a_\ell + n + t) = -\cot(\pi t)\,\frac{\Gamma(k+\frac12+t)}{\Gamma(k+1+t)}\,\frac{\Gamma(j+\frac12+t)}{\Gamma(j+1+t)}.$$

The diagonal has multiplicity one and the two off-diagonal half-lattices have multiplicity two. They therefore form a full square lattice. Define the meromorphic bilateral series and its regularization

$$\begin{aligned} B_r(t) &= \sum_{n\in\mathbb{Z}} \left[\frac{\Gamma(n+\frac12+t)}{\Gamma(n+1+t)}\right]^r, \\ \Omega_r(t) &= \cos^r(\pi t) B_r(t). \end{aligned} \tag{335}$$

The index sums are absolute for $r > 2$. Direct reflection or the square lattice just described gives

$$\Xi_r(t) = (-1)^r \Omega_r(t)^2 \qquad (d = 2). \tag{336}$$

For $r = 4$, the growth and reflection argument behind (311) now yields only two modes:

$$\Omega_4(t) = a + b\cos\bigl(2\pi(t+\tfrac14)\bigr). \tag{337}$$

There is no fitting in this statement: Ω_4 is entire, one-periodic, even about $-1/4$, and has growth $O(e^{4\pi|\Im t|}|\Im t|^{-1})$, which excludes the boundary second mode.

Let $M_{\mathrm{e}} = \int_0^1 \mathbf{K}(k)^2\,dk$ and $N_{\mathrm{e}} = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')\,dk$, with the modulus convention of Section 9.2. The established elliptic identities give

$$M_{\mathrm{e}} = \frac{B_- + B_+}{64}, \qquad N_{\mathrm{e}} = \frac{B_- - B_+}{32}. \tag{338}$$

Here is a short analytic determination of both coefficients in (337). At $t = 0$ the reflected half of Ω_4 vanishes to fourth order. Hence

$$a = \pi^2\, {}_4F_3\left(\begin{matrix}\frac12,\frac12,\frac12,\frac12\\ 1,1,1\end{matrix};1\right) = \frac{8N_{\mathrm{e}}}{\pi}.$$

For $N(\nu) = \int_0^1 k^\nu \mathbf{K}(k)\mathbf{K}(k')\,dk$, the mixed moment beta integral of [25] gives

$$N(\nu) = \frac{\pi^2}{8}\sum_{n\ge0} \frac{(\frac12)_n^2}{(n!)^2}\frac{\Gamma(n+(\nu+1)/2)^2}{\Gamma(n+(\nu+2)/2)^2}. \tag{339}$$

Differentiation in this convergent *proof* representation gives $\Omega_4'(0) = 32N'(0)/\pi$. The already proved Stieltjes–Landen identity (101) says $N'(0) = -\pi M_{\mathrm{e}}/2$. As $\Omega_4'(0) = -2\pi b$, it follows that

$$a = \frac{8N_{\mathrm{e}}}{\pi}, \qquad b = \frac{8M_{\mathrm{e}}}{\pi}. \tag{340}$$

No derivative remains in these coefficients. Substitute $\Xi_4 = (a + b\cos(2\pi(t+1/4)))^2$ into (318). At $q = 1/2$ the only contributing coefficient is $c_{4,1} = 2ab$, and $P_4(2) = 8ab/\pi = 512M_{\mathrm{e}}N_{\mathrm{e}}/\pi^3$. Equation (333) follows immediately.

18.5.2 A second determination of b, without the Stieltjes identity

The value $a = 8N_{\mathrm{e}}/\pi$ above uses only Euler integration. The value of b above rests on the mixed-moment series (339) and on (101), whose proof uses the Stieltjes identity (100) of [25]. The following derivation uses neither. It was written by the Claude Code session that wrote the third and fourth proofs of (270) in Section 18.7 (package `LRI_C5_d2_simplify/`, folder `proof/`).

Put $c_n = (\frac12)_n/n!$, so that $\Gamma(n+\frac12)/\Gamma(n+1) = \sqrt{\pi}\,c_n$. At $t = 0$ the reflected half of Ω_4 vanishes to fourth order and $\cos^4\pi t$ has zero derivative, so

$$-2\pi b = \Omega_4'(0) = 4\pi^2 \sum_{n\ge0} c_n^4\bigl[\psi_0(n+\tfrac12) - \psi_0(n+1)\bigr].$$

Item (ii) of the pairing proposition (368) below evaluates this single sum by Parseval's identity on the unit circle. It uses only the generating function (366) (first line) and the circle values (367), and gives $b = 4V_0/\pi$, with $V_0 = \int_0^1 \mathbf{K}(k)\mathbf{K}(k')\,dk/k'$ as defined before (96). The imaginary part of the upper-semicircle contour that proves (106) gives $V_0 = 2M_{\mathrm{e}}$; that contour uses the boundary-value Landen relation and not (100). Hence $b = 8M_{\mathrm{e}}/\pi$, which is (340). The Fourier representation of Section 18.7.9 gives the same value independently, $b = v_1(\frac{1}{2})/(4\pi) = 4V_0/\pi$. For a the same two routes give $a = 4U_0/\pi$, and $U_0 = 2N_{\mathrm{e}}$ is proved without (100) in Section 18.7.7. Since (338) uses only $V_0 = 2M_{\mathrm{e}}$ and the real Landen substitution (the reduction to the two ${}_4F_3$ values after (107)), (333) is proved without the Stieltjes identity and without the mixed-moment series. As a check, not a proof step, the 70-digit single sums give $8ab/\pi - (B_-^2 - B_+^2)/(4\pi^3) = -1.1 \times 10^{-67}$ (`LRI_C5_d2_simplify/proof/synth/synth_checks.log`).

18.5.3 The five-loop square and the retained five values

Put

$$\begin{aligned} G(s) &= (2s + \tfrac{1}{2}) \left[\frac{\Gamma(s + \frac{1}{2})}{\Gamma(s + 1)}\right]^7, \\ \Omega_{\mathrm{a}}(t) &= \cos^7(\pi t) \sum_{n\in\mathbb{Z}} (-1)^n G(n + t). \end{aligned} \tag{341}$$

At $q = 1/2$ the measure in (303) is $\ell^2 - 4s^2$. On the square lattice it becomes $-(2k+\frac{1}{2}+2t)(2j+\frac{1}{2}+2t)$ and $(-1)^\ell = (-1)^{k-j}$. Consequently

$$\Xi_7(t) = \Omega_{\mathrm{a}}(t)^2 \qquad (d = 2). \tag{342}$$

The sign is plus: the negative quadratic measure and the odd seventh power of the Gamma-reflection sign cancel.

Most usefully, $G(s) = 2f(s)$ for the *one-dimensional* function in (134). Thus

$$\Omega_{\mathrm{a}}(t) = 2\Xi^{(1)}(t) = 2\big[\alpha \cos 2\pi t + \beta \sin 4\pi t + \gamma \cos 6\pi t\big], \tag{343}$$

where α, β, γ are exactly those in (144). Applying the finite residue polynomial gives

$$P_5(2) = \frac{8}{\pi}\left(\alpha^2 - 14\alpha\gamma + 8\beta^2 + 113\gamma^2\right). \tag{344}$$

One convenient exact algebra check is that at $q = 1/2$

$$(B_{7,0}, B_{7,1}, \ldots, B_{7,6}) = \pi^6(20, 0, 28, 0, 84, 0, 924).$$

Expanding the square in (343) and using the normalization $P_5 = \mathcal{E}_7/2$ gives (344); inserting (144) gives (334). This proves the quadratic factorization without a numerical integer-relation search.

18.6 Numerical checks of the finite periodization

These checks were run by OpenAI GPT-6 astra in the calculation that introduced the periodization (package `lri_general_periodization_update/`; scripts in `code/general_periodization/`, records `wheel_periodization.json`, `horn_template_checks.json`, `p5_periodization.json`, `d2_factorization.json`, `d2_control.json`, `d2_inherited_periods.json` and `exact_algebra.json` in `verification/general_periodization/`). They are numerical consistency checks, not interval bounds.

Function counts. The finite evaluation (319) with the arrays of Section 18.4 needs far fewer special-function values than the unweighted Horn arrays obtained by applying the rules of Section 17.6 to the wheel series (277) and to the derivative-free double-residue series (281) of $P_5(d)$:

	Horn values	variables	periodization values, at most	variables
Q_d	20	4	15	2
$P_4(d)$	56	5	20	2
$P_5(d)$	15060	8	56	2

For $P_5(d)$ the Horn count is 6024 for the first channel of (281) and 9036 for the second. The sampling counts are the five blocks of (323) at each of the r nodes of (312) ($r = 3, 4$) and the eight blocks of (329) at the seven nodes for $r = 7$. The counts include the one-variable subblocks, and taking the real part of a complex value counts as one evaluation. They are upper bounds and assert neither minimality nor independence. The gain is the elimination of the auxiliary indices that encoded residue polygammas, not a proof that the multivariable functions reduce to ordinary ones.

Fixed-spin controls. At 35 working digits, fixed-spin wheel samples were evaluated as ordinary hypergeometric functions and compared with real-axis Barnes integrals. The cases $(q, r, \ell) = (0.6, 3, 0)$, $(0.7, 3, 2)$, $(0.6, 4, 1)$ and $(0.25, 4, 0)$ differ by less than 3×10^{-30}, and further off-grid samples check the finite Fourier polynomial (311). These test building blocks of the theorem, not a rederived graph sum.

Parameter arrays and finite transforms. At 65 working digits, 44 coefficients of the Horn blocks, including the signed-index middle blocks, were compared with the Gamma kernels at $q = 1/4, 1/2, 0.63, 3/4$; the largest normalized residual is 3.81×10^{-65}. The cosine transforms (313) were checked on every basis mode for $r = 3, 4, 7$, with residuals below 4.75×10^{-66}. The exact derivation of the arrays is (305)–(306) and the finite row rules; these are coefficient checks.

Noninteger dimensions. The spin-summed finite-sampling expression for $P_5(d)$ was compared with spectral quadrature of (216), with 640 nodes and 160 spin terms; the sampling computation used 128 spin terms, Euler acceleration and doubled radial cutoffs. The difference is -1.39×10^{-9} at $d = 2.4$ and $+7.38 \times 10^{-8}$ at $d = 3$, and the off-grid periodization residuals are below 2.3×10^{-11}. These runs use double-precision Gamma kernels inside wider accumulations, so their residuals are not promoted to high precision because the finite weights were computed at 50 digits. The five-loop residue functional is sensitive to cancellation. Imaginary-axis Chebyshev nodes far from the real axis are particularly ill conditioned; the grid (312) controls this amplification much better, but it is not an error certificate.

Two dimensions. The square factorizations of Section 18.5 were evaluated first at 55 digits, using Gamma-ratio series with a Hurwitz-zeta asymptotic tail and a finite Euler transform for the alternating kernel. A 70-digit rerun changed the direct cutoff from 96 to 144, the tail order from 44 to 58 and the Euler settings from $(20, 260)$ to $(32, 340)$, and used the grid (312). The Q_2 value agrees with $\Gamma(1/4)^8/(24\pi^4)$ to about 67 decimal places, and $P_4(2)$ and $P_5(2)$ are the values quoted in Section 17.2. The retained values $F_\pm$ and the five $\mathcal{Y}_r$ were then recomputed with literal ordinary hypergeometric calls at 60 working digits. The formulas (333) and (334) differ from the 70-digit square-factorization values by -5.97×10^{-59} and $+1.59 \times 10^{-58}$. The quadratic change of basis from (344) to (334) was checked by exact symbolic algebra. No stored period value was an input to either route.

What the finite Fourier theorem does not give. The theorem (311) concerns the Barnes-shift variable t. It does not imply finite dependence on an independent rim exponent b, so it cannot replace the first and second b derivatives that define $\mathcal{L}_d$ and Z_d in (244) by a finite sampling grid in b; that would be an unjustified interpolation. Sections 20 and 21 treat the rim exponent by a separate periodization (Section 20.1) and keep low-order parameter derivatives.

18.7 Proof of the closed form of $\mathcal{L}_2$

We prove (270) for the series (256). The rim exponent of the wheel lattice of Section 18.5 is deformed away from one, the two-mode statement (337) is proved for every nearby rim exponent, and the two derivative sums that appear are evaluated with the elliptic lemmas of Section 9.2 and one further contour integral. This is the one-dimensional contour proof of Section 9.2, transferred to the square lattice. Throughout, $d = 2$, $q = \frac{1}{2}$, $c_k = (\frac{1}{2})_k/k!$ (not the constants c_j of (274)), and

$$\alpha_k = \psi_0(k+\tfrac{1}{2}) - \psi_0(k+1), \qquad \beta_k = \psi_0(k+\tfrac{1}{2}) - \psi_0(\tfrac{1}{2}),$$

that is, $\alpha_k = \Delta_0(\frac{1}{2}, 1; k)$ and $\beta_k = H(\frac{1}{2}; k)$ in the notation of (274). The letters a, b denote the coefficients of (337). The functions f, g, the weight w and the moments U_0, R are those of Section 9.2, whose M, N are $M_{\mathrm{e}}, N_{\mathrm{e}}$ of (338).

The first proof (Sections 18.7.1–18.7.5) and the second (Section 18.7.6) were written by a Claude Code session with prover and reviewer subagents (package `LRI_C5_d2_proof/`, folders `L2_singlesums/` and `L2_structural/`). Another Claude Code session, with prover, verifier and synthesizer subagents, wrote the half-strip evaluation of Section 18.7.7, the third and fourth proofs (Sections 18.7.8 and 18.7.9) and the remark on the integer-relation form of the collapse inside the first proof (package `LRI_C5_d2_simplify/`, folder `proof/`; the third proof is in its subfolder `L2-elliptic/`, the half-strip evaluation and the fourth proof in `L2-fourier/`). That session had first found (270) and the relations (356) numerically, in the top level of the same package.

18.7.1 The series as a rim derivative

Let $I = [\frac{2}{5}, \frac{3}{5}]$. For $x \in I$ and $m, n, k \geq 0$ put

$$\varphi_x(s) = \frac{\Gamma(s+x)}{\Gamma(s+1)}, \qquad N_x = \frac{\Gamma(1-x)}{\Gamma(x)}, \qquad c_k(x) = \frac{(x)_k}{k!}, \qquad T_{mn}(x) = c_m(x)c_n(x),$$
$$\lambda_j(x) = \Delta_{j-1}(x, 1; m) + \Delta_{j-1}(x, 1; n) + \mathtt{c}_j(x), \qquad \sum_{j\geq 1} \mathtt{c}_j(x)\frac{u^j}{j!} = \log\frac{\pi u}{\sin \pi u} + \log\frac{\sin \pi(x+u)}{\sin \pi x}, \tag{345}$$

so that $\mathtt{c}_1 = \pi\cot \pi x$, $\mathtt{c}_2 = \frac{\pi^2}{3} - \frac{\pi^2}{\sin^2 \pi x}$ and $\mathtt{c}_3 = \frac{2\pi^3 \cos \pi x}{\sin^3 \pi x}$; at $x = \frac{1}{2}$ these are the constants c_j of (274). Let $F_4(x)$ be the polynomial F_4 of (254) evaluated at these $\lambda_j(x)$.

One summand as a residue. Gamma reflection [26] gives $N_x \sin(\pi x)\varphi_x(m)\varphi_x(n)/\pi = T_{mn}(x)$, and $\partial_u^j \log \varphi_x(k+u)|_{u=0} = \Delta_{j-1}(x, 1; k)$. Hence, as Laurent expansions at $u = 0$,

$$N_x \frac{\sin \pi(x+u)}{\sin \pi u}\, \varphi_x(m+u)\varphi_x(n+u) = \frac{T_{mn}(x)}{u} \exp\Big(\sum_{j\geq 1} \lambda_j(x)\frac{u^j}{j!}\Big). \tag{346}$$

Since F_4 is the complete Bell coefficient $[u^3]\exp(4\sum_j \lambda_j u^j/j!)$, the residue at $u = 0$ of the fourth power of the left side of (346) is $T_{mn}^4 F_4$. Equation (346) is the two-dimensional form of (262) for rim exponent $2x$.

Identification at $x=\frac12$. Put $m=\ell+n$. Then $T_{mn}(\frac12)=T_{\ell n}$ of (251), and the $\lambda_j(\frac12)$, $j\le 3$, are the λ_j of (252), since $\mathsf{c}_1(\frac12)=\mathsf{c}_3(\frac12)=0$ and $\mathsf{c}_2(\frac12)=-\frac{2\pi^2}{3}=2[\psi_1(1)-\psi_1(\frac12)]$. Moreover $\frac12\partial_x\log T_{mn}|_{x=1/2}=\mu_1$ and $\frac12\partial_x\lambda_j|_{x=1/2}=\eta_j$ of (253), because $\partial_x\Delta_{j-1}(x,1;k)=\psi_j(k+x)$ and $\frac12\mathsf{c}_j'(\frac12)=(-\frac{\pi^2}{2},0,-\pi^4)=\frac12((-1)^j-1)\psi_j(\frac12)$. As $F_{4,1}=\sum_j(\partial F_4/\partial\lambda_j)\,\eta_j$, the summand of (256) is

$$T_{\ell n}^4(4\mu_1F_4+F_{4,1})=\tfrac12\,\partial_x\big[T_{mn}(x)^4F_4(x)\big]_{x=1/2}.$$

These identifications are exact algebra in independent polygamma symbols (machine checked in `LRI_C5_d2_proof/L2_singlesums/verify_symbolic.py`, §§1–3 and 5).

The square lattice. For $x\in I$ and $k\ge0$, Stirling's formula [27] gives $0<c_k(x)\le C(k+1)^{x-1}$. The quantities $\Delta_{j-1}(x,1;k)$, $\psi_j(k+x)$ ($j\ge1$), c_j and c_j' are bounded, and $|\psi_0(k+x)-\psi_0(x)|\le C\log(k+2)$. Hence $T_{mn}^4F_4$ and its x-derivative are bounded by $Cc_m(x)^4c_n(x)^4\log(m+2)\log(n+2)$, which is summable over $\mathbb{N}_0^2$ uniformly on I because $4(x-1)\le-\frac85$. The summands are symmetric in m,n, and at $d=2$ the multiplicities of (240) are $d_0=1$ and $d_{\ell\ge1}=2$. Therefore $\sum_\ell d_\ell\sum_n y_{\ell+n,n}=\sum_{m,n\ge0}y_{mn}$ for every symmetric absolutely summable array (y_{mn}). Define

$$\mathcal{W}(x)=-\sum_{m,n\ge0}T_{mn}(x)^4F_4(x),\qquad x\in I.$$

By (255), $\mathcal{W}(\frac12)=P_4(2)$. By (256) and termwise differentiation (Weierstrass M-test),

$$\mathcal{L}_2=\tfrac12\,\mathcal{W}'(\tfrac12).\tag{347}$$

The function $\mathcal{W}(x)$ equals the wheel $W_4(2;2x)$ of (242) (see the Status paragraph of Section 17.2); the proof below does not use this.

Residue form. Put $\mathcal{H}_x(u)=\sum_{k\ge0}\varphi_x(k+u)^4$. On the circle $|u|=\frac{1}{10}$ we have $|\varphi_x(k+u)|\le C(k+1)^{x-1}$ uniformly, so the double series of fourth powers of the left side of (346) converges uniformly there, to $N_x^4\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2/\sin^4\pi u$. Integrating term by term over the circle,

$$\mathcal{W}(x)=-N_x^4\,\mathrm{Res}_{u=0}\,\frac{\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2}{\sin^4\pi u}.\tag{348}$$

18.7.2 Two Fourier modes for every rim exponent

For $x\in I$ define

$$\Omega(t;x)=\sin^4\pi(t+x)\,\mathcal{H}_x(t)+\sin^4(\pi t)\sum_{j\ge0}\Big[\frac{\Gamma(j+1-t)}{\Gamma(j+2-x-t)}\Big]^4.$$

At $x=\frac12$ this is $\Omega_4(t)$ of (335). We claim

$$N_x^2\,\Omega(t;x)=\mathcal{A}(x)+\mathcal{B}(x)\cos2\pi(t+\tfrac{x}{2}),\qquad \mathcal{A}(\tfrac12)=a,\quad \mathcal{B}(\tfrac12)=b,\tag{349}$$

with $\mathcal{A},\mathcal{B}$ real analytic on the interior of I.

Gamma reflection gives, for $j\ge0$ and $m\in\mathbb{Z}$,

$$\sin^4(\pi t)\Big[\frac{\Gamma(j+1-t)}{\Gamma(j+2-x-t)}\Big]^4=\sin^4\pi(t+x)\,\varphi_x(t-j-1)^4,$$
$$\sin^4(\pi t)\,\varphi_x(-m-x-t)^4=\sin^4\pi(t+x)\,\varphi_x(m+t)^4.$$

By the first identity, $\Omega(t;x)=\sin^4\pi(t+x)\sum_{n\in\mathbb{Z}}\varphi_x(n+t)^4$ away from the poles, which is one-periodic. By the second, with $n\mapsto-n$, $\Omega(-x-t;x)=\Omega(t;x)$. The two series in the definition

converge locally uniformly away from $t \in -x - \mathbb{N}_0$ and $t \in 1 + \mathbb{N}_0$, respectively. Their poles there have order at most four and are canceled by the zeros of $\sin^4 \pi(t+x)$ and $\sin^4 \pi t$, so Ω is entire. For $\Re z \geq 0$, $|z| \geq 1$ and $x \in I$, Stirling's formula with its error bound in the closed right half-plane [27] gives, uniformly in x,

$$\Big|\frac{\Gamma(z+x)}{\Gamma(z+1)}\Big| \leq C|z|^{x-1}, \qquad \Big|\frac{\Gamma(z)}{\Gamma(z+1-x)}\Big| \leq C|z|^{x-1}.$$

With $z = n + t$ and $z = j + 1 - t$, for $t = \sigma + \mathrm{i}y$, $0 \leq \sigma \leq 1$, $|y| \geq 1$, this gives $|\Omega(t;x)| \leq Ce^{4\pi|y|} \sum_{n\geq 0}(n^2+y^2)^{2x-2} \leq C' e^{4\pi|y|}|y|^{4x-3}$, and $4x - 3 < 0$. The argument of (311) now applies: shifting the contour of the k-th Fourier coefficient to $\Im t = \mp Y$ removes every mode with $|k| \geq 2$, including $|k| = 2$ because of the decaying factor $|y|^{4x-3}$. The reflection $\Omega(-x-t) = \Omega(t)$ gives $\omega_1 = e^{2\pi \mathrm{i} x}\omega_{-1}$ for the two remaining nonconstant coefficients, which yields the form (349), and $\mathcal{A}, \mathcal{B}$ are real because Ω is real on the real axis. Finally $\mathcal{A}(x) = N_x^2 \int_0^1 \Omega(\sigma + \mathrm{i}; x)\, d\sigma$ and $\mathcal{B}(x) = 2e^{\mathrm{i}\pi x} N_x^2 \int_0^1 \Omega(\sigma+\mathrm{i};x) e^{2\pi\mathrm{i}(\sigma+\mathrm{i})} d\sigma$. On $\Im t = 1$ both series converge locally uniformly for complex x near I, so $\mathcal{A}$ and $\mathcal{B}$ are real analytic. At $x = \frac{1}{2}$, $N_{1/2} = 1$ and (349) is (337).

18.7.3 The residue functional and an exact reduction

By the definition of Ω, $\Omega(u;x) - \sin^4 \pi(x+u)\mathcal{H}_x(u)$ equals $\sin^4(\pi u)$ times a function analytic for $|u| < 1$, and $\sin \pi(x+u) \neq 0$ near $u = 0$. In (348) the factor $N_x^2 \sin^4 \pi(x+u)\mathcal{H}_x(u)$ may therefore be replaced by (349), since the difference contributes only terms regular at $u = 0$. Hence

$$\begin{gathered}\mathcal{W}(x) = -\big[\rho_0 \mathcal{A}^2 + 2\rho_1 \mathcal{A}\mathcal{B} + \rho_2 \mathcal{B}^2\big], \qquad \rho_p(x) = \mathrm{Res}_{u=0} \frac{\cos^p(2\pi u + \pi x)}{\sin^4 \pi u \, \sin^4 \pi(u+x)}, \\ \rho_0 = -\frac{4(\cos 2\pi x + 4)\cos \pi x}{\pi \sin^7 \pi x}, \qquad \rho_1 = \frac{4(4\sin^2 \pi x - 5)}{\pi \sin^7 \pi x}, \qquad \rho_2 = \frac{4(\sin^2 \pi x - 5)\cos \pi x}{\pi \sin^7 \pi x}.\end{gathered} \tag{350}$$

At $x = \frac{1}{2}$ one has $\rho_0 = \rho_2 = 0$, $\rho_1 = -\frac{4}{\pi}$, $\rho_0' = 12$, $\rho_1' = 0$ and $\rho_2' = 16$. So $\mathcal{W}(\frac{1}{2}) = 8ab/\pi$, the value of $P_4(2)$ found after (340), and by (347)

$$\mathcal{L}_2 = -6a^2 - 8b^2 + \frac{4}{\pi}\big[\mathcal{A}'(\tfrac{1}{2})\, b + a\, \mathcal{B}'(\tfrac{1}{2})\big].$$

At $t = 0$ the second series in Ω carries the factor $\sin^4 \pi t$. Evaluating (349) and its t-derivative at $t = 0$ gives, for $x \in I$,

$$\begin{gathered}N_x^2 \sin^4(\pi x)\, \mathcal{H}_x(0) = \mathcal{A} + \mathcal{B}\cos \pi x, \\ N_x^2\big[4\pi \sin^3 \pi x \cos \pi x\, \mathcal{H}_x(0) + \sin^4 \pi x\, \partial_u \mathcal{H}_x(0)\big] = -2\pi \mathcal{B} \sin \pi x.\end{gathered}$$

The series $\mathcal{H}$ is jointly analytic in (u,x) near $(0, \frac{1}{2})$ and may be differentiated termwise. At $x = \frac{1}{2}$ we have $N_{1/2} = 1$, $(N_x^2)' = -4\psi_0(\frac{1}{2})$, $\varphi_{1/2}(k) = \sqrt{\pi}\, c_k$ and

$$\begin{aligned}\mathcal{H} &= \pi^2 \sum_k c_k^4, & \partial_u \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \alpha_k, \\ \partial_x \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \psi_0(k+\tfrac{1}{2}), & \partial_x \partial_u \mathcal{H} &= 4\pi^2 \sum_k c_k^4 \big[4\psi_0(k+\tfrac{1}{2})\alpha_k + \psi_1(k+\tfrac{1}{2})\big].\end{aligned}$$

The two relations and their x-derivatives at $\frac{1}{2}$ then give $a = \pi^2 \sum_k c_k^4$ and $b = -2\pi \sum_k c_k^4 \alpha_k$; the terms in $\psi_0(\frac{1}{2})$ cancel, and

$$\begin{gathered}\mathcal{A}'(\tfrac{1}{2}) = \pi b + 4\pi^2 \mathcal{S}_1, \qquad \mathcal{B}'(\tfrac{1}{2}) = 2\pi a - 2\pi \mathcal{S}_2, \\ \mathcal{S}_1 = \sum_{k\geq 0} c_k^4 \beta_k, \qquad \mathcal{S}_2 = \sum_{k \geq 0} c_k^4 \big[4\alpha_k \beta_k + \psi_1(k+\tfrac{1}{2})\big].\end{gathered} \tag{351}$$

All four sums converge absolutely, since $c_k^4 \sim (\pi k)^{-2}$. Consequently

$$\boxed{\mathcal{L}_2 = 2a^2 - 4b^2 + 16\pi b\,\mathcal{S}_1 - 8a\,\mathcal{S}_2, \qquad P_4(2) = \frac{8ab}{\pi}.} \tag{352}$$

This is an exact identity; it uses no relation among the single sums.

Remark (the integer-relation form of the collapse). The integer-relation search that first found (270) returned, in the present normalization, $\mathcal{L}_2 = 3a^2 - 2b^2 + 8\pi b\,\mathcal{S}_1 - 12a\,\mathcal{S}_2$. It differs from (352) by $a^2 + 2b^2 - 8\pi b\,\mathcal{S}_1 - 4a\,\mathcal{S}_2$, which vanishes only by the single-sum relations (356) below. The four numbers a^2, b^2, $\pi b\,\mathcal{S}_1$ and $a\,\mathcal{S}_2$ therefore satisfy a $\mathbb{Z}$-linear relation, and the searched form is one member of a one-parameter family of representations. Only (352) follows from the structure without those relations, and it is the form used here; both give the same value. The algebra is checked exactly by `LRI_C5_d2_simplify/proof/synth/synth_checks.py`.

18.7.4 Generating functions and elliptic moments

For $0 < k < 1$,

$$\begin{aligned}
\sum_{n\ge 0} c_n^2 \alpha_n k^{2n} &= -g(k) - \frac{2}{\pi} f(k) \log k, \\
\sum_{n\ge 0} c_n^2 \beta_n k^{2n} &= -\frac{2}{\pi} f(k) \log k', \\
\sum_{n\ge 0} c_n^2 \bigl[4\alpha_n\beta_n + 2\psi_1(n+\tfrac12)\bigr] k^{2n} &= 4g(k)\log k' + 2\pi f(k) + \frac{8}{\pi} f(k) \log k \, \log k'.
\end{aligned} \tag{353}$$

To prove these, put $z = k^2$, $\Phi(x,z) = {}_2F_1(x,x;1;z)$ and

$$\Lambda_x(z) = \sum_{n\ge 0} \frac{(x)_n^2}{n!^2} \bigl[2\psi_0(n+1) - 2\psi_0(n+x) - \log z\bigr] z^n = \frac{\Gamma(x)^2}{\Gamma(2x)}\, {}_2F_1(x,x;2x;1-z),$$

the logarithmic case $m = 0$ of the connection formula 15.8.10 of [23]. At $x = \frac12$ it gives $\Lambda_{1/2}(z) = \pi\, {}_2F_1(\frac12,\frac12;1;1-z) = 2g(k)$, which is the first line, since $\sum_n c_n^2 k^{2n} = \frac{2}{\pi} f(k)$. Euler's transformation $\Phi(x,z) = (1-z)^{1-2x}\Phi(1-x,z)$ (15.8.1 of [23]), differentiated at $x = \frac12$, gives $\partial_x \Phi(\frac12, z) = -\log(1-z)\,\Phi(\frac12,z)$. Termwise, $\partial_x \Phi(\frac12,z) = 2\sum_n c_n^2 \beta_n z^n$, which is the second line. The connection formula 15.8.4 of [23] for the parameters $(x,x;1)$, with its Gamma coefficients simplified by reflection, reads

$$\Lambda_x(z) = (1-z)^{1-2x}\Lambda_{1-x}(z) + 2\pi\cot(\pi x)\,\Phi(x,z), \qquad 0 < x < 1,\ x \ne \tfrac12.$$

For fixed $|z| < 1$ the series converge locally uniformly in x, so both sides are analytic on $0 < x < 1$ and the identity also holds at $x = \frac12$. Its x-derivative there is $\partial_x\Lambda|_{1/2} = -\log(1-z)\,\Lambda_{1/2}(z) - \pi^2\Phi(\frac12,z)$. Termwise, $\partial_x\Lambda|_{1/2} = -\sum_n c_n^2[4\alpha_n\beta_n + 2\psi_1(n+\frac12)]z^n - 2\log z \sum_n c_n^2\beta_n z^n$, and with the second line this is the third.

Write $\mathrm{FG}[\phi] = \int_0^1 f(k)g(k)\phi(k)\,dk$ and $\mathrm{GG}[\phi] = \int_0^1 g(k)^2\phi(k)\,dk$. Section 9.2 gives the following values: $\mathrm{GG}[1] = 2M_{\mathrm{e}}$ by (97); $\mathrm{FG}[w] = -\log 2\, N_{\mathrm{e}}$ and $\mathrm{GG}[w] = -2R - 4\log 2\, M_{\mathrm{e}}$ by (98); $\mathrm{FG}[\log k] = -\frac{\pi}{2} M_{\mathrm{e}}$ by (101); $R = \frac{\pi}{4} N_{\mathrm{e}} - \log 2\, M_{\mathrm{e}}$ by (105); and $U_0 = 2N_{\mathrm{e}}$. One further input is needed:

$$\mathrm{GG}[\log k] = \int_0^1 \mathbf{K}(k')^2 \log k\, dk = -\pi N_{\mathrm{e}}. \tag{354}$$

It is the real part of the first-quadrant identity (104) with $W(z)$ replaced by $\mathrm{Log}\, z$. The arc estimates are unchanged: the integrand is $O(\log^3|z|/|z|^2)$ at infinity and has integrable logarithmic singularities at 0 and 1. After $x = 1/k$ on $(1,\infty)$ the boundary value $\mathrm{Log}(x + \mathrm{i}0) = -\log k$

multiplies $(f+\mathrm{i}g)^2$, so the two contributions $\pm\int_0^1 f^2\log k\,dk$ cancel and the real part of the left side is $\mathrm{GG}[\log k]$. On the imaginary axis $\mathrm{Log}(\mathrm{i}y)=\log y+\frac{\mathrm{i}\pi}{2}$, so after $y=k/k'$ the real part of the right side is $-\frac{\pi}{2}U_0=-\pi N_\mathrm{e}$. Since $\log k'=\frac12(\log k-w)$, it follows that

$$\mathrm{FG}[\log k']=-\frac{\pi}{4}M_\mathrm{e}+\frac{\log 2}{2}N_\mathrm{e},\qquad \mathrm{GG}[\log k']=\log 2\,M_\mathrm{e}-\frac{\pi}{4}N_\mathrm{e}. \tag{355}$$

18.7.5 The single sums and the conclusion

The complementary moment of Section 9, $\int_0^1 k^r g(k)\,dk=\frac{\pi}{4}\Gamma(\frac{r+1}{2})^2/\Gamma(\frac{r+2}{2})^2$ for $r>-1$, and its r-derivative at $r=2n$ (dominated convergence) give

$$c_n^2=\frac{4}{\pi^2}\int_0^1 k^{2n}g(k)\,dk,\qquad c_n^2\alpha_n=\frac{4}{\pi^2}\int_0^1 k^{2n}\log k\;g(k)\,dk.$$

In each single sum write one factor c_n^2, or $c_n^2\alpha_n$, in this form, and sum the remaining factor with (353). For $\mathcal{S}_2$ use

$$c_n^4\big[4\alpha_n\beta_n+\psi_1(n+\tfrac12)\big]=\tfrac12\big\{c_n^2\big[4\alpha_n\beta_n+2\psi_1(n+\tfrac12)\big]\cdot c_n^2+2c_n^2\alpha_n\cdot 2c_n^2\beta_n\big\}.$$

The interchanges of sum and integral are justified by Tonelli's theorem, since $c_n^2>0$, $\alpha_n<0$ and $\beta_n\ge 0$ have fixed signs, and, for the first product above, by Fubini's theorem, since that product is absolutely summable. The two contributions $\pm\frac{16}{\pi^3}\int_0^1 fg\log k\log k'\,dk$ cancel, and

$$\sum_n c_n^4=\frac{8N_\mathrm{e}}{\pi^3},\qquad \sum_n c_n^4\alpha_n=-\frac{4M_\mathrm{e}}{\pi^2},\qquad \mathcal{S}_1=-\frac{8}{\pi^3}\,\mathrm{FG}[\log k'],\qquad \mathcal{S}_2=\frac{8}{\pi^2}\,\mathrm{GG}[\log k']+\frac{4}{\pi}N_\mathrm{e}.$$

Hence $a=8N_\mathrm{e}/\pi$ and $b=8M_\mathrm{e}/\pi$, in agreement with (340), and (355) gives the two single-sum relations

$$\boxed{\mathcal{S}_1=\frac{\pi b-2a\log 2}{4\pi^2},\qquad \mathcal{S}_2=\frac{a}{4}+\frac{b\log 2}{\pi}.} \tag{356}$$

With $\hat a=a/\pi^2=\sum_k c_k^4$ and $\hat b=b/\pi^2=-\frac{2}{\pi}\sum_k c_k^4\alpha_k$ they read $\mathcal{S}_1=\frac14(\pi\hat b-2\hat a\log 2)$ and $\mathcal{S}_2=\frac{\pi^2}{4}\hat a+\pi\hat b\log 2$, the two relations first found numerically. Inserting (356) into (352), the a^2 and b^2 terms cancel:

$$\mathcal{L}_2=2a^2-4b^2+\frac{4b}{\pi}(\pi b-2a\log 2)-8a\Big(\frac{a}{4}+\frac{b\log 2}{\pi}\Big)=-\frac{16\log 2}{\pi}\,ab=-2\log 2\,P_4(2).$$

With $P_4(2)=(B_-^2-B_+^2)/(4\pi^3)$ from (333), this proves (270). Equivalently, (351) and (356) say $\mathcal{A}'(\frac12)=2\pi b-2\log 2\,a$ and $\mathcal{B}'(\frac12)=\frac{3\pi}{2}a-2\log 2\,b$, that is, $(4^x\mathcal{A})'=2\pi\,4^x\mathcal{B}$ and $(4^x\mathcal{B})'=\frac{3\pi}{2}\,4^x\mathcal{A}$ at $x=\frac12$.

Checks. These are not proof steps. The closed forms of $\sum c_k^4$, $\sum c_k^4\alpha_k$, $\mathcal{S}_1$ and $\mathcal{S}_2$ agree with 70-digit Euler–Maclaurin values of the single sums to 3×10^{-70}. With $\mathcal{A}'$ and $\mathcal{B}'$ obtained from the termwise x-derivative of $\Omega(t;x)$ at generic t, (351) and the chain $\frac12\mathcal{W}'(\frac12)=\mathcal{L}_2$ hold to 10^{-82}. The generating functions (353) and the moments (354), (355) hold to about 35 digits.

18.7.6 A second proof by rim–spoke reflection

An independent proof works directly with the wheel derivative of (244). This subsection states the result and its consequences and then proves them in full. For $\frac14<x<\frac34$ it shows that

$$W_4(2;2x)=\frac{2^{4-4x}\Gamma(1-x)^4\Gamma(x)^6}{\pi^{11/2}\Gamma(2x-\frac12)}\big[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)\big],$$

where $\mathfrak{a}(x)+\mathfrak{b}(x)\cos 2\pi\tau$ is the two-mode form of the entire function

$$\pi^4\sum_{k\in\mathbb{Z}}\bigl[\Gamma(1-\tfrac{x}{2}+k+\tau)\,\Gamma(1-\tfrac{x}{2}-k-\tau)\bigr]^{-4},$$

with $\mathfrak{a}(\frac12)=a$ and $\mathfrak{b}(\frac12)=b$. The inputs are the square-lattice factorization of the wheel for every rim exponent, the growth argument of (311), and a Barnes integral of well-poised type. That integral is proved in its terminating case from the Pfaff–Saalschütz sum (16.4.3 of [22]), Whipple's transformations (16.4.14 and 16.4.15 of [22]) and Barnes' first lemma (DLMF 5.13.3), and is extended by Carlson's theorem. The bracket is invariant under $x\to 1-x$, which exchanges rim and spoke exponents. Hence, with $\gamma_C(z)=\Gamma(z)/\Gamma(1-z)$ as in (264),

$$\frac{W_4(2;b)}{W_4(2;2-b)}=2^{4-4b}\frac{\gamma_C(b/2)^2}{\gamma_C(b-\frac12)},\qquad \frac12<b<\frac32,$$

and at the fixed point $b=1$ only the Gamma prefactor contributes to the logarithmic derivative. This gives $\partial_b\log W_4(2;b)|_{b=1}=\frac12(-4\log 2-4\psi_0(\frac12)+6\psi_0(\frac12)-2\psi_0(\frac12))=-2\log 2$. As a check, the factorized form agrees with a direct spectral evaluation of $W_4(2;2x)$ at $x=\frac25,\frac12,\frac35$ to 2×10^{-70}.

The machine checks of the proof below are `LRI_C5_d2_proof/L2_structural/checks_all.wl` (log `checks_all.log`); the 70-digit spectral check just quoted is recorded in `LRI_C5_d2_proof/review_L2_structural_numerics_r0/`.

Notation. Throughout this subsubsection $d=2$, the rim exponent is $b=2x$, and $N_x=\Gamma(1-x)/\Gamma(x)=\gamma_C(1-x)$ and $\varphi_x(s)=\Gamma(s+x)/\Gamma(s+1)$ are as in (345). Since $\Gamma(d/2)=1$, the eigenvalue (242) is $t_{\ell,2x}(\nu)=N_x\,\mathfrak{g}_\ell(\mathrm{i}\nu)$ with

$$\mathfrak{g}_\ell(s)=\frac{\Gamma(\frac{\ell}{2}+\frac{x}{2}+s)\,\Gamma(\frac{\ell}{2}+\frac{x}{2}-s)}{\Gamma(\frac{\ell}{2}+1-\frac{x}{2}+s)\,\Gamma(\frac{\ell}{2}+1-\frac{x}{2}-s)},$$

the kernel $g_\ell(s;2x)$ of (460) at $d=2$, and $W_4(2;2x)=\langle t_{\ell,2x}^4\rangle_2$ with $d_0=1$, $d_{\ell\ge1}=2$. For $0<x<\frac34$ let

$$\Omega(t;x)=\pi^4\sum_{k\in\mathbb{Z}}\bigl[\Gamma(1+k+t)\,\Gamma(1-x-k-t)\bigr]^{-4}=\sin^4\pi(t+x)\sum_{k\in\mathbb{Z}}\varphi_x(k+t)^4,$$

where the second form follows termwise from Gamma reflection [26]. For $x\in I$ this is the function $\Omega(t;x)$ of (349), and $\Omega(t;\frac12)=\Omega_4(t)$ of (335). The function displayed above in the variable τ is $\Omega(\tau-\frac{x}{2};x)$, and its two modes are written $\mathfrak{a}(x)$, $\mathfrak{b}(x)$. The statement to be proved is

$$W_4(2;2x)=\mathfrak{p}(x)\bigl[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)\bigr],\qquad \mathfrak{p}(x)=\frac{2^{4-4x}\Gamma(1-x)^4\Gamma(x)^6}{\pi^{11/2}\Gamma(2x-\frac12)},\qquad \tfrac14<x<\tfrac34. \tag{357}$$

At $x=\frac12$, $\mathfrak{p}(\frac12)=4/\pi$ and (357) is $P_4(2)=8ab/\pi$.

Lemma (two modes of a reciprocal-Gamma kernel). *Let $0<\kappa_1,\dots,\kappa_4<1$ with $\kappa=\kappa_1+\cdots+\kappa_4>\frac52$, and let $\mathfrak{k}(s)=\prod_{j=1}^4[\Gamma(\kappa_j+s)\Gamma(\kappa_j-s)]^{-1}$, an entire even function. Then*

(i) $|\mathfrak{k}(s)|\le C\,e^{4\pi|\Im s|}(1+|s|)^{4-2\kappa}$ *for all $s\in\mathbb{C}$;*

(ii) $E_{\mathfrak{k}}(\tau)=\sum_{k\in\mathbb{Z}}\mathfrak{k}(k+\tau)$ *converges absolutely and locally uniformly, and $E_{\mathfrak{k}}$ is entire, even and one-periodic;*

(iii) $E_{\mathfrak{k}}(\tau)=\int_{\mathbb{R}}\mathfrak{k}(s)\,ds+2\cos(2\pi\tau)\int_{\mathbb{R}}\mathfrak{k}(s)\cos(2\pi s)\,ds$.

Proof. (i) By reflection, $[\Gamma(\kappa_j+s)\Gamma(\kappa_j-s)]^{-1} = \pi^{-1}\sin\pi(\kappa_j-s)\,\Gamma(s+1-\kappa_j)/\Gamma(s+\kappa_j)$. For $\Re s \ge 0$ the Gamma ratio has no poles, because $1-\kappa_j>0$, and it is $O((1+|s|)^{1-2\kappa_j})$ by the Gamma-ratio asymptotics [27] for $|s|\ge 1$ and by continuity on the compact rest; moreover $|\sin\pi(\kappa_j-s)| \le e^{\pi|\Im s|}$. For $\Re s \le 0$ use $\mathfrak{k}(-s)=\mathfrak{k}(s)$. (ii) On a compact set K, (i) gives $|\mathfrak{k}(k+\tau)| \le C_K(1+|k|)^{4-2\kappa}$, and $2\kappa-4>1$; the Weierstrass test applies. Evenness and periodicity follow by re-indexing. (iii) For $\tau=u+\mathrm{i}y$ with $0\le u\le 1$, (i) and comparison with $\int_{\mathbb{R}}(1+|v+\mathrm{i}y|)^{4-2\kappa}\,dv$ give $|E_{\mathfrak{k}}(u+\mathrm{i}y)| \le C' e^{4\pi|y|}(1+|y|)^{5-2\kappa}$, with $5-2\kappa<0$. For an entire one-periodic function the coefficient $e_m=\int_0^1 E_{\mathfrak{k}}(u+\mathrm{i}y)e^{-2\pi\mathrm{i}m(u+\mathrm{i}y)}\,du$ does not depend on y, so $|e_m| \le C' e^{2\pi m y+4\pi|y|}(1+|y|)^{5-2\kappa}$. Letting $y\to-\infty$ for $m\ge 2$ and $y\to+\infty$ for $m\le -2$ gives $e_m=0$; for $|m|=2$ this uses $(1+|y|)^{5-2\kappa}\to 0$. Evenness gives $e_{-1}=e_1$. Finally $\int_{\mathbb{R}}|\mathfrak{k}|<\infty$, so by Fubini $e_0=\sum_k\int_0^1 \mathfrak{k}(k+u)\,du=\int_{\mathbb{R}}\mathfrak{k}$ and $e_1=\int_{\mathbb{R}}\mathfrak{k}(s)e^{-2\pi\mathrm{i}s}\,ds=\int_{\mathbb{R}}\mathfrak{k}(s)\cos(2\pi s)\,ds$. □

This is the growth argument of (311) applied to a reciprocal-Gamma kernel. Two cases are used.

(a) $\kappa_j=1-\frac{x}{2}$ for all j, with $0<x<\frac34$, so that $\kappa=4-2x>\frac52$. Then $\pi^4 E_{\mathfrak{k}}(\tau)=\Omega(\tau-\frac{x}{2};x)$, so that for $0<x<\frac34$

$$\Omega(t;x)=\mathfrak{a}(x)+\mathfrak{b}(x)\cos 2\pi\big(t+\tfrac{x}{2}\big),\qquad \mathfrak{a}(x)=\pi^4\int_{\mathbb{R}}\frac{ds}{\mathcal{G}_x(s)^4},\qquad \mathfrak{b}(x)=2\pi^4\int_{\mathbb{R}}\frac{\cos(2\pi s)\,ds}{\mathcal{G}_x(s)^4},\tag{358}$$

with $\mathcal{G}_x(s)=\Gamma(1-\frac{x}{2}+s)\Gamma(1-\frac{x}{2}-s)$. On I, (349) gives $\mathfrak{a}=\mathcal{A}/N_x^2$ and $\mathfrak{b}=\mathcal{B}/N_x^2$, so $\mathfrak{a}(\frac12)=a$ and $\mathfrak{b}(\frac12)=b$. Evaluating at $t=-\frac{x}{2}$ and $t=\frac12-\frac{x}{2}$, the two combinations $\mathfrak{a}\pm\mathfrak{b}$ are well-poised bilateral series: $\mathfrak{a}(x)+\mathfrak{b}(x)=\sin^4\frac{\pi x}{2}\sum_{k\in\mathbb{Z}}[\Gamma(k+\frac{x}{2})/\Gamma(k+1-\frac{x}{2})]^4$ and $\mathfrak{a}(x)-\mathfrak{b}(x)=\cos^4\frac{\pi x}{2}\sum_{k\in\mathbb{Z}}[\Gamma(k+\frac12+\frac{x}{2})/\Gamma(k+\frac32-\frac{x}{2})]^4$.

(b) $\kappa_j=\frac12+p_j$ with $0<p_j<\frac12$ and $p=p_1+\cdots+p_4>\frac12$, so that $\kappa=2+p>\frac52$. Then

$$E^{\sharp}(\tau)=\sum_{k\in\mathbb{Z}}\prod_{j=1}^{4}\big[\Gamma(\tfrac12+p_j+k+\tau)\,\Gamma(\tfrac12+p_j-k-\tau)\big]^{-1}$$

is a constant plus a multiple of $\cos 2\pi\tau$. If $p_j=\frac{x}{2}$ for all j, then $\pi^4E^{\sharp}(\tau)=\Omega(\tau-\frac{1-x}{2};1-x)$, because $1-\frac{1-x}{2}=\frac12+\frac{x}{2}$.

Proposition (the lattice step). *For $0<x<\frac34$ put $\mathcal{F}_x(s)=[\Gamma(\frac{x}{2}+s)\Gamma(\frac{x}{2}-s)]^4$ and*

$$\mathsf{J}_0(x)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\mathcal{F}_x(s),\qquad \mathsf{J}_1(x)=\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\cos(2\pi s)\,\mathcal{F}_x(s),$$

which converge absolutely because $|\Gamma(\frac{x}{2}+\mathrm{i}y)|^2\sim 2\pi|y|^{x-1}e^{-\pi|y|}$. Then

$$\begin{aligned}W_4(2;2x)&=-N_x^4\,\mathrm{Res}_{u=0}\frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)}=\frac{N_x^4}{\pi^4}\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\Omega(s-\tfrac{x}{2};x)\,\mathcal{F}_x(s)\\&=\frac{N_x^4}{\pi^4}\big[\mathfrak{a}(x)\mathsf{J}_0(x)+\mathfrak{b}(x)\mathsf{J}_1(x)\big].\end{aligned}\tag{359}$$

In particular, with $\mathcal{W}$ of (348),

$$W_4(2;2x)=\mathcal{W}(x)\qquad(x\in I).\tag{360}$$

Proof. (i) *Spectral form.* Since $t_{\ell,2x}(\nu)$ is even in ν, (241) gives

$$W_4(2;2x)=N_x^4\sum_{\ell\ge0}d_\ell\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\mathtt{g}_\ell(s)^4.$$

The double sum and integral converge absolutely for $x<\frac34$: at fixed ℓ, $t_{\ell,2x}(\nu)^4=O(\nu^{8x-8})$, and $\int_0^\infty t_{\ell,2x}^4\,d\nu=O(\ell^{8x-7})$.

(ii) *Lattice identity.* For $\ell \geq 0$ and $n \in \mathbb{Z}$, Gamma reflection applied to $\Gamma(-n-u)$ and to $1/\Gamma(1-x-n-u)$ gives

$$\mathsf{g}_\ell(\tfrac{\ell}{2}+\tfrac{x}{2}+n+u) = -\frac{\sin\pi(x+u)}{\sin\pi u}\,\varphi_x(\ell+n+u)\,\varphi_x(n+u),$$

so $-N_x\,\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)$ is the left side of (346) with $m=\ell+n$. With $(k,j)=(\ell+n,n)$ the weights $d_0=1$, $d_{\ell\geq 1}=2$ turn $\sum_{\ell\geq 0} d_\ell \sum_{n\in\mathbb{Z}}$ into the sum over all $(k,j)\in\mathbb{Z}^2$, because the summand $\varphi_x(k+u)^4\varphi_x(j+u)^4$ is symmetric in k,j. Hence

$$\mathsf{S}(u) = \sum_{\ell\geq 0} d_\ell \sum_{n\in\mathbb{Z}} \mathsf{g}_\ell(\tfrac{\ell}{2}+\tfrac{x}{2}+n+u)^4 = \frac{\sin^4\pi(x+u)}{\sin^4\pi u}\Big[\sum_{k\in\mathbb{Z}}\varphi_x(k+u)^4\Big]^2 = \frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)}.$$

On a circle $|u|=r$ with $0<r<\min(x,1-x)$ one has $|\varphi_x(k+u)|\leq C(1+|k|)^{x-1}$ uniformly, and $4x-4<-1$, so the double series converges absolutely and uniformly there. This is (539) at $d=2$.

(iii) *Residues of the two-dimensional integral.* For each ℓ the poles of g_ℓ in $\Re s>0$ are at $s=\frac{\ell}{2}+\frac{x}{2}+n$, $n\geq 0$, from $\Gamma(\frac{\ell}{2}+\frac{x}{2}-s)$; those of $\Gamma(\frac{\ell}{2}+\frac{x}{2}+s)$ lie in $\Re s<0$. By reflection,

$$\mathsf{g}_\ell(s) = \frac{\Gamma(s+\frac{\ell}{2}+\frac{x}{2})}{\Gamma(s+\frac{\ell}{2}+1-\frac{x}{2})}\cdot\frac{\Gamma(s-\frac{\ell}{2}+\frac{x}{2})}{\Gamma(s-\frac{\ell}{2}+1-\frac{x}{2})}\cdot\frac{\sin\pi(\frac{x}{2}+s-\frac{\ell}{2})}{\sin\pi(\frac{x}{2}-s+\frac{\ell}{2})}.$$

On the arcs $|s-\frac{\ell}{2}-\frac{x}{2}|=N+\frac{1}{2}$ in $\Re s\geq 0$, which pass midway between consecutive poles, the sine ratio is bounded uniformly in N, and for large N both Gamma ratios are $O(|s|^{x-1})$. So $\mathsf{g}_\ell^4=O(|s|^{8x-8})$ there, and since the arcs have length $O(N)$ and $8x-7<0$ their contributions vanish. Closing the contour to the right, which is clockwise, $\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\mathsf{g}_\ell^4 = -\sum_{n\geq 0}\mathrm{Res}_{u=0}\,\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)^4$. For $n<0$ the function $\mathsf{g}_\ell(\frac{\ell}{2}+\frac{x}{2}+n+u)$ is regular at $u=0$: there $\Gamma(-n-u)$ is regular, and so is $\Gamma(\ell+n+x+u)$, since $x\notin\mathbb{Z}$. Summing over ℓ with the weights d_ℓ, and using the uniform convergence of (ii) on $|u|=r$ to take the sums inside one contour integral, gives $W_4(2;2x)=-N_x^4\,\mathrm{Res}_{u=0}\,\mathsf{S}(u)$, the first equality in (359).

For $k\leq -1$, $1/\Gamma(k+1+u)$ has a simple zero at $u=0$, so $\varphi_x(k+u)^4=O(u^4)$, and all terms of S that contain such a factor are regular at $u=0$. Hence $\mathrm{Res}_{u=0}\,\mathsf{S}(u)=\mathrm{Res}_{u=0}[\sin^4\pi(x+u)\,\mathcal{H}_x(u)^2/\sin^4\pi u]$ with $\mathcal{H}_x$ of (348), and for $x\in I$ that equation gives (360). This is the per-spin residue evaluation of (261) at rim exponent $2x$; it is also the case $d=2$ of (479) with (539).

(iv) *Reflection.* For $k\in\mathbb{Z}$, $1/\Gamma(k+1+u)=(-1)^{k+1}\sin(\pi u)\Gamma(-k-u)/\pi$, so $\varphi_x(k+u)^4=\pi^{-4}\sin^4(\pi u)\,\mathcal{F}_x(\frac{x}{2}+k+u)$. With $\mathcal{P}_x(u)=\sum_{k\in\mathbb{Z}}\mathcal{F}_x(\frac{x}{2}+k+u)$ this gives $\Omega(u;x)=\pi^{-4}\sin^4\pi(x+u)\sin^4(\pi u)\,\mathcal{P}_x(u)$, hence

$$\frac{\Omega(u;x)^2}{\sin^4\pi u\,\sin^4\pi(u+x)} = \pi^{-4}\,\Omega(u;x)\,\mathcal{P}_x(u).$$

(v) *Residues of the one-dimensional integral.* The function $\Omega(s-\frac{x}{2};x)=\mathfrak{a}+\mathfrak{b}\cos 2\pi s$ is entire, with $|\Omega(s-\frac{x}{2};x)|\leq(|\mathfrak{a}|+|\mathfrak{b}|)e^{2\pi|\Im s|}$. Writing $\Gamma(\frac{x}{2}-s)=\pi/[\sin\pi(\frac{x}{2}-s)\Gamma(1-\frac{x}{2}+s)]$ gives $|\mathcal{F}_x(s)|\leq C|s|^{4x-4}e^{-4\pi|\Im s|}$ on the arcs $|s-\frac{x}{2}|=N+\frac{1}{2}$ in $\Re s\geq 0$. So the integrand is $O(|s|^{4x-4})$ there, and the arc contributions vanish because $4x-3<0$. The poles of $\mathcal{F}_x$ in $\Re s>0$ are at $s=\frac{x}{2}+n$, $n\geq 0$, and therefore

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,\Omega(s-\tfrac{x}{2};x)\mathcal{F}_x(s) = -\sum_{n\geq 0}\mathrm{Res}_{u=0}\big[\Omega(n+u;x)\,\mathcal{F}_x(\tfrac{x}{2}+n+u)\big] = -\,\mathrm{Res}_{u=0}\big[\Omega(u;x)\,\mathcal{P}_x(u)\big],$$

by the periodicity of Ω and the uniform convergence of $\mathcal{P}_x$ near $u=0$; the terms with $n<0$ are regular at $u=0$. With (iv) this is the second equality of (359), and (358) gives the third. □

Lemma (a well-poised Barnes integral). *Let* $\Re \mathsf{a}_j > 0$ *(*$j = 1,2,3$*),* $\mathsf{a} = \mathsf{a}_1 + \mathsf{a}_2 + \mathsf{a}_3$ *and* $\Re(\mathsf{a} + \frac{1}{2} - \mathsf{b}) > 0$*. Then*

$$\begin{aligned}\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}} \frac{\prod_{j=1}^{3}\Gamma(\mathsf{a}_j+s)\Gamma(\mathsf{a}_j-s)}{\Gamma(\mathsf{b}+s)\Gamma(\mathsf{b}-s)} \\ = \frac{2\sqrt{\pi}\,2^{2\mathsf{b}-2\mathsf{a}}\prod_{i<j}\Gamma(\mathsf{a}_i+\mathsf{a}_j)\prod_i\Gamma(2\mathsf{a}_i)}{\Gamma(\mathsf{a}+\frac{1}{2}-\mathsf{b})\prod_i\Gamma(\mathsf{a}_i+\mathsf{b})}\,{}_4F_3\!\left(\begin{matrix}2\mathsf{b}-1,\ \mathsf{b}-\mathsf{a}_1,\ \mathsf{b}-\mathsf{a}_2,\ \mathsf{b}-\mathsf{a}_3\\ \mathsf{b}+\mathsf{a}_1,\ \mathsf{b}+\mathsf{a}_2,\ \mathsf{b}+\mathsf{a}_3\end{matrix};1\right).\end{aligned} \tag{361}$$

The ${}_4F_3$ *is well-poised and converges, since its parameter excess is* $1 + 2\mathsf{a} - 2\mathsf{b}$*.*

At $\mathsf{b} = \mathsf{a}_3$ this is Barnes' first lemma (DLMF 5.13.3). At $\mathsf{b} = \frac{1}{2}$ it is the de Branges–Wilson integral (DLMF 5.13.5) with one parameter equal to 0, since $\Gamma(s)\Gamma(-s)/[\Gamma(2s)\Gamma(-2s)] = 4\cos\pi s = 4\pi/[\Gamma(\frac{1}{2}+s)\Gamma(\frac{1}{2}-s)]$. Write I_{L} and I_{R} for the two sides of (361).

Proof. (a) *Terminating case* $\mathsf{a}_3 = \mathsf{b} + n$*,* $n \in \mathbb{N}_0$*.* Take $\mathsf{a}_1, \mathsf{a}_2 > 0$ and $\mathsf{b} > \frac{1}{2}$ real and generic ($\mathsf{b} - \mathsf{a}_1, \mathsf{b} - \mathsf{a}_2 \notin \mathbb{Z}$). Then $\Gamma(\mathsf{a}_3 \pm s)/\Gamma(\mathsf{b} \pm s) = (\mathsf{b}+s)_n(\mathsf{b}-s)_n$. The Pfaff–Saalschütz sum (16.4.3 of [22]), with its parameters taken as $\mathsf{a}_2 + s$, $\mathsf{a}_2 - s$, $\mathsf{a}_2 + \mathsf{b}$ and $1 + \mathsf{a}_2 - \mathsf{b} - n$, gives

$$(\mathsf{b}+s)_n(\mathsf{b}-s)_n = (\mathsf{a}_2+\mathsf{b})_n(\mathsf{b}-\mathsf{a}_2)_n \sum_{m=0}^{n} \frac{(-n)_m(\mathsf{a}_2+s)_m(\mathsf{a}_2-s)_m}{m!\,(\mathsf{a}_2+\mathsf{b})_m(1+\mathsf{a}_2-\mathsf{b}-n)_m}.$$

Since $(\mathsf{a}_2 \pm s)_m\Gamma(\mathsf{a}_2 \pm s) = \Gamma(\mathsf{a}_2 + m \pm s)$, Barnes' first lemma evaluates each term: $\int \frac{ds}{2\pi\mathrm{i}}\Gamma(\mathsf{a}_1 + s)\Gamma(\mathsf{a}_1 - s)\Gamma(\mathsf{a}_2 + m + s)\Gamma(\mathsf{a}_2 + m - s) = \Gamma(2\mathsf{a}_1)\Gamma(\mathsf{a}_1 + \mathsf{a}_2 + m)^2\Gamma(2\mathsf{a}_2 + 2m)/\Gamma(2\mathsf{a}_1 + 2\mathsf{a}_2 + 2m)$. With $(2c)_{2m} = 4^m(c)_m(c+\frac{1}{2})_m$ this gives

$$I_{\mathrm{L}} = (\mathsf{a}_2+\mathsf{b})_n(\mathsf{b}-\mathsf{a}_2)_n \frac{\Gamma(2\mathsf{a}_1)\Gamma(2\mathsf{a}_2)\Gamma(\mathsf{a}_1+\mathsf{a}_2)^2}{\Gamma(2\mathsf{a}_1+2\mathsf{a}_2)}\,F_{\mathrm{L}},$$
$$F_{\mathrm{L}} = {}_4F_3\!\left(\begin{matrix}-n,\ \mathsf{a}_1+\mathsf{a}_2,\ \mathsf{a}_2,\ \mathsf{a}_2+\frac{1}{2}\\ \mathsf{a}_2+\mathsf{b},\ 1+\mathsf{a}_2-\mathsf{b}-n,\ \mathsf{a}_1+\mathsf{a}_2+\frac{1}{2}\end{matrix};1\right).$$

On the right, $F_{\mathrm{R}} = {}_4F_3(2\mathsf{b}-1, \mathsf{b}-\mathsf{a}_1, \mathsf{b}-\mathsf{a}_2, -n; \mathsf{b}+\mathsf{a}_1, \mathsf{b}+\mathsf{a}_2, 2\mathsf{b}+n; 1)$ is the terminating very-well-poised ${}_7F_6$ with first parameter $2\mathsf{b}-1$ and further parameters $\mathsf{b}-\mathsf{a}_1$, $\mathsf{b}-\mathsf{a}_2$, $\mathsf{b}-\frac{1}{2}$, b, $-n$: the pair $\mathsf{b}-\frac{1}{2}$, b cancels the very-well-poised factor and itself. Whipple's transformation (16.4.15 of [22], with $f = -n$, $(b,c) \to (\mathsf{b}-\mathsf{a}_1, \mathsf{b}-\mathsf{a}_2)$ and $(d,e) \to (\mathsf{b}-\frac{1}{2}, \mathsf{b})$) gives

$$F_{\mathrm{R}} = \frac{(2\mathsf{b})_n(\frac{1}{2})_n}{(\mathsf{b}+\frac{1}{2})_n(\mathsf{b})_n}\,{}_4F_3\!\left(\begin{matrix}\mathsf{a}_1+\mathsf{a}_2,\ \mathsf{b}-\frac{1}{2},\ \mathsf{b},\ -n\\ \mathsf{b}+\mathsf{a}_1,\ \mathsf{b}+\mathsf{a}_2,\ \frac{1}{2}-n\end{matrix};1\right),$$

and Whipple's transformation of a balanced ${}_4F_3$ (16.4.14 of [22], with $(a;b,c;d,e,f) \to (\mathsf{a}_1+\mathsf{a}_2; \mathsf{b}-\frac{1}{2}, \mathsf{b}; \mathsf{b}+\mathsf{a}_2, \mathsf{b}+\mathsf{a}_1, \frac{1}{2}-n)$) turns the last ${}_4F_3$ into $\frac{(\mathsf{b}-\mathsf{a}_2)_n(\frac{1}{2}-n-\mathsf{a}_1-\mathsf{a}_2)_n}{(\mathsf{b}+\mathsf{a}_1)_n(\frac{1}{2}-n)_n}F_{\mathrm{L}}$. Hence $F_{\mathrm{R}} = K_n F_{\mathrm{L}}$ with

$$K_n = \frac{(2\mathsf{b})_n(\frac{1}{2})_n(\mathsf{b}-\mathsf{a}_2)_n(\frac{1}{2}-n-\mathsf{a}_1-\mathsf{a}_2)_n}{(\mathsf{b}+\frac{1}{2})_n(\mathsf{b})_n(\mathsf{b}+\mathsf{a}_1)_n(\frac{1}{2}-n)_n}.$$

The prefactor of I_{R} times K_n equals the prefactor of I_{L}. Indeed, with $\mathsf{a}_3 = \mathsf{b}+n$: $\Gamma(\mathsf{a}_1+\mathsf{b}+n) = \Gamma(\mathsf{a}_1+\mathsf{b})(\mathsf{b}+\mathsf{a}_1)_n$, $\Gamma(\mathsf{a}_2+\mathsf{b}+n) = \Gamma(\mathsf{a}_2+\mathsf{b})(\mathsf{a}_2+\mathsf{b})_n$, $\Gamma(2\mathsf{b}+2n)(2\mathsf{b})_n/\Gamma(2\mathsf{b}+n) = (2\mathsf{b})_{2n} = 4^n(\mathsf{b})_n(\mathsf{b}+\frac{1}{2})_n$, and $(\frac{1}{2})_n(\frac{1}{2}-n-c)_n/(\frac{1}{2}-n)_n = (\frac{1}{2}+c)_n$ for $c = \mathsf{a}_1+\mathsf{a}_2$, which cancels against $\Gamma(\mathsf{a}+\frac{1}{2}-\mathsf{b}) = \Gamma(c+\frac{1}{2})(c+\frac{1}{2})_n$. What remains is $2\sqrt{\pi}\,2^{-2c}\Gamma(2c)/[\Gamma(c)\Gamma(c+\frac{1}{2})] = 1$, by duplication [26]. So $I_{\mathrm{L}} = I_{\mathrm{R}}$ when $\mathsf{a}_3 = \mathsf{b}+n$.

(b) *Carlson's theorem.* Keep $\mathsf{a}_1, \mathsf{a}_2 > 0$ and a generic $\mathsf{b} \in (\frac{1}{2}, \frac{1}{2}+\mathsf{a}_1+\mathsf{a}_2)$, put $\mathsf{a}_3 = \mathsf{b}+z$ with $\Re z \geq 0$, and let $f(z) = (I_{\mathrm{L}} - I_{\mathrm{R}})/\Gamma(\mathsf{b}+z)^2$. Then f is analytic for $\Re z > -\varepsilon$, and $f(n) = 0$ for $n \in \mathbb{N}_0$ by (a). It is bounded on $\Re z \geq 0$. First, by the Gamma bound below with $w = \mathsf{b}+z$,

$$\Big|\frac{I_{\mathrm{L}}}{\Gamma(\mathsf{b}+z)^2}\Big| \leq \frac{e^{2/3}}{2\pi}\int_{\mathbb{R}} \frac{|\Gamma(\mathsf{a}_1+\mathrm{i}y)|^2|\Gamma(\mathsf{a}_2+\mathrm{i}y)|^2}{|\Gamma(\mathsf{b}+\mathrm{i}y)|^2}\,dy < \infty.$$

Second, by duplication, $I_{\mathrm{R}}/\Gamma(\mathsf{b}+z)^2$ is a constant times $\frac{\Gamma(\mathsf{a}_1+\mathsf{b}+z)\Gamma(\mathsf{a}_2+\mathsf{b}+z)\Gamma(\mathsf{b}+\frac12+z)}{\Gamma(\mathsf{a}_1+\mathsf{a}_2+\frac12+z)\Gamma(2\mathsf{b}+z)\Gamma(\mathsf{b}+z)}\,F_{\mathrm{R}}(z)$, where $F_{\mathrm{R}}(z)$ is the ${}_4F_3$ of (361) at $\mathsf{a}_3=\mathsf{b}+z$. The Gamma ratio has equal total weights in numerator and denominator and no poles for $\Re z\ge 0$, so it is bounded there [27]. For $\Re z\ge 0$ and $k\ge 0$, $|k-z|\le|k+z|\le|k+2\mathsf{b}+z|$, so $|(-z)_n/(2\mathsf{b}+z)_n|\le 1$ and $|F_{\mathrm{R}}(z)|\le\sum_n|\frac{(2\mathsf{b}-1)_n(\mathsf{b}-\mathsf{a}_1)_n(\mathsf{b}-\mathsf{a}_2)_n}{n!\,(\mathsf{b}+\mathsf{a}_1)_n(\mathsf{b}+\mathsf{a}_2)_n}|<\infty$, the terms being $O(n^{2\mathsf{b}-2-2\mathsf{a}_1-2\mathsf{a}_2})$ with exponent below -1. A function that is analytic and bounded on $\Re z\ge 0$ and vanishes at every nonnegative integer is identically zero (Carlson's theorem [36]). So $f\equiv 0$.

(c) *Continuation.* Both sides of (361) are analytic on the convex domain $\{\Re\mathsf{a}_j>0,\ \Re(\mathsf{a}+\frac12-\mathsf{b})>0\}\subset\mathbb{C}^4$. On the left, the imaginary axis separates the poles of $\Gamma(\mathsf{a}_j+s)$ from those of $\Gamma(\mathsf{a}_j-s)$ and the integrand decays like $e^{-2\pi|\Im s|}$, locally uniformly. On the right, $1/[\Gamma(\mathsf{b}+\mathsf{a}_j)(\mathsf{b}+\mathsf{a}_j)_n]=1/\Gamma(\mathsf{b}+\mathsf{a}_j+n)$ is entire and the series converges locally uniformly. By (b) the two sides agree for real $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{b})$ in an open subset of $\mathbb{R}^3$ and $\Re\mathsf{a}_3\ge\mathsf{b}$. Continuation in a_3, and then in $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{b})$ (an analytic function that vanishes on an open subset of $\mathbb{R}^3$ vanishes identically), gives equality on the whole domain. □

Lemma (a uniform Gamma bound). *For $\Re w\ge\frac12$ and real y, $|\Gamma(w+\mathrm{i}y)\Gamma(w-\mathrm{i}y)|\le e^{2/3}|\Gamma(w)|^2$.*

Proof. Binet's first formula [37] gives, for $\Re\zeta>0$, $\log\Gamma(\zeta)=(\zeta-\frac12)\log\zeta-\zeta+\frac12\log 2\pi+\int_0^\infty\varpi(t)e^{-t\zeta}\,\frac{dt}{t}$ with $\varpi(t)=\frac12-\frac1t+\frac{1}{e^t-1}=\frac12[\coth\frac t2-\frac2t]$. The expansion $\coth v=\frac1v+2v\sum_{n\ge1}(v^2+n^2\pi^2)^{-1}$ (DLMF 4.36.3) gives $0<\coth v-\frac1v<\frac v3$ for $v>0$, hence $0<\varpi(t)/t<\frac{1}{12}$, and the remainder integral has modulus at most $1/(12\Re\zeta)\le\frac16$. In $\log|\Gamma(w+\mathrm{i}y)\Gamma(w-\mathrm{i}y)/\Gamma(w)^2|$ the terms $-\zeta$ and $\frac12\log 2\pi$ cancel. The main terms give $\mathfrak{g}(y)+\mathfrak{g}(-y)-2\mathfrak{g}(0)$ with $\mathfrak{g}(t)=\Re[(w+\mathrm{i}t-\frac12)\log(w+\mathrm{i}t)]$, and, with $\zeta=w+\mathrm{i}t$, $\mathfrak{g}''(t)=-\Re[\zeta^{-1}+\frac12\zeta^{-2}]\le(\frac12-\Re w)/|\zeta|^2\le 0$. So $\mathfrak{g}$ is concave and $\mathfrak{g}(y)+\mathfrak{g}(-y)-2\mathfrak{g}(0)\le 0$. The four remainders contribute at most $\frac23$. □

Proposition (Barnes–reciprocal duality). *Let $0<p_j<\frac12$ $(j=1,\dots,4)$, $p=p_1+\cdots+p_4>\frac12$, and*

$$\mathfrak{l}(p_1,\dots,p_4)=\frac{2^{2-2p}\prod_{i<j}\Gamma(p_i+p_j)\prod_i\Gamma(2p_i)}{\pi^{9/2}\,\Gamma(p-\frac12)}.$$

Then, with $E^\sharp$ of case (b) above, for all $\tau\in\mathbb{C}$

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}[\cos 2\pi s+\cos 2\pi\tau]\prod_{j=1}^4\Gamma(p_j+s)\Gamma(p_j-s)=4\pi^7\,\mathfrak{l}(p_1,\dots,p_4)\,E^\sharp(\tau).\tag{362}$$

Proof. The left side is a constant plus a multiple of $\cos 2\pi\tau$, and so is the right side, by case (b). Both sides are symmetric in the p_j. It therefore suffices to prove equality at $\tau_4=\frac12+p_4$ when $p_3\ne p_4$: by symmetry it then holds at $\tau_3=\frac12+p_3$ as well, and $\cos 2\pi\tau_j=-\cos 2\pi p_j$ takes different values at $j=3,4$, because $\cos 2\pi(\cdot)$ is injective on $(0,\frac12)$; a constant plus a multiple of $\cos 2\pi\tau$ is fixed by its values at two different values of $\cos 2\pi\tau$. The case $p_3=p_4$ follows by continuity of both sides in the p_j (dominated convergence on the left, uniform convergence of $E^\sharp$ on the right).

Left side at τ_4. By reflection and $\cos 2\pi s-\cos 2\pi p_4=2\sin\pi(p_4+s)\sin\pi(p_4-s)$, $(\cos 2\pi s-\cos 2\pi p_4)\Gamma(p_4+s)\Gamma(p_4-s)=2\pi^2/[\Gamma(1-p_4+s)\Gamma(1-p_4-s)]$. So the left side is $2\pi^2$ times the left side of (361) with $(\mathsf{a}_1,\mathsf{a}_2,\mathsf{a}_3,\mathsf{b})=(p_1,p_2,p_3,1-p_4)$; the condition of that lemma is $p>\frac12$.

Right side at τ_4. For $k\ge0$ the factor $j=4$ contains $1/\Gamma(-k)=0$. For $k=-1-n$ with $n\ge0$ the summand is $\prod_j[\Gamma(p_j+p_4-n)\Gamma(1+p_j-p_4+n)]^{-1}$, and $1/\Gamma(z-n)=(-1)^n(1-z)_n/\Gamma(z)$ gives

$$E^\sharp(\tau_4)=\prod_{j=1}^4\frac{1}{\Gamma(p_j+p_4)\Gamma(1+p_j-p_4)}\;{}_4F_3\!\left(\begin{matrix}1-2p_4,\ 1-p_1-p_4,\ 1-p_2-p_4,\ 1-p_3-p_4\\ 1+p_1-p_4,\ 1+p_2-p_4,\ 1+p_3-p_4\end{matrix};1\right),$$

the ${}_4F_3$ of (361) at $\mathfrak{b} = 1 - p_4$. The prefactors agree: after cancelling $\prod_{j\le 3}\Gamma(p_j + p_4)$ and $\Gamma(2p_4)$, $4\pi^7\mathfrak{l}/\prod_j[\Gamma(p_j + p_4)\Gamma(1 + p_j - p_4)]$ reduces to $4\pi^{5/2}2^{2-2p}\prod_{i<j\le 3}\Gamma(p_i + p_j)\prod_{i\le 3}\Gamma(2p_i)/[\Gamma(p - \frac{1}{2})\prod_{j\le 3}\Gamma(1 + p_j - p_4)]$, which is $2\pi^2$ times the prefactor of (361) at $\mathfrak{b} = 1 - p_4$. □

Corollary (the symmetric case). *For* $\frac{1}{4} < x < \frac{3}{4}$ *and* $\mathfrak{l}_x = \mathfrak{l}(\frac{x}{2}, \frac{x}{2}, \frac{x}{2}, \frac{x}{2})$, *that is* $\mathfrak{l}_x = 2^{2-4x}\Gamma(x)^{10}/(\pi^{9/2}\Gamma(2x - \frac{1}{2}))$,

$$\mathsf{J}_0(x) = 4\pi^3\,\mathfrak{l}_x\,\mathfrak{b}(1 - x), \qquad \mathsf{J}_1(x) = 4\pi^3\,\mathfrak{l}_x\,\mathfrak{a}(1 - x).$$

Proof. Take all $p_j = \frac{x}{2}$ in (362), so that $p = 2x > \frac{1}{2}$. The left side is $\mathsf{J}_1(x) + \mathsf{J}_0(x)\cos 2\pi\tau$. By case (b) and (358) at $1 - x$, which lies in $(0, \frac{3}{4})$, the right side is $4\pi^7\mathfrak{l}_x\pi^{-4}\Omega(\tau - \frac{1-x}{2}; 1 - x) = 4\pi^3\mathfrak{l}_x[\mathfrak{a}(1 - x) + \mathfrak{b}(1 - x)\cos 2\pi\tau]$. Compare the two modes. □

Proof of (357). By (359) and the corollary, $W_4(2; 2x) = N_x^4\pi^{-4}\,4\pi^3\mathfrak{l}_x[\mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)]$, and $\frac{4}{\pi}N_x^4\mathfrak{l}_x = \mathfrak{p}(x)$. □

Corollary (rim–spoke reflection). *For* $\frac{1}{2} < b < \frac{3}{2}$,

$$\frac{W_4(2; b)}{W_4(2; 2 - b)} = \frac{\mathfrak{p}(b/2)}{\mathfrak{p}(1 - b/2)} = 2^{4-4b}\,\frac{\gamma_C(b/2)^2}{\gamma_C(b - \frac{1}{2})}.$$

Proof. The bracket $\mathfrak{s}(x) = \mathfrak{a}(x)\mathfrak{b}(1-x)+\mathfrak{b}(x)\mathfrak{a}(1-x)$ is invariant under $x \to 1-x$, and $W_4(2; 2x) > 0$ because its spectral integrand is positive, so $\mathfrak{s} \neq 0$. Hence $W_4(2; 2x)/W_4(2; 2-2x) = \mathfrak{p}(x)/\mathfrak{p}(1-x) = 2^{4-8x}\Gamma(x)^2\Gamma(\frac{3}{2} - 2x)/[\Gamma(1 - x)^2\Gamma(2x - \frac{1}{2})] = 2^{4-8x}\gamma_C(x)^2/\gamma_C(2x - \frac{1}{2})$; put $b = 2x$. □

Corollary (the logarithmic wheel). $\partial_b W_4(2; b)|_{b=1} = -2\log 2\,P_4(2)$, *and* $\mathcal{L}_2$ *of* (256) *equals* $-2\log 2\,P_4(2)$.

Proof. The kernel $\mathcal{G}_x(s)^{-4}$ of (358) is entire in (x, s). Part (i) of the two-mode lemma holds verbatim for complex κ_j with $0 < \Re\kappa_j < 1$; for $\kappa_j = 1 - \frac{x'}{2}$ it gives $|\mathcal{G}_{x'}(s)|^{-4} \le C(1 + |s|)^{4\Re x' - 4}$ for real s, locally uniformly in x'. Cauchy's estimate on a circle $|x' - x| = r$ then gives $|\partial_x\mathcal{G}_x(s)^{-4}| \le C(1 + |s|)^{4(x+r)-4}$, which is integrable when $x + r < \frac{3}{4}$. By dominated convergence $\mathfrak{a}, \mathfrak{b} \in C^1(0, \frac{3}{4})$, so $\mathfrak{s} \in C^1(\frac{1}{4}, \frac{3}{4})$, and $\mathfrak{s}(1 - x) = \mathfrak{s}(x)$ gives $\mathfrak{s}'(\frac{1}{2}) = 0$. Therefore

$$\frac{d}{dx}W_4(2; 2x)\Big|_{x=\frac{1}{2}} = \mathfrak{p}'(\tfrac{1}{2})\,\mathfrak{s}(\tfrac{1}{2}) = (\log\mathfrak{p})'(\tfrac{1}{2})\,W_4(2; 1),$$
$$(\log\mathfrak{p})'(x) = -4\log 2 - 4\psi_0(1 - x) + 6\psi_0(x) - 2\psi_0(2x - \tfrac{1}{2}),$$

and $(\log\mathfrak{p})'(\frac{1}{2}) = -4\log 2$, the three $\psi_0(\frac{1}{2})$ terms cancelling. With $b = 2x$, $\partial_b W_4(2; b)|_{b=1} = -2\log 2\,W_4(2; 1)$, and $W_4(2; 1) = P_4(2)$ by (243). This is the wheel-derivative reading (244) of $\mathcal{L}_2$. For the series reading, (360) and (347) give $\mathcal{L}_2 = \frac{1}{2}\mathcal{W}'(\frac{1}{2}) = \partial_b W_4(2; b)|_{b=1}$. □

Remarks. (1) The reflection $b \to 2 - b$ exchanges the rim and spoke exponents of the two-dimensional wheel. It is not a graph symmetry, since planar duality maps the wheel to itself with the same exponents. It is a hypergeometric identity, (361) and (362), between a Barnes integral of Gamma products and a periodization of reciprocal Gamma products. $\mathcal{L}_2$ is the derivative at the fixed point $b = 1$, where only the Gamma prefactor $\mathfrak{p}$ contributes. (2) This route gives only the combination of single sums that $\mathcal{L}_2$ needs, not the two relations (356) separately. (3) The duality (362) is proved for four different parameters, but the lattice step with four different rim exponents was not written out; the resulting four-exponent ratio is the numerical statement listed under the remaining gaps of Section 17.2. (4) Checks, not proof steps: `checks_all.log` records the residue

functional (350) in exact algebra; (357) at $x = \frac{1}{2}, \frac{2}{5}, \frac{1}{3}, \frac{3}{10}, 0.27, 0.6, 0.7$ to all 35 working digits; $(\log \mathfrak{p})'(\frac{1}{2}) = -4\log 2$ and the reflection ratio exactly; (362) at complex τ and (361) at complex parameters to about 20 digits; and the identities $F_{\mathrm{R}} = K_n F_{\mathrm{L}}$ and the prefactor ratio of step (a) exactly for $n = 0, \dots, 6$. The script `extra_checks.wl` of the same folder tests the Gamma bound on random samples.

18.7.7 The elliptic moments by half-strip contours

The two further proofs below use the elliptic moments U_0, V_0, U, V, R defined before (96). This subsubsection evaluates all of them by one method, which does not use the Stieltjes identity (100). The notation is the one introduced with these moments: the modulus convention for $\mathbf{K}$, $f(k) = \mathbf{K}(k)$, $g(k) = \mathbf{K}(k')$, the weight $w(k) = \log \frac{k}{1-k^2}$, and the moments M, N, which are $M_{\mathrm{e}}, N_{\mathrm{e}}$ of (338); $T(k) = \frac{1-k}{1+k}$ is the Landen map of (97).

Proposition (elliptic moments).

$$\begin{aligned} U_0 = 2N_{\mathrm{e}}, \qquad V_0 = 2M_{\mathrm{e}}, \qquad R = \frac{\pi}{4} N_{\mathrm{e}} - \log 2\, M_{\mathrm{e}}, \\ U + \log 2\, U_0 = -\pi M_{\mathrm{e}}, \qquad V + \log 2\, V_0 = -\frac{\pi}{2} N_{\mathrm{e}}. \end{aligned} \tag{363}$$

Equivalently (103), (106), (105), $U_0 = 2N_{\mathrm{e}}$ *and* $V_0 = 2M_{\mathrm{e}}$ *hold, and they are proved here without* (100).

Proof. Angle form. With $k = \sin\beta$ one has $dk/k' = d\beta$ and $kk' = \frac{1}{2}\sin 2\beta$. So, with $F(\beta) = \mathbf{K}(\sin\beta)$, $G(\beta) = \mathbf{K}(\cos\beta)$ and $L(\beta) = \log\sin 2\beta$,

$$\begin{aligned} U_0 &= \int_0^{\pi/2} F^2\, d\beta, & V_0 &= \int_0^{\pi/2} FG\, d\beta, \\ U + \log 2\, U_0 &= \int_0^{\pi/2} F^2 L\, d\beta, & V + \log 2\, V_0 &= \int_0^{\pi/2} FGL\, d\beta. \end{aligned}$$

(1) *Holomorphy.* On the strip $\mathsf{T} = \{0 < \Re\beta < \frac{\pi}{2}\}$ the functions F, G and L (principal branch) are holomorphic. For $\beta = \alpha + \mathrm{i}\eta$ with $0 < \alpha < \frac{\pi}{2}$, $\sin\beta = \sin\alpha\cosh\eta + \mathrm{i}\cos\alpha\sinh\eta$ has positive real part and is real only for $\eta = 0$, where it lies in $(0,1)$; hence $\sin^2\beta$ is real only there and avoids the cut $[1,\infty)$ of $\mathbf{K}$ as a function of k^2. The same holds for $\cos\beta = \cos\alpha\cosh\eta - \mathrm{i}\sin\alpha\sinh\eta$. Next, $\sin 2\beta = \sin 2\alpha\cosh 2\eta + \mathrm{i}\cos 2\alpha\sinh 2\eta$ is real only if $\eta = 0$ or $\alpha = \frac{\pi}{4}$, and then it is positive.

(2) *Boundary values.* For $y > 0$ put $\kappa = \tanh y$ and $\kappa' = \operatorname{sech} y$. The imaginary-modulus transformation $\mathbf{K}(\mathrm{i}k/k') = k'\mathbf{K}(k)$ and the reciprocal-modulus transformation $\mathbf{K}(X + \mathrm{i}0) = X^{-1}[\mathbf{K}(X^{-1}) + \mathrm{i}\mathbf{K}(\sqrt{1-X^{-2}})]$ for $X > 1$ [34] give, as limits from inside T,

$$\begin{aligned} F(\mathrm{i}y) &= \kappa' f(\kappa), & F(\tfrac{\pi}{2} + \mathrm{i}y) &= \kappa'[g(\kappa) + \mathrm{i}f(\kappa)], & L(\mathrm{i}y) &= w(\kappa) + \log 2 + \tfrac{\mathrm{i}\pi}{2}, \\ G(\mathrm{i}y) &= \kappa'[g(\kappa) - \mathrm{i}f(\kappa)], & G(\tfrac{\pi}{2} + \mathrm{i}y) &= \kappa' f(\kappa), & L(\tfrac{\pi}{2} + \mathrm{i}y) &= w(\kappa) + \log 2 - \tfrac{\mathrm{i}\pi}{2}. \end{aligned} \tag{364}$$

Indeed $\sin(\epsilon + \mathrm{i}y) \to \mathrm{i}\sinh y$ with $\mathrm{i}\sinh y = \mathrm{i}\kappa/\kappa'$; $\sin(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to \cosh y + \mathrm{i}0$; $\cos(\epsilon + \mathrm{i}y) \to \cosh y - \mathrm{i}0$; $\cos(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to -\mathrm{i}\sinh y$, and $\mathbf{K}$ is even; $\sin 2(\epsilon + \mathrm{i}y) \to \mathrm{i}\sinh 2y$ and $\sin 2(\frac{\pi}{2} - \epsilon + \mathrm{i}y) \to -\mathrm{i}\sinh 2y$, both from the right half-plane; finally $\log\sinh 2y = \log(2\kappa/\kappa'^2) = w(\kappa) + \log 2$.

(3) *Bounds.* Let $\Theta \in \{F^2, FG, F^2L, FGL\}$. With the boundary values included, $|\Theta(\beta)| \le C(1 + |\log|\beta||^3 + |\log|\beta - \frac{\pi}{2}||^3)$ for $0 \le \Im\beta \le 1$, because $\mathbf{K}(k) = O(1 + |\log(1-k^2)|)$ uniformly as $k^2 \to 1$ in $\mathbb{C} \setminus [1,\infty)$ (from the expansion in powers of k'^2, DLMF 19.12.1) and $L(\beta) = O(1 + |\log|\beta|| + |\log|\beta - \frac{\pi}{2}||)$. For $\Im\beta \ge 1$, $|\Theta(\beta)| \le C(1 + \Im\beta)^3 e^{-2\Im\beta}$, because $|\sin\beta|, |\cos\beta| \ge \sinh\Im\beta$, $\mathbf{K}(k) = O(|k|^{-1}\log|k|)$ as $|k| \to \infty$ [34], and $L = O(1 + \Im\beta)$. These two uniform estimates, which include the one-sided boundary values of (2), are standard and are stated here without further detail.

(4) *Half-strip identity.* Cauchy's theorem on the rectangle $[\epsilon, \frac{\pi}{2}-\epsilon]\times[0,Y]$, followed by $\epsilon\to 0$ (dominated convergence with the bounds of (3) and locally uniform convergence to the boundary values) and $Y\to\infty$ (the top edge is $O(Y^3e^{-2Y})$), gives

$$\int_0^{\pi/2}\Theta(\beta)\,d\beta=-\mathrm{i}\int_0^{\infty}\bigl[\Theta(\tfrac{\pi}{2}+\mathrm{i}y)-\Theta(\mathrm{i}y)\bigr]\,dy. \tag{365}$$

(5) *Evaluation.* Write f,g for $f(\kappa),g(\kappa)$ and $\omega=w(\kappa)+\log 2$; all integrals below are over $\kappa\in(0,1)$ and converge, since the integrands have at most $\log^3$ endpoint singularities. With $dy=d\kappa/\kappa'^2$ and (364), (365) gives

$$U_0=2\int fg-\mathrm{i}\int(g^2-2f^2),\qquad V_0=2\int f^2,$$
$$U+\log 2\,U_0=\int\bigl[2fg\,\omega-\tfrac{\pi}{2}g^2\bigr]-\mathrm{i}\int\bigl[(g^2-2f^2)\omega+\pi fg\bigr],\qquad V+\log 2\,V_0=\int\bigl[2f^2\omega-\pi fg\bigr].$$

The left sides are real, so the imaginary parts vanish: $U_0=2N_{\mathrm{e}}$, $\int g^2=2M_{\mathrm{e}}$, $V_0=2M_{\mathrm{e}}$, $\int(g^2-2f^2)\omega=-\pi N_{\mathrm{e}}$, and $U+\log 2\,U_0=2\int fg\,\omega-\pi M_{\mathrm{e}}$, $V+\log 2\,V_0=2\int f^2\omega-\pi N_{\mathrm{e}}$.

(6) *Landen.* By (97), $f(T)g(T)|T'|=fg$ and $g(T)^2|T'|=2f^2$, and $w(T(\kappa))=-w(\kappa)-2\log 2$ gives $\omega(T(\kappa))=-\omega(\kappa)$. Substituting $\kappa\to T(\kappa)$ gives $\int fg\,\omega=-\int fg\,\omega=0$ and $\int g^2\omega=-2\int f^2\omega$. With (5), $-4\int f^2\omega=-\pi N_{\mathrm{e}}$, that is $\int f^2\omega=\frac{\pi}{4}N_{\mathrm{e}}$. Therefore $U+\log 2\,U_0=-\pi M_{\mathrm{e}}$, $V+\log 2\,V_0=\frac{\pi}{2}N_{\mathrm{e}}-\pi N_{\mathrm{e}}=-\frac{\pi}{2}N_{\mathrm{e}}$ and $R=\int f^2w=\int f^2\omega-\log 2\,M_{\mathrm{e}}=\frac{\pi}{4}N_{\mathrm{e}}-\log 2\,M_{\mathrm{e}}$. With $U_0=2N_{\mathrm{e}}$ and $V_0=2M_{\mathrm{e}}$ these are $U=-\pi M_{\mathrm{e}}-2\log 2\,N_{\mathrm{e}}$, $V=-\frac{\pi}{2}N_{\mathrm{e}}-2\log 2\,M_{\mathrm{e}}$ and $R=\frac{\pi}{4}N_{\mathrm{e}}-\log 2\,M_{\mathrm{e}}$, that is (103), (106) and (105). □

The half-strip contours thus reprove the three logarithmic evaluations of the one-dimensional wheel by a single method. Conversely, (363) follows from (103), (106), (105) and $U_0=2N_{\mathrm{e}}$, $V_0=2M_{\mathrm{e}}$ as proved in Section 9.2; that second proof uses (100). As a check, not a proof step, a 60-digit run (symmetry, Slurm job 567960; `LRI_C5_d2_simplify/proof/L2-fourier/check_strip_dps60.log`) confirms the boundary values (364) to 4×10^{-62}, the four strip identities and the Landen steps to 5×10^{-63}, and the identities for $U+\log 2\,U_0$ and $V+\log 2\,V_0$ to 2×10^{-63} and 2×10^{-65}.

18.7.8 A third proof: Parseval pairings on the unit circle

This proof keeps the rim derivative (347), the residue functional (350), the jets (351) and the reduction (352) of Section 18.7. It evaluates the four single sums $a=\pi^2\sum_k c_k^4$, $b=-2\pi\sum_k c_k^4\alpha_k$, $\mathcal{S}_1$ and $\mathcal{S}_2$ differently: each is a Parseval pairing of two power series on the unit circle, and the circle values turn the pairings into the elliptic moments U_0, V_0, U, V defined before (96). With (363) no step uses (100). Here c_n, α_n, β_n are as in Section 18.7,

$$y_1(z)={}_2F_1(\tfrac12,\tfrac12;1;z)=\sum_{n\ge0}c_n^2z^n,\qquad y_3(z)={}_2F_1(\tfrac12,\tfrac12;1;1-z),$$

and, as in the upper-semicircle contour that proves (106), $f_\vartheta=\mathbf{K}(\sin\vartheta)$ and $g_\vartheta=\mathbf{K}(\cos\vartheta)=f_{\pi/2-\vartheta}$ for $0<\vartheta<\frac{\pi}{2}$.

Generating functions on the slit disk. For $|z|<1$, $z\notin(-1,0]$, with principal logarithms,

$$\sum_{n\ge0}c_n^2\alpha_nz^n=-\frac{\pi}{2}\,y_3(z)-\frac12\log z\;y_1(z),$$
$$\sum_{n\ge0}c_n^2\beta_nz^n=-\frac12\log(1-z)\,y_1(z), \tag{366}$$
$$\sum_{n\ge0}c_n^2\bigl[4\alpha_n\beta_n+2\psi_1(n+\tfrac12)\bigr]z^n=\pi\log(1-z)\,y_3(z)+\pi^2y_1(z)+\log z\,\log(1-z)\,y_1(z).$$

At $z = k^2$ with $0 < k < 1$, where $y_1 = \frac{2}{\pi} f(k)$ and $y_3 = \frac{2}{\pi} g(k)$, these are the three lines of (353). The left sides are power series with polynomially bounded coefficients, hence analytic for $|z| < 1$. The right sides are analytic on the slit disk, since y_1 is analytic off $[1, \infty)$, y_3 off $(-\infty, 0]$ and $\log z$ off $(-\infty, 0]$. The slit disk is connected and the two sides agree on $(0, 1)$, so they agree on the whole slit disk.

Lemma (values on the circle). *For $0 < \vartheta < \frac{\pi}{4}$,*

$$y_1(e^{4\mathrm{i}\vartheta}) = \frac{1}{\pi}\, e^{-\mathrm{i}\vartheta}[g_\vartheta + \mathrm{i} f_\vartheta], \qquad y_3(e^{-4\mathrm{i}\vartheta}) = \frac{2}{\pi}\, e^{\mathrm{i}\vartheta} f_\vartheta. \tag{367}$$

Proof. The function y_1 is analytic on $\mathbb{C} \setminus [1, \infty)$, and $y_1(e^{4\mathrm{i}\vartheta}) = \frac{2}{\pi}\mathbf{K}(e^{2\mathrm{i}\vartheta})$. The boundary-value Landen relation $e^{\mathrm{i}\vartheta}\mathbf{K}(e^{2\mathrm{i}\vartheta}) = \frac{1}{2}[\mathbf{K}(\cos\vartheta) + \mathrm{i}\mathbf{K}(\sin\vartheta)]$, proved with the upper-semicircle contour behind (106), gives the first identity. (The coefficients 1 and i of the reciprocal-modulus value used there can be confirmed at the endpoint. For $s > 1$, $y_1(s + \mathrm{i}0)$ solves the hypergeometric equation on $(1, \infty)$, whose solutions are spanned by $s^{-1/2} y_1(1/s)$ and $s^{-1/2} y_1(1 - 1/s)$. As $s \downarrow 1$ the first behaves like $\pi^{-1} \log \frac{16}{s-1}$ and the second tends to 1, while $y_1(z) = \pi^{-1}[4 \log 2 - \log(1 - z)] + o(1)$ as $z \to 1$ [23] gives $y_1(s + \mathrm{i}0) = \pi^{-1}[\log \frac{16}{s-1} + \mathrm{i}\pi] + o(1)$. So the coefficients are 1 and i.) For the second identity, both sides of ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 1 - w^2) = \frac{2}{1+w}\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; (\frac{1-w}{1+w})^2)$ are analytic for $\Re w > 0$, where $w^2 \notin (-\infty, 0]$ and $|\frac{1-w}{1+w}| < 1$, and they agree on $(0, 1)$, where this is the real Landen identity $f(T(k)) = \frac{1+k}{2} g(k)$ behind (97). Take $w = e^{-2\mathrm{i}\vartheta}$. Then $1 - w^2 = 1 - e^{-4\mathrm{i}\vartheta}$, $(1 - w)/(1 + w) = \mathrm{i} \tan\vartheta$ and $2/(1 + w) = e^{\mathrm{i}\vartheta}/\cos\vartheta$, and Pfaff's transformation [23] gives ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; -\tan^2\vartheta) = \cos\vartheta\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \sin^2\vartheta) = \frac{2}{\pi}\cos\vartheta\, f_\vartheta$. □

Proposition (pairings).

$$a = \frac{4U_0}{\pi}, \qquad b = \frac{4V_0}{\pi}, \qquad \mathcal{S}_1 = -\frac{2}{\pi^3}(U + 2\log 2\, U_0), \qquad \mathcal{S}_2 = \frac{a}{2} + \frac{4}{\pi^2}(V + 2\log 2\, V_0). \tag{368}$$

Proof. If $\mathsf{p}(z) = \sum_n p_n z^n$ and $\mathsf{q}(z) = \sum_n q_n z^n$ have real square-summable coefficients, they lie in the Hardy space H^2, have radial limits almost everywhere, and $\sum_n p_n q_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathsf{p}(e^{\mathrm{i}\theta})\, \mathsf{q}(e^{-\mathrm{i}\theta})\, d\theta$. All coefficient sequences used below are square summable: $c_n^2 = O(1/n)$, $c_n^2 \beta_n = O(\log n/n)$, $c_n^2 \alpha_n = O(n^{-2})$ and $c_n^2[4\alpha_n\beta_n + 2\psi_1(n + \frac{1}{2})] = O(\log n/n^2)$. For $\theta \in (-\pi, \pi) \setminus \{0\}$ the radius to $e^{\mathrm{i}\theta}$ lies in the slit disk, and the right sides of (366) are continuous up to $e^{\mathrm{i}\theta}$, so the radial limits are the values of the closed forms. On the circle $z = e^{\mathrm{i}\theta}$ one has $\log \bar{z} = -\mathrm{i}\theta$ and $\log(1-z) + \log(1-\bar{z}) = 2\log|2\sin\frac{\theta}{2}|$; $\Im \log(1 - \bar{z})$ is odd in θ; $|y_1(z)|^2$ is even; and $y_1(e^{-\mathrm{i}\theta}) y_3(e^{\mathrm{i}\theta}) = \overline{y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta})}$, since y_1, y_3 have real Taylor coefficients. All integrands below have only integrable logarithmic singularities. For $0 < \theta < \pi$ and $\theta = 4\vartheta$, (367) gives $|y_1(e^{\mathrm{i}\theta})|^2 = \pi^{-2}(f_\vartheta^2 + g_\vartheta^2)$ and $y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta}) = 2\pi^{-2}(f_\vartheta g_\vartheta + \mathrm{i} f_\vartheta^2)$. The fold $\vartheta \mapsto \frac{\pi}{2} - \vartheta$ exchanges f and g and fixes $\sin 2\vartheta$; so for any weight $\chi(\sin 2\vartheta)$ it turns $\int_0^{\pi/4}(f^2 + g^2)\chi\, d\vartheta$ into $\int_0^{\pi/2} f^2 \chi\, d\vartheta$ and $\int_0^{\pi/4} fg\, \chi\, d\vartheta$ into $\frac{1}{2}\int_0^{\pi/2} fg\, \chi\, d\vartheta$. Finally, by the angle form in the proof of (363) (with $\beta = \vartheta$), $\int_0^{\pi/2} f_\vartheta^2 \log(2\sin 2\vartheta)\, d\vartheta = U + 2\log 2\, U_0$ and $\int_0^{\pi/2} f_\vartheta g_\vartheta \log(2\sin 2\vartheta)\, d\vartheta = V + 2\log 2\, V_0$.

(i) $a/\pi^2 = \sum c_n^2 \cdot c_n^2 = \frac{1}{2\pi}\int |y_1|^2\, d\theta = \frac{4}{\pi^3}\int_0^{\pi/4}(f_\vartheta^2 + g_\vartheta^2)\, d\vartheta = \frac{4U_0}{\pi^3}$.

(ii) $-b/(2\pi) = \sum c_n^2 \cdot c_n^2 \alpha_n$ is the pairing of y_1 with the first line of (366). The integrand is $-\frac{\pi}{2} y_1(z) y_3(\bar{z}) + \frac{\mathrm{i}\theta}{2}|y_1(z)|^2$, and the second term is odd. So $b = \pi\Re\int_0^{\pi} y_1(e^{\mathrm{i}\theta}) y_3(e^{-\mathrm{i}\theta})\, d\theta = \frac{8}{\pi}\int_0^{\pi/4} f_\vartheta g_\vartheta\, d\vartheta = \frac{4V_0}{\pi}$.

(iii) $\mathcal{S}_1 = \sum c_n^2 \cdot c_n^2 \beta_n = -\frac{1}{4\pi}\int \log(1 - \bar{z})\, |y_1(z)|^2\, d\theta$, by the second line of (366). Dropping the odd part and setting $\theta = 4\vartheta$,

$$\mathcal{S}_1 = -\frac{2}{\pi^3}\int_0^{\pi/4} \log(2\sin 2\vartheta)(f_\vartheta^2 + g_\vartheta^2)\, d\vartheta = -\frac{2}{\pi^3}\int_0^{\pi/2} \log(2\sin 2\vartheta) f_\vartheta^2\, d\vartheta = -\frac{2}{\pi^3}(U + 2\log 2\, U_0).$$

(iv) Since $c_n^4[4\alpha_n\beta_n+\psi_1(n+\frac12)] = (2c_n^2\beta_n)(c_n^2\alpha_n) + c_n^2\cdot\frac12 c_n^2[4\alpha_n\beta_n+2\psi_1(n+\frac12)]$, $\mathcal{S}_2$ is the sum of two pairings. By (366) the combined integrand is

$$\pi\log|2\sin\tfrac{\theta}{2}|\;y_1(z)y_3(\bar z)+\frac{\pi^2}{2}|y_1(z)|^2-\mathrm{i}\theta\log|2\sin\tfrac{\theta}{2}|\;|y_1(z)|^2.$$

The last term is odd. The middle one contributes $\frac{\pi^2}{2}\cdot\frac{a}{\pi^2}=\frac{a}{2}$. The first one contributes $\frac{2}{\pi^2}\int_0^\pi \log(2\sin\frac{\theta}{2}) f_{\theta/4}g_{\theta/4}\,d\theta = \frac{4}{\pi^2}\int_0^{\pi/2}\log(2\sin 2\vartheta) f_\vartheta g_\vartheta\,d\vartheta = \frac{4}{\pi^2}(V+2\log 2\,V_0)$. □

Conclusion. Inserting (363) into (368) gives $a=8N_{\mathrm e}/\pi$ and $b=8M_{\mathrm e}/\pi$, which reproves (340), and

$$\mathcal{S}_1=\frac{2M_{\mathrm e}}{\pi^2}-\frac{4\log 2\,N_{\mathrm e}}{\pi^3}=\frac{\pi b-2a\log 2}{4\pi^2},\qquad \mathcal{S}_2=\frac{2N_{\mathrm e}}{\pi}+\frac{8\log 2\,M_{\mathrm e}}{\pi^2}=\frac{a}{4}+\frac{b\log 2}{\pi},$$

which are (356). With (352) this gives $\mathcal{L}_2=-2\log 2\,P_4(2)$ exactly as at the end of Section 18.7, and with (333) it proves (270). The inputs are Euler's and Pfaff's transformations, the connection formulas of [23] behind (353), the Landen transformations, Parseval's identity in H^2, the boundary-value Landen relation and the half-strip contours of Section 18.7.7; the Stieltjes identity is not used.

Remarks. (1) The pairings (368) use only the circle values. Given $U_0=2N_{\mathrm e}$ and $V_0=2M_{\mathrm e}$, the first relation of (356) is therefore equivalent to $U+\log 2\,U_0=-\frac{\pi}{2}V_0$, that is to (103), and the second to $V+\log 2\,V_0=-\frac{\pi}{4}U_0$, that is to (106). The two single-sum relations are the unit-circle images of the logarithmic evaluations of the one-dimensional wheel. (2) Two relations among second-order logarithmic moments were found numerically while this route was developed; they are not used. With $\langle\chi\rangle=\int_0^1 f(k)g(k)\chi(k)\,dk$, $2\langle\log^2 k\rangle-\langle\log k\log(1-k)\rangle-3\langle\log k\log(1+k)\rangle=\frac{\pi^2}{2}N_{\mathrm e}+\pi\log 2\,M_{\mathrm e}$, which is equivalent to the second relation of (356) through a second-order moment representation that has only been sketched, and $32\langle\log^2(1+k)\rangle=22\langle\log^2 k\rangle-7\pi^2N_{\mathrm e}+8\log^2 2\,N_{\mathrm e}$. Both come from an integer-relation search at 40 digits (`LRI_C5_d2_simplify/proof/L2-elliptic/probe_moments.py`) and are numerical statements only. (3) Checks, not proof steps: the scripts `check_parseval.py`, `collapse_sympy.py`, `check_collapse.py` and `check_rimderivative.py` of the same folder test the generating functions, the circle values and the pairings at 30 digits, the rim derivative (347) with the 70-digit single sums, and the collapse algebra exactly (no logs of these runs are shipped); the synthesis check `LRI_C5_d2_simplify/proof/synth/synth_checks.log` gives, with the 70-digit single sums, residuals 9×10^{-72} and 1.1×10^{-69} for the two relations (356), 7×10^{-68} for (352), and 1.7×10^{-67} for $\mathcal{L}_2+16\log 2\,ab/\pi$.

18.7.9 A fourth proof: the Fourier representation of the two modes

This proof keeps the rim derivative (347) and the residue functional (350), but obtains the modes $\mathcal{A}$, $\mathcal{B}$ and their x-derivatives at $x=\frac12$ without single sums and without growth estimates. Cauchy's beta integral writes the Fourier coefficients of $\Omega(t;x)$ as convolutions of $(2\cos\frac{\theta}{2})^{-x}$ on $(-\pi,\pi)$, which have compact support, and at $x=\frac12$ these convolutions are complete elliptic integrals. Throughout, $I_{\mathrm F}=(\frac38,\frac58)$, which contains I, and $D_{\mathrm F}=\{x\in\mathbb{C}:|x-\frac12|<\frac18\}$. For real $x\in I_{\mathrm F}$ put

$$h_x(\sigma)=\frac{1}{\Gamma(1-\frac{x}{2}+\sigma)\,\Gamma(1-\frac{x}{2}-\sigma)},\qquad \mathsf{P}_x(\theta)=(2\cos\tfrac{\theta}{2})^{-x}\,\mathbf{1}_{(-\pi,\pi)}(\theta),$$
$$\mathsf{Q}_x=\mathsf{P}_x*\mathsf{P}_x,\qquad v_m(x)=(\mathsf{Q}_x*\mathsf{Q}_x)(2\pi m).$$

Thus $h_x=\mathcal{G}_x^{-1}$ in the notation of (358). By reflection, $h_x(\sigma)=\sin\pi(1-\frac{x}{2}-\sigma)\,\Gamma(\frac{x}{2}+\sigma)/[\pi\Gamma(1-\frac{x}{2}+\sigma)]$, so h_x is real, even and continuous on $\mathbb{R}$ with $|h_x(\sigma)|\le C(1+|\sigma|)^{x-1}$; hence $h_x^2\in L^2(\mathbb{R})$ and $h_x^4\in L^1(\mathbb{R})$. The terms of $\Omega(t;x)$ satisfy $\pi^4[\Gamma(1+s)\Gamma(1-x-s)]^{-4}=\pi^4h_x(\sigma)^4$ at $s=\sigma-\frac{x}{2}$.

Lemma (Cauchy's beta integral). *For real $c > -1$ and real σ,*

$$\int_{-\pi}^{\pi} (2\cos\tfrac{\theta}{2})^c \, e^{\mathrm{i}\sigma\theta}\, d\theta = \frac{2\pi\,\Gamma(c+1)}{\Gamma(1+\frac{c}{2}+\sigma)\,\Gamma(1+\frac{c}{2}-\sigma)}.$$

Proof. Put $s = \sigma - \frac{c}{2}$ and $w = e^{\mathrm{i}\theta}$. For $|\theta| < \pi$, $1 + w = 2\cos\frac{\theta}{2}\, e^{\mathrm{i}\theta/2}$, so with principal powers $(2\cos\frac{\theta}{2})^c e^{\mathrm{i}\sigma\theta} = (1+w)^c w^s$, and the integral equals $-\mathrm{i}\oint_{|w|=1}(1+w)^c w^{s-1}\,dw$. Let $\Re s > 0$. The integrand is holomorphic in the disk slit along $(-1, 0]$ and integrable at $w = 0$. For $0 < \rho < 1$, collapsing the circle $|w| = \rho$ onto the two sides of the slit $[-\rho, 0]$, where $w^{s-1} = r^{s-1}e^{\pm\mathrm{i}\pi(s-1)}$ at $w = -r$, gives $\oint_{|w|=\rho}(1+w)^c w^{s-1}\,dw = 2\mathrm{i}\sin(\pi s)\int_0^\rho (1-r)^c r^{s-1}\,dr$. As $\rho \to 1$ the right side tends to $2\mathrm{i}\sin(\pi s)\,\mathrm{B}(s, c+1)$; the left side tends to the integral over the unit circle by dominated convergence, with the majorant $|1+\rho e^{\mathrm{i}\theta}|^c \le (4\rho)^{c/2}|\cos\frac{\theta}{2}|^c$ for $-1 < c < 0$ (since $|1+\rho e^{\mathrm{i}\theta}|^2 - 4\rho\cos^2\frac{\theta}{2} = (1-\rho)^2$) and $|1+\rho e^{\mathrm{i}\theta}|^c \le 2^c$ for $c \ge 0$. Hence the integral is $2\sin(\pi s)\Gamma(s)\Gamma(c+1)/\Gamma(s+c+1) = 2\pi\Gamma(c+1)/[\Gamma(1-s)\Gamma(1+c+s)]$, which is the claim. Both sides are entire in σ, so the restriction $\Re s > 0$ is removed by continuation. □

With $c = -x$ the lemma reads $h_x(\sigma) = (2\pi\Gamma(1-x))^{-1}\int_{\mathbb{R}} \mathsf{P}_x(\theta) e^{\mathrm{i}\sigma\theta}\,d\theta$.

Lemma (the twofold convolution). *For $0 < |\phi| < 2\pi$ put $k = \cos\frac{\phi}{4}$ and $k' = |\sin\frac{\phi}{4}|$. For real $x \in I_{\mathrm{F}}$,*

$$\mathsf{Q}_x(\phi) = 2\int_0^{\pi/2} \frac{(2k\cos\eta)^{1-2x}}{\sqrt{1-k^2\sin^2\eta}}\,d\eta = (2k)^{1-2x}\,\mathrm{B}(\tfrac12, 1-x)\,{}_2F_1(\tfrac12, \tfrac12; \tfrac32 - x; k^2). \tag{369}$$

Both expressions define holomorphic functions of $x \in D_{\mathrm{F}}$ that agree there, and, uniformly for $x \in D_{\mathrm{F}}$,

$$|\mathsf{Q}_x(\phi)| \le C\,|\sin\tfrac{\phi}{2}|^{-1/4}, \qquad |\partial_x \mathsf{Q}_x(\phi)| \le C\,|\sin\tfrac{\phi}{2}|^{-1/3}. \tag{370}$$

At $x = \frac12$,

$$\mathsf{Q}_{1/2}(\phi) = 2\mathbf{K}(k), \qquad \partial_x \mathsf{Q}_x(\phi)\big|_{x=1/2} = \pi\mathbf{K}(k') - 2\mathbf{K}(k)\log(4kk'). \tag{371}$$

Proof. By evenness take $0 < \phi < 2\pi$. The integrand of $\mathsf{P}_x * \mathsf{P}_x$ is supported on $\theta \in (\phi - \pi, \pi)$, and $4\cos\frac{\theta}{2}\cos\frac{\phi-\theta}{2} = 2[\cos\psi - \cos\psi_0]$ with $\psi = \theta - \frac{\phi}{2}$ and $\psi_0 = \pi - \frac{\phi}{2}$. So $\mathsf{Q}_x(\phi) = \int_{-\psi_0}^{\psi_0}[2(\cos\psi - \cos\psi_0)]^{-x}\,d\psi$. The substitution $\sin\frac{\psi}{2} = k\sin\eta$ with $k = \sin\frac{\psi_0}{2} = \cos\frac{\phi}{4}$ gives $2(\cos\psi - \cos\psi_0) = 4k^2\cos^2\eta$ and $d\psi = 2k\cos\eta\,d\eta/\sqrt{1-k^2\sin^2\eta}$, which is the η-integral. Expanding $(1-k^2\sin^2\eta)^{-1/2}$ and integrating termwise with $\int_0^{\pi/2}\sin^{2n}\eta\cos^{1-2x}\eta\,d\eta = \Gamma(n+\frac12)\Gamma(1-x)/[2\Gamma(n+\frac32 - x)]$ gives the ${}_2F_1$. For fixed ϕ both expressions are holomorphic in $x \in D_{\mathrm{F}}$ and agree for real x, hence on D_{F}.

Bounds: for $x \in D_{\mathrm{F}}$, $\xi = \Re(1-2x) \in (-\frac14, \frac14)$, and since $k\cos\eta \le 1$, $|(2k\cos\eta)^{1-2x}| = (2k\cos\eta)^\xi \le 2(k\cos\eta)^{-1/4}$ and $|\partial_x(2k\cos\eta)^{1-2x}| \le 4|\log(2k\cos\eta)|(k\cos\eta)^{-1/4} \le C(k\cos\eta)^{-1/3}$. For $0 < s' < 1$, $\int_0^{\pi/2}(\cos\eta)^{-s'}(1-k^2\sin^2\eta)^{-1/2}\,d\eta \le C_{s'}k'^{-s'}$: on $[0, \frac{\pi}{4}]$ the integrand is bounded, and on $[\frac{\pi}{4}, \frac{\pi}{2}]$ one uses $(1-k^2\sin^2\eta)^{1/2} \ge \max(\cos\eta, k'/\sqrt{2})$ and the variable $\cos\eta$. Since $kk' = \frac12|\sin\frac{\phi}{2}|$, (370) follows, and differentiation under the η-integral is justified by the same majorants.

At $x = \frac12$ the η-integral is $2\mathbf{K}(k)$, and $\partial_x\mathsf{Q}_x|_{x=1/2} = -4\log(2k)\mathbf{K}(k) - 4\int_0^{\pi/2}\log\cos\eta\,(1-k^2\sin^2\eta)^{-1/2}\,d\eta$. The next lemma gives $-4\log(2k)\mathbf{K}(k) - 2\mathbf{K}(k)\log\frac{k'}{k} + \pi\mathbf{K}(k')$, which is (371). □

Lemma (a logarithmic moment). *For $0 < k < 1$,*

$$\int_0^{\pi/2} \frac{\log\cos\eta}{\sqrt{1-k^2\sin^2\eta}}\,d\eta = \frac12\mathbf{K}(k)\log\frac{k'}{k} - \frac{\pi}{4}\mathbf{K}(k').$$

Proof. With $\eta = \mathrm{am}(u, k)$ the left side is $\int_0^{\mathbf{K}}\log\mathrm{cn}\,u\,du$, $\mathbf{K} = \mathbf{K}(k)$. The theta-quotient form of the Jacobian functions [35] gives $\mathrm{cn}\,u = \sqrt{k'/k}\,\theta_2(\zeta, q)/\theta_4(\zeta, q)$ with $\zeta = \pi u/(2\mathbf{K})$ and $q = e^{-\pi\mathbf{K}(k')/\mathbf{K}(k)}$,

and the product formulas [35] give

$$\log\frac{\theta_2(\zeta,q)}{\theta_4(\zeta,q)} = \log(2q^{1/4}) + \log\cos\zeta + \sum_{n\ge1}\bigl[\log(1+2q^{2n}\cos 2\zeta + q^{4n}) - \log(1-2q^{2n-1}\cos 2\zeta + q^{4n-2})\bigr].$$

For $0 \le r < 1$, $\log(1 \pm 2r\cos 2\zeta + r^2) = 2\Re\log(1 \pm re^{2\mathrm{i}\zeta})$ has mean zero over $[0,\pi]$, hence over $[0,\frac{\pi}{2}]$ by the symmetry $\zeta \mapsto \pi - \zeta$, and the series converges uniformly. The mean of $\log\cos\zeta$ over $[0,\frac{\pi}{2}]$ is $-\log 2$. So the mean of $\log\mathrm{cn}$ over $[0,\mathbf{K}]$ is $\frac12\log\frac{k'}{k} + \frac14\log q$, and $\int_0^{\mathbf{K}}\log\mathrm{cn}\,u\;du = \frac{\mathbf{K}}{2}\log\frac{k'}{k} - \frac{\pi}{4}\mathbf{K}(k')$. □

Proposition (two modes from compact Fourier support). *For real $x \in I_{\mathrm{F}}$ and all $t \in \mathbb{C}$, $\Omega(t;x) = \mathfrak{a}(x) + \mathfrak{b}(x)\cos 2\pi(t+\frac{x}{2})$, with*

$$\mathfrak{a}(x) = \pi^4\int_{\mathbb{R}} h_x^4\,d\sigma = \frac{\pi\,v_0(x)}{8\,\Gamma(1-x)^4}, \qquad \mathfrak{b}(x) = 2\pi^4\int_{\mathbb{R}} h_x^4\cos(2\pi\sigma)\,d\sigma = \frac{\pi\,v_1(x)}{4\,\Gamma(1-x)^4}. \tag{372}$$

Consequently, on I, $\mathcal{A}(x) = \sin^2(\pi x)\,v_0(x)/(8\pi)$ and $\mathcal{B}(x) = \sin^2(\pi x)\,v_1(x)/(4\pi)$.

Proof. Let $\omega_m = \int_0^1 \Omega(t;x)e^{-2\pi\mathrm{i}mt}\,dt$. The terms of Ω are integrable on the real line, so unfolding (Fubini) and $s = \sigma - \frac{x}{2}$ give $\omega_m = \pi^4 e^{\mathrm{i}\pi mx}\int_{\mathbb{R}} h_x(\sigma)^4 e^{-2\pi\mathrm{i}m\sigma}\,d\sigma$. By the Cauchy beta integral with $c = -x$ and Fubini ($\mathsf{P}_x \in L^1$), $h_x(\sigma)^2 = (2\pi\Gamma(1-x))^{-2}\int_{\mathbb{R}}\mathsf{Q}_x(\phi)e^{\mathrm{i}\sigma\phi}\,d\phi$. Since $\mathsf{Q}_x \in L^1\cap L^2$ by (370), Plancherel's theorem gives, with $\widehat{\chi}(\theta) = \int\chi(\sigma)e^{-\mathrm{i}\theta\sigma}\,d\sigma$, the identity $\widehat{h_x^2} = \mathsf{Q}_x/(2\pi\Gamma(1-x)^2)$ in L^2. Parseval's identity for the L^2 functions h_x^2 and $h_x^2 e^{\mathrm{i}\xi\sigma}$ then gives

$$\int_{\mathbb{R}} h_x^4 e^{-\mathrm{i}\xi\sigma}\,d\sigma = \frac{1}{2\pi}\int_{\mathbb{R}}\widehat{h_x^2}(\theta)\,\overline{\widehat{h_x^2}(\theta-\xi)}\,d\theta = \frac{(\mathsf{Q}_x * \mathsf{Q}_x)(\xi)}{8\pi^3\Gamma(1-x)^4}.$$

Thus $\omega_m = \pi e^{\mathrm{i}\pi mx}v_m(x)/(8\Gamma(1-x)^4)$. Now Q_x is even, nonnegative and supported in $[-2\pi,2\pi]$, so $\mathsf{Q}_x * \mathsf{Q}_x$ is even and supported in $[-4\pi,4\pi]$, and at $\xi = \pm4\pi$ the convolution integral runs over a single point. Hence $v_{-1} = v_1$ and $v_m = 0$ for $|m| \ge 2$. A continuous one-periodic function whose Fourier coefficients vanish for $|m| \ge 2$ equals the corresponding trigonometric polynomial (Fejér), so for real t, $\Omega(t;x) = \frac{\pi}{8\Gamma(1-x)^4}[v_0(x) + 2v_1(x)\cos(2\pi t + \pi x)]$, and by analyticity for all t. The integral forms follow from ω_0 and ω_1. Finally $N_x^2/\Gamma(1-x)^4 = [\Gamma(x)\Gamma(1-x)]^{-2} = \sin^2(\pi x)/\pi^2$ and (349) give the expressions for $\mathcal{A}$ and $\mathcal{B}$. □

For $x \in I$ this is a second proof of (349): the modes with $|m| \ge 2$ are removed by the compact support of $\mathsf{Q}_x * \mathsf{Q}_x$, without growth estimates. The same argument should work for every real $x < \frac34$, where Q_x behaves like $|\phi|^{1-2x}$ near 0 and is still square integrable; this extension is a sketch, is not proved here and is not used (the general case is (358)).

Lemma (the convolutions at $x = \frac12$). *v_0 and v_1 extend holomorphically to D_{F}, with $v_m'(x) = \int_{\mathbb{R}}\partial_x[\mathsf{Q}_x(\phi)\mathsf{Q}_x(2\pi m - \phi)]\,d\phi$. At $x = \frac12$,*

$$v_0 = 32\,U_0, \qquad v_1 = 16\,V_0, \qquad v_0' = 32\bigl[\pi V_0 - 2U - 4\log 2\,U_0\bigr], \qquad v_1' = 16\bigl[\pi U_0 - 2V - 4\log 2\,V_0\bigr]. \tag{373}$$

Proof. Use the holomorphic extension (369). By (370), $|\mathsf{Q}_x(\phi)\mathsf{Q}_x(2\pi m-\phi)|$ and its x-derivative are bounded, uniformly for $x \in D_{\mathrm{F}}$, by $C|\sin\frac{\phi}{2}|^{-7/12}$ on the support ($|\phi| < 2\pi$ for $m = 0$, $0 < \phi < 2\pi$ for $m = 1$), which is integrable. Holomorphy and differentiation under the integral follow. By evenness, $v_0' = 4\int_0^{2\pi}(\partial_x\mathsf{Q}_x)\mathsf{Q}_x\,d\phi$ and $v_1' = 2\int_0^{2\pi}(\partial_x\mathsf{Q}_x)(\phi)\,\mathsf{Q}_x(2\pi-\phi)\,d\phi$. Substitute $\phi = 2\pi - 4\beta$ with $\beta \in (0,\frac{\pi}{2})$: then $\cos\frac{\phi}{4} = \sin\beta$, $\cos\frac{2\pi-\phi}{4} = \cos\beta$ and $4kk' = 2\sin 2\beta$, so (371) gives $\mathsf{Q}_{1/2}(\phi) = 2F(\beta)$, $\mathsf{Q}_{1/2}(2\pi-\phi) = 2G(\beta)$ and $\partial_x\mathsf{Q}_x(\phi)|_{1/2} = \pi G(\beta) - 2F(\beta)[\log 2 + \log\sin 2\beta]$, with F, G as in the proof of (363). With $d\phi = 4\,d\beta$, the angle forms given there and $\int_0^{\pi/2}G^2\,d\beta = U_0$, this gives (373). □

Conclusion. By (372), since $\sin^2\pi x$ has zero derivative at $x=\frac12$, and by (373),

$$a=\frac{v_0(\frac12)}{8\pi}=\frac{4U_0}{\pi},\qquad b=\frac{v_1(\frac12)}{4\pi}=\frac{4V_0}{\pi},\qquad \mathcal{A}'(\tfrac12)=\frac{v_0'(\frac12)}{8\pi},\qquad \mathcal{B}'(\tfrac12)=\frac{v_1'(\frac12)}{4\pi}.$$

With (363), $a=8N_{\mathrm e}/\pi$, $b=8M_{\mathrm e}/\pi$, $-2U-4\log 2\,U_0=2\pi M_{\mathrm e}-4\log 2\,N_{\mathrm e}$ and $-2V-4\log 2\,V_0=\pi N_{\mathrm e}-4\log 2\,M_{\mathrm e}$, so that

$$\mathcal{A}'(\tfrac12)=16M_{\mathrm e}-\frac{16\log 2}{\pi}N_{\mathrm e}=2\pi b-2\log 2\,a,\qquad \mathcal{B}'(\tfrac12)=12N_{\mathrm e}-\frac{16\log 2}{\pi}M_{\mathrm e}=\frac{3\pi}{2}a-2\log 2\,b. \tag{374}$$

Inserted into the expression for $\mathcal{L}_2$ displayed after (350),

$$\mathcal{L}_2=-6a^2-8b^2+\frac{4}{\pi}\Big[(2\pi b-2\log 2\,a)\,b+a\Big(\frac{3\pi}{2}a-2\log 2\,b\Big)\Big]=-\frac{16\log 2}{\pi}\,ab=-2\log 2\,P_4(2).$$

The rotation part $(2\pi b,\frac{3\pi}{2}a)$ of (374) cancels $-6a^2-8b^2$ exactly, and the diagonal part $-2\log 2$ leaves $-2\log 2$ times the coefficient $8/\pi$ of ab in $P_4(2)=8ab/\pi$. With (351), (374) is equivalent to (356), which this route therefore proves once more; (374) are the relations quoted after (356) and in (540). The inputs are Cauchy's beta integral, Plancherel's theorem, Stirling's formula, the theta-function form of cn and the half-strip contours of Section 18.7.7; the Stieltjes identity is not used.

Remark (a real-line form). Put $\mathfrak{m}_j=\int_{\mathbb R}h_{1/2}(\sigma)^4\cos(2\pi j\sigma)\,d\sigma$, so that $\mathfrak{m}_0=a/\pi^4$ and $\mathfrak{m}_1=b/(2\pi^4)$, and $\Psi(\sigma)=\psi_0(\frac34+\sigma)+\psi_0(\frac34-\sigma)$. Then

$$\int_{\mathbb R}h_{1/2}^4\,\Psi\,d\sigma=2\pi\mathfrak{m}_1-(5\log 2+2\gamma)\mathfrak{m}_0,\qquad \int_{\mathbb R}h_{1/2}^4\,\Psi\cos(2\pi\sigma)\,d\sigma=\frac{3\pi}{8}\mathfrak{m}_0-(5\log 2+2\gamma)\mathfrak{m}_1.$$

To see this, differentiate the integral forms (372) under the integral. One has $\partial_x h_x^4=2h_x^4\Psi_x$ with $\Psi_x(\sigma)=\psi_0(1-\frac{x}{2}+\sigma)+\psi_0(1-\frac{x}{2}-\sigma)$, which is entire in σ because the poles of $\psi_0(1-\frac{x}{2}-\sigma)$ sit at the fourth-order zeros of h_x^4; the reflected form of h_x gives the majorant $|\partial_x h_x^4|\le C\log(2+|\sigma|)(1+|\sigma|)^{4x-4}$, uniformly for x near $\frac12$. So $\mathfrak{a}'(\frac12)=2\pi^4\int h_{1/2}^4\Psi$ and $\mathfrak{b}'(\frac12)=4\pi^4\int h_{1/2}^4\Psi\cos 2\pi\sigma$. Since $\mathfrak{a}=\mathcal{A}/N_x^2$, $\mathfrak{b}=\mathcal{B}/N_x^2$ and $(N_x^{-2})'(\frac12)=4\psi_0(\frac12)$, (374) gives $\mathfrak{a}'(\frac12)=2\pi b-(10\log 2+4\gamma)a$ and $\mathfrak{b}'(\frac12)=\frac{3\pi}{2}a-(10\log 2+4\gamma)b$, which are the displayed identities. By duplication and reflection, $\Psi(\sigma)=\psi_0(\frac12+2\sigma)+\psi_0(\frac12-2\sigma)-2\log 2+\pi\sec 2\pi\sigma$, where the combination is regular; so the first identity also reads $\int_{\mathbb R}h_{1/2}^4[\psi_0(\frac12+2\sigma)+\psi_0(\frac12-2\sigma)+\pi\sec 2\pi\sigma]\,d\sigma=2\pi\mathfrak{m}_1-(3\log 2+2\gamma)\mathfrak{m}_0$. This form is not used.

Checks. These are not proof steps. A 60-digit run (symmetry, Slurm job 567960; logs `LRI_C5_d2_simplify/proof/L2-fourier/check_reduction_dps60.log` and `check_strip_dps60.log`, script `check_reduction.py`) gives: the Cauchy beta integral to 1.4×10^{-42}; $\mathsf{Q}_{1/2}=2\mathbf{K}$ and (371) to 6×10^{-65}; $a=4U_0/\pi$ and $b=4V_0/\pi$ to 4×10^{-65}; the derivatives from (373) against (351) with the 70-digit single sums to 6×10^{-63}; (374) to 5×10^{-63} and 3×10^{-64}; and the collapse with these elliptic values of a, b, $\mathcal{A}'$, $\mathcal{B}'$ against the 70-digit $\mathcal{L}_2$ to 7×10^{-62}. The direct quadrature of Q_x at $x=0.6$ is limited by its endpoint singularity (2×10^{-26}). The symbolic collapse is checked exactly by `collapse.py` (log `collapse.log`).

18.8 Proof of the closed form of $\mathcal{K}_2$

We prove (271) for $\mathcal{K}_2=[t^2](J_t-J_t^{\Gamma})$ of (238) at $d=2$. Write $t=2\tau$. Since $h_2=1/\pi$, the path master is the kite (671) at $\boldsymbol{p}=(1,1,1,2-2\tau,1)$:

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb C^2}\frac{d^2x\,d^2y}{|x|\,|1-x|\,|y|\,|1-y|^{2-2\tau}\,|x-y|},\qquad 0<\tau<1.$$

We use the parameter convention $K(m)$ of Section 1.4, the moments

$$E(a,b)=\int_0^1 x^{a-1}(1-x)^{b-1}K(x)\,dx,$$

the six moments $I_1,\dots,I_6$ listed before (77), evaluated at $t=\tau$, and the functions $\mathfrak{F},\mathfrak{G}$ of (75). Put $s_\tau=\sin\pi\tau$ and $c_\tau=\cos\pi\tau$. At $d=1$ the master is a linear functional of the six moments, (60); at $d=2$ it is a quadratic form in them. This proof uses none of the periodization machinery of this Section; it is placed next to the proof for $\mathcal{L}_2$ for convenience. It was written by the Claude Code session that wrote the first proof of Section 18.7 (package `LRI_C5_d2_proof/`, folder `K2_selberg/`); the closed form (271) itself was first found numerically by the other session named there.

18.8.1 Real cells and elliptic moments

For $\xi=(\xi_1,\xi_2)\in\mathbb{R}^2$ let

$$(L_1,\dots,L_5)=(\xi_1,\,1-\xi_1,\,\xi_2,\,1-\xi_2,\,\xi_2-\xi_1),\qquad(\kappa_1,\dots,\kappa_5)=(-\tfrac12,-\tfrac12,-\tfrac12,\tau-1,-\tfrac12),$$

and $\Upsilon(\xi)=\prod_f|L_f(\xi)|^{\kappa_f}$. Then $\pi^2J_{2\tau}=\int_{\mathbb{C}^2}\prod_f|L_f(x,y)|^{2\kappa_f}d^2x\,d^2y$, where $L_f(x,y)$ is the same affine form at complex arguments. The five lines $L_f=0$ cut $\mathbb{R}^2$ into twelve open cells R: for ξ_2 in one of $(-\infty,0)$, $(0,1)$, $(1,\infty)$, the points $0,1,\xi_2$ cut the ξ_1-line into four intervals. On each cell the sign vector $\varsigma(R)=(\operatorname{sgn}L_f)_f$ is constant, and distinct cells have distinct sign vectors. For $0<\tau<1$,

$$\Upsilon_R:=\int_R\Upsilon(\xi)\,d\xi=2I_{j(R)},\tag{375}$$

with $j(R)$ given by

cell	j	cell	j	cell	j
$\xi_1<\xi_2<0$	6	$0<\xi_1<\xi_2<1$	1	$\xi_2<0,\ 1<\xi_1$	5
$\xi_2<\xi_1<0$	5	$0<\xi_2<\xi_1<1$	2	$0<\xi_2<1<\xi_1$	1
$\xi_1<0<\xi_2<1$	2	$0<\xi_1<1<\xi_2$	3	$1<\xi_1<\xi_2$	4
$\xi_1<0,\ 1<\xi_2$	4	$\xi_2<0<\xi_1<1$	6	$1<\xi_2<\xi_1$	3

Indeed $\Upsilon\ge0$, so by Tonelli we may integrate over ξ_1 first. For $e_1<e_2<e_3$ and $P(\xi_1)=|(\xi_1-e_1)(\xi_1-e_2)(\xi_1-e_3)|^{-1/2}$, the substitutions $\xi_1=e_3-(e_3-e_1)/\sin^2\theta$, $e_1+(e_2-e_1)\sin^2\theta$, $e_3-(e_3-e_2)\sin^2\theta$ and $e_1+(e_3-e_1)/\sin^2\theta$ on the four intervals give

$$\int_{-\infty}^{e_1}P=\int_{e_2}^{e_3}P=2rK(m),\qquad\int_{e_1}^{e_2}P=\int_{e_3}^{\infty}P=2rK(1-m),$$
$$m=\frac{e_3-e_2}{e_3-e_1},\qquad r=(e_3-e_1)^{-1/2}.$$

With $\{e_i\}=\{0,1,\xi_2\}$, the outer weight $|\xi_2|^{-1/2}|1-\xi_2|^{\tau-1}r$ becomes the Euler weight $v^{-1/2}(1-v)^{\tau-1}$, $v^{-1/2}(1-v)^{-\tau}$ or $v^{-\tau}(1-v)^{\tau-1}$ after $v=\xi_2$ on $(0,1)$, $v=-\xi_2/(1-\xi_2)$ on $(-\infty,0)$ or $v=1/\xi_2$ on $(1,\infty)$, respectively. The arguments of K become v or $1-v$, and the table follows. All twelve integrals are finite for $0<\tau<1$.

18.8.2 The rotation lemma

For each form write $|L_f(x,y)|^2=\alpha_f+\beta_f^2$, where $x=x_1+\mathrm{i}x_2$, $y=y_1+\mathrm{i}y_2$, $\alpha_f=L_f(x_1,y_1)^2$, and β_f is the linear part of L_f evaluated at (x_2,y_2); this holds because the L_f have real coefficients. Put

$\xi=(x_1-x_2,y_1-y_2)$ and $\eta=(x_1+x_2,y_1+y_2)$. Then $\alpha_f-\beta_f^2=L_f(\xi)L_f(\eta)$ and $dx_1\,dx_2\,dy_1\,dy_2=\frac14\,d\xi\,d\eta$. We claim

$$\pi^2 J_{2\tau}=-\frac14\sum_{R,R'}e^{\mathrm{i}\pi\phi(R,R')}\,\Upsilon_R\Upsilon_{R'},\qquad \phi(R,R')=\sum_{f:\ \varsigma_f(R)\neq\varsigma_f(R')}\kappa_f, \tag{376}$$

the sum running over all 144 ordered pairs of cells.

(a) *Convergence.* Since $\alpha+\beta^2\geq|\alpha-\beta^2|$ and all $\kappa_f<0$, the integrand $\prod_f(\alpha_f+\beta_f^2)^{\kappa_f}$ is at most $\Upsilon(\xi)\Upsilon(\eta)$, whose integral is $\frac14(\int_{\mathbb{R}^2}\Upsilon)^2=4(\sum_j I_j)^2<\infty$ by (375). In particular $J_{2\tau}$ converges for $0<\tau<1$.

(b) *An analytic family.* For $\Re z>0$ put $\Psi(z)=z^2\int_{\mathbb{R}^4}\prod_f(\alpha_f+z^2\beta_f^2)^{\kappa_f}\,dx_1\,dx_2\,dy_1\,dy_2$, with principal powers; off a null set $\alpha_f+z^2\beta_f^2\notin(-\infty,0]$. If $|\arg z|\leq\theta_0<\frac{\pi}{2}$ and $r_1\leq|z|\leq r_2$, the inequality $|\alpha+Be^{\mathrm{i}\vartheta}|\geq(\alpha+B)\cos\frac{\vartheta}{2}$ for $\alpha,B\geq0$ gives $|\alpha_f+z^2\beta_f^2|\geq\cos\theta_0\,\min(1,r_1^2)\,(\alpha_f+\beta_f^2)$. The integrand is therefore dominated by a constant multiple of the integrable function of (a), and Ψ is holomorphic on $\Re z>0$ by Morera's theorem, Fubini's theorem and dominated convergence. For real $z>0$ the substitution $(x_2,y_2)\to(x_2,y_2)/z$ gives $\Psi(z)=\pi^2J_{2\tau}$, so Ψ is constant by the identity theorem.

(c) *The boundary value.* Let $z=e^{\mathrm{i}\theta}$ with $\theta\uparrow\frac{\pi}{2}$. Since $|\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2|\geq|\alpha_f-\beta_f^2|$, the integrand is dominated by $\Upsilon(\xi)\Upsilon(\eta)$ for every θ. Since $\Im(\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2)>0$ when $\beta_f\neq0$, almost everywhere $(\alpha_f+e^{2\mathrm{i}\theta}\beta_f^2)^{\kappa_f}\to|L_f(\xi)L_f(\eta)|^{\kappa_f}e^{\mathrm{i}\pi\kappa_f[L_f(\xi)L_f(\eta)<0]}$. Dominated convergence and $z^2\to-1$ give $\pi^2J_{2\tau}=-\frac14\iint\Upsilon(\xi)\Upsilon(\eta)\exp(\mathrm{i}\pi\sum_f\kappa_f[\varsigma_f(\xi)\neq\varsigma_f(\eta)])\,d\xi\,d\eta$. The phase is constant on each product of two cells, which proves (376).

This is the Kawai–Lewellen–Tye (KLT) factorization [43] behind (674), proved here directly at the point needed.

18.8.3 The master path as a sum of two squares

With $\omega_\tau=e^{\mathrm{i}\pi\tau}$ the phase in (376) is $(-\mathrm{i})^n(-\omega_\tau)^m$, where n counts the forms with exponent $-\frac12$ whose signs differ on R and R', and $m=1$ if the sign of $1-\xi_2$ differs and $m=0$ otherwise. Collecting the 144 terms with the table of (375), a finite exact computation (machine checked in `LRI_C5_d2_proof/K2_selberg/region_sum_matrix.py`), gives, with $\omega=\omega_\tau$,

$$\begin{aligned}\pi^2J_{2\tau}=4\big[&\mathrm{i}I_1I_2-\mathrm{i}\omega I_1I_4-\mathrm{i}I_1I_5+I_1I_6-\mathrm{i}\omega I_2I_3-\omega I_2I_4+I_2I_5\\&+\mathrm{i}I_3I_4+\mathrm{i}\omega I_3I_5-\omega I_3I_6+\mathrm{i}\omega I_4I_6+\mathrm{i}I_5I_6\big].\end{aligned}$$

Now insert the moment relations (77) at $t=\tau\in(0,1)$ and $\omega_\tau=c_\tau+\mathrm{i}s_\tau$. (The second contour in the proof of (77) also needs, on $x<0$, the boundary value $K(1-x-\mathrm{i}0)=(1-x)^{-1/2}[K(\frac{1}{1-x})-\mathrm{i}K(\frac{-x}{1-x})]$. It follows from the displayed formula for $K(x+\mathrm{i}0)$, $x>1$, by Schwarz reflection, and that proof uses it without displaying it.) All terms except $s_\tau^2(I_3^2+I_4^2)$ cancel; for instance, the coefficient of I_3I_4 becomes $2\mathrm{i}-2\omega_\tau(s_\tau+\mathrm{i}c_\tau)=0$. Euler's integral gives $I_3=\pi^2\mathfrak{F}(1-\tau)/(2s_\tau)$ and $I_4=\pi^2\mathfrak{F}(\tau)/(2s_\tau)$, because $B(\tau,1-\tau)=\pi/s_\tau$. Therefore, for $0<\tau<1$,

$$\boxed{J_{2\tau}=\pi^2\big[\mathfrak{F}(\tau)^2+\mathfrak{F}(1-\tau)^2\big]=\frac{4}{\pi^2}s_\tau^2(I_3^2+I_4^2)=\frac{4}{\pi^2}\big[s_\tau I_1I_4+I_2I_5\big].} \tag{377}$$

The last form is (674) at this point: with $a=b=A'=c=-\frac12$ and $B'=\tau-1$, the four factors $\mathcal{O}(a,b,A',B',c)$, $\mathcal{M}(a,b,A',B',c)$, $\mathcal{O}(A',B',a,b,c)$ and $\mathcal{M}(A',B',a,b,c)$ take the values $2I_1$, $2I_4$, $2s_\tau I_3=2I_2$ and $2I_5$. That identity is thereby proved here; it is not an input. Equation (377) is the two-dimensional partner of (78), with no cross term. It is symmetric under $t\to2-t$, as completion covariance requires. At $\tau=\frac12$ Dixon's sum [22] gives $\mathfrak{F}(\frac12)=\Gamma(\frac14)^4/(4\pi^3)$, so $J_1=\Gamma(\frac14)^8/(8\pi^4)=3Q_2$, in agreement with (267).

18.8.4 Laurent extraction

Write $\mathfrak{F}(\tau) = 1 + u\tau + v\tau^2 + O(\tau^3)$ and $\mathfrak{G}(\tau) = 1 + \sigma\tau^3 + \varrho\tau^4 + O(\tau^5)$, where ϱ is the coefficient called τ in (86) and $\mathfrak{G} - 1 = O(\tau^3)$ because every term $n \geq 1$ of $\mathfrak{G}$ carries $(\tau)_n^3$. By (82), $\pi\mathfrak{F}(1-\tau) = \tau^{-1}\mathcal{Q}(\tau)\mathfrak{G}(\tau)$ for $0 < \tau < 1$, with $\mathcal{Q}(\tau) = \pi\Gamma(1+\tau)/[\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)^2]$. The series of $\mathfrak{F}$ converges locally uniformly on $\Re\tau < 1$, its terms being $O(n^{\Re\tau-2})$; $\mathfrak{G}$ is holomorphic on $|\tau| < \frac{1}{2}$; and $\mathcal{Q}$ is holomorphic and zero-free there, with $\mathcal{Q}(0) = 1$. Hence

$$J_{2\tau} = \pi^2\mathfrak{F}(\tau)^2 + \tau^{-2}\mathcal{Q}(\tau)^2\mathfrak{G}(\tau)^2$$

is meromorphic on $|\tau| < \frac{1}{2}$, with a double pole at $\tau = 0$ only. By the identity theorem it is the continuation in which (238) takes its Laurent coefficient. At $d = 2$, (230) gives

$$J_{2\tau}^{\Gamma} = \mathcal{V}(\tau)^2 + \frac{\mathcal{V}(\tau) - \widetilde{\mathcal{V}}(\tau)}{\tau}, \qquad \mathcal{V} = \frac{\Gamma(\tau)\Gamma(\frac{1}{2}-\tau)}{\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)}, \quad \widetilde{\mathcal{V}} = \frac{\Gamma(\tau)\Gamma(\frac{1}{2}+\tau)\Gamma(\frac{1}{2}-2\tau)}{\Gamma(1-\tau)\Gamma(\frac{1}{2}-\tau)\Gamma(\frac{1}{2}+2\tau)},$$

which are $B_2(1, 2-2\tau)$ and $B_2(1+4\tau, 2-2\tau)$. The series $\log\Gamma(1+\tau) = -\gamma_E\tau + \sum_{k\geq 2}(-1)^k\zeta(k)\tau^k/k$ and $\log\Gamma(\frac{1}{2}+\tau) = \frac{1}{2}\log\pi + \sum_{k\geq 1}\psi_{k-1}(\frac{1}{2})\tau^k/k!$, with $\psi_0(\frac{1}{2}) = -\gamma_E - 2\log 2$, $\psi_1(\frac{1}{2}) = \frac{\pi^2}{2}$, $\psi_2(\frac{1}{2}) = -14\zeta(3)$ and $\psi_3(\frac{1}{2}) = \pi^4$ [32], give

$$\log\mathcal{Q}(\tau)^2 = 8\log 2\,\tau - \pi^2\tau^2 + 8\zeta(3)\tau^3 - \frac{\pi^4}{6}\tau^4 + O(\tau^5),$$
$$J_{2\tau}^{\Gamma} = \frac{e^{8\tau\log 2}}{\tau^2} - 20\zeta(3)\tau - 48\zeta(3)\log 2\,\tau^2 + O(\tau^3).$$

The last coefficient is four times the coefficient of t^2 in J_t^{Γ} stated after (250), evaluated at $q = \frac{1}{2}$. Write $\mathcal{Q}_k = [\tau^k]\mathcal{Q}(\tau)^2$; in particular $\mathcal{Q}_3 = \frac{256}{3}\log^3 2 - 8\pi^2\log 2 + 8\zeta(3)$ and $\mathcal{Q}_4 = \frac{512}{3}\log^4 2 - 32\pi^2\log^2 2 + 64\zeta(3)\log 2 + \frac{\pi^4}{3}$. The Laurent coefficients are

order	$J_{2\tau}$	$J_{2\tau}^{\Gamma}$
τ^{-2}	1	1
τ^{-1}	$8\log 2$	$8\log 2$
τ^0	$\pi^2 + \mathcal{Q}_2 = 32\log^2 2$	$32\log^2 2$
τ^1	$2\pi^2 u + 2\sigma + \mathcal{Q}_3$	$\frac{256}{3}\log^3 2 - 20\zeta(3)$
τ^2	$\pi^2(u^2 + 2v) + 2\varrho + 16\sigma\log 2 + \mathcal{Q}_4$	$\frac{512}{3}\log^4 2 - 48\zeta(3)\log 2$

The orders $\tau^{-2}, \tau^{-1}, \tau^0$ cancel identically. At order τ the difference is $2[\sigma + \pi^2 u - 4\pi^2\log 2 + 14\zeta(3)]$, which vanishes by (87); this is the two-dimensional instance of $J_t - J_t^{\Gamma} = O(t^2)$ in (237), here derived rather than assumed. At order τ^2 the coefficients v and ϱ enter only through $2(\varrho + \pi^2 v) = \pi^2 u^2 - \frac{\pi^4}{3}$, which is (86). Since $t = 2\tau$, $[t^2] = \frac{1}{4}[\tau^2]$, and

$$\mathcal{K}_2 = \frac{\pi^2}{2}u^2 + 4\sigma\log 2 - 8\pi^2\log^2 2 + 28\zeta(3)\log 2.$$

With $u = 4\log 2 - 8\mathsf{G}/\pi$ and $\sigma = 8\pi\mathsf{G} - 14\zeta(3)$ from (87), this is $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$. Eliminating σ with the order-τ relation instead gives $\mathcal{K}_2 = \frac{\pi^2}{2}(u-A)^2 + \frac{1}{2}AU$ with $\pi(A-u) = 8\mathsf{G}$, the second form of (271). All coefficient algebra here is exact and machine checked (`LRI_C5_d2_proof/K2_selberg/laurent_K2_sympy.py` and `LRI_C5_d2_proof/K2_selberg/laurent_K2.wl`).

The one-dimensional inputs (82), (86) and (87) concern $\mathfrak{F}$ and $\mathfrak{G}$ only and are proved in Section 8.3. In that proof (82) is also used at $-t$, with $\mathfrak{F}(1+t)$ continued by (80). This is legitimate: both sides of (82) are meromorphic in a punctured disk around $t = 0$ and agree for $0 < t < \frac{1}{2}$, so they agree there identically.

18.8.5 Further proofs and checks

Section 18.9 gives a second position-space proof of (377), by Riemann's bilinear relation (Section 18.9.8). It also proves at $d = 2$ that the spectral form (248) and the series (260) equal (271), and it proves the mixed-trace step for $0 < \tau < \frac{1}{2}$; its items (i)–(v) say where each statement is proved. Together with Section 18.8.4, these results show that at $d = 2$ the definition (238), the spectral form and the series agree; the agreement of the definition with the series does not use the mixed-trace step.

Checks. These are not proof steps. The closed form (377) agrees with a two-dimensional quadrature of the y-integral, with the inner star (398) inserted, to between 38 and 68 digits at seven values of τ. Direct nested quadratures of the twelve cell integrals agree with $2I_j$ to between 18 and 24 digits, and (250), evaluated with an independent spectral code, reproduces (377) to 54 digits at $\tau = 0.1$ and $\tau = 0.23$. At $\tau = \frac{3}{10}$ the twelve-term form above, with the moments computed by direct quadrature, agrees with $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2] = 47.790785413226103671\ldots$ to 37 digits.

18.9 Spectral and series forms of $\mathcal{K}_2$, the mixed-trace step, and a second position-space proof

Section 18.8 proves (271) for the master-path definition (238). This subsection proves, at $d = 2$, the statements that connect that definition with the spectral form (248) and the series (260).

(i) For $0 < \tau < \frac{1}{2}$ the two regulated traces of (250) are evaluated in closed form: $C(\tau)\langle t_\ell^2 R_\tau\rangle_2 = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ and $\langle t_\ell^2 S_\tau\rangle_2 = C(\tau)\mathcal{X}_2(\tau)^2$ (Section 18.9.5).

(ii) With (377) this proves the mixed-trace representation $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_2$ stated before (250), at $d = 2$, for $0 < \tau < \frac{1}{2}$ and in the full-Mellin-line reading (Section 18.9.5).

(iii) The coefficient $[\tau^2]$ may be taken under the trace, and the right side of (248) at $d = 2$ equals $32\mathsf{G}^2 - 28\zeta(3)\log 2$; as a by-product, $\langle t_\ell^2 E_2\rangle_2 = -21\zeta(3)$ (Sections 18.9.2 and 18.9.6).

(iv) Each inner n sum of (260) at $d = 2$ equals the corresponding spin term of (248), and the spin sum converges absolutely. Hence (260) equals (271), and the strip limit used for the 70-digit evaluation of Section 17.2 computes exactly this ordered sum (Section 18.9.7).

(v) A second proof of (377) from the position-space integral evaluates the inner star by Riemann's bilinear relation and the outer integral by a Stokes pairing on the slit plane (Section 18.9.8).

All five statements are proved here. Besides standard theorems (Stirling's formula, Gamma reflection, Gauss's and Thomae's unity-argument identities, the residue theorem, Liouville's theorem, Green's theorem, dominated convergence and the identity theorem), the proofs use only one-dimensional lemmas of Section 8: the Euler integral for $E(a, b)$, the moment relations (77), the Thomae relation (82), and the Taylor data (87) and (86). These concern the elliptic moments and the two functions $\mathfrak{F}, \mathfrak{G}$ only, and do not depend on the dimension. No numerical identification enters. The results are specific to $d = 2$: at other dimensions the mixed-trace step remains input (M) of Section 21.1, and the interchange of $[\tau^2]$ with the trace is not proved.

The material comes from two packages, both written by Claude Code sessions:

- package `LRI_C5_d2_simplify/` (folders `proof/` and `proof/K2-parseval/`), by the session that wrote the third and fourth proofs of (270): the regularized spectral route, Sections 18.9.2–18.9.6, apart from the parts listed in the next item;

- package `LRI_C5_d2_proof/`, by the session that wrote the first proof of Section 18.7 and the proof of Section 18.8: in folder `K2_gauss/`, the sharper estimate at the end of Section 18.9.2, the second evaluation of the single sums (Section 18.9.4), a second derivation of the Laurent extraction of Section 18.9.6 and the per-spin identity of Section 18.9.7 with its arc bounds; in folders `K2_d1template/` and `K2_d1template_fix1/`, the revised contour bounds used in Section 18.9.3, the third evaluation of the single sums, the mixed-trace corollary of Section 18.9.5 and the Riemann bilinear proof of Section 18.9.8.

Both packages contain the absolute convergence of the series and the strip limit of Section 18.9.7.

18.9.1 Conventions at $d = 2$

Throughout, $d = 2$ and $q = \frac{1}{2}$. Then $a_\ell = \frac{\ell}{2} + \frac{1}{4}$, $b_\ell = \frac{\ell}{2} + \frac{3}{4}$, $d_0 = 1$, $d_\ell = 2$ for $\ell \geq 1$, and $\Gamma(2q) = 1$, so $t_\ell(\nu) = g_\ell(\mathrm{i}\nu)$ with the kernel g_ℓ of (300). At fixed spin we write $a = a_\ell$, $b = b_\ell$; then $b - a = \frac{1}{2}$ and $a + b = \ell + 1$. The quantities of (249) are

$$C(\tau) = \frac{\Gamma(\tau)}{\Gamma(1-\tau)} = \frac{1}{\tau} + c_* + O(\tau), \qquad c_* = 2\psi_0(1), \qquad \mathcal{X}_2(\tau) = \Big[\frac{\Gamma(\frac{1}{2} - \frac{\tau}{2})}{\Gamma(\frac{1}{2} + \frac{\tau}{2})}\Big]^2,$$

where $\mathcal{X}_2$ is $\mathcal{X}_d$ of (555) at $d = 2$. The constants of (221) are $A = 4\log 2$ and $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$, so $-AU/4 = 14\zeta(3)\log 2$. We write

$$w_{\mathcal{K}} = (c_* - p)E_2 - \frac{hk}{2} - \frac{5r}{32}$$

for the weight in (248), with p, h, j, k, r, E_2 of (247). Since h, k are odd in ν and p, j, r are even, E_2 and $w_{\mathcal{K}}$ are even in ν.

The kernel $R_\tau(s)$ is not even in s. As in (554), every trace in this subsection is read on the full Mellin line,

$$\langle f \rangle_2 = \frac{1}{2\pi} \sum_{\ell \geq 0} d_\ell \int_{\mathbb{R}} f(\ell, \nu)\, d\nu.$$

For f even in ν this is (241). For real τ, $R_\tau(-\mathrm{i}\nu) = \overline{R_\tau(\mathrm{i}\nu)}$, so $\langle t_\ell^2 R_\tau \rangle_2$ is $\frac{1}{\pi}\sum_\ell d_\ell \int_0^\infty \Re\{t_\ell^2 R_\tau\}\, d\nu$, the real-part reading used in (250). We put $s_\tau = \sin \pi\tau$ and $c_\tau = \cos \pi\tau$, as in Section 18.8, and $\xi = \frac{\ell}{2} + \mathrm{i}\nu$, $\bar{\xi} = \frac{\ell}{2} - \mathrm{i}\nu$ (not the real points ξ of Section 18.8), so that $\Re\xi = \ell/2 \geq 0$, $a \pm \mathrm{i}\nu = \frac{1}{4} + \xi$ or $\frac{1}{4} + \bar{\xi}$, and $b \pm \mathrm{i}\nu = \frac{3}{4} + \xi$ or $\frac{3}{4} + \bar{\xi}$. The functions $\mathfrak{F}, \mathfrak{G}$ are those of (75), with the Taylor data of Section 18.8.4:

$$\begin{aligned} &\mathfrak{F}(\tau) = 1 + u\tau + v\tau^2 + O(\tau^3), \qquad \mathfrak{G}(\tau) = 1 + \sigma\tau^3 + \varrho\tau^4 + O(\tau^5), \\ &u = 4\log 2 - \frac{8\mathsf{G}}{\pi}, \qquad \sigma = 8\pi\mathsf{G} - 14\zeta(3), \qquad \varrho + \pi^2 v = \frac{\pi^2 u^2}{2} - \frac{\pi^4}{6}. \end{aligned} \tag{378}$$

Here ϱ is the coefficient called τ in (86); the letter τ is the regulator throughout this subsection.

18.9.2 Analytic regularization in the rim exponent

Lemma (uniform Gamma ratios). *Let $\Lambda_0 \subset \mathbb{C}$ be compact and $c_0 > 0$. There is C_0 such that for all $\alpha, \beta \in \Lambda_0$ and all z with $\Re z \geq 0$ and $\Re(z + \alpha) \geq c_0$,*

$$\Big|\frac{\Gamma(z+\alpha)}{\Gamma(z+\beta)} - (1+z)^{\alpha-\beta}\Big| \leq \frac{C_0\,|(1+z)^{\alpha-\beta}|}{1+|z|}, \qquad \Big|\frac{\Gamma(z+\alpha)}{\Gamma(z+\beta)}\Big| \leq C_0(1+|z|)^{\Re(\alpha-\beta)}.$$

The same holds for z in a closed sector $|\arg z| \leq \pi - \delta$, $|z| \geq R$, after enlarging C_0 and R.

Proof. For $|z|$ large, $z + \alpha$ and $z + \beta$ lie in a sector $|\arg w| \leq \pi - \delta'$, where $\log\Gamma(w) = (w - \frac{1}{2})\log w - w + \frac{1}{2}\log 2\pi + O(|w|^{-1})$ holds uniformly [27]. Expanding gives $\log\Gamma(z+\alpha) - \log\Gamma(z+\beta) = (\alpha - \beta)\log(1+z) + O(|z|^{-1})$ uniformly in $\alpha, \beta \in \Lambda_0$. On the remaining compact set of z both sides are continuous, because $1/\Gamma$ is entire and $\Re(z + \alpha) \geq c_0$. □

Lemma (majorant). *Fix $0 < r < \frac{1}{2}$. There is C_r such that for all $\ell \geq 0$, $\nu \in \mathbb{R}$ and complex $|\tau| \leq r$,*

$$\left|t_\ell(\nu)^2 \left[R_\tau(\mathrm{i}\nu) - S_\tau(\mathrm{i}\nu)\right]\right| \leq C_r(1+|\xi|)^{-3+2r}. \tag{379}$$

Hence $\mathcal{D}(\tau) = \langle t_\ell^2(R_\tau - S_\tau)\rangle_2$ converges absolutely and uniformly on $|\tau| \leq r$, and $\mathcal{D}$ is holomorphic in $|\tau| < \frac{1}{2}$. The function $\mathcal{D}$ is $F(\tau)$ of (555) at $d = 2$.

Proof. With $\phi(x) = \Gamma(\frac{1}{4}+x)/\Gamma(\frac{3}{4}+x)$ one has $t_\ell = \phi(\xi)\phi(\bar{\xi})$,

$$R_\tau(\mathrm{i}\nu) = \frac{\Gamma(\frac{1}{4}+\xi)}{\Gamma(\frac{1}{4}+\tau+\xi)} \cdot \frac{\Gamma(\frac{3}{4}-\tau+\bar{\xi})}{\Gamma(\frac{3}{4}+\bar{\xi})}, \qquad S_\tau(\mathrm{i}\nu) = \vartheta_\tau(\xi)\vartheta_\tau(\bar{\xi}),$$
$$\vartheta_\tau(x) = \Big[\frac{\Gamma(\frac{1}{4}-\frac{\tau}{4}+x)\,\Gamma(\frac{3}{4}+x)}{\Gamma(\frac{1}{4}+x)\,\Gamma(\frac{3}{4}+\frac{\tau}{4}+x)}\Big]^2.$$

Every Gamma function in a numerator has an argument with real part at least $\frac{1}{4} - \frac{r}{4} > 0$. The lemma on uniform Gamma ratios gives $|t_\ell|^2 \leq C(1+|\xi|)^{-2}$ and

$$R_\tau(\mathrm{i}\nu),\ S_\tau(\mathrm{i}\nu) = (1+\xi)^{-\tau}(1+\bar{\xi})^{-\tau}\Big(1 + O\Big(\frac{1}{1+|\xi|}\Big)\Big)$$

uniformly in $|\tau| \leq r$. Since $\arg(1+\bar{\xi}) = -\arg(1+\xi)$, the product $(1+\xi)^{-\tau}(1+\bar{\xi})^{-\tau}$ equals $|1+\xi|^{-2\tau}$, whose modulus is at most $(1+|\xi|)^{2r}$. Hence

$$R_\tau(\mathrm{i}\nu) - S_\tau(\mathrm{i}\nu) = |1+\xi|^{-2\tau}\, O\big((1+|\xi|)^{-1}\big) \qquad \text{uniformly in } |\tau| \leq r, \tag{380}$$

and (379) follows. The majorant is summable: $\int_{\mathbb{R}}(1+|\frac{\ell}{2}+\mathrm{i}\nu|)^{-3+2r}\,d\nu \leq C(1+\ell)^{-2+2r}$ and $\sum_\ell (1+\ell)^{-2+2r} < \infty$. The integrands are holomorphic in τ, so dominated convergence and Morera's theorem give holomorphy. □

Corollary (regularized form). *(i) $\mathcal{D}(\tau) = \mathcal{D}_2\tau^2 + \mathcal{D}_3\tau^3 + O(\tau^4)$, with $\mathcal{D}_2 = \langle t_\ell^2 E_2\rangle_2$ and $\mathcal{D}_3 = \langle t_\ell^2[-pE_2 - \frac{hk}{2} - \frac{5r}{32}]\rangle_2$. (ii) $C(\tau)\mathcal{D}(\tau)$ is holomorphic in $|\tau| < \frac{1}{2}$, and $[\tau^2]\{C(\tau)\mathcal{D}(\tau)\} = \langle t_\ell^2 w_{\mathcal{K}}\rangle_2$. Hence the right side of* (248) *at $d = 2$ is*

$$-\frac{AU}{4} + \frac{1}{4}\langle t_\ell^2 w_{\mathcal{K}}\rangle_2 = 14\zeta(3)\log 2 + \frac{1}{4}[\tau^2]\{C(\tau)\mathcal{D}(\tau)\}. \tag{381}$$

(iii) For real $0 < \tau < \frac{1}{2}$, $\mathcal{D}(\tau) = \langle t_\ell^2 R_\tau\rangle_2 - \langle t_\ell^2 S_\tau\rangle_2$, and both traces converge absolutely.

Proof. (i) By Cauchy's formula on $|\tau| = r$ and Fubini's theorem, justified by (379), $[\tau^n]\mathcal{D} = \langle t_\ell^2\,[\tau^n](R_\tau - S_\tau)\rangle_2$. Taylor expansion of the Gamma functions gives

$$\log R_\tau(s) = \sum_{k\geq 1} \frac{(-\tau)^k\psi_{k-1}(b-s) - \tau^k\psi_{k-1}(a+s)}{k!},$$
$$\log S_\tau(s) = 2\sum_{k\geq 1} \frac{(-\frac{\tau}{4})^k\{\psi_{k-1}(a+s)+\psi_{k-1}(a-s)\} - (\frac{\tau}{4})^k\{\psi_{k-1}(b+s)+\psi_{k-1}(b-s)\}}{k!}.$$

Expanding the exponentials to third order and symmetrizing under $s \to -s$ gives the finite polynomial identity

$$\tfrac{1}{2}\big[(R_\tau - S_\tau)(s) + (R_\tau - S_\tau)(-s)\big] = E_2\tau^2 + \Big[-pE_2 - \frac{hk}{2} - \frac{5r}{32}\Big]\tau^3 + O(\tau^4),$$

with p, h, j, k, r of (247) written in the polygamma values at $a \pm s$, $b \pm s$. This is the expansion displayed after (250). It is checked in exact computer algebra, at $q = \frac{1}{2}$ in `LRI_C5_d2_simplify/proof/K2-parseval/symbolic_checks.py` (part A) and for arbitrary real a, b in `LRI_C5_d2_proof/K2_gauss/check_expansion.wl`. The parts odd in ν integrate to zero in the full-line trace.

(ii) $C(\tau)=\tau^{-1}\Gamma(1+\tau)/\Gamma(1-\tau)$ has a simple pole with residue 1, so $[\tau^2](C\mathcal{D})=\mathcal{D}_3+c_*\mathcal{D}_2=\langle t_\ell^2 w_{\mathcal{K}}\rangle_2$. (iii) For real $\tau\in(0,\frac12)$ the lemma on uniform Gamma ratios gives $|t_\ell^2 R_\tau|, |t_\ell^2 S_\tau|\le C(1+|\xi|)^{-2-2\tau}$, which is summable. □

Part (i) supplies, at $d=2$, the interchange of $[\tau^2]$ with the trace that the derivation of (248) after (250) uses.

A sharper estimate for the even part. The bound (379) does not use the cancellation in the even part $D^{\mathrm{e}}_\tau(\nu)=\frac12[R_\tau(\mathrm{i}\nu)+R_\tau(-\mathrm{i}\nu)]-S_\tau(\mathrm{i}\nu)$. Fix $\rho_0=\frac{1}{32}$ and write $z=a+\mathrm{i}\nu$. There is C such that, for all $\ell\ge0$, $\nu\in\mathbb{R}$ and $|\tau|\le\rho_0$,

$$|t_\ell(\nu)|\le C(1+|z|)^{-1},\qquad |D^{\mathrm{e}}_\tau(\nu)|\le C|\tau|^2(1+|z|)^{-2+2\rho_0}. \tag{382}$$

Proof. The first bound is $|\Gamma(w)/\Gamma(w+\frac12)|\le C|w|^{-1/2}$ for $\Re w\ge\frac14$ [27], with compactness. For the second, all arguments $w\in\{a\pm s, b\pm s\}$ have $\Re w\ge\frac14>\rho_0$, so the series $\log\Gamma(w+\alpha\tau)-\log\Gamma(w)=\sum_{k\ge1}\psi_{k-1}(w)(\alpha\tau)^k/k!$ with $|\alpha|\le1$ converge. The polygamma functions obey $|\psi_{k-1}(w)|\le(k-1)!\sum_{n\ge0}|w+n|^{-k}\le(k-1)!\,4^kC_0(1+|w|)^{-2}$ for $k\ge3$, and $\psi_1(w)-\psi_1(w+\frac12)=O(|w|^{-2})$, $\psi_{k-1}(w)=O(|w|^{1-k})$, $\psi_0(w)=\log w+O(|w|^{-1})$. Put $A_k=\psi_{k-1}(a+s)+\psi_{k-1}(a-s)$, $B_k=\psi_{k-1}(b+s)+\psi_{k-1}(b-s)$, $\Delta_{\mathrm{e}}=\frac12[\log R_\tau(s)+\log R_\tau(-s)]-\log S_\tau(s)$ and $\varepsilon_{\mathrm{o}}=\frac12[\log R_\tau(s)-\log R_\tau(-s)]$. The two expansions above give

$$[\tau^k]\Delta_{\mathrm{e}}=\frac{1}{k!}\Big[-\big(\tfrac12+2(-4)^{-k}\big)A_k+\big(\tfrac{(-1)^k}{2}+2\cdot4^{-k}\big)B_k\Big].$$

The coefficient vanishes for $k=1$; it is $-\frac{5}{16}(A_2-B_2)=O(|z|^{-2})$ for $k=2$ and $-\frac{15}{192}(A_3+B_3)=O(|z|^{-2})$ for $k=3$; and it is $O((8^k/k)|z|^{-2})$ for $k\ge4$. Hence $|\Delta_{\mathrm{e}}|\le C|\tau|^2|z|^{-2}$. Similarly $|\varepsilon_{\mathrm{o}}|\le C|\tau||z|^{-1}$; its τ^1 coefficient is $\frac12[(\psi_0(b+s)-\psi_0(a+s))-(\psi_0(b-s)-\psi_0(a-s))]$. Finally $|\log S_\tau|\le2|\tau|\log(2+|z|)+C|\tau|$. Since $D^{\mathrm{e}}_\tau=S_\tau(e^{\Delta_{\mathrm{e}}}\cosh\varepsilon_{\mathrm{o}}-1)$, it is bounded by $C(2+|z|)^{2\rho_0}(|\Delta_{\mathrm{e}}|+|\varepsilon_{\mathrm{o}}|^2)$, which proves (382) for $|z|\ge z_0$. For $|z|\le z_0$ (finitely many spins and bounded ν) compactness and $D^{\mathrm{e}}_\tau=O(\tau^2)$ suffice. □

Consequently $C(\tau)D^{\mathrm{e}}_\tau$, whose pole at $\tau=0$ is cancelled, is holomorphic on $|\tau|\le\rho_0$, and by the maximum principle $|t_\ell^2C(\tau)D^{\mathrm{e}}_\tau|\le C(1+|z|)^{-4+2\rho_0}$ there. Since $[\tau^2]\{C(\tau)D^{\mathrm{e}}_\tau\}=w_{\mathcal{K}}$ (by the expansion in the proof of part (i) of the corollary and $C(\tau)=\tau^{-1}+c_*+O(\tau)$), Cauchy's estimate gives $|t_\ell^2 w_{\mathcal{K}}|\le C(1+|z|)^{-4+2\rho_0}$, so the spin term of (248) at $d=2$ is $O(\ell^{-3+2\rho_0})$, consistent with the decay $O(\ell^{-3}\log\ell)$ stated after (248).

18.9.3 Residue families on the square lattice

For $0<\tau<\frac12$ define the lattice functions (the sans-serif letters are local to this subsection)

$$\begin{aligned}
\mathsf{h}(x)&=\frac{\Gamma(x+\frac12)^3}{\Gamma(x+1)^2\,\Gamma(x+\frac12+\tau)},&\quad \mathsf{k}(x)&=\frac{\Gamma(x+1-\tau)^3}{\Gamma(x+\frac32-\tau)^2\,\Gamma(x+1)},\\
\mathsf{m}(x)&=\Big[\frac{\Gamma(x+\frac{1-\tau}{2})}{\Gamma(x+1)}\Big]^2,&\quad \mathsf{n}(x)&=\frac{\Gamma(x+\frac12)^2\,\Gamma(x+1-\tau)}{\Gamma(x+1)^3}.
\end{aligned} \tag{383}$$

For $x\ge0$, Stirling's formula gives $\mathsf{h},\mathsf{k},\mathsf{m},\mathsf{n}=O((1+x)^{-1-\tau})$. Since h'/h and m'/m are combinations of digamma differences, which are $O(1/(1+x))$, also $\mathsf{h}',\mathsf{m}'=O((1+x)^{-2-\tau})$. Note the shift relation $\mathsf{h}(x+\frac12-\tau)=\mathsf{k}(x)$. The single sums

$$\begin{aligned}
\Sigma_{\mathsf{h}}&=\sum_{m\ge0}\mathsf{h}(m),&\quad \Sigma'_{\mathsf{h}}&=\sum_{m\ge0}\mathsf{h}'(m),&\quad \Sigma_{\mathsf{k}}&=\sum_{m\ge0}\mathsf{k}(m),\\
\Sigma_{\mathsf{m}}&=\sum_{m\ge0}\mathsf{m}(m),&\quad \Sigma'_{\mathsf{m}}&=\sum_{m\ge0}\mathsf{m}'(m),&\quad \Sigma_{\mathsf{n}}&=\sum_{k\ge0}\mathsf{n}(k)
\end{aligned}$$

therefore converge absolutely for $\tau > 0$. At fixed spin the two integrands are

$$\Theta^{\mathrm{R}}_\ell(s) = g_\ell(s)^2 R_\tau(s) = \frac{\Gamma(a+s)^3\,\Gamma(a-s)^2\,\Gamma(b-\tau-s)}{\Gamma(b+s)^2\,\Gamma(b-s)^3\,\Gamma(a+\tau+s)}, \qquad \Theta^{\mathrm{S}}_\ell(s) = g_\ell(s)^2 S_\tau(s) = g_\ell(s;1-\tau)^2,$$

where $g_\ell(s;1-\tau)$ is the balanced kernel of rim exponent $1-\tau$ of (557), that is $\Gamma(a'+s)\Gamma(a'-s)/[\Gamma(b'+s)\Gamma(b'-s)]$ with $a' = a - \frac{\tau}{4}$ and $b' = b + \frac{\tau}{4}$. Thus $\langle t_\ell^2 R_\tau\rangle_2 = \sum_\ell d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}} \Theta^{\mathrm{R}}_\ell(s)$, and similarly for S_τ.

Lemma (per-spin residues). *Let $\ell \geq 0$, $0 < \tau < \frac{1}{2}$ and $m = \ell + n$. Then, with absolutely convergent series,*

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{R}}_\ell(s) = \sum_{n\geq 0}\big[\rho_{\mathsf{h}}(m,n) + \rho_{\mathsf{k}}(m,n)\big], \qquad \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{S}}_\ell(s) = \sum_{n\geq 0}\rho_{\mathsf{m}}(m,n), \tag{384}$$

$$\rho_{\mathsf{h}}(m,n) = -\frac{1}{\pi^2 c_\tau}\Big[\mathsf{h}'(m)\mathsf{h}(n) + \mathsf{h}(m)\mathsf{h}'(n) + \pi\tan(\pi\tau)\,\mathsf{h}(m)\mathsf{h}(n)\Big], \qquad \rho_{\mathsf{k}}(m,n) = \frac{s_\tau^3}{\pi c_\tau^2}\,\mathsf{k}(m)\mathsf{k}(n),$$

$$\rho_{\mathsf{m}}(m,n) = -\frac{1}{\pi^2}\Big[\cos^2(\tfrac{\pi\tau}{2})\big(\mathsf{m}'(m)\mathsf{m}(n) + \mathsf{m}(m)\mathsf{m}'(n)\big) + \pi s_\tau\,\mathsf{m}(m)\mathsf{m}(n)\Big].$$

Proof. Poles. In $\Re s > 0$ the poles of Θ^{R}_ℓ are the double poles $s = a+n$, from $\Gamma(a-s)^2$, and the simple poles $s = b-\tau+n$, from $\Gamma(b-\tau-s)$, $n \geq 0$; the other factors are finite and nonzero there because $\tau \notin \frac{1}{2}+\mathbb{Z}$ and $\tau \notin \mathbb{Z}$. The two families are disjoint because $b-\tau-a = \frac{1}{2}-\tau \notin \mathbb{Z}$, and none lies on $\Re s = 0$, since $a > 0$ and $b-\tau \geq \frac{3}{4}-\tau > 0$. The factors with left poles are holomorphic on $\Re s \geq 0$.

Arcs. Gamma reflection of the three factors with right poles, together with $\sin\pi(b-s) = \cos\pi(a-s)$, gives the identity $\Theta^{\mathrm{R}}_\ell = G_\ell\,T_\tau$, with

$$G_\ell(s) = \Big[\frac{\Gamma(a+s)\Gamma(1-b+s)}{\Gamma(b+s)\Gamma(1-a+s)}\Big]^2 \frac{\Gamma(a+s)\Gamma(1-b+s)}{\Gamma(a+\tau+s)\Gamma(1-b+\tau+s)},$$
$$T_\tau(s) = \cot^2\pi(a-s)\,\frac{\cos\pi(a-s)}{\sin\pi(b-\tau-s)}.$$

For fixed ℓ, G_ℓ is holomorphic on $\{\Re s \geq 0,\ |s| \geq b\}$; there every Gamma argument stays in a sector $|\arg| \leq \frac{3}{4}\pi$, and Stirling's formula gives $|G_\ell(s)| \leq C_\ell |s|^{-2-2\tau}$. Inside $|s| < b$ the factors have poles and zeros that cancel pairwise in Θ^{R}_ℓ: the triple poles of $\Gamma(1-b+s)^3$ at $s = b-1-k$ against the triple zeros of $\cot^2\pi(a-s)\cos\pi(a-s)$; the double poles of $\cot^2\pi(a-s)$ at $s = a-k$, $k \geq 1$, against the double zeros of $1/\Gamma(1-a+s)^2$; and the poles of $1/\sin\pi(b-\tau-s)$ at $s = b-\tau-k$, $k \geq 1$, against the zeros of $1/\Gamma(1-b+\tau+s)$. The residue theorem is therefore applied to Θ^{R}_ℓ itself. The factor T_τ is 1-periodic and bounded on $|\Im s| \geq 1$, and it is bounded on the circles $|s| = \rho_N := a + N + \frac{3}{4} - \frac{\tau}{2}$, which stay at distance at least $\frac{1}{4} - O(\rho_N^{-1})$ from both pole families. So $|\Theta^{\mathrm{R}}_\ell| \leq C\rho_N^{-2-2\tau}$ on the right half-circles, and the arcs, together with the parts of the Mellin line beyond ρ_N (where $|\Theta^{\mathrm{R}}_\ell(\mathrm{i}\nu)| \leq C(1+|\nu|)^{-2-2\tau}$), contribute $O(\rho_N^{-1-2\tau})$. The right half-discs are traversed clockwise, so

$$\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\Theta^{\mathrm{R}}_\ell = -\sum_{n\geq 0}\Big[\mathrm{Res}_{s=a+n} + \mathrm{Res}_{s=b-\tau+n}\Big]\Theta^{\mathrm{R}}_\ell,$$

the residue series converging absolutely by the bounds below. (The number of poles of each family inside a given arc is N or $N+1$; this does not matter, since the series converges absolutely.)

Residues. At $s = a+n+u$ one has $a+s = m+\frac{1}{2}+u$, $b+s = m+1+u$ and $a+\tau+s = m+\frac{1}{2}+\tau+u$, while reflection gives

$$\frac{\Gamma(a-s)^2\,\Gamma(b-\tau-s)}{\Gamma(b-s)^3} = \frac{\cos^3\pi u}{\sin^2\pi u\,\cos\pi(\tau+u)}\,\mathsf{h}(n+u),$$

the signs $(-1)^{2n}$, $(-1)^n$ and $(-1)^{3n}$ of the three reflected factors cancelling. Hence $\Theta^{\mathrm{R}}_\ell = (\pi u)^{-2}\Psi_\tau(u)\,\mathsf{h}(m+u)\mathsf{h}(n+u)$ with $\Psi_\tau(u) = (\pi u)^2\cos^3\pi u/[\sin^2\pi u\cos\pi(\tau+u)] = c_\tau^{-1}(1+\pi u\tan\pi\tau+O(u^2))$, and the residue is $\pi^{-2}[c_\tau^{-1}(\mathsf{h}'(m)\mathsf{h}(n)+\mathsf{h}(m)\mathsf{h}'(n))+\pi s_\tau c_\tau^{-2}\mathsf{h}(m)\mathsf{h}(n)] = -\rho_{\mathsf{h}}(m,n)$. At $s = b-\tau+n+u$ only $\Gamma(b-\tau-s) = \Gamma(-n-u)$ is singular, with residue $-(-1)^n/n!$ in s. At $u=0$ the remaining factors are $\Gamma(a+s)^3/[\Gamma(b+s)^2\Gamma(a+\tau+s)] = \mathsf{k}(m)$ and

$$\frac{\Gamma(a-s)^2}{\Gamma(b-s)^3} = \frac{\Gamma(\tau-\frac12-n)^2}{\Gamma(\tau-n)^3} = (-1)^n\frac{s_\tau^3}{\pi c_\tau^2}\,\frac{\Gamma(n+1-\tau)^3}{\Gamma(n+\frac32-\tau)^2},$$

so the residue is $-(s_\tau^3/\pi c_\tau^2)\mathsf{k}(m)\mathsf{k}(n) = -\rho_{\mathsf{k}}(m,n)$.

The S family. The function $\Theta^{\mathrm{S}}_\ell = g_\ell(s;1-\tau)^2$ has in $\Re s>0$ only the double poles $s = a'+n$. Put $x_\tau = \frac{1-\tau}{2}$; then $2a' = \ell+x_\tau$, $a'+b' = \ell+1$ and $b'-a' = 1-x_\tau$. At $s = a'+n+u$, reflection gives

$$g_\ell(s;1-\tau) = -\frac{\sin\pi(x_\tau+u)}{\sin\pi u}\,\frac{\Gamma(m+x_\tau+u)\,\Gamma(n+x_\tau+u)}{\Gamma(m+1+u)\,\Gamma(n+1+u)},$$
$$\Theta^{\mathrm{S}}_\ell = \frac{\sin^2\pi(x_\tau+u)}{\sin^2\pi u}\,\mathsf{m}(m+u)\mathsf{m}(n+u).$$

Since $\sin^2\pi(x_\tau+u)/\sin^2\pi u = (\pi u)^{-2}[\sin^2\pi x_\tau+\pi u\sin 2\pi x_\tau+O(u^2)]$, $\sin\pi x_\tau = \cos\frac{\pi\tau}{2}$ and $\sin 2\pi x_\tau = s_\tau$, the residue is $\pi^{-2}[\pi s_\tau\mathsf{m}(m)\mathsf{m}(n)+\cos^2(\frac{\pi\tau}{2})(\mathsf{m}'(m)\mathsf{m}(n)+\mathsf{m}(m)\mathsf{m}'(n))] = -\rho_{\mathsf{m}}(m,n)$. On the circles $|s| = a'+N+\frac12$, at distance $\frac12$ from the poles, reflection and Stirling's formula give $|g_\ell(s;1-\tau)|\le C|s|^{-1-\tau}$, so the arcs vanish and the second identity of (384) follows as the first. Absolute convergence follows from $|\rho_{\mathsf{h}}|,|\rho_{\mathsf{k}}|,|\rho_{\mathsf{m}}|\le C(1+m)^{-1-\tau}(1+n)^{-1-\tau}$. □

Proposition (factorized traces). *For $0<\tau<\frac12$,*

$$\begin{aligned}\langle t_\ell^2 R_\tau\rangle_2 &= -\frac{1}{\pi^2 c_\tau}\Big[2\Sigma_{\mathsf{h}}\Sigma'_{\mathsf{h}}+\pi\tan(\pi\tau)\,\Sigma^2_{\mathsf{h}}\Big]+\frac{s_\tau^3}{\pi c_\tau^2}\,\Sigma^2_{\mathsf{k}},\\ \langle t_\ell^2 S_\tau\rangle_2 &= -\frac{1}{\pi^2}\Big[2\cos^2(\tfrac{\pi\tau}{2})\,\Sigma_{\mathsf{m}}\Sigma'_{\mathsf{m}}+\pi s_\tau\,\Sigma^2_{\mathsf{m}}\Big].\end{aligned} \tag{385}$$

Proof. If ρ is symmetric on $\mathbb{Z}^2_{\ge0}$ and absolutely summable, then $\sum_{\ell\ge0}d_\ell\sum_{n\ge0}\rho(\ell+n,n) = \sum_{m,n\ge0}\rho(m,n)$: the spin ℓ labels the diagonal $m-n=\ell$, $d_0=1$ counts the main diagonal, and $d_\ell = 2$ counts the two diagonals $m-n=\pm\ell$, which carry equal sums. Apply this to $\rho_{\mathsf{h}}+\rho_{\mathsf{k}}$ and to ρ_{m}, which are symmetric and absolutely summable for $\tau>0$, and factor the absolutely convergent double sums. □

This is the spectral form, at $\tau>0$, of the square-lattice factorization (336). With $\hat{\mathsf{h}}(z) = \sum_m \mathsf{h}(m)z^m$ and similarly for the other sequences, the Fourier series $\sum_{m,n}(\rho_{\mathsf{h}}+\rho_{\mathsf{k}})(m,n)e^{\mathrm{i}(m-n)\theta}$ is the same bilinear combination of $\hat{\mathsf{h}},\hat{\mathsf{h}}',\hat{\mathsf{k}}$ and their conjugates at $z=e^{\mathrm{i}\theta}$. For $\tau>0$ all coefficient sequences are absolutely summable, these functions are continuous on the closed disk, and the strip limit $\theta\to0$ is evaluation at $z=1$. At $\tau=0$ the same limit needs the logarithmic singular parts of the single sums (Section 18.9.7).

18.9.4 The single sums in closed form

Lemma (two-mode periodization). *For $0<\tau<\frac12$,*

$$\Sigma'_{\mathsf{h}} = -\pi s_\tau\,\Sigma_{\mathsf{n}},\qquad \Sigma_{\mathsf{h}} = c_\tau\,\Sigma_{\mathsf{n}}+s_\tau^2\,\Sigma_{\mathsf{k}}. \tag{386}$$

Proof. Bilateral sum. For $k\ge0$, reflection gives identically in η

$$\mathsf{h}(-k-1+\eta) = T_{\mathsf{h}}(\eta)\,\mathsf{n}(k+\tfrac12-\eta),\qquad T_{\mathsf{h}}(\eta) = \frac{\sin^2(\pi\eta)\,\cos\pi(\eta+\tau)}{\cos^3(\pi\eta)},$$

since the three factors $\Gamma(-k-\frac12+\eta)^3$, $\Gamma(-k+\eta)^{-2}$ and $\Gamma(-k-\frac12+\tau+\eta)^{-1}$ produce the signs $(-1)^{3k+3}$, $(-1)^{2k}$ and $(-1)^{k+1}$, which cancel. The series $\sum_{m\ge0}\mathsf{h}(m+\eta)$ and $\sum_{k\ge0}\mathsf{n}(k+\frac12-\eta)$ converge locally uniformly and are holomorphic for $\Re\eta>-\frac12$ and $\Re\eta<1$ respectively, because $\mathsf{h},\mathsf{n}=O(|x|^{-1-\tau})$ and the poles of n lie at $-\frac12-\mathbb{Z}_{\ge0}$ and $-1+\tau-\mathbb{Z}_{\ge0}$. Hence the bilateral sum $\Sigma^{\mathbb{Z}}_{\mathsf{h}}(\eta)=\sum_{m\in\mathbb{Z}}\mathsf{h}(m+\eta)$ satisfies, on the strip $-\frac12<\Re\eta<1$,

$$\Omega_{\mathsf{h}}(\eta):=\cos^3(\pi\eta)\,\Sigma^{\mathbb{Z}}_{\mathsf{h}}(\eta)=\cos^3(\pi\eta)\sum_{m\ge0}\mathsf{h}(m+\eta)+\sin^2(\pi\eta)\cos\pi(\eta+\tau)\sum_{k\ge0}\mathsf{n}(k+\tfrac12-\eta),$$

and the right side is holomorphic there. *Entire and antiperiodic.* $\Sigma^{\mathbb{Z}}_{\mathsf{h}}$ is 1-periodic and $\cos^3(\pi\eta)$ changes sign under $\eta\mapsto\eta+1$. Since the strip has width $\frac32>1$, Ω_{h} extends to an entire function with $\Omega_{\mathsf{h}}(\eta+1)=-\Omega_{\mathsf{h}}(\eta)$. *Growth.* For $0\le\Re\eta\le1$ and $|\Im\eta|=|y|\ge1$, the lemma on uniform Gamma ratios in the sector $|\arg|\le\pi-\delta$ for $|y|\ge R$, and compactness for $1\le|y|\le R$ (there are no poles off the real axis), give $|\sum_{m\ge0}\mathsf{h}(m+\eta)|,|\sum_{k\ge0}\mathsf{n}(k+\frac12-\eta)|\le C\sum_{k\ge0}|k+\mathrm{i}y|^{-1-\tau}\le C'|y|^{-\tau}$; for the term with real part in $[-\frac12,0)$ the argument stays in such a sector because $|y|\ge1$. Hence $|\Omega_{\mathsf{h}}(\eta)|\le Ce^{3\pi|y|}|y|^{-\tau}$. *Two modes.* An entire function of period 2 that changes sign under $\eta\mapsto\eta+1$ has an expansion $\sum_{k\ \mathrm{odd}}\hat\omega_k e^{\mathrm{i}\pi k\eta}$, with $\hat\omega_k=\frac12\int_0^2\Omega_{\mathsf{h}}(x+\mathrm{i}y)e^{-\mathrm{i}\pi k(x+\mathrm{i}y)}\,dx$ for every y. Thus $|\hat\omega_k|\le Ce^{3\pi|y|}|y|^{-\tau}e^{\pi ky}$. Letting $y\to-\infty$ for $k\ge3$ and $y\to+\infty$ for $k\le-3$ gives $\hat\omega_k=0$ for $|k|\ge3$; for $|k|=3$ this uses the factor $|y|^{-\tau}$, that is $\tau>0$. Therefore $\Omega_{\mathsf{h}}(\eta)=\alpha_0\cos\pi\eta+\beta_0\sin\pi\eta$. *Three evaluations.* At $\eta=0$ (value and derivative; the second term of Ω_{h} has a double zero there, and termwise differentiation is legitimate), $\alpha_0=\Sigma_{\mathsf{h}}$ and $\pi\beta_0=\Sigma'_{\mathsf{h}}$. At $\eta=\frac12$, $\beta_0=\Omega_{\mathsf{h}}(\frac12)=\cos(\frac{\pi}{2}+\pi\tau)\Sigma_{\mathsf{n}}=-s_\tau\Sigma_{\mathsf{n}}$. At $\eta=\frac12-\tau$ the second term vanishes, $\cos^3(\frac{\pi}{2}-\pi\tau)=s_\tau^3$, and $\sum_m\mathsf{h}(m+\frac12-\tau)=\Sigma_{\mathsf{k}}$ by the shift relation, so $\alpha_0 s_\tau+\beta_0 c_\tau=s_\tau^3\Sigma_{\mathsf{k}}$. These three relations give (386). □

Lemma (one-mode periodization). *For $0<\tau<\frac12$ and $x_\tau=\frac{1-\tau}{2}$,*

$$\Sigma_{\mathsf{m}}=\Gamma(\tau)\,\mathcal{X}_2(\tau),\qquad \Sigma'_{\mathsf{m}}=-2\pi\tan(\tfrac{\pi\tau}{2})\,\Sigma_{\mathsf{m}},\qquad \sum_{m\in\mathbb{Z}}\mathsf{m}(m+\eta)=\Sigma_{\mathsf{m}}\,\frac{\sin^2\pi x_\tau}{\sin^2\pi(x_\tau+\eta)}.\tag{387}$$

Proof. Gauss's summation [4] gives $\Sigma_{\mathsf{m}}=\Gamma(x_\tau)^2{}_2F_1(x_\tau,x_\tau;1;1)=\Gamma(x_\tau)^2\Gamma(1-2x_\tau)/\Gamma(1-x_\tau)^2=\Gamma(\tau)\mathcal{X}_2(\tau)$, since $1-2x_\tau=\tau>0$. Reflection gives $\mathsf{m}(-k-1+\eta)=\frac{\sin^2\pi\eta}{\sin^2\pi(x_\tau+\eta)}\tilde{\mathsf{m}}(k-\eta)$ with $\tilde{\mathsf{m}}(x)=[\Gamma(x+1)/\Gamma(x+2-x_\tau)]^2$. Hence

$$\sin^2\pi(x_\tau+\eta)\sum_{m\in\mathbb{Z}}\mathsf{m}(m+\eta)=\sin^2\pi(x_\tau+\eta)\sum_{m\ge0}\mathsf{m}(m+\eta)+\sin^2(\pi\eta)\sum_{k\ge0}\tilde{\mathsf{m}}(k-\eta)$$

is holomorphic on $-x_\tau<\Re\eta<1$, a strip of width larger than 1; it is 1-periodic, hence entire, and it is $O(e^{2\pi|y|}|y|^{-\tau})$. Its Fourier modes $e^{2\pi\mathrm{i}k\eta}$ with $k\ne0$ vanish by the argument of the previous proof (for $|k|=1$ through the factor $|y|^{-\tau}$), so it is a constant c_{m}. The reflected part has a double zero at $\eta=0$, so $c_{\mathsf{m}}=\sin^2(\pi x_\tau)\Sigma_{\mathsf{m}}$, which is the last identity, and $\Sigma'_{\mathsf{m}}/\Sigma_{\mathsf{m}}=\frac{d}{d\eta}\log\sin^{-2}\pi(x_\tau+\eta)\big|_0=-2\pi\cot\pi x_\tau=-2\pi\tan\frac{\pi\tau}{2}$. □

The last identity in (387) is an instance of Dougall's bilateral ${}_2H_2$ summation; the proof above does not assume it.

Lemma (Euler and Thomae). *For $0<\tau<\frac12$,*

$$\Sigma_{\mathsf{n}}=\pi\Gamma(1-\tau)\,\mathfrak{F}(1-\tau)=\frac{\pi\Gamma(\tau)}{\Gamma(\frac12+\tau)^2}\,\mathfrak{G}(\tau),\qquad \Sigma_{\mathsf{k}}=\frac{\Gamma(1-\tau)^3}{\Gamma(\frac32-\tau)^2}\,\mathfrak{G}(1-\tau)=\Gamma(\tau)\Gamma(1-\tau)^2\,\mathfrak{F}(\tau).\tag{388}$$

Proof. The first form of each is the definition of the series: $\Sigma_{\mathsf{n}}=\sum_k\Gamma(k+\frac12)^2\Gamma(k+1-\tau)/k!^3$ and $\Sigma_{\mathsf{k}}=\sum_k\Gamma(k+1-\tau)^3/[\Gamma(k+\frac32-\tau)^2k!]$. The second forms follow from (82) at $t=\tau$ and at

$t = 1-\tau$; the latter reads $\mathfrak{G}(1-\tau) = \Gamma(\frac{3}{2}-\tau)^2\Gamma(\tau)\Gamma(1-\tau)^{-1}\mathfrak{F}(\tau)$, and its ${}_3F_2(1)$ has parametric excess $\tau > 0$. □

In particular $s_\tau\Sigma_{\mathsf{k}} = \pi\Gamma(1-\tau)\mathfrak{F}(\tau)$, since $\Gamma(\tau)\Gamma(1-\tau) = \pi/s_\tau$.

Second evaluation: a bilateral Laurent lemma. The derivative sums can also be obtained without the growth argument for Ω_{h}. Let $f \in \{\mathsf{h}, \mathsf{m}\}$, with $\beta_f = \frac{1}{2}$ and pole order $p_f = 3$ for h, and $\beta_f = x_\tau$, $p_f = 2$ for m.

(a) For each $\eta_0 > 0$ there is C_{η_0} with $|f(K)| \le C_{\eta_0}(1+|K|)^{-1-\tau}$ whenever $\operatorname{dist}(K, -\beta_f - \mathbb{Z}_{\ge 0}) \ge \eta_0$. For $\Re K \ge 0$ this is Stirling's formula. For $\Re K < 0$ use the reflection identities

$$\mathsf{h}(K) = \frac{\sin^2(\pi K)\,\sin\pi(K+\frac{1}{2}+\tau)}{\sin^3\pi(K+\frac{1}{2})}\,\frac{\Gamma(-K)^2\,\Gamma(\frac{1}{2}-\tau-K)}{\Gamma(\frac{1}{2}-K)^3},$$
$$\mathsf{m}(K) = \frac{\sin^2(\pi K)}{\sin^2\pi(K+x_\tau)}\,\frac{\Gamma(-K)^2}{\Gamma(1-x_\tau-K)^2}\,:$$

the trigonometric factors are 1-periodic, bounded away from the poles, and have finite limits as $\Im K \to \pm\infty$, and the Gamma ratios are $O(|K|^{-1-\tau})$ by Stirling's formula in $\Re(-K) > 0$.

(b) $P_f(u) = \sum_{k\in\mathbb{Z}} f(k+u)$ is meromorphic and 1-periodic, with poles only in $-\beta_f + \mathbb{Z}$, of order at most p_f; its Laurent coefficients a_{-j} at $u = -\beta_f$ are the sums of the termwise Laurent coefficients (by (a), the convergence is absolute and locally uniform off the poles and uniform on small circles around $-\beta_f$). Moreover $\sup_{0\le\Re u\le 1}|P_f(u)| \le C\sum_k |k+u|^{-1-\tau} \le C'|\Im u|^{-\tau} \to 0$ as $|\Im u| \to \infty$.

(c) $P_f(u) = a_{-2}\,\pi^2/\sin^2\pi(u+\beta_f) + a_{-3}\,\pi^3\cos\pi(u+\beta_f)/\sin^3\pi(u+\beta_f)$, with $a_{-3} = 0$ for m. Indeed, since $\pi^2/\sin^2\pi\epsilon = \epsilon^{-2}+O(1)$, $\pi^3\cos\pi\epsilon/\sin^3\pi\epsilon = \epsilon^{-3}+O(\epsilon)$ and $\pi\cot\pi\epsilon = \epsilon^{-1}+O(\epsilon)$, the function $P_f - a_{-2}\pi^2\sin^{-2}\pi(u+\beta_f) - a_{-3}\pi^3\cos\pi(u+\beta_f)\sin^{-3}\pi(u+\beta_f) - a_{-1}\pi\cot\pi(u+\beta_f)$ is entire, 1-periodic and bounded, hence constant by Liouville's theorem. Its limits at $\pm\mathrm{i}\infty$ are $\pm\mathrm{i}\pi a_{-1}$, so $a_{-1} = 0$ and the constant vanishes.

(d) $\sum_{k\ge 0} f'(k) = P_f'(0)$: for $k \ge 1$, $f(u-k)$ has a double zero at $u = 0$, from $1/\Gamma(K+1)^2$, while the other factors are finite at $K = -k$ because $\frac{1}{2}-k$, $\frac{1}{2}+\tau-k$ and $x_\tau - k$ are not poles; termwise differentiation of the two halves of P_f is justified by local uniform convergence.

For m, only $\mathsf{m}(-j-x_\tau+\epsilon) \simeq \epsilon^{-2}/[j!^2\Gamma(1-x_\tau-j)^2]$ contributes at $u = -x_\tau$, and $1/\Gamma(1-x_\tau-j) = (-1)^j\sin(\pi x_\tau)\Gamma(x_\tau+j)/\pi$, so $a_{-2} = \pi^{-2}\sin^2(\pi x_\tau)\Sigma_{\mathsf{m}}$. Then (c) is the last identity of (387), and (d) gives $\Sigma'_{\mathsf{m}} = -2\pi\cot(\pi x_\tau)\Sigma_{\mathsf{m}}$. For h, only the terms $\mathsf{h}(-\frac{1}{2}-j+\epsilon)$ have a pole at $u = -\frac{1}{2}$. With $\Gamma(-j+\epsilon)^3 = (-1)^j\epsilon^{-3}/j!^3+\ldots$, $1/\Gamma(\frac{1}{2}-j)^2 = \Gamma(\frac{1}{2}+j)^2/\pi^2$ and $1/\Gamma(\tau-j) = (-1)^j s_\tau\Gamma(1+j-\tau)/\pi$,

$$a_{-3} = \frac{s_\tau}{\pi^3}\sum_{j\ge 0}\frac{\Gamma(\frac{1}{2}+j)^2\,\Gamma(1+j-\tau)}{j!^3} = \frac{s_\tau}{\pi^3}\,\Sigma_{\mathsf{n}},$$

which converges since $\tau > 0$. With $\beta_f = \frac{1}{2}$, (c) reads $P_{\mathsf{h}}(u) = a_{-2}\pi^2\sec^2\pi u - a_{-3}\pi^3\tan(\pi u)\sec^2(\pi u)$, so (d) gives $\Sigma'_{\mathsf{h}} = P'_{\mathsf{h}}(0) = -\pi^4 a_{-3} = -\pi s_\tau\Sigma_{\mathsf{n}}$, the first identity of (386). (As a by-product, $P_{\mathsf{h}}(0) = a_{-2}\pi^2 = \Sigma_{\mathsf{h}}$, because $\mathsf{h}(-k) = 0$ for $k \ge 1$.) The undifferentiated sum follows from the Euler integral of Section 8: with the moments I_j listed before (77), at $t = \tau$, termwise integration with $c_n^2 B(n+\frac{1}{2},\tau) = \Gamma(\tau)\mathsf{h}(n)/\pi$, where $c_n = (\frac{1}{2})_n/n!$, gives

$$\Sigma_{\mathsf{h}} = \frac{\pi^{3/2}}{\Gamma(\frac{1}{2}+\tau)}\,{}_3F_2\left(\begin{matrix}\frac{1}{2},\frac{1}{2},\frac{1}{2}\\ 1,\frac{1}{2}+\tau\end{matrix};1\right) = \frac{2I_1}{\Gamma(\tau)} = \frac{\pi c_\tau\Gamma(\tau)}{\Gamma(\frac{1}{2}+\tau)^2}\,\mathfrak{G}(\tau) + \frac{\pi^2}{\Gamma(\tau)}\,\mathfrak{F}(\tau), \tag{389}$$

the last step by $I_1 = c_\tau I_3 + s_\tau I_4$ of (77) and the values of I_3, I_4 stated after it. By (388) the two terms are $c_\tau\Sigma_{\mathsf{n}}$ and $s_\tau^2\Sigma_{\mathsf{k}}$, which reproves the second identity of (386).

Third evaluation: the logarithmic expansion of K. The derivative sum is also a single elliptic moment. For $0<x<1$ the logarithmic connection formula [23] gives

$$K(1-x)=\frac{1}{2}\sum_{n\ge 0}c_n^2x^n\big[2\psi_0(n+1)-2\psi_0(n+\tfrac12)-\log x\big],$$

with positive terms. Since $\int_0^1 x^{n-1/2}(1-x)^{\tau-1}\log x\,dx=B(n+\frac12,\tau)[\psi_0(n+\frac12)-\psi_0(n+\frac12+\tau)]$ and $\mathsf{h}'(n)/\mathsf{h}(n)=3\psi_0(n+\frac12)-2\psi_0(n+1)-\psi_0(n+\frac12+\tau)$, Tonelli's theorem gives

$$I_2=E(\tau,\tfrac12)=\int_0^1 x^{-1/2}(1-x)^{\tau-1}K(1-x)\,dx=-\frac{\Gamma(\tau)}{2\pi}\,\Sigma_{\mathsf{h}}',$$

so $\Sigma_{\mathsf{h}}'=-2\pi I_2/\Gamma(\tau)$. By (77), $I_2=s_\tau I_3=\frac{\pi^2}{2}\mathfrak{F}(1-\tau)$ and $I_1=c_\tau I_3+s_\tau I_4=\frac{\pi^2}{2}[\mathfrak{F}(\tau)+\cot(\pi\tau)\mathfrak{F}(1-\tau)]$. Hence

$$\Sigma_{\mathsf{h}}=\frac{\pi^2}{\Gamma(\tau)}\big[\mathfrak{F}(\tau)+\cot(\pi\tau)\,\mathfrak{F}(1-\tau)\big],\qquad \Sigma_{\mathsf{h}}'=-\frac{\pi^3}{\Gamma(\tau)}\,\mathfrak{F}(1-\tau),\qquad \Sigma_{\mathsf{k}}=\Gamma(\tau)\Gamma(1-\tau)^2\,\mathfrak{F}(\tau),$$

in agreement with (386) and (388). This route uses the elliptic moments and the moment relations in place of periodization.

18.9.5 The regulated traces and the mixed-trace step

Theorem (regulated traces). *For $0<\tau<\frac12$ both traces converge absolutely, and*

$$\boxed{C(\tau)\langle t_\ell^2R_\tau\rangle_2=\pi^2\big[\mathfrak{F}(\tau)^2+\mathfrak{F}(1-\tau)^2\big],\qquad \langle t_\ell^2S_\tau\rangle_2=C(\tau)\,\mathcal{X}_2(\tau)^2.}\tag{390}$$

Proof. Insert (386) into (385). Since $-2\Sigma_{\mathsf{h}}\Sigma_{\mathsf{h}}'=2\pi s_\tau\Sigma_{\mathsf{n}}\Sigma_{\mathsf{h}}$ and $\Sigma_{\mathsf{h}}=c_\tau\Sigma_{\mathsf{n}}+s_\tau^2\Sigma_{\mathsf{k}}$,

$$-\frac{1}{\pi^2c_\tau}\Big[2\Sigma_{\mathsf{h}}\Sigma_{\mathsf{h}}'+\pi\frac{s_\tau}{c_\tau}\Sigma_{\mathsf{h}}^2\Big]=\frac{s_\tau}{\pi c_\tau^2}\,\Sigma_{\mathsf{h}}\big(2c_\tau\Sigma_{\mathsf{n}}-\Sigma_{\mathsf{h}}\big)=\frac{s_\tau}{\pi c_\tau^2}\big(c_\tau^2\Sigma_{\mathsf{n}}^2-s_\tau^4\Sigma_{\mathsf{k}}^2\big).$$

Adding the k family, the factors c_τ^{-2} cancel:

$$\langle t_\ell^2R_\tau\rangle_2=\frac{s_\tau}{\pi}\big[\Sigma_{\mathsf{n}}^2+s_\tau^2\Sigma_{\mathsf{k}}^2\big]=\pi s_\tau\Gamma(1-\tau)^2\big[\mathfrak{F}(1-\tau)^2+\mathfrak{F}(\tau)^2\big],$$

by (388). Multiplication by $C(\tau)$, with $C(\tau)\,\pi s_\tau\Gamma(1-\tau)^2=\pi s_\tau\Gamma(\tau)\Gamma(1-\tau)=\pi^2$, gives the first identity. For the second, (387) in (385) gives

$$\langle t_\ell^2S_\tau\rangle_2=\frac{\Sigma_{\mathsf{m}}^2}{\pi}\Big[4\cos^2(\tfrac{\pi\tau}{2})\tan(\tfrac{\pi\tau}{2})-s_\tau\Big]=\frac{s_\tau\Gamma(\tau)^2\mathcal{X}_2(\tau)^2}{\pi}=C(\tau)\,\mathcal{X}_2(\tau)^2.\qquad\square$$

The second identity is (556) at $d=2$, which Section 21.2 derives at every dimension from the two-wheel convolution $W_2(d;b')=B_d(d-b',2b')$; here it is obtained from Gauss's sum and the periodization (Dougall) identity of (387). The second and third evaluations of the single sums give the same result: with $\mathcal{A}_\tau=\Gamma(\tau)\mathfrak{G}(\tau)/\Gamma(\frac12+\tau)^2$ and $\mathcal{B}_\tau=\pi\mathfrak{F}(\tau)/\Gamma(\tau)$ one has $\Sigma_{\mathsf{h}}=\pi(c_\tau\mathcal{A}_\tau+\mathcal{B}_\tau)$, $\Sigma_{\mathsf{h}}'=-\pi^2s_\tau\mathcal{A}_\tau$ and $\Sigma_{\mathsf{k}}=\pi\mathcal{B}_\tau/s_\tau^2$, and (385) becomes $\langle t_\ell^2R_\tau\rangle_2=\pi s_\tau\mathcal{A}_\tau^2+\pi\mathcal{B}_\tau^2/s_\tau$, that is

$$C(\tau)\langle t_\ell^2R_\tau\rangle_2=\pi^2\mathfrak{F}(\tau)^2+s_\tau^2\frac{\Gamma(\tau)^4}{\Gamma(\frac12+\tau)^4}\,\mathfrak{G}(\tau)^2,$$

which is (390) by (82). The assembly with the third evaluation is checked in exact algebra in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (part S2), and the periodization route numerically (Section 18.9.9). At $d=1$ the regulated master is linear in $\mathfrak{F}$ and $\mathfrak{G}$ (78); at $d=2$ it is a sum of two squares, which is where G^2 comes from: $u=4\log2-8\mathsf{G}/\pi$ enters the

extraction below through $2\pi^2u^2$. The two-mode lemma is the second-power analogue of (337); here the decay $|y|^{-\tau}$ supplied by the regulator is what removes the third harmonics. The periodization route uses neither elliptic integrals nor the moment relations, only residues, Gauss's sum, Thomae's relation and periodization, whereas the position-space proofs of (377) relate the same single sums to the elliptic moments. At $\tau > 0$ the only single sums are $\Sigma_{\mathsf{h}}, \Sigma'_{\mathsf{h}}, \Sigma_{\mathsf{k}}, \Sigma_{\mathsf{m}}, \Sigma'_{\mathsf{m}}$ and Σ_{n}, all absolutely convergent; the divergent single sums of the factorization at $\tau = 0$, and their finite parts, are never evaluated.

Corollary (the mixed-trace step at $d = 2$). *For $0 < \tau < \frac{1}{2}$,*

$$J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_2, \tag{391}$$

with the trace read on the full Mellin line, equivalently as the real part of the half-line integral (241)*. This is input (M) of Section 21.1 at $d = 2$ on this interval.*

Proof. By (377), $J_{2\tau} = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ for $0 < \tau < 1$, and by (390) the right side of (391) has the same value for $0 < \tau < \frac{1}{2}$. □

The proofs of (377) in Section 18.8 and in Section 18.9.8 use neither the trace nor (M), and the proof of (390) does not use (M); so (391) is proved, not assumed. With the second identity of (390), which is the two-wheel identity at $d = 2$, it gives (250) at $d = 2$ for $0 < \tau < \frac{1}{2}$, and with part (i) of the corollary of Section 18.9.2 the derivation of (248) from (238) given after (250) is complete at $d = 2$. As a normalization check, at $\tau = \frac{1}{2}$ one has $R_{1/2} = t_\ell$ and $C(\frac{1}{2}) = 1$, so the right side of (391) is $\langle t_\ell^3\rangle_2 = 3Q_2$ by (243), while (377) gives $J_1 = 3Q_2$ by Dixon's sum. The point $\tau = \frac{1}{2}$ lies outside the interval of the corollary, because the two pole families of Θ_ℓ^{R} merge there, so this is a consistency check only. The reading of (M) for complex τ, and (M) at other dimensions, are not addressed here.

18.9.6 The spectral form at $d = 2$

Theorem. *The right side of* (248) *at $d = 2$ equals* $32\mathsf{G}^2 - 28\zeta(3)\log 2$.

Proof. By part (iii) of the corollary of Section 18.9.2 and by (390), for $0 < \tau < \frac{1}{2}$,

$$C(\tau)\mathcal{D}(\tau) = \pi^2\mathfrak{F}(\tau)^2 + \pi^2\mathfrak{F}(1-\tau)^2 - C(\tau)^2\mathcal{X}_2(\tau)^2. \tag{392}$$

As in Section 18.8.4, (82) gives $\pi\mathfrak{F}(1-\tau) = \tau^{-1}\mathcal{Q}(\tau)\mathfrak{G}(\tau)$ with $\mathcal{Q}(\tau) = \pi\Gamma(1+\tau)/[\Gamma(1-\tau)\Gamma(\frac{1}{2}+\tau)^2]$, holomorphic and zero-free on $|\tau| < \frac{1}{2}$ with $\mathcal{Q}(0) = 1$; and $\tau C(\tau)\mathcal{X}_2(\tau)$ is holomorphic there with value 1 at 0. So the right side of (392) is meromorphic on $|\tau| < \frac{1}{2}$. The left side is holomorphic there by the corollary of Section 18.9.2. Two functions meromorphic on a punctured disk that agree on $(0, \frac{1}{2})$ agree identically, so their Laurent coefficients at 0 coincide. From $\log\Gamma(c+x) - \log\Gamma(c) = \sum_k \psi_{k-1}(c)x^k/k!$, $\psi_0(\frac{1}{2}) - \psi_0(1) = -2\log 2$, $\psi_k(1) = (-1)^{k+1}k!\,\zeta(k+1)$ and $\psi_k(\frac{1}{2}) = (2^{k+1}-1)\psi_k(1)$ [32],

$$\log\mathcal{Q}(\tau)^2 = 8\log 2\,\tau - \pi^2\tau^2 + 8\zeta(3)\tau^3 - \frac{\pi^4}{6}\tau^4 + O(\tau^5),$$
$$\log\big[\tau C(\tau)\mathcal{X}_2(\tau)\big]^2 = 8\log 2\,\tau + \zeta(3)\tau^3 + O(\tau^5).$$

The first is the series of Section 18.8.4. Hence $C\mathcal{D} = \pi^2\mathfrak{F}(\tau)^2 + \tau^{-2}\{\mathcal{Q}(\tau)^2\mathfrak{G}(\tau)^2 - [\tau C(\tau)\mathcal{X}_2(\tau)]^2\}$, and with (378) and the coefficients $\mathcal{Q}_k = [\tau^k]\mathcal{Q}(\tau)^2$ of Section 18.8.4,

$$\begin{aligned}
[\tau^{-2}]\,C\mathcal{D} &= 1 - 1 = 0, \qquad [\tau^{-1}]\,C\mathcal{D} = 8\log 2 - 8\log 2 = 0,\\
[\tau^0]\,C\mathcal{D} &= \pi^2 + \mathcal{Q}_2 - 32\log^2 2 = 0,\\
[\tau^1]\,C\mathcal{D} &= 2\pi^2 u + 2\sigma + 7\zeta(3) - 8\pi^2\log 2,\\
[\tau^2]\,C\mathcal{D} &= 2(\varrho + \pi^2 v) + \pi^2u^2 + \frac{\pi^4}{3} + 56\zeta(3)\log 2\\
&\qquad - 32\pi^2\log^2 2 + 16\sigma\log 2.
\end{aligned} \tag{393}$$

In the last line the τ^2 coefficient of $\pi^2\mathfrak{F}^2$ is $\pi^2(u^2+2v)$, that of $\tau^{-2}\mathcal{Q}^2\mathfrak{G}^2$ is $\mathcal{Q}_4+2\sigma\mathcal{Q}_1+2\varrho$ with $\mathcal{Q}_1=8\log 2$ and $\mathcal{Q}_4$ of Section 18.8.4, and that of $\tau^{-2}[\tau C\mathcal{X}_2]^2$ is $\frac{512}{3}\log^4 2+8\zeta(3)\log 2$; in the line before, $\mathcal{Q}_3$ of Section 18.8.4 and $[\tau^3][\tau C\mathcal{X}_2]^2=\frac{256}{3}\log^3 2+\zeta(3)$ enter. The vanishing of the first three coefficients is the consistency required by $\mathcal{D}=O(\tau^2)$. The cancellation rule in (378) removes v, ϱ and π^4: $[\tau^2]\,C\mathcal{D}=2\pi^2u^2-32\pi^2\log^2 2+16\sigma\log 2+56\zeta(3)\log 2$. With u and σ from (378), $2\pi^2u^2=32\pi^2\log^2 2-128\pi\mathsf{G}\log 2+128\mathsf{G}^2$ and $16\sigma\log 2=128\pi\mathsf{G}\log 2-224\zeta(3)\log 2$, so $[\tau^2]\,C\mathcal{D}=128\mathsf{G}^2-168\zeta(3)\log 2$. Finally (381) gives

$$14\zeta(3)\log 2+\tfrac14\big(128\mathsf{G}^2-168\zeta(3)\log 2\big)=32\mathsf{G}^2-28\zeta(3)\log 2. \qquad \square$$

Corollary (a spectral moment).

$$\langle t_\ell^2E_2\rangle_2=\mathcal{D}_2=[\tau^1]\,C\mathcal{D}=2\pi^2u+2\sigma+7\zeta(3)-8\pi^2\log 2=-21\zeta(3). \tag{394}$$

This is (567) at $d=2$, $\frac32\psi_2(\frac12)=-21\zeta(3)$, proved here without (M) or (J), and with the interchange of $[\tau^2]$ and the trace justified by the corollary of Section 18.9.2.

Relation to the definition. With (391), $J_{2\tau}-J_{2\tau}^{\Gamma}=C(\tau)\mathcal{D}(\tau)-[J_{2\tau}^{\Gamma}-C(\tau)^2\mathcal{X}_2(\tau)^2]$, and by (563) at $d=2$ the bracket is $\frac32U\tau+AU\tau^2+O(\tau^3)$. So the order-$\tau$ coefficient of $J_{2\tau}-J_{2\tau}^{\Gamma}$ is $[\tau^1]C\mathcal{D}-\frac32U=-21\zeta(3)+21\zeta(3)=0$, which is the vanishing found in Section 18.8.4, and $\frac14[\tau^2](J_{2\tau}-J_{2\tau}^{\Gamma})=\frac14[\tau^2]C\mathcal{D}-\frac{AU}{4}$ is the right side of (381). At $d=2$ the definition (238), the spectral form (248) and the regulated trace therefore give the same expression identically, not only the same number. The exact algebra of both routes is checked in `LRI_C5_d2_simplify/proof/K2-parseval/symbolic_checks.py` (part B) and `LRI_C5_d2_simplify/proof/synth/synth_checks_K2.py`.

18.9.7 The series at $d=2$

At $q=\frac12$ the data of (251) and (260) are $M^2=16$, $T_{\ell n}=c_mc_n$ and $S_{\ell n}=e_me_n$, with $m=\ell+n$, $c_k=(\frac12)_k/k!=\Gamma(k+\frac12)/[\sqrt{\pi}\,\Gamma(k+1)]$ and $e_k=k!/(\frac32)_k$. Write

$$D(\ell)=\sum_{n\ge0}\Big[\frac{M^2S_{\ell n}^2}{8}-\frac{T_{\ell n}^2\mathcal{R}_{\ell n}}{4}\Big],$$

so that (260) at $d=2$ reads $-\frac{AU}{4}+\sum_\ell d_\ell D(\ell)$.

Lemma (local forms). *Let $u\to0$. At $s=a+n+u$,*

$$g_\ell(s)=-\pi\cot(\pi u)\,c(m+u)\,c(n+u),\qquad c(x)=\frac{\Gamma(x+\frac12)}{\sqrt{\pi}\,\Gamma(x+1)},$$

with $\psi_0(a+s)=\psi_0(m+\frac12+u)$, $\psi_0(b+s)=\psi_0(m+1+u)$, $\psi_0(a-s)=\psi_0(n+1+u)+\pi\cot\pi u$ and $\psi_0(b-s)=\psi_0(n+\frac12+u)-\pi\tan\pi u$. At $s=b+n+u$,

$$g_\ell(s)=\tan(\pi u)\,\frac{\Gamma(m+1+u)\,\Gamma(n+1+u)}{\Gamma(m+\frac32+u)\,\Gamma(n+\frac32+u)},$$

with $\psi_0(a+s)=\psi_0(m+1+u)$, $\psi_0(b+s)=\psi_0(m+\frac32+u)$, $\psi_0(a-s)=\psi_0(n+\frac32+u)-\pi\tan\pi u$ and $\psi_0(b-s)=\psi_0(n+1+u)+\pi\cot\pi u$.

Proof. Use $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$ and $\psi_0(1-z)-\psi_0(z)=\pi\cot\pi z$ [26, 32]; for example $\Gamma(-n-u)/\Gamma(\frac12-n-u)=[\sin\pi(\frac12-n-u)/\sin\pi(-n-u)]\,\Gamma(n+\frac12+u)/\Gamma(n+1+u)=-\cot(\pi u)\,\Gamma(n+\frac12+u)/\Gamma(n+1+u)$. $\square$

The products $c(m)c(n) = T_{\ell n}$ and $\Gamma(k+1)/\Gamma(k+\frac{3}{2}) = 2e_k/\sqrt{\pi}$ connect these local forms with (251). Inserting them into the holomorphic weight $F(s)$ of (263) and expanding in u gives the $q = \frac{1}{2}$ case of the residues computed in Section 16.6:

$$\mathrm{Res}_{s=a+n}\,\frac{g_\ell^2 F}{4} = \frac{T_{\ell n}^2 \mathcal{R}_{\ell n}}{4}, \qquad \mathrm{Res}_{s=b+n}\,\frac{g_\ell^2 F}{4} = -2e_m^2 e_n^2 = -\frac{M^2 S_{\ell n}^2}{8}. \tag{395}$$

Both are identities between Laurent coefficients, polynomial in the independent symbols $\psi_j(m+\frac{1}{2}), \psi_j(m+1), \psi_j(n+\frac{1}{2}), \psi_j(n+1)$ (or their shifts by $\frac{1}{2}$) with exact π, $\psi_j(1)$ and $\psi_j(\frac{1}{2})$; they are verified exactly in `LRI_C5_d2_proof/K2_gauss/check_residue_series.wl`.

Proposition (the per-spin identity). *For every $\ell \geq 0$ the inner series $D(\ell)$ converges, and*

$$D(\ell) = \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\,\frac{g_\ell(s)^2 F(s)}{4} = \frac{1}{4\pi}\int_0^\infty t_\ell(\nu)^2\, w_{\mathcal{K}}\, d\nu. \tag{396}$$

Proof. On $s = \mathrm{i}\nu$ one has $g_\ell'/g_\ell = \psi_0(a+s) - \psi_0(a-s) - \psi_0(b+s) + \psi_0(b-s) = 2\mathrm{i}h$, its derivative is $2j$, and $p(\mathrm{i}\nu) = p$, $p'(\mathrm{i}\nu) = \mathrm{i}k$, $p''(\mathrm{i}\nu) = r$ for the function $p(s)$ of Section 16.6. Hence $F(\mathrm{i}\nu) = (c_* - p)(-\frac{h^2}{2} - \frac{5j}{8}) - \frac{hk}{2} - \frac{5r}{32} = w_{\mathcal{K}}$, which is real and even in ν; so is $g_\ell(\mathrm{i}\nu) = t_\ell(\nu)$. This gives the second equality. For the first, close the contour to the right along $|s| = \rho_N := a + N + \frac{3}{4}$; these arcs stay at distance at least $\frac{1}{4}$ from both pole families $a + \mathbb{Z}_{\geq 0}$ and $b + \mathbb{Z}_{\geq 0}$. On the arcs $|g_\ell(s)| \leq C|s|^{-1}$: indeed $\Gamma(a+s)/\Gamma(b+s) = O(|s|^{-1/2})$ by Stirling's formula for $\Re s \geq 0$, and

$$\frac{\Gamma(a-s)}{\Gamma(b-s)} = \frac{\sin\pi(b-s)}{\sin\pi(a-s)} \cdot \frac{\Gamma(1-b+s)}{\Gamma(1-a+s)},$$

where the trigonometric ratio is bounded at distance at least $\frac{1}{4}$ from $a + \mathbb{Z}$ and has finite limits as $\Im s \to \pm\infty$, and the Gamma ratio is $O(|s|^{-1/2})$. In the same way $\psi_0(a-s) = \psi_0(1-a+s) - \pi\cot\pi(a-s)$, and the analogous formula for $\psi_0(b-s)$, give $|F| \leq C\log^2|s|$ on the arcs. The arc integrals are therefore $O(\rho_N^{-1}\log^2\rho_N) \to 0$. At fixed ℓ each residue family converges absolutely, with terms $O(n^{-2}\log^4 n)$, and the residues are summed in increasing n. Closing upward to the right is clockwise, so (395) gives $\int \frac{ds}{2\pi\mathrm{i}}\frac{g_\ell^2 F}{4} = -\sum_n[\mathrm{Res}_{a+n} + \mathrm{Res}_{b+n}] = D(\ell)$. □

This is the per-spin identity behind (260), at $d = 2$, with the contour estimates that Section 16.6 does not write out.

Proposition (the series and the strip limit). *(a) For each $r \in (0, \frac{1}{2})$ there is C_r' with $|t_\ell(\nu)^2 w_{\mathcal{K}}(\ell,\nu)| \leq C_r'(1+|\xi|)^{-3+2r}$; in particular $\langle|t_\ell^2 w_{\mathcal{K}}|\rangle_2 < \infty$. (b) $\sum_\ell d_\ell|D(\ell)| < \infty$, and* (260) *at $d = 2$ equals $-\frac{AU}{4} + \frac{1}{4}\langle t_\ell^2 w_{\mathcal{K}}\rangle_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$. (c) Let $g(m,n)$ be the symmetric square-lattice summand with $g(\ell+n,n)$ the summand of $D(\ell)$, written as a finite sum $\sum_i \alpha_i A_i(m)B_i(n)$ of products of real sequences of bounded variation that are $O(\log^3(m+2)/(m+1))$. Put $\hat{A}_i(z) = \sum_m A_i(m)z^m$. Then $h(\theta) = \sum_i \alpha_i \hat{A}_i(e^{\mathrm{i}\theta})\overline{\hat{B}_i(e^{\mathrm{i}\theta})}$ is continuous on $(0, 2\pi)$, its Fourier coefficients are $D(|\ell|)$, and $\lim_{\theta\to 0} h(\theta) = \sum_{\ell\geq 0} d_\ell D(\ell)$.* Such a factorization, with 27 distinct single sequences, is the exact symbolic one behind the 70-digit evaluation (`LRI_C5_d2_simplify/factor_K2.py`, output `K2_factorized.json`); each sequence is c_k^2 or e_k^2 times polygamma differences with at most one factor $\psi_0(k+1)$.

Proof. (a) Let $H(\tau) = \frac{1}{2}C(\tau)[(R_\tau - S_\tau)(\mathrm{i}\nu) + (R_\tau - S_\tau)(-\mathrm{i}\nu)]$. It is holomorphic in $|\tau| < \frac{1}{2}$, since the pole of C is cancelled by the double zero of the even part, and $[\tau^2]H = w_{\mathcal{K}}$ by the corollary of Section 18.9.2. By (380), $|H| \leq C(1+|\xi|)^{-1+2r}$ on $|\tau| = r$, and Cauchy's estimate gives $|w_{\mathcal{K}}| \leq r^{-2}C(1+|\xi|)^{-1+2r}$; multiply by $t_\ell^2 \leq C(1+|\xi|)^{-2}$. (b) By (396), $|D(\ell)| \leq \frac{1}{8\pi}\int_{\mathbb{R}}|t_\ell^2 w_{\mathcal{K}}|\,d\nu$, so (a) gives absolute convergence of the spin sum, and the value follows from the theorem of Section 18.9.6. (c) By Dirichlet's test (bounded variation and $A_i(m) \to 0$) the series $\hat{A}_i(e^{\mathrm{i}\theta})$ converge locally uniformly on $(0, 2\pi)$, so h is continuous there. Since $A_i, B_i \in \ell^2$, the product $\hat{A}_i\overline{\hat{B}_i}$

is integrable on the circle, and its ℓth Fourier coefficient is the correlation $\sum_n A_i(n+\ell)B_i(n)$, with $A_i(k)=0$ for $k<0$. Summing over i, the coefficient is $\sum_n g(n+\ell,n)=D(\ell)$ for $\ell\ge 0$, and $D(|\ell|)$ for $\ell<0$ by the symmetry of g. By (b) these coefficients are absolutely summable, so the Fourier series converges uniformly to a continuous function equal to h almost everywhere, hence everywhere on $(0,2\pi)$. Letting $\theta\to 0$ gives $\sum_{\ell\in\mathbb{Z}} D(|\ell|)=\sum_{\ell\ge 0} d_\ell D(\ell)$. □

At $d=2$ the inner series of (260) therefore sum in an absolutely convergent spin order, although the unrestricted double series is not absolutely convergent: its square-lattice single sums diverge like $\sum_k \log^p k/k$. Part (c) shows that the strip limit $\theta\to 0$ of the lattice sum with weight $e^{\mathrm{i}(m-n)\theta}$, used for the 70-digit evaluation of Section 17.2 (`LRI_C5_d2_simplify/eval_K2.py`), computes exactly this ordered sum. The sharper bound (382) gives $D(\ell)=O(\ell^{-3+2\rho_0})$.

Remark (why the single sums at $\tau=0$ are avoided). This is a numerical observation and is not used. At $\tau=0$ the $\mathcal{R}$ part of the bilinear form $g(m,n)$ has rank 9, and the $S_{\ell n}$ part adds one more. In the variables $\psi_{j-1}(k+\frac12)-\psi_{j-1}(\frac12)$ and $\psi_{j-1}(k+1)-\psi_{j-1}(1)$, the span of all parameter derivatives of ${}_2F_1(\frac12+\alpha,\frac12+\beta;1+\gamma;z)$ through total order four has dimension 22, and the row space of the form has three directions outside it, at weights 2, 3 and 4 (`LRI_C5_d2_proof/K2_gauss/rowspace.py`). An integer-relation search on the 70-digit single sums (`LRI_C5_d2_proof/K2_gauss/pslq_singles.py`) indicates that individual sums contain constants absent from the final answer, for example

$$\sum_k c_k^2\,[\psi_2(k+\tfrac12)-\psi_2(k+1)] \overset{\text{num}}{=} 3\pi^3+4\pi\log^2 2-64\mathsf{G}\log 2-128\,\Im\,\mathrm{Li}_3(\tfrac{1+\mathrm{i}}{2}),$$

while the weight-five sums are not identified in a small basis. The regulator $\tau>0$ makes every single sum convergent, and the only non-Gamma functions left are $\mathfrak{F}$ and $\mathfrak{G}$, whose unknown coefficients v,ϱ cancel by (86).

18.9.8 A second position-space proof: Riemann's bilinear relation

This subsubsection proves (377) from the integral

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb{C}^2}\frac{d^2x\,d^2y}{|x|\,|1-x|\,|y|\,|1-y|^{2-2\tau}\,|x-y|},\qquad 0<\tau<1,$$

of Section 18.8, without the cell decomposition and rotation of that section. Here d^2x is Lebesgue measure on $\mathbb{C}\cong\mathbb{R}^2$, so $dx\wedge d\bar{x}=-2\mathrm{i}\,d^2x$; $K(m)$ is the parameter convention, holomorphic on $\mathbb{C}\setminus[1,\infty)$; $I_1,\dots,I_6$ are the moments listed before (77), at $t=\tau$; $\mathbb{C}_{\mathrm{cut}}=\mathbb{C}\setminus((-\infty,0]\cup[1,\infty))$; all powers are principal; and the letters ω,Ω_y,Π_j below are local (they are not the ω_τ of Section 18.8 or the Ω_r of (335)).

Lemma (properties of K). (i) For $m\in\mathbb{C}\setminus[1,\infty)$, $K(m)=\frac12\int_0^1 ds/\sqrt{s(1-s)(1-ms)}$, by $s=\sin^2\theta$ in the definition of Section 1.4. (ii) For $m\in\mathbb{C}\setminus[1,\infty)$, $K(m)=(1-m)^{-1/2}K(\frac{m}{m-1})$ [23]; in particular $K(x)=(1-x)^{-1/2}K(\frac{x}{x-1})$ for $x<0$, with $\frac{x}{x-1}\in(0,1)$, and $K(1-x)=x^{-1/2}K(1-\frac1x)$ for $x>1$. (iii) For $x>1$, $K(x\pm\mathrm{i}0)=x^{-1/2}[K(\frac1x)\pm\mathrm{i}K(1-\frac1x)]$. (iv) For $|1-m|<1$, $m\notin[1,\infty)$, $K(m)=\frac12\sum_n c_n^2(1-m)^n[2\psi_0(n+1)-2\psi_0(n+\frac12)-\log(1-m)]$ [23]. (v) There is C such that, for all $m\in\mathbb{C}\setminus[1,\infty)$ and for the boundary values (iii), $|K(m)|\le C(1+|\log|1-m||)$ for $|m|\le 3$ and $|K(m)|\le C|m|^{-1/2}(1+\log|m|)$ for $|m|\ge 3$.

Proof. (iii) In (i) put $m=x\pm\mathrm{i}\eta$ with $\eta\downarrow 0$. For $s>1/x$, $1-ms\to(1-xs)\mp\mathrm{i}0$ with $1-xs<0$, so $(1-ms)^{-1/2}\to\pm\mathrm{i}(xs-1)^{-1/2}$. Since $|1-ms|\ge|1-xs|$ and $(s(1-s)|1-xs|)^{-1/2}\in L^1(0,1)$, dominated convergence applies. On $(0,1/x)$ substitute $s=\sigma/x$, and on $(1/x,1)$ substitute $s=1-(1-\frac1x)\sigma$; the two pieces become $x^{-1/2}K(1/x)$ and $\pm\mathrm{i}x^{-1/2}K(1-1/x)$. The restriction of

K to the upper (lower) half-plane continues analytically across $(1,\infty)$, because the hypergeometric equation has analytic coefficients there, so K extends continuously from each side with these values (the + case is the formula used in the proof of (77)). (v) If $|1-m|\le\frac12$, use (iv) and $|\log(1-m)|\le|\log|1-m||+\pi$. If $|m|\le3$ and $|1-m|\ge\frac12$, use (i) and $|1-ms|^2=(1-s\Re m)^2+s^2(\Im m)^2$: for $\Re m\le\frac34$, $|1-ms|\ge\frac14$; for $\Re m\ge\frac54$, $|1-ms|\ge|1-s\Re m|$ and $\int_0^1(s(1-s)|1-xs|)^{-1/2}ds$ is bounded for $x\in[\frac54,3]$; for $\frac34<\Re m<\frac54$, $(\Im m)^2\ge\frac{3}{16}$, so $|1-ms|\ge\frac38$ for $s\le\frac12$ and $|1-ms|\ge\frac{\sqrt3}{8}$ for $s\ge\frac12$. If $|m|\ge3$, (ii) gives $K(m)=(1-m)^{-1/2}K(\frac{m}{m-1})$ with $|1-\frac{m}{m-1}|=|m-1|^{-1}\le\frac12$, which reduces to the first case. The boundary values obey the same bounds by continuity. □

Green's theorem on slit domains. Below, Green's theorem $\int_V d\alpha=\oint_{\partial V}\alpha$, or Cauchy's theorem, is applied to a domain $V=\{r<|z-e|$ for the marked points e, $|z|<R\}\setminus\Sigma$, where Σ is a finite union of segments (slits) and $\alpha=\mathsf{g}\,d\bar z$ or $\mathsf{g}\,dz$. In every case g is real-analytic and bounded on V and extends continuously to each open slit from each side, and $d\alpha$ is integrable. Delete the η-neighborhood of Σ; on the closure of what remains α is C^1, so Green's theorem applies. As $\eta\to0$, the area integral converges by dominated convergence; the integrals over the two curves parallel to a slit converge to the integrals of the one-sided boundary values, by uniform convergence on compact parts of the open slits and boundedness; and the remaining boundary pieces have length $O(\eta)$. So the slit edges enter with the positive orientation of V: each edge is traversed with V on its left.

Reduction to the inner star. The integrand is positive, so by Tonelli's theorem

$$J_{2\tau}=\frac{1}{\pi^2}\int_{\mathbb C}d^2y\,|y|^{-1}|1-y|^{2\tau-2}\,\mathcal I(y),\qquad \mathcal I(y)=\int_{\mathbb C}\frac{d^2x}{|x|\,|1-x|\,|x-y|}\in(0,\infty],\tag{397}$$

with both sides in $[0,\infty]$. For $y\notin\{0,1\}$, $\mathcal I(y)<\infty$: the three factors are locally integrable in $\mathbb R^2$, and the integrand is $O(|x|^{-3})$ at infinity.

Lemma (the inner star is a Riemann bilinear relation). *For $y\in\mathbb C_{\rm cut}$,*

$$\boxed{\mathcal I(y)=8\,\Re\big[K(y)\,\overline{K(1-y)}\big].}\tag{398}$$

Proof. Fix $y\in\mathbb C_{\rm cut}$ and put $\Pi_y(x)=x(x-1)(x-y)$.

(a) The cut. Let $\Gamma_y=[0,y]\cup[y,1]\cup[1,+\infty)$, oriented from 0 to ∞. Since $y\notin(-\infty,0]\cup[1,\infty)$, the segment $[0,y]$ does not meet $[1,\infty)$; $[y,1]$ meets $[1,\infty)$ only at 1, and $[0,y]$ meets $[y,1]$ only at y (for $y\in(0,1)$ the three pieces are collinear). So Γ_y is a simple arc from 0 to ∞, and $\mathbb C\setminus\Gamma_y$ is simply connected. Π_y has no zero there, so there is a holomorphic ω on $\mathbb C\setminus\Gamma_y$ with $\omega^2=1/\Pi_y$, and $|\omega|^2=1/(|x||1-x||x-y|)$.

(b) Edge values and jumps. Near any point of an open piece of Γ_y, Π_y has two holomorphic square roots on a small disk, and ω equals ±1/(that root) on each side. Hence ω has continuous boundary values ω_L (left of Γ_y) and ω_R (right of Γ_y) on each open piece, and the ratio $\omega_R/\omega_L\in\{\pm1\}$ is a constant $\varepsilon_1,\varepsilon_2,\varepsilon_3$ on $(0,y)$, $(y,1)$, $(1,\infty)$. A small circle around 0 crosses Γ_y once, on $(0,y)$; since Π_y has a simple zero inside, $\Pi_y^{-1/2}$ changes sign along the circle, so $\varepsilon_1=-1$. A small circle around y crosses $(0,y)$ and $(y,1)$ once each, so $\varepsilon_1\varepsilon_2=-1$ and $\varepsilon_2=+1$. A small circle around 1 crosses $(y,1)$ and $(1,\infty)$, so $\varepsilon_3=-1$. The direction of crossing does not matter, because $\varepsilon_k^{-1}=\varepsilon_k$.

(c) Primitive. $|\omega|\le C|x-e|^{-1/2}$ near $e\in\{0,y,1\}$, and $|\omega|\le C|x|^{-3/2}$ for large $|x|$. Let Ω_y be the primitive of ω on $\mathbb C\setminus\Gamma_y$ with $\Omega_y(x)\to0$ as $x\to0$; the limit exists by the Cauchy criterion, since radial and circular paths near 0 give $|\Omega_y(x')-\Omega_y(x'')|\le C(|x'|^{1/2}+|x''|^{1/2})$. Ω_y is bounded: it is continuous up to the edges from each side, has limits at $0,y,1$ within each adjacent sector, and for $|x|\ge R_0:=2+|y|$ it differs from its value on the circle $|x|=R_0$ by the integral along a

radial segment, which does not meet Γ_y (for $\arg x \neq 0$) and is $O(R_0^{-1/2})$. The edge values satisfy $d\Omega_{y,L} = \omega_L\,dx$ and $d\Omega_{y,R} = \omega_R\,dx$ along Γ_y. At the corners y and 1 the left edges of the two adjacent pieces border the same sector, and so do the right edges; $|\omega|$ is integrable there, so the edge values are continuous through the corners. Put

$$\Pi_1 = \int_0^y \omega_L\,dx, \qquad \Pi_2 = \int_y^1 \omega_L\,dx, \qquad \Pi_3 = \int_1^\infty \omega_L\,dx \qquad (\text{along } \Gamma_y).$$

By (b) the edge values are: on $(0,y)$, $\Omega_{y,L} = \int_0^x \omega_L$ and $\Omega_{y,R} = -\Omega_{y,L}$; at y, $\pm\Pi_1$; on $(y,1)$, $\Omega_{y,L} = \Pi_1 + \int_y^x \omega_L$ and $\Omega_{y,R} = -\Pi_1 + \int_y^x \omega_L$; at 1, $\Pi_1+\Pi_2$ and $-\Pi_1+\Pi_2$; on $(1,\infty)$, $\Omega_{y,L} = \Pi_1+\Pi_2+\int_1^x \omega_L$ and $\Omega_{y,R} = -\Pi_1 + \Pi_2 - \int_1^x \omega_L$. Thus $\Omega_{y,L} + \Omega_{y,R} = 0$ on $(0,y)$, $\Omega_{y,L} - \Omega_{y,R} = 2\Pi_1$ on $(y,1)$, and $\Omega_{y,L} + \Omega_{y,R} = 2\Pi_2$ on $(1,\infty)$.

(d) Stokes. On $\mathbb{C}\setminus\Gamma_y$, $d(\Omega_y\bar\omega\,d\bar x) = \omega\bar\omega\,dx\wedge d\bar x = -2\mathrm{i}|\omega|^2 d^2x$. Apply Green's theorem on slit domains to $\{x\notin\Gamma_y:\ |x|<R,\ |x-e|>r,\ e=0,y,1\}$. The positively oriented boundary runs along the left edge of Γ_y in its direction and along the right edge against it, so

$$-2\mathrm{i}\int |\omega|^2 d^2x = \int_{\Gamma_y} (\Omega_{y,L}\bar\omega_L - \Omega_{y,R}\bar\omega_R) d\bar x + (\text{circle terms}).$$

The circle terms vanish in the limit: on $|x|=r$, $|\Omega_y| \le Cr^{1/2}$ and $|\omega| \le Cr^{-1/2}$, so the term is $O(r)$; on $|x-y|=r$ and $|x-1|=r$, Ω_y is bounded and $|\omega| \le Cr^{-1/2}$, so the term is $O(r^{1/2})$; on $|x|=R$, Ω_y is bounded and $|\omega| \le CR^{-3/2}$, so the term is $O(R^{-1/2})$. The edge integrands are integrable. Letting $r\to 0$ and $R\to\infty$ (monotone convergence on the left) gives $-2\mathrm{i}\,\mathcal{I}(y) = \int_{\Gamma_y}(\Omega_{y,L}\bar\omega_L - \Omega_{y,R}\bar\omega_R)\,d\bar x$. By (b) and (c) the integrand is 0 on $(0,y)$, $2\Pi_1\bar\omega_L$ on $(y,1)$ and $2\Pi_2\bar\omega_L$ on $(1,\infty)$. Hence $-2\mathrm{i}\,\mathcal{I}(y) = 2\Pi_1\bar\Pi_2 + 2\Pi_2\bar\Pi_3$.

(e) Cauchy. The same exhaustion applied to $\omega\,dx$ gives $\int_{\Gamma_y}(\omega_L-\omega_R)\,dx = 0$, that is $2\Pi_1+0+2\Pi_3 = 0$, so $\Pi_3 = -\Pi_1$. Therefore $-2\mathrm{i}\,\mathcal{I}(y) = 2(\Pi_1\bar\Pi_2 - \bar\Pi_1\Pi_2) = 4\mathrm{i}\,\Im(\Pi_1\bar\Pi_2)$, that is $\mathcal{I}(y) = 2\,\Im(\bar\Pi_1\Pi_2)$. This is Riemann's bilinear relation for the holomorphic differential dx/w on the elliptic curve $w^2 = \Pi_y(x)$, whose periods are $2\Pi_1$ and $2\Pi_2$; the factor $\frac12$ in $\mathcal{I} = \frac12\Im(\overline{2\Pi_1}\,2\Pi_2)$ is the inverse of the degree 2 of the covering $(x,w)\mapsto x$.

(f) The periods. On $[0,y]$ put $x = ys$. Then $\Pi_y(ys) = y^2 s(1-s)(1-ys)$, and $s(1-s)(1-ys)\notin(-\infty,0]$ for $s\in(0,1)$ because $ys\notin[1,\infty)$. Hence $\omega_L(ys) = \eta_1/(y\sqrt{s(1-s)(1-ys)})$ with a constant $\eta_1\in\{\pm1\}$, and by (i) of the lemma on K, $\Pi_1 = 2\eta_1 K(y)$. On $[y,1]$ put $x = 1-(1-y)\sigma'$. Then $\Pi_y = -(1-y)^2\sigma'(1-\sigma')(1-(1-y)\sigma')$, with $\sigma'(1-\sigma')(1-(1-y)\sigma')\notin(-\infty,0]$, so $\omega_L = \eta_2/(\mathrm{i}(1-y)\sqrt{\sigma'(1-\sigma')(1-(1-y)\sigma')})$ and, with $dx = -(1-y)d\sigma'$, $\Pi_2 = -2\mathrm{i}\eta_2 K(1-y)$. So (e) gives $\mathcal{I}(y) = 8\epsilon_y\Re[K(y)\overline{K(1-y)}]$ with $\epsilon_y = -\eta_1\eta_2\in\{\pm1\}$.

(g) The sign. $\mathcal{I}(y) > 0$, so $\Re[K(y)\overline{K(1-y)}]\neq 0$ on $\mathbb{C}_{\mathrm{cut}}$. This function is continuous on the connected set $\mathbb{C}_{\mathrm{cut}}$ and equals $K(\frac12)^2 > 0$ at $y=\frac12$, so it is positive everywhere and $\epsilon_y\equiv 1$. □

For real $y\in(0,1)$ the sign can also be read off directly: continuation along the upper edge gives $\omega_L = |\Pi_y|^{-1/2}$ on $(0,y)$, $\mathrm{i}|\Pi_y|^{-1/2}$ on $(y,1)$ and $-|\Pi_y|^{-1/2}$ on $(1,\infty)$, hence $\Pi_1 = 2K(y)$, $\Pi_2 = 2\mathrm{i}K(1-y)$ and $\Pi_3 = -2K(y)$ (using $\int_1^\infty dx/\sqrt{\Pi_y} = 2K(y)$ via $x = 1/s$), which confirms $\Pi_3 = -\Pi_1$ and $\mathcal{I}(y) = 8K(y)K(1-y)$. The algebra of (d)–(f) is checked exactly in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (part S0a).

Edge values of the outer integrand. Let $\varphi_\tau(y) = y^{-1/2}(1-y)^{\tau-1}$, $\mathsf{u} = \varphi_\tau K(y)$ and $\mathsf{v} = \varphi_\tau K(1-y)$. These are holomorphic on $\mathbb{C}_{\mathrm{cut}}$, and $|\varphi_\tau|^2 = |y|^{-1}|1-y|^{2\tau-2}$, so by (398)

$$|y|^{-1}|1-y|^{2\tau-2}\,\mathcal{I}(y) = 4(\mathsf{u}\bar{\mathsf{v}} + \bar{\mathsf{u}}\mathsf{v}) = 8\Re(\mathsf{u}\bar{\mathsf{v}}), \qquad y\in\mathbb{C}_{\mathrm{cut}}. \tag{399}$$

Define the positive functions

$$f_-(x) = |x|^{-1/2}(1-x)^{\tau-1}K(x), \quad g_-(x) = |x|^{-1/2}(1-x)^{\tau-3/2}K(\tfrac{1}{1-x}) \quad (x<0),$$
$$f_+(x) = x^{-1}(x-1)^{\tau-1}K(\tfrac1x), \qquad g_+(x) = x^{-1}(x-1)^{\tau-1}K(1-\tfrac1x) \qquad (x>1).$$

The principal powers have the one-sided values $(x \pm \mathrm{i}0)^{-1/2} = \mp \mathrm{i}|x|^{-1/2}$ for $x < 0$ and $(1-(x \pm \mathrm{i}0))^{\tau-1} = -(x-1)^{\tau-1}e^{\mp \mathrm{i}\pi\tau}$ for $x > 1$. With (ii) and (iii) of the lemma on K the edge values are

$$\begin{aligned} x<0: &\qquad \mathsf{u}(x \pm \mathrm{i}0) = \mp \mathrm{i} f_-(x), \qquad \mathsf{v}(x \pm \mathrm{i}0) = -f_-(x) \mp \mathrm{i} g_-(x), \\ x>1: &\qquad \mathsf{u}(x \pm \mathrm{i}0) = -e^{\mp \mathrm{i}\pi\tau}(f_+(x) \pm \mathrm{i} g_+(x)), \qquad \mathsf{v}(x \pm \mathrm{i}0) = -e^{\mp \mathrm{i}\pi\tau} g_+(x). \end{aligned} \tag{400}$$

For the v entry at $x < 0$: $1 - y = (1-x) \mp \mathrm{i}0$ with $1 - x > 1$, so (iii) gives $K(1-y) = (1-x)^{-1/2}[K(\frac{1}{1-x}) \mp \mathrm{i}K(\frac{x}{x-1})]$, and $K(\frac{x}{x-1}) = (1-x)^{1/2}K(x)$ by (ii). For $x > 1$, $K(1-x) = x^{-1/2}K(1-\frac{1}{x})$ by (ii), so v jumps only through φ_τ. The substitutions $w = \frac{x}{x-1}$, $w = \frac{1}{1-x}$, $w = \frac{1}{x}$ and $w = \frac{1}{x}$ give (exact check S0f)

$$\int_{-\infty}^{0} f_- = I_5, \quad \int_{-\infty}^{0} g_- = I_6, \quad \int_1^{\infty} f_+ = I_3, \quad \int_1^{\infty} g_+ = I_4, \quad \int_0^1 \mathsf{u} = I_1, \quad \int_0^1 \mathsf{v} = I_2. \tag{401}$$

By (v) of the lemma on K: near 0, $|\mathsf{u}| \le C|y|^{-1/2}$ and $|\mathsf{v}| \le C|y|^{-1/2}(1 + |\log|y||)$; near 1, $|\mathsf{u}| \le C|1-y|^{\tau-1}(1 + |\log|1-y||)$ and $|\mathsf{v}| \le C|1-y|^{\tau-1}$; for $|y| \ge 4$, $|\mathsf{u}|, |\mathsf{v}| \le C|y|^{\tau-2}(1 + \log|y|)$; and on compact subsets of $\mathbb{C} \setminus \{0,1\}$ both are bounded, uniformly up to the edges. Hence $\mathsf{u}\bar{\mathsf{v}} \in L^1(\mathbb{C})$ for $0 < \tau < 1$, and $J_{2\tau} < \infty$ by (397) and (399). The moment relations (77) at $t = \tau$ are the real and imaginary parts of Cauchy's theorem for u and v on the upper half-plane, $\int_{\mathbb{R}} \mathsf{u}(x + \mathrm{i}0)\,dx = \int_{\mathbb{R}} \mathsf{v}(x + \mathrm{i}0)\,dx = 0$, read with (400) and (401); this is the contour proof given after (77).

Lemma (the outer integral). *For $0 < \tau < 1$,*

$$\int_{\mathbb{C}} |y|^{-1}|1-y|^{2\tau-2}\,\mathcal{I}(y)\,d^2y = 4\big[I_5^2 + 2s_\tau I_1 I_4 - I_4^2\big]. \tag{402}$$

Proof. (a) Primitive. $\mathbb{C}_{\mathrm{cut}}$ is star-shaped with respect to every point of $(0,1)$: a segment from $c \in (0,1)$ to $y \in \mathbb{C}_{\mathrm{cut}} \setminus \mathbb{R}$ meets $\mathbb{R}$ only at c. Put $\mathsf{u}_1(y) = \int_{[0,y]} \mathsf{u}(z)\,dz$. The integral converges since $|\mathsf{u}| \le C|z|^{-1/2}$ near 0, and it differs from $\int_{[1/2,y]} \mathsf{u}$ by a constant (Cauchy's theorem for triangles), so u_1 is a holomorphic primitive of u on $\mathbb{C}_{\mathrm{cut}}$. Along $[0,y]$, $|1 - \rho e^{\mathrm{i}\theta}| \ge |1-\rho|$; with the bounds above, $|\mathsf{u}(\rho e^{\mathrm{i}\theta})| \le C\rho^{-1/2}|1-\rho|^{\tau-1}(1 + |\log|1-\rho||)$ for $\rho \le 3$ and $\le C\rho^{\tau-2}(1+\log\rho)$ for $\rho \ge 3$, so u_1 is bounded on $\mathbb{C}_{\mathrm{cut}}$. Moreover $|\mathsf{u}_1(y)| \le C|y|^{1/2}$ near 0, and $\mathsf{u}_1(y) = I_1 + \int_{[1,y]} \mathsf{u}$ (Cauchy's theorem for the triangle $0,1,y$), so $\mathsf{u}_1(y) \to I_1$ as $y \to 1$.

(b) Edge values of u_1. With $\hat{f}_-(x) = \int_x^0 f_-$, $\hat{f}_+(x) = \int_1^x f_+$ and $\hat{g}_+(x) = \int_1^x g_+$,

$$\mathsf{u}_1(x \pm \mathrm{i}0) = \pm \mathrm{i}\hat{f}_-(x) \quad (x<0), \qquad \mathsf{u}_1(x \pm \mathrm{i}0) = I_1 - e^{\mp \mathrm{i}\pi\tau}(\hat{f}_+(x) \pm \mathrm{i}\hat{g}_+(x)) \quad (x>1).$$

(c) Stokes. On $\mathbb{C}_{\mathrm{cut}}$, $d(\mathsf{u}_1\bar{\mathsf{v}}\,d\bar{y}) = \mathsf{u}\bar{\mathsf{v}}\,dy \wedge d\bar{y} = -2\mathrm{i}\,\mathsf{u}\bar{\mathsf{v}}\,d^2y$. Apply Green's theorem on slit domains to $\{y \in \mathbb{C}_{\mathrm{cut}} : r < |y| < R,\ |y-1| > r\}$. The upper edges of both slits are traversed in the $+x$ direction and the lower edges in the $-x$ direction. With $[\mathsf{g}]_-^+ = \mathsf{g}(x + \mathrm{i}0) - \mathsf{g}(x - \mathrm{i}0)$,

$$-2\mathrm{i}\int \mathsf{u}\bar{\mathsf{v}}\,d^2y = \int_{-R}^{-r} [\mathsf{u}_1\bar{\mathsf{v}}]_-^+\,dx + \int_{1+r}^{R} [\mathsf{u}_1\bar{\mathsf{v}}]_-^+\,dx + (\text{circle terms}).$$

(d) Circle terms. On $|y| = r$: $|\mathsf{u}_1| \le Cr^{1/2}$ and $|\mathsf{v}| \le Cr^{-1/2}(1 + |\log r|)$, so the term is $O(r(1 + |\log r|))$. On $|y-1| = r$: $|\mathsf{u}_1| \le C$ and $|\mathsf{v}| \le Cr^{\tau-1}$, so the term is $O(r^\tau)$. On $|y| = R$: $|\mathsf{u}_1| \le C$ and $|\mathsf{v}| \le CR^{\tau-2}(1 + \log R)$, so the term is $O(R^{\tau-1}(1 + \log R))$. All three vanish in the limit, because $0 < \tau < 1$.

(e) Jumps. From (400) and (b),

$$\begin{aligned} x<0: &\quad [\mathsf{u}_1\bar{\mathsf{v}}]_-^+ = (\mathrm{i}\hat{f}_-)(-f_- + \mathrm{i}g_-) - (-\mathrm{i}\hat{f}_-)(-f_- - \mathrm{i}g_-) = -2\mathrm{i}\,\hat{f}_- f_-, \\ x>1: &\quad [\mathsf{u}_1\bar{\mathsf{v}}]_-^+ = -2\mathrm{i}s_\tau I_1\, g_+ + 2\mathrm{i}\,\hat{g}_+ g_+. \end{aligned}$$

(f) Limit. Since $\mathsf{u}\bar{\mathsf{v}} \in L^1$, the area integral converges. On $(-\infty, 0)$, $\hat{f}_-' = -f_-$, $\hat{f}_-(0) = 0$ and $\hat{f}_-(-\infty) = I_5$, so $\int_{-\infty}^0 (-2\mathrm{i}\hat{f}_- f_-)\,dx = -\mathrm{i}I_5^2$. On $(1, \infty)$, $\hat{g}_+' = g_+$, $\hat{g}_+(1) = 0$ and $\hat{g}_+(\infty) = I_4$, so $\int_1^\infty (-2\mathrm{i}s_\tau I_1 g_+ + 2\mathrm{i}\hat{g}_+ g_+)\,dx = -2\mathrm{i}s_\tau I_1 I_4 + \mathrm{i}I_4^2$. Hence

$$\int_{\mathbb{C}_{\mathrm{cut}}} \mathsf{u}\bar{\mathsf{v}}\, d^2y = \frac{\mathrm{i}}{2}(-\mathrm{i}I_5^2 - 2\mathrm{i}s_\tau I_1 I_4 + \mathrm{i}I_4^2) = \frac{1}{2}(I_5^2 + 2s_\tau I_1 I_4 - I_4^2).$$

(g) By (399), the left side of (402) is $8\Re \int \mathsf{u}\bar{\mathsf{v}}\, d^2y = 4(I_5^2 + 2s_\tau I_1 I_4 - I_4^2)$; the complement of $\mathbb{C}_{\mathrm{cut}}$ has measure zero. □

Second evaluation. Take instead the primitive $\mathsf{v}_1(y) = \int_{[1,y]} \mathsf{v}$ of v, based at 1. On $x < 0$ it has edge values $\mathsf{v}_1(x \pm \mathrm{i}0) = -I_2 + \hat{f}_-(x) \pm \mathrm{i}\hat{g}_-(x)$, with $\hat{g}_-(x) = \int_x^0 g_-$, and the same steps give $\int \mathsf{v}\bar{\mathsf{u}}\, d^2y = \frac{1}{2}(I_4^2 + 2I_2 I_5 - I_5^2)$ (exact check S0c). Since $\int \mathsf{v}\bar{\mathsf{u}} = \overline{\int \mathsf{u}\bar{\mathsf{v}}}$ and both values are real, the two evaluations must agree; given (77) they do, the difference of the two brackets being $2I_4^2(1 - s_\tau^2 - c_\tau^2) = 0$. Their average, $\int \mathsf{u}\bar{\mathsf{v}}\, d^2y = \frac{1}{2}(s_\tau I_1 I_4 + I_2 I_5)$, is an identity of the bracket polynomials that needs no relation among the I_j; it is the bilinear form of (674) at this point, the last form of (377).

Assembly. By (397), (402), (77) and the values $I_3 = \pi^2\mathfrak{F}(1-\tau)/(2s_\tau)$, $I_4 = \pi^2\mathfrak{F}(\tau)/(2s_\tau)$ of Section 18.8,

$$J_{2\tau} = \frac{4}{\pi^2}(I_5^2 + 2s_\tau I_1 I_4 - I_4^2) = \frac{4}{\pi^2}s_\tau^2(I_3^2 + I_4^2) = \pi^2\left[\mathfrak{F}(1-\tau)^2 + \mathfrak{F}(\tau)^2\right] < \infty,$$

which is (377); the other forms there follow from $I_2 = s_\tau I_3$, $I_6 = s_\tau I_4$ and the average above. All of this algebra is checked exactly in `LRI_C5_d2_proof/K2_d1template_fix1/exact_algebra_v2.wl` (parts S0a–S0f). This proof uses neither the cell table (375), nor the rotation lemma (376), nor (674).

Remarks. (1) *Power counting.* Convergence already follows from the bounds above; for completeness, the integrand of $J_{2\tau}$ in $\mathbb{R}^4$ has the following strict margins: $x \to 0$, $x \to 1$, $x \to y$ and $y \to 0$, margin 1; $y \to 1$, margin 2τ; $x, y \to 0$ jointly, $4 - 3 = 1$; $x, y \to 1$ jointly, $4 - (4 - 2\tau) = 2\tau$; $x \to \infty$ alone, $2 - 3 = -1$; $y \to \infty$ alone, $2 - (4 - 2\tau) = 2\tau - 2$; $x, y \to \infty$ jointly, $4 - (6 - 2\tau) = 2\tau - 2$; $x, y \to \infty$ with $x - y$ bounded, total $2\tau - 3$, with $|x-y|^{-1}$ integrable. Mixed configurations factorize into these cases. For $0 < \tau < 1$ every margin has the required sign, so the point $(1, 1, 1, 2 - 2\tau, 1)$ lies in the interior of the convergence domain in all five exponents. (2) *Relation to $d = 1$.* At $d = 1$, integrating x over $\mathbb{R}$ gives the full-line elliptic star, which is linear in K, and folding gives the linear functional (60). At $d = 2$, integrating x over $\mathbb{C}$ gives the bilinear star (398), and the y integration becomes the bilinear Stokes pairing (402). The six Euler moments, Euler integration and the moment relations are common to both dimensions. The higher Taylor coefficients enter the extraction of Section 18.8.4 as $2(\varrho + \pi^2 v)$, against $4(\varrho + \pi^2 v)$ in (88) at $d = 1$.

18.9.9 Status and checks

Statement	Status	Inputs
Majorant (379), regularized form (381), $[\tau^2]$ under the trace; sharper bound (382)	proved	Stirling [27]; the expansion after (250), checked exactly
Per-spin residues (384), factorized traces (385)	proved	reflection, Stirling
Single sums (386), (387), (388), (389) (three evaluations)	proved	Gauss [4], (82), (77), Euler integral, [23]
Regulated traces (390)	proved	the rows above
Mixed-trace step (391) at $d = 2$, $0 < \tau < \frac{1}{2}$, full-line reading	proved	(390), (377)
Right side of (248) at $d = 2$ equals $32\mathsf{G}^2 - 28\zeta(3)\log 2$; $\langle t_\ell^2 E_2\rangle_2 = -21\zeta(3)$	proved	(87), (86)
Per-spin identity (396); (260) at $d = 2$ equals the same; strip limit	proved	(395) (exact algebra), the factorization of `factor_K2.py` for part (c)
Inner star (398), outer integral (402), second proof of (377)	proved	(77), Euler integral, [23]
(M) for complex τ; (M) and $[\tau^2]$ under the trace at $d \neq 2$	not addressed	input (M) of Section 21.1

Checks. None of the steps above uses numerics; the following are checks only, each with its record in the package named.

- *Periodization and traces* (`check_periodization.py` in `LRI_C5_d2_simplify/proof/K2-parseval/`): at $\tau = 0.13$, 0.3 and 0.41 both identities (386), the first two of (387), the Euler–Thomae step (388) and the first identity of (390) hold to 10^{-34} or better; for example $\langle t_\ell^2 R_{0.3}\rangle_2 = 20.73663823644757695029316\ldots$ from both sides.

- *Residues* (`check_perspin.py` in the same folder): (384) at $\tau = 0.3$, $\ell = 0, 3$, Barnes integral against residue sum, to 6×10^{-19}; (396) at $\ell = 0, 1$ to 10^{-21}; the factorized summand of `factor_K2.py` against (259) at four lattice points to 4×10^{-20}. The residues of (384) by Cauchy circles agree with the formulas to 6×10^{-54}, and the identity $\Theta_\ell^{\mathrm{R}} = G_\ell T_\tau$ holds at a test point to 10^{-78} (`LRI_C5_d2_proof/review_K2_d1template_numerics_r0/logs/a01_9_20.568115.log`).

- *Single sums*: the third evaluation, and (385) computed from direct sums, agree with the closed forms and with $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ to 3.3×10^{-67} at $\tau = \frac{1}{7}, \frac{3}{10}, \frac{9}{20}$ (`LRI_C5_d2_proof/review_K2_d1template_numerics_r0/logs/a4b_*.log`). In the second evaluation, Gauss's sum, the Thomae step and the undifferentiated single sums hold to about 10^{-50}, the derivative identities to 10^{-42} (limited by finite differences), and the formula of part (c) for P_{h} at six points (τ, u) to about 24 digits (`LRI_C5_d2_proof/K2_gauss/logs/check_singles.log`, `check_dLambda.log`, `check_bilateral.log`).

- *Algebra*: the expansion after (250), both Laurent extractions and (395) are exact (`K2-parseval/symbolic_checks.out`, `synth/synth_checks_K2.log`, `K2_gauss/logs/check_expansion.log`, `check_extraction.log`, `check_residue_series.log`); the Taylor inputs (378) hold to 10^{-23} (`check_taylor_inputs.out`).

- *Spectral moment*: a direct spectral evaluation of $\langle t_\ell^2 E_2\rangle_2$ on symmetry (spins $\ell < 4096$, Richardson extrapolation in ℓ; job 567967, `K2-parseval/e2check-567967.out`) gives -25.243194966351479993, equal to $-21\zeta(3)$ within 7.3×10^{-24}. A direct spectral integration of $C(\tau)\langle t_\ell^2(R_\tau - S_\tau)\rangle_2$ at $\tau = \frac{1}{10}, \frac{1}{5}$ agrees with the right side of (392) to about 14 digits (`K2_gauss/logs/check_endtoend.log`).

- *Extraction without the cancellation rule*: a 128-point Cauchy integral on $|\tau| = \frac{1}{8}$, at 90 digits, of the true ${}_3F_2$ functions in (392), which does not use (86), gives $-\frac{AU}{4} + \frac{1}{4}[\tau^2]C\mathcal{D}$ equal to

$32\mathsf{G}^2 - 28\zeta(3)\log 2$ to 88 digits, $[\tau^1]C\mathcal{D} = -21\zeta(3)$ to 20 digits, and agreement with the 70-digit series value to 2×10^{-65}, the truncation of that value (`K2_gauss/logs/check_K2_cauchy.log`).

- *Position space*: the edge values (400) hold to 7×10^{-30}, and the Stokes expressions and moment relations to 2×10^{-28}, at $\tau = 0.3, 0.45, 0.6$ (`LRI_C5_d2_proof/K2_d1template_fix1/check_boundary_values_mp.out`); (398) agrees with an independent Feynman-parameter representation to 56–58 digits at fourteen points (`check_lemma1_star.out`); a fresh composite run at $\tau = \frac{37}{100}$, inserting (398) into (397) and integrating over y, gives $J_{0.74} = 41.788097966990073475823724927016212 7\ldots$, equal to $\pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$ to 38 digits (`composite_tau037.out`).

- *The tetrahedron point*: at $\tau = \frac{1}{2}$, (377) gives $J_1 = 2\pi^2\mathfrak{F}(\frac{1}{2})^2 = 3Q_2$ (Dixon's sum, checked to 578 digits), and its second derivative, $\frac{\pi^2}{2}[\mathfrak{F}'(\frac{1}{2})^2 + \mathfrak{F}(\frac{1}{2})\mathfrak{F}''(\frac{1}{2})]$, equals $Z_2 = 4\mathsf{G}Q_2$ of (265) to 77 digits (`review_K2_d1template_numerics_r0/logs/c6b.568116.log`). By (239), a change of the single exponent f by δ has squared completion variation $4\delta^2$, so this is a check of (377) against the complex Selberg evaluation of Z_2.

18.10 Scope of the primitive reduction

The finite sample formulas use at most two-variable hypergeometric functions for generic dimension; at $d = 2$ the factorizations of Section 18.5 use the same scalar values as at $d = 1$. For generic dimension the nonprimitive functions are the explicit finite multivariable combinations of Section 17; Sections 20 and 21 extend the periodization of this section to a general rim exponent and to the family R_τ of (249), and reduce them to two-variable samples and their low-order parameter jets. At $d = 2$ all three are known in closed form: $Z_2 = 4\mathsf{G}Q_2$ by (265), and $\mathcal{L}_2$, $\mathcal{K}_2$ by (270) and (271) (Sections 18.7 and 18.8). Their physical occurrence can be collected as

$$\mathcal{N}_d = \mathcal{K}_d + \frac{3}{8}\mathcal{L}_d - \frac{149}{32}Z_d. \tag{403}$$

This is a change of basis, not an additional evaluation. In this combination (223) reads

$$\boxed{\begin{aligned} 46656C_5(d) = {} & -1536\big(P_5(d) + \mathcal{N}_d\big) + 384AP_4(d) - 1728Q_d^2 + 720A^2Q_d + 1008Q_dR \\ & - 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X, \end{aligned}} \tag{404}$$

since $1536 \cdot \frac{3}{8} = 576$ and $1536 \cdot \frac{149}{32} = 7152$. Thus $\mathcal{K}_d$, $\mathcal{L}_d$ and Z_d enter $C_5(d)$ only through $\mathcal{N}_d$, which appears together with $P_5(d)$. This is a remark on the structure of (223); Sections 20 and 21 reduce the three constituents separately. The finite Fourier spectra used in Sections 20 and 21 belong to the one-parameter families through which the masters are defined, the rim exponent in (244) and τ in (249); for $\mathcal{K}_d$ this route also rests on the inputs of Section 21.1.

19 Three dimensions: half-shift determinants and rank-two functions

The half-shift reduction and the rank-two identification of this section, with their checks, were written by OpenAI GPT-6 astra, in the packages `lri_d3_reduction_update/` and `lri_d3_matrix_update/` (Section 33). The integer dimension $d = 3$ admits a further reduction of the primitive periods Q_3, $P_4(3)$ and $P_5(3)$, but not the square factorization of $d = 2$ (Section 18.5). Its multiplicity $d_\ell = 2\ell + 1$ produces an antisymmetric determinant on two lattices offset by one half (Section 19.4). This eliminates one Fourier-parity sector of the samples (311) in favor of ordinary hypergeometric values. The complementary sector is not fixed by that identity. The formulas below evaluate it by a

small, explicitly specified set of two-variable Horn values, which are identified, from Section 19.6 on, with real rank-two matrix-argument hypergeometric functions. The formulas contain no parameter derivatives, Barnes integrals or external infinite sums. They do not prove that the remaining two-variable functions are finite combinations of ordinary ${}_pF_q$ values.

The tetrahedron takes the form

$$\boxed{Q_3 = 4\pi^2 \mathfrak{H}_3(0) - \frac{\Gamma(\frac{1}{4})^{10}}{48\pi^{7/2}}.} \tag{405}$$

Here Q_3 means $Q_{d=3}$, not the one-dimensional three-loop coefficient. The fixed *two-variable* function $\mathfrak{H}_3(0)$ is fully specified in (414) below. In particular (405) is not a Gamma-only evaluation of the tetrahedron.

For the real formulas with positive rational parameters proved below, the numbers of distinct function evaluations are bounded as in the following table. For comparison, its second column counts the values of functions of at most two variables in the generic-dimensional samples (323) and (329) at the nodes (312): five blocks at each of the r nodes for $r = 3, 4$, and eight blocks at each of the seven nodes for $r = 7$, one-variable blocks included and the real part of a complex value counted once.

Period	Generic-d samples, at most	Two-variable values	Ordinary values	Total
Q_3	15	1	0	1
$P_4(3)$	20	2	3	5
$P_5(3)$	56	5	18	23

Gamma factors and finite arithmetic are not counted. The counts are upper bounds, not independence or minimality statements. The fixed complex nodes of Section 19.5 reduce the two-variable counts further, to one for $P_4(3)$ and two for $P_5(3)$, at the price of fixed complex parameters.

19.1 Ordinary hypergeometric moments

Write $h(x) = \Gamma(x + \frac{1}{8})/\Gamma(x + \frac{7}{8})$. For $\omega = \pm 1$ and nonnegative integer m define the ordinary function

$$\mathfrak{U}^{(\omega)}_{r,m}(u) = u^m h(u)^r \, {}_{r+m+1}F_{r+m}\left(\begin{matrix} 1, (u+\frac{1}{8})^{\times r}, (u+1)^{\times m} \\ (u+\frac{7}{8})^{\times r}, u^{\times m} \end{matrix}; \omega\right). \tag{406}$$

Repeated parameters with multiplicity zero are absent. Identical fixed upper and lower parameters may be canceled. Its series is $\sum_{n\geq 0} \omega^n (n+u)^m h(n+u)^r$. The required pairs are $(r, m) = (3, 0), (3, 1), (4, 0), (4, 1), (7, 1), (7, 3)$; all satisfy $3r/4 - m > 1$, so the displayed ordinary series converge absolutely at both arguments $\omega = 1$ and -1 [22].

Use the following finite combination, also entirely in ordinary functions:

$$\boxed{\begin{aligned} \Psi^{(\omega)}_{r,m}(u) = {} & \sin^r(\pi(u+\tfrac{1}{8}))\mathfrak{U}^{(\omega)}_{r,m}(u) \\ & + \omega(-1)^m \sin^r(\pi(\tfrac{1}{8}-u))\mathfrak{U}^{(\omega)}_{r,m}(1-u). \end{aligned}} \tag{407}$$

In all real formulas below $0 < u < 1$; all upper and lower parameters of the uncancelled ordinary functions are positive rational numbers. At $u = 1/8$ the second term vanishes exactly, so its function need not be computed. The symbol ω is a hypergeometric argument, not the physical expansion parameter ϵ.

The finite-mode identities needed here are

$$\boxed{\begin{aligned}
&\Psi^{(+)}_{3,0}(u) = \mathfrak{a}_3 \cos \pi u, && \Psi^{(+)}_{3,1}(u) = \mathfrak{b}_3 \sin \pi u,\\
&\Psi^{(+)}_{4,0}(u) = \mathfrak{a}_4 + \mathfrak{c}_4 \cos 2\pi u, && \Psi^{(+)}_{4,1}(u) = \mathfrak{b}_4 \sin 2\pi u,\\
&\Psi^{(-)}_{7,m}(u) = \sum_{j=1}^{3} h_{m,j} \sin(2\pi j u), && m = 1, 3.
\end{aligned}} \tag{408}$$

The coefficient $\mathfrak{c}_4$ will cancel from every answer. The useful amplitudes have the fixed-value definitions

$$\begin{aligned}
\mathfrak{a}_3 &= \frac{\Psi^{(+)}_{3,0}(1/8)}{\cos(\pi/8)}, \quad \mathfrak{b}_3 = \frac{\Psi^{(+)}_{3,1}(1/8)}{\sin(\pi/8)},\\
\mathfrak{a}_4 &= \Psi^{(+)}_{4,0}(1/4), \quad \mathfrak{b}_4 = \frac{\Psi^{(+)}_{4,1}(1/8)}{\sin(\pi/4)}.
\end{aligned} \tag{409}$$

For $m = 1, 3$ let $Y_{m,-} = \Psi^{(-)}_{7,m}(1/8)$, $Y_{m,0} = \Psi^{(-)}_{7,m}(1/4)$ and $Y_{m,+} = \Psi^{(-)}_{7,m}(3/8)$. The three coefficients $h_{m,j}$, unrelated to the function $h(x)$ above, are explicitly

$$\begin{aligned}
h_{m,1} &= \frac{Y_{m,0}}{2} + \frac{Y_{m,-} + Y_{m,+}}{2\sqrt{2}},\\
h_{m,2} &= \frac{Y_{m,-} - Y_{m,+}}{2},\\
h_{m,3} &= -\frac{Y_{m,0}}{2} + \frac{Y_{m,-} + Y_{m,+}}{2\sqrt{2}}.
\end{aligned} \tag{410}$$

This is a finite sine transform on a proved three-mode space, not numerical interpolation of an arbitrary function.

19.1.1 Two Gamma evaluations within the ordinary sector

Dixon's sum and the Rogers–Dougall sum [22, eqs. 16.4.4 and 16.4.9] give

$$\boxed{\mathfrak{a}_3 \mathfrak{b}_3 = \frac{\Gamma(\frac{1}{4})^{10}}{16\sqrt{2}\,\pi^4}, \qquad \mathfrak{b}_4 = \frac{\Gamma(\frac{1}{4})^6}{16\pi^{5/2}}.} \tag{411}$$

For clarity the particular sums used in this step are

$$\begin{aligned}
{}_3F_2\left(\begin{matrix}(\frac{1}{4})^{\times 3}\\ 1, 1\end{matrix}; 1\right) &= \frac{\Gamma(\frac{9}{8})\Gamma(\frac{5}{8})}{\Gamma(\frac{5}{4})\Gamma(\frac{7}{8})^2\Gamma(\frac{3}{4})},\\
{}_4F_3\left(\begin{matrix}\frac{9}{8}, (\frac{1}{4})^{\times 3}\\ \frac{1}{8}, 1, 1\end{matrix}; 1\right) &= \frac{\Gamma(\frac{5}{8})\Gamma(\frac{1}{8})}{\Gamma(\frac{5}{4})\Gamma(\frac{3}{4})\Gamma(\frac{3}{8})^2},\\
{}_5F_4\left(\begin{matrix}\frac{9}{8}, (\frac{1}{4})^{\times 4}\\ \frac{1}{8}, 1, 1, 1\end{matrix}; 1\right) &= \frac{\sqrt{\pi}}{\Gamma(\frac{5}{4})\Gamma(\frac{3}{4})^3}.
\end{aligned} \tag{412}$$

For the second identity insert a canceled pair 5/8 into the Rogers–Dougall ${}_5F_4$, with parameters $(a, b, c, d) = (1/4, 1/4, 1/4, 5/8)$ in the notation of [22, eq. 16.4.9]; this d is not the dimension. For the third take $(a, b, c, d) = (1/4, 1/4, 1/4, 1/4)$. Their convergence conditions hold. Substitution into (409), followed by reflection and duplication [26], proves (411). Thus the Gamma powers in (405) are analytic evaluations, not PSLQ fits.

19.2 The remaining two-variable functions and complete sample formulas

Use the Horn convention (320). Put $\boldsymbol{e}=(1,0)$, $\boldsymbol{f}=(0,1)$, $\boldsymbol{c}=(1,1)$ and $\boldsymbol{v}=(1,2)$, and set

$$\begin{aligned}\mathsf{U}_r(t)&=\{(1,\boldsymbol{e}),(1,\boldsymbol{f}),(\tfrac{3}{2},\boldsymbol{e}),(\tfrac{3}{4}+t,\boldsymbol{c})^{\times r},(\tfrac{1}{4}+t,\boldsymbol{f})^{\times r}\},\\ \mathsf{V}_r(t)&=\{(\tfrac{1}{2},\boldsymbol{e}),(\tfrac{3}{2}+t,\boldsymbol{c})^{\times r},(1+t,\boldsymbol{f})^{\times r}\}.\end{aligned}\tag{413}$$

For $r=3,4$ define the fixed two-variable function

$$\boxed{\mathfrak{H}_r(t)=\mathfrak{H}[\mathsf{U}_r(t);\mathsf{V}_r(t);(1,1)].}\tag{414}$$

For $r=7$ adjoin three upper/lower row pairs:

$$\begin{aligned}x_t&=t+\tfrac{5}{8}, & y_t&=t+\tfrac{1}{8}, \quad v_t=2t+\tfrac{3}{4},\\ \mathfrak{H}_7(t)&=\mathfrak{H}\Big[\mathsf{U}_7(t)\cup\{(1+x_t,\boldsymbol{c}),(1+y_t,\boldsymbol{f}),(1+v_t,\boldsymbol{v})\};\\ &\qquad \mathsf{V}_7(t)\cup\{(x_t,\boldsymbol{c}),(y_t,\boldsymbol{f}),(v_t,\boldsymbol{v})\};(-1,1)\Big].\end{aligned}\tag{415}$$

Every row is printed here; there is no unevaluated coefficient-extraction instruction. The two upper rows with parameter 1 cancel the two factorials of the Horn convention. The ratio $(3/2)_\ell/(1/2)_\ell=2\ell+1$ supplies the three-dimensional angular multiplicity. The three additional pairs supply x, y and $x+y$. All real nodes used below have positive rational rows.

For compact scalar prefactors write

$$\begin{aligned}G_r(t)&=\left[\frac{\Gamma(\frac{3}{4}+t)\Gamma(\frac{1}{4}+t)}{\Gamma(\frac{3}{2}+t)\Gamma(1+t)}\right]^r,\\ V_r(t)&=G_r(t)\mathfrak{H}_r(t), & r&=3,4,\\ V_7(t)&=4x_ty_tv_tG_7(t)\mathfrak{H}_7(t).\end{aligned}\tag{416}$$

The symbol $V_r(t)$ here is local to this section; it is not the polygamma constant V in the general-dimensional coefficient formula.

Let $\mathsf{X}_r(t)=\Xi_r(t)|_{q=3/4}$, with Ξ_r as in (304), and define

$$\begin{aligned}t^\dagger&=\tfrac{1}{4}-t, & f(t)&=-\tfrac{1}{2}\cos 2\pi t,\\ D(t)&=\sin\pi t\sin\pi(t+\tfrac{3}{4}), & T(t)&=\tan\pi(\tfrac{3}{4}+2t), \quad z_t=\tfrac{7}{8}-t.\end{aligned}\tag{417}$$

A sample in a wheel is now just

$$\boxed{\begin{aligned}\mathsf{X}_r(t)&=f(t)^rV_r(t)+f(t^\dagger)^rV_r(t^\dagger)\\ &\qquad+2D(t)^r\big[\mathfrak{U}^{(+)}_{r,1}(x_t)\mathfrak{U}^{(+)}_{r,0}(z_t)+\mathfrak{U}^{(+)}_{r,0}(x_t)\mathfrak{U}^{(+)}_{r,1}(z_t)\big], \qquad r=3,4.\end{aligned}}\tag{418}$$

The corresponding five-loop sample is

$$\boxed{\begin{aligned}\mathsf{X}_7(t)=T(t)\Big\{&f(t)^7V_7(t)-f(t^\dagger)^7V_7(t^\dagger)\\ &+8D(t)^7\big[\mathfrak{U}^{(-)}_{7,3}(x_t)\mathfrak{U}^{(-)}_{7,1}(z_t)-\mathfrak{U}^{(-)}_{7,1}(x_t)\mathfrak{U}^{(-)}_{7,3}(z_t)\big]\Big\}.\end{aligned}}\tag{419}$$

Only fixed ordinary and two-variable functions appear. In particular the middle triangular part of the (ℓ,n) lattice of (304) factorizes into ordinary products. At $t=0$ the left and middle pieces vanish, giving the especially simple nonsingular formulas

$$\boxed{\begin{aligned}\mathsf{X}_3(0)&=-2\sqrt{2}\,\pi^{3/2}\mathfrak{H}_3(0),\\ \mathsf{X}_4(0)&=4\pi^2\mathfrak{H}_4(0),\\ \mathsf{X}_7(0)&=\frac{15\sqrt{2}}{8}\pi^{7/2}\mathfrak{H}_7(0).\end{aligned}}\tag{420}$$

These are values of already regularized functions, not singular parameter limits to be taken by the reader.

19.3 Primitive formulas with only real rational parameters

Let $c_8 = \cos(\pi/8)$ and $s_8 = \sin(\pi/8)$, as in (331). For the four-loop answer use

$$\begin{aligned} X_0 &= 4\pi^2 \mathfrak{H}_4(0), \\ X_h &= \frac{1}{32}\left[\frac{\Gamma(\frac{7}{8})\Gamma(\frac{3}{8})}{\Gamma(\frac{13}{8})\Gamma(\frac{9}{8})}\right]^4 \mathfrak{H}_4(\tfrac{1}{8}) + 4s_8^8 \mathfrak{U}_{4,0}^{(+)}(\tfrac{3}{4})\mathfrak{U}_{4,1}^{(+)}(\tfrac{3}{4}), \\ \mathfrak{a}_4 &= c_8^4 \mathfrak{U}_{4,0}^{(+)}(\tfrac{1}{4}) + s_8^4 \mathfrak{U}_{4,0}^{(+)}(\tfrac{3}{4}), \qquad \mathfrak{b}_4 = \frac{\Gamma(\frac{1}{4})^6}{16\pi^{5/2}}. \end{aligned} \tag{421}$$

Here $X_0 = \mathsf{X}_4(0)$ and $X_h = \mathsf{X}_4(1/8)$; they are not the one-dimensional X_0, X_+, X_- of Section 18. The answer is

$$\boxed{P_4(3) = 2\pi X_0 - 10\pi X_h + (14\sqrt{2} - 20)\pi \mathfrak{a}_4 \mathfrak{b}_4.} \tag{422}$$

This uses two $\mathfrak{H}_4$ values, two ordinary ${}_5F_4(1)$ values, and one ordinary ${}_6F_5(1)$ value, plus Gamma factors. No claim that the two $\mathfrak{H}_4$ values have become ordinary functions is intended.

For $P_5(3)$ form the two ordinary-product constants

$$\begin{aligned} \alpha &= h_{3,2}h_{1,1} - h_{3,1}h_{1,2}, \\ \beta &= h_{3,2}h_{1,3} - h_{3,3}h_{1,2}, \\ \theta(t) &= 2\pi(t + \tfrac{3}{8}), \qquad w(t) = \cos\theta(t), \\ \mathcal{O}(t) &= 16w(t)(1 - w(t)^2)\big[-\alpha - 3\beta + 4\beta w(t)^2\big]. \end{aligned} \tag{423}$$

All $h_{m,j}$ were given by the finite ordinary evaluations (410). Use three regularized samples, with $u_j = \cos(4\pi(t_j + 3/8))$ the variable of (435) below:

$$\begin{aligned} t_0 &= 0, & t_+ &= \tfrac{1}{16}, & t_- &= -\tfrac{1}{16}, \\ u_0 &= 0, & u_+ &= \tfrac{1}{\sqrt{2}}, & u_- &= -\tfrac{1}{\sqrt{2}}, \\ E_j &= \frac{\mathsf{X}_7(t_j) - \mathcal{O}(t_j)}{1 - u_j}, & & j = 0, +, -. \end{aligned} \tag{424}$$

Equations (419) and (420) specify each sample without derivatives. The result is

$$\boxed{\begin{aligned} P_5(3) = 4\sqrt{2}\,\pi^{3/2}\left[-35E_0 + \left(22 + \frac{13}{\sqrt{2}}\right)E_+ + \left(22 - \frac{13}{\sqrt{2}}\right)E_-\right] \\ - 448\pi^{3/2}(2\beta - \alpha). \end{aligned}} \tag{425}$$

The only two-variable values are

$$\mathfrak{H}_7(0), \quad \mathfrak{H}_7(\tfrac{1}{16}), \quad \mathfrak{H}_7(-\tfrac{1}{16}), \quad \mathfrak{H}_7(\tfrac{3}{16}), \quad \mathfrak{H}_7(\tfrac{5}{16}). \tag{426}$$

All their uncancelled parameters are positive rational, and their arguments are $(-1, 1)$. The ten ordinary values for $h_{1,j}, h_{3,j}$ use $u = 1/8, 1/4, 3/4, 3/8, 5/8$ and $m = 1, 3$. Eight further ordinary values in the two middle determinants use $u = 9/16, 11/16, 13/16, 15/16$ and $m = 1, 3$. In total $P_5(3)$ therefore uses eighteen ordinary values. The excess of every one of these ordinary series is $17/4 - m > 0$.

19.4 Proof of the dimension-three determinant

19.4.1 The half-offset lattice, not a dimension-two square

At $q = 3/4$ set $k = \ell + n$, $j = n$ in the bilateral sums (304), with g_ℓ and a_ℓ as in (300), and put

$$x = k + t + 5/8, \qquad y = j + t + 1/8, \qquad u = t + 1/8.$$

Gamma reflection gives, as a meromorphic identity,

$$g_\ell(a_\ell + n + t) = -\frac{\sin \pi(3/4 - t)}{\sin \pi t}\, h(x)h(y), \qquad d_\ell = 2\ell + 1 = 2(x - y). \tag{427}$$

Here $k \geq j$, with both indices otherwise bilateral. The two sequences are $h(k + u + 1/2)$ and $h(j + u)$; unlike $d = 2$, they are not the same sequence on the same integer lattice.

The five-loop Gamma measure becomes elementary:

$$\boxed{\frac{\Gamma(\frac{1}{2} + v)\Gamma(\frac{1}{2} - v)}{\Gamma(v)\Gamma(-v)} = -v \tan \pi v.} \tag{428}$$

This follows directly from the two reflection formulas in [26]. At a sampled pole-family shift $v = 2s = x + y$,

$$(\ell + \tfrac{1}{2})^2 - v^2 = -4xy, \qquad (-1)^\ell = (-1)^{k-j}.$$

Multiplying by d_ℓ gives the polynomial $8xy(x^2 - y^2)$ times the common tangent $T(t)$ and the alternating sign. Splitting the bilateral lattice into its two infinite tails and its middle part now proves (418)–(419). For example the middle part has $\ell = j + k + 1$, $j, k \geq 0$, and $x = k + t + 5/8$, $y = j + 7/8 - t$. Its weight is $2(x + y)$ in a wheel and the rank-two polynomial $8(x^3y - xy^3)T(t)$ in the primitive. These separate into the displayed ordinary products.

19.4.2 Completing only the antisymmetric half

A shift $t \mapsto t + 1/2$ interchanges the two lattices. In a wheel the factor $x - y$ changes sign. In the primitive the extra alternating sign also changes sign. After reindexing the two complementary half-planes, the full-square sums factor into determinants. The result is

$$\begin{aligned}
&\mathsf{X}_r(t) - (-1)^r \mathsf{X}_r(t + \tfrac{1}{2}) \\
&\quad = 2(-1)^r \big[\Psi^{(+)}_{r,1}(u + \tfrac{1}{2})\Psi^{(+)}_{r,0}(u) - \Psi^{(+)}_{r,0}(u + \tfrac{1}{2})\Psi^{(+)}_{r,1}(u)\big], \quad r = 3, 4, \\
&\mathsf{X}_7(t) - \mathsf{X}_7(t + \tfrac{1}{2}) \\
&\quad = -8T(t)\big[\Psi^{(-)}_{7,3}(u + \tfrac{1}{2})\Psi^{(-)}_{7,1}(u) - \Psi^{(-)}_{7,1}(u + \tfrac{1}{2})\Psi^{(-)}_{7,3}(u)\big].
\end{aligned} \tag{429}$$

The factors are obtained from $\sin \pi(t + 1/4) \sin \pi(t + 3/4) = \cos(2\pi t)/2$. All moments involved obey $3r/4 - m > 1$, so the product sums used in the reindexing are absolutely convergent. The formulas extend across removable trigonometric singularities by equality of the complete meromorphic expressions, not by discarding a pole family.

For the proof of the finite modes, reflection gives

$$h(-x) = \frac{\sin \pi(x + 1/8)}{\sin \pi(1/8 - x)} h(x).$$

It turns (407) into the regularized bilateral moment $\sin^r \pi(u + 1/8) \sum_{n \in \mathbb{Z}} \omega^n (n + u)^m h(n + u)^r$. This proof representation has all poles canceled, and satisfies

$$\begin{aligned}
\Psi(u + 1) &= (-1)^r \omega \Psi(u), & \Psi(-u) &= (-1)^m \Psi(u), \\
|\Psi(x + \mathrm{i}y)| &= O\Big(e^{r\pi|y|} |y|^{m+1-3r/4}\Big) & &(|y| \to \infty).
\end{aligned} \tag{430}$$

The growth estimate follows from centered Gamma-ratio asymptotics and absolute integral comparison [27]. Periodicity or antiperiodicity supplies integer or half-integer Fourier frequencies. Vertical contour translation excludes their boundary frequencies when $m + 1 - 3r/4 < 0$. Reflection then leaves exactly the modes in (408). This proves finite-dimensionality, rather than assuming it from a few samples.

Substituting those modes in (429) yields

$$\boxed{\begin{aligned}&\mathsf{X}_3(t)+\mathsf{X}_3(t+\tfrac12)=-2\mathfrak{a}_3\mathfrak{b}_3,\\&\mathsf{X}_4(t)-\mathsf{X}_4(t+\tfrac12)=4\mathfrak{a}_4\mathfrak{b}_4\cos\theta(t),\\&\mathsf{X}_7(t)-\mathsf{X}_7(t+\tfrac12)=2\mathcal{O}(t).\end{aligned}} \tag{431}$$

In the cosine convention (311), the first two lines prove $c_{3,2}=0$, $c_{3,0}=-\mathfrak{a}_3\mathfrak{b}_3$, $c_{4,3}=0$, and $c_{4,1}=2\mathfrak{a}_4\mathfrak{b}_4$. The last line fixes

$$c_{7,1}=-4\alpha-4\beta,\qquad c_{7,3}=4\alpha+8\beta,\qquad c_{7,5}=-4\beta. \tag{432}$$

Of these, $c_{3,2}=0$ and $c_{4,3}=0$ hold in every dimension, by the top-mode theorem (473) of Section 20.1.2; the identifications $c_{3,0}=-\mathfrak{a}_3\mathfrak{b}_3$ and $c_{4,1}=2\mathfrak{a}_4\mathfrak{b}_4$ and the coefficients (432) are the additional restrictions special to three dimensions. The identifications of $c_{3,0}$ and $c_{4,1}$ also hold at a general rim exponent b, by the same proof (Section 20.6); the $d=3$ forms of $\mathcal{L}_d$ and Z_d in Section 20.6 use them.

19.4.3 Finite residue algebra and the remaining even polynomial

For a self-contained normalization check, the weights $B_{r,k}/\pi^{r-1}$ in (318) at $q=3/4$ are

$$\begin{aligned}&r=3:\quad(8,-6\sqrt2,12),\\&r=4:\quad(32,-24\sqrt2,40,-40\sqrt2),\\&r=7:\quad(2624,-1904\sqrt2,2912,-2352\sqrt2,4032,-3696\sqrt2,7392).\end{aligned} \tag{433}$$

Together with (319) at $q=3/4$, that is $Q_3=\Gamma(3/2)^3\mathcal{E}_3/3$, $P_4(3)=\Gamma(3/2)^4\mathcal{E}_4$, and $P_5(3)=\Gamma(3/2)^5\mathcal{E}_7/2$, these weights give (405) and (422) immediately. The clockwise right closure supplies the minus sign in $\mathcal{E}_r$.

For $r=7$ there are also exact zeros

$$\mathsf{X}_7(\tfrac18)=\mathsf{X}_7(\tfrac58)=0. \tag{434}$$

Indeed $v=\ell+3/4+2n+2t$ is an integer at either point, and the measure in (428) vanishes. No Gamma pole of g_ℓ occurs there. The termwise vanishing is legitimate in the absolutely convergent regularized sum. Hence the even part is exactly

$$\mathsf{X}_7(t)-\mathcal{O}(t)=(1-u)\big(p_0+p_1u+p_2u^2\big),\qquad u=\cos(4\pi(t+3/8)). \tag{435}$$

The even-mode contribution to the $r=7$ residue weight is $-32(9p_0+13p_1+22p_2)$; the odd contribution is $1792\sqrt2(2\beta-\alpha)$. For every quadratic polynomial p,

$$9p_0+13p_1+22p_2=-35p(0)+\left(22+\frac{13}{\sqrt2}\right)p(1/\sqrt2)+\left(22-\frac{13}{\sqrt2}\right)p(-1/\sqrt2). \tag{436}$$

This is verified by comparing three coefficients. It proves (425) without a derivative, infinitesimal step, or numerical interpolation error.

Crucially, (431) does not determine the complementary parity sector. It leaves one coefficient for $r=3$, two for $r=4$, and three for the even $r=7$ polynomial. Equations (414) and (415) evaluate that remaining information. This explains why the dimension-two square proof of Section 18.5 cannot simply be repeated with the replacement $d_\ell=2\ell+1$. It does not prove that some different identity could never reduce the residual Horn values further.

19.5 A variant with fixed complex nodes

The real rational formulas above are the preferred presentation when parameter simplicity matters. Two further exact finite-node identities give smaller two-variable counts. Define

$$\begin{aligned} t_4^* &= -\tfrac{3}{8} + \frac{\mathrm{i}\log 2}{4\pi}, \qquad & t_7^* &= -\tfrac{3}{8} + \frac{\mathrm{i}}{4\pi}\operatorname{arcosh}\frac{22}{13}, \\ E_0^* &= \mathsf{X}_7(0) - \mathcal{O}(0), & E_*^* &= \mathsf{X}_7(t_7^*) - \mathcal{O}(t_7^*), \end{aligned} \tag{437}$$

with $\mathcal{O}$ as in (423); E_0^* is the E_0 of (424). Then

$$\boxed{\begin{aligned} P_4(3) &= 8\pi\left[3\sqrt{2}\,\mathfrak{a}_4\mathfrak{b}_4 - \mathsf{X}_4(t_4^*)\right], \\ P_5(3) &= \frac{2\sqrt{2}\,\pi^{3/2}}{11}\left[29E_0^* - \frac{2197}{9}E_*^*\right] - 448\pi^{3/2}(2\beta-\alpha). \end{aligned}} \tag{438}$$

Both nodes lie on the line $\operatorname{Re} t = -\frac{3}{8}$. There every $\cos(2\pi k(t+\frac{3}{8}))$ is real, so the samples (311), whose coefficients are real, are real.

Proof. Since Ξ_r is entire, (311) and (435) are identities of entire functions whose right sides are trigonometric polynomials, so they hold at complex t. By (311) at $q = \frac{3}{4}$, with $c_{4,3} = 0$ and $c_{4,1} = 2\mathfrak{a}_4\mathfrak{b}_4$ (Section 19.4), $\mathsf{X}_4(t) = c_{4,0} + 2\mathfrak{a}_4\mathfrak{b}_4\cos\theta(t) + c_{4,2}\cos 2\theta(t)$ with $\theta(t)$ as in (423). The weights (433), inserted in (318) and (319), give $P_4(3) = -\pi\big(8c_{4,0} - 6\sqrt{2}\,c_{4,1} + 10c_{4,2}\big)$. At t_4^* one has $\cos\theta = \cosh(\frac{1}{2}\log 2) = \frac{3}{2\sqrt{2}}$ and $\cos 2\theta = \cos(4\pi(t_4^*+\frac{3}{8})) = \frac{5}{4}$, which gives the first line. For the second, (435) gives $E_0^* = p(0)$ and, since $u = \cos(4\pi(t_7^*+\frac{3}{8})) = \frac{22}{13}$ and $1 - \frac{22}{13} = -\frac{9}{13}$, $E_*^* = -\frac{9}{13}\,p(\frac{22}{13})$. The rule

$$9p_0 + 13p_1 + 22p_2 = \frac{29}{22}\,p(0) + \frac{169}{22}\,p(\tfrac{22}{13}),$$

exact for every quadratic p (compare the coefficients of 1, u and u^2), replaces (436) in the proof of (425); $169 \cdot 13 = 2197$. □

A sample on this line needs only one Horn value. By (413) and (416), with $x = x_t + k$, $y = y_t + n$ and $k = \ell + n$ as in (427), $V_r(t) = \sum_{k\ge n\ge 0} 2(x-y)h(x)^r h(y)^r$ for $r = 3, 4$ and $V_7(t) = 8\sum_{k\ge n\ge 0}(-1)^{k-n}xy(x^2-y^2)h(x)^7h(y)^7$. A shift $t \mapsto t+1$ moves x and y by one and so removes the boundary row $n = 0$; the sum of that row is denoted $V_r^{(0)}(t)$:

$$\begin{aligned} V_r(t+1) &= V_r(t) - V_r^{(0)}(t), \\ V_r^{(0)}(t) &= 2h(y_t)^r\left[\mathfrak{U}_{r,1}^{(+)}(x_t) - y_t\,\mathfrak{U}_{r,0}^{(+)}(x_t)\right], \qquad r = 3, 4, \\ V_7^{(0)}(t) &= 8h(y_t)^7\left[y_t\,\mathfrak{U}_{7,3}^{(-)}(x_t) - y_t^3\,\mathfrak{U}_{7,1}^{(-)}(x_t)\right]. \end{aligned} \tag{439}$$

Let $t = -\frac{3}{8} + \mathrm{i}\sigma$ with $\sigma > 0$. Then $t^\dagger = 1 + \bar{t}$, and since V_r is real on the real axis, $V_r(t^\dagger) = \overline{V_r(t) - V_r^{(0)}(t)}$. The other ordinary factors reduce by $\mathfrak{U}_{r,m}^{(\omega)}(u+1) = \omega\big[\mathfrak{U}_{r,m}^{(\omega)}(u) - u^m h(u)^r\big]$. This proves the counts of one $\mathfrak{H}_4$ value for $P_4(3)$ and two $\mathfrak{H}_7$ values for $P_5(3)$, with no second complex Horn value hidden in the samples. The nodes are fixed constants, not regulators. They do introduce fixed complex parameters, some with negative real part (at t_4^* the row $\frac{1}{4} + t$ of (413) has real part $-\frac{1}{8}$); there are no denominator poles at the stated nodes. No minimality claim is made for this variant. Its numerical agreement with (422) and (425) is given in Section 19.10.

19.6 The real rank-two hypergeometric normalization

This subsection and the next two identify the residual functions $\mathfrak{H}_r$ with standard real matrix-argument hypergeometric functions of rank two; Section 19.9 then records a matrix transformation

that maps the tetrahedron to itself and so does not reduce it. The identification is exact and coefficientwise, not a numerical fit. It does not reduce them to ordinary one-variable ${}_pF_q$ values: the ordinary-hypergeometric endpoint reached for the primitive periods at $d = 2$ (Section 18.5) is not reached at $d = 3$. What it adds is a classification: the residual objects are low-order functions on a rank-two symmetric cone, not unclassified Horn arrays. The rational-node formulas above remain the preferred explicit specification of the periods.

For a partition $\kappa = (m, n)$ with $m \geq n \geq 0$ (here m is a part of κ, not the moment order m of (406)), write

$$[a]_\kappa = (a)_m(a - \tfrac{1}{2})_n, \qquad |\kappa| = m + n. \tag{440}$$

From here to the end of this section the Jack parameter is $\alpha = 2$, the matrix size is two, and the physical dimension is $d = 3$. This α is unrelated to the constant α of (423). These three numbers refer to different notions. Let $C_\kappa^{(2)}$ denote the C-normalized real zonal polynomial, so that $\sum_{|\kappa|=k} C_\kappa^{(2)}(X) = (\operatorname{tr} X)^k$. We use the standard definition [38, 39]

$${}_pF_q^{(\alpha=2)}\left(\begin{matrix}\boldsymbol{a}\\ \boldsymbol{b}\end{matrix}; zI_2\right) = \sum_{m\geq n\geq 0} \frac{\prod_i [a_i]_{(m,n)}}{\prod_j [b_j]_{(m,n)}} \frac{C^{(2)}_{(m,n)}(I_2)}{(m+n)!}\, z^{m+n}. \tag{441}$$

This definition fixes the conventional special function; it is not a claim that writing a summation as a function symbol has removed its rank-two structure.

The identity that matches the angular multiplicity is

$$\boxed{\frac{C^{(2)}_{(m,n)}(I_2)}{(m+n)!} = \frac{2(m-n)+1}{(\frac{3}{2})_m n!} = \frac{2(m-n)+1}{[\frac{3}{2}]_{(m,n)}}.} \tag{442}$$

Here is an explicit hook-product verification, including normalization. If $a(s)$ and $l(s)$ are the arm and leg lengths of a box of κ, the Jack hook product is

$$\begin{aligned} j_{(m,n)}(2) &= \prod_{s\in(m,n)} (2a(s)+l(s)+1)(2a(s)+l(s)+2) \\ &= 2^{2(m+n)} \frac{(\frac{3}{2})_m\, m!\, n!\, (\frac{1}{2})_n}{2(m-n)+1}. \end{aligned} \tag{443}$$

The specialization formula $C_\kappa^{(2)}(I_2)/|\kappa|! = 2^{2|\kappa|}[1]_\kappa / j_\kappa(2)$ then gives (442). In particular the factor $2\ell + 1$ with $\ell = m - n$ is exactly the rank-two real-zonal weight after canceling one $[3/2]_\kappa$. This is why a real matrix-argument function appears.

19.7 A standard ${}_3F_2$ of matrix argument for the tetrahedron

Retain the residual functions $\mathfrak{H}_r(t)$ of (414) and (415) and put

$$a = t + \frac{3}{4}, \qquad b = t + \frac{3}{2}. \tag{444}$$

For $r = 3, 4$ their exact identification is

$$\boxed{\mathfrak{H}_r(t) = {}_{r+1}F_r^{(\alpha=2)}\left(\begin{matrix} a^{\times r},\ \frac{3}{2} \\ b^{\times r}\end{matrix}; I_2\right).} \tag{445}$$

To verify it, use $m = \ell + n$ in the explicit Horn arrays. The summand of either side is

$$(2\ell+1)\left[\frac{(a)_{\ell+n}(a-\frac{1}{2})_n}{(b)_{\ell+n}(b-\frac{1}{2})_n}\right]^r. \tag{446}$$

On the matrix side this follows from canceling $[3/2]_{(m,n)}$ against (442). Thus no summation theorem or exchange of a conditionally convergent sum is being assumed. The endpoint sums here are absolutely convergent: along $m \to \infty$ with n fixed the majorant is $m^{1-3r/4}$, and the remaining n factors are summable for $r = 3, 4$.

At $t = 0$ one upper parameter $3/2$ cancels a lower parameter. The tetrahedron formula (405) therefore becomes

$$\boxed{Q_3 = 4\pi^2 {}_3F_2^{(\alpha=2)}\left(\begin{matrix}\frac{3}{4},\frac{3}{4},\frac{3}{4}\\ \frac{3}{2},\frac{3}{2}\end{matrix}; I_2\right) - \frac{\Gamma(\frac{1}{4})^{10}}{48\pi^{7/2}}.} \tag{447}$$

The same value is the rim sample $\mathfrak{t}_3(b)$ of (495) at $d = 3$ and $b = \frac{3}{2}$: there $\sigma = g = \frac{3}{4}$, and the summand $d_\ell T_{\ell n}(\frac{3}{2})^3$ of (490) is (446) with $r = 3$, $t = 0$. Its second b-derivative enters Z_3 (Section 20.6). Both remaining four-loop values likewise have short conventional arrays:

$$\boxed{\begin{aligned}\mathfrak{H}_4(0) &= {}_4F_3^{(\alpha=2)}\left(\begin{matrix}(\frac{3}{4})^{\times 4}\\ (\frac{3}{2})^{\times 3}\end{matrix}; I_2\right),\\ \mathfrak{H}_4(\tfrac{1}{8}) &= {}_5F_4^{(\alpha=2)}\left(\begin{matrix}(\frac{7}{8})^{\times 4},\frac{3}{2}\\ (\frac{13}{8})^{\times 4}\end{matrix}; I_2\right).\end{aligned}} \tag{448}$$

Substitution in (422) gives an equivalent matrix-argument presentation of $P_4(3)$ without changing its ordinary moment factors. Neither of these matrix functions is to be evaluated by an ordinary scalar `hyper` call.

A local coefficient check makes the distinction unambiguous. With the same upper and lower parameters as in (447),

$$\begin{aligned}{}_3F_2^{(\alpha=2)}(\tfrac{3}{4},\tfrac{3}{4},\tfrac{3}{4};\tfrac{3}{2},\tfrac{3}{2};zI_2) &= 1 + \frac{3}{8}z + \frac{2769}{12800}z^2 + \frac{15113}{102400}z^3 + O(z^4),\\ {}_3F_2(\tfrac{3}{4},\tfrac{3}{4},\tfrac{3}{4};\tfrac{3}{2},\tfrac{3}{2};z) &= 1 + \frac{3}{16}z + \frac{1029}{12800}z^2 + \frac{9317}{204800}z^3 + O(z^4).\end{aligned} \tag{449}$$

The first coefficients already differ. This checks the normalization; it does not preclude an identity with some *other* scalar hypergeometric functions or parameters.

19.8 Five loops: an exact three-term standard-function dictionary

The polynomial insertion in $\mathfrak{H}_7$ can also be expressed using ordinary parameter shifts of standard rank-two functions, rather than a bespoke extra Horn row. Set (this v is local; it is not v_t of (415))

$$v = t + \frac{5}{8}, \qquad \delta_v = v(v - \tfrac{1}{2}). \tag{450}$$

Define three standard matrix-argument functions, all at $-I_2$:

$$\begin{aligned}\mathcal{J}_A(t) &= {}_9F_8^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+2\\ b^{\times 7},v\end{matrix}; -I_2\right),\\ \mathcal{J}_B(t) &= {}_{10}F_9^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+1,v+1\\ b^{\times 7},v,v\end{matrix}; -I_2\right),\\ \mathcal{J}_C(t) &= {}_9F_8^{(\alpha=2)}\left(\begin{matrix}a^{\times 7},\frac{3}{2},v+1\\ b^{\times 7},v\end{matrix}; -I_2\right).\end{aligned} \tag{451}$$

Then the exact dictionary is

$$\boxed{\mathfrak{H}_7(t) = \frac{\delta_{v+1}\mathcal{J}_A(t) - \delta_v\mathcal{J}_B(t) - \mathcal{J}_C(t)}{2v - \frac{1}{2}}.} \tag{452}$$

The five arguments required by (425), listed in (426), are $t = 0, \pm 1/16, 3/16, 5/16$. They all have $v > 1/2$, and all generalized Pochhammer factors in these arrays have positive real starting arguments. At $t = 0$, canceling one common $3/2$ changes the three orders to ${}_8F_7$, ${}_9F_8$ and ${}_8F_7$, still *of matrix argument*. No coalescence limit or parameter differentiation is used.

Here is the coefficientwise proof, which also fixes the sign of the matrix argument. For $\kappa = (m, n)$ put $x = m + v$, $y = n + v - 1/2$. Direct Pochhammer cancellation gives

$$\begin{aligned} \frac{[v+1]_\kappa}{[v]_\kappa} &= \frac{xy}{\delta_v}, \\ \frac{[v+2]_\kappa}{[v]_\kappa} &= \frac{xy(x+1)(y+1)}{\delta_v \delta_{v+1}}. \end{aligned} \tag{453}$$

The only polynomial identity needed is

$$xy(x+y) = xy(x+1)(y+1) - (xy)^2 - xy. \tag{454}$$

Consequently the right-hand side of (452) multiplies the common coefficient (446), with $r = 7$, by $xy(x+y)/(\delta_v(2v-1/2))$, exactly the three polynomial ratios in the Horn array (415) of $\mathfrak{H}_7$. Finally $(-1)^\ell = (-1)^{m-n} = (-1)^{m+n}$, so the matrix argument is $-I_2$, *not* $\mathrm{diag}(-1, 1)$. Each of the three endpoint series separately converges absolutely. For example, the largest polynomial insertion has degree two in each of m, n; its absolute summand is bounded by a constant times $(m+1)^{-9/4}(n+1)^{-13/4}$ on $m \geq n$, which is summable.

This dictionary standardizes the function class, but increases the number of displayed functions per Horn block from one to three. It is not advertised as a more economical numerical formula. In particular, it does not eliminate the complementary ordered-lattice information isolated by the half-shift determinant.

19.9 A checked transformation that does not finish the tetrahedron

The matrix Kummer transformation [39, Eq. (35.8.5)] is a useful diagnostic. For $s_{\mathrm{K}} = b_1 + b_2 - a_1 - a_2 - a_3$ it writes the ${}_3F_2^{(\alpha=2)}$ at I_2 with the parameter array $(a_1, a_2, a_3; b_1, b_2)$ as the one with the array

$$(a_1, a_2, a_3; b_1, b_2) \longmapsto (b_1 - a_1, b_1 - a_2, a_3; b_1, s_{\mathrm{K}} + a_3), \tag{455}$$

times the prefactor $\Gamma_2(b_2)\Gamma_2(s_{\mathrm{K}})/[\Gamma_2(b_2 - a_3)\Gamma_2(s_{\mathrm{K}} + a_3)]$. Here $\Gamma_2(a) = \sqrt{\pi}\,\Gamma(a)\Gamma(a - \frac{1}{2})$ is the multivariate Gamma function of order two, and the conditions $\mathrm{Re}\, b_2, \mathrm{Re}\, s_{\mathrm{K}} > \frac{1}{2}$ hold below. For the array of (447), $s_{\mathrm{K}} = \frac{3}{4}$: the transformed array is identical, and the prefactor equals one. Parameter permutations add nothing, because the three upper parameters and the two lower parameters are separately equal. Transformations generated by this map and these permutations therefore expose no canceling parameter and no terminating series. This is a statement about this explicit route, not a proof that a different reduction to ordinary ${}_pF_q$ values cannot exist. The script `code/rank_two_checks.py` of `lri_d3_matrix_update/` checks the fixed point exactly.

The determinantal formulas for *Hermitian* matrix-argument functions have Jack parameter $\alpha = 1$, whereas (442) fixes $\alpha = 2$ here. Transplanting a Hermitian formula would change the coefficients. The distinction between real and Hermitian matrix-argument transformations is made explicitly in [40]; it explains why the familiar determinant machinery is not itself the missing scalar identity.

19.10 Three-dimensional values and limits of the reduction

The fixed rational formulas give

$$\begin{aligned} Q_3 &= 5.017102564553855369590854845690 18\ldots, \\ P_4(3) &= 62.058327943406672466463388584096 7\ldots, \\ P_5(3) &= 240.340317759083549816366546388004\ldots. \end{aligned}$$

The digits shown are correctly rounded; ... marks omitted digits. Fresh integrations of the Gamma kernels on the real axis, which do not use these formulas, agree with these values to about 10^{-51}, 10^{-52} and 10^{-53}, respectively. For Q_3 and $P_4(3)$ there is a stronger independent check: the 100-digit spectral evaluation of Section 16.8 agrees with $3Q_3$ and $P_4(3)$ to 3.5×10^{-79} and 2.1×10^{-86}. For $P_5(3)$ the kernel integration is the only independent check. The formulas themselves, evaluated at 55 and at 90 working digits with changed truncation and tail order, differ by 2.0×10^{-53}, 2.9×10^{-53} and 1.6×10^{-49}; the 55-digit run is the less accurate one, since the 90-digit values agree with a further 100-digit evaluation to better than 10^{-74}. These are numerical residuals, not interval bounds. The rank-two identity is a coefficientwise identity with Jack parameter $\alpha=2$: neither its superscript nor its matrix argument may be dropped. No complete reduction of these remaining values, or of $\mathcal{N}_3$ ((403) at $d=3$), to scalar ordinary ${}_pF_q$ values is asserted.

How the values were obtained and checked. The formula side sums the ordinary and two-variable series of the formulas above to a truncation point and adds an asymptotic tail built from the Stirling expansion of the Gamma ratios and Hurwitz zeta values. The runs at 55 and 90 working digits used truncation 80 with tail order 36 and truncation 160 with tail order 64 (`code/general_d3/verify_d3.py`, record `verification/general_d3/period_values.json` of `lri_d3_reduction_update/`). The 100-digit run used truncation 220 and tail order 80 (`code/recompute_d3_periods.py`, record `verification/fresh_primitive_values.json` of `lri_d3_matrix_update/`); it repeats the formula side and is not an independent integral. The tails are numerical accelerations, not interval bounds and not fits.

The independent check (`code/general_d3/check_spectral.py`, record `verification/general_d3/fresh_spectral.json`) integrates the real-axis Gamma kernels afresh, without reading stored period values, at 45 and 60 working digits with different quadrature breakpoints and spin cutoffs. Its wheel spin tail comes from analytically integrated centered-Stirling coefficients, not from a fit to the answer. With the symbols λ_j, e_j and $\mathfrak{w}_{r,h}$ local to this paragraph, put $e_0=1$ and

$$\begin{aligned}\lambda_j &= -\frac{rB_{2j+1}(\frac18)}{j(2j+1)}, \qquad e_j=\frac1j\sum_{k=1}^{j}k\lambda_k e_{j-k},\\ \mathfrak{w}_{r,h} &= 2\Gamma(\tfrac32)^r\,\Gamma(\tfrac{3r}{2}+2h-1)\sum_{j=0}^{h}\frac{e_je_{h-j}}{\Gamma(\frac{3r}{4}+2j)\,\Gamma(\frac{3r}{4}+2h-2j)},\end{aligned} \tag{456}$$

where $B_n(x)$ are the Bernoulli polynomials. In $\langle t_\ell^r\rangle_3$ of Section 1.1, which is $3Q_3$ for $r=3$ and $P_4(3)$ for $r=4$, the spins $\ell<\ell_c$ are integrated directly, and the sum over the spins $\ell\ge\ell_c$ is approximated by the truncated asymptotic series $\sum_{h=0}^{h_{\max}}\mathfrak{w}_{r,h}\,\zeta(\frac{3r}{2}-2+2h,\ell_c+\frac12)$, with the Hurwitz zeta function; the 60-digit run used $\ell_c=72$ and $h_{\max}=32$. This is a numerical acceleration, not an extra operation in (405) or (422). The five-loop kernel uses an alternating spin sum, accelerated separately by an Euler transformation (232 spins in the 60-digit run). In that run the formula values minus the kernel values are -1.47×10^{-51} for Q_3, -1.53×10^{-52} for $P_4(3)$ and -2.41×10^{-53} for $P_5(3)$. These residuals are the differences of `period_values.json` (90 digits) and `fresh_spectral.json` (60 digits), also listed in the delivered record `verification/general_d3/comparison_summary.json`.

At 75 working digits, evaluations at the complex points $t=0.063+0.041\mathrm{i}$ and $t=-0.082+0.027\mathrm{i}$ confirm the half-shift determinants (429), with largest residual 2.42×10^{-67}, and evaluations at two complex points u confirm the finite modes (408) to 5×10^{-70} (`verification/general_d3/identities.json`). In the same run the complex-node formulas (438) differ from (422) and (425) by 7.0×10^{-71} and 2.7×10^{-65}. Twenty-five exact symbolic checks of the finite residue algebra, the sample rules and the polynomial identities, and 288 comparisons of rational Horn coefficients, pass (`code/general_d3/check_exact.py`, record `verification/general_d3/exact_algebra.json`). Six literal ordinary-hypergeometric evaluations at 40 working digits agree with the tail evaluator to 2.06×10^{-38} (`verification/general_d3/literal_hypergeometric.json`). All these residuals

are observed values, not bounds. They are master-level checks, not a new three-dimensional full-graph calculation or a rigorous enclosure.

Checks of the rank-two identification. The script `code/rank_two_checks.py` of `lri_d3_matrix_update/` (record `verification/rank_two_checks.json`) builds the Jack hook products independently and compares them with (442) on all 91 partitions $\kappa = (m, n)$ with $m \le 12$. On the same partitions it compares the coefficients of (445) for $r = 3, 4$ at $t = 0, \frac{1}{8}, \frac{2}{7}$, and those of (452) at the five nodes (426), with the Horn coefficients. With the polynomial identity (454), the Kummer fixed point of Section 19.9 and the normalization $\sum_{|\kappa|=k} C_\kappa^{(2)}(I_2)/k! = 2^k/k!$ for $k \le 12$, these are 1107 exact checks, all passing. Direct expansions of the matrix functions and of the Horn series agree through total degree 90 at $z = \frac{1}{4}$ (for $\mathfrak{H}_3(0)$ and $\mathfrak{H}_4(\frac{1}{8})$) and $z = -\frac{1}{4}$ (for $\mathfrak{H}_7(\pm\frac{1}{16})$), with z as in (441), at 90 working digits. These are coefficient and interior-series checks, not interval bounds on the endpoint values.

A bounded search for a scalar form of the tetrahedron. Consider the 6479 convergent ordinary values $F = {}_3F_2(\boldsymbol{a}; \boldsymbol{b}; 1)$ and $F = {}_4F_3(\boldsymbol{a}; \boldsymbol{b}; 1)$ whose upper parameters lie in $\{\frac{1}{8}, \frac{2}{8}, \dots, 1\}$ and lower parameters in $\{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$, with positive excess and no upper parameter equal to a lower one. For each of them PSLQ was run on the vector $\big(\mathfrak{H}_3(0), 1, \Gamma(\frac{1}{4})^{10}/\pi^{11/2}, F\big)$ at 85 working digits, with tolerance 10^{-57}, coefficients up to 10^6 and at most 800 iterations. No relation with a nonzero coefficient of $\mathfrak{H}_3(0)$ was returned (`code/scalar_catalog_search.py` of `lri_d3_matrix_update/`, record `verification/scalar_catalog_search.json`). This finite search is not an independence result and does not rule out other parameter arrays, arguments, prefactors or nonlinear identities. It concerns the tetrahedron only.

The open primitive problem. The smallest open primitive question is to evaluate the single rank-two ${}_3F_2^{(\alpha=2)}$ of (447) by finitely many ordinary scalar hypergeometric values. The four- and five-loop complements need the related rank-two functions (448) and (452). No such scalarization is proved here. Reducing these primitive functions would still leave $\mathcal{N}_3$ for $C_5(3)$.

Coefficients through fourth order. At $d = 3$, that is $q = \frac{3}{4}$, the constants (221) are

$$\begin{aligned} &A = 2 + 4\log 2 - \pi, \quad R = \frac{\pi^2}{3} - 4, \qquad U = 2\pi^3 - 56\zeta(3), \\ &V = 16 - 16\zeta(3), \qquad X = \frac{14\pi^4}{15} - 96, \end{aligned} \tag{457}$$

by reflection, recurrence and the integer and half-integer polygamma values [32, 33]. Here V and X are these constants, not $V_r(t)$ of (416), $\mathsf{X}_r(t)$ or X_0, X_h of (421). Then (222) reads

$$\boxed{\begin{aligned} C_0(3) = \tfrac{3}{2}, \qquad C_1(3) = -\tfrac{1}{6}, \qquad C_2(3) = \tfrac{A}{9}, \\ 108\, C_3(3) = A^2 + \tfrac{\pi^2}{3} - 4 - 8Q_3, \\ 648\, C_4(3) = 16P_4(3) - A^3 - 8AQ_3 - \pi^2 A + 12A + 53\pi^3 - 1436\zeta(3) - 48, \end{aligned}} \tag{458}$$

with Q_3 from (405) and $P_4(3)$ from (422). This gives $C_4(3) = 1.22806354532762307997\ldots$; it is a reevaluation of the coefficient formula, not a new four-loop graph integration.

With these values and those of the $d = 3$ nonprimitive masters $\mathcal{L}_3$, Z_3 and $\mathcal{K}_3$ (Section 16.8), (223) gives

$$C_5(3) = -5.85035142075132722921794423049774430011099734637 83\ldots$$

to the 50 digits shown (`LRI_C5_d3/c5_value.json`). The formula-side values of Q_3, $P_4(3)$ and $P_5(3)$ are stable to 10^{-74} or better, and the independent checks above and in Section 16.8 confirm

Q_3, $P_4(3)$, $P_5(3)$ and the nonprimitive masters to about 10^{-79}, $2\cdot10^{-86}$, $2\cdot10^{-53}$ and $1.2\cdot10^{-93}$ or better. Propagated through (223), these differences change $C_5(3)$ by less than 10^{-54}: the largest part, about 8×10^{-55}, comes from $P_5(3)$, since $\partial C_5/\partial P_5 = -1536/46656 \approx -0.033$. $\mathcal{K}_3$ is the 430-digit value of (260); the two-fold Barnes integral (588), which uses none of the inputs of Section 21.1, reproduces it to $4\cdot10^{-100}$ (Section 21.11), and the generic-dimensional route of Sections 20 and 21 reproduces $C_5(3)$ to $5\cdot10^{-45}$ (Section 20.9).

20 Generic-dimensional reduction of $\mathcal{L}_d$ and Z_d

The nonprimitive masters $\mathcal{L}_d$ and Z_d of (223) are rim jets of the balanced wheels, $\mathcal{L}_d = \partial_b W_4(d;b)|_{b=d/2}$ and $Z_d = \frac{1}{9}\partial_b^2 W_3(d;b)|_{b=d/2}$ (244). Section 18 periodizes the wheels only at $b=d/2$. This section extends the finite periodization to a general rim exponent b (Section 20.1) and proves that the top Fourier modes of every product of balanced kernels satisfying the convergence condition (C) of (469) vanish in every dimension, in particular for every balanced wheel (Section 20.1.2). With these two statements it gives, for $\mathcal{L}_d$ and Z_d, in place of the six-variable Horn representation of (298) (244 and 318 values), samples at three nodes, real for real b: five two-variable Horn functions at unit argument for $\mathcal{L}_d$, and two ${}_7F_6(1)$ values and three two-variable Horn values for Z_d, together with their first and second b-derivatives (Sections 20.2–20.7; the unweighted, derivative-free form (298) remains valid). $\mathcal{K}_d$ is treated in Section 21; $P_5(d)$ is not treated here. Section 20.2 states the sample formulas of the two wheels, and Sections 20.3 and 20.4 prove them for the four-wheel and the three-wheel and write them out in Horn values. The package `LRI_C5_general_d/` holds the code and records: `foundation/` for Section 20.1, `L/` for Section 20.3, `Z/` for Section 20.4 and `specialize/` for the rest; the review records are in `review/`.

Provenance. This section, with the code and records in `foundation/`, `L/`, `Z/` and `specialize/`, was written by a Claude Code session. Its complete proofs, with the detailed checks, were added on 16 October 2026 by the same session, from that session's separate notes, which are not part of the repository. A second Claude Code session, which wrote Section 21, supplied two inputs: the explicit constants of (465), (466) and (467), taken from its growth proof (Section 21.6; package `LRI_C5_general_d/K_general/`), and the spectral reference values `LRI_C5_general_d/references/reference_values.json`, computed with its code `LRI_C5_general_d/refvals/hp_general_d.py`. The $d=3$ identities of Section 20.6 state their own provenance. Section 20.10 says which parts the reviews covered.

Status. The general-b statements of Section 20.1, the lattice identities (539) and the sample formulas of Section 20.2, with the two-node formula (496), have written proofs here (Sections 20.1, 20.3, 20.4 and 20.6). So do the $d=3$ identities (544) and (545) of Section 20.6, whose printed derivation has not been reviewed. The formulas (492)–(553) are *derived*: exact algebra on top of these theorems, confirmed numerically (Section 20.9). The reviews and checks named in this section and in Section 21 were made by separate AI sessions with their own codes; there was no human referee. Statements that a further reduction does not exist are *numerical only*. Section 20.10 lists the status of every statement, the conditional inputs and what each review covered. The evaluation of $\mathcal{K}_d$ quoted in Section 20.7 is that of Section 21, which has its own status list (Section 21.12) and was not part of the review `LRI_C5_general_d/review/VERIFICATION.md`.

Notation. In this section $q=d/4$. Except in Section 20.1.2, where σ is a free offset and g in f_g a free exponent, $\sigma=b/2$ and $g=(d-b)/2=2q-\sigma$ is the spoke half-exponent; the Barnes kernel g_ℓ carries its spin index. Further $s_b=\sin\frac{\pi b}{2}$, $\varsigma_b=\cos\frac{\pi b}{2}$, and in Sections 20.2–20.7, at $b=d/2$,

$$S=\sin\pi q,\qquad C=\cos\pi q,\qquad D=\pi\cot\pi q=\psi_0(1-q)-\psi_0(q). \tag{459}$$

In Section 20.1 and in the bounds of the proofs of Sections 20.3 and 20.4.1, C (also C_K) denotes a generic constant; (C) is the condition (469), and $C_r(t)$ is a block of Section 20.1.3. This D is not the dimension D of Section 10 and not the functions $D_b(t)$, $D_T(t)$ of (462), (470). A, R, U, V, X are the polygamma values (221); at $q = \frac{1}{4}$, $A = \mathsf{A}$ and $U = \mathsf{U}$ of (31), and $D = \mathsf{D} = \pi$. The calligraphic $\mathcal{A}, \mathcal{B}$ are the mode amplitudes of (349), not polygamma values. Likewise the samples $\mathcal{X}_T, \mathcal{X}_0, \mathcal{X}_{1/2}$ of (489) are unrelated to X, the R family of $\mathcal{K}_d$ is unrelated to R, and the simplex variable V of Section 20.1.2 is unrelated to V. A prime denotes ∂_b. Functions of b such as $\mathfrak{t}_3(b)$ of (495) also depend on d; where d is fixed it is written first, as in $\mathfrak{t}_3(1; b)$. Some letters have meanings here that are local to this section: g is the spoke half-exponent, not the coupling; $\sigma = b/2$ is a rim offset, not a spin; s_b is a sine, not the exponent s; S and C are the trigonometric values (459), not C_n, $C(\tau)$ or the condition (C); and $\omega(\rho)$ of (465) is a constant, unrelated to the numbers ω_j of Step 1 of the proof of (471).

20.1 Finite periodization at a general rim exponent

Kernel. For $0 < d < 4$ and a rim exponent b (complex in general) put

$$N_b = \frac{\Gamma(2q)\Gamma(g)}{\Gamma(\sigma)}, \qquad a_\ell = \frac{\ell + \sigma}{2}, \qquad \beta_\ell = a_\ell + g, \qquad g_\ell(s; b) = \frac{\Gamma(a_\ell + s)\Gamma(a_\ell - s)}{\Gamma(\beta_\ell + s)\Gamma(\beta_\ell - s)},$$
$$W_r(d; b) = \langle t_{\ell,b}^r \rangle_d = N_b^r \sum_{\ell \geq 0} d_\ell \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \frac{ds}{2\pi\mathrm{i}}\, g_\ell(s; b)^r. \tag{460}$$

The eigenvalue of (242) is $t_{\ell,b}(\nu) = N_b\, g_\ell(\mathrm{i}\nu; b)$ for real b; for complex b, W_r and $\langle t_{\ell,b}^r \rangle_d$ denote the analytic continuation (460), with $|\cdot|^2$ replaced by $\Gamma(a_\ell + \mathrm{i}\nu)\Gamma(a_\ell - \mathrm{i}\nu)/(\Gamma(\beta_\ell + \mathrm{i}\nu)\Gamma(\beta_\ell - \mathrm{i}\nu))$. Here $2a_\ell = \ell + \sigma$ and $a_\ell + \beta_\ell = \ell + 2q$. At $b = d/2$ one has $\sigma = g = q$, $N_{d/2} = \Gamma(2q)$, β_ℓ is the b_ℓ of (300), and (460) is (302). The multiplicities (301) obey $|d_\ell| \leq C(1+\ell)^{d-2}$ and

$$\sum_{\ell \geq 0} d_\ell z^\ell = (1+z)(1-z)^{1-d}, \qquad |z| < 1. \tag{461}$$

For $\operatorname{Re} b > 0$ the imaginary axis separates the two pole families of every kernel, and the integral in (460) converges absolutely for $2r \operatorname{Re} g > \max(d, 1)$. So (460) represents W_r on $0 < \operatorname{Re} b < b_{\mathrm{spec}} = d - \max(d,1)/r$.

On the contour $|g_\ell(\mathrm{i}\nu; b)|$ is comparable to $(1 + \ell + |\nu|)^{-2\operatorname{Re} g}$. The ν-integral therefore converges absolutely for $2r \operatorname{Re} g > 1$ and the spin sum for $2r \operatorname{Re} g > d$; together these are the condition just stated. The integral also converges at some $\operatorname{Re} b \leq 0$, but there it no longer represents W_r. The function $W_r(d; b)$ is singular at $b = 0$, where the $\ell = 0$ contour is pinched (a pole of order $r - 1$), and at $b = b_{\mathrm{spec}}$, where the spin or the ν integration diverges. For b near $\frac{d}{2}$ the disc $|b - \frac{d}{2}| < \min(\frac{d}{2}, b_{\mathrm{spec}} - \frac{d}{2})$ lies in this strip of analyticity. The bound $|d_\ell| \leq C(1+\ell)^{d-2}$ holds for every $0 < d < 4$ and is proved below, (466); for $d < 1$ the d_ℓ with $\ell \geq 2$ are negative.

Periodization and strip. Define

$$S_r(t; b) = \sum_{\ell \geq 0} d_\ell \sum_{n \in \mathbb{Z}} g_\ell(a_\ell + n + t; b)^r, \qquad D_b(t) = \sin(\pi t)\sin\big(\pi(t+\sigma)\big), \qquad \Xi_r(t; b) = D_b(t)^r S_r(t; b). \tag{462}$$

At $b = d/2$ these are S_r, D_q, Ξ_r of (304); the subscript of D_q there is $\sigma = q$, so D_q is D_b at $b = 2q$, not at $b = q$. Put $\alpha = r \operatorname{Re} g$ and

$$b_{\max}(r, d) = \min\Big\{ d - \frac{d}{r},\ d - \frac{2(d-1)}{r},\ d - \frac{1}{r} \Big\}, \tag{463}$$

that is $d - \frac{1}{r}$ for $d \leq 1$, $\frac{(r-1)d}{r}$ for $1 \leq d \leq 2$ and $d - \frac{2(d-1)}{r}$ for $d \geq 2$. The condition $\operatorname{Re} b < b_{\max}$ is the conjunction of (i) $2\alpha > d$, (ii) $\alpha > d - 1$ and (iii) $2\alpha > 1$. One has $b_{\max} \leq b_{\mathrm{spec}}$, with equality for $d \leq 2$, and $b = d/2$ lies in $0 < b < b_{\max}(r, d)$ iff $\frac{2}{r} < d < 4$: $d > \frac{1}{2}$ for W_4 and $d > \frac{2}{3}$ for W_3.

Lattice split and estimates. Let $f(z)=\Gamma(z)/\Gamma(z+g)$. For $\ell\ge 0$, $n\in\mathbb{Z}$, $j,k\ge 0$ and t off the classes $t\equiv 0$, $t\equiv -\sigma \pmod 1$, Gamma reflection [26] gives

$$\begin{aligned} g_\ell(a_\ell+n+t) &= f(\ell+\sigma+n+t)\,f(-n-t)\\ &= -\frac{\sin\pi(g-t)}{\sin\pi t}\,\frac{\Gamma(\ell+n+\sigma+t)\,\Gamma(n+1-g+t)}{\Gamma(\ell+n+2q+t)\,\Gamma(n+1+t)},\\ g_{j+k+1}(a_{j+k+1}-j-1+t) &= \frac{\Gamma(k+\sigma+t)\,\Gamma(j+1-t)}{\Gamma(k+2q+t)\,\Gamma(j+1+g-t)},\\ g_\ell(a_\ell+n+t) &= g_\ell\big(a_\ell+(-\ell-n-1)+(1-\sigma-t)\big). \end{aligned} \tag{464}$$

At $b=d/2$ the first two are (305) and (306). The lattice splits into the right part $n\ge 0$, the middle part $-\ell\le n\le -1$ and the left part $n\le -\ell-1$; by the last identity the left part is the right part at the mirror point $1-\sigma-t$. The Gamma-ratio bound $|\Gamma(z+u)/\Gamma(z+v)|\le C|z|^{\mathrm{Re}(u-v)}$ [27] gives, on compact sets avoiding the pole classes, $|g_\ell(a_\ell+n+t)|^r\le C(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ in the right part and $C(1+k)^{-\alpha}(1+j)^{-\alpha}$ in the middle part. For $t=x+\mathrm{i}y$ with x bounded and $|y|\ge y_0>|\operatorname{Im}\sigma|+1$ the same holds with $1+\ell+n, 1+n, 1+k, 1+j$ replaced by $\ell+n+|y|,\dots,j+|y|$. The constants are uniform for b in compact subsets of $\operatorname{Re} b>0$. Finally $\sum_{\ell,n\ge 0}(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}<\infty$ iff $\alpha>\max(\frac{d}{2},d-1,\frac12)$, that is iff $\operatorname{Re} b<b_{\max}$; under this condition the version with Y in place of the 1's is $O(Y^{\max(d,1)-2\alpha}(1+\log Y))$, and the same holds for the middle sum. The paragraph *Proofs of the estimates* below proves them in full, with explicit constants in (465) and (467).

Proofs of the estimates. *The identities* (464). For $s=a_\ell+n+t$ one has $a_\ell+s=\ell+\sigma+n+t$, $a_\ell-s=-n-t$, $\beta_\ell+s=\ell+\sigma+n+t+g$ and $\beta_\ell-s=g-n-t$, which is the factorized form. Gamma reflection [26] gives, for every integer n,

$$\frac{\Gamma(-n-t)}{\Gamma(g-n-t)}=-\frac{\sin\pi(g-t)}{\sin\pi t}\,\frac{\Gamma(n+1-g+t)}{\Gamma(n+1+t)},$$

and $\ell+\sigma+n+t+g=\ell+2q+n+t$; this is the second line. The substitution $\ell=j+k+1$, $n=-j-1$ in the factorized form gives the third line. Finally g_ℓ is even in s, and $-(a_\ell+n+t)=a_\ell+(-\ell-n-1)+(1-\sigma-t)$ because $2a_\ell=\ell+\sigma$.

The Gamma-ratio bound. Stirling's formula with its error bound [27] states that for $|\arg w|<\pi$, with principal branches, $\operatorname{Ln}\Gamma(w)=(w-\frac12)\operatorname{Ln} w-w+\frac12\ln 2\pi+\epsilon_\Gamma(w)$ with $|\epsilon_\Gamma(w)|\le \sec^2(\frac12\arg w)/(12|w|)$. From it: let $\rho\ge 0$ and $\omega(\rho)=\rho+\frac32\rho^2+\frac{11}{3}\rho^3+\frac43$. If $\operatorname{Re} z\ge 0$, $|z|\ge\max(1,2\rho)$ and $|u|,|v|\le\rho$, then $\Gamma(z+u)$ and $\Gamma(z+v)$ are finite and nonzero, and

$$\begin{aligned} \frac{\Gamma(z+u)}{\Gamma(z+v)} &= z^{u-v}e^{\theta},\qquad |\theta|\le\frac{\omega(\rho)}{|z|},\\ e^{-\omega(\rho)/|z|}e^{-\frac{\pi}{2}|\operatorname{Im}(u-v)|}|z|^{\mathrm{Re}(u-v)} &\le \Big|\frac{\Gamma(z+u)}{\Gamma(z+v)}\Big| \le e^{\omega(\rho)/|z|}e^{\frac{\pi}{2}|\operatorname{Im}(u-v)|}|z|^{\mathrm{Re}(u-v)}. \end{aligned} \tag{465}$$

Proof. Put $\xi=u/z$, so $|\xi|\le\frac12$, and $w=z+u=z(1+\xi)$. Then $|w|\ge|z|/2\ge\frac12$ and $|\arg w|\le|\arg z|+|\arg(1+\xi)|\le\frac{\pi}{2}+\frac{\pi}{6}$, so w is neither a pole nor a zero of Γ, and $|\epsilon_\Gamma(w)|\le\sec^2(\pi/3)/(12|w|)\le 2/(3|z|)$. The imaginary part of $\operatorname{Ln} z+\operatorname{Ln}(1+\xi)$ lies in $[-\frac{2\pi}{3},\frac{2\pi}{3}]$, so $\operatorname{Ln} w=\operatorname{Ln} z+\operatorname{Ln}(1+\xi)$. Write $\operatorname{Ln}(1+\xi)=\xi-\xi^2/2+\xi_3$ with $|\xi_3|\le|\xi|^3/(3(1-|\xi|))\le\frac23|\xi|^3$. Multiplying out with $z\xi=u$,

$$(z+u-\tfrac12)\operatorname{Ln}(1+\xi)=u+h(z,u),\qquad h(z,u)=\frac{u^2}{2z}-\frac{u}{2z}-\frac{u^3}{2z^2}+\frac{u^2}{4z^2}+(z+u-\tfrac12)\xi_3.$$

With $|u|\le\rho\le|z|/2$, $|z|\ge 1$ and $|z+u-\frac12|\le 2|z|$, each term is bounded, and $|h(z,u)|\le(\frac{\rho}{2}+\frac34\rho^2+\frac{11}{6}\rho^3)/|z|$. Hence $\operatorname{Ln}\Gamma(z+u)=(z+u-\frac12)\operatorname{Ln} z-z+\frac12\ln 2\pi+h(z,u)+\epsilon_\Gamma(z+u)$. Subtracting the same identity for v gives $\operatorname{Ln}\Gamma(z+u)-\operatorname{Ln}\Gamma(z+v)=(u-v)\operatorname{Ln} z+\theta$ with $|\theta|\le 2(\frac{\rho}{2}+\frac34\rho^2+\frac{11}{6}\rho^3)/|z|+\frac{4}{3|z|}=\omega(\rho)/|z|$. Exponentiating gives the first line of (465). The

second follows from $|z^{u-v}| = |z|^{\mathrm{Re}(u-v)} e^{-\mathrm{Im}(u-v)\arg z}$ and $|\arg z| \le \frac{\pi}{2}$. □ The bound is a version of the large-z asymptotics of a Gamma ratio [27] that is uniform in u, v and in the closed right half-plane, with an explicit constant. It is used below for arguments $m + c + t$ with an integer $m \ge 0$ and a bounded constant c: one puts $z = m + t + X$, where $X \ge 0$ bounds $|\mathrm{Re}\, t|$, so that $\mathrm{Re}\, z \ge m$, and absorbs $c - X$ into u or v; arguments $m + c - t$ are treated in the same way.

The multiplicities. For $q \ne \frac{1}{4}$, (301) gives $d_{k+1} = \frac{2}{\Gamma(4q-1)}(2q+k)\frac{\Gamma(k+4q-1)}{\Gamma(k+2)}$. By (465) with $z = k \ge 6$ and $\rho = 3$, $\Gamma(k+4q-1)/\Gamma(k+2) = k^{4q-3}e^{\theta}$ with $|\theta| \le \omega(3)/k$, uniformly in $q \in (0,1)$. The finitely many $k < 6$ are continuous in q, and so is $2/\Gamma(4q-1)$, also at $q = \frac{1}{4}$, where it vanishes; near $q = \frac{1}{4}$ one uses the form (301), with $(4q-1)_k = (4q-1)(4q)_{k-1}$, and at $q = \frac{1}{4}$ the multiplicities are $(1, 1, 0, 0, \dots)$. Hence

$$|d_\ell| \le C_q\,(1+\ell)^{d-2}, \qquad d_\ell\,(1+\ell)^{2-d} \to \frac{2}{\Gamma(d-1)} \quad (\ell \to \infty), \tag{466}$$

with $C_q = \sup_\ell |d_\ell|(1+\ell)^{2-d}$ finite and bounded on compact subsets of $0 < q < 1$, across $d = 1$ included (the limit is 0 at $d = 1$).

The right and middle bounds. Let $\mathrm{Re}\, b > 0$ and let K be a compact set off the classes $t \equiv 0$ and $t \equiv -\sigma$. In the second line of (464), (465) bounds $\Gamma(\ell+n+\sigma+t)/\Gamma(\ell+n+2q+t)$ by $C(1+\ell+n)^{-\mathrm{Re}\, g}$ and $\Gamma(n+1-g+t)/\Gamma(n+1+t)$ by $C(1+n)^{-\mathrm{Re}\, g}$, since $\sigma - 2q = -g$, for all indices with $\ell + n$, respectively n, above a bound fixed by K and b. In the third line it bounds $\Gamma(k+\sigma+t)/\Gamma(k+2q+t)$ and $\Gamma(j+1-t)/\Gamma(j+1+g-t)$ by $C(1+k)^{-\mathrm{Re}\, g}$ and $C(1+j)^{-\mathrm{Re}\, g}$. The prefactor $\sin\pi(g-t)/\sin\pi t$ is bounded on K. Each ratio depends on a single combination of the summation indices ($\ell + n$, n, k or j). A ratio with a small index therefore occurs for infinitely many lattice points but takes only finitely many values; these are continuous on K and bounded there one by one. Raising to the power r gives the bounds stated above, $(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ and $(1+k)^{-\alpha}(1+j)^{-\alpha}$ with $\alpha = r\,\mathrm{Re}\, g$. Now let $t = x + \mathrm{i}y$ with x in a fixed bounded interval and $|y| \ge y_0 > |\mathrm{Im}\,\sigma| + 1$. The numerator arguments $\ell + n + \sigma + t$, $n + 1 - g + t$ and $k + \sigma + t$ have imaginary parts of modulus at least $|y| - |\mathrm{Im}\,\sigma| > 1$ (note $\mathrm{Im}\, g = -\mathrm{Im}\,\sigma$), and $j + 1 - t$ has $|\mathrm{Im}| = |y| > 1$; so no Gamma function in the numerators meets a pole, and the prefactor is bounded. With z chosen as above, $|z|$ is comparable to the index plus $|y|$. For $|y|$ or the index above $2\rho + 1$, where ρ bounds the shifts, (465) applies directly; the remaining factor values, with small index and bounded $|y| \ge y_0$, are finitely many continuous functions on a compact set that meets no pole. This gives the bounds with $1+\ell+n$, $1+n$, $1+k$, $1+j$ replaced by $\ell+n+|y|$, $n+|y|$, $k+|y|$, $j+|y|$. For complex b this is why y_0 must exceed $|\mathrm{Im}\,\sigma|+1$: the class $-\sigma+\mathbb{Z}$ lies on $\mathrm{Im}\, t = -\mathrm{Im}\,\sigma$. All quantities are continuous in b, so the constants are uniform for b in a compact set B of $\mathrm{Re}\, b > 0$, provided K avoids the pole classes of every $b \in B$, $y_0 > 1 + \max_B |\mathrm{Im}\,\sigma|$, and α is replaced by $r \min_B \mathrm{Re}\, g$.

The lattice sums. Let $0 < d < 4$ and $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$, and put $\bar\alpha = 2\alpha - \max(d-1, 0)$, so that $\bar\alpha > 1$ and $1 - \bar\alpha = \max(d,1) - 2\alpha$. Then for $Y \ge 1$

$$\begin{aligned} \Sigma^{\mathrm{r}}(Y) &= \sum_{\ell,n\ge 0} (1+\ell)^{d-2}(\ell+n+Y)^{-\alpha}(n+Y)^{-\alpha} \le C^{\mathrm{r}}\, Y^{\max(d,1)-2\alpha}(1+\log Y), \\ \Sigma^{\mathrm{m}}(Y) &= \sum_{j,k\ge 0} (1+j+k)^{d-2}(k+Y)^{-\alpha}(j+Y)^{-\alpha} \le C^{\mathrm{m}}\, Y^{d-2\alpha}, \end{aligned} \tag{467}$$

where, with $C_1 = 1 + 2^{\alpha-d+1}/(\alpha-d+1)$,

$$C^{\mathrm{r}} = C_1\Big[1 + \frac{1}{\bar\alpha - 1} + \frac{1}{(\bar\alpha-1)^2}\Big], \qquad C^{\mathrm{m}} = \begin{cases} 2^{d-1}C_1(1+\frac{1}{\alpha-1}), & d \ge 2, \\ \Big[\frac{2}{d} + \frac{2^{\alpha-d/2}}{\alpha-d/2}\Big]^2, & d < 2. \end{cases}$$

The two expressions for C^{m} are each continuous on their closed ranges, and C_1 and C^{r} are continuous on the open set of (d, α) defined by the condition; so all the constants are bounded on its compact

subsets, across $d = 1$ and $d = 2$ included. Since $Y^{d-2\alpha} \le Y^{\max(d,1)-2\alpha}(1+\log Y)$ for $Y \ge 1$, the middle sum obeys the bound of the right sum as well. The left part of a lattice sum, with k in place of n, has the form of the right one. *Proof.* (i) For $M \ge 1$ and an exponent $e \in (-2, 2)$ with $\alpha > e+1$ put $\Sigma_e(M) = \sum_{\ell\ge 0}(1+\ell)^e(\ell+M)^{-\alpha}$ and $u = 1+\ell$. Since $\ell + M \ge \max(u, M)$, $\Sigma_e(M) \le M^{-\alpha}\sum_{1\le u\le M} u^e + \sum_{u>M} u^{e-\alpha}$. Here $\sum_{1\le u\le M} u^e \le M^{\max(e+1,0)}(1+\log M)$: for $e > 0$ each term is at most M^e; for $-1 \le e \le 0$, $u^e \le M^{e+1}/u$; for $e < -1$, $u^e \le 1/u$; and $\sum_{u\le M} 1/u \le 1 + \log M$. Since $u^{e-\alpha}$ decreases, $\sum_{u>M} u^{e-\alpha} \le \int_{\lfloor M\rfloor}^{\infty} v^{e-\alpha}dv \le 2^{\alpha-e-1}M^{e+1-\alpha}/(\alpha-e-1)$, using $\lfloor M\rfloor \ge M/2$. With $e = d-2$ this gives $\Sigma_{d-2}(M) \le C_1 M^{\max(d-1,0)-\alpha}(1+\log M)$, and for $d \ge 2$ the same without the logarithm. (ii) The right sum is $\Sigma^{\mathrm{r}}(Y) = \sum_{n\ge 0}(n+Y)^{-\alpha}\Sigma_{d-2}(n+Y) \le C_1\sum_{n\ge 0}(n+Y)^{-\bar\alpha}(1+\log(n+Y))$. The function $v \mapsto v^{-\bar\alpha}(1+\log v)$ decreases on $[1,\infty)$ because $\bar\alpha \ge 1$, so the sum is at most

$$Y^{-\bar\alpha}(1+\log Y) + \int_Y^{\infty} v^{-\bar\alpha}(1+\log v)\,dv = Y^{-\bar\alpha}(1+\log Y) + Y^{1-\bar\alpha}\Big[\frac{1+\log Y}{\bar\alpha-1} + \frac{1}{(\bar\alpha-1)^2}\Big],$$

which is at most $[1 + \frac{1}{\bar\alpha-1} + \frac{1}{(\bar\alpha-1)^2}]\,Y^{1-\bar\alpha}(1+\log Y)$; times C_1 this is the bound with C^{r}. (iii) The middle sum for $d \ge 2$: by $(1+j+k)^{d-2} \le 2^{d-2}[(1+j)^{d-2}+(1+k)^{d-2}]$, $\Sigma^{\mathrm{m}}(Y) \le 2^{d-1}\Sigma_{d-2}(Y)\sum_k (k+Y)^{-\alpha}$, with $\Sigma_{d-2}(Y) \le C_1 Y^{d-1-\alpha}$ and $\sum_k (k+Y)^{-\alpha} \le (1+\frac{1}{\alpha-1})Y^{1-\alpha}$, because $\alpha > d-1 \ge 1$. (iv) The middle sum for $d < 2$: here $1+j+k \ge \sqrt{(1+j)(1+k)}$, so $(1+j+k)^{d-2} \le [(1+j)(1+k)]^{(d-2)/2}$ and $\Sigma^{\mathrm{m}}(Y) \le \Sigma_{(d-2)/2}(Y)^2$. Since $(d-2)/2 \in (-1,0)$, $\sum_{1\le u\le Y} u^{(d-2)/2} \le Y^{d/2}/(d/2)$, and the tail needs $\alpha > d/2$; this gives the stated C^{m}. □

The condition is also necessary. At $Y = 1$, (467) gives $\sum_{\ell,n\ge 0}(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha} < \infty$ for $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$; the converse also holds. Split the sum into $n \le \ell$ and $n > \ell$. For $n \le \ell$, $1+\ell+n$ is comparable to $1+\ell$, and $\sum_{n\le\ell}(1+n)^{-\alpha}$ to 1, $1+\log(1+\ell)$ or $(1+\ell)^{1-\alpha}$ for $\alpha > 1$, $\alpha = 1$, $\alpha < 1$; this part converges iff $\alpha > d-1$ (when $\alpha > 1$), $d < 2$ (when $\alpha = 1$) or $2\alpha > d$ (when $\alpha < 1$). For $n > \ell$, $1+\ell+n$ is comparable to $1+n$, and $\sum_{\ell<n}(1+\ell)^{d-2}$ to n^{d-1}, $1+\log n$ or 1 for $d > 1$, $d = 1$, $d < 1$; this part converges iff $2\alpha > \max(d, 1)$. The conjunction of the two conditions is $\alpha > \max(\frac{d}{2}, d-1, \frac{1}{2})$: the second gives the first and last entries, and if it holds, the first part gives $\alpha > d-1$ in each of its three cases (for $\alpha < 1$ the condition $2\alpha > d$ forces $d < 2$, so $\alpha > \frac{d}{2} > d-1$).

The entries of $b_{\max}$. With $\alpha = r\,\mathrm{Re}\,g = r(d - \mathrm{Re}\,b)/2$ the condition reads $\mathrm{Re}\,b < b_{\max}(r,d)$. Its three entries come from three regions of the lattice: (i) $2\alpha > d$ from the region where ℓ and n are both large, (ii) $\alpha > d-1$ from the end $\ell \gg n$, and (iii) $2\alpha > 1$ from the end $n \gg \ell$. For $r = 4$, (463) is $d - \frac{1}{4}$, $\frac{3}{4}d$, $\frac{d+1}{2}$ on the ranges $d \le 1$, $1 \le d \le 2$, $d \ge 2$; for $r = 3$ it is $d - \frac{1}{3}$, $\frac{2}{3}d$, $\frac{d+2}{3}$. Some values, with the spectral bound of (460):

d	0.8	1	1.25	1.5	2	2.5	2.75	3
$b_{\max}(3,d)$	0.4667	0.6667	0.8333	1	1.3333	1.5	1.5833	1.6667
$b_{\mathrm{spec}}(3,d)$	0.4667	0.6667	0.8333	1	1.3333	1.6667	1.8333	2
$b_{\max}(4,d)$	0.55	0.75	0.9375	1.125	1.5	1.75	1.875	2
$b_{\mathrm{spec}}(4,d)$	0.55	0.75	0.9375	1.125	1.5	1.875	2.0625	2.25

The two bounds differ only for $d > 2$ (paragraph *Beyond* $b_{\max}$ of Section 20.1.4). At $d = 2$ and $r = 4$, $b_{\max} = \frac{3}{2}$ is $x = \frac{b}{2} < \frac{3}{4}$ for the one-variable sum $\Omega(t;x)$ of (349), a range that contains the interval used in Section 18.7; at $d = 3$, $b_{\max}(4,3) = 2$ is the range "every $b < 2$" of the $d = 3$ paragraph of Section 20.6. For $r = 3$, $b_{\max}(3,2) = \frac{4}{3}$ is the range of (264), and $b_{\max}(3,3) = \frac{5}{3}$. The exponent η of part (c) of Section 20.1.1 is $2\alpha - \max(d,1) > 0$ throughout the strip, by (i) and (iii), so the exclusion of the boundary mode needs no extra condition.

20.1.1 Finite Fourier space

Let $r \ge 2$, $0 < d < 4$ and $0 < \mathrm{Re}\,b < b_{\max}(r,d)$, $\sigma \notin \mathbb{Z}$. Then:

(a) $S_r(t;b)$ converges absolutely and locally uniformly off the classes $t \equiv 0$ and $t \equiv -\sigma$; it is one-periodic and meromorphic, with poles of order at most r in these two classes only.

(b) $\Xi_r(t;b)$ is entire, $\Xi_r(t+1;b) = \Xi_r(t;b)$ and $\Xi_r(-\sigma - t;b) = \Xi_r(t;b)$.

(c) For $0 \le x \le 1$ and $|y| \ge 2 + |\operatorname{Im} b|$, $|\Xi_r(x + \mathrm{i}y;b)| \le C\, e^{2\pi r|y|}|y|^{-\eta}(1 + \log|y|)$ with $\eta = r(d - \operatorname{Re} b) - \max(d,1) > 0$ (a decay exponent, unrelated to the weights η_m of (253)).

(d) There are coefficients $c_{r,k}(q,b)$, analytic in b on the strip and real for real b, with

$$\Xi_r(t;b) = \sum_{k=0}^{r-1} c_{r,k}(q,b)\cos\bigl(2\pi k(t + \tfrac{b}{4})\bigr). \tag{468}$$

$\Xi_r(t;b)$ is jointly analytic in (t,b), so every sample at a fixed node is analytic in b.

Proof. (a) By the right and middle bounds and (467), the right and middle parts converge absolutely and uniformly on compact sets off the pole classes, and so does the left part, which is the right part at the mirror point. Each term is meromorphic in t. By (464) poles come only from $\Gamma(-n-t)$ ($t \in \mathbb{Z}$) and $\Gamma(\ell + \sigma + n + t)$ ($t \in -\sigma + \mathbb{Z}$), each to order at most r. If t_0 is such a point, $(t - t_0)^r$ times each term is analytic in a small disc around t_0; the series of these functions converges uniformly on the boundary circle, hence, by the maximum principle, in the disc, so $(t - t_0)^r S_r$ is analytic near t_0. Periodicity follows from $n \mapsto n + 1$ in an absolutely convergent sum. (b) D_b^r vanishes to order r on both classes, so Ξ_r is entire, and $D_b(t+1) = D_b(t)$. Evenness of g_ℓ gives $g_\ell(a_\ell + n - \sigma - t) = g_\ell(a_\ell + (-\ell - n) + t)$, and $n \mapsto -\ell - n$ is a bijection of $\mathbb{Z}$, so $S_r(-\sigma - t) = S_r(t)$; also $D_b(-\sigma - t) = \sin\pi(\sigma + t)\sin\pi t = D_b(t)$. (c) $|D_b(x + \mathrm{i}y)| \le Ce^{2\pi|y|}$. For $|y| \ge 2 + |\operatorname{Im} b| = 2 + 2|\operatorname{Im}\sigma|$ the large-$|y|$ bounds apply at t and, for the left part, at the mirror point $1 - \sigma - t$, whose imaginary part has modulus at least $|y| - |\operatorname{Im}\sigma| \ge \max(|y|/2,\, 2 + |\operatorname{Im}\sigma|)$. With (467) at $Y = |y|$ (or $|y|/2$), each of the three parts of S_r is bounded by $C|y|^{\max(d,1)-2\alpha}(1 + \log|y|)$, and $\eta = 2\alpha - \max(d,1)$. (d) As for (311), the coefficient $\gamma_m = \int_0^1 \Xi_r(x + \mathrm{i}y)e^{-2\pi\mathrm{i}m(x+\mathrm{i}y)}dx$ does not depend on y (Cauchy's theorem on a rectangle, the vertical sides cancelling by periodicity). For $m \ge r$ take $y = -Y$ with $Y \ge 2 + |\operatorname{Im} b|$; by (c), $|\gamma_m| \le Ce^{2\pi(r-m)Y}Y^{-\eta}(1 + \log Y) \to 0$, also for $m = r$ since $\eta > 0$. For $m \le -r$ take $y = +Y$. So $\Xi_r = \sum_{|m| \le r-1} \gamma_m e^{2\pi\mathrm{i}mt}$. The reflection in (b), $\Xi_r(-\sigma - t) = \sum_m \gamma_{-m}e^{2\pi\mathrm{i}m\sigma}e^{2\pi\mathrm{i}mt}$, gives $\gamma_m = \gamma_{-m}e^{2\pi\mathrm{i}m\sigma}$. With $c_{r,0} = \gamma_0$ and $c_{r,k} = 2\gamma_k e^{-\mathrm{i}\pi k\sigma} = 2\gamma_{-k}e^{\mathrm{i}\pi k\sigma}$ one has $\gamma_k e^{2\pi\mathrm{i}kt} + \gamma_{-k}e^{-2\pi\mathrm{i}kt} = c_{r,k}\cos(2\pi k(t + \frac{b}{4}))$, which is (468). For the analyticity in b let B be a compact subset of the strip and $y_B = 2 + \max_B |\operatorname{Im} b|$. On the segment $t = x + \mathrm{i}y_B$, $0 \le x \le 1$, the series for S_r converges uniformly in $(x,b) \in [0,1] \times B$, by the large-$|y|$ bounds with constants uniform on B and (467) with $\alpha = r\min_B \operatorname{Re} g > \max(\frac{d}{2}, d-1, \frac{1}{2})$. Its terms are analytic in b there, since the segment stays at distance at least one from both pole classes, and $D_b(t)^r$ is entire in (t,b). Hence $\Xi_r(x + \mathrm{i}y_B;b)$ is analytic in b, and so is $\gamma_m(b) = \int_0^1 \Xi_r(x + \mathrm{i}y_B;b)e^{-2\pi\mathrm{i}m(x+\mathrm{i}y_B)}dx$ on the interior of B. The same argument, with the Cauchy formula in t on small circles around the pole classes, shows that $\Xi_r(t;b)$ is jointly analytic in (t,b). For real b, Ξ_r is real on the real axis, so $\gamma_{-m} = \overline{\gamma_m}$ and the $c_{r,k}$ are real. □ (The point $\sigma \in \mathbb{Z}$ lies in the strip only at $b = 2$ when $r \ge 3$ and $d > (2r-2)/(r-2)$, never for $r = 3$ or for $r = 4$, $d \le 3$; it is treated in the paragraph *Integer* σ below. The masters use only b near $\frac{d}{2}$, where $0 < \sigma < 1$.)

Integer σ. In the strip, $\sigma \in \mathbb{Z}$ can occur only at the real point $b = 2$, because $b_{\max} \le d - \frac{d}{r} < d < 4$. The point $b = 2$ lies in the strip iff $b_{\max}(r,d) > 2$, that is iff $r \ge 3$ and $d > (2r-2)/(r-2)$. This never happens for $r = 3$; it happens for $d > 3$ when $r = 4$, for $d > \frac{8}{3}$ when $r = 5$, and for $d > (2r-2)/(r-2)$, which decreases to 2, for larger r. At $b = 2$ the two pole classes merge. Ξ_r is still entire: the poles of S_r in $\mathbb{Z}$ have order at most $2r$, and $D_2(t)^r = (-1)^r \sin^{2r}\pi t$. The analyticity argument in (d) does not use $\sigma \notin \mathbb{Z}$, so the $c_{r,k}$ are analytic across $b = 2$, and the finite Fourier statement holds there as well. The top-mode statement (473) holds there by continuity, and also directly, since (471) allows every $\sigma \in \mathbb{C}$. Only the weights of (479) and of (484), which carry

$(\sin\pi\sigma)^{-r}$, are singular at $b=2$. W_r is analytic there, so these poles cancel in the sum (paragraph *Radius of the sample jets* of Section 20.1.4). None of the cases used in this note ($r=3,4$ with $d\le 3$) contains $b=2$ in its strip.

20.1.2 The top modes vanish

The top-mode statement holds for every product of balanced kernels that satisfies the convergence condition (C) below, with different spoke exponents and independent shifts of the left and right Gamma arguments; the wheels are the case of equal factors. Write $f_g(w)=\Gamma(w)/\Gamma(w+g)$, so that $f=f_g$ in (464). In this subsection g_i,α_i,β_i ($i=1,\dots,r$) are numbers attached to the i-th factor: the kernel $g_\ell(s;b)$ is always written with its arguments, and α_i,β_i are unrelated to $\alpha=r\,\mathrm{Re}\,g$ and to β_ℓ in (460). Let $0<d<4$, $r\ge 2$ and $\sigma\in\mathbb{C}$, and let $g_i,\alpha_i,\beta_i\in\mathbb{C}$ satisfy

$$\text{(B)}\quad \sigma+\alpha_i+\beta_i+g_i=\frac{d}{2}\ \ (i=1,\dots,r),\qquad \text{(C)}\quad \varkappa:=\sum_{i=1}^{r}\mathrm{Re}\,g_i>\max\Big(\frac{d}{2},\,d-1,\,\frac{1}{2}\Big). \tag{469}$$

Put

$$\begin{aligned} T_\ell(z)&=\prod_{i=1}^{r} f_{g_i}(\ell+\sigma+\alpha_i+z)\,f_{g_i}(\beta_i-z),\qquad S_T(t)=\sum_{\ell\ge 0} d_\ell \sum_{n\in\mathbb{Z}} T_\ell(n+t),\\ D_T(t)&=\prod_{i=1}^{r}\sin\pi(t+\sigma+\alpha_i)\,\sin\pi(t-\beta_i),\qquad \Xi_T=D_T S_T. \end{aligned} \tag{470}$$

Here $T_\ell(z)$, with one index, is unrelated to $T_{\ell n}(b)$ of (490). Then S_T converges absolutely and locally uniformly off the $2r$ pole classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$; Ξ_T is entire and one-periodic, $\Xi_T=\sum_{|m|\le r-1}\gamma_m e^{2\pi i m t}$, with γ_m analytic in (σ,α_i,g_i) on the convex set defined by (C) (β_i fixed by (B)); and

$$\boxed{\gamma_{1-r}=\gamma_{r-1}=0,\qquad \Xi_T(t)=\sum_{|m|\le r-2}\gamma_m\,e^{2\pi i m t},} \tag{471}$$

equivalently $S_T(t)=O(e^{-4\pi|\mathrm{Im}\,t|})$ as $|\mathrm{Im}\,t|\to\infty$. Individual $\mathrm{Re}\,g_i$ may be zero or negative; only their sum is constrained. It is proved below, and Section 21.5 applies it to the R family of $\mathcal{K}_d$.

Mixed wheels and wheels. A product $\prod_i g_\ell(s+\delta_i;b_i)$ of kernels (460) with rim exponents b_i and shifts δ_i, sampled on the lattice $s=\frac{\ell+\sigma}{2}+n+t$ with any offset σ, is $T_\ell(n+t)$ with

$$g_i=\frac{d-b_i}{2},\qquad \alpha_i=\frac{1}{2}\Big(\frac{b_i}{2}-\sigma\Big)+\delta_i,\qquad \beta_i=\frac{1}{2}\Big(\frac{b_i}{2}-\sigma\Big)-\delta_i, \tag{472}$$

since $g_\ell(s;b)=f_{(d-b)/2}(\frac{\ell}{2}+\frac{b}{4}+s)\,f_{(d-b)/2}(\frac{\ell}{2}+\frac{b}{4}-s)$. Then (B) holds automatically, (C) says that the mean rim exponent satisfies $\frac{1}{r}\sum_i\mathrm{Re}\,b_i<b_{\max}(r,d)$, and the shifts are free. For the wheels, $b_i=b$, $\delta_i=0$ and $\sigma=b/2$, one has $\alpha_i=\beta_i=0$, $g_i=g$, $T_\ell(n+t)=g_\ell(a_\ell+n+t;b)^r$ and $D_T=D_b^r$, so $\Xi_T=\Xi_r$, $\varkappa=\alpha$, and (C) is $\mathrm{Re}\,b<b_{\max}$. With $c_{r,r-1}=2\gamma_{r-1}e^{-i\pi(r-1)\sigma}$ this gives, under the hypotheses of Section 20.1.1,

$$\boxed{c_{r,r-1}(q,b)=0,\qquad \Xi_r(t;b)=\sum_{k=0}^{r-2}c_{r,k}(q,b)\cos\big(2\pi k(t+\tfrac{b}{4})\big),} \tag{473}$$

and equivalently $S_r(t;b)=O(e^{-4\pi|\mathrm{Im}\,t|})$. This wheel statement is the one used in the rest of this section; Section 21 uses the product case $r=3$ and the case $r=2$ of (471).

The R family of $\mathcal{K}_d$. Section 21.5 applies (471) to the R family of $\mathcal{K}_d$, a product of three balanced kernels that is not a wheel. On the τ-circle used for $\mathcal{K}_d$ that application rests in addition on the continuation hypothesis of Section 21.8; the separate lattice-sum code `LRI_C5_general_d/review/lemmaG-numeric` of Section 20.9 confirms it at two points of that kind (top modes below $2\cdot10^{-47}$ at 54 digits).

Proof of (471). Its steps are as follows; the paragraph *Details of the top-mode proof* after the proof gives in full the estimates and computations that the steps state in short form. Let $Y_0 = 1+\max_i \max(|\operatorname{Im}(\sigma+\alpha_i)|, |\operatorname{Im}\beta_i|)$, so that all pole classes lie in $|\operatorname{Im} t| \le Y_0 - 1$.

Step 0 (estimates and finite modes). Split the lattice into the right part $n\ge 0$, the middle part $\ell = j+k+1$, $n=-j-1$, and the left part $n=-\ell-1-k$ ($j,k\ge 0$); there is no mirror map in general. Gamma reflection writes the factors with large negative argument as $f_g(\beta-n-t) = \frac{\sin\pi(\beta+g-t)}{\sin\pi(\beta-t)}\frac{\Gamma(n+1-\beta-g+t)}{\Gamma(n+1-\beta+t)}$ in the right part and $f_g(\sigma+\alpha-1-k+t) = \frac{\sin\pi(\sigma+\alpha+g+t)}{\sin\pi(\sigma+\alpha+t)}\frac{\Gamma(k+2-\sigma-\alpha-g-t)}{\Gamma(k+2-\sigma-\alpha-t)}$ in the left part. The Gamma-ratio bound applied to every factor gives a power $-\operatorname{Re} g_i$ of a base common to all factors, so the exponents add up to $-\varkappa$, and the lattice sums of Section 20.1 apply with $\alpha=\varkappa$. Step 0 of the paragraph *Details of the top-mode proof* carries this out: S_T converges absolutely and locally uniformly off the pole classes; for $0\le x\le 1$ and $|y|\ge\max(Y_0,2)$, $|S_T(x+\mathrm{i}y)| \le C|y|^{\max(d,1)-2\varkappa}(1+\log|y|)$; Ξ_T is entire and one-periodic, its modes $|m|\ge r$ vanish, and the γ_m are analytic in the parameters.

Step 1 (two half-planes). On $\operatorname{Im} t > Y_0$, S_T is analytic, one-periodic and tends to zero, so $S_T = \sum_{m\ge1} s_m^+ w^m$ with $w=e^{2\pi\mathrm{i}t}$; on $\operatorname{Im} t < -Y_0$, $S_T = \sum_{m\ge1} s_m^- w^{-m}$. Let ω_j run over the $2r$ numbers $\sigma+\alpha_i$ and $-\beta_i$. Since $\sin\pi(t+\omega) = \frac{\mathrm{i}}{2}e^{-\mathrm{i}\pi(t+\omega)}(1-e^{2\pi\mathrm{i}\omega}w) = -\frac{\mathrm{i}}{2}e^{\mathrm{i}\pi(t+\omega)}(1-e^{-2\pi\mathrm{i}\omega}w^{-1})$, one has $D_T = \kappa_+ w^{-r}\prod_j(1-e^{2\pi\mathrm{i}\omega_j}w) = \kappa_- w^r\prod_j(1-e^{-2\pi\mathrm{i}\omega_j}w^{-1})$ with $\kappa_\pm = (-\frac14)^r e^{\mp\mathrm{i}\pi\sum_j\omega_j}$, and comparing coefficients gives

$$\gamma_{1-r} = \kappa_+ s_1^+, \qquad \gamma_{r-1} = \kappa_- s_1^-.$$

For the wheels the reflection in Section 20.1.1(b) relates the two. In general there is no such symmetry at complex parameters, and the two coefficients are treated separately.

Step 2 (unfolding). For $Y > Y_0$ the double series converges absolutely on the segment $x+\mathrm{i}Y$, $0\le x\le 1$ (this uses (C)), so

$$s_1^+ = \int_0^1 S_T(x+\mathrm{i}Y)\,e^{-2\pi\mathrm{i}(x+\mathrm{i}Y)}\,dx = \sum_{\ell\ge0} d_\ell I_\ell, \qquad I_\ell = \int_{\mathbb{R}+\mathrm{i}Y} T_\ell(z)\,e^{-2\pi\mathrm{i}z}\,dz, \qquad \sum_\ell |d_\ell||I_\ell| < \infty.$$

Step 3 (Laplace form and I_ℓ). Assume first $\operatorname{Re} g_i > 0$ for all i (removed in Step 7). Rotating the ray of Euler's integral $\Gamma(g)f_g(w) = \int_0^\infty e^{-wu}(1-e^{-u})^{g-1}du$, with $u = \mp\mathrm{i}v$ through the first or the fourth quadrant, gives, first for $\operatorname{Re} w>0$ and then by analytic continuation,

$$\begin{aligned} f_g(w) &= \frac{-\mathrm{i}}{\Gamma(g)}\int_0^\infty e^{\mathrm{i}wv}\big(1-e^{\mathrm{i}v}\big)^{g-1}dv \qquad (\operatorname{Im} w>0),\\ f_g(w) &= \frac{\mathrm{i}}{\Gamma(g)}\int_0^\infty e^{-\mathrm{i}wv}\big(1-e^{-\mathrm{i}v}\big)^{g-1}dv \quad (\operatorname{Im} w<0), \end{aligned} \tag{474}$$

with principal powers. The singularities at $v\in 2\pi\mathbb{N}$ are integrable, since $|(1-e^{\pm\mathrm{i}v})^{g-1}| \le e^{\pi|\operatorname{Im} g|/2}|2\sin\frac{v}{2}|^{\operatorname{Re} g-1}$. On $\operatorname{Im} z = Y$ every left argument $\ell+\sigma+\alpha_i+z$ has imaginary part at least 1 and every right argument $\beta_i - z$ at most -1. Let $a_\pm$ be the convolutions on $[0,\infty)$ of $\mathsf{h}_i^+(v) = e^{\mathrm{i}\alpha_i v}(1-e^{\mathrm{i}v})^{g_i-1}$ and of $\mathsf{h}_i^-(v) = e^{-\mathrm{i}\beta_i v}(1-e^{-\mathrm{i}v})^{g_i-1}$, that is $a_\pm(V) = \int_{\Delta_V}\prod_i \mathsf{h}_i^\pm(v_i)\,d\mu$ with $\Delta_V = \{v_i\ge0, \sum_i v_i = V\}$. By Fubini, $\prod_i f_{g_i}(\ell+\sigma+\alpha_i+z) = c_+\int_0^\infty e^{\mathrm{i}(\ell+\sigma+z)V}a_+(V)\,dV$ and $\prod_i f_{g_i}(\beta_i - z) = c_-\int_0^\infty e^{\mathrm{i}zV}a_-(V)\,dV$, where $c_\pm = \prod_i(\mp\mathrm{i})/\Gamma(g_i)$ and $c_+c_- = \kappa_\Gamma := \prod_i\Gamma(g_i)^{-2}$.

Endpoint behavior. Fix $0<\delta<\frac{\pi}{2}$, and let $\lambda(V) = 1+|\log|2\pi-V||$ if $\varkappa=1$ and $\lambda=1$ otherwise (this λ is local to Step 3 and unrelated to the weights λ_m of Section 20.5). Then

$$|a_\pm(V)| \le \begin{cases} C\,V^{\varkappa-1}, & 0<V\le\pi,\\ C\,|2\pi-V|^{\min(\varkappa-1,0)}\lambda(V), & \pi\le V\le 2\pi+\delta. \end{cases} \tag{475}$$

The paragraph *Details of the top-mode proof* proves (475) from $|\mathsf{h}_i^{\pm}(v)| \le C|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$ and Dirichlet's integral. For $\varkappa > 1$ the bound at 2π is attained in general, and at $\varkappa = 1$ it is at most logarithmic. Hence, near $V = 0$,

$$a_+(V)\, a_-(2\pi - V) = O\big(V^{\,\varkappa - 1 + \min(\varkappa - 1, 0)}\big) \qquad (\text{times at most } 1 + |\log V| \text{ at } \varkappa = 1), \tag{476}$$

and symmetrically near $V = 2\pi$. The exponent exceeds -1 iff $2\varkappa > 1$.

Along $z = x + \mathrm{i}Y$ the two products are $c_\pm$ times the Fourier transforms $\widehat{a_\pm^Y}(x) = \int e^{\mathrm{i}xV} a_\pm^Y(V)\, dV$ of $a_+^Y(V) = e^{\mathrm{i}(\ell+\sigma)V - YV} a_+(V)$ and $a_-^Y(V) = e^{-YV} a_-(V)$, extended by zero to $V \le 0$; both lie in $L^1(\mathbb{R})$. So $T_\ell(z) = \kappa_\Gamma \widehat{a_+^Y * a_-^Y}(x)$. Since $|f_{g_i}(w)| \le C(1 + |w|)^{-\operatorname{Re} g_i}$ on horizontal lines at distance at least one from the real axis, $\widehat{a_+^Y * a_-^Y} = O(|x|^{-2\varkappa})$ is integrable by $2\varkappa > 1$. On $[0, 2\pi + \delta]$ the functions $a_\pm^Y$ are square integrable by (475) and $2\varkappa > 1$, so $a_+^Y * a_-^Y$ is continuous at $V = 2\pi$. Fourier inversion at this point gives

$$I_\ell = 2\pi\kappa_\Gamma \int_0^{2\pi} e^{\mathrm{i}(\ell+\sigma)V} a_+(V)\, a_-(2\pi - V)\, dV = 2\pi\kappa_\Gamma \int_{\widetilde{\Delta}} e^{\mathrm{i}(\ell+\sigma)V} \prod_{i=1}^{r} \mathsf{h}_i^+(v_i)\, \mathsf{h}_i^-(v_i')\, d\mathrm{vol}, \tag{477}$$

where $\widetilde{\Delta} = \{(v, v') \in [0,\infty)^{2r} : V + V' = 2\pi\}$, $V = \sum_i v_i$, $V' = \sum_i v_i'$, and $d\mathrm{vol} = dv_1 \cdots dv_r\, dv_1' \cdots dv_{r-1}'$ is its Lebesgue measure, invariant under permutations of the $2r$ coordinates.

Step 4 (spin sum). By Step 2 and Abel's theorem, $s_1^+ = \lim_{\rho\uparrow 1} \sum_\ell d_\ell \rho^\ell I_\ell$. For $\rho < 1$ the sum may be taken inside (477), since $a_+(V)a_-(2\pi - V) \in L^1(0, 2\pi)$ by (476), and it produces (461) at $z = \rho e^{\mathrm{i}V}$. For $\frac{1}{2} \le \rho < 1$, $|1 - \rho e^{\mathrm{i}V}| \ge \sqrt{2}|\sin\frac{V}{2}|$, so the integrand is dominated by $C|\sin\frac{V}{2}|^{-\max(d-1,0)}|a_+(V)a_-(2\pi - V)|$. This function is bounded on compact subsets of $(0, 2\pi)$ and is $O(V^{\,\varkappa - 1 + \min(\varkappa-1,0) - \max(d-1,0)})$ at both ends (with $2\pi - V$ at the upper end and at most a logarithm at $\varkappa = 1$). The bound is integrable iff $\varkappa + \min(\varkappa - 1, 0) > \max(d - 1, 0)$, which is exactly (C): for $\varkappa \ge 1$ it reads $\varkappa > d - 1$, and for $\varkappa < 1$ it reads $2\varkappa > \max(d, 1)$, which forces $d < 2$ and hence $\varkappa > \frac{d}{2} > d - 1$. For the wheels this is the full hypothesis $b < b_{\max}$, not only $b < b_{\mathrm{spec}}$ (Step 2 uses it as well). Dominated convergence gives

$$s_1^+ = 2\pi\kappa_\Gamma \int_{\widetilde{\Delta}} \Psi\, d\mathrm{vol}, \qquad \Psi = e^{\mathrm{i}\sigma V}(1 + e^{\mathrm{i}V})(1 - e^{\mathrm{i}V})^{1-d} \prod_{i=1}^{r} \mathsf{h}_i^+(v_i)\, \mathsf{h}_i^-(v_i').$$

Step 5 (balance and exchange). For $0 < v < 2\pi$, $1 - e^{\pm\mathrm{i}v} = 2\sin\frac{v}{2}\, e^{\pm\mathrm{i}(v-\pi)/2}$, so that $(1 - e^{\mathrm{i}v})^{g-1} = e^{\mathrm{i}(g-1)(v-\pi)}(1 - e^{-\mathrm{i}v})^{g-1}$ for principal powers, and by (B)

$$\mathsf{h}_i^+(v) = e^{-\mathrm{i}\pi(g_i - 1)}\, e^{\mathrm{i}(\frac{d}{2} - 1 - \sigma)v}\, \mathsf{h}_i^-(v) \qquad (0 < v < 2\pi), \tag{478}$$

with the same rate $\frac{d}{2} - 1 - \sigma$ for every factor. In the interior of $\widetilde{\Delta}$ all coordinates lie in $(0, 2\pi)$. With $1 + e^{\mathrm{i}V} = 2\cos\frac{V}{2}\, e^{\mathrm{i}V/2}$ and $(1 - e^{\mathrm{i}V})^{1-d} = (2\sin\frac{V}{2})^{1-d} e^{\mathrm{i}(1-d)(V-\pi)/2}$ the rates in V cancel, and

$$\Psi = e^{\mathrm{i}\pi(d-1)/2 - \mathrm{i}\pi\sum_i (g_i - 1)}\, 2\cos\frac{V}{2}\Big(2\sin\frac{V}{2}\Big)^{1-d} \prod_{i=1}^{r} \mathsf{h}_i^-(v_i)\, \mathsf{h}_i^-(v_i').$$

The exchange $\iota(v, v') = (v', v)$ maps $\widetilde{\Delta}$ onto itself, preserves $d\mathrm{vol}$ and sends V to $2\pi - V$. Every factor of Ψ is ι-invariant except $\cos\frac{V}{2}$, which changes sign. So $\Psi \circ \iota = -\Psi$, $s_1^+ = 0$ and $\gamma_{1-r} = 0$. Integrated over the fibers of V, this is the odd one-variable integral $s_1^+ = 2\pi\kappa_\Gamma e^{\mathrm{i}\pi(d-1)/2 - \mathrm{i}\pi\sum_i (g_i-1)} \int_0^{2\pi} a_-(V)a_-(2\pi - V)(2\sin\frac{V}{2})^{1-d} 2\cos\frac{V}{2}\, dV$. For the wheels, $a_-(V) = e^{-\mathrm{i}(g-1)(V - r\pi)/2}\mathsf{p}(V)$ with $\mathsf{p}(V) = \int_{\Delta_V} \prod_i (2\sin\frac{v_i}{2})^{g-1} d\mu$, positive for real b, and the prefactor becomes $2\pi e^{\mathrm{i}\pi(\sigma + 1/2)}/\Gamma(g)^{2r}$: the integrand is real up to a constant phase. Equivalently, writing $\Psi = 2\cos\frac{V}{2}(2\sin\frac{V}{2})^{1-d}\, \Pi\, e^{\mathrm{i}\Theta}$ with $\Pi = \prod_i (2\sin\frac{v_i}{2} \cdot 2\sin\frac{v_i'}{2})^{g_i - 1}$, the phase obeys $\Theta - \Theta \circ \iota =$

$\sum_i(\sigma+\alpha_i+\beta_i+g_i-\frac{d}{2})(v_i-v_i')$ for all (v,v'). So per-factor balance, not total balance, makes it ι-invariant.

Step 6 (the other top mode). Since d and the d_ℓ are real, $\overline{\Xi_T(\bar t;p)}=\Xi_T(t;\bar p)$ for the parameter point $p=(\sigma,\alpha_i,g_i)$, and $\bar p$ again satisfies (B) and (C). Hence $\gamma_{r-1}(p)=\overline{\gamma_{1-r}(\bar p)}=0$ by Steps 1–5 at $\bar p$. The direct computation on $\operatorname{Im} z=-Y$ gives the same: the two forms of (474) exchange their roles, every exponential has i replaced by $-$i, and the integrand is again odd under ι.

Step 7 (continuation). $\operatorname{Re} g_i>0$ was used only in Steps 3–5. The γ_m are analytic on the convex, hence connected, set (C), and $\gamma_{\pm(r-1)}$ vanish on its nonempty open part where all $\operatorname{Re} g_i>0$; so they vanish on all of it. □

Details of the top-mode proof. The steps above state some estimates and computations in short form; this paragraph gives them in full, and with it the proof of (471) is complete. Throughout, β_i is given by (B). The set (C) is an open half-space of $\mathbb{C}^{2r+1}\cong\mathbb{R}^{4r+2}$, hence convex and connected. For a compact set B in it, let Y_B be the maximum of Y_0 and $\varkappa_B$ the minimum of $\varkappa$ over B; then $\varkappa_B>\max(\frac{d}{2},d-1,\frac{1}{2})$. Let P_T be the union of the $2r$ classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$.

Step 0. In the right part the Gamma-ratio bound (465) is applied to the left factors $f_{g_i}(\ell+\sigma+\alpha_i+n+t)$ as they stand (base $1+\ell+n$) and to the right factors in the first reflected form (base $1+n$); in the middle part to all factors as they stand, $f_{g_i}(k+\sigma+\alpha_i+t)$ and $f_{g_i}(j+1+\beta_i-t)$ (bases $1+k$ and $1+j$); in the left part to the left factors in the second reflected form (base $1+k$) and to the right factors as they stand, $f_{g_i}(\ell+k+1+\beta_i-t)$ (base $1+\ell+k$). Factor i contributes the power $-\operatorname{Re} g_i$ of a base comparable, uniformly in i, to one of these, so the exponents add up to $-\varkappa$ whatever the signs of the individual $\operatorname{Re} g_i$. On a compact set K off P_T the sine ratios of the reflected forms are bounded, and, as in the paragraph *Proofs of the estimates*, the finitely many factor values with small arguments are bounded on K one by one, although each occurs for infinitely many lattice points. This gives $|T_\ell(n+t)|\le C_K(1+\ell+n)^{-\varkappa}(1+n)^{-\varkappa}$, $C_K(1+k)^{-\varkappa}(1+j)^{-\varkappa}$ and $C_K(1+k)^{-\varkappa}(1+\ell+k)^{-\varkappa}$ in the three parts. For $|y|\ge Y_0$: by (B), $\operatorname{Im}(\beta_i+g_i)=-\operatorname{Im}(\sigma+\alpha_i)$ and $\operatorname{Im}(\sigma+\alpha_i+g_i)=-\operatorname{Im}\beta_i$. So every Gamma argument in a numerator, namely $\ell+n+\sigma+\alpha_i+t$, $n+1-\beta_i-g_i+t$, $k+\sigma+\alpha_i+t$, $j+1+\beta_i-t$, $k+2-\sigma-\alpha_i-g_i-t$ and $\ell+k+1+\beta_i-t$, and every sine argument in a denominator, has imaginary part of modulus at least $|y|-Y_0+1\ge1$. No numerator Gamma function then meets a pole, the sine ratios are bounded, and the bounds hold with each base $1+\cdots$ replaced by $\cdots+|y|$. The constants are uniform for parameters in B, with $\varkappa_B$ in place of $\varkappa$ and $|y|\ge Y_B$, provided K avoids P_T for every parameter point of B. For the wheels the right and middle bounds are those of the paragraph *Proofs of the estimates*. With $|d_\ell|\le C(1+\ell)^{d-2}$ and (467) at $\alpha=\varkappa$ (the left sum has the form of the right one, with k in place of n) the three parts converge absolutely and uniformly on compact subsets off P_T. Each term is meromorphic in t; its poles come from $\Gamma(\ell+\sigma+\alpha_i+n+t)$ and $\Gamma(\beta_i-n-t)$, are simple, and lie on the classes $-\sigma-\alpha_i+\mathbb{Z}$ and $\beta_i+\mathbb{Z}$. At a point t_0 of P_T every term therefore has a pole of order at most the number of the $2r$ classes that contain t_0; multiplied by $(t-t_0)$ to that power, the series converges uniformly on a small circle around t_0, so the product with S_T is analytic near t_0, as in part (a) above. Periodicity follows from $n\mapsto n+1$. Each factor of D_T has simple zeros exactly on one class, so D_T vanishes at t_0 to the same order, and Ξ_T is entire and one-periodic. For $0\le x\le1$ and $|y|\ge\max(Y_0,2)$, the large-$|y|$ bounds and (467) give $|S_T(x+\mathrm{i}y)|\le C|y|^{\max(d,1)-2\varkappa}(1+\log|y|)$, and $|D_T(x+\mathrm{i}y)|\le Ce^{2\pi r|y|}$. Taking $y=-Y$ for $m\ge r$ and $y=Y$ for $m\le-r$ in the Fourier coefficient gives $|\gamma_m|\le Ce^{2\pi(r-|m|)Y}Y^{\max(d,1)-2\varkappa}(1+\log Y)\to0$, because $2\varkappa>\max(d,1)$ by (C). For the analyticity, on the segment $t=x+\mathrm{i}Y_B$, $0\le x\le1$, which stays at distance at least one from P_T for every parameter point of B, the series of S_T converges uniformly in x and in the parameters, its terms are analytic in the parameters, and D_T is entire in t and the parameters. Hence $\gamma_m=\int_0^1\Xi_T(x+\mathrm{i}Y_B)e^{-2\pi\mathrm{i}m(x+\mathrm{i}Y_B)}dx$ is analytic on the interior of B.

Step 1. The coefficients of the expansion in the upper half-plane with $m\le0$ vanish: the coefficient integral $\int_0^1 S_T(x+\mathrm{i}y)e^{-2\pi\mathrm{i}m(x+\mathrm{i}y)}dx$ does not depend on $y>Y_0$, its exponential is bounded by one

for $m \le 0$, and $S_T \to 0$ uniformly as $y \to \infty$. The relation $\gamma_{1-r} = \kappa_+ s_1^+$ compares the coefficients of the lowest power w^{1-r} in $\Xi_T = D_T S_T$ with the expansion of Step 0; the lower half-plane is the same.

Step 3: the Laplace forms (474). For $\operatorname{Re} w > 0$, $\Gamma(g) f_g(w) = B(w,g) = \int_0^\infty e^{-wu}(1-e^{-u})^{g-1}du$ (put $x = e^{-u}$ in Euler's integral). Substitute $u = -\mathrm{i}v$, with v on the positive imaginary axis, and rotate the ray to $[0,\infty)$ through the first quadrant. If also $\operatorname{Im} w > 0$, then $|e^{\mathrm{i}wv}| = e^{-\operatorname{Im}(wv)}$ decays exponentially in the closed quadrant, since $\arg w + \arg v \in (0,\pi)$ there. Moreover $1 - e^{\mathrm{i}v}$ has nonnegative real part in the quadrant and vanishes only at $v \in 2\pi\mathbb{N}_0$, where the integrand is $O(|v - 2\pi k|^{\operatorname{Re} g - 1})$, which is integrable. These points lie on the boundary of the quadrant: $v = 0$ at its corner and $v = 2\pi k$, $k \ge 1$, on its real side. A small quarter circle of radius ϵ at 0 and small semicircles at $2\pi k$, $k \ge 1$, contribute $O(\epsilon^{\operatorname{Re} g})$ each and nothing in the limit. This gives the first formula for $\operatorname{Re} w > 0$, $\operatorname{Im} w > 0$. The substitution $u = \mathrm{i}v$ and the fourth quadrant give the second formula for $\operatorname{Re} w > 0$, $\operatorname{Im} w < 0$. Both right sides are analytic on the respective half-planes, because $|(1-e^{\pm\mathrm{i}v})^{g-1}| \le e^{\pi|\operatorname{Im} g|/2}|2\sin\frac{v}{2}|^{\operatorname{Re} g - 1}$ is 2π-periodic and locally integrable; so (474) holds on the whole half-planes. For real g the second formula is the complex conjugate of the first; for complex g it is the separate rotation just described. On $\operatorname{Im} z = Y > Y_0$ one has $|e^{\mathrm{i}(\ell+\sigma+\alpha_i+z)v}| = e^{-(Y+\operatorname{Im}(\sigma+\alpha_i))v} \le e^{-v}$ and $|e^{-\mathrm{i}(\beta_i - z)v}| = e^{-(Y-\operatorname{Im}\beta_i)v} \le e^{-v}$, so the two products of Step 3 are absolutely convergent integrals and Fubini applies.

Step 3: the endpoint bound (475). On $[0,3\pi]$ the exponentials in $\mathsf{h}_i^\pm$ and the factors $e^{-\operatorname{Im} g_i \arg(1-e^{\pm\mathrm{i}v})}$ are bounded, so $|\mathsf{h}_i^\pm(v)| \le C|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$ there. The modulus $|2\sin\frac{v}{2}|$ lies between cv and v on $(0, \pi+\delta]$ and between $c|2\pi - v|$ and $|2\pi - v|$ on $[\frac{\pi}{2}, 3\pi)$. Hence, whatever the sign of $\operatorname{Re} g_i - 1$, $|\mathsf{h}_i^\pm(v)| \le C v^{\operatorname{Re} g_i - 1}$ on the first interval and $|\mathsf{h}_i^\pm(v)| \le C|2\pi - v|^{\operatorname{Re} g_i - 1}$ on the second. For $V \le \pi$ every coordinate of Δ_V is at most π, and Dirichlet's integral gives $|a_\pm(V)| \le C\int_{\Delta_V}\prod_i v_i^{\operatorname{Re} g_i - 1}d\mu = CV^{\varkappa-1}\prod_i \Gamma(\operatorname{Re} g_i)/\Gamma(\varkappa)$. For $\pi \le V \le 2\pi + \delta$ at most one coordinate of a point of Δ_V exceeds $V/2$. Where none does, all coordinates are below $\pi + \delta$, and the same Dirichlet bound gives $O(1)$. Where $v_j > V/2$, write $s = V - v_j < V/2$ and $\epsilon_V = 2\pi - V \in [-\delta, \pi]$, so that $2\pi - v_j = \epsilon_V + s$. Bounding factor j by $C|\epsilon_V + s|^{\operatorname{Re} g_j - 1}$ and integrating the other $r-1$ factors over their simplex of size s (Dirichlet again, with exponent $\varkappa - \operatorname{Re} g_j > 0$) leaves

$$C\int_0^{V/2} |\epsilon_V + s|^{\operatorname{Re} g_j - 1} s^{\varkappa - \operatorname{Re} g_j - 1} ds.$$

For $\operatorname{Re} g_j \ge 1$ this is bounded. For $\operatorname{Re} g_j < 1$, the part $s \le 2|\epsilon_V|$ is $|\epsilon_V|^{\varkappa-1}$ times the convergent integral $\int_0^2 |1 \pm u|^{\operatorname{Re} g_j - 1} u^{\varkappa - \operatorname{Re} g_j - 1}du$ (substitute $s = |\epsilon_V|u$), and on $s > 2|\epsilon_V|$ one has $|\epsilon_V + s| \ge s/2$, which leaves $C\int_{2|\epsilon_V|}^{2\pi} s^{\varkappa-2}ds$. Both are $O(|\epsilon_V|^{\min(\varkappa-1,0)}\lambda)$. Summing over j proves (475). At $\varkappa = 1$ the bound is at most logarithmic: when $\operatorname{Im}\sum_i g_i \ne 0$, $a_\pm(2\pi - \epsilon_V)$ stays bounded and oscillates like $\epsilon_V^{\mathrm{i}\operatorname{Im}\sum_i g_i}$ (checked numerically in the review `LRI_C5_general_d/review/lemmaG-proof` for $r = 2$, $g = (0.45 + 0.3\mathrm{i}, 0.55)$), and the logarithm can occur only when $\operatorname{Im}\sum_i g_i = 0$. Only the upper bound is used. For the wheels at real b, by dominated convergence, $a_-(2\pi^-) = e^{-\mathrm{i}(g-1)(2-r)\pi/2}\mathsf{p}(2\pi)$ is finite and nonzero, so for $\varkappa > 1$ the exponent in (476) is $\varkappa - 1$, not $2\varkappa - 2$.

Step 3: the formula (477). The functions $a_\pm^Y$ lie in $L^1(\mathbb{R})$: $|a_+^Y(V)|$ is at most the convolution of the functions $Ce^{-(Y+\operatorname{Im}(\sigma+\alpha_i))v}|2\sin\frac{v}{2}|^{\operatorname{Re} g_i - 1}$, each integrable on $[0,\infty)$, and likewise for a_-^Y. The bound $|f_{g_i}(w)| \le C(1+|w|)^{-\operatorname{Re} g_i}$ on horizontal lines at distance at least one from the real axis follows from Gamma reflection and (465). For $U < 2\pi + \delta$ the convolution $(a_+^Y * a_-^Y)(U) = \int_0^U a_+^Y(V)a_-^Y(U-V)dV$ involves $a_\pm^Y$ only on $[0, 2\pi+\delta]$, where they are square integrable by (475) and $2\varkappa > 1$, and a convolution of two square-integrable functions is continuous. Since the convolution and its Fourier transform are both integrable, Fourier inversion holds at this point of continuity, $\int_{\mathbb{R}} \widehat{a_+^Y * a_-^Y}(x)e^{-2\pi\mathrm{i}x}dx = 2\pi(a_+^Y * a_-^Y)(2\pi)$. Together with $e^{-2\pi\mathrm{i}z} = e^{2\pi Y}e^{-2\pi\mathrm{i}x}$, and the factor $e^{-2\pi Y}$ in $(a_+^Y * a_-^Y)(2\pi)$, this is the first form of (477). The second form is Fubini; the absolute integrand obeys the same bounds. The measure $d\mathrm{vol}$ is $(2r)^{-1/2}$ times the surface measure of the hyperplane $V + V' = 2\pi$, hence invariant under every permutation of the $2r$ coordinates.

Step 4. For $\frac{1}{2} \le \rho < 1$, $|1-\rho e^{\mathrm{i}V}|^2 = (1-\rho)^2 + 4\rho\sin^2\frac{V}{2} \ge 2\sin^2\frac{V}{2}$, so $|(1+\rho e^{\mathrm{i}V})(1-\rho e^{\mathrm{i}V})^{1-d}| \le C|\sin\frac{V}{2}|^{-\max(d-1,0)}$. Dominated convergence gives first the one-variable form $s_1^+ = 2\pi\kappa_\Gamma\int_0^{2\pi} e^{\mathrm{i}\sigma V}(1+e^{\mathrm{i}V})(1-e^{\mathrm{i}V})^{1-d}a_+(V)a_-(2\pi-V)\,dV$, and the form on $\widetilde{\Delta}$ is Fubini, since after integration over the fibers of V the function $|\Psi|$ is dominated by the same integrable function. For $d>2$ the entry $\varkappa > d-1$ of (C) is stronger than the spectral condition $2\varkappa > d$, and by (476) the argument does not extend past $b_{\max}$.

Step 6. Conjugating the expansion of Step 1 at $\bar p$ gives the expansion in the lower half-plane at p, so $s_m^-(p) = \overline{s_m^+(\bar p)}$ and $\gamma_{r-1}(p) = \kappa_- s_1^-(p) = \kappa_-\overline{s_1^+(\bar p)} = 0$; this is equivalent to the relation $\gamma_m(\bar p) = \overline{\gamma_{-m}(p)}$ used above. At real parameters it is the reality $\gamma_{-m} = \overline{\gamma_m}$; at complex parameters it relates two different parameter points and is not a symmetry of a single Ξ_T.

Step 7. The part of (C) where all $\operatorname{Re} g_i > 0$ is open and nonempty; it contains the points with all $\operatorname{Re} g_i = \varkappa/r$.

The equivalent form. On $\operatorname{Im} t > Y_0$, $D_T = \kappa_+ w^{-r}(1+O(w))$ with $\kappa_+ \ne 0$, and $\Xi_T = \sum_{|m|\le r-2}\gamma_m w^m = O(e^{2\pi(r-2)\operatorname{Im} t})$; so $S_T = \Xi_T/D_T = O(e^{-4\pi\operatorname{Im} t})$ as $\operatorname{Im} t \to +\infty$. Conversely, this decay forces $s_1^+ = 0$: let $Y\to\infty$ in its coefficient integral. The lower half-plane is the same with κ_-. For the wheels, with $c_{r,r-1} = 2\gamma_{r-1}e^{-\mathrm{i}\pi(r-1)\sigma}$, this is the equivalent form stated after (473).

The wheel case and the earlier proof. For the wheels this proof is the earlier, wheel-only proof that the review `LRI_C5_general_d/review/VERIFICATION.md` checked (proof review in `LRI_C5_general_d/review/foundation-rigor`; Section 20.10), with Steps 3–4 as corrected after that review, up to three changes: the parameters are complex throughout (the earlier version reduced to real b by analyticity in b), the second top mode comes from Step 6 instead of the reflection of part (b) of Section 20.1.1, and Step 5 is written through (478).

Two ingredients enter: the balance of every single factor (Step 5) and the factor $1+z$ in (461), which is odd about $V=\pi$. Both are essential: controls that break either one have nonzero top modes. The numerical tests of the theorem and of its proof, with these controls, are collected in Section 20.9 (item *Top-mode theorem and its product form*).

Relation to the periodization at $b=d/2$ and to the half-shift reduction. At $b=d/2$ and $\frac{2}{r} < d < 4$, (473) sharpens (311) for the wheels: $c_{3,2}(q)=0$ and $c_{4,3}(q)=0$ in every such dimension, and the sum in (318) stops at $k=r-2$. It says nothing about Ξ_7: the kernel K_ℓ of (303) carries the extra factor $[(\ell+p)^2-4s^2]\Gamma(p+2s)\Gamma(p-2s)/(\Gamma(2s)\Gamma(-2s))$ and the multiplicities $(-1)^\ell d_\ell$, so it is not of the form (470). In Section 19 the relations $c_{3,2}=0$ and $c_{4,3}=0$ come out of the half-shift determinant (431); they are the $d=3$ cases of (473), which holds for every d and every b in the strip. What is special to three dimensions is the identification of the surviving modes with products of ordinary moment amplitudes, $c_{3,0} = -\mathfrak{a}_3\mathfrak{b}_3$ and $c_{4,1} = 2\mathfrak{a}_4\mathfrak{b}_4$, and the odd $r=7$ coefficients (432). At $d=1$ and $d=2$, (473) also follows from the one-variable lattices (539) below (at $d=2$, $b=1$ and $r=4$ it is (337)); at generic d only the proof of Section 20.1.2 gives it.

20.1.3 Residue functional, explicit weights and blocks

For $0 < \operatorname{Re} b < b_{\max}(r,d)$ and $\sigma\notin\mathbb{Z}$,

$$W_r(d;b) = -N_b^r\operatorname{Res}_{t=0}\frac{\Xi_r(t;b)}{D_b(t)^r} = -\Big[\frac{N_b}{\pi\sin\pi\sigma}\Big]^r\sum_{k=0}^{r-2}B_{r,k}(\sigma)\,c_{r,k}(q,b), \tag{479}$$

where $B_{r,k}$ is (318) with the logarithmic coefficients (314), both with q replaced by σ. The proof follows the one after (319); it is given in full in the paragraph *The residue functional in full* below.

The weights depend on b only, not on q. For $r = 3, 4$,

$$\boxed{\begin{aligned} W_3(d;b) &= -\frac{2N_b^3}{\pi s_b^5}\Big[(1+2\varsigma_b^2)\,c_{3,0} + 3\varsigma_b\,c_{3,1}\Big],\\ W_4(d;b) &= \frac{4N_b^4}{\pi s_b^7}\Big[\varsigma_b(3+2\varsigma_b^2)\,c_{4,0} + (1+4\varsigma_b^2)\,c_{4,1} + 5\varsigma_b\,c_{4,2}\Big]. \end{aligned}} \tag{480}$$

The weights of the omitted top modes, $B_{3,2} = 6\pi^2/s_b^2$ and $B_{4,3} = -20\pi^3/s_b^3$, are nonzero, so the formulas below depend on (473).

The residue functional in full. *Proof of* (479). Let $0 < \operatorname{Re} b < b_{\max}(r,d)$ and $\sigma \notin \mathbb{Z}$; then $b_{\max} \le b_{\text{spec}}$, so (460) represents W_r. For each spin close the Barnes contour to the right by the arcs of the circles $|s - a_\ell| = n_0 + \frac12$, $n_0 \in \mathbb{N}$, that lie in $\operatorname{Re} s \ge 0$. The part of the disc $|s - a_\ell| < n_0 + \frac12$ in $\operatorname{Re} s > 0$ contains the right poles $a_\ell + n$, $0 \le n \le n_0$, and no left pole, since the left poles $-a_\ell - n$ have negative real part. The arcs stay at distance at least $\frac12$ from $a_\ell + \mathbb{Z}$, because $|s - a_\ell - m| \ge |n_0 + \frac12 - |m|| \ge \frac12$ for every integer m, so the sine ratio produced by reflecting $\Gamma(a_\ell - s)/\Gamma(\beta_\ell - s)$ is bounded on them, and $g_\ell(s)^r = O(|s|^{-2r\operatorname{Re} g})$ there; since $2r\operatorname{Re} g > 1$, the arcs contribute nothing in the limit. (The package's own proof closes the contour on the circles $|s| = a_\ell + N + \frac12$, whose radius is real only for real b. The circles centred at a_ℓ keep the distance $\frac12$ from $a_\ell + \mathbb{Z}$ for every complex b. This change was made when the proof was merged into this note, on 16 October 2026, and has not been reviewed.) The closure is clockwise, so the ℓ-th integral is $-\sum_{n\ge0}\operatorname{Res}_{s=a_\ell+n} g_\ell(s)^r$, and $\operatorname{Res}_{s=a_\ell+n} g_\ell(s)^r = \operatorname{Res}_{t=0} g_\ell(a_\ell+n+t)^r$. In S_r the point $t=0$ is a pole only of the terms with $n \ge 0$: for $n < 0$, $\Gamma(-n-t)$ is regular at $t=0$, and $t = 0 \equiv -\sigma$ would need $\sigma \in \mathbb{Z}$. By the uniform convergence on a small circle $|t| = \epsilon$ (paragraph *Proofs of the estimates*), $\sum_\ell d_\ell \sum_{n\ge0}\operatorname{Res}_{s=a_\ell+n} g_\ell^r = \operatorname{Res}_{t=0} S_r = \operatorname{Res}_{t=0} \Xi_r/D_b^r$, which is the first equality of (479). Next,

$$\log\frac{(\pi t\sin\pi\sigma)^r}{D_b(t)^r} = r\log\frac{\pi t}{\sin\pi t} - r\log\frac{\sin\pi(t+\sigma)}{\sin\pi\sigma} = \sum_{h\ge1} u_h\frac{t^h}{h!},$$

with u_h the coefficients (314) at $q \to \sigma$, $u_h(\sigma) = r[(1+(-1)^h)\psi_{h-1}(1) - (-1)^{h+1}\psi_{h-1}(1-\sigma) + \psi_{h-1}(\sigma)]$. Indeed $\log(\pi t/\sin\pi t) = \sum_{m\ge1}\zeta(2m)t^{2m}/m$, with $2(h-1)!\,\zeta(h) = (1+(-1)^h)\psi_{h-1}(1)$ for even h, and $\partial_\sigma^h \log\sin\pi\sigma = -\psi_{h-1}(\sigma) - (-1)^h\psi_{h-1}(1-\sigma)$ [32]. With the polynomials E_m of (316), which are the Taylor coefficients of $\exp\sum_h x_h t^h/h!$, therefore $\operatorname{Res}_{t=0} e^{2\pi imt} D_b(t)^{-r} = (\pi\sin\pi\sigma)^{-r} E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1})$. Inserting $\Xi_r = \sum_{|m|\le r-2}\gamma_m e^{2\pi imt}$, which is (473), gives the exponential form

$$W_r(d;b) = -\Big[\frac{N_b}{\pi\sin\pi\sigma}\Big]^r \sum_{|m|\le r-2}\gamma_m\,E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1}), \tag{481}$$

and $\gamma_{\pm k} = \frac12 c_{r,k} e^{\pm i\pi k\sigma}$ $(k \ge 1)$, $\gamma_0 = c_{r,0}$ turn it into (479) with $B_{r,k}(\sigma) = \frac12[e^{i\pi k\sigma}E_{r-1}(u_1 + 2\pi i k, u_2, \dots) + e^{-i\pi k\sigma}E_{r-1}(u_1 - 2\pi i k, u_2, \dots)]$, which is (318) at $q \to \sigma$. The functional depends on b only, through σ; the dimension enters only through N_b and the $c_{r,k}(q,b)$. By (315) at $q \to \sigma$ the u_h are elementary, $u_1/(r\pi) = -\varsigma_b/s_b$, $u_2/(r\pi^2) = \varsigma_b^2/s_b^2 + \frac43$ and $u_3/(r\pi^3) = -2(\varsigma_b/s_b)(1 + \varsigma_b^2/s_b^2)$, and for the b-derivatives of the weights $\partial_\sigma u_h = u_{h+1} - r(1-(-1)^h)\psi_h(1)$. For $r = 3, 4$,

$$\begin{gathered} B_{3,0} = \frac{2\pi^2(1+2\varsigma_b^2)}{s_b^2}, \qquad B_{3,1} = \frac{6\pi^2\varsigma_b}{s_b^2}, \qquad B_{3,2} = \frac{6\pi^2}{s_b^2},\\ B_{4,0} = -\frac{4\pi^3\varsigma_b(3+2\varsigma_b^2)}{s_b^3}, \qquad B_{4,1} = -\frac{4\pi^3(1+4\varsigma_b^2)}{s_b^3}, \qquad B_{4,2} = -\frac{20\pi^3\varsigma_b}{s_b^3}, \qquad B_{4,3} = -\frac{20\pi^3}{s_b^3}, \end{gathered} \tag{482}$$

and (480) follows. □ At $d = 2$ and $r = 4$, with $x = \frac{b}{2}$, N_b equals $\mathcal{N}_x$ of Section 20.6, and inserting $c_{4,0}, c_{4,1}, c_{4,2}$ of the $d = 2$ paragraph there into (480) reproduces the weights ρ_0, ρ_1, ρ_2 of (350). At $b = 1$ ($\varsigma_b = 0$, $s_b = 1$, $N_1 = 1$) it reduces to $P_4(2) = 4c_{4,1}/\pi = 8\mathcal{A}(\frac12)\mathcal{B}(\frac12)/\pi$.

Each sample is a sum of five blocks, (483) below. The blocks are (321) with the arrays (322), where the rim parameter q is replaced by σ in $q+t$, $q+1+t$ and in the mirror point, and the spoke parameter q by g in $1-q+t$, $1+q-t$ and $\Gamma(q-t)$. Every parameter is affine in (q,b), and all five blocks converge absolutely exactly for $\operatorname{Re} b < b_{\max}$ (paragraph *Proof of the block decomposition* below). The sections below do not use generic nodes. They sample at the pole node and at the reflection centers, where the blocks collapse to fewer functions.

Proof of the block decomposition. Write A_r^0, A_r^+, C_r for the blocks (321) with the substitutions just stated; at $b=\frac{d}{2}$ they are the blocks of (323). The claim is

$$\Xi_r(t;b) = D_b(t)^r\,[A_r^0(t)+A_r^+(t)+A_r^0(1-\sigma-t)+A_r^+(1-\sigma-t)+C_r(t)] \qquad (t\notin\mathbb{Z}\cup(-\sigma+\mathbb{Z})). \tag{483}$$

In the right part with $\ell=0$, the factorized form in (464) and

$$f(\sigma+n+t)=\frac{\Gamma(\sigma+t)}{\Gamma(2q+t)}\frac{(\sigma+t)_n}{(2q+t)_n}, \qquad f(-n-t)=\frac{\Gamma(-t)}{\Gamma(g-t)}\frac{(1-g+t)_n}{(1+t)_n},$$

which use $\sigma+g=2q$, give the series of $A_r^0(t)$ term by term; the upper parameter 1 of the ${}_{2r+1}F_{2r}$ cancels $n!$. For $\ell=j+1$, $f(j+1+\sigma+n+t)=[\Gamma(1+\sigma+t)/\Gamma(1+2q+t)]\,(1+\sigma+t)_{j+n}/(1+2q+t)_{j+n}$, and the multiplicity $d_{j+1}=4q\,(2q+1)_j(4q-1)_j/((2q)_j(2)_j)$ of (301) supplies the prefactor $4q$ and the rows $(2q+1,\boldsymbol{e}),(4q-1,\boldsymbol{e});(2q,\boldsymbol{e}),(2,\boldsymbol{e})$; the rows $(1,\boldsymbol{e})$ and $(1,\boldsymbol{f})$ cancel the factorials of (320). This gives $A_r^+(t)$. In the middle part, the third line of (464),

$$\frac{\Gamma(k+\sigma+t)}{\Gamma(k+2q+t)}\,\frac{\Gamma(j+1-t)}{\Gamma(j+1+g-t)}=\frac{\Gamma(\sigma+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+g-t)}\,\frac{(\sigma+t)_k(1-t)_j}{(2q+t)_k(1+g-t)_j},$$

with the multiplicity $d_{j+k+1}=4q\,(2q+1)_{j+k}(4q-1)_{j+k}/((2q)_{j+k}(2)_{j+k})$, gives $C_r(t)$, with the multiplicity rows along $\boldsymbol{c}$. By the last line of (464) the left part is the right part at $1-\sigma-t$. All the series converge absolutely by (467), so the rearrangement is justified, and multiplying by $D_b(t)^r$ gives (483). Each block also converges directly under part of the condition: the ${}_{2r+1}F_{2r}(1)$ has parameter excess $r(2q+t)+r(1+t)-1-r(\sigma+t)-r(1-g+t)=2rg-1$, so it needs condition (iii); A_r^+ needs (i)–(iii), and C_r needs (i) and (ii). Together this is exactly $\operatorname{Re} b < b_{\max}$. At integer d the multiplicity rows terminate: $d_{j+1}/(4q)$ is δ_{j0} at $d=1$, 1 at $d=2$ and $1+\frac{2}{3}j$ at $d=3$.

Removable points of individual blocks. As functions of t the five blocks are sub-sums of S_r, so each is analytic off $\mathbb{Z}\cup(-\sigma+\mathbb{Z})$. In the factorized form, a Gamma prefactor times a Horn value, an individual Horn value has poles where one of its lower parameters is a non-positive integer, cancelled by a zero of the same order of its prefactor. This happens for $A_r^0(t)$ at $t\in -2q-\mathbb{N}_0$, for $A_r^0(1-\sigma-t)$ at $t\in 1+g+\mathbb{N}_0$, for $A_r^+(t)$ at $t\in -1-2q-\mathbb{N}_0$, for $A_r^+(1-\sigma-t)$ at $t\in 2+g+\mathbb{N}_0$, and for $C_r(t)$ at $t\in -2q-\mathbb{N}_0$ and $t\in 1+g+\mathbb{N}_0$. At these points (483) holds by continuity. For real b the points are real, and the nodes of (484) below, which have $\operatorname{Im} t_j=\eta_r>0$, never meet them.

20.1.4 Fixed nodes, jets, evaluation of the samples, and the continuation beyond $b_{\max}$

The formulas of this subsection are exact algebra on Sections 20.1.1–20.1.3; the continuation beyond $b_{\max}$ is derived, not proved (Section 20.10). Sections 20.2–20.7 do not use this subsection. It describes the form implemented by the general-b code `LRI_C5_general_d/foundation/wheel_general_b.py` (function `W(r,d,b,dps)`), which Section 20.9 tests.

Nodes that do not move with b. Let N be either $N_r=2r-1$ of (312) or the reduced value $N_r'=2r-3$ permitted by (473), and $M=(N-1)/2$. Put

$$t_j=-\frac{q}{2}+\frac{j}{N}+\mathrm{i}\eta_r, \qquad j=0,\dots,N-1,$$

with $\eta_r = 1/(8\pi r)$ of (312); for $N = N_r$ the first r of them are the nodes of (312). Since Ξ_r is a trigonometric polynomial with exponential modes $|m| \le M$, the discrete Fourier transform of the N complex samples recovers all modes without aliasing, $\gamma_m = N^{-1}\sum_j \Xi_r(t_j;b)e^{-2\pi i m t_j}$. Inserted into (481) this gives, for $0 < \mathrm{Re}\, b < b_{\max}(r,d)$ and $\sigma \notin \mathbb{Z}$,

$$\boxed{\begin{aligned} W_r(d;b) &= N_b^r \sum_{j=0}^{N-1} \mathfrak{w}_{r,j}(b)\, \Xi_r(t_j;b), \\ \mathfrak{w}_{r,j}(b) &= -\frac{1}{N(\pi \sin \pi\sigma)^r} \sum_{|m|\le M} E_{r-1}(u_1 + 2\pi i m, u_2, \dots, u_{r-1})\, e^{-2\pi i m t_j}. \end{aligned}} \tag{484}$$

The weights are elementary: by (315) at $q \to \sigma$ they are polynomials in π, $\cot\frac{\pi b}{2}$ and $1/\sin\frac{\pi b}{2}$, times $e^{-2\pi i m t_j}$. The nodes do not depend on b, and every factor is analytic in b on the strip (the samples by part (d) of Section 20.1.1); so the product rule and (244) give, with $\partial_b \log N_b = -\frac{1}{2}[\psi_0(g) + \psi_0(\sigma)]$,

$$\begin{aligned} \mathcal{L}_d &= \sum_j \Big[\partial_b(N_b^4 \mathfrak{w}_{4,j})\, \Xi_4(t_j;b) + N_b^4 \mathfrak{w}_{4,j}\, \partial_b \Xi_4(t_j;b)\Big]_{b=d/2}, \\ Z_d &= \frac{1}{9}\sum_j \Big[\partial_b^2(N_b^3 \mathfrak{w}_{3,j})\, \Xi_3 + 2\partial_b(N_b^3 \mathfrak{w}_{3,j})\, \partial_b \Xi_3 + N_b^3 \mathfrak{w}_{3,j}\, \partial_b^2 \Xi_3\Big]_{b=d/2}. \end{aligned} \tag{485}$$

Here $\partial_b^k \Xi_r(t_j;b) = \sum_{p\le k} \binom{k}{p} \partial_b^{k-p}[D_b(t_j)^r]\, \partial_b^p[\cdots]$, with $\partial_b D_b(t)^r = \frac{r\pi}{2}\cot\pi(t+\sigma)\, D_b(t)^r$, and the bracket is the sum of the five blocks of (483). The b-derivatives of the blocks at fixed t_j are first and second derivatives of the ${}_{2r+1}F_{2r}(1)$ values and of the two-variable Horn values with respect to their parameters, which are all affine in b with slopes $\pm\frac{1}{2}$, together with polygamma values from the Gamma prefactors; in the mirrored blocks the argument $1-\sigma-t_j$ moves with b, which is again a parameter derivative. In this form $\mathcal{L}_d$ and Z_d are finite combinations, with elementary coefficients, of one- and two-variable hypergeometric values at unit argument and their first and second parameter derivatives. With the reduced nodes W_3 needs 3 complex samples and W_4 needs 5. The b-dependent symmetric nodes $-\frac{\sigma}{2} + \frac{j}{N_r} + i\eta_r$ with the real cosine transform (313) are equally valid, and need 2 and 3 real samples; their b-derivatives also contain t-derivatives of the samples, which is why the fixed nodes are used for the jets. The code implements the symmetric nodes as a cross-check (option `nodes='note'`).

Radius of the sample jets. $W_r(d;b)$ is analytic up to b_{spec}, but the individual samples $\Xi_r(t_j;b)$ and the coefficients $c_{r,k}(q,b)$ with $k \le r-2$ have a simple pole at $b = b_{\max}$ when $b_{\max} < b_{\mathrm{spec}}$, that is for $d > 2$ (nonzero residue in every case computed; paragraph *Beyond* $b_{\max}$ below). Moreover, at a fixed node the individual blocks have poles in b where $\sigma + t_j \in \mathbb{Z}$ (at complex b, since $\mathrm{Im}\, t_j = \eta_r \neq 0$); they are removable in $\Xi_r(t_j;b)$ because of the zero of $D_b(t_j)^r$. The derivatives in (485) at $b = \frac{d}{2}$ are unaffected. When they are computed from Taylor series or Cauchy circles of the samples, however, the radius must stay below $b_{\max} - \frac{d}{2}$; for $d \ge 2$ this is $\frac{1}{2}$ for $r = 4$ and $\frac{4-d}{6}$ for $r = 3$. A second restriction concerns the weights: $\mathfrak{w}_{r,j}(b)$ carries $(\sin\pi\sigma)^{-r}$ and has a pole of order r at $b = 2$. The disc $|b - \frac{d}{2}| < b_{\max} - \frac{d}{2}$ contains $b = 2$ exactly when $b_{\max} > 2$ (paragraph *Integer* σ of Section 20.1.1): for $r = 4$ when $d > 3$, and for $r = 5$ when $d > \frac{8}{3}$. There the samples and W_r are analytic, and the poles of the weights cancel only in the sum (484). Cauchy circles of the weighted products $N_b^r \mathfrak{w}_{r,j}\Xi_r(t_j;b)$, or of the weights alone, must then stay inside $|b - \frac{d}{2}| < |2 - \frac{d}{2}|$. The jets (485) at $b = \frac{d}{2} \neq 2$ are unaffected, and the cases used in this note ($r = 3, 4$ with $d \le 3$) do not meet $b = 2$.

Evaluation of the Horn values. The module `LRI_C5_general_d/foundation/hypsums.py` evaluates the sums of the five blocks as an exact finite part plus an asymptotic tail, and `wheel_general_b.py` assembles (483) and (484). Every sequence that occurs is a Gamma-ratio sequence $\mathtt{y}_n$, a constant times a product of integer powers of Pochhammer ratios $(e)_n/(e')_n$. Its large-n expansion

$\mathsf{y}_n \sim \sum_k \upsilon_k n^{-p_\mathsf{y}-k}$ follows from the Bernoulli-polynomial expansion of $\log\Gamma(n+e)-\log\Gamma(n+e')$ [27]. *Single sums:* $\sum_{n\ge 0}\mathsf{y}_n = \sum_{n<N_\mathrm{H}}\mathsf{y}_n + \sum_k \upsilon_k\,\zeta(p_\mathsf{y}+k, N_\mathrm{H})$, with Hurwitz zeta values [28] computed by Euler–Maclaurin summation in which every term carries the factor $N_\mathrm{H}^{-p_\mathsf{y}-k}$, so that they have relative precision for complex exponents. (The routine `mpmath.zeta(s,a)` has an absolute error floor of about 10^{-P} at P working digits for complex s and is not used.) The same formula, with the Hurwitz values continued in the exponent, is the analytic continuation (the zeta-regularized value) when the series diverges. *Two-variable sums:* A_r^+ and C_r have the form $\sum_{m\ge 0}\mathsf{x}_m\mathsf{q}_m$ with a Cauchy convolution $\mathsf{q}_m = \sum_{n=0}^{m}\mathsf{y}_{m-n}\mathsf{z}_n$. For A_r^+, $\mathsf{x}_m = [(1+\sigma+t)_m/(1+2q+t)_m]^r$, $\mathsf{y}_j = d_{j+1}/(4q)$ and $\mathsf{z}_n = [(1-g+t)_n/(1+t)_n]^r$; for C_r, $\mathsf{x}_m = d_{m+1}/(4q)$, $\mathsf{y}_j = [(1-t)_j/(1+g-t)_j]^r$ and $\mathsf{z}_k = [(\sigma+t)_k/(2q+t)_k]^r$. The code computes q_m exactly for $m < M_\mathrm{H}$ and uses its large-m expansion for $m \ge M_\mathrm{H}$. If $\mathsf{y}_j \sim \sum_a \upsilon_a j^{-p_\mathsf{y}-a}$ and $\mathsf{z}_n \sim \sum_c \upsilon'_c n^{-p_\mathsf{z}-c}$, the generating function of y at $x = e^{-w} \to 1$ has the singular part $\sum_a \upsilon_a\Gamma(1-p_\mathsf{y}-a)w^{p_\mathsf{y}+a-1}$ and the regular part $\sum_k \mathsf{m}_k^\mathsf{y}(-w)^k/k!$, where m_k^y is the zeta-regularized moment "$\sum_j j^k\mathsf{y}_j$" (Mellin asymptotics of harmonic sums [29]; transfer at the singularity $x=1$ [30]), and likewise for z. Multiplying and transferring back gives three families,

$$\begin{aligned}
\mathsf{q}_m &\sim \sum_i \big[\mathfrak{e}_i^{(1)} m^{-p_\mathsf{y}-i} + \mathfrak{e}_i^{(2)} m^{-p_\mathsf{z}-i} + \mathfrak{e}_i^{(3)} m^{1-p_\mathsf{y}-p_\mathsf{z}-i}\big],\\
\mathfrak{e}_i^{(1)} &= \sum_{a+k=i} \upsilon_a \frac{(p_\mathsf{y}+a)_k}{k!}\mathsf{m}_k^\mathsf{z}, \qquad \mathfrak{e}_i^{(2)} = \sum_{c+k=i} \upsilon'_c \frac{(p_\mathsf{z}+c)_k}{k!}\mathsf{m}_k^\mathsf{y},\\
\mathfrak{e}_i^{(3)} &= \sum_{a+c=i} \upsilon_a\upsilon'_c \frac{\Gamma(1-p_\mathsf{y}-a)\Gamma(1-p_\mathsf{z}-c)}{\Gamma(2-p_\mathsf{y}-p_\mathsf{z}-a-c)}.
\end{aligned} \tag{486}$$

The tail $\sum_{m\ge M_\mathrm{H}}\mathsf{x}_m\mathsf{q}_m$ is then a finite combination of three batches of Hurwitz values. For the multiplicity sequence at non-integer d the regularized moments are known in closed form. By (461), $\sum_j (d_{j+1}/4q)e^{-jw} = [(1+e^{-w})(1-e^{-w})^{1-d}-1]e^w/(4q)$, whose first term has only non-integer powers of w; hence

$$\mathsf{m}_k^\mathsf{y} = \frac{(-1)^{k+1}}{4q} \qquad (\mathsf{y}_j = d_{j+1}/(4q),\ d\notin\mathbb{Z}), \tag{487}$$

which the numerical moments reproduce. The other moments are computed with a head $N_\mathrm{H} \le M_\mathrm{H}/2$, so that the cancellation in $\sum_{n<N_\mathrm{H}} n^k\mathsf{z}_n$ is suppressed by M_H^{-k} in the tail. *Exceptional parameters:* for non-integer d the coefficients in (486) have removable poles when $r(d-b)/2 \in \mathbb{Z}$ (the $\log m$ resonance of two families). At such points `W` returns the mean over 8 points of a circle of radius about $10^{-(P+10)/8}$ around b, which is exact for analytic W_r up to the eighth power of the radius, with extra working digits. At $d = 1, 2, 3$ the multiplicities are polynomial in j (or δ_{j0}); then only convergent moments enter, the coefficient of the third family is $\upsilon_a\upsilon'_c(-p_\mathsf{y}-a)!/(1-p_\mathsf{z}-c)_{1-p_\mathsf{y}-a}$, the middle block is a finite sum of products of single moments, and no resonance occurs inside the strip. *Parameters and cost:* the defaults are $M_\mathrm{H} = \max(80, 1.6P')$, $K_\mathrm{H} = \max(30, 0.8P'+10)$ asymptotic terms and $N_\mathrm{H} = M_\mathrm{H}/2$, with $P' = P+15$ guard digits at P requested digits. One W_4 at 50 digits takes about 45 s on one Intel core of the symmetry cluster, and all 846 evaluations of the checks of the general-b code in Section 20.9 took 6 minutes on one 64-core AMD node. The accuracy of this method is established by convergence in $(M_\mathrm{H}, K_\mathrm{H}, N_\mathrm{H})$ and by the independent checks, not by interval bounds. *Independent spectral code:* `LRI_C5_general_d/foundation/spectral.py` evaluates (460) on the Mellin line, by tanh–sinh quadrature for the spins $\ell < 40$ and, for $\ell \ge 40$, by the Stirling expansion in $z_\ell = \frac{\ell}{2} + \mathrm{i}\nu$ with $\frac{1}{2\pi}\int_\mathbb{R} z_\ell^{-e}\bar z_\ell^{-e'}\,d\nu = \ell^{1-e-e'}\Gamma(e+e'-1)/(\Gamma(e)\Gamma(e'))$ and a Hurwitz-zeta spin tail. It shares no code with the finite-sample formula except the Hurwitz routine.

Beyond $b_{\max}$. For $d > 2$ one has $b_{\max} < b_{\mathrm{spec}}$; the ℓ-endpoint condition (ii) binds. The continuation meant here is the continuation in b of each block value at a fixed node t. Each block of (483) is a finite part, plus Hurwitz zeta values with exponents affine in b, plus convergent remainders, and the evaluation above computes exactly this continuation (zeta regularization of the tails). So the samples $\Xi_r(t_j; b)$ and the $c_{r,k}$ continue meromorphically in b. On $b_{\max} \le b < b_{\mathrm{spec}}$

their only pole is at $b_{\max}$: the next candidates $b_{\max}+2k/r$ exceed $b_{\rm spec}$, since $b_{\rm spec}-b_{\max}=(d-2)/r<2/r$. Both sides of (484) are meromorphic and agree on the strip, so the identity persists on $b_{\max}\le b<b_{\rm spec}$ with continued block values. The pole at $b_{\max}$ was derived in the review (`LRI_C5_general_d/review/foundation-rigor`); the derivation is an endpoint asymptotic sketch, not a written proof. The divergence at $b_{\max}$ comes from the ends $\ell\to\infty$ of the lattice at fixed n and at fixed $\ell+n$. There $d_\ell\sim 2\ell^{d-2}/\Gamma(d-1)$ by (466), $f(\ell+c)^r\sim\ell^{-rg}$, and $\sum_\ell \ell^{d-2-rg}$ has the pole $1/(rg-d+1)=-2/(r(b-b_{\max}))$. Collecting these ends gives

$$\begin{aligned}\operatorname{Res}_{b=b_{\max}}\Xi_r(t;b)&=-\frac{4}{r\,\Gamma(d-1)}\,D_b(t)^r\big[\Phi_{\max}(-t)+\Phi_{\max}(\sigma+t)\big],\\ \Phi_{\max}(u)&=\sum_{m\in\mathbb{Z}}\Big[\frac{\Gamma(m+u)}{\Gamma(m+u+g)}\Big]^r,\end{aligned}\tag{488}$$

at $\sigma=b_{\max}/2$ and $g=(d-b_{\max})/2$, where $rg=d-1>1$. The residue has three properties. (1) Its top mode is proportional to $\cos(\pi(rg+2-d)/2)$, which vanishes at $b_{\max}$; this follows from Steps 1, 3 and 5 of the proof of (471) applied to $\Phi_{\max}$, with $a_\pm(2\pi)$ finite because $rg>1$, and it is consistent with (473) holding up to the boundary. (2) Its lower modes are nonzero in every case computed, so the continued $c_{r,k}$ with $k\le r-2$ have a simple pole. (3) The functional annihilates it: $\Phi_{\max}(\sigma+t)$ is regular at $t=0$, and $\operatorname{Res}_{t=0}\Phi_{\max}(-t)=-\sum_{n\ge0}\operatorname{Res}_{z=-n}f(z)^r=0$, because f^r has poles only in $-\mathbb{N}_0$ and is $O(|z|^{-rg})$ on the circles $|z|=n_0+\frac12$, with $rg>1$. Hence W_r has no pole at $b_{\max}$, as the spectral representation requires. The same review checked this numerically at 20 digits for $(d,r)=(2.5,4)$, $(2.75,3)$, $(3.5,4)$ and $(3.5,3)$, with its scripts `bmax_residue.py`, `bmax_normalization.py` and `bmax_compare.py` (symmetry job 569252 for the first two): the top mode of (488) is $5\cdot10^{-21}$ to $2\cdot10^{-18}$ of the largest mode, and the lower modes are of order one; the functional applied to (488) vanishes to $2\cdot10^{-21}$–10^{-18} relative; at $d=2.5$, $r=4$ the predicted residues of $c_{4,0},\dots,c_{4,3}$, $(-9.109258,\,-11.819680,\,-2.719024,\,0)$, agree with the estimate $\frac{\delta}{2}[c_{4,k}(b_{\max}+\delta)-c_{4,k}(b_{\max}-\delta)]$ from the continued code to $O(\delta^2)$, at most $4\cdot10^{-5}$ at $\delta=10^{-3}$ and $4\cdot10^{-7}$ at $\delta=10^{-4}$. This also confirms that the zeta-regularized evaluation is the analytic continuation. The continued formula agrees with the spectral values at $(d,b)=(2.5,1.8)$ and $(2.5,1.76)$ for $r=4$ and $(2.5,1.55)$ for $r=3$ to 10^{-14} at 15 digits, and at 50 digits as in Section 20.9. For $d<1$ condition (iii) binds and coincides with the spectral bound. Nothing downstream uses the continuation: every use of these formulas in this note is at $b=\frac{d}{2}$, inside the strip.

20.2 Samples, modes and the two masters

By (473) and (480), W_4 needs three modes and W_3 two. In the variable $\tilde t=t+\frac{b}{4}$ the trigonometric polynomial is $\sum_k c_{r,k}\cos 2\pi k\tilde t$, even about $\tilde t=0$ and $\tilde t=\frac12$. We sample it at the pole node $t=0$ ($\tilde t=\frac{b}{4}$) and at these reflection centers.

Four-wheel. Three real samples fix the three modes:

$$\mathcal{X}_T(b)=N_b^4\Xi_4(0;b),\qquad \mathcal{X}_0(b)=N_b^4\Xi_4(-\tfrac{b}{4};b),\qquad \mathcal{X}_{1/2}(b)=N_b^4\Xi_4(\tfrac12-\tfrac{b}{4};b).\tag{489}$$

With $x=b/4$, $y=d/2-b/4$ (in the definitions of $\mathfrak{h}_u,\mathfrak{s}_u$ only; elsewhere in this section $t=x+\mathrm{i}y$, and x is the argument of $\Omega_r(t;x)$ in Section 20.6) they are five two-variable Horn functions at argument $(1,1)$: $\mathcal{X}_T=(\pi s_b)^4\mathfrak{h}_x$, $\mathcal{X}_0=\kappa_0^4(\mathfrak{s}_0+2\xi_0^4\mathfrak{h}_1)$, $\mathcal{X}_{1/2}=\kappa_h^4(\mathfrak{s}_{1/2}+2\xi_h^4\mathfrak{h}_{1/2})$, where

$$\mathfrak{h}_u=\sum_{\ell,n\ge0}d_\ell\Big[\frac{(u+x)_{\ell+n}}{(u+y)_{\ell+n}}\Big]^4\Big[\frac{(u+1-y)_n}{(u+1-x)_n}\Big]^4,\qquad \mathfrak{s}_u=\sum_{j,k\ge0}d_{j+k+2u}\Big[\frac{(u+x)_j(u+x)_k}{(u+y)_j(u+y)_k}\Big]^4,$$

$\kappa_0=N_b\sin^2(\pi x)\Gamma(x)^2/\Gamma(y)^2$, $\kappa_h=N_b\cos^2(\pi x)\Gamma(\frac12+x)^2/\Gamma(\frac12+y)^2$, $\xi_0=\frac{x(1-y)}{y(1-x)}$, $\xi_h=\frac{1-2y}{1-2x}$. The index u of $\mathfrak{h}_u$ and $\mathfrak{s}_u$ is unrelated to the weight $u(b)$ of (496). In particular $\mathfrak{h}_x(d;b)=\sum d_\ell T_{\ell n}(b)^4$

with

$$T_{\ell n}(b)=\frac{(\sigma)_{\ell+n}(1-g)_n}{(2q)_{\ell+n}\,n!}, \tag{490}$$

which at $b=d/2$ is the $T_{\ell n}$ of (251).

Section 20.3 proves these sample formulas for every r, (510)–(512).

By (473) the samples are $N_b^{-4}\mathcal{X}_0=c_{4,0}+c_{4,1}+c_{4,2}$, $N_b^{-4}\mathcal{X}_{1/2}=c_{4,0}-c_{4,1}+c_{4,2}$ and $N_b^{-4}\mathcal{X}_T=c_{4,0}+\varsigma_b c_{4,1}+c_{4,2}\cos\pi b$. This system has determinant $4\sin^2\pi\sigma\neq 0$. Solving it and inserting into (480) gives, for every b in the strip,

$$W_4(d;b)=\frac{2}{\pi s_b^7}\Big[(1+\varsigma_b)^2(1+2\varsigma_b)\mathcal{X}_0-(1-\varsigma_b)^2(1-2\varsigma_b)\mathcal{X}_{1/2}-2\varsigma_b\mathcal{X}_T\Big]. \tag{491}$$

At $b=d/2$, and differentiating once (the product rule done once; all samples at $b=d/2$),

$$P_4(d)=\frac{2}{\pi S^7}\Big[(1+C)^2(1+2C)\mathcal{X}_0-(1-C)^2(1-2C)\mathcal{X}_{1/2}-2C\mathcal{X}_T\Big], \tag{492}$$

$$\begin{aligned}\mathcal{L}_d=&-\frac{1}{S^8}\Big[(1+C)^2(4+9C+8C^2)\mathcal{X}_0+(1-C)^2(4-9C+8C^2)\mathcal{X}_{1/2}-2(1+6C^2)\mathcal{X}_T\Big]\\&+\frac{2}{\pi S^7}\,\mathfrak{D}_L,\end{aligned} \tag{493}$$

$$\mathfrak{D}_L=(1+C)^2(1+2C)\,\mathcal{X}_0'-(1-C)^2(1-2C)\,\mathcal{X}_{1/2}'-2C\,\mathcal{X}_T'. \tag{494}$$

Written out in the five Horn values and their first b-derivatives, (493) is (519), derived in Section 20.3. The derivative data enter only through the single frozen-weight combination $\mathfrak{D}_L$, the b-derivative of the bracket of (492) with its coefficients held at $b=d/2$. The first line of (493) is the derivative of the coefficients of (491).

Three-wheel. Two real samples suffice (proofs in Section 20.4.2): the pole node $t=0$ and the self-mirror node $t_b=\frac12-\frac{b}{4}$, where $1-\sigma-t_b=t_b$ and $\tilde t=\frac12$. At $t=0$ only the leading coefficients of the triple poles survive, and at t_b the kernel is sampled on the midpoint lattice, because $a_\ell+n+t_b=n+\frac{\ell+1}{2}$. With

$$\mathfrak{t}_3(b)=\sum_{\ell,n\geq 0}d_\ell T_{\ell n}(b)^3,\qquad \widehat{M}_3(b)=\sum_{\ell\geq 0}d_\ell\sum_{s\in\frac{\ell+1}{2}+\mathbb{Z}}[N_b g_\ell(s;b)]^3, \tag{495}$$

one has $N_b^3\Xi_3(0;b)=-(\pi s_b)^3\mathfrak{t}_3$ and $N_b^3\Xi_3(t_b;b)=(\frac{1+\varsigma_b}{2})^3\widehat{M}_3$. Note that $\widehat{M}_3$ is the bare lattice sum: the factor $D_b(t_b)^3=(1+\varsigma_b)^3/8$ carried by the sample is written separately. The two samples are $c_{3,0}+\varsigma_b c_{3,1}$ and $c_{3,0}-c_{3,1}$, and (480) gives

$$W_3(d;b)=u(b)\mathfrak{t}_3(b)+v(b)\widehat{M}_3(b),\qquad u(b)=\frac{2\pi^2(1+2\varsigma_b)}{s_b^2},\qquad v(b)=\frac{\varsigma_b(1+\varsigma_b)^2}{2\pi s_b^3}. \tag{496}$$

$\mathfrak{t}_3$ is a ${}_7F_6(1)$ plus one two-variable Horn value, $\widehat{M}_3$ a ${}_7F_6(1)$ plus two. In the block decomposition of Section 20.1.3, $\widehat{M}_3$ is the sum of a right-part sum (the ${}_7F_6(1)$ and one Horn value) and a middle-part sum (one Horn value); the count of Section 20.7 treats these two as separate functions. In terms of the normalized zero mode $\hat c_0=N_b^3c_{3,0}=\frac{\varsigma_b(1+\varsigma_b)^2}{8}\widehat{M}_3-\frac{\pi^3 s_b^3}{1+\varsigma_b}\mathfrak{t}_3$, (496) reads $W_3=\frac{6\pi^2}{s_b^2}\mathfrak{t}_3+\frac{4}{\pi s_b^3}\hat c_0$. Then, at $b=d/2$,

$$3Q_d=\frac{6\pi^2}{S^2}\mathfrak{t}_3+\frac{4}{\pi S^3}\hat c_0, \tag{497}$$

$$Z_d=\frac{2\pi^2}{3S^2}\mathfrak{t}_3''-\frac{\pi^2\cos(\pi d/2)}{6S^2}Q_d+\frac19\Big[\frac{\pi(1+C^2)}{S^5}\hat c_0-\frac{4C}{S^4}\hat c_0'+\frac{4}{\pi S^3}\hat c_0''\Big]. \tag{498}$$

Here $\partial_b W_3(d;d/2)=0$ has been used to remove $\mathfrak{t}_3'$. Its proof is Pattern T of Section 16.3, (239), together with (244): on the balanced K_5 subspace there is no linear form invariant under the permutation group S_5 (not the lattice sum S_r of (462) at $r=5$), and the balanced three-wheel path is a balanced variation. ((246) records the vanishing only as a check.) The formula (498), and the Z_d part of (553), depend on this input; the full jet formula $9Z_d=(u\mathfrak{t}_3+v\widehat{M}_3)''$ and (497) do not. At $d=1,2$ the input also follows from the closed forms of Section 20.6. The second derivatives enter only through $\frac{1}{9}(u\mathfrak{t}_3''+v\widehat{M}_3'')$ with $u=\frac{2\pi^2(1+2C)}{S^2}$, $v=\frac{C(1+C)^2}{2\pi S^3}$, and the first derivatives only through $\hat{c}_0'$.

Analyticity and termwise derivatives. The five Horn functions, $\mathfrak{t}_3$ and $\widehat{M}_3$ are analytic on the strip, and their b-derivatives are the termwise derivatives. The bounds of Section 20.1 hold uniformly for b in compact subsets of the strip, with α above the threshold $\max(\frac{d}{2},d-1,\frac{1}{2})$ by a fixed margin; the polygamma weights produced by differentiation grow only logarithmically and cost part of that margin (for the five Horn functions, the proof of (518) in Section 20.3; for the three-wheel, Section 20.4.1).

20.3 Proofs of the four-wheel samples and the explicit formula for $\mathcal{L}_d$

The code `LRI_C5_general_d/L/L_general.py` evaluates the formulas of this subsection. It proves the sample formulas stated after (489) for every $r\ge 2$, writes (491) and (493) out in the five Horn values and their first b-derivatives, and collects the checks of that form (Section 20.3.1).

Setting and status. Throughout, $r\ge 2$, $0<d<4$, $0<\operatorname{Re}b<b_{\max}(r,d)$ of (463) and $\sigma\notin\mathbb{Z}$, the hypotheses of Section 20.1.1. As in Section 20.2,

$$x=\frac{b}{4},\qquad y=\frac{d}{2}-\frac{b}{4},\qquad \sigma=2x,\qquad g=y-x,\qquad x+y=2q,\qquad \tilde{t}=t+x, \tag{499}$$

and in this subsection x,y always have this meaning. For $d\neq 2$ the multiplicities (240) have the very-well-poised form

$$d_\ell=\frac{(4q-2)_\ell\,(2q)_\ell}{(2q-1)_\ell\,\ell!}, \tag{500}$$

since $\Gamma(d-1)=(d-2)\Gamma(d-2)$, $(2\ell+d-2)/(d-2)=(2q)_\ell/(2q-1)_\ell$ and $\Gamma(\ell+d-2)/(\Gamma(d-2)\,\ell!)=(4q-2)_\ell/\ell!$; here $(2q-1)_\ell\neq 0$ because $2q-1\in(-1,1)$ and $2q\neq 1$. The form (301) has no $0/0$ at $d=2$, where $d_\ell=2-\delta_{\ell 0}$. Status: the lattice form (504)–(505), the samples (510)–(512) and the derivative series (518) are proved here, from the estimates of Section 20.1 and the finite Fourier theorem of Section 20.1.1. The formulas (514), (515), (517) and (519) are derived: exact algebra on these statements, on the top-mode theorem (473) and on the residue functional (480), checked symbolically. All numbers in Section 20.3.1 are numerical. A separate review with its own codes (`LRI_C5_general_d/review/L-rigor/`) rederived every step of the package's own derivation by hand and symbolically and found no wrong formula (for the steps added in this note, see Section 20.10); its corrections (the continuity step at $t=0$ in the proof of (510), the restriction of the three-dimensional moment statement to $\mathfrak{s}_0,\mathfrak{s}_{1/2}$, and the list of digamma arguments) are incorporated below.

Lattice sums in the variable $\tilde{t}$. Let

$$\varphi(z)=\frac{\Gamma(z+x)}{\Gamma(z+y)},\qquad \bar{\varphi}(z)=\frac{\Gamma(z+1-y)}{\Gamma(z+1-x)}, \tag{501}$$

$$\begin{aligned}\mathcal{R}_+(u)&=\sum_{\ell,n\ge 0} d_\ell\,[\varphi(\ell+n+u)\,\bar{\varphi}(n+u)]^r,\\ \mathcal{R}_{\mathrm{m}}(\tilde{t})=\sum_{j,k\ge 0} d_{j+k+1}\,[\varphi(k+\tilde{t})\,\varphi(j+1-\tilde{t})]^r,&\qquad \mathcal{R}_{\mathrm{s}}(\tilde{t})=\sum_{j,k\ge 0} d_{j+k}\,[\varphi(k+\tilde{t})\,\varphi(j-\tilde{t})]^r,\end{aligned} \tag{502}$$

$$P_+(\tilde t)=\big[\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)\big]^r,\qquad P_-(\tilde t)=\big[\sin\pi(\tilde t-x)\sin\pi(\tilde t+y)\big]^r,$$
$$P_{\rm m}(\tilde t)=\big[\sin\pi(\tilde t-x)\sin\pi(\tilde t+x)\big]^r=D_b(t)^r. \tag{503}$$

For $t=\tilde t-x$ off the pole classes $t\equiv 0$ and $t\equiv-\sigma$,

$$\Xi_r(t;b)=P_+(\tilde t)\,\mathcal R_+(\tilde t)+P_-(\tilde t)\,\mathcal R_+(1-\tilde t)+P_{\rm m}(\tilde t)\,\mathcal R_{\rm m}(\tilde t) \tag{504}$$
$$=P_+(\tilde t)\,\mathcal R_+(1+\tilde t)+P_-(\tilde t)\,\mathcal R_+(1-\tilde t)+P_{\rm m}(\tilde t)\,\mathcal R_{\rm s}(\tilde t), \tag{505}$$

with absolutely convergent series. The second form is manifestly even in $\tilde t$: $P_+(-\tilde t)=P_-(\tilde t)$, $P_{\rm m}$ is even, and $\mathcal R_{\rm s}$ is even (exchange j and k).

Proof. Split the lattice of (462) into the right part $n\ge 0$, the middle part $\ell=j+k+1$, $n=-j-1$ ($j,k\ge 0$) and the left part $n\le-\ell-1$, as in Section 20.1. By the estimates there and Section 20.1.1(a) the three parts converge absolutely off the pole classes, so they may be summed separately and rearranged.

Right part. By the first line of (464), $g_\ell(a_\ell{+}n{+}t)=f(\ell{+}\sigma{+}n{+}t)\,f(-n{-}t)$ with $f(z)=\Gamma(z)/\Gamma(z{+}g)$. Since $\sigma+t=\tilde t+x$ and $x+g=y$, $f(\ell+\sigma+n+t)=\varphi(\ell+n+\tilde t)$. Gamma reflection in the form $\Gamma(z-n)\Gamma(1-z+n)=(-1)^n\pi/\sin\pi z$ [26], used for $z=x-\tilde t$ and $z=y-\tilde t$, gives

$$f(-n-t)=\frac{\Gamma(x-\tilde t-n)}{\Gamma(y-\tilde t-n)}=\frac{\sin\pi(y-\tilde t)}{\sin\pi(x-\tilde t)}\,\bar\varphi(n+\tilde t).$$

Since $D_b(t)=\sin\pi(\tilde t-x)\sin\pi(\tilde t+x)$, the prefactor is $D_b(t)\sin\pi(y-\tilde t)/\sin\pi(x-\tilde t)=\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)$, and the right part is $P_+(\tilde t)\mathcal R_+(\tilde t)$.

Left part. By the last line of (464), $n\mapsto n'=-\ell-n-1\ge 0$ maps the left part bijectively onto the right part at the mirror point $t'=1-\sigma-t$, that is at $\tilde t'=1-\tilde t$. Since $D_b(1-\sigma-t)=\sin\pi(\sigma+t)\sin\pi t=D_b(t)$, the left part is $P_+(1-\tilde t)\mathcal R_+(1-\tilde t)$, and $P_+(1-\tilde t)=P_-(\tilde t)$.

Middle part. In the second line of (464), $k{+}\sigma{+}t=k{+}\tilde t{+}x$, $k{+}2q{+}t=k{+}\tilde t{+}y$, $j{+}1{-}t=j{+}1{-}\tilde t{+}x$ and $j+1+g-t=j+1-\tilde t+y$, so the summand is $[\varphi(k+\tilde t)\varphi(j+1-\tilde t)]^r$ with multiplicity d_{j+k+1}, and the prefactor is $D_b(t)^r=P_{\rm m}(\tilde t)$. This proves (504).

Second form. $\mathcal R_+(\tilde t)-\mathcal R_+(1{+}\tilde t)=\sum_\ell d_\ell[\varphi(\ell{+}\tilde t)\,\bar\varphi(\tilde t)]^r$ is the column $n=0$. From $\sin\pi(\tilde t-y)\,\Gamma(1-y+\tilde t)=-\pi/\Gamma(y-\tilde t)$ and $1/\Gamma(1-x+\tilde t)=\sin\pi(x-\tilde t)\,\Gamma(x-\tilde t)/\pi$,

$$\sin\pi(\tilde t+x)\sin\pi(\tilde t-y)\,\bar\varphi(\tilde t)=\sin\pi(\tilde t+x)\sin\pi(\tilde t-x)\,\varphi(-\tilde t), \tag{506}$$

so $P_+(\tilde t)[\mathcal R_+(\tilde t)-\mathcal R_+(1+\tilde t)]=P_{\rm m}(\tilde t)\sum_\ell d_\ell[\varphi(\ell+\tilde t)\varphi(-\tilde t)]^r$. This is the row $j=0$ of $\mathcal R_{\rm s}(\tilde t)$, and the rows $j\ge 1$ of $\mathcal R_{\rm s}(\tilde t)$ are $\mathcal R_{\rm m}(\tilde t)$ after $j\to j+1$. All rearrangements are of absolutely convergent series. □

The three samples. Normalize the sums by their first terms: for $u\in\{x,\frac12,1\}$ and $u\in\{0,\frac12\}$ respectively,

$$\mathfrak h_u^{(r)}=\sum_{\ell,n\ge0}d_\ell\Big[\frac{(u+x)_{\ell+n}}{(u+y)_{\ell+n}}\Big]^r\Big[\frac{(u+1-y)_n}{(u+1-x)_n}\Big]^r,\qquad \mathfrak s_u^{(r)}=\sum_{j,k\ge0}d_{j+k+2u}\Big[\frac{(u+x)_j\,(u+x)_k}{(u+y)_j\,(u+y)_k}\Big]^r, \tag{507}$$

so that $\mathcal R_+(u)=[\varphi(u)\bar\varphi(u)]^r\mathfrak h_u^{(r)}$, $\mathcal R_{\rm s}(0)=\varphi(0)^{2r}\mathfrak s_0^{(r)}$ and $\mathcal R_{\rm m}(\frac12)=\varphi(\frac12)^{2r}\mathfrak s_{1/2}^{(r)}$. For $r=4$ these are the $\mathfrak h_u$, $\mathfrak s_u$ of Section 20.2, and the superscript is dropped; for $r=3$, $\mathfrak h_x^{(3)}$ is $\mathfrak t_3$ of (495). At $u=x$ one has $u+x=\sigma$, $u+y=2q$, $u+1-y=1-g$ and $u+1-x=1$, so $\mathfrak h_x^{(r)}=\sum_{\ell,n}d_\ell T_{\ell n}(b)^r$ with $T_{\ell n}(b)$ of (490). In the notation (320), with $\boldsymbol e=(1,0)$, $\boldsymbol f=(0,1)$, $\boldsymbol c=\boldsymbol e+\boldsymbol f$, the first summation

index ℓ (or j), the second n (or k), and argument $(1,1)$,

$$\begin{aligned}
\mathfrak{h}_u^{(r)} &= \mathfrak{H}\big[\{(u+x,\boldsymbol{c})^{\times r},(4q-2,\boldsymbol{e}),(2q,\boldsymbol{e}),(1,\boldsymbol{f}),(u+1-y,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(u+y,\boldsymbol{c})^{\times r},(2q-1,\boldsymbol{e}),(u+1-x,\boldsymbol{f})^{\times r}\};(1,1)\big],\\
\mathfrak{s}_0^{(r)} &= \mathfrak{H}\big[\{(4q-2,\boldsymbol{c}),(2q,\boldsymbol{c}),(1,\boldsymbol{e}),(1,\boldsymbol{f}),(x,\boldsymbol{e})^{\times r},(x,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(2q-1,\boldsymbol{c}),(1,\boldsymbol{c}),(y,\boldsymbol{e})^{\times r},(y,\boldsymbol{f})^{\times r}\};(1,1)\big],\\
\mathfrak{s}_{1/2}^{(r)} &= 4q\,\mathfrak{H}\big[\{(2q+1,\boldsymbol{c}),(4q-1,\boldsymbol{c}),(1,\boldsymbol{e}),(1,\boldsymbol{f}),(\tfrac12+x,\boldsymbol{e})^{\times r},(\tfrac12+x,\boldsymbol{f})^{\times r}\};\\
&\qquad \{(2q,\boldsymbol{c}),(2,\boldsymbol{c}),(\tfrac12+y,\boldsymbol{e})^{\times r},(\tfrac12+y,\boldsymbol{f})^{\times r}\};(1,1)\big].
\end{aligned}\tag{508}$$

The first two use (500) and hold for $d\neq 2$; the third uses (301) and holds for every d. The upper pairs $(1,\boldsymbol{e})$, $(1,\boldsymbol{f})$ cancel the factorials of (320) where the summand has none. At $b=d/2$ ($x=\frac{q}{2}$, $y=\frac{3q}{2}$) the parameters are

$$\begin{aligned}
\mathfrak{h}_x &= \sum_{\ell,n} d_\ell \Big[\tfrac{(q)_{\ell+n}}{(2q)_{\ell+n}}\Big]^4\Big[\tfrac{(1-q)_n}{n!}\Big]^4 = \sum_{\ell,n} d_\ell T_{\ell n}^4, \quad \mathfrak{h}_1 = \sum_{\ell,n} d_\ell \Big[\tfrac{(1+\frac{q}{2})_{\ell+n}}{(1+\frac{3q}{2})_{\ell+n}}\Big]^4\Big[\tfrac{(2-\frac{3q}{2})_n}{(2-\frac{q}{2})_n}\Big]^4,\\
\mathfrak{h}_{1/2} &= \sum_{\ell,n} d_\ell \Big[\tfrac{(\frac{1+q}{2})_{\ell+n}}{(\frac{1+3q}{2})_{\ell+n}}\Big]^4\Big[\tfrac{(\frac{3-3q}{2})_n}{(\frac{3-q}{2})_n}\Big]^4, \qquad \mathfrak{s}_0 = \sum_{j,k} d_{j+k} \Big[\tfrac{(\frac{q}{2})_j(\frac{q}{2})_k}{(\frac{3q}{2})_j(\frac{3q}{2})_k}\Big]^4,\\
\mathfrak{s}_{1/2} &= \sum_{j,k} d_{j+k+1} \Big[\tfrac{(\frac{1+q}{2})_j(\frac{1+q}{2})_k}{(\frac{1+3q}{2})_j(\frac{1+3q}{2})_k}\Big]^4,
\end{aligned}\tag{509}$$

with $T_{\ell n}$ of (251). Thus $\mathfrak{h}_x$ is the unweighted version of the residue series (255). With $\kappa_0,\kappa_h,\xi_0,\xi_h$ of Section 20.2, that is $\kappa_0=N_b\sin^2(\pi x)\varphi(0)^2$, $\kappa_h=N_b\cos^2(\pi x)\varphi(\frac12)^2$, $\xi_0=x(1-y)/(y(1-x))$ and $\xi_h=(1-2y)/(1-2x)$, the samples are

$$N_b^r\,\Xi_r(0;b) = (-1)^r(\pi s_b)^r\,\mathfrak{h}_x^{(r)},\tag{510}$$

$$N_b^r\,\Xi_r(-\tfrac{b}{4};b) = (-1)^r\kappa_0^r[\mathfrak{s}_0^{(r)}+2\xi_0^r\mathfrak{h}_1^{(r)}],\tag{511}$$

$$N_b^r\,\Xi_r(\tfrac12-\tfrac{b}{4};b) = \kappa_h^r[\mathfrak{s}_{1/2}^{(r)}+2\xi_h^r\mathfrak{h}_{1/2}^{(r)}].\tag{512}$$

For $r=4$ these are the expressions for $\mathcal{X}_T,\mathcal{X}_0,\mathcal{X}_{1/2}$ given after (489). For $r=3$, (510) is $N_b^3\Xi_3(0;b)=-(\pi s_b)^3\mathfrak{t}_3$, which Section 20.4.2 also proves directly, (521), and comparison of (512) with the midpoint sample (522) gives $(\frac{1+\varsigma_b}{2})^3\widehat{M}_3=\kappa_h^3[\mathfrak{s}_{1/2}^{(3)}+2\xi_h^3\mathfrak{h}_{1/2}^{(3)}]$: the right and left parts, equal at $\tilde t=\frac12$, give $2\kappa_h^3\xi_h^3\mathfrak{h}_{1/2}^{(3)}$ (whose row $\ell=0$ is a ${}_7F_6(1)$), and the middle part gives $\kappa_h^3\mathfrak{s}_{1/2}^{(3)}$.

Proof. (i) The node $t=0$, $\tilde t=x$. It lies in the pole class $t\equiv 0$, so (504) is first extended to it. Assume $g\neq 1$. Written in t, the terms of $\mathcal{R}_+(\tilde t)$ are $[\Gamma(\ell+n+\sigma+t)\Gamma(n+1-g+t)/(\Gamma(\ell+n+2q+t)\Gamma(n+1+t))]^r$, the Gamma ratios of the first line of (464) without their prefactor $-\sin\pi(g-t)/\sin\pi t$, which has moved into P_+. The terms of $\mathcal{R}_+(1-\tilde t)$ are the same ratios at $1-\sigma-t$, $[\Gamma(\ell+n+1-t)\Gamma(n+2-2q-t)/(\Gamma(\ell+n+1+2q-\sigma-t)\Gamma(n+2-\sigma-t))]^r$, and the terms of $\mathcal{R}_{\mathrm m}(\tilde t)$ are the ratios of the second line of (464). All their poles lie in $-\sigma-\mathbb{N}_0$, $g-1-\mathbb{N}_0$, $\mathbb{N}=\{1,2,\dots\}$ and $2-2q+\mathbb{N}_0$. None of these sets contains 0, since $\operatorname{Re}\sigma>0$, $q<1$, $g\neq 1$ and $\operatorname{Re} g<2$. Choose a closed disc $|t|\le\delta$ that avoids these sets and the class $-\sigma+\mathbb{Z}$. On it the shifts range over a compact set on which no Gamma argument meets a pole, so the Gamma-ratio bound of Section 20.1 holds uniformly, and the terms, multiplicities included, are bounded by $C(1+\ell)^{d-2}(1+\ell+n)^{-\alpha}(1+n)^{-\alpha}$ (the first two sums) and $C(1+j+k)^{d-2}(1+k)^{-\alpha}(1+j)^{-\alpha}$ (the middle sum), $\alpha=r\operatorname{Re} g$. By the lattice-sum criterion of Section 20.1, which holds since $\operatorname{Re} b<b_{\max}$, the three series converge uniformly on the disc. Hence both sides of (504) are analytic on the disc (Ξ_r is entire by Section 20.1.1(b), the weights are entire); they agree on the punctured disc, so they agree at $t=0$. At $\tilde t=x$ the weights P_- and $P_{\mathrm m}$ vanish, so only the right sum survives, with $P_+(x)=[\sin\pi\sigma\,\sin\pi(x-y)]^r=(-1)^r[\sin\pi\sigma\sin\pi g]^r$. With $\varphi(x)=\Gamma(\sigma)/\Gamma(2q)$, $\bar\varphi(x)=\Gamma(1-g)$ and $\Gamma(g)\Gamma(1-g)\sin\pi g=\pi$,

$$N_b\sin\pi\sigma\sin\pi g\;\varphi(x)\bar\varphi(x)=\pi\sin\pi\sigma=\pi s_b,$$

which together with $\mathcal{R}_+(x) = [\varphi(x)\bar\varphi(x)]^r \mathfrak{h}_x^{(r)}$ gives (510). At the isolated point $g = 1$, that is $b = d-2$ (in the strip only for $d > 2$), $\bar\varphi(x)$ has a pole. There (510) follows by continuity in b, since both sides are analytic in b: the left side by Section 20.1.1(d), the right side because the terms of $\mathfrak{h}_x^{(r)} = \sum d_\ell T_{\ell n}(b)^r$ are polynomials in b and obey the same majorant uniformly near $b = d-2$. The masters use only b near $\frac{d}{2}$, where $g = q < 1$.

(ii) The node $\tilde t = 0$, $t = -\frac{b}{4}$. It is off the pole classes, since $t \equiv 0$ or $t \equiv -\sigma$ would need $\frac{b}{4} \in \mathbb{Z}$. In (505), $P_+(0) = P_-(0) = (-1)^r[\sin\pi x \sin\pi y]^r$ and $P_{\rm m}(0) = (-1)^r \sin^{2r}\pi x$, so $\Xi_r = (-1)^r\{2[\sin\pi x \sin\pi y\, \varphi(1)\bar\varphi(1)]^r \mathfrak{h}_1^{(r)} + [\sin^2(\pi x)\varphi(0)^2]^r \mathfrak{s}_0^{(r)}\}$.

(iii) The node $\tilde t = \frac12$, $t = \frac12 - \frac{b}{4}$. It is off the pole classes, since either class would need $b \equiv 2 \pmod 4$, that is $\sigma \in \mathbb{Z}$. In (504), $P_+(\frac12) = P_-(\frac12) = [\cos\pi x \cos\pi y]^r$ and $P_{\rm m}(\frac12) = \cos^{2r}\pi x$, so $\Xi_r = 2[\cos\pi x \cos\pi y\, \varphi(\frac12)\bar\varphi(\frac12)]^r \mathfrak{h}_{1/2}^{(r)} + [\cos^2(\pi x)\varphi(\frac12)^2]^r \mathfrak{s}_{1/2}^{(r)}$.

Finally, $\varphi(1) = \frac{x}{y}\varphi(0)$ and $\bar\varphi(1) = \frac{1-y}{1-x}\Gamma(1-y)/\Gamma(1-x)$ with $\Gamma(1-y)/\Gamma(1-x) = (\sin\pi x/\sin\pi y)\varphi(0)$ by reflection, and $\bar\varphi(\frac12) = \Gamma(\frac32 - y)/\Gamma(\frac32 - x) = \frac{1-2y}{1-2x}(\cos\pi x/\cos\pi y)\varphi(\frac12)$ by $\Gamma(\frac12 + z)\Gamma(\frac12 - z) = \pi/\cos\pi z$. Hence

$$\sin\pi x \sin\pi y\ \varphi(1)\bar\varphi(1) = \xi_0 \sin^2(\pi x)\,\varphi(0)^2, \qquad \cos\pi x \cos\pi y\ \varphi(\tfrac12)\bar\varphi(\tfrac12) = \xi_h \cos^2(\pi x)\,\varphi(\tfrac12)^2, \tag{513}$$

and multiplying (ii) and (iii) by N_b^r gives (511) and (512). □

The three-node formula in Horn values. By (473) the three samples are, in units of N_b^4, $c_{4,0} + c_{4,1} + c_{4,2}$ at $\tilde t = 0$, $c_{4,0} - c_{4,1} + c_{4,2}$ at $\tilde t = \frac12$ and $c_{4,0} + \varsigma_b c_{4,1} + \cos(\pi b)\, c_{4,2}$ at $t = 0$. The determinant is $4\sin^2\pi\sigma = 4s_b^2 \neq 0$, and

$$N_b^4 c_{4,1} = \frac{\mathcal{X}_0 - \mathcal{X}_{1/2}}{2}, \qquad N_b^4 c_{4,0} = \frac{\mathcal{X}_0 + \mathcal{X}_{1/2}}{2} - N_b^4 c_{4,2},$$
$$N_b^4 c_{4,2} = \frac{1}{2s_b^2}\Big[\frac{\mathcal{X}_0 + \mathcal{X}_{1/2}}{2} + \varsigma_b N_b^4 c_{4,1} - \mathcal{X}_T\Big].$$

Inserting these into (480) gives (491), and with (510)–(512) at $r = 4$, for every b in the strip,

$$\boxed{\begin{aligned} W_4(d;b) = &-\frac{4\pi^3\varsigma_b}{s_b^3}\,\mathfrak{h}_x + \frac{2(1+\varsigma_b)^2(1+2\varsigma_b)\kappa_0^4}{\pi s_b^7}\big[\mathfrak{s}_0 + 2\xi_0^4\mathfrak{h}_1\big] \\ &- \frac{2(1-\varsigma_b)^2(1-2\varsigma_b)\kappa_h^4}{\pi s_b^7}\big[\mathfrak{s}_{1/2} + 2\xi_h^4\mathfrak{h}_{1/2}\big]. \end{aligned}} \tag{514}$$

For $r = 3$ and $0 < \operatorname{Re} b < b_{\max}(3,d)$ the solve of the two samples at $t = 0$ and $\tilde t = \frac12$ is the proof of (496) in Section 20.4.2; it gives $W_3(d;b) = -\frac{2N_b^3}{\pi s_b^5}[(1+2\varsigma_b)\Xi_3(0;b) - 2\varsigma_b(1-\varsigma_b)\Xi_3(\frac12 - \frac{b}{4};b)]$, and with (510) and (512) at $r = 3$ this is

$$W_3(d;b) = \frac{2\pi^2(1+2\varsigma_b)}{s_b^2}\,\mathfrak{h}_x^{(3)} + \frac{4\varsigma_b(1-\varsigma_b)\kappa_h^3}{\pi s_b^5}\big[\mathfrak{s}_{1/2}^{(3)} + 2\xi_h^3\mathfrak{h}_{1/2}^{(3)}\big], \tag{515}$$

which is (496) with its midpoint sample written in Horn values. Neither formula holds without the top-mode theorem: a mode $c_{4,3}$ would enter W_4 with the nonzero weight $B_{4,3} = -20\pi^3/s_b^3$ of Section 20.1.3 and would not be fixed by three samples. The agreement of (514) with independent values at $b \neq d/2$ (Section 20.3.1) is therefore also a numerical test of (473). The exact solve and the insertion into (480) are checked symbolically by `LRI_C5_general_d/L/sym_functional.py`, which also rederives (480) from (318) with q replaced by σ.

The four-wheel period. At $b = d/2$ one has $x = \frac{q}{2}$, $y = \frac{3q}{2}$, $\sigma = g = q$, $N_{d/2} = \Gamma(2q)$, $s_b = S$, $\varsigma_b = C$, so $\kappa_0 = \Gamma(2q)\sin^2(\frac{\pi q}{2})\,[\Gamma(\frac{q}{2})/\Gamma(\frac{3q}{2})]^2$, $\kappa_h = \Gamma(2q)\cos^2(\frac{\pi q}{2})\,[\Gamma(\frac{1+q}{2})/\Gamma(\frac{1+3q}{2})]^2$, $\xi_0 = \frac{2-3q}{3(2-q)}$ and $\xi_h = \frac{1-3q}{1-q}$. Put, for b near $d/2$,

$$\begin{aligned} \mathsf{k}_0(b) &= \frac{2(1+\varsigma_b)^2\kappa_0^4}{\pi s_b^7}, \qquad \mathsf{k}_h(b) = \frac{2(1-\varsigma_b)^2\kappa_h^4}{\pi s_b^7}, \\ \mathfrak{S}_0 = \mathfrak{s}_0 + 2\xi_0^4\mathfrak{h}_1 &= \frac{\mathcal{X}_0}{\kappa_0^4}, \qquad \mathfrak{S}_h = \mathfrak{s}_{1/2} + 2\xi_h^4\mathfrak{h}_{1/2} = \frac{\mathcal{X}_{1/2}}{\kappa_h^4}, \end{aligned} \tag{516}$$

so that (514) reads $W_4 = -\frac{4\pi^3\varsigma_b}{s_b^3}\mathfrak{h}_x + (1+2\varsigma_b)\mathsf{k}_0(b)\mathfrak{S}_0 + (2\varsigma_b - 1)\mathsf{k}_h(b)\mathfrak{S}_h$. Write $\mathsf{k}_0, \mathsf{k}_h$ for the values at $b = d/2$; with $1 + C = 2\cos^2\frac{\pi q}{2}$, $1 - C = 2\sin^2\frac{\pi q}{2}$ and $S = 2\sin\frac{\pi q}{2}\cos\frac{\pi q}{2}$,

$$\mathsf{k}_0 = \frac{\Gamma(2q)^4}{16\pi}\Big[\frac{\Gamma(\frac{q}{2})}{\Gamma(\frac{3q}{2})}\Big]^8 \frac{\tan\frac{\pi q}{2}}{\cos^2\frac{\pi q}{2}}, \qquad \mathsf{k}_h = \frac{\Gamma(2q)^4}{16\pi}\Big[\frac{\Gamma(\frac{1+q}{2})}{\Gamma(\frac{1+3q}{2})}\Big]^8 \frac{\cot\frac{\pi q}{2}}{\sin^2\frac{\pi q}{2}}.$$

Then, for $\frac{1}{2} < d < 4$,

$$\boxed{P_4(d) = -\frac{4\pi^3 C}{S^3}\,\mathfrak{h}_x + (1+2C)\,\mathsf{k}_0\,\mathfrak{S}_0 + (2C-1)\,\mathsf{k}_h\,\mathfrak{S}_h,} \tag{517}$$

which is (492) with the samples written in Horn values. The symbols $\mathsf{k}_0, \mathsf{k}_h, \mathfrak{S}_0, \mathfrak{S}_h$ and $\mathsf{v}_0, \mathsf{v}_h$ below are local to this subsection.

Derivative series. A prime denotes the total derivative d/db at $b = d/2$; in $\mathfrak{h}_x$ the argument $u = x$ moves with b, while $u = \frac{1}{2}, 1$ in $\mathfrak{h}_u$ and $u = 0, \frac{1}{2}$ in $\mathfrak{s}_u$ are fixed. Let $H(a;m) = \psi_0(a+m) - \psi_0(a)$ as in (274), and write $\mathfrak{h}_u[\ell, n]$ and $\mathfrak{s}_u[j,k]$ for the summands of (509). For $\frac{1}{2} < d < 4$ the series

$$\begin{aligned} \mathfrak{h}_x' &= 2\sum_{\ell,n\ge 0} d_\ell T_{\ell n}^4\big[H(q;\ell+n) + H(1-q;n)\big] = 4\sum_{\ell,n\ge 0} d_\ell T_{\ell n}^4 \mu_1, \\ \mathfrak{h}_u' &= \sum_{\ell,n\ge 0} \mathfrak{h}_u[\ell,n]\big[H(u+\tfrac{q}{2};\ell+n) + H(u+\tfrac{3q}{2};\ell+n) \\ &\qquad + H(u+1-\tfrac{3q}{2};n) + H(u+1-\tfrac{q}{2};n)\big], \\ \mathfrak{s}_u' &= 2\sum_{j,k\ge 0} \mathfrak{s}_u[j,k]\big[H(u+\tfrac{q}{2};j) + H(u+\tfrac{3q}{2};j)\big], \end{aligned} \tag{518}$$

with $u = \frac{1}{2}, 1$ in the second line, $u = 0, \frac{1}{2}$ in the third and μ_1 of (253), converge absolutely and are the b-derivatives at $b = d/2$. For general r the prefactors $2, 1, 2$ become $\frac{r}{2}, \frac{r}{4}, \frac{r}{2}$.

Proof. $x' = \frac{1}{4}$, $y' = -\frac{1}{4}$ and $\partial_a \log(a)_m = H(a;m)$. In $\mathfrak{h}_u$ with u fixed, the parameters $u + x$ and $u + 1 - y$ stand in numerators and move with slope $\frac{1}{4}$, while $u + y$ and $u + 1 - x$ stand in denominators and move with slope $-\frac{1}{4}$, so all four contributions carry the sign $+$, each with weight $\frac{r}{4}$. In $\mathfrak{h}_x$ the numerator parameters $u + x = \sigma$ and $u + 1 - y = 1 - g$ move with slope $\frac{1}{2}$, and $u + y = 2q$, $u + 1 - x = 1$ are fixed; this gives weight $\frac{r}{2}$, and at $b = d/2$ the bracket is $2\mu_1$. In $\mathfrak{s}_u$ the contributions are $\frac{r}{4}[H(u+x;j) + H(u+y;j) + H(u+x;k) + H(u+y;k)]$, and the summand is symmetric in $j \leftrightarrow k$, so the k-terms sum to the j-terms. For convergence, let B be a closed disc around $d/2$ inside the strip $0 < \operatorname{Re} b < b_{\max}(4,d)$ on which $\sigma \notin \mathbb{Z}$; it exists because $d/2$ lies in the strip for $\frac{1}{2} < d < 4$. On B the summands of the five functions are analytic in b (no denominator Pochhammer symbol vanishes, since $\operatorname{Re} y > 0$ and $\operatorname{Re}(1-x) > 0$ there), and by the Gamma-ratio bound, whose constants are uniform for parameters in compact sets, they are bounded by $C(1+\ell)^{d-2}(1+\ell+n)^{-\alpha_B}(1+n)^{-\alpha_B}$ for $\mathfrak{h}_u$ and $C(1+j+k)^{d-2}(1+j)^{-\alpha_B}(1+k)^{-\alpha_B}$ for $\mathfrak{s}_u$, with $\alpha_B = \min_B 4\operatorname{Re} g > \max(\frac{d}{2}, d-1, \frac{1}{2})$. By the lattice-sum criterion of Section 20.1 the five series converge uniformly on B. By Weierstrass' theorem their sums are analytic inside B and

may be differentiated termwise, and Cauchy's estimate on a circle around $d/2$ inside B bounds the differentiated terms by a constant times the same majorant, so the derivative series converge absolutely. The weights $H(a;m)$ grow like $\log(1+m)$, which is the logarithmic loss mentioned in Section 20.2. □

The rim jet in Horn values. For $\frac12 < d < 4$,

$$\boxed{\begin{aligned}\mathcal{L}_d = {}& \frac{2\pi^4(1+2C^2)}{S^4}\,\mathfrak{h}_x - \frac{4\pi^3 C}{S^3}\,\mathfrak{h}'_x \\ &+ \mathsf{k}_0\Big\{\big[(1+2C)\mathsf{v}_0 - \pi S\big]\mathfrak{S}_0 + (1+2C)\big[\mathfrak{s}'_0 + 2\xi_0^4\mathfrak{h}'_1 + 2(\xi_0^4)'\mathfrak{h}_1\big]\Big\} \\ &+ \mathsf{k}_h\Big\{\big[(2C-1)\mathsf{v}_h - \pi S\big]\mathfrak{S}_h + (2C-1)\big[\mathfrak{s}'_{1/2} + 2\xi_h^4\mathfrak{h}'_{1/2} + 2(\xi_h^4)'\mathfrak{h}_{1/2}\big]\Big\},\end{aligned}} \tag{519}$$

with the five derivatives (518),

$$\begin{aligned}\mathsf{v}_0 &= \partial_b \log \mathsf{k}_0(b)\big|_{b=d/2} = -4\psi_0(q) + 2\psi_0(\tfrac{q}{2}) + 2\psi_0(\tfrac{3q}{2}) + \pi\cot\tfrac{\pi q}{2} - \tfrac{3\pi}{2}\cot\pi q,\\ \mathsf{v}_h &= \partial_b \log \mathsf{k}_h(b)\big|_{b=d/2} = -4\psi_0(q) + 2\psi_0(\tfrac{1+q}{2}) + 2\psi_0(\tfrac{1+3q}{2}) - \pi\tan\tfrac{\pi q}{2} - \tfrac{3\pi}{2}\cot\pi q,\\ (\xi_0^4)' &= 2\xi_0^4\Big[\frac{4}{3q} + \frac{1}{2-q}\Big] + \frac{2\xi_0^3}{3(2-q)}, \qquad (\xi_h^4)' = \frac{4(1-2q)\,\xi_h^3}{(1-q)^2}.\end{aligned}$$

Every coefficient is finite for $\frac12 < d < 4$. Written with the samples, (519) is (493): both are the b-derivative at $d/2$ of (491) = (514), and $\mathcal{X}'_0 = (\kappa_0^4)'\mathfrak{S}_0 + \kappa_0^4\mathfrak{S}'_0$, $\mathcal{X}'_{1/2} = (\kappa_h^4)'\mathfrak{S}_h + \kappa_h^4\mathfrak{S}'_h$, $\mathcal{X}'_T = (\pi^4 s_b^4)'\mathfrak{h}_x + \pi^4 S^4\mathfrak{h}'_x$.

Proof. Near $b = d/2$ one has $0 < \sigma < 1$, so (514) holds on a neighborhood of $d/2$. Every factor is analytic there, the five Horn functions by the proof of (518), so the product rule applies to $W_4 = -\frac{4\pi^3\varsigma_b}{s_b^3}\mathfrak{h}_x + (1+2\varsigma_b)\mathsf{k}_0(b)\mathfrak{S}_0 + (2\varsigma_b - 1)\mathsf{k}_h(b)\mathfrak{S}_h$. With $\partial_b\varsigma_b = -\frac{\pi}{2}s_b$ and $\partial_b s_b = \frac{\pi}{2}\varsigma_b$ one finds $\partial_b(-4\pi^3\varsigma_b/s_b^3) = 2\pi^4(1+2\varsigma_b^2)/s_b^4$ and $\partial_b(1+2\varsigma_b) = \partial_b(2\varsigma_b - 1) = -\pi s_b$, which give the first line of (519) and the terms $-\pi S\,\mathsf{k}_0\mathfrak{S}_0$, $-\pi S\,\mathsf{k}_h\mathfrak{S}_h$. Next, with $\partial_b \log N_b = -\frac12[\psi_0(g) + \psi_0(\sigma)]$, $\partial_b \log\varphi(0)^2 = \frac12[\psi_0(x) + \psi_0(y)]$, $\partial_b \log\varphi(\frac12)^2 = \frac12[\psi_0(\frac12 + x) + \psi_0(\frac12 + y)]$, $\partial_b \log\sin^2\pi x = \frac{\pi}{2}\cot\pi x$ and $\partial_b \log\cos^2\pi x = -\frac{\pi}{2}\tan\pi x$,

$$\begin{aligned}\partial_b \log \mathsf{k}_0(b) &= -\frac{\pi s_b}{1+\varsigma_b} - \frac{7\pi\varsigma_b}{2s_b} - 2\psi_0(g) - 2\psi_0(\sigma) + 2\psi_0(x) + 2\psi_0(y) + 2\pi\cot\pi x,\\ \partial_b \log \mathsf{k}_h(b) &= \frac{\pi s_b}{1-\varsigma_b} - \frac{7\pi\varsigma_b}{2s_b} - 2\psi_0(g) - 2\psi_0(\sigma) + 2\psi_0(\tfrac12 + x) + 2\psi_0(\tfrac12 + y) - 2\pi\tan\pi x.\end{aligned}$$

At $b = d/2$, $\pi S/(1+C) = \pi\tan\frac{\pi q}{2}$, $\pi S/(1-C) = \pi\cot\frac{\pi q}{2}$ and $7\pi C/(2S) = \frac{7\pi}{2}\cot\pi q$; with $\pi\cot\frac{\pi q}{2} - \pi\tan\frac{\pi q}{2} = 2\pi\cot\pi q$ this gives the printed v_0 and v_h, and the terms $(1+2C)\mathsf{v}_0\mathsf{k}_0\mathfrak{S}_0$ and $(2C-1)\mathsf{v}_h\mathsf{k}_h\mathfrak{S}_h$. The remaining terms are $(1+2C)\mathsf{k}_0\mathfrak{S}'_0$ and $(2C-1)\mathsf{k}_h\mathfrak{S}'_h$, with $\mathfrak{S}'_0 = \mathfrak{s}'_0 + 2\xi_0^4\mathfrak{h}'_1 + 2(\xi_0^4)'\mathfrak{h}_1$ and likewise for $\mathfrak{S}'_h$. Finally $(\xi_0^4)' = 4\xi_0^4\partial_b\log\xi_0$ with $\partial_b\log\xi_0 = \frac14[\frac1x + \frac1y + \frac{1}{1-x} + \frac{1}{1-y}] = \frac12[\frac{4}{3q} + \frac{1}{2-3q} + \frac{1}{2-q}]$ at $b = d/2$, and $(\xi_h^4)' = 4\xi_h^4\partial_b\log\xi_h$ with $\partial_b\log\xi_h = \frac12[\frac{1}{1-2x} + \frac{1}{1-2y}] = \frac12[\frac{1}{1-q} + \frac{1}{1-3q}]$; the poles at $q = \frac23$ and $q = \frac13$ cancel against $\xi_0 \propto 2 - 3q$ and $\xi_h \propto 1 - 3q$, which gives the printed forms. The coefficients are finite for $0 < q < 1$: $S \neq 0$, the trigonometric functions of $\frac{\pi q}{2}$ and $\cot\pi q$ are finite, the digamma arguments are positive, and $2 - q, 1 - q \neq 0$. □

The closed coefficients were compared with direct symbolic differentiation of (514) at eleven values of d, including $\frac43$, 2 and $\frac83$, to 10^{-40} (`LRI_C5_general_d/L/check_closed_coeffs.py`), and the review of `LRI_C5_general_d/review/L-rigor/` repeated this with its own coefficient functions (Section 20.9).

Moving and fixed nodes. The nodes $\tilde t = 0, \frac12$ are $t = -\frac{b}{4}$ and $t = \frac12 - \frac{b}{4}$, which move with b. Since $\Xi_4(t;b) = \sum_k c_{4,k} \cos 2\pi k\tilde t$ is even about both points, $\partial_t \Xi_4 = 0$ there, and the total b-derivative of these samples equals their b-derivative at fixed t. In the fixed-t form the right half-lattice is differentiated only in its numerator parameters ($\sigma + t$ and $1 - g + t$ move with slope $\frac12$, while $2q + t$ and $1 + t$ are fixed) and the left half-lattice only in its denominator parameters; their average is the symmetric derivative of (518), in which every parameter moves with slope $\pm\frac14$. The node $t = 0$ is fixed by construction. No t-derivative of any sample enters (519).

Relation to the residue series. The series (256) is a residue at $t = 0$: it needs the Laurent data of the right half-lattice at $t = 0$ up to third order, through the polynomials F_4 and $F_{4,1}$ of (254) (with λ_3 and η_3). Formula (519) keeps only the zeroth-order data at $t = 0$, namely $\mathfrak{h}_x = \sum d_\ell T^4_{\ell n}$ and $\mathfrak{h}'_x = 4\sum d_\ell T^4_{\ell n}\mu_1$, and replaces the higher Laurent coefficients by the values at the two reflection centers. Section 20.5 is the intermediate form with the pole node alone, one Laurent order below (256).

Special dimensions. At $b = d/2$ one has $y = 3x$, and the triple-angle formulas give $\sin\pi y = (1{+}2C)\sin\pi x$ and $\cos\pi y = (2C{-}1)\cos\pi x$, so $P_\pm(0) = (1{+}2C)^r P_{\rm m}(0)$ and $P_\pm(\frac12) = (2C{-}1)^r P_{\rm m}(\frac12)$. In (517) and (519) the sample at $\tilde t = 0$ carries the factor $1 + 2C$, and $\mathfrak{h}_1$ carries in addition ξ_0^4 (or ξ_0^3 in $(\xi_0^4)'$), with $\xi_0 \propto 2 - 3q$; the sample at $\tilde t = \frac12$ carries $2C - 1$, and $\mathfrak{h}_{1/2}$ carries in addition ξ_h^4 or ξ_h^3, with $\xi_h \propto 1 - 3q$. Hence at $d = \frac83$ ($q = \frac23$, $C = -\frac12$, $y = 1$) $\mathfrak{h}_1$ drops out of both P_4 and $\mathcal{L}_d$: its coefficient in $W_4(d;b)$ has a zero of order five at $b = d/2$, a simple zero of $1 + 2\varsigma_b$ times the fourth power of the simple zero of ξ_0. There $\mathfrak{s}_0$ enters $\mathcal{L}_d$ only through its value, with coefficient $-\pi S\mathsf{k}_0$. At $d = \frac43$ ($q = \frac13$, $C = \frac12$, $y = \frac12$) the same holds for $\mathfrak{h}_{1/2}$ and $\mathfrak{s}_{1/2}$. At $d = 2$ ($C = 0$) the coefficient of $\mathfrak{h}'_x$ vanishes, and $\mathfrak{h}_x$ drops out of $P_4(2)$. Thus $P_4(\frac43)$ and $P_4(\frac83)$ need three Horn values, $\mathcal{L}_{4/3}$ and $\mathcal{L}_{8/3}$ four values and three derivatives, and $\mathcal{L}_2$ five values and four derivatives. In the code the vanishing coefficients come out zero to working precision.

The five functions at $d = 1, 2, 3$. At $d = 1$, $d_\ell = \delta_{\ell0} + \delta_{\ell1}$, so each $\mathfrak{h}_u$ is a single sum, a ${}_9F_8(1)$ (an ${}_8F_7(1)$ for $u = x$) plus the analogous single sum of the row $\ell = 1$; $\mathfrak{s}_0 = 1 + 2(x/y)^4$, which is $\frac{83}{81}$ at $b = \frac12$ with $\mathfrak{s}'_0 = \frac{64}{243}$, and $\mathfrak{s}_{1/2} = 1$. The samples at $\tilde t = 0$ and $\tilde t = \frac12$ coincide, since $c_{4,1} \equiv 0$ there (Section 20.6). At $d = 2$, $x + y = 1$, so $\bar\varphi = \varphi$, and $d_\ell = 2 - \delta_{\ell0}$. Put $\chi_m = \varphi(m + u)^r$ and $\mathsf{m}(u) = \sum_{n\ge0}[(u + x)_n/(u + y)_n]^r$, so that $\sum_{m\ge0}\chi_m = \varphi(u)^r\mathsf{m}(u)$. In $\mathcal{R}_+(u)$ set $m = \ell + n$; the weight d_{m-n} is 2 for $m > n$ and 1 for $m = n$, so $\mathcal{R}_+(u) = \sum_{m\ge n\ge0} d_{m-n}\chi_m\chi_n = (\sum_m \chi_m)^2$, the symmetrization of the proof of (539). Likewise $d_{j+k+1} = 2$ and $d_{j+k} = 2 - \delta_{j0}\delta_{k0}$ give $\mathcal{R}_{\rm m}(\frac12) = 2\varphi(\frac12)^{2r}\mathsf{m}(\frac12)^2$ and $\mathcal{R}_{\rm s}(0) = 2[\varphi(0)^r\mathsf{m}(0)]^2 - \varphi(0)^{2r}$, and $\mathsf{m}(0) = 1 + (x/y)^r\mathsf{m}(1)$. Hence

$$\mathfrak{h}_u^{(r)} = \mathsf{m}(u)^2, \qquad \mathfrak{s}_{1/2}^{(r)} = 2\,\mathfrak{h}_{1/2}^{(r)}, \qquad \mathfrak{s}_0^{(r)} = 2\big[1 + (x/y)^r\mathsf{m}(1)\big]^2 - 1 \qquad (d = 2),$$

with $\mathsf{m}(x) = {}_rF_{r-1}(\sigma,\dots,\sigma;1,\dots,1;1)$; for real b, $\mathsf{m}(u) > 0$ is the positive square root of $\mathfrak{h}_u^{(r)}$. The $d = 2$ rows of the test data in Section 20.3.1 satisfy the last two relations to the printed precision. At $d = 3$, $d_\ell = 2\ell + 1$ is linear, so $\mathfrak{s}_0$ and $\mathfrak{s}_{1/2}$ are products of one-variable moments: with $\mathsf{n}_i(u) = \sum_{j\ge0} j^i[(u + x)_j/(u + y)_j]^4$, $\mathfrak{s}_0 = 4\mathsf{n}_0(0)\mathsf{n}_1(0) + \mathsf{n}_0(0)^2$ and $\mathfrak{s}_{1/2} = 4\mathsf{n}_0(\frac12)\mathsf{n}_1(\frac12) + 3\mathsf{n}_0(\frac12)^2$. This does not hold for $\mathfrak{h}_x, \mathfrak{h}_{1/2}, \mathfrak{h}_1$: with $m = \ell + n$ they are nested sums over $m \ge n \ge 0$ with weight $2(m - n) + 1$, genuine two-variable values (rank two). The reduction of $\mathcal{L}_3$ in Section 20.6 works instead with the half-shift structure of the full samples. Its Dougall closed form of the amplitude $\mathfrak{b}_4$ rests on the linear multiplicity $2\ell + 1$, which turns the half-shifted lattice sum into a very-well-poised bilateral series. For non-integer d the multiplicity (500) is not polynomial; no Gamma-product amplitude was found there, and none was searched for systematically. The closed forms of $\mathcal{L}_1$, $\mathcal{L}_2$ and the $d = 3$ reduction are obtained in Section 20.6 from the mode form, not from (519); (519) reproduces all three values numerically (Section 20.3.1).

Function count. At generic $\frac{1}{2} < d < 4$, (517) needs the five two-variable Horn values $\mathfrak{h}_x$, $\mathfrak{h}_{1/2}$, $\mathfrak{h}_1$, $\mathfrak{s}_0$, $\mathfrak{s}_{1/2}$ at argument $(1,1)$, and (519) the same five values and their five first parameter derivatives (518). The coefficients need trigonometric functions of πq and $\frac{\pi q}{2}$; $\Gamma(2q)$, $\Gamma(\frac{q}{2})$, $\Gamma(\frac{3q}{2})$, $\Gamma(\frac{1+q}{2})$, $\Gamma(\frac{1+3q}{2})$; ψ_0 at $q, \frac{q}{2}, \frac{3q}{2}, \frac{1+q}{2}, \frac{1+3q}{2}$ (by the duplication formula, equivalently $\psi_0(q), \psi_0(3q), \psi_0(\frac{q}{2}), \psi_0(\frac{3q}{2})$ and $\log 2$); and rational functions of q. In the unweighted Horn class (320) the derivatives cost 19 three-variable values: each weight $H(a;m) = m\sum_{k\ge 0}[(a+k)(a+m+k)]^{-1}$ of (291) adds one summation index, encoded by (292), and the factor m costs one array for $m = n$ or $m = j$ and two for $m = \ell + n$ (294). This gives 3 arrays for $\mathfrak{h}'_x$, 6 each for $\mathfrak{h}'_{1/2}$ and $\mathfrak{h}'_1$, and 2 each for $\mathfrak{s}'_0$ and $\mathfrak{s}'_{1/2}$. So $\mathcal{L}_d$ needs 5 two-variable and 19 three-variable unweighted Horn values, in place of the 244 values with up to six variables of (298). In the split form used by the code (next paragraph) the five functions are four one-variable values, an ${}_8F_7(1)$ in $\mathfrak{h}_x$, two ${}_9F_8(1)$ in $\mathfrak{h}_{1/2}, \mathfrak{h}_1$ and a ${}_7F_6(1)$ in $\mathfrak{s}_0$, and five two-variable values, each with its first derivative. For $P_4(d)$, the sampling (323) at the nodes (312) uses 20 complex block values (A_4^0, A_4^+ at eight complex points and C_4 at four; 15 with (473)); (517) uses five real Horn values. The five derivative values are the new constants of the generic-dimensional problem. Whether they are independent is not decided here; at $d = 1, 2$ they reduce, and at $d = 3$ they reduce, with the identities of Section 20.6, to the two constants $\mathfrak{d}_H, \mathfrak{d}_a$ (Section 20.6). Contiguous relations do not connect the five functions, since their parameters differ by non-integer shifts (x and $\frac{1}{2}$; Section 20.8).

Evaluation. `LRI_C5_general_d/L/L_general.py` evaluates (514), and (515), for any complex b in the strip, in a split form. Let $\hat{d}_m = d_{m+1}/(4q) = (2q+1)_m(4q-1)_m/((2q)_m(2)_m)$ be the ratio of (301). The row $\ell = 0$ of $\mathfrak{h}_u$ is a one-variable sum, and the rows $\ell = j+1 \ge 1$ are $4q[(u+x)/(u+y)]^r \sum_{j,n\ge 0} \hat{d}_j \mathtt{x}_{j+n} \mathtt{z}_n$ with $\mathtt{x}_m = [(u+1+x)_m/(u+1+y)_m]^r$ and $\mathtt{z}_n = [(u+1-y)_n/(u+1-x)_n]^r$, sequences of the kind $\mathtt{x}, \mathtt{z}$ of (486). Further $\mathfrak{s}_{1/2} = 4q\sum_{j,k} \hat{d}_{j+k} a_j a_k$ with $a_j = [(\frac{1}{2}+x)_j/(\frac{1}{2}+y)_j]^r$, and $\mathfrak{s}_0 = 1 + 4q(x/y)^r[\sum_k \hat{d}_k \tilde{a}_k + \sum_{j,k} \hat{d}_{j+k} \tilde{a}_j a_k]$ with $a_j = [(x)_j/(y)_j]^r$ and $\tilde{a}_k = [(1+x)_k/(1+y)_k]^r$. Each sum is evaluated as an exact finite part plus an asymptotic tail by the method of the paragraph *Evaluation of the Horn values* of Section 20.1.4, with the routines of `LRI_C5_general_d/foundation/hypsums.py` and the default truncation parameters and guard digits stated there; at $d = 1, 2, 3$ the multiplicities are used as exact polynomials. The b-derivatives are exact: `LRI_C5_general_d/L/jet.py` implements first-order jets (a value and its derivative, combined by the product rule), and `LRI_C5_general_d/L/hypsums_jet.py` is the same algorithm with every operation jet-valued, including the Hurwitz zeta derivative in its first argument and the Bernoulli-polynomial tail coefficients. Evaluating at the jet based at $b = \frac{d}{2}$ returns the value and the derivative of the evaluated expression, that is the series (518) with their tails (now with $\log m$ terms) accelerated in the same way. No finite difference and no Cauchy circle in b enters $\mathcal{L}_d$. One pair $(P_4(d), \mathcal{L}_d)$ at 50 digits takes about 15 s on one AMD core of the symmetry cluster (5 s at 30 digits on a laptop); the earlier route of `LRI_C5_general_d/foundation/`, a 96-point Cauchy circle in b of the finite-sample W_4, took about 45 s per point at 50 digits. The function `L_explicit` evaluates (519) as printed, with the closed-form coefficients of `coeffs_closed` times the five values and the five derivative series; it agrees with the jet derivative of the whole of (514) to working precision, and `LRI_C5_general_d/L/sym_L.py` does the same with coefficients obtained by symbolic differentiation. Usage: `python3 L_general.py d [dps] [b]`, with d given exactly (for example `4/3`) and $b = d/2$ by default; the script prints W_4, W_3 and their b-derivatives and, at $b = d/2$, the value of (519). Resonant parameters, $r(d-b)/2 \in \mathbb{Z}$ at non-integer d, are refused; at $b = d/2$ they occur for $r = 3$ at $d = \frac{4}{3}, \frac{8}{3}$ and never for $r = 4$.

20.3.1 Checks of the five-Horn formula

All runs were made on the symmetry cluster. The records are in `LRI_C5_general_d/L/results/` (`summary.json`, `val50.json`, `val70.json`, `cfd40.json` and `ref_d*.json`), produced by `run_validation.py`, `refcheck_r4.py`, `check_func_derivs.py`, `analyze.py` and `make_tables.py` in

`LRI_C5_general_d/L/`. References: at $d = 1$ the proved closed forms $P_4 = 13B_- - 3B_+$ (Section 4) and (33); at $d = 2$ (333) and $\mathcal{L}_2 = -2\log 2\, P_4(2)$ (270); at $d = 3$ the 160-digit values of `LRI_C5_d3`; at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ the 41- to 50-digit values of the independent spectral code in `LRI_C5_general_d/references/reference_values.json`; at $d = 0.6, \frac{4}{3}, \frac{8}{3}, 3.5$ the same spectral code, run at 32 digits for $r = 4$ only (`LRI_C5_general_d/L/refcheck_r4.py`; at $d = 0.6$ the point $b = d/2$ lies outside the $r = 3$ strip and the $r = 3$ spectral sum does not converge). The spectral code shares no code with the five-Horn formula.

Values of $\mathcal{L}_d$. $\mathcal{L}_d$ from (519) with exact jet derivatives, at the working precision of the run (50 digits; 70 digits at $d = 1, 1.5, 2, 2.5, 3$). "dev." is the relative deviation from the reference, "conv." the relative change of $\mathcal{L}_d$ when all truncation parameters are enlarged by the factor 1.4.

d	$\mathcal{L}_d$ (five-Horn formula)	reference	dev. P_4	dev. $\mathcal{L}_d$	conv. $\mathcal{L}_d$
0.6	−45086.56690064459550422	spectral, own run (32 d.)	$3.6 \cdot 10^{-32}$	$7.3 \cdot 10^{-33}$	$1.9 \cdot 10^{-44}$
0.8	−14228.69045883150419322995007	spectral reference (47 d.)	$1.0 \cdot 10^{-45}$	$2.7 \cdot 10^{-45}$	$3.2 \cdot 10^{-45}$
1	−5805.301629548802380141	closed form (proved)	$1.0 \cdot 10^{-80}$	$3.4 \cdot 10^{-81}$	$< 10^{-80}$
1.25	−2361.293891754120963656812701	spectral reference (49 d.)	$3.3 \cdot 10^{-45}$	$8.7 \cdot 10^{-46}$	$1.4 \cdot 10^{-47}$
4/3	−1818.786953676526007105	spectral, own run (32 d.)	$3.7 \cdot 10^{-24}$	$5.5 \cdot 10^{-24}$	$2.8 \cdot 10^{-49}$
1.5	−1127.932175171310391303895184	spectral reference (50 d.)	$9.7 \cdot 10^{-47}$	$3.6 \cdot 10^{-45}$	$1.2 \cdot 10^{-58}$
2	−347.2491531232011310585	$-2\log 2\, P_4(2)$ (proved)	$4.7 \cdot 10^{-81}$	$8.9 \cdot 10^{-81}$	$< 10^{-80}$
2.5	−136.796077287838337518533388	spectral reference (49 d.)	$3.8 \cdot 10^{-45}$	$2.5 \cdot 10^{-45}$	$2.4 \cdot 10^{-58}$
8/3	−104.035830813482931953	spectral, own run (32 d.)	$2.6 \cdot 10^{-26}$	$3.5 \cdot 10^{-26}$	$2.2 \cdot 10^{-47}$
2.75	−91.22798596406115896228950228	spectral reference (50 d.)	$5.3 \cdot 10^{-46}$	$1.9 \cdot 10^{-46}$	$2.5 \cdot 10^{-47}$
3	−62.7105849683876693473	`LRI_C5_d3` (160 d.)	$6.0 \cdot 10^{-81}$	$1.7 \cdot 10^{-81}$	$< 10^{-80}$
3.5	−31.81370266461412493531	spectral, own run (32 d.)	$2.8 \cdot 10^{-27}$	$3.5 \cdot 10^{-27}$	$9.5 \cdot 10^{-47}$

Away from the center. Relative deviation of (514), (515) and the two-node formula (496) from the 56-digit values of the independent spectral code, from the exact $W_3(2; b)$ of (264), and, at $d = 2$, of the ratio $W_4(2; 1.05)/W_4(2; 0.95)$ from the rim–spoke reflection $W_4(2; b)/W_4(2; 2-b) = 2^{4-4b}\gamma_C(b/2)^2/\gamma_C(b-\frac{1}{2})$, which follows from the formula for $W_4(2; 2x)$ in Section 18.7.6. The columns (514) and (515) come from `LRI_C5_general_d/L/L_general.py`, at 70 digits for $d = 1.5, 2, 2.5$ and at 50 digits otherwise. The column (496) comes from `LRI_C5_general_d/Z/offdiag_check.py` (record `LRI_C5_general_d/Z/results/offdiag60.json`), at 50 digits against the spectral values and at 60 digits against the finite-sample formula (484) with five fixed nodes ("fixed nodes"), which shares the Horn summation code and is therefore not independent; that code also gives the value column. The two W_3 columns differ only at $d = 1.5$ and 2.5, where the working precisions differ.

d	b	$\mathrm{Re}\,W_3(d;b)$ (24 digits)	reference	dev. (514)	dev. (515)	dev. (496)
0.8	0.37	`266.761267027041907531165`	spectral	$3.5\cdot10^{-51}$	$2.9\cdot10^{-49}$	$2.9\cdot10^{-49}$
0.8	0.43	`268.642244930670511767657`	spectral	$3.5\cdot10^{-50}$	$1.5\cdot10^{-48}$	$1.5\cdot10^{-48}$
1.25	0.575	`107.42914710548489909118`	spectral	$5.8\cdot10^{-53}$	$2.7\cdot10^{-50}$	$2.7\cdot10^{-50}$
1.25	0.675	`108.364244882288165222595`	spectral	$6.8\cdot10^{-53}$	$1.1\cdot10^{-49}$	$1.1\cdot10^{-49}$
1.5	0.7	`72.7303680788392574835365`	spectral	$2.0\cdot10^{-54}$	$1.5\cdot10^{-52}$	$1.5\cdot10^{-50}$
1.5	0.8	`73.0949788822424244641612`	spectral	$2.3\cdot10^{-53}$	$8.1\cdot10^{-52}$	$1.2\cdot10^{-50}$
2.5	1.2	`23.3776070284027512618263`	spectral	$8.1\cdot10^{-57}$	$2.4\cdot10^{-54}$	$2.8\cdot10^{-51}$
2.5	1.3	`23.403771830554642453335`	spectral	$7.9\cdot10^{-56}$	$9.9\cdot10^{-54}$	$3.2\cdot10^{-51}$
2.75	1.325	`18.6817874548469398300141`	spectral	$4.8\cdot10^{-55}$	$1.6\cdot10^{-51}$	$1.6\cdot10^{-51}$
2.75	1.425	`18.6977204798280561759503`	spectral	$2.3\cdot10^{-52}$	$7.5\cdot10^{-51}$	$7.5\cdot10^{-51}$
2	0.95		exact (264)	(ratio, below)	$5.8\cdot10^{-81}$	
2	1.05		exact (264)	(ratio, below)	$5.6\cdot10^{-81}$	
2	1.05/0.95		reflection	$9.8\cdot10^{-81}$ (ratio)		
1	0.37	`216.134210851145764012993`	fixed nodes			$3.4\cdot10^{-62}$
1	$0.3+0.1\,\mathrm{i}$	`223.841209588900262170043`	fixed nodes			$4.1\cdot10^{-62}$
2	0.8	`45.6639529602144152055429`	fixed nodes			$4.5\cdot10^{-63}$
2	$1.2+0.05\,\mathrm{i}$	`49.6299926202374115388838`	fixed nodes			$1.9\cdot10^{-62}$
3	1.3	`16.4598664890623652308798`	fixed nodes			$5.6\cdot10^{-62}$
3	1.6	`15.5022646572795884994581`	fixed nodes			$1.2\cdot10^{-62}$
3	$1.45+0.05\,\mathrm{i}$	`15.058642846628421202012`	fixed nodes			$3.5\cdot10^{-62}$
1.5	$0.9+0.05\,\mathrm{i}$	`86.1127570666953915845079`	fixed nodes			$4.6\cdot10^{-56}$
2.5	1.4	`25.6947810198617030782933`	fixed nodes			$1.8\cdot10^{-55}$
3.5	1.7	`10.2964255108971126897525`	fixed nodes			$6.3\cdot10^{-55}$
0.8	0.42	`263.52442188748393591694`	fixed nodes			$3.5\cdot10^{-55}$

Test data. The five Horn values and their total first b-derivatives at $b=d/2$ ($u=x$ moving in $\mathfrak{h}_x$), 32 of the at least 55 digits computed in the 70-digit runs. At $d=1$, $\mathfrak{s}_0=\frac{83}{81}$, $\mathfrak{s}_0'=\frac{64}{243}$ and $\mathfrak{s}_{1/2}=1$ exactly.

d	function	value at $b=d/2$	∂_b at $b=d/2$
1	$\mathfrak{h}_x$	1.1186363871641870683496192575256	1.2852461541784784274144166577 49
	$\mathfrak{h}_1$	2.5987822357262795715169656086966	7.9424211032565657723245715768249
	$\mathfrak{h}_{1/2}$	1.6764843387459078434810926719847	3.9229772806249080806880221308364
	$\mathfrak{s}_0$	1.0246913580246913580246913580247	0.2633744855967078189300411522633 7
	$\mathfrak{s}_{1/2}$	1.0	0.0
1.5	$\mathfrak{h}_x$	1.1773925910607038758559762936 85	1.2933702591309377564756247705028
	$\mathfrak{h}_1$	2.7285923368887621906692408263826	7.0118518320221630889168328700506
	$\mathfrak{h}_{1/2}$	1.6390896218805694790051124408507	2.9123315039974903478917414402412
	$\mathfrak{s}_0$	1.0548507244990532977247501057129	0.4452440264452722619797249734899 7
	$\mathfrak{s}_{1/2}$	2.2461861367840794529465463415088	3.3311337225326803025747929195969
2	$\mathfrak{h}_x$	1.2513473666877450268884376715415	1.4377231145268788593624919295969
	$\mathfrak{h}_1$	2.8105997380603038789311930383846	6.5768099722236687733793558671552
	$\mathfrak{h}_{1/2}$	1.6274006603148409180018539998063	2.4813324518913317498491423929508
	$\mathfrak{s}_0$	1.0836461096219775566294370388355	0.5495504249104147258401786138706
	$\mathfrak{s}_{1/2}$	3.2548013206296818360037079996126	4.9626649037826634996982847859016
2.5	$\mathfrak{h}_x$	1.3376359713522616550547394154586	1.6441674626487503441319240236304
	$\mathfrak{h}_1$	2.8658700646971037650444991547941	6.3128962181069309575437047132 37
	$\mathfrak{h}_{1/2}$	1.6288526582621043253519113017864	2.2658474252647133894689489695958
	$\mathfrak{s}_0$	1.1103501379648233369119496859969	0.6126086566837367097941991100351 6
	$\mathfrak{s}_{1/2}$	4.1368224751542518108391408828496	5.8497720890012283607483552162793
3	$\mathfrak{h}_x$	1.4335280971237659036402062676254	1.8841954627648666804462998053832
	$\mathfrak{h}_1$	2.9047690244187603057569160926227	6.1245414235188740303623918912455
	$\mathfrak{h}_{1/2}$	1.6357377373331371059669409314044	2.1424411510929036014226183493683
	$\mathfrak{s}_0$	1.1349627686583730036642106378911	0.6502812349965942458213304530189 7
	$\mathfrak{s}_{1/2}$	4.9458028842831626665221750648488	6.3455500640456084878526885322906

Per-function derivative check. For each of the five functions the jet value and derivative at $b=d/2$ were compared, at 40 digits, with the mean and the first Fourier coefficient of its plain (non-jet) values on a 48-point circle of radius $0.05\min(\frac{d}{2}, b_{\max}-\frac{d}{2})$ around $d/2$ (`LRI_C5_general_d/L/check_func_derivs.py`, `LRI_C5_general_d/L/results/cfd40.json`). This checks the jet implementation for every function separately; it is not an independent evaluation. Each cell gives

the relative deviation of the value / of the derivative, the latter relative to $\max(|f'|,1)$ (at $d=1$, $\mathfrak{s}_{1/2}=1$ has $f'=0$).

d	$\mathfrak{h}_x$	$\mathfrak{h}_1$	$\mathfrak{h}_{1/2}$	$\mathfrak{s}_0$	$\mathfrak{s}_{1/2}$
1	$1{\cdot}10^{-45}$ / $3{\cdot}10^{-45}$	$6{\cdot}10^{-46}$ / $4{\cdot}10^{-46}$	$2{\cdot}10^{-45}$ / $1{\cdot}10^{-44}$	$3{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$< 10^{-60}$ / $7{\cdot}10^{-56}$
1.5	$4{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$4{\cdot}10^{-45}$ / $1{\cdot}10^{-44}$	$3{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$4{\cdot}10^{-45}$ / $2{\cdot}10^{-44}$	$5{\cdot}10^{-44}$ / $3{\cdot}10^{-43}$
2	$2{\cdot}10^{-45}$ / $9{\cdot}10^{-45}$	$6{\cdot}10^{-46}$ / $2{\cdot}10^{-45}$	$1{\cdot}10^{-45}$ / $5{\cdot}10^{-45}$	$4{\cdot}10^{-46}$ / $1{\cdot}10^{-44}$	$1{\cdot}10^{-45}$ / $1{\cdot}10^{-45}$
2.5	$2{\cdot}10^{-45}$ / $2{\cdot}10^{-45}$	$1{\cdot}10^{-45}$ / $2{\cdot}10^{-45}$	$8{\cdot}10^{-46}$ / $9{\cdot}10^{-45}$	$2{\cdot}10^{-48}$ / $7{\cdot}10^{-45}$	$3{\cdot}10^{-44}$ / $3{\cdot}10^{-43}$
3	$1{\cdot}10^{-45}$ / $8{\cdot}10^{-45}$	$2{\cdot}10^{-45}$ / $9{\cdot}10^{-46}$	$1{\cdot}10^{-45}$ / $5{\cdot}10^{-45}$	$2{\cdot}10^{-45}$ / $8{\cdot}10^{-45}$	$3{\cdot}10^{-46}$ / $4{\cdot}10^{-45}$

Summary. At $d=1,2,3$, $\mathcal{L}_d$ and $P_4(d)$ agree with the proved or 160-digit values to 10^{-60} or better at 50 digits and to 10^{-80} or better at 70 digits. At the five non-integer dimensions with 41- to 50-digit spectral references the agreement is 10^{-45} to 10^{-47}, that is the accuracy of the references (the internal convergence is 10^{-45} to 10^{-49} at 50 digits and about 10^{-58} at 70 digits); at $d=0.6,\frac{4}{3},\frac{8}{3},3.5$ it is 10^{-24} to 10^{-33}, limited by the 32-digit spectral runs. Away from the center, $W_4(d;b)$ and $W_3(d;b)$ at the ten points $b=d/2\pm0.05$ (±0.03 at $d=0.8$) agree with the 56-digit spectral values to $1.5\cdot10^{-48}$ or better (W_4: $3.5\cdot10^{-50}$ or better), and at $d=2$ with the exact $W_3(2;b)$ and the rim–spoke reflection to 10^{-80}. The $r=3$ by-product $3Q_d$ agrees with the references at the eight dimensions where they exist to 10^{-45} to 10^{-81}, and $\partial_b W_3(d;b)|_{b=d/2}$, which vanishes by (239) with (244), comes out 10^{-47} to 10^{-86} relative to W_3; this tests the jets of the $r=3$ functions. Each jet derivative agrees with the 48-point Cauchy-circle derivative to $3\cdot10^{-43}$ or better at 40 digits (previous table). The accelerated evaluation is an asymptotic-series method; its accuracy rests on the convergence in the truncation parameters and on the independent comparisons above, not on interval bounds.

20.4 The three-wheel at a general rim exponent: proofs

The code of this subsection is in `LRI_C5_general_d/Z/`. A separate review with its own codes rederived the derivations of the package's own text (`LRI_C5_general_d/review/Z-rigor/`; for the steps added in this note, see Section 20.10).

This subsection proves the three-wheel statements of Section 20.2: the two samples, the two-node formula (496) with its zero-mode form, (497) and (498). It also gives the five functions behind $\mathfrak{t}_3$ and $\widehat{M}_3$, the jets of the weights u,v, series forms of the jets and the first-derivative identity. Throughout $r=3$, $0<d<4$, and b lies in the strip

$$0<\operatorname{Re} b<b_{\max}(3,d)=\min\Big\{\frac{2d}{3},\ \frac{d+2}{3},\ d-\frac{1}{3}\Big\}<2$$

of (463), which contains $b=\frac{d}{2}$ if and only if $d>\frac{2}{3}$. On this strip $0<\operatorname{Re}\sigma<1$, so $\sigma\notin\mathbb{Z}$. The inputs are the finite Fourier theorem of Section 20.1.1, the top-mode theorem (473), the residue functional (480), the reflected form (464) and the block decomposition of Section 20.1.3. By (473) the three-wheel sample has two modes,

$$\Xi_3(t;b)=c_{3,0}(q,b)+c_{3,1}(q,b)\cos 2\pi\Big(t+\frac{b}{4}\Big). \tag{520}$$

20.4.1 Analyticity in the rim exponent

Statement. $\mathfrak{t}_3$ and $\widehat{M}_3$ of (495), the five functions of (526) below and the weighted series of (529) are analytic in b on the strip. Every b-derivative is the termwise derivative, and the differentiated series again converge locally uniformly.

Proof. Let K be a compact subset of the strip and $\alpha_K=\min_{b\in K}3\operatorname{Re} g$. Since $\operatorname{Re} b<b_{\max}(3,d)$ on K, the conditions (i)–(iii) stated after (463) hold at every point of K with $\alpha=3\operatorname{Re} g$, and

by compactness $\alpha_K > \max(\frac{d}{2}, d-1, \frac{1}{2})$. The estimates of Section 20.1 (paragraph "Lattice split and estimates") come from the Gamma-ratio bound $|\Gamma(z+u)/\Gamma(z+v)| \le C|z|^{\mathrm{Re}(u-v)}$, which is uniform for u, v in compact sets. Here it is applied to Gamma ratios whose parameters σ, g, $2q$ are affine in b, and only its part for t in a compact set is used, not the large-$|\operatorname{Im} t|$ part. Write $T_{\ell n} = [(\sigma)_{\ell+n}/(2q)_{\ell+n}]\,[(1-g)_n/n!]$ as in (490). It is a product of two polynomials in b, bounded on K by $C_K(1+\ell+n)^{-\operatorname{Re} g}$ and $C_K(1+n)^{-\operatorname{Re} g}$. With $|d_\ell| \le C(1+\ell)^{d-2}$ there is a constant C_K with

$$|d_\ell\, T_{\ell n}(b)^3| \le C_K(1+\ell)^{d-2}(1+\ell+n)^{-\alpha_K}(1+n)^{-\alpha_K} \qquad (\ell, n \ge 0,\ b \in K).$$

The terms of $\widehat{M}_3$ are $N_b^3 g_\ell(a_\ell + n + t_b; b)^3$ with $t_b = \frac{1-\sigma}{2}$, since $a_\ell + n + t_b = n + \frac{\ell+1}{2}$. For $b \in K$ the node t_b stays at a positive distance from both pole classes $t \equiv 0$ and $t \equiv -\sigma \pmod 1$, because $\operatorname{Re} t_b$ and $\operatorname{Re}(t_b + \sigma) = \operatorname{Re} \frac{1+\sigma}{2}$ lie in compact subintervals of $(0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$. So the right part ($n \ge 0$, by the first line of (464)) and the middle part (by its second line) obey the two bounds of Section 20.1 with a constant independent of $b \in K$; the finitely many terms with small arguments are continuous in b on K. The left part equals the right part, because $1 - \sigma - t_b = t_b$ and the last identity of (464) holds. These majorants do not depend on $b \in K$, and they are summable by the criterion at the end of the same paragraph, since $\alpha_K > \max(\frac{d}{2}, d-1, \frac{1}{2})$. Every term is analytic in b on the strip: $T_{\ell n}$ is a polynomial in b, and at the fixed lattice points $s = n + \frac{\ell+1}{2}$ the numerator Gamma functions of $N_b g_\ell(s; b)$ have the arguments $\ell + n + \frac{1+\sigma}{2}$, $\frac{\sigma-1}{2} - n$ and g, none of which is a pole for $0 < \operatorname{Re}\sigma < 1$ and $\operatorname{Re} g > 0$. By the Weierstrass M-test and Weierstrass's theorem on locally uniform limits of analytic functions, $\mathfrak{t}_3$ and $\widehat{M}_3$ are analytic on the strip, every b-derivative is the termwise derivative, and the differentiated series again converge locally uniformly. The same holds for the subseries of (526). For the weighted series of (529) it can also be seen directly: the weights grow at most logarithmically in the indices, and replacing α_K by a slightly smaller exponent, still above $\max(\frac{d}{2}, d-1, \frac{1}{2})$, gives summable majorants that are uniform on K. □

20.4.2 The two samples and the two-node formula

Pole node. For b in the strip,

$$\Xi_3(0; b) = -\Big(\frac{\pi s_b}{N_b}\Big)^3 \mathfrak{t}_3(b). \tag{521}$$

This is the case $r = 3$ of (510), which Section 20.3 proves for every r by extending the lattice form (504) to $t = 0$; the proof below takes the limit term by term instead.

Proof. By Section 20.1.1(a) and its proof, $t^3 S_3(t; b) = \sum_\ell d_\ell \sum_n t^3 g_\ell(a_\ell + n + t; b)^3$ converges uniformly near $t = 0$, and only the terms with $n \ge 0$ are singular at $t = 0$: the poles of the middle part in the class $t \equiv 0$ lie at $t \ge 1$, by the second line of (464), the same holds for the left part by the mirror identity, and the class $t \equiv -\sigma$ does not contain 0 since $\sigma \notin \mathbb{Z}$. Since $D_b(t)/t \to \pi s_b$, the value $\Xi_3(0) = \lim_{t\to 0} D_b(t)^3 S_3(t)$ is the sum over $\ell \ge 0$, $n \ge 0$ of $d_\ell \lim_{t\to 0}[D_b(t) g_\ell(a_\ell + n + t)]^3$. By the first line of (464),

$$\begin{aligned} D_b(t)\, g_\ell(a_\ell + n + t) \ &\to\ -\, s_b \sin(\pi g)\, \frac{\Gamma(\ell + n + \sigma)\Gamma(n + 1 - g)}{\Gamma(\ell + n + 2q)\, n!} \\ &=\ -\, s_b \sin(\pi g)\Gamma(1-g)\frac{\Gamma(\sigma)}{\Gamma(2q)}\, T_{\ell n} = -\frac{\pi s_b}{N_b}\, T_{\ell n}, \end{aligned}$$

using $\sin(\pi g)\Gamma(1-g) = \pi/\Gamma(g)$. Cubing and summing gives (521). □

The function $T_{\ell n}$ is the normalized residue prefactor at a general rim exponent: $N_b g_\ell(a_\ell + n + u; b) = -T_{\ell n}(b)/u + O(1)$ for $n \ge 0$ (as in Section 20.5), so $\mathfrak{t}_3 = -\sum_{\ell,n} d_\ell \lim_{u\to 0} u^3 [N_b g_\ell(a_\ell + n + u; b)]^3$ collects only the *leading* coefficients of the triple poles. The residues of the cube need the full Laurent data, which is the origin of the polygamma polynomial F_3 of (255).

Self-mirror node. Let $t_b = \frac{1-\sigma}{2} = \frac{1}{2} - \frac{b}{4}$. For b in the strip, $1-\sigma-t_b = t_b$, $\cos 2\pi(t_b + \frac{b}{4}) = -1$, $D_b(t_b) = \cos^2 \frac{\pi\sigma}{2} = \frac{1+\varsigma_b}{2}$, and

$$\Xi_3(t_b; b) = \Big(\frac{1+\varsigma_b}{2}\Big)^3 \frac{\widehat{M}_3(b)}{N_b^3}, \qquad \widehat{M}_3(b) = N_b^3\big[2A_3^0(t_b) + 2A_3^+(t_b) + C_3(t_b)\big], \tag{522}$$

with the blocks A_3^0, A_3^+, C_3 of Section 20.1.3, written out in (524) below. In the Horn values of Section 20.3 the same sample is (512) at $r = 3$.

Proof. The first three statements are direct: $D_b(t_b) = \sin\frac{\pi(1-\sigma)}{2} \sin\frac{\pi(1+\sigma)}{2} = \cos^2\frac{\pi\sigma}{2}$ and $2\cos^2\frac{\pi\sigma}{2} = 1 + \cos\pi\sigma = 1 + \varsigma_b$. Since $0 < \operatorname{Re}\sigma < 1$, t_b lies in neither pole class, so $\Xi_3(t_b) = D_b(t_b)^3 S_3(t_b)$ with an absolutely convergent $S_3(t_b)$. Moreover $a_\ell + n + t_b = n + \frac{\ell+1}{2}$; as n runs over $\mathbb{Z}$ this runs over the midpoint lattice $\frac{\ell+1}{2} + \mathbb{Z}$ of (495), hence $N_b^3 S_3(t_b; b) = \widehat{M}_3(b)$. Finally the block decomposition of Section 20.1.3 at a point equal to its own mirror point $1-\sigma-t$ gives the block form. □

The midpoint lattice is invariant under $s \mapsto -s$, as is $g_\ell(s; b)$.

Proof of (496). By (520), $y_0 := \Xi_3(0; b) = c_{3,0} + \varsigma_b c_{3,1}$ and $y_b := \Xi_3(t_b; b) = c_{3,0} - c_{3,1}$. On the strip $1 + \varsigma_b \neq 0$ ($\varsigma_b = -1$ only for $b \in 2 + 4\mathbb{Z}$), so $c_{3,0} = (y_0 + \varsigma_b y_b)/(1+\varsigma_b)$ and $c_{3,1} = (y_0 - y_b)/(1+\varsigma_b)$. Then

$$(1 + 2\varsigma_b^2)c_{3,0} + 3\varsigma_b c_{3,1} = \frac{(1+\varsigma_b)(1+2\varsigma_b)\, y_0 - 2\varsigma_b s_b^2\, y_b}{1+\varsigma_b},$$

and the first line of (480) gives

$$W_3(d; b) = -\frac{2N_b^3(1+2\varsigma_b)}{\pi s_b^5}\, y_0 + \frac{4N_b^3\varsigma_b}{\pi s_b^3(1+\varsigma_b)}\, y_b.$$

Inserting (521) and (522) gives (496) for every b in the strip and every $0 < d < 4$. □ (With the midpoint sample in Horn values the same formula is (515).)

Zero-mode form. With the normalized zero mode $\hat{c}_0(b) = N_b^3 c_{3,0}(q, b) = N_b^3 \int_0^1 \Xi_3(t; b)\, dt$,

$$W_3(d; b) = \frac{6\pi^2}{s_b^2}\mathfrak{t}_3(b) + \frac{4}{\pi s_b^3}\hat{c}_0(b), \qquad \hat{c}_0(b) = \frac{\varsigma_b(1+\varsigma_b)^2}{8}\widehat{M}_3(b) - \frac{\pi^3 s_b^3}{1+\varsigma_b}\mathfrak{t}_3(b). \tag{523}$$

Proof. $\hat{c}_0 = N_b^3(y_0 + \varsigma_b y_b)/(1+\varsigma_b)$ with (521) and (522) gives the second relation. Since $y_0 = c_{3,0} + \varsigma_b c_{3,1}$, one has $(1+2\varsigma_b^2)c_{3,0} + 3\varsigma_b c_{3,1} = 3y_0 - 2s_b^2 c_{3,0}$ identically (no division by ς_b occurs, so $\varsigma_b = 0$ is allowed), and the first line of (480) with (521) gives the first relation. □ This is the form quoted after (496); at $b = \frac{d}{2}$, where $W_3(d; \frac{d}{2}) = 3Q_d$ by (302), it is (497).

At $b = \frac{d}{2}$, (496) is a finite formula for the primitive, $3Q_d = u(\frac{d}{2})\mathfrak{t}_3(\frac{d}{2}) + v(\frac{d}{2})\widehat{M}_3(\frac{d}{2})$, with five real hypergeometric values at the two real nodes 0 and $\frac{1-q}{2}$, whereas (319) uses three complex samples with five blocks each. The extra input is the top-mode theorem (473). For complex b the node t_b is complex; both nodes are real for real b.

20.4.3 The five functions

At $r = 3$ and a general rim exponent the blocks of Section 20.1.3 are

$$\begin{aligned}
A_3^0(t) &= a_3^0(t)\, {}_7F_6\left(\begin{matrix}1, (\sigma+t)^{\times 3}, (1-g+t)^{\times 3} \\ (2q+t)^{\times 3}, (1+t)^{\times 3}\end{matrix}; 1\right), & a_3^0(t) &= \Big[\frac{\Gamma(\sigma+t)\Gamma(-t)}{\Gamma(2q+t)\Gamma(g-t)}\Big]^3, \\
A_3^+(t) &= a_3^+(t)\, \mathfrak{H}_+(t), & a_3^+(t) &= 4q\Big[\frac{\Gamma(1+\sigma+t)\Gamma(-t)}{\Gamma(1+2q+t)\Gamma(g-t)}\Big]^3, \\
C_3(t) &= a_3^c(t)\, \mathfrak{H}_c(t), & a_3^c(t) &= 4q\Big[\frac{\Gamma(\sigma+t)\Gamma(1-t)}{\Gamma(2q+t)\Gamma(1+g-t)}\Big]^3,
\end{aligned} \tag{524}$$

with the two Horn values

$$\begin{aligned}\mathfrak{H}_+(t) &= \sum_{j,n\ge0} \frac{d_{j+1}}{4q}\Big[\frac{(1+\sigma+t)_{j+n}}{(1+2q+t)_{j+n}}\Big]^3\Big[\frac{(1-g+t)_n}{(1+t)_n}\Big]^3,\\ \mathfrak{H}_c(t) &= \sum_{j,k\ge0} \frac{d_{j+k+1}}{4q}\Big[\frac{(1-t)_j}{(1+g-t)_j}\Big]^3\Big[\frac{(\sigma+t)_k}{(2q+t)_k}\Big]^3.\end{aligned} \tag{525}$$

These are (321) and $\mathfrak{H}[\mathsf{U}_+;\mathsf{V}_+;(1,1)]$, $\mathfrak{H}[\mathsf{U}_c;\mathsf{V}_c;(1,1)]$ of (322) at $r=3$ with the rim parameter $q\to\sigma$ and the spoke parameter $q\to g$, as stated in Section 20.1.3. Here $d_{j+1}/(4q)=(2q+1)_j(4q-1)_j/[(2q)_j(2)_j]$ by (301); it equals δ_{j0}, 1 and $1+\frac{2}{3}j$ at $d=1,2,3$.

Split $\mathfrak{t}_3=\mathfrak{t}_{3,0}+\mathfrak{t}_{3,+}$ into the spin $\ell=0$ and $\ell\ge1$, and $\widehat{M}_3=\widehat{M}_{3,0}+\widehat{M}_{3,+}+\widehat{M}_{3,c}$ by (522):

$$\begin{aligned}\mathfrak{t}_{3,0}(b) &= {}_7F_6\left(\begin{matrix}1,\sigma,\sigma,\sigma,1-g,1-g,1-g\\ 2q,2q,2q,1,1,1\end{matrix};1\right), \qquad \mathfrak{t}_{3,+}(b)=4q\Big(\frac{\sigma}{2q}\Big)^3\mathfrak{H}_+(0),\\ \widehat{M}_{3,0}(b) &= 2N_b^3A_3^0(t_b), \qquad \widehat{M}_{3,+}(b)=2N_b^3A_3^+(t_b), \quad \widehat{M}_{3,c}(b)=N_b^3C_3(t_b).\end{aligned} \tag{526}$$

The spin-zero row is the ${}_7F_6(1)$ because $T_{0n}=(\sigma)_n(1-g)_n/[(2q)_n\,n!]$, the upper parameter 1 cancelling one $n!$; $\mathfrak{t}_{3,+}$ follows from $T_{j+1,n}=\frac{\sigma}{2q}\frac{(1+\sigma)_{j+n}(1-g)_n}{(1+2q)_{j+n}\,n!}$ and $d_{j+1}=4q\cdot d_{j+1}/(4q)$. (The subscript 0 of $\mathfrak{t}_{3,0}$ and $\widehat{M}_{3,0}$ names the spin-zero block; these are not Fourier coefficients $c_{3,0}$.) At the self-mirror node, $\sigma+t_b=1-t_b=\frac{1+\sigma}{2}$ and $2q+t_b=1+g-t_b=2q+\frac{1-\sigma}{2}$, so the two sequences in $\mathfrak{H}_c(t_b)$ coincide and

$$\mathfrak{H}_c(t_b)=\sum_{j,k\ge0}\frac{d_{j+k+1}}{4q}\,\mathfrak{q}_j\mathfrak{q}_k, \qquad \mathfrak{q}_k=\Big[\frac{(\frac{1+\sigma}{2})_k}{(2q+\frac{1-\sigma}{2})_k}\Big]^3.$$

The remaining parameters at t_b are $1-g+t_b=\frac{3+\sigma}{2}-2q$ and $1+t_b=\frac{3-\sigma}{2}$. Every parameter is affine in b, so the b-derivatives of the five functions are directional parameter derivatives (table below). The prefactors a_3^0,a_3^+,a_3^c at t_b and N_b contribute only polygamma values at fixed arguments. So $\mathfrak{t}_3$ is one ${}_7F_6(1)$ and one two-variable Horn value, and $\widehat{M}_3$ is one ${}_7F_6(1)$ and two two-variable Horn values, as stated in Section 20.2.

function	upper parameters (slope in b)	lower parameters (slope in b)	at $b=d/2$
$\mathfrak{t}_{3,0}$: ${}_7F_6$ at $t=0$	$\sigma^{\times3}$ $(\frac12)$, $(1-g)^{\times3}$ $(\frac12)$	$(2q)^{\times3}$, $1^{\times3}$ (0)	$q,\ 1-q;\ 2q,\ 1$
$\mathfrak{H}_+(0)$	$1+\sigma$ $(\frac12)$, $1-g$ $(\frac12)$	$1+2q$, 1 (0)	$1+q,\ 1-q;\ 1+2q,\ 1$
$A_3^0(t_b)$: ${}_7F_6$	$\frac{1+\sigma}{2}$ $(\frac14)$, $\frac{3+\sigma}{2}-2q$ $(\frac14)$	$2q+\frac{1-\sigma}{2}$ $(-\frac14)$, $\frac{3-\sigma}{2}$ $(-\frac14)$	$\frac{1+q}{2}$, $\frac{3-3q}{2}$; $\frac{1+3q}{2}$, $\frac{3-q}{2}$
$\mathfrak{H}_+(t_b)$	$\frac{3+\sigma}{2}$ $(\frac14)$, $\frac{3+\sigma}{2}-2q$ $(\frac14)$	$1+2q+\frac{1-\sigma}{2}$ $(-\frac14)$, $\frac{3-\sigma}{2}$ $(-\frac14)$	$\frac{3+q}{2}$, $\frac{3-3q}{2}$; $\frac{3+3q}{2}$, $\frac{3-q}{2}$
$\mathfrak{H}_c(t_b)$	$\frac{1+\sigma}{2}$ $(\frac14)$, twice	$2q+\frac{1-\sigma}{2}$ $(-\frac14)$, twice	$\frac{1+q}{2}$; $\frac{1+3q}{2}$

The table lists the parameters of the five hypergeometric functions of (526) (each parameter repeated three times, $r=3$) and their slopes in b. The multiplicity factor $d_{j+1}/(4q)$ (or $d_{j+k+1}/(4q)$) of the Horn values and the prefactor $4q$ do not depend on b.

20.4.4 Rim jets and their series

Primes denote ∂_b; by Section 20.4.1 every function above is analytic in b on the strip. For $\frac23<d<4$, at $b=\frac{d}{2}$,

$$9Z_d=u''\mathfrak{t}_3+2u'\mathfrak{t}_3'+u\mathfrak{t}_3''+v''\widehat{M}_3+2v'\widehat{M}_3'+v\widehat{M}_3'', \tag{527}$$

with $\mathfrak{t}_3^{(k)}=\mathfrak{t}_{3,0}^{(k)}+\mathfrak{t}_{3,+}^{(k)}$ and $\widehat{M}_3^{(k)}=\widehat{M}_{3,0}^{(k)}+\widehat{M}_{3,+}^{(k)}+\widehat{M}_{3,c}^{(k)}$ the jets of the five functions (526), and, with S,C of (459),

$$\begin{aligned}u&=\frac{2\pi^2(1+2C)}{S^2}, \quad u'=-\frac{2\pi^3(1+C+C^2)}{S^3}, \quad u''=\frac{\pi^4(1+5C+2C^2+C^3)}{S^4},\\ v&=\frac{C(1+C)^2}{2\pi S^3}, \qquad v'=-\frac{(1+C)^2(1+2C)}{4S^4}, \quad v''=\frac{\pi(1+C)^3(2+C)}{4S^5}.\end{aligned} \tag{528}$$

Proof. By (244), $9Z_d = \partial_b^2 W_3(d;b)$ at $b=\frac{d}{2}$, which lies in the strip for $\frac{2}{3}<d<4$. Differentiate (496) twice, termwise by Section 20.4.1, using $\partial_b s_b = \frac{\pi}{2}\varsigma_b$ and $\partial_b \varsigma_b = -\frac{\pi}{2}s_b$; for example $u' = -2\pi^3[s_b^2+\varsigma_b(1+2\varsigma_b)]/s_b^3$ and $s_b^2+\varsigma_b+2\varsigma_b^2 = 1+\varsigma_b+\varsigma_b^2$. □ Formula (527) is exact differentiation and does not use the input $\partial_b W_3(d;\frac{d}{2})=0$.

Series form of the jets. On the strip,

$$\begin{aligned} \mathfrak{t}_3' &= 3\sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3 \mu_1, & \mathfrak{t}_3'' &= \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3 (9\mu_1^2+3\mu_2),\\ \widehat{M}_3' &= 3\sum_{\ell\ge 0} d_\ell \sum_s [N_b g_\ell(s;b)]^3 w_1, & \widehat{M}_3'' &= \sum_{\ell\ge 0} d_\ell \sum_s [N_b g_\ell(s;b)]^3 (9w_1^2+3w_2), \end{aligned} \tag{529}$$

with $s\in\frac{\ell+1}{2}+\mathbb{Z}$ and the explicit weights (arguments $(\ell,n;b)$ of μ_m and $(\ell,s;b)$ of w_m suppressed)

$$\begin{aligned} \mu_1 &= \tfrac12[\psi_0(\sigma+\ell+n)-\psi_0(\sigma)+\psi_0(1-g+n)-\psi_0(1-g)],\\ \mu_2 &= \tfrac14[\psi_1(\sigma+\ell+n)-\psi_1(\sigma)+\psi_1(1-g+n)-\psi_1(1-g)],\\ w_1 &= -\tfrac12[\psi_0(g)+\psi_0(\sigma)]+\tfrac14\textstyle\sum_\pm[\psi_0(a_\ell\pm s)+\psi_0(\beta_\ell\pm s)],\\ w_2 &= \tfrac14[\psi_1(g)-\psi_1(\sigma)]+\tfrac{1}{16}\textstyle\sum_\pm[\psi_1(a_\ell\pm s)-\psi_1(\beta_\ell\pm s)]. \end{aligned} \tag{530}$$

At $b=\frac{d}{2}$, μ_1,μ_2 are those of (253), and w_1,w_2 are the spectral weights of (245) continued to $s=\mathrm{i}\nu$. Consequently

$$9Z_d = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3\big[(9\mu_1^2{+}3\mu_2)\,u{+}6\mu_1\,u'{+}u''\big]+\sum_{\ell\ge 0} d_\ell \sum_{s\in\frac{\ell+1}{2}+\mathbb{Z}} [N_b g_\ell(s;b)]^3\big[(9w_1^2{+}3w_2)\,v{+}6w_1\,v'{+}v''\big], \tag{531}$$

at $b=\frac{d}{2}$, with $u,\dots,v''$ of (528).

Proof. $\mu_1=\partial_b\log T_{\ell n}$ and $\mu_2=\partial_b\mu_1$, since $\partial_b\sigma=\frac12=\partial_b(1-g)$. At fixed s, $w_1=\partial_b\log[N_b g_\ell(s;b)]$ and $w_2=\partial_b w_1$, since $\partial_b a_\ell=\frac14=-\partial_b\beta_\ell$ and $\partial_b\log N_b=-\frac12[\psi_0(g)+\psi_0(\sigma)]$. The midpoint lattice does not move with b, so (529) are the termwise derivatives of (495), which Section 20.4.1 justifies; (531) is (527) with these series. The weights were also checked term by term against numerical differentiation, to 10^{-30} (`LRI_C5_general_d/Z/check_weights.py`). □

Comparison with (257). The series (257) is $9Z_d=\sum d_\ell T_{\ell n}^3[(9\mu_1^2+3\mu_2)F_3+6\mu_1F_{3,1}+F_{3,2}]$, with the index-dependent triple-pole weight F_3 of (254) and its rim derivatives $F_{3,1},F_{3,2}$. In (531) the same structure appears with $F_3,F_{3,1},F_{3,2}$ replaced by the index-independent elementary numbers u,u',u'', at the price of the midpoint-lattice term. In that term $\frac19\widehat{M}_3''$ is the midpoint-lattice sum of the spectral integrand $t_\ell^3(w_1^2+w_2/3)$ of (246), whose integral is Z_d itself.

20.4.5 The first-derivative identity and the simplified formula for Z_d

First rim derivative. $\partial_b W_3(d;\frac{d}{2})=0$ is equivalent to $u'\mathfrak{t}_3+u\mathfrak{t}_3'+v'\widehat{M}_3+v\widehat{M}_3'=0$, that is, at $b=\frac{d}{2}$,

$$8\pi^3S^2(1+2C)\,\mathfrak{t}_3'+2SC(1+C)^2\,\widehat{M}_3' = 8\pi^4S(1+C+C^2)\,\mathfrak{t}_3+\pi(1+C)^2(1+2C)\,\widehat{M}_3, \tag{532}$$

and, in the zero-mode form (523), to

$$3\pi^3S^2\,\mathfrak{t}_3'+2S\,\hat{c}_0' = 3\pi C\,(\pi^3S\,\mathfrak{t}_3+\hat{c}_0). \tag{533}$$

Proof. The first is $4\pi S^4\,\partial_b W_3$ with (528). For the second write $W_3=u_0\mathfrak{t}_3+v_0\hat{c}_0$ with $u_0=6\pi^2/s_b^2$ and $v_0=4/(\pi s_b^3)$, by (523); at $b=\frac{d}{2}$, $u_0'=-6\pi^3C/S^3$ and $v_0'=-6C/S^4$, and (533) is $\frac{\pi S^4}{2}\,\partial_b W_3=0$. □

Given the input $\partial_b W_3(d;\frac{d}{2})=0$, which is proved by Pattern T (239) with (244) (Section 20.2), the first derivative of the residue sum is not independent:

$$\mathfrak{t}_3' = \frac{\pi C}{S}\,\mathfrak{t}_3+\frac{C}{\pi^2S^2}\,\hat{c}_0-\frac{2}{3\pi^3S}\,\hat{c}_0' \qquad (b=\tfrac{d}{2}) \tag{534}$$

for every d. In the basis $(\mathfrak{t}_3, \widehat{M}_3)$, (532) eliminates $\mathfrak{t}_3'$ for $d \neq \frac{8}{3}$ (where $1+2C=0$), or $\widehat{M}_3'$ for $d \neq 2$ (where $C=0$). At $d=1$, (533) reads $\mathfrak{t}_3'(1;\frac{1}{2}) = \pi\,\mathfrak{t}_3(1;\frac{1}{2})$, which Dixon's closed form (546) confirms. At $d=3$, with the Gamma form of $\hat{c}_0(3;b)$ (Section 20.6), it reads $\mathfrak{t}_3'(3;\frac{3}{2}) = -\pi\,\mathfrak{t}_3(3;\frac{3}{2}) + \Gamma(\frac{1}{4})^{10}/(64\pi^{9/2})$, an identity also found in the $d=3$ computation of `LRI_C5_d3/wf_Z3/`.

Proof of (498). Assume $\partial_b W_3(d;\frac{d}{2}) = 0$. At $b=\frac{d}{2}$, with $\hat{c}_0^{(k)} = \partial_b^k \hat{c}_0|_{b=d/2}$,

$$Z_d = \frac{2\pi^2}{3S^2}\,\mathfrak{t}_3'' - \frac{\pi^4\cos(\pi d/2)}{3S^4}\,\mathfrak{t}_3 + \frac{1}{9}\Big[\frac{3\pi}{S^3}\,\hat{c}_0 - \frac{4C}{S^4}\,\hat{c}_0' + \frac{4}{\pi S^3}\,\hat{c}_0''\Big], \tag{535}$$

and, with (497), equivalently (498). Indeed, at $b=\frac{d}{2}$,

$$u_0' = -\frac{6\pi^3 C}{S^3}, \qquad u_0'' = \frac{3\pi^4(1+2C^2)}{S^4}, \qquad v_0' = -\frac{6C}{S^4}, \qquad v_0'' = \frac{3\pi(1+3C^2)}{S^5}.$$

Insert (534) into $9Z_d = W_3'' = u_0''\mathfrak{t}_3 + 2u_0'\mathfrak{t}_3' + u_0\mathfrak{t}_3'' + v_0''\hat{c}_0 + 2v_0'\hat{c}_0' + v_0\hat{c}_0''$. The coefficient of $\mathfrak{t}_3$ becomes $u_0'' + 2u_0'\pi C/S = 3\pi^4(1-2C^2)/S^4 = -3\pi^4\cos(\pi d/2)/S^4$; that of $\hat{c}_0$ becomes $v_0'' + 2u_0' C/(\pi^2 S^2) = 3\pi(1-C^2)/S^5 = 3\pi/S^3$; that of $\hat{c}_0'$ becomes $2v_0' - 4u_0'/(3\pi^3 S) = -4C/S^4$. This is (535). Replacing $\mathfrak{t}_3$ by $\frac{S^2}{6\pi^2}(3Q_d - \frac{4}{\pi S^3}\hat{c}_0)$ from (497) turns the $\mathfrak{t}_3$ term into $-\frac{\pi^2\cos(\pi d/2)}{6S^2}Q_d + \frac{2\pi\cos(\pi d/2)}{9S^5}\hat{c}_0$, and $3S^2 + 2\cos 2\pi q = 1 + C^2$ gives (498). □

The same computation shows that the full jet formula (527) and (535) differ exactly by $-\frac{2\pi C}{9S}\,\partial_b W_3(d;\frac{d}{2})$ (the review `LRI_C5_general_d/review/Z-rigor/` confirmed this symbolically). So (498) depends on the input only through this term, which is absent at $d=2$ where $C=0$. In (535) the second derivatives enter only through $\frac{2\pi^2}{3S^2}\mathfrak{t}_3'' + \frac{4}{9\pi S^3}\hat{c}_0''$, whose second-derivative part is $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$, exactly as in (527). The data needed in (535) are $\mathfrak{t}_3$, $\mathfrak{t}_3''$ and the jets of the zero mode, which is computed from the two nodes by (523); so the complete set of inputs remains $\{\mathfrak{t}_3^{(k)}, \widehat{M}_3^{(k)}\}_{k\le 2}$, that is, at most 15 special-function numbers (two ${}_7F_6(1)$ values and three two-variable Horn values, each with its first and second b-derivatives), subject to the two exact relations $W_3(d;\frac{d}{2}) = 3Q_d$ and (532).

The count. The 15 numbers enter only through the six combinations $\mathfrak{t}_3^{(k)}, \widehat{M}_3^{(k)}$, $k \le 2$, and the second derivatives only through the single combination $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$. After the two exact relations, Z_d is an elementary multiple of Q_d plus an elementary combination of three further numbers, for instance $\mathfrak{t}_3$ (or $\hat{c}_0$), $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$ at $b=\frac{d}{2}$; this is the count of Section 20.7. The choice works at every d, because the coefficients $6\pi^2/S^2$ and $4/(\pi S^3)$ of $\mathfrak{t}_3$ and $\hat{c}_0$ in (497), and $3\pi^3 S^2$ and $2S$ of $\mathfrak{t}_3'$ and $\hat{c}_0'$ in (533), never vanish for $0<d<4$. At $d=1$ ($\hat{c}_0 \equiv 0$) and at $d=2$ ($\widehat{M}_3 \equiv 0$) only the second-derivative combination remains, and Section 20.6 evaluates it in closed form. All functions used here belong to the classes of (321) and (322), with the rim parameter σ and the spoke parameter g separated.

20.5 The pole node alone: residue series of one lower Laurent order

At $t=0$ the left and middle half-lattices carry $[\sin\pi t]^r$, so $\Xi_r^{(k)}(0;b)$ for $k \le r-1$ comes from the right half-lattice alone. With $r-1$ modes, $k \le r-2$ suffices. Let $\lambda_m(\ell,n;b)$ be (252) with the rim and spoke separated,

$$\lambda_m = \psi_{m-1}(\ell+n+\sigma) - \psi_{m-1}(\ell+n+2q) - (-1)^m\psi_{m-1}(g-n) + (1+(-1)^m)\psi_{m-1}(1) - \psi_{m-1}(n+1),$$

so that $N_b g_\ell(a_\ell + n + u; b) = -\frac{T_{\ell n}(b)}{u}\exp\sum_m \lambda_m \frac{u^m}{m!}$ as in (262). Then, with $\kappa_b = \pi\cot\frac{\pi b}{2}$,

$$\boxed{W_3(d;b) = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^3\big[2\pi^2 - 6\kappa_b\lambda_1\big], \qquad W_4(d;b) = \sum_{\ell,n\ge 0} d_\ell T_{\ell n}^4\Big[4\kappa_b(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1\Big].} \tag{536}$$

At $b = d/2$, $\kappa_b = D$ and the λ_m are those of (252):

$$3Q_d = \sum d_\ell T_{\ell n}^3 (2\pi^2 - 6D\lambda_1), \qquad P_4(d) = \sum d_\ell T_{\ell n}^4 \Big[4D(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1\Big], \tag{537}$$

to be compared with F_3 (quadratic, λ_2) and F_4 (cubic, λ_3) of (255). Differentiating (536) in b (termwise, as for (256); $\partial_b\lambda_m = \eta_m$, $\partial_b\eta_m = \theta_m$, $\partial_b \log T_{\ell n} = \mu_1$, $\partial_b\mu_1 = \mu_2$ of (253); $\kappa_b' = -\frac{\pi^2}{2S^2}$, $\kappa_b'' = \frac{\pi^3 C}{2S^3}$):

$$\begin{aligned}
&\mathcal{L}_d = \sum d_\ell T_{\ell n}^4 [4\mu_1 \widehat{F}_4 + \widehat{F}_4'], && 9Z_d = \sum d_\ell T_{\ell n}^3 [(9\mu_1^2 + 3\mu_2)\widehat{F}_3 + 6\mu_1 \widehat{F}_3' + \widehat{F}_3''],\\
&\widehat{F}_4 = 4D(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) - 8\pi^2\lambda_1, && \widehat{F}_4' = -\tfrac{2\pi^2}{S^2}(4\lambda_1^2 + \lambda_2 - \tfrac{\pi^2}{3}) + 4D(8\lambda_1\eta_1 + \eta_2) - 8\pi^2\eta_1,\\
&\widehat{F}_3 = 2\pi^2 - 6D\lambda_1, && \widehat{F}_3' = \tfrac{3\pi^2}{S^2}\lambda_1 - 6D\eta_1, \qquad \widehat{F}_3'' = -\tfrac{3\pi^3 C}{S^3}\lambda_1 + \tfrac{6\pi^2}{S^2}\eta_1 - 6D\theta_1.
\end{aligned} \tag{538}$$

Here $\widehat{F}_3$, $\widehat{F}_4$ are the weights of (537), the analogues of F_3, F_4 of (254), and primes are their b-derivatives. These replace (256) and (257): the polygamma weights drop by one order (λ_3, η_3 no longer occur in $\mathcal{L}_d$, and $\lambda_2, \eta_2, \theta_2$ no longer occur in Z_d). In this form $P_4, \mathcal{L}_d$ need one Horn family and its parameter derivatives of order ≤ 2 (shift) and ≤ 1 (rim); Q_d, Z_d need one family with orders ≤ 1 and ≤ 2. Status: derived (exact algebra, `LRI_C5_general_d/specialize/sym_taylor.py`); (536) is confirmed numerically, and (538) through b-derivatives of (536).

20.6 Specialization to $d = 1, 2, 3$

The one-variable function behind $d = 1$ and $d = 2$. Let $\Omega_r(t; x) = \sin^r \pi(t + x) \sum_{n\in\mathbb{Z}} [\Gamma(n + t + x)/\Gamma(n + t + 1)]^r$. At $x = \frac{1}{2}$ this is the $\Omega_r(t)$ of (335), and $\Omega_4(t; x)$ is the $\Omega(t; x)$ of (349). For every b in the strip,

$$d = 1: \ \Xi_r(t; b) = (-1)^r 2^{-rb}\, \Omega_r(2t; b), \qquad d = 2: \ \Xi_r(t; b) = (-1)^r\, \Omega_r(t; \tfrac{b}{2})^2. \tag{539}$$

Proof at $d = 1$. Here $d_0 = d_1 = 1$, $d_\ell = 0$ for $\ell \ge 2$, $2q = \frac{1}{2}$ and $1 - g = \frac{1}{2} + \sigma$. In the reflected form (464), valid for every $n \in \mathbb{Z}$, Legendre's duplication formula $\Gamma(w)\Gamma(w + \frac{1}{2}) = 2^{1-2w}\sqrt{\pi}\,\Gamma(2w)$ [26] turns the $\ell = 0$ and $\ell = 1$ Gamma ratios at $z = n + t$ into $2^{1-2\sigma}\Gamma(2z + b)/\Gamma(2z + 1)$ and $2^{1-2\sigma}\Gamma(2z + 1 + b)/\Gamma(2z + 2)$. These are the terms $m = 2n$ and $m = 2n + 1$ of the bilateral sum of $\Omega_r(2t; b)$, so the two rows merge into one lattice. The prefactor is $D_b(t) \sin \pi(g - t)/\sin \pi t = \sin \pi(t + \sigma) \cos \pi(t + \sigma) = \frac{1}{2} \sin \pi(2t + b)$. The sign of (464) gives $(-1)^r$, and the powers of two combine to $2^{-r}2^{r(1-2\sigma)} = 2^{-rb}$.

Proof at $d = 2$. Here $d_0 = 1$, $d_\ell = 2$ for $\ell \ge 1$ (by (461)), $2q = 1$ and $g = 1 - \sigma$. With $\phi(z) = \Gamma(z + \sigma)/\Gamma(z + 1)$ (local to this proof; not the φ of (501)) the factorized form in (464) and Gamma reflection give $f(\ell + \sigma + n + t) = \phi(\ell + n + t)$ and $f(-n - t) = -[\sin \pi(\sigma + t)/\sin \pi t]\, \phi(n + t)$, so $D_b(t)\, g_\ell(a_\ell + n + t) = -\sin^2 \pi(t + \sigma)\, \phi(n + t)\phi(\ell + n + t)$. With $\chi_m = \phi(m + t)^r$ the multiplicities turn $\sum_{n\in\mathbb{Z}} \sum_{\ell\ge 0} d_\ell \chi_n \chi_{n+\ell}$ into the symmetrization $(\sum_{m\in\mathbb{Z}} \chi_m)^2$. All sums converge absolutely, since $|\chi_m| \asymp |m|^{-r \operatorname{Re} g}$ and $r \operatorname{Re} g > 1$ is $\operatorname{Re} b < b_{\max}(r, 2)$. This is the second identity. Both identities are first obtained off the pole classes and then hold everywhere, since both sides are entire. □

Since $\Omega_r(t + 1; x) = (-1)^r \Omega_r(t; x)$ and Ω_r grows like $e^{r\pi|\operatorname{Im} t|}$ times a negative power, the frequencies of Ω_r lie in $\mathbb{Z} + \frac{r}{2}$ and have modulus at most $\frac{r}{2} - 1$. So at $d = 1, 2$ (539) gives (473) directly, and at $d = 1$ more: only the modes $k \equiv r \pmod 2$ occur, so $c_{4,1} \equiv 0$ and $c_{3,0} \equiv 0$.

Two amplitudes and their first-derivative relations. By (349), $\mathcal{N}_x^2 \Omega_4(t; x) = \mathcal{A}(x) + \mathcal{B}(x) \cos 2\pi(t + \frac{x}{2})$ with $\mathcal{N}_x = \Gamma(1 - x)/\Gamma(x)$, the N_x of (345), and Section 18.7 proves, at the single point $x = \frac{1}{2}$,

$$\mathcal{A}' = 2\pi\mathcal{B} - 2\log 2\, \mathcal{A}, \qquad \mathcal{B}' = \tfrac{3\pi}{2}\mathcal{A} - 2\log 2\, \mathcal{B}, \qquad \mathcal{A}(\tfrac{1}{2}) = \frac{8N_{\mathrm{e}}}{\pi}, \quad \mathcal{B}(\tfrac{1}{2}) = \frac{8M_{\mathrm{e}}}{\pi}, \tag{540}$$

with $M_{\rm e}=(B_-+B_+)/64$, $N_{\rm e}=(B_--B_+)/32$ of (338) and x-derivatives. These are first-derivative relations at $x=\frac12$ (from (351) and the elliptic and Landen identities (356)), not a differential system in x; equivalently $(4^x\mathcal{A})'=2\pi\,4^x\mathcal{B}$ and $(4^x\mathcal{B})'=\frac{3\pi}{2}\,4^x\mathcal{A}$ there. They are what makes the rim derivative collapse at $d=1$ and $d=2$: both dimensions sample the *same* Ω_4 at the *same* point $x=\frac12$ ($b=\frac12$ at $d=1$, $b/2=\frac12$ at $d=2$).

$d=1$**.** By (539) with $x=b$: $c_{4,0}=2^{-4b}\mathcal{A}(x)/\mathcal{N}_x^2$, $c_{4,1}\equiv0$, $c_{4,2}=2^{-4b}\mathcal{B}(x)/\mathcal{N}_x^2$, so $\mathcal{X}_0\equiv\mathcal{X}_{1/2}$ and only two samples are independent. At $b=\frac12$ ($N_b=\sqrt{\pi}$, $\varsigma_b=s_b=2^{-1/2}$, $\partial_b\log N_b=-\psi_0(\frac14)$), (540) gives

$$\begin{aligned}P_4(1)&=32\pi\mathcal{A}+40\pi\mathcal{B},\\ \mathcal{L}_1&=-2(\pi+\log2)\,P_4(1)+52\pi^2\mathcal{A}+64\pi^2\mathcal{B}=-(96\pi+512\log2)N_{\rm e}-(128\pi+640\log2)M_{\rm e},\end{aligned}\tag{541}$$

which is (107), that is (33): $\mathcal{L}_1=-\frac{3\mathsf{A}-\mathsf{D}}{6}P_4-\frac{2\mathsf{D}}{3}B_-$ with $\mathsf{A}=\pi+4\log2$, $\mathsf{D}=\pi$ and $B_-=8\pi(c_{4,0}+2c_{4,2})$. The frozen derivative is elementary in the two amplitudes, $\mathfrak{D}_L=\sqrt2\,(5\mathcal{X}_0'-\mathcal{X}_T')=\frac{\sqrt2\pi^2}{4}\big[(\frac{31\pi}{2}-8\log2)\mathcal{A}+(19\pi-10\log2)\mathcal{B}\big]$, and the single Horn derivative is $\mathfrak{h}_x'=4\sum_k c_k(\frac12)^4[\psi_0(k+\frac12)-\psi_0(\frac12)]=4\mathcal{S}_1$ of (351), with $c_k(x)$ of (345) and $\mathfrak{h}_x(1;b)={}_4F_3(b,b,b,b;1,1,1;1)$.

$d=2$**.** By (539) with $x=b/2$: $c_{4,0}=(\mathcal{A}^2+\frac12\mathcal{B}^2)/\mathcal{N}_x^4$, $c_{4,1}=2\mathcal{A}\mathcal{B}/\mathcal{N}_x^4$, $c_{4,2}=\mathcal{B}^2/(2\mathcal{N}_x^4)$, that is $\mathcal{X}_T=\mathcal{A}^2$, $\mathcal{X}_0=(\mathcal{A}+\mathcal{B})^2$, $\mathcal{X}_{1/2}=(\mathcal{A}-\mathcal{B})^2$ at $b=1$. Here $C=0$, so $D=0$ and ς_b passes through zero: the weight of $\mathcal{X}_T'$ in (494) vanishes, and $\partial_b\log(N_b/\mathcal{N}_x)=0$. Then

$$\mathcal{L}_2=-(6\mathcal{A}^2+8\mathcal{B}^2)+\frac{2}{\pi}(\mathcal{X}_0'-\mathcal{X}_{1/2}')=-(6\mathcal{A}^2+8\mathcal{B}^2)+\frac{4}{\pi}\big[\mathcal{A}'\mathcal{B}+\mathcal{A}\mathcal{B}'\big]=-\frac{16\log2}{\pi}\mathcal{A}\mathcal{B}=-2\log2\,P_4(2),$$

with x-derivatives on $\mathcal{A},\mathcal{B}$ (so $\partial_b=\frac12\partial_x$), which is (350) and (270). The motion of the trigonometric weights ($\partial_b\varsigma_b=-\frac{\pi}{2}$) produces $-(6\mathcal{A}^2+8\mathcal{B}^2)$, which the rotation terms $2\pi\mathcal{B}$, $\frac{3\pi}{2}\mathcal{A}$ of (540) cancel exactly; only the normalization 4^x survives. At $d=1$ the same relations leave $-\frac{2\mathsf{D}}{3}B_-$. If A and D in (33) are replaced by their general-d values A of (221) and D of (459), the B_- term vanishes at $d=2$ and the formula becomes exactly $\mathcal{L}_2=-\frac{A}{2}P_4(2)$ with $A=4\log2$, which is (270). At $d=3$ no analogue of B_- with elementary coefficients in the samples was found (an integer-relation search by lattice reduction at 155 digits, repeated at 125 digits, in `LRI_C5_d3`, found no relation of $\mathcal{L}_3$ with the $P_4(3)$ blocks; a numerical statement). In (537) the same $D=0$ leaves $3Q_2=2\pi^2\sum d_\ell T_{\ell n}^3$ (one unweighted Horn value, the complex Selberg value of (264)) and $P_4(2)=-8\pi^2\sum d_\ell T_{\ell n}^4\lambda_1$.

The half-shift identities at $d=3$ for a general rim exponent. The $d=3$ forms of $\mathcal{L}_d$ and Z_d below need two identities at rim exponents $b\neq\frac32$: $c_{4,1}(b)=2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$, and a Gamma form of $c_{3,0}(b)$. Both follow from the half-shift proof of Section 19.4, repeated at a general b. The derivation is this project's own work: a Claude Code session and its subagents made it in the package `LRI_C5_d3/` (2 October 2026), where it was recorded as a sketch that follows that proof step by step, and three separate AI sessions per statement tried to refute the statements, re-deriving steps by hand or with codes of their own, without finding an error. The derivation is written out here; as printed it has not been reviewed. At $d=3$ one has $q=\frac34$, $\sigma=\frac{b}{2}$ and $g=\frac32-\sigma=\frac{3-b}{2}$. In this paragraph x, y, u and v are local, as in Section 19.4, and

$$h_b(z)=\frac{\Gamma(z+\frac{1-g}{2})}{\Gamma(z+\frac{1+g}{2})},\qquad u=t+\frac{1-g}{2}.$$

At $b=\frac32$, $\frac{1-g}{2}=\frac18$, h_b is the function h of (406) and $u=t+\frac18$.

Lattice. Put $k=\ell+n$, $j=n$, $x=k+u+\frac12$ and $y=j+u$. The first line of (464) gives

$$g_\ell(a_\ell+n+t;b)=-\frac{\sin\pi(g-t)}{\sin\pi t}\,h_b(x)\,h_b(y),\qquad d_\ell=2\ell+1=2(x-y),\tag{542}$$

which is (427) at $b=\frac32$. Since $g=\frac32-\sigma$, the prefactor of Ξ_r in (462) combines as $D_b(t)\cdot(-\sin\pi(g-t)/\sin\pi t)=\frac12\sin2\pi(t+\sigma)=-\sin\pi(u+\frac{1-g}{2})\sin\pi(u+\frac12+\frac{1-g}{2})$, so that

$$\Xi_r(t;b)=(-1)^r\Big[\sin\pi\big(u+\tfrac{1-g}{2}\big)\sin\pi\big(u+\tfrac12+\tfrac{1-g}{2}\big)\Big]^r\sum_{k\ge j}2(x-y)\,h_b(x)^r\,h_b(y)^r,$$

summed over all integers $k\ge j$. At $b=\frac32$ the bracket is $\sin\pi(t+\frac14)\sin\pi(t+\frac34)=\frac12\cos2\pi t$, the factor used in Section 19.4.

Half shift. For $m=0,1$ define the regularized bilateral moments

$$\Psi_{r,m}(v;b)=\sin^r\pi\big(v+\tfrac{1-g}{2}\big)\sum_{n\in\mathbb{Z}}(n+v)^m\,h_b(n+v)^r.$$

By the reflection $h_b(-z)=h_b(z)\sin\pi(z+\frac{1-g}{2})/\sin\pi(\frac{1-g}{2}-z)$ they are the functions (407) with $\omega=+1$ and $\frac18,\frac78$ replaced by $\frac{1-g}{2},\frac{1+g}{2}$; at $b=\frac32$ they are $\Psi^{(+)}_{r,m}$. Their series converge absolutely for $r\,\mathrm{Re}\,g>m+1$, since $|h_b(z)|$ is comparable to $|z|^{-\mathrm{Re}\,g}$. The shift $t\mapsto t+\frac12$ moves u by $\frac12$, exchanges the lattices $\mathbb{Z}+u+\frac12$ of x and $\mathbb{Z}+u$ of y, and changes the sign of the trigonometric bracket. If $r\,\mathrm{Re}\,g>2$, all sums involved converge absolutely, and the reindexing of Section 19.4 gives (429) at general b:

$$\Xi_r(t;b)-(-1)^r\,\Xi_r(t+\tfrac12;b)=2(-1)^r\big[\Psi_{r,1}(u+\tfrac12;b)\Psi_{r,0}(u;b)-\Psi_{r,0}(u+\tfrac12;b)\Psi_{r,1}(u;b)\big].$$

Finite modes. As in (430), $\Psi_{r,m}(v+1;b)=(-1)^r\Psi_{r,m}(v;b)$, $\Psi_{r,m}(-v;b)=(-1)^m\Psi_{r,m}(v;b)$, and $|\Psi_{r,m}(v;b)|=O(e^{r\pi|\mathrm{Im}\,v|}|\mathrm{Im}\,v|^{m+1-r\,\mathrm{Re}\,g})$ as $|\mathrm{Im}\,v|\to\infty$, now with the exponent $r\,\mathrm{Re}\,g$ in place of $\frac{3r}{4}$. Contour translation excludes the boundary frequencies when $m+1<r\,\mathrm{Re}\,g$. For $m\le1$ this is $\mathrm{Re}\,g>\frac12$, that is $\mathrm{Re}\,b<2=b_{\max}(4,3)$, for $r=4$, and $\mathrm{Re}\,g>\frac23$, that is $\mathrm{Re}\,b<\frac53=b_{\max}(3,3)$, for $r=3$, by (463); these are also the conditions of the half shift. Reflection then leaves the modes of (408),

$$\Psi_{4,0}=\mathfrak{a}_4(b)+\mathfrak{c}_4(b)\cos2\pi v,\quad \Psi_{4,1}=\mathfrak{b}_4(b)\sin2\pi v,\quad \Psi_{3,0}=\mathfrak{a}_3(b)\cos\pi v,\quad \Psi_{3,1}=\mathfrak{b}_3(b)\sin\pi v,\tag{543}$$

which define the amplitudes at general b; at $b=\frac32$ they are those of (408).

Fourier coefficients. By (468), $\Xi_r(t;b)=\sum_k c_{r,k}(\frac34,b)\cos2\pi k(u+\frac14)$, since $t+\frac b4=u+\frac14$. Inserting (543) in the half-shift identity gives $\Xi_4(t;b)-\Xi_4(t+\frac12;b)=-4\mathfrak{a}_4\mathfrak{b}_4\sin2\pi u$ and $\Xi_3(t;b)+\Xi_3(t+\frac12;b)=-2\mathfrak{a}_3\mathfrak{b}_3$, while the left sides are $-2c_{4,1}\sin2\pi u+2c_{4,3}\sin6\pi u$ and $2c_{3,0}-2c_{3,2}\cos4\pi u$. Hence, for $0<\mathrm{Re}\,b<2$ and $0<\mathrm{Re}\,b<\frac53$ respectively,

$$c_{4,1}(\tfrac34,b)=2\mathfrak{a}_4(b)\mathfrak{b}_4(b),\qquad c_{3,0}(\tfrac34,b)=-\mathfrak{a}_3(b)\mathfrak{b}_3(b),\tag{544}$$

and $c_{4,3}=c_{3,2}=0$, as (473) also gives. At $b=\frac32$ this is (431).

Gamma evaluations. Let $1<b<\frac53$ for $r=3$ and $1<b<2$ for $r=4$, so that $\frac12<g<1$. At the node $v=\frac{1-g}{2}$ the second term of (407) carries the factor $\sin^r\pi(\frac{1-g}{2}-v)=0$, and $h_b(n+\frac{1-g}{2})=\Gamma(n+1-g)/\Gamma(n+1)$. With $\mathfrak{U}_{r,m}=\sum_{n\ge0}(n+\frac{1-g}{2})^m h_b(n+\frac{1-g}{2})^r$, which is $\mathfrak{U}^{(+)}_{r,m}(\frac{1-g}{2})$ of (406) at general b,

$$\mathfrak{a}_3(b)=\frac{\sin^3(\pi g)}{\sin(\pi g/2)}\,\mathfrak{U}_{3,0},\qquad \mathfrak{b}_3(b)=\frac{\sin^3(\pi g)}{\cos(\pi g/2)}\,\mathfrak{U}_{3,1},\qquad \mathfrak{b}_4(b)=\sin^3(\pi g)\,\mathfrak{U}_{4,1}.$$

Dixon's sum [22, eq. 16.4.4] with $a=b=c=1-g$, and the Rogers–Dougall sum [22, eq. 16.4.9] with $(a,b,c,d)=(1-g,1-g,1-g,1-\frac g2)$, after inserting the canceled pair $1-\frac g2$, and with $(a,b,c,d)=(1-g,1-g,1-g,1-g)$ (in the notation of [22]; these b,c,d are not the rim exponent or the dimension) give

$$\mathfrak{U}_{3,0}=\frac{\Gamma(1-g)^3\,\Gamma(\frac{3-g}{2})\,\Gamma(\frac{3g-1}{2})}{\Gamma(2-g)\,\Gamma(\frac{1+g}{2})^2\,\Gamma(g)},\qquad \mathfrak{U}_{3,1}=\frac{1-g}{2}\,\frac{\Gamma(1-g)^3\,\Gamma(\frac{2-g}{2})\,\Gamma(\frac{3g-2}{2})}{\Gamma(2-g)\,\Gamma(g)\,\Gamma(\frac g2)^2},$$

$$\mathfrak{U}_{4,1}=\frac{1-g}{2}\,\frac{\Gamma(1-g)^4\,\Gamma(2g-1)}{\Gamma(2-g)\,\Gamma(g)^3}.$$

Their convergence conditions hold: $g > \frac{1}{3}$ for Dixon's sum, $g > \frac{2}{3}$ and $g > \frac{1}{2}$ for the two Rogers–Dougall sums. Duplication and reflection [26] then give

$$\mathfrak{a}_3(b)\mathfrak{b}_3(b) = \frac{\pi^5\,\Gamma(3g-2)}{\Gamma(g)^9}, \qquad \mathfrak{b}_4(b) = \frac{\pi^3\,\Gamma(2g-1)}{2\,\Gamma(g)^6}. \tag{545}$$

Both sides are analytic in b on the strips of (544), where the poles of $\Gamma(3g-2)$ and $\Gamma(2g-1)$ ($\operatorname{Re} b \geq \frac{5}{3}$ and $\operatorname{Re} b \geq 2$) do not occur, so (545) holds there by analytic continuation. At $b = \frac{3}{2}$ it is (411). Consequently $\hat{c}_0(3;b) = N_b^3 c_{3,0} = -N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$, and, with ν_b of the next paragraph, $\nu_b^{1/2}\mathfrak{b}_4 = \pi^3\Gamma(2g-1)/[2\Gamma(\sigma)^2\Gamma(g)^4]$, whose logarithmic b-derivative at $b = \frac{3}{2}$ is $\psi_0(\frac{3}{4}) - \psi_0(\frac{1}{2}) = \frac{\pi}{2} - \log 2$.

Checks (records in `LRI_C5_d3/`; numerical, not interval bounds). A code of one of the verifiers (`wf_vL3num/`, records `modes_b*.json`), which shares no summation routine with the evaluation code `wf_L3/gen3.py`, fits $\Psi_{4,0}$, $\Psi_{4,1}$ and Ξ_4 at 90 digits at $b = 0.4$, 0.8, 1.2, 1.5, 1.6, 1.8 and 1.9: the modes absent from (543), and $c_{4,3}$, are below $3\cdot10^{-69}$, $c_{4,1} - 2\mathfrak{a}_4\mathfrak{b}_4$ is below $2\cdot10^{-68}$, and at the six points other than $b = 0.4$ the function $W_4(3;b)$ assembled from the fitted coefficients agrees with an independent spectral code to $5.1\cdot10^{-50}$ or better. The product $\mathfrak{a}_3\mathfrak{b}_3$ of (545) holds at $b = 1.1$, 1.25, 1.4, 1.5, 1.6 and $1.3 + 0.2\mathrm{i}$ to $2\cdot10^{-60}$ at 60 digits, and, with a second, literal hypergeometric evaluator of the moments (not run at $b = 1.6$), to $6\cdot10^{-29}$ at 30 digits (`wf_Z3_gamma_verify/check_amp.py`, logs `check_dps30.log`, `check_dps60.log`). The checks of the separate review at $b = 0.8$, outside the range of the Gamma evaluation, and at other points are quoted after (551), and the $d = 3$ dictionary below is checked against the 160-digit values in Section 20.9.

$d = 3$. By (544), $c_{4,1}(b) = 2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$ for $0 < b < 2$, with the moment amplitudes of (543), and (545) gives $\mathfrak{b}_4(b)$ in Gamma form, so that $\partial_b[\nu_b^{1/2}\mathfrak{b}_4] = (\frac{\pi}{2} - \log 2)\mathfrak{b}_4$ at $b = \frac{3}{2}$, where $\nu_b = [\Gamma(\frac{3}{2} - \frac{b}{2})/\Gamma(\frac{b}{2})]^4 = N_b^4/\Gamma(\frac{3}{2})^4$ equals one there. Let $X_0 = \Xi_4(0;\frac{3}{2})$; this is the X_0 of (421), not the sample $\mathcal{X}_0$ of (489). The dictionary to (489) is

$$\mathcal{X}_T = N_b^4 X_0, \qquad \mathcal{X}_0 - \mathcal{X}_{1/2} = 4N_b^4\mathfrak{a}_4\mathfrak{b}_4, \qquad \mathfrak{d}_H = \partial_b\big[\nu_b(4c_{4,0}+5c_{4,2})\big] = \frac{16}{\pi^2}\partial_b\big[N_b^4(4c_{4,0}+5c_{4,2})\big],$$
$$\mathfrak{d}_a = \partial_b\big[\nu_b^{1/2}\mathfrak{a}_4\big] = \frac{8}{\pi^2}\frac{\partial_b(N_b^4 c_{4,1})}{\mathfrak{b}_4} - \Big(\frac{\pi}{2} - \log 2\Big)\mathfrak{a}_4,$$

where the first equalities for $\mathfrak{d}_H$ and $\mathfrak{d}_a$ are their definitions. Inserting the dictionary into (493) gives $\mathcal{L}_3 = 4\pi P_4(3) - 2\pi^2 X_0 + 6\sqrt{2}\pi(\pi - 2\log 2)\mathfrak{a}_4\mathfrak{b}_4 - 2\pi\mathfrak{d}_H + 12\sqrt{2}\pi\mathfrak{b}_4\mathfrak{d}_a$. The derivative constants occur only in the combination $-2\pi\mathfrak{d}_H + 12\sqrt{2}\pi\mathfrak{b}_4\mathfrak{d}_a$. It has the same derivative content as $\widehat{P}'$, the b-derivative at $b = \frac{3}{2}$ of $\widehat{P}(b) = -2\pi\nu_b(4c_{4,0}+5c_{4,2}) + 6\sqrt{2}\pi\nu_b c_{4,1}$, in terms of which the 160-digit evaluation (`LRI_C5_d3/wf_L3/`) writes $\mathcal{L}_3 = 4\pi P_4(3) - 2\pi^2 X_0 + \widehat{P}'$. The two differ by value terms:

$$\frac{2}{\pi S^7}\mathfrak{D}_L = \widehat{P}' - 8\sqrt{2}\,N_b^4 c_{4,1} + 32\,N_b^4 c_{4,2} \qquad (b = \tfrac{3}{2}),$$

numerically -1332.48 against $\widehat{P}' = 274.55$; the identity holds to 10^{-57} with the 160-digit $d = 3$ values stored in `LRI_C5_d3/wf_L3/L3_final_values.json`. So $\mathfrak{D}_L$ is not literally $\widehat{P}'$. The split into a Horn-sector and an ordinary-sector constant follows the rank-two structure. The collapse fails at $d = 3$ because the lattice no longer reduces to a one-variable bilateral sum: the multiplicity $2\ell + 1$ gives rank two, the Dougall factor $\mathfrak{b}_4$ closes but $\mathfrak{a}_4'$ and the Horn-sector derivative do not (lattice reduction at 155 digits, `LRI_C5_d3`; numerical), and $D = -\pi \neq 0$.

Z_d at $d = 1, 2, 3$. In (498) the inputs are $\mathfrak{t}_3(b)$ and $\hat{c}_0(b)$. At $d = 1$, (539) makes Ξ_3 odd under $t \to t + \frac{1}{2}$, so $\hat{c}_0 \equiv 0$; with $\cos\frac{\pi}{2} = 0$ and $\mathfrak{t}_3(1;b) = {}_3F_2(b,b,b;1,1;1)$, which is Dixon's sum [22], this gives $Z_1 = \frac{4\pi^2}{3}\mathfrak{t}_3''(\frac{1}{2}) = QH$ of (224). At $d = 2$ the same symmetrization as in the proof

of (539) gives $\widehat{M}_3 \equiv 0$ (the midpoint sum is the square of a bilateral sum that is odd under $m \to -1-m$) and $\mathfrak{t}_3(2;b) = [{}_3F_2(\sigma,\sigma,\sigma;1,1;1)]^2$, Dixon's sum squared, because $T_{\ell n} = \chi_{\ell+n}\chi_n$ with $\chi_m = (\sigma)_m/m!$. Hence $\hat{c}_0 = -\pi^3 s_b^3 \mathfrak{t}_3/(1+\varsigma_b)$ and $u\mathfrak{t}_3 = \frac{1}{2}\gamma_C(b/4)^3/\gamma_C(3b/4)$, the complex Selberg value (264), here obtained from Dixon's theorem; $Z_2 = 4\mathsf{G}Q_2$ follows from it alone (proofs in the next paragraph, (548)–(550)). At $d=3$, $\hat{c}_0(b) = -N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$, by (544) and (545). Then $\hat{c}_0, \hat{c}_0', \hat{c}_0''$ are elementary, the zero-mode part of (498) is $\sqrt{2}\,(4\pi + 32\mathsf{G}/\pi)\,\hat{c}_0(\frac{3}{2})$, and $Z_3 = \frac{4\pi^2}{3}\mathfrak{t}_3''(\frac{3}{2}) - (4\mathsf{G} + \frac{\pi^2}{2})\frac{\Gamma(1/4)^{10}}{16\pi^{7/2}}$. The collapse stops at $\mathfrak{t}_3''$, because $\mathfrak{t}_3(3;b)$ is a genuine two-variable Horn function.

Proofs of the three-wheel statements at $d = 1, 2, 3$. $d=1$. For $0 < b < \frac{2}{3} = b_{\max}(3,1)$,

$$\mathfrak{t}_3(1;b) = {}_3F_2\left(\begin{matrix}b,b,b\\1,1\end{matrix};1\right) = \frac{\Gamma(1+\frac{b}{2})\,\Gamma(1-\frac{3b}{2})}{\Gamma(1+b)\,\Gamma(1-\frac{b}{2})^2\,\Gamma(1-b)}, \qquad W_3(1;b) = \frac{6\pi^2}{s_b^2}\,\mathfrak{t}_3(1;b). \tag{546}$$

Proof. At $d=1$ only the modes $k \equiv r \pmod 2$ occur (after (539)), so $c_{3,0} \equiv 0$, $\hat{c}_0 \equiv 0$, and (523) gives $W_3 = 6\pi^2\mathfrak{t}_3/s_b^2$. At $d=1$, $d_0 = d_1 = 1$ and $d_\ell = 0$ for $\ell \geq 2$ (the factor $(4q-1)_j$ in (301) vanishes for $j \geq 1$), $2q = \frac{1}{2}$ and $1 - g = \frac{1}{2} + \sigma$. Legendre's duplication $(x)_n(x+\frac{1}{2})_n = 4^{-n}(2x)_{2n}$ [26] gives $T_{0n} = (b)_{2n}/(2n)!$ and $T_{1n} = (b)_{2n+1}/(2n+1)!$. Hence $\mathfrak{t}_3(1;b) = \sum_{m\geq 0}[(b)_m/m!]^3$, which is Dixon's well-poised ${}_3F_2$ with $a=b=c$ [22]; it converges for $\operatorname{Re} b < \frac{2}{3} = b_{\max}(3,1)$, and Dixon's theorem gives the Gamma product. □ Equivalently, the vanishing zero mode is the relation $\widehat{M}_3(1;b) = 8\pi^3 s_b^3\,\mathfrak{t}_3(1;b)/[\varsigma_b(1+\varsigma_b)^3]$ between the two lattices.

Z_1 from Dixon's theorem. Since $\hat{c}_0 \equiv 0$ for all b, $W_3(1;b) = 6\pi^2\mathfrak{t}_3(1;b)/s_b^2$ on $0 < b < \frac{2}{3}$. At $b = \frac{1}{2}$ one has $s_b = \varsigma_b = 2^{-1/2}$, and the logarithmic derivatives of the Gamma product (546), with the recurrences $\psi_m(z+1) = \psi_m(z) + (-1)^m m!\, z^{-m-1}$ and $\psi_1(\frac{1}{2}) = \frac{\pi^2}{2}$, are

$$(\log \mathfrak{t}_3)'(\tfrac{1}{2}) = \psi_0(\tfrac{3}{4}) - \psi_0(\tfrac{1}{4}) = \pi, \qquad (\log \mathfrak{t}_3)''(\tfrac{1}{2}) = \tfrac{5}{2}\psi_1(\tfrac{1}{4}) - \tfrac{1}{2}\psi_1(\tfrac{3}{4}) - \pi^2 = \pi^2 + 24\mathsf{G},$$

using $\psi_1(\frac{1}{4}) = \pi^2 + 8\mathsf{G}$ and $\psi_1(\frac{3}{4}) = \pi^2 - 8\mathsf{G}$. Further $\mathfrak{t}_3(1;\frac{1}{2}) = \Gamma(\frac{5}{4})\Gamma(\frac{1}{4})/[\Gamma(\frac{3}{2})\Gamma(\frac{3}{4})^2\Gamma(\frac{1}{2})] = \Gamma(\frac{1}{4})^4/(4\pi^3)$ by (546) and $\Gamma(\frac{3}{4}) = \sqrt{2}\,\pi/\Gamma(\frac{1}{4})$, so that (497) gives $Q_1 = 4\pi^2\mathfrak{t}_3(1;\frac{1}{2}) = \Gamma(\frac{1}{4})^4/\pi$. The first derivative of $W_3(1;b)$ at $b = \frac{1}{2}$ is $12\pi^2\mathfrak{t}_3[(\log\mathfrak{t}_3)' - \pi] = 0$, so the input $\partial_b W_3 = 0$ holds here by Dixon's theorem; the second derivative is $24\pi^4\mathfrak{t}_3 - 24\pi^3\mathfrak{t}_3' + 12\pi^2\mathfrak{t}_3'' = 12\pi^2\mathfrak{t}_3''$. Hence, with $\mathfrak{t}_3''/\mathfrak{t}_3 = (\log\mathfrak{t}_3)'' + [(\log\mathfrak{t}_3)']^2 = 2\pi^2 + 24\mathsf{G}$,

$$Z_1 = \frac{4\pi^2}{3}\,\mathfrak{t}_3''(1;\tfrac{1}{2}) = Q_1\Big(\frac{2\pi^2}{3} + 8\mathsf{G}\Big), \tag{547}$$

the value $Q_1 H$ of (224), here obtained from Dixon's theorem without Pattern T.

$d=2$. For $0 < \operatorname{Re} b < \frac{4}{3} = b_{\max}(3,2)$, $\widehat{M}_3(2;b) = 0$, and $\mathfrak{t}_3(2;b)$ is the square of Dixon's sum at $\sigma = \frac{b}{2}$:

$$\mathfrak{t}_3(2;b) = \Big[{}_3F_2\left(\begin{matrix}\sigma,\sigma,\sigma\\1,1\end{matrix};1\right)\Big]^2 = \Big[\frac{\Gamma(1+\frac{b}{4})\,\Gamma(1-\frac{3b}{4})}{\Gamma(1+\frac{b}{2})\,\Gamma(1-\frac{b}{4})^2\,\Gamma(1-\frac{b}{2})}\Big]^2 = \frac{s_b^2}{4\pi^2(1+2\varsigma_b)}\,\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}, \tag{548}$$

with $\gamma_C(x) = \Gamma(x)/\Gamma(1-x)$ as in (264). Hence $c_{3,0} = c_{3,1}$ and

$$W_3(2;b) = u(b)\,\mathfrak{t}_3(2;b) = \frac{1}{2}\,\frac{\gamma_C(b/4)^3}{\gamma_C(3b/4)}, \tag{549}$$

which is the complex Selberg value (264). Note also $\mathfrak{t}_3(2;b) = \mathfrak{t}_3(1;\frac{b}{2})^2$.

Proof. At $d=2$: $g = 1-\sigma$, $d_0 = 1$, $d_\ell = 2$ for $\ell \geq 1$. Put $\varrho = \frac{1+\sigma}{2}$ and $\varphi_m = \Gamma(m+\varrho)/\Gamma(m+2-\varrho)$, $m \in \mathbb{Z}$ (both local to this proof). For $s = n + \frac{\ell+1}{2}$ one has $a_\ell + s = \ell + n + \varrho$, $a_\ell - s = \varrho - 1 - n$ and

$\beta_\ell \pm s = a_\ell \pm s + 2 - 2\varrho$. Gamma reflection [26] gives $\Gamma(\varrho-1-n)/\Gamma(1-\varrho-n) = -\varphi_n$, because $\sin\pi(1-\varrho-n)/\sin\pi(\varrho-1-n) = -1$. So $g_\ell(n+\frac{\ell+1}{2}) = -\varphi_{\ell+n}\varphi_n$ and

$$\frac{\widehat{M}_3}{N_b^3} = -\sum_{\ell\ge 0} d_\ell \sum_{n\in\mathbb{Z}} \varphi_{\ell+n}^3\varphi_n^3 = -\sum_{m\ge n}(2-\delta_{mn})\,\varphi_m^3\varphi_n^3 = -\Big(\sum_{m\in\mathbb{Z}}\varphi_m^3\Big)^2.$$

The same reflection gives $\varphi_{-1-m} = -\varphi_m$, so the bilateral sum vanishes. All sums converge absolutely: $|\varphi_m| \asymp |m|^{\mathrm{Re}\,\sigma-1}$, and $3(1-\mathrm{Re}\,\sigma) > 1$ is $\mathrm{Re}\,b < \frac{4}{3}$. Then $y_b = c_{3,0} - c_{3,1} = 0$ by (522).

At the pole node the same rearrangement applies. At $d=2$ one has $2q=1$ and $1-g=\sigma$, so $T_{\ell n} = \chi_{\ell+n}\chi_n$ with $\chi_m = (\sigma)_m/m!$, and $\mathfrak{t}_3 = \sum_{m\ge n\ge 0}(2-\delta_{mn})\chi_m^3\chi_n^3 = (\sum_{m\ge 0}\chi_m^3)^2$. The inner sum is Dixon's ${}_3F_2(\sigma,\sigma,\sigma;1,1;1)$, as at $d=1$ with $b\to\sigma$ [22]; it converges absolutely for $\mathrm{Re}(2-3\sigma)>0$, again $\mathrm{Re}\,b<\frac{4}{3}$. This is the first equality of (548), and Dixon's theorem gives the second. For the third put $x=\frac{b}{4}$. Gamma reflection and $\Gamma(1+x)=x\Gamma(x)$ give

$$\Big[\frac{\Gamma(1+x)\Gamma(1-3x)}{\Gamma(1+2x)\Gamma(1-x)^2\Gamma(1-2x)}\Big]^2\frac{\gamma_C(3x)}{\gamma_C(x)^3} = \frac{x^2\,\Gamma(3x)\Gamma(1-3x)}{[\Gamma(1+2x)\Gamma(1-2x)]^2\,\Gamma(x)\Gamma(1-x)} = \frac{\sin\pi x\,\sin^2 2\pi x}{4\pi^2\sin 3\pi x},$$

and $\sin 3\pi x = \sin\pi x\,(1+2\cos 2\pi x)$ turns the right side into $s_b^2/(4\pi^2(1+2\varsigma_b))$. Finally (496) with $\widehat{M}_3 = 0$ gives $W_3(2;b) = u(b)\mathfrak{t}_3(2;b)$, and $u(b)\,s_b^2/(4\pi^2(1+2\varsigma_b)) = \frac{1}{2}$. □

This is the square structure of the two-dimensional lattice (539) at the two nodes: at the self-mirror node the square vanishes, and at the pole node it is the square of Dixon's sum. It derives the complex Selberg value (264) from (496) and Dixon's theorem, without the complex Selberg integral. Its agreement with (264), which Section 16.7 obtains from that integral, is an exact check of (496) and of the theorems it rests on (Sections 20.1.1, 20.1.2, 20.1.3) at $d=2$ on the range $0<b<\frac{4}{3}$ where both derivations hold.

Z_2 from Dixon's theorem. Since $\widehat{M}_3 \equiv 0$, (527) gives $9Z_2 = (u\mathfrak{t}_3)''(1)$ with $u\mathfrak{t}_3$ of (549). Its logarithmic derivative vanishes at $b=1$ and its logarithmic second derivative there is $\frac{3}{4}[\psi_1(\frac{1}{4})-\psi_1(\frac{3}{4})] = 12\mathsf{G}$, as in Section 16.7, so $(u\mathfrak{t}_3)''(1) = 12\mathsf{G}\,(u\mathfrak{t}_3)(1)$; with $3Q_2 = u(1)\mathfrak{t}_3(2;1) = 2\pi^2\,\mathfrak{t}_3(1;\frac{1}{2})^2 = \Gamma(\frac{1}{4})^8/(8\pi^4)$ this gives

$$Z_2 = 4\mathsf{G}\,Q_2, \tag{550}$$

which is (265). Here it follows from Dixon's theorem alone, without the complex Selberg integral and without the input $\partial_b W_3 = 0$ (which holds here because the logarithmic derivative vanishes).

$d=3$. At $d=3$, $\Gamma(2q) = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and $N_b^3 = \frac{\pi^{3/2}}{8}\Gamma(g)^3/\Gamma(\sigma)^3$. With $c_{3,0}(b) = -\mathfrak{a}_3(b)\mathfrak{b}_3(b) = -\pi^5\Gamma(3g-2)/\Gamma(g)^9$, the Gamma form (544), (545), one has $\frac{4}{\pi s_b^3}\hat{c}_0 = -\frac{\pi^{11/2}\Gamma(3g-2)}{2s_b^3\Gamma(\sigma)^3\Gamma(g)^6}$ with $3g-2 = \frac{5-3b}{2}$, and (523) becomes

$$W_3(3;b) = \frac{6\pi^2\,\mathfrak{t}_3(3;b)}{\sin^2(\pi b/2)} - \frac{\pi^{11/2}\,\Gamma(\frac{5-3b}{2})}{2\sin^3(\pi b/2)\,\Gamma(\frac{3-b}{2})^6\,\Gamma(\frac{b}{2})^3}, \tag{551}$$

the general-b formula of the $d=3$ computation (`LRI_C5_d3/wf_Z3/`). By Section 20.6 the Gamma form holds on the whole strip $0<\mathrm{Re}\,b<\frac{5}{3} = b_{\max}(3,3)$. A run of the separate review with the block code of `LRI_C5_general_d/Z/` confirms it numerically at $b=0.8$, below the range $1<b<\frac{5}{3}$ of the original Gamma evaluation in `LRI_C5_d3/`, at $b=1.2$, 1.35, 1.6 and at the complex point $b=1.45+0.05\mathrm{i}$ (`LRI_C5_general_d/review/Z-rigor/d3_gamma_offcenter.log`); an own code of that review confirms it at $b=1.2$ and $b=1.45+0.05\mathrm{i}$ (Section 20.9).

Z_3. Here $S=\frac{1}{\sqrt{2}}$, $C=-\frac{1}{\sqrt{2}}$ and $\cos\frac{3\pi}{2}=0$. With the Gamma form, $\log(-\hat{c}_0) = \mathrm{const} + \log\Gamma(3g-2) - 3\log\Gamma(\sigma) - 6\log\Gamma(g)$, so at $b=\frac{3}{2}$ ($g=\sigma=\frac{3}{4}$)

$$(\log(-\hat{c}_0))' = \tfrac{3}{2}[\psi_0(\tfrac{3}{4})-\psi_0(\tfrac{1}{4})] = \tfrac{3\pi}{2}, \qquad (\log(-\hat{c}_0))'' = \tfrac{9}{4}[\psi_1(\tfrac{1}{4})-\psi_1(\tfrac{3}{4})] = 36\mathsf{G},$$

and $\hat c_0(\frac32) = -\Gamma(\frac14)^{10}/(2^{15/2}\pi^{5/2})$. In (535) the coefficients are $3\pi/S^3 = 6\sqrt2\pi$, $-4C/S^4 = 8\sqrt2$ and $4/(\pi S^3) = 8\sqrt2/\pi$, so the zero-mode part is $\frac{\sqrt2}{9}[6\pi + 12\pi + 8(\frac{9\pi}{4} + \frac{36\mathsf G}{\pi})]\hat c_0 = \sqrt2\,(4\pi + \frac{32\mathsf G}{\pi})\,\hat c_0(\frac32)$ and

$$Z_3 = \frac{4\pi^2}{3}\,\mathfrak t_3''(3;\tfrac32) + \sqrt2\Big(4\pi + \frac{32\mathsf G}{\pi}\Big)\hat c_0(\tfrac32) = \frac{4\pi^2}{3}\,\mathfrak t_3''(3;\tfrac32) - \Big(4\mathsf G + \frac{\pi^2}{2}\Big)\frac{\Gamma(1/4)^{10}}{16\pi^{7/2}}, \tag{552}$$

the formula of the paragraph above. It uses the Gamma form (545) and the input $\partial_b W_3 = 0$ (through (535)). At $d = 1, 2$ the whole function $\mathfrak t_3(d;b)$ is a Gamma product (Dixon's sum and its square); at $d = 3$, $\mathfrak t_3''(3;\frac32)$ remains as the one new constant.

	$\mathcal L_d$ (four-wheel)	Z_d (three-wheel)
$d = 1$	duplication to $\Omega_4(\cdot;\frac12)$; relations (540); B_- term $\propto D$ remains	$\hat c_0 \equiv 0$; $\mathfrak t_3$ Dixon
$d = 2$	square of $\Omega_4(\cdot;\frac12)$; relations (540); $D = 0$ leaves only $-2\log 2$	$\widehat{M}_3 \equiv 0$; $\mathfrak t_3$ Dixon squared (= Selberg)
$d = 3$	rank two; only $\mathfrak b_4$ (Dougall) closes; $\mathfrak D_L$ not reduced	$\hat c_0$ Gamma; $\mathfrak t_3''$ not reduced
generic	$\mathfrak D_L$ not reduced	$\hat c_0$, $\hat c_0'$, $\frac19(u\mathfrak t_3'' + v\widehat{M}_3'')$ not reduced

20.7 The five-loop coefficient

Inserting (492), (493), (497), (498) into (223) (exact algebra; A, R, U, V, X of (221)) gives

$$\begin{aligned} 46656\,C_5(d) = {} & -1536\big[P_5(d) + \mathcal K_d\big] - 1728Q_d^2 + \Big[720A^2 + 1008R - \frac{1192\pi^2\cos(\pi d/2)}{S^2}\Big]Q_d \\ & + k_0\mathcal X_0 + k_{1/2}\mathcal X_{1/2} + k_T\mathcal X_T - \frac{1152}{\pi S^7}\mathfrak D_L + \frac{4768\pi^2}{S^2}\mathfrak t_3'' \\ & + \frac{2384}{3}\Big[\frac{\pi(1+C^2)}{S^5}\hat c_0 - \frac{4C}{S^4}\hat c_0' + \frac{4}{\pi S^3}\hat c_0''\Big] \\ & - 15A^4 - 90A^2R + 414AU - 108AV - 117R^2 - 75X, \end{aligned} \tag{553}$$

$$\begin{aligned} k_0 &= (1+C)^2\Big[\frac{768A(1+2C)}{\pi S^7} + \frac{576(4+9C+8C^2)}{S^8}\Big], \\ k_{1/2} &= (1-C)^2\Big[-\frac{768A(1-2C)}{\pi S^7} + \frac{576(4-9C+8C^2)}{S^8}\Big], \\ k_T &= -\frac{1536AC}{\pi S^7} - \frac{1152(1+6C^2)}{S^8}, \qquad Q_d = \frac{2\pi^2}{S^2}\mathfrak t_3 + \frac{4}{3\pi S^3}\hat c_0. \end{aligned}$$

The term $384AP_4 - 576\mathcal L_d$ of (223) is $\sum_i k_i\mathcal X_i - \frac{1152}{\pi S^7}\mathfrak D_L$, and $7152Z_d$ contributes the $\mathfrak t_3''$, $\hat c_0$ and $\cos(\pi d/2)Q_d$ terms. The masters $P_5(d)$ and $\mathcal K_d$ enter (553) only through $-1536[P_5(d) + \mathcal K_d]$, as in (223). $P_5(d)$ is (319). $\mathcal K_d$ is defined by (238). It is evaluated by (260), whose derivation uses the mixed-trace step (input (M) of Section 21.1) and the extraction of $[\tau^2]$ under the trace, by the Mellin–Barnes form (588), which uses none of the inputs of Section 21.1, or by (565); any of these may be substituted in this term without changing the rest of (553). With (565),

$$-1536\,\mathcal K_d = 384AU - 576c_*U - 384(\mathsf r_3 - \mathsf s_3),$$

where $c_* = \psi_0(1) + \psi_0(2q)$ is that of (247) and $\mathsf r_3, \mathsf s_3$ are the Laurent coefficients of (555), with $\mathsf r_3 = [\tau^3]\langle t_\ell^2 R_\tau\rangle_d$ (sans-serif; not the r_3 of (259) or of (637)). This is exact given (M) and (J); (J) is the path form of (237), on which (223) itself rests. When $\mathsf r_3$ is taken from three-node samples on the jet circle, as in (586), hypothesis (H) of Section 21.8 is needed as well. On the range $\frac23 < d < 4$ of (553) the caveat for $d \le \frac47$ of Section 21.1 does not arise.

master	before	with (553)
$P_4(d)$	(323): 20 complex block values	3 samples: 5 two-variable Horn values $\mathfrak{h}_x, \mathfrak{h}_1, \mathfrak{h}_{1/2}, \mathfrak{s}_0, \mathfrak{s}_{1/2}$
$\mathcal{L}_d$	(298): 244 values, ≤ 6 variables	the same 5 functions and their first b-derivatives, in the one combination $\mathfrak{D}_L$
Q_d	(319): 3 complex samples	$\mathfrak{t}_3$, $\widehat{M}_3$: 2 one-variable ${}_7F_6(1)$ and 3 two-variable Horn values
Z_d	(298): 318 values, ≤ 6 variables	the 5 functions of the Q_d row with first and second b-derivatives, through $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$
$\mathcal{K}_d$	(298): 386 values, ≤ 6 variables	Section 21: a polygamma part plus $\mathfrak{r}_3$, a third-order τ-jet of three-node samples of three two-variable Horn functions (586), under the inputs of Section 21.1; or one two-fold Barnes integral (588)
$P_5(d)$	(329)	not treated in this section

Apart from P_5 and $\mathcal{K}_d$, $C_5(d)$ needs eight hypergeometric functions with at most two summation indices at unit argument: five for $r = 4$ ($\mathfrak{h}_x, \mathfrak{h}_1, \mathfrak{h}_{1/2}, \mathfrak{s}_0, \mathfrak{s}_{1/2}$) and three for $r = 3$ ($\mathfrak{t}_3$ and the right-part and middle-part sums of $\widehat{M}_3$, Section 20.2). Split into one-variable rows and two-variable remainders, they give six one-variable rows, namely ${}_8F_7$, two ${}_9F_8$ and ${}_7F_6$ for $r = 4$ and two ${}_7F_6$ for $r = 3$, plus eight two-variable Horn values. $C_5(d)$ also needs the first b-derivatives of all eight functions, the second b-derivatives of the three functions for $r = 3$, and elementary numbers (π; $\sin, \cos$ of $\pi q, \pi q/2$; Γ, ψ_0, ψ_1 at $q/2, q, 3q/2, 2q, (1 \pm q)/2, (1 + 3q)/2, (3 \pm q)/2$). After the exact relations they enter as eight numbers per dimension: $\mathcal{X}_T, \mathcal{X}_0, \mathcal{X}_{1/2}, \mathfrak{D}_L$ and $\mathfrak{t}_3, \hat{c}_0, \hat{c}_0', \frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$. For Q_d and Z_d together these are four numbers. Since Q_d is one combination of $\mathfrak{t}_3$ and $\hat{c}_0$, (497), Z_d needs three numbers beyond the value of Q_d: one of $\mathfrak{t}_3, \hat{c}_0$, then $\hat{c}_0'$ and $\frac{1}{9}(u\mathfrak{t}_3'' + v\widehat{M}_3'')$.

20.8 Further reductions tried

- *Frozen weights* (exact): one derivative constant for $\mathcal{L}_d$, one first- and one second-derivative constant for Z_d (above).
- *Pole node only* (exact): (536)–(538), one Horn family per wheel, one Laurent order below (255)–(257). It trades functions for parameter derivatives, so in the unweighted Horn class it is not shorter than (553).
- *Node choices.* $t = 0$ is the only node with a single block; the reflection centers need two; a general node three. A real two-node form of W_4 using $t = 0$ exists only for $|\cos\frac{\pi b}{2}| \le \frac{1}{2}$ (second node $\cos 2\pi\tilde{t}_2 = 3\varsigma_b/(1 + 2\varsigma_b^2)$, a general point moving with b), so it does not shorten $\mathcal{L}_d$.
- *Summation theorems for the zero-spin rows.* The row of $\mathfrak{h}_x$ (and of $\mathfrak{t}_3$) is a well-poised ${}_{2r}F_{2r-1}(1)$ with $a = \sigma$, pairs $(\sigma, 1)$, $(1 - g, 2q)$, not very-well-poised; the rows at the reflection centers are not well-poised for generic b. Dixon and Dougall do not apply for $r \ge 3$; at $d = 1$ only the sum of the $\ell = 0, 1$ rows is summable. Contiguous relations do not connect the functions (shifts $x, \frac{1}{2}, 1$ are non-integer apart). Negative.
- *Gamma form of* $\hat{c}_0$ (true at $d = 1, 2, 3$): PSLQ at 80 digits at $d = \frac{4}{5}, \frac{8}{5}, \frac{12}{5}, \frac{16}{5}$ of $\log|\hat{c}_0|$ against log of $\pi, 2, 5, \frac{1+\sqrt{5}}{2}, \Gamma(\frac{1}{5}), \Gamma(\frac{2}{5})$ (all Gamma and trigonometric values at multiples of $\frac{1}{10}$): no relation with coefficients below 10^{11}. Negative, numerical.
- *Elementary closure of* $\mathfrak{D}_L$ (true at $d = 1, 2$): PSLQ at $d = \frac{4}{3}, \frac{8}{3}$ of $\mathcal{L}_d$ against $\{\mathcal{X}_T, \mathcal{X}_0, \mathcal{X}_{1/2}\} \times \{1, \sqrt{3}, \frac{\log 2}{\pi}, \frac{\log 3}{\pi}, \frac{\sqrt{3}\log 2}{\pi}, \frac{\sqrt{3}\log 3}{\pi}\}$ (95 digits): no relation stable under a change of precision; none with coefficients below about 10^4. Negative, numerical.

- *Polygamma closure of Z_d/Q_d and $\mathfrak{t}_3''/\mathfrak{t}_3$* (true at $d = 1, 2$, (547), (550)). A bounded PSLQ probe at 70 digits, on the 90-digit values of the two-node item of Section 20.9, at $d = 1.25, 1.5, 2.5$, against the 15 constants 1, π^2, $\psi_1(q)$, $\psi_1(2q)$, $\psi_1(\frac{q}{2})$, $\psi_1(1-\frac{q}{2})$, $\psi_1(\frac{3q}{2})$, $\psi_1(3q)$, $\psi_0(q)^2$, $\psi_0(q)\psi_0(2q)$, $\psi_0(2q)^2$, $\pi\psi_0(q)$, $\pi\psi_0(2q)$, $\pi^2/\sin^2\pi q$, $\pi\cot\pi q$ (coefficient bound 10^4): at $d = 1.5$ only a relation internal to the basis (target coefficient zero), at $d = 1.25$ and 2.5 only relations with coefficients of size 10^3–10^4, the size expected of spurious relations at this precision (`LRI_C5_general_d/Z/pslq_quick.py`, `LRI_C5_general_d/Z/results/pslq_quick.log`). Negative, numerical; not an independence proof. No closed form of $\hat{c}_0$ is known at generic d: it vanishes at $d = 1$, equals $-\pi^3 s_b^3 \mathfrak{t}_3/(1+\varsigma_b)$ with $\mathfrak{t}_3$ the square of Dixon's sum at $d = 2$, and is a Gamma product in b at $d = 3$. The $d = 2$ and $d = 3$ Gamma forms have different shapes, so a uniform Gamma product in (d, b) is not expected.

20.9 Checks

Computations on the symmetry computing cluster of the Perimeter Institute (`LRI_C5_general_d/specialize/results/`) and the separate review (`LRI_C5_general_d/review/`). The package replay, `make -f public.mk replay`, compares with the records only the numbers listed in `GFF/results/closed_forms_general/reduction_d/expected.json`. These do not include the numbers of Sections 20.1.4, 20.3 and 20.4, nor those of the items below on the general-b periodization, on the review probes and the product form in more detail, and on the two-node formula; those numbers are read from the records named there, and no replay target compares them with the text.

- (539) against the five-Horn samples: $d = 1$, $b = \frac{1}{2}, 0.45$ and $d = 2$, $b = 1, 1.1$, $r = 3, 4$: 10^{-44}–10^{-46}. Relations (540) and $\mathcal{A}, \mathcal{B}$ against $B_\pm$: $5 \cdot 10^{-46}$. $P_4(1), \mathcal{L}_1, P_4(2), \mathcal{L}_2$ from the modes: $\leq 2 \cdot 10^{-45}$.

- $d = 3$ dictionary against the 160-digit values of `LRI_C5_d3/wf_L3`: $c_{4,k}$, X_0, $\mathfrak{a}_4$, $\mathfrak{b}_4$, $\mathfrak{d}_H$, $\mathfrak{d}_a$, $P_4(3)$, $\mathcal{L}_3$ all to $1.2 \cdot 10^{-51}$ (50 digits); $\mathcal{L}_3$ formula $5 \cdot 10^{-64}$.

- (536) against the multi-node values ($d = 1.5, 2.5$; $b = d/2$, $d/2 + 0.04$; $r = 3, 4$): $5 \cdot 10^{-21}$–$3 \cdot 10^{-23}$ (Cauchy circle in the shift).

- (492), (493), (497), (498) against the five-Horn and two-node codes at $d = 1$, 2, 3, 0.8, 1.25, 1.5, 2.5, 2.75 (50 digits): $\leq 2.1 \cdot 10^{-48}$; the block $384AP_4 - 576\mathcal{L} - 1728Q^2 + (720A^2 + 1008R)Q + 7152Z$ and its reduced form (553): $\leq 1.2 \cdot 10^{-48}$; against closed forms ($d = 1, 2$) $\leq 4 \cdot 10^{-50}$, against the independent spectral references $4 \cdot 10^{-50}$ (50-digit reference values, 60-digit samples; `LRI_C5_general_d/review/joint/numerics-end-to-end`), at $d = 3$ $6 \cdot 10^{-34}$ (limited by the 33-digit Q_3 used).

- The explicit five-Horn forms (514) and (519) against the proved values at $d = 1, 2$, the 160-digit values at $d = 3$, the independent spectral references at nine non-integer dimensions and off-center spectral values at ten points, with per-function checks of the jet derivatives: Section 20.3.1.

- End to end, with $P_5(d)$ from (29), (334) and (425) (values in Section 16.8) and $\mathcal{K}_d$ from (565) (r_3 from the jets of Section 21.11):

$$\begin{aligned} C_5(1) &= -731.139476624539825853869126582527231, \\ C_5(2) &= -38.833645896225360552936924675329232, \\ C_5(3) &= -5.850351420751327229217944230497744300111. \end{aligned}$$

  They agree with $C_5(1)$ of (36), printed to 33 decimals in Section 14, to $3 \cdot 10^{-34}$ (all printed digits), with (272) to $2 \cdot 10^{-50}$ and with `LRI_C5_d3` to $5 \cdot 10^{-45}$ (`LRI_C5_general_d/review/joint/numerics-end-to-end`).

- *Top-mode theorem and its product form* (Section 20.1.2). The balance of every single factor and the factor $1+z$ in (461) are both essential. An unbalanced wheel with the same pole classes ($d=1$, $r=4$, $\sigma=0.2$, spoke half-exponent 0.25 instead of 0.3) has $|\gamma_{\pm 3}|$ equal to 1.7% of the largest mode, against 10^{-41} for the balanced kernel at 40 digits; an independent 50-digit code (`LRI_C5_general_d/review/topmode-numeric`) finds unbalanced top modes of $6.7\cdot 10^{-4}$ to $4.6\cdot 10^{-2}$, linear in the imbalance, and $1.9\cdot 10^{-3}$ when the factor $1+z$ is dropped. For balanced wheels the same code finds $|c_{r,r-1}|/\max_k |c_{r,k}|$ between $2.4\cdot 10^{-51}$ and $1.6\cdot 10^{-49}$ at 22 configurations with $r=2,\dots,6$, $0.8\le d\le 3.2$ and complex b, against surviving modes of $1.6\cdot 10^{-3}$ to 1. For products of different factors, a separate 48-digit lattice-sum code (`LRI_C5_general_d/review/lemmaG-numeric`; reruns at 54 digits on other compute nodes) finds top modes between $2.8\cdot 10^{-48}$ and $4.7\cdot 10^{-44}$ of the largest mode, at the level of the vanishing boundary modes $\gamma_{\pm r}$, at 12 mixed wheels ($r=3,4$; $d=1.3$, 2.4, 3.1; unequal b_i, real and complex shifts and offsets, one with $\operatorname{Re} g_1<0$), $6.5\cdot 10^{-47}$–$5.2\cdot 10^{-45}$ at four R kernels with real τ and $3.0\cdot 10^{-46}$ at one wheel; its reruns give at most $9.3\cdot 10^{-54}$. The two top modes vanish separately, also where $|\gamma_{r-2}|$ and $|\gamma_{2-r}|$ differ by up to 79% of the largest mode, and they stay below $2\cdot 10^{-47}$ (54 digits) at four analytically continued points on the boundary of (C) or outside it, two of them in the R family. A per-factor imbalance δ_{bal} at fixed total balance switches them on linearly, with relative top modes 0.047 down to 0.0013 for $\delta_{\mathrm{bal}}=0.15$ down to 0.0015. The per-spin code of Section 21.11 (`LRI_C5_general_d/K_general/note_fragment/topmode/perspin_topmode.py`; its modes agree with those of the 48-digit lattice-sum code to 10^{-15}) finds top modes of at most $6.5\cdot 10^{-16}$ of the largest mode, at the level of its extrapolation, for five R kernels (real and complex τ) and two random mixed kernels ($r=3,4$), against $3\cdot 10^{-3}$ to $6\cdot 10^{-2}$ for controls that keep the total balance but break (B), use multiplicities of another dimension, or shift one right argument. The short check `LRI_C5_general_d/foundation/check_topG.py` (30 digits, about 20 s on a laptop) finds 10^{-24} to $9\cdot 10^{-16}$, falling like the truncation error, at four mixed points with $r=3,4$ and $d=0.6$, 1.7, 3.3, three of them with one $\operatorname{Re} g_i<0$ (Step 7), against $9\cdot 10^{-10}$ to $2\cdot 10^{-5}$ for controls with right shifts ± 0.1 at fixed total balance; it also confirms (478) and (472) to 10^{-29}. No further mode vanishes in general: at the points tested, $c_{5,3}$ and $c_{6,4}$ are $1.6\cdot 10^{-3}$ to 0.16 of the largest mode. The proof review `LRI_C5_general_d/review/lemmaG-proof` tested Steps 1–6 numerically in unbalanced cases, and the endpoint exponents of (475) (item on the product form below).

- Independent review, own codes: top modes as quoted in the item above; the printed formulas of Sections 20.2–20.6 and (553) rederived symbolically with difference 0 ((538) by hand); off-center W_3, W_4 from independent samples (exact big-integer convolutions) to 44–53 digits; P_4 and $\mathcal{L}_d$ against spectral references to 31–39 and 31–38 digits; the coefficient functions of the five-Horn formula for $\mathcal{L}_d$ at 11 values of d to $1.2\cdot 10^{-38}$; Z_1, Z_2, Z_3 to 30.7, 35.2, 29.5 digits; the reduced C_5 block against the reference-master block to 30.4–35.6 digits at 8 values of d.

- *The general-b periodization* (Section 20.1). The checks use the code `wheel_general_b.py` with `hypsums.py` of `LRI_C5_general_d/foundation/`, driven by `run_validation.py` on the symmetry cluster (job 568940 on one 64-core AMD node, 50 requested digits and 15 guard digits; supplementary job 568959); records `foundation/results/val50.json` and `valx.json`, with the comparison `val50_analysis.txt` and `val50_summary.json` written by `analyze_val.py`. References: the closed forms of Section 1.1 at $d=1,2$ (for P_4, Q, $\mathcal{L}$ and Z), the archived $d=3$ values (≥ 90 digits), and at non-integer d the independent spectral values of `LRI_C5_general_d/references/reference_values.json` (41–50 digits at $b=\frac{d}{2}$ and 56 digits off it when these runs were made; the file was later rewritten with 50 significant digits).

  - At $b=\frac{d}{2}$, with the fixed nodes of (484) ($N=N_r$, complex discrete Fourier transform) and with the b-dependent nodes and the real cosine transform (313) (which at $b=\frac{d}{2}$ is

literally the rule of Section 18), $W_4 = P_4(d)$ and $W_3 = 3Q_d$ agree with the references to $2.2 \cdot 10^{-66}$–$1.5 \cdot 10^{-63}$ at $d = 1, 2, 3$ (the working precision) and to $9.7 \cdot 10^{-47}$–$3.8 \cdot 10^{-45}$ at $d = 0.8, 1.25, 1.5, 2.5, 2.75$, which is the accuracy of the references there (42–50 digits).

– At $b \neq \frac{d}{2}$ ($b = \frac{d}{2} \pm 0.05$, at $d = 0.8$ $b = \frac{d}{2} \pm 0.03$; $r = 3, 4$; the five non-integer d, 20 points), the fixed-node values agree with the spectral references to $3.1 \cdot 10^{-55}$–$3.0 \cdot 10^{-49}$, and the b-dependent nodes (not run at $d = 0.8$) to $3.3 \cdot 10^{-54}$–$7.7 \cdot 10^{-50}$. A second spectral code, `foundation/spectral.py`, which shares only the Hurwitz routine, agrees at six of these points ($d = 1.5$, $b = 0.7, 0.8$, $r = 3, 4$; $d = 2.5$, $b = 1.2$, $r = 3, 4$) to $8.2 \cdot 10^{-54}$–$1.9 \cdot 10^{-50}$. At $d = 2$ the exact $W_3(2; b)$ of (264) is reproduced at $b = 0.8$ and 1.2 to $1.3 \cdot 10^{-65}$ and $7.0 \cdot 10^{-65}$.

– Structure: $|c_{r,r-1}|/\max_k |c_{r,k}|$ lies between $4 \cdot 10^{-66}$ and $9 \cdot 10^{-49}$ at every point; at an extra point $t_* = 0.2371 + 0.0917\mathrm{i}$ the trigonometric polynomial reproduces $\Xi_r(t_*; b)$ to at most $1.0 \cdot 10^{-62}$ at $d = 1, 2, 3$ and $1.2 \cdot 10^{-49}$ at non-integer d; at $d = 1$ also $c_{4,1} = c_{4,3} = 0$ and $c_{3,0} = c_{3,2} = 0$, and at $d = 2$ $c_{3,0} = c_{3,1}$, to working precision, as (539) requires. The reduced nodes $N = N_r'$ agree with $N = N_r$ to $6.4 \cdot 10^{-65}$ ($d = 2$, $b = 1$, $r = 4$), $4.1 \cdot 10^{-63}$ ($d = 3$, $b = 1.5$, $r = 3$) and $1.2 \cdot 10^{-53}$–$2.5 \cdot 10^{-50}$ at four non-integer points.

– The b-dependence: Cauchy integrals of the finite-sample $W_r(d; b)$ over 96 points of the circle $|b - \frac{d}{2}| = \rho$ (phases $2\pi(k + \frac{1}{2})/96$, conjugate symmetry used), with ρ one third of the distance to the nearest singularity of W_r or to $b_{\max}$, so that the truncation error is below about 3^{-96}, give $\mathcal{L}_d = \partial_b W_4$ and $Z_d = \frac{1}{9}\partial_b^2 W_3$ at $b = \frac{d}{2}$ to $3.0 \cdot 10^{-63}$–$1.0 \cdot 10^{-46}$ relative at $d = 1, 2, 3$ and to $1.2 \cdot 10^{-46}$–$3.5 \cdot 10^{-45}$ at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ (references of 41–50 digits). The first derivative $\partial_b W_3(d; \frac{d}{2})$, which vanishes by (239) with (244), comes out between 10^{-61} and 10^{-44}. The sentence after the table that follows this list compares these values with the other evaluations.

– The jet formula (485) evaluated literally at 30 digits (`foundation/sample_jets.py`, job 568967; the per-block values and b-derivatives at the nodes are stored in `foundation/results/sample_jets.json`), with the sample derivatives at the reduced fixed nodes taken from Cauchy circles of 48 points of radius one sixth of the distance to the nearest singularity of the samples: $\mathcal{L}_d$ and Z_d at $d = 1, 1.5, 2, 2.5, 3$ to $2.4 \cdot 10^{-31}$–$2.8 \cdot 10^{-30}$, and $W_r(d; \frac{d}{2})$ to $5.3 \cdot 10^{-32}$–$2.7 \cdot 10^{-30}$.

– Beyond $b_{\max}$ (Section 20.1.4): the continued Horn values against `spectral.py` at $(d, b, r) = (2.5, 1.8, 4)$, $(2.5, 1.55, 3)$, $(2.75, 1.95, 4)$ and $(2.75, 1.65, 3)$ to $4.7 \cdot 10^{-45}$, $4.9 \cdot 10^{-48}$, $2.1 \cdot 10^{-41}$ and $1.7 \cdot 10^{-49}$, limited by the cancellation near the pole of the $c_{r,k}$ at $b_{\max}$. At the resonant points $(d, b, r) = (\frac{4}{3}, \frac{2}{3}, 3)$ and $(2.5, 1.5, 4)$, where $r(d - b)/2 \in \mathbb{Z}$ and the circle mean is used, to $3.5 \cdot 10^{-61}$ and $1.5 \cdot 10^{-61}$.

d	$P_4(d) = W_4(d; \frac{d}{2})$	$3Q_d = W_3(d; \frac{d}{2})$
0.8	`4403.5659705473683681`	`259.83164202536385403`
1.25	`1120.1530930090638933`	`104.22818434903994884`
1.5	`632.7128141805458393`	`71.175347423099135692`
2.5	`118.19407871190995074`	`23.177096659039552764`
2.75	`84.737926139821720537`	`18.545946016773332171`

The table gives the finite-sample $P_4(d)$ and $3Q_d$ (fixed nodes), within $3.8 \cdot 10^{-45}$ of the spectral references. The values of $\mathcal{L}_d$ and Z_d from the b-circles at these five dimensions (`foundation/results/val50_summary.json`) are within $3.5 \cdot 10^{-45}$ of the references and agree in every printed digit with the table of $\mathcal{L}_d$ in Section 20.3.1 and the table of Z_d in the two-node item below.

- *Caution on the structural checks.* The periodicity, reflection and off-grid checks test the structure of Section 20.1.1, not the accuracy of the samples. They cannot see a sample

error that is itself a reflection-symmetric trigonometric polynomial of degree at most r; a t-independent error in a tail amplitude, multiplied by $D_b(t)^r$ and the Gamma prefactors, is of this kind. Only the top and boundary modes and the comparisons with independent references test accuracy. Errors of this kind occurred in the reviewer's own code and were caught only by the top and boundary modes and the references.

- *The review probes of Sections 20.1.1–20.1.3, in more detail.* `LRI_C5_general_d/review/topmode-numeric` (own lattice sums by two-dimensional Euler–Maclaurin summation with tanh–sinh quadrature, 50 digits, symmetry jobs 569215 and 569260–569313): besides the top modes quoted above, the boundary modes are $4 \cdot 10^{-51}$ to $2.1 \cdot 10^{-49}$; at $r = 2$, $d = 1.3$, $b = 0.6 + 0.1\mathrm{i}$, where $\alpha = 0.7$ barely exceeds the bound 0.65, Ξ_2 is constant to $2.7 \cdot 10^{-51}$; the weights $B_{r,k}$ of (479) agree with a direct series residue to $1.7 \cdot 10^{-51}$–$7.8 \cdot 10^{-49}$; W_r agrees with the spectral references to 47–50 digits; the stored $c_{r,k}$ of `foundation/results/val50.json` agree to $1.4 \cdot 10^{-50}$–$4.2 \cdot 10^{-50}$; the unbalanced controls quoted above stand against a floor of 10^{-30}. `LRI_C5_general_d/review/foundation-rigor` (double-precision brute-force lattice sums, `brute_xi.py`, 17–25 nodes, symmetry jobs 569242 and 569252): the five blocks of (483) agree with the brute-force sums to $2.5 \cdot 10^{-16}$–$1.4 \cdot 10^{-14}$, and the top modes are $1.9 \cdot 10^{-15}$ to $2.7 \cdot 10^{-14}$ in the fast cases ($d = 3.5$; $d = 0.9$, with negative multiplicities; complex b); in the slow cases the residuals halve when the cutoffs are doubled, so they are truncation error. These checks also cover $\sigma > 1$ at $d = 3.9$.

- *The checks of the product form, in more detail.* The proof check `LRI_C5_general_d/review/lemmaG-proof` (record `review.json`; the test scripts `t1_laplace.py` to `t5b_r4.py` saved no logs) tested: the Laplace forms (474) for complex g, also at $\mathrm{Re}\, w < 0$, to 10^{-20}–10^{-26}; the relation $\gamma_{1-r} = \kappa_+ s_1^+$ with s_1^+ from Step 4 and all its constants, against lattice sums in unbalanced cases, where the top mode is nonzero, so that the derivation and not only the final cancellation is tested: $7 \cdot 10^{-11}$ ($r = 2$, real), $2 \cdot 10^{-8}$ ($r = 2$, complex; limited by the lattice truncation), $2 \cdot 10^{-13}$ ($r = 2$, total balance only), 10^{-12} ($r = 3$, $g = (1, 2, 3)$, complex) and $3 \cdot 10^{-12}$ ($r = 4$, $g = (1, 1, 2, 2)$, $d = 2.9$, total balance only), and 10^{-22}–$2 \cdot 10^{-27}$ in balanced cases; the lower half-plane of Step 6, by the direct formula and by the conjugation, to $3 \cdot 10^{-12}$ at a complex unbalanced point with $r = 2$, where $|\gamma_1| \neq |\gamma_{-1}|$; and the endpoint exponents of (475), with fitted slopes -0.300 at $\varkappa = 0.7$ and -0.600 at $\varkappa = 0.4$ on both sides of 2π, bounded at $\varkappa = 1.2$, and bounded and oscillating at $\varkappa = 1$ with complex g_i. As a check that the mechanism does not prove too much, its formal extension to $r = 1$ at $d = 2$ predicts $\sum_{m \in \mathbb{Z}} \Gamma(m + a)/\Gamma(m + a + g) = 0$ for $\mathrm{Re}\, g > 1$, a known corollary of Dougall's bilateral sum. The 48-digit code `LRI_C5_general_d/review/lemmaG-numeric` (record `probe_report.json`) shares nothing with the general-b code, with `check_topG.py` or with the $\mathcal{K}_d$ code. It samples Ξ_T at $2r + 1$ points on a horizontal line and Fourier transforms; the boundary modes $\gamma_{\pm r}$, which vanish by Step 0, set the noise floor. It splits S_T into its right, middle and left parts, each a sum over a quadrant, evaluated by exact head sums, exact Abel–Plana tails and the corner double integral (tanh–sinh rules matched to the endpoint exponents, Gauss–Legendre panels); Gamma ratios come from its own asymptotic series, and `mpmath.zeta` is not used. Main runs are at 48 digits; every case inside (C) and every continued point was rerun at 54 digits with finer settings and a disjoint set of nodes (symmetry jobs 569336–569347 and 569461). Validation: closed-form quadrant sums to $1.1 \cdot 10^{-52}$; main against fine runs to $5 \cdot 10^{-44}$; the 50-digit modes of `topmode-numeric` at one wheel reproduced to all 40 printed digits, and the modes of the per-spin code of Section 21.11 at three of its points to its accuracy of 10^{-15}. Controls: a per-factor imbalance at fixed total balance ($\beta_1 \to \beta_1 + \delta_{\mathrm{bal}}$, $\beta_2 \to \beta_2 - \delta_{\mathrm{bal}}$) at a real mixed wheel with $r = 3$, $d = 1.3$ gives relative top modes 0.047, 0.030, 0.0116, 0.0041, 0.00128 for $\delta_{\mathrm{bal}} = 0.15$, 0.05, 0.015, 0.005, 0.0015, with $\gamma_{-2}/\delta_{\mathrm{bal}}$ converging to about $1.72 + 1.30\mathrm{i}$ with an $O(\delta_{\mathrm{bal}})$ correction, so the onset is linear, while $\gamma_{\pm 3}$ stay at 1.4–$1.6 \cdot 10^{-44}$; at the complex version of that point and complex δ_{bal} the two top modes are nonzero and unrelated (0.025 and 0.015 at $\delta_{\mathrm{bal}} = 0.05 + 0.03\mathrm{i}$, 0.0032

and 0.0018 at $\delta_{\mathrm{bal}} = 0.005 + 0.003\mathrm{i}$); an R kernel with right shifts ± 0.15, a control of the per-spin code, gives 0.0135 and 0.0029, as that code does; outside (C), at two mixed wheels with $2\varkappa - d = -0.06$ and (B) exact, the plain lattice sum diverges (largest $|\gamma_m|$ about 10^{102} at truncation depth 26 and 10^{134} at depth 36), while the analytically continued sum (the leading power of the divergent corner integral subtracted and restored analytically) stays a trigonometric polynomial of degree $r-1$ with vanishing top modes ($6.8\cdot 10^{-42}$–$3.7\cdot 10^{-41}$ at 48 digits, at most $6.1\cdot 10^{-50}$ at 54). The continuation reproduces a synthetic closed form outside its convergence region to $7.6\cdot 10^{-17}$, and continued and plain evaluations agree to $8\cdot 10^{-44}$ at a point inside (C). So (C) acts as a convergence hypothesis, as Step 7 suggests. One complex case was also evaluated on the lines $\operatorname{Im} t = 0.2$ and -0.25, with modes agreeing to $1.2\cdot 10^{-42}$. All error bars of this probe are empirical (convergence ladders, independent reruns, must-vanish modes), with no interval bounds. Endpoint exponents near 0.1 converge slowly in the quadrature step; the cases avoid that regime or were cross-checked. The continued points test only failures of $2\varkappa > d$ by less than one; the τ-circle of Section 21.8 at $d = 2.5$, where also $\varkappa > d - 1$ fails, was not tested.

- *Two-node formula for the three-wheel* (Section 20.4). Code in `LRI_C5_general_d/Z/`, records in `LRI_C5_general_d/Z/results/`. `z_blocks.py` evaluates the five functions of (526) at any complex b with the accelerated Horn sums of `LRI_C5_general_d/foundation/hypsums.py` and `wheel_general_b.py` (exact finite part, Stirling tails, Euler–Maclaurin Hurwitz zeta values with relative precision, three-family convolution asymptotics, exact regularized multiplicity moments) and assembles (527), its reduced forms and (535). The main engine `z_jets.py` reruns the same algorithms with every parameter in the ring $\mathbb{C}[\varepsilon]/(\varepsilon^3)$, $b = \frac{d}{2} + \varepsilon$, so the b-derivatives of the five functions are exact derivatives of the accelerated evaluation, with no numerical differentiation; at jet order 0 it reproduces the block values of the general-b code to working precision. `z_cauchy.py` computes the same jets from Cauchy circles in b; it is a cross-check, and it is used at the resonant dimensions $\frac{3d}{4} \in \mathbb{Z}$ ($d = \frac{4}{3}, \frac{8}{3}$), where the accelerated sums have removable singularities at $b = \frac{d}{2}$. `Z_general.py` is the entry point (Z_d to 25 digits in about 5 s on one laptop core); the full 90-digit set of the table below took 12 s on one 64-core node of the symmetry cluster.

  References: at $d = 1, 2$ the closed forms (224) and (265); at $d = 3$ the 250-digit value of `LRI_C5_d3/wf_Z3/`; at $d = 0.8$, 1.25, 1.5, 2.5, 2.75 the independent spectral values of `LRI_C5_general_d/references/reference_values.json` (spectral code `LRI_C5_general_d/refvals/hp_general_d.py`); at $d = \frac{4}{3}$, 1.75, $\frac{8}{3}$, 3.5, 3.75 new runs of the same spectral code, copied unchanged to `LRI_C5_general_d/Z/spectral_ref/` (45 digits, 64-point b-circle). The spectral code shares no code with the finite-sample evaluation. At $d = 0.7$, just above the threshold $d = \frac{2}{3}$, the spectral run did not converge: its own $\partial_b W_3$ came out as -83, and its b-circle passes within 0.013 of the spectral convergence bound $b_{\mathrm{spec}} = d - \frac{1}{3} = 0.367$ of Section 20.1. There only a consistency check is available: a b-circle of the general-b finite-sample formula with five fixed complex nodes and all five blocks per node (`LRI_C5_general_d/Z/fd_circle_check.py`), which shares the Horn summation code and is therefore not independent.

d	Z_d (two-node formula, 36 digits shown)	reference	rel. dev.	$\lvert\partial_b W_3\rvert$	engine, digits
1	764.931030199588577040161778769488397	(224)	$4.0{\cdot}10^{-91}$	$1.1{\cdot}10^{-97}$	jets, 90
2	46.792583072329696656738825877981 0842	(265)	$3.9{\cdot}10^{-91}$	$3.5{\cdot}10^{-98}$	jets, 90
3	8.820924893060092485536867328174080 87	$d=3$, 250 digits	$4.4{\cdot}10^{-91}$	$5.3{\cdot}10^{-98}$	jets, 90
0.8	1870.912407752717377849941062290725 11	spectral, 46 digits	$2.0{\cdot}10^{-45}$	$4.7{\cdot}10^{-70}$	jets, 90
1.25	312.281876942721713720205721011622893	spectral, 41 digits	$1.9{\cdot}10^{-46}$	$1.7{\cdot}10^{-73}$	jets, 90
1.5	149.908309291859708760442421585801485	spectral, 44 digits	$2.9{\cdot}10^{-45}$	$4.3{\cdot}10^{-73}$	jets, 90
2.5	18.787028196469216775295577525 90324	spectral, 44 digits	$1.2{\cdot}10^{-45}$	$2.1{\cdot}10^{-74}$	jets, 90
2.75	12.672126145166654840113185919 3957078	spectral, 44 digits	$1.2{\cdot}10^{-46}$	$1.8{\cdot}10^{-74}$	jets, 90
4/3	240.897293321182080101413442492954567	new spectral run	$1.0{\cdot}10^{-39}$	$4.3{\cdot}10^{-44}$	Cauchy, 60
1.75	80.431085634404387403291158005793 2935	new spectral run	$1.4{\cdot}10^{-40}$	$1.7{\cdot}10^{-73}$	jets, 90
8/3	14.394625919159569839228806546254 5566	new spectral run	$1.0{\cdot}10^{-39}$	$4.3{\cdot}10^{-43}$	Cauchy, 60
3.5	4.604077324137569309169079543257723 11	new spectral run	$1.4{\cdot}10^{-40}$	$1.4{\cdot}10^{-74}$	jets, 90
3.75	3.426911330730984835562506781385969 49	new spectral run	$1.4{\cdot}10^{-40}$	$1.5{\cdot}10^{-74}$	jets, 90
0.7	3193.828118855016790427553349977467 35	fixed-node circle*	$7.9{\cdot}10^{-40}$	$7.8{\cdot}10^{-70}$	jets, 90

The table gives Z_d from (527) against the references (records `LRI_C5_general_d/Z/results/Z_values.json`, `zj90.json`, `zc60x.json`, `compare_refs.log`). "rel. dev." is relative to the reference. The spectral references are good to the number of digits stated, or to about 10^{-39} for the new runs. The deviations of the five rows marked "spectral" were computed with the reference file as it stood before it was rebuilt with 50 printed digits (`compare_refs.log` predates the rebuild); the stated accuracies of the references did not change. $|\partial_b W_3|$ is the first rim derivative at $b=\frac{d}{2}$ computed from the same jets; it must vanish. *At $d=0.7$ the reference is the fixed-node b-circle described above, a consistency check only.

The two-node identity (496) itself was tested away from $b=\frac{d}{2}$, including complex b (`LRI_C5_general_d/Z/offdiag_check.py`, record `LRI_C5_general_d/Z/results/offdiag60.json`), against the 56-digit spectral values of `LRI_C5_general_d/references/reference_values.json` (evaluated at 50 digits) and against the five-fixed-node finite-sample formula (60 digits; not independent, as above). The values and deviations are in the last column of the table in the paragraph *Away from the center* of Section 20.3.1, next to those of the five-Horn formulas.

Block-level checks with the 90-digit exact jets (value, first and second b-derivative at $b=\frac{d}{2}$; `LRI_C5_general_d/Z/check_blocks.py`, `check_formulas.py`, record `results/check_formulas.log`): at $d=1$, $\mathfrak{t}_3$ against (546) to $1.0\cdot10^{-91}$, $1.5\cdot10^{-91}$, $7.3\cdot10^{-91}$, and $\hat{c}_0$ (which must vanish) at most $4.1\cdot10^{-90}$, $5.9\cdot10^{-89}$, $1.7\cdot10^{-87}$ in absolute value; at $d=2$, $\mathfrak{t}_3$ against (548) to $1.3\cdot10^{-92}$, $7.0\cdot10^{-91}$, $8.5\cdot10^{-91}$, and $\widehat{M}_3$ (which must vanish) at most $7.0\cdot10^{-105}$, $9.0\cdot10^{-98}$, $8.0\cdot10^{-97}$; at $d=3$, $\mathfrak{t}_3$ against the values of $\mathfrak{t}_3(3;b)$ and its first two derivatives at $b=\frac{3}{2}$ from the independent $d=3$ code (`LRI_C5_d3/wf_Z3/`) to $1.3\cdot10^{-90}$, $1.1\cdot10^{-90}$, $1.1\cdot10^{-90}$, and $\hat{c}_0$ against $-N_b^3\pi^5\Gamma(3g-2)/\Gamma(g)^9$ to $1.2\cdot10^{-90}$, $1.8\cdot10^{-90}$, $1.4\cdot10^{-90}$; at all d, the jets (528) against numerical differentiation to $5.2\cdot10^{-110}$. Also, at all twelve dimensions of the 90-digit runs, (532), (533), (534), (535) and (498) hold to about 10^{-73} and (497) to $3\cdot10^{-87}$; the $d=1$ relation between the two lattices holds to $4.6\cdot10^{-91}$, and the $d=3$ identity for $\mathfrak{t}_3'(3;\frac{3}{2})$ to $4.5\cdot10^{-91}$.

Summary. (i) The two-node identity holds away from $b=\frac{d}{2}$, also for complex b, to 10^{-48}–10^{-51} against the spectral values and to 10^{-55}–10^{-63} against the fixed-node formula. (ii) Z_1, Z_2, Z_3 agree with their references to $4\cdot10^{-91}$ with the exact-jet engine at 90 digits, and to about 10^{-60} at 60 digits. (iii) At the five dimensions with spectral references the agreement is 10^{-45}–10^{-46}, the stated precision of those references; at the five dimensions of the new spectral runs it is 10^{-39}–10^{-40}, the precision of those runs. (iv) The first rim derivative computed from the same jets vanishes to 10^{-70}–10^{-98} (exact jets, 90 digits) and to $4\cdot10^{-43}$ (Cauchy engine). (v) The exact-jet values at 60 and 90 working digits agree to 10^{-53}–10^{-61} at 12 dimensions, and the Cauchy-circle engine (64 points; its error is the aliasing of the circle) agrees with the 90-digit jets to 10^{-42}–10^{-55} at 10 dimensions. (vi) The block-level identities above hold to about 10^{-90}. The Cauchy circles in b use the same blocks and are a consistency check, not an independent evaluation. The separate review (`LRI_C5_general_d/review/Z-rigor/`, own

codes) found in addition: the direct midpoint-lattice sum at $d = 1$ against $8\pi^3 s_b^3 \mathfrak{t}_3/[\varsigma_b(1+\varsigma_b)^3]$ to 10^{-37}–10^{-42} at $b = 0.2, 0.3, 0.5$; $\mathfrak{t}_3(2; b)$ against (548) to about 10^{-30} at five points including complex b, and the reflection identities behind $\widehat{M}_3(2; b) = 0$ to about 10^{-30}; at $d = 3$ an own evaluation of $\mathfrak{t}_3$ and $\widehat{M}_3$ reproducing the Gamma form of $\hat{c}_0$ to $3 \cdot 10^{-30}$ ($b = 1.2$) and $3 \cdot 10^{-28}$ ($b = 1.45 + 0.05\mathrm{i}$). That review did not check the resonant dimensions $d = \frac{4}{3}, \frac{8}{3}$, where the only reference is the spectral code. All these numbers are numerical evidence from asymptotic-series evaluations, not interval bounds; together with the top-mode tests above they also support (473) at $r = 3$, and at $d = 2$ the identity (549) is an exact check of it.

20.10 Status, limits and open points

- *Proved, with written proofs:* the finite Fourier theorem (Section 20.1.1), the top-mode theorem (473) and its product form (471), the residue functional (479) and the block decomposition; the identities (539); the sample formulas of Section 20.2 (Section 20.3, (504)–(512), with the derivative series (518); for the three-wheel, (521) and (522) of Section 20.4.2) and the analyticity of the samples in b. Checks: a separate review with its own codes (`LRI_C5_general_d/review/VERIFICATION.md`, proof review in `LRI_C5_general_d/review/foundation-rigor`) rederived by hand the proofs of Section 20.1 in the form they had before they were merged into this note: the reflected forms and the estimates with unspecified constants, the finite Fourier theorem, the top-mode proof in its earlier wheel-only form, the residue functional, the block decomposition and the fixed-node formula. It also rederived the identities (539); the proof of the $d = 2$ identity printed here was written after it. It did not cover the explicit constants of (465), (466) and (467), which come from Section 21.6 and have had no independent review, nor the contour in the proof of (479), which was changed at the merge (paragraph *The residue functional in full* of Section 20.1.3), nor the derivation of the $d = 3$ identities in Section 20.6. Within the review a second reader checked parts of the top-mode proof, and a 50-digit code of its own (`LRI_C5_general_d/review/topmode-numeric`) tested the statements. The review found no printed formula wrong and one wrong intermediate estimate (the endpoint exponent in Steps 3–4 of the top-mode proof); Steps 3–4 as printed here use exactly the hypothesis $b < b_{\max}$. A second separate check of the proof in product form, which contains the wheel case, is in `LRI_C5_general_d/review/lemmaG-proof`; it also tested the endpoint exponents of (475) numerically. Beyond the wheel case, the product form needs the product estimates with $\varkappa = \sum_i \operatorname{Re} g_i$, the endpoint bound in product form, the kernel identity (478) with the exchange on $\widetilde{\Delta}$, the conjugation step for the second top mode and the continuation in the exponents. The check in `LRI_C5_general_d/review/lemmaG-proof` rederived every step by hand and tested Steps 1–6 numerically in unbalanced cases; it found no fatal or major issue and seven minor wording and documentation items, which are applied here. A separate 48-digit lattice-sum code (`LRI_C5_general_d/review/lemmaG-numeric`; Section 20.9) confirms (471) numerically. The check in `LRI_C5_general_d/review/lemmaG-proof` covered the statements (471) and (473), including the R-family parameters used in Section 21.5 and the equivalence of (C) with $\tau \in \mathbb{H}_q$ there; Section 21.5 uses these statements. An earlier separate proof of the case $r = 3$, written for the $\mathcal{K}_d$ calculation, is superseded by the proof printed here; where the two differ, that review found the earlier threshold $|\operatorname{Im} t| \geq 2$ insufficient for complex parameters, which is why the proof here uses Y_0. In the case $r = 3$, Step 0 of the proof gives parts (a) (sufficiency), (b) and (c) of the growth theorem of Section 21.4; the sharpness of $\mathbb{H}_q$ and the explicit constants are proved in Section 21.6, which that review did not cover. The use of (471) for the R family of $\mathcal{K}_d$ at continued points also needs the continuation of the regularized samples in τ (hypothesis (H) of Section 21.8). The reviews `LRI_C5_general_d/review/L-rigor/` and `LRI_C5_general_d/review/Z-rigor/` read the derivations of Sections 20.3 and 20.4 in the package's own text. Details that this note writes out where that text only states them, for instance the majorants in the proof of (518) and the regularity of the middle and left parts at $t = 0$ in the proof of (521), were added at the

merge and have not been reviewed.

- *Derived*, meaning exact algebra on the above: (491), (492)–(498), (536)–(538), the specializations of Section 20.6 at $d = 1, 2, 3$ (at $d = 3$ with the identities (544) and (545) of Section 20.6), and (553). All were rederived symbolically by the review of `LRI_C5_general_d/review/VERIFICATION.md`, the $d = 3$ specializations with the two identities as inputs, and are confirmed numerically (Section 20.9); (538) only through b-derivatives of (536), not by literal summation. The explicit forms (514), (515), (517) and (519) of Section 20.3 are derived in the same sense; their coefficients were also compared with direct symbolic differentiation (Section 20.3).

- *Fixed-node form and continuation.* The fixed-node formula (484) and the jets (485) are exact algebra on (479), confirmed numerically (Section 20.9); Sections 20.2–20.7 do not use them. The continuation of the samples beyond $b_{\max}$ (Section 20.1.4) is derived as the continuation of each block value at a fixed node; the residue (488) is an endpoint asymptotic derivation made in the review `LRI_C5_general_d/review/foundation-rigor`, not a written proof, and it is confirmed numerically and not used downstream.

- *Three-wheel, Section 20.4.* Proved here, given the theorems of Sections 20.1.1–20.1.3: the analyticity statement of Section 20.4.1, the two samples (521), (522), the two-node formula (496) for every b in the strip and every $0 < d < 4$, and its zero-mode form (523). Derived (exact differentiation and algebra, rederived symbolically by the review `LRI_C5_general_d/review/Z-rigor/`): (527), (529), (531), the equivalence of $\partial_b W_3(d; \frac{d}{2}) = 0$ with (532) and with (533), (534) and (535); of these only (534) and (535), hence (498), use $\partial_b W_3(d; \frac{d}{2}) = 0$. Proved here at $d = 1, 2$, from Dixon's theorem and without Pattern T: (546), (547), $\widehat{M}_3(2; b) = 0$, (548), (549) and (550). At $d = 3$, (551) and (552) follow with the Gamma form of $\hat{c}_0(3; b)$, proved in Section 20.6, whose printed derivation has not been reviewed. Open: whether $\mathfrak{t}_3''$, $\widehat{M}_3'$, $\widehat{M}_3''$ (equivalently the zero-mode jets) reduce further at generic d.

- *Conditional inputs.* (498) and the Z_d part of (553) use $\partial_b W_3(d; d/2) = 0$. This input is proved, by Pattern T (239) with (244); it is listed here only to record that (498) depends on it, while (497) and the full jet formula $9Z_d = (u\mathfrak{t}_3 + v\widehat{M}_3)''$ do not. The $d = 3$ specializations of Z_d and $\mathcal{L}_d$ use the Gamma form of $\hat{c}_0$ and the identity $c_{4,1}(b) = 2\mathfrak{a}_4(b)\mathfrak{b}_4(b)$. Both are proved in Section 20.6, which repeats the half-shift proof of Section 19.4 at a general rim exponent; separate AI sessions re-derived both statements, and the printed derivation has not been reviewed. They are confirmed numerically: the Gamma form also at $b = 0.8$ and complex b, and the $d = 3$ dictionary to $1.2 \cdot 10^{-51}$ (Section 20.9). If $\mathcal{K}_d$ is taken from (565), (553) is conditional on (M), and on (H) when r_3 is sampled on the jet circle (Section 21.12); with (588) it is not.

- *Numerical:* all values of Section 20.9, the evaluation of the Horn values and their derivatives (asymptotic series, no interval bounds), and every statement that a further reduction does not exist (Section 20.8; PSLQ bounds in stated bases, not independence proofs).

- *Ranges.* $\frac{1}{2} < d < 4$ for $P_4, \mathcal{L}_d$ and $\frac{2}{3} < d < 4$ for Q_d, Z_d, where $b = d/2$ lies in the strip; the continuation below these thresholds was not tested. At $d = \frac{4}{3}, \frac{8}{3}$ the accelerated evaluation of the Z_d jets has removable resonances and uses Cauchy circles in b. The continuation of the samples beyond $b_{\max}$ for $d > 2$ (a simple pole at $b_{\max}$ whose residue the functional annihilates; Section 20.1.4) is not used anywhere. For $C_5(d)$ the range is $\frac{2}{3} < d < 4$.

- *Outside this section:* $\mathcal{K}_d$ (Section 21, status in Section 21.12) and $P_5(d)$. Since $P_5(d)$ was evaluated to high precision only at $d = 1, 2, 3$ (in double precision also at $d = 2.4$ and $\frac{13}{5}$, Sections 18.6 and 17.7), $C_5(d)$ itself was evaluated only at $d = 1, 2, 3$ ($C_5(1) = -731.139476624\ldots$, $C_5(2) = -38.8336458962\ldots$, $C_5(3) = -5.85035142075\ldots$; full values in Section 20.9); the reduced block of (553) without $P_5, \mathcal{K}_d$ was evaluated at eight dimensions.

- *Open:* whether $\mathfrak{D}_L$, $\hat{c}_0$, $\hat{c}_0'$ and $\frac{1}{9}(u\mathsf{t}_3''+v\widehat{M}_3'')$ reduce further at generic d, as they do at $d=1,2$ and partly at $d=3$.

21 The nonprimitive master $\mathcal{K}_d$ at generic dimension

This section evaluates the nonprimitive master $\mathcal{K}_d$ of (223) for generic $0<d<4$ (for $d\le\frac{4}{7}$ conditional on the separating-contour construction, Section 21.1). For $\mathcal{K}_d$ it gives, in place of the 386 Horn values of (298) with up to six variables each, a low-order parameter jet of two-variable functions, obtained by periodizing the family R_τ of (249) as in Sections 18 and 20.1. There are three steps.

1. The S family of (249) is elementary in every dimension (Section 21.2).
2. Given the inputs (M) and (J) of Section 21.1, the Gamma subtraction of (238) fixes the R family through order τ^2. The only non-elementary number in $\mathcal{K}_d$ is then the Laurent coefficient $\mathsf{r}_3=[\tau^3]\langle t_\ell^2R_\tau\rangle_d$ (Section 21.3).
3. $\langle t_\ell^2R_\tau\rangle_d$ is a three-node sample of an entire periodic function with the Fourier modes $|k|\le1$, on the half-plane $\mathbb{H}_q$ of (571) with $\operatorname{Re}\tau<\frac{3}{2}q$, $\tau\neq q$, and, by the continuation hypothesis (H) of Section 21.8, on the circle that gives the jet. Each sample consists of three two-variable Horn functions at argument $(1,1)$ (Sections 21.4–21.8).

By a second route, which uses none of these inputs, $\mathcal{K}_d$ is a polygamma part plus one two-fold Barnes integral (Section 21.10). The specializations to $d=1,2,3$ reproduce (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$ (Section 21.9). Section 21.11 collects the numerical checks, and Section 21.12 states what is proved, what is conditional and what is numerical. Codes and logs are in the package folder `LRI_C5_general_d/K_general/`. The material of this section is this project's own work, done in Claude Code sessions in that folder; the proof of the growth theorem in Section 21.6 was written there as a separate proof file and is merged here in full.

21.1 Conventions and two inputs from the master path

Put $q=d/4$. We use A,R,U,V,X of (221), $c_*=\psi_0(1)+\psi_0(2q)$ of (247), $C(\tau)$, R_τ, S_τ of (249), and d_ℓ,a_ℓ,b_ℓ of (240). For a rim exponent b (not to be confused with b_ℓ), $g_\ell(s;b)$ and N_b are those of (460); we write $a_\ell(b)=\ell/2+b/4$ and $g_b=(d-b)/2$ for the a_ℓ and g there (so $a_\ell(b)+g_b$ is its β_ℓ), and keep a_ℓ,b_ℓ for their values at $b=d/2$, as in (240). For real b, $t_{\ell,b}(\nu)=N_b\,g_\ell(\mathrm{i}\nu;b)$ in (242); for complex b, $t_{\ell,b}$ and $\langle t_{\ell,b}^r\rangle_d$ denote the analytic continuation (460), in which $|\cdot|^2$ of (242) is replaced by $\Gamma(a+\mathrm{i}\nu)\Gamma(a-\mathrm{i}\nu)/(\Gamma(\beta+\mathrm{i}\nu)\Gamma(\beta-\mathrm{i}\nu))$ with $a=a_\ell(b)$, $\beta=a_\ell(b)+g_b$. Further, $g_\ell(s;d/2)=g_\ell(s)$ is the kernel of (300). We write $t_\ell(s)=\Gamma(2q)g_\ell(s)$, so that $t_\ell(\mathrm{i}\nu)$ is (240). The kernel R_τ is not even in s, so every trace is read on the full Mellin line,

$$\langle t_\ell^2R_\tau\rangle_d=\Gamma(2q)^2\sum_{\ell\ge0}d_\ell\int_{-\mathrm{i}\infty}^{\mathrm{i}\infty}\frac{ds}{2\pi\mathrm{i}}\,g_\ell(s)^2R_\tau(s).\tag{554}$$

For real τ the integrand of (554) takes complex-conjugate values at $s=\pm\mathrm{i}\nu$, so this is the real-part reading of (241) used in (248) and in (391) (Section 18.9.5); for complex τ it is the continuation of that trace, in the sense fixed in (M) below. For even kernels such as $t_\ell^2S_\tau$ it coincides with (241). Put

$$\begin{gathered}\mathcal{X}_d(\tau)=\Big[\frac{\Gamma(q-\tau/2)}{\Gamma(q+\tau/2)}\Big]^2,\qquad F(\tau)=\langle t_\ell^2(R_\tau-S_\tau)\rangle_d=\sum_k F_k\tau^k,\\ \mathsf{r}_k=[\tau^k]\langle t_\ell^2R_\tau\rangle_d,\qquad \mathsf{s}_k=[\tau^k]\langle t_\ell^2S_\tau\rangle_d.\end{gathered}\tag{555}$$

The Laurent coefficients r_k are not the polygamma combinations r_j of (258). In this section $C(\tau)$ is that of (249) and is written with its argument; a bare C, also with a subscript such as C_ℓ, is a generic constant; $C_5(d)$ is the coefficient of (223). The constants S, C, D of (459) are not used, and $\mathcal{X}_d(\tau)$ is unrelated to the samples $\mathcal{X}_0, \mathcal{X}_{1/2}, \mathcal{X}_T$ of Section 20.2.

Two statements of Section 16 about the master path J_t of (238) enter below.

(M) *Mixed trace.* $J_{2\tau} = C(\tau)\langle t_\ell^2 R_\tau\rangle_d$, as stated before (250), as an identity of meromorphic functions of τ. The expression (554) converges absolutely, with the Mellin line separating the left and right pole families of every spin, only for $\max(0, \frac{1}{2} - 2q) < \operatorname{Re}\tau < \frac{3}{2}q$. Outside this strip, and in particular near $\tau = 0$, $\langle t_\ell^2 R_\tau\rangle_d$ denotes its meromorphic continuation, which by (M) is $J_{2\tau}/C(\tau)$; the r_k of (555) are its Laurent coefficients at $\tau = 0$. The strip is nonempty iff $d > \frac{4}{7}$. For $d \le \frac{4}{7}$ there is no strip to continue from: the trace must then be defined with contours that separate the pole families for finitely many spins, which is only sketched (Section 21.7), and the statements below that involve $\langle t_\ell^2 R_\tau\rangle_d$ or r_k are conditional on that construction. (M) is proved at $d = 2$ for $0 < \tau < \frac{1}{2}$ ((391) in Section 18.9.5). At other d it is confirmed numerically (Section 21.11).

(J) *Jet.* $J_t - J_t^\Gamma = \mathcal{K}_d t^2 + O(t^3)$ on the path of (238). This is (237) restricted to that path, and at $d = 1$ it is (59). Section 16 derives (237) from completion symmetry and analyticity; it is listed here as an input because its lower orders t^0, t^1 on the path are used explicitly, in (564) and (589). At $d = 2$ its low orders are rederived in Section 18.8.4, and at non-integer d it is confirmed numerically (Section 21.11).

21.2 The S family is elementary

Closed form of the S family. Let $0 < d < 4$ and $\max(0, \frac{1}{2} - 2q) < \operatorname{Re}\tau < 2q$; this strip is nonempty for $d > \frac{1}{2}$. Then

$$\boxed{\langle t_\ell^2 S_\tau\rangle_d = C(\tau)\,\mathcal{X}_d(\tau)^2 = \frac{\Gamma(2q)\,\Gamma(\tau)\,\Gamma(q-\tau/2)^4}{\Gamma(2q-\tau)\,\Gamma(q+\tau/2)^4},} \tag{556}$$

and everywhere else $\langle t_\ell^2 S_\tau\rangle_d$ denotes the meromorphic continuation of the right side in τ and d. Equivalently $C(\tau)\langle t_\ell^2 S_\tau\rangle_d = C(\tau)^2\mathcal{X}_d(\tau)^2$, the first term of (250); (556) makes that term an explicit statement about the S family.

Proof. Put $b' = d/2 - \tau$. Then $a_\ell(b') = a_\ell - \tau/4$, $a_\ell(b') + g_{b'} = b_\ell + \tau/4$ and $N_{b'} = \Gamma(2q)\Gamma(q + \tau/2)/\Gamma(q - \tau/2)$, so (249) gives the kernel identity

$$t_\ell(s)^2 S_\tau(s) = \Gamma(2q)^2 g_\ell(s;b')^2 = \mathcal{X}_d(\tau)\,\big[N_{b'}\,g_\ell(s;b')\big]^2. \tag{557}$$

Hence $\langle t_\ell^2 S_\tau\rangle_d = \mathcal{X}_d(\tau)\langle t_{\ell,b'}^2\rangle_d = \mathcal{X}_d(\tau)W_2(d;b')$, the balanced two-wheel of (242). A two-wheel is a single convolution, so $W_2(d;b') = B_d(d-b', 2b')$ by (230); this is the identity used before (250), and spectrally it is Parseval's identity for the dilatation-invariant operator with eigenvalue $t_{\ell,b'}$. For complex b' it extends because both sides are holomorphic in τ on the strip, the trace by locally uniform absolute convergence. With $(d-b', 2b') = (2q+\tau, 4q-2\tau)$, (230) gives $B_d = C(\tau)\mathcal{X}_d(\tau)$. Finally, $|g_\ell(\mathrm{i}\nu;b')| \asymp (1+\ell+|\nu|)^{-2\operatorname{Re} g_{b'}}$ with $g_{b'} = q + \tau/2$, so the ν-integral and the spin sum converge absolutely iff $4\operatorname{Re} g_{b'} > \max(d, 1)$; together with $\operatorname{Re} b' > 0$, which keeps the $\ell = 0$ contour between its two pole families, this is the stated strip. □

Laurent form. Expanding $\log\Gamma$ gives $\log[\tau\langle t_\ell^2 S_\tau\rangle_d] = \sum_{h\ge1} x_h\tau^h/h!$ with

$$\begin{gathered} x_h = \psi_{h-1}(1) - (-1)^h\psi_{h-1}(2q) - [h \text{ odd}]\,2^{3-h}\psi_{h-1}(q), \\ (x_1, x_2, x_3, x_4) = (c_* - 4\psi_0(q),\ -R,\ V-U,\ -X). \end{gathered} \tag{558}$$

Thus $\mathsf{s}_{m-1} = [u^m]\exp\sum_h x_h u^h/h!$, which is the polynomial $E_m(x_1,\dots,x_m)$ of (316):

$$\mathsf{s}_{-1}=1,\qquad \mathsf{s}_0=x_1,\qquad \mathsf{s}_1=\tfrac12(x_1^2+x_2),\qquad \mathsf{s}_2=\tfrac16(x_1^3+3x_1x_2+x_3),$$
$$\mathsf{s}_3=\tfrac{1}{24}(x_1^4+6x_1^2x_2+4x_1x_3+3x_2^2+x_4). \tag{559}$$

Remarks. (i) At $d=2$, (556) is the value of $\langle t_\ell^2 S_\tau\rangle_2$ obtained in (390) (Section 18.9.5) from Gauss's sum and the bilateral (Dougall) identity of (387) on the square lattice. At generic d no Dixon, Dougall or Thomae evaluation is needed. (ii) By (557) the spin-summed periodization of $t_\ell^2 S_\tau$ is a balanced two-wheel of rim exponent $d/2-\tau$. For $\tau\in\mathbb{H}_q$ of (571) its Ξ is constant in t, by the case $r=2$ of (471) (here r counts the factors of the product; it is unrelated to the Laurent coefficients r_k) with $\alpha_i=\beta_i=-\tau/4$ and $g_i=q+\tau/2$, that is (472) with $b_i=d/2-\tau$; (556) does not use this. (iii) For a rim exponent b with $0<\operatorname{Re} b<b_{\max}(2,d)$ put $\sigma=b/2$, as in Section 20, so that $g_b=2q-\sigma$. By the case $r=2$ of (473), $\Xi_2(t;b)=c_{2,0}(q,b)$ does not depend on t. The residue functional (479) at $r=2$, where $B_{2,0}(\sigma)=-2\pi\cot\pi\sigma$, together with $W_2(d;b)=B_d(d-b,2b)$ from (230) and N_b of (460), gives the constant:

$$\Xi_2(t;b)=c_{2,0}(q,b)=\frac{\pi\sin^3(\pi\sigma)\,W_2(d;b)}{2\cos(\pi\sigma)\,N_b^2}=\frac{\pi\sin^3(\pi\sigma)\,\Gamma(\sigma)^4\,\Gamma(2q-2\sigma)}{2\cos(\pi\sigma)\,\Gamma(2q)\,\Gamma(g_b)^4\,\Gamma(2\sigma)}. \tag{560}$$

At $b=d/2-\tau$, that is $\sigma=q-\frac{\tau}{2}$, with $\tau\in\mathbb{H}_q$ and $\operatorname{Re}\tau<2q$, this reads $\Gamma(2q)^2\Xi_2=\pi\sin^3(\pi\sigma)\langle t_\ell^2 S_\tau\rangle_d/(2\cos\pi\sigma)$, by (556). With the Gamma-reflected form of $g_\ell(a_\ell+n+t)$ in (464), $D_b^2 S_2=c_{2,0}$ becomes a bilateral identity, for $t\notin\mathbb{Z}\cup(-\sigma+\mathbb{Z})\cup(g_b+\mathbb{Z})$:

$$\sum_{\ell\ge0}d_\ell\sum_{n\in\mathbb{Z}}\Big[\frac{\Gamma(\ell+n+\sigma+t)}{\Gamma(\ell+n+2q+t)}\Big]^2\Big[\frac{\Gamma(n+1-g_b+t)}{\Gamma(n+1+t)}\Big]^2=\frac{c_{2,0}(q,b)}{\sin^2\pi(t+\sigma)\,\sin^2\pi(g_b-t)}. \tag{561}$$

At $d=2$ the left side is the square of a single bilateral sum, and (561) is the square of Dougall's ${}_2H_2$ sum, an instance of which is the last identity of (387) (with $x_\tau=\sigma$ and $\eta=t$); at generic d it is its analogue for the multiplicities d_ℓ. The package record `LRI_C5_general_d/K_general/S_part/onemode.jsonl` (code `onemode_check.py` there) confirms the constancy of Ξ_2 at $b=d/2-\tau$ and the value (560), at four nodes each, to $4\cdot10^{-17}$ at seven points $0.8\le d\le3.5$ and to $2\cdot10^{-39}$ at two points at $d=1$.

21.3 Low orders of the R family and the single new number

At $t=2\tau$, (230) gives $B_d(2q,4q-2\tau)=C(\tau)\Gamma(q-\tau)/\Gamma(q+\tau)$ and $B_d(2q+4\tau,4q-2\tau)=C(\tau)\Gamma(q-2\tau)\Gamma(q+\tau)/[\Gamma(q+2\tau)\Gamma(q-\tau)]$, so the Gamma interpolation of (238) is

$$J^\Gamma_{2\tau}=C(\tau)^2\frac{\Gamma(q-\tau)^2}{\Gamma(q+\tau)^2}+\frac{C(\tau)}{\tau}\Big[\frac{\Gamma(q-\tau)}{\Gamma(q+\tau)}-\frac{\Gamma(q-2\tau)\Gamma(q+\tau)}{\Gamma(q+2\tau)\Gamma(q-\tau)}\Big]. \tag{562}$$

Expanding $\log\Gamma$ gives the exact Laurent series

$$\Delta(\tau):=J^\Gamma_{2\tau}-C(\tau)^2\mathcal{X}_d(\tau)^2=\frac32 U\tau+AU\tau^2+O(\tau^3). \tag{563}$$

The orders $\tau^{-2},\tau^{-1},\tau^0$ cancel identically, and $[\tau^2]\Delta=AU$ is -4 times the difference $-AU/4$ of the t^2 coefficients stated after (250). A symbolic check for general q is `LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_delta_symbolic.wls`.

Low orders. Assume (M) and (J). Then $F(\tau)=\frac32U\tau^2+F_3\tau^3+O(\tau^4)$; that is,

$$\mathsf{r}_{-1}=1,\qquad \mathsf{r}_0=\mathsf{s}_0,\qquad \mathsf{r}_1=\mathsf{s}_1,\qquad \mathsf{r}_2=\mathsf{s}_2+\tfrac32U, \tag{564}$$

and

$$\boxed{\mathcal{K}_d = \frac{U}{8}\big[\psi_0(1) + \psi_0(2q) + 4\psi_0(q)\big] + \frac{\mathsf{r}_3 - \mathsf{s}_3}{4},} \tag{565}$$

with s_3 the polygamma polynomial (559). Equivalently,

$$\mathcal{K}_d = -\frac{AU}{4} + \frac{1}{4}[\tau^2]\Big\{C(\tau)\langle t_\ell^2 R_\tau\rangle_d - C(\tau)^2\mathcal{X}_d(\tau)^2\Big\}. \tag{566}$$

Equation (566) uses only (M), (563) and the definition (238), since $[\tau^2](J_{2\tau} - J^\Gamma_{2\tau}) = 4\mathcal{K}_d$ for the Laurent coefficients whatever the lower orders are; here (J) enters only (564) and (565).

Proof. By (M) and (556), $J_{2\tau} = C(\tau)^2\mathcal{X}_d(\tau)^2 + C(\tau)F(\tau)$. By (J) at $t = 2\tau$, $J_{2\tau} - J^\Gamma_{2\tau} = 4\mathcal{K}_d\tau^2 + O(\tau^3)$. Subtracting and using (563),

$$C(\tau)F(\tau) = \tfrac{3}{2}U\tau + (AU + 4\mathcal{K}_d)\tau^2 + O(\tau^3).$$

This is (566). Since $C(\tau) = \tau^{-1} + c_* + O(\tau)$, it follows that $F(\tau) = \frac{3}{2}U\tau^2 + [AU + 4\mathcal{K}_d - \frac{3}{2}c_*U]\tau^3 + O(\tau^4)$, which is (564). With $F_3 = \mathsf{r}_3 - \mathsf{s}_3$ and $A = c_* - 2\psi_0(q)$ this gives (565). □

The spectral moment. On the Mellin line $\mathrm{Re}(R_\tau - S_\tau) = E_2\tau^2 + O(\tau^3)$, with the weight $E_2 = -h^2/2 - 5j/8$ of (247), as displayed after (250) (h, j the weights of (247), not the index h of (558) or the node index j of (580); this E_2 is not the polynomial E_2 of (316), which gives s_1 in (559)). If $[\tau^2]$ may be taken under the trace (the step used in the derivation of (248), justified only at $d = 2$, by the corollary of Section 18.9.2), then $F_2 = \langle t_\ell^2 E_2\rangle_d$, and (564), which assumes (M) and (J), gives

$$\boxed{\langle t_\ell^2 E_2\rangle_d = \frac{3}{2}\,\psi_2\Big(\frac{d}{4}\Big).} \tag{567}$$

At $d = 2$ this is the value $-21\zeta(3)$ of (394); by (73) it is $-3\pi^3 - 84\zeta(3)$ at $d = 1$ and $3\pi^3 - 84\zeta(3)$ at $d = 3$.

Consequence. Equation (565) puts all non-elementary content of $\mathcal{K}_d$ into the single number r_3. The following sections evaluate $\langle t_\ell^2 R_\tau\rangle_d$, and hence r_3, by finite sampling.

21.4 The R family: lattice form, growth and finite Fourier space

Lattice form. Let $f_\ell^{\mathrm{R}}(s) = \Gamma(2q)^2 g_\ell(s)^2 R_\tau(s)$ be the summand of (554). In Sections 21.4–21.9, t is the periodization variable, not the path parameter of J_t. Direct substitution gives, for $\ell \geq 0$ and $n \in \mathbb{Z}$,

$$\begin{gathered} f_\ell^{\mathrm{R}}(a_\ell + n + t) = \Gamma(2q)^2\,\Phi_t(\ell + n)\,\Psi_t(n), \\ \Phi_t(m) = \frac{\Gamma(m+q+t)^3}{\Gamma(m+2q+t)^2\Gamma(m+q+\tau+t)}, \qquad \Psi_t(n) = \frac{\Gamma(-n-t)^2\Gamma(q-\tau-n-t)}{\Gamma(q-n-t)^3}. \end{gathered} \tag{568}$$

Reflection gives $\Psi_t(n) = T_{\mathrm{R}}(t)\Phi_{t+1-2q}(n)$ with $T_{\mathrm{R}}(t) = \sin^3\pi(q-t)/[\sin^2(\pi t)\sin\pi(q-\tau-t)]$. The summand is therefore one Gamma ratio $\widetilde{\Phi}(w) = \Gamma(w+q)^3/[\Gamma(w+2q)^2\Gamma(w+q+\tau)]$ on the two lattices $w = m+t$ and $w = n+t+1-d/2$, with the Toeplitz weight d_{m-n}, where $\sum_\ell d_\ell z^\ell = (1+z)(1-z)^{1-d}$. A half-shift as in (429) maps this structure to itself only if $2(1-d/2) \in \mathbb{Z}$, that is at $d = 1, 2, 3$; at generic d the half-shift factorization of (429) is not available.

Product structure. In the notation (460) the kernel is a product of three balanced factors:

$$t_\ell(s)^2 R_\tau(s) = \Gamma(2q)^2\, g_\ell(s;\tfrac{d}{2})^2\, g_\ell\Big(s+\frac{\tau-q}{2};\, d-2\tau\Big). \tag{569}$$

Indeed $a_\ell(d-2\tau)\pm(s+\frac{\tau-q}{2})$ equal $a_\ell+s$ and $b_\ell-\tau-s$, and $a_\ell(d-2\tau)+\tau\pm(s+\frac{\tau-q}{2})$ equal $a_\ell+\tau+s$ and $b_\ell-s$. Each factor has rim plus spoke exponent d, but the third has rim exponent $d-2\tau$ and its own shift. Equation (569) is the kernel form of the mixed trace of two critical wheel operators and one operator of rim exponent $d-2\tau$ (Section 16.4), and $C(\tau)=N_{d-2\tau}$.

Periodization. Define

$$\begin{gathered} S_{\mathrm{R}}(t;\tau)=\sum_{\ell\ge 0} d_\ell \sum_{n\in\mathbb{Z}} f_\ell^{\mathrm{R}}(a_\ell+n+t), \qquad \Xi_{\mathrm{R}}=D_{\mathrm{R}}S_{\mathrm{R}}, \\ D_{\mathrm{R}}(t)=\sin^2(\pi t)\sin^3\big(\pi(t+q)\big)\sin\big(\pi(t-q+\tau)\big). \end{gathered} \tag{570}$$

The pole classes are $\mathsf{P}_\tau=\mathbb{Z}\cup(-q+\mathbb{Z})\cup(q-\tau+\mathbb{Z})$, of orders at most $2,3,1$, matching the zeros of D_{R}. Put

$$\begin{gathered} \tau_*(q)=\max\big(0,\,2q-1,\,\tfrac12-2q\big), \qquad \mathbb{H}_q=\{\tau\in\mathbb{C}:\operatorname{Re}\tau>\tau_*(q)\}, \\ \eta_{\mathrm{R}}=2\operatorname{Re}\tau-\max(0,1-4q). \end{gathered} \tag{571}$$

On $\mathbb{H}_q$ one has $\eta_{\mathrm{R}}>0$.

Growth and finite modes. Let $0<d<4$.

(a) For $t\notin\mathsf{P}_\tau\cup(q+\mathbb{Z})$ the double series S_{R} converges absolutely iff $\tau\in\mathbb{H}_q$. For $\tau\in\mathbb{H}_q$ it converges locally uniformly off P_τ, and S_{R} is meromorphic and one-periodic in t, with poles only in P_τ of orders at most those of the zeros of D_{R}.

(b) For every compact set of pairs (q,τ) with $\tau\in\mathbb{H}_q$ there are constants Y and C with $|S_{\mathrm{R}}(x+\mathrm{i}y)|\le C|y|^{-\eta_{\mathrm{R}}}(1+\log|y|)$ for all real x and $|y|\ge Y$. The right part ($n\ge 0$), the middle part ($-\ell\le n\le -1$) and the left part ($n\le -\ell-1$) obey this bound separately for $0\le x\le 1$.

(c) Ξ_{R} is entire and one-periodic in t, jointly holomorphic on $\mathbb{C}\times\mathbb{H}_q$, and $\Xi_{\mathrm{R}}(t;\tau)=\sum_{|k|\le 2}c_k(\tau)e^{2\pi\mathrm{i}kt}$ with c_k holomorphic on $\mathbb{H}_q$.

Proof. Section 21.6 gives the complete proof, with explicit constants. Parts (a) (sufficiency), (b) and (c) are also the case $r=3$ of Step 0 of the proof of (471), with the parameters of Section 21.5, for which $\varkappa=2q+\operatorname{Re}\tau$ in (469) and $\max(d,1)-2\varkappa=-\eta_{\mathrm{R}}$. In outline: Gamma reflection writes each summand of the right, middle and left parts as a trigonometric factor times two Gamma ratios, each of total exponent $-(2q+\tau)$; a Stirling bound that is uniform in the closed right half-plane and two lattice-sum estimates give (b) and the convergence in (a); matching lower bounds give the divergence outside $\mathbb{H}_q$; and once D_{R} has cleared the poles, (b) removes the modes $|k|\ge 3$. □

The exponent η_{R} plays the role of $2q(r-2)$ in (308). The estimate holds on the whole half-plane $\mathbb{H}_q$; only the residue functional of Section 21.7 needs $\operatorname{Re}\tau<\frac32 q$. Periodicity and analyticity improve (b) to $S_{\mathrm{R}}=O(e^{-2\pi|\operatorname{Im}t|})$ on $\mathbb{H}_q$ (Section 21.6), while each part decays only like a power.

21.5 The top modes vanish

The kernel (569) is a product of three balanced kernels with different rim exponents and shifts, so the product form (471) of the top-mode theorem of Section 20.1.2 applies to it; its wheel case (473) alone does not.

The R family has the modes $|k| \le 1$. For $0 < d < 4$ and $\tau \in \mathbb{H}_q$, the coefficients $c_{\pm 2}$ in part (c) of the growth theorem of Section 21.4 vanish:

$$\boxed{\Xi_{\mathrm{R}}(t;\tau) = c_0(\tau) + c_1(\tau)e^{2\pi \mathrm{i}t} + c_{-1}(\tau)e^{-2\pi \mathrm{i}t},} \tag{572}$$

equivalently $S_{\mathrm{R}} = O(e^{-4\pi|\operatorname{Im} t|})$.

Proof. By (569), $f^{\mathrm{R}}_\ell(a_\ell + n + t) = \Gamma(2q)^2 T_\ell(n+t)$ with T_ℓ of (470) for $r = 3$, $\sigma = q$, $(g_1, g_2, g_3) = (q, q, \tau)$, $\alpha_i = 0$ and $(\beta_1, \beta_2, \beta_3) = (0, 0, q-\tau)$; this is (472) with $b = (\frac{d}{2}, \frac{d}{2}, d-2\tau)$ and $\delta = (0, 0, \frac{\tau - q}{2})$. Then D_T, S_T and Ξ_T of (470) are D_{R}, $\Gamma(2q)^{-2}S_{\mathrm{R}}$ and $\Gamma(2q)^{-2}\Xi_{\mathrm{R}}$, so $\gamma_m = \Gamma(2q)^{-2}c_m$. Condition (B) of (469) holds factor by factor, and (C) reads $2q + \operatorname{Re}\tau > \max(2q, 4q-1, \frac{1}{2})$, that is $\tau \in \mathbb{H}_q$. □

A three-wheel has the same exponential modes; there reflection pairs $k = \pm 1$ into the single cosine mode $c_{3,1}$, while R_τ is not even and $c_{\pm 1}$ are independent. For complex τ no reflection relates c_2 and c_{-2}; Step 6 of the proof of (471) treats them separately.

Remarks. (i) Read as a wheel with the single pair $(\sigma, g_3) = (q, \tau)$, the third factor is off balance by $\tau - q$; only the per-factor balance (B) holds, which is why the product form is needed; with its own shift $\beta_3 = q - \tau$ the factor is balanced and (478) applies. (ii) Per-factor balance is needed: total balance does not suffice, and the multiplicities must be those of the dimension d in (B), as the following two controls show numerically. At $d = 1.7$, $\tau = 0.55 + 0.3\mathrm{i}$, a control kernel with right shifts $\beta = (0.15, -0.15, q - \tau)$, which keeps the total balance, has $|c_{-2}|/\max_k |c_k| = 1.4 \cdot 10^{-2}$, and the R kernel with the multiplicities of $d' = 1.9$, for which the phase Θ of Step 5 of the proof of (471) obeys $\Theta - \Theta \circ \iota = (d - d')(V - \pi)$, has $6 \cdot 10^{-2}$; for the balanced kernel the ratio is below 10^{-15}, within the spread of the spin-sum extrapolation. (iii) (471) also contains the S family ($r = 2$, rim exponent $d/2 - \tau$, by (557)). The consequences of (473) for Section 19 are stated in Section 20.1.2, paragraph "Relation to the periodization at $b = d/2$ and to the half-shift reduction".

21.6 Proof of the growth theorem

This subsection proves part (b) of the growth theorem of Section 21.4 with explicit constants, derives (a) and (c) from Step 0 of the proof of (471), and proves the converse in (a), that is the sharpness of $\mathbb{H}_q$, and the exponential decay of S_{R}. It uses three estimates proved in Section 20.1: the Gamma-ratio bound (465), the multiplicity bound (466) and the lattice sums (467). Throughout, $0 < q < 1$, $d = 4q$, $t = x + \mathrm{i}y$ and $\varkappa = 2q + \operatorname{Re}\tau$; log, arg $\in (-\pi, \pi]$ and complex powers are principal; and, besides the generic constants C of Section 21.1, a bare lowercase c is a positive generic constant. In terms of $\varkappa$, $\tau \in \mathbb{H}_q$ means

$$\varkappa > 2q \ \ (\text{that is, } \operatorname{Re}\tau > 0), \qquad \varkappa > 4q - 1, \qquad 2\varkappa > 1, \tag{573}$$

which is the condition of (467) with $\alpha = \varkappa$; its three entries come from the region where both lattice indices are large, from the end $\ell \to \infty$ and from the end $n \to \infty$ (paragraph *The entries of* $b_{\max}$ of Section 20.1). So $\tau_* = \frac{1}{2} - 2q$ for $q \le \frac{1}{4}$, $\tau_* = 0$ for $\frac{1}{4} \le q \le \frac{1}{2}$ and $\tau_* = 2q - 1$ for $q \ge \frac{1}{2}$, and $\eta_{\mathrm{R}} = 2\varkappa - \max(d, 1)$.

Reflected form of the three parts. Put

$$\begin{aligned} \mathsf{F}_1(z) &= \frac{\Gamma(z+q)^3}{\Gamma(z+2q)^2\,\Gamma(z+q+\tau)}, & \mathsf{F}_2(z) &= \frac{\Gamma(z+1-q)^3}{\Gamma(z+1)^2\,\Gamma(z+1-q+\tau)}, \\ \mathsf{F}_3(z) &= \frac{\Gamma(z)^2\,\Gamma(z+q-\tau)}{\Gamma(z+q)^3}, & \mathsf{F}_4(z) &= \frac{\Gamma(z+1-2q)^2\,\Gamma(z+1-q-\tau)}{\Gamma(z+1-q)^3}, \end{aligned} \tag{574}$$

so that $\mathsf{F}_1 = \widetilde{\Phi}$ and $\Phi_t(m) = \mathsf{F}_1(m+t)$ in (568) (the F_i are not the Laurent coefficients F_k of (555)). The reflection formula $\Gamma(w)\Gamma(1-w) = \pi/\sin\pi w$ [26] and $\sin\pi(c-n) = (-1)^n \sin\pi c$ give,

for integers $n, j, p \ge 0$,

$$\Psi_t(n) = T_{\mathrm R}(t)\,\mathsf F_2(n+t), \qquad \Psi_t(-j-1) = \mathsf F_3(j+1-t), \qquad \Phi_t(-1-p) = T_{\mathrm L}(t)\,\mathsf F_4(p+1-t),$$

with $T_{\mathrm R}$ of Section 21.4 and $T_{\mathrm L}(t) = \sin^2\pi(2q-1+t)\,\sin\pi(q+\tau-1+t)/\sin^3\pi(q-1+t)$; the reflection signs combine to $(-1)^{4n} = (-1)^{6p} = 1$. Hence the right part ($n \ge 0$), the middle part ($n = -j-1$, $\ell = j+k+1$) and the left part ($n = -\ell-1-p$) of $S_{\mathrm R}$ are

$$\begin{aligned} S^{\rightarrow}(t) &= \Gamma(2q)^2\,T_{\mathrm R}(t)\sum_{\ell,n\ge0} d_\ell\,\mathsf F_1(\ell+n+t)\,\mathsf F_2(n+t),\\ S^{|}(t) &= \Gamma(2q)^2\sum_{j,k\ge0} d_{j+k+1}\,\mathsf F_1(k+t)\,\mathsf F_3(j+1-t),\\ S^{\leftarrow}(t) &= \Gamma(2q)^2\,T_{\mathrm L}(t)\sum_{\ell,p\ge0} d_\ell\,\mathsf F_4(p+1-t)\,\mathsf F_3(\ell+p+1-t), \end{aligned} \tag{575}$$

that is $\Gamma(2q)^2 P_{\mathrm R}H_{\mathrm R}$, $\Gamma(2q)^2 P_{\mathrm M}H_{\mathrm M}$ and $\Gamma(2q)^2 P_{\mathrm L}H_{\mathrm L}$ of (581), and $S_{\mathrm R} = S^{\rightarrow} + S^{|} + S^{\leftarrow}$ wherever the double series converges absolutely. Since $R_\tau(-s) \ne R_\tau(s)$, the left part is not the right part at a reflected point, unlike the wheels of Section 20.1, and it is estimated separately. For $0 \le x \le 1$ every argument z in (575) is $m+t$ or $m+1-t$ with an integer $m \ge 0$ (one of $\ell+n$, n, k, j, p, $\ell+p$), so $\operatorname{Re} z \ge m$ and $|z| \ge \max(m, |y|)$.

Gamma ratios and multiplicities. Let $\rho \ge 2 + |\tau|$. Every parameter in (574) then has modulus at most ρ, and each $\mathsf F_i$ is a product of three ratios of (465),

$$\begin{aligned} \mathsf F_1 &= \Big[\frac{\Gamma(z+q)}{\Gamma(z+2q)}\Big]^2\frac{\Gamma(z+q)}{\Gamma(z+q+\tau)}, \quad \mathsf F_2 = \Big[\frac{\Gamma(z+1-q)}{\Gamma(z+1)}\Big]^2\frac{\Gamma(z+1-q)}{\Gamma(z+1-q+\tau)},\\ \mathsf F_3 &= \Big[\frac{\Gamma(z)}{\Gamma(z+q)}\Big]^2\frac{\Gamma(z+q-\tau)}{\Gamma(z+q)}, \quad \mathsf F_4 = \Big[\frac{\Gamma(z+1-2q)}{\Gamma(z+1-q)}\Big]^2\frac{\Gamma(z+1-q-\tau)}{\Gamma(z+1-q)}, \end{aligned}$$

whose exponents $u - v$ add up to $-2q-\tau$. Since $z^a z^b = z^{a+b}$ for principal powers, (465) gives, for $\operatorname{Re} z \ge 0$, $|z| \ge 2\rho$ and $i = 1, \dots, 4$,

$$\begin{gathered} \mathsf F_i(z) = z^{-2q-\tau}e^{\theta_i(z)}, \qquad |\theta_i(z)| \le \frac{3\omega(\rho)}{|z|},\\ e^{-3\omega(\rho)/|z| - \frac{\pi}{2}|\operatorname{Im}\tau|}\,|z|^{-\varkappa} \le |\mathsf F_i(z)| \le e^{3\omega(\rho)/|z| + \frac{\pi}{2}|\operatorname{Im}\tau|}\,|z|^{-\varkappa}. \end{gathered} \tag{576}$$

By (466), $|d_\ell| \le C_q(1+\ell)^{d-2}$ with C_q bounded for q in compact subsets of $(0,1)$, across $d = 1$, and $d_\ell(1+\ell)^{2-d} \to 2/\Gamma(d-1)$; the limit vanishes only at $q = \frac14$, so for $q \ne \frac14$ also $|d_\ell| \ge c\,\ell^{d-2}$ for large ℓ. By (301), with $(4q-1)_k = (4q-1)(4q)_{k-1}$, all $d_\ell \ge 0$ for $q \ge \frac14$, while for $q < \frac14$ ($d < 1$) $d_\ell < 0$ for every $\ell \ge 2$.

Trigonometric factors. *Lemma.* Let $c_{\mathrm T} = 8/(1-e^{-2\pi})^3 \approx 8.045$. If $|y| \ge 1$ and $|y + \operatorname{Im}\tau| \ge 1$, then $|T_{\mathrm R}(t)|, |T_{\mathrm L}(t)| \le c_{\mathrm T} e^{\pi|\operatorname{Im}\tau|}$. For all t, $|D_{\mathrm R}(t)| \le e^{6\pi|y| + \pi|\operatorname{Im}\tau|}$, and if $|y| \ge 1 + |\operatorname{Im}\tau|$, then $|D_{\mathrm R}(t)| \ge \big(\frac{1-e^{-2\pi}}{2}\big)^6 e^{6\pi|y| - \pi|\operatorname{Im}\tau|}$.

Proof. From $|\sin\pi w|^2 = \sin^2(\pi\operatorname{Re} w) + \sinh^2(\pi\operatorname{Im} w)$ one has $\sinh(\pi|\operatorname{Im} w|) \le |\sin\pi w| \le e^{\pi|\operatorname{Im} w|}$, and $\sinh(\pi b) \ge \frac12(1-e^{-2\pi})e^{\pi b}$ for $b \ge 1$. The sines of $T_{\mathrm R}$ have imaginary parts $-y$ (numerator, three times) and y, y, $-(y+\operatorname{Im}\tau)$ (denominator); those of $T_{\mathrm L}$ have y, y, $y+\operatorname{Im}\tau$ (numerator) and y three times (denominator); those of $D_{\mathrm R}$ have y five times and $y+\operatorname{Im}\tau$ once. Insert these bounds and use $|y| - |\operatorname{Im}\tau| \le |y+\operatorname{Im}\tau| \le |y| + |\operatorname{Im}\tau|$. □

Proof of (b), with constants. With $\alpha = \varkappa$, the sums $\Sigma^{\mathrm{r}}(Y)$ and $\Sigma^{\mathrm{m}}(Y)$ of (467) obey $\Sigma^{\mathrm{r}}(Y) \le C^{\mathrm{r}} Y^{-\eta_{\mathrm{R}}}(1+\log Y)$ and $\Sigma^{\mathrm{m}}(Y) \le C^{\mathrm{m}} Y^{-2\operatorname{Re}\tau}$ for $Y \ge 1$, since $\max(d,1) - 2\varkappa = -\eta_{\mathrm{R}}$ and $d - 2\varkappa = -2\operatorname{Re}\tau$. Let $\mathfrak{K}$ be a compact set of pairs (q,τ) with $0 < q < 1$ and $\tau \in \mathbb{H}_q$, and take maxima over $\mathfrak{K}$. Put $\rho_{\mathfrak{K}} = 2 + \max|\tau|$ and $Y_{\mathfrak{K}} = 2\rho_{\mathfrak{K}}$, which exceeds $1 + \max|\operatorname{Im}\tau|$; let $c_{\mathfrak{K}} = 2^{\varkappa}\exp\bigl(3\omega(\rho_{\mathfrak{K}})/Y_{\mathfrak{K}} + \tfrac{\pi}{2}|\operatorname{Im}\tau|\bigr)$, a function of (q,τ), and

$$C_{\mathfrak{K}} = \max \Gamma(2q)^2\, C_q\, c_{\mathfrak{K}}^2\, \bigl[c_{\mathrm{T}} e^{\pi|\operatorname{Im}\tau|} C^{\mathrm{r}} + 2^{\max(d-2,0)} C^{\mathrm{m}}\bigr],$$

with C^{r}, C^{m} at $\alpha = \varkappa$; it is finite because C_q, C^{r} and C^{m} are bounded on compact sets. *For $(q,\tau) \in \mathfrak{K}$ and $|y| \ge Y_{\mathfrak{K}}$ the three series* (575) *converge absolutely for $0 \le x \le 1$, $|S^{\rightarrow}|$ and $|S^{\leftarrow}|$ are at most $C_{\mathfrak{K}}|y|^{-\eta_{\mathrm{R}}}(1+\log|y|)$, $|S^{|}|$ is at most $C_{\mathfrak{K}}|y|^{-2\operatorname{Re}\tau}$, and for all real x*

$$\begin{aligned} |S_{\mathrm{R}}(x+\mathrm{i}y)| &\le 3C_{\mathfrak{K}}\,|y|^{-\eta_{\mathrm{R}}}(1+\log|y|), \\ |\Xi_{\mathrm{R}}(x+\mathrm{i}y)| &\le 3C_{\mathfrak{K}}\, e^{\pi|\operatorname{Im}\tau|} e^{6\pi|y|}\,|y|^{-\eta_{\mathrm{R}}}(1+\log|y|). \end{aligned} \tag{577}$$

This is (b) with $Y = Y_{\mathfrak{K}}$ and $C = 3C_{\mathfrak{K}}$, the bound on the parts holding for $0 \le x \le 1$.

Proof. Fix $(q,\tau) \in \mathfrak{K}$, $0 \le x \le 1$ and $|y| \ge Y_{\mathfrak{K}}$. Every argument z in (575) has $\operatorname{Re} z \ge 0$ and $|z| \ge \max(m,|y|) \ge \frac{1}{2}(m+|y|)$, in particular $|z| \ge Y_{\mathfrak{K}} = 2\rho_{\mathfrak{K}}$, so (576) with $\rho = \rho_{\mathfrak{K}}$ gives $|\mathsf{F}_i(z)| \le c_{\mathfrak{K}}(m+|y|)^{-\varkappa}$. The lemma on trigonometric factors applies, since $|y| \ge 1 + |\operatorname{Im}\tau|$. With $|d_\ell| \le C_q(1+\ell)^{d-2}$ and $|d_{j+k+1}| \le C_q(2+j+k)^{d-2} \le 2^{\max(d-2,0)} C_q (1+j+k)^{d-2}$, the series (575) are dominated termwise:

$$|S^{\rightarrow}|,\ |S^{\leftarrow}| \le \Gamma(2q)^2 c_{\mathrm{T}} e^{\pi|\operatorname{Im}\tau|} C_q\, c_{\mathfrak{K}}^2\, \Sigma^{\mathrm{r}}(|y|), \qquad |S^{|}| \le \Gamma(2q)^2\, 2^{\max(d-2,0)} C_q\, c_{\mathfrak{K}}^2\, \Sigma^{\mathrm{m}}(|y|);$$

for the left part the dominating sum $\sum_{\ell,p\ge 0}(1+\ell)^{d-2}(\ell+p+|y|)^{-\varkappa}(p+|y|)^{-\varkappa}$ is again $\Sigma^{\mathrm{r}}(|y|)$. This gives the bounds on the parts. Their absolute convergence also justifies $S_{\mathrm{R}} = S^{\rightarrow} + S^{|} + S^{\leftarrow}$ for $0 \le x \le 1$; for other x one uses $S_{\mathrm{R}}(t+1) = S_{\mathrm{R}}(t)$, the shift $n \mapsto n+1$ of the absolutely convergent double series. Finally $|\Xi_{\mathrm{R}}| \le |D_{\mathrm{R}}||S_{\mathrm{R}}|$ and the bound on D_{R}. □

The constant is explicit but crude, because $\omega(\rho)$ grows like $\frac{11}{3}\rho^3$. For $|y| \ge Y \ge Y_{\mathfrak{K}}$ one may replace $Y_{\mathfrak{K}}$ by Y in $c_{\mathfrak{K}}$; as $Y \to \infty$ the constant decreases to the same expression with $c_{\mathfrak{K}}$ replaced by $2^{\varkappa} e^{\pi|\operatorname{Im}\tau|/2}$ (Section 21.11, item Growth constants).

Proof of (a) and (c). Sufficiency in (a), and (c), are the case $r = 3$ of Step 0 of the proof of (471), carried out in full in the paragraph *Details of the top-mode proof* of Section 20.1.2, with the parameters of Section 21.5: $\sigma = q$, $(g_1,g_2,g_3) = (q,q,\tau)$, $\alpha_i = 0$ and $(\beta_1,\beta_2,\beta_3) = (0,0,q-\tau)$. Then $f_\ell^{\mathrm{R}}(a_\ell + n + t) = \Gamma(2q)^2 T_\ell(n+t)$, $D_T = D_{\mathrm{R}}$, $\varkappa = 2q + \operatorname{Re}\tau$, and condition (C) of (469) is $\tau \in \mathbb{H}_q$. The six pole classes of Step 0 are $-q+\mathbb{Z}$ three times, $\mathbb{Z}$ twice and $q-\tau+\mathbb{Z}$ once, so $\mathsf{P}_T = \mathsf{P}_\tau$: through $\Gamma(\ell+n+q+t)^3$, $\Gamma(-n-t)^2$ and $\Gamma(q-\tau-n-t)$ a term of S_{R} has poles of orders at most 3 on $-q+\mathbb{Z}$, 2 on $\mathbb{Z}$ and 1 on $q-\tau+\mathbb{Z}$, added where classes meet, which are the zero orders of D_{R}. Step 0 gives absolute and locally uniform convergence off P_τ; uniform convergence of $(t-t_0)^o$ times the series on a small circle around each $t_0 \in \mathsf{P}_\tau$, with o the order of the zero of D_{R} at t_0, hence the pole orders in (a); periodicity by $n \mapsto n+1$; an entire one-periodic Ξ_{R}; the vanishing of the modes $|k| \ge 3$, the boundary modes included because $\eta_{\mathrm{R}} = 2\varkappa - \max(d,1) > 0$; and coefficients $c_k = \Gamma(2q)^2\gamma_k$ that are holomorphic in the parameters, hence in $\tau \in \mathbb{H}_q$. As a trigonometric polynomial in t with holomorphic coefficients, Ξ_{R} is jointly holomorphic on $\mathbb{C} \times \mathbb{H}_q$. The converse in (a) is proved in the paragraph *Sharpness of* $\mathbb{H}_q$ below. □

Exponential decay of the total. For $y > Y_{\mathfrak{K}}$ there are no poles, since P_τ lies on the lines $\operatorname{Im} t = 0$ and $\operatorname{Im} t = -\operatorname{Im}\tau$; so $S_{\mathrm{R}}(t) = \widehat{S}(e^{2\pi\mathrm{i}t})$ with $\widehat{S}$ holomorphic on $0 < |w| < e^{-2\pi Y_{\mathfrak{K}}}$. By (577), $\widehat{S}(w) \to 0$ as $w \to 0$, so $\widehat{S}$ extends holomorphically with $\widehat{S}(0) = 0$ (Riemann's theorem on removable singularities). The maximum principle for $\widehat{S}(w)/w$ on $|w| \le e^{-2\pi(Y_{\mathfrak{K}}+1)}$ then gives $|S_{\mathrm{R}}(x+\mathrm{i}y)| \le e^{2\pi(Y_{\mathfrak{K}}+1)} M_{\mathfrak{K}}\, e^{-2\pi y}$ for $y \ge Y_{\mathfrak{K}} + 1$, where $M_{\mathfrak{K}} = 3C_{\mathfrak{K}}(Y_{\mathfrak{K}}+1)^{-\eta_{\mathrm{R}}}(1+\log(Y_{\mathfrak{K}}+1))$

bounds $|S_{\mathrm{R}}|$ on $\operatorname{Im} t = Y_{\mathfrak{K}} + 1$; for $y \le -Y_{\mathfrak{K}} - 1$ one uses $e^{-2\pi i t}$ in place of w. So $S_{\mathrm{R}} = O(e^{-2\pi|\operatorname{Im} t|})$ and $\Xi_{\mathrm{R}} = O(e^{4\pi|\operatorname{Im} t|})$, uniformly in x and on $\mathfrak{K}$, while numerically the parts decay only like powers (Section 21.11, item Growth): the exponential decay of the total is a cancellation between the parts. Equation (572) improves this to $S_{\mathrm{R}} = O(e^{-4\pi|\operatorname{Im} t|})$.

Sharpness of $\mathbb{H}_q$. *Let $0 < q < 1$ and $\tau \notin \mathbb{H}_q$. For every $t \notin \mathsf{P}_\tau \cup (q + \mathbb{Z})$ the double series $S_{\mathrm{R}}(t)$ does not converge absolutely.* So the lattice sum, and with it any estimate of the type (b), exists exactly on $\mathbb{H}_q$, up to the discrete set of exceptional t; this is the converse in (a).

Proof. Fix t, and put $\rho = 2 + |\tau|$ and $m_0 = \lceil 2|\operatorname{Re} t| + 2 + 2\rho \rceil$. For an integer $m \ge m_0$ the arguments $z = m + t$ and $z = m + 1 - t$ of (575) have $\operatorname{Re} z \ge m - |\operatorname{Re} t| \ge \max(2\rho, \frac{1}{2}(1+m))$, so (576) gives $c\,|z|^{-\varkappa} \le |\mathsf{F}_i(z)| \le C|z|^{-\varkappa}$, and $\mathsf{F}_i(z) \ne 0$ there; $T_{\mathrm{R}}(t) \ne 0$ because $t \notin q + \mathbb{Z}$; and for $q \ne \frac{1}{4}$, $|d_\ell| \ge c\,\ell^{d-2}$ for large ℓ. One of the conditions (573) fails.

- $2\varkappa \le 1$. The terms of the right part with $\ell = 0$, $d_0 \Gamma(2q)^2 T_{\mathrm{R}}(t) \mathsf{F}_1(n+t) \mathsf{F}_2(n+t)$, have modulus at least $c\,n^{-2\varkappa}$ for large n and are not summable. This case contains $q = \frac{1}{4}$, where $\tau_* = 0 = \frac{1}{2} - 2q$.
- $2\varkappa > 1$ and $\varkappa \le 4q - 1$, so $q > \frac{3}{8}$. Fix $j = j_0 \ge m_0$ in the middle part. Its terms $d_{j_0+k+1} \Gamma(2q)^2 \mathsf{F}_1(k+t) \mathsf{F}_3(j_0 + 1 - t)$, with $\mathsf{F}_3(j_0 + 1 - t) \ne 0$ fixed, have modulus at least $c\,k^{4q-2-\varkappa}$ for large k, and $4q - 2 - \varkappa \ge -1$.
- $2\varkappa > 1$, $\varkappa > 4q - 1$ and $\varkappa \le 2q$, so $q > \frac{1}{4}$ and $\operatorname{Re}\tau \le 0$. In the middle part the block $N \le j, k < 2N$, for large N, has N^2 terms, each of modulus at least $c\,N^{4q-2} N^{-2\varkappa}$, because $j + k + 1$ lies between N and $4N$ and $|k + t|, |j + 1 - t| \le 3N$. The block contributes at least $c\,N^{-2\operatorname{Re}\tau} \ge c$, and the sum over the blocks $N = 2^i$ diverges. □

Is the exponent η_{R} attained? For $q \ge \frac{1}{4}$, where all $d_\ell \ge 0$ and $\eta_{\mathrm{R}} = 2\operatorname{Re}\tau$, the bound on the right and left parts is attained numerically: $\max_x |S^{\rightarrow}||y|^{\eta_{\mathrm{R}}}$ tends to a constant at $d = 1.25$, 1.5, 2.5 and 3 (records `n5_case0.log` and `n5_case2.log` to `n5_case5.log` in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`). Numerically it is not attained at $d = 0.8$ ($q < \frac{1}{4}$), the only such point tested: there the right and left parts decay like $|y|^{-2\operatorname{Re}\tau}$, faster than $|y|^{-\eta_{\mathrm{R}}}$ (`n5_case1.log`). The absolute values taken in the proof lose the cancellation $\sum_\ell d_\ell = \lim_{z\to 1}(1+z)(1-z)^{1-4q} = 0$ of the negative multiplicities.

21.7 Residue functional, three-node rule and the Horn samples

Three-node rule. Let $\tau \in \mathbb{H}_q$ with $\operatorname{Re}\tau < \frac{3}{2}q$ and $\tau \ne q$. Then

$$\langle t_\ell^2 R_\tau \rangle_d = -\operatorname{Res}_{t=0} \frac{\Xi_{\mathrm{R}}}{D_{\mathrm{R}}} - \operatorname{Res}_{t=q-\tau} \frac{\Xi_{\mathrm{R}}}{D_{\mathrm{R}}} = \sum_{|k|\le 1} \rho_k c_k, \tag{578}$$

$$\rho_k = -\frac{2\pi i k - \pi\big[3\cot \pi q + \cot \pi(\tau - q)\big]}{\pi^2 \sin^3(\pi q) \sin \pi(\tau - q)} - \frac{e^{2\pi i k(q-\tau)}}{\pi \sin^2 \pi(q-\tau)\, \sin^3 \pi(2q - \tau)}, \tag{579}$$

and for every real t_0 with $t_0 + \frac{j}{3} \notin \mathsf{P}_\tau$,

$$\boxed{\langle t_\ell^2 R_\tau \rangle_d = \sum_{j=0}^{2} w_j(\tau)\, \Xi_{\mathrm{R}}\big(t_0 + \tfrac{j}{3}; \tau\big), \qquad w_j(\tau) = \frac{1}{3} \sum_{|k|\le 1} \rho_k(\tau)\, e^{-2\pi i k(t_0 + j/3)}.} \tag{580}$$

Proof. The left poles of f_ℓ^{R} are $s = -a_\ell - n$, and the right poles are $s = a_\ell + n$ and $s = b_\ell - \tau + n$, $n \ge 0$, that is $t = 0$ and $t = q - \tau$ on the lattice $s = a_\ell + n + t$. The imaginary axis separates the

two families because $a_\ell \geq \frac{q}{2} > 0$ and $\mathrm{Re}(b_\ell - \tau) \geq \frac{3}{2}q - \mathrm{Re}\,\tau > 0$. The two right families are disjoint because $q - \tau \notin \mathbb{Z}$, which on the strip excludes only $\tau = q$. Moreover $2q - \tau \notin \mathbb{Z}$ holds automatically: $2q - \tau = m \in \mathbb{Z}$ would give $\mathrm{Re}\,\tau = 2q - m$, which is either $\geq 2q > \frac{3}{2}q$ or $\leq 2q - 1 \leq \tau_*(q)$.

The trace converges absolutely. Put $\varkappa = 2q + \mathrm{Re}\,\tau$ (the $\varkappa$ of (469) for this kernel). On the axis, the Gamma-ratio bound (465), which gives $|\Gamma(z+u)/\Gamma(z+v)| \leq C|z|^{\mathrm{Re}(u-v)}$ for $\mathrm{Re}\,z \geq 0$ and $|z|$ large, applied with $z = a_\ell \pm \mathrm{i}\nu$, gives $|f_\ell^{\mathrm{R}}(\mathrm{i}\nu)| \leq C(a_\ell + |\nu|)^{-2\varkappa}$ for $a_\ell + |\nu|$ large. So each ν-integral converges ($2\varkappa > 1$), and with $|d_\ell| \leq C(1+\ell)^{d-2}$, $\sum_\ell |d_\ell| \int |f_\ell^{\mathrm{R}}(\mathrm{i}\nu)|\, d\nu \leq C \sum_{\ell \geq 1} \ell^{4q-2}\ell^{1-2\varkappa} < \infty$, since $4q - 1 - 2\varkappa = -1 - 2\,\mathrm{Re}\,\tau$ and $\mathrm{Re}\,\tau > 0$.

Close each Barnes contour of (554) to the right, on arcs $|s| = R_N \to \infty$, $\mathrm{Re}\,s \geq 0$, with radii chosen so that the arcs keep a distance at least $\frac{1}{8}$ from both right lattices. On the arcs, reflection of $\Gamma(a_\ell - s)^2\Gamma(b_\ell - \tau - s)/\Gamma(b_\ell - s)^3$ [26] gives a ratio of three sines over three sines, bounded away from the lattices and for large $|\mathrm{Im}\,s|$, times two Gamma ratios of the type above. Hence $|f_\ell^{\mathrm{R}}(s)| \leq C_\ell |s|^{-4q-2\,\mathrm{Re}\,\tau}$ with $4q + 2\,\mathrm{Re}\,\tau > 1$, so the arcs do not contribute, and the integral is minus the sum of the right residues. The terms of S_{R} with $n < 0$ are holomorphic at $t = 0$ and $t = q - \tau$, since a pole there would need $q \in \mathbb{Z}$, $q - \tau \in \mathbb{Z}$ or $2q - \tau \in \mathbb{Z}$. Uniform convergence on small circles (Section 21.6) therefore identifies the sums of residues with the residues of $S_{\mathrm{R}} = \Xi_{\mathrm{R}}/D_{\mathrm{R}}$, and (572) gives $\sum_{|k| \leq 1} \rho_k c_k$. The three-point discrete Fourier transform is exact for modes $|k| \leq 1$. □

Part (c) of the growth theorem alone gives a five-node rule, with the nodes $t_0 + j/5$ and the modes $|k| \leq 2$. By (572), three nodes give the same $\mathcal{K}_d$ at three fifths of the sampling cost (Section 21.11). On the jet circle at $d = 2$, ρ_k has a triple pole at $\tau = 0$, from $\sin^3 \pi(2q - \tau)$, which cancels in the sum; there $\tau = 0$ is also a spurious pole of individual pieces of the continued samples (Section 21.8), and the exceptional set E of hypothesis (H), stated in Section 21.8, still meets the closed disk $|\tau| \leq r_\tau$ only in $\{0\}$.

Separating contours (sketch). The strip of (578), $\tau \in \mathbb{H}_q$ with $\mathrm{Re}\,\tau < \frac{3}{2}q$, is nonempty iff $d > \frac{4}{7}$. For $\mathrm{Re}\,\tau \geq \frac{3}{2}q$ the imaginary axis no longer separates the pole families for the finitely many spins with $\mathrm{Re}(b_\ell - \tau) \leq 0$. For $d \leq \frac{4}{7}$ this happens at every $\tau \in \mathbb{H}_q$, because then $\tau_*(q) = \frac{1}{2} - 2q \geq \frac{3}{2}q$. For those spins one replaces the axis by a contour that separates the left pole family from the two right ones. The proof above then gives (578), for $\tau \notin q + \mathbb{Z}$, for the resulting separating-contour trace. That trace is jointly meromorphic in (q, τ). Its poles lie at $\tau \in 2q + \mathbb{N}_0$, where the left family $s = -a_\ell - n$ and the right family $s = b_\ell - \tau + n'$ pinch the contour ($\tau = \ell + 2q + n + n'$). There the factor $1/\sin^3 \pi(2q - \tau)$ of ρ_k in (579) is singular as well, and $\tau = 2q$, which has $\mathrm{Re}\,\tau \geq \frac{3}{2}q$, lies in $\mathbb{H}_q$ for $q > \frac{1}{8}$. The trace agrees with the axis trace wherever the latter is defined, so it is the meromorphic continuation of the trace in τ and, for $d \leq \frac{4}{7}$, also in q. This is a sketch: the construction is not carried out in full in this note (Section 21.12).

The samples as Horn functions. Put

$$\widehat{\Phi}(m) = \frac{(q+t)_m^3}{(2q+t)_m^2 (q+\tau+t)_m}, \qquad \widehat{\Psi}(n) = \frac{(1-q+t)_n^3}{(1+t)_n^2 (1-q+\tau+t)_n},$$
$$\widehat{\Psi}_{\mathrm{M}}(m) = \frac{(1-t)_m^2 (1+q-\tau-t)_m}{(1+q-t)_m^3}, \quad \widehat{\Phi}_{\mathrm{L}}(p) = \frac{(2-2q-t)_p^2 (2-q-\tau-t)_p}{(2-q-t)_p^3}.$$

For $\tau \in \mathbb{H}_q$ and $t \notin \mathsf{P}_\tau$,

$$\boxed{\Xi_{\mathrm{R}}(t;\tau) = D_{\mathrm{R}}(t)\, \Gamma(2q)^2 \big[P_{\mathrm{R}} H_{\mathrm{R}} + P_{\mathrm{M}} H_{\mathrm{M}} + P_{\mathrm{L}} H_{\mathrm{L}}\big],} \tag{581}$$

with $P_{\mathrm R}=\Phi_t(0)\Psi_t(0)$, $P_{\mathrm M}=\Phi_t(0)\Psi_t(-1)$, $P_{\mathrm L}=\Phi_t(-1)\Psi_t(-1)$ from (568), and

$$H_{\mathrm R}=\sum_{\ell,n\ge 0} d_\ell\widehat\Phi(\ell+n)\widehat\Psi(n),\qquad H_{\mathrm M}=\sum_{j,k\ge 0} d_{j+k+1}\widehat\Psi_{\mathrm M}(j)\widehat\Phi(k),$$
$$H_{\mathrm L}=\sum_{\ell,p\ge 0} d_\ell\widehat\Psi_{\mathrm M}(\ell+p)\widehat\Phi_{\mathrm L}(p).$$

In the convention of (320), with $\boldsymbol e=(1,0)$, $\boldsymbol f=(0,1)$ and $\boldsymbol c=\boldsymbol e+\boldsymbol f$, these are $H_{\mathrm R}=\mathfrak H[\mathsf U_{\mathrm R};\mathsf V_{\mathrm R};(1,1)]$, $H_{\mathrm M}=4q\,\mathfrak H[\mathsf U_{\mathrm M};\mathsf V_{\mathrm M};(1,1)]$ and $H_{\mathrm L}=\mathfrak H[\mathsf U_{\mathrm L};\mathsf V_{\mathrm L};(1,1)]$, where

$$\begin{aligned}
\mathsf U_{\mathrm R}&=\{(4q-2,\boldsymbol e),(2q,\boldsymbol e),(q+t,\boldsymbol c)^{\times 3},(1,\boldsymbol f),(1-q+t,\boldsymbol f)^{\times 3}\},\\
\mathsf V_{\mathrm R}&=\{(2q-1,\boldsymbol e),(2q+t,\boldsymbol c)^{\times 2},(q+\tau+t,\boldsymbol c),(1+t,\boldsymbol f)^{\times 2},(1-q+\tau+t,\boldsymbol f)\},\\
\mathsf U_{\mathrm M}&=\{(4q-1,\boldsymbol c),(2q+1,\boldsymbol c),(1,\boldsymbol e),(1-t,\boldsymbol e)^{\times 2},(1+q-\tau-t,\boldsymbol e),\\
&\qquad (1,\boldsymbol f),(q+t,\boldsymbol f)^{\times 3}\},\\
\mathsf V_{\mathrm M}&=\{(2q,\boldsymbol c),(2,\boldsymbol c),(1+q-t,\boldsymbol e)^{\times 3},(2q+t,\boldsymbol f)^{\times 2},(q+\tau+t,\boldsymbol f)\},\\
\mathsf U_{\mathrm L}&=\{(4q-2,\boldsymbol e),(2q,\boldsymbol e),(1-t,\boldsymbol c)^{\times 2},(1+q-\tau-t,\boldsymbol c),\\
&\qquad (1,\boldsymbol f),(2-2q-t,\boldsymbol f)^{\times 2},(2-q-\tau-t,\boldsymbol f)\},\\
\mathsf V_{\mathrm L}&=\{(2q-1,\boldsymbol e),(1+q-t,\boldsymbol c)^{\times 3},(2-q-t,\boldsymbol f)^{\times 3}\}.
\end{aligned}\tag{582}$$

Proof. On the right part ($n\ge 0$) the Pochhammer form of (568) gives $\Phi_t(\ell+n)\Psi_t(n)=P_{\mathrm R}\widehat\Phi(\ell+n)\widehat\Psi(n)$; the reflection signs combine to $(-1)^{6n}=1$. On the middle part put $n=-j-1$ and $\ell=j+k+1$, and on the left part $n=-\ell-1-p$. The multiplicities are $d_\ell=(4q-2)_\ell(2q)_\ell/[(2q-1)_\ell\,\ell!]$ (Section 16.5) for the $\boldsymbol e$ rows and (301) for the $\boldsymbol c$ rows of $H_{\mathrm M}$, and the rows $(1,\boldsymbol e)$, $(1,\boldsymbol f)$ cancel the factorials of (320). Absolute convergence on $\mathbb H_q$ (Section 21.4) justifies the regrouping. □

The subscripts $\mathrm R,\mathrm M,\mathrm L$ of $H,P,\mathsf U,\mathsf V$ and of $T_{\mathrm R},T_{\mathrm L}$ name the right, middle and left parts of the lattice; on S,D,Ξ,η,Ω the subscript R names the R family.

All parameters are affine in (q,t,τ). The series $H_{\mathrm R}$ and $H_{\mathrm L}$ converge absolutely on $\mathbb H_q$, and $H_{\mathrm M}$ on $\operatorname{Re}\tau>\max(0,2q-1)$. At $d=2$ the rows $(4q-2,\boldsymbol e)/(2q-1,\boldsymbol e)$ are $0/0$; there one splits off $\ell=0$ as in (321). At $d=3$ and $\tau=q$, where $R_q=g_\ell$, the rows of $H_{\mathrm R}$ reduce to those of $\mathfrak H_3(t)$ in (413).

21.8 Continuation in τ and the jet

The jet of (565) is taken on a circle $|\tau|=r_\tau$ around the pole $\tau=0$ of $\langle t_\ell^2R_\tau\rangle_d$. Since $\operatorname{Re}\tau>0$ is one of the conditions defining $\mathbb H_q$, no such circle lies in $\mathbb H_q$; for $d=0.8$, 2.5, 2.75 and 3 the whole circle used for the values of Section 21.11 ($r_\tau=0.04$ or 0.08) lies outside it. The samples are continued as follows.

Each of $\widehat\Phi,\widehat\Psi,\widehat\Psi_{\mathrm M},\widehat\Phi_{\mathrm L}$ behaves as a constant times m^{a-1} with $a=1-2q-\tau$. With $\mathsf b_\alpha(m)=\Gamma(m+\alpha)/\Gamma(m+1)$, subtract an I-term model, $X(m)=\sum_{i<I}x_i\mathsf b_{a-i}(m)+\delta X(m)$ with $\delta X=O(m^{a-1-I})$. Here X, Y stand for any of the four factors above and x_i, y_i for their model coefficients; they are not the constant X of (221) or the x_h of (558). The coefficients x_i follow from the Bernoulli-polynomial expansion of $\log X(m)$: each is the Gamma ratio x_0 below times a polynomial in Bernoulli polynomials of the parameters. Each factor has the form $X(m)=\prod_{i=1}^{3}(u_i)_m/(v_i)_m$, with upper and lower parameters u_i,v_i read off above (local names) and $\sum_i(u_i-v_i)=a-1$. As $m\to\infty$ [27],

$$X(m)\sim x_0\,m^{a-1}\exp\sum_{k\ge 1}\lambda_k m^{-k},\qquad x_0=\prod_{i=1}^{3}\frac{\Gamma(v_i)}{\Gamma(u_i)},$$
$$\lambda_k=\frac{(-1)^{k+1}}{k(k+1)}\sum_{i=1}^{3}[B_{k+1}(u_i)-B_{k+1}(v_i)],\tag{583}$$

with the Bernoulli polynomials B_{k+1} (not the B_d of (230) or the weights $B_{r,k}$ of (479)). The basis function b_{a-i} has the same expansion with the single pair $(u,v)=(a-i,1)$ and the constant 1. Matching the coefficients of m^{a-1-j}, $j<I$, gives $x_1,\dots,x_{I-1}$ by a triangular solve with unit diagonal, starting from x_0. The model sums are closed Gamma forms, by Gauss's sum [4] after $(2q)_\ell/(2q-1)_\ell = 1+\ell/(2q-1)$, and by Chu–Vandermonde:

$$\begin{aligned}
\mathcal{G}_\alpha(n) &:= \sum_{\ell\ge 0} d_\ell\, \mathsf{b}_\alpha(\ell+n) = \frac{\Gamma(n+\alpha)\Gamma(2-4q-\alpha)\,(2n+2-4q+\alpha)}{\Gamma(n+3-4q)\Gamma(1-\alpha)},\\
\mathcal{S}(\alpha,\beta) &:= \sum_{n\ge 0} \mathcal{G}_\alpha(n)\,\mathsf{b}_\beta(n) = \frac{\Gamma(\alpha)\Gamma(\beta)\Gamma(2-4q-\alpha-\beta)\,(2-4q+\alpha-\beta)}{\Gamma(1-\alpha)\Gamma(3-4q-\beta)},\\
\widetilde{\mathcal{G}}_\alpha(j) &:= \sum_{k\ge 0} d_{j+k+1}\mathsf{b}_\alpha(k) = \frac{d_{j+1}\Gamma(\alpha)\Gamma(j+2)\Gamma(2-4q-\alpha)}{\Gamma(3-4q)\Gamma(j+2-\alpha)}\Big[2-4q-\alpha+\frac{(4q-1+j)\alpha}{2q+j}\Big],\\
\mathcal{V}(\alpha,\beta) &:= \sum_{j,k\ge 0} d_{j+k+1}\mathsf{b}_\beta(j)\,\mathsf{b}_\alpha(k) = \frac{2(1-2q)\Gamma(\alpha)\Gamma(\beta)\Gamma(2-4q-\alpha-\beta)\,(4q-\alpha-\beta)}{\Gamma(3-4q)\Gamma(2-\alpha-\beta)},
\end{aligned} \tag{584}$$

valid where the sums converge and by continuation elsewhere. For the Toeplitz blocks, H_{R} with $(X,Y)=(\widehat{\Phi},\widehat{\Psi})$ and H_{L} with $(X,Y)=(\widehat{\Psi}_{\mathrm{M}},\widehat{\Phi}_{\mathrm{L}})$, and for the middle block H_{M} with $(X,Y)=(\widehat{\Phi},\widehat{\Psi}_{\mathrm{M}})$, this gives

$$\begin{aligned}
\sum_{\ell,n\ge 0} d_\ell X(\ell+n)Y(n) &= \sum_{\ell,n\ge 0} d_\ell\,\delta X(\ell+n)Y(n) + \sum_{n\ge 0}\Big[\sum_i x_i\mathcal{G}_{a-i}(n)\Big]\delta Y(n)\\
&\quad + \sum_{i,i'} x_i y_{i'}\mathcal{S}(a-i,a-i'),\\
\sum_{j,k\ge 0} d_{j+k+1}Y(j)X(k) &= \sum_{j,k\ge 0} d_{j+k+1}\,\delta Y(j)\,\delta X(k) + \sum_{j\ge 0}\delta Y(j)\sum_i x_i\widetilde{\mathcal{G}}_{a-i}(j)\\
&\quad + \sum_{k\ge 0}\delta X(k)\sum_i y_i\widetilde{\mathcal{G}}_{a-i}(k) + \sum_{i,i'} x_i y_{i'}\mathcal{V}(a-i,a-i').
\end{aligned} \tag{585}$$

For $I\ge 1$ the two-variable remainders converge absolutely in a neighborhood of $\tau=0$ for every $0<q<1$, and so do the one-variable remainders. For $d\neq 2$ the only singularity at $\tau=0$ is $\Gamma(2\tau)$, in $\mathcal{S}(a,a)$ and $\mathcal{V}(a,a)$; at $d=2$ the two spurious families below also contain $\tau=0$, and these poles of individual pieces cancel in the sum. Spurious poles of individual pieces, at $\tau\in 1-2q+\mathbb{Z}$, at $\tau\in 2q-1-\mathbb{Z}$ and at $\tau=q$, bound the admissible radius of the circle. On $\mathbb{H}_q$ the right sides of (585) equal the lattice sums, by an absolutely convergent rearrangement.

Hypothesis (H) and transfer. (H) states that there are a bounded connected open set $\mathcal{U}$ containing the closed disk $|\tau|\le r_\tau$ and meeting $\mathbb{H}_q$, and a closed discrete set $\mathsf{E}\subset\mathcal{U}$ with $\mathsf{E}\cap\{|\tau|\le r_\tau\}\subseteq\{0\}$, such that the continued samples (581), (585) are jointly holomorphic in (t,τ) for $\tau\in\mathcal{U}\setminus\mathsf{E}$ and $t\notin\mathsf{P}_\tau$, locally uniformly. The construction above gives this, except that the uniform remainder bounds of the inverse-factorial expansions [27] are not written out. The radius r_τ is also taken so small that the disk contains no zero of $\sin\pi(\tau-q)$ or $\sin\pi(2q-\tau)$ other than $\tau=0$, so that the weights w_j of (580) have no other singularity there. Given (H), the Fourier coefficients $c_k(\tau)=\int_0^1 \Xi_{\mathrm{R}}(x+\mathrm{i}y_1;\tau)e^{-2\pi\mathrm{i}k(x+\mathrm{i}y_1)}dx$, with $y_1>1+\sup_{\mathcal{U}}|\operatorname{Im}\tau|$, so that the segment avoids P_τ, are holomorphic on $\mathcal{U}\setminus\mathsf{E}$: the integrand is jointly continuous and holomorphic in τ, so by Fubini's theorem its integral over the boundary of every small triangle in $\mathcal{U}\setminus\mathsf{E}$ vanishes, and Morera's theorem applies. For $\tau\in(\mathcal{U}\cap\mathbb{H}_q)\setminus\mathsf{E}$, $c_k(\tau)$ is the k-th Fourier coefficient of the entire one-periodic function $\Xi_{\mathrm{R}}(\cdot\,;\tau)$, which does not depend on the line $\operatorname{Im}t=y_1$. The set $\mathcal{U}\setminus\mathsf{E}$ is connected, since a closed discrete subset does not disconnect a planar domain, and $(\mathcal{U}\cap\mathbb{H}_q)\setminus\mathsf{E}$ is a nonempty open subset of it. The coefficients with $|k|\ge 2$ vanish on that subset by (572), hence identically by the identity theorem. Both sides of $\Xi_{\mathrm{R}}(t;\tau)=\sum_{|k|\le 1}c_k(\tau)e^{2\pi\mathrm{i}kt}$ are holomorphic on the set of (t,τ) with $\tau\in\mathcal{U}\setminus\mathsf{E}$ and $t\notin\mathsf{P}_\tau$, which is connected because its base $\mathcal{U}\setminus\mathsf{E}$ and its fibers

$\mathbb{C} \setminus \mathsf{P}_\tau$ are; they agree for $\tau \in (\mathcal{U} \cap \mathbb{H}_q) \setminus \mathsf{E}$, so they agree on all of it, at the real nodes included. The right side of (580) is then holomorphic on the connected set $(\mathcal{U} \setminus \mathsf{E}) \cup \mathbb{H}_q$ off the poles of the ρ_k, and equals the trace on the residue strip, so it is the continuation of $\langle t_\ell^2 R_\tau \rangle_d$, and

$$\boxed{\begin{aligned} \mathcal{K}_d &= -\frac{AU}{4} + \frac{1}{4}[\tau^2]\Big\{C(\tau)\sum_{j=0}^{2} w_j(\tau)\,\Xi_{\mathrm{R}}\big(t_0+\tfrac{j}{3};\tau\big) - C(\tau)^2\mathcal{X}_d(\tau)^2\Big\}, \\ \mathsf{r}_3 &= [\tau^3]\sum_{j=0}^{2} w_j(\tau)\,\Xi_{\mathrm{R}}\big(t_0+\tfrac{j}{3};\tau\big). \end{aligned}} \tag{586}$$

Thus, for $d \neq 2$, r_3 is a finite combination, with elementary coefficients, of the τ-derivatives of order at most three at $\tau = 0$ of nine two-variable remainder sums (three blocks at three nodes), of one-variable remainder sums, and of Gamma values. At $d = 2$ the triple pole of ρ_k and the spurious poles of individual pieces at $\tau = 0$ raise the orders of the derivatives that enter; see the open item on a form adapted to $d = 2$ (Section 21.12). In the computations below the jet is taken with a Cauchy circle; explicit polygamma-weighted two-variable series for these derivatives have not been written out.

No part-by-part estimate on the jet circle. The transfer cannot be replaced by a direct estimate of the lattice parts on the circle. Heuristically, for large $|\operatorname{Im} t|$ the right part behaves like a t-dependent factor times $\sum_{n\geq 0}(n+t)^{-1-2\tau} \approx t^{-2\tau}/(2\tau)$: on $\mathbb{H}_q$ it decays like $|\operatorname{Im} t|^{-2\operatorname{Re}\tau}$ (item Growth of Section 21.11, for $q \geq \frac{1}{4}$), and for $\operatorname{Re}\tau < 0$ it grows like $|\operatorname{Im} t|^{2|\operatorname{Re}\tau|}$. The total decays fast: given (H), $|\Xi_{\mathrm{R}}(x+\mathrm{i}y;\tau)| \leq \sum_{|k|\leq 1}|c_k(\tau)|\,e^{2\pi|y|}$, and since $|\sin \pi w| \geq \sinh \pi |\operatorname{Im} w|$ for each of the six sine factors of D_{R}, $S_{\mathrm{R}} = \Xi_{\mathrm{R}}/D_{\mathrm{R}} = O(e^{-4\pi|y|})$ for $|y| \geq 1 + |\operatorname{Im}\tau|$. At six points of the jet circles ($d = 0.8, 1.25, 1.5, 2.5, 3$; five of the six outside $\mathbb{H}_q$) the largest of the continued right, middle and left parts, times $|y|^{2\operatorname{Re}\tau}$, is constant to a few percent for $y = 1, 2, 3$, while $|\Xi_{\mathrm{R}}|e^{-2\pi|y|}$ is constant; at $y = 3$ the total is $1.2 \cdot 10^{15}$ to $1.6 \cdot 10^{16}$ times smaller than the largest part (records `n6_case0.log` to `n6_case5.log`, with the `.json` files of the same names, in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`). On the circle the decay of the total is thus a cancellation between the parts, which no part-by-part bound can give; this is why (H) is a continuation statement.

Status of (H) and a route to a proof. The construction (585) is expected to satisfy (H). Its kernels (584) depend on (q, τ) only and are meromorphic in τ; their poles, and the points $\tau \in q + \mathbb{Z}$ where pole classes merge, belong to E. By (583) its model coefficients are the Gamma ratios x_0 times polynomials in Bernoulli polynomials of the parameters, hence holomorphic in (t, τ) wherever the x_0 are. Each model term gains one power in the remainder sums (the package uses $I = 26$ to 30). Only the locally uniform remainder bound [27] is missing. An independent proof of (H), not carried out, could proceed as follows for the right block, with t in a compact strip. (i) Subtract from each of the two Gamma ratios of the right part, $\widetilde{\Phi}(z)$ at $z = \ell + n + t$ and $\widetilde{\Phi}(z + 1 - 2q)$ at $z = n + t$ (Section 21.4), the single ratio $\mathfrak{m}_c(z) = \Gamma(z+c)/\Gamma(z+c+2q+\tau)$ with a constant c. The Gamma-ratio bound (465) makes the differences $O(|z|^{-\varkappa-1})$, $\varkappa = 2q + \operatorname{Re}\tau$, and the remainder double sums then converge absolutely for $\operatorname{Re}\tau > \max(-\frac{1}{2}, 2q-2, -2q)$. (ii) The spin sum of $\mathfrak{m}_c$ is a Beta integral against $\sum_\ell d_\ell x^\ell = (1+x)(1-x)^{1-d}$ of (461), which gives two Beta functions. (iii) The remaining n-sum of $\mathfrak{m}_c$ times these Beta functions is a ${}_3F_2(1)$ with one upper parameter equal to 1 and parameter excess 2τ. A two-term Thomae relation, [22, eq. 16.4.11] applied twice, maps a ${}_3F_2(1)$ with upper parameters $(1, u, u')$, lower parameters (v, v') and excess $s = v + v' - 1 - u - u'$ (local names) to $\Gamma(v)\Gamma(v')\Gamma(s)/[\Gamma(s+u)\Gamma(s+u')]$ times the ${}_3F_2(1)$ with upper parameters $(v-1, v'-1, s)$ and lower parameters $(s+u, s+u')$, whose excess is 1. With $s = 2\tau$ this continues the n-sum as $\Gamma(2\tau)$ times a ${}_3F_2(1)$ that converges for all τ. The left block is a Toeplitz sum of the same type; the middle block needs the two-variable kernels $\widetilde{\mathcal{G}}$ and $\mathcal{V}$ of (584).

21.9 Specializations to $d = 1, 2, 3$

$d = 1$. Here $d_\ell = (1, 1, 0, \dots)$ and the offset is $1 - d/2 = \frac{1}{2}$. The two spins merge into one sum over consecutive pairs $(w, w+\frac{1}{2})$ of the lattice $t+\frac{1}{2}\mathbb{Z}$, which has period $\frac{1}{2}$ in t, while $D_{\mathrm{R}}T_{\mathrm{R}} = -\frac{1}{8}\cos^3(2\pi t)$ is antiperiodic under $t \to t+\frac{1}{2}$. Hence $\Xi_{\mathrm{R}}(t+\frac{1}{2}) = -\Xi_{\mathrm{R}}(t)$: only $c_{\pm 1}$ survive, and $c_0 = c_{\pm 2} = 0$. Numerically, $C(\tau)$ times the sampled $\langle t_\ell^2 R_\tau\rangle_1$ reproduces (78) with the path parameter of J_t set to 2τ. With $\mathcal{K}_1 = \mathcal{K}$ of (76) (see (224)), (565) gives, conditionally on (M) at $d = 1$ (confirmed numerically above), the closed form

$$\begin{aligned}\mathsf{r}_3(1) &= \mathsf{s}_3(1) + 4\mathcal{K}_1 - 2(\pi^3 + 28\zeta(3))(3\gamma_E + 7\log 2 + \pi)\\ &= 767.24520944465678609047981573762 24\dots,\end{aligned}$$

because $\frac{U}{8}[\psi_0(1) + \psi_0(\frac{1}{2}) + 4\psi_0(\frac{1}{4})] = \frac{1}{2}(\pi^3 + 28\zeta(3))(3\gamma_E + 7\log 2 + \pi)$ at $q = \frac{1}{4}$. No further condition enters: (J) at $d = 1$ is (59), on which (76) already rests.

$d = 2$. Here the offset vanishes and $d_\ell = (1, 2, 2, \dots)$. Then

$$\Psi_t(n) = \frac{\cos^3(\pi t)}{\sin^2(\pi t)\cos\pi(t+\tau)}\,\Phi_t(n), \qquad \Xi_{\mathrm{R}}(t) = -\Omega_{\mathrm{R}}(t)^2, \qquad \Omega_{\mathrm{R}}(t) = \cos^3(\pi t)\sum_{m\in\mathbb{Z}}\Phi_t(m),$$

because the lattice sum becomes the square of $\sum_m \Phi_t(m)$. The function Ω_{R} is antiperiodic, and, being entire with $|\Omega_{\mathrm{R}}|^2 = |\Xi_{\mathrm{R}}| = O(e^{2\pi|\operatorname{Im} t|})$ by (572), it has only the two modes $e^{\pm i\pi t}$. This is the square-lattice structure of Sections 18.9.3–18.9.5, where the trace is evaluated in closed form, $C(\tau)\langle t_\ell^2 R_\tau\rangle_2 = \pi^2[\mathfrak{F}(\tau)^2 + \mathfrak{F}(1-\tau)^2]$, that is (377); the generic samples reproduce it numerically. At $d = 2$ both (M) and (J) are proved, (M) for $0 < \tau < \frac{1}{2}$ in Section 18.9.5 and (J) in Section 18.8.4, so (565) and (271) give the exact value

$$\begin{aligned}\mathsf{r}_3(2) &= \mathsf{s}_3(2) + 128\mathsf{G}^2 - 168\,\zeta(3)\log 2 - 42\gamma_E\zeta(3)\\ &= 35.63714944308140680727039972168179\dots.\end{aligned}$$

$d = 3$. Here the offset is $\frac{1}{2}$ and $d_\ell = 2\ell + 1$. The half-shift argument of (429), repeated for this kernel, gives the following for $\frac{1}{2} < \tau < \frac{9}{8}$; the derivation is not written out here, and the identity is checked numerically at six points to $1.1 \cdot 10^{-24}$ (Section 21.11).

$$\Xi_{\mathrm{R}}(t) + \Xi_{\mathrm{R}}(t+\tfrac{1}{2}) = -\frac{\pi}{16}\cos^3(2\pi t)\,[\mathcal{M}^1_{\mathrm{o}}\mathcal{M}^0_{\mathrm{e}} - \mathcal{M}^0_{\mathrm{o}}\mathcal{M}^1_{\mathrm{e}}], \qquad \mathcal{M}^k_{\mathrm{e,o}} = \sum_{n\in\mathbb{Z}}(n+u)^k\,\widetilde{\Phi}(n+u-\tfrac{5}{8})\big|_{q=3/4},$$

with $u = t + \frac{1}{8}$ (e) or $u = t + \frac{5}{8}$ (o) and $\widetilde{\Phi}$ of Section 21.4, so that $\widetilde{\Phi}(w-\frac{5}{8})|_{q=3/4} = \Gamma(w+\frac{1}{8})^3/[\Gamma(w+\frac{7}{8})^2\Gamma(w+\frac{1}{8}+\tau)]$; at $\tau = q = \frac{3}{4}$ this is the cube of the function $\Gamma(x+\frac{1}{8})/\Gamma(x+\frac{7}{8})$ of (406). By (572) the left side is the constant $2c_0$. So c_0 is an ordinary 2×2 determinant of one-variable bilateral moments, while this identity does not reduce $c_{\pm 1}$; this is the R analogue of (431). No closed form of $\mathcal{K}_3$ is known. The residue series (260), whose derivation away from $d = 2$ uses (M) and the extraction of $[\tau^2]$ under the trace (Section 21.12), summed in ball arithmetic, gives 430 digits (`LRI_C5_d3/wf_K3/K3_value.json`), $\mathcal{K}_3 = 1.14859464952717933655664631081067318763\dots$; the position-space value of Section 21.10, which uses neither, agrees with it to $3.6\cdot 10^{-100}$ (Section 21.11). With this value, (565), conditionally on (M) and (J) at $d = 3$, gives $\mathsf{r}_3(3) = 4.51175935823600578025662089796157 5\dots$. A PSLQ search for $\mathcal{K}_3$ at 420 digits found no integer relation in the bases listed in `LRI_C5_d3/wf_K3/pslq_K3_results_420.json`.

Consistency. At $d = 1, 2, 3$ the five-node version of (586) (nodes $t_0 + j/5$, modes $|k| \le 2$) agrees with (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$ to $5 \cdot 10^{-37}$, $5 \cdot 10^{-48}$ and $1.4 \cdot 10^{-43}$ (relative), and the position-space value of Section 21.10 agrees with them to about 10^{-100} (Section 21.11). The Laurent coefficients $\mathsf{r}_3(1)$ and $\mathsf{r}_3(2)$ of the sampled jets agree with the closed forms above to 10^{-32}.

21.10 Position-space representation

By completion symmetry $J_t = J_d(d/2, d/2, d/2, d-t, d/2) = J_d(d-t, d/2, d/2, d/2, d/2)$. The y-integral is then the one-loop star with indices (q,q,q), which has a two-fold Mellin–Barnes representation, and the x-integral is a bubble. With $\mathsf{N}_q = \Gamma(2q)^2/\Gamma(q)^4$ and $z_{12} = z_1 + z_2$,

$$J_t = \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\Gamma(-z_1)^2\Gamma(-z_2)^2\Gamma(q+z_{12})^2\frac{\Gamma(q+z_2)}{\Gamma(q-z_2)}\,\frac{\Gamma(\frac{t}{2}+z_1)\Gamma(q-\frac{t}{2}-z_{12})}{\Gamma(2q-\frac{t}{2}-z_1)\Gamma(q+\frac{t}{2}+z_{12})}, \quad (587)$$

on $-\frac{t}{2} < \operatorname{Re} z_1 < 0$, $-q < \operatorname{Re} z_2 < 0$, $\operatorname{Re} z_{12} > -q$. Moving $\operatorname{Re} z_1$ to the left of $-\frac{t}{2}$ crosses one simple pole. By Barnes' first lemma its residue is $J_t^{\mathrm{pole}} = \Gamma(2q)\Gamma(\frac{t}{2})^2\Gamma(q-\frac{t}{2})^4/[\Gamma(q)^4\Gamma(2q-t)]$, and the remaining integral is holomorphic at $t=0$. The coefficient $[t^2]$ of $J_t - J_t^{\Gamma}$ is therefore the sum of the polygamma part $\mathcal{K}_d^{\Gamma} := [t^2](J_t^{\mathrm{pole}} - J_t^{\Gamma})$ and the coefficient $\mathcal{I}_d$ of t^2 in the remaining integral:

$$\boxed{\mathcal{K}_d = \mathcal{K}_d^{\Gamma} + \mathcal{I}_d,} \quad (588)$$

$$\mathcal{K}_d^{\Gamma} = \frac{A^2\vartheta}{2} + \frac{A}{2}\big[\psi_2(2q) - \psi_2(q)\big] - \frac{R\vartheta}{4} + \frac{\vartheta^2}{8} + \frac{2\psi_3(q) - 7\psi_3(2q)}{48}, \qquad \vartheta = 2\psi_1(q) - \psi_1(2q),$$

$$\mathcal{I}_d = \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0\,\frac{\varpi_1^2 + \varpi_2}{2},$$

$$\mathfrak{M}_0 = \frac{\Gamma(-z_1)^2\Gamma(z_1)}{\Gamma(2q-z_1)}\,\frac{\Gamma(-z_2)^2\Gamma(q+z_2)}{\Gamma(q-z_2)}\,\Gamma(q+z_{12})\Gamma(q-z_{12}),$$

$$\varpi_1 = \tfrac{1}{2}\big[\psi_0(z_1) + \psi_0(2q-z_1) - \psi_0(q-z_{12}) - \psi_0(q+z_{12})\big],$$

$$\varpi_2 = \tfrac{1}{4}\big[\psi_1(z_1) - \psi_1(2q-z_1) + \psi_1(q-z_{12}) - \psi_1(q+z_{12})\big],$$

on $-q < \operatorname{Re} z_1, \operatorname{Re} z_2 < 0$, $\operatorname{Re} z_{12} > -q$ (for example $\operatorname{Re} z_1 = \operatorname{Re} z_2 = -q/3$). By (J) the orders t^0 and t^1 of $J_t - J_t^{\Gamma}$ vanish, which gives two Barnes identities:

$$\mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0 = \psi_1(2q) - 2\psi_1(q), \qquad \mathsf{N}_q \iint \frac{dz_1\,dz_2}{(2\pi\mathrm{i})^2}\,\mathfrak{M}_0\,\varpi_1 = -\Big[A\vartheta + \frac{\psi_2(2q)}{2} - \psi_2(q)\Big]. \quad (589)$$

This route uses only the definition (238), which defines $\mathcal{K}_d$ as the coefficient $[t^2]$; it uses none of (M), (J), (H), and not the spectral kernel. Its agreement with (586) at the level of 10^{-41} to 10^{-47} (Section 21.11) is therefore a numerical test of (M), and of (H) on the jet circle, at non-integer d. The completed graph of J_t can be put into this star-and-bubble form in 60 ways (choice of the vertex sent to infinity, the two external vertices, the star vertex and the bubble vertex). A computer enumeration shows that each of the 60 two-fold Mellin–Barnes integrands contains a squared Gamma function, so this framework has no derivative-free residue expansion of J_t. The kite with general indices can also be written as a sum of four Kampé de Fériet series $F^{2:2;2}_{2:1;1}(1,1)$ and two Horn double series. This form is not derived or used in this note; it is recorded, with its convergence conditions, in `LRI_C5_general_d/K_general/position_space/position_space_summary.json`.

21.11 Numerical confirmation

Values. The first table gives $\mathcal{K}_d$ from (588), evaluated to 100 digits by the trapezoid rule in the ball arithmetic of the Arb library [48] on $\operatorname{Re} z_1 = \operatorname{Re} z_2 = -q/3$ (`LRI_C5_general_d/K_general/position_space/K_d_values_mb_D100.json`), and r_3 obtained from it through (565). That column assumes (M) and (J); it agrees with the τ^3 coefficients of the 34-digit Laurent tables of the directly sampled five-node jets to $3\cdot10^{-34}$ (relative). It is printed to 34 digits, so that (565) with the printed r_3 reproduces the printed $\mathcal{K}_d$ to $3\cdot10^{-32}$ (relative; 10^{-33} against the 100-digit values). The digits shown are correctly rounded; … marks omitted digits.

d	$\mathcal{K}_d$, position space (588)	$\mathsf{r}_3 = [\tau^3]\langle t_\ell^2 R_\tau\rangle_d$ by (565)
0.8	92.372770239922771632645391899036…	1894.360867513991151448939858484147…
1	39.331569128259039057503187151598…	767.245209444656786090479815737622 4…
1.25	17.203345729921688327085720106549…	302.137977543021229758469687902 9900…
1.5	9.018297401594820547084743104 6793…	136.568544321226036920660372299842 3…
2	3.518109138590252241649090170 3077…	35.637149443081406807270399721681 79…
2.5	1.840501295428588030730181860 2723…	11.467317760441838785767751689501 30…
2.75	1.429249188660312000099175536 8839…	6.984719816576732523926795215898772…
3	1.148594649527179336556646310 8107…	4.511759358236005780256620897961575…

Routes. The second table gives relative deviations from the position-space values. "Spectral reference" is the τ-circle evaluation of (248) in `LRI_C5_general_d/references/reference_values.json` (50 printed digits, of which that file states 45 to 47 to be correct). "Abel–Plana" is (248) with an Abel–Plana spin tail at 60 digits, equal to the residue series (289) to better than 10^{-53} for $d \le 2.5$ and to $5\cdot10^{-49}$ at $d = 2.75$ (`LRI_C5_general_d/K_general/refcheck/`). The two jets are the first line of (586), that is (566), evaluated with the five nodes $t_0+j/5$ (modes $|k| \le 2$) and with the three nodes of (580), on a Cauchy circle of 48 points and radius 0.08 (0.04 at $d = 0.8$), at 72 digits. The three-node circles at $d = 1.5$, 2.5 and 3 were computed by the separate check code of an independent AI review of Sections 20 and 21 (`LRI_C5_general_d/review/joint/numerics-end-to-end/k3_jets.json`); through (565) instead of (566) they give $3.8\cdot10^{-45}$, $5.0\cdot10^{-44}$ and $-8.1\cdot10^{-43}$ (`LRI_C5_general_d/K_general/note_fragment/test_build_joint/check_k3_routes.log`). The circles at $d = 0.8$, 2.5, 2.75 and 3 lie entirely outside $\mathbb{H}_q$. The last column has the opposite sign convention: it gives the position-space value minus the reference, relative to the reference, for the references (76), (271) and the 430-digit (260) value of $\mathcal{K}_3$, the last conditional on (M) and on the extraction of $[\tau^2]$ under the trace.

d	spectral reference	Abel–Plana	five-node jet	three-node jet	position space vs (76), (271), (260)
0.8	$-1.9\cdot10^{-49}$	$8.2\cdot10^{-57}$	$-8.0\cdot10^{-47}$	$-8.1\cdot10^{-47}$	–
1	–	–	$-4.8\cdot10^{-37}$	–	$5.2\cdot10^{-101}$
1.25	$1.8\cdot10^{-49}$	$-2.4\cdot10^{-55}$	$-1.0\cdot10^{-41}$	$-1.0\cdot10^{-41}$	–
1.5	$1.9\cdot10^{-49}$	$2.1\cdot10^{-56}$	$-9.6\cdot10^{-46}$	$-1.4\cdot10^{-45}$	–
2	–	–	$5.2\cdot10^{-48}$	–	$-1.5\cdot10^{-100}$
2.5	$5.2\cdot10^{-51}$	$-1.2\cdot10^{-55}$	$5.4\cdot10^{-44}$	$4.1\cdot10^{-44}$	–
2.75	$2.1\cdot10^{-50}$	$-8.3\cdot10^{-56}$	$6.0\cdot10^{-44}$	$-3.5\cdot10^{-44}$	–
3	–	–	$-1.4\cdot10^{-43}$	$-9.5\cdot10^{-43}$	$3.6\cdot10^{-100}$

Further checks. All are numerical, with empirical error control and no interval enclosures.

- *S family.* (556) agrees with the spectral trace at 36 points ($0.8 \le d \le 3.5$, real and complex τ) to $2\cdot10^{-38}$ or better, with an independent real-axis lattice route at 18 points to $4\cdot10^{-37}$ or better, and with the continued spectral trace on τ-circles to $8\cdot10^{-29}$.
- *Low orders.* With the 34-digit Laurent tables of the five-node jets, $|\mathsf{r}_0 - \mathsf{s}_0|$, $|\mathsf{r}_1 - \mathsf{s}_1|$ and $|\mathsf{r}_2 - \mathsf{s}_2 - \frac{3}{2}U|$ are at most $5\cdot10^{-32}$ at all eight d, and (565) reproduces $\mathcal{K}_d$ to 10^{-33}. In the 72-digit jets $|F_2 - \frac{3}{2}U|$ lies between 10^{-47} and $4\cdot10^{-35}$.
- *Inputs (M) and (J).* The mixed trace (M) holds pointwise at three complex τ on the τ-circles of the spectral reference (paragraph Routes) at $d = 0.8, 1.25, 1.5, 2.5, 2.75$ to $4\cdot10^{-32}$. For (J), at $d = 1.25$ the full quadratic jet of $J_d - (G_2 + G_1 - G_0 + \frac{2}{p_1}\Delta_B)$ of (237) at its base point $(d/2, d/2, d/2, d, d/2)$, reconstructed from its restrictions to 18 random lines through that point, has only the $z_1 z_2$ coefficient, equal to $\mathcal{K}_d$ to $2\cdot10^{-12}$; the other 14 coefficients are below $3\cdot10^{-12}$. The orders t^0, t^1, that is (589), hold to 10^{-101} at all eight d.

- *Samples.* (581) with (585) agrees with the spectral trace at eight (d,τ) points, three of them outside $\mathbb{H}_q$, to $1.1\cdot10^{-19}$ or better (relative; at $d=0.8,1.25$ against the spectral trace with an analytic ν-tail). The termwise identities behind (568) and (582), the kernels (584) and ρ_k of (579) hold to $6\cdot10^{-39}$ (`LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_blocks.py`). At $d=1,2$ the generic samples reproduce (78) and (377) at two values of τ each, to 10^{-21} and $2\cdot10^{-19}$ (absolute), and at $d=3$ the determinant form of Section 21.9 holds to $1.1\cdot10^{-24}$ at six points.

- *Growth.* At six points of $\mathbb{H}_q$ ($0.8\le d\le3$, real and complex τ) the three parts obey the bound of Section 21.4, which is attained for $q\ge\frac14$, while the total $S_{\mathrm R}$ is up to 10^{25} times smaller and decays like $e^{-4\pi|\operatorname{Im}t|}$.

- *Growth constants.* These test the estimates used in Section 21.6. Stirling's error bound, the input of (465), holds at 4000 random points with $|\arg w|\le0.97\pi$ and $0.05\le|w|\le10^4$, the largest ratio of $|\epsilon_\Gamma(w)|$ to the bound being 0.99999973; in (465), $|\theta||z|/\omega(\rho)\le0.29$ on random samples for $\rho=0.25,\dots,3$, the corner $|z|=\max(1,2\rho)$, $\arg z\to\pm\frac{\pi}{2}$ included. For $0.02\le q\le0.99$ the constant C_q of (466), evaluated over $\ell\le2\cdot10^4$, lies between 1.04 and 7.9, and $d_\ell(1+\ell)^{2-d}$ approaches $2/\Gamma(d-1)$. At ten pairs (q,α), some close to the thresholds (573), and $Y=2,\dots,512$, the sums $\Sigma^{\mathrm r}(Y)$ and $\Sigma^{\mathrm m}(Y)$ of (467) (FFT convolutions of 2^{22} terms with a fitted tail) stay below their bounds, the largest ratio being 0.81, and the exact limits of $Y^{2\alpha-\max(d,1)}\Sigma^{\mathrm r}(Y)$ (for $d\ne1$) and $Y^{2\alpha-d}\Sigma^{\mathrm m}(Y)$ as $Y\to\infty$ lie below $C^{\mathrm r}$ and $C^{\mathrm m}$. The Horn-block code of (581), used for the item Growth, reproduces the three parts (575), computed as Richardson-extrapolated direct lattice sums in ball arithmetic, at twelve complex t with $|\operatorname{Im}t|\le4$ ($d=0.8,1.5,2.5,3.4$, $\operatorname{Re}\tau=2.37$ or 2.6) to between $2\cdot10^{-17}$ and $5\cdot10^{-9}$ (relative), within the error of the direct sums. At the six points of the item Growth, $C_{\mathfrak K}$ of Section 21.6 for the single point $\mathfrak K=\{(q,\tau)\}$ is $4\cdot10^{31}$ to $4\cdot10^{44}$; its limit for large $|y|$ is $2.6\cdot10^3$ to $2.7\cdot10^4$, which is 200 to 1000 times the observed maximum of $|S^{\rightarrow}|$, $|S^{|}|$, $|S^{\leftarrow}|$ times $|y|^{\eta_{\mathrm R}}/(1+\log|y|)$. Records: `n1_stirling.log`, `n2_mult.log`, `n3_lattice.log`, `n4_crossval.log` and `n8_constants.log` in `LRI_C5_general_d/K_general/note_fragment/growth/checks/`.

- *Top modes.* With the Horn-block code at eleven points, $0.3\le d\le3.9$, including two continued points, $|c_{\pm2}|/\max_k|c_k|$ lies between $6\cdot10^{-33}$ and $3\cdot10^{-27}$ and falls with the truncation error. An independent per-spin code gives ratios below 10^{-15} at five points of $\mathbb{H}_q$, where single spins contribute $O(1)$, each below the spread of its spin-sum extrapolation (at most $8.5\cdot10^{-16}$); three unbalanced controls give between $3\cdot10^{-3}$ and $6\cdot10^{-2}$. At six points of the jet circles the continued samples have $|c_{\pm2}|/\max_k|c_k|\le1.3\cdot10^{-37}$.

- *Three nodes.* At $d=0.8,1.25,2.75$ the three-node $\mathcal K_d$ equals the five-node value to $3\cdot10^{-49}$, $3.5\cdot10^{-47}$ and $9.5\cdot10^{-44}$, at 0.5 to 0.65 of the wall time per circle point. At the two points of the item Top modes nearest to $d=4$, $d=3.7$ and 3.9, the three-node and five-node traces differ by $8\cdot10^{-24}$ and $1.5\cdot10^{-21}$ (relative), more than the top modes, which are about 10^{-27} of the largest mode. The cause is the conditioning of the residue functional, not the rule: at $d=3.9$, $\tau=1.2{+}0.3\mathrm{i}$, the weights (579) have $|\rho_k|\approx2{\cdot}10^4$, through the factor $\cot(\pi q)/\sin^3(\pi q)$, against a trace of modulus about 9, so (578) cancels about five digits and amplifies the truncation error of the samples, for both rules alike; at $d=3.9$ the difference is $5\cdot10^{-14}$ with the coarser of the two truncations (`modes_d3.7_t0.95+0.15j.json` and `modes_d3.9_t1.2+0.3j.json` in `LRI_C5_general_d/K_general/note_fragment/topmode/results/`).

- *Position space.* Two contours and step sizes agree to $3{\cdot}10^{-55}$ (relative) at $d=0.8,1.25,1.5,2.5$. An independent double-precision evaluation of (588) and (589), written from the formulas as printed here, reproduces $\mathcal K_d$ to $7\cdot10^{-14}$ (relative) and the two identities to 10^{-12} absolute ($3\cdot10^{-15}$ relative) at all eight d (`LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_barnes.py`).

- *A caution.* The literal bracket of (260), summed with the Abel–Plana tails of `LRI_C5_general_d/K_general/refcheck/routeB_residue.py`, fails numerically at $d = 0.8$, the only $d < 1$ tested: it gives 70.3 at 40 digits and 2539.5 at 60 digits in place of $\mathcal{K}_{0.8} = 92.37\ldots$, while the canceled form (289), summed by the same code, works. For $1.25 \le d \le 3$ the two brackets agree to the working precision. The cause is numerical. The two brackets agree term by term, also at non-integer and complex indices, but the literal one is a difference of two nearly equal terms of order n^{-d} in the radial index n, and that code fixes its constants at 20 digits beyond its working precision, so the difference keeps no correct digit beyond $n \approx 10^{80}$ at 60 digits (10^{60} at 40 digits). At $d = 0.8$ the radial tail of that code reaches $n \approx 10^{394}$ ($3.5 \cdot 10^{279}$), where n times the corrupted term is not small; at $d = 1.25$ it stops near $n \approx 10^{93}$ (10^{67}), and n times the error stays below 10^{-70} (10^{-50}), far below the working precision (`check_kbracket_d08.log` and `check_kbracket_d08_huge.log` in `LRI_C5_general_d/K_general/note_fragment/test_build_joint/`). This does not bear on the convergence statement after (260).

21.12 Status

Proved. The following have written proofs, in this section or in Section 20. Here an independent review is a separate AI session that re-derived the statement and tried to refute it, and a numerical probe is a separate code. Unless stated otherwise, a statement has had no independent review.

- The closed form (556) and its Laurent form (559): Gamma algebra, the kernel identity (557), and the two-wheel convolution $W_2(d; b') = B_d(d - b', 2b')$ of (230), also used for (250). Three separate AI sessions re-derived it independently and tried to refute it; none found an error.
- The constant (560) and the bilateral identity (561) (Section 21.2, remark (iii)): consequences of the case $r = 2$ of (473), of (479) and of (230); the constancy and the value at $b = d/2 - \tau$ are confirmed numerically there. No independent review.
- The Laurent series (563): exact Gamma algebra, checked symbolically for general q.
- The lattice and product forms (568) and (569), the sample formula (581) with the arrays (582), and the kernels (584): exact derivations. Separate AI sessions re-derived the lattice and product forms, the sample formula and the arrays and tried to refute them, without finding an error; all of them, the kernels included, are rechecked numerically to $6 \cdot 10^{-39}$ (Section 21.11, item Samples).
- The growth theorem of Section 21.4, proved in Section 21.6: growth bound with explicit constants, finite Fourier space with $|k| \le 2$, and sharpness of $\mathbb{H}_q$. Parts (a) (sufficiency), (b) and (c) are the case $r = 3$ of (471); an independent AI review re-derived every step of that proof, Step 0 included, by hand (Section 20.10). The sharpness of $\mathbb{H}_q$ and the explicit constants of Section 21.6 have had no independent review.
- The theorem (572) (R family, modes $|k| \le 1$) on $\mathbb{H}_q$. It is the case $r = 3$ of the product form (471) of the top-mode theorem, which has a written proof (Section 20.1.2). An independent AI review re-derived every step of that proof by hand and found no fatal or major issue (record `LRI_C5_general_d/review/lemmaG-proof`); it also checked the R-family parameters of Section 21.5 and the equivalence of condition (C) of (469) with $\tau \in \mathbb{H}_q$. A separate 48-digit lattice-sum code finds the top modes of four R kernels below $5.2 \cdot 10^{-45}$ of the largest mode (Section 20.9; `LRI_C5_general_d/review/lemmaG-numeric`).
- The residue functional (578) and the three-node rule (580) for $\tau \in \mathbb{H}_q$, $\operatorname{Re}\tau < \frac{3}{2}q$, $\tau \ne q$.
- The representation (587) and (588): exact derivation in Section 21.10 from (238) (completion symmetry, the two-fold Mellin–Barnes form of the one-loop star, Barnes' first lemma). It

uses none of (M), (J), (H), since (238) defines $\mathcal{K}_d$ as the coefficient $[t^2]$. Separate AI sessions re-derived the statement and tried to refute it, without finding an error; the derivation printed in Section 21.10 has not been reviewed. A separate code, written from the formulas as printed here, re-evaluates them (Section 21.11, item Position space; `LRI_C5_general_d/K_general/note_fragment/fragment_checks/check_barnes.py`).

Conditional.

- (564) and (565) are exact given (M) and (J), and (566) given (M) alone. The note proves (M) only at $d = 2$; at other d it is numerical. Its strongest test is the agreement of (586) with the position-space route, which uses none of (M), (J), (H); on the jet circle this agreement tests (M) and (H) together. (J) is the path form of (237), derived in Section 16, on which (223) itself rests. At $d = 2$ both inputs are proved, (M) for $0 < \tau < \frac{1}{2}$ in Section 18.9.5 and (J) in Section 18.8.4, so there these statements and the value of $\mathsf{r}_3(2)$ in Section 21.9 are proved. Elsewhere, including the values of $\mathsf{r}_3(1)$ and $\mathsf{r}_3(3)$ in Section 21.9 and the r_3 column of the first table of Section 21.11, they are conditional on (M) and (J). At $d = 1$, (J) is (59), on which (76) itself rests, so there $\mathsf{r}_3(1)$ is conditional on (M) alone.
- The two Barnes identities (589) hold given (J).
- (567) needs, in addition, $[\tau^2]$ under the trace. The derivation of (248) uses the same step, which is justified only at $d = 2$ (corollary of Section 18.9.2); there (567) is (394).
- On the jet circle, (572), (580) and (586) need (H), the analyticity in τ of the continued samples; the first line of (586) needs in addition only (M), through (566). None of the circles used in Section 21.11 lies in $\mathbb{H}_q$.
- For $d \le \frac{4}{7}$ the strip of (M) is empty; there the residue functional, and with it the definition of $\langle t_\ell^2 R_\tau \rangle_d$ and of the r_k, is only sketched. For $\mathrm{Re}\,\tau \ge \frac{3}{2}q$ the residue functional is only sketched (Section 21.7, paragraph "Separating contours").
- For $C_5(d)$ these inputs reduce to (M), plus (H) when r_3 is sampled on the jet circle as in (586): (J) adds no condition, since (223) already rests on (237), and with $\mathcal{K}_d$ from (588) none of (M), (J), (H) enters. The range of (553) is $\frac{2}{3} < d < 4$, so the caveat for $d \le \frac{4}{7}$ never affects $C_5(d)$.

Numerical only. All digit agreements of Section 21.11, and the $d = 3$ determinant form of c_0 in Section 21.9. The 100-digit values are not interval enclosures: ball arithmetic bounds the rounding error, while discretization and truncation are controlled empirically. The tested range is $0.8 \le d \le 3$ for $\mathcal{K}_d$ and $0.3 \le d \le 3.9$ for the top modes. A reduction of r_3 to known constants was not attempted at generic d; at $d = 3$ the PSLQ search was negative. The half-shift factorization of (429) is not available at generic d, and there is no derivative-free Mellin–Barnes residue expansion of J_t (Section 21.10).

Open.

- An independent review of the sharpness of $\mathbb{H}_q$ and of the explicit constants of the growth theorem (Section 21.6).
- A proof of (M) at generic d, and a written proof of (H) with uniform remainder bounds.
- Explicit polygamma-weighted two-variable series for r_3 (the τ-derivatives of the remainders in (585)), and a form adapted to $d = 2$, where individual pieces have removable singularities.

- A PSLQ search on r_3 or $\mathcal{K}_d$ with the 100-digit values, keeping $n \log_{10}(\max|c|)$ at least ten digits below the working precision and confirming any relation at a second precision.
- Dimensions $d < 0.8$, and the five-loop primitive kernel, which (471) does not cover.
- A confluent series for J_t on the path of (238) for $d \geq 2$ at small t. The confluent form recorded in `LRI_C5_general_d/K_general/position_space/position_space_summary.json` holds only for $0 < d < 2 + 2\,\mathrm{Re}\,t$; the confluent limit of the formula with four Kampé de Fériet and two Horn series mentioned at the end of Section 21.10, which would cover $d \geq 2$, has not been written.

Part IV

Numerical sixth- to eighth-order coefficients

22 The sixth-order coefficient in one dimension

22.1 Result, convention, and scope

This section extends the one-dimensional calculation by one perturbative order. It does not extend the arbitrary-dimensional formula to sixth order and does not alter any fifth-order identity of the analytic parts of this note. The regulator and normalization remain

$$\epsilon = 2s - 1, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + \sum_{n=2}^{6} C_n(1)\epsilon^n + O(\epsilon^7). \tag{590}$$

The result of the graph-based numerical calculation is

$$\boxed{C_6(1) = 14331.68703277 \pm 0.00000049,} \tag{591}$$

that is, $C_6(1) = 14331.6870328(5)$. The value is the mean of the two infrared-cutoff evaluations at quadrature degree $N_* = 22$. Both are computed by exact summation over every nonzero six-loop graph and every gap ordering, as described in Section 22.5. The quoted uncertainty is the pre-registered empirical allowance of Section 22.7. It adds the last observed change with degree, the residual difference between the two cutoffs, a measured floating-point allowance, and the convergence of the lower-loop inputs. It is not a rigorous interval enclosure or a probability guarantee, and it is not a bound on the omitted orders of the ϵ series.

This is a six-loop graph integration. It is not a Padé prediction, large-order fit, interpolation in dimension, or extrapolation from the five previously known physical coefficients, which do not determine C_6. No Monte Carlo sampling enters the final value. An earlier evaluation combined exhaustive degree-six sums, deterministic hard-sector refinements and Monte Carlo corrections for the higher degrees. It gave 14331.77075 with sampling standard error 0.14344, quoted as 14331.8 ± 0.5. The exact sums show that this estimate was high by 0.084, or 0.58 of its standard error; its sampled level differences are consistent with the exact ones (Section 22.8).

Section 22.9 compares (591) with a finite special-function representation of a reduced analytic expression for $C_6(1)$, derived separately; the two agree to 1.0×10^{-8}. Section 1.2 quotes that representation, 14331.68703275805, as the value of $C_6(1)$, and (591) is its independent numerical confirmation.

22.2 Exactly which new amplitudes are required

Write the beta function and mass anomalous dimension in the minimal-subtraction conventions of Sections 2 and 3:

$$\beta(g) = -\epsilon g + g^2 B(g), \quad B(g) = 3 + b_2 g + \cdots + b_6 g^5, \qquad \gamma_r(g) = g + c_2 g^2 + \cdots + c_6 g^6. \tag{592}$$

Since $\epsilon = g_* B(g_*)$, Lagrange inversion gives, for $n \geq 2$,

$$C_n = \frac{1}{n}[g^{n-1}]\gamma_r'(g) B(g)^{-n}. \tag{593}$$

With $A = \pi + 4\log 2$, $b_2 = -6A$, and $c_2 = -A$, the exact sixth-order assembly is

$$\begin{aligned} 2187 C_6 = {} & 3c_6 - b_6 + 2016A^5 - 448A^3 b_3 + 672A^3 c_3 - 84A^2 b_4 + 168A^2 c_4 \\ & + \frac{49}{3} A b_3^2 - 42A b_3 c_3 - 12A b_5 + 30A c_5 + \frac{7}{3} b_3 b_4 - 4 b_3 c_4 - 3 b_4 c_3. \end{aligned} \tag{594}$$

The script `LRI_C6_1D_HPC/reference/verify_algebra.py` is an independent exact-rational polynomial check of (594). It also checks the sensitivities $\partial C_6/\partial c_6 = 1/729$ and $\partial C_6/\partial b_6 = -1/2187$.

Let $\mathcal{A}_L^\chi(\epsilon)$ and $\mathcal{B}_L^\chi(\epsilon)$ denote the bare coupling and mass amplitudes for the cutoff χ. These are amplitudes, not the scalar A or the function $B(g)$ in (592). Define

$$H_6^\chi = [\epsilon^{-1}](3\mathcal{B}_6^\chi - \mathcal{A}_6^\chi), \qquad J_5^\chi = [\epsilon^0](8\mathcal{B}_5^\chi - 3\mathcal{A}_5^\chi). \tag{595}$$

The only new five- and six-loop inputs to the physical assembly are these two numbers. Specifically,

$$\boxed{C_6 = R_6^\chi + \frac{4}{243} J_5^\chi - \frac{2}{729} H_6^\chi.} \tag{596}$$

Here R_6^χ is obtained from the same Laurent minimal-subtraction recursion, setting the five-loop finite terms and six-loop simple poles to zero, while retaining all previously required poles. Two-loop amplitudes are now needed through ϵ^3, three-loop amplitudes through ϵ^2, and four-loop amplitudes through ϵ^1. Thus R_6^χ contains newly calculated higher Laurent data even though it contains no new six-loop master.

For clarity, the new contributions to the renormalization coefficients are $b_6 = 108 a_{5,0} - 6 a_{6,-1} + \cdots$ and $c_6 = 96 e_{5,0} - 6 e_{6,-1} + \cdots$, where a, e are the coefficients of $\mathcal{A}, \mathcal{B}$. The combination $3c_6 - b_6 = 36 J_5 - 6 H_6 + \cdots$ proves the factors and signs in (596). Unit shifts of J_5 and H_6 were also checked through the full high-precision recursion.

22.3 Exhaustive graph enumeration and exact weights

At L loops the ordered-coordinate representation has $L+1$ vertices and $2L$ internal edges. An integer multiplicity $m_{ij} \in \{0, 1, 2, 3\}$ is assigned to each pair. Tadpoles are omitted, as in the five-loop graph set (12); every vertex has degree between two and four, and every nontrivial cut has at least two crossing edges. The latter requirement enforces connectivity and one-particle irreducibility.

If $e_i = 4 - \deg(i)$, the rational Wick weights (13) are, with $w_A = w_4$ and $w_B = w_2$,

$$w_A(G) = \frac{24}{\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_B(G) = \frac{w_A(G)}{12} \sum_i e_i(e_i - 1), \qquad w_H = 3 w_B - w_A. \tag{597}$$

All weights are stored exactly as integers in units of $1/1728$. The fresh enumeration gives

	Five loops	Six loops
Labeled 1PI graphs	59,145	2,427,005
Zero w_H graphs	9,540	289,590
Nonzero w_H graphs	49,605	2,137,415
$\sum_G w_A$	197,887.5	7,716,476.25
$\sum_G w_B$	21,562.5	732,471.25
$\sum_G w_H$	$-133,200$	$-5,519,062.5$

These are labeled graph counts, not counts of topological isomorphism classes or master integrals. Every surviving six-loop w_H is negative; zero-weight graphs are removed by exact integer arithmetic before integration. The 720 gap orderings reduce to 360 after spatial reversal; this is an exact symmetry reduction, not a sampled subset of orderings.

22.4 Meromorphic quadrature and the two cutoffs

Let $D = L - 1$ and order the L consecutive gaps by magnitude. If R is the largest gap, the smaller ones are $Rt_0, Rt_0t_1, \ldots, Rt_0 \cdots t_{D-1}$, with $0 \le t_j \le 1$. For an edge e, factor its distance as

$$|x_i - x_j| = R\left(\prod_{j<r_e} t_j\right) Q_e(t), \qquad Q_e(t) > 0, \quad b_j = \sum_{e:r_e>j} m_e, \qquad a_j = L - 2 - j - \frac{b_j}{2}. \tag{598}$$

Here r_e is the rank of the largest gap crossed by that edge. Each Q_e is a sum of products of consecutive t_j, including a constant term 1. Define

$$H(t) = 1 + t_0 + t_0t_1 + \cdots + t_0 \cdots t_{D-1}. \tag{599}$$

The two infrared prescriptions are $R < 1$ (maxgap) and $RH(t) < 1$ (span). Integrating the common scale exactly, a weighted graph-sector contributes

$$\frac{w_G}{L\epsilon} \int_{[0,1]^D} \prod_j dt_j\, t_j^{a_j+\epsilon b_j/2} \prod_e Q_e(t)^{-m_e/2} \exp\left\{\epsilon\left[\frac{1}{2}\sum_e m_e \log Q_e(t) - L\,\mathbf{1}_{\chi=\text{span}} \log H(t)\right]\right\}. \tag{600}$$

The integral is meromorphically continued in ϵ before taking its Laurent coefficients. It is not an ordinary convergent integral at $\epsilon = 0$ in every sector.

Each smooth factor is tensor-interpolated on the degree-N Chebyshev grid. Integration of the interpolation polynomial is then exact as a meromorphic function. For one monomial,

$$\int_0^1 t^{a+\epsilon b/2+k}\, dt = \frac{1}{a + k + 1 + \epsilon b/2}. \tag{601}$$

If $a + k + 1 = 0$, its contribution is the explicit pole $2/(b\epsilon)$; otherwise the usual geometric expansion gives every needed Laurent term. This implements endpoint subtraction without estimating a regulated integral by a small nonzero ϵ.

The quadrature weights are generated with 100-decimal-digit arithmetic and stored both in binary64 and as pairs of binary64 numbers whose sum is accurate to about 10^{-32}. The binary64 tables agree bitwise with an independent 80-digit generation, apart from entries that vanish identically, and they reproduce the exact Laurent coefficients of polynomial test integrands at every degree used. The arithmetic of the sums themselves is described in Section 22.5.

22.5 Exact summation at increasing degree

Let $\boldsymbol{H}^{(N)}$ collect the twelve six-loop Laurent coefficients at quadrature degree N: powers -6 through -1 for maxgap, followed by the same powers for span. Each entry is a finite sum over all

2,137,415 nonzero six-loop graphs, all 360 reversal classes, and all $(N+1)^5$ tensor nodes of the interpolated integrand (600). These sums are evaluated in full for $N = 6, 8, \ldots, 22$; no graph, sector or node is sampled. Two exact reorganizations make this affordable. Graphs with the same bucket vector (b_j) in a given sector share their Laurent weight tensor, and the edge factors $\prod_e Q_e^{-m_e/2}$ and the logarithms in (600) factor into two halves of the non-adjacent edges. The complete sequence through $N = 22$ used about 45 node-hours on 64-core processors.

Each graph-sector term is assigned to one of two strata according to its exponents a_j in (598). Let p be the number of axes with integral $a_j \le -1$, which carry a Laurent pole in (601). The hard stratum is

$$\mathcal{S} = \{p \ge 2\} \ \cup \ \{\text{some } a_j < -1\}, \tag{602}$$

and the light stratum is its complement, in which every axis satisfies $a_j \ge -1$ and at most one axis produces a pole. The strata are summed in different arithmetic.

- The light stratum uses binary64 vector arithmetic on blocks of eight nodes. Within each block the contributions of all weight groups are accumulated with error-free two-sum compensation; the blocks are then reduced in extended precision in a fixed order. Each sector value is therefore independent of the thread count, the division of sectors among jobs, and restarts. Its sensitivity to random one-unit perturbations in the last place of every node datum and weight is at most 7×10^{-10} in C_6 and does not grow with N.
- The hard stratum uses 80-bit extended precision (64-bit significand) throughout, with the weights read from the double-length tables. In binary64 the hard-stratum error grows with the degree because of cancellation within the stratum: the difference between binary64 and extended precision is 5×10^{-9}, 7.6×10^{-8} and 1.5×10^{-7} in C_6 at $N = 6$, 8 and 10.

Plain binary64 accumulation of the sector sums, as in the earlier deterministic degree-six evaluation, shifts C_6 by about 10^{-5} at $N = 6$ and 2×10^{-5} at $N = 8$. The earlier degree-six value differs from the present one by about 10^{-5} for this reason.

Before production, the implementation was required to reproduce the earlier deterministic degree-six sums and the earlier hard-stratum sums at $N = 6, 8, 10$ within those records' own summation errors, to give the complete sums as light plus hard stratum, and to agree between binary64 and extended precision within the measured errors. Identical executables give bitwise identical binary64 results on AMD and Intel processors; extended-precision results agree to 1.7×10^{-16} relative, the difference coming from vendor-specific x87 transcendental instructions.

22.6 Five-loop and lower-loop inputs

The five-loop data in (596) are obtained by the same exact summation over the five-loop graphs, with the 120 gap orderings reduced to 60 by spatial reversal; a graph is removed exactly when its weight in the combination being summed vanishes, $w_H = 3w_B - w_A$ for $\mathcal{H}_5$ (9,540 of the 59,145 graphs) or $8w_B - 3w_A$ for $\mathcal{J}_5$, in extended precision, at degrees $M = 12, 16, \ldots, 40$. The coupling and mass amplitudes are separated through $\mathcal{A}_5 = 8\mathcal{H}_5 - 3\mathcal{J}_5$ and $\mathcal{B}_5 = 3\mathcal{H}_5 - \mathcal{J}_5$, where $\mathcal{H}_5$ and $\mathcal{J}_5$ are the combinations $3\mathcal{B}_5 - \mathcal{A}_5$ and $8\mathcal{B}_5 - 3\mathcal{A}_5$. Their poles enter R_6^χ, and the finite part of $\mathcal{J}_5$ is J_5^χ. From $M = 28$ onward, successive changes of C_6 are at most 4.4×10^{-9}, below the measured five-loop rounding sensitivity of 1.0×10^{-8}. The two-, three- and four-loop amplitudes, through the orders listed after (596), are evaluated at degree 28; the change from degree 22 is -1.54×10^{-9} (maxgap) and $+4.14 \times 10^{-10}$ (span) in C_6. Recomputing them in binary64 instead of extended precision changes C_6 by 6.0×10^{-8}, which implies an extended-precision rounding error of about 3×10^{-11}. The inputs used in the final assembly are

	maxgap	span
R_6^χ, lower degree 28, five-loop poles at $M = 40$	39814.47028245011	81364.48627171610
J_5^χ, $M = 40$	−231603.7671545272	−1836182.1032583138
$K_\chi = R_6^\chi + \frac{4}{243} J_5^\chi$	36002.06259266366	51139.26646499489

so that $C_6^\chi(N) = K_\chi - \frac{2}{729} H_6^\chi(N)$ for each cutoff.

22.7 Degree sequence, stopping rule and uncertainty

N	$C_6^{\mathrm{maxgap}}(N)$	$C_6^{\mathrm{span}}(N)$	maxgap−span	$\rho(N)$
6	14334.549146861	14332.554509427	1.99	5.5×10^{-10}
8	14331.816869671	14331.694925294	0.122	1.9×10^{-10}
10	14331.694513177	14331.686134511	8.4×10^{-3}	7.4×10^{-10}
12	14331.687501161	14331.686917319	5.8×10^{-4}	1.9×10^{-9}
14	14331.687061994	14331.687022076	4.0×10^{-5}	4.1×10^{-9}
16	14331.687034432	14331.687031833	2.6×10^{-6}	2.9×10^{-9}
18	14331.687032845	14331.687032694	1.5×10^{-7}	3.3×10^{-8}
20	14331.687032736	14331.687032727	9.1×10^{-9}	4.5×10^{-9}
22	14331.687032770	14331.687032767	2.5×10^{-9}	8.8×10^{-8}

The convergence is geometric. Each step of two degrees reduces the change by a factor of about 0.06 for maxgap and 0.1 for span, and the two cutoffs, which differ by 2.0 at $N = 6$, approach a common value. The estimator, the uncertainty and the stopping rule were fixed before any degree beyond $N = 8$ was computed. Degrees 6 to 16 were unconditional. From $N = 16$ on, the next degree was computed unless both last changes $\Delta^\chi(N) = C_6^\chi(N) - C_6^\chi(N-2)$ and the cutoff difference were below $2\rho(N)$, where $\rho(N)$ is the floating-point estimate described below. At $N = 22$ the changes are $+3.3 \times 10^{-8}$ and $+4.0 \times 10^{-8}$ and the cutoff difference is 2.5×10^{-9}, all below $2\rho(22) = 1.8 \times 10^{-7}$, so the sequence stopped at $N_* = 22$.

The headline value is $\frac{1}{2}[C_6^{\mathrm{maxgap}}(N_*) + C_6^{\mathrm{span}}(N_*)]$, with the allowance

$$U = \mathrm{ceil}_2\Big(2 \max_\chi |\Delta^\chi(N_*)| + |C_6^{\mathrm{maxgap}} - C_6^{\mathrm{span}}| + 4\rho(N_*) + U_5 + U_{5,\mathrm{round}} + U_{\mathrm{low}}\Big), \tag{603}$$

where ceil_2 rounds up to two significant figures. The terms are 8.0×10^{-8}, 2.5×10^{-9}, 3.5×10^{-7}, 8.7×10^{-9}, 4.1×10^{-8} and 3.1×10^{-9}, so that $U = 4.9 \times 10^{-7}$.

- $\rho(N)$ is a rounding estimate. A fixed subset of 36 sectors is recomputed with every node datum and weight moved by one unit in the last place, in random directions, separately for the two strata. The resulting changes are scaled to all 360 sectors and multiplied by a calibration factor of 1.03. That factor is the largest observed ratio of the true binary64 error, measured against extended precision, to the same perturbation estimate in binary64 at $N = 6, 8, 10$.
- U_5 is twice the change of C_6 between $M = 36$ and $M = 40$. $U_{5,\mathrm{round}}$ is four times the calibrated perturbation estimate for the $M = 40$ five-loop data. U_{low} is twice the lower-degree change quoted above.

The allowance is dominated by the rounding term. Increasing the degree further would not reduce it. The separately measured rounding of the lower-loop amplitudes, about 3×10^{-11}, does not change U.

22.8 Independent checks

The poles ϵ^{-6} through ϵ^{-2} of $3\mathcal{B}_6 - \mathcal{A}_6$ are fixed by lower-order renormalization, so their errors test the whole six-loop pipeline at every degree. For maxgap, the double-pole error is 9.2×10^{-2}, 8.0×10^{-4}, 5.2×10^{-6} and 6.9×10^{-8} at $N = 10, 14, 18, 22$, and the triple-pole error at $N = 22$ is 2.7×10^{-9}; span behaves in the same way. The two cutoffs, whose six-loop integrands and subtraction constants differ, agree to 2.5×10^{-9} at N_*.

The earlier Monte Carlo evaluation sampled the light-stratum differences between degrees. Comparing its level estimates with the exact differences, in units of its standard errors, gives -0.57 and -0.60 for $8-6$, $+0.86$ and $+1.33$ for $10-8$, and $+0.09$ and $+1.26$ for the all-graph $12-10$ diagnostic (maxgap and span). That estimator was therefore consistent within its errors; its final value was 0.58 standard errors above the present one.

The seventh-order calculation of Section 27 produces C_6 as a by-product, in a different representation: coordinate forests, randomized integration, and radial quadrature through degree six. Its degree-six value 14332.43593 ± 0.30500 lies $+2.46$ of its standard errors above (591). Adding its measured degree-six to degree-seven check step gives 14331.98210, at $+0.97$; adding further a geometric tail model gives 14331.62055, at -0.22. The degree-six tension is the radial truncation of that representation, discussed in Section 27; agreement beyond the measured step is conditional on the tail model.

22.9 A finite special-function representation

A separate analytic calculation, recorded in the package `C6_research_note/`, starts from an analytic reduction of the six-loop coefficient to explicitly defined periods and series. It converts that expression into a finite formula in nine special-function values. The reduction itself is derived neither in this note nor in the package. We state it as the input (604) and compare its value with (591). Two of its leading Laurent coefficients, $R_{147,-3} = R_{148,-3} = \frac{2}{5}$ in the package's numbering, were missing from the first version of the input and were added by a correction; their contribution is given below.

In this subsection, for $n \geq 0$,

$$a_n = \binom{2n}{n} 4^{-n}, \qquad h_n = \log 2 + H_n - H_{2n}, \qquad \delta_n = \psi_1(n+\tfrac{1}{2}) - \psi_1(n+1), \qquad y_n = n + \tfrac{1}{4},$$

with $H_n^{(p)} = \sum_{j=1}^{n} j^{-p}$ and $H_n = H_n^{(1)}$; $\delta_n > 0$ is a trigamma difference, not a Kronecker delta. With $A = \pi + 4\log 2$, $Q = \Gamma(\frac{1}{4})^4/\pi$ and Catalan's constant G, the input reads

$$2187\, C_6(1) = E + S_{\mathrm{pol}} + S_{\mathrm{low}} + S_6 + S_8 - 393\pi^3 W_0 + 24\pi^3 M_7 + 96 P_a, \tag{604}$$

with the known polynomial

$$\begin{aligned} E = {} & \frac{Q}{8}\left[63A^3 + 120A\mathsf{G} + 79A\pi^2 + 2013\pi^3 + 54204\zeta(3)\right] + Q^2(29\pi - 249A) + \frac{21}{640}A^5 + \frac{7}{64}A^3\pi^2 \\ & + \frac{1149}{64}A^2\pi^3 + \frac{7863}{16}A^2\zeta(3) + 192A\mathsf{G}(16\mathsf{G} + 3\pi^2) + \frac{14853}{640}A\pi^4 - \frac{544}{3}\mathsf{G}\pi^3 \\ & + \frac{37631}{128}\pi^5 + \frac{722279}{48}\pi^2\zeta(3) - \frac{805959}{8}\zeta(5), \end{aligned} \tag{605}$$

which already contains the restored coefficients, and the polylogarithmic sector

$$S_{\mathrm{pol}} = 4352\left[(32\mathsf{G} + 9\pi^2)D_3 - 64 D_5\right], \qquad D_3 = \operatorname{Im}\operatorname{Li}_{2,1}(\mathrm{i}, 1), \quad D_5 = \operatorname{Re}\operatorname{Li}_{2,1,2}(\mathrm{i}, 1, 1), \tag{606}$$

where $\mathrm{Li}_{s_1,\dots,s_k}(z_1,\dots,z_k)=\sum_{n_1>\cdots>n_k\ge1}\prod_j z_j^{n_j}n_j^{-s_j}$. The three ordinary sectors are

$$\begin{aligned}
S_{\mathrm{low}} &= 4\pi^3 C_d\varpi_4+2\pi^4 C_s\varpi_6,\\
C_d &= 96Q-42A^2-18A\pi-1227\pi^2, \qquad C_s=60Q-\tfrac{105}{4}A^2-9A\pi-\tfrac{2469}{4}\pi^2,\\
S_6 &= 24\Big[\frac{6Q^3}{\pi}(2\Phi_- -1)-104\pi^5\Phi_0+\frac{2^{18}\pi^5}{Q^3}\Phi_+\Big],\\
S_8 &= \pi^6(4932-5004\sqrt2)\varpi_8+556\pi^2\big[(17+12\sqrt2)Y_1-(17-12\sqrt2)Y_7\big],
\end{aligned}\tag{607}$$

with the hypergeometric values at unit argument (all parametric excesses are positive)

$$\begin{gathered}
\varpi_4={}_4F_3\big((\tfrac12)^{\times4};1^{\times3};1\big),\quad \varpi_6={}_7F_6\big(\tfrac54,(\tfrac12)^{\times6};\tfrac14,1^{\times5};1\big),\quad \varpi_8={}_9F_8\big(\tfrac54,(\tfrac12)^{\times8};\tfrac14,1^{\times7};1\big),\\
\Phi_-={}_7F_6\big(1,(\tfrac14)^{\times6};(\tfrac34)^{\times6};1\big),\quad \Phi_0={}_6F_5\big((\tfrac12)^{\times6};1^{\times5};1\big),\quad \Phi_+={}_7F_6\big(1,(\tfrac34)^{\times6};(\tfrac54)^{\times6};1\big),\\
Y_r=t\Big[\frac{\Gamma(t+\frac14)}{\Gamma(t+\frac34)}\Big]^8{}_{10}F_9\big(1,1+t,(t+\tfrac14)^{\times8};t,(t+\tfrac34)^{\times8};1\big),\qquad t=\frac{r}{8}.
\end{gathered}$$

The three remaining sectors are sums with no known evaluation in standard constants:

$$\begin{aligned}
W_0 &= \sum_{n\ge0}a_n^4\big[8+5\pi(4n+1)a_n^2\big]\delta_n, \qquad M_7=\sum_{n\ge0}(-1)^n a_n^7F_n,\\
F_n &= (98H_{2n}+74\log2-6\pi)T_n-98\pi\kappa_n-98(11u_n-13\pi y_n)\Big(\frac{\pi^2}{3}+4H_{2n}^{(2)}\Big)\\
&\quad -8624y_n\big[\zeta(3)-H_{2n}^{(3)}\big],\\
T_n &= 80\pi^2y_n-26\pi u_n+11v_n, \qquad \kappa_n=28\pi^2y_n-10\pi u_n+3v_n,\\
u_n &= 1-14y_nh_n, \qquad v_n=196y_nh_n^2-28h_n+7y_n(\delta_n-\pi^2),
\end{aligned}\tag{608}$$

and

$$P_a=\pi^4(15I_{22}+24I_{11}-4I_{00}),\qquad I_{ij}=\int_0^1\frac{p_i(x)\,p_j(1-x)}{\pi^{i+j}\sqrt{x(1-x)}}\,dx,\tag{609}$$

with $p_0=v(x)^2$, $p_1=\sqrt{Q}\,v(x)\,w(1-x)$, $p_2=Q\,w(1-x)^2$, $v(x)={}_2F_1(\frac14,\frac14;1;x)$ and $w(x)={}_2F_1(\frac14,\frac14;\frac12;x)$. The sum W_0 includes $n=0$, whose term is $(8+5\pi)\pi^2/3$. Its terms decay like $14/(\pi^2n^4)$; those of M_7 alternate in sign and are $O(n^{-5/2}\log n)$; the pairings have integrable endpoint singularities, so all three converge absolutely.

Special-function form. Appendix B converts every sector of (604) into one value at unit argument, using gamma reflection, sine addition, beta integration and Mellin–Barnes transforms. The result is the nine-value formula (2), whose values and parameters Section 1.2 defines. In particular $D_3=-\frac12\,\mathrm{Ls}_3(\frac{\pi}{2})-\frac{7\pi^3}{192}$, $D_5=\mathrm{Cl}_{2,1,2}(\frac{\pi}{2})$, $W_0=\sqrt{89}\,\mathcal{H}_W/\pi^2$, $\pi^{3/2}M_7=\rho_7\,\mathcal{G}_7+196\pi\sqrt{185}\,\mathcal{H}_7$ and $P_a=80\pi^3\mathcal{H}_*$. The count of nine values describes how the terms are packaged, not the number of independent constants: sine addition merges two G values into one with an angle parameter, and α_4 depends on Q, A and π.

Value and comparison with the graph sum. The representation evaluates to

$$C_6^{\mathrm{rep}}(1)=14331.687032758054320445463255246217118317 32\dots,$$

with the following sector contributions, in units of C_6:

$E/2187$	-1334.069267305	$S_8/2187$	4446.133424647
$S_{\mathrm{pol}}/2187$	-30.841591973	$-393\pi^3W_0/2187$	-437.775914461
$S_{\mathrm{low}}/2187$	-944.671764724	$24\pi^3M_7/2187$	-5217.069054363
$S_6/2187$	3148.152054246	$96P_a/2187$	14701.829146691

It lies 1.03×10^{-8} below the unrounded value 14331.68703276838 of (591), which is 0.021 of the allowance U. The two added Laurent coefficients contribute +659.278983 to C_6; without them (604) would give 13672.408. Because (591) is computed from the graphs directly, without (604), the agreement tests the reduction and its correction numerically at the level of 3×10^{-11} relative. It is not a proof of the reduction.

The package encloses $C_6^{\mathrm{rep}}(1)$ in an outward-rounded interval of width below 1.2×10^{-145}, using explicit bounds on the truncated tails of the series for W_0, M_7 and P_a. That enclosure is rigorous for (604) as stated, conditional on the identities the package uses (Section B.8.5); the tail of M_7 is bounded by (956). An independent review (`LRI_C6_closed_form_review/`) re-evaluated (604) from the definitions above to about 65 digits, recomputed the nine values with separate codes (the two-fold integrals directly, to 9–14 digits), and checked the algebra of the conversions. It found no error. Some conversions rest on further identities. The coefficient series for S_6 and S_8 and the two periodization identities for the wheel are derived in Appendix B, the latter from three elliptic-moment evaluations that the package takes over without derivation, (911)–(913); the Barnes representation of Bailey type [14] and a hypergeometric symmetry are published identities. All were confirmed numerically, the series and periodization identities to better than 10^{-63}. For W_0, M_7 and P_a no reduction to standard constants is known, and a PSLQ search in the package found no relation in the three bases it tested.

22.10 Result in the physical series

Combining the retained analytic lower-order coefficients with the sixth-order value of Section 22.9 gives

$$\begin{aligned}\Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} &+ 0.657131263981\,\epsilon^2 - 3.719856322902\,\epsilon^3 \\ &+ 45.42488010356\,\epsilon^4 - 731.13947662454\,\epsilon^5 + 14331.68703275805\,\epsilon^6 + O(\epsilon^7).\end{aligned} \tag{610}$$

The graph sum of this section gives the sixth-order coefficient as 14331.6870328(5), in agreement to 1.0×10^{-8}. Neither number is a bound on the omitted $O(\epsilon^7)$ term.

23 Finite Schwinger kernels and local subtraction

Unless stated otherwise, this section fixes $d = 2$, $A = 4\log 2$, $a = (2+z)/4$ and $r(2,z) = 1$. The kernel normalization is exactly (195). For a primitive L-loop graph define

$$a_G = [z^{-1}]\mathcal{I}_G(2,z), \qquad \Pi_G = \frac{L}{2}a_G = \frac{1}{\pi^L}\int d\Omega\,\frac{\prod_e x_e^{-1/2}}{U_G(x)}. \tag{611}$$

The perturbative sign is not included in a_G. At six loops every primitive mass weight vanishes, so

$$C_{6,\mathrm{prim}}(2) = \frac{2}{729}\sum_{G\ \text{primitive}} w_A(G)a_G.$$

The following local one- and two-subgraph constructions define the finite kernels that the previous two-dimensional estimator (Section 26.13) retained as numerical inputs, (741). The 2026 two-dimensional production integrates all six-loop kernels in every replicate (Section 26.7). Their MS conversions are given separately so that their finite-scheme values are not confused with partial MS contributions. The numerical values quoted in this section are archived records of that superseded estimator. They and the programs that produced them are kept in the research archive `LRI_complete_research_bundle.zip` (packages `lri_C6_d2_primitive_update/`, `lri_C6_d2_subtraction_update/` and `lri_C6_d2_two_forest_update/`), which is not part of this repository; no current coefficient depends on them.

23.1 The sixth-order assembly at general dimension

The material of this subsection was written by OpenAI GPT-6 astra in the earlier sixth-order calculations at $d = 2$ and $d = 3$ (package `LRI_C6_D2_HPC/reference/`), with one exception: the value of $\mathfrak{B}_{\rm kn}/2187$ below was computed from the closed forms by a Claude Code session on 16 October 2026, when this subsection was merged into the note; the earlier calculation had evaluated it from numerical master values. The same session rechecked the polynomial (615) and the six-loop totals.

Assembly. The sixth-order formula (594) is the case $n = 6$ of the Lagrange formula (593), an identity between polynomials in the renormalization-group coefficients. In the bubble-normalized MS prescription (174), with

$$\beta(g) = -\epsilon g + g^2 B(g), \quad B(g) = 3 - 6A_d g + b_3 g^2 + \cdots + b_6 g^5, \qquad \gamma_r(g) = g - A_d g^2 + \sum_{j=3}^{6} c_j g^j,$$

it therefore holds at every d with A replaced by A_d of (176); in particular $A_2 = 4\log 2$ and $A_3 = 2 + 4\log 2 - \pi$. All b_j, c_j in one assembly must belong to the same prescription. The sensitivities to c_6 and b_6 quoted after (594) hold at every d as well, and the physical $C_0, \ldots, C_5$ at a fixed dimension cannot supply the independent six-loop combination $3c_6 - b_6$. For the bare amplitudes (193) put

$$J_5(d) = [z^0](8\mathcal{B}_5(d,z) - 3\mathcal{A}_5(d,z)), \qquad H_6(d) = [z^{-1}](3\mathcal{B}_6(d,z) - \mathcal{A}_6(d,z)). \tag{612}$$

The Laurent recursion (204) then gives, exactly as (596) at $d = 1$,

$$\boxed{C_6(d) = R_6(d) + \frac{4}{243} J_5(d) - \frac{2}{729} H_6(d),} \tag{613}$$

where $R_6(d)$ is the result of the same recursion with the five-loop finite parts and the six-loop simple poles set to zero and all other inputs retained. The new bare terms are $b_6 = 108a_{5,0} - 6a_{6,-1} + \cdots$ and $c_6 = 96e_{5,0} - 6e_{6,-1} + \cdots$ in the notation (202), and their combination in (594) is exactly $(36J_5 - 6H_6)/2187$. The two-, three-, four- and five-loop amplitudes are needed through z^3, z^2, z^1 and z^0, and the six-loop ones through z^{-1}, each with all its higher poles; the one-dimensional span and maxgap finite parts belong to a different prescription and are not inputs at $d \neq 1$. These identities were checked by exact rational polynomial arithmetic; the checker is in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository.

A fifth-order channel that reenters. At fifth order 28 vertex-isomorphism classes have $w_4 = 3w_2$ and drop out of $3\mathcal{B}_5 - \mathcal{A}_5$, whose weight w_H is (211). In J_5 their weight is $8w_2 - 3w_4 = -w_2 \neq 0$, so all 28 reenter at sixth order. Reusing only the fifth-order combined-channel data would omit them. This is consistent with the fifth-order result: these terms cancel there, but not here.

Six-loop inventory. The 2,427,005 labeled six-loop 1PI graphs of the table after (597) fall into 660 vertex-isomorphism classes. With the weights divided by $|\,\mathrm{Aut}\,|$, as in the right side of (196),

$$\sum_G w_A = \frac{97987}{64}, \qquad \sum_G w_B = \frac{83711}{576}, \qquad \sum_G (3w_B - w_A) = -\frac{105125}{96}.$$

Times 7! these are the six-loop totals of the table after (597), and the first two are the zero-dimensional coefficients of (16). Every class has twelve separately indexed internal edges, and its first Symanzik polynomial is stored as a list of complementary spanning-tree masks; the six-loop inventory contains 140,781 such monomials.

MS bookkeeping at $d = 2$. The MS-converted sector values below are quoted together with the five-loop mass terms $-6Ac_5/2187$. The reason is the following exact identity. Use the MS values of b_3, c_3 (179), b_4, c_4 (226)–(227) and $D_5 = 3c_5 - b_5$ (229), and eliminate $b_5 = 3c_5 - D_5$ in (594); since $-12Ab_5 + 30Ac_5 = 12AD_5 - 6Ac_5$, at $d = 2$

$$\boxed{2187\, C_6(2) = \mathfrak{B}_{\rm kn} + 3c_6(2) - b_6(2) - 6A\, c_5(2), \qquad A = 4\log 2.} \tag{614}$$

With the dictionary $R = 0$, $M = -4$, $N = 4 - A$, $U = -14\zeta(3)$, $V = -4\zeta(3)$ and $X = 0$ of (221) and (225) at $q = 1/2$, writing $Q = Q_2$ and $P_j = P_j(2)$,

$$\begin{aligned}\mathfrak{B}_{\rm kn} = &-\frac{14385}{16}A^5 - 1719A^3Q + 186A^3 - \frac{771}{2}A^2P_4 + \frac{164985}{16}A^2\zeta(3) + 174A^2\\ &- 288A\mathcal{K}_2 - 108A\mathcal{L}_2 - 288AP_5 - 852AQ^2 + 240AQ + 1341AZ_2 - \frac{16}{3}A\\ &- 84P_4Q + 32P_4 + \frac{5511}{2}Q\zeta(3) + 144Q - 718\zeta(3).\end{aligned} \tag{615}$$

Every coefficient follows by exact expansion of $\mathcal{R}(b_3, c_3, b_4, c_4) + 12AD_5$, with $\mathcal{R}$ of (676). With $Z_2 = 4\mathsf{G}Q_2$ (265), the closed forms (270), (271), and $P_4(2)$, $P_5(2)$ of Section 17.2,

$$\frac{\mathfrak{B}_{\rm kn}}{2187} = -1285.866857775584\ldots, \qquad C_6(2) = -1285.866857775584\ldots + \frac{3c_6 - b_6 - 24\log 2\, c_5}{2187}.$$

The fifth-order physical coefficient fixes D_5 but not c_5 separately, so c_5 is a genuine input of (614). This decomposition was used to organize the partial MS sectors below; the current estimate does not use it, since it computes the physical combination in the finite scheme of Section 25.

Why a different representation was needed at $d = 2$. The same 660 six-loop classes occur in every dimension, so the graph count is not what makes $d = 2$ harder. On the line, seven vertices can be ordered and replaced by six positive gaps; removing the common scale leaves five dimensionless gap variables, every distance is a sum of consecutive gaps, and the collision sectors (598) have an efficient implementation. The two-dimensional Schwinger representation has 12 edge parameters and an 11-dimensional projective simplex. (This describes the representation, not a minimal dimension: in coordinate space seven points have 14 real coordinates, of which translation, rotation and scale remove four.) Without a linear ordering and with many more relative configurations, naive tensor quadrature and untuned sampling are much less effective. An earlier attempt at $d = 2, 3$ sampled the unsubtracted Schwinger sector integrals with Taylor endpoint subtraction, both uniformly and with a subset dynamic-programming proposal. The separate large endpoint subtraction terms caused severe variance and loss of precision: the error propagated to $C_6(2)$ from H_6 alone remained of order 10^5, and no value was accepted. More samples do not repair a poor subtraction representation. The common finite kernels, the exact root-scale coefficients and the regulator-aware fixed-point assembly of this section and of Sections 24 and 25 address that obstacle directly. Two dimensions also have advantages: complex-plane factorization, the complex Selberg integral (191) and the bilinear ${}_3F_2$ form (674), which allow many counterterms to be evaluated analytically.

23.2 Exact primitive inventory and a product reduction

Use the complete 660-class six-loop inventory. A graph is declared primitive only if every proper induced vertex subset S with $|S| \geq 2$ satisfies $e(S) < 2(|S| - 1)$. This selects ten classes. All edge labels, parallel-line labels, rational Wick weights and spanning-tree polynomials come from the unchanged input inventory. The class indices in this section are zero-based indices in the project's 660-class six-loop inventory (archived as `data/inventory6.json` in the research archive),

not indices in any external graph catalog; the same indices identify the six-loop kernels in `LRI_C6_D2_HPC/inputs/inventory.json`.

Completing each selected graph to a four-regular graph on eight vertices partitions the ten classes into five isomorphism classes:

Group	Uncompleted class indices	$\sum w_A$	Period
0	$639, 654, 657$	16	$P_3(2)P_4(2)$
1	648	12	Π_1
2	$649, 651, 655$	48	Π_2
3	$653, 658$	12	Π_3
4	659	1/6	Π_4

Each equality within a group follows from conformal completion, not from numerical comparison. In the present two-dimensional normalization each critical coordinate edge has absolute-distance power one, and each completed vertex has incident power four. Inversion of an integration variable therefore cancels its Jacobian, proving invariance under the choice of the deleted vertex. This is the weighted completion argument; the general completion and product identities are reviewed in [42].

For the representative 639, the three-vertex separator is $\{4, 5, 6\}$. Removing it leaves components $\{0, 1\}$ and $\{2, 3, 7\}$. Adding the uniquely required triangles on the separator completes the components to K_5 and the octahedral graph. Fixing the separator at $0, 1, \infty$ makes their integrations independent. Thus their normalized periods multiply, with no additional measure factor:

$$\Pi_0 = P_3(2)P_4(2) = 3Q_2P_4(2). \tag{616}$$

The explicit separator and child-graph isomorphisms are checked in the archived record

`lri_C6_d2_primitive_update/verification/primitive_reduction_checks.json`

of the research archive. The complete primitive contribution can therefore be written

$$\boxed{C_{6,\mathrm{prim}}(2) = \frac{2}{2187}\left[48Q_2P_4(2) + 12\Pi_1 + 48\Pi_2 + 12\Pi_3 + \frac{1}{6}\Pi_4\right].} \tag{617}$$

Only the last four periods require new numerical integration. The first term is evaluated from the retained ordinary-hypergeometric formula for $P_4(2)$ and the exact Gamma value of Q_2:

$$\begin{aligned} Q_2 &= \frac{\Gamma(1/4)^8}{24\pi^4}, \\ P_3(2)P_4(2) &= 9597.206978165420901472979104294093 3\ldots, \\ \frac{96}{2187}Q_2P_4(2) &= 140.425525057747356583052277703434 38\ldots. \end{aligned} \tag{618}$$

The unreduced ten-class integration is retained as an independent check of this reduction; the product value is not fitted to its output.

23.3 Positive tropical sampling and a stable cycle determinant

At $d = 2$, the primitive residue of the normalized Schwinger integral is

$$a_G = \frac{2}{L\pi^L}\int_{x_e\ge 0} d\Omega\, \frac{\prod_{e=1}^{2L} x_e^{-1/2}}{U_G(x)}. \tag{619}$$

There is no endpoint analytic continuation in this primitive integral. We use tropical importance sampling of the positive projective integrand, in the framework of [41]. Here its specialized construction is spelled out to fix the normalization.

In an edge ordering σ, set the largest edge parameter to one and use $x_{\sigma(j)} = \prod_{k=j}^{2L-1} t_k$ for $j < 2L$. If $h(S)$ is the cycle rank of the first $|S|$ edges, write

$$\alpha(S) = \frac{|S| - 2h(S)}{2}, \qquad U_G = \left(\prod_{j=1}^{2L-1} t_j^{h(S_j)} \right) \widetilde{U}_G. \tag{620}$$

All proper-prefix $\alpha(S_j)$ are strictly positive for the selected graphs. The integral over a sector has the form $\int \prod_j t_j^{\alpha(S_j)-1} dt_j / \widetilde{U}_G$. The exact sector-normalization recursion is

$$Z_G(\varnothing) = 1, \qquad Z_G(S) = \frac{\sum_{e \in S} Z_G(S \setminus \{e\})}{\alpha(S)} \quad (S \neq E), \qquad Z_G(E) = \sum_{e \in E} Z_G(E \setminus \{e\}). \tag{621}$$

At each backward step choose the removed edge with probability proportional to $Z_G(S \setminus \{e\})$. Independently sample $t_j = U_j^{1/\alpha(S_j)}$, with U_j uniform on $(0, 1)$. For graph weights w_G, sample graphs proportionally to $w_G Z_G(E)$. If $W = \sum_G w_G Z_G(E)$, each sample of the weighted bare residue is simply

$$Y = \frac{2W}{L\pi^L} \frac{1}{\widetilde{U}_G}. \tag{622}$$

Thus the estimator is positive and bounded; it does not sample a difference of divergent contributions. This positivity does not extend to the omitted nonprimitive subtraction problem.

A direct evaluation of $\widetilde{U}_G$ sums all spanning-tree monomials. For the production runs we instead construct a minimum spanning tree in the sampled edge order. Let f_i be its L chords and let e range over its L tree edges. Define

$$V_{ie} = \begin{cases} \sigma_{ie} \sqrt{x_e / x_{f_i}}, & e \text{ lies in the fundamental cycle of } f_i, \\ 0, & \text{otherwise,} \end{cases} \qquad \sigma_{ie} \in \{+1, -1\}. \tag{623}$$

The matrix-tree identity gives the scaled form

$$\boxed{\widetilde{U}_G = \det(I_L + V V^{\mathsf{T}}).} \tag{624}$$

To see the normalization, the fundamental-cycle matrix has one unit entry on its own chord and zero entries on all other chords. Factoring $\prod_i x_{f_i}$ from its Gram determinant leaves $\det(I + VV^{\mathsf{T}})$. The minimum-spanning-tree property makes that product the leading Hepp monomial and ensures $x_e \le x_{f_i}$ on each fundamental cycle. Hence $|V_{ie}| \le 1$ and every eigenvalue of $I + VV^{\mathsf{T}}$ is at least one. Cholesky evaluation avoids the poor conditioning of an unscaled Laplacian near a sector boundary.

An independent 60-digit check compares the monomial sum with this determinant on 180 configurations, including exponentially small sector variables. The largest relative difference is 4.49×10^{-61}. Production runs also perform pointwise checks before accumulating the batch estimates. The graph input and both C++ implementations are in the research archive (package `lri_C6_d2_primitive_update/`), which is not part of this repository.

23.4 Normalization and the exact set of graphs

Work at $d = 2$, $\epsilon = 2s - 2$. With the regulator z used for Laurent coefficients, $a = \frac{1}{2} + z/4$ and the loop normalization (195) becomes

$$r(2, z) = 1, \qquad I_G(z) = \frac{\Gamma(Lz/2)}{\Gamma(\frac{1}{2} + z/4)^{2L}} \int_{\Delta_{2L-1}} \frac{\prod_e x_e^{a-1}}{U_G(x)} \, d\Omega, \quad U_G = \sum_T \prod_{e \notin T} x_e. \tag{625}$$

Here every internal line is a separate Schwinger variable, including parallel lines. The weights w_A, w_B are the rational vertex-isomorphism weights of the six-loop inventory used throughout this section; set $w_H = 3w_B - w_A$. No additional vertex factorial is introduced in this section.

Let $\mathcal{S}_{26}$ consist of the classes with exactly one connected, proper, one-particle-irreducible logarithmic subgraph γ, and no power-divergent subgraph. Both γ and G/γ are primitive. The class IDs and aggregate weights are as follows; a number j in the second column denotes the unchanged inventory label `L6Gj`, not its zero-based array index.

$h = \ell(\gamma)$	Class numbers	Number	$\sum w_H$
1	524, 529, 539, 563, 575, 587, 588, 596, 599, 607, 613, 614, 617, 619, 620, 621, 627	17	$-279/2$
3	642	1	-4
5	638, 639, 641, 647, 648, 651, 653, 657	8	$-93/4$

The total weight is $-667/4$. The archived machine-readable record gives every edge list, the subgraph edge mask, the rational weights, and the exact isomorphism to the primitive subgraph and quotient. A separate checker also reconstructs their spanning-tree polynomials; its 158 exact checks include both aggregate conversion identities below.

23.5 A common-integrand subtraction with no large endpoint difference

Let $h = \ell(\gamma)$, $k = L - h$, and use $\rho = G/\gamma$ for the quotient. Define

$$X_\gamma = \sum_{e\in\gamma} x_e, \quad X_\rho = \sum_{e\notin\gamma} x_e, \quad S = X_\gamma + X_\rho, \quad U_0 = U_\gamma U_\rho, \quad U_\times = U_G - U_0. \tag{626}$$

The edge variables of the quotient are the original edges outside γ. The monomials of U_0 are a subset of those of U_G; therefore $U_\times$ has nonnegative coefficients. This is checked exactly, not by evaluating the polynomials at selected points.

The auxiliary amplitude is

$$\boxed{D_G(z) = I_G(z) - \frac{k}{L} I_\gamma(z) I_\rho(z).} \tag{627}$$

Its common projective kernel is

$$\boxed{\frac{1}{U_G} - \frac{X_\rho}{S\,U_0} = \frac{X_\gamma U_0 - X_\rho U_\times}{S\,U_0\,U_G}.} \tag{628}$$

In simplex gauge $S = 1$; retaining S makes the homogeneity correct in any Hepp gauge. This is not a replacement of the product by a local approximation: its integral is exactly the product term in (627).

Proof of the factor k/L. In the product integral, separate the two sets of variables by $x_\gamma = \lambda u$ and $x_\rho = (1-\lambda)v$, with u, v each on a unit simplex. Their scale factors are $\lambda^{hz/2-1}(1-\lambda)^{kz/2-1}$. Without a scale weight the beta integral exactly combines the two Gamma prefactors into $\Gamma(Lz/2)$. Inserting $X_\rho/S = 1 - \lambda$ multiplies the beta integral by

$$\frac{B(hz/2, 1 + kz/2)}{B(hz/2, kz/2)} = \frac{k}{L}. \tag{629}$$

The identity holds in the ordinary convergence region and then by meromorphic continuation. Hence the subtraction has the correct normalization at nonzero z, not just at leading pole order.

Endpoint cancellation. When γ shrinks, $U_\times$ has at least one extra power of the subgraph scale compared with U_0. The first numerator term in (628) has this power through X_γ. The artificial singularity when the complementary factor shrinks is removed similarly, using the fact that it is not a proper subdivergence of the original G. Since the two factors are primitive, these are the only zero exponents introduced by the product denominator. The cancellation is applied to the monomial exponents before floating point evaluation. The implementation never forms two nearly equal, individually singular reciprocal polynomials.

More explicitly, order the $2L$ edges, write the ordered Schwinger variables as products of $t_j \in (0,1)$, and let E_j be the first j edges. If

$$h_0(E_j) = \min_{m \in \operatorname{supp} U_0} |m \cap E_j|, \qquad \alpha_j = \frac{j}{2} - h_0(E_j) + \mathbf{1}_{E_j = E_\gamma \text{ or } E_j = E_\rho}, \tag{630}$$

then every proper-sector α_j is positive for these graphs. The indicated powers are stripped from the numerator and t_j is sampled with density $\alpha_j t_j^{\alpha_j - 1}$. Sector probabilities are proportional to the corresponding product of $1/\alpha_j$ and are generated by a subset dynamic program. The normalization for the simple residue is $2/(L\pi^L)$. The remaining rational kernel is bounded on each closed sector after these structural powers have been removed; this distinguishes it from an earlier estimator of the same graphs, which sampled the unsubtracted integrands with importance weights taken from their logarithmic moments and was much noisier.

Define the quantity directly measured by this sampler as

$$Y_{26} = \sum_{G \in \mathcal{S}_{26}} w_H(G)[z^{-1}]D_G(z). \tag{631}$$

Y_{26} is not an MS residue. Its conversion is indispensable.

23.6 Conversion to the same MS coefficient as the six-loop assembly

Write the primitive factors as

$$I_\gamma = \frac{r_\gamma}{z} + f_\gamma + O(z), \qquad I_\rho = \frac{r_\rho}{z} + f_\rho + O(z). \tag{632}$$

For a single proper primitive subdivergence the incomplete MS operation is $R'_G I_G = I_G - (r_\gamma/z) I_\rho$. Taking only the simple pole gives

$$\boxed{[z^{-1}]R'_G I_G = [z^{-1}]D_G - \frac{h}{L} r_\gamma f_\rho + \frac{k}{L} f_\gamma r_\rho.} \tag{633}$$

Higher poles of $R'_G I_G$ are not assumed to vanish. Equation (633) is a Laurent-coefficient identity and does not confuse this auxiliary kinematic subtraction with MS.

There are three primitive five-loop inventory classes. In zero-based array notation they are 123, 125 and 126, with coupling weights $2, 3/2, 12$, respectively. Let f_i denote their individual unweighted finite parts, evaluated in the massive normalization (625), and put

$$a^{\mathrm{p}}_{5,0} = 2f_{123} + \frac{3}{2} f_{125} + 12 f_{126}. \tag{634}$$

The finite parts of conformally equivalent completed graphs are not identified: completion identities fix their residues, not their mass-dependent finite terms. All three terms in (634) were integrated separately.

The one-loop bubble is exactly $I_1 = 2/z$, with zero finite part. For the $h = 3$ class the two primitive factors are isomorphic, so its conversion terms in (633) cancel. Exact aggregation over the other 25 classes gives

$$\boxed{M_{26} := \sum_{G \in \mathcal{S}_{26}} w_H [z^{-1}] R'_G I_G = Y_{26} + \frac{5}{2} a^{\mathrm{p}}_{5,0}.} \tag{635}$$

For example, the three finite-part coefficients on the right are $5, 15/4, 30$, which are exactly $5/2$ times the weights in (634). The direct contribution to the physical assembly is

$$\boxed{\Theta_{26} = \frac{(3c_6 - b_6)_{26}}{2187} = -\frac{2}{729} M_{26}.} \tag{636}$$

The last sign follows from the even-loop sign in the bare amplitudes: $(3c_6 - b_6)_{26} = -6M_{26}$.

That earlier estimator measured the same weighted simple-pole coefficient of the unsubtracted amplitudes,

$$H_{26}^{\mathrm{bare}} = \sum_{G \in \mathcal{S}_{26}} w_H(G)[z^{-1}] I_G(z).$$

For a comparison with it, let r_3, f_3 be the unweighted primitive three-loop residue and finite part. A second exact aggregate identity is

$$\boxed{H_{26}^{\mathrm{bare}} = Y_{26} - \frac{31}{2} a_{5,0}^{\mathrm{p}} - 4 r_3 f_3, \qquad r_3 = 2Q_2.} \tag{637}$$

Comparing Y_{26} directly to H_{26}^{bare} would therefore give a spurious normalization discrepancy.

23.7 The two-subgraph classes and a common finite kernel

There are 40 six-loop classes with exactly two proper logarithmic subgraphs and no power-divergent subgraph: 23 have a nested pair and 17 have a vertex-disjoint pair. Every decomposition used here has two one-loop primitive leaves and one four-loop primitive leaf. At five loops there are 12 analogous classes (seven nested, five disjoint); only five of them carry a nonzero mass weight. Their primitive leaves are two bubbles and the three-loop tetrahedron.

For G let $U = U_G$ and let U_1, U_2 be its leading polynomials when respectively γ_1, γ_2 shrink. For a subgraph of loop number h_i, U_i consists exactly of the monomials of U containing h_i edges from γ_i. The common face is $W = U_{12}$. Set

$$X = U_1 - W, \qquad Y = U_2 - W, \qquad Z = U - U_1 - U_2 + W. \tag{638}$$

Thus $U = W + X + Y + Z$, $U_1 = W + X$, $U_2 = W + Y$. The four polynomials partition the spanning-tree monomials and have nonnegative coefficients. For nested subgraphs put

$$\sigma_1 = \sum_{e \in \gamma_1} x_e, \quad \sigma_2 = \sum_{e \in \gamma_2 \setminus \gamma_1} x_e, \quad \sigma_3 = \sum_{e \notin \gamma_2} x_e, \qquad S = \sigma_1 + \sigma_2 + \sigma_3. \tag{639}$$

For disjoint subgraphs, the three sums run over γ_1, γ_2 and their complement.

The two auxiliary forest kernels, including the common projective scale weight but not the common propagator measure, are

$$K_{\mathrm{n}} = \frac{1}{U} - \frac{\sigma_2 + \sigma_3}{S U_1} - \frac{\sigma_3}{S U_2} + \frac{\sigma_3}{S W}, \tag{640}$$

$$K_{\mathrm{d}} = \frac{1}{U} - \frac{\sigma_2 + \sigma_3}{S U_1} - \frac{\sigma_1 + \sigma_3}{S U_2} + \frac{\sigma_3}{S W}. \tag{641}$$

These are auxiliary subtractions, not MS graph contributions; the conversion follows below. For stable evaluation write either kernel as

$$K_\star = \frac{N_\star}{S\, U\, U_1\, U_2\, W}, \tag{642}$$

where the numerators are the following explicit finite polynomials:

$$\begin{aligned} N_{\rm n} &= \sigma_1 W(W+X)(W+Y) - \sigma_2(Y+Z)W(W+Y) \\ &\quad + \sigma_3\left[XY(2W+X+Y+Z) - ZW^2\right], \end{aligned} \tag{643}$$

$$\begin{aligned} N_{\rm d} &= -\sigma_1 W(W+X)(X+Z) - \sigma_2 W(W+Y)(Y+Z) \\ &\quad + \sigma_3\left[XY(2W+X+Y+Z) - ZW^2\right]. \end{aligned} \tag{644}$$

Equations (643)–(644) are exact rational identities. The code evaluates these polynomials, not a floating-point subtraction of the four singular fractions.

Why the subtracted integrals converge. In a Hepp sector, choose the largest edge equal to one and write $x_{\pi(j)} = \prod_{k=j}^{2L-1} t_k$. For a proper edge subset s let $h_P(s)$ be the smallest degree of the polynomial P in the edges of s. For a scale sum σ_i this is one if every edge in its group belongs to s, and zero otherwise. Let $h_N(s)$ be the minimum degree among the nonzero products displayed in (643) or (644). No cancellation of leading terms is used to increase this minimum. The residual exponent is

$$\alpha_s = \frac{|s|}{2} + h_N(s) - h_U(s) - h_{U_1}(s) - h_{U_2}(s) - h_W(s). \tag{645}$$

An exact integer check over all proper edge subsets gives $\alpha_s \geq 1/2$ for each of the 40 six-loop and 12 five-loop kernels: there are $40(2^{12}-2) + 12(2^{10}-2) = 176024$ checks. This controls all nested endpoint limits, not just separate limits of individual edges. It establishes a common analytic neighborhood of $z = 0$ for the angular integral after removing the overall $\Gamma(Lz/2)$ pole.

For completeness, the sector weights are computed by

$$D(\varnothing) = 1, \qquad D(s) = \frac{1}{\alpha_s}\sum_{e\in s} D(s\setminus\{e\}), \qquad \alpha_{\rm all} := 1. \tag{646}$$

A descending edge choice has probability proportional to $D(s\setminus\{e\})$; each continuous coordinate has density $\alpha_s t^{\alpha_s - 1}$. Graphs are sampled proportionally to $|w_G| D(\text{all})$. After dividing numerator and denominator polynomials by their exact leading Hepp monomials, the residual factor has no endpoint pole. The overall residue prefactor is $2/(L\pi^L)$. The numerical estimates below include the graph sign; positivity of the primitive estimator is not asserted for the forest estimator.

23.8 Exact conversion of the auxiliary residues to MS

The conversion needs finite parts, and at this order also the coefficient linear in z of a primitive graph. They cannot be replaced by period residues. Use the unweighted amplitudes

$$I_1 = \frac{2}{z}, \qquad I_p = \frac{r_p}{z} + f_p + a_p z + O(z^2), \qquad p = 3, 4, \tag{647}$$

where

$$r_3 = 2Q_2, \qquad r_4 = P_4(2)/2, \qquad Q_2 = \frac{\Gamma(1/4)^8}{24\pi^4}. \tag{648}$$

For a graph with one logarithmic child γ of loop number h, define

$$D_G(z) = I_G(z) - \frac{L-h}{L} I_\gamma(z) I_{G/\gamma}(z) = \frac{d_G}{z} + f_G + O(z). \tag{649}$$

For the two-loop auxiliary graph this gives

$$D_2 = \frac{A}{z} + f_2 + O(z), \qquad f_2 = -4.836812282568919475092936478902267785676845800388 3\ldots. \tag{650}$$

The value in (650) was checked with the two-${}_3F_2(1/4)$ formula (199) and two Cauchy grids; it is a finite part in the specified massive prescription, not a physical epsilon coefficient.

General nested and disjoint identities. For a nested pair, write $\gamma_1 = G_a$, $\gamma_2 = J$, $J/\gamma_1 = G_b$, $G/\gamma_2 = G_c$, $G/\gamma_1 = T$, with primitive loop numbers h, k, m and $L = h + k + m$. Beta integration of the relative scale variables gives the exact all-z identity

$$D_G = I_G - \frac{k+m}{L} I_a I_T - \frac{m}{L} I_J I_c + \frac{m}{L} I_a I_b I_c. \tag{651}$$

Meanwhile

$$R' I_G = I_G - \frac{r_a}{z} I_T + Z_J I_c, \qquad Z_J = -\,\mathrm{PP}\left(I_J - \frac{r_a}{z} I_b\right). \tag{652}$$

where PP keeps the negative powers of z (the pole part), as in (20). For a disjoint pair with primitive children G_a, G_b and primitive cograph G_c, the analogous identities are

$$D_G = I_G - \frac{L-h}{L} I_a I_{G/G_a} - \frac{L-k}{L} I_b I_{G/G_b} + \frac{m}{L} I_a I_b I_c, \tag{653}$$

$$R' I_G = I_G - \frac{r_a}{z} I_{G/G_a} - \frac{r_b}{z} I_{G/G_b} + \frac{r_a r_b}{z^2} I_c. \tag{654}$$

The factors m/L and $(L-h)/L$ arise from the beta mean of the remaining scale fraction, not from fitted subtraction constants. Substituting (649) into (651)–(654) and extracting z^{-1} proves each graph's conversion. All graph identifications, including the inherited subgraph vertex labels, are checked against the six-loop inventory before the rational weights are collected.

Let $\mathcal{S}_{6,2}$ be the set of the 40 six-loop classes with exactly two proper logarithmic subgraphs, let $Y_{40} = \sum_{G \in \mathcal{S}_{6,2}} w_H(G)[z^{-1}] D_G$, and abbreviate $f_{5,j} = f_{\mathrm{L5G}j}$, $d_{5,j} = d_{\mathrm{L5G}j}$. Define

$$F_5 = \frac{15}{4} f_{5,052} + \frac{15}{2} f_{5,079} + 13 f_{5,100} + 26 f_{5,114} + \frac{13}{4} f_{5,125} + \frac{13}{2} f_{5,129}. \tag{655}$$

The exact aggregate result is

$$\boxed{\begin{aligned} M_{40} &:= \sum_{G \in \mathcal{S}_{6,2}} w_H(G)[z^{-1}] R' I_G \\ &= Y_{40} + F_5 + 9A f_4 - 15 a_4 - 18 f_2 r_4, \\ \Theta_{40} &= -\frac{2}{729} M_{40}. \end{aligned}} \tag{656}$$

Thus the primitive four-loop coefficient a_4, not just its finite part, is essential. The same integrations give the entire one-subdivergence five-loop mass sector:

$$\boxed{c_5^{(1)} = \frac{5}{2} d_{5,052} + 5 d_{5,079} + 3 f_4.} \tag{657}$$

Only the first two of the six fifth-order classes have nonzero mass weights, although all six finite parts enter (655).

A further five-loop mass sector. The mass-carrying two-subgraph classes are exactly L5G053, L5G054, L5G056, L5G077, L5G080, with respective $w_B = (1, 1, 1/2, 1/2, 1)$. All five have nested subdivergences. Put $Y_{5B} = \sum w_B(G)[z^{-1}] D_G$ over this set, and define $f_{4,j}$ using (649) for the indicated fourth-order class. The independent rational collection gives

$$\boxed{c_5^{(2)} = 5\left[Y_{5B} + \frac{2A}{5} f_3 - a_3 - \frac{3}{5} f_2 r_3 - \frac{4}{5} f_{4,013} + \frac{6}{5} f_{4,024}\right].} \tag{658}$$

These classes are disjoint from the one-subgraph and closed Gamma classes. Their contribution to the sixth-order combination is $-6A c_5^{(2)}/2187$, not a sixth-order primitive period.

23.9 Finite-part integrands and numerical results

For any convergent auxiliary kernel D_G, differentiation of the common propagator measure at $z = 0$ gives the logarithmic multiplier

$$\mathcal{J}_L = L \log 2 + \frac{1}{4} \sum_{j=1}^{2L-1} j \log t_j - \frac{L}{2} \log S. \tag{659}$$

Multiplying its residue integrand by (659) gives f_G. For primitive jets through a_p, the code uses the full Taylor coefficients of $z\Gamma(pz/2)/\Gamma(1/2 + z/4)^{2p}$ and the exponential of the logarithmic measure, retaining the exact factorials. No new period identity for the massive finite parts is assumed.

Each of the two main production runs uses three independent streams: 64 000 000 observations for Y_{40}, the six fifth-order auxiliary jets, and the primitive four-loop jets, respectively. Each stream is split into 128 batches. Runs A and B use distinct seeds. Primitive residues are fixed-expectation controls for their finite and linear coefficients; the control slopes are fitted only to excluded one-million-sample pilots. The fourth-order slopes are $(-8.0444695540, 40.8894797679)$ for (f_4, a_4).

Quantity	Run A	Run B
Y_{40}	761.536908 ± 42.870149	790.704780 ± 44.081215
Θ_{40}	220.727969 ± 0.224779	220.412578 ± 0.227748
$c_5^{(1)}$	1651.795516 ± 1.874636	1649.773937 ± 1.603267
$\Theta_{40} - 6Ac_5^{(1)}/2187$	208.163498 ± 0.218856	207.863484 ± 0.223344

The displayed errors are sampling standard errors, not certified uncertainties. Like every partial sum of this section, Θ_{40} is neither an estimate nor a bound for $C_6(2)$; graph-sector contributions are defined only in the stated MS prescription. The main runs agree within one combined standard error for the MS six-loop result. Equal weighting, without fitting to a target value, gives

$$\begin{aligned} Y_{40} &= 776.1208440 \pm 30.7449312, & F_5 &= 6208.4320415 \pm 48.4413942, \\ f_4 &= -957.9657216 \pm 0.0502308, & a_4 &= 4958.8036029 \pm 0.6170590, \\ M_{40} &= -80397.8646408 \pm 58.3185778, & c_5^{(1)} &= 1650.7847264 \pm 1.2333617. \end{aligned} \tag{660}$$

The auxiliary residue Y_{40} and the MS residue M_{40} differ substantially; interchanging them would give an incorrect coefficient. Keeping the within-stream covariance of finite parts and residues yields

$$\boxed{\Theta_{40} - \frac{6A}{2187} c_5^{(1)} = 208.01349115\ldots, \qquad \widehat{\sigma} = 0.15634950\ldots.} \tag{661}$$

Without the primitive controls, the same raw observations give $\Theta_{40} = 220.5459148 \pm 0.1606925$ and the joint combination $207.9871304 \pm 0.1571710$. Controls were not fitted to an assumed value of C_6 or to equality of the two runs.

The additional five-loop calculation uses two independently seeded runs, each with 32 000 000 observations in each of its three streams (Y_{5B}, the two fourth-order auxiliary finite parts, and primitive three-loop jets), again with 128 batches. It gives

$$\begin{aligned} c_5^{(2)} &= 2431.5047824 \pm 1.8267989, \\ -\frac{6A}{2187} c_5^{(2)} &= -18.49537102 \pm 0.01389564. \end{aligned} \tag{662}$$

The two independent estimates of $c_5^{(2)}$ differ by 0.053 combined standard errors. The analysis retains the covariance of f_3 and a_3, as well as that of every residue/finite-part pair from the same stream.

Checks of the two-subgraph kernels. These checks were made by OpenAI GPT-6 astra with the kernels above (research-archive package `lri_C6_d2_two_forest_update/`; the result record is `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/verification/C6_d2_continuation_result.json`). The rational numerator identities (643) and (644) were checked symbolically. An independent scalar implementation compares the normalized floating-point numerator with the original four fractions at 500 digits; over 312 configurations, including a coordinate as small as 10^{-140}, the largest normalized error is 2.49×10^{-16}. This is a pointwise test of the kernel, not an integration error bound.

The third-order nested and disjoint instances of the kernel have the unweighted auxiliary residues

$$[z^{-1}]D_{\mathrm{L3G004}} = A^2, \qquad [z^{-1}]D_{\mathrm{L3G007}} = \frac{A^2 + 4f_2}{3},$$

which follow from the Gamma counterterms of Section 24.1 and the exact two-loop finite part (650) through the conversion of Section 23.8. A sixteen-million-observation control, with the graph weights inserted, deviates by -0.185 and -0.180 standard errors; a separate four-million-observation run by -0.192 and 1.542. For the finite-part extension a sixteen-million-observation two-loop run gives $[z^{-1}]D_2 = 2.77218438 \pm 0.00048568$ and $[z^0]D_2 = -4.83492035 \pm 0.00210898$, at -0.833 and 0.897 standard errors from A and (650). The reference $r_4 = P_4(2)/2$ was evaluated afresh from (333) at 65 digits, and two Cauchy evaluations of f_2 differ by 2.0×10^{-64}; this is consistency of the numerical special-function evaluation, not an interval certificate.

24 Analytic power-sector compensation at $d = 2, 3$

24.1 An exact Gamma-reducible sector

The momentum-space normalization at $d = 2$ has $r(2, z) = 1$. With one nonzero momentum p, a convolution of n parallel propagators with indices α_i reduces to a single propagator of index $\eta = \sum_i \alpha_i - (n - 1)$ and multiplier

$$\mathcal{B}_n(\boldsymbol{\alpha}) = \frac{\Gamma(\eta)}{\Gamma(1-\eta)} \prod_{i=1}^{n} \frac{\Gamma(1-\alpha_i)}{\Gamma(\alpha_i)}, \qquad I_{\|}(p) = \mathcal{B}_n(\boldsymbol{\alpha})(p^2)^{-\eta}. \tag{663}$$

The integration convention in this formula is d^2k/π per loop. Successive use of the two-propagator convolution proves the displayed n-propagator version by telescoping Gamma factors. A degree-two vertex with zero momentum injection adds the two adjacent indices. Initially every line has index $1/2 + z/4$; the final line of an accepted L-loop reduction has index $Lz/2$. Thus the bare amplitude is a finite product of Gamma functions at affine half-integer arguments.

This rearrangement is deliberately restricted. All original degree-two vertices must be among the two momentum terminals. Both terminals are inherited under every contraction. An internal index reaching the infrared boundary $\alpha = 1$ at $z = 0$, an identically singular Gamma factor, a tadpole quotient, a scaleless quotient, or an unsolved cograph causes that reduction to be rejected. Power-divergent mass subgraphs are analytically continued in the existing critical-mass prescription; they are not assigned an extra logarithmic vertex counterterm. These restrictions distinguish the accepted rearrangements from an unrestricted infrared rearrangement that would require additional R^* counterterms [21, 2].

As in (652), PP projects a Laurent series onto its negative powers of z. The UV counterterm is computed recursively as

$$Z_G = -\,\mathrm{PP}\left[I_G + \sum_{S \neq \varnothing \text{ proper spinney}} \left(\prod_{\gamma \in S} Z_\gamma \right) I_{G/S} \right]. \tag{664}$$

Spinney components here are vertex-disjoint. Nested subdivergences are already included recursively in Z_γ. Confusing this convention with the edge-compatible scale-derivative forests of the finite kernels of Section 23 changes individual graph counterterms. Formula (664) is the usual MS R-operation; the special graph selection, Gamma reductions and exact arithmetic are the present calculation [21].

If a graph is a one-vertex join of two or more logarithmic blocks, its amplitude factorizes. Induction in (664), or the multiplicativity of the counterterm character for this join, gives

$$Z_{G_1\vee\cdots\vee G_k} = \prod_{j=1}^{k} Z_{G_j}, \qquad k \geq 2. \tag{665}$$

Every factor on the right has strictly negative Laurent degree, so the product has no simple pole. This establishes the simple-pole zero even when the individual finite amplitudes are not evaluated. Only joins whose blocks are all logarithmic are used in this additional zero test.

Exact arithmetic and closed counts. Gamma recurrence, reflection and half-integer polygamma values reduce the Laurent arithmetic to an exact rational polynomial ring in $\ell = \log 2$ and zeta values [26, 32]. Euler's constant cancels explicitly. No integer-relation fit is used. Requiring both the original graph and every necessary cograph/counterterm to close gives

Loop order	1	2	3	4	5	6
Closed Gamma-sector classes	1	2	6	15	47	137

The bare two-terminal graph being reducible by itself is not sufficient: unsupported cograph finite parts are never filled with zero.

Define b_L^Γ, c_L^Γ by precisely this closed class set, with all original rational graph weights. Two useful exact results are

$$c_5^\Gamma = \frac{12692}{3}\ell^4 - 208\ell^2 + \frac{1085}{4}\ell\zeta(3) - \frac{224}{3}\ell - \frac{73}{6}, \tag{666}$$

$$\boxed{\begin{aligned} 2187\,\Xi_\Gamma &:= (3c_6^\Gamma - b_6^\Gamma) - 24\ell c_5^\Gamma \\ &= \frac{5518904}{75}\ell^5 - \frac{216256}{45}\ell^3 + \frac{5763}{20}\ell^2\zeta(3) - 2784\ell^2 + \frac{6368}{15}\ell - 69\zeta(3) + \frac{83397}{640}\zeta(5). \end{aligned}} \tag{667}$$

Numerically,

$$\frac{3c_6^\Gamma - b_6^\Gamma}{2187} = 12.175885035555361667474529\ldots, \qquad \Xi_\Gamma = 4.27467950247038413834713 0\ldots. \tag{668}$$

Neither expression includes the missing graph classes. The full exact counterterm of every accepted graph, not just the weighted totals, is recorded in `sp_exact.json` of `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/analytic/verification/`.

For example, the closed third-order sector gives $b_3^\Gamma = 336\ell^2 - 4$ and $c_3^\Gamma = 52\ell^2 - 4/3$. Restoring the retained wineglass residue $A^2/3$ with its original weights, and the primitive tetrahedron for the beta function, recovers

$$c_3 = \frac{7}{2}A^2 - \frac{4}{3}, \qquad b_3 = \frac{87}{4}A^2 - 4 + 6Q_2. \tag{669}$$

This check takes the wineglass residue and the tetrahedron as inputs; it is not a new independent evaluation of either. The two-loop checks $b_2 = -6A, c_2 = -A$ are entirely within the new Gamma calculation.

Checks of the Gamma sector. These checks were made by OpenAI GPT-6 astra (code `sp_exact.py`, `sp_ms.py` and `check_gamma_coverage.py` in, and records `sp_exact.json` and `sp_ms_90.json` under, `lri_C6_d2_complete_update/reproduction/lri_C6_d2_two_forest_update/analytic/`). The 793 pole coefficients of the exact table were compared with 90-digit Gamma and polygamma arithmetic; the largest absolute difference is below 4.5×10^{-78}. There are 214 accepted terminal-routing instances through six loops. Wherever alternative routings are admitted their complete pole parts agree, and changing $\log p^2$ of the terminal momentum from zero to one leaves the pole parts unchanged; exact arithmetic checks both cancellations before numerical specialization, and the 90-digit residuals are below 3.8×10^{-85} and 2.0×10^{-85}.

24.2 An analytic advance special to two dimensions

Write d^2x/π for the normalized complex-plane measure. Riesz convolution gives

$$\mathcal{B}(u,v) = \frac{\Gamma(1-u/2)\Gamma(1-v/2)\Gamma((u+v)/2-1)}{\Gamma(u/2)\Gamma(v/2)\Gamma(2-(u+v)/2)}. \tag{670}$$

It replaces two distance powers u, v joined at an integrated vertex by the power $u+v-2$. This formula is used meromorphically, with all normalizing factors retained.

For the five-edge two-variable integral define

$$\mathcal{J}(\boldsymbol{p}) = \frac{1}{\pi^2}\int_{\mathbb{C}^2} \frac{d^2x\, d^2y}{|x|^{p_1}|1-x|^{p_2}|y|^{p_3}|1-y|^{p_4}|x-y|^{p_5}}. \tag{671}$$

Put $a = -p_1/2, b = -p_2/2, A' = -p_3/2, B' = -p_4/2, c = -p_5/2$, and

$$\begin{aligned}\mathcal{O}(a,b,A',B',c) &= B(a+1,c+1)B(a+A'+c+2,B'+1)\\ &\quad\times {}_3F_2\left(\begin{matrix}-b, a+1, a+A'+c+2\\ a+c+2, a+A'+B'+c+3\end{matrix};1\right),\end{aligned} \tag{672}$$

$$\begin{aligned}\mathcal{M}(a,b,A',B',c) &= B(a+1,-a-b-c-1)B(B'+1,-A'-B'-c-1)\\ &\quad\times {}_3F_2\left(\begin{matrix}-c, a+1, B'+1\\ -b-c, -A'-c\end{matrix};1\right).\end{aligned} \tag{673}$$

Here A', B' are temporary exponent names, not the RG constant $A = 4\log 2$. A split-contour identity gives

$$\boxed{\begin{aligned}\mathcal{J}(\boldsymbol{p}) = \frac{1}{\pi^2}\big[&\sin(\pi a)\sin(\pi B')\,\mathcal{O}(a,b,A',B',c)\mathcal{M}(a,b,A',B',c)\\ &+\sin(\pi A')\sin(\pi b)\,\mathcal{O}(A',B',a,b,c)\mathcal{M}(A',B',a,b,c)\big].\end{aligned}} \tag{674}$$

Thus the needed nonsymmetric kite is evaluated by four ordinary ${}_3F_2(1)$ values, not by a noisy four-real-dimensional integration. This is a bilinear identity. For complex regulator values the products in (674) are not replaced by products involving complex conjugation.

For clarity, the contour argument starts in a common convergence region. Split the two antiholomorphic contours so that one lies left of zero and the other right of one. The two possible orderings of the remaining holomorphic coordinates give $\mathcal{O}$ and its exchanged version. The discontinuities of the two outer contours give the sine factors in (674). In the mixed domain the changes $x = -u/(1-u)$, $y = 1/(1-v)$ give $\mathcal{M}$ by two Euler beta integrations. The one-variable normalization is fixed by $-\sin(\pi b)B(a+1,b+1)B(-a-b-1,b+1)$; applying the construction twice fixes the positive sign and the π^{-2} in (674). Analytic continuation then proves the meromorphic identity away from its polar divisors. This is the two-integration specialization of the usual KLT contour factorization

[43]; the relation between two-dimensional massless Feynman integrals and bilinear hypergeometric expressions is also established in [45].

The symmetric cases reduce independently to the complex Selberg Gamma product [6]. In particular, setting all five powers to one recovers $\mathcal{J} = 3Q_2$. Three nonsingular symmetric tests, five nontrivial completion transformations at a nonsymmetric point, and a changed-tail hypergeometric calculation verify the normalization separately from the graph recursion. These are consistency checks of the analytic formula, not a fit for its coefficients.

Counterterm closure and infrared conditions. A graph is accepted by the coordinate reduction only when its raw graph, all required subgraphs, and every inherited-root quotient have a declared reduction. A root pair is rejected if it is contained in a proper induced collision set with $e \geq 2(v-1)$. This prevents a spurious infrared singularity being interpreted as a UV counterterm. An unanchored one-attachment massless block is assigned its analytic scaleless value zero only when its superficial degree at $d = 2$ is the nonzero odd number 1 or -1. No logarithmic scaleless factor is assigned zero by this rule. Its logarithmic subtractions are still performed before the graph counterterm is extracted.

Cauchy extraction of the regular function $z^2\mathcal{J}(z)$ uses two settings: 32 nodes at radius 0.02 with 45-digit function evaluation, and 48 nodes at radius 0.016 with 55-digit evaluation. The two runs compare 2699 pole coefficients; their maximum absolute discrepancy is 7.38×10^{-30}. The maximum discrepancy between distinct admissible roots in the higher-precision run is 7.97×10^{-45}. The Gamma-only counterterms of Section 24.1 provide additional independent checks. No zero is inserted because a nonzero result is numerically inconvenient.

The closed graph/subgraph/quotient reductions and the two contractions below cover the nonzero power-sector target. The other logarithmic graph contributions are evaluated by the finite-scheme calculation, without requiring separate numerical values for every individual MS finite part.

24.3 The complete power sector is explicitly accounted for

The only two nonzero-target power graphs not closed by the preceding procedure are L6G106 and L6G117. Both are a sunset insertion on the four-loop primitive L4G027. Put $p = 1 - z/2$. Integrating the chain with powers $p, 3p, p$ gives

$$\mathcal{S}(p) = \frac{\Gamma(1-p/2)^2\Gamma(1-3p/2)\Gamma(5p/2-2)}{\Gamma(p/2)^2\Gamma(3p/2)\Gamma(3-5p/2)}, \qquad \mathcal{S}(1) = -4. \tag{675}$$

Its effective edge power is $5p - 4$. The remaining graph is primitive, so the deformation of this edge does not affect the required simple pole. Each graph has $w_A = 2, w_B = 0$, giving $16P_4(2)$ to b_6. Together they contribute $32P_4(2)$ to b_6 and zero to c_6. This contraction was checked by exact graph isomorphism and not inferred from an expected numerical coefficient.

Let a superscript p denote the sum of graphs containing at least one proper power-divergent subgraph. Define the lower-order polynomial of the constant-coefficient MS assembly, that is, the terms of (594) that contain only A, b_3, c_3, b_4 and c_4, with A at the present dimension (Section 23.1 shows that (594) holds at every dimension with this A):

$$\begin{aligned}\mathcal{R}(b_3, c_3, b_4, c_4) = {} & 2016A^5 - 448A^3b_3 + 672A^3c_3 - 84A^2b_4 + 168A^2c_4 \\ & + \frac{49}{3}Ab_3^2 - 42Ab_3c_3 + \frac{7}{3}b_3b_4 - 4b_3c_4 - 3b_4c_3.\end{aligned} \tag{676}$$

Subtracting the constant-coefficient MS assemblies for the full and power-free graph sets gives the

exact bookkeeping identity

$$\boxed{\begin{aligned} 2187\,\Delta_p &= (3c_6 - b_6)^p - 12Ab_5^p + 30Ac_5^p \\ &\quad + \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p), \\ C_6^{\text{full}}(2) &= C_6^{\text{power-free}}(2) + \Delta_p. \end{aligned}} \tag{677}$$

Only the complete combination $(3c_6 - b_6)^p$, rather than its two individual summands, is needed. Any remaining individual sixth-order power graph with $3w_B = w_A$ has zero weight in this combination and is excluded by an exact rational identity.

The actual MS sums include

$$\begin{aligned} b_3^p &= -4, & c_3^p &= -\frac{4}{3}, \\ b_4^p &= 16A + 24, & c_4^p &= \frac{8}{3}A + 8, \\ b_5^p &= -1540.38932947914290838\ldots, & c_5^p &= -251.621917341000466476\ldots. \end{aligned} \tag{678}$$

The evaluated higher-order part of (677), divided by 2187, is $1.535501191157190785360393154334\ldots$; the lower-order difference is its negative at the computed precision. The two Cauchy settings give $\Delta_p = -8.02 \times 10^{-34}$ and $\Delta_p = 6.91 \times 10^{-47}$, respectively. These residuals are numerical consistency checks, not a proof of an all-orders power-subgraph cancellation theorem. The calculation retains the explicitly evaluated correction (677); it does not omit the power graphs by assumption. Their residual numerical uncertainty is negligible on the scale of the reported $C_6(2)$ uncertainty.

24.4 Power subgraphs: a separate three-dimensional audit

It would not be justified to infer cancellation of power graphs from the two-dimensional result. The correction is instead evaluated in the MS prescription before being added to the finite-scheme power-free result. The coordinate convolution with distance exponents a, b is

$$B_d(a,b) = \Gamma(d/2)\frac{\Gamma((d-a)/2)\Gamma((d-b)/2)\Gamma((a+b-d)/2)}{\Gamma(a/2)\Gamma(b/2)\Gamma((2d-a-b)/2)}. \tag{679}$$

The dimension-three implementation uses affine edge powers $m(3-z)/2$, not the two-dimensional powers with a numerical prefactor changed. It applies the Taylor reductions of the kite master (231) at its critical patterns A, B and T only to their declared orders. Gamma factors, logarithmic counterterms, contractions and safe nonzero-distance roots are retained recursively. A requested coefficient beyond the available master Taylor order is rejected, not assigned zero. Nonlogarithmic one-attachment massless factors are removed only in the specified analytic critical-mass prescription; this test is not used to discard a logarithmic scaleless UV counterterm.

There are 1,5,31 relevant power-containing classes at loops 3,4,5. At six loops all 121 power classes with $3w_B - w_A \neq 0$ are included; the other 52 power classes have exactly zero weight in the sixth-order target. A zero sixth-order target weight does not license dropping a lower-order input. In particular the five-loop graph described next remains necessary.

Two sunset contractions. The graphs L6G106 and L6G117 are sunset insertions on the four-loop primitive. The chain of three effective edges with critical powers $p, 3p, p$, $p = d/2$, has factor

$$M_d = \frac{\Gamma(-q)\Gamma(2q)^2}{\Gamma(3q)}, \qquad M_3 = -\frac{16\pi}{15}. \tag{680}$$

Each graph contributes $-4M_3P_4(3)$ to b_6 and zero to c_6. Only the primitive residue is needed after the contraction; no unknown edge-deformation finite term is inserted. The inherited graph identification and its weights are checked in the audit.

24.5 An exact cancellation avoids two unknown core integrals

The nontrivial extra obstruction in dimension three is a four-loop two-terminal core. Using the terminal labels $2, 3$, its seven internal edges are

$$(01), (02), (03), (03), (12), (12), (13). \tag{681}$$

Every edge has distance power $p = (3 - z)/2$. Equivalently its two-integrated- vertex kite powers are $(p, 2p, 2p, p, p)$. The required expansion has the form

$$J_{\text{core}}(z) = \frac{8}{z^2} + \frac{2A}{z} + u_0 + u_1 z + O(z^2), \qquad A = 2 + 4\log 2 - \pi. \tag{682}$$

The pole coefficients follow from its proper logarithmic subgraphs. The finite coefficients u_0, u_1 are not computed. Their cancellation from the physical target is proved below, rather than assumed.

Let $C(z)$ be the product of the two external chain convolutions and $F(z) = C(z)J_{\text{core}}(z)$. For reference,

$$C(0) = M_3, \qquad C'(0) = \frac{7}{4}M_3(A+N), \qquad N = \psi_0(-q)+\psi_0(3q)-\psi_0(1)-\psi_0(2q) = \frac{62}{15}-\pi-4\log 2. \tag{683}$$

Denote by F_0, F_1 the unknown-dependent regular coefficients. Known pole-dependent terms are retained elsewhere in the audit. Write $\doteq$ for equality only of the dependence on F_0, F_1. If $r_G = [z^{-1}]R'I_G$, then the three affected graphs satisfy

$$\begin{aligned} r_{\text{L5G086}} &\doteq \frac{2}{5}F_0, \\ r_{\text{L6G332}} &\doteq \frac{A}{3}F_0 + \frac{2}{15}F_1, \\ r_{\text{L6G333}} &\doteq \frac{A}{3}F_0 - \frac{2}{15}F_1. \end{aligned} \tag{684}$$

These expressions follow by applying $B_d(d - kz, d/2 - lz) = 2/(kz) + A + O(z)$ to the attached convolutions. For L6G333 the external bubble counterterm contributes $-4F_1/5$; omitting it would spoil the identity.

The five-loop weights are $(w_A, w_B) = (3/4, 1/4)$ and both six-loop weights are $(3/2, 1/4)$. Since $b_L = (-1)^{L+1}Lw_A r_G$ and $c_L = (-1)^{L+1}Lw_B r_G$, the unknown part of the physical sixth-order MS numerator is precisely

$$\boxed{\frac{9}{2}(r_{\text{L6G332}} + r_{\text{L6G333}}) - \frac{15}{2}A\, r_{\text{L5G086}} \doteq 0.} \tag{685}$$

This is an exact rational-polynomial identity in A, F_0, F_1. It eliminates only the two unknown core coefficients, not the other parts of these graphs. Substituting $(u_0, u_1) = (0, 0)$ and $(101, -37)$ in the full graph recursion changes the three individual residues exactly as predicted by (684), to better than 4×10^{-73}. Thus those trial assignments are algebraic checks, not estimates of u_0, u_1. The saved auxiliary individual b_5, b_6 sums obtained with a trial assignment must not be reported as separately evaluated physical MS coefficients.

24.6 Assembled power correction and completeness

The constant-MS lower-order polynomial is the same function $\mathcal{R}$ as in (676); only A changes, to its $d = 3$ value $2 + 4\log 2 - \pi$ of (682). It is repeated here for reference:

$$\begin{aligned} \mathcal{R}(b_3, c_3, b_4, c_4) = {} & 2016A^5 - 448A^3b_3 + 672A^3c_3 - 84A^2b_4 + 168A^2c_4 \\ & + \frac{49}{3}Ab_3^2 - 42Ab_3c_3 + \frac{7}{3}b_3b_4 - 4b_3c_4 - 3b_4c_3. \end{aligned} \tag{686}$$

A superscript p denotes the complete power-containing graph sum. Subtracting the full and power-free MS assemblies gives

$$\boxed{\begin{aligned} 2187\Delta_p &= (3c_6 - b_6)^p - 12Ab_5^p + 30Ac_5^p \\ &\quad + \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p), \\ C_6^{\text{full}}(3) &= C_6^{\text{pf}}(3) + \Delta_p. \end{aligned}} \tag{687}$$

For the three- and four-loop power pieces the graph and counterterm recursion of this section agrees with

$$\begin{aligned} b_3^p &= M, & c_3^p &= M/3, \\ b_4^p &= -\tfrac{11}{2}AM - \tfrac{3}{2}MN, & c_4^p &= -\tfrac{7}{6}AM - \tfrac{1}{2}MN, \end{aligned} \tag{688}$$

where $M = M_3$. Setting $R = \psi_1(2q) - \psi_1(1)$, $U = \psi_2(q)$, $V = \psi_2(1) + \psi_2(2q)$, $Q = Q_3$, and $P = P_4(3)$, exact assembly algebra gives the lower-order part of (687), $\delta\mathcal{R} = \mathcal{R}(b_3, c_3, b_4, c_4) - \mathcal{R}(b_3 - b_3^p, c_3 - c_3^p, b_4 - b_4^p, c_4 - c_4^p)$ with the complete lower-order coefficients (179), (226) and (227), in closed form; it is $2187\,\Delta_{p,\text{lower}}$ below:

$$\begin{aligned} \delta\mathcal{R} = \frac{M}{24}\big(&2727A^3 + 603A^2N - 8AM + 72AQ + 63AR \\ &- 216NQ + 27NR - 192P - 318U + 36V\big). \end{aligned} \tag{689}$$

This polynomial alone does not prove the higher graph sum. The latter was separately computed by the graph and counterterm recursion.

At 90 working digits the two parts of (687), including its factor 1/2187, are

$$\begin{aligned} \Delta_{p,\text{higher}} &= +0.073651697044164869469222945561 1\ldots, \\ \Delta_{p,\text{lower}} &= -0.073651697044164869469222945561 1\ldots. \end{aligned} \tag{690}$$

Their sum is -1.02×10^{-89}. A separate 75-digit run with different core assignments gives -1.94×10^{-75}. A third algebraic stress test replaces the values of the masters Q_3, $P_4(3)$, $\mathcal{K}_3$ and Z_3 by the test numbers 7, 11, 5 and 13, and the core assignments by $(-17, 29)$: the combined residual is 1.52×10^{-65} at 65 working digits. This last test is not a physical evaluation of those masters; it checks that common master-dependent terms cancel consistently in the target.

These are high-precision numerical checks of the *complete* power correction, together with an exact proof of the unknown-core cancellation. They are not an all-orders theorem allowing individual power graphs to be dropped. The tiny computed residual is negligible relative to the sampling error, but it is not an interval-certified error bound. Every nonzero-weight sixth-order power graph and every required fifth-order power input has been covered. Two unrelated non-power four-loop graphs rejected by the coordinate reduction are instead included in the finite-kernel integration and are not a missing power contribution.

25 Finite-scheme sixth-order assembly

25.1 A finite zero-momentum scheme avoids separate MS finite parts

In this subsection only, $\widehat{b}_L(z), \widehat{c}_L(z)$ denote the *epsilon-dependent finite-scheme* RG coefficients of the power-free graph set. They are not the constant MS coefficients. Since the coupling and the mass are defined by bare amplitudes at zero external momentum, this scheme is called finite-MOM (finite momentum subtraction) below. Let

$$\mathcal{A}(h, z) = 1 + \sum_{L=1}^{6} (-1)^L \mathcal{A}_L(z) h^L, \qquad \mathcal{B}(h, z) = 1 + \sum_{L=1}^{6} (-1)^L \mathcal{B}_L(z) h^L, \tag{691}$$

where the graph weights (597) and the bubble normalization are those of Sections 23 and 24. Defining $g = h\mathcal{A}(h,z)$, differentiation at fixed bare parameters gives

$$\beta(g,z) = -z\,h\,\partial_h(h\mathcal{A}(h,z)), \qquad \gamma_r(g,z) = -z\,h\,\partial_h \log \mathcal{B}(h,z). \tag{692}$$

The right sides are reexpanded in g. They have the form

$$\beta = -zg + \sum_{L=1}^{6} \widehat{b}_L(z) g^{L+1}, \qquad \gamma_r = \sum_{L=1}^{6} \widehat{c}_L(z) g^L, \qquad \widehat{b}_1 = 3, \quad \widehat{c}_1 = 1. \tag{693}$$

The fixed-point exponent is unchanged by this finite coupling and operator redefinition: the coupling is reparametrized, and the extra operator-anomalous-dimension term is proportional to β, hence vanishes at the fixed point. This assertion is used only for the same power-free amplitudes in the two prescriptions. The correction for the omitted power-containing amplitudes is separately supplied by (677).

For a compatible forest F of proper logarithmic collision sets, let its blocks be the contracted subgraphs and the remaining root block. Write h_R for the loop number of that root block and I_B for a bare block amplitude. The finite scale-derivative combination is

$$D_G(z) = \sum_F (-1)^{|F|} \frac{h_R}{L} \prod_{B \in F \cup \{G\}} I_B(z). \tag{694}$$

The notation on the product means the forest blocks after their children have been contracted, not a repeated product of unreduced parent graphs. Terms with $h_R = 0$ vanish. Complete rational polynomial comparison with independent inversion of (691) proves

$$\widehat{b}_L(z) = (-1)^{L+1} L z \sum_G w_A(G) D_G(z), \qquad \widehat{c}_L(z) = (-1)^{L+1} L z \sum_G w_B(G) D_G(z). \tag{695}$$

This comparison was rerun at all six loop orders. The sixth-order polynomials contain 532 beta and 383 mass monomials; both differences vanish over the rationals. It is not a numerical agreement at selected values of the bare integrals. The use of common parametric forest kernels is consistent with the general parametric renormalization framework [1].

The independent inversion. The comparison uses the Lagrange–Bürmann inversion of $g = h\mathcal{A}(h,z)$ in (692). With primes denoting ∂_h at fixed z and $\mathcal{A}, \mathcal{B}$ the series (691),

$$\begin{aligned} \widehat{b}_L(z) &= -\frac{z}{L+1}\,[h^L](2h\mathcal{A}' + h^2\mathcal{A}'')\mathcal{A}^{-L-1}, \\ \widehat{c}_L(z) &= -\frac{z}{L}\,[h^{L-1}]\,\frac{d}{dh}\left(\frac{h\mathcal{B}'}{\mathcal{B}}\right)\mathcal{A}^{-L}. \end{aligned} \tag{696}$$

Indeed $[g^n]F(h(g)) = \frac{1}{n}[h^{n-1}]F'(h)\mathcal{A}^{-n}$; for $F = -zh(h\mathcal{A})'$ one has $F' = -z(\mathcal{A} + 3h\mathcal{A}' + h^2\mathcal{A}'')$, and $[h^L](\mathcal{A} + h\mathcal{A}')\mathcal{A}^{-L-1} = 0$ for $L \geq 1$. The same comparison of every monomial was made against the forest polynomials (695). For the six-loop inputs the 487 power-free graphs, with 66,198 forests, combine into 395 nonzero rational kernels; 92 graph kernels vanish identically, and the largest retained kernel has 52 distinct denominators. These scale-derivative coefficients carry explicit regulator dependence; they are not the constant MS coefficients and are never inserted into an MS fixed-point assembly [2]. The inversion formulas (696), the comparison and the counts are those of the earlier two-dimensional calculation by OpenAI GPT-6 astra (package `LRI_C6_D2_HPC/reference/`); the derivation of (696) from $[g^n]F(h(g))$ was written out by a Claude Code session on 16 October 2026, when this paragraph was merged into the note.

25.2 A common, locally subtracted integration kernel

For a forest, let U_F be the product of the Symanzik polynomials of its blocks, embedded in the original edge variables, and let $E_R(F)$ be the root-block edge set. With $S = \sum_e x_e$, define

$$\boxed{K_G(x) = \sum_F (-1)^{|F|} \frac{\sum_{e\in E_R(F)} x_e}{S\, U_F(x)}.} \tag{697}$$

It is homogeneous of degree $-L$. All terms are combined before numerical evaluation. At $d = 2$, the normalization $r(2, z) = \Gamma(1 + z/2)/\Gamma(1 + z/2)$ is identically one, and

$$D_G(z) = \frac{\Gamma(Lz/2)}{\Gamma(1/2 + z/4)^{2L}} \int d\Omega \prod_e x_e^{-1/2+z/4} K_G(x). \tag{698}$$

To verify the factor in (694), integrate first over the relative scales of the blocks. Their Dirichlet parameters are $h_B z/2$, so insertion of the root scale fraction has mean $(h_R z/2)/(Lz/2) = h_R/L$. This proves the exact common-integral identity for positive regulator in its initial convergence region, and then by continuation. No fitted subtraction constant is involved.

The forest cancellations remove the proper logarithmic endpoint singularities. The remaining overall pole is isolated by $\Gamma(Lz/2)$. Consequently the integral in (698) and its required logarithmic moments are finite at $z = 0$. The individual terms of (697) need not be separately integrable there and are never sampled as unrelated singular integrals.

There are 1, 5, 16, 76 and 395 nonzero finite kernels at loop orders 2 through 6. For all 493 kernels, an independent exact-rational check compares the co-tree-mask implementation with a product of Kirchhoff cofactor determinants constructed directly from the forest blocks. Every difference is zero. This validates the common-integral mapping separately from the RG polynomial check.

25.3 Retaining the regulator dependence through sixth order

Write $\widehat{b}_L(z) = \sum_j b_{L,j} z^j$ and $\widehat{c}_L(z) = \sum_j c_{L,j} z^j$. Terms through $j = 6 - L$ are required. In a projective sector with one largest edge fixed at one, put

$$s_L(x) = \frac{1}{4} \sum_e \log x_e - \frac{L}{2} \log S, \qquad \mathcal{G}_L(z) = \frac{\pi^L \Gamma(1 + Lz/2)}{\Gamma(1/2 + z/4)^{2L}}. \tag{699}$$

The normalized Gamma factor has $\mathcal{G}_L(0) = 1$ and

$$\log \mathcal{G}_L(z) = \sum_{j\geq 1} a_{L,j} z^j, \quad a_{L,1} = L \log 2, \quad a_{L,j} = \frac{(-1)^j \zeta(j)}{j} \left[\left(\frac{L}{2}\right)^j - \frac{2L(2^j - 1)}{4^j} \right] \quad (j \geq 2). \tag{700}$$

Only $j \leq 4$ is used. If $t = s_L + a_{L,1}$, the required polynomials are

$$E_0 = 1, \qquad E_1 = t, \quad E_2 = \frac{t^2}{2} + a_{L,2},$$
$$E_3 = \frac{t^3}{6} + a_{L,2} t + a_{L,3}, \qquad E_4 = \frac{t^4}{24} + \frac{a_{L,2} t^2}{2} + a_{L,3} t + a_{L,4} + \frac{a_{L,2}^2}{2}. \tag{701}$$

Integrating $E_j K_G$ with the critical projective measure and multiplying by $2(-1)^{L+1} w_{A,B}/\pi^L$ gives $b_{L,j}, c_{L,j}$. This computes Taylor moments of a finite integral, not a numerical subtraction of Laurent poles.

The two-loop jets are evaluated by the convergent two-${}_3F_2(1/4)$ formula (199) for $S_d(z)$ at $d = 2$, rather than by Monte Carlo. Explicitly,

$$\widehat{b}_2(z) = -6z \left[S_2(z) - \frac{2}{z^2} \right], \qquad \widehat{c}_2(z) = \frac{1}{6} \widehat{b}_2(z). \tag{702}$$

Fresh 65-digit evaluations with radii 0.05 and 0.08 and respectively 64 and 80 Cauchy nodes agree with the saved input through z^4 to better than 6.8×10^{-61}. The five beta coefficients begin

$$\begin{aligned} b_{2,0} &= -16.63553233344, \quad & b_{2,1} &= 29.02087369541, \\ b_{2,2} &= -47.11795797162, \quad & b_{2,3} &= 74.14063002868, \\ b_{2,4} &= -114.64528966290. \end{aligned} \tag{703}$$

Let $\widehat{B}(g,z) = \sum_{L\geq 1} \widehat{b}_L(z) g^{L-1}$. The physical fixed point satisfies $\epsilon = g_* \widehat{B}(g_*, \epsilon)$. Lagrange inversion with z held fixed, followed by $z = \epsilon$, gives

$$\boxed{C_n^{\text{power-free}}(2) = -\frac{1}{2}\delta_{n1} + \sum_{k=1}^{n} \frac{1}{k} [g^{k-1} z^{n-k}] \partial_g \gamma_r(g,z) \widehat{B}(g,z)^{-k}, \qquad n \geq 1.} \tag{704}$$

The constant term is $C_0 = 1$. Only finite polynomials enter this extraction. Independent exact-rational fixed-point substitution and bivariate Lagrange extraction agree in 192 checks, including 24 checks of the complete constant-coefficient MS sixth-order assembly.

The regulator jets cannot be discarded. Substituting only $\widehat{b}_L(0), \widehat{c}_L(0)$ into the constant-coefficient MS assembly (the analogue of (594)) would use a different truncated RG problem. All required jets in (704) are retained, including the two-loop z^4, three-loop z^3, four-loop z^2, and fifth-loop z^1 terms. At sixth order only the combination $3c_{6,0} - b_{6,0}$ is required, as in the MS assembly.

25.4 Importance sampling and retained sector controls

The integration uses Hepp sectors and subset dynamic programming, in the spirit of tropical Feynman quadrature [41]. For each proper nonempty edge subset S, choose a positive proposal exponent $\alpha(S)$. Numerical leading-power diagnostics, performed on the *combined* kernel, guide this choice; they are not presented as rigorous valuations. The normalization is computed without enumerating all $(2L)!$ edge orders:

$$Z(\varnothing) = 1, \qquad Z(S) = \frac{1}{\alpha(S)} \sum_{e \in S} Z(S \setminus \{e\}), \qquad Z(E) = \sum_{e \in E} Z(E \setminus \{e\}). \tag{705}$$

An edge is removed with probability proportional to $Z(S \setminus \{e\})$; the next sector ratio is sampled with density $\alpha(S) t^{\alpha(S)-1}$. The exact reciprocal proposal weight is applied to every measurement. Changing the positive exponents changes the variance, not the target integral. Diagram probabilities are fixed from preceding pilot data with a nonzero uniform floor, and likewise corrected by their reciprocal. A stated sample count is the number of random graph–sector–point draws for an entire loop order, not that number of samples for each individual graph.

Co-tree monomials are evaluated relative to their largest logarithm. A cancellation monitor compares the signed forest value with the sum of absolute term contributions. Evaluation starts in extended precision and is repeated in binary128 when that ratio is below 10^{-9}; a 100-digit fallback is available below 10^{-23} at the second stage. No sample is clipped. The primary run, run C below, invokes binary128 21703 times and never invokes the final fallback. These diagnostics are precision controls, not interval arithmetic.

The 100-digit level of that fallback was later found too low. The following account comes from the three-dimensional sixth-order bundle `lri_C6_d3_complete_update/`, written by OpenAI GPT-6 astra (model stated by the author): a deliberately extreme stress test of the graph L3G007 in that package exposed insufficient relative accuracy, and the fallback was raised to 300 digits before the test was repeated. This is the level used by the later sixth-order runs, including the 2026 productions (Sections 26.7 and 26.8). No production run of the three-dimensional package

called the emergency path, so none used the defective branch, and two same-seed 128,000-sample replays at loops three and six agreed batch for batch before and after the change. None of the runs A–D described next reached that branch: the counters of the final fallback in their records (`lri_C6_d2_complete_update/reproduction/mom_prod_*.json` of the research archive) are zero at every loop order, so their values are unaffected.

Runs A and B, which are unrelated to the two runs of (660), integrate all 395 nonzero sixth-order finite kernels. In run C the 10 primitive, 26 one-subgraph and 40 two-subgraph kernels are replaced by their already measured finite-scheme values, with their uncertainties propagated. The correct replacement is

$$\begin{aligned} K_{76}^{\mathrm{MOM}} &= C_{6,\mathrm{primitive}} - \frac{2}{729}(Y_{26} + Y_{40}) \\ &= 528.2946383532544, \qquad \widehat{\sigma}(K_{76}^{\mathrm{MOM}}) = 0.1373774929551. \end{aligned} \tag{706}$$

It is *not* the sum of the separately MS-converted Θ_{26}, Θ_{40}. This distinction avoids double counting finite-scheme conversion terms. Exact zero weights remove another 28 kernels from the sixth-order physical combination, so the remaining sixth-order integration in C has 291 kernels. Its proposal exponents are 0.85 times those of A and B. A fourth run, D, used the same substitution as C with the proposal exponents of A and B, a different seed and 16,000,000 draws per loop order (A and B used 8,000,000 and C 32,000,000). The measurements behind (706) were made before, and independently of, the loop-order samples and are not used to calibrate an unknown C_6. Runs A–D, combined by inverse covariance with the weights 0.041, 0.039, 0.702 and 0.219, gave the first complete two-dimensional estimate, superseded by the 2026 production (Section 26.7); their records are in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository, and the input (706) is also in `LRI_C6_D2_HPC/reference/baseline_previous/baseline/lri_C6_d2_complete_update/reproduction/mom_known_sectors.json`.

25.5 The changes required by dimension three

The long-range regulator is varied at fixed dimension, with $q = d/4$ and edge exponent $a = q + z/4$. The general-dimensional renormalization setting follows [7]; the sixth-order numerical work and its evidence are described here and are not part of that three-loop paper.

For a proper logarithmic forest F of a power-free L-loop graph G, let U_F be the product of the first Symanzik polynomials of its contracted blocks, $E_R(F)$ the edges in its root block, and $S = \sum_e x_e$. The correct common kernel in dimension three is

$$\boxed{K_G^{(3)}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R(F)} x_e}{S\, U_F(x)^{3/2}}.} \tag{707}$$

The fractional power applies to each complete forest denominator; it cannot be obtained by raising an already summed two-dimensional kernel to 3/2. Each term is evaluated on the same point before the sum is sampled. This preserves local cancellation of logarithmic endpoint singularities. General parametric forest renormalization provides the framework [1, 21]; the retained exact forest-to-RG polynomial identity remains dimension independent.

With $r(d,z) = \Gamma(d/2 + z/2)/\Gamma(1 + z/2)$ and $d = 3$, the corresponding scale-derivative amplitude is

$$D_G(3,z) = \frac{r(3,z)^L \Gamma(Lz/2)}{\Gamma(3/4 + z/4)^{2L}} \int d\Omega \prod_e x_e^{-1/4+z/4}\, K_G^{(3)}(x). \tag{708}$$

Relative block-scale integration has Dirichlet parameters $h_B z/2$. The inserted root fraction therefore has mean h_R/L, proving the same forest coefficient as in the two-dimensional construction. The finite RG coefficients, distinguished by hats from constant MS coefficients, are

$$\widehat{b}_L(z) = (-1)^{L+1} Lz \sum_G w_A(G) D_G(3,z), \qquad \widehat{c}_L(z) = (-1)^{L+1} Lz \sum_G w_B(G) D_G(3,z). \tag{709}$$

The inherited inventory contains 660 sixth-order graph classes. There are 487 power-free classes and 173 classes with a proper power subgraph. After exact forest cancellation, the power-free numerical inputs at $L = 2, 3, 4, 5, 6$ contain respectively $1, 5, 16, 76, 395$ nonzero kernels. Their edge lists and rational weights were checked again against the inventory. This is reuse and checking of the project's graph inventory, not a second independent enumeration.

Dimension-correct analytic prefactors. Unlike at $d = 2$, $r(3, z)$ is not identically one. In a largest-edge projective sector define

$$s_L(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S,$$
$$G_{L,d}(z) = \frac{r(d,z)^L \Gamma(1 + Lz/2)}{\Gamma(q + z/4)^{2L}} \frac{\Gamma(q)^{2L}}{\Gamma(2q)^L}, \qquad q = \frac{3}{4}. \tag{710}$$

Then $G_{L,d}(0) = 1$ and $\log G_{L,d}(z) = \sum_{j\geq 1} a_{L,j} z^j$, where

$$\boxed{a_{L,j} = \frac{L2^{-j}\psi_{j-1}(2q) + [(L/2)^j - L2^{-j}]\psi_{j-1}(1) - 2L4^{-j}\psi_{j-1}(q)}{j!}.} \tag{711}$$

The overall critical normalization is

$$n_L = 2(-1)^{L+1} \frac{\Gamma(2q)^L}{\Gamma(q)^{2L}}. \tag{712}$$

Integrating the coefficients of $\exp(s_L z) G_{L,d}(z)$ against the critical projective kernel and multiplying by $n_L w_{A,B}$ gives the required jets. These coefficients are the finite Bell polynomials (701). Their maximum order is four; no pole-fitting operation is used in the Monte Carlo integration. All 20 Gamma logarithmic jets and five normalizations were freshly checked by independent 65-digit Cauchy evaluation of the analytic Gamma product. The maximum absolute discrepancy is 2.27×10^{-59}.

25.6 Regulator dependence and finite-order extraction

The two-loop input is recomputed from the explicit general-dimensional two-${}_3F_2(1/4)$ formula (199) for $S_d(z)$, using

$$\widehat{b}_2(z) = -6z\left(S_3(z) - \frac{2}{z^2}\right), \qquad \widehat{c}_2(z) = \frac{1}{6}\widehat{b}_2(z). \tag{713}$$

Its beta jets through z^4 are

$$\begin{aligned} b_{2,0} &= -9.7859764118999280, \quad & b_{2,1} &= 13.1771369571188313, \\ b_{2,2} &= -16.3759608428540408, \quad & b_{2,3} &= 19.5740514096297382, \\ b_{2,4} &= -22.7913170046010398. \end{aligned} \tag{714}$$

Two independent 60-digit evaluations use radii $0.05, 0.07$ and respectively 48,64 Cauchy nodes. Their full outputs, rather than these rounded entries, are kept in the package records listed in Section 33.

Let $\widehat{B}(g, z) = \sum_{L\geq 1} \widehat{b}_L(z) g^{L-1}$, with $\widehat{b}_1 = 3, \widehat{c}_1 = 1$, and $\gamma_r(g, z) = \sum_L \widehat{c}_L(z) g^L$. The power-free physical fixed point is defined by $\epsilon = g_* \widehat{B}(g_*, \epsilon)$. Keeping z independent during Lagrange inversion gives

$$\boxed{C_n^{\rm pf}(3) = -\frac{1}{2}\delta_{n1} + \sum_{k=1}^{n} \frac{1}{k}[g^{k-1} z^{n-k}] \partial_g \gamma_r(g, z) \widehat{B}(g, z)^{-k}, \qquad n \geq 1,} \tag{715}$$

with $C_0 = 3/2$. The extraction is from finite polynomials, not from an unevaluated master. The needed orders are z^{6-L} at each loop L. Thus the fifth-loop mass and coupling finite-scheme jets are

included separately, while at six loops only $3c_{6,0} - b_{6,0}$ is needed. Replacing every hatted function by its constant term would be incorrect. An independent rational-polynomial implementation of implicit fixed-point substitution and bivariate Lagrange inversion gives 256 exact checks, including 32 checks of the constant-MS sixth-order assembly.

26 Final numerical design, covariance and precision

Current and superseded estimates. This section serves both dimensions. Only the 2026 productions of Sections 26.7 and 26.8 are current. Each has two campaigns, a and b, with 64 replicates at every loop order, and they give $C_6(2) = 362.366 \pm 0.004$ and $C_6(3) = 33.5754 \pm 0.0005$. The other campaigns described in this Part belong to superseded estimates; they are kept because the productions inherit their kernels, allocations and checks. The superseded sixth-order method-refinement calculation at $d = 2, 3$, called the earlier sixth-order calculation in this section, with its refinement campaigns, previous estimators, method benchmarks (Section 26.11), control experiment (Section 26.14) and method assessment (Section 26.16), was run by OpenAI GPT-6 astra (model stated by the author) and written up by it in a methods note delivered with the package; that note is merged into this section. Its code and records are in `LRI_C6_D2_HPC/reference/`; record paths given below for the earlier calculation are relative to that folder. Three parts were added by a Claude Code session when the methods note was merged, on 16 October 2026: formula (738), rederived from the stored covariance, where it reproduces the reported ratios 0.985 and 0.962; the descriptions of the benchmark variants, of the pilot counts and of the trained maps in Section 26.11, taken from the package code (`code/experiment.cpp`, `code/pilot.py`, `code/train_grid.py`); and the proof of (750).

- Runs A–D of Section 25.4 gave the first complete two-dimensional estimate.
- The previous estimators of Section 26.13 pooled, at each dimension, two baseline campaigns A and B with 64 replicates (Sections 26.2 and 26.4) and two refinement campaigns, also labeled A and B, with 32 replicates (Section 26.12). They gave $C_6(2) = 362.3489 \pm 0.1526$ and $C_6(3) = 33.5682 \pm 0.0242$ (sampling standard errors).

Upper-case letters label superseded campaigns; lower-case a and b label the campaigns of the 2026 productions, here and in Sections 28, 29, 31 and 32.

26.1 Integrated RG Taylor vectors

For a power-free L-loop graph G with $E = 2L$ edges, the retained finite forest kernel is

$$K_G^{(d)}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R(F)} x_e}{S(x)\, U_F(x)^{d/2}}, \qquad S(x) = \sum_{e=1}^{E} x_e. \tag{716}$$

The sum is over the compatible forests of (694); U_F is the product of the first Symanzik polynomials of the forest blocks. Terms with identical denominators are combined before floating-point integration. The rational graph weights, normalized edge measure, critical-mass prescription, and signs are those of Section 25. The finite-kernel selection is fixed by exact forest cancellation. There are respectively 5, 16, 76 and 395 nonzero finite kernels at loops 3, 4, 5 and 6. This is a count after exact algebraic cancellation, not the count of all six-loop graph classes.

The dimension-dependent prefactor, including its regulator dependence, is

$$\frac{\Gamma(d/2)^L}{\Gamma(d/4)^{2L}}\, G_{L,d}(z) = \Gamma(1 + Lz/2) \left[\frac{\Gamma(d/2 + z/2)}{\Gamma(1 + z/2)}\right]^L \Gamma(d/4 + z/4)^{-2L}. \tag{717}$$

Here $G_{L,d}$ is the normalized prefactor (710) with $q = d/4$, so $G_{L,d}(0) = 1$; at $d = 2$ it is $\mathcal{G}_L$ of (699). The overall sign and factor are $n_L = 2(-1)^{L+1}\Gamma(d/2)^L/\Gamma(d/4)^{2L}$, as in (712), times the retained coupling or mass graph weight. Put

$$a_{L,j} = [z^j] \log G_{L,d}(z), \qquad s_G(x) = \frac{1}{4}\sum_e \log x_e - \frac{L}{2}\log S(x). \tag{718}$$

At each integration point the Taylor coefficients needed through z^{6-L} are generated by the finite recurrence

$$e_0(x) = 1, \qquad e_n(x) = \frac{1}{n}\sum_{j=1}^{n} j\big(a_{L,j} + \delta_{j1} s_G(x)\big) e_{n-j}(x). \tag{719}$$

The scalar prefactors and their logarithmic Taylor coefficients were recomputed at 70 decimal working digits and checked against the values stored with the superseded estimator. The separately evaluated two-loop data remain inputs. Four additional independently seeded two-loop sampling checks use both ordering maps in both dimensions and a different, constant-exponent importance density. All five required Taylor coefficients agree with the exact two-loop master (199) within 1.46 estimated standard errors. Their total 8,388,608 samples are controls only and do not enter the final estimate.

For every loop the stored vector contains all required beta-function Taylor coefficients followed by all required mass-anomalous-dimension Taylor coefficients:

$$\boldsymbol{v}_L = (b_{L,0}, \ldots, b_{L,6-L}, c_{L,0}, \ldots, c_{L,6-L}). \tag{720}$$

Their covariance is not discarded. Setting $z = \epsilon$ only after forming these Taylor series, the coefficient map $\mathcal{F}_6$, which returns the coefficient C_6 of $\Delta_{\mathcal{E}}$ at dimension d, is defined by

$$0 = -\epsilon g_* + 3g_*^2 + \sum_{L=2}^{6} \widehat{b}_L(\epsilon) g_*^{L+1} + O(\epsilon^8), \tag{721}$$

$$\Delta_{\mathcal{E}} = \frac{d-\epsilon}{2} + g_* + \sum_{L=2}^{6} \widehat{c}_L(\epsilon) g_*^L + O(\epsilon^7). \tag{722}$$

Using only the values of the finite-scheme RG functions at $z = 0$ would not reproduce this calculation. An independent finite two-variable Lagrange expansion checks the fixed-point iteration used by $\mathcal{F}_6$.

26.2 Graph stratification and fixed pilot allocation

The estimator integrates every graph as a separate stratum, using a fixed sample count n_G within each independent randomized replicate. For a generic component of the integral,

$$\widehat{I}_b = \sum_G w_G \frac{1}{n_G} \sum_{i=0}^{n_G-1} f_G(X_{bGi})/p_G(X_{bGi}). \tag{723}$$

Here p_G is the normalized, unchanged importance density for graph G; there is no inverse random-graph selection probability. This removes random fluctuations in the number of times a graph is visited without changing the integral being estimated.

The counts are powers of two, at least four. A separate pilot and two validation calculations estimate the graphwise variance of the linearized C_6 contribution. These estimates determine the allocation, with a uniform floor; the allocation is frozen before either production campaign. No sample of a campaign is reused as a training pilot for that campaign, and no allocation is selected by agreement with a known physical coefficient. This target-aware allocation is a variance-reduction design, not an identity for C_6 and not a claim of mathematically optimal allocation.

Each graph stratum uses a linearly scrambled Sobol net with a separate random digital shift. Powers-of-two net sizes preserve the intended balance; the first point is retained and the sequence is not thinned. The direction matrices are generated with the documented SciPy Sobol construction, then checked against its public point generator [46]. Every baseline campaign has 64 independent randomized replicates at every loop order, as does each campaign of the 2026 productions; the refinement campaigns of Section 26.12 have 32. The same linear scrambling is shared within a replicate while different graph shifts are independently seeded; covariance is measured from the *total* replicate vector.

Campaign A uses 52-bit digital coordinates and the original sequence of conditional draws for the edge ordering. Campaign B uses 60-bit digital coordinates and the equivalent one-coordinate ordering construction below. The integer net coordinate k is mapped to $(k + 1/2)/2^b$ before taking logarithms. This avoids evaluating an endpoint singularity at zero. Finite digital resolution is itself an approximation; comparing 52 and 60 bits is an empirical control, not a proof of a bound on its bias. The production arithmetic begins at long-double precision in both campaigns; the optional guarded-double experimental build is not used.

26.3 An exact compressed map for the edge ordering

Let S be the set of edges not yet selected and let $D(S)$ be the positive dynamic-programming weight used by the importance sampler. The probability of choosing edge $j \in S$ next is

$$p(j \mid S) = \frac{D(S \setminus \{j\})}{\sum_{k \in S} D(S \setminus \{k\})}. \tag{724}$$

One may use a fresh uniform coordinate for every such draw. Alternatively, start with a single $u \in (0, 1)$, choose the interval $[P_j, P_j + p_j)$ containing u, and replace it by

$$u \longmapsto \frac{u - P_j}{p_j}. \tag{725}$$

Conditional on selecting j, the new variable is again uniform on $(0, 1)$. Induction proves that an entire ordering σ has exactly its original probability $\prod p(j \mid S)$ under this map. The gap coordinates are sampled separately, so the continuous density and the importance weight do not change.

For $E = 2L$, the original map uses $2E - 1$ digital coordinates after graph stratification: E ordering draws and $E - 1$ scale draws. The compressed map uses E coordinates: one ordering coordinate and $E - 1$ scales. At six loops the counts are thus 23 and 12. This reduces the dimension of the sampling map, *not* the physical Schwinger integral. It is not claimed to improve every integrand, and the two campaigns test its empirical behavior rather than assuming an improvement. The probability identity also passed exhaustive exact rational checks for 870 edge permutations in smaller examples.

26.4 Baseline campaigns of the previous estimators

d	Run	$L = 3$	$L = 4$	$L = 5$	$L = 6$	C_6	SE
3	A	4,456,448	9,502,720	32,768,000	67,076,096	33.590772	0.057948
3	B	4,456,448	9,502,720	32,768,000	67,076,096	33.560344	0.053265

The two three-dimensional baseline campaigns use 227,606,528 kernel samples over all four loop orders; pilot and validation samples were additional and excluded. They entered only the previous three-dimensional estimator (740), and the two corresponding two-dimensional campaigns, with 115,969,024 samples each, only the previous two-dimensional estimator. Both are superseded by the productions of Sections 26.7 and 26.8.

Counts in this table are sums over graph strata, not counts per graph. They are the actual rounded power-of-two allocations, not nominal budgets. Both campaigns use the same frozen graph allocation and importance exponents; seeds, net precision and ordering maps differ. The two campaigns are pooled with the fixed weights $1/2, 1/2$ at the level of the RG Taylor vectors. The weights were not optimized after observing the production central values or their empirical variances.

Separate campaigns have independent sampled points.

26.5 Replicate covariance and fixed-point propagation

For campaign t, loop L and $R_{L,t}$ independent randomized replicates, let $v_{L,t,r}$ be the complete vector of coupling, mass and regulator coefficients. Define

$$\begin{aligned} \mu_{L,t} &= \frac{1}{R_{L,t}} \sum_r v_{L,t,r}, \\ \widehat{\Sigma}_{L,t} &= \frac{1}{R_{L,t}(R_{L,t}-1)} \sum_r (v_{L,t,r} - \mu_{L,t})(v_{L,t,r} - \mu_{L,t})^{\mathsf{T}}. \end{aligned} \tag{726}$$

This is the estimated covariance of the mean. The unit of statistical independence is a randomized replicate, not a point within one Sobol net [47]. Independent loops have independent randomizations; within each loop all beta/mass and Taylor-order covariances are retained. The fixed weights, shared-input terms and final Jacobian propagation are given in (740)–(744) below.

No known physical coefficient is used to calibrate the estimator and no outlying replicate is discarded; the quoted central value is the direct pooled estimator. For the previous four-campaign estimators the nonlinear fixed-point map was checked by an independent finite Lagrange expansion, and a delete-one-replicate calculation diagnosed the leading finite-sample bias. These archived checks concern the inherited map and estimator, not the replicate data of the 2026 productions. A replicate bootstrap checks variance propagation. In the previous two-dimensional estimator it drew each shared retained input once per complete bootstrap sample; modeling that input by a normal draw does not prove normality or give an interval certificate. The 2026 productions have no shared retained input.

26.6 The quantity being integrated

For each loop order L, the sampled vector contains the coupling and mass Taylor coefficients through regulator order $6 - L$:

$$\boldsymbol{\mu}_L = (b_{L,0}, \dots, b_{L,6-L}, c_{L,0}, \dots, c_{L,6-L}), \qquad L = 3, 4, 5, 6. \tag{727}$$

The one-loop input is exact, and the two-loop input is the special-function evaluation of Section 25 at each dimension. The physical coefficient is evaluated as a finite polynomial

$$C_6 = \mathcal{F}_6(\boldsymbol{\mu}_3, \boldsymbol{\mu}_4, \boldsymbol{\mu}_5, \boldsymbol{\mu}_6) + \eta. \tag{728}$$

Here $\mathcal{F}_6$ solves the regulator-dependent finite-MOM fixed-point problem before setting $z = \epsilon$; η accounts only for separately retained numerical kernels. Only the previous two-dimensional estimator used such kernels, so $\eta = 0$ in both current estimators. In particular,

$$\frac{\partial C_6}{\partial(b_{6,0}, c_{6,0})} = \frac{(-1, 3)}{2187}. \tag{729}$$

One must not replace the Taylor vectors by their values at $z = 0$ and then apply the constant-coefficient MS assembly. The field-theoretic derivation of these finite vectors and their power-sector compensation is unchanged.

The finite kernel retains the pointwise forest sum (716): every numerator and denominator is evaluated on the same integration point before accumulating the graph estimate. Tropical importance sampling uses the known endpoint geometry of these parametric integrals [41]. More samples cannot in general repair a representation which postpones subtraction of divergent terms until after separate numerical integrations.

26.7 The two-dimensional production of September 2026

At $d = 2$ the current estimate comes from a single new production with two independent campaigns, a and b. It integrates the finite forest kernels (716) of all 5, 16, 76 and 395 nonzero classes at loops 3–6, generates the regulator jets by (719), and applies the fixed-point map (722) with the retained analytic power-sector compensation of Section 24. The code is an optimized, restartable reimplementation of the same calculation. Its kernel keeps the addition order of every polynomial sum and the binary128 and 300-decimal fallbacks at the unchanged thresholds 10^{-9} and 10^{-23}. In its legacy summation mode, replays of all 492 kernels against the original executable agree to a normalized difference below 9×10^{-20}, with identical promotion counts; production uses compensated summation within each task.

Both campaigns have 64 independent randomized replicates at every loop order and use the conditional-draw ordering map with $2E - 1$ digital coordinates. Campaign a uses 52-bit and campaign b 60-bit coordinates, and only campaign a uses, at $L = 6$, the pilot-frozen separable map (737). Every stratum receives the frozen per-replicate count of the refinement allocation (739), raised to at least four, multiplied by 2^{13}; the floor affects only the 28 six-loop kernels whose C_6 projection vanishes exactly. Per replicate this gives 1,107,296,256, 2,164,260,864 and 9,487,515,648 points at $L = 3, 4, 5$, and 38,076,809,216 points for the 385 nonprimitive six-loop kernels. The ten primitive six-loop kernels, which the previous estimator retained as P_{10}, are integrated in every replicate with 5,645,533,184 points per replicate, divided among them by a variance-based (Neyman) allocation frozen from archived paired records of the earlier calculation (`paired_new_a_d2_L6.json`, `paired_new_b_d2_L6.json`, `paired_prod_a_d2_L6.json` and `paired_prod_b_d2_L6.json` in `LRI_C6_D2_HPC/reference/records/`, read by `tools/primitive_allocation.py`). The two campaigns contain 7,229,621,141,504 kernel samples. Group seeds are derived by SHA-256 from a fresh base seed, the campaign, the loop and the role. No production sample was used to choose the allocation, and pilot, smoke and timing samples are excluded. Every required kernel thus has expectation weight one in every replicate, no numerical input is retained, and

$$\boxed{\widehat{C}_6(2) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^a + \boldsymbol{\mu}^b}{2}\right), \qquad \widehat{\sigma}_{\text{common}} = 0,} \tag{730}$$

with the covariance (744) at $w_a = w_b = 1/2$.

The production ran on SDSC Expanse as 428 independent 64-thread array elements, 12,535 core-hours in total. Sample identities do not depend on node, thread count or checkpoint boundaries: two blocks of 1024 six-loop tasks run at 64 and at 16 threads gave identical rows, including 152,717 binary128 promotions. The campaigns give

Campaign	$C_6(2)$	Sampling SE
a	362.36672022	0.00097561
b	362.36622331	0.00126747
Combined	362.36647176	0.00079973

They differ by 0.31 combined standard errors. The combined standard error is about half the value 0.0015 projected from the earlier campaigns under Monte Carlo scaling, in which the variance falls as $1/n$ and the standard error as $1/\sqrt{n}$. This single comparison does not determine a convergence

rate; the replicate covariance, not the projection, is reported. The new estimator lies 0.12 previous standard errors from the previous estimator $362.3489138 \pm 0.1526445$.

The reported allowance is four sampling standard errors, 0.0032, rounded up to a multiple of 0.001, which gives $C_6(2) = 362.366 \pm 0.004$. The four-standard-error rule was written down before the production result was available; the rounding step was refined afterwards from the 0.01 first proposed to 0.001, which changes only the rounding of the allowance. An independent two-dimensional calculation through seventh order, in which the mixed local-mass and logarithmic forests are integrated explicitly instead of through the analytic compensation, gives as a by-product $C_6(2) = 362.3727 \pm 0.0086$ (sampling standard error), $+0.73$ combined standard errors from (730). It shares the graph and renormalization framework and therefore does not test that framework. The previous two-dimensional baseline and refinement campaigns, the retained groups (741) and the estimator built from them are superseded; their records are in `LRI_C6_D2_HPC/reference/` (Section 33).

26.8 The three-dimensional production of September 2026

At $d = 3$ the current estimate likewise comes from a single new production with two independent campaigns, a and b. It integrates the finite forest kernels (716) of all 5, 16, 76 and 395 nonzero classes at loops 3–6, the ten primitive six-loop kernels included, generates the regulator jets by (719), and applies the fixed-point map (722) with the retained analytic power-sector compensation of Section 24. The code is the three-dimensional counterpart of the one in Section 26.7, with the binary128 and 300-decimal fallbacks at the same thresholds 10^{-9} and 10^{-23}. In its legacy summation mode it replays the archived three-dimensional refinement streams, 385,744,896 samples on their archived nets and seeds, with fallback counts equal to the archive in every stream, and reproduces the refinement-only estimate 33.56084035 bit for bit; production uses compensated summation within each task. Section 25.4 explains why the last fallback has 300 decimal digits.

Both campaigns have 64 independent randomized replicates at every loop order and use the conditional-draw ordering map with $2E - 1$ digital coordinates and the baseline importance map. Campaign a uses 52-bit and campaign b 60-bit coordinates. Every stratum receives the frozen per-replicate count of the three-dimensional refinement allocation (739), multiplied by 2^{13}. Per replicate this gives 1,241,513,984, 2,499,805,184, 9,743,368,192 and 35,891,576,832 points at $L = 3, 4, 5, 6$, and the two campaigns contain 6,320,161,816,576 kernel samples. Group seeds are derived by SHA-256 from a fresh base seed, the campaign, the loop and the role. No production sample was used to choose the allocation, and the samples of the thread-identity check below are excluded. Every required kernel has expectation weight one in every replicate, no numerical input is retained, and

$$\boxed{\widehat{C}_6(3) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^a + \boldsymbol{\mu}^b}{2}\right), \qquad \widehat{\sigma}_{\text{common}} = 0,} \tag{731}$$

with the covariance (744) at $w_a = w_b = 1/2$.

The production ran on SDSC Expanse on 29 September 2026 as 458 independent 64-thread array elements. Sample identities do not depend on node, thread count or checkpoint boundaries: two blocks of 1024 six-loop tasks run at 64 and at 16 threads gave identical rows, including 141,994 binary128 promotions. The campaigns give

Campaign	$C_6(3)$	Sampling SE
a	33.57562307	0.00015062
b	33.57523021	0.00018311
Combined	33.57542664	0.00011855

They differ by 1.66 combined standard errors. The combined standard error is 0.54 of the value

0.00022 projected from the refinement campaigns by plain Monte Carlo scaling, in which the standard error falls as $1/\sqrt{n}$; the replicate covariance, not the projection, is reported. The estimator lies 0.30 standard errors of the superseded estimator (740) from its value 33.5681968 ± 0.0241840, and 0.52 refinement standard errors from the refinement-only estimate.

The reported allowance is four sampling standard errors, 0.00047, rounded up to a multiple of 0.0001, which gives $C_6(3) = 33.5754 \pm 0.0005$. The estimator, the four-standard-error rule and the consistency checks were written down and confirmed before any production result was available; the rounding step was refined afterwards from the 0.001 first registered to 0.0001, which changes only the rounding of the allowance. An independent three-dimensional calculation through seventh order (Section 29), in which the power-containing sector is integrated explicitly instead of through the analytic compensation, gives as a by-product $C_6(3) = 33.57387 \pm 0.00198$ (sampling standard error), -0.79 combined standard errors from the estimator. It shares the graph and renormalization framework and therefore does not test that framework. The previous three-dimensional baseline and refinement campaigns and the estimator (740) built from them are superseded. Their records are in the reference tree that the two- and three-dimensional sixth-order packages share and that is shipped once, in `LRI_C6_D2_HPC/reference/` (`records/`, `baseline_previous/records/` and `verification/method_precision_results.json`); the replay of Section 33 recomputes them from there.

26.9 Result and interpretation of the errors

The current estimates are

$$\boxed{\begin{aligned} C_6(2) &= 362.366 \pm 0.004, \\ C_6(3) &= 33.5754 \pm 0.0005. \end{aligned}} \tag{732}$$

The allowances are four estimated sampling standard errors rounded upward, to a multiple of 0.001 at $d = 2$ and of 0.0001 at $d = 3$. They are empirical numerical allowances, not rigorous enclosures, guaranteed confidence intervals, or truncation errors for the physical ϵ expansion. The underlying estimator digits are recorded for reproducibility, not as independently certified physical digits.

d	Estimator	Sampling standard error
2	362.36647176	0.00079973
3	33.57542664	0.00011855

At $d = 2$ and $d = 3$ the estimates are the 2026 productions of Sections 26.7 and 26.8 alone. Fresh pilot and method-validation samples are excluded. The lower-loop allocations were planned in part from the variances measured in the superseded baseline campaigns (Section 26.4); since the allocation is fixed and the production samples are independently randomized, this does not change the expectation of the estimate. No known C_3, C_4, C_5 is used as a fitted control or imposed as a constraint.

26.10 A faster evaluator with the same integral

Each monomial in a loop-L forest polynomial has exactly L distinct edge factors. Associate a bit mask m to a monomial and let j be one of its set bits. Precompute a directed acyclic product graph obeying

$$p_0 = 1, \qquad p_m = x_j p_{m\setminus\{j\}}. \tag{733}$$

All forest terms share these products. This replaces repeated $\exp(\sum_{e\in m} \log x_e)$ evaluations by one exponential per edge and a shared set of multiplications. For very small coordinates the

same common-monomial scaling as the original implementation is used, with h the largest of the monomial logarithms $\sum_{e\in m}\log x_e$ in the forest sum, as in Section 25.4:

$$p_m = \prod_{e\in m}\exp\left(\log x_e - \frac{h}{L}\right) = \exp\left(\sum_{e\in m}\log x_e - h\right). \tag{734}$$

The common factor is restored outside the forest sum. The direct unscaled path is used only in a range checked against the exponent range of extended precision. The original binary128 and 300-decimal fallbacks for severe cancellation are retained. The numerical kernels are not evaluated at lower precision to obtain the speedup.

An identical-point comparison used 1,095,680 six-loop evaluations for each dimension, with identical seeds, digital matrices, allocations, and importance distributions:

d	Original evaluator (s)	Shared products (s)	Speedup
2	7.88237	2.77982	2.836
3	7.74461	2.69407	2.875

These are elapsed times on one four-CPU machine, not hardware-independent throughput claims. The maximum normalized per-graph replicate difference was below 9.0×10^{-13}. The improvement is in cost per evaluation, not in the variance of an individual sample. Independently constructed Kirchhoff cofactor integrands, built from cofactor determinants rather than from the sampler's list of spanning-tree monomials, supplied 1,038 additional checks over both $d=2$ and $d=3$ and all finite kernels, with reference values at 70 digits at ordinary points and at 650 digits at extreme ones, including endpoint coordinates down to 10^{-130}; the largest normalized discrepancy was 6.88×10^{-18}. Neither comparison is an interval certificate for the full integral.

26.11 Sampling methods tested at matched budgets

The baseline sampler is the one of Section 26.2: independently randomized Sobol nets inside each fixed graph stratum, with power-of-two counts chosen separately for every stratum. The uncertainty is always estimated between independent randomized nets, never from the dependent points within one net [46, 47]. The benchmark target is the six-loop influence (729): with every six-loop finite kernel sampled, let $y_{G,r} = (3c_{6,0} - b_{6,0})_{G,r}/2187$ be the contribution of graph G in replicate r. Its variance is estimated by

$$\widehat{V}_{\mathrm{bench}} = \frac{1}{R}\sum_{G}\widehat{V}_G, \qquad \widehat{V}_G = \frac{1}{R-1}\sum_{r=1}^{R}\left(y_{G,r} - \bar{y}_G\right)^2, \tag{735}$$

the sum of the graph-stratum replicate variances divided by the number R of replicates. It contains no covariance between strata; the production analysis instead keeps the full covariance of the total replicate vectors, (726).

Each pilot comparison used $R=16$ randomizations and 1,095,680 six-loop kernel evaluations. The per-graph counts are powers of two, at least 64, the same at both dimensions, and were fixed in advance from the graph probabilities of an earlier, independent pilot, not from this benchmark (`LRI_C6_D2_HPC/reference/code/pilot.py`). Antithetic sampling used half as many points with two kernel evaluations each. The table gives $\widehat{V}_{\mathrm{bench}}$ relative to the baseline at the same number of kernel evaluations; smaller is better.

Method	$d = 2$	$d = 3$
Randomized Sobol nets (baseline)	1.000	1.000
Independent pseudo-random Monte Carlo	1.899	1.499
Radial coordinates placed first	1.105	1.126
One-coordinate edge-order encoding	1.083	1.055
Endpoint rate 0.6κ	1.667	2.301
Endpoint rate 0.5κ	4.194	6.194
Endpoint rate 1.0κ	4.452	1.797
Radial antithetic pairs	1.584	1.629

The variants are those of `code/experiment.cpp` in the same package. In the baseline layout the E ordering draws and the $E-1$ scale draws of the conditional-draw map alternate in the $2E-1$ net coordinates. "Radial coordinates placed first" puts the $E-1$ scale (radial) coordinates in the first net dimensions and the ordering draws after them. The one-coordinate encoding is the compressed map (725), with E coordinates. The antithetic variant evaluates each point together with its reflection $v \mapsto 1-v$ in the radial coordinates only, keeping the ordering coordinates, and averages the pair. The endpoint rates are defined in the next paragraph. The endpoint-rate and antithetic variants were run in the radial-first layout, so their ratios include the layout effect of the third row; for pseudo-random sampling the layout is immaterial.

The pilot differences are empirical and subject to finite-replicate uncertainty; they are not theorems about the methods. Coordinate reordering, and compression of all order choices into one uniform coordinate, gave no convincing improvement. The compressed map was used only by baseline campaign B (Section 26.2); both refinement campaigns, like both campaigns of the 2026 productions, use the conditional-draw map. Independent pseudo-random sampling was worse than randomized Sobol sampling.

Why a smoother endpoint map was not the winner. Write $\kappa(S)$ for the nominal endpoint power of the sector variable that closes the edge subset S. The proposal exponents $\alpha(S)$ of the baseline sampler, used as in (705), correspond to the nominal endpoint rate $\alpha(S) = 0.8\,\kappa(S)$. The rows "endpoint rate" multiply all of them by 0.75, 0.625 and 1.25, which gives $\alpha = 0.6\kappa$, 0.5κ and 1.0κ; the dynamic-programming normalization is recomputed with the changed exponents and the exact reciprocal proposal weight is applied, so the target integral is unchanged. An illustrative endpoint integral is

$$\int_0^1 t^{\kappa-1} f(t)\,dt = \frac{1}{\alpha}\int_0^1 u^{\kappa/\alpha-1} f(u^{1/\alpha})\,du, \qquad t = u^{1/\alpha}; \tag{736}$$

sampling t with the density $\alpha t^{\alpha-1}$ is this substitution with uniform u. Reducing α makes the endpoint factor $u^{\kappa/\alpha-1}$ smoother, but it simultaneously changes the normalized sampling distribution in every radial coordinate and can move variance into a narrow central region. The variants 0.6κ and 0.5κ lost to the baseline in both dimensions, and so did 1.0κ. Smoothness alone is not a sufficient optimization criterion for these finite, signed, high-dimensional kernels.

A pilot-frozen separable importance map. A separate training stream estimates the conditional second moment in 32 bins for each radial coordinate. Its square root is smoothed, clipped, and blended equally with a uniform density. Inverting this positive piecewise-linear cumulative distribution defines $u = T(v)$. The exact change of measure is

$$\int_{[0,1]^s} f(u)\,du = \int_{[0,1]^s} f(T(v)) \prod_{j=1}^{s} T_j'(v_j)\,dv. \tag{737}$$

All Jacobian factors are included in the kernel. This is a small VEGAS-inspired separable transport, not an implementation or benchmark of the full VEGAS+ algorithm. VEGAS+ additionally

uses adaptive stratification and has documented advantages for some multi-peak and diagonal structures [49].

Fresh validation used 32 randomizations and 4,382,720 evaluations per map. Relative variance per evaluation was

Frozen transport	$d = 2$	$d = 3$
None	1.0000	1.0000
Three trained axes	0.9533	0.9871
All eleven trained axes	0.8985	0.9905

Thus the all-axis map supplied a modest approximately 10% variance reduction at $d = 2$, but no material improvement at $d = 3$. Among the refinement campaigns only campaign A at $d = 2, L = 6$ uses this map; refinement campaign B keeps the baseline map. In the 2026 production only campaign a uses it (Section 26.7). Both three-dimensional campaigns use the baseline map. No map is adapted using its production samples. The mathematical map has endpoints $0, 1$. Delivered decimal maps have endpoint roundoff below 7.8×10^{-16}; a same-point replay with endpoints snapped exactly to $0, 1$ changed the six-loop mean by -2.95×10^{-13}. This is a floating-point consistency check, not a bound on all digital-discretization effects.

The trained maps are built by `code/train_grid.py` of the same package. At six loops each graph has $E - 1 = 11$ radial coordinates. On every axis the training histogram of the second moment is smoothed with the five-bin filter $(1, 2, 3, 2, 1)/9$; its square root is normalized to mean one on that axis and clipped to $[0.15, 4]$, and the density is half this clipped score and half uniform. "Three trained axes" transports, for each graph, the three radial coordinates whose normalized score varies most across the 32 bins and leaves the others uniform; "all eleven" transports every radial coordinate.

Conditional stratification and linear controls. Exact stratification of the first edge-order choice was also tested. The first edge j is chosen with the probability $p(j \mid E)$ of (724); in the stratified variant each choice receives its own randomized net, with a count proportional to $p(j \mid E)$, rounded to a power of two and at least four, and its estimate enters with the weight $p(j \mid E)$. On fresh validation streams with 32 randomizations, its variance times the number of kernel evaluations, relative to the same product for the matched unstratified validation run, was 1.043 at $d = 2$ and 1.062 at $d = 3$. It was not adopted.

Separately, let Σ be the covariance of the estimates $(\widehat{C}_3, \widehat{C}_4, \widehat{C}_5, \widehat{C}_6)$ of the baseline campaigns, and Σ_{low} its 3×3 block for C_3, C_4, C_5. The covariance-optimal linear control by the known C_3, C_4, C_5 would reduce the standard error of $\widehat{C}_6$ by the factor

$$\frac{\widehat{\sigma}_6^{\mathrm{lin}}}{\widehat{\sigma}_6} = \Big(1 - \sum_{k,l=3}^{5} \Sigma_{6k} \big(\Sigma_{\mathrm{low}}^{-1}\big)_{kl} \Sigma_{l6} / \Sigma_{66}\Big)^{1/2}, \tag{738}$$

which is 0.985 at $d = 2$ and 0.962 at $d = 3$. (Formula (738) reproduces both ratios from the covariance stored for the baseline-campaign estimator, field `previous` of `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`.) This is a diagnostic within that particular linear span, not a universal bound, and those controls were neither fitted nor applied. They cannot directly remove six-loop sampling variation, since C_3, C_4, C_5 contain no six-loop input.

26.12 Refinement campaigns and cost-aware allocation

Let a_g/n_g model the variance of the target contribution from graph g, and let c_g denote the measured cost of one sample. Minimizing this model at fixed cost gives

$$\min_{n_g>0} \sum_g \frac{a_g}{n_g}, \qquad \sum_g c_g n_g = T \quad \Longrightarrow \quad n_g = \lambda \sqrt{\frac{a_g}{c_g}}. \tag{739}$$

The derivative of C_6, rather than raw beta-function variance, defines the pilot target. New six-loop allocations use the independent method-validation streams, and lower-loop allocations use the variances measured in the baseline campaigns of Section 26.4; no sample of a refinement campaign sets its own allocation. Each count is rounded to a power of two and frozen before production integration. This is an MC-variance-model allocation used for RQMC, not a claim that the RQMC variance is exactly proportional to $1/n_g$ or that the allocation is globally optimal.

The baseline samples were used twice by the previous estimators: to plan the lower-loop refinement allocations, and as two of the four pooled campaigns of Section 26.13. This does not change the expectation of the pooled jet means. The allocation only sets the variance of the independently randomized refinement samples, and the combination weights, 1/4 for each campaign and, at $d = 2$, 1/2 for the 66 nonprimitive kernels added separately, were fixed before the refinement production (`LRI_C6_D2_HPC/reference/inputs/frozen_production_plan.json`). They are neither inverse-variance weights nor chosen from any result.

Each refinement campaign has 32 independently randomized replicates at each loop order. The per-replicate point counts are

d	$L = 3$	$L = 4$	$L = 5$	$L = 6$
2	135,168	264,192	1,158,144	4,647,936
3	151,552	305,152	1,189,376	4,381,184

The refinement campaigns comprise 385,744,896 kernel evaluations at $d = 3$ and 397,148,160 at $d = 2$. Both are superseded; each row of the table is the frozen allocation that the corresponding 2026 production multiplies by 2^{13} (Sections 26.7 and 26.8). Of the refinement-stage runs, only these two campaigns entered the previous estimators, together with the two baseline campaigns. The 34,144,256 two-dimensional evaluations of the dimension-coupled control experiment of Section 26.14 are counted separately; that experiment entered neither the previous nor the current estimators. Pilot, method-validation, speed-test, and endpoint-replay samples are not included. Campaigns A and B use different seeds and 52-bit and 60-bit digital coordinates, respectively. The source and input hashes were frozen before production and rechecked afterward.

The three-dimensional refinement-campaign determinations, before combination with the baseline campaigns, were

	$C_6(3)$	Sampling SE
Refinement A	33.59417512	0.040029
Refinement B	33.52750475	0.039502
Refinement only	33.56084035	0.028119

At $d = 3$ the campaigns share no retained input, so the standard error of the difference of two estimates is the quadrature sum of their standard errors. Refinement A minus refinement B is +1.19 such standard errors, and the refinement-only value minus the estimator of the two baseline campaigns alone, 33.57555796 ± 0.039354, is -0.30 such standard errors (fields `fresh_campaign_pull` and `fresh_minus_old_pull` of the record `verification/method_precision_results.json` of

the earlier calculation). Pooling all four campaigns in the previous estimator (740) lowered the three-dimensional sampling standard error from 0.039354, for the baseline campaigns alone, to 0.024184, a factor 1.63, which also reflects the larger number of production samples. The 2026 code reproduces the refinement-only value bit for bit by replaying the archived streams (Section 26.8).

At $d = 2$ the corresponding determinations were

	$C_6(2)$	Sampling SE
Refinement A	362.41502790	0.275726
Refinement B	362.46528830	0.273306
Refinement only	362.44015668	0.196310

The refinement campaigns integrated the 66 nonprimitive kernels of the groups (741) but not the ten primitive ones. Each value in this table therefore adds P_{10} once, and each standard error contains $\widehat{\sigma}_P$ once. Because P_{10} is common to the three rows, its variance is removed from campaign-difference tests but not from the uncertainty of each complete value. With it removed, campaigns A and B differ by -0.13 standard errors of the difference, $(\widehat{\sigma}_A^2 + \widehat{\sigma}_B^2 - 2\widehat{\sigma}_P^2)^{1/2}$. The refinement-only value lies $+0.62$ standard errors of the difference, again with the P_{10} variance removed, from the estimator of the baseline campaigns alone, $362.25767208 \pm 0.226352$, which contains P_{10} too (fields `fresh_campaign_pull` and `fresh_minus_old_pull` of `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`). Pooling the four campaigns lowered the sampling standard error of the two-dimensional estimator from 0.226352, for the baseline campaigns alone, to 0.152644, a factor 1.48. This gain also uses more production samples; it is not attributed to the change of sampling algorithm alone. The speedup of the shared-product evaluator (733) lowers the cost per evaluation, a different quantity from the variance per evaluation, and the two are not combined.

26.13 Campaign pooling, retained inputs and covariance

The previous three-dimensional estimator pooled the four campaign jet means, from two baseline and two refinement campaigns, with fixed weights 1/4; every vector is complete:

$$\boxed{\widehat{C}_6(3) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^{\mathrm{old,A}} + \boldsymbol{\mu}^{\mathrm{old,B}} + \boldsymbol{\mu}^{\mathrm{new,A}} + \boldsymbol{\mu}^{\mathrm{new,B}}}{4}\right).} \tag{740}$$

There is no separately retained numerical kernel at $d = 3$.

The previous two-dimensional estimator pooled the same four campaign generations, but replaced the 76 six-loop finite kernels of Section 23 by three independently measured groups. In the finite-MOM normalization these were

$$\begin{aligned} P_{10} &= 718.7129769223133, & \widehat{\sigma}_P &= 0.0414043282783, \\ S_{26} &= -188.2890632768604, & \widehat{\sigma}_{26} &= 0.1002179257784, \\ S_{40} &= -2.129275292198507, & \widehat{\sigma}_{40} &= 0.0843482336693. \end{aligned} \tag{741}$$

These are not the separately MS-converted partial sectors Θ_{26}, Θ_{40} of Section 23. Its baseline campaigns omitted all 76 kernels and its refinement campaigns only the ten primitive ones, so it added $P_{10} + \frac{1}{2}S_{66}$, with $S_{66} = S_{26} + S_{40}$, and the common variance

$$\widehat{\sigma}^2_{\mathrm{common}} = \widehat{\sigma}_P^2 + \frac{1}{4}(\widehat{\sigma}_{26}^2 + \widehat{\sigma}_{40}^2). \tag{742}$$

The 2026 productions (730) and (731) integrate every kernel in every replicate. Both current estimators therefore have $\widehat{\sigma}_{\mathrm{common}} = 0$, and no retained input is averaged away as though independent.

The previous two-dimensional estimator. With the baseline campaigns labeled old and the refinement campaigns new, it was

$$\widehat{C}_6^{\mathrm{prev}}(2) = \mathcal{F}_6\left(\frac{\boldsymbol{\mu}^{\mathrm{old,A}} + \boldsymbol{\mu}^{\mathrm{old,B}} + \boldsymbol{\mu}^{\mathrm{new,A}} + \boldsymbol{\mu}^{\mathrm{new,B}}}{4}\right) + P_{10} + \frac{1}{2}S_{66}. \tag{743}$$

The six-loop vectors of the two baseline campaigns omit all 76 kernels of (741), those of the two refinement campaigns omit only the ten primitive ones, and all lower-loop vectors are complete. Inside the average the 66 nonprimitive kernels therefore have total weight $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and the ten primitive kernels weight zero, so adding $P_{10} + \frac{1}{2}S_{66}$ gives every required contribution total expectation weight one. The groups P_{10}, S_{26}, S_{40} were measured independently, so the added term has variance $\widehat{\sigma}_P^2 + \frac{1}{4}(\widehat{\sigma}_{26}^2 + \widehat{\sigma}_{40}^2)$, which is (742), and it enters (744) once. The baseline records also contain four-point diagnostic evaluations of the 76 omitted kernels; they are dropped and never enter as measurements. The estimators of the refinement campaigns alone add P_{10} once, with variance $\widehat{\sigma}_P^2$, and that of the baseline campaigns alone adds $P_{10} + S_{66}$. The last sum is the replacement used in run C of Section 25.4: $P_{10} + S_{26} + S_{40} = 528.2946383532544$ and $(\widehat{\sigma}_P^2 + \widehat{\sigma}_{26}^2 + \widehat{\sigma}_{40}^2)^{1/2} = 0.13737749$ agree with K_{76}^{MOM} and its standard error in (706) to all printed digits.

The sampling standard error of (743), propagated with (744), was 0.152644, made of 0.005635, 0.037539, 0.080576 and 0.096763 from loops 3 to 6 and 0.077485 from the common term, combined in quadrature. That of the previous three-dimensional estimator (740) was 0.024184, made of 0.001145, 0.005894, 0.012458 and 0.019839 from loops 3 to 6, with no common term, combined in quadrature (`verification/method_precision_results.json` of the earlier calculation).

For each campaign and loop, let $\widehat{\Sigma}_{L,t}$ be the covariance of its mean estimated from independent randomized replicates. With $J_L = \partial\mathcal{F}_6/\partial\boldsymbol{\mu}_L$, the propagated sampling variance is

$$\widehat{\mathrm{Var}}(\widehat{C}_6) = \sum_{L,t} w_t^2 J_L \widehat{\Sigma}_{L,t} J_L^{\mathsf{T}} + \widehat{\sigma}^2_{\mathrm{common}}. \tag{744}$$

All coupling/mass and Taylor-order covariance is retained. The error budget, in units of C_6, is

d	$L=3$	$L=4$	$L=5$	$L=6$	Common	Total
2	0.000005	0.000113	0.000365	0.000702	0.000000	0.000800
3	0.000001	0.000021	0.000058	0.000101	0.000000	0.000119

Both rows are the 2026 productions, and the common term is zero in both. Contributions in this table combine in quadrature, not by addition of standard errors.

26.14 A dimension-coupled primitive control experiment

This control experiment belongs to the earlier calculation. It concerns only the previous three-dimensional estimator (740), and it entered neither that estimator nor the 2026 production (731).

The Sobol matrices and graph shifts of a three-dimensional campaign can also evaluate the ten primitive six-loop kernels at $d = 2$, on the same randomized net points; each dimension keeps its own importance exponents. This uses common random numbers; it assumes no interpolation in d and no relation between the two physical coefficients. Let $\widehat{P}^{\mathrm{pair}}(2)$ be this paired estimate, normalized as P_{10} in (741), and $\widehat{P}^{\mathrm{ref}}(2)$ an independent estimate of the same quantity with standard error σ_{ref}. For a fixed coefficient θ_{cv},

$$\widehat{C}_6^{\mathrm{cv}}(3) = \widehat{C}_6(3) - \theta_{\mathrm{cv}}(\widehat{P}^{\mathrm{pair}}(2) - \widehat{P}^{\mathrm{ref}}(2)). \tag{745}$$

The difference in parentheses has expectation zero. The paired estimate is correlated with the sampled three-dimensional primitive block; the reference is independent of both and not exact, so

its variance enters once:

$$\operatorname{Var}\widehat{C}_6^{\mathrm{cv}}(3) = \operatorname{Var}\widehat{C}_6(3) + \theta_{\mathrm{cv}}^2(\operatorname{Var}\widehat{P}^{\mathrm{pair}}(2) + \sigma_{\mathrm{ref}}^2) - 2\theta_{\mathrm{cv}}\operatorname{Cov}(\widehat{C}_6(3), \widehat{P}^{\mathrm{pair}}(2)),$$
$$\theta_{\mathrm{cv}}^{\mathrm{opt}} = \frac{\operatorname{Cov}(\widehat{C}_6(3), \widehat{P}^{\mathrm{pair}}(2))}{\operatorname{Var}\widehat{P}^{\mathrm{pair}}(2) + \sigma_{\mathrm{ref}}^2}. \tag{746}$$

The optimum assumes a known population covariance; it does not justify fitting production coefficients without error control.

An independent paired pilot fitted one coefficient for each of the groups P_{10}, S_{26} and S_{40} of (741), the primitive, one-subgraph and two-subgraph blocks: the least-squares slope of the three-dimensional block on its two-dimensional partner over the pilot replicates, that is, the plain ratio of covariance to variance, without the term σ_{ref}^2 of (746). On independent validation streams with 32 randomizations the residual-variance ratios were 0.234, 0.743 and 0.597; these ratios also leave out the reference term. The control doubles the kernel evaluations of a block, so the chosen equal-cost rule kept a block only if twice its ratio was below one (`inputs/cross_primitive_production_plan.json`). Only the primitive block passed, with the frozen values

$$\theta_{\mathrm{cv}} = 0.09955611941934894, \qquad \widehat{P}^{\mathrm{ref}}(2) = P_{10}, \qquad \sigma_{\mathrm{ref}} = \widehat{\sigma}_P. \tag{747}$$

No production sample was used to fit θ_{cv} or to choose the block. Evaluating the primitive block at $d = 2$ on the stored net points of all four campaigns of (740) added 34,144,256 two-dimensional evaluations and no three-dimensional sample, with identical seeds and digital matrices verified (`verification/cross_production_manifest.json`). An independent three-dimensional replay of 894,464 primitive evaluations, the ten primitive graphs in the first two replicates of baseline campaign B and of refinement campaign A, reproduced the stored graph-replicate values with maximum difference zero (`verification/pairing_checks.json`). With the campaign weights 1/4 of (740) the controlled result is

$$\widehat{C}_6^{\mathrm{cv}}(3) = 33.5681256836, \qquad \widehat{\sigma}_{\mathrm{cv}} = 0.0243339897 \tag{748}$$

(`verification/controlled_precision_result.json`), against 33.5681968 ± 0.0241840 for the plain estimator.

The validation variance of the primitive block fell by a factor of about 4.3, but this block carries only about 5.8% of the plain variance. In the production its sampling error fell from 0.00584436 to 0.00325730; the reference adds $\theta_{\mathrm{cv}}\widehat{\sigma}_P = 0.00412205$, giving 0.00525370 in all (`verification/cross_control_budget.json`). In variance, the control lowered the sampling part of the complete coefficient by 9.7×10^{-6} (its six-loop error fell from 0.019839 to 0.019593), while the reference added $(\theta_{\mathrm{cv}}\widehat{\sigma}_P)^2 = 1.70 \times 10^{-5}$. Because the reference term exceeds the gain, the controlled error is slightly larger than the plain 0.02418404; this small difference is not evidence of a general disadvantage of controls. With the reference term included, the optimum of (746) estimated after the production is 0.0579 (same record), a diagnostic that was not used. The experiment shows that reducing the variance of a strongly correlated subintegral need not reduce the uncertainty of the complete coefficient, and that an uncertain auxiliary value cannot be treated as an exact control mean. The plain estimator (740) remained the previous three-dimensional result; the controlled value served only as a consistency check and a method benchmark.

26.15 Final checks and evidence limits

The same uncalibrated samples give the lower-order discrepancy checks

d	C_3 pull	C_4 pull	C_5 pull
2	-0.97	-1.54	-1.07
3	-1.69	-1.09	+0.92

Both rows are the 2026 productions. The reference special-function values are checks only; their numerical values are not refitted or replaced by the Monte Carlo estimates. For the previous estimators, an archived independent finite two-variable Lagrange expansion agreed with the fixed-point iteration to 7×10^{-13} at $d = 2$ and 5×10^{-14} at $d = 3$, and the delete-one-replicate bias estimates were below 3.4×10^{-7} and 6.5×10^{-8} in C_6 units; these check the inherited map $\mathcal{F}_6$, not the 2026 replicate data. Bootstrap resampling of the independent replicate vectors gives sampling errors 0.00080124 at $d = 2$ and 0.00011534 at $d = 3$ (4096 resamples each). The bootstrap is a diagnostic, not an interval certification. The bootstrap means differ from the direct estimates by -1.3×10^{-5} at $d = 2$ and $+2.1 \times 10^{-8}$ at $d = 3$.

For the previous two-dimensional estimator (743), the same uncalibrated samples gave the pulls $+0.09$, $+0.50$ and $+0.60$ for C_3, C_4 and C_5. Its replicate bootstrap, which draws the common inputs once per resample according to (742), gave the sampling error 0.14987068 from 4096 resamples, against the propagated 0.15264446.

For the previous three-dimensional estimator (740) the pulls were $+0.44$, $+0.80$ and -0.21 for C_3, C_4 and C_5, and its replicate bootstrap gave the sampling error 0.02376720 from 4096 resamples, against the propagated 0.02418404 (`verification/method_precision_results.json` and `verification/extra_checks.json` of the earlier calculation).

Verification scope. The final numbers retain the analytic power-sector compensation and the same graph/renormalization framework; narrower sampling errors do not bound shared errors in that framework. The reference C_3–C_5 values were used for comparisons only, not imposed as constraints. The reported results are $C_6(2) = 362.366 \pm 0.004$ and $C_6(3) = 33.5754 \pm 0.0005$; all earlier, less precise full-coefficient estimates and partial sums are superseded.

The method comparison of Sections 26.11 and 26.16 does not bound a shared error in graph generation, renormalization or analytic continuation, and it did not rerun the analytic power-sector compensation. Faster arithmetic and smaller sampling errors are neither an independent field-theoretic validation nor rigorous error enclosures.

26.16 Method selection and methods not adopted

Best supported choice for these integrands. The strongest tested practical combination is local finite forest subtraction (716), tropical importance sampling (705), graph-stratified randomized Sobol nets (Section 26.2), pilot-frozen cost-aware allocation (739), and the shared polynomial products (733) with monitored high-precision fallback. A separately trained transport (737) is only modestly useful at $d = 2$ and unnecessary at $d = 3$ in these tests. The runtime gain and the variance gains are different quantities and are not conflated, and the total precision gain of the refinement campaigns also comes from more production samples (Section 26.12).

Methods not benchmarked here. Rank-one shifted lattice rules with smooth inverse-CDF importance fits are an established alternative for multi-loop integrals, including CPU and CUDA implementations interfaced to pySecDec [50]. That implementation was studied but not benchmarked, so it is not declared better or worse than the present Sobol method. Global tropical sampling across primitive graphs has recently reached much higher primitive loop orders [51]; those positive primitive examples do not by themselves establish its performance for the signed, regulator-dependent renormalized forest combinations of this calculation.

Conditional preintegration. A promising next structural improvement is conditional preintegration: a subset of the coordinates is integrated exactly and only the rest, u_{s}, is sampled. If Y is

an unbiased single-point estimator, the law of total variance,

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid u_{\mathrm{s}})] + \mathrm{Var}(\mathbb{E}[Y \mid u_{\mathrm{s}}]), \tag{749}$$

has a nonnegative first term, so

$$\mathrm{Var}(\mathbb{E}[Y \mid u_{\mathrm{s}}]) \le \mathrm{Var}(Y). \tag{750}$$

This is useful computationally only when the cost of the exact inner integration and any deterministic quadrature error are controlled. No integral was replaced by an unchecked low-order quadrature, and no favorable variance reduction is assumed. The first-edge stratification of Section 26.11, which lost at equal cost, shows why a conditionalization has to be benchmarked at equal cost.

27 Full seventh order in one dimension: coordinate forests and binary-cluster integration

27.1 Result, normalization, and scope

This section continues the numerical calculation to seventh order at $d = 1$, with the unchanged convention

$$\epsilon = 2s - 1, \qquad \Delta_{\mathcal{E}} = \frac{1}{2} - \frac{\epsilon}{6} + \sum_{n=2}^{7} C_n(1)\epsilon^n + O(\epsilon^8). \tag{751}$$

The full graph-based numerical estimate is

$$\boxed{C_7(1) = -322550, \qquad \text{pre-registered empirical allowance: } \pm 41.} \tag{752}$$

The unrounded estimator is -322549.889, with propagated sampling standard error 10.088; a bootstrap over the replicates of every stream gives 10.111. The estimator uses the radial quadrature through degree six. The allowance is the output of a policy fixed before production: the largest of four sampling errors, twice the last refinement plus four of its errors, twice the degree six to seven check step plus four of its errors, and the half-split displacement, rounded up to two significant figures. It is not a rigorous enclosure, a probability guarantee, or an estimate of the error of the truncated physical ϵ series.

The policy value is not a demonstrated total accuracy. It takes the largest of its candidates rather than combining the estimated truncation with the sampling error. The measured degree six to seven step is $+10.40 \pm 0.43$, and a geometric tail beyond degree seven, extrapolated with the last step ratio 0.600, adds $+15.59$. Treating that tail as uncertain by 100% and neglecting the covariance between the estimate and the check level, the truncation-aware estimate is

$$C_7(1) \approx -322524 \pm 38, \tag{753}$$

with the allowance taken as twice the combined error $10.09 \oplus 0.43 \oplus 15.59 = 18.6$. This estimate is post hoc and depends on the tail model, whose ratio at seventh order is larger than at fifth and sixth order (Section 27.9). The intervals of (752) and (753) overlap. Establishing the total error at this level would require radial data at degree eight or an independently validated tail model. The long decimal strings in the machine-readable files specify the estimator, not certified physical digits. In particular, no independent determination of the common graph and renormalization framework is claimed.

Both logarithmically and power-subdivergent diagrams are included. No seventh-order power correction is assigned zero, and no Padé, ratio, large-order, or interpolation estimate is used to fill a missing graph class. The 44 seventh-order primitive classes are integrated directly in a separate campaign, giving -696179.472 ± 0.867. That campaign replaces the primitive contribution

-696147.446 ± 25.368 of the first, smaller seventh-order calculation (23 September 2026, $C_7(1) = -323405 \pm 5862$ sampling standard error; record `GFF/results/C7_1/v1_46batches/archive_original.json`), which it confirms within 1.26 combined standard errors. The primitive number alone is not the physical C_7.

27.2 Complete inventory and exact algebraic coverage

The eight-vertex enumeration was rebuilt from integer edge multiplicities. There are 3986 vertex-isomorphism classes, representing 125894580 labeled 1PI graphs. Of these, 1091 contain an unmarked power-divergent two-point subgraph. The seventh-order $3c_7 - b_7$ target has nonzero weight for 845 of these classes. The primitive sector has 44 classes.

For a graph with vertex automorphism group $\mathrm{Aut}(G)$, internal multiplicities m_{ij}, and external half-edge counts $e_i = 4 - \sum_j m_{ij}$, the class weights are

$$w_A(G) = \frac{24}{|\,\mathrm{Aut}(G)|\prod_i e_i! \prod_{i<j} m_{ij}!}, \qquad w_B(G) = \frac{w_A(G)}{12} \sum_i e_i(e_i - 1). \tag{754}$$

They include the vertex orbit factor and use the full coordinate domain; there is no second vertex-factorial division in the numerical integral. All weights and all forest signs are generated exactly before numerical conversion. The inventory reproduces the previously recorded class and labeled counts.

The weighted coordinate forests were independently compared with formal bare-series inversion, retaining every graph amplitude as an independent symbol and canonically factoring articulation products. The comparison is a coefficientwise rational-polynomial identity, not a test at a few numerical inputs:

Loop order	Coupling monomials	Mass monomials	Nonzero differences
1	1	1	0
2	2	2	0
3	8	6	0
4	29	25	0
5	134	103	0
6	696	525	0
7	4164	2935	0

These counts are taken after articulation products are identified; without that factorization the counts are larger.

An additional graph-by-graph exact reduction gives zero RG polynomials for 691 of the seventh-order classes. Among the 845 power classes with nonzero channel weight, 151 are such exact zeros; the remaining 694 are integrated. For the fifth- and sixth-order power inputs, respectively 24 and 132 nonzero classes remain after the same algebraic zero test. These exclusions are identities, not variance-based pruning. All other nonzero classes have positive production allocation.

The new 44 primitive representatives were matched by completed-graph isomorphism to all 14 periods used by the first calculation. Every aggregated rational weight agrees, including the three lower-period products. The final production integrates these 44 representatives directly; the agreement of that integration with those periods is quoted in Section 27.8.

27.3 One common coordinate-space finite prescription

Computing the power-free and power-containing parts in two different finite prescriptions would require a conversion between them. Here both are computed in one coordinate-space prescription.

An edgewise exponential infrared cutoff is used:

$$C_\mu(x;z)=\frac{2^{z/2}}{\sqrt{\Gamma(1+z)}}|x|^{-(1-z)/2}e^{-\mu|x|}. \tag{755}$$

The one-loop bubble is exactly

$$\int_{\mathbb{R}} C_\mu(x;z)^2\,dx=\frac{2}{z}\mu^{-z}. \tag{756}$$

The edgewise cutoff factorizes on products of subgraphs; replacing it by an independently assigned cutoff on the span of each whole graph would change finite parts and is not done here.

Let the dimensionless bare coupling be x, and write the weighted bare amplitude series as

$$\mathcal{A}(x,z)=1+\sum_{L\ge1}(-1)^L a_L(z)x^L,\qquad \mathcal{B}(x,z)=1+\sum_{L\ge1}(-1)^L e_L(z)x^L. \tag{757}$$

The finite coupling is $g=x\mathcal{A}(x,z)$. The two RG functions follow from differentiating with respect to the common scale:

$$\widehat{\beta}(g,z)+zg=-zx^2\partial_x\mathcal{A}(x,z),\qquad \widehat{\gamma}(g,z)=-zx\partial_x\log\mathcal{B}(x,z),\quad x=x(g,z). \tag{758}$$

Thus the regulator-dependent functions, not just their $z=0$ values, are required. The physical substitution is $z=\epsilon$ after the Taylor jets and finite fixed point have been assembled.

For clarity, the independent algebraic check extracts the coupling and mass coefficients directly from

$$\widehat{b}_L(z)=-\frac{z}{L+1}[x^L]\left(\sum_{j\ge1}j(j+1)(-1)^j a_j(z)x^j\right)\mathcal{A}(x,z)^{-L-1}, \tag{759}$$

$$\widehat{c}_L(z)=-\frac{z}{L}[x^{L-1}]\partial_x\left(\frac{x\partial_x\mathcal{B}(x,z)}{\mathcal{B}(x,z)}\right)\mathcal{A}(x,z)^{-L}. \tag{760}$$

The factor z is carried symbolically before Laurent cancellation. The code does not estimate these expressions by subtracting independently sampled divergent amplitudes.

27.4 Local full-graph forest integrands

For every compatible proper logarithmic forest F, replace its blocks by their quotient graphs and glue the quotients on the original labeled vertices. Let G_F be the resulting edge multigraph. Terms whose quotient is a tadpole are set to zero in the same analytically continued critical mass prescription as the preceding sections. Unmarked power subgraphs are not dropped: their coordinate integrals are continued meromorphically as described below.

For projective coordinates u, put

$$H_F(u)=\sum_{e\in G_F}|u_{e_1}-u_{e_2}|,\qquad R_F(u)=\sum_{e\in\mathrm{root}(F)}|u_{e_1}-u_{e_2}|. \tag{761}$$

The common finite RG integral for an L-loop graph is generated by

$$\mathcal{D}_G(z)=\frac{2^{Lz}\Gamma(1+Lz)}{\Gamma(1+z)^L}\int d\Omega\sum_F(-1)^{|F|}\frac{R_F(u)}{H_F(u)}H_F(u)^{-Lz}\prod_{e\in G_F}|u_{e_1}-u_{e_2}|^{-(1-z)/2}. \tag{762}$$

Then $\widehat{b}_L=(-1)^{L+1}\sum_G w_A(G)\mathcal{D}_G$ and $\widehat{c}_L=(-1)^{L+1}\sum_G w_B(G)\mathcal{D}_G$. The factor R_F/H_F is produced by the scale derivative on the root block, followed by the common scale integral. Explicitly the latter is $\int_0^\infty\lambda^{Lz}\exp(-\lambda H_F)d\lambda$. The exact bare-polynomial check above independently verifies all forest

and root weights. This is a local representation of renormalization in the spirit of scale-dependent forest formulas[1], but the specific kernel and its combinatorial certificate here are part of the present calculation.

All terms in a forest combination are evaluated on the same coordinates. For a fixed forest term the Taylor variable appears through

$$\exp\left\{z\left[\frac{1}{2}\sum_{e}\log|u_{e_1}-u_{e_2}|-L\log H_F(u)\right]\right\}. \tag{763}$$

The common Gamma factor in (762) has logarithmic Taylor coefficients

$$\ell_1 = L\log 2, \qquad \ell_j = \frac{L^j - L}{j!}\psi_{j-1}(1), \quad j \geq 2. \tag{764}$$

These are included through the required order, with all products and Laurent poles retained before the finite coefficients are extracted.

27.5 Binary vertex clusters instead of fully ranked gaps

Order the $L+1$ vertices along the line. A binary tree is obtained by splitting at the largest gap, then repeating inside each of the two subintervals. This partitions the positive-gap domain except for measure-zero ties. The internal nodes are contiguous vertex clusters; there are L of them. Let the root scale be λ, and let $t_C \in [0,1]$ be the ratio between each nonroot node's gap and its parent's gap. Then

$$\delta_C = \lambda \prod_{D\preceq C,\ D\neq\text{root}} t_D, \qquad \left|\frac{\partial(\delta_1,\dots,\delta_L)}{\partial(\lambda,(t_C))}\right| = \lambda^{L-1}\prod_{C\neq\text{root}} t_C^{|C|-2}. \tag{765}$$

The coordinate of every vertex distance is formed as a sum of positive gaps, never by subtracting two nearly equal vertex positions.

Unlike a full ranking of all gaps, this construction does not impose an unnecessary relative ordering between gaps in disjoint clusters. It has $(L+1)!\,\mathrm{Cat}_L$, with $\mathrm{Cat}_L = (2L)!/[L!(L+1)!]$, ordered tree sectors rather than $(L+1)!L!$ ranked-gap sectors, while keeping $L-1$ continuous projective variables. At seven loops there are six continuous cluster ratios. The sector sum is sampled, not exhaustively enumerated.

The tree probabilities are computed without enumerating all trees. For a vertex subset S, let $W_{\{i\}} = 1$ and

$$W_S = \frac{1}{\alpha_S}\sum_{\substack{A\sqcup B=S,\ A,B\neq\varnothing\\ \min S\in A}} 2W_AW_B, \qquad \alpha_{\text{root}} = 1. \tag{766}$$

At a split, choose the unordered pair with probability proportional to W_AW_B and then choose its orientation uniformly. The normalization is $Z = W_{\text{root}}$. A particular ordered tree has probability $1/(Z\prod_{C\neq\text{root}}\alpha_C)$.

For the continuous power-free calculation, the proposal sees every forest term, not only the original graph. With $e_F(C)$ the number of edges internal to C in G_F, choose

$$p_C = |C| - 1 - \frac{1}{2}\max_F e_F(C), \qquad \alpha_C = r\max(\tfrac{1}{2}, p_C), \qquad t_C = u_C^{1/\alpha_C}. \tag{767}$$

Here $r = 0.8$ or 0.7 in the two independent campaigns. The combined tree and radial importance weight is simply

$$Z\prod_{C\neq\text{root}} t_C^{|C|-1-\alpha_C}. \tag{768}$$

Using only the original graph's edge count can miss a more singular forest term and can give infinite sampling variance; this happens for seven-loop forests. The present construction examines the whole finite forest kernel.

The Jacobian is checked by exact rational determinants for 84 random cluster geometries. As an independent normalization check, set $\alpha_C = |C| - 1$ and remove all edge factors. The dynamic program then gives $Z = (L+1)!L$ for $L = 1, \dots, 7$, exactly the full projective volume. Explicit enumeration of the small trees confirms the proposal probabilities with arbitrary rational subset rates.

27.6 Power subgraphs: meromorphic radial integration and paired refinement

In a power-containing graph, a cluster may have negative radial exponent. It is not sampled as an ordinary convergent endpoint integral. For each sampled tree, the regular factor is interpolated on a tensor Chebyshev–Lobatto grid of degree N, and the singular powers are integrated analytically. If $L_i(t) = \sum_k \ell_{ik} t^k$, the one-variable moment is

$$\int_0^1 t^{a+bz} L_i(t)\, dt = \sum_k \frac{\ell_{ik}}{a+k+1+bz}, \tag{769}$$

under meromorphic continuation. A zero denominator at $z = 0$ supplies its Laurent pole explicitly; a negative half-integer power is continued by the same rational expression. No endpoint cutoff is fitted or sent to zero numerically. Tensor products of these moments are formed before the finite Taylor coefficient is selected.

For this discrete tree sampling only, the rates use $\alpha_C = r \max(1/2, |p_C|)$. The tree probability is positive and the inverse probability is $Z \prod_C \alpha_C$. It is a probability on a finite set of trees, not a claim that the unregularized radial power integral converges.

Let Y_N denote the complete per-tree Laurent/Taylor vector at degree N. The production estimator uses the exact telescoping organization

$$\mathbb{E}[Y_6] = \mathbb{E}[Y_1] + \mathbb{E}[Y_3 - Y_1] + \mathbb{E}[Y_5 - Y_3] + \mathbb{E}[Y_6 - Y_5]. \tag{770}$$

Within every difference, both resolutions use the same graph, binary tree, and random stream. Different levels use independent streams. This is the multilevel variance-reduction principle[52]; it reduces the noise of the refinement differences without extrapolating from lower physical coefficients. The residual difference between finite degree six and the meromorphic integral is assessed by the successive paired refinements, including the last degree-five/six comparison, not assumed to be zero by definition.

Negative Laurent coefficients of the fully subtracted RG functions vanish. Fixed linear combinations of these exact-zero quantities are used as control variates. The control matrices were fitted to separate coarse pilot samples, with a fixed regularization, and then frozen before production; they were not chosen to match any physical reference. A separate first seventh-order degree-one/three refinement run was used to freeze a cost-aware graph allocation; that training stream is excluded from the final seventh-order refinement mean. Every nonzero remaining stratum keeps positive coverage. The same matrix is applied at every level in (770). Consequently any finite-grid error in a control is also refined by the telescoping calculation. Neither C_5, C_6, nor a conjecture that the seventh-order power correction vanishes is imposed as a constraint.

27.7 Seventh-order assembly and covariance

Write

$$\widehat{\beta}(g,\epsilon) = -\epsilon g + \sum_{L=1}^{7} \widehat{b}_L(\epsilon) g^{L+1}, \qquad \widehat{\gamma}(g,\epsilon) = \sum_{L=1}^{7} \widehat{c}_L(\epsilon) g^L, \tag{771}$$

with $\widehat{b}_1 = 3$, $\widehat{c}_1 = 1$, and retain each jet through ϵ^{7-L}. Solving the finite fixed point is equivalent to

$$C_7 = [\epsilon^7] \sum_{k=1}^{7} \frac{\epsilon^k}{k} [g^{k-1}] \partial_g \widehat{\gamma}(g, \epsilon) \left[3 + \sum_{L=2}^{7} \widehat{b}_L(\epsilon) g^{L-1} \right]^{-k}. \tag{772}$$

In a regulator-independent MS scheme this reduces to $\frac{1}{7}[g^6]\gamma_r'(g)B(g)^{-7}$, but that shorter expression cannot be used on just the constant terms of the present finite-scheme jets.

For fixed one- through four-loop inputs, C_5, C_6, C_7 are affine in all fifth-, sixth-, and seventh-loop jets. Two such high-loop insertions would first contribute beyond order seven. Thus, if v_L is the vector of coupling and mass coefficients (with negative Laurent orders kept for controls),

$$\boldsymbol{C} = \boldsymbol{K}_{\le 4} + \sum_{L=5}^{7} J_L (v_L^{\text{free}} + v_L^{\text{power}}). \tag{773}$$

Here $\boldsymbol{C}$ is the vector of the coefficients C_n, $\boldsymbol{K}_{\le 4}$ its value from the one- through four-loop inputs alone, and J_L the exact matrix of its dependence on v_L. The complete matrices J_L and the lower-loop vectors are in `LRI_C7_HPC/reference/verification/full_affine_assembly.json`, and the frozen control matrices in `LRI_C7_HPC/reference/verification/power_control_training.json`. Thirty-two exact rational tests verify both (772) against an independently iterated fixed point and the affine identity (773).

All coupling/mass and Taylor-order covariances are retained. Each random replicate is projected through the full matrix before computing its covariance. The two power-free campaigns are averaged with equal weights fixed independently of their observed values. The direct primitive campaign is added only once, with variance $(0.8667)^2$. The two independent coarse power campaigns are combined with preassigned sample-count weights. Independent refinement-level covariance matrices are then added. Independent production batches of one frozen design are combined with equal weights, and their replicate covariances are propagated. The finite high-loop assembly itself has no nonlinear Monte Carlo bias, because it is affine.

27.8 Production results and refinement

The final production consists of 399 independent batches of one frozen design, each with its own seeds. The design was fixed before production from a separate pilot: the replicate counts and sample multipliers of every campaign were chosen by minimum-variance allocation at the measured pilot variances and costs, with every multiplier a power of two. Each batch contains the two independent power-free campaigns at five, six and seven loops (128 randomized replicates each); the two independent coarse power campaigns (128 and 64 replicates); the paired refinements $1 \to 3$, $3 \to 5$ and $5 \to 6$ of (770) (128, 128 and 64 replicates); and a paired $6 \to 7$ check level (64 replicates), which is never part of the estimate. The 44 primitive classes are integrated once, in a separate campaign with 2^{23} points per graph and 64 replicates. It gives -696179.472 ± 0.867, which agrees with the period-based value -696147.446 ± 25.368 of the first calculation within 1.26 combined standard errors. The production used about 2.5×10^5 core-hours.

The telescoping resolution sequence for the complete coefficient is

Power-grid degree	Full C_7 estimate	Sampling standard error
1	−331,309.98	6.14
3	−322,614.21	9.21
5	−322,567.22	10.01
6	−322,549.89	10.09

The individual refinement corrections, in absolute units of C_7, are

Paired degrees	Correction	Sampling standard error
$1 \to 3$	+8,695.78	6.86
$3 \to 5$	+46.99	3.93
$5 \to 6$	+17.33	1.21
$6 \to 7$ (check only)	+10.40	0.43

The corrections decrease, but the ratio of successive steps grows, from 0.37 for $5 \to 6$ over $3 \to 5$ to 0.60 for $6 \to 7$ over $5 \to 6$. No limit has been replaced by a fitted extrapolation in (752). The allowance candidates are $4 \times 10.09 = 40.35$, $2 \times 17.33 + 4 \times 1.21 = 39.50$, $2 \times 10.40 + 4 \times 0.43 = 22.49$ and the half-split displacement 12.58, which gives ± 41.

The scatter of the 399 batch estimates about their mean gives $\chi^2 = 348.9$ for 398 degrees of freedom, so the replicate-based errors describe the batch-to-batch variation. A bootstrap over the replicates of every stream gives the standard error 10.11, compared with the covariance value 10.09. Splitting the replicates of every stream into halves gives -322537.30 and -322562.47. The two independent power-free campaigns agree at each loop order to within 0.42 of their combined sampling errors. The largest absolute standardized residual among the 24 monitored spurious Laurent poles is 2.31. With the zero-pole controls the estimate is 71.6 ± 41.0 above the uncontrolled estimator, -322621.5 ± 41.1 (the same convention as at eighth order, Section 30.4).

27.9 Lower-order checks and the radial truncation

The same calculation gives the following uncalibrated lower-order values:

Coefficient	Degree-six value	Sampling error	Reference
$C_5(1)$	-731.174185	0.008355	-731.139477 (exact)
$C_6(1)$	14,332.435930	0.305002	14,331.687033 (deterministic)

The differences are -4.15 and $+2.46$ sampling errors. At this precision the lower orders resolve the truncation of the radial quadrature at degree six. Adding the measured degree six to seven step and a geometric tail with each coefficient's own last step ratio gives

	Degree six	+ step $6 \to 7$	+ geometric tail
$C_5(1)$, difference in errors	-4.15	-1.88	-0.30
$C_6(1)$, difference in errors	$+2.46$	$+0.97$	-0.22
$C_7(1)$, value	-322549.9	-322539.5	-322523.9

The tail ratios are 0.408 for C_5, 0.443 for C_6 and 0.600 for C_7. The lower-order agreement is consistency with the tail model at fifth and sixth order. It does not validate the seventh-order tail, which converges more slowly, and it does not exclude other systematic errors. This is the origin of the truncation-aware estimate (753). Deterministic low-loop assembly also gives $C_3 = -3.71985632290349$ and $C_4 = 45.4248801371166$; their discrepancies from the retained higher-precision values are respectively 1.5×10^{-12} and 3.4×10^{-8}.

The lower-order reference coefficients are used only as checks. They are not fitted into the estimator; in particular, $C_6(1)$ is not an input to the mean in (773). The one- through four-loop inputs are evaluated by deterministic coordinate quadrature. Changing the four-loop degree from 16 to 20 moves the seventh-order assembly by less than 0.001 with the higher-loop inputs held fixed; the preceding degree changes decrease rapidly. This is a convergence observation, not an interval bound.

The tree point evaluator was compared with an independent positive-gap reconstruction at 650 decimal digits in 60 cases, including cluster ratios exponentially smaller than the production minima.

The maximum normalized discrepancy was 1.18×10^{-14}; the extreme-endpoint subset had maximum 3.45×10^{-18}. Continuous integration uses extended precision, binary128 on strongly canceling samples, and a 100-digit fallback for more severe cancellations. The 650-digit comparison tests the values that the evaluator actually returns, including the chosen fallbacks (extended precision in 51 of the 60 cases, binary128 in one and 100 digits in eight), rather than only the uncancelled summands (`LRI_C7_HPC/reference/verification/tree_point_checks.json`). The meromorphic weights are generated at 100 working digits and stored in extended precision. Independent monomial-moment tests of the stored weights include the Laurent-pole and negative-power cases; the largest normalized error in 1152 tests is 1.1×10^{-11} (`LRI_C7_HPC/reference/verification/meromorphic_weight_checks.json`). The production implementation evaluates the radial tensor quadrature in bounded tiles with compensated accumulation. Replayed against binary128 on 112 degree-seven elements, its aggregate projected error in C_7 is 2.8×10^{-12}, compared with 1.9×10^{-11} for the original dense implementation. The worst individual elements, which cancel strongly, reach 4.2×10^{-14} relative. These errors are many orders of magnitude below the sampling error.

The two point streams are randomized differently. In the continuous power-free integration every graph stratum receives a power-of-two number of points of one linearly scrambled Sobol net, and the uncertainty is estimated between independent randomized replicates, not between the correlated points of one net [46]. The export of the direction matrices first reconstructs the sample values of the public SciPy generator and checks them exactly. The power-sector quadrature instead draws its finite tree sectors from independent 64-bit Mersenne Twister streams, seeded per batch and per graph; the paired degrees of (770) evaluate the same sampled trees, which is what makes the paired refinements small in variance. The meromorphic weight files are native extended-precision binaries; on a platform with a different extended-precision format they are regenerated from the shipped code, not read. For the first, smaller calculation, a direct fixed-point assembly of the raw Taylor coefficients, followed by the recorded zero-pole control, agreed with the affine assembly (773) to 1.8×10^{-10}.

28 Full seventh order in two dimensions: mixed local-mass forests

28.1 Result and scope

At $d=2$ the seventh-order coefficient uses the same physical regulator and finite zero-momentum prescription as the sixth-order calculation of Sections 23–26, with $\epsilon=2s-2$,

$$\Delta_{\mathcal{E}}(2,\epsilon)=1-\frac{\epsilon}{2}+\gamma_r(g_*,\epsilon),\qquad \beta(g,z)=-zg+\sum_{L=1}^{7}\widehat{b}_L(z)g^{L+1},\qquad \gamma_r(g,z)=\sum_{L=1}^{7}\widehat{c}_L(z)g^{L},\tag{774}$$

$\widehat{b}_1=3$, $\widehat{c}_1=1$ and $\beta(g_*,\epsilon)=0$. Every loop order keeps its Taylor polynomial through z^{7-L} before $z=\epsilon$ is set. Unlike the sixth-order calculation, the power-containing sector is integrated explicitly instead of being taken from an analytic compensation. Write $f_L=(b_L^{\mathrm{pf}},c_L^{\mathrm{pf}})$ and $p_L=(b_L^{\mathrm{p}},c_L^{\mathrm{p}})$ for the power-free and power-containing jets in the same prescription, and $\mathcal{F}$ for the fixed-point series extraction. Then

$$\boxed{C_7(2)=C_7^{\mathrm{pf}}(2)+\Delta_pC_7(2),\qquad \Delta_pC_7=\mathcal{F}_7(f+p)-\mathcal{F}_7(f).}\tag{775}$$

The difference includes lower-loop nonlinear changes; it is not the simple-pole sum of the seven-loop power graphs. The production of Section 28.7 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_7^{\mathrm{pf}}(2)$	−3876.0700590	0.5224502
Power correction $\Delta_pC_7(2)$	−0.0560338	0.0458637
Complete coefficient $C_7(2)$	−3876.1260928	0.5244497

The quoted uncertainty is the empirical reporting allowance

$$E_{\mathrm{emp}} = \max\Big(4\widehat{\sigma},\ \max_{j}|C_{7,j}^{\mathrm{half}} - C_7|\Big), \quad \text{rounded up,} \tag{776}$$

where $\widehat{\sigma}$ is the sampling standard error and $C_{7,j}^{\mathrm{half}}$, $j = 1, 2$, are the half-stream estimates: the estimator recomputed from the first and from the second half of the replicates of every campaign, loop and sector. The rule is that of an earlier, lower-precision integration of $C_7(2)$ with the same graph set and prescription, called the earlier $C_7(2)$ calculation, or the earlier calculation for short, in this section. The rule was applied to the 2026 production, and its rounding step chosen, after the result was available. Here $4\widehat{\sigma} = 2.098$ exceeds both half-stream deviations, 0.36, and rounding up to a multiple of 0.1 gives

$$\boxed{C_7(2) = -3876.1 \pm 2.1, \qquad \epsilon = 2s - 2.} \tag{777}$$

The earlier calculation rounded the same rule to a multiple of 10 at its precision and reported -3950 ± 310 (estimator -3953.6336215, sampling standard error 76.7310362); the new estimator lies 1.01 of its standard errors from it. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

The earlier calculation, its method and its frozen design were written by OpenAI GPT-6 astra (model stated by the author); its code and records are in `LRI_C7_D2_HPC/reference/` (Sections 28.7 and 33). When this method was merged into this section, on 16 October 2026, a Claude Code session wrote out or rechecked four parts: the three-loop check (783)–(785), for which it wrote out the graph weight, the forest cancellation (786), the bound $\mathtt{p}_3/\mathtt{p}_2 \le 1/9$ and the series identity used for the regulator derivative, and checked the quadrature and the Euler and Pfaff steps numerically; the two-loop entries (790)–(791) and the polynomial (792), which it rechecked by a separate Cauchy evaluation and an exact symbolic expansion; the primitive classes of Section 28.6, for which it supplied the argument for $w_B = 0$ and the normalization (793) and compared the 40-digit values with the closed forms; and the variance decomposition (797), which it evaluated with the 2026 record. It also checked that the group configurations of the 2026 production hold the sample counts of Section 28.7.

28.2 Graph weights and complete coverage

An L-loop graph has $L + 1$ vertices and $2L$ individually labeled lines. With edge multiplicities m_{ij} and $e_i = 4 - \deg(i)$,

$$w_A = \frac{24}{|\operatorname{Aut} G| \prod_{i<j} m_{ij}! \prod_i e_i!}, \qquad s_G = \sum_i e_i(e_i - 1), \qquad w_B = \frac{s_G}{12} w_A, \tag{778}$$

so that $3w_B - w_A = (s_G - 4)w_A/4$. This rational weight, not a sampled value, determines the zero-channel exclusions at seven loops; at lower loops the coupling and mass jets are kept separately even when the final-order channel weight vanishes. A proper one-particle-irreducible subgraph with v vertices and e internal lines is logarithmic if $e = 2v-2$ and an unmarked power-divergent two-point subgraph if $e = 2v - 1$. Its loop number is $h = e - v + 1$, so a power subgraph has $h = v$. All proper edge subgraphs, including those not induced by their vertex set, were enumerated independently; their mixed forest maps agree exactly with the production maps for all 1,301 power-containing classes at loops three to seven.

L	All classes	Power-containing	Nonzero mixed kernels	Integrated power classes
3	8	1	1	1
4	27	5	3	3
5	129	31	20	20
6	660	173	107	107
7	3986	1091	709	590

At seven loops 459 classes have $3w_B - w_A = 0$ and 639 further classes have an exactly vanishing mixed forest map. The remaining 2,254 power-free nonprimitive, 44 primitive and 590 power-containing classes are all integrated in every replicate of the production below; there is no omitted nonzero class. At loops three to six the integrated power-free and power-containing classes number 5, 16, 76, 395 and 1, 3, 20, 107.

A graph is power-free if it has no unmarked power-divergent proper two-point subgraph; nonprimitive graphs whose subdivergences are all logarithmic are power-free. The power-free sector partitions as

L	Power-free classes	Nonzero finite kernels	Primitive among them	Zero kernels
2	2	1	0	1
3	7	5	1	2
4	22	16	1	6
5	98	76	3	22
6	487	395	10	92
7	2895	2441	44	454

and at seven loops the 3,986 classes split by sector, kernel and channel weight as

Seven-loop classes	$3w_B - w_A \neq 0$	$3w_B - w_A = 0$	All
Power-free, nonzero kernel	2298	143	2441
Power-free, zero kernel	384	70	454
Power-containing, nonzero kernel	590	119	709
Power-containing, zero kernel	255	127	382
All	3527	459	3986

The first column contains the integrated classes, $2{,}298 = 2{,}254 + 44$ power-free and 590 power-containing, and the $639 = 384 + 255$ classes with a vanishing mixed forest map; the 845 power-containing classes of nonzero channel weight are $590 + 255$. The class-by-class assignment is in `LRI_C7_D2_HPC/reference/verification/full_graph_coverage.json`. The $262 = 143 + 119$ classes with a nonzero kernel and zero channel weight are not integrated here; they first contribute at eighth order (Section 31.2).

The production builders take their subgraphs from induced vertex sets. The independent enumeration of all edge subgraphs also contains the non-induced ones, for example a two-line logarithmic bubble inside a triple-line power insertion, and the exact agreement of the forest maps for all 1,301 power-containing classes shows that the forest terms of such subgraphs cancel and were not dropped by assumption; the three-loop example (786) below shows the cancellation explicitly, and Section 32.7 proves it for every graph of the family. After graph and line labels are matched, all 492 nonzero power-free kernels at loops three to six are identical to the kernels of the sixth-order calculation (Section 25; record `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/legacy_mapping.json`). A second construction from contracted graph Laplacians and exact rational Kirchhoff determinants, which does not use the monomial selection, reproduces 24 selected seven-loop power-free forest kernels exactly; the analogous construction for the power-containing kernels is the check at 66 configurations quoted below.

28.3 Local mass subtraction and the mixed forest kernel

The Schwinger infrared factor does not perform critical mass tuning: an unmarked self-energy insertion must have its local zero-momentum part subtracted, as for the three-loop graph I_3 in [7, Appendix C.3.4]. In two dimensions its superficial degree at $z = 0$ is one in momenta; the amplitude is even in the external momentum, so the constant Taylor subtraction leaves a remainder of quadratic order. Put

$$U_G(x) = \sum_T \prod_{e \notin T} x_e, \qquad S(x) = \sum_e x_e, \qquad d\Omega = \delta(1 - S(x)) \prod_e dx_e, \tag{779}$$

the sum running over spanning trees. A compatible mixed forest $F = F_{\log} \cup F_{\mathrm{p}}$ consists of nested subgraphs, or edge-disjoint subgraphs meeting in at most one vertex; nested blocks are taken after contracting their children, and U_F is the product of the block Symanzik polynomials. Equivalently, a complement-of-tree monomial m of U_G is kept in U_F if and only if $|m \cap E_\gamma| = h_\gamma$ for every $\gamma \in F$. With the logarithmic-root edge set $E_R^{\log}(F) = E_G \setminus \bigcup_{\gamma \in F_{\log}} E_\gamma$, the finite RG kernel is

$$\boxed{K_G^{\mathrm{full}}(x) = \sum_F (-1)^{|F|} \frac{\sum_{e \in E_R^{\log}(F)} x_e}{S(x)\, U_F(x)}.} \tag{780}$$

Only logarithmic subgraphs are removed from the root numerator. A power subgraph changes the denominator through its local Taylor subtraction but does not remove its interaction order from the root amplitude; excluding its lines from the numerator would implement a different scale derivative and already changes the three-loop answer. To derive (780), apply $1 - T_\gamma^0$, with T_γ^0 setting the external momentum to zero, to every compatible local two-point insertion. Gaussian integration replaces U_G by U_F for the selected power subgraphs, while the vertex interaction order and the common Schwinger mass factor are unchanged. The finite coupling and mass inversion of these critically subtracted amplitudes then gives the same logarithmic root derivative as in the power-free calculation, distributed over the projective block scales as the root fraction $\sum_{e \in E_R^{\log}} x_e / S$. Expanding both operations gives one sign for each selected subgraph. This is a local implementation of parametric forest renormalization [1]. The scale exponents of the blocks are derived in the next paragraph. Identical selected denominators are grouped with integer numerator coefficients and identically zero vectors are removed before any floating-point arithmetic. An independent rational Kirchhoff-determinant implementation agrees exactly at 66 configurations, including endpoint-scaled points.

Scale exponents of the forest blocks. This bookkeeping fixes the numerator of (780). Rescale the lines of a block B of loop number h_B by a common factor λ. The critical line measure $\prod_e x_e^{-1/2+z/4} dx_e$ and the block Symanzik polynomial, of degree h_B, give $\lambda^{|E_B|(1/2+z/4)-h_B}$. A logarithmic block has $|E_B| = 2h_B$ and exponent $h_B z/2$, the Dirichlet parameter of its relative scale in the proof of (698); this produces the root fraction and the sign of each logarithmic forest element. A power block has $|E_B| = 2h_B - 1$ and exponent $-1/2 + (2h_B - 1)z/4$ before its Taylor remainder. Its complement, with $2L - 2h_B + 1$ lines and loop number $L - h_B$, has exponent $+1/2 + (2L - 2h_B + 1)z/4$, the opposite engineering dimension, and the two add to $Lz/2$, the exponent of the whole graph. A mass Taylor operation therefore splits off no logarithmic scale: it carries no further factor h_B/L and introduces no second coupling-inversion root factor. After the logarithmic inversion the root keeps its original lines, including those of the power Taylor factors, and these form the edge set $E_R^{\log}(F)$ of (780). Nested blocks obey the same argument after their children are contracted. In the monomial selection, m is the complement-of-tree mask of the individually labeled lines, so parallel lines are separate variables.

In the normalization $r(2,z)=1$ the power-containing jets are

$$\boxed{\begin{pmatrix}\widehat{b}_L^{\mathrm{p}}(z)\\ \widehat{c}_L^{\mathrm{p}}(z)\end{pmatrix}=\frac{2(-1)^{L+1}\Gamma(1+Lz/2)}{\Gamma(\frac12+z/4)^{2L}}\sum_{G\in\mathrm{p}}\begin{pmatrix}w_A(G)\\ w_B(G)\end{pmatrix}\int d\Omega\prod_e x_e^{-1/2+z/4}K_G^{\mathrm{full}}(x),}\tag{781}$$

with the same prefactor and finite coupling prescription as the power-free data. Before local subtraction a connected power face has radial exponent $e/2-h=-1/2$ at $z=0$; the Taylor remainder raises it to $+1/2$. Logarithmic faces start at zero and their combined subtraction raises the exponent; nested limits are handled by the compatible forest. Regulator derivatives add finite powers of logarithms without removing these margins. The importance maps use the forest denominator valuations and the positive half-integer endpoint floor after subtraction.

Equation (781) is (695) with D_G of (698) and K_G replaced by K_G^{full}, since $Lz\,\Gamma(Lz/2)=2\Gamma(1+Lz/2)$: the prefactor is the logarithmic scale derivative of $\Gamma(Lz/2)$, and at $z=0$ it equals $2(-1)^{L+1}\pi^{-L}$. The power-free jets f_L are given by the same formula with K_G of (697), so no regulator-independent MS coefficient enters the input jets of (775).

Importance rates. Let $\nu_S(P)$ be the smallest degree of a polynomial P in the variables of a proper line subset S, and $N_F=\sum_{e\in E_R^{\log}(F)}x_e$. When the lines of S shrink by a common factor λ, the forest term F of (780), times the line measure at $z=0$, scales as $\lambda^{|S|/2+\nu_S(N_F)-\nu_S(U_F)}$. The builders store the rate

$$\alpha_G(S)=\tfrac12\max(r_G(S),p_S),\qquad r_G(S)=\min_F\big[\,|S|+2\nu_S(N_F)-2\nu_S(U_F)\big],\tag{782}$$

with $p_S=1$ for odd and $p_S=2$ for even $|S|$. The termwise bound $r_G(S)/2$ can be nonpositive where the forest sum cancels. Since the kernel is rational and the measure contributes $|S|/2$, the exponent of the summed kernel lies in $|S|/2+\mathbb{Z}$, and its positivity after subtraction implies the floor $p_S/2$. The floor therefore sharpens a nonpositive termwise bound without changing the integral; Section 31.5 uses the same rule at eighth order and shows that the weight then stays bounded. At seven loops the floor strengthened the rates of 81,632 even subsets in the power-free files (record `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/parity_rate_update.json`); at loops three to six the power-free rates are the frozen tables of the sixth-order calculation, mapped exactly to the new labels. The campaign factors 0.8 and 0.9 multiply these rates. All such choices change the variance, not the integral, and no interval error is derived from these power-counting arguments.

Routes not used. Subtracting the local part of a self-energy insertion is not the same as discarding the graph, and the absence of wave-function renormalization at this order is not used to delete power amplitudes: in the long-range prescription the local subtraction of a power divergence leaves a finite two-point amplitude [7]. Two other routes to the power sector were tried. A coordinate-space routing audit through the Gamma convolutions, complex Selberg specializations and five-edge kite formulas of Section 24, with all required logarithmic subgraphs and contraction quotients, closed structurally for 740 of the 845 power-containing seven-loop classes of nonzero channel weight and did not close for the other 105. It was an audit of routes, not an evaluation, and no class was assigned zero. An exploratory radial integration by parts did not implement the local subtraction: it recovered a constant term, but its regulator derivatives still produced divergent endpoint integrals. It was rejected, and none of its samples enters a result. The kernel (780) instead performs the local subtraction and the logarithmic forest combinations before any sampling.

28.4 An independent three-loop check

The first power graph G has three external-chain parameters with sum R and internal parameters y_1,y_2,y_3. With $\mathsf{P}_2=y_1y_2+y_1y_3+y_2y_3$, $\mathsf{P}_3=y_1y_2y_3$, $U_G=R\mathsf{P}_2+\mathsf{P}_3$ and $U_0=R\mathsf{P}_2$, its mixed

kernel (780) is exactly

$$K_G^{\mathrm{full}} = \frac{1}{U_G} - \frac{1}{U_0} = -\frac{\mathsf{P}_3}{R\mathsf{P}_2(R\mathsf{P}_2 + \mathsf{P}_3)}, \tag{783}$$

a rational remainder that can be integrated and differentiated at $z = 0$. Let $a = \frac{1}{2} + z/4$, $k = 3a - 1$, and use simplex variables u with $\mathsf{p}_2 = u_1u_2 + u_1u_3 + u_2u_3$ and $\mathsf{p}_3 = u_1u_2u_3$. Integrating the external parameters and the radial variable, and applying a Pfaff transformation, gives

$$\widehat{b}_3^{\mathrm{p}}(z) = -\frac{\Gamma(1 + 3z/2)\,\mathrm{B}(k,k)}{\Gamma(a)^3\Gamma(3a)} \int d\Omega_u\, \mathsf{p}_3^z \mathsf{p}_2^{-3a}\, {}_2F_1\!\left(\frac{3z}{2}, k; 2k; 1 - \frac{\mathsf{p}_3}{\mathsf{p}_2}\right), \tag{784}$$

with $w_A = 1/2$ included. At $z = 0$ the hypergeometric function is one and $\int d\Omega_u\, \mathsf{p}_2^{-3/2} = 2\pi$, so $\widehat{b}_3^{\mathrm{p}}(0) = -4$ and $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$, in agreement with the subtracted I_3 normalization of [7]. The first regulator coefficient is

$$[z]\widehat{b}_3^{\mathrm{p}}(z) = -\frac{2}{\pi} \int d\Omega_u\, \mathsf{p}_2^{-3/2} \left[-\frac{3}{2} + \log \mathsf{p}_3 - \frac{3}{4}\log \mathsf{p}_2 + 3\log \frac{2}{1 + \sqrt{\mathsf{p}_3/\mathsf{p}_2}}\right] = 18.6743207750, \tag{785}$$

from two independently tightened deterministic quadratures differing by 4.6×10^{-10}; this reference uses neither the graph sampler nor its monomial arrays. The production of Section 28.7 gives $\widehat{b}_3^{\mathrm{p}}(0) = -3.99999979 \pm 0.00000042$ and $[z]\widehat{b}_3^{\mathrm{p}} = 18.6743200 \pm 0.0000021$, at $+0.50$ and -0.36 sampling standard errors from these values.

Derivation of (783)–(785)**.** The polynomial $U_G = R\mathsf{P}_2 + \mathsf{P}_3$ fixes the topology of the single three-loop power-containing class. Four vertices lie on a ring. One adjacent pair is joined by the three lines y_1, y_2, y_3, the sunset γ, a power subgraph with $v = 2$, $e = 3$ and $h = 2$. The other three lines are in series, so their parameters enter U_G only through their sum R, and each of their two inner vertices carries two external legs. Hence $|\operatorname{Aut} G| = 2$, the triple line contributes 3! and the legs 2! 2!, so $w_A = 24/48 = 1/2$, and $s_G = 4$ gives $w_B = w_A/3$, which is the statement $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$.

The kernel. The only power subgraph is γ. The logarithmic subgraphs are the three bubbles $\beta_{ij} = \{y_i, y_j\}$ inside γ, which are not induced. Two bubbles share a line and are not compatible, so the compatible forests are $\varnothing$, $\{\gamma\}$, $\{\beta_{ij}\}$ and $\{\beta_{ij}, \gamma\}$. For $\varnothing$ and $\{\gamma\}$ the numerator is S, since a power subgraph removes no line, and the terms are $1/U_G$ and $-1/U_0$ with $U_0 = U_\gamma U_{G/\gamma} = \mathsf{P}_2 R$. For a bubble, $U_{\beta_{ij}} = y_i + y_j$, and G/β_{ij} is the one-vertex join of the loop y_k and the contracted chain, $\{i, j, k\} = \{1, 2, 3\}$. Since γ/β_{ij} is the loop y_k and G/γ the chain,

$$U_{\{\beta_{ij}\}} = (y_i + y_j)\, y_k R = U_{\{\beta_{ij}, \gamma\}}, \tag{786}$$

which are also the monomials of U_G with exactly one of y_i, y_j. Both forests have the numerator $R + y_k$ and they carry opposite signs, so they cancel, and (783) remains.

The integral. With $a = \frac{1}{2} + z/4$ the line measure is $\prod_e x_e^{a-1} dx_e$. Integrating the chain parameters at fixed R gives $R^{3a-1}\Gamma(a)^3/\Gamma(3a)$. Put $R = r$ and $y_i = (1 - r)u_i$ with $\sum_i u_i = 1$. Then $\mathsf{P}_2 = (1 - r)^2 \mathsf{p}_2$, $\mathsf{P}_3 = (1 - r)^3 \mathsf{p}_3$, the Jacobian is $(1 - r)^2$, $\prod_i y_i^{a-1} = (1 - r)^{3a-3} \prod_i u_i^{a-1}$, and $K_G^{\mathrm{full}} = -\mathsf{p}_3/[r(1 - r)\mathsf{p}_2(r\mathsf{p}_2 + (1 - r)\mathsf{p}_3)]$. With $k = 3a - 1$ the r integral is Euler's,

$$-\frac{\mathsf{p}_3}{\mathsf{p}_2} \int_0^1 \frac{r^{k-1}(1 - r)^{k-1}}{r\mathsf{p}_2 + (1 - r)\mathsf{p}_3}\, dr = -\frac{\mathrm{B}(k,k)}{\mathsf{p}_2}\, {}_2F_1\!\left(1, k; 2k; 1 - \frac{\mathsf{p}_2}{\mathsf{p}_3}\right). \tag{787}$$

Its argument is at most -8, because $\mathsf{p}_3/\mathsf{p}_2 = \left(\sum_i u_i^{-1}\right)^{-1} \le 1/9$. The Pfaff transformation ${}_2F_1(1, k; 2k; w) = (1 - w)^{-k}\, {}_2F_1(2k - 1, k; 2k; w/(w - 1))$, with $1 - w = \mathsf{p}_2/\mathsf{p}_3$, $w/(w - 1) = 1 - \mathsf{p}_3/\mathsf{p}_2 \in [8/9, 1)$ and $2k - 1 = 3z/2$, turns the u integrand into $-\mathrm{B}(k,k)\, \mathsf{p}_2^{-1-k} \mathsf{p}_3^k \prod_i u_i^{a-1}\, {}_2F_1(3z/2, k; 2k; 1 - \mathsf{p}_3/\mathsf{p}_2)$. This equals $-\mathrm{B}(k,k)\, \mathsf{p}_3^z \mathsf{p}_2^{-3a}\, {}_2F_1(3z/2, k; 2k; 1 - \mathsf{p}_3/\mathsf{p}_2)$ because $\prod_i u_i = \mathsf{p}_3$ and $4a - 2 = z$. With the prefactor $2\Gamma(1 + 3z/2)/\Gamma(a)^6$ of (781) at $L = 3$, the weight $w_A = 1/2$ and the chain factor $\Gamma(a)^3/\Gamma(3a)$, this is (784).

The two coefficients. At $z=0$ one has $a=k=\frac12$, $\mathrm{B}(\frac12,\frac12)=\pi$ and $\Gamma(\frac12)^3\Gamma(\frac32)=\pi^2/2$, so the scalar prefactor is $-2/\pi$, and $\int d\Omega_u\,\mathsf{p}_2^{-3/2}=2\pi$ gives $\widehat{b}_3^{\mathrm{p}}(0)=-4$. The logarithmic derivative of the scalar prefactor at $z=0$ is $\frac34[\psi(\frac12)-\psi(\frac32)]=-\frac32$, and $\mathsf{p}_3^z\mathsf{p}_2^{-3a}$ contributes $\log\mathsf{p}_3-\frac34\log\mathsf{p}_2$. Since $(3z/2)_n=O(z)$ for $n\ge1$, the z dependence of k enters the hypergeometric function only at order z^2, and for $0\le\xi<1$

$$ {}_2F_1\Big(\frac{3z}{2},\frac12;1;\xi\Big)=1+\frac{3z}{2}\sum_{n\ge1}\frac{(\frac12)_n}{n!\,n}\,\xi^n+O(z^2)=1+3z\log\frac{2}{1+\sqrt{1-\xi}}+O(z^2), $$

because $\sum_{n\ge1}(\frac12)_n\xi^n/(n!\,n)=\int_0^\xi[(1-t)^{-1/2}-1]\,dt/t=2\log[2/(1+\sqrt{1-\xi})]$. At $\xi=1-\mathsf{p}_3/\mathsf{p}_2$ the three contributions give (785).

Quadrature. The integrand is symmetric in u. On the sector $u_1\ge u_2\ge u_3$ put $u=(1,t,tv)/(1+t+tv)$ with $t,v\in(0,1)$, and $t=\rho^2$. Then $d\Omega_u\,\mathsf{p}_2^{-3/2}=2(1+v+\rho^2v)^{-3/2}d\rho\,dv$, so the six sectors give the weight $12(1+v+\rho^2v)^{-3/2}d\rho\,dv$, free of the algebraic corner singularity, and the logarithms of (785) remain integrable. The adaptive error estimates of the two deterministic quadratures are not interval enclosures. In these variables a separate tanh-sinh quadrature at 20 digits reproduces 2π and 18.6743207750, and (787) and the Pfaff step were confirmed numerically at a sample point.

28.5 Regulator jets and physical assembly

On a projective chart put $h(x)=\frac14\sum_e\log x_e-\frac{L}{2}\log S(x)$. The scalar normalization logarithm has Taylor coefficients

$$ a_{L,j}=\frac{(L/2)^j\psi_{j-1}(1)-2L(1/4)^j\psi_{j-1}(\frac12)}{j!}, \tag{788} $$

and the integrand jets follow from $e_0=1$ and

$$ e_n(x)=\frac1n\sum_{j=1}^{n}j\big(a_{L,j}+\delta_{j1}h(x)\big)e_{n-j}(x),\qquad n\le7-L, \tag{789} $$

so the maximum orders at loops two to seven are $5,4,3,2,1,0$. The two-loop input is the unchanged exact-hypergeometric finite-scheme input and the one-loop input is exact. With all jets present, $g(\epsilon)=\frac13[\epsilon-\sum_{L=2}^{7}\widehat{b}_L(\epsilon)g(\epsilon)^L]$ mod ϵ^8 is solved by eight iterations from $g=0$ and substituted in (774). An independent bivariate Lagrange extraction agrees exactly in 24 regulator-dependent rational cases, and 24 further cases agree with the constant-coefficient MS seventh-order polynomial; these test the assembly, not the sampled data. No higher-order coefficient is extrapolated from $C_0,\dots,C_6$.

By (702) and (199) at $d=2$, where $r(2,z)=1$ and $\Gamma(d/2)=1$,

$$ [z^j]\,\widehat{b}_2(z)=-6\,[z^{j+1}](4F_1(2,z)-2W(2,z)F_2(2,z)),\qquad \widehat{c}_2=\tfrac16\widehat{b}_2,\qquad j=0,\dots,5. \tag{790} $$

Seventh order needs, beyond (703),

$$ b_{2,5}=175.39598710648998744915099527723636\ldots. \tag{791} $$

Two Cauchy evaluations at 75 working digits, one with radius 0.05 and 80 nodes and one with radius 0.08 and 96 nodes, agree to 1.75×10^{-68} over the six entries (`LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/two_loop_extension.json`; the stored input is `inputs/two_loop.json` of the same folder). This is a numerical consistency check, not an interval bound. A separate 60-digit Cauchy evaluation, with radii 0.05 and 0.08 and 48 and 64 nodes, reproduces $b_{2,0},\dots,b_{2,6}$, including the eighth-order entry of Section 31.2.

For comparison only: in a regulator-independent MS scheme with $b_2 = -6A$, $c_2 = -A$ and $A = A_2 = 4\log 2$ (176), the Lagrange formula (593) at $n = 7$ gives the seventh-order counterpart of (594),

$$\begin{aligned} 6561\, C_7 = {} & 3c_7 - b_7 + 12672A^6 - 3360A^4 b_3 + 4320A^4 c_3 - 672A^3 b_4 + 1152A^3 c_4 \\ & + 192A^2 b_3^2 - 432A^2 b_3 c_3 - 112A^2 b_5 + 240A^2 c_5 + \frac{128}{3} A b_3 b_4 - 64A b_3 c_4 \\ & - 48A b_4 c_3 - 14A b_6 + 36A c_6 - \frac{4}{3} b_3^3 + 4b_3^2 c_3 + \frac{8}{3} b_3 b_5 - 5 b_3 c_5 \\ & + \frac{4}{3} b_4^2 - 4 b_4 c_4 - 3 b_5 c_3. \end{aligned} \tag{792}$$

It is the polynomial of the 24 constant-coefficient test cases above. An exact symbolic expansion of (593) reproduces all 23 monomials of (792) and, at $n = 6$, (594). The polynomial must not be applied to the constant terms $\widehat{b}_{L,0}, \widehat{c}_{L,0}$ of the finite-scheme jets alone, which define a different truncated problem; the finite-scheme coefficient is extracted from the full jets, and independently by (772), whose form does not depend on d. In both schemes the seven-loop constants enter linearly, with $\partial C_7/\partial(b_7, c_7) = (-1, 3)/6561$, so that $C_7(2) \supset (3\widehat{c}_{7,0} - \widehat{b}_{7,0})/6561$ and the classes with $3w_B = w_A$ drop out at this order.

28.6 Primitive classes: completed periods and three products

A primitive class has $w_B = 0$: a vertex with $e_i \geq 2$ would leave the other L vertices with at least $2L - 2$ lines, a logarithmic subgraph, so each of four vertices carries one external leg and $s_G = 0$. For a primitive graph the forest sum (694) has only its root term, so by (695) and (611), with $a_G = (2/7)\Pi_G$,

$$C_{7,\mathrm{prim}}(2) = \frac{(3c_7 - b_7)_{\mathrm{prim}}}{6561} = -\frac{2}{6561} \sum_{G \text{ primitive}} w_A(G)\, \Pi_G. \tag{793}$$

Completing each of the 44 classes to a four-regular graph on nine vertices partitions them into 14 isomorphism classes, with periods $\Pi_0^{(7)}, \ldots, \Pi_{13}^{(7)}$; the equality of periods within a class is the completion invariance of Section 23. In the zero-based order of the completion dictionary `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/reference/primitive_completion.json`, the summed weights $\bar{w}_j = \sum w_A$ are

$$(\bar{w}_0, \ldots, \bar{w}_{13}) = \Big(27, \tfrac{27}{2}, 27, 18, 3, 108, 18, \tfrac{27}{4}, 108, 12, 108, 54, 54, 12\Big).$$

Three completed graphs have three-vertex separators: $\Pi_6^{(7)}$ has one, splitting it into two octahedra (four-loop factors); $\Pi_{12}^{(7)}$ has one, splitting it into K_5 and the completed five-loop primitive; and $\Pi_7^{(7)}$ has two, splitting it into three copies of K_5. The factorization behind (616), applied once or twice, gives

$$\Pi_6^{(7)} = P_4(2)^2, \qquad \Pi_7^{(7)} = P_3(2)^3 = 27 Q_2^3, \qquad \Pi_{12}^{(7)} = P_3(2) P_5(2) = 3 Q_2 P_5(2), \tag{794}$$

and therefore

$$\boxed{C_{7,\mathrm{prim}}(2) = -\frac{2}{6561}\Big[18 P_4(2)^2 + \frac{729}{4} Q_2^3 + 162\, Q_2 P_5(2) + \sum_{j \notin \{6,7,12\}} \bar{w}_j \Pi_j^{(7)}\Big].} \tag{795}$$

The remaining sum has eleven periods. The representatives, the weights and the map from the 44 classes to the 14 periods are in the completion dictionary and in `verification/primitive_completion_checks.json` of the same folder. At 40 working digits

$$\begin{aligned} Q_2 &= 12.771381198646957657749763684524008 4\ldots, \\ P_4(2) &= 250.487315581867285531851756887911407\ldots, \\ P_5(2) &= 1548.082640233120615132439073945789 34\ldots, \end{aligned}$$

and the three product terms of (795) contribute $-1436.355740622534051652319212811748 43\ldots$ to $C_7(2)$. These values of Q_2 and $P_4(2)$ agree with (268) and with the fifty-digit $P_4(2)$ of Section 17.2, and solving (272) for $P_5(2)$ with the 36-digit $C_5(2)$ of Section 20.9 reproduces $P_5(2)$ to 34 digits.

The earlier calculation used (795) as follows. Two independently randomized integrations of all 44 classes, each with 16 replicates of 4,096 points per class, measured the three product blocks; all six comparisons with the exact references lie within 1.88 sampling standard errors. The measured blocks were then replaced by the references, which gave the primitive contribution $-8657.4262196 \pm 1.2165092$ (the equal-weight mean of the two integrations, with its sampling standard error). It entered the earlier estimate once, as a retained input. The 2026 production integrates the 44 classes directly in every replicate and does not use the reduction, so (795) remains an exact identity available as a check. The product references have negligible numerical error on the Monte Carlo scale; no interval certification is asserted.

28.7 The 2026 production

The production uses an optimized, restartable reimplementation of the same calculation. Its kernel keeps the reference addition order and the extended-precision first pass, with promotions to binary128 and to the final tier at the unchanged thresholds 10^{-9} and 10^{-23}. The final tier uses 350-decimal GMP arithmetic instead of the original 300-decimal tier; this raises the working precision and was fixed before the production. On the production host both backends reproduce the original executable at all 3,511 graph kernels with zero normalized difference and identical promotion counts, and pass the independent high-precision point checks. Results do not depend on the thread count: 5,776 seven-loop rows run at 64 and at 16 threads were identical, including 31,663 promotions to the GMP tier.

The original executable of the earlier calculation was itself checked against independently ordered high-precision monomial sums: 69 point comparisons for power-free and 174 for power-containing kernels, with maximum errors, normalized by one plus the sum of the absolute forest terms, of 1.64×10^{-32} and 2.0×10^{-32} in binary128 and of 1.85×10^{-302} and 1.3×10^{-302} in the 300-decimal tier. It used compensated accumulation and strict floating-point flags, without fast-math, and shared monomial products in place of repeated exponentials, as in (733), which changes the rounding order but neither the precision nor the expression. These are pointwise arithmetic checks, not bounds on an integral.

Two independent campaigns a and b have 64 randomized replicates for every loop and sector, 52- and 60-bit digital coordinates, and importance rates multiplied by 0.8 and 0.9. Every stratum receives its frozen per-graph allocation from the earlier pilots, at least four points, multiplied by 2^{12}. The 44 primitive classes, which the earlier calculation retained from a separate integration, are integrated with 2^{26} points each per replicate. Counts are powers of two, the base seed is fresh, and a cost model from an independent pilot affects only the placement of tasks. The two campaigns contain 1,094,688,178,176 kernel samples; they ran on the symmetry cluster on 14 AMD and 14 Intel nodes in about 6.5 hours, and every merge and the assembly used the package's own tools.

The frozen design. The per-graph counts were fixed from pilot runs that never entered a result. Each stratum receives a count proportional to the square root of its projected contribution to the variance of C_7 per unit cost, the rule (739) with C_7 in place of C_6, rounded upward to a power of two and frozen before production, separately in the power-free and in the power-containing sector. Per replicate the frozen counts are

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	Total
Power-free, nonprimitive	10816	24704	97600	188480	251472	573072
Power-containing	32768	163840	226304	130688	240384	793984

together with 4,096 points for each primitive class (records `LRI_C7_D2_HPC/reference/LRI_C7_d2_progress/verification/frozen_allocation.json` and `LRI_C7_D2_HPC/reference/power_mass/inputs/production_plan.json`). With the factor 2^{12} and the 2^{26} primitive points of the 2026 production, a replicate has $2^{12}(573{,}072 + 793{,}984) + 44 \cdot 2^{26}$ points, and the two campaigns of 64 replicates contain the 1,094,688,178,176 samples quoted above; the group configurations `GFF/data/C7_2/run1-20260930-m4096/groups/*/config.json` hold these counts. The earlier calculation used the frozen counts themselves, with 16 replicates in each of two campaigns per sector, rate multipliers 0.8 and 0.9, and 24,105,472 power-free and primitive and 25,407,488 power-containing kernel evaluations. Its power-free coefficient $-3955.383753 \pm 76.541680$ (sampling standard error) includes the primitive contribution of Section 28.6 once; adding the power correction 1.750131 ± 5.424390 (sampling standard error) gave its estimate. There $4\widehat{\sigma} = 306.9$ exceeded both half-stream deviations, 89.3, and rounding up to a multiple of 10 gave the allowance 310 of Section 28.1.

The campaign means are combined with fixed weights 1/2. With J_L and $J_{f,L}$ the Jacobians of $\mathcal{F}$ at $f + p$ and at f, and $V_{f,L}, V_{p,L}$ the covariances of the campaign-mean input vectors, the complete coefficient and the correction have covariances

$$V_C = \sum_L J_L(V_{f,L} + V_{p,L})J_L^{\mathsf{T}}, \qquad V_\Delta = \sum_L (J_L - J_{f,L})V_{f,L}(J_L - J_{f,L})^{\mathsf{T}} + \sum_L J_L V_{p,L} J_L^{\mathsf{T}}. \tag{796}$$

The earlier calculation added a retained primitive variance once; the 2026 production has no shared retained input. The two campaigns and the two half streams give

Estimate	$C_7(2)$	Sampling SE
Campaign a	-3875.3202	0.7266
Campaign b	-3876.9320	0.7565
First half stream	-3876.4894	
Second half stream	-3875.7627	
Combined	-3876.1261	0.5244

The campaigns differ by 1.54 combined standard errors. The error is dominated by the power-free seven-loop graphs:

Sector	$L = 3$	$L = 4$	$L = 5$	$L = 6$	$L = 7$
Power-free	0.0007	0.0047	0.0200	0.0590	0.5187
Power	0.0000	0.0001	0.0013	0.0130	0.0440

Entries are in units of C_7 and combine in quadrature.

28.8 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6, with the two-dimensional production of Section 26.7:

Order	Estimate	Sampling SE	Pull
3	-0.8748513621	1.25×10^{-6}	-1.08
4	5.0488792961	3.01×10^{-5}	$+0.36$
5	-38.8344913344	5.07×10^{-4}	-1.67
6	362.3727456	0.0085778	$+0.73$

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. The measured power corrections at orders four to six are $(5.4 \pm 15.1) \times 10^{-8}$, $(1.10 \pm 1.85) \times 10^{-5}$ and $(3.3 \pm 11.8) \times 10^{-4}$; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at seventh order, where the correction is -0.056 ± 0.046. A bootstrap of 4,096 replicate resamples gives standard error 0.52219, against 0.52445 from covariance propagation. The new value lies 1.01 standard errors of the earlier calculation from its estimator, and its power correction lies 0.33 of that calculation's standard errors from its value 1.750131 ± 5.424390 (sampling standard error). Because the power sector is integrated explicitly here, the sixth-order by-product also tests the analytic power-sector compensation used at sixth order. These checks share the graph and renormalization framework and do not validate it. The eighth-order calculation of Section 31 gives $C_7(2) = -3875.55 \pm 0.29$ (sampling standard error) as a by-product, +0.97 combined standard errors from (777), and the independent code of Section 31.7 gives -3875.47 ± 0.35, +1.04.

Besides (796), the linearization $\delta C_7 = J\,\delta(f+p) = J_f\,\delta f + [(J - J_f)\,\delta f + J\,\delta p]$ gives the power-free variance and the cross-covariance of the power-free coefficient with the correction,

$$V_{\rm pf} = \sum_L J_{f,L} V_{f,L} J_{f,L}^{\mathsf{T}}, \qquad V_{f\Delta} = \sum_L J_{f,L} V_{f,L} (J_L - J_{f,L})^{\mathsf{T}}, \qquad V_C = V_{\rm pf} + V_\Delta + V_{f\Delta} + V_{f\Delta}^{\mathsf{T}}, \tag{797}$$

where the last relation is an exact identity. The record `GFF/results/C7_2/C7_result.json` gives $V_{f\Delta} = -5.09 \times 10^{-6}$ for C_7, a correlation of -2×10^{-4}, and $0.5224502^2 + 0.0458637^2 - 2 \cdot 5.09 \times 10^{-6} = 0.5244497^2$ reproduces the sampling standard error of Section 28.1. In the earlier calculation a retained primitive variance $\sigma^2_{\rm prim}$ entered V_C and $V_{\rm pf}$ once and canceled from V_Δ, so neither the old power-free error nor the common primitive error was averaged down by its two power campaigns.

29 Full seventh order in three dimensions

29.1 Result and scope

At $d = 3$, with $\epsilon = 2s - 3$, the seventh-order coefficient follows the construction of Section 28 with the three-dimensional kernel and normalization of Section 25. With β and γ_r as in (774), $\widehat{b}_1 = 3$ and $\widehat{c}_1 = 1$,

$$\Delta_{\mathcal{E}}(3, \epsilon) = \frac{3}{2} - \frac{\epsilon}{2} + \gamma_r(g_*, \epsilon), \tag{798}$$

and every loop order keeps its Taylor polynomial through z^{7-L} before $z = \epsilon$ is set. As at $d = 2$, and unlike the three-dimensional sixth-order production of Section 26.8, the power-containing sector is integrated explicitly, and $C_7(3)$ is split as in (775). The production of Section 29.3 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_7^{\rm pf}(3)$	-220.3723574	0.0277073
Power correction $\Delta_p C_7(3)$	0.0178769	0.0156478
Complete coefficient $C_7(3)$	-220.3544805	0.0318188

An earlier three-dimensional seventh-order calculation with the same worker, whose code and records are kept in `LRI_C7_D3_HPC/reference/` (Section 33), reported four sampling standard errors rounded upward, with its two campaigns and its two half streams (the estimates from the first and from the second half of the replicates of every group) as diagnostics. The same rule, rounded up to a multiple of 0.01, was written down before the production result was available and confirmed unchanged afterwards. Here $4\widehat{\sigma} = 0.127$, which gives

$$\boxed{C_7(3) = -220.35 \pm 0.13, \qquad \epsilon = 2s - 3.} \tag{799}$$

Both half-stream estimates deviate from $C_7(3)$ by 0.034, less than $4\widehat{\sigma}$, so the two-dimensional rule (776) gives the same allowance. The earlier $C_7(3)$ calculation reported -228 ± 16 (estimator -227.6181095, sampling standard error 3.8273969); the present estimator lies 1.90 of its standard errors from it. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

29.2 Kernel, normalization and coverage

The mixed forest kernel is (780) with every forest denominator U_F replaced by $U_F^{3/2}$, as in (707); the fractional power is applied to each forest term before the terms are summed. The edge measure is $\prod_e x_e^{-1/4+z/4}$ and the normalization is that of (708) and (710)–(712), now with Gamma jets through order $7-L$. The integrand jets follow (789) with $a_{L,j}$ now taken from (711); the logarithmic function $h(x)$ coincides with $s_L(x)$ of (710). In three dimensions an unmarked self-energy insertion has superficial degree 3/2 in momenta at $z = 0$. The amplitude is even in the external momentum, so the constant Taylor subtraction again leaves a remainder of quadratic order and the local mass subtraction of Section 28 applies unchanged. Before subtraction a connected power face has radial exponent $3e/4 - 3h/2 = -3/4$ at $z = 0$; the Taylor remainder raises it to $+1/4$.

The graph classes, rational weights, zero-channel exclusions and forest maps are those of Section 28. The integrated classes at loops three to seven number $5, 16, 76, 395$ and 2,298 (2,254 nonprimitive and 44 primitive) in the power-free sector and $1, 3, 20, 107, 590$ in the power-containing sector, 3,511 kernels in all. For every one of them the denominator polynomials and weights were checked to be unchanged from the two-dimensional inventory; only the dimension-dependent exponents differ. The two-loop input is (713), now through z^5, with $b_{2,5} = 26.0214190551330710$ in addition to (714); two 65-digit evaluations with radii $0.04, 0.07$ and 64,80 Cauchy nodes agree to 3.3×10^{-57}, and with the sixth-order input to 1.3×10^{-54}. Thirty independent Cauchy evaluations of the Gamma factors agree to 8.3×10^{-54}. The fixed point is solved as in Section 28 and substituted in (798); 64 exact rational cases test the assembly. The seven-loop constants enter only as $C_7(3) \supset (3\widehat{c}_{7,0} - \widehat{b}_{7,0})/6561$.

The single three-loop power graph has the closed form $\widehat{b}_3^{\mathrm{p}}(0) = -16\pi/15 = -3.3510322$, recorded with the earlier calculation, and $\widehat{c}_3^{\mathrm{p}} = \widehat{b}_3^{\mathrm{p}}/3$. The production below gives $\widehat{b}_3^{\mathrm{p}}(0) = -3.3510653 \pm 0.0000262$, -1.26 sampling standard errors from this value.

29.3 The 2026 production

The production worker is byte-identical to that of the earlier calculation. It keeps the extended-precision first pass, with promotions to binary128 and to 350-decimal GMP arithmetic at the unchanged thresholds 10^{-9} and 10^{-23}. Before the campaign, the worker replayed all 168,528 archived production rows on their archived nets and seeds. Of these, 96% reproduced bit for bit and the rest within 1.0×10^{-11} in normalized units, with identical promotion counts. The rebuilt estimator of the earlier calculation was -227.61810947469922, against its recorded value -227.61810947469945. An independent high-precision reimplementation of the sampler and integrand reproduced 13 archived rows across loops three to seven and both sectors. Results do not depend on the thread count: 5,776 seven-loop rows run at 64 and at 16 threads were identical, including 255,396 binary128 and 20,511 GMP promotions. The same rows computed on AMD and on Intel processors differ by at most 2.0×10^{-11} in normalized units, with identical promotion counts; this is about 10^{-7} of the relative statistical error.

Two independent campaigns a and b have 64 randomized replicates for every loop and sector, 52- and 60-bit digital coordinates, and importance rates multiplied by 0.8 and 0.7. Every stratum receives the frozen per-graph allocation of the earlier calculation, at least 64 points, multiplied by 2^{12}. The exception is the 44 seven-loop primitive classes. They are integrated with 2^{22} points

each per replicate, a flat allocation in place of their frozen counts, which changes only the variance. Counts are powers of two and the base seed is fresh. The two campaigns contain 867,784,720,384 kernel samples. They ran on the symmetry cluster on 14 AMD and 14 Intel nodes and completed on 1 October 2026; every merge and the assembly used the package's own tools. The covariances follow (796), and there is no shared retained input. The two campaigns and the two half streams give

Estimate	$C_7(3)$	Sampling SE
Campaign a	-220.3220	0.0431
Campaign b	-220.3870	0.0468
First half stream	-220.3207	
Second half stream	-220.3883	
Combined	-220.3545	0.0318

The campaigns differ by 1.02 combined standard errors. The error is dominated by the seven-loop graphs of both sectors:

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$
Power-free	0.0000	0.0006	0.0034	0.0119	0.0248
Power	0.0000	0.0002	0.0014	0.0058	0.0145

Entries are in units of C_7 and combine in quadrature.

29.4 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6, with the three-dimensional production of Section 26.8, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.3535806960	3.59×10^{-7}	$+2.30$
4	1.2280727249	8.52×10^{-6}	$+1.08$
5	-5.8503454757	1.52×10^{-4}	$+0.04$
6	33.5738672	0.0019782	-0.79

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. A pull above three in magnitude would have been reported without changing $C_7(3)$; none occurred. The measured power corrections at orders four to six are $(2.0 \pm 2.1) \times 10^{-6}$, $(4.0 \pm 6.4) \times 10^{-5}$ and $(-3.7 \pm 10.2) \times 10^{-4}$; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at seventh order, where the correction is 0.018 ± 0.016. A bootstrap of 4,096 replicate resamples gives standard error 0.03204, against 0.03182 from covariance propagation. The power correction lies 1.92 standard errors of the earlier calculation from its value -3.836223 ± 2.007117. Because the power sector is integrated explicitly here, the sixth-order by-product also tests the analytic power-sector compensation used for $C_6(3)$. These checks share the graph and renormalization framework and do not validate it.

30 Full eighth order in one dimension

30.1 Result and scope

The eighth-order coefficient uses the coordinate-forest construction of Section 27, extended by one order:

$$\boxed{C_8(1) = 8059767, \qquad \text{empirical allowance: } \pm 17000,} \tag{800}$$

that is, $C_8(1) = (8.060 \pm 0.017) \times 10^6$. The propagated sampling standard error is 4210; a bootstrap over the replicates of all independent streams gives 4140. The estimator is the mean of fourteen independent production batches of one frozen design (Section 30.3). Their individual values lie between 8041935 and 8089624, each with standard error between 15090 and 16343, and give $\chi^2 = 8.49$ for 13 degrees of freedom about the combined value. The allowance rule was fixed before production and was not changed: the largest of four sampling errors, twice the last refinement plus four of its errors, twice the degree eight to nine check step plus four of its errors, the half-split displacement and twice the last degree step of the exact loop-five input, rounded up to two significant figures. The four-error candidate, $4 \times 4209.58 = 16838.3$, is the largest. The number of production batches was not fixed in advance. Under the stopping rule set first, nine batches give 8061480 ± 5300 with allowance ± 22000; all fourteen planned batches were completed after interim results of five to eight batches had been seen (Section 30.3). As for the lower orders, the allowance is not a rigorous enclosure and does not bound the omitted orders of the ϵ series. The ratio $C_8(1)/C_7(1) = -25.0$ continues the sequence $-12.2, -16.1, -19.6, -22.5$ of the preceding ratios.

30.2 Changes relative to seventh order

The kernel, the binary cluster trees, the meromorphic radial quadrature, the zero-pole controls and the affine assembly (773) are unchanged. Each loop vector carries eight Laurent and Taylor slots per channel instead of seven, and loops up to eight are integrated. Since two insertions of loops five or higher first contribute beyond eighth order, $C_2, \dots, C_8$ are again affine in the loop-five to loop-eight jets. The affine map shows that C_8 depends on loop eight only through the combination $3c_8(0) - b_8(0)$: its two coefficients are in the ratio -3 to rounding, so the classes with $3w_B = w_A$ contribute exactly zero.

The nine-vertex inventory has 26,540 classes. Of these, 7,231 contain a power-divergent subgraph, 248 are primitive and power free, and 3,908 have an identically vanishing forest polynomial. The weighted forests have 2,507,176 coordinate terms. The enumerator and the forest builder are the seven-loop programs generalized to nine vertices with 128-bit packed multiplicity keys. Regenerating loops one to seven reproduces the archived inventories, forest arrays and exact-zero lists byte for byte. The coefficientwise comparison with bare-series inversion holds as a rational polynomial identity through eight loops. The exact class weights satisfy the zero-dimensional sum rules (16) at every loop order through eight. Every power-free class with an articulation vertex is an exact zero, so no class is excluded beyond the exact zeros and the vanishing top-loop channel.

Loops two to five are no longer sampled. Their vectors are deterministic sums over all $(L+1)!\,\mathrm{Cat}_L/2$ reflection-reduced ordered trees, each integrated with the full tensor quadrature: loops two to four at degree 24 for the constant K, and loop five at degrees $6, 8, 10, 12, 14$. The constant reproduces C_2 exactly and C_3 and C_4 to 1.3×10^{-12} and 1.1×10^{-9}. Since C_5 depends only on loops up to five, it is now a deterministic check: at the five degrees it differs from the exact value by -0.94, -0.13, -0.019, -0.0032 and -5.8×10^{-4}. The loop-five contribution to C_8 at degree 14 is -1297392.1, and its last degree step is $+2.8$.

Loops six to eight are sampled. The power-free sector uses the two independent randomized Sobol campaigns of Section 27. The power sector is telescoped as in (770), but to degree eight:

$$\mathbb{E}[Y_8] = \mathbb{E}[Y_1] + \mathbb{E}[Y_3 - Y_1] + \mathbb{E}[Y_5 - Y_3] + \mathbb{E}[Y_6 - Y_5] + \mathbb{E}[Y_7 - Y_6] + \mathbb{E}[Y_8 - Y_7], \tag{801}$$

with a paired degree eight to nine level that is never added to the estimate. With one tree per active eight-loop power graph, one replicate costs about 5, 14, 32 and 67 core-hours at degrees

six, seven, eight and nine. Degree eight is therefore affordable, whereas the seventh-order estimate stopped at degree six, where its truncation dominated.

30.3 Pilot, frozen design and production

An independent pilot of all thirty sampled campaigns fixed the design and trained the zero-pole controls on its coarse power campaigns; it never enters the estimate. It also exposed a sampling hazard. The seventh-order production gives 16372.8 for the non-primitive contribution to the C_7 row. With 64 points per graph and 16 replicates, the two independent seven-loop power-free campaigns of the pilot gave -16793.9 ± 9359.6 and -13230.8 ± 7406.2 (sampling standard errors), that is 33166.7 and 29603.7 below the production value, or 3.5 and 4.0 of their own standard errors. A repetition of the first campaign with 4096 points per graph and 32 replicates gives 16838.1 ± 724.1. That this value agrees to the unit with the four-error candidate 16838.3 of Section 30.1 is a coincidence: the two are unrelated quantities. The randomized estimator is unbiased, but at small point counts its replicate distribution is strongly skewed. The production therefore uses at least 4096 points per graph in every power-free campaign. Per-graph sample counts follow a Neyman allocation from the pilot, and the per-campaign multipliers are split into at most 128 replicates and a power-of-two sample multiplier.

The production consists of fourteen batches with fresh seeds, each of 498 work elements. The first four ran on SDSC Expanse in elements of 16 AMD EPYC 7742 cores and give 8066723 ± 8015 (sampling standard error). Ten further batches of the same design were then planned together, at first under a rule that started no new work after 48 hours and, once the running work had finished, included the batches then complete. That rule gives the first nine batches, 8061480 ± 5300 with allowance ± 22000. After interim results of five to eight batches had been seen, it was decided, on grounds of precision, to complete all fourteen planned batches; from then on every planned batch was to be included once complete, and all were. The design and the allowance rule are unchanged. Seven batches ran on Intel Xeon Gold 6148 nodes of the symmetry cluster and seven on Expanse; each batch ran on one processor type. The means of the two groups of seven differ by 7037 ± 8410. All campaigns are verified against the frozen design, including its counts, multipliers, numerical input hashes and the loop-five input. No seed of the pilot or of the control training is reused. A coverage guard requires every active class of loops six to eight to be integrated with total weight one at every level, including the check level. The calculation used about 9×10^4 physical core-hours (1.4×10^5 as charged, counting hyperthreads on the symmetry cluster), including development, the exact low-loop stage and the pilot.

The kernel was compared on identical tasks with the seventh-order implementation. In power mode the seven common slots agree bitwise at loops five to seven and degrees one to six; the randomized point evaluations agree to 8×10^{-17} relative. A binary128 replay of the contraction gives 2.5×10^{-17} of the row scale at loop eight.

30.4 Composition and lower-order checks

The contributions to C_8 are

Piece	Contribution to C_8	Sampling SE
Loops one to four, K	1531302.35	
Loop five, exact at degree 14	−1297392.1	
Power-free sector, loops six to eight	7879086.8	790.7
Power sector, degree one	316544.2	1690.3
Refinement $1 \to 3$	−364981.8	2887.4
Refinement $3 \to 5$	−4590.2	2346.3
Refinement $5 \to 6$	−26.5	528.4
Refinement $6 \to 7$	−202.2	321.0
Refinement $7 \to 8$	26.3	117.8
Check $8 \to 9$, not added	−113.3	68.1

The error is dominated by the eight-loop refinements $1 \to 3$ and $3 \to 5$ and by the eight-loop coarse level. The last two refinements and the check step are small and statistically consistent with zero. Adding the check step would move the estimate by −113, less than one per cent of the allowance; since it exceeds the last refinement in magnitude, no geometric tail beyond degree nine is fitted. The radial truncation is therefore negligible at the present precision, in contrast to seventh order. With the frozen zero-pole controls the estimate is 11078 above the uncontrolled estimator, 8048689, and its standard error falls from 23730 to 4210. The controls are fixed combinations of 30 Laurent pole coefficients that vanish exactly. Their standardized sample means are at most 2.18 in magnitude; the four above 2 form two pairs of columns that move together, and the squares sum to 39.8, which is only descriptive because the columns are correlated. Projected onto C_8, the control correction, $+11078 \pm 24734$, is consistent with zero, but these residuals alone do not exclude a nonzero pole expectation at the scale of the allowance. Errors of the controls at finite degree are refined with the levels and so enter the refinement terms of the allowance; a failure of the exact vanishing itself would be an error of the framework, which the allowance does not cover. The half-split estimates are 8055451 and 8064082.

The same samples give the lower-order coefficients, uncalibrated:

Coefficient	Estimate	Sampling SE	Reference	Difference
$C_5(1)$	−731.140052	deterministic	−731.139477 (exact)	-5.8×10^{-4}
$C_6(1)$	14332.8222	2.52	14331.68703 (deterministic)	+0.45 SE
$C_7(1)$	−322495.6	74.4	-322549.9 ± 10.1 (Sec. 27)	+0.72 SE

The C_7 comparison combines both sampling errors. Against the truncation-aware seventh-order value (753) the difference is 0.4 combined standard errors. The references are checks only and are not imposed. As at lower orders, these comparisons share the graph and renormalization framework and do not validate it.

31 Full eighth order in two dimensions

31.1 Result and scope

At $d = 2$, with $\epsilon = 2s-2$, the eighth-order coefficient extends the mixed local-mass forest calculation of Section 28 by one order. The renormalization functions of (774) are summed through eight loops,

$$\beta(g,z) = -zg + \sum_{L=1}^{8} \widehat{b}_L(z) g^{L+1}, \qquad \gamma_r(g,z) = \sum_{L=1}^{8} \widehat{c}_L(z) g^L, \tag{802}$$

with $\widehat{b}_1 = 3$, $\widehat{c}_1 = 1$ and $\Delta_{\mathcal{E}}(2,\epsilon) = 1 - \epsilon/2 + \gamma_r(g_*,\epsilon)$, and every loop order keeps its Taylor polynomial through z^{8-L} before $z = \epsilon$ is set. Both sectors are integrated explicitly, and in the notation of (775)

$$C_8(2) = C_8^{\mathrm{pf}}(2) + \Delta_p C_8(2), \qquad \Delta_p C_8 = \mathcal{F}_8(f+p) - \mathcal{F}_8(f). \tag{803}$$

The production of Section 31.4 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_8^{\mathrm{pf}}(2)$	45991.2684796	7.6797436
Power correction $\Delta_p C_8(2)$	−1.2658435	2.9224044
Complete coefficient $C_8(2)$	45990.0026361	8.2168210

The estimate with its sampling standard error is $C_8(2) = 45990.0 \pm 8.2$; the production record quotes this, together with the weight bounds and tail diagnostics of Section 31.5. As for the lower orders, the note quotes instead the reporting allowance of the rule (776) of Section 28, with half streams defined there. Here $4\widehat{\sigma} = 32.87$ exceeds both half-stream deviations, 2.39, and rounding up to a multiple of 1, two significant figures as for $C_7(2)$ and $C_7(3)$, gives

$$\boxed{C_8(2) = 45990 \pm 33, \qquad \epsilon = 2s-2.} \tag{804}$$

The rule was chosen after the result was available. No allowance was registered with the production decision, unlike the four-standard-error rules of $C_6(2)$ and $C_6(3)$, which were written down before their results (Sections 26.7 and 26.8). There is no earlier eighth-order calculation at $d = 2$. A separately developed code, run after this production, gives 46006.1 ± 13.4, $+1.03$ combined standard errors from it (Section 31.7). The exact-path pilot of Section 31.3, with an independent seed, gave 45620 ± 214, 1.73 combined standard errors from the production. The ratio $C_8(2)/C_7(2) = -11.86$ continues the sequence $-5.77, -7.69, -9.33, -10.70$ of the preceding ratios. The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework; the additional estimator digits are given for reproduction only.

31.2 Changes relative to seventh order

The mixed forest kernel (780), the power-sector normalization (781), the graph weights (778), the importance sampler and the default precision policy of Section 28.7 are unchanged. The native integrator is the seven-loop one generalized to sixteen propagators and 31-dimensional digital nets. It keeps an order-seven mode, in which it reproduces the seventh-order production executables bit for bit (Section 31.3).

Regulator jets and low-loop inputs. The integrand jets follow (789) with $n \le 8 - L$, so the maximum orders at loops two to eight are $6, 5, 4, 3, 2, 1, 0$. The Gamma table gains the column

$a_{L,6}$, used only by a two-loop control, and an eight-loop row with normalization $-2/\pi^8$, whose logarithms are not used. The two-loop input gains

$$\widehat{b}_{2,6} = -266.52096796135938173225219261908980094902\ldots, \tag{805}$$

from the same exact-hypergeometric numerator; its first six entries are unchanged. Two Cauchy contours at 75 digits agree to 3.1×10^{-66}, and a direct 130-digit Taylor series confirms the stored value to 7.7×10^{-68}. Every printed Gamma jet is the correctly rounded value of two independent Cauchy contours at 110 digits. The two-loop control graph, integrated by the production code, reproduces the seven entries of $\widehat{b}_2$ within 1.22 sampling standard errors.

Assembly. The physical assembly is a d-general eighth-order module shared with the three-dimensional calculation, used byte for byte and pinned by hash in both. Its floating-point inversion agrees with two exact rational inversions, the bivariate Lagrange formula and the coefficientwise solution of $\beta(g_*, \epsilon) = 0$, to 7.0×10^{-16} of the coefficient scale in 64 random eighth-order cases. With z-independent jets it reproduces the constant-coefficient MS polynomials through eighth order; the eighth-order polynomial, $19683\, C_8 = 3c_8 - b_8 + 42Ac_7 - 16Ab_7 + \cdots$ with 34 monomials after $b_2 = -6A$ and $c_2 = -A$, was derived independently twice and agrees exactly. At order seven the module reproduces the assembly of the seventh-order production (Section 28.7) to 2 ulp. The inversion fixes how the new data enter:

- the top slot of every loop enters additively, $\partial C_8/\partial(\widehat{b}_{L,8-L}, \widehat{c}_{L,8-L}) = (-1, 3)/3^{L+1}$ for $L = 3, \ldots, 8$, without changing a lower C_n. The eight-loop constants therefore enter only as $C_8(2) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$, through the channel weight $3w_B - w_A$;
- loop seven enters through both channels, $\partial C_8/\partial(\widehat{b}_{7,0}, \widehat{c}_{7,0}) = (-16A_2, 42A_2)/19683$, with $A_2 = -\widehat{b}_{2,0}/6 = 4\log 2$ of (176). This is not proportional to $(-1, 3)$, so the seven-loop classes with $3w_B = w_A$, which Section 28 excludes, now contribute through $-2A_2\widehat{b}_{7,0}/19683$;
- $C_2, \ldots, C_8$ are affine in the jets of loops five to eight taken together, since two insertions of loops five or higher first contribute beyond eighth order; loops two to four enter nonlinearly.

At the production means the three rules are hard gates: the top-slot and loop-seven derivatives agree with their exact values to 6×10^{-16} relative, and the response of C_8 to the jets of loops five to eight is linear to rounding.

The eight-loop sums are complete for $C_8(2)$ but not individually. Since the eight-loop classes with $3w_B = w_A$ are not integrated, the stored $\widehat{b}_{8,0}$ and $\widehat{c}_{8,0}$ omit contributions with $3\,\delta\widehat{c}_{8,0} = \delta\widehat{b}_{8,0}$, which cancel in $3\widehat{c}_{8,0} - \widehat{b}_{8,0}$. They are therefore not the eight-loop coefficients of β and γ_r, and a quantity that depends on them separately, such as the ϵ^8 coefficient of g_*, is not an output of this calculation. At loops three to seven no nonzero class is omitted, so those jets are complete. For the restored seven-loop classes the top-slot combination $3\widehat{c}_{7,1} - \widehat{b}_{7,1}$ vanishes by the weights, but both jets are integrated, as for every other class. The independent code of Section 31.7 also leaves out the eight-loop classes with $3w_B = w_A$, so the caveat applies equally to both codes.

Graph inventory and complete coverage. The nine-vertex inventory has the 26,540 classes of Section 30.2. Its $d = 2$ per-graph input files, called payloads below (line masks, forest terms, weights and importance rates), come from the seven-loop builders of Section 28, generalized to nine vertices with 128-bit class keys. The only dimension-dependent part, the rate table, uses the parity floor of the seven-loop power-free files: every proper edge subset has rate at least $1/2$ if its size is odd and at least 1 if it is even. Regenerating loops three to seven reproduces every production file byte for byte, except the rate lines of the power-free files at loops three to six, whose earlier rate tables are kept unchanged. The same payloads, with the rate line rewritten, serve the three-dimensional calculation.

L	All classes	Power-containing	Nonzero mixed kernels	Integrated classes power-free	power
3	8	1	1	5	1
4	27	5	3	16	3
5	129	31	20	76	20
6	660	173	107	395	107
7	3986	1091	709	2441	709
8	26540	7231	4891	16064	4336

At eight loops 2,254 classes have $3w_B - w_A = 0$ and 3,886 further classes have an exactly vanishing mixed forest map. The remaining 15,816 power-free nonprimitive, 248 primitive and 4,336 power-containing classes are integrated in every replicate. At seven loops all 3,150 classes with a nonvanishing map (2,397 nonprimitive, 44 primitive and 709 power-containing) are integrated, 262 more than in Section 28. In all, 24,173 kernels are integrated, about seven times the 3,511 of the seventh-order calculation, and there is no omitted nonzero class. An eight-loop kernel has up to 4,032 complement-of-tree monomials and 280 forest terms. The exactly vanishing kernels are exactly the classes with an articulation vertex (2,511 power-free and 2,340 power-containing at eight loops), and the exact class weights satisfy the zero-dimensional sum rules (16) at every loop order. An independent edge-subgraph enumeration rederives the mixed forest map of every power-containing class (1,091 at seven and 7,231 at eight loops) without a failure; exact rational Kirchhoff determinants agree with the stored kernels at random rational points for 144 seven- and eight-loop kernels; and an independent count of the compatible families of logarithmic subgraphs (nested, or sharing at most one vertex; the empty family included) gives 11,254,373 over all eight-loop classes, the total recorded with the inventory of the one-dimensional calculation, and agrees with the builders' count for each of the 19,309 power-free classes (`LRI_C8_shared/inventory/inventory8_d2/gates/k_forests.json`).

Power sector and campaigns. The power-containing jets follow (781) through eight loops. Their reporting follows a protocol registered before any eighth-order data. The differences $\Delta_p C_n$ at orders three to seven are null rows, that is, quantities expected to vanish: through sixth order the power sector cancels in the exact coefficients (Section 24), and at seventh order the measured difference is consistent with zero (Section 28). They are flagged when a pull exceeds three in magnitude but never used as gates, and the eighth-order correction is reported and never assumed to vanish. Two independent campaigns a and b again use importance rates multiplied by 0.8 and 0.9 and 52- and 60-bit digital coordinates, with group seeds derived by hashing from a fresh base seed. The campaign means are combined with fixed weights 1/2, and the covariances follow (796) through eight loops; there is no retained input.

31.3 Pilot, design and numerics

An exact-path pilot integrated all 24,173 kernels in both campaigns with 8 replicates of 4,096 points per graph, on three nodes of the symmetry cluster for 227 core-hours. It gave $C_8(2) = 45620 \pm 214$, with campaigns 0.64 standard errors apart and lower-order pulls at most 2.11 (at C_6). It fixed the design and never enters the estimate; its seed is excluded from the production. With its uniform allocation the power-free seven-loop graphs carried 71% of the variance at 2.4% of the cost, and the power-free eight-loop graphs 18% at 71%.

Per-graph counts follow a Neyman allocation, $n_g \propto \sigma_g/\sqrt{c_g}$ with the pilot's per-point standard deviations σ_g and measured costs c_g, on a power-of-two lattice with at least 4,096 points per graph and 128 replicates in every campaign, sector and loop. After the pilot the author chose, from a table of costs at four targets, a predicted standard error of 2.5×10^{-4} of $|C_8| \approx 4.6 \times 10^4$, that is 11.50, which includes a safety factor 1.5 on the raw prediction 7.67. The design, fixed before

the production, gives every power-free eight-loop graph at least 32,768 points per replicate and contains 779,836,456,960 kernel samples in one batch.

Exact path. The default numerics are the precision policy of Section 28.7: a long-double first pass, binary128 below the ratio 10^{-9} and 350-digit GMP arithmetic below 10^{-23}. In order-seven mode the default executables reproduce the seventh-order production executables bit for bit, on both backends, on the two-loop graph and all 3,773 kernels of loops three to seven, including the 262 new seven-loop ones; the order-eight slots common to order seven are identical. Each later kernel speedup (parallel graph loading, and inline correctly rounded binary128 addition, subtraction and multiplication, tested against the compiler library on 2^{32} random operand pairs per operation) had to reproduce the first unoptimized eight-loop build bit for bit on a sample of 1,600 power-free and 400 power-containing eight-loop kernels, on both backends, so the default path is no numerical change. The final executables passed these replays, the 450-digit point checks and the thread-identity tests again on the production cluster.

The faster numerics profile. The production used an opt-in profile, chosen by the author after the pilot and run only by an executable that passed an accuracy gate. Here v is the kernel value at a point and err the same forest sum with every numerator term replaced by its absolute value, so that $|v|/\mathrm{err}$ is the cancellation ratio of Section 25.4. A point that fails the long-double test is first evaluated in double-double arithmetic, with a 106-bit significand, and accepted when $|v| \geq 10^{-20}\,\mathrm{err}$; otherwise, or outside the double-double range, it goes to the unchanged binary128 tier. A point that reaches the multiprecision tier is first evaluated at 77 digits (320 bits) and accepted when $|v| \geq 10^{-63}\,\mathrm{err}$, and otherwise at the unchanged 350 digits. The gate compared the production executable with the exact path on the production hardware in four parts:

(i) on 153,600 seven- and eight-loop points with 450-digit references, every accepted value is within the exact path's bound of 10^{-6} relative, with largest errors 5.2×10^{-12} in double-double and 1.4×10^{-11} in binary128; every 77-digit value equals the 350-digit one after conversion;

(ii) a paired campaign on identical streams, with four replicates of 256 points for every kernel of loops three to eight in both campaigns (193,384 tasks), gives $|\Delta C_8| + 3\,\mathrm{SE}_\Delta = 4.2 \times 10^{-16}$ against the bound 0.062, one per cent of the standard error at the tightest target considered, with per-task normalized differences at most 5.6×10^{-14}.

(iii) the fractions of points promoted to each tier are unchanged;

(iv) the results are identical at 1, 8, 16 and 64 threads and in eight lanes of eight threads.

The profile runs about 2.5 times faster than the exact path at eight loops. The production design keeps the exact pilot's variances and only re-costs the allocation with a cost model measured under the profile, which (ii) justifies; it costs 0.37 of the exact path at the same target.

31.4 The 2026 production

The batch ran on the symmetry cluster on 6 and 7 October 2026, in lanes of eight threads on AMD EPYC 7543 nodes of one runtime identity: eight nodes for 22.4 to 23.0 hours in a first round and six nodes for 4.4 to 5.6 hours after the 24-hour limit. All 1,129 work indices, 6,192,256 tasks in 24,848 checkpoint blocks, ran once without a failure, in 28.7 hours from submission to the final merge and 13,454 allocated core-hours, 0.93 of the design's estimate. Every merge, the assembly and an independent recomputation of the result record used the package's own tools. With development, the pilot and the tail diagnostics the calculation used about 1.6×10^4 core-hours. The two campaigns and the two half streams give

Estimate	$C_8(2)$	Sampling SE
Campaign a	45993.58	12.15
Campaign b	45986.42	11.06
First half stream	45987.61	
Second half stream	45992.39	
Combined	45990.00	8.22

The campaigns differ by 0.44 combined standard errors. The error is dominated by the power-free eight- and seven-loop graphs:

Sector	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	$L=8$
Power-free	0.01	0.07	0.22	0.65	3.67	6.71
Power	0.00	0.01	0.11	0.43	1.02	2.70

Entries are in units of C_8 and combine in quadrature. The realized standard error is 1.07 times the raw design prediction, the size expected from per-graph standard deviations estimated with seven degrees of freedom, and well inside the safety factor; per stratum the realized variances at loops seven and eight are 0.98 to 1.32 times the predictions.

31.5 Weight bounds and tail diagnostics

The package's heavy-tail rule, calibrated on synthetic families at $d=3$, computes a fixed-k ($k=16$) Hill estimate [44] $\widehat{\xi}$ of the tail index of the per-point weights of every graph and campaign. At production scale it confirmed 4,832 eight-loop graphs (4,631 power-free and 201 power-containing) as possible infinite-variance cases, with two-campaign mean $\widehat{\xi} > 0.44$ at 32,768 or more points per replicate. Together they carry 27.6% of the variance of $C_8(2)$, and no single graph in one campaign more than 0.08%. Since an infinite variance would invalidate the replicate standard error, the weights were analyzed directly.

In an ordered sector the sampler draws successive scale gaps $z_i = u_i^{1/(\rho_c \alpha_G(S_i))}$, with u_i uniform, S_i the set of lines below the ith gap, $\alpha_G(S)$ the stored importance rate and $\rho_c = 0.8, 0.9$ the campaign multiplier. The weight is

$$t = Z_c \, K_G(x) \prod_i z_i^{|S_i|/2 - \rho_c \alpha_G(S_i)}, \tag{806}$$

with K_G the rational kernel (780) and Z_c a constant. Let $m_G(S) = |S|/2 + \nu_S(K_G)$ be the exact leading exponent of the line measure times K_G, after all forest cancellations, when the lines of a proper subset S are scaled to zero. Near that face $t \sim z^{m_G(S) - \rho_c \alpha_G(S)}$, so the weight is bounded if $m_G(S) \geq \rho_c \alpha_G(S)$ on every face and has a finite second moment if $2m_G(S) > \rho_c \alpha_G(S)$. The builders set $\alpha_G(S) = \max(r_G(S), p_S)/2$, where $r_G(S)/2$ is the smallest exponent of the individual forest terms and p_S is 1 for odd and 2 for even $|S|$. Where $r_G(S) > 0$ a sum cannot have a lower exponent than its terms, so $m_G(S) \geq \alpha_G(S)$. Where $r_G(S) \leq 0$ the forest sum must cancel; since K_G is rational, $m_G(S) \in |S|/2 + \mathbb{Z}$, and $m_G(S) > 0$ already implies $m_G(S) \geq \alpha_G(S)$. Once these cancellations are confirmed, $m_G(S) \geq \alpha_G(S) > \rho_c \alpha_G(S)$ on every face, and the weight is bounded at both rates. Nested limits are covered because every forest denominator U_F is the basis generating polynomial of a matroid, so that a single monomial dominates all nested limits of a sector at once; this rests on the builders' construction and was checked on random orderings.

The cancellations were verified at all 3,959,753 expanded subsets of the 24,173 kernels: every subset with $r_G(S) \leq 0$ of the kernels with the builders' rate tables, and every subset of the 492 power-free kernels at loops three to six, which keep their earlier tables. At each, $\sum_F N_F/U_F$ is expanded in the scaling parameter in exact arithmetic modulo four primes drawn uniformly from $[2^{61}, 2^{62})$, at two random points per prime and graph. A nonzero residue proves a nonvanishing

coefficient, so an error can only overestimate a valuation. With explicit degree and height bounds and the Schwartz–Zippel lemma, the union bound puts the probability of any wrong valuation below 10^{-53}, assuming independent uniform draws. Every stored rate table is reproduced, every required cancellation holds, and the eight trials agree everywhere. Nine archived controls behave as asserted: deleting a forest term or doubling a numerator breaks the cancellation, and rate tables multiplied by 3 or 2.4 give infinite variance with a finite mean at both rates, or at rate b only. Over all seven- and eight-loop kernels the margins are $m_G - \rho_c\alpha_G \geq 0.1$ and 0.05 and $2m_G - \rho_c\alpha_G \geq 0.6$ and 0.55 at rates a and b. Seventeen power-free kernels at loops five and six that keep their earlier rate tables have unbounded weights of finite variance.

The estimated sampling standard error is 8.22. Boundary analysis, supported by this multi-prime randomized cancellation verification of all 24,173 integrated kernels, supports finite second moments at both production rates and bounded weights for all seven- and eight-loop kernels; it is a randomized verification, not a deterministic certificate in characteristic zero. The 4,832 eight-loop tail warnings are consistent with bounded, sharply peaked weights. The confirmed power-free graphs have large canceling forest sums (median 128 terms, against 16 for the cleared graphs), and the fixed-k Hill estimate falls with the number of points in every class, with log-log slopes from -0.34 to -0.17; such finite-sample behavior cannot by itself prove boundedness. Fresh nested sampling supports variance stability in the selected main groups: over 2^{12} to 2^{17} points the slope of $\log(N\,\mathrm{Var})$ is -0.018 $[-0.051, +0.030]$ for 16 confirmed power-free and -0.062 $[-0.110, +0.060]$ for 8 confirmed power-containing graphs, with 95% intervals conditional on the selected panel. In the 380 power-containing eight-loop graphs with a subset that needs two orders of cancellation ("double lift"; 0.058% of the measured variance), a targeted test of the 16 largest contributors (52% of that class's measured variance) shows rare-peak sampling effects, with per-point variance growing over 2^{12} to 2^{19} points in the largest two. Inflating that class's variance by factors 18.9, 45 or 335 gives standard errors 8.26, 8.32 or 8.97; these variance-inflation calculations are sensitivity scenarios, not error bounds. Every graph stays in the estimand.

31.6 Checks

The same uncalibrated samples give lower-order coefficients, compared with the special-function values of Section 1 and, for C_6 and C_7, with the productions of Sections 26.7 and 28, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.8748487258	1.24×10^{-6}	$+1.04$
4	5.0489759126	6.80×10^{-5}	$+1.58$
5	-38.8332791554	9.37×10^{-4}	$+0.39$
6	362.3625083	0.0116244	-0.34
7	-3875.545222	0.289774	$+0.97$

The squared pulls sum to 4.78 for five checks. The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. The power null rows at orders four to seven are $(1.37 \pm 4.03) \times 10^{-6}$, $(7.7 \pm 23.5) \times 10^{-5}$, $(2.28 \pm 4.86) \times 10^{-3}$ and $(2.54 \pm 9.00) \times 10^{-2}$, with pulls at most 0.47; at order three the difference vanishes by the weights. These are consistent with zero but do not prove a cancellation at eighth order, where the correction is -1.27 ± 2.92. A bootstrap of 4,096 replicate resamples gives standard error 8.09, against 8.22 from covariance propagation. Because both sectors are integrated explicitly, the sixth-order by-product again tests the analytic power-sector compensation used for $C_6(2)$, and the seventh-order by-product, -3875.55 ± 0.29, is an independent sampling of the seventh-order integrands of Section 28. These checks share the graph and renormalization framework and do not validate it.

31.7 An independent code

A separately developed eighth-order code extends the seventh-order package of Section 28 independently of the package used above. It generates its own nine-vertex classes with 128-bit canonical keys and its own eight-loop payloads, which are different files from those of Section 31.2. It extends the kernel and the regulator jets to eight loops on its own, derives the two-loop and Gamma inputs separately, and solves for the eighth-order fixed point with its own assembly. It integrates the same 24,173 kernels: 16,064 power-free and 4,336 power-containing at eight loops, and 3,150 at seven. It shares with the calculation above the mixed forest kernel (780) and its precision policy, the importance sampler with rates multiplied by 0.8 and 0.9, the digital nets and the seed hashing, the finite renormalization scheme and, byte for byte, the payloads of loops three to seven. It runs only the exact path of Section 31.3 (long double, binary128 and 350-digit GMP), without the double-double and 77-digit tiers.

The independent code, its derivation of the eighth-order extension and the construction checks under the next four headings were written by OpenAI GPT-6 astra (model stated by the author), as were the caveat on the separate eight-loop sums in Section 31.2, the reading of the bootstrap and the reporting rule described below. They were delivered as the package `LRI_C8_D2_HPC_optimized/`, together with an original package that holds the graph payloads, the class generator and its check records (hosted, see `GFF/data/C8_2_check/DATA.md`). A Claude Code session ported the code to the clusters and ran the pilot and the production described afterwards. On 16 October 2026, when the delivered derivation was merged into this subsection, a Claude Code session also compared the shipped inputs of the two packages: the Gamma table of the independent code agrees with that of Section 31.2 to 65 digits, the precision of that table, and its two-loop jets $\widehat{b}_{2,0},\dots,\widehat{b}_{2,6}$ agree with those of the calculation above to 70 digits (largest relative difference 2.9×10^{-70}, at $\widehat{b}_{2,6}$).

Inputs. The Gamma table `inputs/gamma_d2_8.txt` has one row for each $L=2,\dots,8$. A row holds the order $8-L$, the normalization $2(-1)^{L+1}/\pi^L$ and the logarithms $a_{L,0},\dots,a_{L,6}$ of (700), with $a_{L,0}=0$; every row carries all seven logarithms, whether or not its loop uses them.

For the two-loop input, at $d=2$ the parameters of (198) are $a=\frac12+z/4$ and $b=\frac12+3z/4$. The jets are (790) of Section 28, extended to $j=6$: the numerator $z^2S_2(z)=4F_1(2,z)-2W(2,z)F_2(2,z)$ of (199) is analytic at $z=0$, where it equals 2, and the new entry $\widehat{b}_{2,6}$ needs its seventh Taylor coefficient. At $d=2$ the upper parameter $2a=1+z/2$ of F_2 equals its lower parameter $1+z/2$, and the series reduces to

$$F_2(2,z)={}_2F_1\left(\begin{matrix}z,\ b\\ b+\frac12\end{matrix};\frac14\right). \tag{807}$$

The code expands $F_1(2,z)$ and (807), both convergent at the argument $1/4$, in finite Taylor arithmetic in z at 120 decimal digits, where 183 terms of the series suffice, and it expands $\log W(2,z)$ analytically in polygamma values. The result is

$$\widehat{b}_{2,6}=-266.52096796135938173225219261908980094902003629215748\dots.$$

Two evaluations of the unreduced representation on Cauchy contours, of radius 0.05 with 128 nodes and of radius 0.08 with 160 nodes, differ from the Taylor coefficients by at most 5.26×10^{-112}. The first six coefficients are identical as binary64 numbers to the two-loop input of the seventh-order calculation and differ from it by at most 4.06×10^{-70} at high precision. The identity $\widehat{b}_{2,0}=-24\log2$, that is $A_2=4\log2$ in (176), holds exactly and reproduces $C_2(2)=4\log2/9$ of (192). The 100-digit values are in `inputs/two_loop.json`, written by `tools/analytic_inputs.py`; its check record is `verification/analytic_inputs.json` of the original package. These checks concern the special-function input, not the sampled integrals.

Graphs and payloads. The code enumerates the one-particle-irreducible classes with its own canonical keys of 128 bits. At eight loops the keys need up to 69 bits, and 5,068 of them exceed

64. Its class files of loops one to seven are byte for byte those of the seventh-order calculation. At every loop through eight the exact sums of w_A and w_B of (778) over the classes equal the zero-dimensional values, which the code computes separately from normal-ordered Gaussian moments and a Legendre transform. For each of the 4,810 classes of loops three to seven, in both sector inventories (9,620 rows), its forest builder reproduces the weights, the monomial masks, the forest maps and the exact-zero lists of the seventh-order files. The importance-rate arrays are left out of this comparison, because the archived power-free rate arrays were changed after they were generated; the archived rates of loops three to seven are used as supplied, and the rates of the new eight-loop payloads follow the rate formulas of the seventh-order builder. The rates affect only the efficiency of the sampling, since the integral is defined by the complete weighted kernel. At eight loops an edge-subgraph enumeration, which includes the subgraphs not induced by their vertex sets, rederives the mixed forest maps of all 7,231 power-containing classes, 10,451,732 forests in all, without a failure. Thirty-six exact rational evaluations on 18 eight-loop kernels agree with Kirchhoff determinants of the contracted forest blocks.

The census agrees with Section 31.2, with the exclusions taken in the other order. Of the 19,309 power-free and 7,231 power-containing eight-loop classes, 2,511 and 2,340 have an exactly vanishing kernel. Of the remaining 16,798 and 4,891 nonzero kernels, 734 and 555 have $3w_B = w_A$, and 16,064 and 4,336 are integrated (`inputs/coverage.json`). The 4,851 exact zeros and 1,289 channel exclusions are the $2{,}254 + 3{,}886 = 6{,}140$ excluded classes of Section 31.2. At seven loops all 2,441 nonzero power-free kernels, 44 of them primitive, and all 709 nonzero power-containing kernels are integrated, including the 262 classes that the seventh-order production excluded by $3w_B = w_A$ (the split table of Section 28); they are already in the seventh-order reference inventories, although its production payloads omit them. The exact-zero list is computed from the mixed kernel (780) itself. The 3,908 classes with an identically vanishing coordinate-forest polynomial in the one-dimensional calculation (Section 30.2) belong to a different kernel and are not used; the one-dimensional inventory serves only as a check of class counts and topology.

Kernel arithmetic. Besides the 350-digit GMP final tier used in the production, the kernel has an alternate final tier in Boost decimal arithmetic at 300 digits. In order-seven mode, at eight points in each of two streams on all 3,773 kernels of loops three to seven, including the 262 restored ones, all 60,368 point contributions reproduce the arithmetic of the seventh-order kernel exactly on both backends. At 37 points per backend, among them physical eight-loop kernels and cancellation fixtures, the kernel values agree with an independent 450-digit evaluator to 4.55×10^{-304} (decimal) and 2.22×10^{-354} (GMP), in units of the absolute-term scale err of Section 31.3. These are arithmetic comparisons, not interval bounds on the integrals. Restart, chunking, complete coverage, immutable inputs and idempotent merges were tested on native runs. The records are `verification/kernel_checks.json` and `generator/reports/` of the original package.

Assembly. The assembly evaluates the bivariate Lagrange formula (704) through $n = 8$, with $\widehat{B}(g,z) = 3+\sum_{L=2}^{8}\widehat{b}_L(z)g^{L-1}$ and γ_r of (802). Separately, it solves $g = \frac{1}{3}\big[\epsilon-\sum_{L=2}^{8}\widehat{b}_L(\epsilon)g^L\big] \bmod \epsilon^9$ by nine iterations from $g = 0$ and substitutes the result in $\Delta_{\mathcal{E}} = 1-\epsilon/2+\gamma_r(g,\epsilon)$. In 24 exact rational cases with regulator-dependent jets the two constructions agree exactly, and the floating-point assembly used by the production agrees with them to a normalized difference of 2.7×10^{-15}. The same test, `tests/assembly_formula.py` (record `verification/assembly_formula.json` of the original package), confirms $\partial C_8/\partial\widehat{b}_{8,0} = -1/19683$ and $\partial C_8/\partial\widehat{c}_{8,0} = 1/6561$, the contribution at eighth order of the seventh-order classes with $3w_B = w_A$, the dependence of C_8 on the new entry $\widehat{b}_{2,6}$, and the affine dependence on the jets of loops five to eight, all as derived in Section 31.2.

Development runs (superseded). Before the port, OpenAI GPT-6 astra ran the delivered code in three complete-coverage runs of its own, each with two campaigns a and b and small point counts. A pilot gave 48626 ± 2633. A refinement with 8 replicates per group and at least 32 points

per eight-loop graph and replicate gave 47068 ± 1854. A second refinement with 16 replicates and at least 128 power-free and 64 power-containing points per eight-loop graph and replicate gave 45512 ± 524, quoted with the code as 45510 ± 520 (sampling standard errors throughout). In the last run the two campaigns differed by 2.26 combined standard errors, the power correction was 153 ± 122, and in 21 of the 25 graphs with the largest variances a single replicate carried at least 80% of the squared deviations in its campaign. This is the small-sample hazard also met at one dimension (Section 30.3). The three values lie $+0.99$, $+0.57$ and -0.94 combined standard errors from the production below. They are superseded by it and enter neither its allocation, which was computed from the pilot below alone, nor its estimate. The records of the first two runs are in the original package; those of the third were not delivered.

Port, pilot and production. The code was adapted to the Expanse cluster without a change to its numerical files. Its optimized kernel reproduced the package's 52 frozen fixtures bit for bit on the production hardware. It used a fresh base seed, checked against the seeds of this section. A pilot with 32 replicates per campaign gave 46178 ± 417 and fixed a Neyman allocation at 28,500 kernel core-hours. The allocation has 64 replicates in every campaign, sector and loop, at least 8,192 points per replicate for every power-free and 4,096 for every power-containing eight-loop graph, and a predicted standard error of 14.97. The production ran on 9 and 10 October 2026 on eleven whole AMD EPYC 7742 nodes for 16.3 to 20.1 hours each. All 342 work indices ran once without a failure, in 207 node-hours (26,526 core-hours), 96.5% of it kernel time.

Estimate	$C_8(2)$	Sampling SE
Campaign a	46012.55	19.70
Campaign b	45999.73	18.23
First half stream	46008.16	
Second half stream	46004.11	
Combined	46006.14	13.42

The campaigns differ by 0.48 combined standard errors, and a bootstrap of 4,096 replicate resamples gives standard error 13.38. The power-free part is 46004.84 ± 12.90 and the explicitly integrated power correction $+1.29 \pm 3.71$. The power-free eight- and seven-loop graphs carry 78% and 14% of the variance, and no single graph more than 0.2%. At orders three to seven the lower-order by-products have pulls -0.01, $+1.75$, $+0.57$, -1.18 and $+1.04$ against the references of Section 31.6; the squared pulls sum to 5.88 for five checks. The power null rows at orders four to six have pulls of at most 0.76.

The bootstrap resamples the 64 replicates of each group. For a mean of R replicates the plug-in bootstrap variance is $(R-1)/R$ times the unbiased estimate (726). To first order the bootstrap standard error is therefore $\sqrt{(R-1)/R}$ times the propagated one: 0.992 here, against the observed $13.38/13.42 = 0.997$, and 0.996 for the 128 replicates of the calculation above, against $8.09/8.22 = 0.984$ (Section 31.6). The finite number of resamples and the nonlinearity of $\mathcal{F}_8$ also enter these ratios. The bootstrap mean differs from the estimator by -0.11, that is -0.008 standard errors (`GFF/results/C8_2_check/prod1/C8_result.json`). Neither a failure of the first-order propagation (796) nor a finite-sample bias of the nonlinear map is therefore visible at this precision. Resampling adds no integration points and cannot reveal regions that no replicate sampled.

Comparison with the calculation above. The combined estimate differs from the estimator of (804) by $+16.1 \pm 15.7$, that is $+1.03$ combined standard errors, well inside the allowance ± 33. The power-free parts differ by 0.90 and the power corrections by 0.54 combined standard errors. The agreement tests the parts that were developed independently: the eight-loop graph generation and payloads, the eighth-order kernel and jet extension, the analytic inputs, the assembly and

the sampling. It does not test what the two codes share. An error in the seventh-order kernel construction, the precision policy, the renormalization scheme or the payloads of loops three to seven would be common to both. The independent estimate is not combined with (804), which is unchanged.

No reporting allowance is assigned to the independent estimate, and its result record carries none. Under the reporting rule that came with the independent code, an allowance and its rule are fixed only after fresh campaigns, half streams, successive refinements, precision-tier replays and the lower-order by-products have been compared, because a sampling standard error alone does not establish final digits; no allowance was fixed for the run above. The estimate therefore enters this note only through its sampling standard error, as a check. The weight bounds of Section 31.5 were established for the rate tables of the payloads of the calculation above. The independent eight-loop payloads carry their own rate tables, built from the seventh-order rate formulas, and the bounds are not claimed for them here.

31.8 Records

The result record, the production, tail and pilot reports and the package are listed in Section 33. The production's plan, design bundle and the merged record of each of its 24 groups, from which the estimator and every number of this section are recomputed, are in `GFF/data/C8_2/`. The full campaign directories of the production `prod1-2.5e-4-ddmp256-20261006-ba68f401e46e` and of the pilot `pilot1-20261006-ba68f401e46e`, with every checkpoint, are archived as listed in `GFF/data/C8_2/DATA.md`. The records of the independent code of Section 31.7 are in `GFF/data/C8_2_check/` and `GFF/results/C8_2_check/`; its full campaign directory is hosted (`GFF/data/C8_2_check/DATA.md`).

32 Full eighth order in three dimensions

32.1 Result and scope

At $d = 3$, with $\epsilon = 2s - 3$, the eighth-order coefficient extends the calculation of Section 29 by one order, as Section 31 extends Section 28. The renormalization functions are summed through eight loops, $\Delta_{\mathcal{E}}(3, \epsilon)$ is (798), and every loop order keeps its Taylor polynomial through z^{8-L} before $z = \epsilon$ is set. Both sectors are integrated explicitly and $C_8(3)$ is split as in (803). The production of Section 32.4 gives

Contribution	Estimate	Sampling standard error
Power-free coefficient $C_8^{\mathrm{pf}}(3)$	1601.9999290	0.7279030
Power correction $\Delta_p C_8(3)$	0.6052940	0.6129303
Complete coefficient $C_8(3)$	1602.6052230	0.9516489

The quoted uncertainty is an allowance, not a standard error. It follows the rule (776), also used for $C_7(3)$ and $C_8(2)$: the larger of four sampling standard errors and the deviations of the two half-stream estimates (computed from the first and from the second half of the replicates of every group), rounded up, here to a multiple of 0.1, which keeps two significant figures as for $C_7(3)$ and $C_8(2)$. The rules for $C_6(3)$ and $C_7(3)$ were written down before their production results. This one was chosen after the result was available and confirmed by the author; no allowance was registered with the production decision. Here $4\widehat{\sigma} = 3.81$ exceeds both half-stream deviations, 0.046, and the rule gives

$$\boxed{C_8(3) = 1602.6 \pm 3.9, \qquad \epsilon = 2s - 3.} \tag{808}$$

There is no earlier eighth-order calculation at $d = 3$. The pilot of Section 32.3, with independent seeds, gave 1619 ± 16, 1.05 combined standard errors from the production. The ratio $C_8(3)/C_7(3) = -7.27$ continues the sequence $-1.95, -3.47, -4.76, -5.74, -6.56$ of the preceding ratios. A separately developed code gives 1600.98 ± 3.97 (sampling standard error), -0.40 combined standard errors from the estimator (Section 32.7). The allowance is empirical. It is not a probability guarantee and does not bound an error shared by the graph and renormalization framework, and its coverage is conditional on the finite variance discussed in Section 32.5. The additional estimator digits are given for reproduction only. The production record quotes the sampling standard error alone, $C_8(3) = 1602.6 \pm 1.0$, together with the tail diagnostics of Section 32.5; the reported uncertainty is the allowance ± 3.9 of (808).

32.2 Changes relative to seventh order

The three-dimensional kernel, edge measure and normalization of Section 29 are unchanged: every forest denominator is raised to $U_F^{3/2}$ before the forest terms are summed, and the Gamma jets (711) now run through order $8 - L$. The graph inventory, the mixed forest maps, the weights, the zero-channel rule and the shared eighth-order assembly are those of Section 31.2, used with the three-dimensional rate lines; the integrated classes at loops three to eight number $5, 16, 76, 395, 2{,}441$ and 16,064 in the power-free sector and $1, 3, 20, 107, 709$ and 4,336 in the power-containing sector, 24,173 kernels in all. The worker is the three-dimensional seven-loop worker generalized to sixteen propagators and 31-dimensional digital nets, with the extended-precision first pass and promotions to binary128 and to 350-digit GMP arithmetic at the unchanged thresholds 10^{-9} and 10^{-23}. In its order-seven mode it reproduces the unchanged seventh-order worker bit for bit on the two-loop graph and all 3,511 kernels of loops three to seven, and it replays the archived seventh-order production. Results do not depend on the thread count and differ between processor types only in the last bits, at most 2.0×10^{-11} in normalized units between AMD and Intel processors; every production job ran on one processor model, AMD EPYC 7543.

Low-loop inputs. The two-loop input gains

$$\widehat{b}_{2,6}(3) = -29.258124480916806558671227221989158045345\ldots, \tag{809}$$

from (713), now through z^6; its first six entries are those of Section 29. Two Cauchy contours at 65 digits agree to 2.7×10^{-56}, and a 95-digit reference contour confirms the stored entries to 1.7×10^{-57}. Every printed Gamma jet, 42 in all, is the correctly rounded value of two independent Cauchy contours at 100 digits. The single three-loop power graph has the closed form $\widehat{b}_3^{\mathrm{p}}(0) = -16\pi/15$; the production gives -3.3509158 ± 0.0000894, $+1.30$ sampling standard errors from it.

Assembly gates. At the production means the top slots enter with $\partial C_8/\partial(\widehat{b}_{L,8-L}, \widehat{c}_{L,8-L}) = (-1, 3)/3^{L+1}$ to 4×10^{-16} relative, so the eight-loop constants enter only as $C_8(3) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$; loop seven enters through $(-16A_3, 42A_3)/19683$ with $A_3 = -\widehat{b}_{2,0}/6 = 1.6309960686$, to 8×10^{-16}; and the response of C_8 to the jets of loops five to eight is linear to 3×10^{-13}. These three gates are hard; all pass.

32.3 Pilot and design

A staged pilot fixed the design and never enters the estimate; its seeds are excluded from the production. A screen at 4,096 points per replicate (2,732 eight-loop graphs) selected the heavy-tail candidates. A full pilot integrated all 24,173 kernels in both campaigns with 8 replicates of 4,096 points per graph, 16 replicates for the power-containing eight-loop graphs, and gave $C_8(3) = 1619 \pm 16$. A confirmation pilot integrated 710 selected eight-loop graphs at 32,768 points

per replicate, within a cap of a quarter of the full pilot's eight-loop cost. The pilot used 422 core-hours.

Per-graph counts follow the Neyman allocation of Section 31.3 on a power-of-two lattice with at least 4,096 points per graph and 128 replicates in every campaign, sector and loop. For the 710 confirmed graphs the per-point standard deviation is the larger of the full-pilot and confirmation estimates. After the pilot the author chose a target standard error of 1, about 6×10^{-4} of $C_8(3)$, and the cheapest of three power-of-two designs, with a predicted standard error of 0.87 (1.31 with the safety factor 1.5 of Section 31.3) and an estimated cost of 15,492 core-hours, to be topped up if the realized error exceeded 1. Two independent campaigns a and b use importance rates multiplied by 0.8 and 0.7, as in Section 29, and 52- and 60-bit digital coordinates, with group seeds derived from a fresh base seed. The design contains 365,441,318,912 kernel samples.

32.4 The 2026 production

The production ran on the symmetry cluster from 9 to 12 October 2026, in lanes of eight threads on whole AMD EPYC 7543 nodes, at most eight at a time, on ten nodes in all. All 1,941 work indices and 25,714 checkpoint blocks completed. The lanes used 14,970 core-hours, 0.97 of the design's estimate, and the merges 106; with the plan job and the rate probes of Section 32.5 the production used 15,102 core-hours. Every merge, the assembly and an independent re-verification of the result record used the package's own tools. The two campaigns and the two half streams give

Estimate	$C_8(3)$	Sampling SE
Campaign a	1604.684	1.239
Campaign b	1600.527	1.445
First half stream	1602.651	
Second half stream	1602.559	
Combined	1602.605	0.952

The campaigns differ by 2.18 combined standard errors. Linearized with the assembly Jacobian, the difference 4.16 ± 1.90 is spread over loops and sectors, $+1.36$ and $+1.39$ from the power-free and power-containing eight-loop graphs, $+0.90$ from the power-free seven-loop graphs and $+0.19 \pm 0.10$ from the power-free five-loop graphs, with no term above two of its own standard errors. Over the eight-loop graphs the per-graph campaign differences have χ^2 per graph 0.994 (16,064 power-free) and 1.010 (4,336 power-containing), the ten largest carry $+0.19$ and -0.07, and no class of tail index carries the difference. No concentrated discrepancy is therefore seen. This is consistent with a fluctuation, but it does not exclude a small coherent effect of the importance rates spread over many graphs, and it does not by itself establish that the per-graph standard errors are calibrated. The error is dominated by the eight- and seven-loop graphs of both sectors:

Sector	$L = 3$	$L = 4$	$L = 5$	$L = 6$	$L = 7$	$L = 8$
Power-free	0.001	0.010	0.048	0.160	0.362	0.609
Power	0.000	0.003	0.018	0.079	0.220	0.566

Entries are in units of C_8 and combine in quadrature. The realized standard error is 1.09 times the design prediction.

32.5 Tail diagnostics

The fixed-k ($k = 16$) Hill estimate $\widehat{\xi}$ of Section 31.5 listed 10,034 eight-loop graphs with two-campaign mean $\widehat{\xi} > 0.44$ at production length. No graph carries more than 1.65% of the variance of

its group, and none was flagged above 3%. The listed graphs carry 62% of the power-free and 99.5% of the power-containing eight-loop variance, so the finiteness of their variance matters for the quoted error. The exact weight bound of Section 31.5 does not carry over, because the three-dimensional kernel contains $U_F^{3/2}$. Two empirical tests replace it.

First, the importance rates were re-derived on 48 graphs: those with the largest variance shares, representatives of the double-lift classes, two controls and the 16 power-containing eight-loop graphs whose $\widehat{\xi}$ did not fall in the pilot. No asymptotic rate deficit was found. Five apparent deficits were crossovers of the integrand from $t^{1.5}$ to $t^{1.0}$, with a zero of the integrand in between, and the asymptotic exponent 1.0 equals the table rate. This is a probe at one base point, not a certificate.

Second, a true heavy tail keeps $\widehat{\xi}$ constant as the number N of points per replicate grows, while bounded, peaked weights make it fall. For the 441 graphs confirmed in the pilot as possible infinite-variance cases (246 power-free, 195 power-containing), $\widehat{\xi}$ fell from $N = 4{,}096$ to $N = 32{,}768$ as about $N^{-0.23}$ and $N^{-0.18}$. The production, with independent seeds, mostly at intermediate lengths $N = 8{,}192$ and 16,384, reproduces this dependence: the medians of $\widehat{\xi}$ lie at 0.996 to 1.031 of the falling trend interpolated between the pilot lengths, 13 to 38% above the pilot's value at $N = 32{,}768$, which a true heavy tail would keep. The 13 graph-campaigns run beyond the pilot's range, at $N = 65{,}536$, continue to fall, to medians of 0.82 to 0.87 of their $N = 32{,}768$ values, at 0.94 to 1.02 of the extrapolated trend. At matched sample length the campaign shift of $\widehat{\xi}$ in the power-free sector is positive, +0.03 to +0.05, opposite in sign to the shift a rate-limited tail would produce; in the power-containing sector it is inconclusive. The data therefore behave as bounded, peaked weights would. This is empirical evidence, not a bound.

32.6 Checks

The same samples give lower-order coefficients, compared with the special-function values of Section 1 and with the productions of Sections 26.8 and 29, with both uncertainties combined:

Order	Estimate	Sampling SE	Pull
3	-0.3535809354	1.38×10^{-6}	$+0.43$
4	1.2280445577	3.88×10^{-5}	-0.49
5	-5.8499922821	4.86×10^{-4}	$+0.74$
6	33.5763740	0.0050853	$+0.19$
7	-220.3847628	0.0528950	-0.49

The reference values are checks only; no known coefficient is imposed and no power correction is forced to vanish. A pull above three in magnitude would have been reported without changing $C_8(3)$; none occurred. The measured power corrections at orders four to seven are $(8.6 \pm 7.3) \times 10^{-6}$, $(9.2 \pm 13.1) \times 10^{-5}$, $(3.3 \pm 19.5) \times 10^{-4}$ and 0.009 ± 0.024; at order three the difference is at the level of rounding. These are consistent with zero but do not prove a cancellation at eighth order, where the correction is 0.61 ± 0.61. A bootstrap of 4,096 replicate resamples gives standard error 0.940, against 0.952 from covariance propagation. These checks share the graph and renormalization framework and do not validate it.

32.7 An independent code

The eighth-order code of this subsection, `LRI_C8_D3_OPT/`, and its derivation, given below, were written by OpenAI GPT-6 astra (model stated by the author), which was given the seventh-order package of Section 29 and an earlier version of this note; the cluster port and the production were done within this project. The code extends that package independently of the package used above. It regenerates the graph inputs at every loop, restores the seven-loop classes excluded at

seventh order, extends the integrator and the regulator jets to eight loops on its own, derives its own low-loop inputs, and solves for the eighth-order fixed point with its own assembly; it integrates the same 24,173 kernels. It shares with the calculation above the three-dimensional mixed forest kernel and the precision policy of the seventh-order integrator, the importance sampler with rates multiplied by 0.8 and 0.7, the digital nets and the finite renormalization scheme. An error in those shared parts would not be detected.

Graph inputs. The code enumerates its own classes of the graph family of Section 28: connected one-particle-irreducible multigraphs without self-lines, with $L+1$ vertices, $2L$ individually labeled lines and at most four lines at each vertex. Classes are identified by a canonical form from partition refinement, keyed by multiplicity strings, and their counts agree at every loop with those of Section 31.2. An independent nauty computation on two-color incidence graphs (vertices and labeled lines) confirms all 26,540 eight-loop classes and their automorphism orders, and reproduces the archived lower-loop inventories with their exact weights. At eight loops the class weights sum to $\sum_G w_A = 49{,}985{,}915/768$ and $\sum_G w_B = 33{,}994{,}099/6{,}912$, the zero-dimensional four-point and mass-insertion values of (16) in class-weight form.

Its census of exact zeros and channel exclusions is that of Sections 31.2 and 31.7, which also explain why its 4,851 eight-loop zeros differ from the 3,908 of Section 30.2; it integrates $26{,}540 - 4{,}851 - 1{,}289 = 20{,}400$ eight-loop kernels, all exclusions being exact. At seven loops the code restores the 143 power-free and 119 power-containing nonzero classes with $3w_B = w_A$ that Section 29 omits, giving 2,441 and 709 kernels. Its builders reproduce all 3,511 payloads of the seventh-order calculation (loops three to seven), that is, line masks, denominator lists, integer numerators and importance rates, byte for byte after the header line, whose weights are checked against the exact rational values. A coverage record binds the census to the active inventory by its SHA-256 hash, and the assembly refuses to run without it.

Non-induced divergent subgraphs. A subgraph is induced if it contains every line of G between its vertices. The code builds its forests from induced divergent subgraphs only, as do the production builders whose maps the edge-subgraph enumerations of Sections 28 and 31.2 confirm. With parallel lines a graph can also have proper divergent subgraphs that are not induced. The forest terms that contain them cancel exactly, by the following argument. It uses only the graph family and the structure of the forest sum (780) with any fixed power of U_F in its denominators, so it holds in every dimension in which this sum is used, at $d = 2$ and, with $U_F^{3/2}$, at $d = 3$.

Let W be a proper nonempty subset of the vertices of G, $E(W)$ the set of lines with both ends in W and ∂W the set of lines with exactly one end in W. Since G is connected and has no bridge, $|\partial W| \geq 2$, and the degree bound gives

$$2|E(W)| + |\partial W| = \sum_{u \in W} \deg_G(u) \leq 4|W|, \qquad \text{hence} \qquad |E(W)| \leq 2|W| - 1. \tag{810}$$

A proper divergent subgraph does not contain every vertex: on all $L+1$ vertices a logarithmic subgraph would need all $2L$ lines, and a power subgraph one line more than G has. A power subgraph on v vertices has $2v-1$ lines, the maximum allowed by (810), so every proper power subgraph is induced. A proper divergent subgraph γ that is not induced is therefore logarithmic, with $2v-2$ lines on its v vertices, and misses exactly one line ℓ_0 of its induced closure $\bar{\gamma}$, the subgraph of all $2v-1$ lines between the vertices of γ. Adding a line to the one-particle-irreducible γ keeps it one-particle irreducible, so $\bar{\gamma}$ is a proper power subgraph, with loop number $h_{\bar{\gamma}} = v$.

Let m be a complement-of-tree monomial kept by γ, that is $|m \cap E_\gamma| = h_\gamma = v - 1$. The complementary spanning tree then has $v-1$ lines in γ, which form a spanning tree of the vertices of γ, so it cannot also contain ℓ_0, which would close a cycle. Hence $\ell_0 \in m$ and $|m \cap E_{\bar{\gamma}}| = v = h_{\bar{\gamma}}$: the constraint of $\bar{\gamma}$ follows from that of γ, and adding $\bar{\gamma}$ to a forest that contains γ does not change U_F.

Now take a compatible forest F with at least one non-induced logarithmic member, let γ be an inclusion-maximal one, chosen by a fixed rule, and toggle $\bar\gamma$: remove it from F if it is a member and add it otherwise. The result is again a compatible forest. By the definition of compatibility before (780), two members are either nested or edge-disjoint and meeting in at most one vertex. Members inside γ lie inside $\bar\gamma$. Members not nested with γ meet its vertex set, which is that of $\bar\gamma$, in at most one vertex, so they share no line with $\bar\gamma$. Members containing γ are induced, by the maximality of γ if they are logarithmic and by (810) if they are power subgraphs, so they contain $\bar\gamma$. The toggle does not change the set of non-induced logarithmic members, so it selects the same γ again and is an involution without fixed points. It leaves U_F unchanged, by the implied constraint, and the root numerator $\sum_{e\in E_R^{\log}(F)} x_e$, because $\bar\gamma$ is a power member and only logarithmic members remove lines from it; it reverses the sign $(-1)^{|F|}$. The forest terms with a non-induced logarithmic member therefore cancel in pairs, and the sum over forests of induced subgraphs equals the full forest sum. □

By the argument above, the closure of a non-induced divergent subgraph is a proper power subgraph, so power-free graphs have no such subgraph. The code also checks the cancellation by direct enumeration, which therefore needs only the power-containing classes. Its all-edge audit visits every proper subset of labeled lines of all 8,532 power-containing classes of loops three to eight, 492,490,392 subsets, tests one-particle irreducibility, builds the 11,123,924 unreduced mixed forests and reproduces every grouped integer map, the zero maps and the maps with a vanishing top channel included. At eight loops alone it covers the 7,231 power-containing classes, with 40,679 non-induced logarithmic subgraphs and 10,451,732 forests.

Assembly and analytic inputs. The truncated fixed point is obtained by nine iterations of $g \leftarrow \frac{1}{3}\big[\epsilon - \sum_{L=2}^{8} \widehat{b}_L(\epsilon) g^L\big] \bmod \epsilon^9$ from $g=0$, and in 64 exact rational cases, regulator-dependent and constant-coefficient ones included, it agrees exactly with the Lagrange formula (715) carried to eight loops. The response to the jets of loops five to eight has a closed form. Hold the jets of loops one to four fixed and put

$$\Phi_{\le 4}(g,z) = 3g + \sum_{L=2}^{4} \widehat{b}_L(z) g^L, \qquad \gamma_{r,\le 4}(g,z) = g + \sum_{L=2}^{4} \widehat{c}_L(z) g^L, \qquad \Phi_{\le 4}\big(g_{\le 4}(\epsilon),\epsilon\big) = \epsilon, \quad (811)$$

so that $g_{\le 4}(\epsilon) = \epsilon/3 + O(\epsilon^2)$ is the fixed point of the functions (802) truncated after four loops. A change of the jets of loops five to eight changes the exponent by

$$\delta\Delta_{\mathcal{E}}(3,\epsilon) = \sum_{L=5}^{8} g_{\le 4}^L \Big[\delta\widehat{c}_L(\epsilon) - \frac{\partial_g \gamma_{r,\le 4}(g_{\le 4},\epsilon)}{\partial_g \Phi_{\le 4}(g_{\le 4},\epsilon)}\, \delta\widehat{b}_L(\epsilon)\Big] \pmod{\epsilon^9}, \quad (812)$$

with $g_{\le 4} = g_{\le 4}(\epsilon)$. This first variation is exact through order eight, because two insertions of loops five or higher first contribute at order $5+5-1=9$. With $\widehat{b}_L(z) = \sum_j \widehat{b}_{L,j} z^j$ and likewise for $\widehat{c}_L$, the Jacobian of C_8 is therefore, for $L = 5,\dots,8$ and $0 \le j \le 8-L$, whatever the jets of loops five to eight,

$$\frac{\partial C_8}{\partial \widehat{c}_{L,j}} = [\epsilon^{8-j}]\, g_{\le 4}^L, \qquad \frac{\partial C_8}{\partial \widehat{b}_{L,j}} = -[\epsilon^{8-j}]\, \frac{\partial_g \gamma_{r,\le 4}(g_{\le 4},\epsilon)}{\partial_g \Phi_{\le 4}(g_{\le 4},\epsilon)}\, g_{\le 4}^L. \quad (813)$$

For $j = 8-L$ it reduces to the top-slot rule $(-1,3)/3^{L+1}$ of Section 32.2, so the eight-loop constants enter only as $C_8(3) \supset (3\widehat{c}_{8,0} - \widehat{b}_{8,0})/19683$; for $(L,j) = (7,0)$, with $A_3 = -\widehat{b}_{2,0}/6$ (Section 32.2) and $\widehat{c}_{2,0} = \widehat{b}_{2,0}/6$, it gives $(-16A_3, 42A_3)/19683$. A seven-loop class with $3w_B = w_A$ has $\delta\widehat{b}_{7,j} = 3\delta\widehat{c}_{7,j}$. Its top slot cancels, but through the mixing with the two-loop input its constant terms give $\delta C_8 = -2A_3\,\delta\widehat{b}_{7,0}/19683 = \widehat{b}_{2,0}\,\delta\widehat{b}_{7,0}/59049$, which is why the code restores these classes.

The code derives its own two-loop and Gamma inputs. At 80 digits, Cauchy contours of radii 0.04 and 0.07 with 80 and 96 nodes agree on all seven two-loop entries to 3.5×10^{-72}; the six entries shared with Section 29 agree with that calculation's input to 4.5×10^{-59}; and a 112-node Cauchy

extraction of the logarithm of the Gamma product checks all 42 Gamma-jet coefficients (loops two to eight, orders one to six) to 1.2×10^{-73}. These are high-precision cross-checks, not interval bounds. By direct comparison of the files `inputs/two_loop.json` and `inputs/gamma_d3_8.txt` of the two packages, the two-loop entries and the Gamma table differ from those of the calculation above by at most 1.7×10^{-57}, the precision of the stored two-loop entries there (Section 32.2), and 4.1×10^{-58}.

Worker. The production used the code's optimized worker, with the 350-digit GMP tier. Its one change to the arithmetic is in the binary128 additions inside the positive denominator polynomials: the 113-bit integer significands are aligned with guard, round and sticky bits and rounded to nearest, ties to even, so each sum is the correctly rounded one; zeros, subnormal and negative operands, exceptional values and possible overflow go to the compiler's addition. Signed forest sums, the addition order, the promotion thresholds and the sampling are unchanged. A test of 12,051,840 additions agrees bit for bit (NaN cases by outcome); on 179 graphs at 40,448 fixed points the optimized and the unoptimized worker give identical coefficient vectors, maximum contributions and promotion counts; 2,432 evaluations at shared points reproduce the jets and promotion counts of the seventh-order worker of Section 29 exactly; and 48 points of 12 eight-loop graphs agree with a 450-digit reference within the tolerances of their precision tiers. Three builds on Expanse gave one binary hash and passed all 14 build and test steps.

Pilot and allocation. The code was delivered with a pilot on its development host: both campaigns with 8 replicates each and 32,768, 8,192, 2,048, 512, 64 and 16 points per graph and replicate at loops three to eight, 21,342,208 evaluations in all. It gave $C_8(3) = 1514.6 \pm 161.7$ (sampling standard error), an exploratory estimate. Its per-graph variances and timings fixed the allocation of the production by the code's Neyman rule: the points per graph are proportional to the square root of the ratio of per-point variance to per-point cost, rounded up to a power of two, at least 4,096, and scaled to the budget. The replay reproduces the allocation file byte for byte. The pilot is not part of the estimate.

Production. The code was adapted to the Expanse cluster without a change to its numerical files. With a fresh base seed, an allocation of about 2,000 core-hours of its development host, 16 replicates per campaign and at least 4,096 points per graph, it ran on 10 and 11 October 2026 on six whole AMD EPYC 7742 nodes, the last 5 of its 1,994 jobs in a separate one-node job, with 3.7×10^{10} evaluations and about 9,800 service units in all. It gives

$$C_8(3) = 1600.98 \pm 3.97 \quad \text{(sampling standard error)}, \tag{814}$$

-0.40 combined standard errors from the estimator. Its campaigns, 1603.53 ± 4.91 and 1598.42 ± 6.24, differ by 0.64 combined standard errors; its lower-order coefficients agree with the references within 1.73 standard errors (pulls -1.41, -1.55, -0.10, $+1.73$ and -0.21 at orders three to seven; the squared pulls sum to 7.40 for these five checks), and its sixth- and seventh-order by-products are 33.599 ± 0.014 and -220.40 ± 0.19. Its power-free part is 1602.61 ± 2.96 and its power correction -1.63 ± 2.64, with covariance 1.9×10^{-4}; they differ from those of Section 32.1 by $+0.20$ and -0.82 combined standard errors. The half streams give 1599.79 and 1602.17, deviations of -1.19 and $+1.19$ from the estimator, and a bootstrap of 4,096 replicate resamples gives standard error 3.78. The eight-loop graphs carry 42.1% (power-free) and 31.5% (power-containing) of the variance, the seven-loop graphs 12.2% and 12.6%, and the six-loop graphs 1.4%. The power null rows at orders four to seven are $(-4.66 \pm 2.22) \times 10^{-5}$, $(-1.91 \pm 4.57) \times 10^{-4}$, $(-1.21 \pm 5.62) \times 10^{-3}$ and 0.072 ± 0.126, with pulls -2.10, -0.42, -0.22 and $+0.57$, whose squares sum to 4.95 for these four null rows.

The quoted error is the covariance of the replicates propagated through the fixed point. The code assigns no allowance, and its result record marks the reportable precision as not established. There

were no tail diagnostics like those of Section 32.5 and no convergence test beyond the campaign and half-stream splits, and neither the delivered science code nor this number had an independent review: the reviews of the port covered only its cluster layer. The independent estimate is not combined with the reported value.

32.8 Records

The production result, the frozen plan and the merged replicate data that reproduce the estimator are in `GFF/results/C8_3/`; the per-graph replicate tables are hosted in `GFF/data/C8_3/`. The verification, the job records and the analyses of Sections 32.4 and 32.5 are in `LRI_C8_D3_HPC/benchmarks/symmetry_2026-10-c8/production/prod1/`. The pilot report is in `pilot/evidence/` and the rate probes in `production/margin_*` of the same `symmetry_2026-10-c8/` folder, and the independent code's production is in `GFF/results/C8_3_check/`. `LRI_C8_D3_HPC/public.mk` recomputes every number of Sections 32.1–32.6 from these records. The numbers of Section 32.7 come from the independent package. `LRI_C8_D3_OPT/public.mk` recomputes those of its production from `GFF/results/C8_3_check/`, except the half streams and the bootstrap error, which need the replicate matrices and are read from its result record. The census, audit, input, worker and pilot checks are the package's delivered evidence (`LRI_C8_D3_OPT/verification/`, `docs/` and `README.md`), its builds are summarized in its `public/REVIEWS.md`, and the differences between the input files follow from the files `inputs/two_loop.json` and `inputs/gamma_d3_8.txt` of the two packages. The complete run tree of the production of Section 32.4 is archived in two copies (Section 33); the native outputs of the independent production are not archived.

33 Reproduction and data dependencies

The code and records behind this note are in the folder `GFF/` of the public repository of this work. Its folder `GFF/code/` holds the packages named in the table below, each at the commit that produced the quoted results; package paths in this note, in the text and in the table, are relative to `GFF/code/`. The records from which the quoted numbers are recomputed are in `GFF/data/`, and the published outputs in `GFF/results/`; these paths are written in full. Every package has a file `public.mk`: `make -f public.mk test` runs its tests, and `make -f public.mk replay` recomputes from that package's shipped records the numbers this note quotes for the results the package produced, and compares them with the quoted digits. Numbers of earlier calculations that a package keeps in `reference/`, checks made when the separate notes were merged into this note, and construction checks whose records are hosted are not all replayed; the rows below say which. Large raw files (checkpoints, graph packs, payload archives) are hosted outside the repository and fetched with `make -f public.mk fetch-data`; the package's `DATA.md` says which target needs them, and its review summary, `REVIEWS.md`, says which reviews it received. The present note is a mathematical condensation, not a production integration campaign. A package's finite parts and graph ordering must be used with that same package's code; files with similar names from different schemes are not interchangeable. Packages of the earlier research archive `LRI_complete_research_bundle.zip` that hold only superseded estimates are not part of the repository. The separate TEX notes that came with several packages are merged into this note and are left out of the repository: their derivations, methods, results and caveats are in the sections cited in the table, and Section 33 says where each block came from. Markdown records of derivation steps that ship inside a package stay there as its records; for example, the delivered stage scripts of `C6_research_note/` read or write such files under `code/stages/` and `inputs/source_snapshots/`. No step of this note cites them; the arguments are written out here, for $C_6(1)$ in Appendix B.

Provenance of the merged material. The table lists the blocks that were merged into this note from separate notes and records on 16 October 2026, with their source and author. None of

these sources names its author: their bylines read "Research calculation record" or "Calculation record", or are empty. The entry "OpenAI GPT-6 astra" records the author's statement that the source was written by that model (for the research bundle and the delivered packages, 13 October 2026; for `C6_research_note/`, 16 October 2026); "author not named" marks a source for which there is no such statement. Claude Code sessions wrote the merged text itself and adapted it to the notation of this note.

Block	Source	Author
Appendix B (derivation of the $C_6(1)$ representation)	`C6_research_note/`: its note and input records	OpenAI GPT-6 astra
Section 27: method, design and checks of the seventh order	`LRI_C7_HPC/reference/`: the two seventh-order notes	OpenAI GPT-6 astra
Sections 1 (lattice model), 1.4 (the caveat on repeated parameters), 3 (zero-dimensional benchmark), 8.3 (checks of $\mathcal{K}$), 9 (decimal value of $\mathcal{L}$), 10 (100-digit value of P_5), 14	the earlier complete note in `LRI_C7_HPC/reference/`	OpenAI GPT-6 astra; the resummation of Section 14.2 recomputed at the merge by a Claude Code session
Sections 34.9, 34.10; the $d = 1$ checks of Section 34.12	`lri_ope_reproducibility/` and [10]	written at the merge by a Claude Code session
Section 16.8, values	the note's closed forms; `LRI_C5_d3/` records	a Claude Code session
Section 16.8, spectral route to $d = 2$	`lri_d3_reduction_update/`: its complete note	OpenAI GPT-6 astra
Section 17.4 (double residue series of $P_5(d)$)	`lri_d3_reduction_update/`: the general-dimensional addition	OpenAI GPT-6 astra; steps the source left implicit written out at the merge by a Claude Code session
(404) and the measure-pole bound	`lri_d3_reduction_update/`: the periodization addition	OpenAI GPT-6 astra
Section 19: opening, count table, Section 19.5, values and dictionary of Section 19.10, catalogue search	`lri_d3_reduction_update/`: the three-dimensional addition	OpenAI GPT-6 astra
Section 19.9, rank-two checks, open problem	`lri_d3_matrix_update/`: its complete note	OpenAI GPT-6 astra
Section 19: the link to $\mathfrak{t}_3(3; \frac{3}{2})$	derived at the merge	a Claude Code session
Section 20.6 (half-shift identities at $d = 3$)	`LRI_C5_d3/`	a Claude Code session (written out at the merge)
Section 21, including the growth proof and the continuation	`LRI_C5_general_d/K_general/`	a Claude Code session
(560)	`LRI_C5_general_d/K_general/`; constant derived at the merge	a Claude Code session
Section 25.4, the 300-digit fallback	`lri_C6_d3_complete_update/`	OpenAI GPT-6 astra
Section 26.14, $d = 3$ rows of the method comparison	`LRI_C6_D2_HPC/reference/`: the numerical-methods note	OpenAI GPT-6 astra; the explanation of the failed control added at the merge by a Claude Code session
Section 32.7	`LRI_C8_D3_OPT/`: its derivation and calculation notes	OpenAI GPT-6 astra; cluster port and production by a Claude Code session
Part V, general d, $d = 3$ and the bootstrap comparison	`lri_ope_reproducibility/`: its note	author not named; supplied by the author on 16 October 2026; the checks of Section 34.12 by Claude Code sessions
Section 17.2: numerical discovery of $\mathcal{L}_2$, $\mathcal{K}_2$ and the single-sum relations	`LRI_C5_d2_simplify/`	a Claude Code session

Block	Source	Author
Sections 18.7 (first and second proofs of $\mathcal{L}_2$) and 18.8 (proof of $\mathcal{K}_2$); in Section 18.9 the sharper estimate, the per-spin identity, the second and third single-sum evaluations, the mixed-trace corollary and the Riemann bilinear proof	`LRI_C5_d2_proof/`	a Claude Code session with prover and reviewer subagents
Sections 18.5.2, 18.7.7–18.7.9; the regularized spectral route of $\mathcal{K}_2$ in Section 18.9	`LRI_C5_d2_simplify/proof/`	a Claude Code session with prover, verifier and synthesizer subagents
Section 20: the generic-dimensional reduction of $\mathcal{L}_d$, Z_d and its proofs	`LRI_C5_general_d/`	a Claude Code session; the explicit constants of (465), (466), (467) come from the growth proof of Section 21.6
Sections 23–25: the blocks merged from the earlier complete note	`LRI_C6_D2_HPC/reference/`: the complete note	OpenAI GPT-6 astra; one closed-form value and one derivation added at the merge by a Claude Code session
Checks merged from the same complete note into Sections 3.4 (what the higher-pole identities cannot detect), 15.8, 16.8 (controls of the master evaluations), 17.7 (checks of the finite multivariable construction) and 18.6	`LRI_C6_D2_HPC/reference/`: the complete note	OpenAI GPT-6 astra
Sections 26.11–26.16: method blocks	`LRI_C6_D2_HPC/reference/`: the methods note	OpenAI GPT-6 astra; the variant descriptions read from the code, the linear-control formula and one proof added at the merge by a Claude Code session
Section 28: merged blocks	`LRI_C7_D2_HPC/reference/`: the method and progress notes	OpenAI GPT-6 astra; the three-loop check, the two-loop entries, the primitive classes and the variance decomposition written out or rechecked at the merge by a Claude Code session
Section 31.7: merged blocks	`LRI_C8_D2_HPC_optimized/`: the calculation note and results fragment	OpenAI GPT-6 astra; the 65- and 70-digit comparison of the two packages' inputs made at the merge by a Claude Code session
Section 34.8, the $d = 2$ rows of Section 34.11 and (863)	written for this note from `lri_ope_reproducibility/`	a Claude Code session; checks in `ope_checks/d2/`

Result or dependency	Package and principal record
One-dimensional analytic C_5	`lri_numerical_resolution/`: graph audit in `verification/records/resolution_audit.json`, with the raw five-loop and lower-loop Laurent records in `verification/records/`, the residue certificate in `verification/inputs/residue_certificate.json` and the kernels that produced the records in `verification/source/`. The packages `lri_psi_complete_update/` and `lri_fixed_hypergeometric_update/` supply the checks of the polygamma evaluation of $\mathcal{K}$ and of the five-value P_5, with the records `verification/numerical_checks.json` and `verification/fixed_hypergeometric_checks.json`. The quoted values are collected in `GFF/results/closed_forms_1/C5_1_quoted.json`, and `make -f public.mk replay` in each package folder recomputes them from the shipped files in under a minute on one core.
General C_0–C_5	`lri_general_special_functions_update/` (with the general-dimensional C_0–C_3, graph-inventory, tetrahedron and two-loop files of Section 15), `lri_general_closed_update/`, `lri_general_periodization_update/`; master evaluators, finite arrays, residue and assembly checks. The quoted values and formulas, with their records, are in `GFF/results/closed_forms_general/C0_C5/expected.json`; `make -f public.mk replay` in each package recomputes them.
Closed forms of $\mathcal{L}_2$, $\mathcal{K}_2$ and $C_5(2)$	`LRI_C5_d2_proof/` (proof routes with their machine checks; the computations behind the quoted checks in `review_*/`; reviews summarized in `REVIEWS.md`) and `LRI_C5_d2_simplify/` (discovery, 70-digit single sums `confirm_dps70.json`, scripts and logs of the three further proofs in `proof/`, whose $\mathcal{L}_2$ parts are written out in Sections 18.7.7–18.7.9; reviews summarized in `REVIEWS.md`); the independent spectral evaluation is `LRI_C5_d3/wf_d2check/d2check_results.json`. Expected values: `GFF/results/closed_forms_2/closed_forms_2.json`; `make -f public.mk replay` in either package recomputes every quoted number.

Result or dependency	Package and principal record
Generic-dimensional reduction of $\mathcal{L}_d$, Z_d and $\mathcal{K}_d$	`LRI_C5_general_d/`: the code and validation records of the general-b periodization (Section 20.1) in `foundation/`; the specialization code and records in `specialize/`; records of the checks in `K_general/note_fragment/growth/checks/`, `K_general/note_fragment/topmode/results/`, `specialize/results/`, `L/results/` (with the code of Section 20.3 in `L/`), `Z/results/` and the end-to-end $C_5(1)$, $C_5(2)$, $C_5(3)$ in `review/joint/numerics-end-to-end/e2e_report.json`; the review records `review/VERIFICATION.md`, `review/lemmaG-proof/` and the check code of `review/` (`lemmaG-numeric/`, `topmode-numeric/`, `joint/numerics-end-to-end/`), with all reviews of both sections, including those of `K_general/note_fragment/`, summarized in `REVIEWS.md` (the review transcripts are not included); independent spectral references `references/reference_values.json`; 100-digit $\mathcal{K}_d$ in `K_general/position_space/K_d_values_mb_D100.json`. `make -f public.mk replay` compares with these records every quoted number of Section 21 (expected values `public/expected_K_d.json`) and the numbers of Section 20 listed in `GFF/results/closed_forms_general/reduction_d/expected.json`, and reruns the fast checks among them. It does not compare the numbers of Sections 20.1.4, 20.3 and 20.4, nor those of the items of Section 20.9 on the general-b periodization, the review probes, the product form in detail and the two-node formula; these were added on 16 October 2026 and are read from the records named there. `make -f public.mk replay-full` recomputes part of those records (`L/results/val50.json`, `L/results/val70.json`, `Z/results/zj*.json`, `Z/results/zc*.json`, `Z/results/compare_refs.log`, `Z/results/check_formulas.log`) and compares them with the shipped records, not with the text; the records in `foundation/results/`, `Z/results/offdiag60.json` and the other files of `L/results/` are recomputed by no target. The $d = 3$ inputs are in `LRI_C5_d3/` (`wf_L3/`, `wf_K3/K3_value.json`), with the quoted values in `GFF/results/closed_forms_3/`.

Result or dependency	Package and principal record
Three-dimensional specialization	`lri_d3_reduction_update/`: `code/general_d3/check_exact.py` (residue algebra, polynomial identities and rational Horn coefficients; `verification/general_d3/exact_algebra.json`, with the eight real-node Horn arrays in `horn_arrays.json` of the same folder), `code/general_d3/verify_d3.py` (formula-side values at 55 and 90 digits, the 75-digit determinant, mode and complex-node checks and the literal hypergeometric checks; `period_values.json`, `identities.json`, `literal_hypergeometric.json`) and `code/general_d3/check_spectral.py` (fresh real-axis Gamma kernels; `fresh_spectral.json`; the residuals are its differences from `period_values.json`, also listed in the delivered record `comparison_summary.json`). `lri_d3_matrix_update/`: `code/rank_two_checks.py` (the 1107 exact checks and the degree-90 interior sums; `verification/rank_two_checks.json`), `code/recompute_d3_periods.py` (the 100-digit values; `verification/fresh_primitive_values.json`) and `code/scalar_catalog_search.py` (the catalogue search of Section 19.10; `verification/scalar_catalog_search.json`, rerun by `make -f public.mk search`). Section 19 quotes these records. The quoted values and their tolerances are in `GFF/results/closed_forms_3/d3_specialization_quoted.json`.
$C_6(1)$	`GFF/results/C6_1/C6_final.json`, with the degree sequence in `assembled.json`, the rounding estimates in `rounding.json`, the stopping decisions in `decisions/`, the pre-registered plan `PLAN.md` and the report `REPORT.md` of the same folder. The results-review checks are `C7_crosscheck.json` and `lower_amplitude_precision/` in that folder. The code is `LRI_C6_1D_HPC/` at git commit `cef6fa7`; the production ran at `8680864`. The merged level records and the engine outputs are in `GFF/data/C6_1/`. The target `replay` of `public.mk` in the code folder regenerates these records and checks every quoted number. The superseded Monte Carlo estimator is not included in the public repository.
Analytic representation of $C_6(1)$	`C6_research_note/`: the reassembly `outputs/reassembly.json`, the interval certificate in `outputs/rerun/complete_refined/` and the task runner `code/reproduce_final.py`, which runs the 22 tasks of the final method (Section B.8). The derivation is Appendix B. The target `replay` of `public.mk` in that folder recomputes the quoted numbers from these records and compares them with `GFF/results/C6_1/analytic/quoted_numbers.json`. The independent review is `LRI_C6_closed_form_review/REVIEW.md`, with its check scripts and their recorded outputs in `checks/` of the same folder.

Result or dependency	Package and principal record
Inherited $d = 2$ finite kernels	Superseded inputs of the previous two-dimensional estimator (the primitive and locally subtracted kernels of Section 23), not additional contributions to add again and not needed to reproduce any current coefficient. The kernels, rates, two-loop input and fixed-point map that the 2026 production uses are byte-identical copies in `LRI_C6_D2_HPC/reference/`, bound by `LRI_C6_D2_HPC/inputs/manifest.json`. The construction records (`lri_C6_d2_primitive_update/`, `lri_C6_d2_subtraction_update/`, `lri_C6_d2_two_forest_update/`) are in the research archive `LRI_complete_research_bundle.zip`, which is not part of this repository.
Power compensation and finite kernels	`lri_C6_d2_complete_update/`: the part of the research-archive package that the two-dimensional power-sector compensation and the audits of Sections 24 and 25 need, namely the saved counterterms, the check scripts and the counterterm chain. `make -f public.mk replay` reruns the checks and compares them with the records in `GFF/results/C6_2/power_compensation/`, among them `GFF/results/C6_2/power_compensation/verification/power_sector_audit.json`; `make -f public.mk regen` regenerates the counterterms from the graph inventory. The three-dimensional compensation is `lri_C6_d3_complete_update/`, with its full-coverage, analytic-extension, power-sector and fixed-point audits; its quoted values are in `GFF/results/C6_3/power_compensation_quoted.json`. The superseded first full sampling run of $C_6(3)$ in that package is not included.
$C_6(2)$	`GFF/results/C6_2/C6_result.json`, with the production log, reporting rule and checks in `GFF/results/C6_2/CAMPAIGN.md`. The frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records are in `GFF/data/C6_2/run-20260928-m8192/`, and the complete run directory with the checkpoints is hosted outside the repository and fetched by `make fetch-data`. The code is `LRI_C6_D2_HPC/` at git commit `6aaf656`; `make -f public.mk replay` there recomputes every number quoted here for $C_6(2)$ from the merged records, and `make -f public.mk remerge` re-merges the checkpoints. The seventh-order cross-check is `GFF/results/C7_2/C7_result.json`. The superseded estimator is `LRI_C6_D2_HPC/reference/verification/method_precision_results.json`.

Result or dependency	Package and principal record
Superseded sixth-order estimators at $d = 2, 3$ and the method benchmarks	`LRI_C6_D2_HPC/reference/`, the earlier method-refinement calculation shared by the two sixth-order packages: the refinement, pilot, validation, speed and endpoint-replay records in `records/` (the baseline campaigns in `baseline_previous/records/`); the per-graph counts, binary Sobol direction matrices, frozen transport tables `grid_d2_top3.txt`, `grid_d2_top11.txt` (and their $d = 3$ counterparts) and the frozen plan `frozen_production_plan.json` in `inputs/`; the source and input hashes in `verification/production_source_hashes.json`. Without new integration, `code/analyze.py` reassembles the previous estimators and their campaign comparisons (`verification/method_precision_results.json`; the rounded reporting convention is stored separately in `verification/reported_values.json`), `code/summarize_methods.py` the benchmarks of Section 26.11 (`verification/method_benchmarks.json`) and `code/extra_checks.py` the finite Lagrange, bootstrap and delete-one checks (`verification/extra_checks.json`). These archived scripts rewrite their output files in `verification/` in place, and on another NumPy build the last bits can differ; `LRI_C6_D2_HPC/tools/reassemble_reference.py` runs the archived `code/analyze.py` for $d = 2$ without writing and compares with the stored result to relative 10^{-12}. The integrator is rebuilt by `code/build.sh` (C++17 with OpenMP, libquadmath and Boost.Multiprecision) and replayed from the shipped direction matrices by `code/replay.py`. `make -f public.mk replay` does not run these archived scripts; the method numbers of Sections 26.11–26.13 that are derived from stored values, such as the linear-control ratios and the refinement pulls, are recomputed by no script.
$C_6(3)$	`GFF/results/C6_3/results_run-20260929-m8192/C6_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records in the same folder. The production log, reporting rule and checks are in `GFF/results/C6_3/CAMPAIGN.md`, and the complete run directory is `GFF/data/C6_3/run-20260929-m8192_full.tgz`, hosted outside the repository and fetched by `make fetch-data`. The code is `LRI_C6_D3_HPC/` at git commit `1c6dfb3`; its archived-stream replay is `benchmarks/expanse_2026-09/logs/validation_54510646/archived_stream_replay.json` in that folder, and `make -f public.mk replay` there recomputes every number quoted here for $C_6(3)$ from the merged records. The superseded estimator is `reference/verification/method_precision_results.json` in the reference tree that the package shares with `LRI_C6_D2_HPC/`.

Result or dependency	Package and principal record
Full $C_7(1)$	`GFF/results/C7_1/C7_final.json`, with the frozen design `design_v1.json`, the batch inclusion list `inclusion.txt`, the combined assembly `combined_assembly.json` and the report `REPORT.md` in the same folder; the earlier 46-batch combination is in `v1_46batches/` of that folder. The code is `LRI_C7_HPC/`, exported from git commit `68dc6be`, whose code files equal those of commit `7320afc`, which ran the final combination; the production workers ran commit `3213ce0`, which differs from it only in the verification, assembly, combination and design tools and their tests. Full forest arrays, the retained primitive input and the earlier assembly are in `LRI_C7_HPC/reference/`. The target `replay` of `public.mk` in the code folder recomputes the quoted numbers from these records. The merged replicate records of the 399 batches and of the primitive campaign are hosted data, described in `GFF/data/C7_1/DATA.md`; from them the target `replay-full` reruns the combination.
Full $C_7(2)$	`GFF/results/C7_2/C7_result.json`, with the frozen plan `manifest.json` and the per-group `merged.json` records in `GFF/data/C7_2/run1-20260930-m4096/`; `make -f public.mk replay` in `LRI_C7_D2_HPC/` recomputes the quoted numbers from them for the 2026 production, together with the earlier estimator and power correction (reassembled from `reference/` by `tools/reassemble_reference.py`), the three-loop reference and the coverage tables; the other numbers of the earlier calculation in Section 28 are records in `reference/`, reassembled by `reference/code/reproduce.py`, and are not replayed. The production log and checks are in `GFF/results/C7_2/CAMPAIGN.md`; the code is `LRI_C7_D2_HPC/` at git commit `fb64c47`. The code and records of the earlier calculation (Section 28), with its power-free and primitive campaigns in `reference/LRI_C7_d2_progress/` and its power-sector completion in `reference/power_mass/`, are in `LRI_C7_D2_HPC/reference/`. The checkpoints and task tables of the run are archived, not included.
Full $C_7(3)$	`GFF/results/C7_3/run1-20260930-m4096/C7_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records in the same folder. The production log, reporting rule, errata and checks are in `GFF/results/C7_3/CAMPAIGN.md`, and the complete run directory is the hosted `GFF/data/C7_3/run1-20260930-m4096_full.tgz`. The production used `LRI_C7_D3_HPC/`, the export of git commit `4de9412`; the re-verification tool `LRI_C7_D3_HPC/reverify_a6ce987/tools/report.py`, from commit `a6ce987`, reproduces the result. The previous calculation and its construction checks are in `LRI_C7_D3_HPC/reference/`, with its raw records in the hosted `GFF/data/C7_3/reference_records.tar.gz`; the archived-stream replay is in `GFF/results/C7_3/evidence/replay_report.json`, and the reviews are summarized in `GFF/results/C7_3/REVIEWS.md`.

Result or dependency	Package and principal record
Full $C_8(1)$	`GFF/results/C8_1/2026-10_ext1/C8_final_A3/C8_final.json`, the fourteen-batch result, with the report `REPORT.md` one level up, the nine-batch interim in `C8_A1_close/` and the records of both clusters in `records_symmetry/` and `records_expanse/`; the extension rules are `LRI_C8_HPC/docs/EXTENSION_A1.md`, `EXTENSION_A2.md` and `EXTENSION_A3.md`, and the independent reviews are summarized in `LRI_C8_HPC/REVIEWS.md`. The four-batch result `GFF/results/C8_1/2026-10_expanse/C8_final.json` has the report `REPORT.md`, the frozen design `frozen_design/`, the controlled affine map `full_affine8.json`, the batch plans and the combined assembly in the same folder. The merged records of all fourteen batches and of the seven-loop test with 4096 points per graph are in `GFF/data/C8_1/`; `make -f public.mk replay` in `LRI_C8_HPC/` recombines them with the production tools and checks every quoted number. The production code is `LRI_C8_HPC/` at git commit `c0ffa63` (tools of the extension at `15a110e` and `bd13a63`; the public copy is commit `1282e37`, whose code equals `bd13a63`); its nine-vertex inventory is `inputs8/` (the eight-loop coordinate file compressed in `GFF/data/C8_1/inputs8/`), generated by `gen/`, and its kernel regressions against the seventh-order binary are in `validation/`.
Full $C_8(2)$	`GFF/results/C8_2/C8_final.json`, with the tail diagnostics in `GFF/results/C8_2/tailcheck/` and the production's replicate records (plan, design bundle, merged groups) in `GFF/data/C8_2/`. The code is `LRI_C8_D2_HPC/` at git commit `9df5b1c`, whose code equals that of `ba68f40`, the export the production ran; its folder `benchmarks/symmetry_2026-10-c8/` holds the result report (`production/result/README.md`), the tail-check report (`production/tailcheck/README.md`) and the pilot analysis (`pilot/analysis/`), and `make -f public.mk replay` recomputes every number of Sections 31.1–31.6 from the records. Its plan reviews and code inspections are summarized in `LRI_C8_D2_HPC/REVIEWS.md`. The shared nine-vertex inventory is `LRI_C8_shared/inventory/inventory8_d2/`, and the shared eighth-order assembly `LRI_C8_shared/assembly/` is vendored byte for byte in `vendor/c8_assembly/`. The full campaign directories are archived as listed in `GFF/data/C8_2/DATA.md`.

Result or dependency	Package and principal record
Independent check of $C_8(2)$	`GFF/results/C8_2_check/prod1/C8_result.json`, with the frozen plan, the group configurations and merged records in `GFF/data/C8_2_check/prod1/`, the pilot in `GFF/data/C8_2_check/pilot1/` and the analysis in `GFF/results/C8_2_check/prod1_analysis/`. The code is `LRI_C8_D2_HPC_optimized/` at git commit `a791a1f`; `make -f public.mk replay` there recomputes the numbers of the paragraphs on the port, pilot and production and on the comparison in Section 31.7; the construction checks under Inputs, Graphs and payloads, Kernel arithmetic and Assembly, and the development runs, rest on the records of the hosted original package, and the comparison of the two packages' inputs, the bootstrap ratios and the bootstrap mean are not replayed. Its reviews are summarized in `LRI_C8_D2_HPC_optimized/REVIEWS.md`. The full campaign directory and the original package with its graph packs are hosted (`GFF/data/C8_2_check/DATA.md`).
Full $C_8(3)$	`GFF/results/C8_3/C8_result.json`, with the frozen plan `manifest.json`, the group configurations and the per-group `merged.json` records that reproduce the estimator in the same folder; the per-graph replicate tables `graph_replicates.csv` are hosted in `GFF/data/C8_3/`. The production report `README.md`, the tail diagnostics `tail_ntrend/` and the campaign split `ab_split/` are in `LRI_C8_D3_HPC/benchmarks/symmetry_2026-10-c8/production/prod1/`, the rate probes in `production/margin_*`, the priced designs in `production/price/` and the pilot report in `pilot/evidence/` of the same `symmetry_2026-10-c8/` folder. The code is `LRI_C8_D3_HPC/`: the production ran git commit `4fbd1bae`, which is the code shipped, and the evidence is taken from commit `6dd4a75a`; `public.mk` replays every quoted number. The plan, its review log and the results review are not included; they are summarized in `public/REVIEWS.md`. The shared nine-vertex inventory and eighth-order assembly are those of the $C_8(2)$ row. The campaign directories (13 GB) are not included; they are archived in two copies.

Result or dependency	Package and principal record
Independent check of $C_8(3)$	`GFF/results/C8_3_check/runs/prod1/C8_result_54838171.json`, with the frozen plan `manifest.json` and the per-graph statistics `graph_statistics_54838171.json` in the same folder, the frozen allocation `allocations/prod1_2000.json` and the production summary `README.md` in `GFF/results/C8_3_check/`. The code is `LRI_C8_D3_OPT/` at git commit `d5deeb9`; the target `replay` of `public.mk` in that folder recomputes the estimator, its standard error and the quoted checks from the means and covariances stored in the result (`public/replay_check.py`). The port to the clusters, its decisions and its reviews are summarized in `public/PROVENANCE.md` and `public/REVIEWS.md` of the same folder; the logs, the admission ledger and the lane summaries are not included. The graph payloads and the delivered pilot are hosted data, and the native outputs of the campaign are not archived; both are described in `GFF/data/C8_3_check/DATA.md`.
Bulk one-point coefficient (Part V)	`lri_ope_reproducibility/`, supplied by the author on 16 October 2026: the engines in `code/`, the graph inventories and master inputs in `data/`, the coefficient table `results/coefficients.json` and the verification reports in `results/verification/`; the tasks of its runner `run.py` are listed in Section 34.13. `make -f public.mk test` checks the release hashes and runs the quick task in a work folder, `make -f public.mk replay` also reruns the checks of Section 34.12, which are in `ope_checks/` (folders `d1/`, `d2/`, `d3/`, with their recorded outputs), and `make -f public.mk full` reruns the remaining tasks (about ten minutes on one core). The package's own note is merged into Part V and is not part of the repository.

The mathematical definitions and intermediate steps above specify the quantities being computed; raw numerical replicate files are evidence for their estimators, not substitutes for missing master definitions. The file `GFF/REPRODUCE.md` distinguishes the inexpensive replay from saved records from graph regeneration and expensive integration. The numerical allowances throughout this note are empirical and are not upgraded by checksum or source-preservation checks.

Part V

Bulk one-point data

34 The bulk one-point coefficient through sixth order

This part records a second observable built from the graphs and master functions of Parts I–III: the bulk one-point coefficient a_{ϕ^2} of the composite $:\Phi^2:$ of the auxiliary bulk field, through ϵ^6, at fixed $0 < d < 4$. It is the horizontal coordinate of the numerical-bootstrap bounds of [8]. The calculation, its code and its checks were supplied by the author on 16 October 2026 as a reproducibility package; this part contains its derivations, results, conditions and caveats, so that the package is needed only to rerun the computations. The package is `lri_ope_reproducibility/`; package paths in

this part are relative to `GFF/code/`, as in Section 33. Sections 34.1–34.6 hold for general d, and Sections 34.7–34.10 specialize to $d = 1, 2, 3$. Section 34.11 compares with the bootstrap, and Section 34.12 lists what was checked independently for this note.

34.1 Status

The material has four logically different layers.

1. *Definitions and the two lowest orders.* The observable, its normalization and $a_{\phi^2}^{(2)}$, $a_{\phi^2}^{(3)}$ agree with the published values of [8] (Section 34.6.1).

2. *The calculation of* $a_{\phi^2}^{(4)}, a_{\phi^2}^{(5)}, a_{\phi^2}^{(6)}$. It uses the graph set, local subtractions and master functions of this note. Its checks are internal to that framework: the five- and six-vertex graphs are reconstructed from shared master jets and checked with alternative regulator directions, not integrated one by one. The four-vertex finite parts were also evaluated at finite regulator.

3. *The conversion to the thermal coefficients* $C_j(d)$ (Section 34.4). It is exact algebra between two calculations that share their inputs; it does not validate either of them independently.

4. *Finite-coupling estimates* (Section 34.11). The Borel–Padé values and the comparison with published bootstrap figures are exploratory. No bootstrap problem was solved, the bootstrap data are readings of published figures, and the spread between resummation prescriptions is not an error bar.

Residuals of the internal checks, however small, do not bound the omitted orders of the series, the resummation bias, or an error shared by the calculation and its inputs. The coefficients are numbers of a finite-order calculation; $a_{\phi^2}^{(7)}$ and higher are not determined by the thermal coefficients C_6, C_7, C_8 of Part IV.

34.2 The observable

Write the long-range Ising model as the d-dimensional defect of a free scalar Φ in a bulk of dimension $d + 2 - s = 2 + 2\Delta_\phi$ and codimension $2 - s$, the local description of [8, 9]. In the notation of [8] the physical dimension is p, which is our d; the bulk dimension is its d and the codimension its q, whereas here $q = d/4$ throughout. The bulk propagator and the defect interaction are

$$\langle \Phi(X)\Phi(X')\rangle_0 = |X - X'|^{-2\Delta_\phi}, \qquad \phi(x) = \Phi(x, 0), \qquad S_{\text{int}} = \frac{\lambda_B}{4!}\int d^d x\, \phi(x)^4, \tag{815}$$

with $\Delta_\phi = (d - \epsilon)/4 = q - \epsilon/4$ and $\epsilon = 2s - d$, $q = d/4$, as in (220). The critical local mass is tuned away, and no wave-function counterterm arises in the long-range regime, so Δ_ϕ is protected. For a bulk point at distance $|y|$ from the defect, define

$$\boxed{a_{\phi^2}(d, \epsilon) = |y|^{2\Delta_\phi} \lim_{X' \to X} \Big[\langle \Phi(X)\Phi(X')\rangle - |X - X'|^{-2\Delta_\phi}\Big], \qquad \langle :\Phi^2:(x, y)\rangle = \frac{a_{\phi^2}(d, \epsilon)}{|y|^{2\Delta_\phi}}.} \tag{816}$$

This is the coincident-limit convention of Appendix B.4.2 of [8] (equation, figure and appendix numbers of [8] refer to its arXiv version 6). For the unit-normalized composite $:\Phi^2:/\sqrt{2}$, divide by $\sqrt{2}$. After tadpole and critical-mass subtraction there is no term of order λ, so

$$a_{\phi^2}(d, \epsilon) = \sum_{n=2}^{6} a_{\phi^2}^{(n)}(d)\, \epsilon^n + O(\epsilon^7). \tag{817}$$

The superscript (n) is an order of ϵ; it is not an index of the Laurent coefficients $a_{L,k}$ of Sections 3 and 15. Below, $a^{(n)}$ abbreviates $a^{(n)}_{\phi^2}(d)$.

Three quantities must be kept apart. The thermal dimension $\Delta_{\mathcal{E}} = \sum_n C_n(d)\epsilon^n$ has the free value $d/2$; the squared defect OPE coefficient $\lambda^2_{\phi\phi\mathcal{E}}$ has the free value 2; and a_{ϕ^2} has the free value 0. Only the last is computed here. For the second, only the one-loop relation is recorded:

$$\lambda^2_{\phi\phi\mathcal{E}} = 2 - 6C_2(d)\,\epsilon + O(\epsilon^2), \tag{818}$$

in agreement with the tree-level and one-loop terms $\sqrt{2}$ and $-A\epsilon/(3\sqrt{2})$ of $\lambda_{\phi\phi\mathcal{E}}$ in Eqs. (3.24) and (3.33) of [8]. It is a separate low-order observable, and the higher-order formulas of this part must not be used for it. For the bootstrap, Eq. (2.16) of [8] allows at least $-0.575408 \le a_{\phi^2} \le 0$ at $d = 3$ and $-\frac{7}{8} \le a_{\phi^2} \le 0$ at $d = 2$. This range contains the generalized-free end, $a_{\phi^2} = 0$, and the short-range end of its Eq. (2.15), where Δ_ϕ is the short-range Ising value, 0.5181489 at $d = 3$ and $\frac{1}{8}$ at $d = 2$; it is compatible with bulk–defect crossing with nonnegative squared defect-channel coefficients.

34.3 The result

With A, R, U, V, X of (221), the masters Q_d, $P_4(d)$, $P_5(d)$, $\mathcal{K}_d$, $\mathcal{L}_d$, Z_d of Section 16, and the reflected polygamma differences

$$\mathsf{e}_j = \psi_j(-q) - \psi_j(q) = \psi_j(1-q) - \psi_j(q) + \frac{j!}{q^{j+1}}, \qquad \mathsf{e}_0 = \pi\cot(\pi q) + \frac{1}{q}, \tag{819}$$

the calculation gives

$$\boxed{a_{\phi^2}(d,\epsilon) = \epsilon^2\,\frac{\Gamma(2q)^3\,\Gamma(-q-\epsilon/4)}{54\,\Gamma(4q)\,\Gamma(q-\epsilon/4)}\Big[1 + \mathsf{H}_1\epsilon + \mathsf{H}_2\epsilon^2 + \mathsf{H}_3\epsilon^3 + \mathsf{H}_4\epsilon^4\Big] + O(\epsilon^7),} \tag{820}$$

with

$$\begin{aligned} \mathsf{H}_1 &= \frac{4}{3}A, \qquad \mathsf{H}_2 = \frac{25}{36}A^2 + \frac{R}{6} - \frac{4}{9}Q_d, \\ \mathsf{H}_3 &= \frac{17}{108}A^3 + \frac{AR}{12} - \frac{4}{9}AQ_d + \frac{4}{27}P_4 + \frac{71}{216}U - \frac{V}{36}, \end{aligned} \tag{821}$$

$$\boxed{\begin{aligned} \mathsf{H}_4 = {} & -\frac{7A^4}{5184} - \frac{7A^2R}{288} - \frac{7A^2Q_d}{162} + \frac{14AP_4}{81} + \frac{301AU}{864} - \frac{7AV}{144} - \frac{28\mathcal{K}_d}{81} - \frac{7\mathcal{L}_d}{81} - \frac{16P_5}{81} \\ & - \frac{103R^2}{5184} + \frac{7RQ_d}{54} - \frac{14Q_d^2}{81} - \frac{61X}{3888} + \frac{175Z_d}{162}. \end{aligned}} \tag{822}$$

Here $P_4 = P_4(d)$, $P_5 = P_5(d)$, and $6Q_d$ is the corrected tetrahedron $\mathcal{T}_d$ of (177). The leading coefficient is

$$a^{(2)} = \frac{\Gamma(-q)\Gamma(2q)^3}{54\,\Gamma(q)\Gamma(4q)} = -\frac{\Gamma(1-q)\Gamma(2q)^3}{54\,\Gamma(1+q)\Gamma(4q)} < 0. \tag{823}$$

Expanding the Gamma ratio, $\Gamma(-q-\epsilon/4)\Gamma(q)/[\Gamma(q-\epsilon/4)\Gamma(-q)] = \exp\sum_{k\ge1}(-\frac{1}{4})^k\mathsf{e}_{k-1}\epsilon^k/k! = \sum_j \mathsf{f}_j\epsilon^j$, with

$$\begin{aligned} &\mathsf{f}_0 = 1, \quad \mathsf{f}_1 = -\frac{\mathsf{e}_0}{4}, \quad \mathsf{f}_2 = \frac{\mathsf{e}_0^2 + \mathsf{e}_1}{32}, \quad \mathsf{f}_3 = -\frac{\mathsf{e}_0^3 + 3\mathsf{e}_0\mathsf{e}_1 + \mathsf{e}_2}{384}, \\ &\mathsf{f}_4 = \frac{\mathsf{e}_0^4 + 6\mathsf{e}_0^2\mathsf{e}_1 + 3\mathsf{e}_1^2 + 4\mathsf{e}_0\mathsf{e}_2 + \mathsf{e}_3}{6144}, \end{aligned} \tag{824}$$

the ordinary coefficients are the finite convolution

$$\boxed{a^{(n)} = a^{(2)}\sum_{j=0}^{n-2}\mathsf{H}_j\,\mathsf{f}_{n-2-j}, \qquad \mathsf{H}_0 = 1, \quad 2 \le n \le 6.} \tag{825}$$

For example $a^{(3)}/a^{(2)} = \frac{4}{3}A - \frac{\mathsf{e}_0}{4} = \frac{4}{3}A - \frac{\pi}{4}\cot(\pi q) - \frac{1}{d}$. Equations (819)–(825) and the master values suffice to evaluate the coefficients; no graph has to be rebuilt.

The form (820) is a statement about finite orders. Keeping the Gamma ratio unexpanded generates terms beyond ϵ^6; those terms are not predicted. The organization itself is derived: all dependence on the e_j comes from derivatives of the external-leg integral along one direction (Section 34.6.1), and the remaining coefficients H_j contain only A, R, U, V, X and the masters.

34.4 Relation to the thermal coefficients

Eliminating Q_d, P_4 and P_5 with (222)–(223) gives

$$\mathsf{H}_2 = 6C_3 + \frac{23}{36}A^2 + \frac{R}{9}, \qquad \mathsf{H}_3 = 6C_4 + 5AC_3 + \frac{13}{108}A^3 + \frac{7AR}{108} + \frac{U}{12}, \tag{826}$$

$$\mathsf{H}_4 = 6C_5 + 5AC_4 + \frac{5A^2 - R}{6}C_3 + 9C_3^2 - \frac{A^4}{5184} + \frac{7A^2R}{2592} + \frac{235AU}{2592} - \frac{5AV}{432} - \frac{7R^2}{1728} - \frac{47X}{7776} + \frac{13Z_d - \mathcal{L}_d - 12\mathcal{K}_d}{81}. \tag{827}$$

The logarithm of the normalized series is shorter. Define

$$\log \frac{a_{\phi^2}(d,\epsilon)}{a^{(2)}\epsilon^2} = \mathsf{I}_1\epsilon + \mathsf{I}_2\epsilon^2 + \mathsf{I}_3\epsilon^3 + \mathsf{I}_4\epsilon^4 + O(\epsilon^5), \tag{828}$$

a formal Taylor series of a ratio that tends to one (no logarithm of a negative number is taken). Then

$$\boxed{\begin{aligned}
&\mathsf{I}_1 = 12C_2 - \frac{\mathsf{e}_0}{4}, \qquad \mathsf{I}_2 = 6C_3 - \frac{A^2}{4} + \frac{R}{9} + \frac{\mathsf{e}_1}{32},\\
&\mathsf{I}_3 = 6C_4 - 3AC_3 + \frac{19A^3}{324} - \frac{AR}{12} + \frac{U}{12} - \frac{\mathsf{e}_2}{384},\\
&\mathsf{I}_4 = 6C_5 - 3AC_4 + \Big(A^2 - \frac{5}{6}R\Big)C_3 - 9C_3^2 - \frac{11A^4}{576} + \frac{37A^2R}{864} - \frac{53AU}{2592} - \frac{5AV}{432} - \frac{53R^2}{5184}\\
&\qquad - \frac{47X}{7776} + \frac{13Z_d - \mathcal{L}_d - 12\mathcal{K}_d}{81} + \frac{\mathsf{e}_3}{6144},
\end{aligned}} \tag{829}$$

where $A = 9C_2$. Exponentiation gives

$$\begin{aligned}
&\frac{a^{(3)}}{a^{(2)}} = \mathsf{I}_1, \qquad \frac{a^{(4)}}{a^{(2)}} = \mathsf{I}_2 + \frac{\mathsf{I}_1^2}{2}, \qquad \frac{a^{(5)}}{a^{(2)}} = \mathsf{I}_3 + \mathsf{I}_1\mathsf{I}_2 + \frac{\mathsf{I}_1^3}{6},\\
&\frac{a^{(6)}}{a^{(2)}} = \mathsf{I}_4 + \mathsf{I}_1\mathsf{I}_3 + \frac{\mathsf{I}_2^2}{2} + \frac{\mathsf{I}_1^2\mathsf{I}_2}{2} + \frac{\mathsf{I}_1^4}{24}.
\end{aligned} \tag{830}$$

Through $a^{(5)}$ the one-point coefficients are therefore fixed by $C_2, \dots, C_4$, the polygamma values $A = 9C_2$, R, U and $\mathsf{e}_0, \mathsf{e}_1, \mathsf{e}_2$, and the Gamma functions of (823). At sixth order they are not fixed by $C_2, \dots, C_5$: the combination $13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$ is independent of the combination $-1536\mathcal{K}_d - 576\mathcal{L}_d + 7152Z_d$ that enters C_5 in (223), so $a^{(6)}$ needs one more master value at a generic dimension. At $d = 2$ the closed forms of Section 17.2 remove it (Section 34.8). These relations eliminate shared integrals. They do not identify scaling dimensions with OPE data, and they are checked by substitution, not by a new loop calculation.

34.5 Renormalization-group input

The amplitudes of this part are massless position-space integrals, and ϵ itself is their regulator. The coupling is $g = \lambda/h_d = \pi^{d/2}\lambda/\Gamma(d/2)$, with λ the quartic coupling of (815), so that every vertex

integration carries the normalized measure $h_d\, d^d x$ of Section 16.3. It is renormalized by minimal subtraction of the poles in ϵ (coordinate-space MS, with the forest counterterms of Section 34.6.5), with the critical-mass prescription for local power subgraphs, and

$$\beta(g) = -\epsilon g + 3g^2 - 6Ag^3 + b_3 g^4 + b_4 g^5 + b_5 g^6 + \cdots . \tag{831}$$

This is not the massive momentum-space prescription of (173) and (174), which has its own regulator z. The coefficients b_L, c_L of the two prescriptions nevertheless coincide where they were compared: b_3, c_3 and b_4, c_4 of this prescription are those of (179), (226) and (227), with M and N of (225); its $3c_5 - b_5$ is D_5 of (229); and its b_5 agrees numerically with five-loop data of this note at $d = 1, 2, 3$ (Section 34.12, items "Exact algebra" and "Five-loop split"). Finite parts are a different matter: finite parts of a massive-regulator prescription and of the massless position-space prescription must not be interchanged. The package's low-order matching relates a massive-regulator coupling $g_{\rm alt}$ to g through the regulator-dependent factor $\mathsf{F}_{\rm sch}$; with both couplings minimally subtracted,

$$\begin{aligned} \mathsf{F}_{\rm sch}(\epsilon) &= 2^{-\epsilon}\frac{\Gamma(2q+\epsilon/2)}{\Gamma(2q)\Gamma(1+\epsilon/2)}\frac{\Gamma(q-\epsilon/4)^2}{\Gamma(q+\epsilon/4)^2} = 1 + \mathsf{f}_1^{\rm sch}\epsilon + \mathsf{f}_2^{\rm sch}\epsilon^2 + \cdots , \\ g_{\rm alt} &= \mathsf{F}_{\rm sch}\, g - \big[3\mathsf{f}_1^{\rm sch} + 3(\mathsf{f}_2^{\rm sch} + (\mathsf{f}_1^{\rm sch})^2)\epsilon + \cdots\big] g^2 + \big[9(\mathsf{f}_1^{\rm sch})^2 + 6A\,\mathsf{f}_1^{\rm sch} + O(\epsilon)\big] g^3 + \cdots . \end{aligned} \tag{832}$$

The package checks this matching at the fixed point through the order used there (`lri_ope_reproducibility/code/a4/symbolic_check.py`). The higher-order assembly below uses the coordinate-space prescription and its forest counterterms throughout; it imports no unmatched finite amplitude. The fixed point is

$$\begin{aligned} g_* = \frac{\epsilon}{3} + \frac{2A}{9}\epsilon^2 + \Big(\frac{8A^2}{27} - \frac{b_3}{81}\Big)\epsilon^3 + \Big[\frac{40A^3}{81} - \frac{10Ab_3 + b_4}{243}\Big]\epsilon^4 \\ + \frac{672A^4 - 84A^2 b_3 - 12Ab_4 + b_3^2 - b_5}{729}\epsilon^5 + O(\epsilon^6). \end{aligned} \tag{833}$$

Since a_{ϕ^2} starts at g^2, $a^{(6)}$ needs b_5 but not b_6; it needs b_5 itself, not only the thermal combination $D_5 = 3c_5 - b_5$ of (229). With $\widetilde{N} = \psi_1(-q) - \psi_1(3q)$, the five-loop forest sums of the calculation are

$$\begin{aligned} b_5 = {} & \frac{64341A^4}{128} + \frac{2145A^2M}{64} - \frac{17001A^2R}{64} + \frac{3159A^2Q_d}{8} + \frac{393AMN}{32} + 63AP_4 \\ & + \frac{14907AU}{64} - \frac{603AV}{32} + 87\mathcal{K}_d + \frac{51}{4}\mathcal{L}_d + \frac{3}{4}M^2 + \frac{81MN^2}{64} - \frac{9}{8}MR + 6MQ_d \\ & + \frac{81M\widetilde{N}}{64} + 24P_5 + \frac{1839R^2}{128} - \frac{459RQ_d}{8} + 63Q_d^2 + \frac{565X}{128} - \frac{1311Z_d}{8}, \end{aligned} \tag{834}$$

$$\begin{aligned} c_5 = {} & \frac{10549A^4}{128} + \frac{411A^2M}{64} - \frac{3561A^2R}{64} + \frac{267A^2Q_d}{8} + \frac{99AMN}{32} + 3AP_4 \\ & + \frac{2531AU}{64} - \frac{75AV}{32} + 21\mathcal{K}_d + \frac{5}{4}\mathcal{L}_d + \frac{1}{4}M^2 + \frac{27MN^2}{64} - \frac{1}{8}MR \\ & + \frac{27M\widetilde{N}}{64} + \frac{463R^2}{128} - \frac{63RQ_d}{8} + \frac{415X}{384} - \frac{139Z_d}{8}. \end{aligned} \tag{835}$$

Their combination $3c_5 - b_5$ is exactly D_5 of (229); with the Lagrange formula (593), which holds at any d, the coefficients (179)–(227), (834) and (835) give $C_2, \ldots, C_5$ of (222)–(223). The split into b_5 and c_5 is not visible in D_5: zero-weight vertex classes, for example, cancel from D_5 but not from b_5. It was checked separately against this note's own data at $d = 1, 2, 3$ (Section 34.12). The sensitivity is $\partial a^{(6)}/\partial b_5 = -\frac{2}{243}a^{(2)}$, about 2×10^{-4} at $d = 3$.

For $a_{\phi^2} = \sum_{n\geq 2} g^n \mathsf{a}_n(\epsilon)$, sixth order needs the regular Taylor orders

target	a_2	a_3	a_4	a_5	a_6
$a^{(4)}$	ϵ^2	ϵ^1	ϵ^0	–	–
$a^{(5)}$	ϵ^3	ϵ^2	ϵ^1	ϵ^0	–
$a^{(6)}$	ϵ^4	ϵ^3	ϵ^2	ϵ^1	ϵ^0

together with g_* through ϵ^{n-1}. Extending the fixed point alone does not give the next coefficient: each order needs new finite amplitudes.

34.6 Graphs, kernels and finite parts

34.6.1 External legs and the two lowest orders

Put the bulk point at unit distance from the defect. Both bulk legs end on defect vertices x_0, x_1, and every two-point graph reduces to an internal kernel $\propto |x_0 - x_1|^{-2b}$ followed by the external-leg integral

$$\mathsf{J}_d(a,b) = \int d^d x\, d^d y\, \frac{1}{(1+x^2)^a (1+y^2)^a |x-y|^{2b}} = \pi^d \frac{\Gamma(\frac{d}{2}-b)\Gamma(2a+b-d)\Gamma(a+b-\frac{d}{2})^2}{\Gamma(a)^2\Gamma(\frac{d}{2})\Gamma(2a+2b-d)}. \quad (836)$$

The closed form follows from Schwinger parameters (integrate the parameter of $|x-y|$ first, then the two others as a Gamma and a Beta integral) in the region of convergence; at the physical exponents it denotes the critical-mass continuation. A convergent normalization check is $\mathsf{J}_1(1,\frac{1}{4}) = \pi^2$. For an n-vertex graph the $n-2$ internal integrations fix $b = (2n-1)\Delta_\phi - (n-2)\frac{d}{2}$, and with h_d of Section 16

$$\mathsf{L}_n(\epsilon) = h_d^2\, \mathsf{J}_d\Big(\Delta_\phi, (2n-1)\Delta_\phi - \frac{(n-2)d}{2}\Big) = \frac{\Gamma(2q)\Gamma(-q+\frac{2n-1}{4}\epsilon)\Gamma(q-\frac{2n+1}{4}\epsilon)\Gamma(2q-\frac{n}{2}\epsilon)^2}{\Gamma(q-\frac{\epsilon}{4})^2\,\Gamma(4q-n\epsilon)}. \quad (837)$$

All $\mathsf{L}_n(0)$ are equal, to $54a^{(2)}$. Along the direction $(a,b) = (q+t, 3q-t)$, $\mathsf{J}_d = \pi^d\Gamma(2q)\Gamma(-q+t)/[\Gamma(4q)\Gamma(q+t)]$, so the kth derivative of $\log \mathsf{J}_d$ along $\partial_a - \partial_b$ at $(q,3q)$ is exactly e_{k-1} of (819). The Gamma ratio of (820) is, up to a constant, J_d on this line at $t = -\epsilon/4$; this shift is the origin of the factors $(-\frac{1}{4})^k$ in (824). When the assembly is carried out with these outer derivatives kept as independent symbols, they enter only through the combination $\mathsf{H}_4 - \frac{\mathsf{H}_3}{4}\frac{\partial_-\mathsf{J}_d}{\mathsf{J}_d} + \frac{\mathsf{H}_2}{32}\frac{\partial_-^2\mathsf{J}_d}{\mathsf{J}_d} - \frac{\mathsf{H}_1}{384}\frac{\partial_-^3\mathsf{J}_d}{\mathsf{J}_d} + \frac{1}{6144}\frac{\partial_-^4\mathsf{J}_d}{\mathsf{J}_d}$, $\partial_- = \partial_a - \partial_b$ at $(a,b) = (q,3q)$, which is the origin of (820); no all-order statement about the interacting amplitude is assumed.

The first graph is the two-vertex sunset with Wick weight $\frac{1}{6}$. With the normalized bubble

$$\mathsf{Y}(\epsilon) = \Gamma(\tfrac{d}{2})\frac{\Gamma(\frac{\epsilon}{2})^2\Gamma(\frac{d}{2}-\epsilon)}{\Gamma(\frac{d-\epsilon}{2})^2\Gamma(\epsilon)} = \frac{4}{\epsilon} + R\epsilon - \frac{V}{2}\epsilon^2 + \Big(\frac{R^2}{8} + \frac{7X}{48}\Big)\epsilon^3 + O(\epsilon^4), \quad (838)$$

whose constant term vanishes, the renormalized amplitudes of order g^2 and g^3 (at $\mu|y| = 1$) are

$$\mathsf{a}_2 = \frac{\mathsf{L}_2}{6}, \qquad \mathsf{a}_3 = \frac{\mathsf{L}_2}{\epsilon} - \frac{1}{4}\mathsf{Y}\mathsf{L}_3. \quad (839)$$

The pole of a_3 cancels because $\mathsf{L}_2(0) = \mathsf{L}_3(0)$. With $g_* \simeq \epsilon/3$ the sunset gives (823); the sign is that of $\Gamma(-q) < 0$ and involves no positivity statement about a divergent integral. The second arguments of L_2 and L_3 differ by $\epsilon/2$, so the finite part of a_3 is $\frac{1}{2}h_d^2\,\partial_b\mathsf{J}_d$ at $(a,b) = (q,3q)$. Adding the ϵ-dependence of L_2, the finite part of a_3 and the $O(\epsilon^2)$ term of g_* gives $a^{(3)}/a^{(2)} = \frac{4}{3}A + \frac{1}{4}[\psi(q) - \psi(-q)]$. At $d = 2$ and $d = 3$,

$$\begin{aligned} a_{\phi^2}(2,\epsilon) &= -\frac{\epsilon^2}{27} + \Big(\frac{1}{54} - \frac{8\log 4}{81}\Big)\epsilon^3 + O(\epsilon^4), \\ a_{\phi^2}(3,\epsilon) &= -\frac{\pi^{5/2}\epsilon^2}{576\sqrt{2}\,\Gamma(\frac{7}{4})^2} + \frac{4\sqrt{2\pi}}{6075}(13\pi - 28 - 64\log 2)\Gamma(\tfrac{9}{4})^2\epsilon^3 + O(\epsilon^4), \end{aligned} \quad (840)$$

which are the values of Eq. (3.36) of [8] (Section 34.12). The same two coefficients also follow from the exact bulk–defect relation of [8] and the two-loop ratio of its defect-channel coefficients,

without any higher-order one-point input. Let $b_0^{\Phi,-}/b_0^{\Phi,+} = \mathsf{r}_1\epsilon + \mathsf{r}_2\epsilon^2 + \cdots$ be the ratio of the bulk-to-defect coefficients of the two scalar defect modes of Φ, in the notation and normalization of [8]. Its Eqs. (3.25) and (3.34) give

$$\mathsf{r}_1 = \frac{\Gamma(1-q)\Gamma(2q)}{3\sqrt{6}\,\Gamma(1+q)}, \qquad \mathsf{r}_2 = -\frac{\mathsf{r}_1}{12}\big[13\psi(q) + 3\psi(-q) - 8\psi(2q) - 8\psi(1)\big], \tag{841}$$

and expanding the exact relation, its Eq. (2.11), through cubic order gives

$$a^{(2)} = \frac{\Gamma(2q)\Gamma(q)}{\Gamma(4q)\Gamma(-q)}\,\mathsf{r}_1^2, \qquad a^{(3)} = \frac{\Gamma(2q)\Gamma(q)}{\Gamma(4q)\Gamma(-q)}\Big(2\mathsf{r}_1\mathsf{r}_2 + \frac{\mathsf{e}_0}{4}\,\mathsf{r}_1^2\Big). \tag{842}$$

These are (823) and the value of $a^{(3)}/a^{(2)}$ above. The agreement is not independent of all low-order renormalization-group input: the diagrammatic route of this subsection still uses the known two-loop β function, through g_*.

Read literally, two general formulas of [8], its Eqs. (3.35) and (3.47), disagree in places with these values (Section 34.9; the package records the literal evaluations in `lri_ope_reproducibility/results/verification/low_order_checks.json`). These are apparent printed-formula inconsistencies, not conclusions about the authors' numerical code. The conventions used here are fixed by the operational definition (816) and the explicit low-order checks, not by a transcription of those formulas.

34.6.2 Four vertices

A tadpole-free two-point graph with n quartic vertices is a symmetric multiplicity matrix $m_{ij} \in \{0,1,2,3\}$ with zero diagonal, degrees 3 at the two roots, which carry the bulk legs, and 4 elsewhere, and a connected underlying graph. Up to root exchange and permutations of the $n-2$ internal vertices, its Wick weight is

$$w(m) = \frac{|\mathrm{orb}(m)|}{(n-2)!\prod_{i<j} m_{ij}!}, \tag{843}$$

and the sign $(-1)^n$ is kept separately. Reducible graphs are included. Graphs with both bulk legs on one vertex give only local mass and tadpole terms. The numbers of graphs for $n = 2,\dots,6$ are $1, 1, 5, 14, 64$. With edges listed as $(01, 02, 03, 12, 13, 23)$ and squared-distance exponents, the four-vertex graphs are

edge word	weight	internal reduction
$(0,0,3,3,0,1)$	$1/36$	chain $\mathsf{C}(3\Delta_\phi, \Delta_\phi, 3\Delta_\phi)$
$(0,1,2,2,1,1)$	$1/4$	crossed kite $\mathsf{K}_\times$
$(1,0,2,2,0,2)$	$1/8$	chain $\mathsf{C}(2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi)$
$(1,1,1,1,1,2)$	$1/4$	symmetric kite $\mathsf{K}_\parallel$
$(2,0,1,1,0,3)$	$1/12$	chain $\mathsf{C}(\Delta_\phi, 3\Delta_\phi, \Delta_\phi)$

where the normalized three-link chain is

$$\mathsf{C}(v_1,v_2,v_3) = \Gamma(\tfrac{d}{2})^2 \prod_{i=1}^{3} \frac{\Gamma(\frac{d}{2}-v_i)}{\Gamma(v_i)}\,\frac{\Gamma(v_1+v_2+v_3-d)}{\Gamma(\frac{3d}{2}-v_1-v_2-v_3)}. \tag{844}$$

The two kites are, through the needed order,

$$\mathsf{K}_\times = \frac{8}{\epsilon^2} + \frac{2A}{\epsilon} + 4R + O(\epsilon), \qquad \mathsf{K}_\parallel = \frac{4}{\epsilon^2} + \frac{2A}{\epsilon} + \frac{A^2}{2} + \frac{7}{2}R + O(\epsilon). \tag{845}$$

The symmetric kite is the master (231) at the absolute exponents $(2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi, 2\Delta_\phi, 4\Delta_\phi)$, that is, Pattern A with the coordinates (233) $t = \epsilon$, $p_0 = p_1 = 2\epsilon$ and $u = v = 0$; its expansion through

ϵ^0 is the Laurent expansion of $\mathcal{P}_A$ in (234), because the remainder (235) is $O(\epsilon)$. The crossed kite is not one of the patterns A, B, T: both of its internal vertices have absolute exponent sum $2d-2\epsilon$. Its finite part comes from contour deformation of a two-fold Mellin–Barnes representation, in which the crossed residues and the single-contour pieces give the displayed terms and the remaining double contour is $O(\epsilon)$. Keeping only the poles would give the wrong $a^{(4)}$. Summing the weights gives the internal kernel

$$\mathsf{S}_4 = \frac{9}{2\epsilon^2} + \frac{A}{\epsilon} + \frac{A^2}{8} + 3R + \frac{M}{9} + \epsilon\Big[\frac{M(A+N)}{18} + AR + \frac{U}{8} - \frac{9}{4}V\Big] + O(\epsilon^2), \tag{846}$$

and the renormalized amplitude

$$\mathsf{a}_4 = \Big(\frac{9}{2\epsilon^2} - \frac{A}{\epsilon}\Big)\mathsf{L}_2 - \frac{9}{4\epsilon}\mathsf{Y}\mathsf{L}_3 + \mathsf{S}_4\mathsf{L}_4. \tag{847}$$

Its finite limit is

$$\mathsf{a}_4(0) = h_d^2\Big[\frac{9}{8}\,\partial_b^2\mathsf{J}_d - A\,\partial_b\mathsf{J}_d + \Big(\frac{A^2}{8} + \frac{3}{4}R + \frac{M}{9}\Big)\mathsf{J}_d\Big]_{(a,b)=(q,3q)}. \tag{848}$$

With a_2 through ϵ^2, a_3 through ϵ and g_* through ϵ^3, the M terms cancel and H_2 of (821) follows. The full finite-ϵ Mellin–Barnes values of both kites and the exact Gamma chains reproduce the assembled $a^{(4)}$ at $d=1,2,3$ by extrapolation $\epsilon\to 0$ to better than 10^{-13}; at $d=2$ the symmetric kite also has the exact complex-Selberg form $2\gamma(a)^2\gamma(a+c)^2\gamma(2c)/[\gamma(2a+c)\gamma(2a+2c)\gamma(c)]$ with $\gamma(z)=\Gamma(z)/\Gamma(1-z)$, $a=1-\Delta_\phi$, $c=-\Delta_\phi$.

34.6.3 Five vertices

The fourteen five-vertex graphs, with edges ordered $(01,02,03,04,12,13,14,23,24,34)$ and roots $0,1$, have the finite internal coefficients below (before the Wick weight). Graphs 1 and 13 are scaleless mass terms. Graph 7 has no finite part but a pole: it is a marginal bubble times the regular tetrahedron, $4Q_d/\epsilon + O(\epsilon)$, and the vanishing balanced first derivative of the tetrahedron (239) is what removes its finite part. At $d=2$ the package also evaluates this graph as an exact three-variable complex-Selberg integral at finite ϵ; with the pole removed, its extrapolation to $\epsilon=0$ is 1.3×10^{-36}, consistent with a zero finite part (task `integrals`, record `lri_ope_reproducibility/results/verification/a5_numeric/verification_results.json`).

graph	edge word	weight	finite internal coefficient
1	0003111300	1/36	0
2	0003120210	1/12	$-M(A+N)$
3	0012102300	1/24	$3M(A+N)$
4	0012111210	1/2	$5A^3/18+65AR/6+43U/36-163V/18$
5	0012120120	1/4	$-7A^3/18+77AR/6+7U/36-247V/18$
6	0012210111	1/4	$-A^3/3+11AR+11U/6-13V/3$
7	0111111111	1/6	0 (simple pole $4Q_d/\epsilon$)
8	1002011310	1/6	$2M(A+N)$
9	1002020220	1/16	$-40V$
10	1002110211	1/4	$2A^3/9+26AR/3+11U/9-166V/9$
11	1011011220	1/8	$8A^3/9+32AR/3+23U/9-160V/9$
12	1011101211	1/2	$3A^3/2+13AR/2+23U/12-17V/2$
13	2001001311	1/24	0
14	2001010221	1/8	$5M(A+N)$

The weighted sum gives

$$\begin{aligned}\mathsf{S}_5 = {}& \frac{18}{\epsilon^3}+\frac{11A}{\epsilon^2}+\frac{b_3/9+3A^2/2+45R/2+4M/3}{\epsilon}\\ &+\frac{7}{8}A^3+M(A+N)+\frac{145}{8}AR+\frac{43}{16}U-\frac{181}{8}V+O(\epsilon),\end{aligned} \tag{849}$$

whose finite term is new integral data; pole cancellation does not fix it. The renormalized amplitude is

$$\mathsf{a}_5=\Big[\frac{18}{\epsilon^3}-\frac{10A}{\epsilon^2}+\frac{b_3}{9\epsilon}\Big]\mathsf{L}_2+\Big[-\frac{27}{2\epsilon^2}+\frac{9A}{4\epsilon}\Big]\mathsf{Y}\mathsf{L}_3+\frac{12}{\epsilon}\mathsf{S}_4\mathsf{L}_4-\mathsf{S}_5\mathsf{L}_5. \tag{850}$$

The $O(\epsilon)$ term of S_4 reaches the finite part through $12/\epsilon$. After the lower Taylor terms and g_* are inserted, all M and MN terms cancel and H_3 of (821) remains.

34.6.4 Six vertices

The 64 six-vertex graphs give the kernel

$$\begin{aligned}\mathsf{S}_6 = {}& \frac{135}{2\epsilon^4}+\frac{153A}{2\epsilon^3}+\frac{399A^2/8+23M/2+1053R/8+9Q_d}{\epsilon^2}\\ &+\frac{1}{\epsilon}\Big[\frac{365}{16}A^3+\frac{283}{24}AM+\frac{2979}{16}AR+\frac{9}{2}AQ_d+\frac{81}{8}MN+\frac{1}{2}P_4+\frac{985}{32}U-\frac{2733}{16}V\Big]\\ &+\mathsf{w}_6+O(\epsilon).\end{aligned} \tag{851}$$

All four pole coefficients agree with the counterterm constraints, including the parts multiplying derivatives of the external-leg integral; this checks the subtractions and does not determine w_6.

The new data at sixth order are $\mathsf{w}_4 = [\epsilon^2]\mathsf{S}_4$, $\mathsf{w}_5 = [\epsilon]\mathsf{S}_5$, $\mathsf{w}_6 = [\epsilon^0]\mathsf{S}_6$ and b_5:

$$\begin{aligned}\mathsf{w}_4 = {}& \frac{A^4}{384} + \frac{7A^2M}{288} + \frac{7A^2R}{64} + \frac{7AMN}{144} + \frac{11AU}{192} - \frac{47AV}{96} \\ & + \frac{\mathsf{c}_\times}{4} + \frac{7MN^2}{288} - \frac{M\widetilde{N}}{288} + \frac{103R^2}{128} + \frac{85X}{96}, \end{aligned} \tag{852}$$

$$\begin{aligned}\mathsf{w}_5 = {}& \frac{113A^4}{576} + \frac{55A^2M}{96} + \frac{713A^2R}{96} + \frac{55AMN}{48} + \frac{619AU}{288} - \frac{2071AV}{144} + \frac{9}{2}\mathsf{c}_\times + \frac{55MN^2}{96} + \frac{7MR}{12} \\ & + \frac{7M\widetilde{N}}{96} + \frac{643R^2}{64} + \frac{3}{2}RQ_d + \frac{2179X}{192} - \frac{5Z_d}{12}, \end{aligned} \tag{853}$$

$$\begin{aligned}\mathsf{w}_6 = {}& \frac{1139A^4}{128} + \frac{1697A^2M}{192} + \frac{9057A^2R}{64} + \frac{9}{8}A^2Q_d + \frac{1513AMN}{96} + \frac{2549AU}{64} \\ & - \frac{6545AV}{32} + \frac{189}{4}\mathsf{c}_\times + 5\mathcal{K}_d + \frac{1}{4}\mathcal{L}_d + \frac{11M^2}{108} + \frac{459MN^2}{64} + \frac{89MR}{8} + \frac{123M\widetilde{N}}{64} \\ & + \frac{11345R^2}{128} + \frac{231}{8}RQ_d + \frac{40211X}{384} - \frac{89Z_d}{8}. \end{aligned} \tag{854}$$

Here $\mathsf{c}_\times$ is the quadratic Taylor coefficient of the crossed-kite pattern along its one undetermined direction. It is kept as a free symbol and cancels:

$$10 \cdot \frac{1}{4} - \frac{4}{3} \cdot \frac{9}{2} + \frac{2}{27} \cdot \frac{189}{4} = 0, \tag{855}$$

because the scalar assembly is

$$\begin{aligned}\mathsf{H}_4 = {}& \frac{4811A^4}{1296} + \frac{23A^2M}{162} - \frac{839A^2R}{216} + \frac{253A^2Q_d}{81} - \frac{2AMN}{81} + \frac{56AP_4}{81} + \frac{521AU}{324} + \frac{2}{3}AV - \frac{M^2}{729} \\ & - \frac{MR}{18} + \frac{4MQ_d}{81} - \frac{161R^2}{144} - \frac{13RQ_d}{27} + \frac{28Q_d^2}{81} - \frac{35X}{24} + 10\mathsf{w}_4 - \frac{4}{3}\mathsf{w}_5 + \frac{2}{27}\mathsf{w}_6 - \frac{2}{243}b_5. \end{aligned} \tag{856}$$

Substituting (852)–(854) and (834) removes M, N, $\widetilde{N}$ and $\mathsf{c}_\times$ and gives (822). The masters survive only through $\mathcal{K}_d, \mathcal{L}_d, Z_d, P_5$; no further master enters.

34.6.5 Finite parts and local subtractions

Graphs that are not reduced by convolution (230) and star–triangle are completed conformally as in (232). Their edge exponents are written $\alpha_e = q\,r_e + \epsilon(v_e + \vartheta\,t_e)$ with rational r_e, v_e, t_e; the auxiliary coordinate ϑ along the direction (t_e) separates the logarithmic collision poles. For a UV cluster γ the degree $(|\gamma|-1)\frac{d}{2} - \sum_{e\subset\gamma}\alpha_e$ is ϵ times a linear function of ϑ; IR poles are treated after inversion. At each critical hyperplane the residue factorizes into the normalized anchored subgraph period and its contracted cograph, and the poles are subtracted before the analytic remainder is reconstructed. With m remaining integrations, sixth order needs exponent jets through degree $4-m$. The pole-subtracted remainder is a polynomial of that degree in the deformations; its restrictions to reference hyperplanes with a vanishing completion edge reduce by star–triangle, and exact rational linear algebra checks that these restrictions determine it. The jets used are those of the patterns A, B and T, (235), (237) and (239), together with the wheel derivatives (244); Gamma jets are expanded in abstract polygammas, and no fit in the dimension is made. The one direction not fixed by these restrictions is $\mathsf{c}_\times$.

The local counterterms are the recursive forest formula of Section 16, $Z_G = -\,\mathrm{PP}[I_G + \sum_{\mathcal{F}\neq\varnothing}(\prod_{\gamma\in\mathcal{F}} Z_\gamma) I_{G/\mathcal{F}}]$ over disjoint proper logarithmic subgraphs, with radial factor $2/(L\epsilon)$ for an L-loop vertex graph. With the enumeration's sign convention,

$$b_L = L(-1)^L \sum_G w_4(G)\,[\epsilon^{-1}]Z_G, \qquad c_L = L(-1)^L \sum_G w_2(G)\,[\epsilon^{-1}]Z_G, \tag{857}$$

over the 129 five-loop vertex classes of Section 16, with w_4 and w_2 the four-point and mass weights. The mass-weighted sum is computed separately; c_5 is not obtained by fitting a thermal coefficient. The reconstruction presumes the master identities and the analytic pole structure of Parts II–III. Alternative rational directions test its consistency inside that framework, not its independence from those inputs.

34.7 Values at $d = 1, 2, 3$

With the closed forms of Section 1.1 at $d = 1, 2$ and the $d = 3$ values of Sections 19.10 and 21.11 (with $\mathcal{L}_3$ and Z_3 from the same evaluation as $C_5(3)$), (825) gives

	$d = 1$	$d = 2$	$d = 3$
$a^{(2)}$	−0.139409946645025	−0.0370370370370370	−0.0254241874995486
$a^{(3)}$	−0.850425350705174	−0.118399443073569	−0.0667823808153469
$a^{(4)}$	1.67987388417139	0.0717109289555818	−0.0279145897482122
$a^{(5)}$	−26.2720127156913	−0.751719824443267	−0.156637777571136
$a^{(6)}$	450.850442331799	6.09946795238583	0.507382663942981
$13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$	15277.4261926043	913.335423400404	163.599472783843

The digits shown are exact roundings: the master values are known to at least 40 digits at all three dimensions. The supplied package evaluates the $d = 3$ nonprimitive masters by double-precision spectral quadrature with analytic spin tails; its $\mathcal{K}_3$, $\mathcal{L}_3$, Z_3 agree with this note's values to about 12 significant digits (largest difference 1.4×10^{-11}, in $\mathcal{L}_3$, a relative 2.2×10^{-13}), its $a^{(6)}(3) = 0.507382663943009$ differs from the table by 2.8×10^{-14}, and its $d = 2$ value before the closed forms of Section 17.2 differed by 2.7×10^{-13}. The table entries are coefficients, not estimates of a_{ϕ^2} at finite ϵ. At $d = 1, 2$ they alternate in sign from $a^{(3)}$ on, and at all three dimensions they grow quickly, so a truncated series at moderate ϵ is not reliable (Section 34.11).

34.8 Two dimensions

At $d = 2$, $q = \frac{1}{2}$ and $A = 4\log 2$, $R = X = 0$, $U = \psi_2(\frac{1}{2}) = -14\zeta(3)$, $V = 2\psi_2(1) = -4\zeta(3)$, $\mathsf{e}_j = j!\,2^{j+1}$. The external-leg ratio is elementary, $\Gamma(-\frac{1}{2} - \frac{\epsilon}{4})/\Gamma(\frac{1}{2} - \frac{\epsilon}{4}) = -2/(1 + \frac{\epsilon}{2})$, so (820) reads

$$a_{\phi^2}(2, \epsilon) = -\frac{\epsilon^2}{27}\,\frac{1 + \mathsf{H}_1\epsilon + \mathsf{H}_2\epsilon^2 + \mathsf{H}_3\epsilon^3 + \mathsf{H}_4\epsilon^4}{1 + \epsilon/2} + O(\epsilon^7). \tag{858}$$

Every master in $\mathsf{H}_1, \ldots, \mathsf{H}_4$ is known in closed form at $d = 2$: Q_2 of (267); $P_4(2)$ and $P_5(2)$ of (333), (334); $\mathcal{L}_2 = -2\log 2\, P_4(2)$ of (270), proved in Section 18.7; $\mathcal{K}_2 = 32\mathsf{G}^2 - 28\zeta(3)\log 2$ of (271), proved in Section 18.8 for the master-path definition (238), which is the definition used in (822); and $Z_2 = 4\mathsf{G}Q_2$ of (265), proved in Section 16.7. Given the supplied calculation of $\mathsf{H}_1, \ldots, \mathsf{H}_4$

(Section 34.1), the coefficients are therefore exact. With Q_2, $P_4 = P_4(2)$ and $P_5 = P_5(2)$,

$$\begin{aligned}
a^{(3)}(2) &= \frac{1}{54} - \frac{16\log 2}{81}, \qquad -27\,a^{(4)}(2) = \frac{1}{4} - \frac{8\log 2}{3} + \frac{100\log^2 2}{9} - \frac{4}{9}Q_2,\\
-27\,a^{(5)}(2) &= -\frac{1}{8} + \frac{4\log 2}{3} - \frac{50\log^2 2}{9} + \frac{272\log^3 2}{27} - \frac{485}{108}\zeta(3)\\
&\quad + \Big(\frac{2}{9} - \frac{16\log 2}{9}\Big)Q_2 + \frac{4}{27}P_4,\\
-27\,a^{(6)}(2) &= \frac{1}{16} - \frac{2\log 2}{3} + \frac{25\log^2 2}{9} - \frac{136\log^3 2}{27} - \frac{28\log^4 2}{81}\\
&\quad + \frac{485}{216}\zeta(3) - \frac{2933}{324}\zeta(3)\log 2 - \frac{896}{81}\mathsf{G}^2\\
&\quad + \Big(\frac{350}{81}\mathsf{G} - \frac{56\log^2 2}{81} + \frac{8\log 2}{9} - \frac{1}{9}\Big)Q_2 - \frac{14}{81}Q_2^2 + \Big(\frac{70\log 2}{81} - \frac{2}{27}\Big)P_4 - \frac{16}{81}P_5,
\end{aligned} \tag{859}$$

with $a^{(2)}(2) = -\frac{1}{27}$. So $a^{(6)}(2)$ needs no constant beyond those of $C_5(2)$ in (272): $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the five values $\mathcal{Y}_r$. Its value is $a^{(6)}(2) = 6.0994679523858305821\ldots$

The thermal map at $d = 2$. The extra combination of (829) is

$$13Z_2 - \mathcal{L}_2 - 12\mathcal{K}_2 = 52\mathsf{G}Q_2 + 2\log 2\,P_4(2) - 384\mathsf{G}^2 + 336\,\zeta(3)\log 2 = 913.335423400404160696 29\ldots, \tag{860}$$

and (192), (269) give, with C_j standing for $C_j(2)$ in this paragraph,

$$Q_2 = 2\log^2 2 - \frac{27}{2}C_3, \qquad P_4(2) = \frac{81}{2}C_4 + 8\log^3 2 - 27\log 2\,C_3 + \frac{359}{16}\zeta(3). \tag{861}$$

Inserting these into (829) removes (860):

$$\begin{aligned}
\mathsf{I}_1(2) &= \frac{16\log 2}{3} - \frac{1}{2}, \qquad \mathsf{I}_2(2) = 6C_3 - 4\log^2 2 + \frac{1}{8},\\
\mathsf{I}_3(2) &= 6C_4 - 12\log 2\,C_3 + \frac{304}{81}\log^3 2 - \frac{7}{6}\zeta(3) - \frac{1}{24},\\
\mathsf{I}_4(2) &= 6C_5 - 11\log 2\,C_4 + \frac{46\log^2 2 - 26\mathsf{G}}{3}C_3 - 9C_3^2 - \frac{380}{81}\log^4 2\\
&\quad + \frac{104}{81}\mathsf{G}\log^2 2 - \frac{128}{27}\mathsf{G}^2 + \frac{1303}{216}\zeta(3)\log 2 + \frac{1}{64}.
\end{aligned} \tag{862}$$

At $d = 2$ the one-point coefficients through $a^{(6)}$ thus follow from $C_3(2)$, $C_4(2)$, $C_5(2)$ and the constants $\log 2$, G, $\zeta(3)$. Numerically $\mathsf{I}_1, \ldots, \mathsf{I}_4$ are 3.19678496299, −7.04591213159, 37.3758123361 and −277.339425824.

Checks at $d = 2$. The following were checked with code of their own (`ope_checks/d2/ope_d2_check.py`, record `ope_d2_check.json`), which takes every $d = 2$ input from this note. Exactly: the logarithm of (858), with $\mathsf{H}_1, \ldots, \mathsf{H}_4$ of (821), (822), the closed forms above and $C_3(2)$, $C_4(2)$, $C_5(2)$ from (269), gives (862), and also (829) at $d = 2$; (859) and the second form of (860) follow in the same way, and the two forms of $C_5(2)$, (269) and (272), agree. All residuals are zero. Numerically, at 60 digits: the closed-form masters reproduce the supplied package's $6Q_2$, $P_4(2)$, $P_5(2)$ and Z_2 to 2×10^{-58}, and (862) with the numerical $C_3(2)$, $C_4(2)$, $C_5(2)$ reproduces (825) to 10^{-59}. These checks take H_4 as given. They test the two-dimensional specialization and the shared masters, not the six-vertex calculation.

The five-loop coefficient b_5 enters $a^{(6)}$ on its own, not only through $D_5 = 3c_5 - b_5$ of (229) (Section 34.5). Our two-dimensional productions use the finite-MOM coefficients $\widehat{b}_L(z) = \sum_j b_{L,j} z^j$

of (693). The conversion of Section 34.12 (item "Five-loop split") to regulator-independent coefficients gives, explicitly,

$$b_5 = b_{5,0} + \frac{9}{4}b_{4,1} + \frac{9}{2}b_{3,2} + \frac{27}{4}b_{2,3} - \frac{1}{2}b_{2,1}b_{3,0} + \frac{1}{3}b_{2,0}b_{3,1} + \frac{1}{2}b_{2,0}b_{2,2} - \frac{3}{8}b_{2,1}^2, \tag{863}$$

where $b_{2,j}$ are the exact two-loop jets. With D_5 held at (229), the jets of the seventh-order production (Section 28.7) and of the eighth-order production (Section 31), which use independent random streams, give $b_5(2) = 135708.65 \pm 0.27$ and 135709.17 ± 0.42, together 135708.80 ± 0.23; (834) gives $135708.425\ldots$. Without that constraint the two productions give 135709.43 ± 0.55 and 135709.64 ± 0.96, because their five-loop jets fluctuate along D_5 (by -2.1 and -0.7 standard errors). The same conversion reproduces b_4 and c_4 of (226), (227) within 2.0 standard errors. Since $a^{(6)}(2)$ changes by 3.05×10^{-4} per unit of b_5, this checks the five-loop input of $a^{(6)}(2)$ at the level of 10^{-4} in $a^{(6)}(2)$, against data of this note only (`ope_checks/d2/ms_b5_from_jets.py`, records `ms_b5_c7d2.json`, `ms_b5_c8d2.json`).

34.9 One dimension

At $d = 1$ every master in (822) is a closed form proved in Parts II–III: $Q = \Gamma(\frac{1}{4})^4/\pi$, $P_4 = 13B_- - 3B_+$, P_5 in the five values $\mathcal{Y}_r$ (Section 1.1), $\mathcal{K}$ (76), $\mathcal{L}_1$ (541) and $Z_1 = QH$ (224). The coefficients $a^{(2)}(1), \ldots, a^{(6)}(1)$ of Section 34.7 are therefore exact expressions in the same fixed values as $C_5(1)$: $\Gamma(\frac{1}{4})$, π, $\log 2$, G, $\zeta(3)$, $F_\pm$ and the $\mathcal{Y}_r$. They are conditional only on the calculation itself (Section 34.1). The first two are

$$a_{\phi^2}(1,\epsilon) = -\frac{2\pi^{3/2}\Gamma(\frac{3}{4})}{27\,\Gamma(\frac{1}{4})}\,\epsilon^2\Big[1 + \Big(\frac{13\pi}{12} + \frac{16}{3}\log 2 - 1\Big)\epsilon\Big] + O(\epsilon^4). \tag{864}$$

Both agree with the general formulas of [8]: $a^{(2)}$ with its quadratic coefficient for the Ising case, Eq. (3.35), and $a^{(3)}$ with its cubic coefficient for the long-range $O(N)$ model at $N = 1$, Eq. (3.47). Read literally, its Ising cubic coefficient in Eq. (3.35) and the sign of its $O(N)$ quadratic coefficient in Eq. (3.47) disagree with these values and with its own Eq. (3.36).

Two checks exist only at $d = 1$. First, the package's reconstruction of all six-vertex kernels and five-loop forest sums at 60 digits, with rational regulator and reference directions other than those of the derivation and $\mathsf{c}_\times = 17$, runs at $d = 1$. It differs from (822) by 7.6×10^{-54} in $a^{(6)}(1)$, with residual poles below 1.4×10^{-52}; it shares the master identities and the subtraction framework (Section 34.1). Second, the split of D_5 into b_5 and c_5, which $a^{(6)}$ needs and (229) does not fix, is compared at $d = 1$ with numerical graph integrations that do not use the master reduction (Section 34.12). The residue certificate of Appendix A cannot serve for this: it omits the 28 five-loop classes of zero weight in $3c_5 - b_5$, which is correct for D_5 but not for b_5 alone. Since b_5 enters (856) as $-\frac{2}{243}b_5$, the coefficient $a^{(6)}(1)$ moves by $-\frac{2}{243}a^{(2)}(1) = 1.15 \times 10^{-3}$ per unit of b_5.

34.10 A one-dimensional functional cross-check

In one dimension the super sum rule of [10] gives a determination of the low-order thermal coefficients that is independent of the graphs. Its expansion parameter $\epsilon = 1 - 4\Delta_\phi$ is ours at $d = 1$. Writing $\Delta_{\mathcal{E}} = 2\Delta_\phi + \gamma^{(1)}\epsilon + \gamma^{(2)}\epsilon^2 + \cdots$, the functional $\widehat{\beta}_0 = -\frac{\epsilon}{4}\widetilde{\beta}_0$ gives $-\gamma^{(1)} + 3(\gamma^{(1)})^2 = 0$, so $\gamma^{(1)} = \frac{1}{3} = C_1 + \frac{1}{2}$ on the nontrivial branch. At the next order, assuming that only $\mathcal{E}$ contributes, a high-precision numerical evaluation of the functional gives $\gamma^{(2)} = \frac{1}{9}(\pi + 4\log 2) = C_2(1)$. The cross-check therefore concerns C_1 and $C_2(1)$; it does not test the one-point coefficients. At the next order the sums over double-trace operators and over the twist-four family, and the ϕ^4 operator, contribute at $O(\epsilon^3)$ although each term starts at $O(\epsilon^4)$, for example through

$$\epsilon^4 \sum_{n\geq 1} n^{-1-\nu\epsilon} = \frac{\epsilon^3}{\nu} + \gamma_{\mathrm{E}}\epsilon^4 + \cdots; \tag{865}$$

their cancellation requires an exponent that [10] does not determine, so [10] does not determine $C_3(1)$, and no functional determination of it is made here. In that reference a_{ϕ^2} denotes the squared defect OPE coefficient of ϕ^2 in the four-point function of ϕ, not the bulk one-point coefficient of this part.

34.11 Comparison with the numerical bootstrap

At fixed d and Δ_ϕ, the bootstrap of [8] constrains a region in the plane $(a_{\phi^2}, \Delta_{\mathcal{E}})$; a_{ϕ^2} is its horizontal coordinate. Figure 11 of [8] shows allowed regions at $d = 3$ for four external dimensions; their upper right corners (kinks) are the candidate long-range Ising points. The comparison below inserts the series of this note into one resummation prescription and places the result next to readings of that figure. It does not rerun the bootstrap, it has no access to the underlying numerical data, and it does not provide an excluding functional. The package draws both comparisons, with its pixel traces of the published boundaries and kinks, in `lri_ope_reproducibility/results/figures/`: `3d_sixth_order_comparison.png` and `2d_sixth_order_comparison.png`, and, for the successive truncations and the two resummations at the weakest coupling, `3d_weak_point_orders.png` and `2d_weak_point_orders.png`. These plots use the older C_6 inputs; `results/bootstrap/current/` holds the same plots with the current ones. The traces are approximate readings, not bootstrap data, and the plots are not exclusion certificates.

Prescription. For a truncated series $F(\epsilon) = \sum_k c_k \epsilon^k$ use the Borel transform $\sum_k c_k t^k/k!$, its $[L/M]$ Padé approximant, and the Laplace integral $\int_0^\infty e^{-t}[L/M](\epsilon t)\, dt$. Horizontally, resum $a_{\phi^2}/\epsilon^2 = a^{(2)} + \cdots + a^{(6)}\epsilon^4$ with [3/1] and [1/3], whose denominators have no positive real root at these points; [2/2] has one at every point and is not used. Vertically, resum $\Delta_{\mathcal{E}}$ through C_6 with [4/2]; the package also records [3/2] through C_5 as an order diagnostic. Factoring out ϵ^2 is part of the prescription. No bootstrap point and no short-range endpoint is fitted, and varying the Borel–Leroy parameter, the approximant or the factored power would add uncertainty that has not been quantified.

Three dimensions. The four external dimensions of Figure 11 are equally spaced between $\Delta_\phi = \frac{3}{4}$ and the short-range end $\Delta_\phi = 0.5181489$, the short-range Ising value used by [8] (Section 34.2), so $\epsilon = 3 - 4\Delta_\phi$:

Δ_ϕ	ϵ	a_{ϕ^2}, [3/1]	a_{ϕ^2}, [1/3]	$\Delta_{\mathcal{E}}$, [4/2]	kink of Fig. 11
0.70363	0.18548	−0.00135602	−0.00134382	1.4738699	(−0.00187, 1.4700)
0.65726	0.37096	−0.00797138	−0.00713938	1.4541700	(−0.00292, 1.4500)
0.61089	0.55644	−0.0253361	−0.0185657	1.4383505	(−0.0110, 1.4318)
0.56452	0.74192	−0.0613651	−0.0352226	1.4248531	(−0.0420, 1.4154)

The [4/2] column uses $C_6(3) = 33.5754$ of Section 26.8; with the older input 33.57 it is lower by 4.6×10^{-8}, 1.2×10^{-6}, 7.6×10^{-6} and 2.6×10^{-5}. The kinks are read from the vector data of the published figure (`ope_checks/d3/check_fig11.py`, record `ope_checks/d3/check_fig11.json`; Section 34.12). The figure file of the arXiv source, converted with `pdftocairo -svg`, draws each region boundary as a polyline; its vertices are mapped to data coordinates by the axis frame, and the kink is the vertex with the largest turning angle in axis-normalized coordinates. Vertices that should lie on round values, such as $a_{\phi^2} = -0.005$ or $\Delta_{\mathcal{E}} = 1.45$, are recovered to within 7×10^{-7}, so a kink is limited by the spacing of the polyline vertices around it, 3×10^{-4} to 4×10^{-3} in a_{ϕ^2}, not by the reading. The supplied package used pixel readings that agree to about one pixel.

At the weakest coupling, $\Delta_\phi = 0.70363$, the drawn region at $\Delta_{\mathcal{E}} = 1.4739$ requires $a_{\phi^2} \lesssim -0.0032$, while both resummations give -0.0013 to -0.0014: this resummed point lies outside the drawn

region. The truncated sums through ϵ^4, ϵ^5 and ϵ^6 are -0.00133384, -0.00136823 and -0.00134757, and the cubic-order point that [8] itself plots, $(-0.0013004, 1.473078)$, is also on the excluded side of its own boundary. The mismatch is therefore not created by the higher orders; equally, it does not test them. At this point the thermal terms $C_4\epsilon^4$, $C_5\epsilon^5$, $C_6\epsilon^6$ are about $+0.00145$, -0.00128, $+0.00137$, so the vertical coordinate is not known to six digits from apparent Padé stability.

What the mismatch can and cannot mean. An excluding functional must be positive on every admissible exchanged dimension and spin. The bounds of [8] impose the long-range OPE relations on a grid of exchanged dimensions (its App. A.2: 3000 points, and up to 12000 in its refined two-dimensional runs, where the kinks move up as the grid is refined). A functional positive on the grid can be negative between grid points, which would make a finite-grid boundary too restrictive. The two numerical truncations have different logical effects: keeping fewer crossing derivatives, with positivity still imposed on the continuum, only restricts the available functionals and weakens the exclusions, so it cannot by itself make a true solution inconsistent; sampling the exchanged dimension instead can miss a negative interval. Near the generalized-free end the spectrum approaches double-twist values at a rate set by ϵ^2; the package estimates the leading spin-2 shift as $-4\epsilon^2/[9d(d+2)] \approx -0.0010$ at the first point, smaller than a grid spacing of order 0.007 (this estimate was not checked here). This is a possible mechanism, not a demonstrated one. The decisive test would be to check the candidate functional between grid points, including the smallest eigenvalue of each matrix sector and the large-dimension tail. If positivity held and both coordinates had controlled errors, the discrepancy would have to be resolved in the perturbation theory, the resummation or the spectral assumptions. None of this has been done.

Two dimensions. Figure 7 of [8] shows, at $d = 2$ and eight external dimensions between $\Delta_\phi = 0.13333$ and 0.425, the kinks of runs with 3000, 6000 and 12000 grid points and the authors' own two-sided Padé estimates; it shows kinks, not allowed regions. With $\epsilon = 2 - 4\Delta_\phi$, the four weakest couplings give

Δ_ϕ	ϵ	a_{ϕ^2}, [3/1]	a_{ϕ^2}, [1/3]	$\Delta_\mathcal{E}$, [4/2]	kink of Fig. 7	$\Delta_\mathcal{E}$, [5/3] to C_8
0.425	0.3	-0.00654824	-0.00601150	0.9678196	$(-0.0060, 0.9600)$	0.9679055
0.35	0.6	-0.0414334	-0.0277046	0.9578452	$(-0.0430, 0.9602)$	0.9590956
0.275	0.9	-0.134446	-0.0621047	0.9629664	$(-0.1420, 0.9660)$	0.9680003
0.2	1.2	-0.325205	-0.105209	0.9803317	$(-0.3750, 0.9878)$	0.9929583

The [4/2] column uses $C_6(2) = 362.366$ of Section 26; with the older input 362.3 it is lower by 1.7×10^{-6}, 2.6×10^{-5}, 1.1×10^{-4} and 2.7×10^{-4}, and the allowance ± 0.004 of $C_6(2)$ moves the first row by 10^{-7}. As at $d = 3$, [2/2] has a positive Borel pole at every point. The kinks are those of the 12000-point runs. Each is the centre of its cross marker, read from the marker's placement in the vector figure file of the arXiv source of [8] (`ope_checks/d2/check_fig7.py`, record `check_fig7.json`), which places it to about 3×10^{-9} in a_{ϕ^2} and 2×10^{-10} in $\Delta_\mathcal{E}$; the supplied package's pixel readings agree to 2×10^{-4} in a_{ϕ^2}, except at $\Delta_\phi = 0.425$, where its -0.00676 is 7.6×10^{-4} off, and to 8×10^{-5} in $\Delta_\mathcal{E}$.

At the weakest coupling, $\Delta_\phi = 0.425$, the kink at $a_{\phi^2} = -0.0060$ lies between the two horizontal resummations, while the resummed $\Delta_\mathcal{E} = 0.96782$ lies 0.0078 above it. The truncated sums $\sum_{n \le k} a^{(n)}(2)\,\epsilon^n$ at this point are -0.00594925977, -0.00777593895 and -0.00332942681 for $k = 4, 5, 6$, so the last available term is not an error estimate. The kinks of the same figure move up as the grid is refined, from 0.9575 at 3000 points to 0.9592 and 0.9600, and the authors' own Padé estimates at this Δ_ϕ are 0.9695 with all their perturbative data and 0.9723 without the order ϵ^3; [8] itself notes that the kinks of best agreement lie noticeably below its perturbative line. So the gap was present before the orders computed here, as at $d = 3$. At the three stronger couplings the resummed points lie 0.0023 to 0.0074 below the kinks. The last column is a diagnostic only: it adds

$C_7(2)$ of Section 28 and $C_8(2)$ of Section 31 in [5/3], which has no positive Borel pole at the four points ([4/3], [3/4], [4/4] and [3/5] have one at $\epsilon = 0.3$). It moves $\Delta_{\mathcal{E}}$ at $\epsilon = 0.3$ by 8.6×10^{-5}, and by at most 7.4×10^{-6} more when $C_7(2)$ and $C_8(2)$ vary over their allowances; at the three stronger couplings it moves it by 1.3×10^{-3}, 5.0×10^{-3} and 1.3×10^{-2}, comparable to the distances from the kinks. Neither column is an error estimate (`ope_checks/d2/ope_d2_check.py`).

34.12 Checks made for this note

The following were done independently of the supplied code, with this note's inputs; the scripts and their outputs are in `ope_checks/`.

1. *Rerun.* All tasks of the package except the TEX build (low-order checks, symbolic reconstruction of the five- and six-vertex kernels and of the 129-class forest sums, the $d = 1$ alternative-regulator reconstruction at 60 digits, the four-vertex finite-regulator checks, the $d = 2, 3$ spectral masters, and the comparison plots) pass on a clean extraction on the symmetry cluster (Intel Xeon Gold 6148, Python 3.13.3, the pinned library versions). Every output equals the shipped one byte for byte, apart from recorded run times.

2. *Exact algebra* (32 identities, own code): the package's $C_2, \dots, C_5$ equal (222)–(223); its b_3, c_3, b_4, c_4 equal (179), (226), (227); $3c_5 - b_5$ from (834)–(835) equals (229), and Lagrange inversion with these coefficients gives (223); (833) solves (831); (856) with (852)–(854) and (834) gives (822); (824), (826), (827), (829) and (830) are exact.

3. *Low orders.* $a^{(2)}$ and $a^{(3)}$ at $d = 2, 3$ equal Eq. (3.36) of [8], (840), to 35 digits. At $d = 1$ they equal the quadratic coefficient of Eq. (3.35) and the $N = 1$ cubic coefficient of Eq. (3.47) of [8], and the ratios $R^{(1)}$, $R^{(2)}$ of its Eq. (3.34), to 80 digits; the cubic coefficient of its Eq. (3.35) read literally gives -1.50734 instead of $a^{(3)}(1) = -0.850425$, and the quadratic coefficient of its Eq. (3.47) has the opposite sign (`ope_checks/d1/`, with its recorded run).

4. *Amplitudes.* With (837), (838), (839), (847), (850), the kernels (846), (849) and g_*, the amplitudes a_3, a_4, a_5 are finite and the re-expanded $\sum_n g_*^n \mathsf{a}_n$ reproduces $a^{(2)}, \dots, a^{(5)}$ of (825) to 10^{-31} at $d = 3$ and at $d = 2.6$. The bubble expansion (838) and the closed form (836) (by direct quadrature at $d = 1$) were checked numerically.

5. *Kernels.* The symmetric kite (845) through ϵ^0 follows from (234) at $d = 1$, 2.6 and 3, including its R term; the four-vertex table with the exact chains (844) gives the pole and constant terms of (846); the five-vertex table sums to the finite part of (849). The crossed kite (845) enters the package through a two-fold Mellin–Barnes representation, whose finite-ϵ values reproduce the assembled $a^{(4)}$ (Section 34.6). At $d = 2$ this note's kite formula (674), evaluated at $\epsilon = 0.02, 0.04, 0.06, 0.1, 0.15$ (the slowly convergent ${}_3F_2(1)$ after Thomae's transformation), reproduces those finite-ϵ values of $\mathsf{K}_\times$ to 2×10^{-14} or better, and those of $\mathsf{K}_\parallel$ to 10^{-16}. At $d = 1$ the full-line Selberg formula of Section 5 gives $\mathsf{K}_\parallel$ to all 15 digits of the package's values; for $\mathsf{K}_\times$ at $d = 1$ only a double-precision direct quadrature was made, which agrees to 0.3%, within its own error (it misses the exact $\mathsf{K}_\parallel$ by 0.7%). Since $R = 0$ at $d = 2$, the $4R$ term of $\mathsf{K}_\times$ is tested only through the package's representation.

6. *Five-loop split.* At $d = 1$, b_5 and c_5 of (834)–(835) agree to about 10^{-12} relative with the two-channel five-loop graph sums of Section 14 at quadrature degree 18, renormalized with the degree-28 lower-loop inputs of the $C_6(1)$ calculation (records `amplitude5_maxgap_18.json` and `amplitude5_span_18.json` in `lri_numerical_resolution/verification/records/`, assembled in `LRI_C6_1D_HPC/reference/assembly_lower6.json`). These are numerical graph integrations that do not use the master reduction (Section 34.9). At $d = 2$, the M-dependent parts of (834) and (835) equal the minimal-subtraction power sums b_5^p and c_5^p of (678) to all 21 printed digits, and their b_3, b_4 parts equal b_3^p, b_4^p there. At $d = 2$

and $d = 3$ the complete b_5 was also extracted from the finite-scheme Taylor jets of the seventh- and eighth-order productions. A regular redefinition $g = g' + \sum_k \rho_k(z) g'^{k+1}$ with an operator normalization brings the finite-scheme β and γ_r to regulator-independent coefficients. Because of the $-zg'$ term this redefinition is unique, so the result is the minimal-subtraction form used here. It needs b_2 through z^3, b_3 through z^2, b_4 through z^1 and b_5 at z^0, and likewise for c_L. Holding D_5 at its proved value (229), the $C_7(3)$ and $C_8(3)$ productions give $b_5(3) = 18822.676 \pm 0.097$ and 18822.42 ± 0.28, against 18822.627 from (834). The same conversion reproduces b_3, c_3, b_4, c_4 of (179)–(227) within the sampling errors. At $d = 2$ it gives $b_5(2) = 135708.80 \pm 0.23$, against 135708.43. The individual b_5 sums of Section 24 cannot serve here, because they depend on trial core assignments.

7. *Values.* The table of Section 34.7 was computed from this note's master values, which reproduce $C_2, \ldots, C_5$ of the coefficient table of Section 1 at $d = 1, 2, 3$. The log map with those $C_2, \ldots, C_5$ and the master value of $13Z_d - \mathcal{L}_d - 12\mathcal{K}_d$ reproduces it to the precision of the displayed C_j.

8. *Bootstrap rows.* An own Borel–Padé evaluation reproduces the four $d = 3$ rows of Section 34.11 to all printed digits and finds the positive Borel pole of [2/2] at every point. The kinks and the cubic-order point of Figure 11 were taken from the vector paths of the figure file in the arXiv source of [8]: the plotted point lies at $(-0.0013004, 1.473078)$, the package's recomputation of it is $(-0.00130080, 1.473065)$, and the package's four kink readings agree with the vector kinks to within 9×10^{-5} in a_{ϕ^2} and 2×10^{-4} in $\Delta_{\mathcal{E}}$.

34.13 Reproduction

The package `lri_ope_reproducibility/` contains the low-order, four-, five- and six-vertex and relation engines, the graph inventories (`data/a6/graphs_n*.json`, `vertex_n*.json`), the master inputs, the bootstrap figure readings, and stored certificates. Its runner stages each engine in a temporary directory and does not overwrite inputs:

task	content (run time on the cluster above)
`quick`	low-order diagrams, the thermal map, the coefficient table, and a regression against stored certificates (4 s)
`symbolic`	fresh graph inventories, five- and six-vertex symbolic kernels, forest sums, pole cancellation, $3c_5 - b_5$, the compact result (2 min)
`numeric`	all sixth-order kernels and five-loop forest sums at 60 digits in $d = 1$ with other rational regulator directions and $\mathtt{c}_\times = 17$; agreement with (822) to about 10^{-53}, residual poles about 10^{-52} (5 min)
`integrals`	finite-regulator four-vertex kite and chain checks at $d = 1, 2, 3$, the $d = 2$ Selberg check, the five-vertex checks (1.5 min)
`masters`	$d = 2, 3$ spectral evaluation of $\mathcal{K}_d, \mathcal{L}_d, Z_d$ with varied quadrature (20 s)
`figures`	Borel–Padé rows and comparison plots, with both C_6 inputs (30 s)

The command `python run.py`, followed by the task names and `--output-dir DIR`, runs them; each writes `verification/run_TASK.json`. The numerical residuals of these tasks are consistency checks within the framework above, not uncertainties of the coefficients.

34.14 Open points

- An evaluation of selected five- and six-vertex finite parts by a method that does not use the pattern jets, for example sector decomposition with all local subtractions, would test the reconstruction beyond its alternative directions. The R term of the crossed-kite finite part (845) is checked only through the package's own Mellin–Barnes representation; an evaluation at $d \neq 2$ to many digits would close this.

- The resummation has no controlled error: varying the Borel–Leroy parameter, the approximants and the factored power, rejecting or treating positive Borel poles, and propagating the input uncertainties remain to be done.
- The bootstrap mismatch needs a continuum-positivity check of the candidate functional, not more coefficients.
- The apparent printed-formula inconsistencies in Eqs. (3.35) and (3.47) of [8] (Sections 34.6.1 and 34.9) should be checked against the authors' ancillary calculation or clarified with them. They should not be used to infer that the authors' numerical bootstrap code shares them.
- $a^{(7)}$ would need $\mathsf{a}_2, \ldots, \mathsf{a}_6$ one order further, a new finite a_7 and b_6; neither C_6 nor the pattern of (829) determines them, and new master derivatives may enter.

A Complete retained residue certificate

This appendix lists the residues that enter the one-dimensional assembly of Sections 11–13: the 26 non-wheel four-loop rows and 92 retained five-loop rows, as exact polynomials, so that the weighted sum can be inspected without the machine-readable certificate. The six wheel rows are in Section 12; the three primitive rows are determined by their periods; the 28 five-loop zero-weight classes contribute nothing to $3c_5 - b_5$. Together these disjoint sets exhaust the stated graph list.

For each row, i is the zero-based canonical class index and the word lists upper-triangular multiplicities in lexicographic vertex-pair order. The columns are *RG weights*, not raw Wick weights:

$$\omega_b = L(-1)^L w_4/|\operatorname{Aut} G|, \qquad \omega_c = L(-1)^L w_2/|\operatorname{Aut} G|.$$

A row with residue $R_j^{(L)}$ contributes $\omega_b R_j^{(L)}$ to b_L and $\omega_c R_j^{(L)}$ to c_L. Equal residues are identified to keep the certificate legible. All expressions use the same $A, Q, \mathsf{G}, \mathcal{K}$ as the main result. Their transcription from the retained JSON is checked by exact symbolic substitution, including $\log 2 = (A - \pi)/4$. No numerical fit enters this table.

A.1 4-loop rows and residue polynomials

Class	Edge word	ω_b	ω_c	Residue
0	0002011310	2	$\frac{2}{3}$	$R_0^{(4)}$
1	0002020220	$\frac{3}{4}$	$\frac{1}{4}$	$R_0^{(4)}$
2	0002021210	3	$\frac{1}{2}$	$R_0^{(4)}$
3	0002110211	3	1	$R_0^{(4)}$
4	0002111201	12	2	$R_0^{(4)}$
5	0002112200	3	$\frac{1}{2}$	$R_0^{(4)}$
6	0011011220	$\frac{3}{2}$	$\frac{1}{2}$	$R_1^{(4)}$
7	0011012210	6	1	$R_2^{(4)}$
8	0011101211	6	2	$R_2^{(4)}$
9	0011102201	6	1	$R_3^{(4)}$
10	0011102210	12	2	$R_4^{(4)}$
11	0011102300	4	$\frac{2}{3}$	$R_5^{(4)}$
12	0011111111	12	2	$R_6^{(4)}$
13	0011111120	24	4	$R_7^{(4)}$
14	0011201102	6	1	$R_8^{(4)}$

Class	Edge word	ω_b	ω_c	Residue
15	0011201111	24	4	$R_9^{(4)}$
16	0011201201	12	2	$R_3^{(4)}$
17	0011202110	3	1	$R_{10}^{(4)}$
18	0011202200	3	$\frac{1}{2}$	$R_{11}^{(4)}$
19	0011211110	6	1	$R_{12}^{(4)}$
20	0011301011	1	$\frac{1}{3}$	$R_0^{(4)}$
21	0011301101	4	$\frac{2}{3}$	$R_{13}^{(4)}$
22	0012102200	6	0	$R_{11}^{(4)}$
23	0012111110	24	0	$R_6^{(4)}$
24	0012111200	24	0	$R_4^{(4)}$
25	0012210101	3	0	$R_0^{(4)}$

$$R_0^{(4)} = 0.$$

$$R_1^{(4)} = -\frac{2A^3}{9} + \frac{\pi^3}{9} - 4\zeta(3) - \frac{2A\pi^2}{9}.$$

$$R_2^{(4)} = -\frac{23A^3}{36} + \frac{25\pi^3}{36} + 19\zeta(3) + \frac{A\pi^2}{36}.$$

$$R_3^{(4)} = -\frac{23A^3}{36} - \frac{11\pi^3}{36} - 9\zeta(3) + \frac{A\pi^2}{36}.$$

$$R_4^{(4)} = -\frac{5A^3}{18} + \frac{5\pi^3}{36} + 3\zeta(3) - \frac{A\pi^2}{9}.$$

$$R_5^{(4)} = 2A\pi - 2\pi^2 - 4\pi.$$

$$R_6^{(4)} = -\frac{3AQ}{2}.$$

$$R_7^{(4)} = -\frac{4A^3}{3} + \frac{5\pi^3}{3} + 48\zeta(3) + \frac{A\pi^2}{3}.$$

$$R_8^{(4)} = -\frac{A^3}{9} + \frac{\pi^3}{18} + 2\zeta(3) - \frac{A\pi^2}{9}.$$

$$R_9^{(4)} = -\frac{4A^3}{3} - \frac{\pi^3}{3} - 8\zeta(3) + \frac{A\pi^2}{3}.$$

$$R_{10}^{(4)} = 4\pi^2 + 8\pi.$$

$$R_{11}^{(4)} = -\frac{A^3}{12} - \frac{\pi^3}{12} - \zeta(3) - \frac{A\pi^2}{12}.$$

$$R_{12}^{(4)} = -\frac{5A^3}{9} + \frac{5\pi^3}{18} - 2\zeta(3) - \frac{2A\pi^2}{9}.$$

$$R_{13}^{(4)} = 2A\pi + 2\pi^2 + 4\pi.$$

A.2 5-loop rows and residue polynomials

Class	Edge word	ω_b	ω_c	Residue
5	000020021201200	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
6	000020021210110	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
7	000020021301100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
8	000020022210100	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
13	000020111111200	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
14	000020111120110	-15	$-\frac{5}{2}$	$R_0^{(5)}$
15	000020111121100	-15	$-\frac{5}{2}$	$R_0^{(5)}$
16	000020111130010	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
17	000020111220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
18	000020112120100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
19	000020112220000	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_0^{(5)}$
20	000020120210011	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
21	000020120211001	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
22	000020120211010	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_0^{(5)}$
23	000021110110111	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_0^{(5)}$
27	000110012201200	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_1^{(5)}$
28	000110012210110	-15	$-\frac{5}{2}$	$R_2^{(5)}$
29	000110012301100	-5	$-\frac{5}{6}$	$R_3^{(5)}$
30	000110012310100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
31	000110013210100	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_0^{(5)}$
32	000110022201100	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_4^{(5)}$
37	000110102111200	-15	$-\frac{5}{2}$	$R_5^{(5)}$
38	000110102120110	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_6^{(5)}$
39	000110102121100	-15	$-\frac{5}{2}$	$R_7^{(5)}$
105	000120111310000	-5	0	$R_8^{(5)}$
106	000120120201001	-15	0	$R_9^{(5)}$
107	000120120202000	$-\frac{15}{4}$	0	$R_{10}^{(5)}$
108	000120120211000	$-\frac{15}{2}$	0	$R_{11}^{(5)}$
109	000120120301000	$-\frac{5}{2}$	0	$R_{12}^{(5)}$
110	000120121210000	$-\frac{15}{4}$	0	$R_{13}^{(5)}$

Class	Edge word	ω_b	ω_c	Residue
111	000121101101200	$-\frac{15}{2}$	0	$R_{14}^{(5)}$
112	000121101110101	-30	0	$R_{15}^{(5)}$
114	000121101110200	-30	0	$R_{16}^{(5)}$
115	000121101200101	-15	0	$R_{6}^{(5)}$
116	000121101210100	$-\frac{15}{2}$	0	$R_{17}^{(5)}$
117	000121102110100	-15	0	$R_{14}^{(5)}$
118	000121110110110	-15	0	$R_{18}^{(5)}$
119	000121110200011	-15	0	$R_{6}^{(5)}$
120	000121110200101	-15	0	$R_{0}^{(5)}$
121	000121120200001	$-\frac{15}{8}$	0	$R_{0}^{(5)}$
122	000131101110100	-5	0	$R_{19}^{(5)}$
127	001111011101200	-30	0	$R_{20}^{(5)}$
40	000110102130010	-5	$-\frac{5}{6}$	$R_{21}^{(5)}$
41	000110102210101	-15	$-\frac{5}{2}$	$R_{22}^{(5)}$
42	000110102211100	-15	$-\frac{5}{2}$	$R_{23}^{(5)}$
43	000110102220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{24}^{(5)}$
44	000110102220010	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{9}^{(5)}$
45	000110102220100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{25}^{(5)}$
46	000110102310001	-5	$-\frac{5}{6}$	$R_{3}^{(5)}$
48	000110103210100	-5	$-\frac{5}{6}$	$R_{21}^{(5)}$
49	000110103220000	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_{12}^{(5)}$
52	000110111112100	-30	-5	$R_{26}^{(5)}$
53	000110111201101	-30	-5	$R_{27}^{(5)}$
54	000110111202100	-15	$-\frac{5}{2}$	$R_{28}^{(5)}$
55	000110111211001	-15	$-\frac{5}{2}$	$R_{29}^{(5)}$
56	000110111211010	-30	-5	$R_{30}^{(5)}$
57	000110111301001	-10	$-\frac{5}{3}$	$R_{31}^{(5)}$
59	000110112201100	-15	$-\frac{5}{2}$	$R_{32}^{(5)}$
60	000110112210001	-15	$-\frac{5}{2}$	$R_{33}^{(5)}$
61	000110112210010	-15	$-\frac{5}{2}$	$R_{7}^{(5)}$
62	000110112210100	-15	$-\frac{5}{2}$	$R_{34}^{(5)}$
63	000110112211000	-15	$-\frac{5}{2}$	$R_{35}^{(5)}$
64	000110112220000	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{11}^{(5)}$
66	000110112310000	-5	$-\frac{5}{6}$	$R_{8}^{(5)}$
67	000110201210002	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{36}^{(5)}$
68	000110201211001	-15	$-\frac{5}{2}$	$R_{37}^{(5)}$
69	000110201220001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{24}^{(5)}$
71	000110202220000	$-\frac{15}{8}$	$-\frac{5}{16}$	$R_{10}^{(5)}$
72	000110211211000	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{38}^{(5)}$
76	000111101101201	-15	$-\frac{5}{2}$	$R_{39}^{(5)}$
77	000111101102200	-15	$-\frac{5}{2}$	$R_{16}^{(5)}$
79	000111101111110	-30	-5	$R_{20}^{(5)}$
80	000111101111200	-30	-5	$R_{40}^{(5)}$

Class	Edge word	ω_b	ω_c	Residue
81	000111101200102	-15	$-\frac{5}{2}$	$R_{41}^{(5)}$
82	000111101201101	-30	-5	$R_{42}^{(5)}$
83	000111101210101	-15	$-\frac{5}{2}$	$R_{43}^{(5)}$
84	000111102200101	-15	$-\frac{5}{2}$	$R_{22}^{(5)}$
86	000111102210100	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{17}^{(5)}$
88	000111111200002	$-\frac{15}{4}$	$-\frac{5}{8}$	$R_{0}^{(5)}$
89	000111111200011	-30	-5	$R_{44}^{(5)}$
91	000111111201100	-15	$-\frac{5}{2}$	$R_{45}^{(5)}$
92	000111111300001	$-\frac{5}{2}$	$-\frac{5}{12}$	$R_{0}^{(5)}$
94	000111201210001	$-\frac{15}{2}$	$-\frac{5}{4}$	$R_{46}^{(5)}$
95	000120012210100	$-\frac{15}{4}$	0	$R_{0}^{(5)}$
96	000120021201100	$-\frac{15}{2}$	0	$R_{47}^{(5)}$
97	000120102120100	-15	0	$R_{9}^{(5)}$
98	000120102220000	$-\frac{15}{8}$	0	$R_{10}^{(5)}$
100	000120111120010	-30	0	$R_{48}^{(5)}$
101	000120111210001	-30	0	$R_{34}^{(5)}$
102	000120111210010	-15	0	$R_{48}^{(5)}$
103	000120111211000	-15	0	$R_{49}^{(5)}$
104	000120111220000	-15	0	$R_{11}^{(5)}$

$$R_0^{(5)} = 0.$$

$$R_1^{(5)} = -\frac{19A^4}{60} + 16A\zeta(3) + \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} + \frac{8A\pi^3}{15}.$$

$$R_2^{(5)} = -\frac{499A^4}{720} + \frac{17A^2\pi^2}{360} - \frac{3\pi^4}{400} + \frac{191A\pi^3}{180} + \frac{141A\zeta(3)}{5}.$$

$$R_3^{(5)} = -\frac{4\pi^3}{3} - \frac{24\pi^2}{5} - \frac{48\pi}{5} + 2A^2\pi.$$

$$R_4^{(5)} = -\frac{47A^4}{720} - \frac{43A^2\pi^2}{360} + \frac{83\pi^4}{1200} + \frac{7A\pi^3}{180} - \frac{3A\zeta(3)}{5}.$$

$$R_5^{(5)} = -\frac{499A^4}{720} - \frac{4\mathcal{K}}{5} - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} + \frac{191A\pi^3}{180} + \frac{141A\zeta(3)}{5}.$$

$$R_6^{(5)} = -\frac{47A^4}{180} - \frac{A^2\pi^2}{18} - \frac{11\pi^4}{900} + \frac{2A\pi^3}{9} + \frac{24A\zeta(3)}{5}.$$

$$R_7^{(5)} = -\frac{499A^4}{720} + \frac{4\mathcal{K}}{5} - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} + \frac{227A\pi^3}{180} + \frac{169A\zeta(3)}{5}.$$

$$R_8^{(5)} = -\frac{4\pi^3}{15} - \frac{12\pi^2}{5} - \frac{24\pi}{5} + \frac{3A^2\pi}{5} + \frac{2A\pi^2}{5} + \frac{4A\pi}{5}.$$

$$R_9^{(5)} = -\frac{A^4}{16} - A\zeta(3) + \frac{3\pi^4}{400} - \frac{5A^2\pi^2}{72} - \frac{A\pi^3}{12}.$$

$$R_{10}^{(5)} = -\frac{A^4}{60} - \frac{4A\zeta(3)}{5} - \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} - \frac{A\pi^3}{15}.$$

$$R_{11}^{(5)} = -\frac{11A^4}{80} - \frac{47A^2\pi^2}{360} - \frac{3\pi^4}{400} + \frac{17A\pi^3}{60} + \frac{33A\zeta(3)}{5}.$$

$$R_{12}^{(5)} = \frac{4\pi^3}{3} + \frac{24\pi^2}{5} + \frac{48\pi}{5} + \frac{6A^2\pi}{5} - \frac{12A\pi^2}{5} - \frac{24A\pi}{5}.$$

$$R_{13}^{(5)} = -\frac{7A^4}{180} - \frac{A^2\pi^2}{18} - \frac{11\pi^4}{900} + \frac{2A\pi^3}{45} + \frac{8A\zeta(3)}{5}.$$

$$R_{14}^{(5)} = -\frac{9A^2Q}{10} - \frac{44\mathsf{G}Q}{5} - \frac{5Q\pi^2}{6}.$$

$$R_{15}^{(5)} = -\frac{39A^2Q}{20} - \frac{22\mathsf{G}Q}{5} - \frac{7Q\pi^2}{60}.$$

$$R_{16}^{(5)} = -\frac{53A^4}{180} + \frac{4\mathcal{K}}{5} + \frac{11\pi^4}{900} - \frac{A^2\pi^2}{6} + \frac{16A\pi^3}{45} + 8A\zeta(3).$$

$$R_{17}^{(5)} = -\frac{7A^4}{48} - A\zeta(3) + \frac{623\pi^4}{3600} - \frac{A^2\pi^2}{8} - \frac{A\pi^3}{12}.$$

$$R_{18}^{(5)} = -\frac{9A^2Q}{20} + \frac{22\mathsf{G}Q}{5} + \frac{7Q\pi^2}{60}.$$

$$R_{19}^{(5)} = \frac{24Q\pi}{5}.$$

$$R_{20}^{(5)} = -\frac{39A^2Q}{20} + \frac{74\mathsf{G}Q}{5} + \frac{89Q\pi^2}{60}.$$

$$R_{21}^{(5)} = \frac{4\pi^3}{15} + \frac{12\pi^2}{5} + \frac{24\pi}{5} + \frac{13A^2\pi}{5} - \frac{14A\pi^2}{5} - \frac{28A\pi}{5}.$$

$$R_{22}^{(5)} = -\frac{499A^4}{720} - 11A\zeta(3) - \frac{3\pi^4}{400} + \frac{17A^2\pi^2}{360} - \frac{61A\pi^3}{180}.$$

$$R_{23}^{(5)} = -\frac{A^4}{3} - \frac{2A^2\pi^2}{9} + \frac{2A\pi^3}{5} + \frac{16A\zeta(3)}{5}.$$

$$R_{24}^{(5)} = -\frac{19A^4}{60} - \frac{88A\zeta(3)}{5} + \frac{19\pi^4}{900} - \frac{A^2\pi^2}{30} - \frac{2A\pi^3}{3}.$$

$$R_{25}^{(5)} = -\frac{62\pi^3}{15} - 16\pi^2 - 32\pi - \frac{2A^2\pi}{5} + 4A\pi^2 + 8A\pi.$$

$$R_{26}^{(5)} = -\frac{3A^2Q}{2} + \frac{76\mathsf{G}Q}{5} + \frac{41Q\pi^2}{30}.$$

$$R_{27}^{(5)} = -\frac{3A^2Q}{2} - 4\mathsf{G}Q - \frac{7Q\pi^2}{30}.$$

$$R_{28}^{(5)} = -\frac{287A^4}{240} - \frac{8\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{37A\pi^3}{20} + \frac{271A\zeta(3)}{5}.$$

$$R_{29}^{(5)} = -\frac{3A^2Q}{2} - \frac{52\mathsf{G}Q}{5} - \frac{23Q\pi^2}{30}.$$

$$R_{30}^{(5)} = -\frac{25A^4}{12} - \frac{16\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5}.$$

$$R_{31}^{(5)} = \frac{4\pi^3}{15} + \frac{12\pi^2}{5} + \frac{24\pi}{5} + \frac{13A^2\pi}{5} + \frac{14A\pi^2}{5} + \frac{28A\pi}{5}.$$

$$R_{32}^{(5)} = -\frac{287A^4}{240} + \frac{4\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{37A\pi^3}{20} + \frac{271A\zeta(3)}{5}.$$

$$R_{33}^{(5)} = -\frac{47A^4}{240} - \frac{11A^2\pi^2}{120} + \frac{3\pi^4}{400} + \frac{19A\pi^3}{60} + \frac{43A\zeta(3)}{5}.$$

$$R_{34}^{(5)} = -\frac{53A^4}{180} + 8A\zeta(3) + \frac{11\pi^4}{900} - \frac{A^2\pi^2}{6} + \frac{16A\pi^3}{45}.$$

$$R_{35}^{(5)} = -\frac{287A^4}{240} - \frac{4\mathcal{K}}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} + \frac{41A\pi^3}{20} + \frac{299A\zeta(3)}{5}.$$

$$R_{36}^{(5)} = -\frac{19A^4}{720} + A\zeta(3) - \frac{3\pi^4}{400} - \frac{23A^2\pi^2}{360} - \frac{A\pi^3}{180}.$$

$$R_{37}^{(5)} = -\frac{287A^4}{240} - \frac{177A\zeta(3)}{5} + \frac{3\pi^4}{400} + \frac{127A^2\pi^2}{360} - \frac{27A\pi^3}{20}.$$

$$R_{38}^{(5)} = -\frac{73A^4}{180} - 8A\zeta(3) - \frac{389\pi^4}{900} - \frac{7A^2\pi^2}{18} + \frac{26A\pi^3}{45}.$$

$$R_{39}^{(5)} = -\frac{3A^2Q}{2} + \frac{44\mathsf{G}Q}{5} + \frac{5Q\pi^2}{6}.$$

$$R_{40}^{(5)} = -\frac{25A^4}{12} + \frac{4\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{52A\pi^3}{15} + \frac{512A\zeta(3)}{5}.$$

$$R_{41}^{(5)} = -\frac{13A^4}{180} - \frac{A^2\pi^2}{6} + \frac{11\pi^4}{900} + \frac{8A\pi^3}{45} + \frac{24A\zeta(3)}{5}.$$

$$R_{42}^{(5)} = -\frac{25A^4}{12} - \frac{4\mathcal{K}}{5} - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} + \frac{28A\pi^3}{15} + \frac{288A\zeta(3)}{5}.$$

$$R_{43}^{(5)} = -\frac{79A^4}{80} - \frac{79A\zeta(3)}{5} - \frac{623\pi^4}{3600} - \frac{43A^2\pi^2}{360} + \frac{A\pi^3}{60}.$$

$$R_{44}^{(5)} = -\frac{25A^4}{12} - 32A\zeta(3) - \frac{11\pi^4}{900} + \frac{5A^2\pi^2}{6} - \frac{4A\pi^3}{3}.$$

$$R_{45}^{(5)} = -\frac{79A^4}{80} - \frac{43A^2\pi^2}{360} - \frac{623\pi^4}{3600} + \frac{17A\pi^3}{12} + \frac{117A\zeta(3)}{5}.$$

$$R_{46}^{(5)} = \frac{62\pi^3}{15} + 16\pi^2 + 32\pi + \frac{2A^2\pi}{5} + 4A\pi^2 + 8A\pi.$$

$$R_{47}^{(5)} = -\frac{11A^4}{80} + A\zeta(3) - \frac{3\pi^4}{400} - \frac{47A^2\pi^2}{360} + \frac{A\pi^3}{12}.$$

$$R_{48}^{(5)} = -\frac{9A^2Q}{10} + 4\mathsf{G}Q + \frac{7Q\pi^2}{30}.$$

$$R_{49}^{(5)} = -\frac{9A^2Q}{10} + \frac{52\mathsf{G}Q}{5} + \frac{23Q\pi^2}{30}.$$

B Derivation of the special-function representation of $C_6(1)$

This appendix reproduces the derivation of the analytic calculation in the package `C6_research_note/`, written by OpenAI GPT-6 astra, adapted to the notation of this note. It starts from the reduced input (604) of Section 22.9 and proves the conversions stated there and in Section 1.2: the ordinary-period sectors (Section B.3), the wheel (Section B.4), the seventh-power sector (Section B.5), the primitive (Section B.6) and the polylogarithmic sector (Section B.7). Sections B.8 and B.9 give the numerical method and its error control. Definitions already given in Sections 1.2 and 22.9 are used without restating them. The package's records of these steps are the reduced setup `C6_research_note/inputs/upstream_reduced_setup.md` (extracted from `C6_research_note/inputs/source_snapshots/C6_central_binomial_wheel_expression.json`), the derivation snapshots `C6_research_note/inputs/source_snapshots/*_derivation.md` with their expression and check files, and the stage folders `C6_research_note/code/stages/` with their `DERIVATION.md` files, inputs and code; the steps below are written out where they are used, so these records are not cited again step by step, and the inputs that remain are named in Section B.1.

B.1 Scope, levels of statements and notation

Scope. The identity (604) is the supplied reduced analytic input of the method; the graph-to-period reduction that produced it is derived neither here nor in the package. The derivations below convert its sectors into the formula (2). The model, the expansion parameter, the conventions and the graph enumeration are those of this note (Sections 1–3), and Section 1.3 gives the status of each statement.

Three levels of statements. The package keeps three kinds of statement apart, and so does this appendix. *Physical identification*: the graph-to-period identities and the renormalization-group assembly behind (604) are inherited inputs; the downstream checks do not establish them independently. *Analytic identity*: a finite algebraic, differential or contour derivation may prove the equality of two representations, conditional on its stated input identities. *Numerical certification*: an outward-rounded interval plus an analytic tail bound encloses a specified expression; agreement of floating-point calculations alone is not a certificate. Input identities that are supplied with the reduced expression, rather than derived from it, are named as such where they are used. After the derivations below, the conversion to the nine values uses three of them, the elliptic-moment evaluations (911)–(913) on which the wheel's periodization identities rest (Section B.4.3), besides published identities. The interval certificate uses more: it encloses an earlier form of the reduced expression (Section B.8.5), which reaches (604) only through further exact identities of the package, among them the inherited wheel normalizations of $B_- + B_+$ and $B_- - B_+$. This appendix does not reproduce those identities; they are inputs of the enclosure.

Symbols. The package writes B_{pol}, B_{low}, B_6, B_8 for the sectors S_{pol}, S_{low}, S_6, S_8 of (604); U_4, U_6, U_8 and H_-, H_0, H_+ for the hypergeometric values $\varpi_4, \varpi_6, \varpi_8$ and Φ_-, Φ_0, Φ_+ of (607); K_n for κ_n of (608), whose sum M_7 it writes with a script letter; $h(x)$ for the Gauss function $w(x)$ of (609); G for G; ℓ for $\log 2$; and $\mathcal{H}_7^{(u)}$ for $\mathcal{H}_7$. It defines P_b by $S_6 = 24P_b$, and it writes $E_{\mathrm{real}} = E - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2)$ for the polynomial part of (2). To avoid clashes with symbols of this note, the following local symbols are also renamed: the Barnes functional (package $\mathcal{B}_c$) is $\mathfrak{B}_c$; the wheel sum that the package writes as a script S_7 is Σ_7; the contour kernel of the primitive (package $\mathcal{K}$) is $\widetilde{\mathcal{N}}$; the primitive Meijer kernels (package M_j) are $M_j^{(3)}$; the Barnes kernel of the lower block (package K) is $\widetilde{M}$; the wheel's parameter point and direction are $\mathbf{p}_0$, $\mathbf{v}$; the coefficient matrices of the primitive are M, N; and the Euler-weighted approximant (package E_N) is $\widehat{E}_N$. Repetition is

written $a^{\times k}$, as in Section 1.2.

Harmonic remainders. The constants $A = \pi + 4\log 2$, $Q = \Gamma(\frac{1}{4})^4/\pi$, G and the sequences a_n, h_n, δ_n, y_n are those of Section 22.9, with $H_0^{(p)} = 0$. Equivalently,

$$a_n = \frac{(1/2)_n}{n!}, \qquad \delta_n = \frac{\pi^2}{3} - 4H_{2n}^{(2)} + 2H_n^{(2)} = \psi_1(n+\tfrac{1}{2}) - \psi_1(n+1) > 0 \tag{866}$$

for every $n \geq 0$. For example $\delta_0 = \pi^2/3$, $\delta_1 = \pi^2/3 - 3$ and $\delta_2 = \pi^2/3 - 115/36$. The empty harmonic sum vanishes; the terms with $n > 0$ do not.

Polylogarithm and Meijer conventions. The multiple polylogarithm has the descending index order of Section 22.9. In the ordinary alphabet of harmonic polylogarithms, $H_0(z) = \log z$ and $H_1(z) = -\log(1-z)$, with differential letters dz/z and $dz/(1-z)$. Thus $H_{011}(z) = \mathrm{Li}_{2,1}(z,1)$ and $H_{01101}(z) = \mathrm{Li}_{2,1,2}(z,1,1)$. Principal branches continued from the stated convergence domains are used.

For an ordinary $G_{p,q}^{m,n}$ we use the inverse Mellin measure $ds/(2\pi\mathrm{i})$ and the argument factor x^{-s}. Its kernel is

$$\frac{\prod_{j=1}^{m}\Gamma(b_j+s)\prod_{j=1}^{n}\Gamma(1-a_j-s)}{\prod_{j=m+1}^{q}\Gamma(1-b_j-s)\prod_{j=n+1}^{p}\Gamma(a_j+s)}. \tag{867}$$

The first n upper and the first m lower parameters are in numerator groups. An identical numerical parameter in two different denominator groups does not cancel. The repetition notation $a^{\times k}$ means a list of k entries, not a power of a function [22, 18].

For contour manipulations we use the functional

$$\mathfrak{B}_c[f] = \frac{1}{2\pi\mathrm{i}}\int_{\mathrm{Re}\,s=c} f(s)\,ds, \tag{868}$$

with the vertical contour directed upward. The kernels and contour majorants are specified wherever this notation is used.

The restored Laurent coefficients. The correction of Section 22.9, $R_{147,-3} = R_{148,-3} = \frac{2}{5}$, contributes to $r_6 = 2187\,C_6(1)$

$$\begin{aligned}\Delta r_6^{\mathrm{repair}} = {}& \frac{1024}{25}A^5 - \frac{1216}{15}A^3\pi^2 + \frac{2672}{5}A^2\pi^3 + \frac{60864}{5}A^2\zeta(3)\\ & - \frac{2738}{75}A\pi^4 + \frac{5963}{15}\pi^5 - \frac{12046}{5}\pi^2\zeta(3) + \frac{711576}{5}\zeta(5).\end{aligned} \tag{869}$$

Divided by 2187 this is the contribution +659.278983 quoted in Section 22.9. It is already included in the polynomial E of (605) and must not be added again. The package describes it as a correction of the symbolic skeleton, not an adjustment of the independent numerical estimate.

B.2 Specification of the nine values

B.2.1 The four ordinary Meijer values

The values $\mathcal{G}_4, \mathcal{G}_6, \mathcal{G}_8, \mathcal{G}_7$ and their parameters are defined in Section 1.2. The package writes $X_4 = -C_d$ and $Y_4 = -C_s$, so that $\rho_4 = \sqrt{X_4^2 + Y_4^2}$ and $\alpha_4 = \pi^{-1}\arctan(Y_4/X_4)$. Both $X_4, Y_4 > 0$ for the specified constants (Section B.3.3); the real principal inverse tangent is used. For the eighth-power block,

$$d_8 = \frac{1}{2\pi}\arctan\frac{556}{1233 - 417\sqrt{2}}, \qquad e_8 = \frac{1}{2\pi}\,\mathrm{arcosh}\,\frac{1233 - 139\sqrt{2}}{\rho_8}, \qquad \alpha_{8,\pm} = d_8 \pm \mathrm{i}e_8. \tag{870}$$

All displayed real square roots and real inverse functions take their positive or principal values. The subscript 8 denotes eighth-power kernel content, not an eighth-order thermal coefficient. In the seventh block the $\eta_{7,\pm}$ use real principal inverse tangents and β_7 uses the principal complex inverse tangent; here $q_{7,2} > 0$, $q_{7,1} > 0$ and $q_{7,0} < 0$, as follows from $2/3 < \log 2 < 7/10$ and $3 < \pi < 22/7$. These are the retained lower-, sixth-, eighth- and seventh-sector reductions.

B.2.2 The bivariate convention: the kernel is part of the specification

The bivariate H convention is that of Andrade, Rathie and Farias [17] (its bivariate Definition B), as adopted by the package. All three values have arguments $(1, 1)$ and unit slopes. In a local variable s, the factor is

$$\Theta(s) = \frac{\prod_{j=1}^{m} \Gamma(b_j + B_j s) \prod_{j=1}^{n} \Gamma(1 - a_j - A_j s)}{\prod_{j=m+1}^{q} \Gamma(1 - b_j - B_j s) \prod_{j=n+1}^{p} \Gamma(a_j + A_j s)}. \tag{871}$$

For the shared group, the present cases have $m_0 = 0$. Its numerator factors are $\Gamma(1 - e_j - E_j s - F_j t)$ for the first n_0 upper entries; its denominator contains the remaining upper factors $\Gamma(e_j + E_j s + F_j t)$ and every lower factor $\Gamma(1 - f_j - A_j s - B_j t)$. The standard measure is $ds\, dt/(2\pi\mathrm{i})^2$, with $x^{-s}y^{-t}$. Ordered pair and triple lists and numerator counts fix each function without naming a residual integral as a new function. Both the lists and the simplified kernels are given, to prevent any ambiguity of convention; the kernels and contours are those displayed in Section 1.2.

B.2.3 The wheel value

With $\alpha_W = \pi^{-1} \arctan(5/8)$ put

$$\boldsymbol{a}_W = ((1/2, 1)^{\times 4}, (\alpha_W, 1)), \qquad \boldsymbol{b}_W = ((0, 1)^{\times 4}, (\alpha_W, 1)). \tag{872}$$

Then

$$\mathcal{H}_W = H^{0,2;2,3;2,2}_{2,2;5,5;2,2}\left[\begin{matrix} (1/2; 1, 1)^{\times 2} : \boldsymbol{a}_W; (1/2, 1), (0, 1) \\ (-1/2; 1, 1)^{\times 2} : \boldsymbol{b}_W; (0, 1), (1/2, 1) \end{matrix} \middle| 1, 1 \right]. \tag{873}$$

Its kernel is $N_{\alpha_W}(s)P(t)/(1/2 - s - t)^2$, the integrand of Section 1.2, where for any angle parameter α

$$\begin{aligned} N_\alpha(s) &= \frac{\Gamma(s)^2 \Gamma(1/2 - s)^3}{\Gamma(1 - s)^2 \Gamma(1/2 + s) \Gamma(s + \alpha) \Gamma(1 - s - \alpha)}, \\ P(t) &= \Gamma(t)\Gamma(t + 1/2)\Gamma(1/2 - t)\Gamma(1 - t), \end{aligned} \tag{874}$$

so that the kernel $N(s)$ of Section 1.2 is $N_{\alpha_W}(s)$. The upward lines $\operatorname{Re} s = 1/4$, $\operatorname{Re} t = 1/8$ separate the poles and give absolute convergence. The shared quotient is $\Gamma(1/2 - s - t)^2/\Gamma(3/2 - s - t)^2$. The normalization $W_0 = \sqrt{89}\, \mathcal{H}_W/\pi^2$, so that $-393\pi^3 W_0 = -393\pi\sqrt{89}\, \mathcal{H}_W$, is derived in Section B.4.4 from the wheel identity (901), which rests on the periodization identities (908) proved in Section B.4.3.

B.2.4 The unit-slope seventh-sector value

With $\beta = \beta_7$, define the lists

$$\begin{aligned} \boldsymbol{a}_7 &= ((-1/4, 1), (1/2, 1)^{\times 7}, (\beta, 1), (\bar{\beta}, 1)), \\ \boldsymbol{b}_7 &= ((0, 1)^{\times 6}, (1, 1), (3/4, 1), (\beta, 1), (\bar{\beta}, 1)), \\ \boldsymbol{c}_7 &= ((0, 1), (1/2, 1), (1/2, 1), (1/4, 1), (3/4, 1)), \\ \boldsymbol{d}_7 &= ((0, 1), (1/2, 1), (-1/2, 1), (1/4, 1), (3/4, 1)). \end{aligned}$$

Then

$$\mathcal{H}_7 = H^{0,1;3,8;2,3}_{1,1;10,10;5,5}\left[\begin{matrix}(1/2;1,1):\boldsymbol{a}_7;\boldsymbol{c}_7\\(-1/2;1,1):\boldsymbol{b}_7;\boldsymbol{d}_7\end{matrix}\middle|1,1\right], \tag{875}$$

with the simplified kernel $\mathcal{A}_7(s)\cot(2\pi t)/[(1/2-t)(1/2-s-t)]$ and the contours $\operatorname{Re} s = 1/4$, $\operatorname{Re} t = 1/16$ of Section 1.2. The second local gamma quotient equals $\cot(2\pi t)/(1/2-t)$; its parameter lists are those in (875). This is a bivariate function, not an ordinary Meijer G value. The normalization $24\pi^3 M_7 = 24\pi^{3/2}[\rho_7\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7]$ is proved in Section B.5.3.

B.2.5 The primitive value

The unique root $\lambda_* \in (15,16)$ of $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250 = 0$ has the explicit form

$$\lambda_* = \frac{28 + \sqrt{409}\cos\left[\frac{1}{3}\arccos\frac{7298}{409\sqrt{409}}\right]}{3}. \tag{876}$$

In the definitions of e_*, χ_*, B_σ, C_σ and β_σ of Section 1.2, both real square-root radicands are positive, and the complex inverse tangent is principal. These parameters depend only on elementary and algebraic constants, not on a numerically fitted primitive value. Define

$$\boldsymbol{a}_\sigma = ((1/2,1)^{\times 4}, (\beta_\sigma,1), (\bar\beta_\sigma,1)),$$
$$\boldsymbol{b}_\sigma = ((0,1)^{\times 3}, (\beta_\sigma,1), (\bar\beta_\sigma,1)).$$

Then

$$\mathcal{H}_* = H^{0,0;3,3;3,3}_{0,1;6,5;6,5}\left[\begin{matrix}-:\boldsymbol{a}_+;\boldsymbol{a}_-\\(0;1,1):\boldsymbol{b}_+;\boldsymbol{b}_-\end{matrix}\middle|1,1\right]. \tag{877}$$

For each local group the first three upper and the first three lower parameters are numerator entries. The kernel is $\Psi_+(s)\Psi_-(t)/\Gamma(1-s-t)$ on the lines $\operatorname{Re} s = \operatorname{Re} t = 1/4$ (Section 1.2). The exact primitive normalizations are $P_a = 80\pi^3\mathcal{H}_*$ and $96P_a = 7680\pi^3\mathcal{H}_*$. The primitive normalization is unchanged. The global contour relation and the factorization producing this value are derived in Section B.6 from the stated Gauss-product input.

B.3 The ordinary-period sectors

B.3.1 The input periods and their weights

The three ordinary sectors of (604) have the hypergeometric input forms (607), with $S_6 = 24P_b$. They specify the analytic quantities that are converted into $\mathcal{G}_4$, $\mathcal{G}_6$ and $\mathcal{G}_8$; their graph identifications are part of the reduced input. The formerly separated contribution $-144Q^3/\pi$ belongs to the complete P_b and is inside S_6 there (it comes from the -1 in $2\Phi_- - 1$); it is not an additional term in E and must not be appended to E. The parameter excesses are positive, so the ordinary series converge at their unit arguments.

B.3.2 A common gamma expansion

The gamma recurrence and the integer and half-integer polygamma values give, for $n \geq 0$ and $\varepsilon \to 0$,

$$\frac{\Gamma(-n+\varepsilon)}{\Gamma(1/2-n+\varepsilon)} = \frac{a_n}{\sqrt{\pi}\,\varepsilon}\exp\left[2h_n\varepsilon + \frac{\delta_n - 2\pi^2/3}{2}\varepsilon^2 + O(\varepsilon^3)\right]. \tag{878}$$

For $n = 0$, the first logarithmic coefficient is $2\log 2$ and the second derivative is $-\pi^2/3$. Moving to $-n$ adds $2(H_n - H_{2n})$ and $-4H^{(2)}_{2n} + 2H^{(2)}_n$ respectively. This gives (878) with exactly the remainders h_n and δ_n of Section 22.9 and (866). The same expansion fixes the logarithmic normalizations of the primitive and of the wheel residues.

B.3.3 The lower block: two fourth-order kernels and sine addition

On the upward line $\operatorname{Re} s = 1/4$, define

$$M_1(s) = \frac{\Gamma(s)\Gamma(1/2-s)^3}{\Gamma(1-s)^3\Gamma(1/2+s)},$$
$$M_2(s) = \frac{\Gamma(s)^2\Gamma(1/2-s)^2}{\Gamma(1-s)^2\Gamma(1/2+s)^2}. \tag{879}$$

Gamma reflection writes these as $R_4(s)\tan^3(\pi s)$ and $R_4(s)\tan^2(\pi s)$, where $R_4(s) = [\Gamma(s)/\Gamma(s+1/2)]^4$. Their contour values are

$$g_{13} = G^{1,3}_{4,4}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 4}\\ 0^{\times 4}\end{matrix}\right), \qquad g_{22} = G^{2,2}_{4,4}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 4}\\ 0^{\times 4}\end{matrix}\right). \tag{880}$$

The residues of M_1 at $s = -n$ are πa_n^4; hence $g_{13} = \pi\varpi_4$. To obtain the other normalization, the Bailey–Mishev Barnes and ${}_7F_6$ representations at $(1/2, 1/2, 1/2, 1/2; 1; 1, 1)$ [14], a published identity, give

$$\frac{\varpi_6}{2\pi} = \frac{1}{\pi^5}\mathfrak{B}_{1/4}[\widetilde{M}], \qquad \widetilde{M}(s) = \frac{\Gamma(s)^2\Gamma(1/2-s)^4}{\Gamma(1-s)^2}. \tag{881}$$

Here and below $\mathfrak{B}_c$ is the upward Barnes functional (868). Set $M_0(s) = \Gamma(1/2-s)^4/\Gamma(1-s)^4$. Reflection gives $\widetilde{M} = \pi^2(M_2 + M_0)$. The contribution $\mathfrak{B}_{1/4}[M_0]$ is zero: under $s \mapsto 1/2 - s$ it is the inverse Mellin value of R_4 at 1. The beta integral identifies $\Gamma(s)/\Gamma(s+1/2)$ as the Laplace transform of

$$b(t) = \frac{1}{\sqrt{\pi}\sqrt{1-e^{-t}}}, \qquad t > 0.$$

Since $b(t) = t^{-1/2}(1 + O(t))/\sqrt{\pi}$, its fourfold additive convolution is $t(1 + O(t))$ at 0 and has value zero there. Inversion is justified by the integrability of $R_4(1/4 + \mathrm{i}u)$. Therefore

$$\varpi_4 = g_{13}/\pi, \qquad \varpi_6 = 2g_{22}/\pi^2. \tag{882}$$

The gamma kernel N_α of (874) obeys

$$N_\alpha = \cos(\pi\alpha)M_1 + \sin(\pi\alpha)M_2. \tag{883}$$

Using $X_4 = -C_d$, $Y_4 = -C_s$ and the angle α_4, (882) and (883) yield

$$S_{\mathrm{low}} = 4\pi^2(C_d g_{13} + C_s g_{22}) = -4\pi^2\rho_4\mathcal{G}_4.$$

There is no contour boundary correction: the kernels are $O((1 + |\operatorname{Im} s|)^{-2})$ on the common line. The branch can be checked without sampling the periods. The inequalities $\pi > 3$ and $\log 2 > 1/2$ give $A > 5$. Splitting Euler's integral for $\Gamma(1/4)$ at $t = 1$, and using $e^{-t} < 1 - t + t^2/2$ on $(0, 1)$ and $t^{-3/4} < 1$ on $(1, \infty)$,

$$\Gamma(1/4) < \int_0^1 t^{-3/4}\Big(1 - t + \frac{t^2}{2}\Big)\,dt + \int_1^\infty e^{-t}\,dt < 4 - \frac{4}{5} + \frac{2}{9} + \frac{2}{5} < 4.$$

Thus $Q < 256/3$. These bounds imply

$$C_s < 5120 - \frac{105}{4}\,25 - 135 - \frac{2469}{4}\,9 = -\frac{2453}{2}, \qquad C_d < 8192 - 42\cdot 25 - 270 - 1227\cdot 9 = -4171.$$

Hence $X_4, Y_4 > 0$, as used above.

B.3.4 The sixth-power block

The sixth-power sector $S_6 = 24P_b$ of (607), with $P_b = \frac{6Q^3}{\pi}(2\Phi_- - 1) - 104\pi^5\Phi_0 + \frac{2^{18}\pi^5}{Q^3}\Phi_+$, has the absolutely convergent coefficient series

$$P_b = 24\pi^3 \sum_{n\geq 0} a_n^6(312h_n^2 + 13\delta_n + 88\pi h_n). \tag{884}$$

It is derived from the three hypergeometric values by a finite Fourier interpolation, not inferred from a numerical value. Define the proof-only function

$$\mathsf{f}_6(t) = \pi^3 \sum_{m\in\mathbb{Z}} \left[\Gamma(\tfrac{3}{4} + m + t)\Gamma(\tfrac{3}{4} - m - t)\right]^{-6}. \tag{885}$$

It is entire, even and one-periodic. Gamma reflection and ratio asymptotics [26, 27] give locally uniform convergence and, on a vertical strip, $\mathsf{f}_6(t) = O(e^{6\pi|\operatorname{Im} t|}(1 + |\operatorname{Im} t|)^{-2})$. Moving Fourier-coefficient contours vertically excludes the frequencies $|k| \geq 3$, including the boundary frequencies because of the factor $(1 + |\operatorname{Im} t|)^{-2}$. Hence f_6 lies in the span of 1, $\cos 2\pi t$ and $\cos 4\pi t$. For every function f in this three-dimensional space,

$$384f(0) - 104f(\tfrac{1}{4}) + 32f(\tfrac{1}{2}) = 312f(\tfrac{1}{4}) - \frac{176}{\pi}f'(\tfrac{1}{4}) + \frac{52}{\pi^2}f''(\tfrac{1}{4}). \tag{886}$$

Gamma reflection fixes the sample normalizations exactly:

$$\mathsf{f}_6(0) = \frac{Q^3}{64\pi^6}(2\Phi_- - 1), \qquad \mathsf{f}_6(\tfrac{1}{4}) = \Phi_0, \qquad \mathsf{f}_6(\tfrac{1}{2}) = \frac{8192}{Q^3}\Phi_+.$$

Consequently P_b equals π^5 times the left side of (886) with $f = \mathsf{f}_6$. At $t = \frac{1}{4}$ each negative-index term of (885) vanishes to order six. All derivatives needed here have order at most two, so this half contributes zero. On the nonnegative half, polygamma reflection and recurrence [32] yield

$$\mathsf{f}_6(\tfrac{1}{4}) = \sum a_n^6, \qquad \mathsf{f}_6'(\tfrac{1}{4}) = -12\sum a_n^6 h_n, \qquad \mathsf{f}_6''(\tfrac{1}{4}) = \sum a_n^6(144h_n^2 + 6\delta_n - 6\pi^2).$$

The constant 312 term cancels the $-6\pi^2$ derivative term, and π^5 times the right side of (886) is (884). This is a positive central-binomial series; it does not give separate finite evaluations of Φ_-, Φ_0 or Φ_+.

Introduce the proof kernels

$$M_j^{(6)}(s) = \left[\frac{\Gamma(s)}{\Gamma(s + 1/2)}\right]^6 \tan^{6-j}(\pi s), \qquad j = 1, \dots, 5.$$

Their reflection is $M_j^{(6)}(1/2 - s) = M_{6-j}^{(6)}(s)$. For the ordinary values

$$X_{35} = G_{6,6}^{3,5}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 6}\\ 0^{\times 6}\end{matrix}\right), \qquad X_{44} = G_{6,6}^{4,4}\left(1 \,\middle|\, \begin{matrix}(\frac{1}{2})^{\times 6}\\ 0^{\times 6}\end{matrix}\right),$$

the corresponding kernels satisfy

$$\begin{aligned} N_{35} &= \frac{\Gamma(s)^3\Gamma(1/2 - s)^5}{\Gamma(1 - s)^3\Gamma(1/2 + s)} = \pi^2(M_1^{(6)} + M_3^{(6)}),\\ N_{44} &= \frac{\Gamma(s)^4\Gamma(1/2 - s)^4}{\Gamma(1 - s)^2\Gamma(1/2 + s)^2} = \pi^2(M_2^{(6)} + M_4^{(6)}). \end{aligned} \tag{887}$$

Expanding by (878) and using the reflection of $M_4^{(6)}$ gives

$$X_{35} = 3\pi^2 \sum a_n^6(24h_n^2 + \delta_n), \qquad X_{44} = 24\pi^3 \sum a_n^6 h_n. \tag{888}$$

Multiplication by 13, 11 and 8π now reproduces every term in (884): $P_b = 8\pi(13X_{35} + 11X_{44})$. The kernel of $\mathcal{G}_6$ is

$$N_6(s) = \frac{\Gamma(s)^4\Gamma(1/2-s)^5}{\Gamma(1-s)^2\Gamma(1/2+s)\Gamma(s+\alpha_6)\Gamma(1-s-\alpha_6)}.$$

Sine addition gives $N_6 = \cos(\pi\alpha_6)N_{35} + \sin(\pi\alpha_6)N_{44}$. Because these two coefficients are $13/\sqrt{290}$ and $11/\sqrt{290}$, respectively, this proves $S_6 = 192\pi\sqrt{290}\,\mathcal{G}_6$. All kernels decay at least cubically on $\operatorname{Re} s = 1/4$, and their residue series converge absolutely at 1, justifying these operations.

B.3.5 The eighth-power block

The eighth-power sector of (607) has the coefficient series, with $y_n = n + 1/4$,

$$S_8 = 8\pi^4 \sum_{n\ge0} a_n^8 \big[(2466 - 834\sqrt{2})\pi^2 y_n - 1112\pi(1 - 16y_n h_n) \\ + 139\sqrt{2}(256y_n h_n^2 - 32h_n + 8y_n\delta_n)\big]. \tag{889}$$

It is derived from ϖ_8, Y_1 and Y_7 as follows. Write $\mathfrak{Y} = (17 + 12\sqrt{2})Y_1 - (17 - 12\sqrt{2})Y_7$, so that $S_8 = \pi^6(4932 - 5004\sqrt{2})\varpi_8 + 556\pi^2\mathfrak{Y}$, and note that $Y_r = \sum_{n\ge0}(n+t)[\Gamma(n+t+\frac14)/\Gamma(n+t+\frac34)]^8$ with $t = r/8$; in particular $Y_2 = \frac{\pi^4}{4}\varpi_8$, so that $S_8 = \pi^2[(19728 - 20016\sqrt{2})Y_2 + 556\mathfrak{Y}]$. Define the proof-only function

$$\mathsf{f}_8(t) = \pi^4 \sum_{m\in\mathbb{Z}} \frac{m+t}{[\Gamma(\frac34 + m + t)\Gamma(\frac34 - m - t)]^8}. \tag{890}$$

It is odd and one-periodic. The same argument gives $\mathsf{f}_8(t) = O(e^{8\pi|\operatorname{Im} t|}(1 + |\operatorname{Im} t|)^{-2})$, so the frequencies $|k| \ge 4$ vanish, including $k = \pm4$. Thus f_8 lies in the span of $\sin 2\pi t$, $\sin 4\pi t$ and $\sin 6\pi t$, and interpolation on these three sine modes gives the exact identity

$$\mathsf{f}_8(\tfrac18) = \frac{5\sqrt{2}}{8}\mathsf{f}_8(\tfrac14) - \frac{1}{4\pi}\mathsf{f}_8'(\tfrac14) + \frac{\sqrt{2}}{32\pi^2}\mathsf{f}_8''(\tfrac14). \tag{891}$$

Gamma reflection, including the sign from the numerator of the negative half, gives

$$Y_2 = \pi^4\mathsf{f}_8(\tfrac14), \qquad \mathfrak{Y} = 64\pi^4\mathsf{f}_8(\tfrac18).$$

The factors 64 are fixed by $\cos^8(\pi/8) = (17 + 12\sqrt{2})/64$ and $\sin^8(\pi/8) = (17 - 12\sqrt{2})/64$. At the center the negative half now vanishes to order eight. Put

$$u_n^{(8)} = 1 - 16y_n h_n, \qquad v_n^{(8)} = 256y_n h_n^2 - 32h_n + 8y_n\delta_n - 8\pi^2 y_n.$$

Then

$$\mathsf{f}_8(\tfrac14) = \sum a_n^8 y_n, \qquad \mathsf{f}_8'(\tfrac14) = \sum a_n^8 u_n^{(8)}, \qquad \mathsf{f}_8''(\tfrac14) = \sum a_n^8 v_n^{(8)}.$$

Substituting these sums and (891) into the expression of S_8 through Y_2 and $\mathfrak{Y}$ gives (889). No eighth-shifted gamma value or parameter derivative occurs on the right. This converts the combined ordinary periods to explicit harmonic sums; it is not a finite elementary evaluation.

To convert this coefficient polynomial, let K_j be the contour value of

$$(1/4 - s)\left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^8 \tan^{8-j}(\pi s).$$

The factor $1/4 - s = \Gamma(5/4 - s)/\Gamma(1/4 - s)$ supplies an ordinary contiguous gamma pair. Equation (878) gives the first three residue polynomials, each multiplied by a_n^8:

$$\begin{aligned} K_1 &: \ \pi^3 y_n, \\ K_2 &: \ \pi^2(16y_n h_n - 1), \\ K_3 &: \ \pi[y_n(128h_n^2 + 4\delta_n - \pi^2) - 16h_n]. \end{aligned} \tag{892}$$

Set $a = 1233 - 278\sqrt{2}$, $b = 556$, $c = 139\sqrt{2}$. Comparing the full polynomial in (889) gives

$$S_8 = 16\pi^3(aK_1 + bK_2 + cK_3). \tag{893}$$

The angle parameters $\alpha_{8,\pm}$ satisfy the exact identity

$$\begin{aligned} a\sin^2(\pi s) + b\sin(\pi s)\cos(\pi s) + c\cos^2(\pi s) \\ = \rho_8 \sin\pi(s+\alpha_{8,+})\sin\pi(s+\alpha_{8,-}). \end{aligned} \tag{894}$$

Indeed $\rho_8^2 = (a-c)^2 + b^2$ and $4ac - b^2 = 685548\sqrt{2} - 618272 > 0$, fixing the real e_8 and the conjugate angles (870). Gamma reflection converts the full kernel to

$$\frac{\Gamma(s)^3\Gamma(1/2-s)^7\Gamma(5/4-s)}{\Gamma(1-s)^5\Gamma(1/2+s)\Gamma(1/4-s)\prod_{\sigma=\pm}\Gamma(s+\alpha_{8,\sigma})\Gamma(1-s-\alpha_{8,\sigma})}. \tag{895}$$

This is exactly the kernel of $\mathcal{G}_8$, proving $S_8 = 16\pi^3\rho_8\mathcal{G}_8$. The common contour has cubic decay, so the evaluation at 1 needs no regularization.

B.4 The wheel: one positive sequence and one bivariate value

B.4.1 Zero-index bookkeeping and direct convergence

The legacy positive wheel and its zero-included version are

$$\begin{aligned} W_c &= \sum_{n=1}^{\infty} a_n^4[8 + 5\pi(4n+1)a_n^2]\delta_n, \\ W_0 &= W_c + \frac{(8+5\pi)\pi^2}{3}. \end{aligned} \tag{896}$$

The zero-included convention is used in (604) and (2). The $n = 0$ contribution to $-393\pi^3 W_0$ is $-1048\pi^5 - 655\pi^6$ and is already present there. The recurrence

$$a_{n+1} = a_n\frac{2n+1}{2n+2}, \qquad \delta_{n+1} = \delta_n - \frac{4}{(2n+1)^2} + \frac{1}{(n+1)^2} \tag{897}$$

requires only one binomial sequence and one reciprocal-square accumulator.

For $x = n + 1/4$, the integral $\psi_1(z) = \int_0^\infty ue^{-zu}(1-e^{-u})^{-1}\,du$ [32] and (866) give $\delta_n = \int_0^\infty ue^{-(n+1/2)u}(1+e^{-u/2})^{-1}\,du$, that is,

$$\delta_n = \frac{1}{2}\int_0^\infty ue^{-xu}\operatorname{sech}(u/4)\,du, \qquad 0 < \delta_n < \frac{1}{2x^2}; \tag{898}$$

the bound follows from $0 < \operatorname{sech}(u/4) < 1$ for $u > 0$. Moreover $a_n^2 \le 1/(3n+1)$ for every $n \ge 0$: equality holds at $n = 0$, and since $a_{n+1}^2 = a_n^2(2n+1)^2/(2n+2)^2$ and $(2n+1)^2(3n+4) = (3n+1)(2n+2)^2 - n$, the bound passes from n to $n+1$. Hence $a_n^2 \le 1/(3x)$ and $(4n+1)a_n^2 \le 4/3$. Together these give, for the literal sum over $1 \le n < N$,

$$0 < W_c - W_{c,N} < \frac{8 + 20\pi/3}{54(N-3/4)^3}. \tag{899}$$

The direct-tail estimate applies to the literal finite sum; the stronger numerical construction of Section B.8.2 controls the corrected tail.

B.4.2 The global fourth/sixth-power identity

Put

$$\tau_n = 12\zeta(3) - 16H_{2n}^{(3)} + 4H_n^{(3)}, \quad D_4 = \sum_{n\geq 0} a_n^4\delta_n, \quad D_6 = \sum_{n\geq 0}(4n+1)a_n^6\delta_n. \tag{900}$$

The key identity of the calculation is

$$\sum_{n\geq 0} a_n^4(8h_n\delta_n + \tau_n) = \pi^2 D_6, \qquad W_0 = 8D_4 + 5\pi D_6. \tag{901}$$

It is a global summation identity, not a termwise identification of fourth and sixth binomial powers. The package derives it from a completed hypergeometric symmetry [14, 15] and the two periodization identities (908) below. The specialized symmetry calculation is given here, and the two periodization identities are proved in Section B.4.3.

Write

$$\begin{aligned} q_n &= (4n+1)a_n^6, \\ A_n &= 4H_{2n} - 3H_n - 2\log 2 + \frac{1}{4n+1}, \\ R_n &= 4H_{2n}^{(2)} - H_n^{(2)} + \frac{2}{(4n+1)^2}. \end{aligned} \tag{902}$$

For parameters satisfying $e+f+g-a-b-c-d = 1$, define the completed function $L(a,b,c,d;e;f,g)$ by the following ${}_7F_6$ representation. Put $\alpha = d+g-e$ and take $\mathrm{Re}(f-d) > 0$:

$$\begin{aligned} L = {} & \frac{\Gamma(1+\alpha)}{\pi\Gamma(g)\Gamma(1+g-e)\Gamma(f-d)\Gamma(1+a+d-e)\Gamma(1+b+d-e)\Gamma(1+c+d-e)} \\ & \times {}_7F_6\left(\begin{matrix}\alpha, 1+\alpha/2, g-a, g-b, g-c, d, 1+d-e \\ \alpha/2, 1+a+d-e, 1+b+d-e, 1+c+d-e, 1+g-e, g\end{matrix}; 1\right). \end{aligned} \tag{903}$$

The input symmetry is invariance under

$$\mathcal{C}(a,b,c,d;e;f,g) = (g-c, a, b, g-d; 1+a+b-f; g, 1+a+b-e). \tag{904}$$

Let $\omega = e^{\mathrm{i}\pi/4}$ and set

$$\begin{aligned} \mathbf{p}_0 &= (1/2, 1/2, 1/2, 1/2; 1; 1, 1), \\ \mathbf{v} &= (\omega^2, \omega, 1, 1-\omega+\omega^2; 1+\omega^2; \omega^2-\omega^3, 1+\omega^3). \end{aligned} \tag{905}$$

Direct substitution gives $\mathcal{C}(\mathbf{p}_0+x\mathbf{v}) = \mathbf{p}_0+\omega x\mathbf{v}$. Hence $L(\mathbf{p}_0+x\mathbf{v}) = L(\mathbf{p}_0+\omega x\mathbf{v})$, and the coefficient of x^2 vanishes. The parameter excess is one at $x=0$ and stays positive in a sufficiently small closed neighborhood; the twice-differentiated coefficients have a summable bound $n^{-2+\varepsilon}(1+\log^2 n)$ for some $\varepsilon < 1$.

Include the gamma prefactor in each summand $l_n(x)$ of (903). At the base point, $l_n(0) = q_n/(2\pi)$. With $\varrho = 1-\omega+\omega^3 = 1-\sqrt{2}$, the sums of slopes and of squared slopes needed for differentiation are

Gamma-factor group	Sum of slopes	Sum of squared slopes
Six half-parameter factors depending on n	4ϱ	$2\varrho^2$
Five unit-parameter denominator factors	2ϱ	0
Half-parameter normalization factors	2ϱ	$2\varrho^2$

The very-well-poised linear factor has slope ϱ. The polygamma recurrence then gives

$$(\log l_n)'(0) = 2\varrho A_n, \qquad (\log l_n)''(0) = -2\varrho^2 R_n.$$

Therefore

$$l_n''(0) = \frac{q_n}{2\pi} 4\varrho^2 (A_n^2 - R_n/2), \qquad \sum_{n\geq 0} q_n (A_n^2 - R_n/2) = 0. \tag{906}$$

Differentiating only the hypergeometric factor and omitting its gamma normalization would not give this identity.

To apply (906), define

$$\begin{aligned} L_n &= 8H_{2n} - 6H_n - 4\log 2 + \frac{2}{4n+1} = 2A_n, \\ M_n &= \frac{5\pi^2}{3} - 16H_{2n}^{(2)} + 6H_n^{(2)} - \frac{4}{(4n+1)^2}, \\ \Sigma_7 &= \sum_{n\geq 0} q_n (L_n^2 + M_n), \qquad V_w = \sum_{n\geq 0} q_n = \varpi_6. \end{aligned} \tag{907}$$

The pointwise identity

$$L_n^2 + M_n - (\pi^2 + 2\delta_n) = 4(A_n^2 - R_n/2)$$

implies $\Sigma_7 = \pi^2 V_w + 2D_6$. The two periodization identities specific to the calculation are, with $B_s = 2\pi^4 V_w$ and $\mathcal{T} = \sum a_n^4 h_n (8h_n^2 + \delta_n)$,

$$\begin{aligned} \Sigma_7 &= \frac{3B_s}{2\pi^2} - \frac{32}{\pi^2}\mathcal{T}, \\ \sum_{n\geq 0} a_n^4 (128h_n^3 + 24h_n\delta_n + \tau_n) &= B_s/2. \end{aligned} \tag{908}$$

They are proved in Section B.4.3 from a periodization of the gamma ratio $\Gamma(z+\frac{1}{2})/\Gamma(z+1)$ and three elliptic-moment evaluations that the package inherits; they do not follow from the symmetry alone. Subtracting $16\mathcal{T}$ from the second line gives

$$\begin{aligned} \sum a_n^4 (8h_n\delta_n + \tau_n) &= \frac{B_s}{2} - 16\mathcal{T} = \frac{\pi^2\Sigma_7}{2} - \frac{B_s}{4} \\ &= \frac{\pi^2}{2}(\pi^2 V_w + 2D_6) - \frac{2\pi^4 V_w}{4} = \pi^2 D_6. \end{aligned}$$

This proves (901) from the symmetry and (908). Every series used in this elimination is absolutely convergent.

B.4.3 Proof of the two periodization identities

The quartic periodization. Put $\mathsf{g}(z) = \Gamma(z+\frac{1}{2})/\Gamma(z+1)$, so that $\mathsf{g}(n)^4 = \pi^2 a_n^4$, and $\mathsf{b}(t) = \sum_{n\geq 0} \mathsf{g}(n+t)^4$. Half-integer recurrence and special values of the polygamma functions [32] give

$$(\log \mathsf{g})'(n) = -2h_n, \qquad (\log \mathsf{g})''(n) = \delta_n, \qquad (\log \mathsf{g})'''(n) = -\tau_n. \tag{909}$$

Gamma reflection [26] gives $\mathsf{g}(-\frac{1}{2} - z) = -\cot(\pi z)\,\mathsf{g}(z)$. Consequently

$$\mathfrak{P}(t) = \cos^4(\pi t)\,\mathsf{b}(t) + \sin^4(\pi t)\,\mathsf{b}(\tfrac{1}{2} - t)$$

is the cosine-multiplied bilateral sum $\cos^4(\pi t)\sum_{n\in\mathbb{Z}} \mathsf{g}(n+t)^4$. It is entire and periodic of period one: the cosine factor removes all fourth-order half-integer poles. In a vertical strip, gamma-ratio asymptotics and reflection [26, 27] give $|\mathfrak{P}(\theta + \mathrm{i}y)| = O(e^{4\pi|y|}/|y|)$; indeed the one-sided sum

of four gamma ratios is bounded by a constant times $\sum_{n\ge0}|n+\theta+\mathrm{i}y|^{-2}=O(1/|y|)$. Shifting the contour of a Fourier coefficient vertically therefore excludes the frequencies ±2 as well as all larger ones. Reflection under $t\mapsto\frac12-t$, which leaves $\mathfrak{P}$ invariant, removes the cosine mode, so $\mathfrak{P}$ is a constant plus a multiple of $\sin(2\pi t)$. Since $\sin^4(\pi t)\mathsf{b}(\frac12-t)$ starts at order four and $\cos^4(\pi t)=1-2\pi^2t^2+O(t^4)$,

$$\mathfrak{P}'(0)=\mathsf{b}'(0),\qquad \mathfrak{P}'''(0)=\mathsf{b}'''(0)-12\pi^2\mathsf{b}'(0).$$

The finite Fourier form gives $\mathfrak{P}'''(0)=-4\pi^2\mathfrak{P}'(0)$. Hence

$$\mathsf{b}'''(0)=8\pi^2\mathsf{b}'(0). \tag{910}$$

Three inherited elliptic-moment evaluations. With $\mathsf{k}(t)={}_2F_1(\frac12,\frac12;1;t)=2K(t)/\pi$ in the parameter convention, put

$$\mathfrak{N}(a,b)=\int_0^1 t^{a-1}(1-t)^{b-1}\mathsf{k}(t)\mathsf{k}(1-t)\,dt,\qquad \mathfrak{M}(s)=\mathfrak{N}(s,1),\qquad \mathfrak{I}(s)=\mathfrak{N}(s,1-s),$$

and $\mathfrak{M}_2(a,b)=\int_0^1 t^{a-1}(1-t)^{b-1}\mathsf{k}(t)^2\,dt$. These integrals and their fixed-order logarithmic derivatives are holomorphic in their common positive-beta domain, and the endpoints have at most logarithmic growth. The package takes three evaluations of these moments from the earlier stages of the calculation and does not rederive them. First, the very-well-poised Mellin series, for $0<\operatorname{Re}s<1$,

$$\mathfrak{I}(s)=\frac{\Gamma(1+s)\Gamma(s)^2\Gamma(1-s)}{\Gamma(s+\frac12)^2}\,{}_7F_6\!\left(\begin{matrix}s,s,s,s,1+\frac{s}{2},\frac12,\frac12\\ \frac{s}{2},1,1,1,s+\frac12,s+\frac12\end{matrix};1\right). \tag{911}$$

Second, the mixed Mellin series

$$\mathfrak{M}(s)=\frac{\Gamma(s)^2}{\Gamma(s+\frac12)^2}\,{}_4F_3\!\left(\begin{matrix}\frac12,\frac12,s,s\\ 1,s+\frac12,s+\frac12\end{matrix};1\right). \tag{912}$$

Third, the upper-half-plane contour relation for the square moments, with $c=2-a-b$,

$$0=\mathfrak{M}_2(a,b)-e^{\mathrm{i}\pi a}\mathfrak{M}_2(a,c)-e^{-\mathrm{i}\pi b}\big[\mathfrak{M}_2(c,b)-\mathfrak{M}_2(b,c)+2\mathrm{i}\,\mathfrak{N}(c,b)\big], \tag{913}$$

whose arcs vanish when $\operatorname{Re}a,\operatorname{Re}b,\operatorname{Re}c>0$.

Consequences of the inputs. At $s=\frac12$ the n-th term of (911), gamma prefactor included, is $\frac{\pi^2}{2}q_n$, and its first and second logarithmic derivatives in s are L_n and M_n. Hence

$$\mathfrak{I}(\tfrac12)=\frac{\pi^2}{2}V_w=\frac{B_s}{4\pi^2},\qquad \Sigma_7=\frac{2}{\pi^2}\,\mathfrak{I}''(\tfrac12). \tag{914}$$

Taking $(a,b,c)=(s,1,1-s)$ in (913) gives $\mathfrak{M}_2(s,1-s)=2\mathfrak{M}(1-s)/\sin\pi s$. Taking $(a,b,c)=(1,s,1-s)$, and its imaginary part, then gives

$$\mathfrak{I}(s)=\frac{\mathfrak{M}(s)-\mathfrak{M}(1-s)}{\cos\pi s}. \tag{915}$$

The singularity at $s=\frac12$ is removable. All derivatives below include the derivative of this denominator, with its sign. By (912), $\mathfrak{M}(\frac12+t)=\pi^{-1}\sum_{n\ge0}\mathsf{g}(n)^2\mathsf{g}(n+t)^2$, and the differentiated series converge locally normally.

The second identity. At its removable point (915) gives $\mathfrak{M}'(\frac12) = -\frac{\pi}{2}\mathfrak{I}(\frac12)$, and termwise differentiation gives $\mathfrak{M}'(\frac12) = \mathsf{b}'(0)/(2\pi)$. With (914),

$$\mathsf{b}'(0) = -\frac{B_s}{4}, \qquad \text{equivalently} \qquad \sum_{n\ge0} a_n^4 h_n = \frac{B_s}{32\pi^2}. \tag{916}$$

By (909),

$$\frac{d^3}{dt^3}\mathsf{g}(n+t)^4\Big|_{t=0} = -4\pi^2 a_n^4(128h_n^3 + 24h_n\delta_n + \tau_n).$$

Equations (910) and (916) now give $\sum_{n\ge0} a_n^4(128h_n^3 + 24h_n\delta_n + \tau_n) = -2\,\mathsf{b}'(0) = B_s/2$, the second line of (908).

The first identity. Write $\langle f\rangle = \sum_{n\ge0}\mathsf{g}(n)^4 f_n$ and $\mathsf{d}_n = (\log\mathsf{g})'(n) = -2h_n$. Expanding (915) about $s=\frac12$ gives $\mathfrak{I}''(\frac12) = -\frac{2}{3\pi}[\mathfrak{M}'''(\frac12) + \pi^2\mathfrak{M}'(\frac12)]$. The third derivative of the series for $\mathfrak{M}(\frac12+t)$ contains $\langle(\log\mathsf{g})'''\rangle$, which (910), written as $\langle 64\mathsf{d}^3 + 48\mathsf{d}\delta + 4(\log\mathsf{g})'''\rangle = 8\pi^2\langle 4\mathsf{d}\rangle$, eliminates. The result is

$$\mathfrak{I}''(\tfrac12) = \frac{16}{\pi^2}\langle\mathsf{d}^3\rangle + \frac{8}{\pi^2}\langle\mathsf{d}\delta\rangle - 12\langle\mathsf{d}\rangle.$$

With $\mathsf{g}(n)^4 = \pi^2 a_n^4$, $\mathsf{d}_n = -2h_n$, (916) and (914), this is the first line of (908):

$$\Sigma_7 = \frac{3B_s}{2\pi^2} - \frac{32}{\pi^2}\sum_{n\ge0} a_n^4 h_n(8h_n^2 + \delta_n).$$

What remains an input. The two periodization identities, and with them (901) and the wheel normalization of Section B.4.4, rest on the three inherited evaluations (911)–(913), besides gamma and polygamma identities and the published symmetry used above. The quartic periodization (910) is proved here. Both periodization identities were also confirmed numerically (Section 22.9); these are floating-point checks, not part of the proof.

B.4.4 From the common Barnes contour to one H value

Use the kernels M_1, M_2 of (879), and put $d(s) = \psi_1(1/2-s) - \psi_1(1-s)$. At $s=-n$, $d(-n) = \delta_n$ and $d'(-n) = \tau_n$. Equation (878) gives

$$\mathrm{Res}_{s=-n}(M_1 d) = \pi a_n^4\delta_n, \qquad \mathrm{Res}_{s=-n}(M_2 d) = a_n^4(8h_n\delta_n + \tau_n).$$

Thus (901) yields

$$W_0 = \frac{1}{\pi}\mathfrak{B}_{1/4}[(8M_1+5M_2)d] = \frac{\sqrt{89}}{\pi}\mathfrak{B}_{1/4}[N_{\alpha_W} d]. \tag{917}$$

The second equality uses (883) and $\cos(\pi\alpha_W) = 8/\sqrt{89}$, $\sin(\pi\alpha_W) = 5/\sqrt{89}$.

To remove the trigamma insertion, start with

$$d(s) = -4\int_0^1 \frac{x^{-2s}\log x}{1+x}\,dx, \qquad \mathrm{Re}\,s < 1/2.$$

The Mellin representation $(1+x)^{-1} = \mathfrak{B}_{1/4}[\Gamma(z)\Gamma(1-z)x^{-z}]$ and $\int_0^1 x^{a-1}(-\log x)\,dx = a^{-2}$ give

$$d(s) = 4\mathfrak{B}_{1/4}\left[\frac{\Gamma(z)\Gamma(1-z)}{(1-2s-z)^2}\right].$$

Substitute $z=2t$ and use gamma duplication,

$$\Gamma(2t)\Gamma(1-2t) = \frac{\Gamma(t)\Gamma(t+1/2)\Gamma(1/2-t)\Gamma(1-t)}{2\pi} = \frac{P(t)}{2\pi}.$$

The Jacobian and the squared denominator then give

$$\mathfrak{B}_{1/8}\Big[\frac{P(t)}{(1/2-s-t)^2}\Big] = \pi d(s). \tag{918}$$

Substituting this identity proves $W_0 = \sqrt{89}\,\mathcal{H}_W/\pi^2$, with precisely the kernel of Section B.2.3. On the specified contours, $|1/2-s-t| \geq 1/8$, $P(t) = 2\pi^2/\sin(2\pi t)$ decays exponentially, and $N_{\alpha_W}(1/4+\mathrm{i}u) = O((1+|u|)^{-2})$. These give an integrable product majorant and justify the Mellin interchanges.

B.5 The seventh-power sector

B.5.1 A complete central series, not just a logarithmic fragment

The sum $M_7 = \sum_{n\geq 0}(-1)^n a_n^7 F_n$ and its coefficients are defined in (608). The $n=0$ term is included. The definition includes the complete ordinary and logarithmic seventh-sector contributions; no separate ordinary correction is added to it. For auditing the initial term,

$$\begin{aligned} F_0 = {} & 39886\log^3 2 - 10906\pi\log^2 2 - 22792\log^2 2 - 2188\pi^2\log 2 + 8156\pi\log 2 \\ & - \frac{1679}{6}\pi^3 + \frac{2330}{3}\pi^2 - 2156\zeta(3). \end{aligned} \tag{919}$$

B.5.2 Matching the complete coefficient with a Barnes kernel

For the Barnes proof, put

$$Y(s) = \tfrac{1}{4} - s, \quad R(s) = \Big[\frac{\Gamma(s)}{\Gamma(s+1/2)}\Big]^7, \quad z(s) = \tan(\pi s), \quad \Psi(s) = \psi(1-2s) + \gamma_{\mathrm{E}}. \tag{920}$$

Here γ_{E} is Euler's constant; it is used in the proof but is not a parameter of the final functions. Define $p(z) = 7z^2 + 13z + 11$ and $q(z) = q_{7,2}z^2 + q_{7,1}z + q_{7,0}$. The common kernel is

$$I(s) = Y(s)R(s)\sec(\pi s)z(s)^4[196p(z(s))\Psi(s) + q(z(s))]. \tag{921}$$

For $j = 1, 2, 3$, the kernels $M_j^{(7)}(s) = Y(s)R(s)\sec(\pi s)z(s)^{7-j}$ have the following residues at $s = -n$, with the common factor $(-1)^n a_n^7$ omitted:

$$\begin{aligned} M_1^{(7)} &: \ \pi^{5/2} y_n, \\ M_2^{(7)} &: \ \pi^{3/2}(14 y_n h_n - 1), \\ M_3^{(7)} &: \ \sqrt{\pi}\,[y_n(98h_n^2 + 7\delta_n/2 - \pi^2/2) - 14h_n]. \end{aligned} \tag{922}$$

They follow from (878) and the tangent and secant expansions. The digamma insertion satisfies

$$\Psi(-n) = H_{2n}, \quad \Psi'(-n) = 2H_{2n}^{(2)} - \pi^2/3, \quad \Psi''(-n) = -8[\zeta(3) - H_{2n}^{(3)}]. \tag{923}$$

Taking the residues of the products with this Taylor expansion reproduces the complete polynomial F_n:

$$\mathrm{Res}_{s=-n} I(s) = \sqrt{\pi}\,(-1)^n a_n^7 F_n. \tag{924}$$

At negative half-integers, $R(s)$ has zeros of order seven. The tangent and secant remove $8-j$ of these orders in $M_j^{(7)}$, leaving order $j-1 \geq 0$; Ψ is analytic there. There is consequently no additional left-pole family. On the initial line $\mathrm{Re}\, s = 1/4$ the kernels decay as $O((1+|\,\mathrm{Im}\, s|)^{-5/2} e^{-\pi|\,\mathrm{Im}\, s|})$, with a logarithmic factor for Ψ. On the closing lines $\mathrm{Re}\, s = -N - 1/4$, reflection gives an integrated

bound $O(N^{-5/2}\log(N+2))$. The horizontal sides vanish exponentially. Thus closing to the left is justified and gives

$$M_7 = \pi^{-1/2}\mathfrak{B}_{1/4}[I]. \tag{925}$$

The parameters $\eta_{7,\pm}$ and $\beta_7, \bar{\beta}_7$ of Section 1.2 factor both quadratics:

$$\begin{aligned} p(\tan\theta) &= \sqrt{185}\,\frac{\sin(\theta+\pi\beta_7)\sin(\theta+\pi\bar{\beta}_7)}{\cos^2\theta}, \\ q(\tan\theta) &= \rho_7\,\frac{\sin(\theta+\pi\eta_{7,+})\sin(\theta+\pi\eta_{7,-})}{\cos^2\theta}. \end{aligned} \tag{926}$$

For example the tangent parameters in the first line have sum 13/7, product 11/7, and normalizing square 185/49. Gamma reflection gives

$$\begin{aligned} YR\sec(\pi s)z^4 p(z) &= \frac{\sqrt{185}}{\pi}N_p(s), \\ YR\sec(\pi s)z^4 q(z) &= \frac{\rho_7}{\pi}N_7(s), \\ N_p(s) &= \frac{\Gamma(s)^3\Gamma(1/2-s)^7\Gamma(5/4-s)}{\Gamma(1-s)^4\Gamma(1/4-s)\prod_{b\in\{\beta_7,\bar{\beta}_7\}}\Gamma(s+b)\Gamma(1-s-b)}. \end{aligned} \tag{927}$$

The kernel N_7 is the same expression with $\beta_7, \bar{\beta}_7$ replaced by $\eta_{7,+}, \eta_{7,-}$, and hence is the ordinary kernel of $\mathcal{G}_7$. Also $\mathcal{A}_7(s) = (-s)N_p(s)$ by the gamma recurrence. These identities fix both the ordinary contribution and the outer kernel of the bivariate contribution.

B.5.3 The inner digamma transform and the unit-slope value

For $\operatorname{Re} s = 1/4$, set

$$J_\pm(s) = \mathfrak{B}_{1/8}\left[\frac{\Gamma(t)\Gamma(1-t)e^{\pm i\pi t}}{(1-t)(1-2s-t)}\right]. \tag{928}$$

Residues at $t=-k$ give $1/[(k+1)(k+1-2s)]$; their partial-fraction sum is

$$J_-(s) = \frac{\psi(1-2s)+\gamma_{\mathrm{E}}}{-2s} = \frac{\Psi(s)}{-2s}. \tag{929}$$

The phase and the alternating gamma residue cancel. Quadratic rational decay in the undamped direction and exponential decay in the other direction justify closing the contour. Gamma reflection gives

$$\Gamma(t)\Gamma(1-t)(e^{-i\pi t}-e^{i\pi t}) = -2\pi i.$$

Thus $J_- - J_+$ is $-2\pi i$ times the contour value of $[(1-t)(1-2s-t)]^{-1}$. Both poles of this rational function are on the right, and its quadratic decay permits closing to the left. The difference is zero. Averaging the two phases and substituting $t = 2v$ now gives

$$J_-(s) = \frac{\pi}{2}\mathfrak{B}_{1/16}\left[\frac{\cot(2\pi v)}{(1/2-v)(1/2-s-v)}\right]. \tag{930}$$

The factor is $\pi/2$: the measure contributes 2, and the two denominators contribute 4. Therefore

$$\mathfrak{B}_{1/16}\left[\frac{\cot(2\pi v)}{(1/2-v)(1/2-s-v)}\right] = -\frac{\Psi(s)}{\pi s}. \tag{931}$$

Gamma reflection at quarter arguments gives the unit-slope quotient

$$\cot(2\pi v) = \frac{\Gamma(v)\Gamma(v+1/2)\Gamma(1-v)\Gamma(1/2-v)}{\Gamma(v+1/4)\Gamma(v+3/4)\Gamma(1/4-v)\Gamma(3/4-v)}. \tag{932}$$

There is no extra π in this identity. Combining it with the two rational factors yields exactly the lists in (875). Inserting $\mathcal{A}_7 = (-s)N_p$ into (931) gives $\mathcal{H}_7 = \mathfrak{B}_{1/4}[N_p\Psi]/\pi$. Consequently

$$M_7 = \frac{\rho_7\mathcal{G}_7}{\pi^{3/2}} + \frac{196\sqrt{185}}{\sqrt{\pi}}\mathcal{H}_7,$$
$$24\pi^3 M_7 = 24\pi^{3/2}[\rho_7\mathcal{G}_7 + 196\pi\sqrt{185}\,\mathcal{H}_7]. \tag{933}$$

This proves the complete normalization of the seventh-power sector in (2).

For $s = 1/4 + \mathrm{i}u$ and $v = 1/16 + \mathrm{i}w$, $|\cot(2\pi v)| \le 1 + \sqrt{2}$. Cauchy–Schwarz bounds the integral of the absolute rational factors by

$$\int_{\mathbb{R}} \frac{dw}{\sqrt{(7/16)^2 + w^2}\sqrt{(3/16)^2 + (u+w)^2}} \le \frac{16\pi}{\sqrt{21}}.$$

The outer factor is $O((1+|u|)^{-3/2}e^{-\pi|u|})$. This gives a product majorant for the bivariate integral and justifies the interchanges. The logarithms of both final arguments $(1,1)$ are zero; conjugation of the parameter pairs and of the imaginary contour coordinates also shows that the specified value is real.

B.6 The primitive: from the Gauss pairing to one bivariate value

B.6.1 The Gauss-product input

The primitive input is (609), with $v(x) = {}_2F_1(\frac{1}{4},\frac{1}{4};1;x)$, $w(x) = {}_2F_1(\frac{1}{4},\frac{1}{4};\frac{1}{2};x)$, the products p_0, p_1, p_2 and the pairings $I_{ij} = I_{ji}$ defined there. The zero-balanced Gauss factor has at most a logarithmic endpoint singularity, so these pairings converge absolutely. The coefficient in r_6 is $96P_a$.

B.6.2 The contour relation used in the final reduction

On the open upper half-plane define

$$f(z) = v(z), \qquad g(z) = \frac{\sqrt{Q}}{\pi}w(1-z) - v(z), \qquad f_* = f(1-z), \quad g_* = g(1-z).$$

The star means reflection, not complex conjugation. Principal Gauss branches and separate principal square roots $\sqrt{z}, \sqrt{1-z}$ are used. The Gauss cut-jump and connection formulas give the exterior boundary data

Region	f	g	f_*	g_*
$x<0,\ z = x + \mathrm{i}0$	u	$r - \mathrm{i}u$	$a - \mathrm{i}b$	$a + \mathrm{i}b$
$x>1,\ z = x + \mathrm{i}0$	$a + \mathrm{i}b$	$a - \mathrm{i}b$	u	$r + \mathrm{i}u$

Here u, r, a, b are real local values. For $x < 0$, the argument $1 - z$ approaches the zero-balanced cut from below; this fixes the minus sign in $g = r - \mathrm{i}u$. For $x > 1$, the connection formula

$$v(z) = \frac{\sqrt{Q}}{2\pi}w(1-z) - \frac{2}{\sqrt{Q}}\sqrt{1-z}\,{}_2F_1\left(\begin{matrix}3/4, 3/4\\ 3/2\end{matrix}; 1-z\right)$$

shows that f and g are conjugate on the upper bank. Reflection supplies the remaining entries.

Use the holomorphic numerator

$$\mathcal{N} = g^2g_*^2 + g^2f_*^2 + f^2g_*^2 - 4fgf_*g_* - 3f^2f_*^2$$
$$+ 4\mathrm{i}(fgf_*^2 - f^2f_*g_*), \qquad \widetilde{\mathcal{N}}(z) = \frac{\mathcal{N}(z)}{\sqrt{z}\sqrt{1-z}}. \tag{934}$$

Substitution of either boundary row gives the polynomial identity

$$\mathcal{N} = 2r(a-b)[(a+b)r - 2(a-b)u] \in \mathbb{R}.$$

On $x < 0$ and $x > 1$, the reciprocal square-root factor is respectively $-\mathrm{i}/\sqrt{(-x)(1-x)}$ and $\mathrm{i}/\sqrt{x(x-1)}$. Therefore $\mathrm{Re}\,\widetilde{\mathcal{N}}(x+\mathrm{i}0) = 0$ outside $[0,1]$. At infinity the Gauss connection expansion gives $\widetilde{\mathcal{N}}(z) = O(|z|^{-2}(1+\log|z|)^4)$. Near 0 and 1 the bounds are of inverse-square-root type with at most quadratic logarithms. The large semicircle and the two endpoint indentations all vanish. Taking the real part of Cauchy's theorem leaves $\int_0^1 \mathrm{Re}\,\widetilde{\mathcal{N}}(x)\,dx = 0$.

On the interior interval, $p_0 = f^2$, $p_1/\pi = f(f+g)$ and $p_2/\pi^2 = (f+g)^2$. Expanding the real numerator in these products, and using $I_{ij} = I_{ji}$, gives

$$\begin{aligned} I_{22} - 4I_{12} + 4I_{02} - 4I_{00} &= 0, \\ P_a &= \pi^4[60(I_{12} - I_{02}) + 24I_{11} + 56I_{00}]. \end{aligned} \tag{935}$$

Thus the logarithm-squared times logarithm-squared pairing I_{22} of (609) is removed. The contour proof, not an equality of individual power-series coefficients, establishes the relation.

B.6.3 Identification of the three ordinary Meijer solutions

All three p_i solve the symmetric-square equation

$$[\theta^3 - x(\theta + \tfrac{1}{2})^3]p = 0, \qquad \theta = x\frac{d}{dx}. \tag{936}$$

Define

$$g_j(x) = G^{j,3-j}_{3,3}\left(x \,\middle|\, \begin{matrix} (\frac{1}{2})^{\times 3} \\ 0^{\times 3} \end{matrix}\right), \quad j = 1,2,3. \tag{937}$$

Their inverse-Mellin kernels are

$$M_j^{(3)}(s) = \frac{\Gamma(s)^j\Gamma(1/2-s)^{3-j}}{\Gamma(1-s)^{3-j}\Gamma(1/2+s)^j} = \left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^3 \tan^{3-j}(\pi s). \tag{938}$$

The standard Meijer differential equation is also (936). To determine the complete normalization, expand at $s = 0$:

$$\left[\frac{\Gamma(s)}{\Gamma(s+1/2)}\right]^3 = \frac{1 + 6s\log 2 + (18\log^2 2 - \pi^2/2)s^2 + O(s^3)}{\pi^{3/2}s^3}.$$

Writing $L = -\log x$, the residues give

$$\begin{aligned} g_1(x) &= \sqrt{\pi} + O(x), \\ g_2(x) &= (L + 6\log 2)/\sqrt{\pi} + O(x|\log x|), \\ g_3(x) &= [(L+6\log 2)^2 - \pi^2]/(2\pi^{3/2}) + O(x\log^2 x). \end{aligned} \tag{939}$$

On the Gauss side, $v(x) = 1 + O(x)$ and the zero-balanced connection formula is $\sqrt{Q}\,w(1-x) = L + \pi + 6\log 2 + O(x|\log x|)$. Matching all constant and logarithmic coefficients gives

$$g_1 = \sqrt{\pi}p_0, \qquad g_2 = \frac{p_1 - \pi p_0}{\sqrt{\pi}}, \qquad g_3 = \frac{p_2 - 2\pi p_1}{2\pi^{3/2}}. \tag{940}$$

The Frobenius recurrence of (936) then fixes every higher coefficient. Matching the differential equation without these boundary polynomials would not suffice.

For the real numerical evaluator, introduce the elliptic coordinate

$$m(x) = \frac{x}{2(1+\sqrt{1-x})}, \qquad K_x = K(m(x)), \qquad K_x^c = K(1-m(x)), \tag{941}$$

with $K(m)$ in the parameter convention of Section 1.4. The argument is the elliptic parameter, not the modulus. The substitution $x = 4m(1-m)$ transforms the elliptic equation $m(1-m)Y'' + (1-2m)Y' - Y/4 = 0$ into $x(1-x)y'' + (1-3x/2)y' - y/16 = 0$. Its regular normalization gives $v(x) = 2K_x/\pi$. At zero the complete logarithmic normalization is $2(K_x + K_x^c) = -\log x + \pi + 6\log 2 + O(x|\log x|)$, which matches the zero-balanced connection formula above; the regular and logarithmic Frobenius data then give $\sqrt{Q}\, w(1-x) = 2(K_x + K_x^c)$. Therefore

$$g_1 = \frac{4}{\pi^{3/2}} K_x^2, \qquad g_2 = \frac{4}{\pi^{3/2}} K_x K_x^c, \qquad g_3 = \frac{2}{\pi^{3/2}} [(K_x^c)^2 - K_x^2]. \tag{942}$$

These are finite kernel identities used to check the primitive numerically. The package also contains a Wolfram implementation of them but claims no run of it; its numerical values come from the equivalent Python implementations.

B.6.4 Beta integration and the four intermediate pairings

Write

$$\langle f, g\rangle = \int_0^1 \frac{f(x)g(1-x)}{\sqrt{x(1-x)}}\, dx, \qquad H_{jk} = \langle g_j, g_k\rangle.$$

Inserting the two kernels (938) and applying Euler's beta integral gives the bivariate kernel

$$\frac{\Gamma(s)^j \Gamma(1/2-s)^{4-j} \Gamma(t)^k \Gamma(1/2-t)^{4-k}}{\Gamma(1-s)^{3-j} \Gamma(1/2+s)^j \Gamma(1-t)^{3-k} \Gamma(1/2+t)^k \Gamma(1-s-t)}. \tag{943}$$

The added factor is $\Gamma(1/2-s)\Gamma(1/2-t)/\Gamma(1-s-t)$. On $\operatorname{Re} s = \operatorname{Re} t = 1/4$, each $M_j^{(3)}$ is $O((1+|\operatorname{Im} s|)^{-3/2})$. Before integrating in x, the absolute three-variable integral is bounded by $B(1/4, 1/4)$ times the product of the two integrable local majorants. This justifies the interchanges and the absolute convergence of (943).

Inverting (940) gives

$$p_0 = g_1/\sqrt{\pi}, \quad p_1 = \sqrt{\pi}(g_1 + g_2), \quad p_2 = 2\pi^{3/2}(g_1 + g_2 + g_3).$$

Substituting these into (935) gives

$$\begin{aligned} P_a/\pi^3 = {} & 120(H_{11} + 2H_{12} + H_{13} + H_{22} + H_{23}) \\ & - 120(H_{11} + H_{12} + H_{13}) + 24(H_{11} + 2H_{12} + H_{22}) + 56H_{11}. \end{aligned}$$

Thus

$$P_a = 8\pi^3(10H_{11} + 21H_{12} + 18H_{22} + 15H_{23}), \qquad H_{33} + 2H_{13} - H_{22} = 0. \tag{944}$$

The second relation is the first line of (935) in the g_j basis. It is the relation used in the final compression.

B.6.5 From the four pairings to the final single value

The coefficient matrix and the zero-relation matrix are

$$\mathsf{M} = \begin{pmatrix} 10 & 21/2 & 0 \\ 21/2 & 18 & 15/2 \\ 0 & 15/2 & 0 \end{pmatrix}, \qquad \mathsf{N} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \tag{945}$$

Adding $\lambda\mathsf{N}$ leaves the definite pairing unchanged. The cubic $4\lambda^3 - 112\lambda^2 + 909\lambda - 2250$ is $4\det(\mathsf{M}+\lambda\mathsf{N})$. For its root λ_* and the parameters of $\mathcal{H}_*$, $v_\sigma = (1, B_\sigma, C_\sigma)^T$ satisfies

$$\mathsf{M} + \lambda_*\mathsf{N} = 5(v_+ v_-^T + v_- v_+^T). \tag{946}$$

With $f_\sigma = g_1 + B_\sigma g_2 + C_\sigma g_3$, symmetry of the pairing gives

$$10H_{11} + 21H_{12} + 18H_{22} + 15H_{23} = 10\langle f_+, f_-\rangle.$$

The kernel of f_σ contains $P_\sigma(z) = z^2 + B_\sigma z + C_\sigma$, $z = \tan(\pi s)$. Put $D_\sigma = \sqrt{(1-C_\sigma)^2 + B_\sigma^2}$. The angle parameters β_σ give

$$P_\sigma(\tan\theta) = D_\sigma \frac{\sin(\theta + \pi\beta_\sigma)\sin(\theta + \pi\bar{\beta}_\sigma)}{\cos^2\theta}.$$

Gamma reflection consequently converts f_σ/D_σ to one local Meijer kernel. The normalization is exact:

$$P_+(\mathrm{i})P_-(\mathrm{i}) = -\frac{4}{5} - \frac{3}{5}\mathrm{i}, \qquad |P_\sigma(\mathrm{i})| = D_\sigma, \qquad D_+D_- = 1.$$

After beta integration the local kernels are precisely Ψ_+, Ψ_- of Section 1.2, with the shared factor $1/\Gamma(1-s-t)$. It follows that $\langle f_+, f_-\rangle = \mathcal{H}_*$, and hence

$$P_a = 80\pi^3\mathcal{H}_*, \qquad 96P_a = 7680\pi^3\mathcal{H}_*. \tag{947}$$

No term is extracted from an endpoint or transferred into the known polynomial in this conversion.

B.7 Conversion to the real log-sine and Clausen values

The starting block is S_{pol} of Section 22.9, with the descending-index convention. For $l(\theta) = \log(2\sin(\theta/2))$ and $0 < \theta < 2\pi$, the principal logarithm is

$$\log(1 - e^{\mathrm{i}\theta}) = l(\theta) + \frac{\mathrm{i}}{2}(\theta - \pi). \tag{948}$$

The derivative

$$\frac{d}{dz}\operatorname{Li}_{2,1}(z,1) = \frac{\log^2(1-z)}{2z}$$

and $z = e^{\mathrm{i}\theta}$ imply

$$\frac{d}{d\theta}\operatorname{Im}\operatorname{Li}_{2,1}(e^{\mathrm{i}\theta},1) = \frac{1}{2}l(\theta)^2 - \frac{(\theta-\pi)^2}{8}. \tag{949}$$

The initial imaginary part at $\theta = 0$ is zero. The logarithmic singularity is integrable. Integrating to $\pi/2$ gives

$$D_3 = -\frac{1}{2}\operatorname{Ls}_3(\pi/2) - \int_0^{\pi/2}\frac{(\theta-\pi)^2}{8}\,d\theta = -\frac{1}{2}L_3 - \frac{7\pi^3}{192}.$$

This is a differential identity. For weight five, the real coefficient convention $\mathrm{Cl}_{2,1,2}(\sigma) = \sum_{n>m>k\geq 1}\cos(n\sigma)/(n^2mk^2)$ of Section 1.2 gives directly $D_5 = C_{212}$: at odd total weight the multiple Clausen convention used here is the cosine component of the descending multiple polylogarithm. This is a real-function convention, not a new elementary evaluation of D_5. Absolute convergence follows from

$$\sum_{n>m>k\geq 1}\frac{1}{n^2mk^2} \leq \zeta(2)\sum_{n\geq 1}\frac{H_{n-1}}{n^2} < \infty,$$

so the cosine component may be taken term by term. The definitions of L_3 and C_{212} fix established special functions [16]; they are not additional residual operations in the fixed-value formula (2).

Substitute these two identities into the entire block, not just into its individual special values:

$$\begin{aligned} S_{\mathrm{pol}} &= 4352\big[(32\mathsf{G} + 9\pi^2)(-L_3/2 - 7\pi^3/192) - 64C_{212}\big] \\ &= -2176\big[(32\mathsf{G} + 9\pi^2)L_3 + 128C_{212}\big] - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2). \end{aligned} \tag{950}$$

Adding the last polynomial to E gives the polynomial $E_{\mathrm{real}} = E - \frac{476}{3}\pi^3(32\mathsf{G} + 9\pi^2)$ of (2). The fully expanded E has eighteen monomials in $A, Q, \mathsf{G}, \pi, \zeta(3), \zeta(5)$, and only two of them change. Their new coefficients are

$$-\frac{544}{3} - \frac{476 \cdot 32}{3} = -\frac{15776}{3} \quad (\mathsf{G}\pi^3), \qquad \frac{37631}{128} - \frac{476 \cdot 9}{3} = -\frac{145153}{128} \quad (\pi^5).$$

There is no new physical contribution, and the Q-dependent block is unchanged. Together with the ordinary, wheel, seventh and primitive conversions, this completes the derivation of (2) from the stated reduced input.

The formula (2) has no explicit Re or Im operation. Some Meijer G and bivariate H parameter lists still contain complex-conjugate pairs. Removing the projection notation does not remove those parameters, and the multiple Clausen name does not constitute a reduction of arithmetic independence.

B.8 Numerical evaluation and error control

B.8.1 Evaluation of the final expression

A mathematically short special-function formula need not be the fastest way to evaluate the constant. The preferred numerical routes are a rational central-wheel tail correction, Euler-transformed seventh coefficients, a geometric primitive recurrence, and algorithms for harmonic polylogarithms and log-sine values for the real polylogarithmic sector. These compute the same specified values using the proved conversion identities, not by subtracting unknown parts from the target C_6.

B.8.2 Wheel certificate: a finite correction, not an infinite asymptotic sum

For $t \geq N$, put $x = t + 1/4$, $y = x^{-2}$ and

$$\Delta_r q(t) = q(t) - \left(\frac{t+1/2}{t+1}\right)^r q(t+1), \qquad d_j = \frac{E_{2j}(2j+1)}{2\,16^j}. \tag{951}$$

The rational bases are

$$\begin{aligned} \phi^{(4)}_{-1}(t) &= \frac{(t-1/4)^3}{(t-1/2)^4}, & \phi^{(4)}_j(t) &= \frac{t-1/4}{t^j(t-1/2)^{j+4}}, \\ \phi^{(6)}_j(t) &= \frac{(t-1/4)^6}{t^j(t-1/2)^{j+6}}. \end{aligned} \tag{952}$$

To make the matching and the bound explicit, define the Euler numbers by $\operatorname{sech} v = \sum_{j\geq 0} E_{2j} v^{2j}/(2j)!$ near zero. Write

$$q^{(4)}_M = \sum_{j=-1}^{M} \alpha_j \phi^{(4)}_j, \qquad q^{(6)}_L = \sum_{j=0}^{L} \beta_j \phi^{(6)}_j, \qquad L = M + 1.$$

Choose their rational coefficients by matching

$$\Delta_4 q^{(4)}_M = \sum_{k=0}^{M+1} d_k y^{k+1} + O(y^{M+3}), \qquad \Delta_6 q^{(6)}_L = 4\sqrt{y} \sum_{k=0}^{L} d_k y^k + O(\sqrt{y}\, y^{L+1}).$$

The triangular leading multipliers are 3 for the fourth-power index -1, $2j+5$ for its indices $j \geq 0$, and $2j+3$ in the sixth-power construction. The initial coefficients are $\alpha_{-1} = 1/6$, $\alpha_0 = -77/480$ and $\beta_0 = 2/3$, $\beta_1 = -91/120$. The rational differences have the denominator polynomials

$$\begin{aligned} \mathcal{D}_4(y) &= (1-y/16)^M (1-9y/16)^{M+4}, & \Delta_4 q^{(4)}_M &= \mathcal{P}_4(y)/\mathcal{D}_4(y), \\ \mathcal{D}_6(y) &= (1-y/16)^L (1-9y/16)^{L+6}, & \Delta_6 q^{(6)}_L &= \sqrt{y}\,\mathcal{P}_6(y)/\mathcal{D}_6(y). \end{aligned}$$

Let $\mathcal{U}_4(y)=\sum_{k=0}^{M+1} d_k y^{k+1}$ and $\mathcal{U}_6(y)=4\sum_{k=0}^{L} d_k y^k$. Exact polynomial division gives

$$\mathcal{E}_4(y)=\frac{\mathcal{P}_4-\mathcal{D}_4\mathcal{U}_4}{y^{M+3}},\qquad \mathcal{E}_6(y)=\frac{\mathcal{P}_6-\mathcal{D}_6\mathcal{U}_6}{y^{L+1}}.$$

For $Y=(N+1/4)^{-2}$, the constants used below are

$$K_4=|d_{M+2}|+\frac{\sum_j |[y^j]\mathcal{E}_4(y)|Y^j}{\mathcal{D}_4(Y)},$$
$$K_6=4|d_{L+1}|+\frac{\sum_j |[y^j]\mathcal{E}_6(y)|Y^j}{\mathcal{D}_6(Y)}.$$

Both denominators are positive for $0\le y\le Y$ when $N\ge 2$. They imply, for $t\ge N$ and $x=t+1/4$,

$$|\delta(t)-\Delta_4 q_M^{(4)}(t)|\le K_4 x^{-2M-6},\qquad |4x\delta(t)-\Delta_6 q_L^{(6)}(t)|\le K_6 x^{-2L-3}.$$

Here $\delta(t)=\psi_1(t+1/2)-\psi_1(t+1)$ extends the integer remainder, and (898) holds with $x=t+1/4$. The global Taylor estimate used is

$$\left|\operatorname{sech} v-\sum_{k=0}^{K}\frac{E_{2k}v^{2k}}{(2k)!}\right|\le\frac{|E_{2K+2}|v^{2K+2}}{(2K+2)!}\quad(v\in\mathbb{R}).\tag{953}$$

It follows from the positive product $\cosh v=\prod_{k\ge1}(1+4v^2/((2k-1)^2\pi^2))$ [20]. Since $(1+\mu\varsigma)^{-1}$ is the Laplace transform at ς of an exponential variable with mean μ, $\operatorname{sech}\sqrt{\varsigma}=\mathbb{E}\,e^{-\varsigma\mathsf{X}}$, where X is the sum of independent exponential variables with means $4/((2k-1)^2\pi^2)$; these means are summable, with sum $\frac12$. Comparing Taylor coefficients at $\varsigma=0$ gives $\mathbb{E}\,\mathsf{X}^k=k!\,|E_{2k}|/(2k)!$. The finite remainder $|e^{-u}-\sum_{k=0}^{K}(-u)^k/k!|\le u^{K+1}/(K+1)!$ for $u\ge0$, applied at $u=v^2\mathsf{X}$ and averaged, gives (953). Integrated against (898), it gives $\delta(t)=\sum_{k=0}^{K} d_k x^{-2k-2}+\mathsf{r}_K$ with $|\mathsf{r}_K|\le|d_{K+1}|x^{-2K-4}$, which supplies the next-term bounds above. No infinite asymptotic expansion is summed at fixed t.

Rational triangular matching determines $q_M^{(4)}$ and $q_{M+1}^{(6)}$ so that their finite differences match the truncated expansions of $\delta(t)$ and $(4t+1)\delta(t)$. The constants K_4,K_6 above give the finite approximant

$$\begin{aligned}W_{N,M}=&\sum_{n=1}^{N-1}a_n^4[8+5\pi(4n+1)a_n^2]\delta_n\\&+8a_N^4q_M^{(4)}(N)+5\pi a_N^6q_{M+1}^{(6)}(N),\\|W_c-W_{N,M}|\le&\frac{(8/9)K_4+(110/189)K_6}{(2M+7)(N-3/4)^{2M+7}}.\end{aligned}\tag{954}$$

The added terms evaluate the two rational tails completely by telescoping, since $a_{n+1}^r=a_n^r[(n+\frac12)/(n+1)]^r$. The upper boundaries vanish: $q_M^{(4)}$ is $O(t^{-1})$ and $q_{M+1}^{(6)}$ is $O(1)$, whereas the respective gamma weights a_n^4 and a_n^6 are $O(n^{-2})$ and $O(n^{-3})$. Using $a_n^4\le1/(9x^2)$, $a_n^6\le1/(27x^3)$, $\pi<22/7$ and integral comparison then gives (954). Each approximant is a rational cubic in π, but its coefficients depend on N,M and the error is not identically zero. It does not prove that the exact wheel lies in $\mathbb{Q}+\mathbb{Q}\pi+\mathbb{Q}\pi^2+\mathbb{Q}\pi^3$.

The remainder is controlled by a finite global Taylor inequality for sech, obtained from the positive product for cosh, not by assuming convergence of an infinite asymptotic expansion at fixed t. The certificate independently bounds π by Machin's formula and uses exact rational arithmetic plus outward dyadic rounding.

B.8.3 Euler-weighted alternating sums

For alternating coefficients b_n, the evaluator uses the finite weights

$$w_{N,n} = 1 - 2^{-N} \sum_{k=0}^{n} \binom{N}{k}, \qquad \widehat{E}_N = \sum_{n=0}^{N-1} (-1)^n w_{N,n} b_n. \tag{955}$$

For the seventh-sector sum, $b_n = a_n^7 F_n$, write $\widehat{M}_{7,N}$ for $\widehat{E}_N$. The lemma below gives the tail bound

$$|M_7 - \widehat{M}_{7,N}| \le 2^{100-N}, \qquad N \ge 1. \tag{956}$$

Finite arithmetic must still be enclosed; the dyadic interval code does so. This bound applies to the Euler-weighted approximant, not to a raw partial sum. The elementary remainder kernel $2^{-N}(1-x)^N/(1+x)$ and beta-moment bounds underpin this family of evaluations; Section B.8.5 applies them to further alternating series.

Lemma (Euler tail of the seventh-sector sum). For $N \ge 1$, $|M_7 - \widehat{M}_{7,N}| \le 2^{99-N}$; in particular (956) holds with one bit of slack.

Proof. Restrict x and s to the closed bidisk $|x|, |s| \le \frac{1}{64}$, put $t = \frac{1}{4} + s$, $\mathsf{a} = \frac{1}{2} + s + x$, $\mathsf{d} = \frac{1}{2} - 2x$, and define

$$\mathfrak{Z}(x,s) = \left[\frac{\Gamma(1-x)\cos\pi(s+x)}{\Gamma(\frac{1}{2}+x)\cos\pi x}\right]^7 \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a})}{\Gamma(n+\mathsf{a}+\mathsf{d})}\right]^7.$$

By gamma reflection this is the nonnegative-index half of the finite-Fourier family with which the package derives (608) from its earlier five-shift form. Its terms are $O(n^{-5/2+14\,\mathrm{Re}\,x})$, so the series converges absolutely and uniformly on the bidisk, where $\mathfrak{Z}$ is analytic. At $x = s = 0$ the n-th term is $(-1)^n a_n^7 y_n$. Its local logarithmic derivatives there follow from polygamma reflection, recurrence and half-integer values [32]:

$$\partial_s :\ y_n^{-1} - 14h_n, \qquad \partial_s^2 :\ -y_n^{-2} - 7\pi^2 + 7\delta_n, \qquad \partial_x :\ 14H_{2n}, \qquad \partial_x\partial_s :\ R_n^{(7)}, \qquad \partial_x\partial_s^2 :\ S_n^{(7)},$$

with $R_n^{(7)} = -7(\frac{\pi^2}{3} + 4H_{2n}^{(2)})$ and $S_n^{(7)} = -112[\zeta(3) - H_{2n}^{(3)}]$. With u_n, v_n of (608), the six derivatives at $(0,0)$ needed below are

$$\begin{aligned}
\mathfrak{Z} &= \textstyle\sum (-1)^n a_n^7 y_n, & \mathfrak{Z}_x &= \textstyle\sum (-1)^n a_n^7\, 14H_{2n} y_n,\\
\mathfrak{Z}_s &= \textstyle\sum (-1)^n a_n^7 u_n, & \mathfrak{Z}_{xs} &= \textstyle\sum (-1)^n a_n^7 (14H_{2n}u_n + y_n R_n^{(7)}),\\
\mathfrak{Z}_{ss} &= \textstyle\sum (-1)^n a_n^7 v_n, & \mathfrak{Z}_{xss} &= \textstyle\sum (-1)^n a_n^7 (14H_{2n}v_n + 2u_n R_n^{(7)} + y_n S_n^{(7)}).
\end{aligned}$$

The linear combination with the coefficients

Derivative	Coefficient	Derivative	Coefficient
$\mathfrak{Z}_x$	$560\pi^2$	$\mathfrak{Z}$	$5920\pi^2\log 2 - 3224\pi^3$
$\mathfrak{Z}_{xs}$	-182π	$\mathfrak{Z}_s$	$1136\pi^2 - 1924\pi\log 2$
$\mathfrak{Z}_{xss}$	77	$\mathfrak{Z}_{ss}$	$814\log 2 - 360\pi$

reproduces F_n of (608) term by term (an identity of polynomials in π, $\log 2$, y_n, h_n, δ_n, H_{2n}, $H_{2n}^{(2)}$, $H_{2n}^{(3)}$ and $\zeta(3)$), so it equals M_7. Let $\mathfrak{Z}_N$ be $\mathfrak{Z}$ with the n-sum replaced by its Euler approximant with the weights $w_{N,n}$. The weights act termwise, so the same combination of the derivatives of $\mathfrak{Z}_N$ at $(0,0)$ is $\widehat{M}_{7,N}$.

Euler's beta integral [19] gives the sevenfold beta-moment representation

$$\mathfrak{Z}(x,s)=\mathfrak{c}(x,s)\int_{[0,1]^7}\Big[\sum_{n\ge0}(-1)^n(n+t)\xi^n\Big]\prod_{i=1}^{7}u_i^{\mathsf{a}-1}(1-u_i)^{\mathsf{d}-1}\,du_i,\qquad \xi=u_1\cdots u_7,$$

with

$$\mathfrak{c}(x,s)=\left[\frac{\Gamma(1-x)\cos\pi(s+x)}{\Gamma(\frac12+x)\cos\pi x\,\Gamma(\frac12-2x)}\right]^7.$$

The alternating kernel is $\sum_{n\ge0}(-1)^n(n+t)\xi^n=t/(1+\xi)-\xi/(1+\xi)^2$, and its exact remainder after N Euler coefficients is

$$2^{-N}\left[\frac{t(1-\xi)^N}{1+\xi}-\frac{N\xi(1-\xi)^{N-1}}{1+\xi}-\frac{\xi(1-\xi)^N}{(1+\xi)^2}\right].$$

Its absolute value is at most $(|t|+2)2^{-N}$, since $N\xi(1-\xi)^{N-1}\le1$. On the bidisk $\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d}\ge\frac{15}{32}$ and $B(\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d})<16$. Elementary integral bounds give $|\Gamma(1-x)|<3$; reflection and the same integral bounds give $|1/\Gamma(\frac12+x)|<2$ and $|1/\Gamma(\frac12-2x)|<2$. Also $|\cos\pi(s+x)|<2$ and $|1/\cos\pi x|<2$. Thus

$$|\mathfrak{c}(x,s)|\,B(\operatorname{Re}\mathsf{a},\operatorname{Re}\mathsf{d})^7<(48\cdot16)^7<2^{70},$$

and $|\mathfrak{Z}-\mathfrak{Z}_N|<2^{72-N}$ uniformly on the bidisk. Cauchy's coefficient bound yields

$$|\partial_x^\mu\partial_s^\nu(\mathfrak{Z}-\mathfrak{Z}_N)(0,0)|\le\mu!\,\nu!\,64^{\mu+\nu}2^{72-N},\qquad \mu=0,1,\quad \nu=0,1,2.$$

Using $\pi<4$ and $0<\log2<1$, the sum of the Cauchy-weighted absolute coefficients of the table is less than $2^{20}+2^{22}+2^{26}+2^{19}+2^{21}+2^{25}<2^{27}$. This proves the bound 2^{99-N}. The proof differentiates a bounded analytic remainder; it makes no assumption that an infinite asymptotic expansion converges.

B.8.4 The primitive's fast recurrence

The primitive evaluator uses integer and half-integer Frobenius branches in a nine-component system with logarithmic degree at most two. Section B.9 states both exact 9×9 matrices and all seed polynomials. The recurrence is

$$\begin{aligned}[(n+\lambda)I-A_0]z_{n,j}&=C_0z_{n-1,j}+[(n-1+\lambda)I+A_0]z_{n-2,j}\\&\quad+(j+1)(z_{n-2,j+1}-z_{n,j+1}),\quad \lambda\in\{0,1/2\}.\end{aligned}\tag{957}$$

The terminal log channel is zero, and j is processed in descending order. The first components are combined with elementary logarithmic moments at $\rho=3-2\sqrt2$. The complete-interval evaluation uses an explicit last-two-state norm majorant; its numerical values are recomputed for each run. The evaluation weights and the last-two-state bound are specified in Section B.9. They are implemented in the package's interval code; no previous primitive value is used to initialize the recurrence.

B.8.5 The complete interval certificate

What is enclosed. The interval of width below 1.183×10^{-145} quoted in Section 22.9 is built by the package's interval stage for the form of the reduced expression that this stage receives,

$$\begin{aligned}2187\,C_6^{\mathrm{rep}}(1)&=E_{\mathrm{central}}+835584\Lambda_{3,2}+391680\mathsf{H}_5+13056\sqrt2(32\mathsf{G}+3\pi^2)\mathsf{H}_3\\&\quad-393\pi^3W_c+96P_a-784\pi^{3/2}\mathsf{L}_7.\end{aligned}\tag{958}$$

Here W_c is (896) and P_a is (609); $\Lambda_{3,2}=\operatorname{Re}\operatorname{Li}_{3,2}(\frac{1+\mathrm{i}}{2},1-\mathrm{i})$, $\mathsf{H}_3=\sum_{n\ge0}a_n2^{-n}(2n+1)^{-3}$ and $\mathsf{H}_5=\sum_{n\ge0}2^{-n}(n+1)^{-5}=2\operatorname{Li}_5(\frac12)$. The sum L_7 is a fixed combination of the five alternating series

$\mathsf{L}(t)$ below, and the polynomial E_{central} contains, besides elementary constants, the ten ordinary series of the table below and the five alternating series $\mathsf{Z}(t)$. The coefficients of E_{central} and the weights of L_7 are those of the package's input expression for this stage. Exact identities established in later stages of the package carry (958) into (604). Among them are closed-form evaluations of two ordinary values in E_{central}, the identity that turns the complete seventh-power contribution into $24\pi^3 M_7$, the conversion of $\Lambda_{3,2}$, H_3 and H_5 into the values D_3, D_5 of S_{pol}, and the inherited wheel normalizations $B_- + B_+ = 2\pi^4\varpi_6$, $B_- - B_+ = 4\pi^3\varpi_4$ that write the lower block through ϖ_4 and ϖ_6. These identities are not reproduced in this appendix. Reassembling the later forms from the same intervals of the unchanged constants, with M_7 enclosed directly through (956), again gives a width below 1.183×10^{-145}; the values D_3 and D_5 receive no separate interval.

Alternating series of the earlier form. For the five shifts $t \in \{\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\}$ put $\mathsf{a}_t = t + \frac{1}{4}$, $\mathsf{b}_t = t + \frac{3}{4}$, $g_t = \Gamma(\mathsf{a}_t)/\Gamma(\mathsf{b}_t)$ and

$$\mathsf{Z}(t) = \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a}_t)}{\Gamma(n+\mathsf{b}_t)}\right]^7, \qquad \mathsf{L}(t) = \sum_{n\ge 0} (-1)^n (n+t) \left[\frac{\Gamma(n+\mathsf{a}_t)}{\Gamma(n+\mathsf{b}_t)}\right]^7 H_{2n}(2\mathsf{a}_t),$$

with $H_m(a) = \sum_{j=0}^{m-1} (j+a)^{-1}$. Their finite Euler approximants $\widehat{E}_N$ use the weights $w_{N,n}$ above; the equivalent form $w_{N,n} = \sum_{k=n}^{N-1} \binom{k}{n} 2^{-k-1}$ follows from finite binomial summation, and the first form avoids a forward-difference table. Euler's beta integral [19] writes $[(\mathsf{a}_t)_n/(\mathsf{b}_t)_n]^7 = \mathbb{E}\,\mathsf{S}^n$, where S is the product of seven independent $\mathrm{Beta}(\mathsf{a}_t, \frac{1}{2})$ variables. For fixed $0 \le \xi \le 1$ the alternating generating function of $(n+t)\xi^n$ is $t/(1+\xi) - \xi/(1+\xi)^2$, and its finite Euler remainder is, with $u = (1-\xi)/2$,

$$\mathsf{r}_N(\xi) = \frac{t u^N}{1+\xi} - \frac{\xi u^N}{(1+\xi)^2} - \frac{N\xi(1-\xi)^{N-1}}{2^N(1+\xi)}, \qquad |\mathsf{r}_N(\xi)| \le (t+2)2^{-N},$$

since $N\xi(1-\xi)^{N-1} \le 1$ for $N \ge 1$, endpoints included. Integrating against the positive beta measure proves

$$|\mathsf{Z}(t) - \widehat{E}_N[\mathsf{Z}(t)]| \le (t+2) g_t^7\, 2^{-N}. \tag{959}$$

For $\mathsf{L}(t)$ deform each beta density to $\mathrm{Beta}(\mathsf{a}_t + x, \frac{1}{2} - 2x)$. Its normalized n-th moment is $(\mathsf{a}_t + x)_n/(\mathsf{b}_t - x)_n$, and for seven copies

$$\frac{d}{dx}\left[\frac{(\mathsf{a}_t + x)_n}{(\mathsf{b}_t - x)_n}\right]^7\Bigg|_{x=0} = 14\left[\frac{(\mathsf{a}_t)_n}{(\mathsf{b}_t)_n}\right]^7 H_{2n}(2\mathsf{a}_t),$$

by the duplication $H_{2n}(2a) = [H_n(a) + H_n(a+\frac{1}{2})]/2$. The logarithmic derivative of one density is $\varphi(\xi) - \mathbb{E}\,\varphi(\xi)$ with $\varphi(\xi) = \log\xi - 2\log(1-\xi)$; its mean absolute value is at most $2\,\mathbb{E}|\varphi|$. With the factor $1/14$, the seven copies give a signed measure whose total variation is at most $\mathbb{E}|\varphi| \le \psi(\mathsf{b}_t) - \psi(\mathsf{a}_t) + 2[\psi(\mathsf{b}_t) - \psi(\frac{1}{2})]$. The polygamma series and an integral comparison give $\psi_1(a) \le a^{-2} + a^{-1}$ [32]. Since $\mathsf{a}_t \ge \frac{3}{8}$, $\mathsf{b}_t \le \frac{13}{8}$ and $\psi_1(\frac{1}{2}) = \pi^2/2 < 5$,

$$\psi(\mathsf{b}_t) - \psi(\mathsf{a}_t) \le \frac{44}{9}, \qquad 2[\psi(\mathsf{b}_t) - \psi(\tfrac{1}{2})] \le \frac{45}{4}, \qquad \frac{44}{9} + \frac{45}{4} < 17.$$

Integrating r_N against this signed measure proves

$$|\mathsf{L}(t) - \widehat{E}_N[\mathsf{L}(t)]| \le 17(t+2) g_t^7\, 2^{-N}. \tag{960}$$

The beta parameters remain positive near $x = 0$ and logarithmic beta moments are integrable, which justifies the differentiation and the finite Euler operation. The derivative serves only the error bound; the evaluator contains none.

Ordinary positive series. The ten positive unit-argument series have rational term ratios $R(n) = t_{n+1}/t_n = \prod_i (n+a_i)/\prod_i (n+b_i)$, $t_0 = 1$, with all shifts positive, equal numerator and denominator degrees, and $\alpha = \sum_i b_i - \sum_i a_i > 1$; gamma-product asymptotics give $t_n = O(n^{-\alpha})$ [27, 22]. For a finite $M \geq 1$ choose $q_M(n) = \sum_{j=0}^{M} c_j n^{1-j}$ with $q_M(n) - R(n) q_M(n+1) = 1 + O(n^{-M-1})$. This is a finite triangular rational calculation: the leading multiplier for c_j is $\alpha + j - 1 > 0$. To turn it into a bound, put $y = 1/n$, $\mathcal{A}_R(y) = \prod_i (1 + a_i y)$, $\mathcal{B}_R(y) = \prod_i (1 + b_i y)$ and $\mathcal{C}_R(y) = \sum_{j=0}^{M} c_j y^j$. The exact polynomial

$$\mathcal{E}(y) = y(1+y)^{M-1}\mathcal{B}_R(y) - (1+y)^{M-1}\mathcal{B}_R(y)\mathcal{C}_R(y) + \mathcal{A}_R(y)\sum_{j=0}^{M} c_j y^j (1+y)^{M-j} \tag{961}$$

has zero coefficients through degree $M+1$, and the exact rational residual is

$$\mathsf{r}(n) = 1 - q_M(n) + R(n) q_M(n+1) = \frac{\mathcal{E}(y)}{y(1+y)^{M-1}\mathcal{B}_R(y)}.$$

Since the denominator factors other than y are at least 1 for $y \geq 0$, for $n \geq N$

$$|\mathsf{r}(n)| \leq \varepsilon_{N,M} = \sum_{k \geq M+2} |[y^k]\mathcal{E}(y)|\, N^{1-k},$$

a finite rational number, not an omitted asymptotic term. The identity $t_n R(n) = t_{n+1}$ telescopes, and $t_n q_M(n) \to 0$ because $\alpha > 1$. For $T_N = \sum_{n \geq N} t_n$ this gives $|T_N - t_N q_M(N)| \leq \varepsilon_{N,M} T_N$, so that, whenever $\varepsilon_{N,M} < 1$ and $q_M(N) > 0$ (both checked exactly),

$$\frac{t_N q_M(N)}{1 + \varepsilon_{N,M}} \leq T_N \leq \frac{t_N q_M(N)}{1 - \varepsilon_{N,M}}. \tag{962}$$

Adding the finite rational partial sum gives an exact rational interval for each series. The parameter ratios, including the factorial denominator, are

Series	Numerator shifts a_i	Denominator shifts b_i	α
${}_4F_3$ factor of B_-	$\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}$	$\frac{3}{4}, \frac{3}{4}, 1, 1$	2
${}_4F_3$ factor of B_+	$\frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$	$\frac{5}{4}, \frac{5}{4}, 1, 1$	2
${}_5F_4$ value	$\frac{1}{4}, (\frac{1}{2})^{\times 4}$	$\frac{5}{4}, 1^{\times 4}$	3
${}_6F_5$ value	$1, 1, (\frac{5}{4})^{\times 4}$	$2, (\frac{7}{4})^{\times 4}, 1$	3
Φ_-	$(\frac{1}{4})^{\times 6}$	$(\frac{3}{4})^{\times 6}$	3
Φ_0	$(\frac{1}{2})^{\times 6}$	$1^{\times 6}$	3
Φ_+	$(\frac{3}{4})^{\times 6}$	$(\frac{5}{4})^{\times 6}$	3
Y_r, $r = 1, 2, 7$, normalized	$t+1, (t+\frac{1}{4})^{\times 8}$	$t, (t+\frac{3}{4})^{\times 8}$	3

Here $t = r/8$, the Y_r are normalized by their $n = 0$ term, which is restored afterwards, $B_- = \Gamma(\frac{1}{4})^4\, {}_4F_3(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}; \frac{3}{4}, \frac{3}{4}, 1; 1)$, $B_+ = 64\pi^4 \Gamma(\frac{1}{4})^{-4}\, {}_4F_3(\frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}; \frac{5}{4}, \frac{5}{4}, 1; 1)$, and the ${}_5F_4$ and ${}_6F_5$ values are the two ordinary values of E_{central} named above. Gamma prefactors and the linear combinations built from these series are evaluated with outward-rounded interval operations.

Elementary constants, polylogarithms and the wheel. All certificate arithmetic uses dyadic endpoints, integers divided by 2^P. Integer floors and ceilings enclose every multiplication and division, square roots use integer square roots, and additions and subtractions of endpoint integers are exact; no binary floating-point special-value call enters the construction. The constant π is enclosed by Machin's arctangent identity with alternating rational tails. Logarithms use range reduction and a finite artanh series with a geometric tail; exponentials use range reduction and a factorial Taylor remainder. For a positive gamma argument a, the argument is shifted to $a + K$, the finite Stirling expansion of $\log\Gamma$ is applied, and the logarithm of the exact product

$a(a+1)\cdots(a+K-1)$ is subtracted; the real-positive remainder has the sign of, and at most the magnitude of, its first omitted Bernoulli term [27]. Exponentiating the resulting interval encloses Γ. This uses a finite remainder theorem, not convergence of Stirling's formal series. Catalan's constant, $\zeta(3)$ and $\zeta(5)$ are enclosed by Euler-transformed alternating Hausdorff-moment sums with nonnegative tails at most 2^{-N}; the zeta values include the exact Dirichlet-eta prefactor, and $\zeta(2) = \pi^2/6$. The trigonometric eighth-angle constants are algebraic expressions in $\sqrt{2}$. The half-argument series H_3 and H_5 are positive, with tails bounded by twice their first omitted term. For $\Lambda_{3,2}$, the substitution $n = m + r$ in the strict double sum and $\frac{1+\mathrm{i}}{2}(1-\mathrm{i}) = 1$ give

$$\Lambda_{3,2} = \operatorname{Re}\sum_{r\ge1} z_\Lambda^r \Lambda_r, \qquad z_\Lambda = \frac{1+\mathrm{i}}{2},$$
$$\Lambda_r = \frac{\zeta(3) - H_r^{(3)}}{r^2} + \frac{3\zeta(2) - 2H_r^{(2)}}{r^3} - \frac{3H_r}{r^4} = \sum_{m\ge1} \frac{1}{m^2(m+r)^3}.$$

Since $\Lambda_r \le \zeta(2)/(r+1)^3$, $|z_\Lambda| < \frac{3}{4}$ and $\zeta(2)/(1-|z_\Lambda|) < 8$, the tail after N terms is less than $8(\frac{3}{4})^{N+1}/(N+2)^3$; the real and imaginary parts of the powers of z_Λ are generated by rational interval recurrences. The wheel W_c is recomputed with the certificate of Section B.8.2 at $N = 256$, $M = 40$ (orders 40 and 41 for the fourth- and sixth-power corrections), using the new π interval; earlier wheel endpoints are read only for an overlap check. The primitive uses (968).

Assembly. Each run evaluates the complete polynomial and all six further weighted contributions of (958) before division by 2187; the source repair is included once, in the polynomial. The three recorded configurations are

Configuration	Arithmetic bits	Ordinary N/M	Primitive N	Full interval width
smoke	640	128/40	140	$< 4.673 \times 10^{-67}$
release	1024	256/80	240	$< 1.986 \times 10^{-136}$
refined	1280	320/96	280	$< 1.183 \times 10^{-145}$

The three intervals are nested, as checked with their exact rational endpoints. All runs use the same wheel cutoff, so its finite tail bound dominates the last run. In the refined run the weighted widths are bounded by 1.1826×10^{-145} (wheel), 6.1870×10^{-166} (polynomial), 2.1146×10^{-205} (primitive), 2.7549×10^{-244} ($\Lambda_{3,2}$), 3.7886×10^{-280} (H_3), 7.1367×10^{-284} (H_5) and 4.9568×10^{-323} (L_7). These are outward-rounded upper bounds, not differences between floating-point approximations. The direct enclosure of M_7 by (956) initializes π by Machin's formula, $\log 2$ by a rational artanh series and $\zeta(3)$ by an Euler-transformed eta series, never starts from an earlier value of M_7, and gives nested intervals of width below 1.245×10^{-60}, 6.110×10^{-151} and 3.000×10^{-241} at $N = 300$, 600 and 900.

B.8.6 Recorded validation of the retained computation

The execution records of the package contain real-function checks, interval assemblies, residual enclosures, exact kernel transformations and separate Barnes comparisons. These records were produced for the consolidation of the package's sources. They are reruns of the supplied implementations, not independent rederivations of every analytic input. The final-method profile of the package (Section 33) selects the tasks needed for the method described here.

The real check used 105 working decimal digits for its 90-digit request. The formula (2) was then freshly assembled from the rerun values and the proved conversions. Its absolute discrepancy from the refined interval midpoint was about 4.2620×10^{-93}. The baseline assembly from complex polylogarithms and series differed by about 3.9697×10^{-94}. Using the independently rerun 40-digit

seventh Barnes block gave a discrepancy in the full coefficient of about 4.1439×10^{-37}. These are floating-point assembly checks, not new interval bounds for the bivariate Barnes integrals.

The refined full interval constructor of Section B.8.5 was also rerun. It uses 1280-bit outward interval arithmetic and returns a width below 1.183×10^{-145} for the retained analytic expression in the form (958). The subsequent assembly in the real basis uses those intervals through exact identities but is not itself advertised as a new nine-function interval implementation.

Quantity	Rerun numerical value (shortened)
L_3	$-2.033576506072054600912068969700518 2\ldots$
C_{212}	$0.032383901800869146760958099191565833\ldots$
W_0	$78.570312136547117160682748886719852 44\ldots$
M_7	$-15332.554194774575276978946534093031 57\ldots$
P_a	$334926.045248051837883222001240077786 61\ldots$
$\mathcal{H}_W$	$82.198372839948704474444779671977025\ldots$
$\mathcal{H}_7$	$-12.961075412927996857090472232198888\ldots$
$\mathcal{H}_*$	$135.023486011160924990848819782168 82\ldots$

The $\mathcal{H}$ entries are obtained through the proved beta, Barnes and series equivalents; they are not three independent fully two-dimensional numerical integrations. Their exact normalization maps are $W_0 = \sqrt{89}\,\mathcal{H}_W/\pi^2$, (933) and (947).

The three residual contributions, in C_6 rather than r_6 units, are

$$\begin{aligned} -\frac{393\pi^3 W_0}{2187} &= -437.775914461168923635033332127009\ldots, \\ \frac{24\pi^3 M_7}{2187} &= -5217.069054363479839305067485280987\ldots, \\ \frac{96 P_a}{2187} &= 14701.829146690890003104395116162537\ldots. \end{aligned} \tag{963}$$

Their sum is $9046.984177866241240164294298754541\ldots$. They are three normalization-consistent contributions, not nine additional unknown constants.

B.9 Primitive recurrence data

This subsection supplies the matrices, seeds, evaluation weights and the a posteriori bound used in (957). They are the computational input recorded in the supplied recurrence and interval implementation. The global constant A still means $\pi + 4\log 2$; the matrix A_0 below is a different object.

$$A_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{964}$$

$$C_0 = \begin{pmatrix} -1 & -4 & 0 & -4 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 2 & -2 & 0 & -4 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 5 & 0 & 0 & -4 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & 0 & 1 & -4 & 0 & -2 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 4 & -2 & 0 & -2 & 0 \\ 0 & 0 & \frac{1}{4} & 0 & \frac{1}{2} & 7 & 0 & 0 & -2 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 3 & -4 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 6 & -2 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 9 \end{pmatrix}. \tag{965}$$

There are two branches $\lambda = 0, 1/2$. At each order n, the nine-component vectors $z_{n,j}^{(\lambda)}$ have logarithmic channels $j = 0, 1, 2$; set $z_{n,3}^{(\lambda)} = 0$. For $n \geq 2$, compute the channels in descending order using (957). The matrix on its left is upper triangular, so each solve is an exact back substitution.

The initial vectors are specified by finite polynomials. Let L be a formal variable, $x = L - A$, and define the three-vectors

$$\begin{aligned} \mathbf{e} &= (1, 0, 0), \\ \mathbf{a} &= (15x^2 - 12\pi x - \pi^2,\ 15x - 6\pi,\ 15), \\ \mathbf{b} &= \left(\frac{15(x^2 + 6x) - 12\pi(x + 3) - \pi^2}{8}, \right. \\ &\qquad \left. \frac{15(x^2 + 8x + 6) - 12\pi(x + 4) - \pi^2}{16}, \frac{15(x + 7) - 6\pi}{8} \right), \\ \mathbf{p} &= (\pi + 6x, 3, 0), \qquad \mathbf{q} = \left(\frac{\pi + 6x + 18}{8}, \frac{\pi + 6x + 24}{16}, \frac{3}{8} \right). \end{aligned}$$

Here $\otimes$ is the Kronecker product with the first vector's index outermost. Form the nine-component polynomials

$$\begin{aligned} U_0 &= Q\,\mathbf{a} \otimes \mathbf{e}, \\ U_1 &= Q[\mathbf{b} \otimes \mathbf{e} + \mathbf{a} \otimes (3/4, 1/8, 0)] - \frac{16\pi^4}{Q}\mathbf{e} \otimes (1, 1/2, 1/4), \\ V_0 &= 8\pi^2 \mathbf{p} \otimes (1, 1/4, 0), \\ V_1 &= 8\pi^2[\mathbf{q} \otimes (1, 1/4, 0) + \mathbf{p} \otimes (1, 1/2, 1/16)]. \end{aligned}$$

With $[L^j]$ denoting coefficient extraction from these finite polynomials, the seeds are

$$\begin{aligned} z_{0,j}^{(0)} &= [L^j]U_0, & z_{1,j}^{(0)} &= [L^j](4U_1 - 3U_0 - 2\partial_L U_0), \\ z_{0,j}^{(1/2)} &= 2[L^j]V_0, & z_{1,j}^{(1/2)} &= [L^j](8V_1 - 8V_0 - 4\partial_L V_0). \end{aligned}$$

The half-integer branch has no degree-two logarithmic channel. The displayed derivatives are derivatives of explicit finite polynomials used to initialize the algorithm, not special-function parameter derivatives.

Put $\rho = 3 - 2\sqrt{2}$, and write $u_{n,j}$ and $v_{n,j}$ for the first components of $z_{n,j}^{(0)}$ and $z_{n,j}^{(1/2)}$. Define

$$J_j(s) = \rho^s \sum_{r=0}^{j} \binom{j}{r} \frac{(-1)^r r! \log^{j-r} \rho}{s^{r+1}}. \tag{966}$$

These are the elementary moments $\int_0^\rho x^{s-1} \log^j x\, dx$. The finite primitive evaluator, retaining $n < N$, is

$$P_{a,N} = 4 \sum_{n=0}^{N-1} \left[\sum_{j=0}^{2} u_{n,j} J_j(n + 1/2) + \sum_{j=0}^{1} v_{n,j} J_j(n + 1) \right]. \tag{967}$$

The recurrence represents P_a as the convergent limit of these approximants. No numerical primitive value is used as a seed.

For completeness, the error bound can be constructed from the final two states. The matrix norms are $\|A_0\|_\infty = 4$ and $\|C_0\|_\infty = 10$. On all logarithmic channels, the operator $(Dz)_j = (j+1)z_{j+1}$ has norm 2. Rearranging the recurrence and applying a Neumann inverse bound gives

$$\|z_n^{(\lambda)}\|_\infty \le \frac{10\|z_{n-1}^{(\lambda)}\|_\infty + (n+\lambda+5)\|z_{n-2}^{(\lambda)}\|_\infty}{n+\lambda-6}.$$

Let $U_n^{(\lambda)} = \rho^n \|z_n^{(\lambda)}\|_\infty$. Since $\rho < 1/5$, the two scalar coefficients are bounded by $\mathfrak{a}_n = 2/(n-6)$ and $\mathfrak{b}_n = (n+5)/(25(n-6))$. For $n \ge 48$,

$$\frac{\mathfrak{a}_n}{1/4} + \frac{\mathfrak{b}_n}{(1/4)^2} = \frac{16n+280}{25(n-6)} \le 1.$$

With $K_\lambda = \max\{U_{N-1}^{(\lambda)}, U_{N-2}^{(\lambda)}/4\}$, induction gives $\sum_{n\ge N} U_n^{(\lambda)} \le K_\lambda/3$. Also $|\log\rho| < 2$ and $j \le 2$ imply, for $s \ge N \ge 48$,

$$\sum_{j=0}^{2} |J_j(s)|/\rho^s \le 7/N + 5/N^2 + 2/N^3 < 8/N.$$

The resulting bound is

$$|P_a - P_{a,N}| \le \frac{32}{3N}(\sqrt{\rho}\,K_0 + \rho K_{1/2}), \qquad N \ge 48. \tag{968}$$

The code uses outward upper norms of the computed states in this inequality and encloses every finite operation in (967). Thus both the arithmetic uncertainty and the remaining infinite tail are included.

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